RESEARCH REPORT

Critical densities for restricted arithmetic progressions

Results, constructions, and obstructions

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4 November 2026

This report assembles the mathematical results obtained in the investigation of prescribed Szemerédi density thresholds. It includes the principal proofs, the subsidiary constructions and exclusion results, and the supporting arguments needed to identify their precise scope. The conjecture for subsets of the positive integers remains open in this report.

Abstract

For a set D of positive integers, let $a_h(D)$ be the largest density of a set containing no h-term arithmetic progression with common difference in D, and put $b(D) = \sup_{h\geq 2} a_h(D)$. The prescribed-threshold conjecture asks whether every $\delta \in (0,1)$ occurs as $b(D)$. We prove an endpoint lemma that makes this formulation equivalent to recurrence at density δ and failure at every smaller positive density. We present fixed-length interpolation, realization of all thresholds in specified ergodic systems and on suitable countable abelian groups, thinning and transfer principles, and restrictions on several possible integer constructions. Further results concern nil-Bohr recurrence, finite-field line avoidance, symbolic constructions, spectral conditions, and translated finite sums. The report preserves distinctions between arbitrary integer systems, one fixed system, and actions of groups with torsion. It records explicit conditional dependencies and withdrawals where an earlier argument did not justify an unconditional conclusion. No intermediate value $0 < b(D) < 1$ is established for an unrestricted integer step set.

Reading conventions

The principal exposition is followed by a thematic guide and a complete technical collection. The identifiers R001–R270 identify the 270 mathematical entries from the prescribed-threshold investigation. Entries in the D-series reproduce earlier supporting arguments used by that investigation. Identifiers are stable even where entries are rearranged thematically. Internal references are clickable.

Each technical entry states its scope and status. A conditional result is asserted only under its displayed hypothesis; a construction verified in a specified action is not an assertion about all actions. An entry may contain both proved lemmas and a discussion of an unresolved extension. Withdrawn claims are marked as such and are not counted as proved results. Proofs are mathematical research arguments; this document is not a machine-checked certificate. The editorial review checked coverage, dependencies, and quantifiers, and incorporated the corrections identified in the research archive. It does not constitute external peer review.

Throughout, density on $\mathbb{N}$ means upper natural density unless another density is explicitly specified. Progressions have positive common difference; their length and their common difference may vary with the density threshold being tested. In an action of a group with torsion, the points of a progression may repeat. This distinction is essential.

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Chapter 1

The problem and the results

Let $D \subseteq \mathbb{N}$. For $0 < \delta < 1$, say that D is δ-*Szemerédi large* if every $A \subseteq \mathbb{N}$ with

$$\overline{d}(A) = \limsup_{N\to\infty} \frac{|A \cap [1, N]|}{N} \geq \delta$$

contains an arithmetic progression of every finite length whose positive common difference belongs to D. The difference is allowed to depend on the set and the requested length. The conjecture investigated here has the following strong form.

> For every $\delta \in (0, 1)$, there is one set $D_\delta \subseteq \mathbb{N}$ that is δ-Szemerédi large and is not $(\delta - \varepsilon)$-Szemerédi large for every $0 < \varepsilon < \delta$.

Thus the set is fixed while the smaller threshold varies. In particular, the conjecture implies the existence of a set that is large at some positive density but not at every positive density, the problem stated as Question 6.1 in Alweiss–Farhangi [2].

For each fixed length $h \geq 2$, define $a_h(D)$ using either integer avoidance or stationary events, as proved equivalent below. Set $b(D) = \sup_{h\geq 2} a_h(D)$. The exact endpoint lemma proves that for a nonempty D with $b(D) > 0$, no finite $a_h(D)$ equals $b(D)$. Consequently the conjecture is exactly the problem of realizing every number in $(0, 1)$ as $b(D)$. An example that realizes only a fixed $a_h(D)$ does not address the common all-length threshold.

1.1 Principal conclusions

The main exposition and technical collection give complete proofs of the following statements. The technical collection retains the further results, special cases, and limitations developed along the way.

1. **Endpoint and compactness.** The stationary extremum at each fixed length is attained, has a finite-subset approximation, and agrees with extremal upper natural density on the integers. The strict endpoint lemma establishes the equivalence just described.

2. **Thinning and limiting values.** Independent products yield $b(E \cup F) \geq b(E)b(F)$, and finite changes do not affect b. Every D with $b(D) < 1$ has countably many disjoint

subsets of the same critical value which are eventually divisible by every integer and have zero upper Banach density. Increasing critical values along a decreasing sequence of step sets have an integer realization. A diffuse finite-partition hypothesis yields an interpolation theorem; the hypothesis itself is not established for an intermediate example.

3. **Fixed-length spectra.** For every $h \geq 2$, the values of $a_h(D)$ for nonempty D are exactly $[0, 1-1/h]$. The largest value is characterized by coprimality of the normalized differences with h, and a finite relation gives a quantitative deficit from that value. An anchored version of interpolation preserves an arbitrary seed set. Further applications preserve nil-Bohr denseness, with the range and dependencies stated in R124.

4. **Exact thresholds in one system.** A single ergodic rotation of the infinite torus realizes every prescribed threshold, including recurrence at equality. A separate Gaussian construction does the same in one weakly mixing transformation (R006). The corresponding unrestricted integer critical values are one. Finite spectral-rank classes can be tested simultaneously, and the step sets can have a prescribed natural density strictly between zero and one.

5. **Exact thresholds on other groups.** Boolean constructions give all values in $(0, 1/2]$. Finite odd-order factors and horizontal extensions give every value in $(0, 1)$ on countable abelian groups. The distinct-prime construction gives a single ambient group with only finite bounded-exponent subgroups, infinite-order allowed steps, and distinct positive integer coordinates for those steps. It controls all measure-preserving actions of that group, including the density endpoint. Its projection to the integers has critical value zero.

6. **Residue classes in fixed systems.** For profinitely open step sets in a fixed ergodic integer system, the exact critical value is determined by the orders of the allowed steps in the rational Kronecker factor. Over all integer systems, these step sets have only the values zero and one.

7. **Translated finite sums.** Every translate of every tail of the Erdős–Taylor finite-sums family is strongly recurrent for pairs and has dense orbits in every minimal compact metric distal system. Further measurable conclusions are established for the specified nilsystem and rank-one classes, without an all-system multiple-recurrence conclusion.

8. **Positive density and nil-Bohr recurrence.** The previously constructed nil-Bohr-recurrent sets with $b = 1$ can be enlarged to syndetic sets with lower density as close to one as desired while retaining $b = 1$. The report includes the supporting construction and the elementary enlargement argument.

1.2 Scope of the realizations

The distinction between the following formulations is essential. For a fixed probability-preserving transformation T, write

$$b_T(D) = \sup\Big\{\mu(A) : \text{for some } h \geq 2,\ \mu\Big(\bigcap_{j<h} T^{-jd}A\Big) = 0 \text{ for every } d \in D\Big\}.$$

The unrestricted quantity is the supremum of $b_T(D)$ over all systems. An upper bound on $b_T(D)$ for one T, even when that transformation is universal for a specified class of factors, does not imply an upper bound on $b(D)$.

Setting	Realized values	Endpoint and limitation
Nonempty integer sets, fixed h	All $a_h \in [0, 1 - 1/h]$	Fixed-length extremum attained
One fixed torus rotation	All $b_T \in (0, 1)$	Endpoint recurrence; global $b = 1$
One fixed weakly mixing system	All $b_T \in (0, 1)$	Endpoint recurrence; global $b = 1$
One suitable mixed group, all actions	All critical values in $(0, 1)$	Endpoint recurrence; torsion remains essential
Integer sets, all actions and lengths	0 and 1; intermediate existence unresolved	The conjecture is open here

1.3 What the technical collection contributes

The subsidiary results serve three purposes. Transfer principles reduce integer questions to finite cyclic or squarefree models while tracking the order of the quantifiers. Recurrence theorems identify action classes in which proposed constructions cannot fail. Avoidance constructions show that a proposed family has $b = 1$, or that an approximation principle cannot yield an all-system upper bound. These negative results concern the stated construction or proof method; they do not disprove the prescribed-threshold conjecture.

The collections concerning marked prime directions, higher digits, relative eigenfields, and ordered query words retain their exact parameters. In particular, dependence on the number of registers, the rank, the nilpotency step, or the progression length is not suppressed. A bound approaching one as one of these parameters grows cannot be used as a uniform threshold below one.

Two dependencies require special care. Some interpolation statements use the additional conditional-density inequality on pronilfactors recorded among the supporting arguments. Their reliance on that input is explicit. A broader automatic-set argument used a rational-spectrum assertion for which the archive later found a counterexample. The unconditional claim is withdrawn; the independent terminal-state criterion and the separately proved relative-cube results retain their stated scopes. These qualifications are attached to the relevant entries rather than left implicit in the general discussion.

The next chapter contains the principal proofs. The following guide organizes the technical results, and the remainder of the report gives their full recorded arguments and dependency statements. A final concordance accounts for every mathematical entry in the research corpus.

Chapter 2

Principal theorems and proofs

2.1 The endpoint

For $D \subseteq \mathbb{N} = \{1, 2, \ldots\}$ and $h \geq 2$, let $a_h(D)$ be the supremum of $\mu(A)$ over invertible probability-preserving systems (X, μ, T) and measurable sets A satisfying

$$\mu\left(\bigcap_{j=0}^{h-1} T^{-jd} A\right) = 0 \qquad (d \in D). \tag{2.1}$$

Put $b(D) = \sup_{h \geq 2} a_h(D)$. A set D is δ-Szemerédi large if each $B \subseteq \mathbb{N}$ with $\limsup_{N \to \infty} |B \cap [1, N]|/N \geq \delta$ contains a progression of every length with positive common difference in D. The difference may depend on the length. This is the notion in Question 6.1 of Alweiss and Farhangi [2].

Every $a_h(D)$ is a maximum. Take the closed shift-invariant space of binary sequences containing no all-one h-term progression of an allowed difference and maximize the mass of the origin cylinder over its invariant probability measures. The name of A gives a measure on this space after discarding a countable union of null sets. Conversely, the origin cylinder in any such measure satisfies (2.1). For an integer set with no forbidden run, take a limit of its empirical shift laws over prefixes realizing its upper natural density, extending its indicator by zero to the left. In the other direction a maximizing ergodic component has generic points whose positive coordinates have density $a_h(D)$. The same compactness argument gives

$$a_h(D) = \inf_{\substack{F \subseteq D \\ F \text{ finite}}} a_h(F).$$

These assertions also hold for empty D, with $a_h(\varnothing) = 1$.

Theorem 2.1.1. *If $D \neq \varnothing$ and $b(D) > 0$, then $a_h(D) < b(D)$ for every h. Consequently for $0 < \delta < 1$, a single D is δ-Szemerédi large and fails $(\delta - \varepsilon)$-density largeness for every $0 < \varepsilon < \delta$ if and only if $b(D) = \delta$.*

Proof. Call the complement of an event satisfying (2.1) an h-hitter. Choose a k-hitter H_0 of mass $p_0 < 1$, possible since $b(D) > 0$. Let H be an h-hitter of minimum mass $p = 1 - a_h(D)$. For each fixed $d \in D$, every h-slot d-run meets H on an invariant conull set. In particular $p \geq 1/h$. Take $W \geq 2h$ such that every subset of $[1, W]$ of size at least $W/(2h)$ contains

a k-term arithmetic progression. This uses only the finite Szemerédi theorem [3]. In every W-slot d-run at least $\lfloor W/h \rfloor$ sites meet H. Some k such slots form a progression in the time index with a stride c between 1 and W.

For each $1 \le c \le W$, independently attach a c-th root of the system of H_0. This is the tower of c equally weighted levels that applies the old transformation at a wrap. Mark H_0 on every level, writing $H_{0,c}$ for its pullback to the product. At stride cd its original state advances at stride d. The k selected slots therefore meet $H_{0,c}$ on the root corresponding to their stride. All selections are on a single conull set by countability. It follows that

$$H \cap \bigcup_{c=1}^{W} H_{0,c}$$

is a W-hitter. Its mass is $p[1-(1-p_0)^W] < p$. Thus $a_W(D) > a_h(D)$. Now δ-Szemerédi largeness is equivalent to $a_h(D) < \delta$ for every h. Failure at each smaller positive density is equivalent, in conjunction with this inequality, to $\sup_h a_h(D) = \delta$. □

2.2 Two ways of realizing limiting values

Independent products give

$$b(E \cup F) \ge b(E)b(F). \tag{2.2}$$

Indeed take the product of avoiding events and use the larger of their forbidden lengths. Any finite set of steps admits a hitter of arbitrarily small mass: in a cyclic clock of prime order exceeding those steps, its zero level hits every complete orbit. Consequently adding or removing finitely many steps preserves b.

Proposition 2.2.1. *Every integer set D with $b(D) < 1$ has a subset E with the same critical value such that, for each $q \ge 1$, all but finitely many elements of E are divisible by q. One can choose countably many pairwise disjoint such subsets of the same D.*

Proof. For each q, the positive integers outside $q\mathbb{Z}$ have critical value one. Indeed, fix L and use the cyclic clock of modulus $qL!$. If $q \nmid d$, the order of d on this clock exceeds L: otherwise $qL! \mid rd$ for some $r \le L$ would imply $q \mid d$. Every orbit meets the consecutive initial segment of length $\lceil qL!/L \rceil$ within $qL!$ steps. Removing this segment and letting L increase proves the assertion. By (2.2), $b(D \cap q\mathbb{Z}) = b(D)$.

Enumerate the pairs $(j,n) \in \mathbb{N} \times \{2,3,\dots\}$. At each pair remove the finitely many steps already used. The part of $D \cap n!\mathbb{Z}$ that remains still has critical value $b(D)$, so it contains a finite set $F_{j,n}$ with $a_n(F_{j,n}) \le b(D) + 1/n$. Set $E_j = \bigcup_{n \ge 2} F_{j,n}$. For each h, $a_h(E_j) \le a_n(F_{j,n})$ whenever $n \ge h$. Letting n increase gives $b(E_j) \le b(D)$. The reverse inequality is monotonicity. For each q, only the finitely many stages with $n < q$ can contribute points outside $q\mathbb{Z}$. □

Thus these subsets have gaps tending to infinity and upper Banach density zero. For each nonzero integer c, the positive part of $E_j + c$ has critical value one: choose $q \nmid c$ and discard finitely many terms to place it outside $q\mathbb{Z}$.

Theorem 2.2.2. *Let $D_1 \supseteq D_2 \supseteq \cdots$ be integer step sets and let $b(D_i) \uparrow t$. Some integer step set has critical value t.*

Proof. For $i \geq 2$ choose a finite $F_i \subseteq D_i$ with $a_i(F_i) \leq t + 1/i$. Put $M_i = \max(F_i \cup \{1\})$. Choose integral bases $B_i \geq 2$ with $\sum_i M_i/B_i < \infty$. Write $\ell_2 = 1$, $\ell_{i+1} = B_i \ell_i$, and put $V = \bigcup_{i \geq 2} \ell_i F_i$. Any stationary h-avoider for V, viewed under T^{ℓ_i}, is an i-avoider for F_i for every $i \geq h$. Hence $b(V) \leq t$.

Fix j. Take an h-avoider A for D_j with mass a arbitrarily close to $b(D_j)$. For a nonnegative integer n let $z_i(n) \in \{0, \ldots, B_i - 1\}$ be its mixed-radix digits and set $\sigma(n) = \sum_i z_i(n)$. For $J \geq \max(j, 2)$ retain only those n whose i-th digit is less than $B_i - hM_i$ for every $i \geq J$. Take J large enough that every such cutoff is positive. Along $[0, \ell_{m+1})$ the first $m - 1$ digits are independent uniform digits and the rest vanish. The fraction removed is therefore at most $h \sum_{i \geq J} M_i/B_i$. Mark a retained n when $T^{\sigma(n)} x \in A$. For $f \in F_i$ and $i \geq J$, a retained start satisfies $\sigma(n + u\ell_i f) = \sigma(n) + uf$ for $0 \leq u < h$, by the digit bound. Almost surely there is no forbidden h-run of these steps. The expected marked fraction on each displayed prefix is at least $a(1 - h \sum_{i \geq J} M_i/B_i)$. The reverse Fatou inequality gives a sample with at least this upper density. The finitely many earlier steps cost arbitrarily little by (2.2). Let J increase, then take the supremum over the chosen avoiders and over j. This gives $b(V) \geq t$. □

Theorem 2.2.3. *Suppose $0 < b(D) < 1$, and for each $\eta > 0$ the set D has a finite partition such that $b(C) > 1 - \eta$ on every nonempty cell. Then every real $t \in [b(D), 1]$ occurs as $b(E)$ for some $E \subseteq D$.*

Proof. On subsets of D put $\phi(F) = -\log b(F)$. It is finite, increasing, and subadditive by (2.2). Choose successive refining finite partitions such that every cell at level i has $\phi \leq 2^{-i}$, using intersections for refinement. Give D the interval $[0, 1]$ and divide each interval equally among its children in any order, introducing an empty child if necessary so that every interval divides in at least two. Each $d \in D$ receives the unique label $x(d)$ belonging to the nested closed intervals of its cells. Set $E_u = \{d : x(d) \leq u\}$. For a fixed level i, any parameter interval shorter than its smallest cell interval meets at most three cell intervals at that level. Hence $\phi(E_v) - \phi(E_u) \leq 3 \cdot 2^{-i}$ for such $u < v$. This proves continuity. The elements labeled zero belong to the first cell at every depth, so $\phi(E_0) = 0$, while $E_1 = D$. Apply the intermediate value theorem. □

The hypothesis in Theorem 2.2.3 has a precise alternative. It fails if and only if some ultrafilter p containing D satisfies $\sup_{E \in p} b(E) < 1$. Indeed any finite cover by sets of $b > 1 - \eta$ can be made a partition by taking successive differences; removing elements can only increase b. If such covers fail, their complements relative to D have the finite intersection property. In the reverse direction one cell of every finite partition belongs to p. The local value $\sup_{E \in p} b(E)$ is itself an attained critical value by Theorem 2.2.2, applied to finite intersections of members with critical values approaching the supremum.

2.3 A finite projective obstruction

Over $\mathbb{F}_p^m$ every nonzero representative of a projective direction traverses every point of each line parallel to it in p steps. Thus one might try first to force complete lines at high density, then select the representative of each direction to prohibit short runs in a low-density set. There is a dimension restriction when the short runs have length two.

Proposition 2.3.1. *Suppose $A \subseteq \mathbb{F}_p^m$ has density at least $a > 0$, and $S \subseteq \mathbb{F}_p^m \setminus (A - A)$. For $0 < \lambda \leq 1$ there is a subspace H of codimension at most $2\sqrt{p/\lambda}/a^{3/2} + 1$ which is zero*

or has less than a fraction λ of its projective directions represented in S. In particular, if every projective direction is represented, $m \le 2\sqrt{p}/a^{3/2} + 1$, and a set of density at most m/p meets every complete affine line.

Proof. Restrict A to a translate of a current subspace of dimension $n > 0$ on which its relative density is $b \ge a$. Identify the translate with $V = \mathbb{F}_p^n$ and let f be the indicator of the slice. If at least a λ fraction of the directions in V are represented, then $R(d) = \mathbb{E}_{x \in V} f(x) f(x+d)$ vanishes on at least a λ/p fraction of V. This uses $(p^n - 1)/(p^n(p-1)) \ge 1/p$, and avoidance is inherited by the slice. For normalized Fourier coefficients,

$$\sum_{\xi \neq 0} |\widehat{f}(\xi)|^4 = \mathbb{E}_{d \in V} |R(d) - b^2|^2 \ge \lambda b^4 / p.$$

Since $\sum_\xi |\widehat{f}(\xi)|^2 = b$, there is a nonzero ξ with $|\widehat{f}(\xi)| \ge b^{3/2}\sqrt{\lambda/p}$. The slice densities b_t on the cosets of its kernel have average b, and $\left|\widehat{f}(\xi)\right| \le \mathbb{E}_t |b_t - b| = 2\mathbb{E}_t (b_t - b)_+$. One coset therefore gains at least $a^{3/2}\sqrt{\lambda/p}/2$ in density. Each gain costs one dimension and the density is bounded by one. This proves the first claim. If every direction is represented use $\lambda = 1$; every positive-dimensional subspace must keep contributing a gain. The union of the m coordinate hyperplanes meets every complete line and has density at most m/p. □

Remark 2.3.2. Taking $\lambda = p^{-1/3}$, the union of the kernels of a basis of the annihilator of H meets every admissible complete line whose direction lies outside H, at density cost $O_a(p^{-1/3})$. Within H the remaining directions can still be numerous. This argument requires a pair witness. It also uses the complete p-point line, so it supplies no uniform short hitter as p increases.

2.4 The spectrum at a prescribed length

At length two there is a direct example with a specified angular window and a prescribed extremal density. Fix $0 < s < 1/2$, choose an irrational α, and set

$$D_s = \{d \ge 1 : \|d\alpha\|_{\mathbb{T}} > s\}.$$

Then $a_2(D_s) = s$ while $b(D_s) = 1$. For the first lower bound take those positive integers whose rotation coordinates lie in $[0, s)$. For the upper bound, let B avoid the pairs and let K be the closure of its rotation coordinates. Then $K - K \subset [-s, s]$ in the circle. For a prime p, let J index the intervals of the regular p-mesh that meet K. Each residue in $J - J$ lies, after division by p, within $1/p$ of $[-s, s]$. Cauchy–Davenport gives $\min(p, 2|J| - 1) \le 2sp + O(1)$. The upper density of B is at most $|J|/p$ by equidistribution, and letting p increase gives the claim. For $s = 1/2$, the positive odd differences give $a_2 = 1/2$.

For completeness, fix $0 < \eta < 1$ and an open circle arc I of length η. Take M divisible by all positive integers at most $2/\eta$. Every rotation of velocity v with $\|Mv\| \ge s$ hits I from every starting point. Irrational rotations have dense orbits and the possible rational velocities have mesh less than η. Compactness in (v, x) supplies one finite hitting time. In the rotation by α/M, the complement of I thus avoids all sufficiently long progressions at the differences in D_s. The same argument with $\alpha = 1/2$ and $\|Mv\| = 1/2$ treats the odd steps.

Theorem 2.4.1. *For every $h \geq 2$ the set of values $a_h(D)$ for nonempty integer step sets D is exactly $[0, 1 - 1/h]$.*

Proof. Fix an irrational α, write U_s for the open circle arc of radius s about $1/h$, and let $D_s = \{n \in \mathbb{N} : n\alpha \in U_s\}$. We claim that $a_h(D_s)$ is continuous on $0 < s \leq 1/2$. Left continuity follows by compactness of stationary binary laws, since $D_v \uparrow D_s$ as $v \uparrow s$.

Fix $0 < s < 1/2$. Take an ergodic h-avoider of measure $a = a_h(D_s)$, with indicator f, and set $g = \mathbb{E}(f \mid Z_{h-1})$. Write $I_g(n) = \int \prod_{j<h} T^{jn} g$. The mean of $1_{U_s}(n\alpha) I_g(n)$ vanishes: the corresponding terms for f are zero, and $I_f - I_g$ tends to zero in uniform density [14, Corollary 4.6]. For any $\xi > 0$, choose $t > 0$ so $B = \{g \geq t\}$ has mass at least $a - \xi$. Its weighted correlation mean is again zero. On a sufficiently fine finite nilfactor Y, the event $A' = \{\mathbb{E}(1_B \mid Y) > 1 - 1/(2h)\}$ has mass at least $a - 2\xi$, by martingale convergence. At each step n, the B-correlation is at least half the A'-correlation, by a union bound over fibers of starting points whose h images belong to A'. Thus A' has zero weighted mean on Y.

Write S for this ergodic nilrotation. Fix $x_0 \in Y$ and take in $Y^h \times \mathbb{T}$ the closure L of the orbit

$$\big((S^{m+jn} x_0)_{0 \leq j < h}, n\alpha\big), \qquad (m, n) \in \mathbb{Z}^2.$$

It is a finite union of compact connected translated subnilmanifolds $L_i = H_i z_i$; the orbit is equidistributed for the measure Λ giving each component its Haar law and a positive weight [13]. This measure is also the time mean in n of the AP tuple measures with x Haar on Y, including the angle $n\alpha$. Indeed for continuous tests first average m at each fixed n by unique ergodicity, and then take rectangular Følner sets with first side tending sufficiently rapidly to infinity. Approximation in the one-slot Haar marginals, including the angular marginal, now gives $\Lambda((A')^h \times U_s) = 0$. Minimality puts every $((S^{jn}x)_j, n\alpha)$ in L for every fixed n. The angle derivative on each connected Lie group H_i is nonzero, since a constant angle on L_i would yield an atom in the Haar angular marginal.

For each i take a tangent vector of angle derivative one and complete it by a basis of the kernel of this derivative. Write $\sigma_{i,\rho}$ for the law of the exponential of a random vector with these coordinates independently uniform on $[-\rho, \rho]$. For sufficiently small $\rho < s$ this is absolutely continuous relative to Lie Haar measure, and the angle increment is uniform on $[-\rho, \rho]$. Retain in B_ρ the points $x \in Y$ such that for every component and slot

$$\Pr_{w \sim \sigma_{i,\rho}} (w_j x \in A') > 1 - \frac{1}{10h}.$$

By continuity of translations in $L^1(Y)$, B_ρ tends in measure to A'. If $n\alpha \in U_{s+\rho/4}$ and all the $S^{jn}x$ belong to B_ρ, their tuple with its angle is a point $z \in L_i$. The probability that wz has angle in U_s is at least $3/8$, and the probability of any missed slot is at most $1/10$. This contradicts the zero Haar measure above, since the law of wz is absolutely continuous on L_i. Thus B_ρ avoids all steps in $D_{s+\rho/4}$. Letting ξ decrease to zero proves right continuity.

As $s \downarrow 0$, the largest gap between the h points at a velocity in U_s is at most $1/h + 2hs$, by comparison with the regular h-gon. Deleting a slightly longer arc gives $a_h(D_s) \to 1 - 1/h$. At $s = 1/2$ every positive integer is allowed, so $a_h(D_{1/2}) = 0$. Continuity now gives every intermediate value. The upper endpoint is attained by the residue class $1 \bmod h$ on the cyclic clock of order h. □

In this construction the window already contains an IP set of integer steps when $s = 1/h$: successively choose small positive phases with sum less than the arc length. For $s < 1/h$ its

separation from phase zero gives $b(D_s) = 1$ by the circle-root sieve above. Thus prescribing a_h by this interpolation does not prescribe b.

Proposition 2.4.2. *For nonempty D, put $g = \gcd D$. Then $a_h(D) = 1 - 1/h$ if and only if $\gcd(d/g, h) = 1$ for every $d \in D$.*

Proof. If equality holds, the hitting event H in a maximizing process has measure $1/h$. For each $d \in D$ the number of hits in h successive slots is at least one and has expectation one. It equals one almost surely. Comparing consecutive windows gives $T^{hd}H = H$ modulo null sets. A finite integer combination of the hd equals hg, so $T^{hg}H = H$. If $v = d/g$ is not coprime to h, put $l = h/\gcd(v, h) < h$. Then ld is a multiple of hg, forcing a second hit l slots after almost every hit at this step, a contradiction. Conversely, on the cyclic clock of order hg, mark the residues $0, \dots, g-1$. For each allowed d, the h-slot orbit visits precisely the coset modulo g of its initial point, and so contains exactly one mark. □

There is a quantitative version. Suppose $ld = h\sum_i n_i d_i$ for $d, d_i \in D$, integers n_i, and $1 \le l < h$. Write $M = \sum_i |n_i|$. Then

$$1 - 1/h - a_h(D) \;\ge\; \frac{1}{h\big(h(1+M)-1\big)}.$$

Indeed let the minimum hitting mass be $p = 1/h + \eta$. At each allowed step the h-window hit count N satisfies $N \ge 1$ and $\mathbb{E}(N-1) = h\eta$. Comparing consecutive windows gives $\mu(H \triangle T^{hd_i}H) \le 2h\eta$ at each d_i. At step d the probability of hits at both offsets 0 and l is at most $h\eta$. On the other hand the relation and the triangle inequality give $\mu(H \cap T^{-ld}H) \ge p - Mh\eta$. Combining the bounds proves the estimate.

Proposition 2.4.3. *Let $E \subseteq \mathbb{N}$ be arbitrary, fix $h \ge 2$, and write $a = a_h(E)$. For every $0 \le t < (1 - 1/h)a$ there is a superset D of E with $a_h(D) = t$.*

Proof. Use the angular windows of Theorem 2.4.1. Left continuity of $a_h(E \cup D_s)$ follows by the same compactness argument. For right continuity, start with an ergodic maximizing event A for $E \cup D_s$ and keep track of its original points. The mass of A outside $\{g \ge u\}$ is at most u, by conditional expectation. Choose the finite nilfactor so its high-conditional-density event A' approximates this superlevel set in measure, by martingale convergence. The smoothed masks B_ρ of the preceding proof then tend in measure to A'. Intersect A with the pullback of B_ρ to the original system. This intersection retains the E-avoidance of A, avoids the enlarged angular window, and loses arbitrarily little mass, by taking u small, then the factor fine and ρ small. This proves right continuity. Independent products at the same length give $a_h(E \cup D_s) \ge a\, a_h(D_s)$. The latter factor tends to $1 - 1/h$ as $s \downarrow 0$. At $s = 1/2$ the cap is zero. The intermediate value theorem now proves the claim. □

2.5 All critical values on a Boolean group

For a countable abelian group use the stationary definition of a_h and b, replacing T^d by the action of d. Write L for the countable direct sum of copies of $\mathbb{F}_2$.

Theorem 2.5.1. *For every $0 < \delta \le 1/2$ there is a set $D \subset L \setminus \{0\}$ such that the maximum mass of an avoiding event is δ for every length $h \ge 2$.*

Proof. All nonzero directions have order two, so it suffices to treat pairs. For an integer $k \geq 1$ set $L_k = \mathbb{F}_2^{2k+1}$, let K_k be the radius-one Hamming ball, and put

$$V_j = \sum_{l=0}^{j} \binom{2k+1}{l}, \qquad c(k) = \frac{V_{k-1}}{V_k} = 1 - \frac{\binom{2k+1}{k}}{2^{2k}}.$$

We first show that $|B| \leq c(k)|B + K_k|$ whenever $|B + K_k| \leq V_k = |L_k|/2$. By Harper's vertex-isoperimetric theorem [4], an initial segment of the simplicial ordering minimizes the neighborhood for each size. Once its size exceeds V_{k-1}, the neighborhood contains the radius-k ball and at least one point of weight $k+1$. At size $V_{j-1} + u\binom{2k+1}{j}$, $1 \leq j \leq k-1$, $0 \leq u \leq 1$, the neighborhood has size at least $V_j + u\binom{2k+1}{j+1}$ by counting upper-shadow incidences from the chosen points in level j. The ratios V_{j+1}/V_j decrease with j by log-concavity of the binomial partial sums. Thus the claimed inequality holds at each end of this interpolation and hence throughout. The empty set and the singleton give the remaining cases directly. The ball of radius $k-1$ attains equality.

Now choose integers $k_i \geq 1$. Take a separate cube for each i and identify their all-ones vectors e_i with a common involution e by quotienting the direct sum by the span of the $e_i - e_j$. Write K_i for the corresponding unit ball and $K_I = \sum_{i \in I} K_i$ for finite I. Take $D = \bigcup_I (e + K_I - K_I)$. This excludes zero: a block difference from $K_i - K_i$ has weight at most two and cannot be e_i. On the subgroup of the first n blocks, avoidance of the differences $e + K_{[n]} - K_{[n]}$ means precisely that $A + K_{[n]}$ is disjoint from its translate by e. Add the K_i successively. Every intermediate expansion remains disjoint from its e-translate. On each coset of the next cube the expanded slice occupies at most half the cube. The inequality just proved increases its size by a factor at least $1/c(k_i)$. The final expansion occupies at most half the group, giving the density bound

$$\frac{1}{2} \prod_{i=1}^{n} c(k_i).$$

For equality take the image of the product of the radius-$(k_i - 1)$ balls. The quotient identifies only even collections of antipodal flips, so it is injective on this product. Its expansion lies in the image of the product of radius-k_i balls, disjoint from its e-translate. Counting against the group order $2^{1-n} \prod_{i=1}^{n} |L_{k_i}|$ gives the asserted density.

In any action, average the finite-subgroup orbit masks to obtain each finite upper bound. Conversely, tile every coset of the first n blocks with an independently uniformly translated copy of a maximizing mask, using any coset representatives. The resulting law on binary configurations is translation invariant, since translating permutes the cosets and only changes the uniform offsets. A weak limit as n increases avoids every specified pair and has mass $\frac{1}{2} \prod_i c(k_i)$. Finally $c(k) \to 1$ from below. For $0 < \delta < 1/2$, write $-\log(2\delta)$ as a sum of terms $-\log c(k)$, allowing repetitions: at successively smaller sizes take as many copies as fit into the remaining sum. The remainder tends to zero. If a finite choice already gives equality, extend the group by unused coordinates. For $\delta = 1/2$ take $D = \{e\}$. Every infinite group constructed here is isomorphic to L. □

Here an avoiding event actually attains the critical density, whereas Theorem 2.1.1 gives strict inequalities at fixed length on the integers. There is also a direct obstruction to transferring the Boolean example through higher binary digits. If $A \subset \mathbb{F}_2^m$ of density a

avoids the specified pairs, mark a vector of $(\mathbb{Z}/2^r\mathbb{Z})^m$ if some binary digit vector lies in A. The marked density is $1-(1-a)^r$. Along a line of any lifted step, successively keep the current time or add 2^i to force digit i outside A. This move goes to the other endpoint of its forbidden binary pair and fixes all lower digits. The final time is at most 2^r-1. Consequently the marked set avoids that common length for all lifted steps, uniformly in the number and sizes of the finite blocks.

2.6 Interpolation on a group with torsion

Theorem 2.6.1. *Write $V_* = \bigoplus_{p\,\text{prime}} \mathbb{F}_p^{(\mathbb{N})}$. For every $0<t<1$ there is $W \subset V_*$ with critical value t. All steps in this example have the same finite order, and the maximum avoiding mass at this order equals t.*

Proof. Take an odd prime p, a finite vector space K over $\mathbb{F}_p$, and a nonempty set $S \subset K\setminus\{0\}$. Let α be the maximum fraction of K occupied by a set containing no affine line parallel to any $s \in S$. These fractions are dense in $(0,1)$. Indeed, fix a target s_0 strictly between zero and one. For any sufficiently large p, choose the dimension so large that the maximum over masks avoiding lines in *all* directions is below s_0, by density Hales–Jewett [5]. Starting with no directions, impose them one at a time. Each new direction costs at most a factor $1-1/p$: delete a uniform translate of a hyperplane transversal from the old mask. Thus some α lies in $[(1-1/p)s_0, s_0]$. Choose the parameters so that $\max(0,2t-1)<\alpha<t$.

Start with the Boolean line $\{0,e\}$ and repeatedly adjoin the antipodal cubes of the preceding section. At each finite stage let L be the resulting group and $D_L = e+\sum_i (K_i - K_i)$ its Boolean directions. Impose the directions $D_L \times S$ on $L\times K$ and write b_L for the largest density avoiding their full orbits. All these orbits have length $2p$. This finite-group maximum also equals the stationary capacity, by averaging orbit masks. Write a mask as sections $B_x \subset K$, for $x \in L$. For each of the finitely many admissible affine lines $l \subset K$, the set $\{x : l \subset B_x\}$ must avoid the pairs of D_L, and these conditions suffice. Indeed, a (d,s)-orbit traverses the entire s-line on both Boolean sections, since p is odd. Initially the two sections have line-free intersection, so $b_L = (1+\alpha)/2$. For equality fill one section and use a line-free maximizer in the other.

Adjoining a cube of parameter k changes the cap from b to b' with $c(k)b \le b' \le b$. The upper bound is restriction and coset averaging. In the new cube keep the balls of radius $k-1$ at the two antipodes, jointly of density $c(k)$, and label them 0 and 1. The antipodal flip exchanges labels. Two kept vertices at Hamming distance at most two have the same label, since the balls are at distance three. Given an old mask on $L\times K$, on these two balls evaluate it at $(x+\text{label}\cdot e, y)$. This rule descends to the quotient identifying antipodes and has density $c(k)b$. Represent the common flip in the old cube factor. On a full retained orbit the new cube positions differ by a vector of weight at most two, so their labels agree and the orbit would give a forbidden old orbit.

Let N_K be the number of admissible lines in K. The pair bound in the preceding section and the union bound give

$$\alpha \le b_L \le \alpha + \tfrac{1}{2} N_K \prod_i c(k_i).$$

Indeed outside the line-containing sections each section has size at most $\alpha|K|$. Thus adjoining sufficiently many copies of any fixed size eventually brings the cap below t. At each of a

succession of sizes k tending to infinity, take the last number of copies before the cap would fall strictly below t. Stop if the cap equals t. At the end of the stage the cap is at most $t/c(k)$. The resulting caps decrease to t. If infinitely many blocks are used, restrict to finite subgroups for the upper bound and take a stationary weak limit of coset fillings by finite maximizers for the lower bound, as in the preceding section. The common order $2p$ ensures that the fixed-length maximum is attained in the limit. Finally embed $L \times K$ in V_*; subgroup restriction and coinduction (independent copies on cosets) preserve the caps. □

Proposition 2.6.2. *Suppose that for each q there is a q-large integer set E_q with $b(E_q) = 1$, where q-largeness means that every q-coloring admits monochromatic progressions of every length at its steps. Then every $0 < t < 1$ has an example on $\mathbb{Z} \times V_*$ consisting entirely of infinite-order steps and forcing every length at the endpoint t itself.*

Proof. Use $W = D_L \times S$ from Theorem 2.6.1, put $V = L \times K$ and $E = E_{|K|}$. On $\mathbb{Z} \times V$ take

$$W' = (\mathbb{N} \times W) \cup (E \times \{0\}).$$

The product of a maximizing vertical event with horizontal E-avoiders of mass approaching one gives $b(W') \geq t$. For the reverse bound let A avoid W' at length h, let T denote the horizontal motion and U the vertical action. For each $w \in W$ the intersection along the entire U_w-orbit has measure zero. If it had positive measure, ordinary multiple recurrence under T on that U_w-invariant intersection would give an A-progression along (n, w) for some $n \geq 1$. Thus A is a vertical full-orbit avoider and $\mu(A) \leq t$.

Assume equality. Let F be the set of points whose entire K-orbit misses A. If $\mu(F) > 0$, add an independent regular K-clock to the vertical action, pulling back A on every clock state, and mark one clock state over F. This raises the mean by $\mu(F)/|K|$. A full (d, s)-orbit using an added point visits, on returning to that same Boolean parity, the p different states of the clock over an originally empty K-orbit. Only one of these states is marked. Therefore the enlarged event still avoids each full orbit, contradicting its excess over t. We conclude $\mu(F) = 0$.

For almost every x and every positive integer m choose a $k_m \in K$ with $T^m U_{k_m} x \in A$. The resulting coloring has a monochromatic h-term E-progression. This contradicts pointwise horizontal avoidance, which holds at all these countably many translates on a common conull set. Compactness attains the supremum at each fixed length, so it is strictly below t. Embedding the vertical subgroup in V_* gives the proposition. □

The horizontal input to Proposition 2.6.2 is treated in the companion research proofs [6]. The distinction between the two groups is essential even though all steps of W' have infinite order. The upper bound by $\alpha + N_K a_2(D_L)$ gives $a_2(D_L) \geq (t - \alpha)/N_K > 0$. The higher binary digit construction therefore supplies arbitrarily dense avoiders for ordinary free lifts of its vertical steps. The remaining horizontal-only steps have the separate near-full E-avoiders. Intersecting the two independent events blocks the union.

2.6.1 What a finite extension can contribute

Let H be any countable abelian group, let K be finite abelian, and let $W \subset H \times K$. Write $D_s = \{d : (d, s) \in W\}$. In this statement a family containing the zero step has critical value zero, and the empty family has critical value one.

Proposition 2.6.3. *Suppose that $b_H(D_s) \in \{0,1\}$ for every $s \in K$. Let $\mathcal{B}$ consist of the subsets of K containing no full coset of $\langle s \rangle$ for any s such that $b_H(D_s) = 0$. Then*

$$b_{H \times K}(W) = \max_{B \in \mathcal{B}} \frac{|B|}{|K|}.$$

In particular this value is rational. An irrational value on a fixed finite extension of a countable torsion-free group would therefore require an interior value already on the torsion-free group.

Proof. Let A avoid W at length h. Denote the two commuting actions by T_d and U_s. For s with $b_H(D_s) = 0$, the event $C_s = \bigcap_{j < \mathrm{ord}(s)} U_s^{-j} A$ has measure zero. Indeed C_s is U_s-invariant. Positive measure would give an h-term T_d-return of C_s at some $d \in D_s$, and hence an h-term (d,s)-return of A. Thus the mask $B_x = \{k \in K : U_k x \in A\}$ satisfies all the stated exclusions almost surely. Its expected relative size is $\mu(A)$.

For the reverse inequality, take a maximizing B in the regular action on K. For each s with $b_H(D_s) = 1$, independently attach an H-system admitting an avoider of mass greater than $1 - \eta$ for D_s at some finite length. Take the product of those events with B. Its mass is at least $(|B|/|K|)(1-\eta)^{|K|}$. Each zero-valued slice is blocked by B within $|K|$ slots, and each other slice is blocked by its horizontal event. Take the maximum of the finitely many lengths and let η decrease to zero. □

2.7 Translates of the Erdős–Taylor finite sums

Let $u_1 = 1$ and $u_{i+1} = i u_i + 1$. Write S_m for the nonempty finite sums of generators u_i with $i \geq m$. Griesmer asks whether $1 + S_1$ gives measurable, topological or strong recurrence, and asks about other translates [7].

Theorem 2.7.1. *For every $e \in \mathbb{Z}$, every $m \geq 1$, every invertible probability preserving system (X, μ, T) and every measurable A of positive measure,*

$$\limsup_{\substack{d \in e + S_m \\ d \to +\infty}} \mu(A \cap T^{-d} A) > 0.$$

On every minimal compact metric distal system the $e + S_m$-orbit of every point is dense.

Proof. Partition the tail into successive pairs $(i, i+1)$. Starting at $B_0 = A$, enlarge B_j on the introduction of each pair by setting

$$B_{j+1} = \bigcup_{v \in \{0, u_i, u_{i+1}, u_i + u_{i+1}\}} T^v B_j, \qquad \eta_j = \mu(B_{j+1}) - \mu(B_j).$$

Each of $T^{u_i} B_j$ and $T^{u_{i+1}} B_j$ differs from B_j on a set of measure at most $2\eta_j$. From $1 = u_{i+1} - i u_i$ we obtain

$$\mu(T B_j \triangle B_j) \leq 2(i+1)\eta_j.$$

The sum of the η_j is finite, whereas the sum of $1/(i+1)$ over the beginnings of the pairs diverges. Thus $(i+1)\eta_j$ tends to zero along a subsequence. The sets B_j increase to B, so $TB = B$ modulo null sets. Their union consequently agrees with the invariant saturation of A. This holds for every tail.

Choose an index $s \ge m$ and apply this assertion to the tail after s. The union B includes the zero sum, and its translate by $e + u_s$ agrees with B. Since $A \subset B$, some $d_0 = e + u_s + v$, with v a possibly empty subsum of this later tail, satisfies $c = \mu(A \cap T^{-d_0}A) > 0$. Here we used the symmetry of pair correlations to select the displayed sign. Put $C = A \cap T^{-d_0}A$. Beyond any chosen index later than this finite support, take M distinct successive partial sums v_j. Cauchy–Schwarz, applied to the sum of their translates of 1_C, gives

$$\sum_{j,l \le M} \mu(T^{-v_j}C \cap T^{-v_l}C) \ge M^2 c^2.$$

For M sufficiently large, some $t = v_l - v_j$, $l > j$, gives $\mu(C \cap T^{-t}C) \ge c^2/2$. This t is a nonempty later subsum. Also $C \cap T^{-t}C \subset A \cap T^{-(d_0+t)}A$, and $d_0 + t \in e + S_m$. Taking arbitrarily late tails proves the limsup.

For the distal assertion let E be the enveloping group of the minimal system, with the topology of pointwise convergence. This is a compact right-topological group by the distal criterion of Ellis [8]. Set

$$K = \bigcap_m \overline{\{T^v : v \in S_m\}} \ \subset E.$$

It is a subsemigroup. Given $p, q \in K$, an open neighborhood U of pq and a prescribed tail, first use right continuity to choose a sum a in that tail for which $T^a q \in U$. Continuity of T^a then gives b supported after a with $T^{a+b} \in U$. The set Kp is a compact subsemigroup for each $p \in K$. By the compact semigroup idempotent lemma it contains an idempotent, which in a group must be the identity. Hence $p^{-1} \in K$.

Take a full-support ergodic invariant measure on X. Applying saturation for T^{-1} to each member of a countable open base shows that almost every starting point has dense orbit along every tail S_m. One may exclude the zero sum by freezing one generator and translating the open target. Fix one such point y. For every $x \in X$, compactness of the nested tail closures inside the closed set $\{p : py = x\}$ gives $k \in K$ with $ky = x$. Thus $Ky = X$. Since K is a group, $Kx = X$ for every x. Translation by any integer preserves the resulting density. □

In particular one may apply the distal conclusion on the minimal orbit closure of a tuple under $(T, T^2, \dots, T^{h-1})$. For an arbitrary measurable event, the AP test samples a diagonal tuple. Its law is generally singular to an invariant law of this product action. The saturation proof above uses invariance to bound the two symmetric differences and does not give the required all-length density bound in that setting.

2.8 Residue classes in a fixed ergodic system

Let $b_T(D)$ denote the supremum in the definition of $b(D)$ when the probability-preserving system is fixed. For an ergodic transformation T, write $\mathcal{Q}(T)$ for the set of orders of its finite cyclic factors, including 1.

Theorem 2.8.1. *Suppose $D \ne \varnothing$ is a union of residue classes in the positive integers. Then*

$$b_T(D) = 1 - \frac{1}{k}, \qquad k = \sup_{q \in \mathcal{Q}(T)} \min_{d \in D} \frac{q}{\gcd(q, d)}.$$

Here $1/\infty = 0$. In particular, for odd differences the answer is $1 - 1/Q$, where Q is the largest order of a 2-primary cyclic factor, with $Q = \infty$ if these orders are unbounded.

Proof. On a cyclic factor of order q, delete the first $M = \max_{d\in D} \gcd(q, d)$ consecutive residues. Every orbit of a permitted step is a residue class modulo $\gcd(q, d)$, so it meets the deleted interval within q terms. This proves the lower bound, even for arbitrary D.

Suppose $k < \infty$ and A of measure $a > 1-1/k$ avoids all permitted h-runs. The normalized correlation mean along each residue class inside D vanishes. We may replace 1_A by $g = \mathbb{E}(1_A \mid Z_{h-1}(T))$ in each such mean. This is the linear characteristic-factor theorem with fixed shifts of the inputs, using the equality of the positive-order Host–Kra factors of T and its powers [9, 10]. Approximate g in L^1 by $f = \mathbb{E}(1_A \mid Y')$ on a finite-dimensional ergodic nilfactor Y'. We will give a lower bound for the f-mean along some permitted class depending only on a, h, k. Since replacing f by g costs at most $h\|f - g\|_1$ on any class, this will give a contradiction.

Let $m \in \mathcal{Q}(T)$ be the number of connected components of Y'. Take $d_0 \in D$ such that $u = m/\gcd(m, d_0) \le k$, and put $t = \gcd(m, d_0)$, $b = d_0/t$. Choose a residue class $d_0 + \ell\mathbb{Z}$ whose positive part is in D, with $m \mid \ell$. Under T^t the components split into cycles of size u. Put $\eta = (a - 1 + 1/k)/2$. A fraction bounded below in terms of a, k of these cycles have normalized mean of f exceeding $1 - 1/k + \eta$, since $0 \le f \le 1$ and $\int f = a$.

Fix such a cycle and one of its components Y with normalized Haar measure. On it $S = T^m$ is an ergodic affine nilrotation on a connected nilmanifold. For $0 \le r < u$ write $f_r(y) = f(T^{rt}y)$. If $E_r = \{f_r \ge \epsilon\}$, then

$$\sum_{r<u}(1 - \mu_Y(E_r)) \le \frac{u(1/k - \eta)}{1 - \epsilon} \le 1 - c$$

for sufficiently small $\epsilon > 0$ and some $c > 0$, each depending only on a, k.

We use a totally ergodic affine map W with W^u factoring onto S by π. Here is the root construction. Present S on the connected simply connected group N with lattice Γ as $x\Gamma \mapsto v\phi(x)\Gamma$. The lattice automorphism ϕ induced by conjugation in the original nilpotent group is rational and unipotent, and its logarithm lowers the Lie algebra's induced central filtration. The real powers $\phi^s = \exp(s \log \phi)$ are automorphisms, since the bracket identity is polynomial in s and holds at all integers. The lattices $\phi^{j/u}\Gamma$ are commensurable by rationality in Mal'cev coordinates. Their intersection for $0 \le j < u$ gives a finite cover preserved by $\phi^{1/u}$. Take the u-th root of $(v, 1)$ in the simply connected nilpotent group $N \rtimes_{\phi^s} \mathbb{R}$. Its affine action W on the cover has W^u lifting S. This lift is ergodic by the horizontal character criterion for affine nilrotations [12]: a multiple of an obstructing character on the cover would descend to one on the original system. Connectedness gives total ergodicity.

The event $C = \bigcap_{r<u} W^r\pi^{-1}E_r$ has measure at least c. For $d = d_0 + \ell n$, let r_j represent $jb \bmod u$. If $W^{jd/t}x \in C$ for all $j < h$, projection gives

$$f_{r_j}\big(S^{(jd/t - r_j)/u}\pi x\big) \ge \epsilon \qquad (0 \le j < h).$$

These are the slot tests of the original progression on Y. For totally ergodic W, the mean of the C-correlations at velocities $b + (\ell/t)n$ equals its ordinary mean. This follows from the arithmetic-progression mean identity [11, Theorem 6.4] after shifting the slot functions by W^{jb}. The ordinary mean has the uniform positive lower bound supplied by finite Szemerédi and Varnavides averaging at h, c. Multiplication by ϵ^h, the fraction of qualifying cycles, and at least $1/k$ for the selected sheet proves the asserted uniform bound. □

Equivalently, k is the least order, possibly infinite, of an element in the closure of D on the maximal rational Kronecker factor. Take a divisibility-increasing cofinal sequence of cyclic factors. If their minimum orders are bounded, the closed sets where each projected order is at most that bound have the finite intersection property. The orders at a point of the intersection eventually stabilize by divisibility. Across all integer systems, profinitely open sets have only the two endpoint values. If 0 belongs to the profinite closure of D, successively choose elements of D divisible by every modulus protecting a previous finite sum. All the finite sums then remain in D, so finite-sums recurrence applies [15]. Otherwise a fixed congruence separates D from zero, and arbitrarily deep cyclic factors give critical value one.

2.9 All thresholds in one fixed torus rotation

Theorem 2.9.1. *Fix a dense cyclic rotation T of $\mathbb{T}^{\mathbb{N}}$ with Haar measure. For each $0 < t < 1$ there is a set $D \subset \mathbb{N}$ such that in this system every event of measure at least t has a positive return of every length at a step in D, while each smaller density threshold fails. The same D has unrestricted integer critical value one.*

Proof. On a finite abelian group K let $c(K, S)$ be the largest fraction occupied by a set containing no full orbit of any direction in a nonempty set S. Suppose $c = c(K, S) < t$, and let C count the distinct allowed affine orbits. Take a prime p coprime to $|K|$, $V = \mathbb{F}_p^m$, and a union R of punctured one-dimensional subspaces of V. Every (s, r)-orbit is the Cartesian product of the old s-orbit and the new r-line, by coprimality. Hence

$$c \le c(K \times V, S \times R) \le c + C\, c(V, R).$$

For the lower bound pull back an old mask. For the upper bound, given an avoiding mask, mark for each old orbit l the $y \in V$ whose K-section contains l. These marked y contain no full R-line. Outside the union of the C marked sets, the old section has density at most c.

Deleting one projective direction from R increases the product cap by at most $1/p$. Indeed removing from a relaxed mask all points above a fresh hyperplane transverse to that direction restores the deleted constraint. When just one direction remains, the complement of such a hyperplane gives product cap at least $1 - 1/p$.

Start with all nonzero directions in an $\mathbb{F}_{p_0}$-space of sufficiently high dimension that its full-line cap is below t, by density Hales–Jewett [5]. At each subsequent stage choose a fresh increasing prime p with $1 - 1/p > t$. When $R = V \setminus \{0\}$, choose m so large that $C\, c(V, R) < t - c$, again by density Hales–Jewett. Remove projective directions until the cap first reaches or exceeds t, and use the immediately preceding family. The new cap is at least c and lies in $[t - 1/p, t)$. We obtain finite vector spaces V_i, nonempty directions R_i excluding zero, and finite product caps increasing to t.

Write $K_\infty = \prod_i V_i$ with Haar measure and $S_\infty = \prod_i R_i$. Finite masks pull back to K_∞ and avoid every direction in S_∞ at one finite length per mask. Conversely let A avoid h-runs at all these velocities. On the first n coordinates keep the tuples whose conditional A-measure exceeds $1 - 1/h$. This set avoids all projected h-runs: lifting a projected direction to S_∞, each slot misses A with conditional probability less than $1/h$, so on a putative projected run their simultaneous intersection has positive conditional probability. Martingale convergence shows that these finite masks have measures tending to $\mu(A)$. Thus $\mu(A) \le t$.

In fact the upper bound for each fixed h is strictly smaller. Choose i with $p_i > 2h$. For an h-avoider of mass a, add a whole slice $x_i = y$ meeting its complement in mass at least $(1-a)/|V_i|$. In almost every $2h$-run of a fixed allowed direction, each half contains an original miss. Their i-coordinates differ, since the velocity there is nonzero and their index difference is less than p_i. At most one is in the added slice. The enlarged mask therefore still avoids $2h$-runs. We have proved

$$a + (1 - a)/|V_i| \leq t.$$

Embed K_∞ in $X = \mathbb{T}^{\mathbb{N}}$ by using disjoint coordinate blocks of p_ith roots of unity. The Haar law on every coset of K_∞ is a translate of its own Haar law. A countable dense set $S_* \subset S_\infty$ suffices to test avoidance, by continuity of translations in L^1. Disintegrate Haar measure over X/K_∞, using a Borel section. Applying the estimates just proved on each coset preserves the upper bounds. To lift a finite mask, partition each of its circle coordinates into p_i half-open intervals and keep the boxes indexed by the finite mask. Translation by a target permutes the boxes exactly as in the finite group. Write A_i for these lifted masks of masses approaching t, and h_i for their respective avoiding lengths.

Let α generate the given rotation, and repeat each element of S_* infinitely often in a sequence v_j. Choose strictly increasing $d_j > 2^j$ so that $d_j\alpha - v_j \longrightarrow 0$ in the product torus. We may also keep the first block of $d_j\alpha$ uniformly separated from zero, since R_0 is finite and excludes zero. An avoider for $D = \{d_j : j \geq 1\}$ avoids all $v \in S_*$ by passing to constant-target subsequences. Hence it satisfies the strict fixed-length bound.

For the converse fix i and trim a margin $\tau > 0$ from each face of its retained boxes. For all sufficiently large j, the first h_i slots at velocity $d_j\alpha$ lie within τ in every observed coordinate of the corresponding slots at velocity v_j, from the same start. An actual run in the trimmed boxes would give an ideal run in A_i, which is impossible. To handle the finitely many earlier j, delete also an arbitrarily short open strip in the first circle coordinate. Each of those finitely many steps induces an irrational rotation there, so it hits the strip from every start in bounded time by compactness. Increasing the common forbidden length now handles all steps. Let the losses tend to zero, then take the supremum over i. This proves the lower bound t. Finally the first-block separation makes D miss a Bohr neighborhood of zero. Its unrestricted critical value is one by the polychromatic coloring theorem of Alweiss and Farhangi [2, Theorem 1.8]: at each number of colors one can omit a color of small lower density while blocking all sufficiently long D-progressions. □

Proposition 2.9.2. *Fix $k \geq 1$ coordinate angles $\beta_1, \ldots, \beta_k$ of the rotation in Theorem 2.9.1. The set in that theorem may simultaneously be chosen to give all positive-measure returns in every ergodic system whose eigenfrequency group misses all nonzero multiples of at least one β_i. Its natural density may be prescribed arbitrarily in $(0, 1)$. In particular the conclusion covers every weakly mixing system and every ergodic system whose eigenfrequency group modulo torsion has rational rank less than k.*

Proof. Fix a coordinate β and a target mass $0 < c < 1$. Choose $\epsilon_j > 0$ with $\sum_j \epsilon_j < 1 - c$. Around the jth rational point take an open arc of length less than $\epsilon_j/2$. By equidistribution its visit frequency in $[1, N]$ is below ϵ_j for $N > M_j$. Shrink the arc to an open arc I_j about the same rational point, excluding $n\beta$ for $1 \leq n \leq M_j$. The visit frequency of I_j is now at most ϵ_j in *every* prefix. The closed complement of all these arcs has measure greater than c and contains no rational points. Intersect it with a fixed closed interval to obtain a closed set C of measure exactly c, by continuity of cumulative Lebesgue measure. Let C_j impose

only the first j arc exclusions inside this interval. The weights $1_{C_j}(n\beta)$ and $1_C(n\beta)$ differ on at most a fraction $\sum_{l>j}\epsilon_l$ of any prefix. In particular $F=\{n : n\beta\in C\}$ has natural density c. Inside the fixed rotation, F has cap one. Every velocity in C hits any given open circle arc from every start and the hitting time is bounded uniformly by compactness of $C\times\mathbb{T}$. Delete the pullback of an arbitrarily short such arc.

In an ergodic system let $u(n)$ be the h-term correlation of a positive-measure event. Its ordinary mean u_* exists and is positive. If every nonzero multiple of β misses the eigenfrequency group, the twisted means of u at all these frequencies vanish by the spectral refinement of Moreira and Richter [1]. Indeed their structured approximation is a nilsequence with spectrum in that group. In a joining with the circle rotation a character at an excluded frequency projects to zero on the nilsystem. The approximation errors are arbitrarily small in mean. By trigonometric approximation and equidistribution, the mean of $1_{C_j}(n\beta)u(n)$ is $|C_j|u_*$. The uniform prefix error just proved then gives

$$\lim_{N\to\infty}\frac{1}{N}\sum_{\substack{n\le N\\ n\in F}}u(n)=cu_*>0.$$

Adjoin to the original set the k auxiliary families so constructed. Their near-full avoiding masks in the fixed rotation can be intersected with its lower witnesses at arbitrarily small total cost, using the largest of their forbidden lengths. Adding steps preserves its upper bounds. To prescribe the natural density ρ, take $c=1-(1-\rho)^{1/k}$ in every coordinate. Joint equidistribution for the finite interval approximations and their summable prefix errors give natural density $1-(1-c)^k=\rho$ for the union. In Theorem 2.9.1 one can take $d_j>2^j$, so adjoining the original set does not change this density. Each auxiliary family misses a circle neighborhood of zero; intersecting these neighborhoods with the one missed by the original set shows that the combined set still has unrestricted cap one. Finally, nonzero multiples of all the β_i would supply k rationally independent classes modulo torsion. This gives the rank assertion. □

The finite rank cutoff in this argument is necessary. If the angles in a countable union of these auxiliary families have unbounded rational rank modulo torsion, that union gives multiple recurrence for *every* positive-measure event. Indeed a common h-avoider in an ergodic system would, by the uniform-density comparison and the two superlevel-set arguments in the proof of Theorem 2.4.1, give a positive event on one finite-dimensional nilfactor whose correlation mean with *each* of the auxiliary weights is zero. The upper bounds there use only that the weights lie in $[0,1]$. The eigenfrequencies of a finite-dimensional ergodic nilsystem form a group of finite rational rank, by its horizontal Kronecker factor. Some angle then has no nonzero multiple in this group, so the displayed positive weighted mean gives a contradiction.

2.10 A mixed group with only finite bounded-exponent subgroups

Theorem 2.10.1. *There are finite integers $m(p)$ for odd primes p such that on the single group*

$$G=\mathbb{Z}\times\bigoplus_{p\text{ odd}}\mathbb{F}_p^{m(p)}$$

every $0 < t < 1$ occurs as an endpoint-inclusive threshold over all probability-preserving G-actions. The allowed directions all have distinct positive integer coordinates.

Proof. Choose $m(p)$ so that avoiding every full line in $\mathbb{F}_p^{m(p)}$ permits density less than $1/p$, by density Hales–Jewett. Use the prime stages of Theorem 2.9.1 with these predetermined ranks. At each stage we can take the next prime arbitrarily large, in particular with $C/p < t - c$ in the notation of that proof. Write K_m and S_m for the first m factors and the product of their velocity sets, with cap $c_m < t$ increasing to t. Choose one $w_i \in R_i$ per factor.

After stage $m \geq 2$ set $\epsilon_m = (t - c_m)/(2|S_m|)$. Choose $M_m \geq m$ such that in any integer system an event of measure at least ϵ_m has an h-return at a step in $[1, M_m]$ for all $2 \leq h \leq m$, using finite Szemerédi and orbit averaging. For each $s \in S_m$ allocate the positive integers $n = q_s j$, $1 \leq j \leq M_m$, taking q_s larger than all previously allocated integers, including those in earlier stages. Include as steps

$$(n,\ s - n(w_1, \ldots, w_m))$$

with zero entries on the unused factors. Take all later primes larger than the largest integer allocated at this stage. The homomorphism $(n, v) \mapsto (nw_i + v_i)_i$ into $\prod_i V_i$ maps every step into $\prod_i R_i$: on the tail nw_i is a nonzero scalar multiple of w_i. Unused prime factors may be ignored. Pulling back the compact action gives the lower bounds approaching t.

For the upper bound write T for the integer generator and U for the finite-support torsion action. Fix h and $\mu(A) \geq t$, and take $m \geq h$. There is $s \in S_m$ for which $C_s = \bigcap_{l<h} U_{ls}^{-1} A$ has measure at least ϵ_m. Otherwise deleting all these start sets leaves an h-avoider of mass greater than c_m in the K_m-action. Averaging its orbit masks over uniform elements of K_m contradicts the finite cap. Apply the definition of M_m to C_s under $(TU_{-(w_1,\ldots,w_m)})^{q_s}$. A return at j gives a return of A at the allocated step with $n = q_s j$, since at slot l one may apply U_{ls} to a point in C_s. This also proves the upper bound at equality. □

The projection to the integers contains homogeneous copies of $[1, M_m]$ for arbitrarily large m. Finite Szemerédi gives critical value zero for that projection, despite its injectivity on the specified steps.

2.11 Positive lower density does not suffice for a threshold

The following observation applies to the nil-Bohr-recurrent example whose all-density avoidance property is proved in the supporting entry on the original Alweiss construction. That entry is included in the D-series, under the title "The original Alweiss example fails density largeness at every density below one."

Proposition 2.11.1. *Suppose $E \subseteq \mathbb{N}$ is nil-Bohr recurrent at every finite order and $b(E) = 1$. For every prime p, the set*

$$R = E \cup (\mathbb{N} \setminus p\mathbb{N})$$

is nil-Bohr recurrent, is syndetic, and has lower Banach density at least $1 - 1/p$. Nevertheless $b(R) = 1$, so R is not δ-Szemerédi large for any $\delta < 1$.

Proof. Recurrence is inherited from E, and the periodic subset $\mathbb{N} \setminus p\mathbb{N}$ gives the asserted lower density and bounded gaps. Fix $\delta < 1$ and choose $\eta > 0$ so that $1 - 2\eta > \delta$. Since

$b(E) = 1$, choose $A \subseteq \mathbb{N}$ with $\overline{d}(A) > 1 - \eta$ avoiding all k-term E-progressions for some $k \geq 2$. Choose m with $p^{-m} < \eta$ and let $B = A \setminus p^m\mathbb{N}$. Then

$$\overline{d}(B) \geq \overline{d}(A) - p^{-m} > 1 - 2\eta > \delta.$$

The set B avoids every progression of length $K = \max(k, p^m)$ with common difference in R. A difference in E is excluded by the definition of A. If $p \nmid d$, the first p^m terms of a d-progression cover all residues modulo p^m, so one term lies in the deleted class. Thus $b(R) = 1$. □

In particular, global positive lower density does not substitute for positive density inside the nil-Bohr neighborhoods relevant to a recurrence problem. In this construction the periodic part contributes no differences divisible by p.

Chapter 3

Guide to the technical results

Exact relative spectra and structural restrictions

The critical value of a difference set depends on the class of actions over which avoidance is tested. This distinction accounts for several of the strongest conclusions in the present collection. For a countable abelian group G, write $a_h^G(D)$ for the supremal measure of an event avoiding every h-term progression with difference in D, over all probability-preserving G-actions, and put $b_G(D) = \sup_{h\geq 2} a_h^G(D)$. If one transformation T is fixed, use $b_T(D)$ for the corresponding supremum in that probability space. The constructions below determine entire spectra for b_T and for some groups containing torsion. They do not produce a value $0 < b_{\mathbb{Z}}(D) < 1$.

Complete spectra in fixed systems There is one ergodic Haar rotation of the infinite torus in which every value $t \in (0, 1)$ occurs as $b_T(D)$, and every event of measure at least t has a return of every length along the corresponding D [*R*005]. The same transformation works for all t. This endpoint assertion is stronger than the equality of suprema: for each fixed h, the maximum possible avoiding mass is bounded strictly below t.

The construction first works with finite regular actions. For a finite abelian group K and nonzero directions S, let $c(K, S)$ be the maximum density of a subset containing no complete cyclic orbit in a direction of S. Adjoin a fresh vector space $V = \mathbb{F}_p^m$, with p coprime to the exponent of K, and let R be a union of nonzero projective directions. An orbit in direction (s, r) is the product of its old orbit and its new affine line. If C is the number of old orbits, then

$$c(K, S) \leq c(K \times V, S \times R) \leq c(K, S) + Cc(V, R).$$

Deleting a single projective direction increases the capacity by at most $1/p$: a hyperplane transverse to that direction meets every newly forbidden orbit. Density Hales–Jewett supplies sufficiently high-dimensional fresh spaces with very small full-line capacity. Successive deletion therefore produces finite projection capacities increasing to t from below.

In the compact product, conditional densities on finite coordinates give the matching upper bound. Strictness at length h comes from a fresh coordinate of prime order greater than $2h$. Adding an entire slice to an h-avoider increases its measure by at least $(1-a)/|V_i|$, where a is its original measure, yet produces a $2h$-avoider: the two necessary misses in the two consecutive h-blocks occupy different values in this coordinate. Consequently

$$a + \frac{1-a}{|V_i|} \le t.$$

Embedding the compact product in a fixed infinite torus and approximating a countable dense family of target translations by integer powers transfers the upper bound. Finite-coordinate masks, small boundary deletions, and the removal of finitely many early directions give the lower bound. The approximating times may all miss one fixed Bohr neighborhood of zero; hence their unrestricted integer critical value is one.

A separate construction realizes every threshold in one fixed weakly mixing Gaussian transformation [*R*006]. A nonatomic measure on a Cantor Kronecker set allows prescribed measurable phase multipliers to be approximated strongly by positive powers of that transformation. Compact order-two graph actions supply relative values up to one half. Their nonattainment at the limiting independence ratio gives recurrence at equality. Odd cyclic tags and products enlarge this to all of $(0, 1)$. This is a statement about measurable events in the fixed Gaussian host, not about all integer systems.

The Gaussian approximation can be strengthened substantially [*R*020]. Finite collections of times simultaneously approximate the required phases and meet prescribed nil-Bohr tests. The resulting set is dense at every integer in every finite-order nil-Bohr topology, and this remains true after deleting any finite initial segment. Its relative threshold is still the chosen value, while its unrestricted value is one. The Gaussian host may depend on the prescribed countable collection, although within that host all relative thresholds are realized. Moreover, the construction can meet every tail of each member of any prescribed countable collection of infinite subsets of $\mathbb{N}$. By choosing those subsets to be popular return-time sets, one obtains full positive multiple recurrence in every system from any prescribed countable family. Thus a countable list of systems cannot detect the unrestricted critical value of every integer difference set, even within this nil-Bohr-dense class.

For finite-dimensional tori there is an exact finite approximation [*R*036]. If $T(x) = x+\alpha$ is ergodic on $\mathbb{T}^m$, set $S = \overline{\{d\alpha : d \in D\}}$, $P = \operatorname{lcm}(1, \dots, M)$, and let c_M be the complete-orbit avoidance ratio on the P-torsion grid, using directions in S on that grid. Then

$$b_T(D) = \lim_{M\to\infty} c_M, \qquad 0 \le c_M - b_T(D) \le \frac{m}{M}.$$

The upper bound is orbit disintegration. For the lower bound, thicken a finite extremizer to cubes and trim their faces. Velocities close to a tested grid direction are then blocked. Every remaining velocity has a coordinate of order greater than M, and a coordinate strip of width just greater than $1/M$ meets all its orbits in a uniformly bounded time. On a circle the formula reduces to $1 - 1/q$, where q is the smallest torsion order present in S. For an infinite torus one first takes the finite-grid limit in each finite factor and then the supremum over factors.

Exact spectra on groups containing torsion On one countable Boolean group, every value in $(0, 1/2]$ is an attained maximum for pair avoidance [*R*021]. Since nonzero directions are involutions, pair avoidance is also avoidance at every longer length. For blocks $\mathbb{F}_2^{2k_i+1}$ whose antipodes are identified, the exact capacity is

$$\frac{1}{2}\prod_i c(k_i), \qquad c(k) = 1 - \frac{\binom{2k+1}{k}}{2^{2k}}.$$

Harper's vertex-isoperimetric inequality gives the upper bound, and products of Hamming balls give equality. Since $c(k)$ tends to one, its infinite products realize every required value. The endpoint is attained by an avoiding event, so this Boolean theorem has different endpoint behavior from the integer conjecture.

A finite cyclic coordinate has an exact affine effect [R040]:

$$b_{mathbbZ\times C_r}(\{(d,1) : d \in D\}) = 1 - \frac{1 - b_{mathbbZ}(D)}{r}.$$

Intersecting the r translates of an avoiding event proves the upper bound. Filling all but one cyclic sheet and putting an old avoider on the last sheet proves the reverse inequality. Replacing the cyclic coordinate by unrestricted integer lifts instead permits further roots and gives critical value one.

Combining Boolean interpolation with an odd finite group gives every threshold on one mixed countable abelian group [R004]. A horizontal extension makes recurrence hold at the endpoint and ensures that all allowed directions have infinite order. This construction uses the earlier finite-color hierarchy. The separate distinct-prime mixed-group construction elsewhere in the report provides a stronger version with less reliance on that input. Neither removes the torsion from the ambient group.

Structural restrictions on possible integer examples Several results show that familiar ways of producing a positive avoiding event force the all-length value all the way to one. Uniformly field-definable fresh-prime direction sets have only the values zero and one [R003]. Bounded formula complexity gives a bounded-size/dense-fiber dichotomy. Dense fibers have vanishing higher Gowers uniformity error and force every fixed length at positive density; bounded fibers admit a character separating all late directions from zero.

Bounded sums of signed powers also have value one [R025]. A digit-square marker handles separated exponents, and clustering reduces the number of terms. Consequently automatic sets with polynomial growth in word length have value one. A different automatic-set theorem gives value zero under a terminal-state hypothesis [R015]: for every modulus there must be an accepted positive multiple whose digit-reading path reaches a terminal component. A padded strongly connected automaton therefore has an endpoint dichotomy, and an automatic set meeting every modulus at positive upper Banach density is fully recurrent. The proof uses finite transformation monoids and ordered query identities; it does not use the disputed rational-cube lemma discussed elsewhere in the report. No dichotomy is claimed for arbitrary automatic languages with transient accepting behavior.

Uniformly bounded-period polynomial trajectory obstructions amplify to value one on torsion-free hosts [R027]. After deleting thin bands, legal binary observation sequences form a compact subshift of periodic sequences and have a common period. Successive root strides synchronize the obstructing observations. The critical near-antipodal Hamming regime also admits arbitrarily dense avoiders, even when the angular tolerance is $O(n^{-1/2})$ and $O(\sqrt{n})$ coordinates are exceptional [R037]. These bounds are uniform in dimension, but require the roots available on the integers or on sufficiently large prime cycles.

Further exclusions are quantitative. A finite paired packet loses at most its total vertical reciprocal-prime mass in the construction of a near-full avoider [$R013$]. Tightness of marked mass on boundedly many physical coordinates forces endpoint one [$R060$]. Boundedly many true marks in the deep quadratic model suffice for the same conclusion [$R043$], as does summability of the reciprocal primes without any bound on mark count [$R055$]. In the dyadic constructions, either of two summable seam costs, or their coordinatewise minimum, gives endpoint one [$R071, R080, R083, R084$]. These results explain why surviving candidates require sufficiently many effective coordinates at relevant scales.

There are also necessary conditions that do not settle a candidate. An intermediate translated-box example with a positive pair obstruction must carry a nonatomic measure on its space of Bohr-cancelling paths, dominated on every cylinder by its local logarithmic cost [$R019$]. Finite L^1 bracketing of an ergodic avoiding orbit yields partitions into pieces of capacity arbitrarily close to one [$R029$]. Its further interpolation consequence invokes the separate interpolation theorem. High-probability Boolean gates below the individual row extremum are schedulable at arbitrary strides [$R022$], and bounded-width forbidden certificates have a dimension-free covering estimate [$R024$]. Boundedly many word maps cannot create a genuinely high-density-only forcing mechanism when every word line contains a sufficiently long allowed progression [$R028$]. Daisy and independent-column rank constructions have additional restrictions [$R041$].

Positive recurrence in specified action classes The paired-coordinate and marked-coordinate constructions have positive short obstructions, but many natural systems cannot witness their failure. Independent finite block cocycles admit an exact matching criterion at each fixed length [$R017$]. It programs the subgroup generated by profiles of divergent fractional matching mass and removes all other profiles with summable cost. Uniformly bounded interval codings then give full odd-support recurrence under divergent reciprocal mass [$R023$]. Arbitrary profinite abelian cocycles over the paired Haar base have full recurrence for the free-horizontal single-mark family [$R011$], while arbitrary compact abelian Lie cocycles have a system-dependent threshold below one [$R012$].

The asymmetric family has a common horizontal characteristic factor [$R014$]. Its unconditional triple avoidance cap is at most one third; a general fixed-h bound tends to one as h grows. Cyclically symmetric ergodic nilsystem joinings yield a stronger all-positive result when the horizontal action is ergodic [$R039$]. Such a joining is a finite cover whose degree has no prime factors outside the cyclic symmetry order. The accompanying audit records why cyclic symmetry of an average does not suffice and supplies countermodels to overly broad component extensions [$R038$]. Bipartite prime graphs have a complementary recurrence theorem when one side of the tested orbit is compact [$R016$].

Finite-rank affine actions on compact abelian groups with Haar measure have positive polynomial-Bohr sets of multiple returns [$R002$]. For automorphism actions a finite-index subgroup suffices; for one acting generator the polynomial degrees can be bounded by $h-1$. Bounded-exponent translations on a connected compact nilspace are compact on each fixed observation [$R026$]. Additive ultrafilters of finite semigroup order supply pair recurrence, a fixed-length mass estimate, and full recurrence on distal and Gaussian classes [$R001$]. These hypotheses remain visible in each theorem: none is an unqualified structure theorem for arbitrary stationary processes.

Support-regular nilorbit-hit sets form another endpoint class [$R059$]. If the sampled in-

dicators admit the stated simultaneous Haar and Cesaro approximation and every actual hit lies in the appropriate essential support, a positive avoidance capacity implies value one. Consequently nil-Bohr recurrence gives full multiple recurrence inside this class. Positive global density alone is not the support hypothesis used here.

Deep quadratic and higher-charge constructions The deep quadratic family on primary cyclic groups and its rational saturation have a fixed nine-slot color obstruction and pass all ordinary circle-valued group-polynomial zero tests [$R049$]. Every positive event has a three-term return, including after the length-preserving transfer to integer differences [$R057, R058$]. The ordinary-polynomial assertion should not be read as the generalized-polynomial assertion; additional nilrotation tests are separate results. A hypothetical avoiding rational action must have large primary orbit depths on the base of its horizontal ergodic components [$R044$].

The explicit affine Haar witness has all-length relative capacity at most one half [$R042$]. It extends to the full rational group on a compact torsor of dimension three [$R046$]. Every affine Haar action of dimension at most two has full recurrence along the same directions [$R050$]. Higher charge r sharpens both statements: affine Haar recurrence holds through length $2r + 1$, a positive obstruction occurs at $2r + 2$, dimension at most r gives recurrence at every horizon, and dimension $r + 1$ can obstruct it [$R047, R053$].

Further positive classes include countably many prescribed nonlinear horizontal symbols, uniform phase-integral correlations, and the indicated scalar or vector array models [$R045, R048, R052$]. Gaussian and Abramov cases occur among these classes. Rooted spatial models have upper bounds when their strides omit a suitable prime [$R056$], but their Fourier laws rule out one bounded-state proof uniformly over all root orders [$R051$]. Per-direction soft blockers give quantitative errors tending to zero on late tails [$R054$]. Such errors rule out a uniform supersaturation argument; they do not create exact simultaneous avoiding events.

Dyadic word methods and their limits The dyadic family separates short-order forcing from the all-length problem. Ordered query identities give four-slot and then five-slot recurrence on sufficiently replicated true integer lifts [$R078, R077$]. A mixed-radix realization preserves a positive six-slot obstruction, producing an integer set that forces five-term progressions at every positive density but fails six-term recurrence [$R081$]. This is a fixed-length separation, consistent with the established hierarchy of recurrence orders.

Ordered words also give polynomial nilorbit recurrence, compact finite-rank nilspace morphism recurrence, and paired representatives of finite IP systems modulo passive components [$R087, R069, R090$]. Tensor tag cancellation is uniform in the number of input variables for each fixed nilmanifold target [$R088$]. Eberlein functions and ordinary polynomial phases can be synchronized through a common idempotent evaluation [$R072$]. The relevant free-query kernels have nilpotent residuals [$R089$]; this is not a nilpotent reduction of the full ambient product of free groups.

The explicit dyadic Haar witness has a half upper bound [$R061$], as do arbitrary modules over the stated native rings [$R085$]. The conclusion extends to arbitrary invariant measures in the single-axis category and to the specified bounded-interaction categories [$R063, R066, R082$]. It can be imposed simultaneously with fixed nilspace tests [$R062$]. Arbitrary compact abelian cocycles over the dyadic product base have bounds $1 - m^{-d}$ at

bounded extra dyadic depth, with the stated extensions to disconnected fibers and solenoids [*R*064]. Wider native rings exhibit a half bound without any bounded-state supported-trajectory argument [*R*070, *R*086]. Carry cocycles supply exact relative thresholds and show why ergodicity of a parity kernel alone does not prevent a short obstruction [*R*067].

Several sufficient dynamical hypotheses give full recurrence: a compact high-subgroup action, finite binary spectral detection on the relevant components, an ergodic finite-rank passive subgroup, or the common scalar invariant algebra specified in the corresponding theorem [*R*079, *R*075, *R*074, *R*068]. Finite-period high derivatives give a weaker all-order bound depending on that period [*R*076]. Energy plateaus control invariant projections and frozen complements but not arbitrary return products chosen after the averaging envelope [*R*073]. The bipartite support-agreement lemma localizes coefficient labels without excluding all binary-tag behavior [*R*065]. These restrictions preserve the distinction between a recurrence theorem for a stated action class and the unrestricted integer threshold sought in the conjecture.

Additional circuit and angular families A finite-field circuit produces a locally quadratic charge with no bounded-degree global polynomial representation [*R*033]. Its own uniformly quadratic trajectories still imply endpoint one. Shearing it by an unconstrained clock retains a positive four-slot obstruction [*R*034], while its actual three-channel Haar model has an all-length upper bound of one sixth [*R*035]. A quadratic criterion gives recurrence in the stated primary Conze–Lesigne category [*R*032]. Charged tensor products have a uniform finite-cover estimate when the complete local velocity family is available [*R*031], and ordered charge identities force prescribed signed integer totals after replication [*R*030]. Neither result licenses a passage to exactly one unsigned exceptional coordinate.

The angular-clock family retains a six-slot obstruction and imposes exact spectral resonance conditions [*R*008]. Binary saturation bounds depend on the rank of the observed low modulation, and can be sharpened under the stated primitive-tag hypotheses [*R*007]. Four-slot relative eigenfields can nevertheless have large matched angular atoms [*R*010]. Finally, the k-knob angular-cube family forces all positive-density returns through length $k+1$ in arbitrary actions while admitting a later short obstruction [*R*009]. Across these families, the unproved requirement remains a single upper density threshold below one valid at every progression length and over every integer action.

Fixed-order interpolation, spectral restrictions, and structured actions

Throughout this chapter, $a_{h,G}(D)$ denotes the supremum of the measures of events avoiding every h-term progression with difference in D, where the supremum is over probability-preserving actions of the indicated countable abelian group G. We write $b_G(D) = \sup_{h\geq 2} a_{h,G}(D)$. A capacity computed in one specified action will always be called an action-relative capacity. Passing from an action-relative statement to the supremum over all actions is a separate problem.

Integer interpolation at one fixed length For every fixed $h \geq 2$, the complete integer fixed-length spectrum is

$$\{a_{h,\mathbb{Z}}(D) : \varnothing \neq D \subseteq \mathbb{N}\} = [0, 1 - 1/h].$$

The upper bound is forced by any single permitted difference. To interpolate, fix an irrational α, let U_s be the circle arc of radius s about $1/h$, and set $D_s = \{n \geq 1 : n\alpha \in U_s\}$. The function $s \mapsto a_h(D_s)$ is continuous. Its limit at zero is $1 - 1/h$, whereas it vanishes once the angular interval contains a suitable IP-set. The nontrivial proof of right continuity replaces an avoiding event by a positive superlevel on a finite nilfactor, preserving a zero weighted correlation mean. On the finite-dimensional progression orbit closure, absolutely continuous small motions can move an angle into a smaller arc while preserving membership in a high-probability core of the event. Compactness gives the other one-sided continuity. These operations concern weighted means and do not require a pointwise assertion at exceptional differences ($R123$).

The interpolation can preserve a fixed direction family. Combining it with a nil-Bohr-dense Gaussian construction gives, for every

$$0 < t < (1 - 1/h)^2,$$

an integer D with $a_h(D) = t$, with every tail nil-Bohr dense at every integer, and with $b(D) = 1$. The square in this range is a limitation of the particular construction: if E is nil-Bohr recurrent, then

$$a_h(E \cup D_s) \leq (1 - 1/h)a_h(D_s).$$

The proof bounds conditional density on finite nilfactors by $1 - 1/h$ and applies the same superlevel and angular-smoothing argument ($R124$). Thus the fixed-length theorem supplies no intermediate value of the all-length capacity.

For odd integer differences, the periodic reduction also gives

$$a_h(2\mathbb{N} - 1) \leq 1 - \frac{1}{\{}h - 1\}$$

when $h \geq 3$ is not a power of two. In particular the exact values at $h = 2^s$ and $h = 2^s + 1$ are $1 - 2^{-s}$. The proof averages odd slopes in a dyadic cyclic group and locates a starting residue whose expected number of holes is less than one. No formula for every intervening length is asserted ($R165$).

Attained capacities and quantitative recurrence obstructions Several constructions preserve a capacity that has already been attained. Increasing limits of capacities along decreasing direction sets can be realized by a single integer set, using finite tests on separated mixed-radix coordinates. For an ultrafilter p, the local value

$$s(p) = \sup_{E \in p} b(E)$$

is consequently attained by an integer direction set. Independent thinning gives a deterministic capacity function almost surely. If its limit as the retention probability tends to zero is one, a finite-partition interpolation argument realizes every capacity between the original value and one ($R145$). These facts do not establish that any intermediate value exists.

There is also a multiplicative closure statement based on a substantial additional input: the conditional-density inequality

$$\|\mathbb{E}(1_A \mid Z_s(T))\|_\infty \leq b(D)$$

for a D-avoiding event. Subject to that inequality and its nilmanifold proof, the lexicographic family

$$(D_1 \times \mathbb{Z}) \cup (\{0\} \times D_2)$$

has capacity $b(D_1)b(D_2)$, and an integer realization is available. The same input gives $a_h(D) \le (1 - 1/h)b(D)$ when D itself has positive upper Banach density. These conclusions must retain their full dependency on the conditional-density inequality; the published Host–Kra characteristic-factor theorem alone is not that inequality ($R145$).

Every attained value also has a realization of the form $E + \mathrm{FS}(q_i)$, with the same capacity as E, and with capacity one for every positive translate. This uses a rigid-system realization with eventually divisible directions, summably accurate rigidity at the selected finite-sums times, and periodic repair of the finitely many initial offsets ($R146$).

Quantitative finite-color witnesses provide a different route. If $N_{r,h}(D)$ is the least size of a finite integer set forcing a monochromatic h-term D-progression in every r-coloring, then

$$N_{r,h}(D)(1 - a_h(D))^r \ge 1.$$

Indeed, r independent avoiding masks would otherwise cover that finite set with positive probability. A common exponential growth bound for these witnesses, uniform in the progression length, would give an all-length capacity below one. No such bound is obtained for the main one-mark digital family. A further endpoint-one criterion is available when every minimal subsystem of the stride-name hull has period dividing one fixed integer ($R150, R173$).

Multiplicative products of ultrafilters Put $u_h(p) = \sup_{E \in p} a_h(E)$, and use the convention that $E \in qp$ precisely when $\{e : E/e \in p\} \in q$. Then

$$u_h(qp) \le \min\{u_h(q), u_h(p)\}, \qquad s(qp) \le \min\{s(q), s(p)\}.$$

More strongly, for $s(q) \le 1 - \gamma$, a finite Szemerédi number $W = W(k, \gamma)$ gives

$$1 - s(qp) \ge \frac{1 - s(p)}{1 - u_k(p)^W}.$$

The proof selects a finite forcing family, finds a short progression inside a long hitting pattern, and intersects the original hitter with a union of independent root-tower hitters. The shrinkage factor is independent of the original forbidden length. Multiplicatively regular ultrafilters therefore have local value zero or one. Products of increasingly many factors whose capacities stay uniformly below one have fixed-order capacities tending to zero ($R130$).

If D is nil-Bohr recurrent and F has positive upper Banach density, then the multiplicative product DF is Szemerédi-large. In fact, for each fixed positive-measure event and progression length, some $d \in D$ has a positive-upper-Banach-density set of successful multipliers in F. The proof shows that the ultrafilters containing every finite-order nil-Bohr zero neighborhood form a two-sided multiplicative ideal, and combines this fact with the nilsequence-plus-null description of multiple-return sets. There are related statements for positive index density along polynomial values and polynomial values at primes. The order of ultrafilter multiplication remains essential: the archive also constructs factors whose two multiplication orders have opposite local endpoint values ($R130$).

Finite factors and exact relative thresholds For $\mathbb{Z} \times K$, with K finite abelian, full-orbit exclusions in K determine exact capacities after suitable horizontal directions are added. A finite-color recurrent horizontal family of capacity one excludes equality at the finite maximum, while product masks approximate that maximum from below. This yields recurrence at an exact endpoint, not merely a supremal avoidance density. Varying finite-field direction families gives a dense set of such exact values across these mixed groups. It does not yield one integer realization of every real threshold ($R121$).

For a general family $W \subset H \times K$, if every horizontal slice has capacity zero or one, the exact answer is a finite optimization: maximize the density of a subset of K containing no full coset in the directions whose slices have capacity zero. Intermediate horizontal slice values yield explicit linear constraints on random finite masks, but exactness is established only for the zero-one slice case ($R122$).

Inside one ergodic integer system, the odd-difference capacity is exactly $1 - 1/Q$, where Q is the largest dyadic cyclic-factor order, with $Q = \infty$ allowed. Finite-rank $\mathbb{Z}^m$-actions admit a joint version: high-mass avoidance of a profinitely open family forces a finite rational factor in which every allowed direction has large order. For parity cosets this determines the exact relative capacity through the divisibility height of the parity character in the joint point spectrum ($R147, R148, R170$).

Infinite rank behaves differently. A weakly mixing action can have parity-coset capacity one without any nonconstant joint eigenfunctions. A stronger construction avoids full-group root spectral mass even after testing dilations, despite arbitrarily high avoiding masses. The parity character still occurs on a suitable progression joining. Thus original-system spectrum, subaction spectrum, and joining spectrum must remain distinct. At three slots the stated weak-mixing parity problem has an exact zero-or-half classification ($R142, R143$).

One fixed torus rotation can be tested together with uncountably many other systems: add all ergodic transformations missing every nonzero multiple of at least one of finitely many prescribed eigenfrequencies. Every threshold occurs over this class; every positive event in the added systems returns, and the failures below the chosen threshold occur in the fixed rotation. The direction set can have any prescribed natural density strictly between zero and one, while its unrestricted integer capacity remains one. Adding infinitely many positive auxiliary windows with unbounded rational eigenfrequency rank instead gives full recurrence in every system ($R151$).

Gaussian positivity and root amplification Gaussian actions permit a useful analysis of limiting progression joinings. Full-rank covariance fibers have positive rectangle probabilities; nullity can consequently be reduced to finite atomic spectral fibers missing the identity phase. Deterministic measurable selectors preserve the relevant nullity. For a power-closed Gaussian action, this leads to an exact parity-height formula in terms of atoms of spectral images under all dilations. Its finite relative capacities may be attained at finite forbidden length, so they should not be confused with the strict integer endpoint phenomenon ($R131, R133$).

Several Gaussian approaches to an intermediate integer capacity instead amplify to one. Summably accurate finite-order displacement programming has this consequence if the fixed-space projections have a uniform variance gap and bounded expected Gaussian supremum. Related angular constructions remain at endpoint one under finitely many nonzero centers, square-root Hamming errors, moving centers, and the stated controlled zero-mode exten-

sions. Character roots and uniform control of the entire allowed family are essential in these arguments ($R115, R117, R132$).

Distinct parity corrections cannot destroy every uniformly positive baseline return when the correlation function is Eberlein, in particular for Gaussian systems. A single marked integral combination of normal moves has an additive ultrafilter image agreeing with a genuine idempotent on all Eberlein functions. These conclusions rule out Gaussian obstructions for several one-mark families, without asserting that arbitrary weak limits of Gaussian event models remain Gaussian ($R134, R174$).

For odd prime sums, disjoint-block Gaussian obstructions have capacity one under an explicit relation between the harmonic mass and the square-sum mass of the block. For unrestricted mixed supports, adaptive programming gives dense cylinder trajectories. Uniform marginal-density estimates strengthen this to a necessary Cameron–Martin affine-pencil nullity condition for measurable avoiders. Neither cylinder density alone nor this necessary condition supplies the missing unrestricted all-length conclusion ($R155, R162, R166$).

Dyadic and mixed-clock direction families The two-odd-center dyadic construction has a genuine six-term positive-density obstruction. Its normal residues are zero and one modulo a low dyadic modulus, and an active coordinate carries an additional high parity mark. Replicating fixed parameter levels and increasing the separation of high and low depths prevents several elementary endpoint-one certificates, but the all-system capacity of the free lift remains unknown ($R107$).

Positive results cover large structured classes. All lengths recur on global phase-polynomial factors, on the stated full-derivative relative extensions, and when the passive action is totally ergodic. Integral nilpotent word identities give simultaneous polynomial recurrence with a uniform coefficient-independent replication number. Native ring programming extends to unbounded interaction arity, and a representation-free version isolates the precise binary high-derivative conditions required. Profinite cocycle extensions of the independent product base also recur ($R091, R093 - -R096, R101 - -R102, R106$).

At fixed levels, ordered trades give native recurrence through five slots and integer recurrence through three slots. The separate four-slot scalar obstruction calculation was subsequently strengthened by the ordered-four-slot result elsewhere in the report. Singularity of the event's high-bit spectral measure forces four- and five-slot returns. Algebraic incidence modules bound the native high-bit rank, but a finite elementary abelian obstruction group does not imply a two-state trajectory ($R103 - -R105$).

For pure high marks, relative increments of even exponent q give the bound $1-1/(1+q/2)$; odd exponent gives full recurrence. Weak mixing of the coarse Boolean component action yields another full-recurrence class for relative eigenfunctions. Odd-block symmetrization produces spectral involutions but does not control arbitrary temporal joinings ($R097 - -R099, R163$).

Mixed-primary clocks replace the native dyadic low coordinates by distinct odd-prime cycles. They retain a short obstruction while forcing a divergent total cost for certain local polynomial repairs. Scalar and two-channel affine registers have full recurrence; three channels admit a half-density bound. Relative registers satisfy the same conclusions under bounded dependency or a suitable independent subsequence. More generally, full-rank circle noise gives a dimension-only sufficient bound $1-r^{-r}$, independent of the number of temporal lags. None of these bounds is uniform over an unbounded number of observable channels

$(R120, R152 - -R154)$.

Algebraic exclusions and combinatorial barriers Unrestricted prime digits accepted by one finite group product have only endpoint capacities after rational saturation; the acceptance group may be nonabelian. Fixed additive finite-palette colorings cannot obstruct directions using arbitrarily many nonconstant sufficiently large prime axes. Scalar characteristic-p filters admit bounded-degree interpolation in the stated one-valuation class, even at unbounded primary rank, while bounded-rank elementary abelian signatures satisfy a separate polynomial classification. These results do not cover arbitrary multivariate Artin rings or bounded low-residue strips with unrestricted deeper digits $(R113 - -R114, R157, R160)$.

Several general proof strategies are excluded. A uniformly bounded list of choices at each word vertex, realizing every combinatorial line, forces full positive-density recurrence. A fixed finite-offset polynomial-IP argument in a torsion-free group does the same under its recurrence hypothesis. Hamming expansion alone does not provide compatible lifts along all points of a prime-field affine line. A one-affine-constraint correlation with freely averaged lower coefficients has a uniform finite-character positivity neighborhood, forcing endpoint-one blockers in the associated polynomial-gate construction $(R116, R118, R135, R172)$.

The source also records rank loss for parity entropy certificates, random-coordinate freezing estimates, endpoint-slope discrepancy bounds, arbitrary-forcing counterexamples, and exact-repair criteria for Euclidean approximate masks. Each result distinguishes uniform simultaneous avoidance from control of one direction or one average. Prime-template computations are finite exhaustive evidence and do not verify an asymptotic uniform template theorem $(R092, R100, R108, R125, R127 - -R128, R159)$.

Odd primes, marked words, and translated finite sums Signed odd supports satisfy an explicit pair-avoidance bound $(\sum_i 1/p_i)^{-1}$. Divergence therefore gives pair recurrence, and the corresponding tail Bohr argument gives all finite niltranslation tests. Multiple recurrence is proved on phase-polynomial systems, class-two circle and torus bundles, and Conze–Lesigne systems. Pro-odd compact towers and inverse limits preserve the relevant recurrence. Paired odd-prime single-mark families likewise recur on phase-polynomial systems, although general actions have short obstructions $(R156, R161 - -R162, R164, R167 - -R169, R171, R179 - -R180)$.

The language of base-b integers with exactly one digit two and all other digits zero or one is recurrent for every finite coloring and every piecewise syndetic set. It is also a set of measurable three-term recurrence in every integer system. All lengths recur in compact metrizable group Haar extensions of the odometer and in Eberlein-correlation classes. The all-length measurable assertion for arbitrary systems remains unresolved. Sparse geometric sublanguages can instead have capacity one, and factorial packet constructions require separate treatment $(R173 - -R178)$.

Finally, all translates of Erdős–Taylor finite-sums tails recur multiply on pronilsystems and on the specified rigid weakly mixing rank-one model. Harmonic panel estimates supply quantitative finite-dimensional Bohr returns and further exact Gaussian relative thresholds. Carry-budget constructions are valid in their fixed product-digit models, but their finite-chart counterexamples prevent an automatic passage to all systems $(R109 - -R112)$. The existing finite-color hierarchy also has unrestricted capacity one, including its least admissible radix choice $(R139)$.

The additional horizontal-bundle results impose necessary prime-spectrum conditions, give finite-index comparison theorems at four terms, and establish model-dependent bounds for finite polynomial fibers. Saturated free-coordinate nilrotation tests and locally compact orbit-map reductions have precisely stated transfer limits. Hard-tail ultrafilters force finite-atomic character constraints, while changing profinite prime ratios leaves explicit unresolved resonances $(R126, R129, R136, R138, R140 - -R141, R144, R149)$. Together these results narrow several possible constructions, while leaving the existence of an intermediate unrestricted integer capacity open.

Structural restrictions, finite models, and endpoint phenomena

For a set of directions D in a countable abelian group, write $a_h(D)$ for the supremum of the measures of events having no positive h-term return at a direction in D, and write $b(D) = \sup_{h\geq 2} a_h(D)$. All statements about a restricted class of actions will identify that class. In particular, an upper bound for compact affine actions, finite-dimensional toral observations, or a single ergodic transformation is not an upper bound for $(b(D))$. Several results below give genuine restrictions on a possible integer example; none constructs an integer set with $0 < b(D) < 1$.

Exact thresholds on other groups and finite criteria on the integers Every prescribed $t \in (0, 1)$ is attained on a single countable mixed group of the form

$$G = \mathbb{Z} \oplus \bigoplus_{p\ \{modd\}} \mathbb{F}_p^{m(p)}, \qquad m(p) < \infty.$$

The directions can all have infinite order and have distinct positive integer coordinates. The construction gives a positive return for every event of measure at least t, including equality, and avoiding events whose measures approach t from below. The latter use compatible finite projective blocks; the former use finite Szemerédi recurrence after conditioning on a finite head. This is an all-action assertion on G. Its integer projection contains arbitrarily long dilated intervals and has capacity zero. Thus neither infinite order of every direction nor injectivity of the integer coordinate transfers the threshold to an integer action $(R204)$.

Periodic suspension gives a precise alternative route. If $D \subset \mathbb{Q}$ is saturated by addition of integers, then its capacity is the supremum of the capacities of its images in the cyclic-kernel quotients. For a squarefree torsion family $V \subset \bigoplus C_{p_i}$, its rational saturation has capacity

$$b_{\mathbb{Q}}(\pi^{-1}V) = \lim_{J\to\infty} b_{\oplus C_{p_i}}(V_J).$$

An integer realization then preserves the relevant all-length value. The tail limit, rather than any single native torsion capacity, is essential $(R241, R247)$.

An exact intermediate integer value δ is equivalent to prime rows $S_i \subset C_{p_i} \setminus \{0\}$ with two properties. Their one-axis i-term avoiding densities tend to at most δ; and, for every r, all sufficiently late finite products have a common k_r-term avoiding mask of density at least $\delta - 1/r$, where k_r is independent of the product length. Separate short masks on individual axes do not establish this joint condition $(R261)$. Suspension also extends to fixed rational rank. Variable-rank criteria remain possible, but the required denominator and progression length depend on rank and must be determined before the prime is chosen $(R269)$. A cyclic forcing row must have exponentially many directions in its forcing length; this necessary size condition remains far from sufficient $(R250)$.

Periodic directions and characteristic reductions Periodic integer direction sets have exact finite cyclic models at each fixed progression length. The period may grow without changing the length; when the base period is prime, prime-power cyclic models suffice ($R194$). For a fixed ergodic transformation and an open union of residue classes, the all-length threshold has the form $(1-1/k)$, where k is determined by the system's cyclic factors and the orders of allowed residues. This fixed-system spectrum coexists with a zero-or-one alternative for the unrestricted integer capacity of the same open periodic families ($R195$).

Several characteristic reductions are relative rather than classical nilsystem statements. For the paired family with unrestricted horizontal normals, an avoiding event can be enlarged to the support of its projection on the horizontal Host–Kra factor; a vertical centralizer action still remains to be analyzed ($R183$). For translated signed boxes, lazy tail averages yield spectral factors and relative cube factors characteristic for nested append-tail progression means. Under countability assumptions these lie in finite Furstenberg–Zimmer distal towers. Measurable compact extensions in such towers need not be finite-dimensional nilsystems. A skew-shift example shows explicitly why the first spectral factor alone does not control triple returns ($R216$).

The proposed unrestricted rational-spectrum reduction in R214 is conditional. Its cited cube lemma is contradicted by an explicit binary odometer extension: an ordinary uniformity seminorm vanishes while the asserted relative-cube quantity does not. The counterexample invalidates that proof input, without proving that every possible replacement theorem is false. The finite-rational-spectrum recovery in that note is unconditional. By contrast, the endpoint theorem for sum boxes whose tails have uniformly positive density in arbitrarily long progressions of a fixed gap uses a separate argument ($R206$). For a fixed event, progression averages on increasingly rough residue classes approach the ordinary average uniformly; the event cannot vary with the modulus in this assertion ($R219$).

Scalar identities, interpolation, and affine actions Exact scalar progression identities impose strong algebraic rigidity. On sufficiently rough cyclic groups, a fixed h-slot identity valid at every starting point forces a tag of degree at most $(h-1)$; the integer version leads to finite or nil-Bohr separation ($R223$). Characteristic-p scalar filters on cyclic prime-power modules admit bounded-degree interpolation on the rational cover, uniformly in the depth. An additional interpolation statement applies in arbitrary coefficient characteristic when sufficiently many distinct first-digit slopes occur ($R203$). Three temporal slots admit a degree-three refinement in the specified primary-rank range ($R252$).

For high-rank elementary primary modules, sparse directions permit homogeneous extraction only under a characteristic bound; a Frobenius example shows the need for that restriction ($R202$). Symmetric multiaffine filters have a stronger rank-uniform polynomial tag, and finitely many exchangeable coordinate types give a bounded list of tags. Choices from that list cannot be assembled independently at different primes without an additional argument ($R248$). Partial bilinear tags, on the other hand, can lack every bounded-degree extension. Their exact noncommutative identities do not yet yield a positive short avoiding event of uniform mass ($R192$).

The distinction between native and saturated actions is substantive. A prime-square affine example is invisible to native group-polynomial tests but has a degree-eight separating polynomial on its rational cover, whose capacity is therefore one ($R205$). Shallow and deep presentations can have identical rational capacities while differing sharply in the native affine

models they admit ($R226$). Affine Haar actions of the squarefree torsion group have return sets containing bounded-degree polynomial neighborhoods; linear Haar actions need only degree one ($R239$). Related rational rank-one linear and affine profinite results appear in R212.

Saturation also makes Bohr recurrence sufficient for finite-dimensional niltranslation tests and, under the stated torsion-quotient hypotheses, affine nilmanifold tests ($R213, R220, R222$). These are statements about genuine actions of the saturated group. A trivial central circle bundle over a connected compact abelian base has compact translation dynamics for an abelian torsion subgroup; dual rank one guarantees the required triviality. Nontrivial rank-two bundles and infinite-order acting subgroups are not covered ($R257$).

Paired primes: obstructions, recurrence classes, and endpoint one Paired prime constructions separate several phenomena. Free horizontal normals together with small-cost odd diagonal marks admit a fixed finite coloring excluding monochromatic four-term progressions, giving a positive lower witness ($R182$). Dense single-mark residue menus admit bounded detectors, while simultaneous many-mark decoding entails quantitative restrictions on its weights ($R193$). A single high dyadic clock with two low shears similarly has a common positive six-term avoiding event. Its unrestricted upper threshold remains open, although substantial classes with restricted clock spectrum, precompact low orbits, or discrete projected spectrum recur ($R237$).

Other paired regimes provably have capacity one. These include bounded total logarithmic marked weight ($R184$), suitable nearby prime matchings ($R185$), one positive vertical mark or vanishing positive marked budget ($R200$), and boundedly many signed marks even with completely free horizontal normals ($R231$). Sparse curvature corrections handle further ratio and mass-shell regimes ($R199$). For fixed small positive budget, an explicit lower bound approaches one at square-root logarithmic rate as the budget tends to zero ($R198$). Statistical signed blockers require stated approximation and curvature conditions; they do not supply those conditions automatically ($R232$).

The strongest matched-ratio exclusion uses Sturmian curvature words. If the two primes tend to a fixed positive ratio θ, satisfy eventual congruence conditions modulo every fixed integer, and the marked budget is below $\min(1, \theta)$, the rational and integer capacities are one ($R245, R215$). This does not cover supports containing a continuum of distinct limiting ratios. Irrational ratios also defeat every finite list of scalar polynomial-curvature tests with summable mean errors, even when different steps use different detecting channels. That limitation is compatible with the Sturmian construction, which uses another mechanism ($R270$).

Positive recurrence is known in several broad but restricted models: independent grids with profinite fibers ($R186$), fixed cyclic outputs ($R187$), pure toral fibers with full Laurent rank($R189$), and correlations satisfying a spectral nonconcentration condition ($R188$). Pair correlations require no such higher-order hypothesis; the one-mark signed family has return supremum at least the square of the event mass. At higher order, its Fourier–Stieltjes closure theorem remains a restricted correlation statement ($R181$).

Quantitative control of toral and moving-stream models For the toral models in which the required quotient filling is established, the recurrence threshold in ambient dimen-

sion r is at most

$$1 - \frac{1}{\{} r!\}.$$

If the even annihilator has Laurent rank s, the bound improves to the smaller of $(1 - 1/r!)$ and $(1 - 1/(s+1)!)$. In particular, one-dimensional models recur at every positive mass and rank-one residual constraints give a half-density threshold. The proof orders independent Haar controls by their leading degrees, compresses the event through successive coordinate sections, and applies a union bound at each completion time. Its probability lower bound is uniform in temporal supports and coefficients. The quotient-filling hypotheses remain indispensable, and the bound tends to one with dimension ($R190, R253$).

Shift-fixed torsion cosets in profile groups have periodic representatives when the fiber dual satisfies the stated finite-generation hypothesis. This gives a model-dependent all-event threshold below one. A solenoid counterexample shows that finite topological dimension alone cannot replace finite generation ($R196$). Independent low axes with a common high clock also admit dimension-dependent toral bounds, while tightness or suitable shell decay of relative eigenfields gives further recurrence criteria ($R235, R236$). Exact prime eigenvalue increments have locality properties that explain some of these restrictions ($R201$). Totally disconnected horizontal characteristic fibers form another positive-recurrence class; a Heisenberg resonance shows why one proposed extension to connected factors fails ($R224, R225$).

Common-stride controls give a half-density finite-window theorem; several strides, full positive sections, and dispersed cylinder certificates give further sufficient conditions ($R242, R244$). Two opposite moving streams have factorial bounds based on spatial completion cohorts, half-density theorems for certain small or scalar-projection windows, and a dimension-free result for the exact even subgroup consisting of constants plus free waves ($R259$). Merely containing those waves is insufficient. Indeed near-full statistical traps exist in larger windows, and polynomial even modes produce abstract shift-fixed order-two cosets with near-full avoidance ($R258$). These cosets have no established arithmetic realization. Signed-star lemmas ($R233$) and the failure of a thickened-ring statistical mask in the presence of free slope streams ($R251$) further delimit the control approach.

Charged products and the cost of replication A finite pairwise-connected progression query law with nonconstant charge coprime to the underlying group has an ordered contradiction at every fixed length. Explicit translation symmetrization eliminates the former characteristic restriction, including binary charges. Exponential cancellation of nonconstant charge characters in tensor powers, combined with density Hales–Jewett, forces charged returns ($R218$). An exact integer-query construction makes this conclusion uniform over all covering periods for a fixed local modulus and law ($R260$).

The number of charged coordinates is unrestricted in these theorems. Conditioning the product law onto bounded mark count or bounded cost can destroy the useful mass, so the conclusion does not resolve the small-budget candidates. Genuine replication is likewise essential when an IP family of unmarked integer queries absorbs sufficiently divisible velocities. Under that availability hypothesis one obtains further endpoint results, including the stated padded-automaton classes ($R217$). Different primes can be handled uniformly for odd weighted boxes of macroscopic relative width: a bounded query pattern is repeated along an infinite subsequence, giving all-length recurrence on lifted free integer axes. Unit-width menus are outside this theorem ($R266$).

Sum boxes, Erdős–Taylor tails, and unsigned odd supports A positive pair witness for a translated symmetric sum box implies that every ultrafilter on that box contains members with capacities arbitrarily close to one. Strong path limits and exact retained-run tracking prove this local statement. It does not imply capacity one for the entire box. A global upper bound below one would combine with this property to give the prescribed spectrum ($R256$). The analogous local-capacity theorem holds for arbitrary unsigned odd finite sums, without lacunarity or unique representations ($R265$).

Signed boxes also admit usable-relation pair tests, a subgroup of forced periods, variable-spacer spectral tests, and an exact Bohr characterization by continuous functions on the compact space of sign paths. Summable successive size ratios imply Bohr separation and capacity one. These are sufficient criteria, not a global dichotomy for all signed boxes ($R255$). Translated signed trace boxes of quadratic Pisot units do satisfy an endpoint dichotomy; a specified conjugate-product inequality extends the result to some higher-degree units, but not to all of them ($R197$).

Every translate of every unsigned Erdős–Taylor finite-sum tail has strong measurable pair recurrence. More precisely, translates of an event along a tail sweep out its invariant saturation. The elementary proof uses divergent reciprocal coefficients in the relations between successive generators. It also gives a broader constant-spacer invariance theorem ($R254, R262$). In compact minimal distal systems, Ellis-group arguments strengthen this to dense tail orbits through every point and simultaneous pointwise recurrence for finite tuples. Measurable multiple recurrence in arbitrary systems does not follow: the diagonal law used for progression tests is generally not invariant under the slope action.

For unsigned odd supports on odd prime axes, the maximum pair-avoiding density is

$$O\left((1+\sum_i 1/p_i)^{-1/3}\right).$$

Thus divergent reciprocal mass gives pair recurrence in every action ($R264$). At higher orders, even and odd tail saturations agree for every invariant slope law, but their difference may still carry the noninvariant diagonal law. Independent finite-cocycle models and certain independent block models have full positive recurrence. The same analysis gives an adaptive selection map with uniformly bounded L^2 density. The signed family can nevertheless fail the spectral van der Corput criterion, so pair recurrence and that criterion remain distinct ($R263$). Fixed finite registers over independent axes furnish another complete restricted model ($R208$).

Geometric, probabilistic, and entropy restrictions Several exclusions concern the shape of finite templates. Positive pair witnesses cannot prohibit too many projective directions without a substantial dimension reduction ($R207$). Arbitrarily small masks hit every full finite-field line whose direction has sufficiently small centered norm, uniformly in the ambient dimension. The final sparse-neighborhood rank estimate ($R228$) closes the earlier logarithmic gap in histogram and random-table methods ($R227, R230$), complementing the dimension-free suspension comparison for short representatives ($R229$). These are full-line results, not fixed-length forcing theorems. Bounded-size prime supports with bounded menu multiplicity have capacity one ($R240$); a common exact lattice lift of every short box edge has a sharp product density bound ($R234$). Fixed finitely many sheets covering all large parameter subspaces force finite-sums patterns of every rank($R246$).

Random prime rows have a noncritical dilation-separator threshold at logarithmic size ($R210$). Independent integer selectors have capacity one in the ($O(1/n)$) regime and capacity zero in the subpolynomially dense regime, leaving intermediate polynomial regimes unresolved ($R211$). Bounded-entropy angle templates admit near-full Hamming blockers ($R249$). Pure-coordinate radix localization preserves all-length capacity without a summability hypothesis ($R209$), while certain square-free graph targets force single-character separation ($R238$).

Finally, varying-parity avoidance forces quantitative shattering and an ℓ^1 sequence of correlation translates. Tame correlations therefore recur. In the independent-parity application, avoidance also forces positive event-specific sequence entropy along specified unmarked directions ($R267$). Pairwise-independent halfspaces supply a universal endpoint compatibility bound but do not solve temporal consistency ($R191$). Finite-VC error families permit exact repair in a fixed product power, preserving positive avoiding mass ($R268$). Root thinning imposes additional dilution constraints on any hypothetical intermediate example ($R243$). Stabilization of joining limits in one witness system is insufficient: a separate countermodel has such stabilization together with near-full avoidance ($R221$). These restrictions identify necessary features of a future construction without supplying the missing unrestricted integer threshold.

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Part I

Technical results and proof details

Chapter 4

Further recurrence criteria and structural restrictions

R001. Additive ultrafilters of finite semigroup order

Scope and status. Result with proof.

Result. Finite additive semigroup-order ultrafilters give pair recurrence, a fixed-h mass bound $(h-2)/(h-1)$, distal recurrence, and Gaussian multiple recurrence.

Scope. The unrestricted multiple-recurrence conclusion for every member of such an ultrafilter is left open.

Let p be an ultrafilter on positive integers whose additive convolution powers satisfy $p^{*n} = p^{*(n+1)}$ for some n. Its eventual power e is an additive idempotent. Non-idempotent examples exist; finite subgroups of βN itself are trivial. See Zelenyuk, Finite semigroups and periodic sums systems, Introduction. The explicit generators in that paper arise by iterating limits and sums starting from idempotents, so (by the finite-sums subsemigroup criterion) they do not give an intermediate threshold.

For any unitary action and any integer j the weak-operator limit $V_j = p - \lim_d U^{jd}$ lies in the abelian von Neumann algebra of U. Separate weak continuity of products with fixed bounded operators gives $V_j^n = V_j^{n+1}$. It is normal, hence an orthogonal projection by spectral calculus. The projection is also the e-limit. In a measure-preserving Koopman action its range consists exactly of the rigid functions $p - \lim ||U^{jd} f - f||_2 = 0$; on bounded functions the range is a closed factor algebra by the usual multiplication and truncation argument.

Three consequences:

- Every pair of slots of the $p-\lim h$-term joining has same-set intersection at least $\mu(A)^2$, because $\langle V_j 1_A, 1_A\rangle = \|V_j 1_A\|^2 \geq \mu(A)^2$ (V_j fixes constants). Hence if p misses h-term returns of A, its mass satisfies $\mu(A) \leq (h-2)/(h-1)$. Indeed let N count the h memberships. A zero common intersection means $N \leq h-1$, so $h(h-1)\mu(A)^2 \leq EN(N-1) \leq (h-2)EN = (h-2)h\mu(A)$. In particular every positive event has a pair return and every event above one half has a three-slot return along p. This bound deteriorates with h.
- In every topologically distal flow the Ellis image of p belongs to a group and its n-th and $(n+1)$-th powers agree, so it is the identity. This gives recurrence at all finite orders

on nilsystems, including for Haar measurable events by fixed-marginal approximation.

- In a Gaussian system each conditional spectral phase measure has Fourier coefficients in $\{0, 1\}$, since the moment multipliers V_j are projections. The set of indices where the moment is one forms a subgroup of Z: each corresponding character equals one almost surely, and the other moments are zero. Uniqueness of Fourier coefficients makes the phase law Haar on the annihilator subgroup of the circle, which contains the identity phase. The Gaussian selector theorem [R133] therefore gives positivity of every same-event multiple return in the limiting joining. These facts also apply to a system whose tested multiple correlation belongs to the uniform algebra spanned by matrix coefficients and evaluations on distal compact flows: its p-limit equals its e-limit. For same-event positivity of the e-limit use finite density IP recurrence, or the usual idempotent IP multiple recurrence with a positive bound.

The statement that all p-members have unrestricted multiple recurrence is not proved here. Equality of two-slot limits for all Koopman actions does not by itself identify the full joining of a diagonal progression. The source's finite-order constructions lying in the closed finite-sums subsemigroup do not establish that every finite-order ultrafilter lies there.

R002. Haar recurrence for finite-rank affine actions on compact abelian groups

Scope and status. Result with proof.

Result. Every positive Haar event in a finite-rank affine compact abelian action has a positive polynomial-Bohr set of multiple returns; automorphism actions admit a finite-index subgroup of returns.

Scope. Finite-rank acting group and Haar measure are essential; in rank one degree $h-1$ suffices.

Let $G=\mathbb{Z}^r$, acting by commuting continuous affine automorphisms $S_g x = A_g x + b_g$ on a compact metrizable abelian group K with Haar probability m_K. For a measurable event B of positive measure and $h \geq 2$, write

$$c(g) = m_K(\{x : S_{jg}x \in B \quad \text{for every} \quad 0 \leq j < h\}).$$

There exist real polynomials $P_1, \ldots, P_t$ in r variables, each vanishing at zero, together with $\epsilon, c_0 > 0$, such that $c(g) \geq c_0$ whenever $\max_i \|P_i(g)\|_{\mathbb{T}} < \epsilon$. For an action by group automorphisms (all $b_g = 0$), a single finite-index subgroup of G already suffices. In particular ordinary polynomial-Bohr recurrence of all finite orders precludes a short witness in any of the displayed affine Haar systems. This applies to arbitrary Haar measurable events and imposes no ergodicity condition. It does not address arbitrary invariant measures on the same topological systems or infinite-rank acting groups.

The linear graph limit For an ultrafilter p on G, let λ_p^0 be the weak limit along p of the laws of $(A_{jg}x)_{j<h}$ with Haar x. Its character coefficients are zero or one, since the coefficient at $(m_0, \ldots, m_{h-1}) \in \widehat{K}^h$ is the indicator of

$$\sum_j (A_g *)^j m_j = 0. \tag{1}$$

The tuples satisfying (1) on a member of p form a subgroup L of the dual; hence the limit law is Haar on $J = L^\perp \subset K^h$, by character orthogonality and uniqueness.

Assume p contains all finite-index subgroups of G. Then J contains the diagonal ΔK. Indeed if $m = \sum_j m_j \neq 0$, the character tuple does not belong to L. Here are two equivalent ways to see this. In the finitely generated module $M = \sum_j \mathbb{Z}[G] m_j$, the nonzero element m survives in a finite module quotient. To check the algebra, choose a maximal ideal $\mathfrak{m}$ containing $\operatorname{ann}(m)$. Then $m/1 \neq 0$ in $M_{\mathfrak{m}}$. By Krull intersection, $m \notin \mathfrak{m}^n M$ for some n. The residue field at a maximal ideal of a finitely generated commutative (mathbb Z)-algebra is finite, and the successive $\mathfrak{m}$-adic layers of this finite module are finite-dimensional over that field. Thus $M/\mathfrak{m}^n M$ is finite. On the finite-index kernel of the action on this quotient, the left side of (1) is the nonzero image of m. Alternatively the dual of M is a Noetherian algebraic G-system with dense periodic points, so a periodic point detects the same nonzero sum; test (1) on its finite-index stabilizer. The latter periodic-point fact appears in Lind, Schmidt and Ward, Lemma 7.2.

Character relations are virtually unipotent on an ultrafilter subgroup Choose a subgroup $H \subseteq G$ of least rank among those belonging to p. A coset of a subgroup $H_0 \subseteq H$ can belong to p only if it is H_0 itself: subgroups of finitely generated abelian groups are profinitely closed, as their quotients are residually finite. Further, such an $H_0 \in p$ must have finite index in H, by minimality of rank.

Fix a tuple satisfying (1) p-often, and now let $M = \sum_j Rm_j$ for the smaller Noetherian group ring $R = \mathbb{Z}[H]$. Write $u^g \in R^\times$ for the group unit. There are a finite-index subgroup $H' \subseteq H$ and an integer $N \geq 1$ such that the augmentation ideal $I = (u^g - 1 : g \in H') \subset R$ satisfies $I^N M = 0$.

For $M \neq 0$, let $\mathfrak{q}$ be any minimal prime over $\operatorname{ann}_R M$. The localization $M_\mathfrak{q}$ is nonzero (use finite generation), and at least one of the images of the generators m_j in $M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q}$ is nonzero by Nakayama's lemma. In this vector space over the residue field $k(\mathfrak{q}) = \operatorname{Frac}(R/\mathfrak{q})$, relation (1) makes the field element $u^g \bmod \mathfrak{q}$ a root of a fixed nonzero vector-valued polynomial of degree at most $(h-1)$. One nonzero scalar component bounds its possible roots by $(h-1)$. Thus on a member of p within H, the homomorphism $g \mapsto u^g \bmod \mathfrak{q}$ has one constant value. The coset observation shows its kernel has finite index in H. Intersect the finitely many such kernels to obtain (H'). Then I lies in the intersection of the minimal primes, the radical of $\operatorname{ann}_R M$. Noetherian finite generation implies $I^N \subseteq \operatorname{ann}_R M$ for some N. The zero module is immediate. Also $H' \in p$, since (H') contains $H \cap L_G$ for some finite-index subgroup $L_G \subseteq G$ (or use the profinite coset observation on the partition of H).

The affine translation and the positive return Suppose now that p sends every ordinary zero-constant real polynomial on G to zero modulo one. This implies the preceding profinite condition by testing rational linear polynomials. For any $m \in M$ let $F_m(g) = m(b_g) \in \mathbb{T}$, writing the pairing additively. From the cocycle identity and commutativity,

$$\Delta_e F_m(g) = F_{(A_e * - 1)m}(g) + F_m(e).$$

On (H'), this shows that every difference of order $(N+1)$ of F_m is zero, and $F_m(0) = 0$. In lattice coordinates for (H'), the binomial-coefficient Newton expansion is valid also for circle-valued functions, by iterated ordinary differences. Lift its finitely many circle coefficients to real numbers, choosing zero for the constant. This represents $F_m \bmod 1$ on (H') by a real polynomial with zero constant. Its lattice coordinates extend to rational linear maps on G, since a basis of (H') is rationally independent in $G \otimes \mathbb{Q}$; extend the corresponding linear functionals from its rational span. Thus each $g \mapsto F_{m_j}(jg)$ on (H') is the restriction of an ordinary zero-constant polynomial on G. Its p-limit is zero.

The affine graph joining now has the same p-limit as the linear one. For a tuple outside L the character coefficient is zero on a member of p. For a tuple in L it equals the phase of $\sum_j F_{m_j}(jg)$ there, whose limit is one by the preceding argument. Therefore its weak limit is Haar measure on $J \supseteq \Delta K$.

Disintegrate Haar on J over $J/\Delta K$. Write each coset by the differences of its last $(h-1)$ coordinates from its first. The resulting difference group is compact and its Haar measure gives positive mass to every neighborhood of zero. Conditional on a difference tuple $z = (z_1, \ldots, z_{h-1})$, the first coordinate is Haar on K. The conditional probability of B^h is

$m_K(B \cap \bigcap_{j=1}^{h-1}(B - z_j))$; it tends to $m_K(B) > 0$ as $z \to 0$, by continuity of translation in L^1. Consequently the limit joining gives positive mass to B^h. Its value there is the p-limit of $(c(g))$, by approximating 1_B by bounded continuous functions in $L^1(m_K)$ and using all the fixed Haar marginals.

Finally the scalar extension $p \mapsto p\text{-}\lim c(g)$ is continuous on βG. The ultrafilters sending all zero-constant polynomial phases to zero form a compact intersection of closed sets. The displayed positive values have a positive minimum on it. By compactness finitely many polynomial zero neighborhoods already enforce, on G, a lower bound of half this minimum. For the linear case apply the same argument to all ultrafilters containing the finite-index subgroups; the compactness argument then produces a single finite-index subgroup. A principal ultrafilter at zero causes no issue. The polynomial degrees in the affine theorem are finite but have not been bounded here in terms of h alone.

Degree at most $(h-1)$ for one acting generator For $G = \mathbb{Z}$ it is sufficient throughout to assume only that p kills ordinary polynomial phases of degrees at most $(h-1)$. If p is principal this condition implies it is at zero. Otherwise the preceding virtual unipotence yields an integer $q \geq 1$ such that $U = A_q^*$ acts unipotently on the module generated by the m_j under H, with $q\mathbb{Z} \subset H$. This module M is finitely generated as an abelian group: use a finite transversal of $q\mathbb{Z}$ in H, then the finitely many powers of $(U-1)$ before nilpotence (negative powers of $U = 1 + (U-1)$ use the same finite span). Let $L = \log U$ on $M \otimes \mathbb{Q}$, a finite nilpotent rational endomorphism. On infinitely many (qn), (1) says $\sum_j \exp(jnL)m_j = 0$, so every coefficient of this polynomial in n vanishes. For $0 \leq k < h$ apply L^{h-1-k} to $\sum_j j^k L^k m_j = 0$. The Vandermonde matrix on the h distinct time indices yields $L^{h-1}m_j = 0$ on the rationalized module for each j. Hence $X^{h-1}m_j$ is finite additive torsion for $X = U-1$, since $X = L$ times an invertible polynomial in L.

Let e kill the finite additive torsion in M. Choose c divisible by (eN!), with $X^N = 0$. All $\binom{c}{k}$ for $1 \leq k < N$ are divisible by e, so $U^c - 1$ is (eX) times an integral polynomial in X. It follows that $(U^c-1)^{h-1}M = 0$: the assertion on each m_j follows from $X^{h-1}m_j$ being e-torsion, and the assertion on M follows by commutativity. On $qc\mathbb{Z}$ the affine cocycle evaluations therefore have degree at most $(h-1)$. Extend these one-variable polynomials to $\mathbb{Z}$ and conclude by exactly the same compactness argument. In higher acting rank the relation can lie on an infinite proper polynomial variety; the one-variable coefficient comparison does not apply there.

R015. Terminal accepting states of a finite digit automaton

Scope and status. Result with proof.

Result. A padded least-significant-first finite automaton whose accepted multiples reach terminal components has full multiple recurrence.

Scope. This corrected terminal-state theorem uses ordered query words and does not depend on the disputed rational-cube lemma. It is not a dichotomy for every automatic set.

Qualifications. Corrected independent proof: dependency is replicated-integer-word-menus.md, not the withdrawn rational-spectrum-tail-boxes.md implication.

Fix a base $b \geq 2$. Read the ordinary nonnegative digits from least to most significant, and assume that appending any number of zero digits at the high end does not change acceptance of a positive integer. All automata below are finite and complete. A terminal strongly connected component of the transition graph is one with no outgoing transition to another component. Suppose that **for every positive modulus q there is an accepted positive multiple of q reaching some terminal component.** The corresponding set of integers D then has measurable multiple recurrence at every positive density and every finite length.

Two special cases follow. For a strongly connected automaton there is a dichotomy: divisibility gives full recurrence, and missing all multiples of one modulus gives critical value one. Also, if $D \cap q\mathbb{N}$ has positive upper Banach density for every q, the terminal condition holds. On uniformly random fixed-length words the proportion ending in transient states tends to zero: from every transient state some path of bounded length enters a terminal component, with uniformly positive probability. In arbitrary long integer intervals the low m digits are uniformly distributed up to an error $O(b^m/\text{length})$; reaching a terminal component there suffices for all higher continuations. Thus the set of integers ending in transient states has zero upper Banach density. The missing-modulus conclusion uses the elementary cyclic-root sieve. This result does not address a divisible set all of whose sufficiently divisible words have a required transient transition, such as the language with exactly one special digit.

The block monoid. Choose $\ell > 0$ for which the transformation e of the state set induced by 0^ℓ is idempotent. Write $Q' = e(Q)$, $q_0 = q_{\text{initial}} \cdot e$, and consider on Q' the transformations induced by all blocks ewe of length a multiple of ℓ (the letter e here denotes the all-zero block). Together with the zero block they form a finite transformation monoid M; unused intervals of digit positions of length divisible by ℓ act identically on Q'. If a state of Q' is in a terminal component of the original graph, its reachable states in the block graph are mutually reachable. Indeed within that original component every path of length a multiple of ℓ can be reversed by a path of such length: if the original return path has length s, follow it and then repeat the resulting loop $\ell - 1$ times. Prefixing and suffixing by e does not change endpoints already in Q'. All further paths stay in the terminal component.

For each q coprime to b, tag a word w by the affine map $z \mapsto [w]_b + b^{|w|}z$ on $\mathbb{Z}/q$. Concatenation multiplies tags in the finite affine group, with the order convention of reading low digits first. Products of transformations together with their tags give a finite lift of the block graph. In a strongly connected part every lifted path can be reversed: follow any base return path and repeat the resulting loop to the order of its affine tag.

There is a block word, of affine tag the identity, taking q_0 to an accepting state in a terminal component. To see this use an accepted integer divisible by $qb^{2\ell}$. Read its first ℓ

zero digits to reach q_0; retain the next ℓ zeros at the start of the remaining word. Its value is divisible by q. Append high-end zeros so that it ends in a block e, with total remaining length divisible by both ℓ and the multiplicative order of $b \bmod q$. Padded acceptance and the terminal component are preserved. This word has the stated tag and is anchored at both ends. For $q = 1$ the tag convention is trivial.

Let $A \in M$ have globally minimum image cardinality. All its positive powers have the same rank, since smaller rank is impossible. A sufficiently large power of a representing word has affine tag the identity; call this word A_q. After the preceding accepted path has reached r, append A_q. In the lifted terminal component, return to $(r, 1)$. The whole word V_q has affine tag the identity, has minimum rank, and maps q_0 to an accepting state. Among these words there is one fixed state transformation V occurring on a divisibility-cofinal sequence of moduli prime to b. For example use successive products of all such moduli up to a growing bound and pass to a subsequence. Let $C = V(Q')$. By minimality of the rank, V permutes C; denote that permutation by γ. A power R of V fixes C pointwise, maps Q' onto C, and satisfies $RV = VR = V$. In particular $q_0V = (q_0R)\gamma$ is accepting, independent of q.

Integer velocities. Unmarked positive blocks may be chosen with state transformation R, in addition to the all-zero option. There exists such a positive block: insert a nonzero word of length divisible by ℓ between two representatives of R, anchoring with e as needed. The sandwich still has minimum rank, induces a permutation of C, and acts on every state through its R-image. Raising it to the permutation order realizes R with positive value. Copies in disjoint aligned blocks give an IP set of unmarked velocities; unused blocks are zero, and every nonempty composition of those transformations again equals R.

Allow marked words with transformation V. If γ has order $m > 1$, all their representatives are positive, since the all-zero aligned word induces the identity on Q'. The tagged values meet every $q\mathbb{Z}$: the chosen V_q's handle the part coprime to b, and prefixing enough low zero blocks absorbs the part supported on primes dividing b. If $m = 1$, $q_0R = q_0V$ is accepting, so already the nonempty unmarked IP family suffices. For $m > 1$, choose any finite list of unmarked and marked challenges whose ordered contradiction is given by the replicated integer velocity lemma [R217], using the IP family and divisibility just proved. All their digit representatives can be padded to one common length divisible by ℓ. Place disjoint copies above an initial block 0^ℓ. Whenever the number of marked choices is $1 \bmod m$, the composed transitions send the state to $(q_0R)\gamma$: zeros do nothing, and the first nonzero block acts through R. Thus every such integer difference belongs to D. The cyclic-character and density Hales–Jewett arguments of the linked lemma supply positive h-term recurrence for each h, as asserted.

For comparison, the automatic-set dichotomy of Byszewski and Konieczny involves shifted finite-sums sets and the combinatorial notion of aridity. The argument here concerns multiple recurrence at positive density and uses the stronger structure of the relevant terminal component. Byszewski and Konieczny, *Automatic sequences and generalised polynomials*

R017. An exact finite matching criterion for independent block cocycles

Scope and status. Result with proof.

Result. A finite matching subgroup exactly computes fixed-length avoidance for independent finite block cocycles.

Scope. The formula concerns the specified stationary block system; it is not an all-system integer capacity formula.

Fix $h \geq 2$, a finite abelian group K and independent finite blocks P_i, each a product of cyclic groups at designated coordinate axes. Haar measure on $Y = \prod P_i$ is independent. On $Y \times K$ use the componentwise product action of the direct sum: on block i translation d changes the register by $\phi_i(x_i + d) - \phi_i(x_i)$, for any functions $\phi_i : P_i \to K$. Legal local directions d are sums of distinct positive standard axes in the block, including zero. Their tag t is the cardinality of their selected axes modulo 2. Global legal directions have total tag one. The argument works for other finite local direction systems with a designated zero letter and a tag additive across independent blocks.

Write $L = C_2 \times K^{h-1}$. A local h-slot edge with start x and direction d has label

$$\ell(x, d) = (t, (\phi_i(x + jd) - \phi_i(x))_{1 \leq j < h}).$$

For label ℓ let $m_i(\ell)$ be the greatest mass of a nonnegative measure on edges of that label whose marginal in each of the h slots is at most Haar on P_i. This is a finite fractional matching LP on the h-partite hypergraph, one copy of P_i per slot (parallel edges are harmless). Define

$$H = \langle \ell \in L : \sum_i m_i(\ell) = \infty \rangle.$$

The subgroup H is unchanged by finite prefix deletions. It projects onto C_2 if each block has at least one odd choice: use the fixed single-axis direction with uniform x, whose partition by the finitely many labels gives for at least one odd label a matching of mass at least $1/|L|$ on each block, then pigeonhole in infinitely many blocks.

Programming all of H Any desired $\ell \in$H can be programmed by finitely many possibly adaptive tail choices, with exactly that total profile and with a uniform bound $C = C(h, K, H)$ on the L^2 norms of the marginal densities of all h resulting base slots. The constant is independent of the initial head; all head coordinates are unchanged. Write ℓ as a sum of labels with divergent matching series, using positive multiplicities (L is finite). If a label is needed repeatedly, assign disjoint infinite subfamilies to its copies, still with divergent matching series each. This can be done simultaneously for the finitely many requirements by sequentially allocating disjoint finite intervals past all previous ones, adding at least one unit of matching mass to every requested subsequence in turn.

For a single such subfamily, draw independent Haar starting coordinates and independent auxiliary random variables. At block i select an edge from an optimal subprobability by

disintegrating over its initial slot x_i: the chance of a mark given x_i is its origin density, at most one. Given a mark use its conditional edge. Otherwise leave the block unchanged. Use only the first marked block I, finite almost surely by the divergent series and independence. The output slot j moves that block by jd, leaving all others fixed.

Here is the uniform marginal estimate. Write $a_i = m_i(\ell)$, $q_i \le 1$ for the origin density of a mark, $r_i \le 1$ for its slot-j density; both integrate to a_i. The output density at y equals $N = \sum_i [\prod_{k<i}(1-q_k(y_k))] r_i(y_i)$, first on cylinders and hence on the product. Its nonnegative summands have squares bounded by themselves and their means sum to one. For $i < l$, the expectation of the product is at most $\Pr[I = i]$ times the chance that l is the first mark starting *after* i. This follows by bounding $(1-q_k)^2 \le (1-q_k)$ before i and $r_i(1-q_i) \le r_i$ at i. Summing over $l > i$ gives at most $\Pr[I = i]$. Consequently $EN^2 \le 3$. For several assigned disjoint subfamilies their slot densities multiply on independent coordinates. This proves the claim, with $C \le 3^{s/2}$ when s labels are used. Select one finite decomposition per element of the finite H to make C uniform. Each outcome selects a genuine finite support, and the collection of possible global directions remains countable.

Simultaneous deletion of the other labels For ℓ outside H, $\sum_i m_i(\ell) < \infty$. In its h-layered hypergraph take a maximal matching of edges disjoint in each corresponding slot. The matching size divided by $|P_i|$ is $\le m_i(\ell)$; every edge meets one of the matching's vertices in its own layer. Collapsing the layers to P_i gives a subset of Haar measure $\le h m_i(\ell)$ such that outside this subset in *all* slots the offending label cannot occur. Union over finitely many ℓ outside H to obtain C_i with $\sum_i Haar(C_i) < \infty$. On every progression avoiding C_i at all its slots, the ith label belongs to H. A far tail of deletions discards arbitrarily little product measure.

If $(1,0) \in H$, all positive events have returns along odd supports from every tail. Indeed program that label, so the initial fiber and head recur at all h slots. For a positive event A, choose a head and fiber atom where its conditional mass is $> 1 - \epsilon$ using martingale convergence. On this atom the h adaptive slot distributions have missing probabilities $\le C\sqrt{\epsilon}$ by the density estimate. Choose ϵ with $hC\sqrt{\epsilon} < 1$. Countability then gives at least one genuine returning difference. If $(1,0) \notin H$, imposing $b = 0$ and excluding every C_i past a finite head makes an h-avoider for all tail directions. A single fixed head point also excludes all moves touching the head and produces an h-avoider of positive mass for all global directions. On odd cycles of length ≥ 3 in particular no positive single head move fixes the head point.

Exact global-system threshold For each m use finite space $(\prod_{i \le m} P_i) \times K$, and write $\Phi_m(x) = \sum_{i \le m} \phi_i(x_i)$. Let α_m be the largest density of a mask containing no entire h-slot word of the form

$$(x + jd_0, b + \Phi_m(x + jd_0) - \Phi_m(x) + v_j), 0 \le j < h,$$

where d_0 is any head choice of tag t_0, $v_0 = 0$ and $(1 - t_0, v) \in H$. Then α_m converges, and its limit equals the supremum density of h-avoiding events in the full block system for all legal global differences.

For one direction, take a maximizing head mask and impose tail avoidance of all sets C_i for $i > m$ on the initial event. Any full progression in it would avoid C_i in every tail slot,

so its tail labels sum into H and it would contradict the optimizing mask. The resulting density is at least $\alpha_m(1 - \sum_{i>m} Haar(C_i))$. This gives the limsup lower bound.

Conversely suppose A is a genuine h-avoider. Fix ϵ with $hC\sqrt{\epsilon} < 1$ and let B_m be the head-and-fiber atoms where A has conditional density $> 1 - \epsilon$. The density of B_m tends to $\mu(A)$. If $\mu(B_m) > \alpha_m$, fix even a single forbidden head word in B_m. Starting at its first head site and register, program the prescribed v on the independent tail. Every output head slot and register are the ones in that word. The conditional chance of missing A is at most $C\sqrt{\epsilon}$ per slot, regardless of the finite head atom's mass; hence with positive probability all slots lie in A. Countably many choices imply an actual fixed-d return of positive measure (the starting head atom has positive Haar mass). This contradiction yields the liminf upper bound and proves convergence.

For the pure tail class with finite-prefix obstructions ignored, the finite target is the explicit K-word family $V = \{v : (1, v) \in H\}$. Its extremal density is the maximum density of a subset of K with no translated trace $\{b, b + v_1, \ldots, b + v_{h-1}\}$ for $v \in V$. Finite-prefix effects are retained in α_m above.

The criterion computes a fixed-length extremum in this class of stationary systems. It supplies no upper bound over arbitrary systems or integer-root lifts. To construct a new native fixed-h obstruction in this class it is necessary and sufficient to find blocks and potentials with $(1, 0) \notin H$. For $K = C_2$ and $h = 3$ this forces some character of the slot-difference profile, hence the parity difference between a fixed pair of its three slots, to equal the support tag on every local edge surviving summably small vertex deletions. The third site may belong to a cover even when the two endpoints do not, which is the remaining difference from an ordinary pair obstruction.

R019. A diffuse necessary condition for a translated-box seed

Scope and status. Result with proof.

Result. An intermediate translated-box seed with a positive pair obstruction would support a nonatomic measure on its Bohr-cancelling path space, controlled by local costs.

Scope. This is a necessary condition; no intermediate seed is constructed.

Let $D = e + \sum_i S_i$ for finite symmetric S_i containing zero, and assume a positive-measure event avoids every pair of difference in D. For a finite word v of site choices let D_v be all finite sums represented by paths starting with v, padding by zero after the finite sum. Write $c(v) = -\log b(D_v)$. The pair witness implies all costs are finite, with the same uniform upper bound. The local path construction [R256], applied directly with the fixed infinite path in place of the ultrafilter, gives $c(v_m) \to 0$ for the length-m prefixes of every path. Indeed its tracked difference family is exactly that prefix cylinder.

Define the compact set of Bohr-cancelling paths by the equations in the signed-box characterization [R255],

$$P_0 = \{\omega : e\theta + \sum_i \omega_i \theta = 0 moduloone \quad \text{for every} \quad \theta \in G_1(S)\}.$$

If $b(D) < 1$ there is a nonzero nonatomic Borel measure μ supported on P_0 such that for every prefix cylinder $[v]$

$$\mu([v]) \le c(v), \mu(pathspace) = -\log b(D).$$

To construct it, the independent-product bound gives $c(v) \le \sum_w c(w)$ over the children w of v, since their difference families cover D_v. Starting with c(empty) at the root, split each assigned mass into child masses no greater than their respective $c(w)$. This gives a consistent finite measure on the compact product path space. No singleton has positive mass, by the local limit. If a path fails a defining equation of P_0, absolute uniform convergence gives an entire prefix neighborhood on which that character stays away from zero. Every finite sum with that prefix has the same separation, hence its cylinder cost is zero by the Bohr sieve. The complement of P_0 is an open union of these zero-mass cylinders on a second-countable space. It follows that $\mu(P_0) = \mu(\text{total})$.

Consequently, if P_0 is countable then $b(D) = 1$. More generally $b(D) = 1$ whenever P_0 admits cylinder covers of arbitrarily small total local cost. For an intermediate threshold with a positive pair obstruction, P_0 must support a perfect subset and a measure satisfying the displayed quantitative control. For example if at level m there are t_m prefixes meeting P_0 and $c(v) \le \eta_m$ for all such prefixes, it is necessary that $t_m \eta_m \ge -\log b(D)$ at every m. No path-counting bound of that kind follows just from compactness.

R020. Bohr-recurrent sets realizing every critical value in one weakly mixing system

Scope and status. Result with proof.

Result. Gaussian packets realize all relative thresholds with every tail nil-Bohr dense and unrestricted critical value one; they can also meet any prescribed countable family of infinite sets.

Scope. The universal Gaussian host may depend on the prescribed countable test family; countable tests do not cover all systems.

The fixed-system theorem in all-thresholds-one-weak-mixing-system.md [R006] can require every integer set D to meet the Bohr neighborhoods of zero on every tail. All the constructions use **the same** weakly mixing Gaussian transformation. The unrestricted critical value of each of these sets is still one, now by Gaussian root amplification rather than a separating Bohr character.

Universal finite packets Use one atomless Cantor spectral probability σ_l in each of countably many disjoint open arcs, with Gaussian channel l the realification of the complex L^2 space of that probability. The integer transformation is multiplication by $e(\theta)$ in every channel. The construction below specifies the Cantor measures simultaneously. The Gaussian action is weakly mixing by atomlessness, as in the companion note.

At stage n, build m_n disjoint candidate children for each of the first n channels, of equal conditional mass in that channel. For each previously treated channel place exactly m_n/m_{n-1} children in each of its previous intervals; choose m_n a sufficiently large multiple of m_{n-1}. In the new channel start with m_n children. Choose all nm_n interior center points rationally independent together with 1. Require m_n so large that $n^2\sqrt{\log(2nm_n)/m_n} < n^{-3}$, and use a net tolerance $\tau_n = m_n^{-2}$ for the center phases. After choosing the following finite packets, shrink the child intervals about their centers while retaining nonempty interiors. They can be made as small as desired and of diameter tending to zero; subsequent branching yields the nonatomic Cantor probabilities.

For the first n channels take **all** target lists with functions

$$f_l(\theta) = \zeta_l e(b_l\theta), \operatorname{ord}(\zeta_l) \le n, |b_l| \le n.$$

There are only finitely many lists at each stage. For each list choose a finite packet of integer times, all beyond every earlier-stage packet, with these two properties:

- Every $\beta \in T^n$ has some time d in the packet with $\|d\beta\|\infty < 1/n$.
- For each d the error $|e(dc) - f_l(c)| < \tau_n$ at all but at most n of the centers c (over the entire collection of channels).

Indeed for a given β discard at most n of the independent centers to leave a family linearly independent over $\operatorname{span}_Q(1, \beta_1, \ldots, \beta_n)$, meaning that the retained images in the quotient are rationally independent. This follows from the dimension of the newly added span modulo $Q \cdot 1$. The compact orbit of (β, c_{good}) has closure equal to $closure(< \beta >) \times T^{good}$ by the

character criterion for Kronecker's theorem: an integer relation involving a retained center would violate that linear independence. Choose an arbitrarily large positive time meeting both requirements with strict inequalities. Its condition on β holds on an open neighborhood. A finite subcover gives the packet. Directions from different packets need not be distinct; each use has its specified label and estimates, and one may also demand disjoint packets by increasing the search bound at each finite selection.

Simultaneous Gaussian estimate. For the unit spectral vector in each channel use its complex Gaussian observation Z_l, formed with the two orthogonal real vectors $1, i$ (choose the sign of the second coordinate so multiplication by a constant unit complex number rotates Z_l by that number). For every time in every packet, all $l \le n$ and $0 \le j \le n$, compare $T^{jd}Z_l$ with the Gaussian observation at the multiplier $f_l(\theta)^j$. Uniformly outside an exceptional event of probability $\le 2 \cdot 2^{-n}$ the error in every such comparison is $o_n(1)$.

In detail, replace each difference of the two multiplier vectors by its step value at the center of the child. The independent real and imaginary normalized child indicators yield standard Gaussians. The maximum of their absolute values over $2nm_n$ entries is at most $C\sqrt{\log(2nm_n) + n}$ outside a set of probability 2^{-n}. The Gaussian absolute size of *every* step-function discrepancy, using m_n equal-weight children per channel, is at most that maximum times

$$C(j\tau_n\sqrt{m_n} + 2n/\sqrt{m_n}).$$

This tends to zero independently of packet sizes. For the residual, the number of tests at this stage is finite and already fixed. By continuity of the finitely many multiplier functions, choose child intervals so small that each residual has Gaussian standard deviation small enough for their joint probability of any error greater than $1/n$ to be $\le 2^{-n}$ by the elementary union bound and normal tail inequality. The two real components of each complex observation are included. For arbitrary fixed spectral vectors there is also strong (nonprobabilistic) convergence to their target multipliers whenever the first channels agree with that target: exceptional children have σ_l-mass at most n/m_n, the pointwise good errors tend uniformly to zero, and arbitrary squared integrable densities are absolutely continuous with respect to σ_l. This carries to the full Gaussian Koopman actions as in the companion note.

Selecting the sets and carrying the masks For a target δ, take exactly the compact products, cyclic tags, and finite lexicographic copies in the companion proof. They use countably many target transformations; all act in each channel by a function $\zeta_l e(b_l\theta)$ as above. At a given target their orders and exponents are bounded across the channels, though the bound may depend on the target. Choose a sequence of stage-target assignments repeating each target infinitely many times, taking only stages large enough for the appropriate menu. At each assigned stage use the packet for its first n target channels. The union D meets every Bohr neighborhood on every tail since each sufficiently late packet already meets all n-frequency tests of tolerance $1/n$. A fixed target is a strong limit along its assigned packets. Hence any actual integer avoiding event has measure at most the family capacity δ by the upper proof of the companion note.

The lower masks of that proof can all be chosen as finite product masks in finitely many Gaussian sign bits and phase sectors. Indeed take finite Boolean independent masks in the corresponding factors, the finite C_q tag and products of these choices. They exactly

avoid the target family at some fixed h: in each lexicographic target, the first factor with a nontrivial finite menu move enforces avoidance in its own leaf regardless of arbitrary powers on the later factors. Choose such an ideal mask with mass arbitrarily close to δ. Delete a small margin around every tested real sign boundary and around every sector boundary in the relevant complex Gaussian planes; this costs arbitrarily little because there are finitely many absolutely continuous observables. Also discard the union of the exceptional tests above on sufficiently late stages. If this retained event contained an h-run at a late packet time, the uniform comparisons at the initial site would give exactly the same sign and sector labels at every target slot (the actual slots lie in the margin-trimmed event), hence a forbidden run of the original untrimmed ideal mask. The finite earlier packets are repaired using the same-system Rokhlin mask in the companion note. This proves the fixed-system lower bound.

Unrestricted cap. The same construction supplies a near-full blocker in other integer actions. In each of the finitely many leaves fix the first Boolean finite graph, or use the pure order-three tag if applicable. The product directions in each compact leaf have a nonzero bit on that first graph or a nonidentity tag. Thus every target in the finite lexicographic recursion has at least one among a fixed finite set of channels where its action is multiplication by a constant nonidentity root of unity. For each packet time assign one such channel and one such phase (subdividing ties consistently). The uniform first-order estimates hold in each of the resulting finitely many classes. If the phase ζ has order q, apply independent root rows of strides $1, q, \dots, q^{r-1}$ to that channel, using the integer Gaussian multipliers $e(\theta/q^j)$. At original time-stride $q^j d$ each row has the same first-order approximation to multiplication by ζ as before. Its event is retained if every complex row is bounded above, every late error test passes, and at least one row lies in a closed small disk about 1; choose disk radius $< |1-\zeta|/4$. The disk has positive Gaussian measure, so by independence the OR event has measure arbitrarily close to one as r increases. After making the bounds and tails large, all further losses are arbitrarily small. A hypothetical sequence of arbitrarily long runs on stages tending to infinity gives bounded bilateral limit rows; on every residue modulo its stride each row is exactly rotated by ζ at the next stride, hence is invariant at q strides. Successively choose times that differ by one of $0, 1, \dots, q-1$ times the current stride so its row is outside the disk at a positive distance. One such time always exists because two successive values cannot both be near the chosen disk (if the value is tiny it is already far away). Later choices preserve all earlier decisions. This contradicts the closed OR condition. Thus there is one finite avoiding length on all sufficiently late stages of the class; repair finitely many earlier packet times as before. The finite product of these classwise high-density avoiding systems gives $b_Z(D) = 1$. The Bohr recurrence of D follows independently from the packet construction and rules out an ordinary global Bohr separation.

Nil-Bohr density at every finite order The universal packets can in fact meet every finite-order nil-Bohr neighborhood of **every integer** on every tail. Use the joint polynomial-orbit argument and compact parameter exhaustion in the earlier nil-dense Gaussian construction [D08], based on Leibman's polynomial orbit theorem and Green, Propositions 3.2–3.3. Here is why arbitrary target phase lists are allowed. Let the independent centers have total count L, let b be a specified integer target, and let $P(k)\Gamma$ be the product nilmanifold polynomial test with $P(0) = 1$. A subsequence $k = qj$ of $(kc \bmod 1, P(k)\Gamma)$ is equidistributed in a connected subnilmanifold through the identity, with surjective torus projection. In its Lie algebra the kernel of the projection to the nil Lie algebra has dimension at least

$L - \dim G$; this is a subspace of $R^L \times \{0\}$, with jointly surjective coordinate projection onto some $L - \dim G$ coordinates. Set these linear coordinates to any representatives of the desired $arg f_l(c)/(2\pi) - bc$. The exponential point has nilcoordinate the identity and the requested shifted phases on those centers. Approaching it along the subsequence gives arbitrarily late $d = b + k$ with both strict target inequalities. No requirement on the phase values at the remaining $\dim G$ centers is needed.

At stage n test the compact coefficient boxes, the finitely many special generalized polynomial shapes and integer targets $|b| \le n$ used in the earlier construction. The dimensions of the unitriangular presentations in that finite list have a common bound t_n; continuous dependence of each fixed-time polynomial orbit on the coefficients gives a finite packet by a finite subcover for each target list. Replace the exceptional-center count n above by $\max(n, t_n)$ and choose m_n large enough after this bound is known. All Gaussian estimates still apply. The special generalized polynomial description of finite-order nil-Bohr neighborhoods gives the announced density. The fixed relative cap and unrestricted endpoint-one conclusions are unchanged.

Any prescribed countable collection of tests can have full recurrence Let $P_1, P_2, \dots$ be **any** prescribed infinite subsets of the positive integers. The universal weakly mixing transformation and packets can additionally be chosen so that *every* D constructed for any δ meets each P_i on every tail. At any layer choose the finitely many child centers generically in their required nonempty open intervals, additionally to the earlier rational-independence condition. For Lebesgue-almost every vector c of centers and each i, the vectors $(dc \bmod 1)_{d \in P_i}$ in increasing order are equidistributed on the whole product torus. For completeness, at any nonzero integer frequency the exponential functions along distinct d are orthonormal in $L^2(dc)$. Their first N normalized sum has squared norm $1/N$; along $N = j^2$ these norms are summable, giving almost-sure convergence to zero along the squares by Chebyshev and Borel–Cantelli. The gap ratio tends to one, so boundedness fills the other times. Use the countable Weyl test and intersect over i. Such vectors exist inside every required product of intervals. For each stage-n list add one arbitrarily large $d \in P_i$ realizing its specified phases on **all** centers for every $i \le n$. The finite residual Gaussian estimates are imposed after these additions and still hold. This refinement is compatible with all the nil-Bohr conditions.

In particular, for any prescribed countable collection of standard probability-preserving integer systems one may require strong positive multiple recurrence for D in every system in the collection. For each system use a countable algebra dense in its measure algebra. For each positive algebra event C and each h, the integers where its h-fold correlation is at least $\kappa(\mu C, h) > 0$ form an infinite set; one may choose $\kappa(a, h)$ nondecreasing in a, uniformly in the system, by finite Szemerédi–Varnavides averaging. List all these sets as the P_i. For any event A of measure $a > 0$, approximate it by C with measure at least $a/2$ and symmetric difference less than $\kappa(a/2, h)/(2h)$. Telescoping shows arbitrarily late D-times with A-correlation at least $\kappa(a/2, h)/2$. Thus no single prescribed countable list of standard systems suffices to detect unrestricted critical density for all integer difference sets, even if the sets are dense in every finite-order nil-Bohr topology. The newly constructed Gaussian host may depend on the prescribed list, but within it all the values δ are realized.

R023. All finite orders for uniformly bounded interval codings on arbitrary block clocks

Scope and status. Result with proof.

Result. Uniformly bounded interval codings on arbitrary block clocks give full odd-support recurrence under divergent reciprocal mass.

Scope. Some varying precompact gauge extensions are only established under the additional stated hypotheses.

Let the native axes have distinct odd prime periods p with divergent reciprocal sum, tending to infinity. Partition them into arbitrary finite blocks P_i. Each block carries an arbitrary character $\alpha_i : P_i \to T$. In the finite abelian block cocycle [R017], suppose the potential is $\phi_i(x) = f_i(\alpha_i(x))$, where f_i on the finite cyclic image has at most J constant arcs in cyclic order, uniformly in i. Both the arc endpoints and values can vary without regularity or uniform separation. Then every positive event in the resulting action has unsigned odd-support returns at every finite length. No restriction is made on the frequencies of the coordinate characters or on the growth of the block sizes.

Fix h and its subgroup H from the matching criterion. Suppose $(1, 0) \notin H$. A character of $C_2 \times K^{h-1}$ vanishing on H can be chosen with value $1/2$ at $(1, 0)$, by extension of the nonzero character on the two-element group generated by its image in the quotient. Consequently there are characters q_j of K, $0 \le j < h$, with sum zero, and summable $\eta_i \ge 0$ such that for every legal local d of tag t and phase $s = \alpha_i(d)$,

$$\sum_j q_j(f_i(y + js)) = t/2 outside a set of measure \le \eta_i. (1)$$

Indeed take $\eta_i = h$ times the probability of the cover C_i from the matching criterion. The union bound controls the h slots for each fixed d. Extend each cyclic step function f_i to T by placing one interval of length equal to the grid mesh at each grid point. Haar statement (1) is then exact, because every s is a mesh translation. By increasing η_i with any positive summable sequence we can assume $\eta_i > 0$.

Finite boundary alignment Write $g_j = q_j \circ f_i$. For large i at least one g_j has long constant arcs with different values, where long means length $> 2\eta_i$. Otherwise all g_j are constant except on sets of total measure at most $2hJ\eta_i$. Their dominant constants sum to zero, by testing at a common y (the characters sum to zero). At any fixed odd support the sum still takes value zero except on mass $\le 2hJ\eta_i$. For large i this contradicts (1).

Fix such a j, and two consecutive long arcs with different values in the cyclic list of long arcs; the physical space between them consists of at most J arcs of length $\le 2\eta_i$. Let a mark one endpoint of this cluster. For an allowed phase s, inspect the strip consisting of this translated cluster in the jth summand and the flanking intervals of length $2\eta_i$ in the two long arcs. If no other summand cuts the strip, the whole sum is constant with different values on its two flanks; at least one flank would disagree with the target in (1) on mass $2\eta_i$, a contradiction. Thus for each s some cut a' of g_k, $k \ne j$, obeys

$$||(k - j)s - (a' - a)|| \le (2J + 4)\eta_i.$$

The lifts of the equations $(k-j)s = a' - a \bmod 1$ give at most $M_0 = M_0(h, J)$ centers; all allowed phases lie in their $O_{h,J}(\eta_i)$ neighborhoods. They can be partitioned into $\leq M_0$ clusters of diameter $D_i \leq C_{h,J}\eta_i$ whose mutual distances exceed twice their maximal diameter. To see this put w equal to the initial neighborhood radius and consider the finitely many pairwise center distances. Some scale r in a list $w, (100M_0 w), (100M_0)^2 w, \ldots$ of length $M_0^2 + 2$ has no such distance in $[r, 100M_0 r)$. Chain together centers at distances $< r$. Each chain has diameter $< M_0 r$, while centers in different chains are at least $100M_0 r$ apart, and thickening by the original w preserves the claimed separation. Since $\eta_i \to 0$ this works on the circular metric for all sufficiently large i. It is harmless to discard empty clusters.

Limited cluster growth forces summable reciprocal mass List a block's axes in any order and inspect the clusters reached by all subset sums of each prefix. The number never decreases. Adding an axis translates the occupied clusters injectively into clusters of the longer prefix: the supports obtained by adding the axis to any prefix sums are all legitimate, points from the same cluster translate to points in the same cluster by the diameter/separation inequality, and distinct clusters cannot merge. At most $M_0 - 1$ axes increase the number of observed clusters. Between consecutive increases, the actual set U of occupied clusters stays fixed since old subset sums remain available. Each of these translations induces a permutation of U.

On any such segment and for any subset of its axes, let z be their sum of phases. Choose a representative partial sum from the prefix just before the segment in each cluster of U. Adding those selected axes to each representative lands in its permuted cluster, within D_i of the representative of that cluster. Summing over U gives $|||U|z|| \leq |U|D_i$, and in particular $||Qz|| \leq QD_i$ with the fixed integer $Q = M_0!$. This bound controls *every* subset of the same segment. For once $QD_i < 1/10$, choose centered representatives of the individual $Q\alpha_i(e)$ coordinates. Their positive subseries and their negative subseries have absolute sums $\leq 2QD_i$: a first crossing of the bound by increments $\leq QD_i$ would still lie below $3QD_i$, contrary to the preceding circle bound. Nonzero $\alpha_i(e)$ is a p-th root with p coprime to Q on the tail, so its corresponding absolute value is $\geq 1/p$. If $\alpha_i(e) = 0$ for a late axis, its single odd query at phase zero already contradicts (1). There are at most M_0 segments, hence

$$\sum_{p \in blocki} 1/p \leq C_{h,J}\eta_i + M_0 \max_{p \in blocki} 1/p.$$

If the sum over blocks of the last maxima converges, summation contradicts divergence of all axis reciprocals. If that sum diverges, choose one maximizing axis per block; the independent-block profile Fourier argument [R263] gives recurrence directly for arbitrary finite abelian block potentials. This proves the theorem.

A complementary flow rigidity estimate for a fixed gauge Let S_t be a strongly continuous ergodic probability-preserving real flow, f a measurable observable valued in a finite abelian group K, and $P(Y) = \sum_j q_j Y^j$, where the characters have common finite cyclic target and $P(1) = 0$. If

$$\liminf_{s \to 0} \mu[P(S_s)f \neq 0]/|s| = 0,$$

then each $q_j(f)$ is constant almost surely. Write $P = (Y-1)^r B$ with first nonzero coefficient $B(1) = \beta$ in its Taylor expansion at 1 (the zero polynomial is harmless). Along a witnessing sequence, $g_s = B(S_s)f \to \beta(f)$ in measure. Choose Q so that $Y^Q - 1$ is divisible by $(Y-1)^r$ over Z/M, where M annihilates all the character values; choosing Q a multiple of `M r!` suffices by the binomial coefficients. Then $\mu[S_{Qs}g_s \neq g_s] = o(|s|)$ by a bounded union of translates of the original defect. For fixed t, take an integer n_s with $n_s Qs \to t$. Telescoping costs $O_t(1/|s|)o(|s|) = o(1)$; strong continuity and convergence in measure imply $\beta(f) \circ S_t = \beta(f)$. Thus $\beta(f)$ is constant. Translate f into $\ker \beta$ by removing a single constant and restrict every q_j to this smaller group, observing $P(1) = 0$. Induction proves the assertion.

For a fixed measurable clock f:$T \to K$, the exact-zero sets $Z_t = \{s : P(S_s)f = t/2 a.e.\}$ are finite whenever some $q_j(f)$ is nonconstant, and near these zeros the defect is bounded below by a positive multiple of distance to Z_t. Indeed at any zero s_0 apply the preceding result to the fixed vector $g = (f(\cdot + js_0))_j$, with polynomial $\sum_j q_j(Y^j - 1)$ acting on its separate components. The defect has positive liminf divided by $|s - s_0|$. Strong continuity makes Z_t compact; a compact set of isolated points is finite. Away from those points the continuous defect has a positive minimum. Thus the same finite-cluster proof applies to arbitrary *fixed* measurable scalar Haar-clock gauges in a model where f is evaluated at an independent continuous Haar phase per block (adjoin an independent uniform initial clock there). A bounded-cut model on the native finite cyclic grid is covered by the previous argument without any assumptions about sampling representatives.

A varying L^1-precompact family of arbitrary gauges is a separate matter. For example, $f_n = (g, (S_{s_n} - 1)g)$ converges to $(g, 0)$ yet satisfies an additional mixed polynomial identity exactly at the scale s_n. The inductive restriction to $\ker \beta$ above requires a fixed f; L^1 convergence of f_n alone does not justify changing f_n by $o(s_n)$. The finite boundary alignment proof is what permits varying bounded-interval codes.

Varying precompact gauges in two instances For $K = C_2$, $h = 3$, L^1-precompactness alone suffices with arbitrary scalar-clock frequencies. Any nonzero character of the two noninitial profile bits compares a fixed pair of slots. Thus (1) becomes, after changing y, $f_i(y + cs) - f_i(y) = t$ off mass η_i, for c either 1 or 2. For two same-tag legal phases s,s', their difference r obeys $\mu[f_i(y + cr) \neq f_i(y)] \leq 2\eta_i$. There cannot be a sequence of such differences tending to zero with $\eta_i/|r| \to 0$. Otherwise choose a convergent subsequence $f_i \to f$ in L^1. As in the telescoping argument, the limit is invariant under all circle translations and hence constant. The existence of any odd query on those blocks contradicts this, since no translate of a near-constant binary function can differ almost everywhere from it. It follows that there are constants ϵ_0,C such that two same-tag legal phases with distance $< \epsilon_0$ have distance $\leq C\eta_i$ on all sufficiently late blocks. Otherwise a sequence violating successively $\epsilon_0 = 1/n$,$C = n$ gives the forbidden differences. Clustering in $T \times C_2$, the reached phases of each tag then have boundedly many clusters of diameter $O(\eta_i)$, separated by ϵ_0: a chain of η_i-size steps cannot first reach an intermediate distance below ϵ_0. The preceding permutation proof works unchanged by keeping the tag as part of the cluster state and summing its circle coordinates. Consequently the matching criterion gives full positive three-term recurrence. The analogous single-pair argument holds at $h = 2$.

For arbitrary h and an elementary abelian 2-group K, L^1-precompactness suffices also when the relevant odd singleton phases tend to 1/2 and the blocks admit even phases $s_i \to 0$ with divergent sum of absolute values. Use (1) separately on such even s_i and an odd singleton

in each same block. On a subsequence with $\eta_i = o(|s_i|)$, take $f_i \to f$. In characteristic two $\beta = P'(1) = \sum_{j\,odd} q_j$. The limiting odd equation at $1/2$ says $\beta(f(\cdot + 1/2)) - \beta(f) = 1$ in C_2, so $\beta(f)$ is nonconstant. Hence $P = (Y-1)B$ with $B(1) = \beta$. Near-zero even queries give $g_i = B(S_{s_i})f_i \to \beta(f)$ and $\mu[g_i(S_{s_i}y) \neq g_i(y)] = o(|s_i|)$. Telescoping on this *single* function shows $\beta(f)$ is constant, a contradiction. In particular the even-phase hypothesis holds for $\alpha(e_p) = 1/2 - 1/(2p)$ whenever the one-maximizing-axis reciprocal series converges: choose even subcollections of small total reciprocal weight, with the cap per block tending to zero so slowly their weights still have divergent sum. The loss from rounding the cap and enforcing even cardinality is bounded by a constant times the summable block maximum. When that maximum series diverges the general one-axis-per-block theorem already applies.

R026. Bounded-exponent translations on a connected compact nilspace

Scope and status. Result with proof.

Result. A bounded-exponent translation group on a connected compact nilspace has finite image on each fixed finite-rank observation and a compact action.

Scope. Fiberwise compactness does not imply global pure point spectrum for a bundle with moving component data.

Let Z be a connected compact finite-step metrizable nilspace, let K be an abelian group of nilspace translations of exponent $m < \infty$, and let $\pi : Z \to Y$ be a fibration onto a compact finite-rank nilspace. Then

$$\#\{\pi \circ k : k \in K\} \leq B(Y, m) < \infty.$$

In particular $\{k : \pi k = \pi\}$ is a finite-index subgroup. The K-action on $L^2(Z, \mu_{\text{Haar}})$ has pure point spectrum and is equicontinuous. No prior K-consistency of the fibration or finite generation of K is assumed. This applies to the native exponent-two high subgroup, not the generally nontorsion high subgroup on a free lift.

Fix $x_0 \in Z$. Given $k, l \in K$, use Candela, González-Sánchez and Szegedy, Theorem 1.6 to refine π to $Z \to Y' \xrightarrow{\{} p\}Y$, with the first fibration consistent for both translations. Both finite-rank factors are connected as images of Z. Thus the connected nilpotent Lie groups $N' = \Theta(Y')^0$ and $N = \Theta(Y)^0$ act transitively on their spaces, as recalled in Remark 5.3 of that paper. The induced finite-order automorphisms $k', l' \in \Theta(Y')$ centralize N'. Indeed each adjoint action on its Lie algebra is unipotent since the full translation group is nilpotent of bounded class: repeated commutators of the given element with N' yield identity after that number of steps, and differentiation at the identity gives $(\mathrm{Ad}(k') - 1)^s = 0$. Its finite order and unipotence in characteristic zero force the identity; connectedness then gives the centralization claim.

Work initially on simply connected covers, and base the homogeneous presentations at the images of x_0. Each stabilizer is closed and cocompact; its identity component is normal by the normalizer argument in the connected graph proof [R106]. Factoring it out gives effective presentations $Y' = N'_{\text{eff}}/\Lambda'$ and $Y = N_{\text{eff}}/\Lambda$, with simply connected nilpotent groups and discrete cocompact lattices. Here the subscript refers to quotienting by the connected stabilizer, and further discrete kernel is harmless. By Gutman–Manners–Varjú, Theorem 1.28, p is equivariant for a surjective continuous homomorphism on the connected groups. It lifts to simply connected covers and descends after removing connected stabilizers, with $p_*(\Lambda') \subset \Lambda$.

Because k' commutes with N'_{eff}, it acts as right translation by a representative b of its image of the basepoint: $k'(a\Lambda') = ab\Lambda'$. In particular $b^m \in \Lambda'$. Hence $\pi(kx_0) = p_*(b)\Lambda$, with $p_*(b)^m \in \Lambda$. For $Y = N_{\text{eff}}/\Lambda$ the set

$$E_m(Y) = \{a\Lambda : a \in N_{\text{eff}}, a^m \in \Lambda\}$$

is finite. To check this choose a rational Mal'cev basis adapted to Λ. The logarithms of lattice elements have coordinates of bounded denominators in this basis: their coordinates of the second kind are integers, and the coordinate conversion is a fixed rational Baker–Campbell–Hausdorff polynomial. Dividing the logarithm by m and converting back thus gives coordinates of bounded denominators for every a with $a^m \in \Lambda$. Reducing these coordinates into a bounded fundamental cell uses successively integer translations. The product law in second-kind coordinates is a fixed rational polynomial, and one can do this reduction in at most $\dim N_{\mathrm{eff}}$ steps by the triangular Mal'cev law. Denominators after the finite number of operations remain bounded in terms of Y, m, independent of all the integer translations. This yields only finitely many possible reduced representatives. Alternatively these elements lie in a common finite-index rational over-lattice of Λ.

If $\pi(kx_0) = \pi(lx_0)$, use the same common refinement for k, l. For every $a \in N'_{\mathrm{eff}}$ the two projected values at $a\Lambda'$ are $p_*(a)\pi(kx_0)$ and $p_*(a)\pi(lx_0)$, respectively. They agree everywhere by transitivity, hence $\pi k = \pi l$ on Z. The number of these maps is at most $|E_m(Y)|$. Equal maps give equal group cosets since K is abelian. Every function of this finite-rank fibration consequently has a finite K-orbit of functions, so belongs to a finite-dimensional K-invariant subspace; all such subspaces have pure point spectrum because the group is abelian. Finite-rank fibrations approximate arbitrary measurable functions by the nilspace inverse-limit theorem, proving the assertion about L^2. Equicontinuity follows from the same argument on continuous finite-stage tests: the finitely many maps πk are simultaneously uniformly continuous, and finite-stage functions approximate continuous functions uniformly on the inverse limit.

Relative and native scope If the whole jointly ergodic system is a bundle over a K-fixed base and K has finite image on almost every entire fiber, then it has a common finite image on the whole system. For every $k \in K$ the property of acting as the identity on a conditional fiber is measurable (test a countable separating algebra) and invariant under the commuting global group. Its almost-sure truth value is therefore constant. The common kernel L agrees with the kernel on almost every fiber. In the native case an infinite-dimensional kernel alone suffices for positive recurrence: arbitrarily many fresh finite singleton tag characters are independent after restriction to L, since their restrictions separate the vectors of the infinite tail subspace. A DHJ baseline built from those axes therefore admits an element of L flipping its flag without altering any point. In particular a native avoiding ergodic action can have only finite kernel, globally and on almost every fiber of any high-fixed factor.

Applied separately to conditional connected compact nilspace fibers, the theorem gives compactness of each restricted high action; this does not in general imply global pure point spectrum. The finite kernels for **chosen observations** need not be preserved by the global translations between the base points, unlike the kernels of the entire fiber actions in the preceding paragraph.

R059. Density-point nilreturn families have only endpoint capacities

Scope and status. Result with proof.

Result. Support-regular density-point nilreturn families have only endpoint integer capacities; in this class nil-Bohr recurrence implies full multiple recurrence.

Scope. The simultaneous Cesaro/Haar approximation and support condition must be stated; arbitrary sampled measurable sets are not covered.

Let Y_i be a finite-dimensional minimal nilsystem with Haar measure m_i, nilrotation S_i, base point z_i, and measurable set C_i. There may be countably many systems of unrelated dimensions and steps. Assume the sampled indicators have the following approximation property: for every $\epsilon > 0$ there is $0 \le g \le 1$ continuous on Y_i with $\int |g - 1_C i| dm_i < \epsilon$ and with the limsup initial Cesàro mean of $|g(S_i^n z_i) - 1_C i(S_i^n z_i)|$ less than ϵ. Assume every actual hit $S_i^n z_i \in C_i$ is a Haar density point of C_i within its connected nilmanifold component. For

$$D = \bigcup_i \{n \ge 1 : S_i^n z_i \in C_i\}$$

the integer all-length avoidance cap is either zero or one. More precisely, if it is positive then a *single finite nilsystem* has a positive event avoiding all D-progressions at a fixed length. The measurable nil-return theorem [D10] together with the nil-Bohr localization from the earlier notes then gives cap one. Thus in this specific class nil-Bohr recurrence at every finite order implies ordinary unrestricted measurable multiple recurrence. Open C_i work without the approximation hypothesis (use continuous nonnegative interior bumps). If only some exceptional set E contains failures of the orbit density-point property, then whenever $b(D) > 0$ one still has $b(D) = b(D \cap E)$, under the same sampling assumption: the proof gives the common finite nilsystem witness outside E, and the independent-product bound applies.

From the avoiding system to a finite nilfactor. Let A avoid D at some length h in an ergodic integer action and have positive mass. Put $f = E(1_A | Z_{h-1})$. The BHK uniform-null comparison (or Wiener for pairs) shows every weighted mean against one of the C_i is still zero after this replacement, since the original nonnegative correlation is zero at every hit. A positive superlevel event $B_0 = \{f > t\}$ on the pronilfactor inherits these zero weighted means. Choose a finite nilfactor Y so that $B = \{E(1_{B_0}|Y) > 1 - 1/(2h)\}$ has positive mass, by martingale convergence. At each individual step, whenever all projected slots lie in B, conditional union bound for the initial probability kernel gives conditional probability at least $1/2$ of membership in B_0 at every slot. Thus the weighted B means on Y vanish for **every** i. The finite factor and event were chosen only once. This is the finite nilmodel reduction [R123].

From zero means to actual steps. For each i take the closure of the two-parameter orbit

$$((T^{m+jn} y_0)_{0 \le j < h}, S_i^n z_i), (m, n) \in Z^2,$$

in $Y^h \times Y_i$, with T its nilrotation. Leibman's multiparameter orbit theorem gives a finite union of compact connected homogeneous subnilmanifolds and normalized Haar laws there with positive weights. Denote the resulting joining by Λ_i. Averaging first in m along progressively long rectangles gives the initial Cesàro mean in n with y integrated against Haar on Y. The Y slots and the external slot have Haar marginals. Continuous tests lift to the measurable B by marginal L^1 approximation; for the external C_i use its stated two approximations. Consequently $\Lambda_i(B^h \times C_i) = 0$. For open C_i one may instead use all continuous nonnegative bump functions supported there and positive on a countable cover.

Let $B\dagger$ denote the Haar density points of B within the connected components of Y; its measure is the measure of B. Fix a hit d in the ith family. For any $y \in Y$ the tuple $((T^{jd}y)_j, S_i^d z_i)$ belongs to the orbit closure by density of the diagonal m-orbit in Y. Suppose all its Y slots are in $B\dagger$. Each connected joining component projects with full differential rank to the connected component of every target slot. Indeed the component Haar marginal is absolutely continuous relative to the full slot-Haar marginal, since the component has positive joining weight. The orbit projection has constant differential rank by homogeneity; a smaller rank would give zero ambient volume for its image. Now move the tuple by Haar probability on successively small Lie-algebra boxes in the connected transitive subgroup of its component. In each projected slot the law is supported in a radius-$O(r)$ chart ball and has density $O(r^{-d_{\text{slot}}})$ there, by the submersion theorem in smooth local coordinates; the constants need not be uniform in i or the component. Therefore in each Y slot it lands in B with probability tending to one, and in the external slot it lands in C_i with probability tending to one by the density-point assumption. In zero-dimensional slots the distributions are point masses and the same conclusion holds. The simultaneous event then has positive probability, although the perturbation law on the joint component is absolutely continuous to its Haar measure. This contradicts the zero mass. Thus $B\dagger$ avoids the whole union pointwise. Countably many different perturbation scales cause no loss of Haar mass because it is the same set of density points for all i.

Fat irrational-only circle examples. The missing-eigenfrequency construction [R151] can enforce the orbit density-point hypothesis at no cost. For a prescribed irrational β, when deleting an open interval I_j centered at a rational q_j, additionally bound its radius by $dist(q_j, n\beta)/4$ and its length by $2^{-j}dist(q_j, n\beta)^2$ for every $1 \le n \le j$. Choose endpoints outside the countable orbit. These finite additional conditions allow a nonempty interval. For a surviving $x = n\beta$ the finitely many intervals with $j < n$ or any prescribed earlier j have closures separated from x. Among the remaining intervals meeting the radius-r ball about x, their centers are at distance at most $2r$, so their total length is $O(r^2)$. Hence every surviving orbit point has density one. When the intervals also obey the per-prefix visit caps in the linked note, the sampled indicators satisfy the stated approximations: the first finitely many arcs have zero-boundary weights, while the remaining error tends to zero uniformly on prefixes and in Haar measure. This covers countably many arbitrarily thin prescribed fat sets regardless of the rational rank of the angles.

Essential support suffices. In fact each external hit need only have positive Haar mass of C_i in *every* neighborhood; its local density may tend to zero. In the joining above the marginal on each pair $(Y - slot, externalY_i)$ is the product of the two Haar measures, by the diagonal m-average for each fixed n and the subsequent external orbit average. On each positive-weight connected component, projection onto this pair thus has full differential rank by the same homogeneity argument. Write H for the connected transitive subgroup of that component, and K^0 for the identity component of the kernel of its projection onto

the external Lie group. Point stabilizers in the ambient nilmanifolds are discrete. Pair-submersivity therefore implies K^0 projects submersively to **each old slot**, while fixing the external point exactly.

At a supposed tuple z with all old slots density points of B, first sample a small absolutely continuous Lie-neighborhood law for $u \in K^0$. Each projected slot of `u z` lies in B with probability $> 1 - 1/(4h)$; the projected slot laws also have L^1 densities. Hence after left translation by any sufficiently small $w \in H$, each projected slot of `w u z` still has probability $> 1 - 1/(2h)$ in B, by total variation continuity of translates of each L^1 density. This holds uniformly on a neighborhood of the identity in H chosen for this tuple. Now sample w independently from an absolutely continuous law positive throughout that neighborhood. The external slot of `w u z` equals that of `w z`, and its marginal density is positive throughout a neighborhood of the original hit by submersivity. It therefore lies in C_i with positive probability by the essential-support assumption. Conditional on every such w, all old slots simultaneously enter B with probability greater than one half. The joint law of `w u z` is absolutely continuous on its component (fix u, then average over w). This contradicts the same zero weighted joining mass. The conclusions, including localization to the exceptionally unsupported hits, hold with this sharper hypothesis.

Shorter route to the capacity statement, and an arbitrary-set reduction. For arbitrary $D \subset N$ define D^+ as the set of integers each of whose finite-order nil-Bohr neighborhoods meets D in positive upper Banach density. If D has a positive avoiding event, Huang–Shao–Ye, Corollary 2.4 supplies a nil-Bohr neighborhood U of zero with $d * (D \cap U) = 0$: a positive-return set contains that neighborhood up to removal of a zero-upper-Banach-density set. Take U open by shrinking. It is disjoint from D^+ by the definition. Both D^+ and $D \setminus U$ therefore have cap one by nil-Bohr localization. Independent intersections of avoiding masks give

$$b(D) = b(D \setminus D^+) = b(D \cap U) \quad \text{whenever} \quad b(D) > 0.$$

In particular any hypothetical intermediate value is already realized by an upper-Banach-density-zero subset of its original difference set. For the support-regular families in this note every included step is in D^+. To see this, join the external orbit with that defining any nil-Bohr open neighborhood of the given integer step. On its one-parameter orbit closure, choose an open neighborhood of the orbit point above that step still meeting the prescribed test. The connected component projects submersively to the external slot by its Haar marginal; an interior bump positive at the orbit point pushes to external density positive on a neighborhood of the hit. Essential support of C_i therefore gives positive joint mass. Equidistribution and the assumed simultaneous Cesàro/Haar approximation give positive lower natural density of steps in the intersection with the nil-Bohr test. This proves the endpoint dichotomy directly; the preceding two-parameter argument additionally exhibits a common actual avoiding nilsystem.

R119. Multiple returns for compact group cocycles over a finite-order-ultrafilter rigid base

Scope and status. Result with proof.

Result. Finite-order additive ultrafilter rigidity lifts to multiple recurrence in compact group Haar cocycle extensions.

Scope. The base must be strongly rigid along the ultrafilter; abelian fibers have the stated uniform lower bound.

Let P be an additive ultrafilter on the integers such that $P^{*n} = P^{*(n+1)}$ for some n, and let S preserve a standard probability space (Y, ν). Suppose $S^d \to I$ in the strong Koopman topology along P, for example for an ergodic compact abelian rotation. For an arbitrary measurable cocycle $\alpha_d(y)$ into a compact metrizable group K, consider $T^d(y,k) = (S^d y, k\alpha_d(y))$ with base probability times Haar. Every positive-measure event A and finite h then satisfy

$$P - \lim_d \mu(\bigcap_{j=0}^{h-1} T^{-jd} A) > 0.$$

The same argument handles compact homogeneous-space fibers by pulling back to their Haar group cover. It is a result for one isometric extension over a P-rigid base, including nonabelian K and arbitrary measurable cocycles.

Write $a_d(y) = (\alpha_{jd}(y))_{0 \le j < h} \in L = K^h$. Let κ_y be the conditional P-limit kernel of the point masses at $a_d(y)$. In precise terms, for each continuous f on L form the weak-star P-limit of $f(a_d)$ in $L^\infty(Y)$. By a countable dense continuous test algebra, positivity, the mass bound, and Riesz representation, these are the moments of a jointly measurable probability kernel. All subsequent tests can be multiplied by any fixed integrable base function. The constant-in-y tests extend to bounded measurable base-dependent continuous fiber tests using simple approximation in $L^1(Y; C(L))$.

The kernel along the additive power P^{*r} is κ_y^{*r} for each r. To justify inductively, every finite power of P is still rigid on the base: the weak Koopman limit remains I by separate operator continuity, and weak convergence to I of unitaries implies strong convergence. Powers j likewise preserve rigidity. For fixed d and a variable e summed on the inside of an iterated limit, the cocycle identity gives

$$\alpha_{j(d+e)}(y) = \alpha_{je}(y)\alpha_{jd}(S^{je}y).$$

The last factor can be replaced by $\alpha_{jd}(y)$ with error tending to zero in measure along the inner ultrafilter, by approximating that fixed measurable K-valued function with a finite partition and applying base rigidity. The conditional inner integral is therefore convolution with the fixed tuple $a_d(y)$. Taking the outer P-limit gives the stated convolution law: for a continuous f the kernel-dependent test $u \mapsto \int f(vu) d\kappa_y^{*(r-1)}(v)$ is measurable in y and uniformly equicontinuous in u. Its finite simple approximations are valid in the kernel formula above. Orientation issues are harmless since the same κ is repeated.

Consequently almost everywhere $\kappa_y^{*n} = \kappa_y^{*(n+1)}$ as measures. We need only that the identity belongs to the support of κ_y. Here is a proof valid for every compact metrizable group. Write $\sigma = \kappa^{*n}$; then $\sigma * \sigma = \sigma$. An idempotent probability on a compact group is Haar on a compact subgroup. One elementary proof: its support is a compact subsemigroup, hence a group H (powers approximate the identity and then inverses). On each finite-dimensional unitary irrep, its Fourier transform is an idempotent contraction, hence an orthogonal projection. A vector in the range is fixed by almost every element of H, by equality in the averaged Hilbert norm, and thus by every element of H by continuity. Conversely every H-fixed vector belongs to the range. These are precisely the Fourier transforms of Haar on H, implying equality by Peter–Weyl.

For each finite-dimensional unitary irrep of L the Fourier matrix $M = hat\kappa$ satisfies $M^n = M^{n+1}$; its eigenvalues are 0 and 1, hence $\operatorname{Tr} M = \operatorname{Tr} M^n$ even if M is not normal. It follows that κ and σ agree on continuous class functions, by character orthogonality and density of their span for central functions (equivalently average Peter–Weyl approximants under conjugation). Every identity neighborhood contains the support of a nonnegative continuous class function equal to 1 at the identity: compact groups have a compatible bi-invariant metric, or one may conjugation-average a bump in a smaller invariant neighborhood. The σ integral of such a bump is positive. So every identity neighborhood has positive κ-mass. In the abelian case κ already equals σ by the scalar Fourier transform.

Finally strong base convergence allows $1_A(S^{jd}y, k)$ to be replaced by $1_A(y, k)$ in integrated L^1 even after the k-translation by $\alpha_{jd}(y)$, using Haar invariance and finite rectangle approximation. Write A_y for the section. The limiting return is

$$\int_Y \int_L \int_K \prod_{j<h} 1_{A_y}(kv_j) dk d\kappa_y(v) d\nu(y).$$

The inner function of v is continuous for almost every y by continuity of translation in $HaarL^1$, is bounded, and equals $m_K(A_y)$ at the identity tuple. It is a measurable base-dependent continuous test as above. On every positive section the κ_y integral is therefore positive. Integrating proves the claim.

A second easy category from the finite-order projection lemma [R001] is any mildly mixing system that is pairwise independently determined, meaning that every self-joining with product two-slot projections is the product joining. The P-limit of each U^{jd} for nonzero j is projection onto constants, since any nonconstant rigid function would give an ordinary rigidity subsequence and a nontrivial rigid factor. Thus the P-progression joining is pairwise independent and hence product. This yields the full product return at every length in that category.

Uniformity for abelian fibers For any compact group K, any closed subgroup $J \le K^h$ containing the full diagonal, and any Haar-measurable $B \subset K$ of mass at least $a > 0$,

$$m_J(B^h) \ge c(a, h) > 0,$$

uniformly in K,J. For the finite density Hales–Jewett number N at density $a/2$, sample independent Haar elements $V^i \in J$ and a Haar $k \in K$. At word w in the length-N cube with

alphabet of size h observe $k \prod_i V^{i_{w_i}}$, in this fixed factor order. Every word has probability at least a of visiting B. With probability at least $a/2$ the visiting set has density $\geq a/2$ and contains a combinatorial line. For each fixed line, the full tuple of observed group elements is Haar in J: the passive factors are diagonal elements in J, and at least one independent factor is Haar on J. Thus a union bound over the finite possible lines proves the estimate.

For compact abelian K the conditional joint kernel κ_y itself is Haar on a subgroup H_y, and translating it by independent diagonal Haar gives Haar on $J_y = \Delta K + H_y$. Apply the bound on sections $m_K(A_y) \geq \mu(A)/2$, whose total probability is bounded below in terms of $\mu(A)$. The resulting P-limit lower bound depends only on $\mu(A)$ and h, not on the compact group, the measurable cocycle, or the rigid base. Thus approximation by such factor events with uniform positive mass also transfers the multiple return to the event being approximated.

R133. Which spectral fibers can support a Gaussian avoidance witness?

Scope and status. Result with proof.

Result. Gaussian joining positivity reduces null rectangles to finite atomic identity-missing spectral fibers via deterministic selectors.

Scope. A theorem for Gaussian actions and their covariance limits, not arbitrary probability-preserving systems.

The higher-order full-rank positivity and finite-atomic necessary condition were already proved in the earlier Gaussian witness note [D09] using semigroup smoothing. This note records an independent Cameron–Martin proof and the conditional-support reduction to the finite-atomic part of the limiting joining for arbitrary ultrafilters.

Let the integer pmp action be a Gaussian action on a standard probability space, generated by a real separable Hilbert space of centered Gaussian observables. Write U for its orthogonal action on that space. Let p be any ultrafilter on positive integer differences. We consider the ultralimit of the joint distributions of this Gaussian system at times $0, d, \dots, (h-1)d$, with $h \geq 2$ fixed. This is about a limiting joining. Its zero correlations need not be exact zeros on any single member of p.

Spectral description. On complexifying the first chaos, use the usual spectral representation of U with maximal spectral type σ on the circle, including the multiplicity spaces. For every integer l the weak operator ultralimit of U^{ld} is a scalar multiplier $\phi_l(z)$ in the abelian von Neumann algebra generated by U. All these multipliers can be chosen so that, σ-almost everywhere, $\phi_0 = 1$, $\phi_{-l} = conj\phi_l$, and the sequence $\phi_l(z)$ is positive definite. Indeed every scalar trigonometric square is positive before taking limits; take rational coefficients, test by nonnegative integrable spectral functions, and use countability and continuity. There is therefore a unique probability ν_z on the circle with moments $\phi_l(z) = \int u^l d\nu_z(u)$. This yields a measurable family and the reality symmetry under conjugation of z.

The limiting joint Gaussian covariance on the time slots is the multiplier with matrix

$$C_h(z) = (\int u^{j-i} d\nu_z(u))_{0 \leq i,j < h}.$$

This assertion follows first for finite linear Gaussian cylinders by their characteristic functions and then for all bounded measurable slot tests by a marginal L^1 approximation. Approximating cylinder tests by bounded continuous cylinder tests is legitimate for every d and for the limiting joining since their one-slot marginals agree. A vector in the kernel of $C_h(z)$ gives a polynomial of degree at most $h-1$ vanishing on $supp\nu_z$ and conversely. Thus $C_h(z)$ is invertible exactly where ν_z has at least h distinct support points.

Full-rank fibers allow positive joint rectangles. First restrict the underlying Gaussian space to the conjugation-invariant spectral set G where C_h is invertible. The increasing sets $G_m = \{z \in G : \lambda_{\min}(C_h(z)) \geq 1/m\}$, with their reflected conjugates automatically present, exhaust G. On each corresponding real Hilbert subspace H_m, every independent tuple of Cameron–Martin shifts in H_m^h belongs to the Cameron–Martin space of the joint law: apply the bounded inverse of the covariance square root on this subspace. In particular independent such shifts preserve joint null sets. No uniform lower bound over G is required.

Choose an orthonormal basis of the real Hilbert space over G in $\bigcup_m H_m$, by Gram–Schmidt on a countable dense subset, and let L be its rational span. Each element of L^h may shift the slots independently preserving joint null sets. For every positive-measure one-slot Gaussian event A, the union of its translates by L has full marginal Gaussian measure. Indeed in the independent standard normal coordinates of this basis the union is invariant under all finite rational translations. Invariance on each finite-dimensional fiber implies a constant there almost everywhere, by equivalence to Lebesgue measure and ergodicity of dense rational translations on Euclidean space. The resulting event is in the tail σ-algebra modulo null sets, which is trivial by independence. If $A_0, \dots, A_{h-1}$ have positive marginal measures, a zero-measure rectangle in the joint law would leave all countably many independent translates of that rectangle null. Their union has full joint measure by the marginal saturation just proved. It follows that the rectangle has strictly positive joint measure.

Reduction. Write $E_h = \{z : |supp\nu_z| < h\}$. The Gaussian spaces generated by G and E_h are independent in the original process and in the limiting h-slot joining; cross covariances vanish. Given any positive event A, put $A_* = \{E(1_A|Gaussian factor over E_h) > 0\}$. Its mass is at least the mass of A. Given a tuple of E_h states in A_*^h, each respective section of A in the G Gaussian space has positive measure. Their limiting G-rectangle has strictly positive probability by the preceding paragraph. Disintegration gives the equivalence

$p\text{-}\lim \mu(\bigcap_{j=0}^{h-1} T^{-jd}A) = 0$ if and only if the limiting E_h joining gives zero to A_*^h.

In particular, an exact avoiding witness for a member of p in a Gaussian system requires a positive-mass obstruction in this limiting joining on fibers with at most $h-1$ spectral phase choices. It does not automatically give an *exact* avoiding witness on the reduced system for one member of p. At $h = 2$ those fibers have $\nu_z = \delta_{\phi_1(z)}$. There the weak limit restricts to a unitary multiplier and the powers converge strongly along p on that entire subspace (expand the squared norms). The limiting two-slot joining is its deterministic graph. If this multiplier acts identically the graph is diagonal and no positive pair obstruction is possible. An arbitrary Gaussian model need not have a common constant value of the phase multiplier on its rigid part, and unitary strong limits need not be uniform on a single member of p.

The criterion treats all Gaussian events, including functions of infinitely many observables and limiting covariances whose positive eigenvalues approach zero. It is restricted to Gaussian systems and supplies neither a uniform all-length density gap for arbitrary systems nor an integer example with prescribed threshold.

Rectangle zero sets and deterministic spectral selectors Here is a further refinement. Put $S_h(z) = supp\nu_z$ if this support has fewer than h points, and put $S_h(z) = T$ otherwise. A selector is a measurable phase $w(z) \in S_h(z)$ respecting conjugation, so multiplication by w is a real orthogonal operator on the first chaos. Let V_w be its associated Gaussian transformation. For any h measurable Gaussian events $A_0, \dots, A_{h-1}$,

$$p - \lim \mu(\bigcap_{j<h} T^{-jd} A_j) = 0$$

$$\iff \mu(\bigcap_{j<h} V_w^{-j} A_j) = 0 \quad \text{for every} \quad selector w. (\dagger)$$

In the implication from right to left it even suffices to test selectors from the actual $supp\nu_z$. This is an equivalence for zero probabilities; it is not equality between probabilities or exact avoidance sets of integer times.

Concentrating a finite phase law on one supported choice. Work first on the part E_h with fewer than h phases. Choose any target branch w in the support. Enumerate the remaining at most $h-2$ phase branches measurably, giving absent branches weight zero. One construction enumerates angles on the original upper semicircle of z and reflects on the conjugate semicircle, so all multipliers are real-compatible. At the real spectral atoms $z=\pm 1$ the ultrafilter phase is a single real sign. For each branch u its channel in the joint Gaussian law may be generated with a separate independent Gaussian input. Its map on the first chaos is $L_U f = (U^j f)_{0\le j<h}$ where U multiplies by $u(z)$; its spectral intensity is multiplication by the branch weight. This gives the displayed covariance, so it gives the same joint law.

Suppose a rectangle is null for this law. Transfer an intensity A from a source branch U to the chosen target W, of current intensity $B>0$ almost everywhere. First restrict the transfer to the spectral subset $J=\{B\ge 1/m\}$. All intensities commute with all branch multipliers. On the Hilbert space over J choose finite-rank orthogonal projections P_n converging strongly to its identity such that for $Q=V_u^{-1}V_w$ **on the first chaos**, $\|[P_n,Q]\|_{HS}\to 0$. Explicitly, start with the first k of a countable dense list of vectors and partition the spectral values of the one circle multiplier Q into l arcs. Project onto the span of the restrictions of the vectors to these arcs. The rank is at most `kl`; the arcwise constant approximation to Q commutes with the projection, so the commutator has Hilbert–Schmidt norm at most a constant times $\sqrt{k/l}$. Let $k\to\infty$ and $l/k\to\infty$. Choose a conjugation-stable partition and real basis vectors so the projections respect the real structure. Extend P_n by zero off J.

Fix $0<\alpha<1$. Decrease the source covariance by $D_n=\alpha A^{1/2}P_nA^{1/2}$ and increase the target covariance by the same finite-rank positive operator, leaving the other channels unchanged. The input law after these changes is absolutely continuous with respect to its initial law. For the source, apply the strictly positive finite-rank perturbation $I-\alpha P_n$ to the covariance of the standardized source noise before weighting by $A^{1/2}$. For the target, the standardized-noise covariance changes by the finite-rank positive operator $B^{-1/2}D_nB^{-1/2}$, bounded on J and zero off it. On the finite-dimensional subspaces affected the two Gaussian laws are equivalent, and their orthogonal complements retain their original laws. Pushing these independent inputs to the joint space keeps the rectangle null.

In time slot j the modified marginal covariance differs from the original identity by

$$W^jD_nW^{-j}-U^jD_nU^{-j}.$$

Its Hilbert–Schmidt norm is at most $\alpha\|A\|\|[P_n,Q^j]\|_{HS}$, tending to zero for each fixed j. Only the ratio Q enters, even if the two branch phases together have high-dimensional joint spectrum. The marginal Gaussian laws converge in total variation: the finite-rank covariance perturbations have operator norm tending to zero and squared Hilbert–Schmidt norm tending to zero, so the Gaussian relative entropy is bounded by a constant times that squared norm, by the finite-dimensional determinant formula and expansion of $x-\log(1+x)$. Pinsker's inequality applies. The joint covariances converge weakly to the law transferring $\alpha 1_J A$ from U to W because $P_n\to 1_J$ strongly. Approximation in each original marginal by bounded continuous Gaussian cylinders, the marginal total-variation bound and finite-

dimensional Gaussian convergence now preserve the rectangle nullity in this limit, even for arbitrary measurable slot events.

Let $m \to \infty$ and $\alpha \uparrow 1$. All the limiting laws in the last paragraph have *exactly* their original one-slot marginals. The same cylinder approximation transfers nullity to the law moving the entire source weight to the target branch. Repeat for each of the finitely many source branches. The final joint law is the deterministic graph of powers of V_w. Thus on E_h every originally null rectangle stays null for every target selector.

For the full joint law, take the positive conditional-support event of each A_j on the factor over E_h. The full-rank positivity result above shows these form a null rectangle whenever the original rectangle is null. Concentrate the law on E_h as just proved. Extending the resulting deterministic action on G by *any* real-compatible phase leaves the original rectangle null, since each A_j is contained, modulo a marginal null set, in the inverse image of its positive conditional support. This gives the forward implication of (†).

Approximating by deterministic actions. At each atom of σ the p-limit of the scalar phase z^d is deterministic and ν_z is its point mass. On the nonatomic part, there are measurable selectors $w_n(z) \in supp\nu_z$ whose powers converge weak-star, for each fixed j, to $\phi_j(z)$. For completeness, choose progressively fine finite partitions there with maximal σ-cell mass tending to zero. On each cell of the upper semicircle independently draw one uniform real parameter b. At each point in the cell choose the circle phase given by the b-quantile of ν_z, with a fixed cut in the circle; reflect the choice on the lower semicircle. Quantiles are jointly measurable, have the specified laws, and take values in the support almost surely, after modification on a null set. For any bounded test f and fixed moment the weighted spectral integral has its prescribed expectation. Its variance is at most a constant times $\|f\|\infty^2$ times the maximal cell mass since distinct cells (and their reflections grouped with them) are independent. Use sufficiently fine partitions for the initial segments of a countable dense family of tests and moments and select successful deterministic realizations. This proves the stated weak-star convergence by bounded approximation. The real points ± 1 have already been covered as atoms, if present.

The Gaussian deterministic graph joinings from w_n converge on rectangles to the initial joining since their covariances converge and all marginals are unchanged. If every such graph assigns zero, the limiting rectangle is null. This proves the converse and (†).

A positive same-set obstruction must omit the identity phase. If at almost every spectral point either $supp\nu_z$ has at least h points or it contains 1, then the identity is a selector, and (†) yields strictly positive limiting correlation for the h copies of every positive Gaussian event. More generally, on the part of E_h where 1 occurs choose it as the target in the transfer argument and leave the other finite phase fibers alone. The resulting null joining for any original same-set obstruction is deterministic diagonal on the identity-bearing fibers. Disintegrating there, and first replacing sections on the full-rank part by their positive supports, gives a single event on the remaining finite, identity-missing spectral part which has mass at least $\mu(A)$ and is null in its limiting self-joining. In detail the mean, over diagonal states, of those support masses is at least $\mu(A)$, and almost every chosen diagonal-state section is null under the residual joining. In general one must fix such a common state; simply taking the whole support of a projection onto the remaining spectral part would lose the shared condition between the slots.

R136. Continuous characters forced by a hard tail ultrafilter

Scope and status. Result with proof.

Result. A hard tail ultrafilter in the specified prime-ratio shell model induces a continuous finite-atomic character with a quantitative variation bound.

Scope. This is a necessary condition under an explicit hardness hypothesis, not existence of a hard ultrafilter.

This is an internal derivation for the positive-unit vertical family in the shell note [R199]. Suppose the primes are eventually congruent to 1 modulo every prescribed modulus, $q_i/p_i \to \theta$ with θ positive irrational, and every step has odd centered $\sum_{iactive} a_i$ and $\sum_{iactive} |a_i|/p_i \leq \tau < 1$. A tail-supported ultrafilter P contains the steps confined to each sufficiently late tail. Call it hard at gap η if $\sup_{E\in P} b(E) \leq 1-\eta$; use capacities for the squared-prime group, or consistently the corresponding tail capacities. This is an extra hypothesis on an ultrafilter; finite partitions of a set of positive critical density can be addressed separately by the interpolation criterion [R145].

Uniform continuity. For a sequence of coefficients $k_i \in F_{p_i}$, let

$$\rho(k) = P - \lim_{d} \sum_{iactive} a_i k_i/p_i \in T.$$

This is an additive character on the full grid product. There exist an integer M depending on η and a modulus of continuity depending on η,τ,θ such that the map $(z_i) \mapsto M\rho(k)$ for $z_i = q_i k_i/p_i \bmod 1$ is uniformly continuous in the tail seminorm $\limsup_i ||z_i||$. The same M works at every neighborhood of zero.

Indeed first fix a star with total site loss $< \eta/2$. The quantitative Weyl test selects M using only the box tolerance and dimension; after that a desired positive angular separation γ determines the hitting time. Choose the correction parameters H,L for γ. When $\limsup_i ||z_i|| < \delta$ for δ smaller than (say) $\theta/(4LH)$, the sparse correction device [R199] proves that the steps with distance $||M\Sigma a_i k_i/p_i|| \geq \gamma$ in some fixed tail admit a common blocker of mean $> 1-\eta/2$. This subset cannot belong to P. Thus $||M\rho(k)|| \leq \gamma$. Additivity proves uniform continuity at every point. The subsets used here are ordinary subsets of the countable direction set, hence define clopen neighborhoods in its ultrafilter space. No measurability of the ultralimit as a function of arbitrary coefficient data is assumed; continuity was proved.

Atomic representation. There are finitely many distinct free ultrafilters U_j on the axes and nonzero integers n_j such that

$$M\rho(k) = \sum_j n_j \lim_{i\to U_j} q_i k_i/p_i (\mathrm{mod} 1)$$

for every coefficient sequence. The total $\Sigma|n_j|$ has a bound depending only on the uniform continuity modulus, hence on η,τ,θ. Equivalently, on the tail where M is invertible modulo the p_i,

$$\rho(k) = \sum_j n_j \lim_{i \to U_j} q_i(M^{-1}k_i)/p_i(\mathrm{mod}1).$$

To prove this put $K = \beta N \setminus N$. Every circle sequence on the natural numbers has a continuous extension to βN, and two such extensions agree on K exactly when their difference tends to zero. Every continuous function from K to the circle lifts continuously to the real line: using zero dimensionality and compactness, partition K into finitely many clopens on which one can choose a branch. Its real lift extends to βN by Tietze. Rounding the resulting sequence to the p_i-grid, whose mesh tends to zero, does not change its restriction to K. Thus the coefficient product modulo tail-distance zero represents $C(K,T)$ with its uniform metric. The preceding character therefore extends to a continuous character on this function group.

Compose with real exponentiation to get a character on the Banach space $C(K,R)$. For each f the character on the line `t f` is $e(tL(f))$ for a unique real slope. Additivity gives linearity of L. Continuity at zero implies boundedness: if $||f||\infty \leq \delta$ ensures the character remains in the arc of radius 1/4, apply this also to $tf, 0 \leq t \leq 1$, yielding $|L(f)| \leq 1/4$. Hence L is integration against a finite signed regular measure. The measure of each clopen is an integer by exponentiation of its indicator. If its support were infinite there would be infinitely many disjoint clopens bearing nonzero variation; inside each choose a clopen with nonzero signed mass by regularity and the clopen approximation of continuous functions. Infinitely many disjoint nonzero integer masses contradict finite variation. Consequently the measure consists of finitely many integer atoms. After choosing disjoint clopens about them, a clopen sign function times $1/(2\Sigma|n_j|)$ has character 1/2; the uniform modulus at 1/4 bounds the total weight.

Parity and signed cost. Define real and mod-2 finitely additive charges on axis subsets by

$$\nu(S) = P - \lim \sum_{i \in S, active} a_i/p_i, \pi(S) = P - \lim \sum_{i \in S, active} a_i \bmod 2.$$

They have the form

$$\nu = \sum_j [b_j + (\theta - 1)n_j/M]\delta_{U_j}, \pi = \sum_j (b_j \bmod 2)\delta_{U_j},$$

where $b_j \in Z$, Σb_j is odd, and $\Sigma|b_j + (\theta - 1)n_j/M| \leq \tau$. For k equal to the indicator of S, the last character formula gives $\nu(S) \equiv (\theta - 1)\sum_{S \in U_j} n_j/M \bmod 1$, by eventual congruence of both primes to 1 modulo M. In particular, clopens disjoint from the finitely many indicated ultrafilters have integer mass of magnitude < 1 and hence mass zero. Decomposing into disjoint clopens about the ultrafilters yields the real formula. Now use $k_i = (p_i + 1)/2$ on S and zero elsewhere. The original character equals $(\pi(S) + \nu(S))/2 \bmod 1$. The represented one equals $(\theta - 1)\sum_{S \in U_j} n_j/(2M) \bmod 1$, since the eventual congruences also hold at 2M. This proves the parity identity. In particular for each fixed gap only finitely many real values of $\nu(N)$ can occur.

The weak finite-atomic conclusion from local signed shell separation alone also has an elementary proof. Its sign-flipped tests imply the range of ν is countable. For any bounded

signed charge on the full power set of N this already forces a finite linear combination of ultrafilter charges. If there are infinitely many disjoint S_n with nonzero mass, pull back to the charge $\xi(J) = \nu(\bigcup_{n\in J} S_n)$, subtract its absolutely summable singleton masses, and take an infinite J of residual variation zero using an uncountable almost-disjoint family (only countably many of its members can have strictly positive variation). On subsets of J the charge consists solely of the point masses, whose subsums contain continuum many values, by choosing a rapidly decreasing nonzero subsequence. If no such disjoint family exists, the regular representing measure on βN has finite support by the disjoint-clopen argument. The uniform character formula above is substantially stronger. It concerns every fixed coefficient array and does not give uniform convergence for arrays subsequently selected as functions of the direction or its starting point.

R137. Iterated Hartman convolutions on translated signed boxes

Scope and status. Result with proof.

Result. Iterated Hartman convolution boxes give multiple recurrence for all translates under the stated tail spectral condition.

Scope. The finite residual-spectrum variant gives an endpoint dichotomy according to the relevant congruence class.

Qualifications. The conclusion requires the stated order of iterated limits and the tail Hartman hypotheses.

Let S_i be finite sets of integers containing zero. Suppose, for every starting index m, that there are finitely supported probability measures $\nu_{m,N}$ supported on finite sums using only indices from the tail $i \geq m$, with

$$\lim_N \hat{\nu}_{m,N}(\theta) = 0 (\theta \in T \setminus \{0\}).$$

Then every integer translate of $\sum_i S_i$ forces arithmetic progressions of every length at every positive upper density. The same conclusion holds on `gZ` after scaling, for translates by elements of `gZ`. The hypothesis is substantially stronger than Bohr denseness of the tail sums. Here all the $\nu_{m,N}$ have mass one and every sampled sum is assigned one of its finite-support representations.

Admissible independent blocks. An average with b blocks is taken successively. Choose the first (outermost) probability measure on a tail. For the second use a tail strictly beyond *all* indices in the finite first support, and iterate, always using a common new starting index for the whole preceding sample. In iterated limits send the innermost N to infinity first. The support of the complete convolution is contained in the original signed sum box at every stage. In statements about convergence below one can take nested upper and lower limits, so individual intermediate limits need not exist. All estimates apply for every fixed tail starting index. Each block has a Hartman probability sequence on its own tail. In particular its averages of $U^n f$ in any unitary representation of the integers converge strongly to the invariant projection, by the spectral theorem and dominated convergence. Replacing n by `k n` or by $k(n - n')$ for independent samples gives the invariant projection of U^k. For the latter assertion the Fourier multipliers are $|\hat{\nu}(k\theta)|^2$.

Pointwise nilsequence smoothing. For a continuous test on an ergodic s-step nilrotation, s such convolution blocks from any starting point whose orbit is dense give the Haar integral in the iterated limit (the nested limsup of the absolute error is zero). More generally for any starting point each block sum translates it inside its own orbit closure, and the limit is the ordinary orbit-mean there. We use the standard Leibman nilorbit theorem: the orbit closure of a nilrotation with its uniquely ergodic measure is again a nilsystem of no larger nilpotence step, after the usual cyclic component decomposition if needed. We also use the standard central-torus tower of an ergodic finite-dimensional nilsystem: at the last nonzero central layer its compact structure group acts freely and transitively on fibers over a factor of smaller step, with Haar fiber conditionals. One may use the presentation generated by the identity component and the translating element, passing to the simply connected cover

of the identity component. Its lower-central rational subgroups give the tower. In a finite cyclic component decomposition one may separate the residues of each sampled time modulo the component count q; they have limiting weight $1/q$, and conditioning on a residue and dividing the time by q still gives a Hartman sequence on the integers, by the finite Fourier expansion of the residue indicator.

For one-step rotations the claim is Fourier approximation on the compact orbit and the Hartman property. For the induction, fix the ergodic orbit space and its compact final central fiber group U. On the quotient every $(s-1)$-stage cluster law has Haar marginal by induction. To clarify the quantifier, fixing a finite outer block fixes a permissible tail for the inner blocks; all weak cluster laws obtained by taking the successive inner upper limits, including subsequence choices at intermediate stages, still have this marginal. Smooth any such law ρ by the same probability kernel on U, with bounded Haar density and support in a sufficiently small neighborhood of the identity. The new law has density at most the kernel sup norm relative to total-space Haar, by disintegration over the quotient. For each translate $f \circ T^n$, the change in its ρ-integral under smoothing is uniformly small, by uniform continuity of f and centrality of U. For a fixed outer probability ν the extremal inner upper limits are bounded by the supremum of $\int (\sum_n \nu(n) f \circ T^n) d\rho$ over such cluster laws. The smoothed contribution differs from the Haar integral by at most the kernel sup norm times $|| \sum_n \nu(n) f \circ T^n - \int f||_1$. This tends to zero for the outermost block by the Hartman mean ergodic assertion. Shrinking the kernel yields the claimed nested limsup and liminf, including when the inner law varies with the outer index. The proof yields the same limit with any larger number of blocks: for fixed outer samples the innermost required blocks already give the common orbit mean. A fixed initial time offset does not affect this mean.

Characteristic estimate for arithmetic progression correlations. Fix $h \geq 2$ and $d = h - 1$. For bounded functions $|f_j| \leq 1$, $0 \leq j < h$, write

$$c(v) = \int \prod_{j<h} T^{jv} f_j d\mu.$$

If any one slot f_r is orthogonal to the full Host–Kra pro-nilfactor in an ergodic integer system, the averages of this correlation along at least d of the above independent convolution blocks tend to zero, also with a fixed offset. Here is the generalized von Neumann estimate. Assign the d other slots distinct increment variables. For the variable assigned to slot j, change the integration origin by the corresponding `j u`, making that slot independent of this variable. Apply Cauchy–Schwarz over the remaining variables and the starting point, discard the bounded independent slot, and introduce two fresh independent copies of this increment. Repeating eliminates all slots except r. At the end, the absolute average to the power 2^d is bounded by the mean cube correlation

$$E_{t_1,\ldots,t_d} \int \Delta_{t_1} \cdots \Delta_{t_d} f_r d\mu,$$

with $t_i = (r - j_i)(u_i - u_i')$, where u_i, u_i' are independent samples of the ith block and $\Delta_t f = (T^t f)\bar{f}$. Complex conjugations may reverse the displayed expression; the fully averaged parallelepiped form is nonnegative. A fixed offset simply translates slot functions. Hold any extra outer blocks fixed. All inner starts are chosen beyond previous finite supports, but

they are deterministic at the time the variables are integrated, so the finite Cauchy–Schwarz argument uses independent variables.

Successive limits of this bound give the ordinary mixed cube seminorm for powers T^{r-j_i}. Indeed after integrating one derivative the squared norm of a conditional projection may be represented as an inner product on the corresponding invariant relative self-product; the next derivative is again a matrix coefficient for the diagonal action on this cube system. This is the standard iterative construction of the box seminorm for commuting transformations. The Hartman mean-ergodic convergence computes each of these matrix coefficients with exactly the same invariant projection as its ordinary Cesàro average. For a direct verification, at the first stage the quantity has the form $\int (T^t F)\bar{F}$, for the other derivatives held fixed; replacing by the invariant projection gives $\int F(x)\bar{F}(y)$ on the relative self-joining over the invariant σ algebra. Apply the same identity on that joining for each succeeding variable. At each fixed outer support only finitely many configurations occur, so ordinary inner limits commute with those finite sums.

This mixed seminorm vanishes here. Put $L = \operatorname{lcm}(1, \dots, h-1)$. In its ordinary Cesàro representation split each t_i into its finitely many residues modulo L. Each resulting generalized cube inner product is a cube form for T^L on translates of f_r; by Gowers–Cauchy–Schwarz it is bounded in modulus by $||f_r||^{2^d}_{U^d(T^L)}$. This last cube seminorm is zero: its ordinary cube average for T^L uses the distinct nonconstant linear forms $L\sum_{i\in\omega} m_i$, and the ordinary polynomial characteristic-factor theorem for T annihilates it if $f_r \perp Z_\infty(T)$ (see Bergelson–Leibman–Moreira, discussion of the characteristic factors). Equivalently use the usual finite-power pro-nilfactor identity. For $d = 1$ one only needs orthogonality to the finite rational invariant factor. This proves the characteristic estimate. The same estimate for ordinary Cesàro averages follows from the usual Host–Kra characteristic-factor theorem.

Return. For a positive event the ordinary same-event h-term mean has a positive lower bound, by multiple recurrence. It suffices to work on one ergodic component of positive mass. Project the indicator to Z_∞ and approximate that projection in L^1, preserving sup norm, by a continuous function on a finite-step ergodic nilfactor. Make the approximation error smaller than the ordinary lower bound divided by $4h$. Choose the total number of independent blocks at least $h-1$ and at least the nilstep of this approximant. Terms involving any original orthogonal error vanish by the preceding estimate; bounded approximation errors cost at most the sum of the L^1 errors in any sampled or ordinary integrated average.

For the continuous nilfactor functions, at an individual starting point x apply pointwise smoothing to the polynomial-time list, i.e. the single nilrotation $T^0 \times T^1 \times \cdots \times T^{h-1}$ starting at $(x, \dots, x)$. For tests shifted by the fixed offset in the difference variable the ordinary orbit mean is unchanged. The nested limsup of the absolute pointwise error vanishes at each x; successive applications of bounded reverse Fatou give the same conclusion after integration. This step does not interchange an arbitrary ultrafilter limit with integration. We conclude that some admissible convolution sample gives a strictly positive same-event return. Countability of the differences transfers the ergodic-component conclusion to the original system. If zero is to be omitted, the sampled measures have vanishing largest atom under the Hartman assumption: Wiener's criterion, or integrating $|\hat{\nu}|^2$ against circle Haar, gives $\sum_n \nu(n)^2 \to 0$. A fixed offset and other finite samples do not affect this exclusion.

Concrete sufficient condition. If $0, \pm a_i \in S_i$ and $\sum_i ||a_i\theta||^2 = \infty$ for every $\theta \neq 0$, use independent $0, \pm a_i$ increments of probabilities $1/2, 1/4, 1/4$. Their Fourier multiplier on a finite interval is $\prod \cos^2(\pi a_i \theta)$, tending to zero on each fixed tail. More generally, if

$G_2(a) = \{\theta : \sum_i ||a_i\theta||^2 < \infty\}$ is finite of order g, eventually each a_i is divisible by g, and the corresponding rescaled tails are Hartman-uniform on the integers. Thus any translate of the full (possibly multivalued) box meeting `gZ` is fully density-large: freeze such a finite signed head, discard a larger finite initial segment if needed, and use only the Hartman selector on the tail. A translate missing `gZ` has critical density one. In particular $a_{i+1} = q_i a_i + c$ with nonzero fixed c and infinitely many bounded q_i has $G_2(a) \subset \{\theta : c\theta = 0\}$ by the recurrence along that bounded subsequence. This gives a genuine zero–one criterion for all lengths in this subcase; full Hartman failure in other translated boxes leaves the integer challenge open.

R138. Finite high-capacity partitions when different axes have different prime ratios

Scope and status. Proved subcases and unresolved extensions.

Result. Compact prime-ratio families have finite high-capacity partitions in the specified residue regime.

Scope. Different profinite ratios admit an explicit unresolved arithmetic resonance; the argument is not uniform across residue regimes.

Retain the eventual congruences $p_i \equiv q_i \equiv 1 \bmod m$ for each fixed m, the odd sum of signed active horizontal residues a_i, and either sign for vertical unit motion. Assume the ratios $r_i = q_i/p_i$ belong to a compact subset K of $(0, \infty)$, and for all legal directions

$$\sum_{\text{active}} |a_i|/p_i \le \tau < 1, \sum_{\text{active}} |a_i|/q_i \le \tau' < 1.$$

For every $\eta > 0$ the rational saturation of this entire heterogeneous family has a finite partition into cells of capacity $> 1-\eta$. This assertion places no bound on how many different ratios occur within a single direction.

Fix $\theta \in K$. The matched irrational [R245] and matched rational [R215] arguments at θ actually give a neighborhood U_θ of θ, a finite head J_θ and a fixed uniform horizon in which the projected family with all its axes in U_θ after J_θ admits a single blocker of mean $> 1-\eta$. To verify this joint locality, keep the probes for θ (words on each actual grid, with the same fixed graphs). At each good limit parameter in the compact parameter space, the profile contradiction proceeds unchanged if the supremum of $|r_i - \theta|$ over the entire support tends to zero. Failure of some fixed parameter neighborhood, ratio neighborhood, support tail and horizon would produce such a sequence by choosing the $1/n$-neighborhoods, nth tail and horizon $2n+1$. The sparse probes are jointly local as their first-coordinate convergence is uniform under the same assumptions. A finite subcover of the parameter space therefore gives one ratio neighborhood by intersection, one tail, and one horizon. Each universal graph is charged only once and the supplementary probes have summable losses. If $\theta < 1$ the second budget ensures an effective first budget at most $(\theta + o(1))\tau' < \theta$ in a sufficiently small ratio neighborhood, exactly as required in the subunit-ratio proof. At $\theta = 1$ use the sparse coefficient there directly. No uniform rate of convergence for the ratios is needed.

Use finitely many such neighborhoods to cover K and assign each axis to exactly one of the neighborhoods containing its ratio. Discard their finite initial heads. In any retained direction, at least one of the bins has odd sum of the centered horizontal residues of its vertically active axes. Partition directions by the first such bin. For a given cell pull back the single-bin blocker along coordinate projection. Directions in that cell are projected to legal odd directions for the bin, so the entire cell is blocked with the already uniform horizon. The squarefree tail formula transfers all finitely many cell capacities to Q. Directions carrying a nonzero coordinate in the finite initial head have a separate cell of capacity one, by finite-prime sieving after the rational lift. Thus the advertised rational partitions exist. In the symmetric unit-move model one may exchange the two axes of each pair to put all ratios ≥ 1, and adjoin the endpoint ∞ to K: ratios tending to infinity have the uniformly sparse correction for $(2M)^{-1}$ from the positive-vertical note, with p_i/q_i as small as needed at the chosen fixed probe.

Conditional implication for integer thresholds. If any such saturated heterogeneous family has capacity strictly between zero and one, apply the nested-label construction of interpolation [R145] *on Q* using the just constructed partitions; only independent products of Q-actions and monotonicity of capacity enter that step. All values from the intermediate threshold to one then occur as capacities of subsets of Q. Transfer each of these subsets individually to Z by the countable torsion-free realization, then apply the integer multiplicative closure below the intermediate threshold. No single integer realization preserving every partition is required. This does not decide whether any heterogeneous family here has intermediate capacity.

A different residue regime If on a support tail the profinite ratio $q_i p_i^{-1}$ converges to a unit v different from 1, the root coefficient $e_i = (2M)^{-1} \bmod p_i$ instead has $q_i e_i / p_i \to (\theta - v_{2M})/(2M)$, where v_{2M} is any integer representative of v modulo 2M. Its scalar output multiplied by M is still $(S-1)/2 \bmod 1$. In the graph branch of the two-clock criterion this changes the slope relation to $S = b + (\theta - v_{2M})k/M$, b odd. There is consequently no general lower bound on $|A - S|$ in terms of θ and the mass budget. For example fix even M, choose $v_{2M} = M - 1$, $k = 1$ and $b = 1$. Then $S = (1+\theta)/M$, $A = 1/M$, $A - S = -\theta/M$. In the positive-mark case this one-sided drift can be countered by the allowed positive jump mass S. The congruence class $M - 1 \bmod 2M$ is a unit. This is an unexcluded arithmetic resonance of these tests, not an example of intermediate capacity; the detecting probe multiplier M is dictated by a fixed graph. Congruences substantially different from 1 are therefore also worth testing directly rather than assuming the foregoing argument uniform in the prime order residues.

R150. Minimal systems inside the stride traces of a positive avoider

Scope and status. Result with proof.

Result. Uniform periodicity of minimal stride subsystems implies endpoint one; finite coloring witnesses bound avoiding densities, and right-continuity passes to b.

Scope. The periodicity hypothesis is uniform; the finite-witness growth criterion is sufficient but unproved for the principal one-mark candidate.

Let a stationary binary integer process have a one-coordinate event of mass $a > 0$ and avoid all runs of `h0` ones in steps D. Pass to its support X in the two-sided binary shift. For each $d \in D$ the excluded cylinders and their translates are open and null, so the exclusion holds pointwise in X. Write Y for the closed shift-invariant hull of all sequences $(x(t+jd))_{j\in Z}$ for $x \in X$, $d \in D$ and $t \in Z$. Every word of Y has a zero in each interval of `h0` consecutive slots.

Criterion. If all minimal subsystems of Y have period dividing the same integer P, then $b_Z(D) = 1$. In particular this holds if the zero fixed point is the only minimal subsystem of the stride hull. Nonminimal traces need not be periodic, approximable by periodic names in Banach density, or have finite orbit closures.

For any r, take independent root towers of the original process at strides $q_i = P^{i-1}$ for $1 \le i \le r$; for $P = 1$ take every stride equal to one. The independent OR of their lifted one-coordinate indicators has mass $1 - (1 - a)^r$. Suppose it allowed arbitrarily long runs at steps in D. Center these runs and extract a two-sided joint shift-name w; take a minimal subsystem of its orbit closure. The OR is identically one on this subsystem. For every i and every time residue modulo q_i, its ith word read along that residue lies in Y, since a step of $q_i d$ on that tower acts as the original d-step. This is a closed, shift-invariant condition, including after taking the minimal subsystem. In a compact minimal system for the shift, each point lies in a minimal component for its qth power: the finitely many translates of any minimal component for the power cover the whole minimal system. Thus on each q_i-residue the ith word is minimal in Y and is P-periodic. The whole ith row is $q_i P$-periodic.

Starting at time zero, move by one of $0, q_1, \dots, (h0 - 1)q_1$ to hit a zero in the first row. At stage i move by one of $0, q_i, \dots, (h0 - 1)q_i$ to hit a zero in row i. These later moves preserve all earlier zeros because their strides are divisible by every earlier full period. The resulting all-zero time contradicts the OR. The maximum run length in this product mask is therefore bounded uniformly over all allowed steps. Taking $r \to \infty$ proves the criterion.

A necessary escape for chromatically recurrent differences. Suppose D forces every length in every finite coloring. In this case every piecewise syndetic integer set contains every length at a D-step, by the usual finite witness and translation argument. The original avoiding support X has only the zero minimal subsystem: a nonzero minimal binary name would have syndetic ones. For each *fixed* d, the compact stride image of X also has only the zero minimal subsystem. Indeed a nonzero minimal stride image would lift to a minimal system for the dth shift on X; its finitely many translates give a nonzero minimal system of the original shift. Thus if D also has $0 < b_Z(D) < 1$, the total stride hull of **every** positive fixed-order avoiding process has a minimal aperiodic subsystem or periodic minimal subsystems with unbounded periods. These must arise from unbounded sequences of differences, not from the individual compact stride factors. This applies in particular to any putative

intermediate example built on the one-mark all-color theorem [R173].

Capacity and finite Ramsey numbers. For fixed h, let $H_h(D)$ be the integer hypergraph with edges the h-term progressions with differences in D. Then $a_h(D)$ is the reciprocal of the supremum of the fractional chromatic numbers of the finite induced subhypergraphs, with the convention $1/\infty = 0$. A stationary avoiding mask of mean a gives a random independent set hitting each vertex with probability a, proving one inequality. Conversely, if every finite hypergraph has fractional chromatic number at most $1/a$, normalize its fractional coloring to such a probability law, fill the rest of the integer variables by zero, and use compactness of binary probability laws to obtain an independent random mask with every vertex marginal at least a. Cesàro averaging its translates gives a stationary law with the same lower marginal bound. Compactness gives the equality also at the supremum.

In particular, if $W_D(h, r)$ denotes the smallest cardinality of a finite integer vertex set on which every r-coloring has a monochromatic D-progression of length h (infinity if none exists), then $W_D(h, r) \geq (1 - a_h(D))^{-r}$. Indeed r independent stationary masks of mass $a_h(D)$ cannot cover this finite set: otherwise one could choose a covering mask as the color at each vertex. A union bound on the uncovered probability proves the inequality. Consequently $\sup_h \liminf_r r^{-1} \log W_D(h, r) < \infty$ would imply $b_Z(D) < 1$, and subexponential growth for every fixed h would give full measurable multiple recurrence. Qualitative piecewise-syndetic recurrence alone gives no such bound.

One-sided continuity. Along any nested increasing real-parameter family D_s, if every $a_h(D_s)$ is right-continuous then so is $b(D_s)$: take one near-maximizing h at s and let the parameter decrease to s from the right. No uniform nilmodel or rate in h enters. Left-continuity does not follow. If the left limit L exceeds the positive value $b(D_s)$, every sufficiently narrow left strip $D_s \, D_{s-u}$ is itself intermediate: monotonicity and the independent-product inequality give its capacity between $b(D_s)$ and $b(D_s)/b(D_{s-u}) < 1$. If $b(D_s) = 0$ and $L > 0$, every such strip has zero capacity. This explains the residual jump mechanism even when every fixed-order cap varies continuously.

R155. Mixed odd supports in the near-antipodal Gaussian model

Scope and status. Proved subcases and unresolved extensions.

Result. The near-antipodal mixed odd-prime Gaussian model has divergent higher-difference suprema and dense programmable cylinder trajectories.

Scope. Adaptive cylinder density does not establish recurrence for every measurable event.

Qualifications. Improved marginal control in the measurable Gaussian model does not by itself establish multiple recurrence in arbitrary systems.

Take the spectral model of the disjoint-block Gaussian construction [R162] on the **unrestricted** family of odd sums of distinct prime axes with $\Sigma 1/p = \infty$. Let $a_i = \pi/p_i$, let ϵ_i be independent fair signs, and let U_i be multiplication by $-\exp(ia_i\epsilon_i)$ on the realification of their complex L^2 space. These operators commute, with $U_i^{p_i} = 1$. Use a real isonormal Gaussian process W on this space; $W(f)$ transforms to $W(U_d f)$. The following claims concern this particular Gaussian representation.

Anchored higher-difference suprema diverge Fix a nonempty odd finite anchor A, write $Z = \exp(i\sum_{i\in A} a_i\epsilon_i)$, and let $m \geq 1$ be fixed. In every sufficiently far tail pair up its axes, allowing F to be any finite union of these pairs. Then almost surely

$$\sup_F W((Z \cdot \exp(i\sum_{i\in F} a_i\epsilon_i) - 1)^m) = \infty.$$

Write $P(z) = (z-1)^m$. For $k \in F$, the Walsh coefficient of the tested vector at the single fresh sign ϵ_k (retain arbitrary dependence on the anchor signs) equals

$$\sum_{j=1}^{m} P_j Z^j i\sin(ja_k) \prod_{l\in F, l\neq k} \cos(ja_l)$$

$$= ia_k Z P'(Z) + O_m(a_k \sum_{l\in tail} a_l^2),$$

where P_j are the polynomial coefficients; for $k \notin F$ the coefficient is zero. The leading anchor vector is nonzero in L^2: Z is not identically 1 by the odd distinct prime denominators. Choose the tail so that the error is smaller than half its norm times a_k. Orthogonality of the different fresh singleton Walsh subspaces shows that the canonical distance of the tested Gaussian variables indexed by F,G is at least $c_A(\sum_{k\in F\triangle G} a_k^2)^{1/2}$.

By the finite Sudakov–Fernique comparison, the expected maximum over all unions of a finite list of pairs is bounded below by the expected maximum of the independent linear Gaussian sum with coefficient $c_A(a_i^2+a_k^2)^{1/2}$ for pair $\{i,k\}$. This latter expectation is the sum of the coefficients times $1/\sqrt{2\pi}$ and goes to infinity. Every finite maximum is 2^m-Lipschitz as a function of the standard Gaussian on its span, hence has uniformly bounded variance by Gaussian Poincaré. It follows that the countable supremum is infinite almost surely. This is an obstruction for the particular polynomial increment mask on every anchored mixed-support cylinder. It is not a nonexistence theorem for other measurable masks.

Every finite Gaussian slot trace can be programmed Fix $h \geq 2$ and finite linearly independent observations $f_1,\ldots,f_r$ in the real Hilbert space, all depending only on a finite list of signs. On any prescribed tail, almost surely the positive-slot vector

$$(W(U_d^j f_l) : 1 \leq j < h, 1 \leq l \leq r)$$

over unrestricted unsigned odd d is dense in its full Euclidean space. Thus the Gaussian system has almost-sure simultaneous topological recurrence at every order for the cylinder topology of finite Walsh observations, including recurrence to distinct prescribed open cylinders at the different slots.

Choose an odd anchor A disjoint from the observation head in the prescribed tail, large enough that U_A takes at least $h-1$ distinct values as a complex multiplier. Distinct sign assignments really give distinct values: equality of two products would force a nontrivial sum of signed reciprocals of distinct odd primes to be an integer, which reduction modulo each prime rules out. The real-Hilbert vectors $v_{jl} = ijU_A^j f_l$ are linearly independent. Holding the head signs fixed, a linear dependence would be a complex polynomial in the multiplier with only terms of degrees 1 to $h-1$; after removing its initial factor the $h-1$ distinct evaluations force every coefficient to be zero. Real-linear independence of the f_l finishes the argument. Let Σ be their positive-definite Gram matrix and $R = (W(U_A^j f_l))_{jl}$.

On a far tail use fresh disjoint pairs and let $Y_i = (W(v_{jl}\epsilon_i))_{jl}$. These are iid real Gaussians with covariance Σ and independent of R. For pair $u = \{i,k\}$, put $b_u = (a_i^2 + a_k^2)^{1/2}$ and $Y_u = (a_i Y_i + a_k Y_k)/b_u$, also iid with covariance Σ. The b_u tend to zero and their sum diverges, while their squared tail sum tends to zero.

To reach a fixed target z set $\Delta = z - R$. Conditional on R, when $\Delta \neq 0$ take a finite panel of pairs with total b-mass tending to $L = (2\pi\Delta^T\Sigma^{-1}\Delta)^{1/2}$, by first crossing this level in successively later tails. Select exactly the pairs with $(\Sigma^{-1}\Delta)\cdot Y_u \geq 0$. Their union F has even cardinality; for $\Delta = 0$ one may use F empty. Since $E[Y_u 1_{(\Sigma^{-1}\Delta)\cdot Y_u \geq 0}] = \Delta/L$, the sum of their $b_u Y_u$ tends to Δ in probability, the conditional variance being bounded by $C_\Sigma \sum_{u\in tail} b_u^2$. Cut off $|R|$ at a finite bound before making uniform estimates, then remove the cutoff.

The true trajectory at $A \cup F$ differs in probability by $o(1)$ from R plus this selected sum. To verify this, expand $U_A^j f_l \prod_{i\in F}(\cos(ja_i) + i\epsilon_i \sin(ja_i))$ in its fresh Walsh variables. The constant term changes R by at most $O_h(\sum_{tail} a_i^2)|R|$. Each singleton coefficient differs from its linearized version $a_i v_{jl}$ by a scalar factor $O_h(a_i \sum_{tail} a_i^2)$ times the same head vector (with constants depending on j). On a bounded-R cutoff, the deterministic conditional panel has bounded total Σa_i by the b-mass bound, so these singleton errors tend to zero in L^1 even after adaptive selection, by bounding with the sum of absolute Gaussian coordinates over the whole panel. Terms involving at least two fresh signs lie in a Gaussian subspace independent of all the data used to select F. Conditional on F and the selecting data, their variance is bounded by $C_h(\sum_{tail} a_i^2)^2 ||f_l||^2$, using the Walsh expansion, or the probability of at least two hits in Bernoulli trials of probabilities $\sin^2(ja_i)$. This proves the trajectory approximation for all the finitely many slots.

Choose an almost surely convergent subsequence of the programmed traces for each rational target z, then intersect the countably many full-measure conclusions. The resulting extreme directions may depend on the Gaussian sample. Applying this to finite sublists of a fixed Walsh real basis, all integer h and all starts of tails gives the asserted topology

statement. In particular open cylinder witnesses cannot block a short progression in this model. A dense trajectory in the cylinder topology does not by itself give a multiple return in every measurable event, since the adaptive transformations can distort the distribution of approximation errors. This argument does not give that stronger conclusion.

R156. Odd numbers of exceptional coordinates and two coprime backgrounds

Scope and status. Proved subcases and unresolved extensions.

Result. Paired odd-prime marked models have fixed short color obstructions and rule out certain low-degree mixed primitives.

Scope. These are candidate constructions; uniform all-system upper bounds as primes grow remain missing.

Removing the common order-two factor This version is a single squarefree torsion model which transfers to an integer critical capacity by the tail formula [R241]. Choose pairwise distinct odd primes p_i, q_i and identify $P_0 = \bigoplus_i C_{p_i q_i}$ with its usual subgroup of $\mathbb{Q}/\mathbb{Z}$. In the i-th cycle permit the normal speeds $0, \pm p_i, \pm q_i$ and special speed $s_i = p_i + q_i$, where the arithmetic is modulo $p_i q_i$. Retain all finite-support directions with an odd number of special coordinates and $\sum_{i \text{ special}} \eta_i < 1/8$, for $\eta_i = 1/(2p_i) + 1/(2q_i) + 1/(2p_i q_i)$. In rescaled CRT coordinates the unsigned speeds are $(0,0), (1,0), (0,1), (1,1)$. Thus special count parity on the unsigned subset is the binary inner product of the two support vectors. There is no need for a factor of even order. Simultaneous reversal of every speed does not affect recurrence. Independent signs on all normal axes are permitted.

On one cycle C_{pq} write S for unit spatial shift. Apply the five-factor spatiotemporal operator $\mathcal{D} = (T - I)(T - S^p)(T - S^{-p})(T - S^q)(T - S^{-q})$ to an array b. If time acts by a normal speed, the output is zero. At the special speed $s = p + q$ the operator is the spatial product of the five differences $S^s - S^w$, $w = 0, p, -p, q, -q$. Up to a commuting shift these are the differences with increments $p + q, q, 2p + q, p, p + 2q$. The three increments $p + q, 2p + q, p + 2q$ are units: here distinctness and oddness of the primes are both used. By the Fourier and torus argument below the image is exactly the torus of arrays with all row and column sums zero modulo one. In particular it contains the explicit half-turn approximation H in the next section. Choose b_i so the product returns H. If only unsigned normals are used one omits the negative factors, obtaining the same image.

The nine-offset coloring below applies word for word, retaining a monochromatic obstruction of length six (four with unsigned normals), independent of the prime list. Alternatively its finite-product restrictions have a largest color class of proportion at least the reciprocal of the fixed number of colors. Compactness gives this uniform positive lower bound for the capacity of every tail, hence also for the saturated rational and integer realizations. The remaining task is to bound the capacities of these tails away from one uniformly over all lengths. The spectral single-mark argument [R181] proves positive recurrence in every tail for Eberlein higher-return functions and for every pair test. The reciprocal sums need not be finite for the coloring, so divergent choices of the primes remain available.

Every finite list of ordinary group-polynomial tests based at zero returns arbitrarily close to zero along these directions. Indeed a torus-valued polynomial of fixed finite-difference degree has no mixed difference between a p-axis and a distinct q-axis, by the coprime torsion argument at the end of this note. Thus on a sum of primitive labels from different prime axes, the vector-valued test is the sum of its one-axis values. Extract an infinite subsequence along which the ordered pair of one-axis values converges to (x, y) in the finite product torus. Choose positive integers a, b with ax, by as close to zero as desired. Arbitrarily far

along the subsequence choose index sets of sizes a, b meeting in exactly one index. Their step is allowed, the only special cost tends to zero, and the test value is arbitrarily close to $ax + by$. This concerns genuine finite-difference polynomial tests on P_0, and does not transfer to arbitrary polynomial tests on its torsion-free cover by bare composition in the reverse direction.

This is a candidate on a torsion-free group with a proved low-density obstruction. No uniform high-density bound is asserted. All group thresholds below use stationary avoiding events.

Stage and coloring Take distinct odd primes p, q and $N = 2pq$. For each coordinate in a countably generated free abelian group a step may be normal (residue $0, \pm p, \pm q \bmod N$) or special (residue $pq \bmod N$). Full integer lifts are allowed. Require an odd number k of specials with

$$k\left(\frac{1}{\{}2p\} + \frac{1}{\{}2q\} + \frac{1}{\{}2pq\}\right) < \frac{1}{8}.$$

In particular arbitrarily many specials can be used as p, q grow. Use disjoint infinite coordinate sets for a sequence of pairs $p, q \to \infty$, and take the union of the pure-stage families. One can also permit mixed stages in a single step under the same bound on the sum of errors over its special coordinates.

On $\mathbb{Z}/(pq) \cong C_p \times C_q$, let $r_u, c_v \in \{\pm\frac{1}{2}\}$ have $\sum_u r_u = \sum_v c_v = \frac{1}{2}$, which is possible since both sizes are odd. Put

$$H(u, v) = \frac{1}{2} + \frac{r_u}{q} + \frac{c_v}{p} - \frac{1}{\{}2pq\} \pmod 1.$$

Every row sum and every column sum of H is zero in $\mathbb{T}$. Every entry is within the displayed error per mark of $1/2$. Write S for the cyclic shift. There exists a $\mathbb{T}$-valued array G on $\mathbb{Z}/(pq)$ with

$$(I - S^p)(I - S^{-p})(I - S^q)(I - S^{-q})G = H.$$

Indeed the real kernel of the transpose of the operator on the left consists exactly of arrays that are a sum of one function of u and one of v, by the finite Fourier basis (the shifts by p and q generate the two component cycles). Every integer array in this kernel admits such a decomposition with integer values, by subtracting its first row and column. The image of the integer matrix on the torus is a connected subtorus whose annihilator is exactly this integer kernel. The row and column sums of H establish its membership. Omitting both negative speeds gives the same image and a shorter four-site obstruction.

Set $g(x) = (-1)^x G(x \bmod pq)$ on C_N. Then $g(x + pq) = -g(x)$. Choose $b : C_N \to \mathbb{T}$ with $b(x + pq) - b(x) = g(x)$, for example by setting $b(x) = 0,\ b(x + pq) = g(x)$ for $0 \le x < pq$. The temporal/spatial filter is

$$(T - I)(T - S^p)(T - S^{-p})(T - S^q)(T - S^{-q}).$$

It is a linear combination of at most 32 signed evaluations at times $0, \dots, 5$ and at spatial offsets $ap+bq,\ a, b \in \{-1, 0, 1\}$. It kills each normal increment, for which one factor vanishes. On a special increment the first factor gives g. In each remaining factor the temporal shift acts by $-I$ on g, including its spatial translates. Because p, q are odd, conjugating spatial shifts by multiplication by $(-1)^x$ negates them. The four resulting signs cancel, giving the special output $(-1)^x H(x \bmod pq)$. It lies within the specified error of $1/2$.

For each offset pair (a, b) color the value

$$F_{a,b}(x) = \sum_i \big(b_i(x_i + ap_i + bq_i) - b_i(ap_i + bq_i)\big) \pmod 1$$

by a fine equal-arc partition of $\mathbb{T}$, and take the tuple of the nine colors. Here b_i denotes the stage word (the second occurrence of b in $ap_i + bq_i$ is an integer offset). Only finitely many summands are nonzero. Constants in each offset cancel because the filter contains $T - I$. On a monochromatic six-term line the filter is within 32 times the cell width of zero, since at each individual spatial offset its coefficients sum to zero. But its true value is within $1/8$ of $1/2$ by oddness, a contradiction for width $< 1/128$. The number of colors is fixed independently of the primes, ranks, and number of active specials. Translating and averaging a fixed color along Følner sets gives a stationary avoiding event of positive measure. The torsion-free integer realization preserves this positive all-length threshold.

Why the single-background predecessor fails For stage $N = 2q$ with q odd, normal $0, \pm 1$, special q, the same odd-count variant admits a bounded-degree quasipolynomial potential. Let $m = (q-1)/2$ and choose integer c with $2c \equiv m \bmod q$. Take

$$b(x) = (-1)^x \frac{c}{q} \binom{x}{2} \pmod 1.$$

It is $2q$-periodic since $\binom{x+q}{2} - \binom{x}{2}$ is divisible by q. The special difference $g = (S^q - I)b = -2b$ is antiperiodic. The remaining filter $(S^q - S)(S^q - S^{-1})g = (2I + S + S^{-1})g$ equals $(-1)^x m/q$, within $1/(2q)$ of $1/2$. Thus it detects all odd special sets with total error $< 1/8$.

For this predecessor the family has critical threshold one. To see this on the free group, at each root R use the three spatial potentials formed by replacing x_i by $\lfloor x_i/R \rfloor$, with basepoint subtraction as above, and add independent Haar constants to the three outputs. On residue times modulo $2R$ along any line, every such potential is a torus-valued ordinary polynomial of degree at most two: each parity is fixed and each binomial is quadratic in the residue index. At stride R the filter applies with the original step, regardless of the starting floors. Hence if a row is active when at least one of the three values has circle distance $> c_0$ from zero, some site in its four-site stencil is active, for a fixed $c_0 > 0$. Independent Haar constants give the row a fixed inactive probability $(2c_0)^3$. Choose finitely many rows so deletion of simultaneous activation costs arbitrarily little. Also delete visits of each scalar to a short open arc, and to a narrow band at c_0, with arbitrarily small additional cost.

For each fixed root the binary gate codes of limits of arbitrarily long legal segments form a compact subshift. Extract polynomial coefficients on every residue modulo $2R$ and for each of the three scalar potentials. The limiting torus polynomials miss the short open arc at all integers; Weyl equidistribution implies all nonconstant coefficients are rational. The boundary margin guarantees that their periodicity gives periodicity of the limiting binary word. A compact subshift all of whose points are periodic has a common period. Recursively take each next root a multiple of the previous row periods. If arbitrarily long legal segments could avoid simultaneous activation, their joint limit could be activated row by row using the four-site stencil while leaving earlier rows unchanged, a contradiction. Thus the surviving marks avoid all sufficiently long allowed lines uniformly. Reverse Fatou along a Følner sequence in the free group, over the random Haar constants, and translate averaging give a stationary avoiding event of arbitrarily near-full mass. This proof works for disjoint or mixed stages. It illustrates that ruling out a single ordinary polynomial of the difference does not suffice.

Initial constraints for the coprime construction Each fixed $2pq$ stage alone has threshold one: the short avoiding coloring factors through an action of exponent $2pq$, and its higher roots amplify any such bounded-exponent witness. The necessary missing estimate is uniform as the primes grow. A single finite cyclic group cannot have two independent 2-components. The torsion-free realization does not assume that it can: the construction above is on independent free coordinates, and any all-systems estimate there would be transferred directly to the integers by finite replication and the mixed-radix realization.

There is no analogous low-degree polynomial primitive on both primary coordinate cycles with a nonzero mixed derivative. A torus-valued function on $C_p \times C_q$ that on every row and every column is an ordinary group polynomial of a fixed degree is a polynomial of at most twice that degree on the product, by expanding mixed differences. Its mixed unit difference is annihilated by both a power of p and a power of q, hence vanishes. For detail, on a cyclic component of order p write $S^p - I = p(S - I) + \sum_{j=2}^{p} \binom{p}{j} (S - I)^j = 0$. When $S - I$ is nilpotent of uniformly bounded order on the function, repeated use of this identity shows $p^D(S-I)$ annihilates it for some finite D. The same argument applies for q, and Bézout kills the mixed difference. A fixed bounded period in those row polynomials also does not help when p, q are eventually larger than, and coprime to, that period. This exclusion concerns separately polynomial primitives. It does not show that every possible root blocker must be such a primitive.

R164. Odd finite group extensions preserve diagonal mass along odd finite sums

Scope and status. Result with proof.

Result. Odd finite Haar extensions preserve diagonal mass along odd finite sums, and recurrence extends through pro-odd towers and inverse limits.

Scope. The finite-extension diagonal hypothesis is stated explicitly; no bound uniform over all tower complexities is inferred.

Let T be a measure-preserving action of a countable abelian group G on a standard probability space (X, μ). Suppose d_i are sums over pairwise disjoint odd finite supports of the designated generators, receding along the tail. Fix a progression length h. Assume the joint laws of $(x, T^{d_i}x, \dots, T^{(h-1)d_i}x$ have a subsequential weak limit which dominates $c\Delta_\mu$, where Δ_μ is the full diagonal law and $c > 0$. Weak limits here may be taken in a compact metrizable model carrying the Borel action and the named measurable functions. A countable algebra, or the compact metric orbit coding of countably many separating events, gives the equivalent convergence on measure-algebra cylinders.

Let K be a finite, possibly nonabelian group of odd exponent q. In a measurable cocycle extension over X with fiber K and Haar conditional measures, there is again a sequence of pairwise disjoint odd finite sums receding on the tails for which the extended h-slot limit dominates

$$c'\Delta_{\mu\times Haar K}, c' = c^q|K|^{(h-1)(1-q)}.$$

The same holds for a homogeneous finite K-space fiber driven by the cocycle. No locality or cylinder assumption is imposed on the cocycle. A fixed choice of left action is used below.

One block. For a direction d define the random joint profile consisting of x, the $h-1$ new base points $T^{jd}x$, and the group vector $v = (\rho(jd, x))_{1\le j<h} \in Q = K^{h-1}$. By compactness and finiteness of Q, extract a limiting law ν. On the base diagonal its restriction decomposes as $\Delta_{f_v\mu}$ for each profile label v, with $\sum_v f_v(x) \ge c$ almost everywhere. These are conditional subprobabilities: their sum may be larger than c.

Composition. Take an iterated subsequential limit of q successively later directions, always letting the newest index tend to infinity first with the earlier indices frozen. For fixed old $\sum g$, the joint profile for $g+d$ is obtained from the profile for d by applying the old maps T^{jg} to its jth new base point y_j and multiplying its jth register on the left by $\rho(jg, y_j)$. This pushforward commutes with the indicated weak limit. For merely measurable old maps and cocycles the claim follows by Lusin approximation against each y_j marginal μ. Indeed the bad sets for these finitely many evaluations have mass at most the sum of their μ-measures, both before and after the limit. All old tests involving the initial base similarly have marginal μ. One may instead use a simultaneous countable continuous orbit coding.

For the newest block keep the part of ν on its diagonal at any given v. Its weight is $f_v(x)d\mu(x)$, and the old fixed transformations now act on common base point x. Iterating this restriction through the older blocks gives a submeasure in which their profiles all equal v, of density $f_v(x)^q d\mu(x)$ on the fully returned base diagonal. Multiplication uses the ordered factors $v_j^q = 1$, so all fiber slots return as well. During this induction weighted

weak convergence for $f_v(x)^a$ follows by bounded measurable approximation against the fixed initial marginal μ. Contributions for different v are disjoint in the law retaining all q successive block profiles; after projecting one may add the resulting submeasures. Therefore the resulting h-slot law dominates the diagonal with density $\sum_v f_v^q \geq |Q|^{1-q}c^q$ pointwise, by convexity. The initial fiber register remains uniform and independent of the sampled base through every step. The proof works uniformly for a homogeneous K-space since group identity fixes every fiber state.

A joint iterated weak limit on a compact metric model is the limit of an ordinary sequence of realized q-tuples: for each finite list of continuous tests choose successive approximating indices in their nesting order, all beyond the previously used supports. Diagonalizing over the tests ensures convergence. Every chosen sum still has odd support cardinality since q is odd, and different packets have disjoint supports. This proves the assertion.

Consequences. Every finite tower of arbitrary measurable extensions by odd finite group cocycles has all-length recurrence along odd finite sums whenever the base admits odd-sum directions tending strongly to the identity on every tail. Such base directions give $c = 1$, and the theorem iterates with positive constants for any fixed finite tower and h. Positive diagonal mass in a limiting h-slot law gives returns for every positive event by approximating its indicator in L^1 with bounded continuous functions, using the common marginal.

In particular let $G = \oplus C_{p_i}$, with distinct odd primes and $\Sigma 1/p_i = \infty$. Every ergodic compact-rotation G-action has a tail sequence of disjoint odd directions tending strongly to identity, by the odd-support Bohr theorem [R169] and a successive approximation on increasingly many characters. Thus all finite odd-group towers over any such rotation have positive recurrence at every order. For the native coordinate base $\prod C_{p_i}$, the singleton generators themselves tend strongly to identity; the conclusion for towers over that base does not depend on reciprocal divergence. The property also passes to factors and to ergodic mixtures by countability of the allowed directions. If a finite extension has a normal fiber subgroup N and the quotient tower is already covered, sets of mass $> 1 - 1/|N|$ have full returns by taking the positive-measure base event on which the entire N-fiber lies in the set. In particular for an arbitrary finite abelian fiber K over the native compact rotation one may use its Sylow 2-subgroup as N.

This addresses finite odd fibers even when the short profile on each generator is globally correlated across all coordinates. The proof does not show that an arbitrary even finite-group extension returns at every positive mass, or that the stated lower bounds persist over an infinite tower.

A last compact pro-odd extension The fiber of a final extension may be an arbitrary metrizable compact pro-odd group K (an inverse limit of finite groups of odd order). Here the proof gives all positive-measure returns; it does not in general give a fixed lower bound for a limiting atom on the extended diagonal. The same argument works for any metrizable compact group K such that every tuple in K^{h-1} has odd positive powers accumulating at the identity. This condition holds for pro-odd K, since every identity neighborhood contains the kernel of a finite odd-order quotient.

Extract the one-block joint law ν with the full compact-register profile, disintegrating its restriction over the base diagonal as $d\mu(x)\eta_x(dv)$. We have $||\eta_x|| \geq c$ almost everywhere. For each positive odd integer q, the same iteration used above gives a realized q-block limiting law dominating on the base diagonal the distribution of the ordered product of q independent

profiles with conditional subprobability law η_x. It is enough to prove that at least one q gives a positive correlation for the requested event A in the full extension.

Let $B_x = \{b : (x, b) \in A\}$ and, for fixed x, let $P_x(s) = \int 1_{B_x}(b) \prod_{j=1}^{h-1} 1_{B_x}(s_j b)\, db$. This is a continuous nonnegative function of $s \in K^{h-1}$ (strong continuity of translations in $HaarL^1$), and $P_x(1) = Haar(B_x)$. Whenever this is positive, choose any v in the nonempty support of η_x. There is an odd q with $P_x(v^q) > 0$ by the assumption on K. The tuple of q copies of v belongs to the support of $\eta_x^{\otimes q}$, and multiplication is continuous, so $\int P_x d\eta_x^{*q} > 0$. These integrals are measurable in x for each q, and A has positive mass. By countability there is one odd q on a positive-measure set of x, giving a positive return in the corresponding q-block limiting law. Approximation by bounded continuous functions in L^1, using the common single-slot marginals, gives an actual positive return for a finite odd sum. Measurable cocycles present no issue in the iteration: for a fixed old transformation their compact-register values are continuous off a base set of arbitrarily small μ-measure by Lusin, and compact group multiplication is continuous.

Thus the preceding finite tower may be followed by any compact pro-odd group cocycle extension. A purely finite-factor argument for this last step would need approximation errors to beat probabilities which may shrink with the quotient size. The conditional support argument avoids that requirement.

Qualitative extension over any returning base In fact all positive-measure recurrence passes through arbitrary odd finite-group extensions without any assumption on limiting diagonal mass. The argument has a stronger formal reach than the preceding quantitative result.

Suppose every tail of the allowed odd finite sums is h-recurrent for all positive-measure events of the given base system. Fix a finite K of odd exponent q. Given a positive event A on a left K-cocycle extension, choose a fixed register b with $E_0 = \{x : (x, b) \in A\}$ positive. Choose the first odd direction d_1 on a fresh tail for which $\cap_{j<h} T^{-jd_1} E_0$ is positive; split the intersection by the finite vector $v_1 = (\rho(jd_1, x))_{1 \le j < h}$ and keep a positive part E_1 on which this profile is constant. Repeat finitely many times with each new odd support beyond all earlier ones, obtaining nested positive E_i, constant labels v_i, and $T^{jd_i} E_i \subset E_{i-1}$ for every j. Choose $M = q|K|^{h-1}$ directions. Some q labels coincide. Let d be the sum of their directions, still on an odd support. Starting at any $x \in E_M$ and evaluating the chosen coordinate shifts in decreasing index order, the current base point at stage i lies in E_i; an unchosen stage is skipped using $E_i \subset E_{i-1}$. For each slot j the q transformations encounter precisely q copies of the same group label v_j. Their product is identity and the base endpoint belongs to E_0. Thus all h visits to A occur, on the positive subevent $E_M \times \{b\}$. This applies likewise to a homogeneous K-space. All selected directions can start beyond any prescribed head.

For completeness, positive h-recurrence on all tails is closed under measure-algebra inverse limits, with **no** uniform bound on the return mass over the approximating factors. Given a positive event A in such a limit, at some stage there is a positive factor event E on which conditional measure of A over that factor is greater than $1 - 1/h$. This follows by L^1 martingale convergence, or approximation by a finite-factor event. Recur E in its factor, obtaining a positive base set where all h slots lie in E. At those x the invariant conditional measures send the respective h membership conditions for A back to the original fiber, each with measure $> 1 - 1/h$. Their intersection has positive conditional measure by the union bound. This argument only needs factor recurrence, and resolves the apparent issue of

shrinking finite-stage return probabilities in the preceding paragraph.

A metrizable compact pro-odd cocycle extension is an inverse limit of its finite odd-group quotient extensions. Consequently any tower of such extensions and inverse limits (at countable or transfinite rank on a standard probability space) over an h-returning base is h-returning. At a limit stage, any fixed measurable event can be approximated in measure by events on a countable collection of earlier factors, with the same conditional-density proof. This argument applies to every h for an all-length returning base. On the direct sum with reciprocal-divergent odd periods the base may, in particular, be any ergodic compact-rotation action. On the native product rotation the divergence hypothesis can be dropped. The quantitative diagonal-minorization assertion above remains separately useful, as a pointwise all-event lower bound for finite towers.

R165. A fixed-length bound for odd integer differences

Scope and status. Result with proof.

Result. Odd integer differences satisfy the fixed-length upper bound $1-1/(h-1)$ off powers of two, with exact caps at powers of two and their successors.

Scope. No formula is claimed for all intervening h; unrestricted all-length capacity is one.

Qualifications. The proof invokes fixed-length periodic-mask reduction, so periodic-fixed-length-cyclic-models.md is a dependency even though the source does not hyperlink it.

Using the fixed-length reduction to periodic masks for periodic direction sets, the odd-difference cap satisfies $a_h \le 1-1/(h-1)$ whenever $h \ge 3$ is not a power of two. In particular its exact values at $h = 2^s$ and $h = 2^s + 1$ are $1 - 2^{-s}$ ($s \ge 1$). This bound concerns fixed lengths, not the unrestricted all-length cap of the odd integers, which is one.

It suffices to work in C_{2^n}: in a cycle with an odd component restrict the step to odd multiples of the order of that component and use a coset average. Any nonempty set of allowed odd steps in the dyadic cycle with all possible odd residues includes each of these directions. When the power of two is absent in a tested cycle, an odd multiple of its order is zero and no positive avoider can exist there.

Choose a consecutive interval $J \subset Z$ of size h centered as closely as possible about zero and containing zero. Let Q be twice the largest power of two dividing any nonzero $j \in J$. For h not a power of two, $Q \le h-1$: in odd length the largest $|j|$ is $(h-1)/2$, and in even length it is $h/2$ with equality inside the dyadic bound possible only at a power of two. If $2^n < Q$, the entire orbit under an odd step has length at most $h-1$, so avoiding density is at most $1 - 1/2^n$.

In the remaining case let the holes B have density $b < 1/(h-1)$. For fixed start x and a uniform odd d in the cycle, the point $x + jd$ is uniform in the single residue class modulo $2^{1+v_2(j)}$ opposite to x above their shared residue modulo $2^{v_2(j)}$. Let $W(x)$ be the sum of the corresponding hole probabilities for the $h-1$ nonzero offsets j. It depends only on $x \bmod Q$, and its average over x is $(h-1)b < 1$. Take a class with $W < 1$; because no class modulo Q can be fully contained in B at the specified hole density, choose some $x \notin B$ in it. A union bound over the other offsets gives a positive probability of missing every hole along the complete progression. This proves the upper estimate. At $h = 2^s$, the elementary single-step bound $a_h \le 1 - 1/h$ is sharp since deleting zero in the cycle of the same order hits every odd orbit. The same example is still an avoider at $h = 2^s + 1$, giving the second exact value. No formula on the intervening lengths is claimed.

R166. Measurable Cameron–Martin controls in the unrestricted odd-prime Gaussian model

Scope and status. Result with proof.

Result. Uniform marginal control upgrades adaptive Gaussian programming to a nullity statement for every Cameron-Martin affine pencil.

Scope. This necessary condition for measurable avoiders is not by itself full recurrence in arbitrary systems.

Use the Gaussian realization and notation of the mixed-odd Gaussian models [R155]. For every fixed odd anchor A and every Cameron–Martin vector z, a measurable event avoiding all odd h-term progressions in the tail of the anchor also satisfies

$$\mu\{x : U_A^j(x + jz) \mathit{belongsto} E \quad \text{for all} \quad 0 \le j < h\} = 0.$$

Here the action notation has the same convention as the Gaussian observations: $U_A^j(x+jz)$ at a test vector f reads $W(U_A^j f)+j\langle U_A^j f, z\rangle$. This genuine measurable restriction is stronger than cylinder-density of the traces. It says neither that the resulting affine pencils recur on every positive measurable event nor that the unrestricted prime system has been fully classified.

A binary partition bound For a function p from subsets of $[N]$ into $[0,1]$, put $w_i = \sup_{S \not\ni i} |p(S+i) - p(S)|$, $W = \sum_i w_i$. If $W < 1$, then

$$Z(p) = \sum_S p(S)^{|S|}(1 - p(S))^{N-|S|} \le 1/(1 - W).$$

Choose a uniform random ordering, with S_k its first k sites. Then $Z(p) = \sum_{k=0}^{N} \mathbb{E}\,\mathbb{P}(\mathrm{Bin}(N, p(S_k)) = k)$. Write the point probability as a difference of adjacent binomial cumulative distribution functions $F_{k,N}$ at the same parameter and telescope in k. This gives 1 plus the sum of the differences $F_{k,N}(p(S_k)) - F_{k,N}(p(S_{k+1}))$, $0 \le k < N$. Since $-F'_{k,N}(u) = N\,\mathbb{P}(\mathrm{Bin}(N-1, u) = k)$, each difference, in absolute value, is bounded by Nw_i times the integral of this $N-1$ point probability along the linear interpolation between the two parameters (without the factor N). Conditional on the next site being i, S_k is a uniform k-subset of the other sites, and the next-site probability is $1/N$. For each interpolation parameter t, summing k thus yields precisely the partition function on the remaining indices for $p_i^t(S) = (1-t)p(S) + tp(S+i)$. Its influences are bounded by w_l, $l \ne i$. Induction gives $Z(p) \le 1 + \sum_i w_i/(1 - W + w_i) \le 1/(1-W)$, including the empty-parameter base case. No monotonicity is necessary.

A smoothed Gaussian selection First let $z \ne 0$ have finite Walsh support in the underlying complex L^2 realification. Put $v_0 = iz/\|z\|$. Pair fresh primes $u = \{i, k\}$, set $a_i = \pi/p_i$, $b_u = (a_i^2 + a_k^2)^{1/2}$, $m_u = (a_i\epsilon_i + a_k\epsilon_k)/b_u$, and $Y_u = W(v_0 m_u)$. These are independent standard real normals. Fix an odd smooth function s, bounded by 1 with bounded derivative,

for example the hyperbolic tangent, and $0 < \sigma < 1$. Conditional on the Gaussian sample, include pair u independently with probability $(1+\sigma s(Y_u))/2$. Set $c_* = \sigma\mathbb{E}[Y s(Y)]/2 > 0$ for this choice. In successively late fresh tails take finite panels with $\sum b_u \to \|z\|/c_*$, possible by reciprocal divergence, vanishing maximum, and convergence of $\sum b_u^2$. The anchor plus the included pairs is an odd finite support.

For every finite Walsh test f and every fixed slot j, its observation at this word converges in probability to $W(U_A^j f) + j\langle U_A^j f, z\rangle$. In the fresh-Walsh expansion, the constant term tends to $W(U_A^j f)$. The singleton terms, to first order, sum $b_u\, W(ijU_A^j f\, m_u)$ over included pairs. They are independent by pairs and have total variance $O(\sum b_u^2)$. Their expectation tends to the stated shift since $\langle ijU_A^j f, v_0\rangle = j\langle U_A^j f, z\rangle/\|z\|$. Errors in singleton coefficients are $O_j(a_i \sum_{\text{tail}} a_k^2)$ times a bounded-norm head observation, so summing their absolute expectations costs $o(1)$ because $\sum b_u$ stays bounded. Terms with two or more fresh signs are in a Gaussian subspace independent of all selectors and selection coins, and conditional on the choice have variance $O_j((\sum_{\text{tail}} a_k^2)^2)$. This is the expansion detailed in the preceding note.

Uniform L^2 bounds on the actual slot marginals Let R_j be the density of the j-th selected point with respect to the original Gaussian law. Such a density exists for each finite panel by summing the selector probabilities over the finitely many inverse transformations. Squaring and using invariance brings in two independent *fair* masks F, G; for their signs $\epsilon_{F,u} = 2\,1_{u\in F} - 1$, the Gaussian integrand for the second moment is

$$\prod_u (1 + \sigma\epsilon_{F,u} s(Y_u))(1 + \sigma\epsilon_{G,u} s(Y'_u)),$$

averaged over fair masks and centered standard Gaussian pairs with within-family covariance identity. Their cross-covariance C is that between the vectors $v_0 m_u$ and their translates by the signed group difference $j(F-G)$; the fixed anchor cancels. For $H = F \triangle G$ put $r_H = \prod_{u=\{i,k\}\in H} \cos(ja_i)\cos(ja_k)$, positive on sufficiently far tails. Independence of the Walsh signs gives, with constants for the fixed j,

$$|C_{uv}| \le C_j b_u b_v (u \ne v), |C_{uu} - r_H| \le C_j b_u^2.$$

In the off-diagonal integral each distinct factor m_u, m_v has mean zero, and expanding its one-pair complex exponential gives a factor $O_j(b_u)$ resp. $O_j(b_v)$; the other factors have modulus at most one. On the diagonal $m_u^2 \le 2$ has mean one, and replacing its one cosine by its mean costs $O_j(b_u^2)$; imaginary terms have zero real part.

Compare with cross-covariance $r_H I$ by Gaussian interpolation. Both endpoint covariance matrices are positive semidefinite. The logarithm of the expectation of the positive product above changes by at most $C_\sigma \sum_{u,v} |C_{uv} - r_H 1_{u=v}|$ along the segment between them: Gaussian integration by parts gives cross mixed derivatives, and each individual factor has logarithmic derivative bounded by $\sigma\|s'\|_\infty/(1-\sigma)$. One may add small independent noise first at singular endpoints and pass by bounded convergence. This comparison costs a multiplicative factor at most $\exp(C_{j,\sigma}[(\sum b_u)^2 + \sum b_u^2])$, uniform in the panel size.

At cross-covariance $r_H I$ the integral factors into $\prod_u (1+\sigma^2(-1)^{1_{u\in H}} u_*(r_H))$, where $u_*(r) = \mathbb{E}[s(Z)s(Z')]$ for correlated standard normals of covariance r. Gaussian interpolation also gives $|u_*'(r)| \le \|s'\|_\infty^2$. As H is uniform, the mask average of this product is $Z(p)$ for $p(H) = (1-\sigma^2 u_*(r_H))/2$. Its influences are $O_{j,\sigma}(b_u^2)$, since adding u changes r_H by at most $O_j(b_u^2)$. Their sum tends to zero. The partition bound proves that $\mathbb{E}R_j^2$ is bounded uniformly on all far panels, by a constant depending on z, j, σ but not on the number of selected axes.

Consequence for measurable events Cylinder observations converge to the affine pencil jointly as above. Approximate each bounded measurable test by continuous cylinder observables in L^2. The error for actual slot marginals is bounded uniformly by Cauchy–Schwarz using the proved density bound. The limiting slot also has L^2 density: it is an orthogonal transformation of a deterministic Cameron–Martin translate of Gaussian measure. Thus the actual correlations converge for every measurable event. If the event avoided all true odd words, each term of every finite mixture is zero even with state-dependent selection, by absolute continuity and countability of the words. The stated affine correlation must vanish. Finite-Walsh shifts are dense in the Cameron–Martin Hilbert space and Gaussian translation is strongly continuous on indicators in their integrals on compact sets of shift parameters, so the same holds for every Cameron–Martin shift (use cylinder approximation and locally bounded L^2 translation densities). The case $z = 0$ also follows by continuity.

R169. A reciprocal-prime obstruction for pair avoidance on signed odd supports

Scope and status. Result with proof.

Result. Signed odd supports have an explicit pair-avoidance bound reciprocal to the sum of inverse prime orders; divergent sums give pair and nilrotation recurrence.

Scope. Pairs in arbitrary actions and all finite lengths only in the niltranslation class are distinct conclusions.

Let $K = \prod_{i=1}^{n} C_{p_i}$ with odd integers $p_i \geq 3$ (the finite estimate does not need primality). Let V consist of all v with entries $0, \pm 1$ and an odd number of nonzero entries, allowing the signs independently. If $A \subset K$ contains no pair with difference in V, then

$$|A|/|K| \leq 1/(\sum_{i=1}^{n} 1/p_i).$$

The usual elementary bound of one half also holds, since a fixed difference has a disjoint translate of A. The displayed estimate records the decay as more odd cycles are added.

Proof. Lift A to the integer grid, periodically modulo the coordinate periods p_i. Work on the real torus $\prod(R/2p_iZ)$ with its quotient maximum metric and Haar probability. Divide the occupied lattice sites into even and odd according to the sum of their integer coordinates. An occupied pair of opposite parities must have maximum-metric distance at least two: otherwise its coordinate differences may all be represented by $-1, 0, 1$, and an odd number are nonzero. Since each $p_i \geq 3$, the projected difference is a forbidden nonzero element of V. Form the two closed sets E_+, E_- by filling the side-one cubes centered on the respective sites. Their mutual maximum-metric distance is at least one. The union has Haar measure $a = |A|/|K|$ and translation by $p_i e_i$ exchanges the two sets, for every i.

If $a > 0$, both sets are nonempty. Define $u_\pm = \min(1, dist_\infty(\cdot, E_\pm))$ and $F = (u_- - u_+)/2$. This is one-Lipschitz in the maximum metric. It takes the values $\pm 1/2$ on $E_\pm$ and satisfies $F(x + p_i e_i) = -F(x)$. Hence $2\int |F| \geq a$. By absolute continuity on coordinate lines, translation invariance of Haar probability, and the triangle inequality,

$$2\int |F| = \int |F(x + p_i e_i) - F(x)| dx \leq p_i \int |\partial_i F|.$$

At almost every point the sum of the absolute coordinate derivatives is at most one: Rademacher's theorem identifies the differential, whose operator norm for the maximum norm is this sum and is at most the Lipschitz constant. Sum the preceding inequalities divided by p_i to get $a\Sigma 1/p_i \leq 1$. The $a = 0$ case is immediate. An arbitrary finite-group action has the same upper bound for pair-avoiding events: its almost-everywhere orbit masks obey the pair constraint and their mean fraction is the event probability.

In particular for $P_0 = \oplus C_{p_i}$ with distinct odd primes and divergent $\Sigma 1/p_i$, every positive-measure event has a pair return along signed odd support, even on every tail of the axes. Restricting to an increasing finite subgroup gives this directly by the estimate.

For comparison if $\Sigma 1/p_i < \infty$, the saturated rational model from squarefree tails [R241] has $b = 1$ for this family. Take a far tail where $\Sigma 1/(2p_i) < 1/4$. The character sending e_i to $(p_i - 1)/(2p_i)$ sends every signed odd support into $[1/4, 3/4]$ mod 1. For any given short arc, take M a multiple of all small positive denominators in the rotation hitting argument, then move the tail further past the primes dividing M so this character has an Mth root there. The root-rotation arc complement blocks all tail differences with an arbitrarily small loss at a fixed length. Use the squarefree-tail transfer lemma. The divergence estimate concerns pairs only; it proves neither a higher-order recurrence theorem nor an intermediate all-length capacity. The argument requires all independent sign patterns and is not a claim for just ordinary unsigned odd finite sums.

Weighted boxes. The same bound holds with $\Sigma L_i/p_i$ in place of $\Sigma 1/p_i$ when $1 \le L_i \le (p_i - 1)/2$ are integers and the allowed directions comprise all vectors whose centered entries are in $[-L_i, L_i]$ and have odd total sum. In the weighted metric $\max_i |x_i - y_i|/L_i$, the thickened opposite-parity sets have distance at least one: two opposite lattice sites differ by at least $L_i + 1$ somewhere, and the unit cubes use up at most one unit of this gap. The differential bound is now $\Sigma L_i |\partial_i F| \le 1$. If the full weighted series converges the same half-character separates every sufficiently late support, since its total phase error is bounded by half that series on the tail. Thus a positive pair-avoider cannot yield an intermediate counterexample in any of these families.

Bohr and nil tests in the divergent prime case. For sharp scope, this paragraph is about P_0-actions, not all actions of its torsion-free cover. In fact even the unsigned odd-support family in the unit case is Bohr recurrent when $\Sigma 1/p_i = \infty$. Let $x_i \in T^m$ be the images of e_i under any finite character test and look at finite subset sums of $(x_i, 1)$ in $T^m \times C_2$. The intersection of the closures of all tail finite-sum sets, with the empty sum allowed, is a closed subgroup L. Indeed addition of two tail limits can be approximated by sums with disjoint finite supports, in that order, yielding a closed subsemigroup; every compact subsemigroup of a compact group containing zero is a group (positive powers approximate the inverse).

If $(0, 1) \notin L$, projection of L into T^m is injective, and the parity character on L extends to an ambient torus character of that projected compact subgroup, by character extension for closed subgroups of compact abelian groups. Thus for some $k \in Z^m$ every $(x, e) \in L$ satisfies $k \cdot x = e/2$ mod 1. The nested compact tail sets approach L uniformly, so for every sufficiently late starting index, all finite sums obey $|| \sum_{i \in F} (k \cdot x_i - 1/2)|| < 1/8$, and the bound tends to zero with the starting index. In particular the centered individual errors tend to zero. Their positive and negative subseries each have finite sum: otherwise a finite subcollection of all one sign in a far tail first crosses $1/4$, with overshoot smaller than $1/8$, contradicting the uniformly small circular norm. On the other hand $k \cdot x_i$ is p_i-torsion, so its distance from $1/2$ is at least $1/(2p_i)$ for every odd p_i. This contradiction proves $(0, 1) \in L$, exactly the required odd-subset Bohr return in every tail.

For this acting group, Bohr recurrence on the tails already suffices for finite-dimensional niltranslation tests. To see this, let G/Γ be a compact nilmanifold acted on by a homomorphism into its group of left niltranslations. Quotient G by the kernel of its action so its action is faithful (the kernel is a discrete normal subgroup of Γ). A finite-order translation acts trivially by conjugation on $\texttt{Lie(G}^0\texttt{)}$: the adjoint action of every element of a nilpotent group is unipotent and a unipotent operator of finite order in characteristic zero is the identity. It therefore commutes with the connected group G^0. For primes larger than the number of connected components of the nilmanifold the translation by e_i also preserves each

component, by its prime order. On a component it lies in the centralizer of the transitive G^0-action. This centralizer is a compact Lie group, explicitly the right-translation group $N_{G^0}(\Lambda)/\Lambda$ for the corresponding homogeneous presentation G^0/Λ: the closed normalizer contains the cocompact lattice, hence its quotient is a closed subset of the compact component. The group of component-preserving equivariant homeomorphisms is thus a finite product of compact Lie groups. The closure of the abelian tail action inside it is compact abelian. In its compact-open topology, returning near identity is a finite-character test. The Bohr recurrence just proved consequently gives simultaneous pointwise nil returns of every fixed length, and measurable returns on such a system for any invariant probability, by continuity in measure (continuous functions are dense in L^1). The general all-length density question on the abstract group still involves noncompact actions.

R171. Odd prime-axis sums recur at all orders on Abramov systems

Scope and status. Result with proof.

Result. Divergent-reciprocal odd prime sums are multiply recurrent on phase-polynomial systems and the specified totally disconnected or explicit non-Abramov extensions.

Scope. Sparse-average characteristicity is not proved for the full Host-Kra hierarchy.

Let $G = \oplus C_{p_i}$, with **distinct odd primes** tending to infinity and $\sum_i 1/p_i = \infty$. Let D comprise all sums of distinct basis vectors over odd finite supports, on any prescribed tail. A phase polynomial of degree $\leq s$ on a G-system is a circle-valued function whose $(s+1)$-fold discrete differences vanish almost surely. If a system is generated by its finite-degree phase polynomials, every positive event has h-term recurrence along D for every h. The same conclusion holds on each Abramov factor of an arbitrary system, and after taking factors and inverse limits. This does not prove characteristicity for the averages sampled by these sparse sets of directions.

The prime decomposition of a phase polynomial Work on an ergodic system, writing f additively in T and $\Delta_i = T^{e_i} - 1$. Finite-order action gives $(1 + \Delta_i)^{p_i} - 1 = 0$. Hence $p_i \Delta_i f$ is a sum of derivatives of order at least two in i. Iterating this relation shows $p_i^s \Delta_i f = 0$ when f has degree $\leq s$. For $i \neq l$ the mixed derivative $\Delta_i \Delta_l f$ is killed by both coprime powers, hence vanishes. For $p_i > s$ one has even $p_i \Delta_i f = 0$: after factoring out $p_i \Delta_i$ the binomial relation on f has constant coefficient one and higher nilpotent terms (all intermediate binomial coefficients are divisible by p_i). Its multiplier is invertible on the finite-degree derivative module. The small primes can always be put in the head.

Thus $\Delta_i f$ is measurable on $Y_i = I(\oplus_{l \neq i} C_{p_l})$. The ergodic finite cyclic action on that factor is a p_i-cycle or a singleton. The collective Y factor is their independent product, since the translations act transitively on every finite product of the finite cyclic factors. Choose an origin per cycle. For any finite family $f_1, \ldots, f_k$ of degree $\leq s$ the trajectory increments at i have local form

$$(T^{je_i} f_l - f_l)(x) = P_{li}(a + j) - P_{li}(a),$$

where a is the local coordinate and P_{li} has zero constant and is an ordinary polynomial of degree $\leq s$ with values in p_i-th roots viewed additively ($P = Q/p_i$, $Q \in F_{p_i}[a]$). Indeed integrate the local difference on a full cyclic orbit starting at zero and apply the discrete Newton expansion in the binomial basis, whose denominators are invertible modp_i. For the singleton case only a constant increment remains, and an affine polynomial gives the same formula if one adjoins a dummy uniform cyclic coordinate. The different axes add independently. The finite-family factor generated by Y and the f_l is therefore represented by the base Y and the register $v = (f_l(x))_l \in T^k$, with an arbitrary invariant ergodic law having Haar base marginal and the displayed additive action at each axis. It contains all translates of its named polynomials.

Finding a Haar increment law through the identity For fixed h, choose a sequence of odd supports F escaping to infinity along which simultaneously

$$\sum_{i\in F} P_{li}(t) \to 0 \in T (1 \le t < h, 1 \le l \le k).$$

This is precisely the odd-support Bohr recurrence theorem [R169] applied to the finitely many characters sending $e_i \mapsto P_{li}(t)$, which are p_i-torsion. Consider the joint register increments across the h slots. The Fourier coefficient at any integer test $\{m_{lt}\}$ is a product of complete averages of the scalar local polynomials

$$Q_i(a) = \sum_{l,t} m_{lt}[P_{li}(a+t) - P_{li}(a)].$$

If every Q_i on the support is constant, its average is its phase at zero and the product tends to one by the Bohr choice. If even one is nonconstant, the product tends to zero along such indices. For an ordinary nonconstant degree-d polynomial over F_p with $p > d$, the absolute complete average is at most $((d-1)/p)^{1/2^{d-1}}$ when $d \ge 2$, and is zero when $d = 1$. To see this apply Cauchy–Schwarz and successive differences $d-1$ times. Unless one of the increments is zero the remaining polynomial is linear with nonzero leading coefficient and so has average zero. This proves the required uniform $o(1)$ without any quantitative equidistribution theorem. Pass to a subsequence so the two alternatives stabilize for each countable test. The limiting increment law ν has every Fourier coefficient zero or one; it is Haar on a closed subgroup of $(T^k)^{h-1}$ and contains the identity in its support.

Arbitrary invariant register laws and measurable events Let Λ consist of those integer register characters q with $e(q \cdot v)$ measurable on Y. If this character's conditional expectation on Y is nonzero, its modulus is invariant since each translation increment is Y-measurable. By ergodicity that modulus is a positive constant and the quotient by the conditional expectation is invariant; hence the character is Y-measurable. Otherwise the conditional expectation vanishes. By Fourier uniqueness, therefore, adding any element of the compact subgroup $H = \Lambda^{\perp} \le T^k$ to v preserves the law μ. For every $q \in \Lambda$, its increments on escaping supports go to zero in measure by approximating its Y-measurable observable on a finite head, which all tail moves fix. It follows that ν is supported on H^{h-1}.

The initial state is independent of the increments in the limit. Test the initial state by a function of Y times a register character. If the latter is outside Λ, conditional expectation on Y shows a zero integral against the Y-measurable increments. For a character in Λ, approximate the whole initial test by a finite-head function of Y; the selected tail profiles are independent of that head. The base coordinates themselves do not move in the product topology in the limit. Thus the limiting full h-slot law is an initial μ sample together with an independent ν-distributed tuple of fiber shifts (zeroth shift zero).

For a positive measurable event A the correlation as a function of these fiber shifts is continuous on H^{h-1}, by invariance of μ and strong L^1 continuity of compact translations; it equals $\mu(A) > 0$ at the identity. Since zero lies in the support of ν its integral is positive. The actual progression correlations converge to that integral: first test bounded continuous approximants and then approximate indicators in L^1, using the common one-slot marginal μ. Some actual selected support gives a positive return. Finite-family factors suffice for all phase-polynomial-generated systems by the conditional-density inverse-limit argument

[R164]; this avoids needing any uniform lower bound as the list of observables increases. Ordinary ergodic decomposition handles mixtures.

Where this applies in Host–Kra theory In particular this applies to finite-order systems satisfying Shalom's splitting condition, which are strongly Abramov. Shalom's Theorem 1.21 also gives an order-< 3 circle extension of a solenoid that is not Abramov of any order, so ordinary finite Host–Kra order alone does not allow substitution in the proof above. See Shalom, Host–Kra theory for $\oplus F_p$ systems, Theorems 1.19–1.26.

There is also a direct route for totally disconnected finite-order systems, even without reciprocal divergence and with repetitions of odd primes. Shalom's Theorem 5.1 gives each totally disconnected structure group as a product of cyclic groups of prime-power orders from the acting group's prime set. With all acting primes odd, these are pro-odd groups, so the elementary odd-extension theorem [R164] applies at every successive level starting from a point. The basis sums are genuine nonzero directions in this application. Neither this observation nor the Abramov proof establishes that the full Host–Kra factor is characteristic along unsigned odd supports.

Shalom's explicit order-< 3 non-Abramov example, Theorem 1.21 and Section 9 of the same paper, also has recurrence along these odd prime sums. His construction partitions the primes into P_1 and P_2. At least one class has divergent reciprocal sum after discarding 2. Suppose this is P_1. Its acting subgroup G_1 moves only the second solenoid U_2, so the first solenoid coordinate (x_1, v_1) can be frozen in a measurable choice of representatives. On each such fiber the circle cocycle restricted to G_1 is cohomologous to a homomorphism. In fact, in Shalom's notation, the generator lift is $\alpha_g = (0, b_g)$. Formula (9.1) gives $\tilde{f}(x_1, x_2 + n) = \tilde{f}(x_1, x_2)e(x_1 n)$, so $B(x_2) = \tilde{f}(x_1, x_2)e(-x_1 x_2)$ is a well-defined function on the circle and hence on U_2. On these generators the $\tilde{f}$-ratio is a B-coboundary times a constant; the other factors in $f \cdot \chi \cdot \phi$ depend only on the frozen x_1,v_1. Removing this one common coboundary leaves rotations of the fiber register, and the constants define a homomorphism since this is a genuine G_1-action. If P_2 has divergent reciprocal sum instead, the symmetric gauge with the corresponding sign works. Thus the restriction disintegrates into compact rotation systems. Bohr recurrence on the selected class and countability of the candidate directions give the assertion for every positive event. The same observation works on finite products of these examples, even with distinct bipartitions: a common finite refinement of the prime classes has a divergent cell, on which all the restrictions are mixtures of compact rotations.

R191. Pairwise-independent halfspaces and compatible endpoint marginals

Scope and status. Result with proof.

Result. A halfspace carrying pairwise-independent bounded variables has uniformly positive product probability; this yields compatible endpoint marginals above a universal density.

Scope. Finite collections of bounded real variables and one-coordinate compatibility.

Qualifications. Tree or endpoint consistency does not establish cyclic or temporal consistency.

Theorem. Let $X_1, \dots, X_r$ be bounded pairwise-independent real random variables, with arbitrary and possibly very different laws. If $\Sigma X_i \geq L$ almost surely, and $X_i^i nd$ are mutually independent with the same respective marginal laws, then

$$P(\Sigma X_i^i nd \geq L) \geq c_0$$

for an absolute positive constant independent of the number of variables and their distributions. In particular $c_0 = 10^{-6}$ works. The result is not true with arbitrary dependence: take one weight-n item chosen with probability ϵ and n weight-one items all chosen exactly when the huge item is absent. The dependent sum is always n, but independent copies succeed with probability $\epsilon + (1-\epsilon)^{n+1}$, which tends to zero for $\epsilon = (\log n)/n$.

Proof. Fix $c = 1/256$ and choose a median m_i of each variable. If $\Sigma P(X_i \neq m_i) \leq c$, there is positive probability under the given law that all variables equal their medians, by the union bound. The same event under the product law has probability at least $1 - c$, and its sum is at least L. Otherwise there exists $T > 0$ with

$$\Sigma P(|X_i - m_i| > T) \leq c \leq \Sigma P(|X_i - m_i| \geq T),$$

by the one-sided tails of a finite measure. Clip X_i to $[m_i - T, m_i + T]$, obtaining Y_i. These variables are still pairwise independent. Write $\mu = \Sigma EY_i$, $\sigma^2 = \Sigma \operatorname{Var}(Y_i)$. Since m_i is a median, $|EY_i - m_i| \leq T/2$. On the event $|X_i - m_i| \geq T$ this entails $|Y_i - EY_i| \geq T/2$. Therefore $\sigma^2 \geq cT^2/4 > 0$.

Let $B_i = \{X_i > m_i + T\}$, $N = \Sigma 1_{B_i}$ and $G = \{N > 0\}$. The square of $N - 1_G$ is at most $N(N-1)$, which has expectation at most c^2 by pairwise independence of the tail indicators. Moreover $\operatorname{Cov}(\Sigma Y_i, N) \geq 0$: cross terms vanish and Y_i takes its largest value on B_i. It follows by Cauchy–Schwarz that

$$\operatorname{Cov}(\Sigma Y_i, 1_G) \geq -c\sigma.$$

On the complement of G clipping can only increase the original sum, so $\Sigma Y_i \geq L$ there. The preceding covariance bound and $P(G) \leq c$ give $L \leq \mu + c\sigma/(1-c)$.

Under mutually independent sampling, standardize the clipped sum to $Z = (\Sigma Y_i^i nd - \mu)/\sigma$. Fourth-moment expansion and $|Y_i - EY_i| \leq 2T$ give $EZ^4 \leq 3 + 4T^2/\sigma^2 \leq 3 + 16/c$. By interpolation $E|Z| \geq 1/\sqrt{EZ^4}$; since Z is centered, $EZ_+ \geq 1/(2\sqrt{3 + 16/c})$. With $t = c/(1-c)$, Cauchy–Schwarz gives

$$P(Z \geq t) \geq (1/(2\sqrt{3+16/c}) - t)^2 > 10^{-5}.$$

Indeed the part of EZ_+ with $0 \leq Z < t$ is at most t, and the remainder is at most $\sqrt{P(Z \geq t)}$. Finally, in the product law condition on having no variable below $m_i - T$, an event of probability at least $1 - c$. The conditional variables remain independent, and each conditional law stochastically dominates the corresponding unconditioned clipped law: removing the lower tail and undoing the upper clipping can only increase the variable in quantile coupling. Thus the product law of the original variables has sum at least L with probability at least $(1 - c)P(Z \geq t) > 10^{-6}$, as required.

A marginal consequence. Let $C \subset \prod K_i$ be a compact set supporting a law whose i-th coordinate has marginal μ_i and whose coordinates are pairwise independent. Let a measurable $B \subset \prod K_i$ have product marginal measure greater than $1 - c_0$, with the K_i compact metrizable. There are probabilities on C and on B with the same i-th marginal for every i. Take a compact subset of B still of mass greater than $1 - c_0$. The compact convex sets of marginal vectors of probabilities on the two compact sets cannot be disjoint: strict separation would yield continuous real functions f_i with $\inf_C \Sigma f_i > \sup_B \Sigma f_i$ (using the chosen compact subset for the supremum). Apply the theorem to the pairwise-independent law on C, taking $L = \inf_C \Sigma f_i$. The product probability of this threshold event is at least c_0, a contradiction.

For the unforced endpoint-shear surrogate take $C = \{(X + jY)_{j \in J}\}$ with $Haar(X, Y)$; distinct j-projections are pairwise Haar. Thus no sum of arbitrary single-arm real potentials can certify a near-one avoiding event by comparing its independent row sum to the sum forced at every endpoint. More strongly, every sufficiently high row predicate admits compatible one-coordinate marginals at the model's row and endpoint constraints. Such marginals give a consistent law on the corresponding *bipartite tree* with one port of each offset type at every vertex (sample each new vertex configuration conditionally on its already sampled incoming port). The physical integer time graph has cycles, so the argument does not furnish an all-times trajectory there. It isolates a genuine compatibility requirement beyond any additive potential or one-edge-marginal obstruction.

R194. A finite cyclic model for periodic differences, with no change of length

Scope and status. Result with proof.

Result. For periodic integer differences, fixed-length capacity equals the supremum over finite cyclic masks; prime periods can be exhausted by prime powers.

Scope. The progression length is unchanged, but the cyclic period is unbounded.

Qualifications. The recorded proof uses the prior fixed-length nilfactor approximation and rational nilmanifold approximation; these substantial dependencies are not replaced by numerical checks.

Let $D = \{n \geq 1 : n \bmod m \in S\}$, where $S \subseteq \mathbb{Z}/m$. At each fixed $h \geq 2$, the integer extremal density $a_h(D)$ equals the supremum over finite cyclic groups of orders divisible by m of the densities of masks avoiding h-runs for every difference whose projection belongs to S. If $m = p$ is prime, it suffices to use orders p^v. Here the progression length is the original h; the period need not be bounded. The lower inequality follows by periodic lifting.

Finite nilmodel. Begin with the maximizing ergodic integer system. The weighted mean of its h-correlations on the residue classes S vanishes. The fixed-length angular proof [R123] supplies, on a finite-dimensional ergodic nilsystem (Y, S_0), an event A of mass arbitrarily near the original density and still with zero residue-weighted mean. To recall the exact-zero step, the conditional expectation onto Z_{h-1} has zero weighted mean by Bergelson–Host–Kra, Corollary 4.6. A positive superlevel event B inherits that zero mean, and its mass can be chosen arbitrarily close to or above the original one. In a fine finite nilfactor use $\{E(1_B \mid Y) > 1 - 1/(2h)\}$. At **each fixed step** d integrate over the starting fiber above y: on those y whose h deterministic images lie in this finite-factor event, the conditional sum of the h individual failure probabilities is less than $1/2$, by equivariance of disintegration. This proves the same zero mean on the finite factor before taking an AP joining. No assertion of conditional independence in an AP joining is used.

A connected ambient group. We can take the basepoint to be the lattice identity in a connected simply connected nilpotent Lie group G, with a rational lattice Γ, even when Y has $k > 1$ components. Indeed Y is the cyclic k-tower over the return to one connected component. As in the root construction [R170] that return has the affine form $a\phi$ on N/Δ, with N connected simply connected, ϕ a unipotent rational lattice automorphism induced by conjugation in the original nilpotent group. Put $G = N \rtimes_{\phi^s} \mathbb{R}$, $\Gamma = \Delta \rtimes_{\phi} \mathbb{Z}$, and take $b = (a, 1)^{1/k}$. Then the orbit of the identity under b stays on the k slices $j/k \bmod 1$, and is dense in their union. The map $b^j x \mapsto S_0^j x$ from the k slices over the initial one intertwines the transformations because b^k is the original affine return on that first slice. Each slice carries its component Haar law; Y below denotes this embedded union.

Work in the simply connected group $P = G^h \times \mathbb{R}$, with lattice $\Gamma^h \times \mathbb{Z}$. Consider the commuting translations

$$g_1 = (b, \ldots, b, 0), \qquad g_2 = (1, b, \ldots, b^{h-1}, 1/m).$$

The closure L of their joint orbit from the rational identity basepoint has finitely many connected components. By Leibman's orbit theorem, its Haar-component measure λ is the mean AP joining with the recorded residue angle, by first averaging the diagonal variable and then taking rectangular Følner limits. It therefore gives zero mass to A^h on the allowed

residue components. Its identity component is $L_0 = H\Gamma_P/\Gamma_P$, with H a rational connected group and Γ_P the ambient lattice.

Rational approximation preserving all residue constraints. Positive powers of g_1, g_2 stabilize L_0. Any such power g_i^l normalizes H: conjugating the connected subgroup H by the stabilizer and evaluating tangents on L_0 preserves its Lie algebra (stabilizers of ambient points are discrete). Equivalently, the identity component of the setwise stabilizer of L_0 is H by the tangent space at the lattice coset. Unipotence then implies g_i itself normalizes H, since $\log \mathrm{Ad}(g_i^l) = l \log \mathrm{Ad}(g_i)$ preserves its Lie algebra. The normalizer $M = N_P(H)$ is connected and rational: in logarithms it is given by the rational linear conditions $[X, \mathfrak{h}] \subseteq \mathfrak{h}$. All components of L are consequently H-orbits, obtained by translating L_0 by elements in the group generated by g_1, g_2. In M/H both $\log(g_i H)$ are rational, since a power g_i^l lies in $H(\Gamma_P \cap M)$.

Write $v = \log b$ and, for a variable $w \in \mathrm{Lie}(G)$, let $g_1(w), g_2(w)$ be the two displayed tuples with $\exp w$ in place of b. The conditions

$$\log g_i(w) - \log g_i(v) \in \mathfrak{h} \quad (i = 1, 2), \qquad \mathrm{time}(w) = 1/k$$

define a nonempty rational affine subspace. Indeed the two logarithm tuples depend rational-affinely on w, and the prescribed cosets modulo the rational subspace $\mathfrak{h}$ are rational by the preceding quotient calculation. Choose rational $w \to v$ in that subspace. Each $b' = \exp w$ has a periodic orbit at the identity coset (some positive power belongs to Γ), and it preserves the union of rational slices. In M/H its diagonal and slope tuples agree exactly with those of b. Hence, for any integers j, n, the tuple $g_1(w)^j g_2(w)^n$ applied at the identity belongs to L on the component carrying the correct residue $n \bmod m$. The uniform measures on the finite b'-orbits tend weakly to the original Haar measure on Y: every limit is b-invariant, using continuity on the compact space, and b is uniquely ergodic on Y.

An open mask. Choose a small absolutely continuous probability on a neighborhood of the identity of H, tending to the identity, and retain points where the translation in each of the h coordinate projections sends that point into A with probability greater than $1 - 1/(2h)$. As in the angular proof these masks tend in measure to A. They can be taken open: projection of H to each of the h groups acts transitively on an individual Y-slice. Indeed it lies in the spatial connected group N because the slices are discrete in time, and has full dimension there since the corresponding component marginal, as part of λ, is absolutely continuous relative to Haar on the slices and cannot be supported on a lower-dimensional smooth image. Thus the projected small-ball laws have L^1 densities on N; convolution of a bounded measurable indicator against them is continuous on each slice. If an allowed tuple of L were in the h-fold mask, a positive fraction of those H-translations would place it in A^h. Their law is absolutely continuous on its H-component, contrary to the zero weighted mean. As the mask is open, the periodic b'-orbits approximate its density from below. Mark the corresponding indices in a period divisible both by the orbit period and by m. Every allowed progression of indices maps to a tuple of L on an allowed residue component, so this finite mask is an exact h-avoider.

Prime period. If the constructed period is $p^v c$ with $(p, c) = 1$, take a densest slice at fixed c-coordinate. Every unit or nonunit residue step on the p^v-group whose reduction belongs to S has a CRT lift leaving that slice fixed, so the slice still avoids all the specified h-runs. This proves the assertion for prime periods, and the same reasoning keeps only prime factors of m in general. It gives a finite cyclic formulation for testing candidate forcing rows, without producing the required rows.

R197. Endpoint dichotomies for translated signed trace boxes of Pisot units

Scope and status. Result with proof.

Result. Translated signed trace boxes of every quadratic Pisot unit have endpoint capacities; a stated conjugate-product inequality gives the same alternative in higher degree.

Scope. The higher-degree result is conditional on an explicit numerical inequality, not a theorem for all Pisot units.

Let λ be a real algebraic unit greater than one, with every other conjugate β_j of modulus less than one. Put $O = Z[\lambda]$, a full order in $K = Q(\lambda)$, and $a_i = \mathrm{Tr}(\lambda^i)$. At any fixed starting $m \geq 0$ consider

$$D = e + \sum_{i \geq m} 0, a_i, -a_i (finitesupport).$$

We obtain an integer all-length zero–one dichotomy for **every quadratic Pisot unit** and for units in any degree with the strict norm condition below. The three small quadratic units need a separate argument because a Bohr-cancelling carry can be genuinely infinite.

Codifferent and carry Write $O^\vee = \{\theta \in K : \mathrm{Tr}(\theta O) \subset \mathbb{Z}\}$ for the codifferent lattice. Its classes modulo one are precisely the frequencies θ for which $||a_i\theta|| \to 0$, and also precisely the frequencies for which these norms are summable. In one direction,

$$a_i\theta - \mathrm{Tr}(\theta\lambda^i) = \sum_{j>1} (\theta - \theta_j)\beta_j^i$$

is exponentially small for $\theta \in O^\vee$. Conversely, the nearest integers k_i to $a_i\theta$ eventually satisfy the monic integer recurrence given by the minimal polynomial of λ: its integer residual tends to zero. The unit constant coefficient extends this recurrence backwards on the integers to $i = 0$. There is a unique $w \in O^\vee$ with $\mathrm{Tr}(w\lambda^i)$=k_i for the first $\deg(\lambda)$ terms, by nondegeneracy of the trace form on the power basis. The recurrence gives this equality for every i. Comparison of the leading λ^i terms with $a_i\theta$ gives $w = \theta$ in the expanding embedding. All statements are invariant under changing a representative by an integer.

If D is Bohr recurrent, the signed-path characterization [R255] gives choices $s_i \in \{0, \pm 1\}$ cancelling e in every codifferent character. Put $U_j = \sum_{i \geq m} s_i \beta_j^i$. The vector $e + \sum_{j>1} U_j, -U_2, \ldots, -U_r$, viewed in real Minkowski space with conjugate complex coordinates paired, pairs integrally under the trace pairing with every $\theta \in O^\vee$, by the displayed formula. Thus it is the embedding of some $v \in O$, since the dual of this codifferent lattice is O. It has Tr(v)=e. For $n \geq m$ set

$$v_n = (v + \sum_{m \leq i < n} s_i\lambda^i)/\lambda^n \in O.$$

Its jth conjugate, $j > 1$, equals minus the tail $\sum_{i \geq n} s_i \beta_j^{i-n}$, of modulus at most $1/1 - |\beta_j|$. Its expanding coordinate has modulus at most $1/(\lambda - 1) + o(1)$. These states are in a fixed bounded lattice region and hence finite in number. In particular cancellation feasibility always reduces to a finite directed carry graph after the initial head (with edges $v_{n+1}{=}v_n + s_n/\lambda$). Every infinite admissible walk there has an eventually periodic alternative.

If

$$(\lambda - 1) \prod_{j>1} (1 - |\beta_j|) > 1,$$

the absolute algebraic norms of v_n are < 1 eventually. Such algebraic integers vanish, so all sufficiently late digits are zero. Taking the trace at a vanishing state gives the genuine finite equality $e + \sum_i s_i a_i = 0$. Conversely a finite equality leaves a full nonempty IP tail in D, which gives all h-term returns at positive density by finite density Hales–Jewett. If D has no finite equality, it is Bohr separated and has all-length critical density one. Fixed early terms of value zero do not affect the argument, and a nonzero return is available on the IP tail.

The three small quadratic units For quadratic units $\beta{=}\pm 1/\lambda$. The strict condition covers $\lambda > (3 + \sqrt{5})/2$. The only smaller or equal quadratic Pisot units are $\lambda = \phi = (1{+}\sqrt{5})/2$, $1{+}\sqrt{2}$, and $\phi^2 = (3{+}\sqrt{5})/2$, by writing their minimal polynomials as $X^2 {-} tX \pm 1$. Set $R = 1/(1 - |\beta|)$. For each of these three choices the complete set of finite balanced expansions contains every $w \in O$ with $|w'| < R$, and is contained in the closed strip $|w'| \leq R$. Here w' is the other real conjugate.

Indeed the closed-strip inclusion follows by summing the geometric series. For the reverse inclusion, the three open intervals $s + (-|\beta|R, |\beta|R)$, $s = -1{,}0{,}1$, overlap and cover $(-R, R)$ since $|\beta|R > 1/2$ and $1 + |\beta|R = R$. Successively choose s with $|(w' - s)/\beta| < R$, replace w by $(w - s)/\lambda$, and stay in O since λ is a unit. Along this trajectory the real coordinates satisfy $|w_n| \leq 1/(\lambda - 1) + o(1)$ and $|w_n'| < R$; hence all late states form a finite set. We next check that those late states have finite expansions.

- At $\lambda = \phi^2$ the two limiting bounds have product one. Finiteness of the states gives a uniform strict margin in $|w_n'| < R$, so all sufficiently late norms have modulus < 1, forcing $w_n = 0$.
- At $\lambda = 1{+}\sqrt{2}$ write $w_n = a{+}b\sqrt{2}$. The bound on $|w_n {-} w_n'|$ is at most $1/\sqrt{2}{+}R{+}o(1) = 1 + \sqrt{2} + o(1) < 2\sqrt{2}$, forcing $b = 0$. Then $|a| < 1$ eventually, again giving zero.
- At $\lambda = \phi$ write $w_n = a + b\phi$. Here $|b|\sqrt{5} \leq \phi + \phi^2 + o(1) < 2\sqrt{5}$, so $|b| \leq 1$. For $b = 0$ the late states are $0{,}\pm 1$. For $b = 1$ the real-coordinate bound gives $a \leq 0$ eventually, and the strict inequality $a + \beta > -\phi^2$ gives $a > -2$. Thus the possible states are $\phi - 1$ and ϕ. The $b = -1$ case is the negative of this list. All have finite balanced expansions; append one after the already chosen prefix.

Multiplying by λ^m, the signed tail expansions in O contain the full strip $|w'| < R|\beta|^m$ and are contained in its closure. The trace map has image gZ, $g = \gcd(2, \operatorname{Tr} \lambda)$. Choose $\xi \in O$ of trace g and a generator ζ of the rank-one kernel of the trace on O. The possible conjugates w' for an element of trace $n \in gZ$ have the form $(n/g)\xi' + k\zeta'$. The ratio ξ'/ζ'

is irrational, since rational proportionality in one field embedding would force proportional trace. Projecting the two strips by the trace shows that the translated integer box D lies between the pullbacks of the open and closed intervals of radius $R|\beta|^m/|\zeta'|$ on this circle rotation, with center shifted by e, in the coset $e + gZ$.

If $e \notin gZ$, or zero lies outside the relevant closed circle interval after imposing that congruence, D misses a Bohr neighborhood, giving endpoint one. Otherwise the interval is either the whole circle or has zero in the closure of its interior, including a possible one-sided boundary. Its open part pulls back to a set of full positive-density multiple recurrence on gZ. To see this, every Nil-Bohr zero neighborhood meets that pullback in positive upper Banach density. Take its nilrotation jointly with the irrational circle, passing to a connected-component multiple if necessary. The joint orbit closure is a homogeneous subnilmanifold, and its projection onto the circle is open: on its identity-component presentation the projection is a surjective Lie group map, since its image has finite index in the full circle and is thus all of it. Therefore the neighborhood of the original point contains an open positive-Haar-measure set whose circle value lies in the specified interval. Unique ergodicity gives the density assertion. The multiple return-time set of any positive event contains a Nil-Bohr zero neighborhood up to removal of a set of zero upper Banach density, by Huang–Shao–Ye, the almost Nil-Bohr return theorem. Hence the open pullback supplies the required return. Equivalently apply this argument after replacing the tested transformation by its gth power. Finite exceptional directions do not affect the positive-density intersection. This completes the quadratic dichotomy, including the nonterminating boundary case.

R204. Exact thresholds on a mixed group using only escaping odd primes

Scope and status. Result with proof.

Result. Every t in $(0,1)$ is attained by an all-action threshold on one mixed group Z plus escaping odd-primary vector spaces, including recurrence at equality.

Scope. The construction has infinite-order directions and an injective positive integer graph coordinate; the host is still the mixed group.

Qualifications. Projection to Z has capacity zero. The graph coordinate does not transfer the threshold to integer differences.

The finite prime stages of the one-Kronecker-system proof [R005] also give the unconditional unrestricted-action threshold on one mixed countable group, with all steps of infinite order. This version uses the ordinary finite Szemerédi and density Hales–Jewett theorems, without a finite-color hierarchy or any order-two components. It is still not an integer solution.

Fix $0 < t < 1$ and construct $V_i = F_{p_i}^{m_i}$, nonempty projectively closed velocity families $R_i \subset V_i \setminus \{0\}$ and the finite stages $K_m = \prod_{i=0}^{m} V_i$, $S_m = \prod_{i=0}^{m} R_i$, as in the compact theorem. Let $w_i \in R_i$. In $G = Z \times \oplus_{i\ge 0} V_i$ set

$$D = \{(n, v) : n \ge 1,\ v \text{ finitely supported, and } nw_i + v_i \in R_i \text{ for every } i\}.$$

For an event in any probability-preserving G-action write T for the integer generator and U for the torsion action. The homomorphism $(n, v) \mapsto (nw_i + v_i)_i$ into $\prod V_i$ pulls back the compact Haar translation action, giving avoiding densities arbitrarily close to t from below. Every element of D has infinite order.

Choose the fresh primes sufficiently large also to satisfy

$$\sum_{i>m} 1/p_i < \kappa_m \quad \text{for every} \quad m \ge 2,$$

where $\kappa_m > 0$ is chosen after stage m from the finite Szemerédi–Varnavides lower bound for all lengths $2 \le h \le m$ and event mass $1/(m|S_m|)$. Specifically one may ensure that in any invertible integer system with such an event the set of positive times having positive h-correlation has lower natural density at least $2\kappa_m$. Inductively require the reciprocal of each later prime to be smaller than each preceding bound times 2^{-i-1}. This leaves all rank and direction-deletion choices in the compact construction available.

For each fixed h, choose j with $p_j > 2h$. The full-orbit caps of all finite stages are $< t$. For $m \ge j$, the fresh-slice argument on that very coordinate in K_m shows every h-avoiding mask there has density at most

$$(|V_j|t - 1)/(|V_j| - 1) = t - \eta_h, \eta_h = (1 - t)/(|V_j| - 1) > 0.$$

This same bound holds in every action of the finite group K_m: sample the orbit of each point over uniform elements of K_m and apply the finite bound, even in the presence of

stabilizers. Now let $\mu A \geq t$ in any G-action. With $m \geq h, j$ sufficiently large that $1/m < \eta_h$, at least one $s \in S_m$ has

$$\mu(C_s) \geq 1/(m|S_m|), C_s = \bigcap_{l<h} U_{ls}^{-1} A.$$

Otherwise delete all of the start cylinders C_s from A. The remainder loses at most $1/m$ of its mass and avoids the finite-stage family in the restricted K_m-action, contradicting its just-proved gap. This is why the equality case $\mu A = t$ is covered.

Use the integer transformation $T_m = TU_{-(w_0,\ldots,w_m)}$. Any positive h-return for C_s under T_m^n gives a positive h-return for A at the direction $(n, s-n(w_0,\ldots,w_m))$, taking all remaining torsion coordinates zero. In fact every retained $T_m^{\ln}$ position belongs to C_s, so its translate by U_{ls} belongs to A. At the head this direction maps to s; in the tail it maps to $(nw_i)_{i>m}$, which lies in $\prod_{i>m} R_i$ as long as no tail prime divides n. The exceptional positive integers have upper natural density at most $\sum_{i>m} 1/p_i < \kappa_m$, by the union bound on positive multiples in each $[1, N]$. Hence the positive lower-density return set supplied by Varnavides meets this coprime set. This proves the unrestricted upper for every finite length at and above t.

All examples fit the same group $Z \times \oplus_{podd} F_p^{m(p)}$ with **every** $m(p)$ **finite**, chosen before the target. For each prime take a rank where all directions have full-line cap $< 1/p$, by density Hales–Jewett. When choosing the next stage also require $1/p < (t-c)/(2C)$ and use this predetermined rank; in the initial stage take $1/p_0 < t$. Restrict any full-group action to the indicated direct summands for the upper bound and pull back along coordinate projection for the lower bound. Consequently every bounded-exponent subgroup of this single group is finite; one can start beyond any prescribed finite set of primes. Only pairwise distinct primes escaping to infinity are used within an example. Every direction can nevertheless be separated from zero by its first finite-stage projection. That projection supports a positive bounded-exponent obstruction, which admits arbitrary-density root amplification on a torsion-free free cover, so this strengthening does not transfer the threshold to Z.

A graph over the positive integers. There is also a sparse subfamily with the same exact threshold on which horizontal projection is injective. This version needs no infinite prime sieve. Construct stages inductively, always requiring each next prime to exceed **all** horizontally assigned packet integers from earlier stages. After fixing stage $m \geq 2$, let $c_m < t$ be its finite full-orbit cap and set $\epsilon_m = (t - c_m)/(2|S_m|) > 0$. Fix an integer $M_m \geq m$ large enough that in any integer system every event of mass $\geq \epsilon_m$ has some positive return of each length $2 \leq h \leq m$ at an integer $j \leq M_m$ (finite Szemerédi plus stationary averaging/compactness). For each $s \in S_m$, choose a separate integer q_s exceeding the greatest endpoint already used. At the distinct positive integers $n = q_s j$, $1 \leq j \leq M_m$, put the unique torsion correction $v(n) = s - n(w_0,\ldots,w_m)$, zero elsewhere. At later prime coordinates nw_i is a nonzero scalar multiple of w_i, so every such direction lies in the compact product menu. This gives the lower bound.

For the upper bound of the graph, consider any event of mass $\geq t$ and horizon h. At a stage $m \geq \max(2, h)$ some $s \in S_m$ satisfies $\mu(C_s) \geq \epsilon_m$, by deleting all the start cylinders otherwise; the finite h-cap is at most c_m. Apply the defining property of M_m to C_s under $(TU_{-w_{\leq}m})^{q_s}$. The successful j yields the required run in A with the graph direction at $n = q_s j$. This proves exact endpoint forcing even if the successive packet gaps are arbitrarily large. The horizontal projection of the graph, considered as an *integer* set, contains dilated

intervals $q_s[1, M_m]$ of unbounded length. Its unrestricted integer cap is therefore zero by ordinary finite Szemerédi applied to T^{q_s}. Injectivity on the steps does not preserve their recurrence problem under projection.

R206. Translated sum boxes with uniformly dense tails on progressions

Scope and status. Result with proof.

Result. Translated sum boxes with uniformly positive density in arbitrarily long fixed-gap progressions on every tail have capacity zero if they meet every finite-index subgroup, and one otherwise.

Scope. The gap may depend on the tail but must be fixed while the progression lengths grow.

Qualifications. The theorem is independent of the disputed rational-spectrum cube lemma in R214.

Let S_i be finite subsets of Z containing zero, P_m the finite-support sum of the S_i with $i \geq m$, and $D = e + P_1$. Assume there is $a > 0$ such that for each m there is a *fixed* nonzero integer λ_m for which arbitrarily long progressions of gap λ_m have at least proportion a in P_m. Their starting points can vary with their lengths. The gaps may be arbitrarily large with m. Symmetry of the S_i is unnecessary.

Dichotomy. If D meets every QZ, its nonzero differences force all finite arithmetic progressions at every positive upper density. If D misses some QZ, its all-length integer critical density is one. The latter follows directly by taking longer periodic masks of density arbitrarily close to one that forbid long runs at every nonzero residue modulo Q, using roots on Z (or the usual finite Bohr-separation lemma). The term about zero in the former statement matters only if zero is deleted from D; the samples below put vanishing mass on any fixed finite set.

Local measures with prescribed residue Fix Q. Choose one residue $r \bmod Q$ for which, at arbitrarily late m, arbitrarily long progressions of one fixed gap λ_m contain at least $b = a/(2Q)$ of their positions in $P_m \cap (r + QZ)$. This is possible by two finite pigeonhole arguments, first on the progressions at each m, then on the indices m. After any given starting index, choose $Q - 1$ elements of P_m of this same residue successively on disjoint later finite supports. After their indices take one of the good m and a sequence of its dense finite samples. Add the previously chosen $Q - 1$ fixed elements to each sample. We thus obtain a sequence of finite probability measures of the following kind, available after *any* finite head:

- Each law is uniform on a finite set of later admissible sums, all divisible by Q. Choose a representation for each sum. Each finite packet therefore uses only finitely many indices.
- A packet comprises at least fraction b of a progression in QZ of length tending to infinity. The progression gap λ' is fixed within this sequence although it may depend on the chosen head. Indeed, restrict the original progression to its r-residue $raisingitsgapto \operatorname{lcm}(|\lambda_m|, Q)$ and translate. The inequality relative to the restricted progression is at least as good.

Write U for a sample. For each irrational θ,

$$\limsup |E \exp(2\pi i \theta U/Q)| \leq \rho(b) < 1.$$

To verify this, the full ambient progression modulo the circle in this phase is a segment of an irrational rotation, equidistributed with discrepancy tending to zero uniformly in the initial phase. Every arc of fixed sufficiently small length contains at most $b/2$ of the ambient progression, eventually, uniformly over its center. At least half of the conditional sample then lies outside an arc of fixed positive angular radius about the argument of its mean. Taking real parts after rotating establishes the uniform contraction. Also the largest atom is at most $1/(bL)$ when the ambient progression has length L.

Smoothing a finite nilorbit Consider any basic nilsequence $f(U_0^n x)$, where the orbit closure is equipped with its uniquely ergodic nilsystem distribution. Choose Q to be a component modulus so the orbit of x under $V = U_0^Q$ is connected and totally uniquely ergodic on its closure Y. The ordinary mean along QZ is its Haar integral. We use Leibman's pointwise nilorbit theorem. Total unique ergodicity follows on the connected component from the nilrotation criterion on its horizontal torus; one can equivalently take the standard arithmetic-progression component decomposition.

Fix $z \in D \cap QZ$ and one finite representation. Beyond it take M independent successive packets as above, always starting later than all possible indices of the preceding finite packet. The corresponding averaging operators on (Y, V) are polynomials in V with time U/Q. For f of Haar mean zero, their convolutions satisfy

$$nested \limsup ||Ef(V^{(U_1+\cdots+U_M)/Q}\cdot)||_2^2$$

$$\leq \rho(b)^{2M}||f||_2^2.$$

The spectral integral has integrand the product of the squared Fourier multipliers. For fixed outer packets, bounded reverse Fatou bounds the innermost limsup by removing the inner factor at cost $\rho(b)^2$, and then one repeats. The spectral measure of f has no mass at nonzero rational frequencies by total ergodicity and none at zero by the mean-zero assumption.

Append one more independent inner packet after these M packets. For each fixed outer support, every limit law of $V^{U/Q}x$ for the final packet is dominated by b^{-1} Haar on Y. Indeed for a nonnegative continuous test the sum over its packet is at most b^{-1} times the average over the containing progression. That progression has fixed gap in QZ throughout its sequence, and the relevant fixed power of V is uniquely ergodic, uniformly in the possibly moving initial point. The domination constant is independent of the outer support. Applied to the continuous function averaged over the outer packets and with the fixed additional translate $V^{z/Q}$, domination bounds the two-sided deviation of the resulting nilsequence mean from its Haar integral by b^{-1} times the L^1 norm of that function after subtracting the integral. In the nested upper absolute limits this is at most $b^{-1}\rho(b)^M||f - \int f||_2$. Thus by choosing the number of packets large the sampled nilsequence mean approaches its ordinary mean along QZ. This argument allows arbitrary finite nilstep and does not assume any single packet is Hartman-uniform.

Multiple returns In an ergodic integer system the same-event h-correlation $c(n)$ is the sum of a uniform limit of basic nilsequences and a sequence null in uniform density, by

Bergelson–Host–Kra. The ordinary mean of c along every QZ has a lower bound $\kappa(h, \mu(A)) > 0$ independent of Q. This is the elementary Varnavides consequence of the finite Szemerédi theorem applied to the transformation T^Q: for bounded length $L = L(h, \mu(A))$, the sites 0,s,. . .,$(L-1)s$ inside A have expected proportion $\mu(A)$, so with probability bounded below they contain a full h-run at a difference ts, $1 \leq t < L$. Integrate over the space and average s up to N/L; bounding multiplicities gives the asserted positive Cesàro lower bound at times $\leq N$. The starting offset of the finite h-run drops from the integral by stationarity.

Choose a basic nilsequence approximating the nil part uniformly within $\kappa/8$ and then choose its component modulus Q. Freeze a head $z \in D \cap QZ$ and apply the preceding packet construction. The null error has mean tending to zero already for the last, innermost packet: for each fixed outer packet and its finitely many outcomes the last packet lies at positive density in intervals on a fixed integer lattice, so its average absolute null error is at most a fixed multiple of an ordinary interval mean tending to zero. The finite gap factor may depend on that outer packet, which causes no problem since the innermost limit is taken first. The null error also has zero mean along the ordinary QZ averages by the same bound. The nil approximation, together with M sufficiently large, gives a positive total sampled h-return. Every sampled difference is admissible, and the final packet's vanishing maximal atom permits deletion of zero or any finite exception. Ergodic decomposition or the usual correspondence principle completes the assertion.

Radix examples and the residual obstruction For an integer base $q \geq 2$ and any fixed finite integer digit alphabet B containing one representative of each residue $\bmod q$ (and with zero available at each site), suppose S_i contains $q^i B$. Then P_m on the progression lattice $q^m Z$ contains q^n distinct integers from n successive positions, in an interval of length at most Cq^n on that lattice, with C depending only on B and q. Thus the hypothesis holds uniformly. It also holds for varying bounded bases and uniformly bounded normalized complete digit alphabets. This allows multivalued translated digit boxes with an infinite carry path and no exact finite cancellation.

More generally, a translated-sum-box seed with positive short obstruction and intermediate integer threshold would have to violate the displayed tail-density property, regardless of how its finite digit sets were chosen: for every $a > 0$ there is a tail which does not occupy fraction a on arbitrarily long progressions of *any fixed gap*. The gap must be fixed within the arbitrarily long sequences in this assertion. It is not enough simply to find one long progression per tail at an unconstrained, varying gap.

R214. Conditional rational-spectrum reduction and a failed cited cube lemma

Scope and status. Proved subcases and unresolved extensions.

Result. A proposed all-system rational-spectrum reduction is conditional; an explicit binary odometer extension disproves the cited cube lemma, while a finite-rational-spectrum recovery is unconditional.

Scope. The full reduction requires a replacement for Kra–Shalom v1 Lemma 4.10. The counterexample does not itself disprove every possible form of the desired theorem.

Qualifications. The unrestricted reduction remains conditional on a replacement for the failed cube lemma. Assertions depending on that reduction have the same conditional status. The explicit counterexample and the finite-rational-spectrum recovery are unconditional.

Status after proof audit. The implication below from the stated Kra–Shalom characteristic-factor theorem and the Holikov–Shalom nilorbit theorem is valid, but we found a counterexample to Lemma 4.10 used to prove the former theorem in arXiv:2506.17771v1. Thus we do **not** regard the all-system conclusion in this note as settled by those inputs. The example disproves the intermediate lemma and has not yet disproved their characteristic-factor theorem. The same dependency appears in the internal earlier automatic-set proof. The issue does not affect the separate relative-cube or progression-dense tail arguments. Details follow the conditional proof.

Suppose finite sets $S_i \subset Z$ contain zero and a positive selected integer a_i on infinitely many distinct sites (reindex the selected sites). If

$$G_2(a) = \{\theta : \sum_i ||a_i\theta||^2 < \infty\} \subset Q/Z,$$

then every translated finite-support box $D = e + \sum_i S_i$ which meets each QZ is fully Szemerédi-large. A box missing QZ has all-length threshold one by the finite-residue sieve. The selected part can be unsigned. With symmetric S_i one may take absolute values of any nonzero selector. This statement includes countably infinite torsion G_2, for which the elementary finite-G_2 reduction does not work.

The condition implies rational spectrum as in Kra–Shalom, Definition 3.7 and Theorem 4.2: for every irrational θ the absolute Fourier multiplier of the uniform 0/1 subset sums on any tail is a product of $|\cos(\pi a_i\theta)|$ tending to zero. We use their linear characteristic-factor theorem and the nilorbit limit formula Holikov–Shalom, Proposition 5.3. These are results of the cited preprints. All averages used below are on ordinary initial intervals of the tail index and count representations with multiplicity.

In an ergodic integer system take an event A of mass $a > 0$ and fix $h \geq 2$. Project 1_A to the ordinary characteristic nilfactor $Z_{h-1}(T)$, taking the unnecessarily high order to avoid indexing issues. For $h \geq 3$ fixed-offset AP correlations averaged on any selected IP tail may replace 1_A by this projection, also when conditioned on the subset $\sum u$ being zero modulo any fixed positive integer L, provided the conditioning probability has a positive limit. To check the rational weight, expand $1_{\{L|u\}}$ in its L characters. Encode any one such character using the extra finite cycle C_L in the product with the original system, with slot 1 carrying its character and slot 0 the opposite character. Orthogonality of a telescoped

error to the required ordinary Host–Kra factor persists upon lifting to the product and multiplying by a cyclic character; the ordinary cube seminorm there factors as the product of the two seminorms by the iterated Cesàro cube formula. A fixed step offset just translates the slot functions and also preserves their orthogonality. To use the characteristic theorem for distinct nonzero slopes, shift the integration origin so the designated error slot has a nonzero slope, and remove the fixed zero-slope test. There are $h-1$ remaining slopes. For $h=2$ the same scalar orthogonal-error assertion follows from the spectral theorem: the error is orthogonal to the Kronecker factor, so its continuous spectral measure gives zero to the countable exceptional rational frequencies and their rational shifts.

Approximate the projection in L^1 by $0 \le b \le 1$ continuous on a finite-dimensional ergodic nilfactor (Y,S), with $\int b \ge a/2$. The ordinary Cesàro h-correlation of b along multiples of every L has lower bound $\kappa(a,h)>0$ independent of L, by finite Szemerédi/Varnavides for S^L on $\{b \ge a/4\}$. Make the approximation error small enough that h times its L^1 error is $< \kappa/8$; this choice takes place *before* L. On the h-fold nilspace use $V = S^0 \times \cdots \times S^{h-1}$, let $x^\Delta = (x,\dots,x)$ and $F(y) = \prod_{j<h} b(y_j)$. For each x the orbit closure of x^Δ is a finite union of connected components cyclically permuted by V. Take L divisible by the number of these components for all x outside a set of Haar mass $< \kappa/4$. This is possible since the component count is a finite measurable integer-valued function; measurability follows also from the compact-valued orbit closures of V^q. At these good x the orbit closure of V^L at x^Δ is connected; call its Haar law m_x. The ordinary L-multiple mean is the integral of its pointwise orbit means. Hence $\int_{\{x \text{ good}\}} m_x(F) \ge 3\kappa/4$.

Choose $z \in D \cap LZ$ and freeze a finite representation. Beyond its final site discard any further initial segment necessary so that every observed residue class of the selected $a_i \bmod L$ recurs infinitely often. The unsigned subset-sum law of this tail tends modulo L to the uniform law on the subgroup generated by the remaining residues, since every character nontrivial on that subgroup contracts infinitely often. Thus its probability p_N of residue zero tends to $p \ge 1/L$. Discarding a finite initial part does not change the rational spectrum condition.

For a good x apply the Holikov–Shalom formula to the product orbit starting at $V^z x^\Delta, 0$ under $V \times (r \mapsto r+1)$ on $Y^h \times C_L$. Within residue zero this joint orbit contains just the connected component supporting m_x, with z divisible by L. The pointwise formula weights the Haar measures of the connected components by their actual limiting IP probabilities. After restricting to zero and dividing by p_N the average of F therefore tends to $m_x(F)$. For other x all integrands are nonnegative. Fatou gives a lower bound $3\kappa/4$ on the nil correlation averaged with the normalized conditional law. The cost of replacing b by the nilfactor projection is at most h times its L^1 error with *any* normalized sampling measure, so no $1/p$ loss occurs there. The orthogonal errors vanish after division by the fixed $p>0$. Some $z+u \in D$ consequently gives a positive return for A. Since a_i are positive and infinitely many, unsigned subset sums tend to infinity in probability before and after residue conditioning, so zero or any finite exception may be omitted. Ergodic decomposition yields the universal conclusion.

Explicit counterexample to Lemma 4.10 of the cited version On the binary odometer $Y = Z_2$ with Haar measure and the uniform bit fiber C_2 take the Morse skew product

$$T(y,s) = (y+1, s+1+v_2(y+1) \bmod 2),$$

ignoring the single exceptional $y = -1$. Its cocycle over adding 2^k changes the fiber sign by $(-1)^{1+J_k(y)}$, where J_k is the length of the consecutive ones in the binary expansion starting at the kth digit. This follows either by telescoping binary digit-sum parity below the first unchanged high position or by adding 2^k in a finite binary truncation past its last carry. For $Haary$ its law is $P[J_k = j] = 2^{-j-1}$ for every k, hence the sign has mean $-1/3$ and positive variance independent of k.

There is no eigenfunction on the vertically odd subspace. Such an eigenfunction would have the form $b(y)(-1)^s$, with $|b|$ constant almost everywhere by ergodicity of the odometer. Normalizing that constant to one, the eigenvalue equation iterated 2^k times would make $(-1)^{1+J_k(y)}$ asymptotically equal in L^2 to the constant eigenvalue raised to 2^k, because $b(y+2^k) \to b(y)$ strongly. Its fixed nonzero variance excludes that. The remaining, vertically even, functions factor through Y. Thus T is ergodic and its full Kronecker factor and its rational Kronecker factor are both exactly Y.

Take $f(y, s) = (-1)^s$. It has ordinary $U^2(T)$ seminorm zero, being orthogonal to the Kronecker factor. In contrast the first joining of Kra–Shalom, Definition 4.6 is the relative self-product over the common y. On it $f(x_0)\bar{f}(x_1) = (-1)^{s_0-s_1}$ is invariant under the diagonal action, since both copies see the same base cocycle. In particular it belongs to that action's rational Kronecker factor. Its squared L^2 norm is one, so the alternating f-cube integral at the second rational-cube stage is exactly one. This contradicts the conclusion of their Lemma 4.10. Its unproved interchange of the joining limit when the measures themselves vary between levels cannot be justified by monotone convergence of the first-level rational invariant factors. An independently proved characteristic-factor statement, or a weaker argument sufficient for translated-box recurrence, would still be needed.

Unconditional recovery for systems with finite rational spectrum For a symmetric box with selected $0,\pm a_i$, the multiple-recurrence conclusion holds in every ergodic action whose rational eigenvalue group is finite, with no appeal to Kra–Shalom or Holikov–Shalom. In particular no totally ergodic system is an obstruction. Here one can also use any of the mixed symmetric selectors of the relative cube note [R216] whose effective spectral subgroup is contained in Q/Z.

Let g be the order of the rational eigenvalue group of T. The ergodic components of T^g are totally ergodic; indeed T^g has the g cyclic components from the order-g eigencharacter, and an additional finite cyclic factor on a component would induce a rational T-eigenvalue of order not dividing g. Work on one such component where A has positive mass, writing $S = T^g$ there.

Two elementary facts are helpful. First, if an invariant law has totally ergodic ergodic components, its relative self-product over the invariant factor again has totally ergodic ergodic components. Each component law lifted is of the form $\mu_y \times \mu_y$. The point spectrum of its diagonal action is contained in pairwise products of eigenvalues of μ_y: the spectral formula on elementary tensors is convolution, and any atom of a convolution comes from two atoms. The eigenvalues of an ergodic system form a group. Thus with totally ergodic μ_y this product has no nontrivial rational eigenvalue, although it can have a large invariant factor. This assertion on the full product implies it for its ergodic components almost surely, by measurable spectral projection onto each of the countably many rational eigenspaces. The same reasoning applies to every nonzero power. Consequently the usual Host–Kra cube joinings of S agree with those of any power S^L: at each level its invariant and power-invariant

factors coincide.

Second, a signed symmetric block law from any fixed tail, now conditioned on $u \in MZ$, has acceptance probability at least $1/M$ since all its Fourier multipliers are nonnegative. Its rescaled multiplier at irrational θ tends to zero as the finite tail endpoint increases. Indeed the numerator is a sum of the original multipliers at $(\theta+j)/M$, all irrational; the denominator is bounded below. The limit applies also to doubled independent differences and nonzero integral multiples of θ. Thus on any system with totally ergodic ergodic components these rescaled block averages and their nonzero powered variants converge strongly to the ordinary invariant projection, by spectral integration.

Approximate the $Z_{h-1}(S)$ projection of 1_A on a finite nilfactor and choose L dividing out the diagonal nilorbit component numbers off a small set, exactly as in the conditional proof. Freeze $z \in D \cap \mathrm{gLZ}$ and use independent successive blocks conditioned to $M = gL$. Their times produce AP steps $(z + \Sigma u)/(gL)$ under S^L. Apply the finite generalized von Neumann estimate of Hartman convolution boxes [R137]: on each usual HK cube of S^L the just-proved conditional averaging gives the same invariant projection as the ordinary Cesàro limit for any of the nonzero slot-difference powers. Thus its limiting mixed cube is the usual HK cube of S^L, which equals that of S. All orthogonal replacement errors vanish.

For the continuous nilfactor approximant, whenever the component count divides L, the relevant V^L diagonal orbit is connected and totally uniquely ergodic. The central smoothing proof in the same note [R137] uses only mixing of the sampled unitary averages down to the mean on the connected orbit and its connected quotients, plus the central compact-group tower. The conditional laws have these limits by the second fact. With at least the nilstep many independent blocks their pointwise nested averages of the nilsequence equal Haar on that component; the argument still works for the initial translate $z/(gL)$. Bounded reverse Fatou and the previous uniform Varnavides lower bound give an actual positive return. This deals with every h. If zero must be omitted, at a fixed starting index the original block's largest atom tends to zero by squared-multiplier integration against circle Haar; division by an acceptance probability $\geq 1/M$ preserves this conclusion.

R215. Freely chosen clocks for matched rational prime ratios

Scope and status. Result with proof.

Result. Artificial irrational clocks extend the matched-prime endpoint-one theorem to every positive limiting ratio with budget below the smaller of one and that ratio.

Scope. Eventual congruence of the primes modulo each fixed integer is assumed.

Qualifications. This is not a theorem for supports mixing a continuum of different limiting ratios.

The full endpoint in the Sturmian argument [R245] also holds for $q_i/p_i \to \theta > 1$ **rational**, with $\tau < 1$. At $\theta = 1$ a single sparse word suffices. For every $0 < \theta < 1$, rational or irrational, the same method works if $\tau < \theta$. Use the same eventual congruences of both primes to 1 modulo every fixed positive integer, centered signed horizontal mass, odd centered sum, and either vertical orientation. The statement is about the squarefree tail limit and its rational saturation and integer realization.

An artificial irrational clock Fix an irrational α in $(0, 1)$ such that $\theta\alpha$ is irrational. Round αq_i to an integer c_i, so $c_i/q_i \to \alpha$, and choose $d_i \equiv -c_i/q_i \bmod p_i$, where division is in the prime field. Then $q_i d_i/p_i \bmod 1 \to \beta = -\theta\alpha \bmod 1$. Write

$$S(d) = \sum_{active} a_i/p_i, U(d) = \sum_{active} a_i d_i/p_i \bmod 1.$$

Use the curvature word $\Delta^D b_i = d_i + \lfloor (v+1)c_i/q_i \rfloor - \lfloor v c_i/q_i \rfloor$. It exists since its integral over the vertical cycle is zero modulo p_i. With arbitrary starts and either vertical orientation the limit at parameters (S, U) is

$$\Delta^{D-1} W(t) = \kappa t + C + G(t\alpha) + e_t \bmod 1, \kappa = U + \alpha S \bmod 1.$$

Indeed the curvature contributed by each axis is $\mu_i(d_i + c_i/q_i)$, with $\mu_i = a_i/p_i$. The fractional parts and their jumps are exactly those of the signed Sturmian proof. In particular $\Sigma|e_t| \le \tau$ and $DG = -S dx + \sigma$, where $||\sigma|| \le \tau$ and $\sigma(T) = S$. If $\kappa \notin Q + Q\alpha$, this universal word and its single fixed graph exclude indefinite retained runs near (S, U), by the independent Haar-coordinate integral there.

A preliminary rational clock When θ is rational and $\theta \neq 1$, use in addition the original word with $d_i^0 = (p_i + 1)/2$ and $c_i^0 = (p_i - q_i)/2$. Then $\alpha_i^0 = c_i^0/q_i \to \alpha_0 = (1/\theta - 1)/2$ is rational. Its limiting slope is $\kappa_0 = (-1 + S/\theta)/2$, which is irrational whenever S is irrational. The jump-error argument needs a minor replacement because a rotation by α_0 repeats points. Write B for its denominator. On each residue modulo B the pointwise limit of the bounded fractional-part contribution $G(t)$ has total variation $\le \tau$ along positive times. In fact, on any fixed finite segment in this residue, the phase increments at one axis are of one sign with total change at most the segment length times $|\alpha_i^0 - \alpha_0|$, uniformly tending to zero on support tails. Thus its fractional part has variation at most $1 + o(1)$ on that segment, since it wraps at most once. Sum weighted by $|\mu_i|$ over both orientations, then pass to a diagonal

pointwise limit. The bound is independent of the finite segment. Bounded variation on a positive-integer sequence implies a limit on each residue. Consequently

$$\Delta^{D-1}W(t) = \kappa_0 t + C + G_p er(t) + o(1) ast \to +\infty,$$

for a B-periodic real function. On the rotation base of the phases $\kappa_0 t, t \bmod B$ and any affine nuisances, the κ-coordinate is independent Haar over the finite coordinate. The integrals *() of the Sturmian note vanish for every nonzero l and every* $H \geq 1$*; all the skew-tower, empirical and graph arguments go through. Thus a* single fixed preliminary probe* succeeds locally around every irrational S. When θ is irrational, the natural word instead succeeds whenever $S \notin Q + Q\theta$, as in the original proof. In both cases it leaves at most a countable list of S values for the next step.

Sparse tests on a residual parameter pair The compact parameter space is $[-\tau, \tau] \times T$ with coordinates (S, U). Beyond the two universal probes only the pairs with S in the just described countable list and $\kappa \in Q + Q\alpha$ remain, hence only countably many pairs. Assign each a separate summable loss. At its assigned loss choose the statistical graph and its sparse test multiplier M in advance of the angular separation, as in the sparse result [R199]. Put $e_i = (2M)^{-1} \bmod p_i$ in the prime field. Eventual congruences give

$$q_i e_i / p_i \bmod 1 \to \lambda = (\theta - 1)/(2M) \bmod 1,$$

$$M \sum_a ctivea_i e_i / p_i = (S(d) - 1)/2 \bmod 1.$$

Consider the compact subgroup of T^2 generated by

(λ,(S−1)/2) and (β,M U).

It surjects onto the first circle since β is irrational. If the vertical kernel is nontrivial it contains a point of distance at least 1/3 from zero in the second coordinate. Choose the separation 1/6 and its sparse horizon and tolerance first. Then choose a finite integer combination (m, n) of the generators whose first coordinate is as small as prescribed and second has distance $> 1/5$. The word test with desired coefficient $me_i + nd_i \bmod p_i$ has sparse curvature corrections on a sufficiently late tail, by the convergence to $m\lambda + n\beta$. Its tested curvature multiplied by M is near $m(S-1)/2 + nMU$ whenever $(S(d), U(d))$ is close to the pair. The signed sparse test therefore gives one uniform high-density avoiding probe on a neighborhood of that pair.

If the kernel is trivial the subgroup is the graph of a continuous circle character $z \mapsto kz$ for an integer k. The two generators force

$$S = b + (\theta - 1)k/M, MU = -k\theta\alpha \bmod 1,$$

with odd b, and hence

$$\kappa = A\alpha + r/M, A = S - k\theta/M = b - k/M$$

for some integer r. The artificial word itself detects this pair by the one-sided occupation estimate. For $\theta > 1$ conjugate signs to take $b \geq 1$. Then $u = A - S = \theta(b-S)/(\theta-1) > 0$ and the negative opposing mass $(\tau - S)/2$ is strictly smaller than u, since $\tau < 1$. For $0 < \theta < 1$ take $b \geq 1$, so $u = \theta(b-S)/(\theta-1) < 0$ and the opposing positive mass is at most $(\tau + S)/2$. Here $|u| \geq \theta(1-\tau)/(1-\theta) > \tau$ exactly when $\tau < \theta$. Conjugating the phase produces a positive drift and the previous proof. The direction of α and the rational additive intercept have no effect on this estimate. At $\theta = 1$ one can dispense with all clocks: $\lambda = 0$ makes the coefficient e_i uniformly sparse and $(S-1)/2$ uniformly separated from the integers since $|S| \leq \tau < 1$.

The successful neighborhoods of both universal words and the countably many supplemental probes cover the compact parameter space. Take a finite subcover, use each universal word at its fixed cost only once, and take the maximal horizon and tail among the finite tests. The squarefree tail identity proves the asserted endpoint. In particular every matched prime schedule with a single limiting ratio in $(0, \infty)$ is excluded at the symmetric budget $\tau < \min(1, \theta)$; this applies a fortiori to the matched small-budget four-slot models within that range. No corresponding statement for a continuum of limiting ratios distributed across a single support has been proved here.

R216. Relative cube factors for a translated signed sum box

Scope and status. Result with proof.

Result. Lazy tail convolutions define characteristic relative cube factors for signed sum boxes; under countable spectrum these lie in a finite distal tower, and a relative weak-mixing class reduces further.

Scope. Nested append-tail limits and measurable relative factors, not ordinary finite-dimensional nilfactors.

Qualifications. The skew-shift example shows that the base spectral factor alone is insufficient for triple returns.

Let $S_i \subset Z$ be finite, with $0, \pm a_i \in S_i$, and let $D = e + \sum_i S_i$ be the finite-support sum box. We use signed differences interchangeably with absolute values. The arguments below concern the whole box as written, including zero if it occurs. If zero is excluded and the selected nonzero integers occur at infinitely many indices, the argument is unchanged, by the atom estimate below.

The one-dimensional tail operator Write $\Sigma = \{\theta \in T : \sum_i ||a_i\theta||^2 < \infty\}$, a Borel subgroup. For indices m through N let ν be the law of the sum of the independent increments taking values 0 with probability 1/2 and $\pm a_i$ with probability 1/4 each. For any pmp invertible transformation T with Koopman action U and nonzero integer k, two successive strong limits give

$$\lim_m \lim_N E_u \; {}_\nu U^{ku} = 1_{k^{-1}\Sigma}(U) =: P_k.$$

Indeed the multipliers for the finite intervals are the products of $\cos^2(\pi k a_i \theta)$. For fixed m, the decreasing limit is $r_m(k\theta)$, and r_m increases to the indicator in the formula as m tends to infinity. Both limits follow by dominated convergence on the spectral measure of any tested vector. The same two limits apply to $u - u'$ for independent samples since its multipliers are the squared products. All inner starts may be chosen beyond any preassigned finite head.

The projection P_k is a Markov conditional expectation. It is a self-adjoint projection preserving positivity, constants, and the integral, by its stochastic approximations. If a bounded real f is fixed, positivity and Jensen give $P_k f^2 \geq f^2$; since the integrals agree, equality holds. Polarization shows that the bounded fixed functions form a unital algebra, stable under conjugation. It is closed under bounded pointwise limits by L^2 closedness, and the bounded fixed functions are dense in the range since bounded tests are dense and P_k preserves sup norms. Thus its range is precisely the L^2-space of a factor $Y_k(U)$, with no countability or topological closedness hypothesis on Σ. On a nonergodic action this factor includes the invariant factor.

One may use countably many signed selectors at once. At each site select the rth one with fixed probability $w_r > 0$, with sum one, then use the same signed increment law. The finite multiplier is the product over sites of $1 - \sum_r w_r \sin^2(\pi a_i^{(r)} \theta)$. Its effective subgroup is the set of θ with $\sum_i \sum_r w_r \|a_i^{(r)}\theta\|^2$ finite, a Borel subgroup contained in the intersection of the selector subgroups. The entire argument below applies to this subgroup verbatim; in particular countability of that intersection suffices for the finite distal tower theorem. For

the atom-removal argument it suffices that at least one of the positive-weight selectors is nonzero on infinitely many indices.

A characteristic factor at each fixed length Fix $h \geq 2$ and write $d = h - 1$. For each slot $0 \leq r < h$ and each ordering $j_1,\ldots,j_d$ of the other slots build d cube joinings, starting from $\mu_0 = \mu$. At stage i take the relatively independent self-product of μ_{i-1} over $Y_{r-j_i}(U_{i-1})$, where U_{i-1} is the full diagonal action of T on that space. Denote the result μ_i with its diagonal action. For every such final joining form all dual functions at the origin, i.e. condition the product of bounded functions on the other corners on the coordinate at the origin. Let F_h be the T-invariant factor generated jointly by these dual functions. All cube integrals with a bounded input orthogonal to F_h at the origin vanish. In particular this applies to the alternating-conjugation cube with the same input at every vertex. We do not assert this is the smallest characteristic factor.

The finite generalized von Neumann estimate is described in Hartman convolution boxes [R137]. With independent tail-block samples u_i, an offset z and bounded inputs, the absolute integrated AP correlation averaged at step $z + \sum_i u_i$, raised to power 2^d, is at most the averaged d-cube correlation of the retained input f_r, with increments $r - j_i u_i - u_i'$. One first fixes all outer choices and cutoffs, with each next one beyond the preceding supports. At each elimination translate the integration origin by the eliminated slot's coefficient times its current increment to make that slot independent of this increment, and apply Cauchy–Schwarz. Any further outer blocks may be conditioned on first. Taking the inner limits $N \to \infty$ then $m \to \infty$ in each block, from innermost block to outermost, yields exactly the preceding relative cube. To see this, integrate the innermost derivative first; it has the matrix coefficient form $\langle U^{kt} g,\mathrm{g}\rangle$, with $t = u - u'$. Averaging gives $\langle P_k g,\mathrm{g}\rangle$, which is the integral of g and its conjugate on the two coordinates of the relative self-product. Repeat there using its diagonal action. The ordering in the definition of F_h allows every innermost-first ordering. All outer sums at each step are finite. Consequently an averaged AP correlation vanishes in nested upper absolute limits whenever any one input is orthogonal to F_h.

Lifting a fixed return. Suppose A has positive measure and put $g = E(1_A|F_h)$, $B = \{g > 0\}$. If B returns along a difference $z = e + \sum_{i\leq l} s_i$ in D, then A returns at *some* difference in D at the same length. More precisely, this new difference can be chosen by appending selected signed increments after l. Choose $\eta > 0$ so that

$$C = intersection_{j<h} T^{-jz} g > \eta$$

has mass $c > 0$. Let M suffice in density Hales–Jewett for word length M, alphabet size h and density $c/2$. For any trial directions $y_1,\ldots,y_M$ the mean, over x, of the density of words $w \in \{0,\ldots,h-1\}^M$ for which $T^{\Sigma w_i y_i} x$ belongs to C is c. A set of x of measure at least $c/2$ has word density at least $c/2$. Summing over the finitely many lines and using invariance to discard a fixed word prefix therefore gives

$$\sum_l ines\mu(intersection_{j<h} T^{-jy_I} C) \geq c/2,$$

where I is the nonempty set of variable positions of the line. Each term is at most η^{-h} times the g-correlation at $z + y_I$. Sample each y_i independently as a sum of d admissible

signed blocks, using disjoint, successively later sets of indices and always excluding the head through l. Every $z + y_I$ is admissible. For each line, telescope the difference between its g-product and its 1_A-product; any resulting term has an input $1_A - g$ orthogonal to F_h. Take the innermost d included blocks for the cube estimate, with any other included blocks temporarily fixed. Unused blocks impose only finite lower bounds on the next starting index and do not affect the limiting projections. The nested upper absolute limits of all these errors are zero. The fixed lower bound for the sum of projected correlations thus gives a positive 1_A-correlation for some admissible sample. If deleting a possible zero difference matters, the largest atom of a block tends to zero: by Parseval its square is bounded by the Haar integral of the squared Fourier multiplier, which tends to zero already at fixed m. In fact Σ is Haar-null (a proper measurable subgroup): Haar-positive would imply $\Sigma = T$, whereas convergence of $||a_i\theta||$ to zero at every θ contradicts its fixed positive Haar mean along the infinitely many nonzero a_i. Independent convolution cannot increase the largest atom. These estimates also permit excluding any finite set of differences.

In particular the supremal mass of an h-avoiding event in this system equals that on its factor F_h. One implication is pullback. For the other, any avoiding event projects to an avoiding B of at least the same mass. This gives full positive recurrence in a system at length h whenever admissible finite heads z_n give $T^{z_n} \to Id$ strongly on F_h. Indeed for positive B their same-event correlations tend to $\mu(B)$. This criterion requires higher-order information, as the next example shows.

A translation already fails in a two-step skew shift Choose α irrational within a small error ϵ_0 of $1/2$ and put $d_0 = e = 1$. Choose summably small $\epsilon_i > 0$ so that $12 \sum_{i \geq 0} \epsilon_i < 1/8$. Successively choose positive integers d_i, growing as rapidly as desired, with

$$||d_i\alpha|| < \epsilon_i, ||d_i^2\alpha - 1/2|| < \epsilon_i,$$

$$\sum_{j<i} ||d_i d_j \alpha|| < \epsilon_i.$$

The ordinary polynomial Weyl criterion gives joint equidistribution of $(n\alpha, n^2\alpha)$ on the two-torus; prescribing $n\alpha$ sufficiently close to 0 ensures every previous mixed constraint as well. Let $a_i = d_i - d_{i-1}$, $i \geq 1$, and S_i=$\{0,\pm a_i\}$. Then $\alpha \in \Sigma$, and even $\sum_i ||a_i\alpha|| < \infty$. The all-plus heads equal d_i, with $d_i\alpha \to 0$.

Every signed difference $z \in D$ is a $\sum \sum_{j=0}^{m} c_j d_j$, with $c_0 = 1 - s_1$, $c_j = s_j - s_{j+1}$ for $j \geq 1$, where $s_i \in \{0, \pm 1\}$ and $s_{m+1} = 0$; the empty head is interpreted separately. Thus $\Sigma c_j = 1$ and $|c_j| \leq 2$. By the estimates,

$$||\alpha z^2 - (\sum_j c_j^2)/2|| < 1/8.$$

Since $\Sigma c_j^2 \equiv \Sigma c_j \equiv 1 \bmod 2$, every αz^2 stays at distance at least $3/8$ from zero.

On the skew shift $T(x, y) = (x + \alpha, y + x)$ on $HaarT^2$, the one-block factor $Y_1(U)$ is exactly the horizontal circle. Its characters have frequencies $m\alpha \in \Sigma$. Every vertical Fourier

mode with nonzero vertical frequency lies in a bilateral shift of the Fourier basis and has Lebesgue spectral type, giving no mass to Σ. Thus the all-plus heads converge strongly to the identity on Y_1. Nevertheless a sufficiently short vertical interval avoids every triple at a step $z \in D$, since the second difference of its three vertical coordinates equals αz^2 mod 1. The failure is visible in the relative cube itself. After the first joining over the common x, the difference $y' - y$ is invariant under diagonal T, so the second cube integral for the vertical character has modulus one. For $h = 3$, all factors Y_k on the original skew shift are the same horizontal circle; $k^{-1}\Sigma$ is Haar-null and contains every $m\alpha$. Base-factor strong convergence alone therefore cannot prove even three-term recurrence.

A subclass where the base factor does suffice Suppose Σ is countable, the system is ergodic, and it is relatively weakly mixing over $Y = Y_1(U)$. Then for every h, $F_h = Y$. First, the spectrum on Y is pure point with its eigenfrequencies in Σ. Its whole L^2-space is retained by P_k for every nonzero integer k since Σ is a subgroup. For every n, the relatively independent n-fold power of X over Y is relatively weakly mixing over Y, and all its eigenfunctions come from Y. This follows, for example, by testing relative weak mixing on elementary product functions: the fiber correlations factor by conditional independence, any zero-conditional-mean coordinate has averaged squared conditional correlations tending to zero, and bounded elementary products are dense. Consequently Y_k on any such power is exactly the common Y, since $k^{-1}\Sigma$ is countable. Inducting in the cube construction shows all its joinings are conditionally independent copies over that common Y. The dual functions then all lie in Y and generate Y (put a Y-test at one vertex and constants at all others). In particular if D is Bohr recurrent, all sets in any such system give h-term returns, at every h: approximate a factor event by functions of finitely many eigencharacters and use the simultaneous return in the corresponding finite torus. The countability hypothesis cannot simply be discarded in this argument, since a continuous spectral type can be supported on an uncountable Haar-null subgroup.

A finite distal characteristic tower for countable Σ Assume Σ is countable and the original system is ergodic. Set W_0 to be the trivial factor and W_{i+1} the maximal relatively compact factor of X over W_i, in the usual Furstenberg–Zimmer sense. Then $F_h \subset W_{h-1}$. In particular the supremal h-avoiding mass in X equals that on this finite-height distal factor. This assertion is about measurable compact extensions; the tower need not consist of nilsystems.

Here is the joining calculation. For any factor W and any joining λ of m copies of W, write λ^X for the conditionally independent lift, taking the product of the m conditional laws of X and then integrating over λ. Let V be the relatively compact factor over W. Every eigenfunction of the diagonal action on λ^X factors through V^m. Indeed the orthogonal complement of $L^2(V^m)$ is spanned densely by bounded simple tensors having at least one input f with E(f|V)=0. This follows by telescoping ordinary simple tensors and their projections to V^m, using independence of the remaining conditional fibers. For the indicated f, the compact/weak-mixing decomposition gives

$$\lim_N (1/N) \sum_{n<N} ||E(f \cdot T^n \bar{f}|W)||_2^2 = 0.$$

This also follows by the mean ergodic theorem on $X \times_W X$ since every invariant vector there lies on V^2, by Furstenberg's invariant-kernel lemma; see Frantzikinakis, Lemma 2.2. The autocorrelation of the simple tensor in λ^X is an integral over λ of a product of conditional correlations over W; its average squared modulus is bounded by a fixed constant times the preceding marginal mean. Wiener's criterion now makes its spectrum nonatomic, proving the eigenfunction assertion. This argument does not require λ to be an ergodic joining.

Inductively, the ith cube joining μ_i is the conditionally independent lift of its projection to W_i raised to the 2^i power. The base $i = 0$ is tautological. Re-express this same μ_i as the independent lift over its projection to V^{2^i}, where $V = W_{i+1}$; conditional independence persists after fixing the finer factors separately. Its Y_k is determined by that projection, by the eigenfunction assertion and countability of $k^{-1}\Sigma$. Taking two copies relatively independently over that Y_k therefore gives an independent lift over the resulting joining on $V^{2^{i+1}}$. The induction follows. On μ_d a dual function at the origin consequently depends just on the W_d-coordinate there, which proves the conclusion.

In particular a Bohr-recurrent translated signed box with a positive *pair* obstruction must have uncountable Σ for every signed selector satisfying the hypotheses: the first level is a compact rotation and every Bohr-recurrent set gives pair returns there. More generally, at countable Σ, an arbitrary relatively weak-mixing extension of any base preserves the h-avoiding supremum. In the same joining argument take $V = W$; then any relative cube upstairs is the independent lift of the corresponding base cube. The argument for ergodic original systems suffices for universal claims by ergodic decomposition.

No claim here identifies these relative cube factors with ordinary nilfactors in general, and the universal critical density of such boxes remains undetermined.

R217. An IP set of unmarked integer queries absorbs every sufficiently divisible velocity

Scope and status. Result with proof.

Result. An IP normal menu absorbs sufficiently divisible velocities in the free query group; charged tensor recurrence follows when genuine copies can be replicated.

Scope. Replication and the stated arithmetic menu hypotheses are essential. Specific padded automata then have endpoint alternatives.

Qualifications. This independent route does not validate arbitrary automatic-set assertions based on R214.

Fix $h \geq 2$. Given an integer velocity r and start x, let $e(x, r)$ be the tuple of letters $(u_{j,x+jr})_{0\leq j<h}$, in the direct product of h free groups, one free group per player with a letter for each integer. Suppose V_0 contains 0 and all finite sums of a sequence of positive integers. Let H_0 be the subgroup generated by $e(x, v)$ for $x \in Z$ and $v \in V_0$. There exists a positive integer q, depending on h,V_0, such that $e(0, r) \in H_0$ for every r divisible by q. The conclusion is exact equality of finite words in free groups. No finite-generation hypothesis on the original query groups is necessary.

Isolation. Choose a positive $v \in V_0$ and set $L = v\,\mathrm{lcm}(1, \ldots, h-1)$. Normalize the letters at player j to $k_{j,y} = u_{j,y}u_{j,[y]}^{-1}$, with $[y]$ the residue representative mod$L$. Let K_j be the subgroup generated by these letters and J_j its subgroup of isolated elements: the elements whose lift supported at j alone lies in H_0. It is normal in K_j since H_0 projects onto all of player $j's$ group. For $i \neq j$, by alternating queries of velocities 0 and v one can connect j-symbols differing by $(j-i)v$, with identity i-projection in the lifted difference. Thus every element of K_j lifts to H_0 with trivial i-projection. Lifting the inputs of an $(h-1)$-fold commutator separately to kill all the other players, and taking conjugates, gives $\gamma_{h-1}(K_j) \subset J_j$. The group $G_j = K_j/J_j$ is nilpotent of class $\leq h-2$.

A reversible mixed recurrence. For the other players take mutually disjoint finite blocks of the IP generators with positive sums u_i divisible by L. Such blocks exist arbitrarily late by the partial-sums pigeonhole principle. Put t_i=sgn$(j-i)u_i$ and $w_i = (j-i)t_i > 0$. Start at the unmarked velocity equal to the sum of the u_i with negative signs t_i; every vertex after adding a subset of the t_i is an allowed unmarked velocity. At the query tuple take successive multiplicative differences under $(x, r)\mapsto x-it_i, r+t_i$, one for each $i \neq j$. After the operation for i the ith slot is trivial: all previous products at this slot depend only on $x + ir$. In G_j the remaining isolated word proves

$$\Delta_{w_{h-1}} \cdots \Delta_{w_1} k_j(y) = 1 \quad \text{for all} \quad y, \Delta_w k(y) = k(y+w)k(y)^{-1},$$

where the positive steps are indexed in their chosen order. Their divisibility by L justifies normalizing after the first difference. In the expanded word each Boolean vertex occurs once. The starting and final indices y and $y + \Sigma w_i$ each occur just once, even if interior offsets coincide. Thus the recurrence computes the last letter from earlier ones and also computes the first from later ones. It follows that G_j is finitely generated, and that the normalized letters in every finite quotient of G_j are periodic in y, by the reversible finite-state recurrence.

Real hull. Put $\Gamma = \prod G_j$. In its torsion-free quotient use unipotent upper triangular coordinates on the real Malcev completion. Each matrix entry of $k_j(y)$ is polynomial on

residue classes of a common integer M divisible by L and all the w_i at all slots. Indeed at each triangular gap, the mixed difference word gives the *ordinary scalar mixed difference* of the relevant entry plus a polynomial in smaller-gap entries at finitely many shifts. This is because multiplication is addition on that gap modulo smaller-gap terms, and inversion is negation modulo such terms. Inductively the right side is polynomial on the M-residue classes. Multiplying the commuting scalar differences by the finite geometric sums taking each w_i to M shows Δ_M^{h-1} of the entry is polynomial on these classes. Finite discrete integration gives the claim.

Let B be the image in Γ of $H_0 \cap \prod K_j$. The kernel of $\prod K_j \to \Gamma$ lies in H_0 by isolation. Let T be the inverse image in Γ of the connected rational Malcev closure of the image of B. Then $[T : B] < \infty$ by the lattice theorem for finitely generated nilpotent groups. For r divisible by M, normalize $e(0,r)e(0,0)^{-1}$; its image has coordinates $k_j(jr)$, since all reference letters are at zero. Its image in the real hull is polynomial in r on this class. Infinitely many $r \in V_0$ are divisible by M, by grouping the IP generators, and at all of them this image belongs to the Malcev closure of B. In logarithms membership is expressed by linear equations, so the polynomial identity extends to every r divisible by M. The normalized tuple therefore belongs to T.

Finitely generated nilpotent groups are subgroup separable (Segal, *Polycyclic Groups*). Separating one representative of every nonidentity B-coset in T and intersecting the finite quotient kernels gives a normal subgroup N of finite index in Γ satisfying $N \cap T \subset B$. The normalized letters are periodic in the finite image of each G_j in Γ/N by the preceding recurrence. Take q divisible by M and all these periods. For q|r the normalized tuple is both in T and in N, hence in B. The kernel observation lifts it to H_0, which also contains $e(0,0)$. This proves the assertion.

Consequences for finite patterns that replicate. If a menu V_1 of tagged velocities meets every qZ, take a tagged $r \in V_1 \cap qZ$. The expression of $e(0,r)$ as a word in unmarked tuples gives a finite ordered contradiction for the game whose target is zero on those tuples and one on this tagged tuple. Give the finitely many required challenges any positive weights. By Braverman–Minzer, Lemma 3.5, the signed correlation against any product of h bounded player functions in n independent copies is $O(\rho^n)$, $\rho < 1$. The same proof works for every nontrivial character of a fixed cyclic group if the tagged move represents its generator: in their gluing proof the final character values are nonconstant since the original ordered contradiction has total label 1, and the only needed operations are multiplication, conjugation and Cauchy–Schwarz. One can also reapply the gluing argument separately for each character.

The density Hales–Jewett theorem converts this estimate to same-event recurrence as follows. Use m blocks, each of n local copies, where m is a density Hales–Jewett dimension for density $\mu(A)/2$ and alphabet $\{0, \dots, h-1\}$. A random trial in each block gives its total physical velocity d_i. For any such velocities, translating the test through all words $x + \Sigma w_i d_i$ has average A-density $\mu(A)$. With probability bounded below over stationary x, these words have a full combinatorial line in A. Thus the sum of the ordinary AP correlations over the fixed finite list of lines is bounded below independently of n, after averaging over the trials and translating away the fixed letters. For any nonempty variable set I the character-weighted version is $O(\rho^{n|I|})$: shift the stationary start by the random sum of the local query starts; then each player's indicator is a bounded function only of that player's letters on its $n|I|$ copies, with the common stationary starting point fixed. Integrate over that point. Fourier expansion now gives a positive correlation for some line with any prescribed cyclic

tag. This needs the actual allowed set to realize all the copy locations and omit unused blocks; it is not a statement about the unreplicated menu V_1.

Odd digit sums. For every even $b \geq 2$ the positive integers $\sum_{i \in F} b^i$ with $|F|$ odd form a set of multiple measurable recurrence at every positive density. This holds on each digit tail. Take V_0 to be the even subsums, including zero; it contains an IP set by pairing axes. Every modulus q has an odd-cardinality subsum divisible by q: on sufficiently late powers the 2-part of q divides each, and one can add an odd multiple of the period-group order on a repeating residue for its odd part. The finitely many challenges in the obstruction use finitely many digit positions; put their copies in disjoint increasingly shifted blocks of digit positions, multiplying each velocity by a suitable b-power. Digits do not overlap and the total mark is exactly the parity of the number of chosen positions. The preceding argument applies to any stationary integer system. If b is odd, every such direction is odd, giving critical value one instead. This is a full multiple-recurrence result for one geometric family, not a resolution of the prescribed intermediate critical density.

Finite permutation automata. The same method works when allowed base-b digits include zero and each such digit acts as a permutation of a finite state set. Disallow other digits. Pad the word length to a multiple of the order of the zero-digit permutation, always using the least such length or any further whole periods. Define $\gamma(d)$ to be the resulting permutation, with $\gamma(0)$=identity; the acceptance convention here tests this padded-word permutation in a fixed set of group elements. If D is divisible, some accepting g has $\gamma^{-1}(g)$ divisible: otherwise one common modulus excludes their finite union. The neutral menu $\gamma^{-1}(1)$ contains 0 and an IP set: take any nonzero allowed word padded to such a length, repeat it ord(γ(word)) times to obtain a nonzero neutral block, then repeat this block at disjoint aligned positions. If all positive digits are forbidden then D is empty, and no divisible positive D exists. For $g \neq 1$ use only menus with permutations 1 and g. The finitely many query words can be padded to one common block width (as multiples of the zero-digit period), and repetitions test powers inside the cyclic subgroup generated by g even if the full permutation group is nonabelian. Use the preceding character argument for the tag 1 in this cyclic group. For $g = 1$ the neutral IP construction gives recurrence directly. Thus a divisible D in this explicitly specified permutation-automaton class has critical value zero. If D misses qZ it has critical value one, by the profinite root sieve. The result does not presume that a general finite automaton acts by permutations.

R219. A fixed vertical return cannot be killed by arbitrarily rough horizontal progressions

Scope and status. Result with proof.

Result. For a fixed event and progression length, averages on residue classes with increasingly rough moduli approach the ordinary average uniformly.

Scope. The event and system remain fixed; the modulus has least prime factor tending to infinity.

Qualifications. Changing the event with the modulus is not covered.

For an integer pmp transformation T, a positive-measure event C and a length h, write $c(n) = \mu(\bigcap_{0\le j<h} T^{-jn}C)$. Denote by $A_{m,r}$ its Cesàro mean on the residue $n = r \bmod m$, and by A its unrestricted Cesàro mean. For this *fixed* event one has

$$\sup_{m:\ \text{all}} \sup_{prime factors \ge L} \sup_{r \bmod m} |A_{m,r} - A| \to 0 as L \to \infty.$$

On an ergodic component the Bergelson–Host–Kra nilsequence and null decomposition gives a nilsequence uniformly approximable by basic nilsequences, plus a sequence of zero uniform-density absolute mean. The latter has zero mean on every fixed residue progression. For each fixed basic approximant, restrict to its orbit closure with s connected components. Its residue means agree with its ordinary mean for every modulus coprime to s: the s components are cyclically permuted and the component return on each connected nilmanifold is totally ergodic, with the corresponding equidistribution of its translation orbit. Uniform approximation now gives the displayed convergence on the ergodic component. For general T, scalar residue limits disintegrate componentwise by bounded convergence; the measurable supremum of the componentwise discrepancies over countably many m, r tends to zero and is bounded. Integrating gives the asserted uniform convergence on the entire system. In particular $A_{m,r}$ is positive for every sufficiently rough m and every residue, by positivity of the ordinary multiple recurrence mean.

In an asymmetric $Z \times V$ model, if for some fixed v the event $C = \bigcap_{j<h} U^{-jv}B$ has positive mass, the horizontal section at that v cannot avoid B if it contains an entire residue progression at arbitrarily rough moduli. Thus independently resampling the horizontal prime indefinitely at the *same* vertical return, while leaving all other residues free, yields full recurrence when combined with vertical density Hales–Jewett for each B, h. It cannot keep any positive short obstruction. Finite prime columns per vertical axis with simultaneous conditions at the other active primes, or vertical events changing with the moduli, are not ruled out by this argument.

R238. A restriction on possible pair witnesses from square-free graph targets

Scope and status. Result with proof.

Result. A square-free graph target for an integer difference graph forces one-character separation, hence capacity one.

Scope. Graph homomorphisms into targets without four-cycles, including the stated odd Kneser examples.

Qualifications. An arbitrary positive pair witness need not admit this kind of graph target.

Let $D \subset N$ be nonempty. Suppose there is a graph homomorphism from the integer difference graph, with edges at the steps D, to a simple undirected graph K containing no simple four-cycle. There is then a single circle character χ on Z satisfying

$$||\chi(d)|| \geq (g-1)/(2g) \quad \text{for all} \quad d \in D,$$

where g is the odd girth of the relevant connected component of K (with 1/2 as the bound for a bipartite component). In particular the integer critical density is one. This extends the elementary three-color winding argument in the earlier pair notes [D12]. The use of squares and fundamental groups is related to Matsushita's graph fundamental-group work and Enami–Matsushita's recent results on circular colorings. No claim of literature priority.

Let F be the free abelian group on symbols e_d, $\psi{:}F \to Z$ the map $e_d \mapsto d$, and R its kernel. Lift the original coloring $c\circ\psi$ from the grid on F to the universal covering tree of its target graph, starting at a chosen lift of $c(0)$. To see path independence, a closed grid path reduces by adjacent commutations and backtracks. Every commutation maps to a closed four-edge walk in the target; this walk cancels by backtracks because there is no simple four-cycle. Write f for the lift. For $r \in R$, $f(x+r) = g_r f(x)$ for a unique deck transformation g_r, by uniqueness of lifts on a connected domain and transitivity of the deck group on each fiber. These transformations commute and form a homomorphic image of R.

If this image is trivial, f descends to $\psi(F)$. The bipartition of the tree shows that every relation in R has even coordinate sum. The homomorphism $F \to C_2$ sending every generator to one then descends, giving $\chi(d) = 1/2$. The same conclusion follows whenever all deck transformations in the image preserve the bipartition.

Otherwise choose a deck transformation reversing the bipartition. It acts by a nonzero translation on a bi-infinite geodesic of the tree, since the deck action is free without edge inversions. Every commuting deck transformation preserves this axis and its orientation. Signed displacement on the axis is an injective homomorphism of this deck subgroup into Z: displacement zero would fix an axis vertex. Its nonzero image is ℓZ. Choose its positive generator. The projected fundamental segment of length ℓ is an odd closed walk in K (an orientation-reversing bipartition element exists, so ℓ is odd). Hence $\ell \geq$g. For each r let $n(r)$ be the corresponding integer translation in units of ℓ. The parity identity is

$$\sum_d r_d \equiv n(r) (\mathrm{mod} 2).$$

Indeed passage in the tree from $f(0)$ to $g_r f(0)$ changes bipartition according to the length of a grid path from 0 to r, and according to the parity of $\ell\ n(r)$ along the axis.

Project normally to the axis and take its signed vertex coordinate. The resulting real function h on F is one-Lipschitz on every generator edge and $h(x+r)-h(x)=\ell\ n(r)$. Choose a translation-invariant mean on F (or use a diagonal Følner limit on the countably many bounded increments). Put $u_d = \text{mean}(h(x+e_d)-h(x)) \in [-1,1]$. Telescoping and translation invariance give $\sum_d r_d u_d=\ell\ n(r)$. Thus $e_d \mapsto 1/2 + u_d/(2\ell)$ modulo one kills R by the parity identity. This defines χ on $\psi(F)$, with the claimed separation, and extends to Z by taking a root in the circle. To obtain $b(D) = 1$, take an open arc of arbitrarily small length η, divide χ by M clearing every denominator $\le 2/\eta$, and delete that arc in the resulting rotation system. Its velocities at the differences are uniformly bounded away from the bad small-denominator velocities after multiplying by M; compactness gives a common finite hitting length.

For instance an integer difference graph that maps to the odd graph KG(2k + 1,k) admits such a character with $||\chi(d)|| \ge k/(2k+1)$. Two distinct k-sets admit at most one common disjoint k-set, so this graph has no four-cycle. Any two-step walk changes a k-set in at most one element, giving odd girth at least 2k+1: after 2l steps a neighbor of the start is reachable only if $l \ge k$. This specializes to proper three-colorability for $k = 1$. A positive-density pair avoider gives a probability distribution of independent sets, and in general does not give an odd-graph coloring. The argument supplies no bound for such arbitrary measure witnesses or for colorings that only prohibit longer progressions.

R242. A half-density lemma for a common root stride

Scope and status. Result with proof.

Result. Common-stride Haar controls give the half-density finite-window theorem; time-dependent masks and several strides have explicit weaker thresholds.

Scope. Freely available Haar control streams with deterministic forcing.

Qualifications. No arithmetic realization is supplied by this control lemma.

This concerns a specific Haar background law suggested by the two-observable construction with varying background centers. It is not a model theorem for general actions. Suppose K is a compact metrizable abelian group, $M \geq 1$, the controls X_t are independent Haar K-variables, and

$$Z_t = \big(X_t,\ X_{t-M} + X_{t+M} + v_t\big),$$

where $v_t \in K$ is any fixed forcing sequence. Random forcing independent of the controls can be conditioned upon. Coordinates with the same stride can be bundled into the group K.

Lemma. If $B \subset K^2$ is measurable with Haar mass $> 1/2$, then, for every h, the event $Z_t \in B,\ 0 \leq t < h$, has positive probability. This is uniform in the sense of positivity over all forcing terms v_t.

Let $b(x)$ be the Haar mass of the section B_x. Since $\int b > 1/2 = \int_0^1 (1-u)\,du$, there is $u \in (0,1)$ with $P = \{x : b(x) > u\}$ of Haar measure exceeding $1-u$. Initialize the two earliest control positions of each residue class modulo M in P; this is a positive-probability event. Process the remaining constraints in increasing time on each residue class. When X_{t-M} and $X_t \in P$ are known, the fresh X_{t+M} can lie in P and make the second coordinate of Z_t lie in B_{X_t} with conditional probability at least $\mu(P) + u - 1 > 0$. Induction or Fubini over the finitely many slots proves the lemma. The first coordinate at the start of each residue also needs to lie in P, as provided by the initial choice.

A time-dependent form holds at threshold $3/4$. For measurable B_t with $\mu(B_t) > 3/4$ for every relevant slot, set $P_t = \{x : \mu((B_t)_x) > 1/2\}$, so $\mu(P_t) > 1/2$. At time t enforce $X_{t+M} \in P_{t+M}$ when that is still a constrained time (otherwise use the whole group), and $Z_t \in B_t$, with a strictly positive intersection of admissible fresh values. The same greedy argument applies.

For r independent such streams with different strides, elementary repeated conditioning gives a sufficient threshold $1-(1/2)4^{-(r-1)}$ for a single common event in the product output space. To append a stream, retain preceding output states whose new-coordinate section has measure $> 3/4$. If the product event has density a, this retained base has density at least $1-4(1-a)$. A positive trajectory there can be lifted using the time-dependent form and independent controls. Start the induction with the common-event $1/2$ result. This estimate deteriorates with the number of distinct strides, and therefore does not give the length-independent density bound for the original question.

For the proposed spectral model this lemma applies only if the background law has actually been shown to include the displayed freely controlled law (positive probability is needed on the relevant projected measurable sets, not merely a set-theoretic list of possible paths). The numerical phase stencil $(S^M - I)((S^{2M} + I)F_0 - S^M F_1)$, with constants on each residue and the exceptional contribution absorbed into v_t, has exactly this form when its annihilator is the full list of background restrictions. The abstract control lemma is independent of that identification.

R244. A Haar-preserving feedback for arbitrary root strides

Scope and status. Result with proof.

Result. Haar-preserving feedback gives finite-window positivity for full positive sections, and dispersed cylinder certificates yield a separate local-lemma criterion.

Scope. The section or certificate hypotheses are essential.

Qualifications. Large event mass by itself does not imply either condition.

Use the stride-control note [R242] for the common-stride half-density theorem. Let K_i be compact metrizable abelian groups, $1 \le i \le r$, and $M_i \ge 1$. Let $X_{i,t}$ be independent Haar controls, and set

$$Y_{i,t} = X_{i,t-M_i} + X_{i,t+M_i} + v_{i,t}.$$

The terms $v_{i,t}$ are arbitrary prescribed elements. Write $K = \prod K_i$, $X_t = (X_{i,t})_i$, and likewise for Y_t.

Full first projection. Let $B_t \subset K \times K$ be measurable. Suppose, for each $0 \le t < h$, that $B_{t,x} = \{y : (x,y) \in B_t\}$ has positive Haar mass for almost every $x \in K$. Then the simultaneous event $(X_t, Y_t) \in B_t$, $0 \le t < h$, has strictly positive probability. There is no restriction on the number or size of the strides and no lower bound on the individual nonzero section masses.

Initialize all registers $X_{i,s}$, $-M_i \le s < M_i$, independently with their Haar laws. Just before checking time t, the state consists of the registers $t - M_i \le s < t + M_i$ in each row. By induction their joint law is product Haar. Draw the joint vector Y_t with normalized Haar law on B_{t,X_t}, using fresh randomness conditionally independent of the registers given the middle vector X_t. Complete the new registers by

$$X_{i,t+M_i} = Y_{i,t} - X_{i,t-M_i} - v_{i,t}.$$

The dropped vector $(X_{i,t-M_i})_i$ has product Haar law independent of *all* retained registers and Y_t because $M_i > 0$ makes the middle vector entirely retained. Given these quantities the newly inserted vector has product Haar law. Thus the next state again has exactly its original Haar law. At each of the finitely many steps the middle vector avoids the exceptional null set of empty sections almost surely, and the constrained event holds almost surely. In the original process the fresh vector at each step has unrestricted product Haar law conditional on the past. The constrained transition is absolutely continuous with respect to this original transition, by its normalized Haar section law and the translation/reflection formula with the past fixed. Finite iteration preserves absolute continuity, proving the assertion.

This is dimension-free for events whose first-coordinate projection has full measure in the positive-section sense. If the positive-section projection has mass $1 - e < 1$, the same construction with unrestricted Haar fallback at the missing sections yields an h-block of probability at least $1 - he$ **in the feedback measure**. This proves original-law positivity when $e < 1/h$. It does not give a threshold uniform over all h when there are positive-measure missing sections. In particular, a mark depending entirely on the first coordinate

should be handled by independence of the raw controls, not by assuming it satisfies the full-projection hypothesis.

Dispersed sufficient conditions. A certificate of support $I \subset [r]$ is a positive-measure cylinder depending only on $((X_i, Y_i) : i \in I)$ which is contained in the common event B up to a Haar-null set. Suppose finitely many certificates admit a probability distribution for which λ_i is the chance that coordinate i is in the certificate and $\sum_i \lambda_i^2 < 1/32$. Then there is positive all-B probability, uniformly in the sense of existence for every h, M_i, v. This is the independent-certificate argument of the palette note [D13], here applied to arbitrary positive cylinder tests of the linear fields.

Select a certificate independently at each time t. A conflict between $t, t+l$ is the selection of supports both containing an i with $l = M_i$ or $l = 2M_i$. Its probability is at most $p_l = \sum_{i:l\in\{M_i,2M_i\}} \lambda_i^2$. Events are independent of the collections of events sharing neither endpoint. Set the local-lemma parameters to $2p_l$, ignoring zero-probability events. The sum of parameters on all events whose endpoints meet either endpoint of a given conflict is at most $8\sum_l p_l \le 16\sum_i \lambda_i^2 < 1/2$. Thus the asymmetric local lemma gives a positive probability of no conflicts. With a conflict-free selection, different cylinder tests use disjoint underlying control variables. Each selected test has positive probability because its one-time input has product Haar law even for the prescribed forcing. Their conjunction also has positive probability.

For disjoint supports of sizes $m_j \ge 1$, choose their probabilities proportional to $1/m_j$. The criterion becomes $\sum_j 1/m_j > 32$. For example, the union of more than 32 one-row events of arbitrarily small positive measures has positive all-event probability at every length and at all choices of strides. Almost-sure cylinder containment suffices, since the finite conjunction removes only one-time Haar-null sets. Projection fullness and dispersed certificates are sufficient conditions, and a general high-mass event need not have either property.

R245. A Sturmian curvature word gives full avoidance for matched prime pairs

Scope and status. Result with proof.

Result. A Sturmian curvature construction gives capacity one for matched irrational prime ratios with marked budget below one.

Scope. The limiting ratio is greater than one and primes satisfy eventual fixed-modulus congruences; R215 covers other positive ratios.

Qualifications. The one-sided occupation estimate gives the full stated budget range below one, including the exceptional resonances omitted by the preliminary half-budget argument.

Assume the signed or unsigned paired-prime family of the shell result [R199], with $q_i/p_i \to \theta > 1$ irrational, with both primes eventually congruent to 1 modulo every fixed modulus, and with budget $\Sigma|a_i|/p_i \le \tau < 1$ and odd Σa_i. Its full rational saturation has capacity one. We first show that when $\tau < 1/2$ there is no tail-supported ultrafilter P on its directions with $\sup_{E\in P} b(E) \le 1-\eta$ at any fixed gap $\eta > 0$. Use the corresponding squarefree group capacity throughout, and require the ultrafilter to contain every support tail. A countable exceptional-set argument then gives the full endpoint, and a one-sided variation bound covers every $\tau < 1$.

Setup. Suppose such a P exists. By the continuous-character theorem [R136] and its signed version from the statistical expander [R232], its limiting signed mass has the form

$$S = P - \lim_d \sum_i a_i/p_i = b + c(\theta - 1),$$

where b is an odd integer and c is rational. Only this consequence is needed. Fix the graph and site test of the signed expander at cost strictly below $\eta/2$, and take any even D at least two more than the number of embedded vertices. On each C_{q_i} use the F_{p_i}-word whose D-th forward difference is prescribed by

$$d_i + \lfloor (v+1)c_i/q_i \rfloor - \lfloor vc_i/q_i \rfloor,$$

$$d_i = (p_i+1)/2, c_i = (p_i - q_i)/2.$$

The integer increment is q_i-periodic, and $q_i d_i + c_i = p_i(q_i+1)/2$ is zero in the prime field. Since $p_i \ne q_i$ this is a D-th difference on that cycle. Put $\alpha_i = c_i/q_i$, $\alpha = (1/\theta - 1)/2$ and write b_i for the resulting word.

If at some fixed horizon the directions permitting a completely retained run are absent on a member of P, the finite-packet compactness argument of the signed expander provides one event of mass $> 1 - \eta/2$ avoiding that P-member. In defining permission for a run one may freely allow arbitrary affine spatial errors in the mirror identity and arbitrary nuisance starts; ruling out these relaxed runs suffices. Thus if P is hard, for successively longer horizons one can select a violating direction d supported after the horizon index, with $\Sigma a_i/p_i \to S$, and initial positions permitting the violation. The test configuration is finite before taking this

sequence. Translate to the midpoint of each run and pass diagonally to an everywhere retained infinite row profile in the limit; all scalar circle trajectories and the coefficients of the affine nuisance terms belong to compact spaces.

The limiting curvature. Write $\mu_i = a_i/p_i$ in centered representatives, and $s_i{=}\pm 1$ on the active axes. The marked scalar in the signed mirror identity is

$$W(t) = \sum_i \mu_i b_i (v_i + s_i t) \bmod 1.$$

Along each finite direction there is an exact identity, with a constant C,

$$\Delta^{D-1} W(t) = \kappa_I t + C + G_I(t) \bmod 1,$$

$$\kappa_I = (-\sum_i a_i + \sum_i a_i/q_i)/2 \bmod 1.$$

Indeed $\Delta^D b_i = d_i + \alpha_i + \{v\alpha_i\} - \{(v+1)\alpha_i\}$. A positive-speed contribution to G_I is $-\mu_i\{v_i\alpha_i + t\alpha_i\}$. At negative speed use $\Delta_t^D b_i(v_i - t) = \Delta^D b_i(v_i - t - D)$ because D is even; its contribution to G_I is $+\mu_i\{(v_i - D + 1)\alpha_i - t\alpha_i\}$. Finally $\mu_i(d_i + \alpha_i) \equiv -a_i/2 + a_i/(2q_i)$. Consequently $\kappa_I \to \kappa = (-1 + S/\theta)/2 \bmod 1$.

For the two orientations separately take weak limits of the positive and negative parts of the finite signed atomic measures having weights μ_i at the just displayed initial phases. Their total variations together are $\le \tau$. If their limiting signed measures are ν_+,ν_-, put

$$G(x) = -\int \{v + x\} d\nu_+(v) + \int \{v - x\} d\nu_-(v).$$

This bounded circle-periodic real function has total variation at most 2τ, since each fractional-part function has that variation before weighting. Its discontinuities are contained in the atoms of the limiting positive measures and thus form a countable set. For each fixed integer t the error between any subsequential limit of $G_I(t)$ and $G(t\alpha)$ is bounded by the total limiting variation mass at the two possible discontinuity locations $-t\alpha$ and $t\alpha$ in their respective measures. To verify this split off arbitrarily small neighborhoods of the jumps and use uniform continuity outside, since the α_i converge uniformly on late tails. The two orientations use separate positive/negative variations, so cancellations are not assumed. As α is irrational, summing these bounds over all integer t gives at most τ. After another diagonal extraction, the limiting profile therefore satisfies

$$\Delta^{D-1} W(t) = \kappa t + C + G(t\alpha) + e_t \bmod 1, \sum_t |e_t| \le \tau.$$

Invariant measures of the primitive. The arithmetic identity essential here is

$$\kappa = A_0\alpha + B_0 \bmod 1, A_0 = b - c = (b\theta - S)/(\theta - 1), B_0 \in Q.$$

It follows by substituting $S = b + c(\theta - 1)$; for example $B_0 = (b-1)/2$ (modulo one). In particular $|A_0| \geq (\theta - \tau)/(\theta - 1) > 1$. Let the compact rotation base be the orbit closure of the phases $t\alpha, t\kappa$ and any finite list of affine nuisance frequencies of the expander graph. Choose Haar on this orbit closure (or its translate for initial phases); the rotation is minimal and uniquely ergodic there. On it the curvature cocycle is $F(z) = z_\kappa + C + G(z_\alpha)$.

For every nonzero integer l and every positive H,

$$|\int e(l\sum_{j=0}^{H-1} F(R^j z))dz| \leq \mathrm{Var}(G)/|A_0| < 1.(*)$$

Here $e(x) = \exp(2\pi i x)$. Choose Q with $A = QA_0 \in Z$ and $QB_0 \in Z$. On each residue class of the rotation orbit modulo Q, the projected Haar law of (z_α, z_κ) is the law of $(Qx + z_1, Ax + z_2)$ for uniform circle x, by irrationality of α. Extra nuisance phases do not affect this integral. Now the leading character in the integrand is of frequency lHA in x. The remaining scalar function is the exponential of a sum of H translates of $G(Qx)$, of total variation at most $2\pi|l|HQ\,\mathrm{Var}(G)$. The elementary integration-by-parts bound for a Fourier coefficient of a BV function gives (*).

In particular $e(lF)$ is not a measurable coboundary on the rotation base. If it were, choose unbounded return times H along which the rotation tends uniformly to the identity as a group translation; the transfer and its translates would approach one another in L^2, making those same integral moduli tend to one. Thus the skew circle extension by F is ergodic for the product of base and circle Haar, by expansion of an invariant function in its fiber characters.

Adjoin the coordinates $Z_0, \ldots, Z_{D-2}$ for the lower differences of W, listed in reverse order: the first increments by $F(z)$ and each subsequent one increments by the preceding coordinate. The product of base Haar and independent Haar on all these coordinates is ergodic. Indeed under the unipotent fiber matrix the L^2 norms of coefficients of any invariant function agree along its dual integer orbits, so infinite orbits have zero coefficients. The only finite orbits are the multiples of the first coordinate character; the preceding coboundary argument eliminates its nonzero multiples. This applies also to each initial segment of the tower. The Haar-extension law is the unique invariant measure projecting to Haar on the base: induct up the tower of one-circle extensions, averaging translates of any invariant law by the *last* circle; translations in that circle commute with the action, and the Haar-extension law is ergodic. Hence almost every translate, and then the original law, is this Haar extension. This uses no continuity of F on its countable exceptional base fibers.

The empirical limits of the actual trajectory obey this conclusion as well. The base rotation projects to Haar, so F and its finite translates are continuous outside a limiting-null set. The equation for the first primitive has one error $e_t \to 0$; the other update equations are exact. For any continuous test function the empirical average of its actual forward values differs by $o(1)$ from that of the skew update. By weak convergence and the null discontinuity set, each empirical limit is invariant, and uniqueness identifies it.

The graph contradiction. Given the finitely many distinct endpoint offsets n_x of the expander proof, the empirical joint law of $(W(t + n_x))_x$ is product Haar conditional on the

rotation phases, including the affine nuisances. To see this expand W at these translates by successive finite differences. It equals the matrix with entries $\binom{n_x}{k}$, $0 \le k \le D-2$, times the vector of lower differences at time t, plus a known function of finitely many translates of F and errors tending to zero. The displayed matrix has full row rank over Q by Vandermonde, so maps independent circle Haar onto full Haar at the endpoints. Bounded tests in the nuisance corrections are continuous in the tracked phases; one can track real half-lifts of the finitely many affine coefficients as additional circle rotations. The known translations of F do not affect the conditional Haar law.

The Fejér-product programming argument of the signed expander now applies with these empirical Haar endpoints. Bounded dissociation at each vertex makes the positive product mean one and its averaged phase defect as small as prescribed. Some finite time translation of the graph therefore cancels the mirror errors on average. But the everywhere retained profile has every central value in the closed set with a missing arc and has small mirror averages, contradicting the expander spectral and Fourier estimates. This contradiction proves the claimed nonexistence of P.

Compactness consequence. Fix $\eta > 0$. The closed subset of the Stone space consisting of ultrafilters P with $\sup_{E \in P} b(E) \le 1 - \eta$ has empty intersection with the nested clopen support tails. By compactness it misses some tail. Every ultrafilter on that tail therefore has a neighborhood of capacity $> 1 - \eta$; compactness provides a finite cover by such cells and disjointization gives a finite partition. This statement allows the initial tail to depend on η. In the rational saturation, each finite head can separately be sieved by a density-arbitrarily-near-one mask, so the same finite-partition property is available there, with the squarefree tail correspondence.

Full endpoint for the matched paired-prime family In fact the full rational saturation has capacity **one** under the hypotheses above. The preceding Stone conclusion is strictly weaker; its conditional intermediate-model possibility is excluded by the following separate argument. The signed expander also makes the positive-vertical shell lemma available with either vertical speed.

Fix $\epsilon > 0$. Use one graph and one Sturmian word per coordinate, exactly as above, at site loss $< \epsilon/2$. Neither the word nor the graph depends on the possible limiting signed mass $S \in [-\tau, \tau]$. Consider the following good parameters:

- If $S \notin Q + Q\theta$, then $1, \alpha, \kappa$ are rationally independent. In the integral (*) the projected two-coordinate Haar law is full torus Haar. Integrating over the κ-coordinate makes the integral zero for all $l \neq 0$ and $H \ge 1$, regardless of any extra nuisance phases on the full rotation base. The empirical Haar and graph contradiction above apply without any change.
- If $S = A + C\theta$, with rational A,C and $|A| > 2\tau$, then $\kappa = A\alpha + B$ for rational $B = (A + C - 1)/2$. Inequality (*) is strict since $\mathrm{Var}(G) \le 2\tau < |A|$, with no need for $A + C$ to be an odd integer. Again the graph contradiction applies.

For each good S, this *same* probe has a neighborhood I_S, a tail J_S and a finite horizon H_S excluding completely retained runs for every direction supported after J_S with mass in I_S, uniformly over all starts and all affine nuisances. Otherwise for each n one could use the $1/n$-neighborhood, the nth tail and horizon $2n + 1$ to select a violating run; recentering

would give precisely the forbidden infinite profile above. The loss $\epsilon/2$ is incurred once, no matter how many good neighborhoods are used.

The remaining parameters form a countable set $S_n = A_n + C_n\theta$, with $|A_n| \le 2\tau$. They are outside the exceptional set of the sparse shell lemma: their unique rational coefficients satisfy

$$|A_n + C_n| \le (1 - 1/\theta)|A_n| + |S_n|/\theta \le 2\tau(1 - 1/\theta) + \tau/\theta < 1,$$

whereas an expression $S_n = b+c(\theta-1)$, with odd b and rational c, would have $A_n+C_n = b$. For each such S_n apply the sparse shell lemma [R199], using its signed expander version, at its own loss $\epsilon/2^{n+2}$, $n \ge 1$. This gives a detecting interval, a tail, and a uniform finite horizon on that interval. Its independent probe may have entirely different spatial fields, dimensions and degree. The countable set costs at most $\epsilon/4$ in total before selecting any subcover.

All these neighborhoods cover the compact real interval $[-\tau, \tau]$. A finite subcover suffices. Take the maximum of its finitely many tails and horizons. On finite packets impose the single universal Sturmian site test and the finitely many distinct additional sparse tests of this subcover. Independent Haar shifts and the union bound give site density $> 1 - \epsilon$; equivalently use independent stationary products. Every direction on that tail whose signed mass lies in $[-\tau, \tau]$ is excluded from a uniform run by at least one retained-site condition. Finite-packet compactness gives $b \ge 1 - \epsilon$ on that tail. The squarefree tail formula [R241] now gives capacity one for the full rational saturation, hence also for its corresponding integer realization. This excludes the entire one-clock matched-ratio family with budget $\tau < 1/2$ as a seed for the conjecture; it does not decide arbitrary integer step sets.

A one-sided variation bound extends the endpoint to all $\tau < 1$ For $\tau < 1$ it suffices to handle by the same universal word all exceptional parameters $S = b+c(\theta-1)$, where b is odd and c rational. Use the universal word at every $S \notin Q + Q\theta$ as above. The members of $Q + Q\theta$ that are not exceptional are still countable and covered by individual sparse probes at summable losses. The good-word neighborhoods always admit the finite uniform horizons already proved by contradiction.

For an exceptional parameter write $A_0 = b - c$ as before. Reflect all signs if necessary, so $b \ge 1$; the associated distributions are conjugated and all moduli of integrals are unchanged. The finite variation function G on the circle has distributional derivative

$$DG = -Sdx + \sigma, \sigma(T) = S, ||\sigma|| \le \tau.$$

Indeed each summand $-\mu\{v + x\}$ or $+\mu\{v - x\}$ contributes $-\mu dx$ plus a signed atom of mass μ. Passing to the measures $\nu_\pm$ gives the stated formula and bound. Thus the negative Jordan mass of σ is at most $(\tau - S)/2$. Set $u = A_0 - S = \theta(b - S)/(\theta - 1) > 0$. In particular

$$(\tau - S)/(2u) < 1.$$

Consider H terms in the rotation sum and a positive integer l; negative l gives conjugation. On each class of the base rotation described in (*), view the real phase lift on $0 \le x \le 1$ with continuous linear part `lHQ u x`, compensating for the term $-Sdx$ in every $G(Qx+\ldots)$. The residual BV derivative has negative mass N at most $lHQ(\tau - S)/2$, and the full lift changes by `lHQ A0` over one x-period. One may shift the integration endpoints to points of continuity of the finitely many translates; circle integration does not depend on that shift.

Here is the one-sided occupation estimate. For a real periodic test $0 \le f \le 1$, continuous, with circle average ρ, and a lift ϕ as just described,

$$lHQu \int_0^1 f(\phi(x))dx \le \rho(lHQ|A_0| + 1) + N.$$

For a smooth residual derivative this follows by integrating $f(\phi)\phi'$, using the fundamental theorem for a primitive of f. The positive part of the residual derivative can only increase the integral, and its negative part costs at most N. For general BV residual derivatives, convolve the periodic part with a smooth approximate identity, keeping the real linear part. The lower bound on negative Jordan mass is not required; its upper bound N persists under smoothing, since convolution contracts the total masses of each positive and negative part. The smooth phases converge at all continuity points of the original lift and at its endpoints, which proves the claim by dominated convergence. The elementary bound on the oriented primitive integral uses full unit periods and at most one leftover period, each costing at most ρ.

Taking a continuous majorant of an arc of length r fixed sufficiently small (with support length $< 2r$), all such arc probabilities have limsup as $H \to \infty$ at most $2r|A_0|/u + (\tau - S)/(2u) < 1$, uniformly in their center and the rotation component. If the cocycle were a coboundary, the real mod-one sums multiplied by l along rigidity times would converge to zero in probability, contradicting this estimate. Thus every nontrivial multiple of the curvature cocycle again fails to be a coboundary, and the Haar primitive argument goes through. The countable-cover proof yields capacity one at every $\tau < 1$.

R251. Thickening the mixed ring leaves freely chosen slope streams

Scope and status. Result with proof.

Result. Thickened mixed-ring profiles retain free slope streams; bounded quotients removing those streams also remove the charge, and those streams defeat one proposed statistical mask.

Scope. An exact description of this filter model and a limitation of a particular blocker.

Qualifications. No general recurrence or threshold theorem follows from this obstruction to a proof method.

Consider the full-spatial mixed module [R056] with top element $p^{2D_p}\varpi^K\epsilon^2$, as opposed to $K = 6$, at sufficiently large primes $p > K + 2$. Truncate at $2D_p(p-1) + K + 1$ in the ramified coefficient ring. In the expression

$$B_M = (T-1)^3(T^M - S)^2(T^M - S^{-1})^2$$

the normal prime still contributes exactly zero. At sign σ the value of B_M lies in the ϵ^2-slice with factor $p^{2D_p}\varpi^5$. On this slice T, S act by V, ζ_p with $V^M = \zeta_p^\sigma$. The full orbit of the filtered observation therefore splits into two univariate contributions in $t + \sigma Ms$ (one can use $V^{t+\sigma Ms}$). Each is polynomial of difference degree at most $K-5$: an additional $K-4$ temporal differences kill the remaining ramified layers. This exact algebraic restriction survives finite packets and weak limits as a shift relation. For the finite-window limits polynomial denominators and common roots of coefficients must still be treated compatibly. Describing $B_M z$ as an arbitrary polynomial in two variables overstates its free parameters.

Inverting the filter on univariate polynomial ridges accounts for forced terms $t^2 C_\sigma(t + \sigma Ms)$, up to freely moving value and slope waves. More explicitly for a real-polynomial lift of a ridge, apply its tangent factor $(T^M - S^\sigma)^2$ to $t^2 Q(t + \sigma Ms)$: the result is $2M^2 Q(t + \sigma Ms + 2M)$. The remaining factors act as ordinary nonzero-step finite differences on that univariate polynomial. Finite-difference integration gives a real polynomial Q, with degree increased by five. For circle polynomials one may use a binomial-basis real lift on the integer parameter. The homogeneous solutions include arbitrary $P(t + \sigma Ms) + tY(t + \sigma Ms)$ for either sign, as well as degree-at-most-two temporal polynomials on each spatial row. This statement about the annihilator by B_M alone does not assert every solution is a profile for every fixed choice of finitely many marked directions.

Suppressing those slope waves by a bounded spatial-difference relation destroys the scalar used by the filter. Indeed in any quotient module stable under the local group action by $u = \zeta(1+\epsilon)$, if $p^D\epsilon(u-1)^J$ vanishes then multiplying by $u^p - 1 = p\epsilon + \binom{p}{2}\epsilon^2$ yields $p^{D+1}\varpi^J\epsilon^2 = 0$. On the ϵ^2-slice $u = \zeta$, so all its cyclotomic coefficient multiples vanish. For fixed J, K and large D, p, $p^{2D}\varpi^K$ is such a multiple because $(D-1)(p-1) + K \geq J$. Hence $p^{2D}\varpi^K\epsilon^2 = 0$. This already follows for invariant additive module quotients, without assuming they are quotient rings.

Why the two-stream statistical mask cannot remove this nuisance Even a single free value-and-slope stream defeats the two-stream empirical Fourier test [R258] for every

deterministic forcing. Here is a pointwise statement. Let $M \geq 1$ and add to arbitrary circle offsets $W_j(t)$ the controls

$$F_j(t) = W_j(t) + P_{t+Mj} + tY_{t+Mj}.$$

Require $F_0(u)$ to equal arbitrary prescribed x_u at all u, e.g. within the compact set omitting a small arc in that test. Put $P_u = x_u - W_0(u) - uY_u$, and start with independent Haar Y_u. At fixed t the variables

$$F_j(t) + F_{-j}(t) = deterministicterm + Mj(Y_{t-Mj} - Y_{t+Mj}), j > 0,$$

are independent uniform circles. Thus for each fixed finite family of nonzero Fourier frequencies, two parities, and any tolerance $\tau > 0$, the probability that either parity's mean of a tested character over $1 \leq j \leq s$ has absolute value above τ is at most Ce^{-cs}. This follows by the elementary bounded-variable exponential estimate separately on real and imaginary parts. The constants depend only on the frequencies' count and τ.

For all $s \geq m_0$ and time centers t consider these bad events simultaneously. At size s' an event shares a slope variable with at most $2(s + s') + 1$ other centers: their difference must be M times an integer of size at most $s + s'$. Disjoint-variable collections are independent, as required for the ordinary dependency graph. Apply the asymmetric Lovász local lemma with $x_s = e^{-cs/2}$. For m_0 sufficiently large,

$$\prod_{s' \geq m_0} (1 - x_{s'})^{\lfloor 2(s+s')+1 \rfloor} \geq e^{-cs/4}, s \geq m_0,$$

by $\log(1 - x) \geq -2x$ for small x and the geometric tail. After increasing m_0, $Ce^{-cs} \leq x_s e^{-cs/4}$. Thus for every finite time horizon and every finite upper spatial cutoff the simultaneous good event has positive probability. In the countably infinite case, the corresponding closed requirements have the finite intersection property on the compact product of slope circles. This supplies a control meeting all requirements. It allows all common offsets and deterministic polynomial forcing, arbitrarily many temporal roots in independent copies, and arbitrary other fixed streams. The cutoff m_0 may be increased freely in the empirical-test proof to secure near-full measure. This argument concerns that particular mask. A different spatial predicate or a suitable arithmetic dependence between the available nuisance streams could behave differently.

R253. A dimension-only threshold for finite toral controls

Scope and status. Result with proof.

Result. Toral control thresholds are at most $1 - 1/r!$, improved to $1 - 1/(s+1)!$ when the annihilator has rank s.

Scope. The earlier pure-fiber or horizontal-polynomial quotient-filling hypotheses remain necessary; the pivot/compression step itself permits arbitrary forcing.

Qualifications. This dimension-dependent conclusion gives no uniform threshold as dimension tends to infinity.

The preceding pure-fiber theorem [R190] and measurable horizontal polynomial fiber theorem [R140] both have the following stronger upper bound in every fixed ambient torus dimension r:

$$\delta_r \le 1 - \frac{1}{\{} r!\}.$$

This is independent of the Laurent modules, coefficient supports, orders of the horizontal polynomial channels, and the array sizes. It gives full positive recurrence when $r = 1$ and the half-density assertion when $r = 2$. The argument uses the quotient-filling statements from the respective earlier notes; it does not establish such filling outside their stated hypotheses. Its elementary control and compression step works for arbitrary forcing.

Let K_f be the connected coefficient subtorus of the Laurent nullspace controls, of dimension $k \le r$. The projected odd coset contains zero in those models. If $k = 0$ a constant admissible profile suffices. Otherwise shift the finitely many integral nullspace-polynomial vectors by monomials so their coefficient supports are disjoint, and add them to form one vector $v(T)$ of polynomials. This stays in the nullspace, and the real span of its coefficients is exactly that of K_f. Use a single iid Haar circle stream U_n, $n \in \mathbb{Z}$, adding $v(T)U$ to the prescribed forcing profile in the chosen coset of K_f.

Use the lattice $\mathbb{Z}^r \cap \mathrm{Lie}(K_f)$ to identify K_f with a k-torus. After a unimodular change of these intrinsic output coordinates, the scalar polynomials $v_1, .., v_k$ have distinct leading degrees. Indeed scan the coefficient columns from latest to earliest and record the k indices at which their rational span increases. In the dual lattice the annihilators of these nested rational spans form a nested flag of saturated sublattices of consecutive ranks, so choose one integral lattice basis adapted to the flag. Each chosen basis row first pairs nontrivially at its own recorded index. Order the resulting observation coordinates so their leading degrees satisfy $d_1 < \cdots < d_k$. Then in a symbol at time t, coordinate j is completed at input time $n = t + d_j$. Its dependence on the newly drawn U_n has nonzero integer coefficient and all its other inputs are earlier. Its conditional marginal at this instant is uniform on the circle, regardless of the previous draws and the forcing. At the same input time there is at most one symbol requesting each coordinate index.

For a measurable set $B \subset \mathbb{T}^k$ define its successive lower rearrangement in the unit cube. First replace each fiber in its last coordinate by an initial interval of equal length. With this last threshold fixed rearrange the next-to-last coordinate the same way, and continue to the first coordinate. Measurable representatives suffice throughout by Fubini. The resulting downset $B^{\downarrow}$ has the original volume. More explicitly, membership of a threshold tuple $(p_1, \ldots, p_k)$ in this downset means the following. Starting with $E_k = B$, put

$$E_{j-1} = \{x_{<j} : \mu\{x_j : (x_{<j}, x_j) \in E_j\} > p_j\}, \qquad j = k, k-1, \ldots, 1.$$

Then E_0 contains the empty prefix. This description with strict inequalities has the same volume, by the elementary identity $\int_0^1 1_{p<a}\,dp = a$ at each compression. It implies that any accepted prefix has more than the specified fraction p_j of choices for its next coordinate.

If $\mu B \geq \beta > 1 - 1/k!$, choose $c > 0$ so small that $1 - (1-c)^k/k! < \beta$. The upper simplex $\sum_j (1-p_j) < 1 - c$ has volume $(1-c)^k/k!$. Hence the downset meets it, giving thresholds with $\sum_j (1-p_j) < 1 - c$. Reveal the input stream in increasing order for the desired finite window. At every pivot time the previously filled prefix of each corresponding symbol is in its designated E_{j-1} on the induction event. Its new coordinate fails to extend that prefix with conditional probability less than $1 - p_j$. The same scalar input may complete different symbols, but at most once for each j, so a union bound gives conditional success greater than c. Non-pivot times may be unrestricted (only a finite interval of inputs is needed). In an h-window there are at most kh pivot times. Thus the probability of filling the whole window in B is at least c^{kh}. In particular the lower bound depends only on h, k, β, irrespective of temporal supports or coefficients.

On a larger ambient torus, average the independent common Haar translation over its K_f-cosets. If an ambient set has mass at least $\beta > 1 - 1/r!$, a uniformly positive fraction of those cosets have relative mass at least a fixed β' strictly above $1 - 1/k!$, using $k \leq r$. Apply the just-proved bound in each such coset, also conditional on the K_f-coordinate of the common translate. This gives the uniform translated-profile lower bound needed for the continuous-test and tilted-pool transfers in the prior notes. In the pure-fiber case approximate the control law on its odd coset by actual finite profile sums. In the horizontal-polynomial system approximate the original event by an early/base cylinder with error chosen from the displayed uniform lower bound and use the pool marginal bound. The threshold still tends to one as the ambient dimension increases; a universal high-density conclusion for the candidate is not obtained this way.

Dependence on the number of constraints. Let s be the rational Laurent rank of the even annihilator and $d = r - s > 0$. A refinement gives at most $m \leq k - d + 1 \leq s + 1$ pivot cohorts. Thus the threshold is at most $1 - 1/m!$, in particular at most the smaller of $1 - 1/r!$ and $1 - 1/(s+1)!$. To see this, do not combine the independent scalar streams. Choose a polynomial nullspace basis whose leading coefficient vectors are linearly independent. Start with an ordinary polynomial basis over $\mathbb{Q}(T)$. If its leading coefficient vectors have a rational dependence, choose a vector of highest degree occurring in that dependence and cancel its leading coefficient by a linear combination of monomial translates of the others. The shifts used have nonnegative degrees. This preserves the rational basis while decreasing the sum of degrees; eventually the leading coefficient vectors are independent. Clear scalar denominators and align the leading degrees by monomial translates. The real coefficient span still gives K_f: a constant rational row kills this new coefficient span exactly when it annihilates the rational nullspace, which is basis-independent.

Scan the coefficient **matrices** of these d vectors from largest input degree to smallest. The first matrix contributes d independent coefficient columns. Each subsequent strict enlargement of their accumulated rational span contributes at least one dimension. Choose an intrinsic output lattice basis adapted to the annihilator flag as before. At each pivot time the entire block of newly completed output coordinates has a jointly Haar marginal, because its leading coefficient matrix on the new independent Haar d-tuple has full row rank. There are at most $k - d + 1$ pivot times per symbol. The same compression argument applies with these output blocks as the successive probability spaces (the sole input in the rearrangement is the Haar measure of each fiber). At an input time there is at most one symbol per cohort;

their simultaneous conditional success has the same union bound based on a simplex of dimension m. This yields the improvement, including the uniform half-density assertion for rank one and the bound $5/6$ for rank two even if r is large.

R255. Global restrictions on translated signed sum boxes

Scope and status. Result with proof.

Result. Translated signed sum boxes satisfy usable-relation pair tests, forced-period and spectral criteria, an exact Bohr path characterization, and endpoint-one lacunarity tests.

Scope. Several separate hypotheses are stated; none gives a global endpoint dichotomy for all symmetric sum boxes.

Qualifications. The signed-box conclusions and the unsigned Erdős–Taylor pair theorem are separate results; the latter is proved in R254 and R262.

Write $D = e + \sum_i \{-a_i, 0, a_i\}$ with finite-support sums and $0 \notin D$; negative steps have the same recurrence obstruction after reversing a progression. The following gives a global test across all paths and a pair-recurrence result for the signed Erdős–Taylor example. Its unsigned counterpart, where the increments are only $0, a_i$, is a different problem: see Griesmer's Warwick seminar, pp. 20–24 for the classical unsigned example.

Divergence of usable relations forces pair recurrence. More generally suppose there are distinct, ordered indices i_j and positive integers q_j with $\sum_j 1/q_j = \infty$ such that one of $q_j a_{i_j}, -q_j a_{i_j}$ is e plus a permitted signed sum on indices outside $\{i_1, \dots, i_{j-1}\}$. Then D is a set of measurable pair recurrence. For broader symmetric S_i, the same conclusion holds if $\pm a_i, 0 \in S_i$ at the expansion sites.

Indeed, suppose a pmp transformation T has A of measure $a > 0$ disjoint in measure from $T^d A$ for every $d \in D$. Before using the jth index let B_j be the union of translates $T^u A$ over the binary sums using $0, a_{i_1}, \dots, a_{i_{j-1}}$, one choice per preceding index. Every difference of two such head sums uses the signed choices there. Therefore $B_j \cap T^{q_j a_{i_j}} B_j$ is null by the given representation and the symmetry of pair disjointness. Telescoping the symmetric-difference norm gives

$$2\mu(B_j) \le q_j \mu(B_j \triangle T^{a_{i_j}} B_j).$$

The next binary expansion has mass at least $(1 + 1/q_j)\mu(B_j)$. Multiplying contradicts $\mu(B_j) \le 1$. Without the divergence assumption this also gives the explicit upper bound $a \le \prod_j (1 + 1/q_j)^{-1}$ (and in fact a factor $1/2$ on the right, because each B_j is disjoint from $T^e B_j$).

In particular if $a_{i+1} = q_i a_i + e$, then $-q_i a_i = e - a_{i+1}$ uses a fresh coordinate and the test applies to every tail with divergent $\Sigma 1/q_i$. For $e = 1$ and the usual Erdős–Taylor sequence omit the first generator (which would otherwise give the zero step in the signed box). Every remaining tail gives pair recurrence. This calculation does not prove multiple recurrence or address the unsigned question.

A subgroup of forced periods. In a general translated symmetric box consider all integers c with the following property. There are disjoint finite blocks I_j and a choice $s_i \in S_i$ at each site used, with integer coefficients such that

$$c = \sum_{i \in I_j} n_i s_i, w_j = \max_{i \in I_j} |n_i| \ge 1, \sum_j 1/w_j = \infty.$$

Let L be the subgroup of Z generated by these integers. A positive pair witness for D implies $D \cap L = \varnothing$. Hence, if L contains even one nonzero element, either D has pair recurrence or the full box misses a finite-index subgroup and has all-length capacity one.

To prove this, for any finitely many listed integers choose disjoint blocks across all their representations with the respective reciprocal sums still divergent. Cycle through the integers, each time using sufficiently late new blocks with reciprocal sum at least one; disjointness in each source family makes this possible beyond any fixed head. A finite head of signed choices can similarly be frozen, changing the translate to e' and removing its indices. Starting at the positive pair-avoiding A, enlarge by the binary sets $\{0, s_i\}$ for all coordinates in each successive chosen block. Denote the events before the blocks by B_j, their union by B, and their successive increases in mass by δ_j. The δ_j have finite sum. For every oriented binary subsum v on the whole current block, the new event contains B_j and $T^v B_j$, so $\mu(B_j \triangle T^v B_j) \le 2\delta_j$. Every signed unit subsum is the difference of two such subsums and has discrepancy at most $4\delta_j$. Split the coefficient vector into w_j signed unit layers; the triangle inequality gives $\mu(B_j \triangle T^c B_j) \le 4 w_j \delta_j$ for the integer represented in this block. In its own divergent subsequence there are arbitrarily late indices with $w_j \delta_j \to 0$. Since $B_j \to B$ in measure, $T^c B = B$ modulo null sets for every one of the finitely many selected integers. But $B \cap T^{e'} B$ is null: the difference between any two of the binary head sums uses only signed allowed increments disjoint from the frozen head. Therefore e' is not in the subgroup generated by the selected integers. This applies after freezing any finite signed choices; if an element of D lay in L, finitely many generators of L would already generate that element, a contradiction.

For example take $a_{i+1} = q_i a_i + c_i$, and let C_* consist of all nonzero values c for which $\sum_{i:c_i=c} 1/q_i = \infty$. For each such c, choose one parity with divergent reciprocal sum and the disjoint two-site blocks $\{i, i{+}1\}$ representing c. Thus $C_* \subset L$. In particular for *every translate* of the signed constant-spacer sequence with divergent reciprocal cuts, meeting `cZ` forces pair recurrence. In the signed Erdős–Taylor case ($c = 1$) this holds for all translates; a zero difference should be omitted or handled separately according to the recurrence convention. A completed finite head can always be chosen away from the tail blocks. If zero is artificially deleted, there are only finitely many representations of zero in the Erdős–Taylor sequence after its first few terms, by dominance of each sufficiently large term over the absolute sum of its predecessors. Frozen tails then differ by only the zero sign word, for which an ordinary nonempty IP subfamily suffices when $e' = 0$.

Variable spacer averages. For a signed unit box, suppose disjoint blocks I_j represent integers $c_j = \sum_{i \in I_j} n_i a_i$, with $w_j = \max |n_i| \ge 1$ and $W_N = \sum_{j \le N} 1/w_j \to \infty$. Suppose the means

$$L(\theta) = \lim_N W_N^{-1} \sum_{j \le N} \exp(2\pi i c_j \theta)/w_j$$

exist for all θ and $\{\theta : L(\theta) = 1\}$ is countable. If this translated box has a positive pair witness, it misses a Bohr neighborhood of zero. For instance one may take $a_{i+1} = i a_i + P(i)$ with a fixed nonconstant integer polynomial P, using either parity of disjoint neighbor pairs. Weyl equidistribution and partial summation give zero limit at irrational θ, while periodicity gives the rational limits. This corollary does not assume boundedness of the spacers.

Fix a sign path ω. At every site choose the binary expansion $0, \sigma_i a_i$, with $\sigma_i = \omega_i$

where nonzero and arbitrary signs at the remaining sites. Enumerate the disjoint blocks in their given order, interleaving singleton blocks exhausting any unused indices. Define the increasing expansions $B_j \to B$ exactly as above; sign reversals only change the coefficient signs in the representations of c_j. On the relation blocks, with mass increases δ_j summable, one has

$$\mu(B \triangle T^{c_j} B) \le 2\mu(B \setminus B_j) + 4w_j\delta_j.$$

The normalized weighted discrepancy therefore tends to zero. In its spectral representation this discrepancy is the integral of $2(1 - \operatorname{Re}\exp(2\pi i c_j\theta))$ against the spectral measure of 1_B. Dominated convergence and nonnegativity imply that the measure is supported on $\{L = 1\}$ (the limit lies in the closed unit disk). Thus it is pure point.

All one-sided finite tail sums v of the oriented steps carry the corresponding finite head approximation inside B, so their B discrepancies tend uniformly to zero. The same is true for arbitrary signed tails by writing a signed tail as the difference of two oriented positive tails and telescoping. Consequently for every individual spectral atom θ all signed tail phase sums tend uniformly to zero. In elementary real terms this implies $\sum_i ||a_i\theta|| < \infty$: otherwise, on an arbitrarily late tail of individually small angles, choose signs to make all centered terms positive and take their partial sum first crossing (say) $1/4$. The resulting circle distance would be bounded below. The tail condition is unchanged by interleaving or reordering the sites because the sums have finite support and exhaust all sites.

For every finite truncation v of the path's signed sum, $T^v A \subset B$ by orientation and $T^e B \cap B$ is null. Since $A \subset B$, this yields $\mu(T^{e+v}B \triangle B) \ge \mu(A)$ for all such truncations. Finitely many spectral atoms already detect this discrepancy. Their absolutely summable coordinate phases vary continuously on the entire compact sign-path space, so at least one detects the original infinite path by a positive angular margin. The resulting open path neighborhoods have a finite subcover; their frequencies and the minimum of their positive margins separate the entire original box.

Why a single useful selector suffices in an arbitrary symmetric box. Suppose the variable-spacer hypothesis holds with one chosen $a_i \in S_i$ on its relation blocks. The Bohr conclusion still holds for the whole translated multivalued box. Pass to the factor Y generated by translates of the full enlarged event $B_{\text{full}} = \bigcup_{u \in \Sigma S_i} T^u A$ in the path lemma [R256]. The event $A' = \{E(1_A|Y) > 0\}$ has positive measure, is contained in B_{full}, and is disjoint from $T^e B_{\text{full}}$. Also $T^u A' \subset B_{\text{full}}$ for each full signed finite sum u, by conditional expectation of the corresponding inclusion for A. Therefore A' is a pair witness for the entire original box. The action on this factor is strongly rigid uniformly over the full signed tail sums.

Fix any path ω, let R_ω be the strong limit of its translated heads, and choose N such that every signed tail sum after N moves A' in symmetric-difference measure by less than $\mu(A')/3$. Choose binary expansions in the first N sites matching ω, discard any of the finitely many relation blocks meeting them, and make the remaining choices on the relation blocks from the useful selector. Expand on these sites (zero choices elsewhere suffice). The weighted-spacer proof gives a Kronecker-measurable event B' with $A' \subset B'$, $T^{\sum_{i \le N} \omega_i} A' \subset B'$, and $B' \cap T^e B'$ null. If R_ω fixed the Kronecker factor, it would send A' inside B', whereas the tail estimate sends all but mass at most $\mu(A')/3$ inside $T^e B'$. Contradiction. Thus some eigenfrequency in Y detects the path. Uniform strong rigidity on *all* signed tails forces every eigenfrequency of

Y into the common absolute summability group $G_1(S)$ defined below, by the signed partial-sum argument on a choice maximizing each individual phase. Compactness again gives a finite list separating the original box.

Absolute lacunarity gives a global gap. Suppose $a_i > 0$ are increasing and $\sum_i a_i/a_{i+1} < \infty$. Every translate as above omitting zero misses one finite-dimensional Bohr neighborhood of zero. In particular its all-length capacity is one. No initial pair witness is needed.

For each sign path $\omega \in \{-1, 0, 1\}^N$ we construct a θ with $\sum_i ||a_i\theta|| < \infty$ and $e\theta + \sum_i \omega_i a_i \theta \neq 0$ on the circle. For large n start with $\theta^{(n)} = 1/(2a_n)$, and for successive $i > n$ round the current value to the nearest multiple of $1/a_i$. The limit θ_n exists, is positive, and differs from $1/(2a_n)$ by at most $\sum_{i>n} 1/(2a_i) = O(1/a_{n+1})$; here the successive ratios eventually exceed two. For $i > n$ it satisfies $||a_i\theta_n|| \leq a_i \sum_{j>i} 1/(2a_j) = O(a_i/a_{i+1})$. In particular it lies in the indicated summability group. If infinitely many path coordinates are nonzero choose such a late n; its own term is within $o(1)$ of $1/2$. All earlier terms and the translation have combined magnitude at most $(|e| + \sum_{i<n} a_i)|\theta_n| = o(1)$, and the total later phase norm is $o(1)$ by the summability assumption. Hence the path value is separated from zero. For a finite path its value is $d\theta_n$, with $d = e + \Sigma\omega_i a_i \neq 0$; arbitrarily late positive $\theta_n \to 0$ ensure that this too is nonzero modulo one.

For each constructed frequency the corresponding circle path function is continuous by uniform tail summability. Thus it separates an open neighborhood of its path by a fixed positive margin. Finitely many such neighborhoods cover the compact path space. Taking the smallest of their margins proves the Bohr assertion on all finite sums. Its capacity is one by the elementary character-root sieve. Consequently for an eventually increasing positive sequence $a_{i+1} = q_i a_i + e$ with $q_i \geq 1$, divergence of $\Sigma 1/q_i$ forces pair recurrence, and convergence of that series gives capacity one whenever the translate omits zero. This entire class cannot supply an intermediate threshold with a positive-measure pair obstruction.

A complete characterization of Bohr recurrence for these boxes. For finite symmetric sets with zero let

$$G_1(S) = \{\theta \in T : \sum_i \max_{s \in S_i} ||s\theta|| < \infty\}.$$

Then 0 lies in the Bohr closure of D exactly when there is a path $\omega \in \prod S_i$ with $e\theta + \sum_i \omega_i \theta = 0 \bmod 1$ for every $\theta \in G_1(S)$. The necessity is compactness of path space and continuity of the path functions indexed by $G_1(S)$: if no path satisfied the equations, finitely many open angular separations would cover all paths.

For sufficiency fix any finite frequency vector $\alpha \in T^r$. Let K_m be the torus closure of all signed finite sums on indices $\geq m$ mapped by multiplication by α, and $K = \bigcap K_m$. This is a compact subgroup: to approximate a sum of two elements inside any prescribed tail, approximate the first by a finite subsum and then approximate the second on the tail past its final index; symmetry gives inverses. A character $k \in Z^r$ annihilates K exactly when its phase distances tend uniformly to zero over all signed tail sums, by nested compactness. In turn this is exactly $k \cdot \alpha \in G_1(S)$: use the positive centered representatives of maximizing summands, changing signs by symmetry, to prove necessity. Thus each cluster point of the images of the translated heads of ω is annihilated by every character annihilating K. By torus duality it lies in K. Given such a cluster point, approximate its negative by a finite

sum beyond the chosen sufficiently close head; the two disjoint sums combine into an element of D near zero for the frequencies α. This proves the claimed equivalence.

A spectral obstruction test. For a general symmetric sum box possessing the pair witness of the path lemma [R256], denote by B its enlarged event there. For every sequence $s_i \in S_i$, the spectral measure σ_B of 1_B is supported on

$$G_2(s) = \{\theta : \sum_i ||s_i\theta||^2 < \infty\}.$$

Indeed uniformly over signed tail sums the L^2 discrepancy of B from its translate tends to zero. In the spectral representation average over independent increments each chosen as $0, \pm s_i$ with probabilities $1/2, 1/4, 1/4$. The mean multiplier on a finite tail interval is $\prod \cos^2(\pi s_i\theta)$. Off G_2 these products tend to zero on every tail by the elementary bound on $1 - \cos^2(\pi x)$. Passing to the infinite tail and then sending its starting index to infinity shows the complement has measure zero. This is the elementary support mechanism of Aaronson–Hosseini–Lemańczyk.

If some countable family of such selectors has countable intersection of its G_2 sets, then σ_B is pure point. Since the path lemma supplies $\mu(T^d B \triangle B) \geq \mu(A) > 0$ over all $d \in D$, finitely many atoms of σ_B already detect every d (choose a finite set whose complementary spectral mass is less than $\mu(A)/8$). This finite list of frequencies separates D from zero, and again its capacity is one. More generally the same finite-cover argument applies if every strong path limit is nonidentity on the spectral portion supported by $\{\theta : \sum_i \max_{s \in S_i} ||s\theta|| < \infty\}$, since each such frequency defines a continuous function of the path and agrees there with the strong path limit for almost every such θ. All of these are sufficient conditions rather than a global dichotomy for arbitrary signed sum boxes.

R256. Local thresholds for a translated symmetric sum of finite sets

Scope and status. Result with proof.

Result. A positive pair witness for a translated symmetric sum box makes every ultrafilter on it contain members with capacities arbitrarily close to one.

Scope. A local-capacity theorem using exact path tracking and roots.

Qualifications. The assertion does not say the capacity of the whole box is one; a global upper bound below one would still be needed for interpolation.

Let e be an integer and S_i finite symmetric subsets of Z containing zero. Put $D = e + \sum_i S_i$, using sums with finite support; signed steps may be replaced by their absolute values. Suppose a probability-preserving integer shift has an event A of positive mass such that A and $T^d A$ are disjoint in measure for all $d \in D$. Then for every ultrafilter P on D, its member capacities approach one:

$$\sup_{E \in P} b(E) = 1.$$

In particular D has the finite partitions in the interpolation criterion. If such a D also has $b(D) < 1$, the entire prescribed spectrum follows by the interval and product results in interpolation [R145]. A positive-measure higher-order avoiding event alone is not assumed to imply this assertion.

A compact family of nonidentity transformations. Choose an ergodic component with positive mass for A. Write $P_m = \sum_{i \le m} S_i$, $B_m = \bigcup_{u \in P_m} T^u A$ and $B = \bigcup_m B_m$. Let $E_m = B \setminus B_m$, whose measure tends to zero. For every symmetric finite tail $\sum s$ on indices $> m$, $T^{\pm s} B_m \subset B$. Consequently, at any two positions x and $T^s x$ outside E_m the membership bits of B coincide. These are simultaneous exact statements after discarding a common null set. Also $\mu(T^s B \triangle \mathrm{B}) \le 2\mu(E_m)$. For every head $u \in P_m$, $A \subset T^{-u} B$ while $A \cap T^e B$ is null, so $\mu(T^{e+u} B \triangle \mathrm{B}) \ge \mu(A)$.

Pass to the standard ergodic factor generated by all integer translates of B. For every choice $\omega = (s_i^*) \in \prod_i S_i$, the powers T^{l_m} with $l_m = e + \sum_{i \le m} s_i^*$ form a Cauchy sequence in the strong topology on its measure algebra. Indeed changes on B and on its finitely many translates have discrepancy at most $2\mu(E_m)$, and finite unions/intersections give a dense algebra. The inverses are Cauchy by commutativity and invariance. The strong limit R is an invertible measure-algebra automorphism, represented on this standard factor by a transformation. It commutes with T and is not the identity by the preceding uniform separation on B. Its fixed sets for each nonzero power are invariant under the ergodic T. Thus R is aperiodic or has a common finite order $q \ge 2$ with almost every orbit of size q.

Lift P to an ultrafilter on the finite-support choices representing D (choose any representative for each d). Its first m coordinates are fixed on a P-member, consistently defining a path ω. On this member, $d = l_m + s$ for a tail $\sum s$; $d \mapsto |d|$ causes no issue since an ultrafilter chooses one sign and forbidding positive and negative runs is equivalent. Only one head at each m needs to be considered.

Exact tracking on a retained run. For a set F depending on the finitely many bits $1_B(T^l x)$, $l \in L$, remove from its ambient space the union of translates $T^{-l-jl_m} E_m$ for $l \in L$,

$0 \le j \le H$. This costs at most $|L|(H+1)\mu(E_m)$. If the sites x,$T^d x$,...,$T^{(H-1)d}x$ all avoid those defects, the F-bit at $T^{td}x$ equals the one at $T^{tl_m}x$ for $0 \le t < H$. Indeed the two entries at positions nd+$(j+1)l_m$ and $(n+1)d + jl_m$ differ by s, and their L-translates avoid E_m. Compare along diagonals with $n + j \le H$. This holds for all the tail sums in the member at once. The number of tested translates is independent of m despite the size of its head.

Aperiodic R. Given $\epsilon > 0$, by a Rokhlin tower choose C of mass $> 1-\epsilon$ and H such that C has no H-run under R. For example on a tower with large height and small uncovered mass, omit the base and the complement of the tower. Approximate C in measure by a finite-bit event F with error arbitrarily small compared to ϵ/H. For large m, the mass of $Z_m = \bigcap_{t<H} T^{-tl_m} F$ is less than ϵ by strong convergence and telescoping. Remove Z_m and the tracking defects from F. Any H-run for d in the selected member would start in Z_m by exact tracking, a contradiction. Its retained mass exceeds $1 - 4\epsilon$ for large m.

Periodic R. Choose C of measure $1/q$ containing one point from each R-orbit. Under R its word has period q and no adjacent ones. Fix r, put $c_i = q^i$ for $0 \le i < r$, and take independent c_i-root towers of the original T. Put $H = 1 + \sum_i c_i$. Approximate C by one finite-bit F so closely that the following total errors are arbitrarily small. For large m, remove from each original base the preceding tracking defects as well as the points at which, for some $0 \le j \le H$, the F-bit at T^{jl_m} differs from the C-bit at R^j. The latter cost is at most $(H+1)\mu(\mathrm{F}\triangle\mathrm{C})+o_m(1)$. Pull these defects to every root level. Mark a product site if no row has a defect and at least one row has its F-bit equal to one. The mass can be made arbitrarily close to or above $1 - (1-1/q)^r$, using independence.

In an entirely marked length-H d-run, each row on each residue class of the time index modulo c_i advances its original base by T^d per sample. From the first slot of that residue the tracked F-word agrees with an R-word throughout the available stretch. Starting with $t = 0$, for row i keep t or add c_i to put that row at an unmarked F-bit, since two consecutive R-bits cannot both be one. Later changes are divisible by qc_i, so all earlier zero bits are preserved. There is room for every change, and the final site has no F-mark on any row. This contradicts its retention. Thus a P-member has capacity arbitrarily near one, first choosing r large and then the approximations and m. The factor argument uses a single path depending on P; it does not establish $b(D) = 1$ for the full union of paths.

R258. A near-full trap for two oppositely moving circle streams

Scope and status. Result with proof.

Result. Two oppositely moving streams admit near-full finite-window traps; polynomial even modes realize analogous traps uniformly on an abstract shift-fixed order-two coset.

Scope. An abstract profile model with freely chosen forcing; the profile dimension and even subgroup vary with the target density.

Qualifications. No realization of the requisite coset in the paired-prime arithmetic family is proved. The finite binary examples are separate computer checks, not dependencies of the analytic theorem.

This is an obstruction in the surrogate finite-window model, where the forcing is arbitrary. It resolves the question about an absolute high-density bound for that model. It gives no integer critical threshold, and does not assert that the required forcing word belongs to an arithmetically admissible odd coset.

Theorem. For every $\epsilon > 0$ there are an integer L, a compact $B \subset T^{[-L,L]}$ of Haar measure greater than $1 - \epsilon$, and a finite forcing word $w(t)$ such that there are no circle streams P, Q and spatial common offset $c \in T^{[-L,L]}$ for which all rows

$$(P_{t+j} + Q_{t-j} + c_j + w_j(t))_{-L \le j \le L}$$

belong to B. With no common offset, one may take the forcing identically zero. The proof uses both signs through their symmetric sum.

Write $e(x) = \exp(2\pi i x)$. Choose a closed circle set A missing a nonempty open arc and of Haar measure greater than $1 - \epsilon/4$. On the compact space of Borel probability measures carried by A, Haar measure does not occur. Because Fourier coefficients determine measures, there are $K \ge 1$ and $\kappa > 0$ such that

$$\max_{1 \le k \le K} \left| \int e(kx)\, d\nu(x) \right| \ge \kappa$$

for every such probability measure. Indeed otherwise a weak limit as the list grows would have all its nontrivial coefficients zero. Put $\tau = \kappa^2/4$.

For a row $z = (z_j)$, require $z_0 \in A$ and all of the following for $m_0 \le s \le L$, both parities $e_0 \in \{0, 1\}$, and $1 \le k \le K$:

$$\left| \frac{1}{\{} N_{s,e_0}\} \sum_{\substack{1 \le j \le s \\ j \equiv e_0 \pmod 2}} e\big(k(z_j + z_{-j})\big) \right| \le \tau, \qquad N_{s,e_0} = \#\{1 \le j \le s : j \equiv e_0 \pmod 2\}.$$

Take $m_0 \ge 2$. Under Haar row measure the $z_j + z_{-j}$ for $j \ge 1$ are independent uniform circles. For each parity and character the average therefore converges to zero almost surely. The probability that some average after the cutoff violates the threshold tends to zero as $m_0 \to \infty$, simultaneously for the finitely many characters and parities, even when supremizing over **all** $s \ge m_0$. Choose m_0 making this loss less than $\epsilon/4$ independently of L. The resulting set B_0 is closed and has measure exceeding $1 - \epsilon/2$ for every $L \ge m_0$.

Suppose all unforced rows at centers $t = 0, \dots, L-1$ were in B_0. Let $X_u = P_u + Q_u$, and let I be the even integers of $[0, L-1]$ of cardinality N. The empirical law $\nu = N^{-1} \sum_{u \in I} \delta_{X_u}$ is supported on A by the slot-zero condition. For each tested positive character, its coefficient squared is

$$\left(\frac{1}{N} \sum_{u \in I} e(kX_u) \right)^2 = \frac{1}{N^2} \sum_{u,v \in I} e(k(X_u + X_v)).$$

At midpoint $t = (u+v)/2$, nondiagonal pairs have half-difference $j = |u - v|/2 \geq 1$, parity $j \equiv t \bmod 2$, and upper bound $s_t = \min(t, L - 1 - t)$. Conversely each such positive j gives two ordered pairs in I^2. The raw row at that center satisfies exactly

$$z_j(t) + z_{-j}(t) = X_{t+j} + X_{t-j}.$$

Whenever $s_t \geq m_0$ the contributing pair sum after grouping by t is consequently bounded in absolute value by τ times its number of ordered pairs. The diagonal contributes at most N. The boundary centers $s_t < m_0$ contribute at most $4m_0^2$ terms. It follows that every tested coefficient has squared modulus at most

$$\tau + 1/N + 4m_0^2/N^2.$$

Choose L sufficiently large to make this strictly below κ^2. This contradicts the finite Fourier test. In particular B_0 obstructs unforced motion, regardless of the actual two streams.

For the common offset, compactness allows a uniform margin. In the finite torus of controls for the length-L test, the closed set of raw row arrays and B_0^L are disjoint, hence have strictly positive distance. Choose $\eta > 0$ so that perturbing every tested raw row by sup-distance less than η cannot put them all in B_0. Concatenate blocks of length L with constant forcing minus each point of a finite $\eta/2$-net of the entire spatial torus. For every c the appropriate block cancels it to within $\eta/2$. The unforced contradiction applies on that block with arbitrary input starting indices. Thus $B = B_0$ works with mass greater than $1 - \epsilon$.

The reasoning works for every infinite compact metrizable abelian alphabet permitting closed near-full sets whose complement contains a nonempty open set. Replace the finite Fourier test by finitely many nontrivial characters. For a particular finite alphabet of size q, taking A to omit one symbol gives traps as close to $1 - 1/q$ as desired by increasing the spatial and temporal windows. Alternately for characteristic two the symmetric sum is an endpoint difference, and one can count equal pairs using Cauchy–Schwarz instead of Fourier tests. The argument applies to masks in the full moving-stream model through the row map $z \mapsto (z_0, (z_j + z_{-j})_{j \geq 1})$; its one-slot Haar law is product Haar, although its temporal rows are highly dependent.

A shorter finite-alphabet illustration with binary *full* rows is reproducible in two_stream_counterexamples.js. On raw, unquotiented input registers the masks of masses 33/64 and 67/128 die after respectively 15 and 29 unforced slots. They are separate computer-checked examples; the theorem above does not rely on them.

Shift-fixed odd cosets can carry the same obstruction The free-wave subgroup by itself has the periodicity property proved in two-stream windows [R259]. Adding polynomial even modes changes the conclusion: for each $\epsilon > 0$ a genuine shift-fixed order-two coset containing all freely moving waves and all common offsets has an avoiding event of mass

greater than $1-\epsilon$, uniformly for every profile in its odd coset. This is an abstract profile system, without an arithmetic realization in the paired primes.

Use the compact trap and robustness radius from above. For its finite set of spatial indices choose distinct integers $d_j \geq L$ with successive differences at least L. For an integer M chosen below set $\kappa = 1/(2M)$ and prescribe $w_j(t) = \kappa\binom{t}{d_j}$ in the circle. This whole vector sequence has some finite common period since its rational denominators and degrees are fixed. Write U for the compact two-wave image, C for all constant spatial profiles, and let L_0 be their sum together with the finite shift group generated by $(T-I)w, 2w$ and their shifts. The coset of w is shift-fixed modulo L_0 and has order dividing two there. Every member consists of a free wave, arbitrary constants, and $P = a(T)w$ for an integer Laurent polynomial with odd $a(1)$. Indeed such multipliers constitute exactly the coset of the ideal $(T-I, 2)$, and the polynomial-mode module is finite. The highest binomial coefficient of every P_j is $\alpha = a(1)\kappa$, so $M\alpha = 1/2$.

The vector $(P_j(t+s) + c_j)_{j,0\leq s<L}$ enters the robustness box uniformly within a bounded time for every such odd multiplier and every common offset. Apply the torus polynomial criterion of Green–Tao, Theorem 2.9 with these fixed degrees, dimension, and a Lipschitz bump in the box; failure gives a nonzero character in a fixed finite list and bounded binomial $C\infty[H]$ norm. Choose its highest nonzero spatial degree d_j. The first nonzero moment $b_l = \sum_s \lambda_{j,s}\binom{s}{l}$ occurs at some $l < L$ by the binomial evaluation matrix. The surviving coefficient in binomial degree $d_j - l \geq 1$ is $b_l\alpha$, regardless of the lower coefficients in this polynomial or all lower spatial components. Choose M divisible by every possible nonzero $|b_l|$ from the finite list. A bounded norm for a sufficiently long interval would imply $||M\alpha|| \leq (M/|b_l|)O(H^{-(d_j-l)}) < 1/2$, contradicting $M\alpha = 1/2$. The coefficient bounds are fixed before choosing M.

Thus every odd profile has a length-L block in which its polynomial and constant contribution has uniformly small size. The statistical trap on the two free waves precludes a full run through that block, which proves the claim and in particular shows that the odd coset really is nontrivial. The ambient group, spatial window and even subgroup vary with the target mass, consistently with the bound below one in each fixed compact Lie profile model.

R259. Two oppositely moving Haar streams

Scope and status. Proved subcases and unresolved extensions.

Result. Two-stream windows satisfy half-density, full-section, factorial-cohort, compatible-coset and scalar-projection recurrence theorems; small binary windows are computationally classified.

Scope. Each theorem has its own structural or dimension hypothesis. Exact finite computations are separately reproducible.

Qualifications. The near-full traps in R258 rule out a uniform threshold below one across arbitrary spatial windows. The compatible-coset theorem here requires equality of the even subgroup with the sum of constants and free waves, not merely containment.

Let P, Q be independent iid Haar streams on a compact metrizable abelian group K, indexed by the integers. At spatial site j and time t put

$$F_j(t) = P_{t+j} + Q_{t-j} + w_j(t),$$

where the array w is deterministic and arbitrary. A free spatial offset can be conditioned on in all the assertions below. All positivity assertions concern finitely many time slots.

Four equally spaced spatial sites Proposition. If J is contained in $\{a, a+M, a+2M, a+3M\}$ with $M > 0$, every measurable B of Haar measure $> 1/2$ on K^J has positive probability of being visited simultaneously for all $0 \le t < h$, for every h and every forcing.

Enlarge the event by unrestricted missing coordinates. The middle two processes $X = F_{a+M}$ and $Y = F_{a+2M}$ are independent iid Haar streams. Indeed each of these two spatial sites picks out a perfect matching between the two sets of input times, and the union has no finite cycle. An alternating two-edge walk in a component translates its input time by $2M$. On every finite forest the sums of independent Haar vertex labels on the edges are independent Haar, by successively integrating out leaves. Translation by deterministic forcing preserves this law.

Writing $U = F_a, V = F_{a+3M}$, the wave equation gives

$$U_t + Y_t = X_{t-M} + X_{t+M} + v_t, \qquad V_t + X_t = Y_{t-M} + Y_{t+M} + v'_t$$

for deterministic errors. The spatial change $(U, X, Y, V) \to ((X, Y), (U + Y, V + X))$ preserves product Haar. The transformed row is the common-stride model on K^2 from stride-control.md [R242]. For completeness, if the section density of a transformed event at z is $b(z)$ and its mean exceeds $1/2$, there is u with $P = \{z : b(z) > u\}$ of mass $> 1 - u$. Otherwise integrating its tail would give mean at most $1/2$. On each residue class modulo M, start with the first two consecutive controls in P. Inductively the next fresh control must lie both in P and in a translate of a section of density $> u$. This is possible with positive conditional probability; finitely many choices suffice.

Any consecutive window with full interior projection Proposition. Let $J = \{0, \dots, r-1\}$ with $r \ge 3$. If the conditional Haar measure of the allowed endpoint pair (F_0, F_{r-1}), given the interior row, is positive for almost all interior values, every finite forced run has positive probability.

At time t the old input register consists of $P_t, \dots, P_{t+r-2}$ and $Q_{t-r+1}, \dots, Q_{t-1}$. The retained register removes P_t, Q_{t-r+1}. The interior row depends only on the retained register, and its Haar law is full Haar because the P labels at that time are distinct. The fresh input registers are P_{t+r-1}, Q_t. Conditional on retained labels, each endpoint is one dropped plus one fresh label and a prescribed constant. Initially choose the entire old register with product Haar. At each step draw an endpoint pair according to Haar restricted to its nonnull admissible section, then determine the fresh registers using the two dropped labels. Those dropped labels are independent Haar given the retained registers, hence the fresh pair is independent Haar of the retained registers, regardless of how the admissible section varies. The next complete register therefore still has product Haar distribution. This procedure generates a finite input law absolutely continuous relative to product Haar: conditional on the already drawn full input the fresh pair is a translate of normalized Haar in a positive section. Its trajectories satisfy all constraints with probability one, proving the claimed positivity.

For arbitrary B, at a fixed retained tuple the set of reachable fresh pairs is an affine sum of the slice of reachable dropped pairs and the endpoint section of B at the determined interior. Null endpoint sections can progressively cut down the set of reachable retained tuples. The argument above does not bound this effect.

Exact small binary windows For $K = F_2$, a register at time t is two binary words of length $r-1$. From it append two bits to get the full P-word and the time-reversed Q-word, XOR them to obtain the current row, then shift both words one position. Simultaneously flipping both input streams on even times or on odd times fixes all output labels and induces a bisimulation of the register graph. For $r \geq 2$ the quotient therefore has 2^{2r-4} states. With A the set of allowed rows, a force w allows exactly edges whose label lies in `A XOR w`. Starting from the full set of register states, run breadth-first search on the power set, using every w. A forcing word rules out all trajectories exactly when the empty set is reachable.

For $r = 3$ the largest binary mask killed by some forcing has 2 of the 8 words. For $r = 4$ it has 6 of the 16 words. For example, integer mask 845 (row x allowed iff its bit at x is 1), and forcing $[0, 0, 0, 4, 0, 4, 0, 4, 6, 0, 4, 0, 0, 0, 2, 4, 2, 4, 0]$ kill all paths for $r = 4$. These calculations test arbitrary time-dependent translates of a common event. The near-full traps in R258 [R258] show that no threshold bounded away from one works uniformly over all spatial windows.

Shells in any spatial window There is a direct improvement and simplification of the four-site proposition. Pick two integers a, b, which may be taken as the extremes of the window, and reveal at stage u the independent pair P_{u+b}, Q_{u-a}. For a spatial offset j of the row centered at t, its completion stage is

$$u = t - d_j, \qquad d_j = \min(b - j, j - a).$$

At a given value of d_j at most two spatial sites occur. If there are two, one is completed by the fresh P and the other by the fresh Q; their other summands are earlier. A singleton at the center may depend on both fresh variables, but its sum is Haar too. Thus the entire cohort for any one completing row has jointly Haar conditional law given all earlier stages. Distinct cohorts completing at the same stage can be highly dependent.

Let m be the number of distance values. Then **any** measurable event on the spatial row of Haar mass at least $\beta > 1 - 1/m!$ has a simultaneous finite-window probability bounded below by c^{mh} for some $c = c(\beta, m) > 0$, uniformly in the forcing and in the spatial positions. To verify this order the spatial cohorts from largest distance to smallest, in their completion order for each fixed row. Sweep the event backwards through its coordinate sections: for proposed tolerances $p_1, \ldots, p_m$, keep at each prefix exactly those fibers having more than the selected tolerance fraction of next-coordinate choices leading to a previously retained prefix. The set of tolerance tuples in the unit cube for which the empty prefix survives has volume equal to the original event mass, by Fubini and the layer-cake identity, even for atomic compact alphabets. For small enough $c > 0$, the upper simplex $\sum_j (1 - p_j) < 1 - c$ has volume $(1 - c)^m/m! > 1 - \beta$, so some successful tuple belongs to it. At a reveal stage, at most one row requests each cohort. Conditional on all maintained prefixes, a union bound gives conditional success at least c. There are at most `mh` nontrivial reveal stages. A fixed translate of the event preserves its mass and its rearranged tolerance region, so an independent common Haar offset may freely be included.

For every arbitrary triple there are at most two cohorts on taking its extremes. The same holds for **any** symmetric four-point set $a, a + j, b - j, b$, not just four equal steps. Any spatial positions sharing an arbitrary chosen reflection center can be paired by choosing a, b with that center (enlarging the window as needed). This proof avoids any alphabet torsion qualifications.

Compatible odd forcing There is a dimension-free conclusion when the even profile subgroup really is all constants plus freely generated wave backgrounds. Let $X = (K^J)^Z$ and write T for shift. Suppose $L = C + U$ with C a subgroup of constant profiles and U a closed subgroup on which each of $T - I, T^2 - I$ is surjective. If $(T - I)w \in L, 2w \in L$, its coset $w + L$ has a 2-periodic representative. Indeed write $(T - I)w = c + u$ and solve $(T - I)h = u$ in U. For $v = w - h$, one has $(T - I)v = c$ and $2v \in L$. Hence $2c = (T - I)2v \in U$. Solve $(T^2 - I)h' = 2c$ in U, and $v - h'$ has period dividing two. This holds for arbitrary compact abelian groups. Every moving-average image of unrestricted Haar streams has both surjections by the bi-infinite scalar input recursion, also for the compact subgroup closure of sums of such images. If C includes all spatially constant-in-time translations of the full output group, any event of mass above half is hit along this coset in every finite window with positive Haar-background probability. First uniformly translate its two representative points, obtaining simultaneous membership with probability at least $2\mu(B) - 1$. The Haar-averaged membership probability is continuous as a function of the finite translated trajectory by translation continuity in measure. A sufficiently small neighborhood of the representative thus retains positivity, and a neighborhood in the compact Haar control group has positive measure. This result for compatible forcing uses the precise equality $L = C + U$; mere inclusion of freely generated waves is insufficient for this argument.

The exact binary calculations above are reproducible with two_stream_binary.js. It checks all 46 and 4336 mask-translation classes for three and four bits respectively.

A dimension-free class of arbitrary events Suppose the spatial alphabet is a torus or a prime field, and the forbidden subset of a spatial row is a countable union of restrictions on single scalar linear projections. Precisely on the torus these are inverse images of measurable circle sets under nonzero characters of the spatial torus. Combine proportional characters into one primitive character and one measurable bad target set. On a prime field combine

directions on the same projective line. If the allowed mass exceeds one half, every finite forced run has positive probability in arbitrarily large spatial windows.

Indeed the distinct remaining projections are independent in pairs for a Haar spatial row. On a torus their character rows are independent over the rationals, so the compact image is the whole two-torus. Write p_i for the bad masses. On a finite sublist with $L = \Sigma p_i > 0$ the bad count has second moment at most $L + L^2$. Thus the bad union has mass at least $L/(1 + L)$. If the total bad mass is $e < 1/2$, this gives $\Sigma p_i \le e/(1 - e) < 1$ on the entire countable list.

The scalar value for the row centered at t and the ith projection is a finite weighted sum of the two input streams. It is completed at a single stage $t + b_i$ in any fixed chronological input ordering simultaneously exposing one new label from each stream. At least one of those labels has a nonzero coefficient in the projection; it supplies a conditionally uniform scalar because every nonzero torus character is onto the circle and nonzero coefficients on a prime field are invertible. At a stage there is at most one request for each projection. Conditional on all earlier inputs, the probability of any failure is at most the sum of the p_i, for arbitrary prescribed translations. Only finitely many stages occur for a finite time window since all spatial supports lie in the given finite window, even if the character list is countable. This argument also works for a single moving stream and for any finite number of moving streams with these scalar completion properties.

Why the projective-plane example is not realized by scalar factors Pairwise independent rows alone need not give a universal threshold. On a finite field of order q, the scalars $A + tB$, for finite field slopes t, and B for the slope at infinity, are pairwise independent when A, B are independent uniform. No such trajectory has all $q + 1$ scalars nonzero. The one-slot allowed density is $1 - 1/q$.

An exact pencil with four distinct slopes cannot be a rowwise scalar factor of the two-wave model over a prime field. First, any equation $f_2(\pi_2) = af_0(\pi_0) + bf_1(\pi_1)$ with nonzero a, b forces the scalar functions f_0, f_1 to be affine linear. Here π_i denotes the map on the entire input group giving the finite row at the indicated time. Maps at different times are jointly surjective by the same two-matching forest argument: now vertices are the two raw streams and the matchings join them along the diagonals centered at two fixed times. Applying the equation to a shift of the inputs by an element of $\ker \pi_2$ and subtracting it gives one function of π_0 plus one of π_1 identically zero. Joint surjectivity makes both functions constant, and $\pi_j(\ker \pi_2)$ fills the whole row group for $j = 0, 1$. Thus every first difference of each function is constant. A second pencil equation forces the remaining row functions affine as well.

In the dual Laurent representation reverse the spatial index on the first raw input component. A scalar linear observation based on time t now represents a vector on the line of slope S^{2t} in the rational function plane over $F_q(S)$. The ratio of its components has this form because the original two components, for a spatial weight $f(S)$, are $S^t f(S)$ and $S^t f(S^{-1})$. Four distinct nonzero vectors lying in a two-dimensional pencil over the **constant field** would have their four line slopes with constant cross-ratio. For exponents $a < b < c < d$, the ratio

$$\frac{(S^a - S^c)(S^b - S^d)}{(S^a - S^d)(S^b - S^c)}$$

has constant term one in its Laurent expansion at zero (shift all exponents nonnegative if necessary), but its difference from one is a nonzero rational function, since its numerator

after subtraction is up to sign $(S^a - S^b)(S^c - S^d)$. The contradiction rules out this particular pairwise-independent counterexample. This does not establish a universal threshold for arbitrary predicates.

R263. Odd unsigned supports at higher orders

Scope and status. Proved subcases and unresolved extensions.

Result. Unsigned odd supports have equal even/odd saturation for invariant laws, full positive recurrence in independent finite-cocycle models, and a uniform adaptive-selection L2 bound; the signed family fails the vdC criterion.

Scope. All-action higher-order avoidance must live on a boundary null for every invariant slope law; local and block cocycle conclusions retain their independence hypotheses.

Qualifications. The null boundary can carry the diagonal joining. It is not a contradiction and does not settle unrestricted multiple recurrence.

Let $G = \oplus C_{p_i}$, where the p_i are odd, tend to infinity, and $\sum_i 1/p_i = \infty$. The allowed directions are the sums of the standard e_i over finite sets of odd cardinality. The pair transfer [R264] and the signed harmonic pair estimate [R169] do not on their own give multiple recurrence in arbitrary actions. Two higher-order consequences follow.

A slope joining obstruction must be supported on a universally null boundary In fact, for any pmp G-action and event B, the tail-even and tail-odd saturation events agree almost everywhere. These are intersections over m of unions of translates $T^{\sum_{i\in F} e_i} B$ on finite supports $F \subset [m, \infty)$ of the indicated parity; the even union includes the empty support. To see this use $S_i{=}T^{e_i}, w$ on the product with a uniform bit, w the flip, and let L be the saturation of $B \times \{0\}$ over supports of either parity. For any finite tail panel its subset-sum set C has $\mu(L + C) - \mu(L)$ tending uniformly to zero with the start m. If $\mu(L\, wL) > 0$, the small-growth subset and the exact tracked set of the pair proof [R264] produce a positive pair-avoider for all signed odd moves in that panel, with a uniform positive lower bound on its mass. The signed harmonic estimate contradicts this on sufficiently large panels. Thus $L = wL$, which gives the saturation assertion in the untagged space. This argument uses no advance pair assumption on B.

Take any pmp action T with A h-avoiding all the unsigned odd directions. Model the binary event by a clopen origin event on the orbit subshift if desired. In X^h let V_g apply T^{jg} at slot j, $0 \le j < h$. The set

$$Y_bad = (tail - evensaturationofA^h)\ (tail - oddsaturationofA^h)$$

is null for **every** V-invariant probability law, including every invariant slope self-joining, by the preceding result. Nevertheless the diagonal joining λ_0, the law of $(x, \dots, x)$, gives it measure at least $\mu(A)$: the empty even sum supplies the diagonal positive piece, and every odd translate of A^h is λ_0-null. Hence absolute continuity of the diagonal with respect to even one invariant joining on this saturation algebra would force a return. This condition is not automatic. Graph joinings at distinct offsets can be mutually singular, and for a free action the diagonal itself is slope-transient. In particular one cannot simply replace λ_0 by the product joining in the pair argument.

Unconditional all-h positive recurrence for arbitrary local finite-group cocycles Let K be any fixed finite abelian group. On the compact-product base $Y = \prod C_{p_i}$ with independent Haar coordinates, take the $HaarK$-extension in which e_i acts by

$$(x, b) \mapsto (x + e_i, b + \ell_i(x_i + 1) - \ell_i(x_i))$$

for arbitrary functions $\ell_i : C_{p_i} \to K$. The actions commute and have the prescribed periods. The family of unsigned odd supports gives all h-term returns at every positive measure in this system, with no regularity on the local functions. A common finite group can include any fixed finite tuple of such cocycles.

For $h \geq 2$ define the local profile

$$Z_i(x_i) = (\ell_i(x_i + j) - \ell_i(x_i))_{1 \leq j < h} \in K^{h-1}.$$

Its sum over $x_i \in C_{p_i}$ is zero. There are finitely many possible support sets $P \subset K^{h-1}$, so on one subsequence of divergent reciprocal sum every support equals one fixed P. Discard the finitely many p_i not coprime to $|K|$. A character constant on P must take value one there: the product of the character values over all x_i is one and its constant value has order coprime to p_i. For any character nonconstant on P, the modulus of its local mean is at most $1 - c(K, h)/p_i$. Indeed the character assumes two different values each on at least one atom of the p_i-point space, and one of its values has probability at least $1/\exp(K)$; in the identity $1 - |E\chi|^2 = E(1 - \mathrm{Re}(\chi(x)\chi(y)^{-1}))$ one distinct value contributes a fixed multiple of $1/p_i$.

On any far tail choose longer and longer finite subsets I of odd size in this subsequence exhausting unbounded harmonic mass. The independent profiles' sum tends in law to Haar on the subgroup generated by P, by the preceding Fourier test. In particular one can arrange

$$\Pr[\sum_{i \in I} Z_i(x_i) = 0] \geq 1/(2|K|^{h-1}),$$

where the bound is independent of the finite head. Approximate any event A in measure by A' depending only on that head and the fiber register, with error chosen in advance much smaller than $\mu(A)/h|K|^{h-1}$. At the odd direction $d = \sum_{i \in I} e_i$, the head does not move and, on the displayed independent tail event, the entire fiber register is unchanged at all h times. The A' correlation is thus at least $\mu(A')/2|K|^{h-1}$, and the integrated error costs at most h times the approximation discrepancy. This proves the claim.

For comparison, for any finite-fiber extension of the same compact base, with fibers of fixed size m and arbitrary cocycle, events of mass $> 1 - 1/m$ already have all returns: the subevent downstairs with the *entire* fiber in A has positive measure and the unsigned odd steps are Bohr recurrent on this compact base. The local-cocycle argument proves recurrence at every positive mass in its smaller class.

The same positive recurrence conclusion remains true for block potentials ℓ_B on arbitrarily dependent coordinates *inside* mutually independent finite blocks B, provided one axis $i(B)$ per block can be selected with $\sum_B 1/p_i(B) = \infty$. Write $Q = K^{h-1}$ and use the profile $Z_B = (\ell_B(x + je_i) - \ell_B(x))_j$. Choose a maximally frequent v_B in Q and restrict to one reciprocal-divergent subsequence on which $v_B = v$. Every character χ has probability $\geq 1/|Q|$ at $\chi(v)$. If $\chi(v) \neq 1$, the orbitwise product of $\chi(Z_B)$ must be 1 on each p_i-cycle, so at least one of the p_i entries differs from $\chi(v)$, since p_i is coprime to $|K|$. Thus $1 - |E\chi(Z_B)| \geq c_Q/p_i$. If $\chi(v) = 1$, the finite-root separation and the mass at one show $|1 - E\chi(Z_B)| \leq C_Q(1 - |E\chi(Z_B)|)$. Split

all characters into those with divergent contraction sum (whose product along sufficiently long tails tends to zero) and those with finite contraction sum. The latter all satisfy $\chi(v) = 1$, and their products on sufficiently far tails are uniformly close to one. By Fourier inversion, arbitrarily far odd finite panels therefore give zero total profile with probability at least $1/(2|Q|)$; the preceding approximation argument applies. This condition permits arbitrary block sizes but does not cover every partition into blocks, as the reciprocal mass of one representative from each block can converge.

The signed family fails the van der Corput criterion Assume $\Sigma 1/p_i^2 < \infty$, as holds for distinct primes. Set $r_i = \cos(\pi/p_i)$ and let ν_i^- be uniform on the two p_i-th roots nearest -1. Its first Fourier coefficient is $-r_i$. Take $a_i = (1-r_i)/(1+r_i)$ and $\nu_i^+ = (1-a_i)\delta_1 + a_i\nu_i^-$, with first coefficient r_i. On the dual product group the probability $\sigma = (\otimes\nu_i^- + \otimes\nu_i^+)/2$ has atom at the identity of mass at least $\frac{1}{2}\prod(1-a_i) > 0$. For every odd signed finite support, its Fourier coefficient vanishes because its two summands are opposite products of the same r_i over the active coordinates. Thus even the signed family fails the spectral van der Corput criterion. This coexists with the exact pair recurrence above when $\Sigma 1/p_i$ diverges. In particular a positive mass atom cannot force a correlation bound by Fourier vanishing alone.

Choosing a coordinate from rare local configurations Let $E_i \subset C_{p_i}$ satisfy $\Sigma a_i = \infty$, where a_i is its Haar probability. Under independent Haar coordinates, let I be the first i in a prescribed tail with $x_i \in E_i$ (finite almost surely). For any j the random map $x \mapsto x + je_I$ has pushforward density N_j relative to independent Haar; $||N_j||_2 \le \sqrt{3}$ uniformly in j, all starting indices, all seam sets and all periods. At y this density equals the sum of the indicators of $y_k \notin E_k$ for $k < i$ and $y_i - j \in E_i$. Its mean is 1. For indices $i < l$ the joint term additionally requires $y_i \notin E_i$. Its probability is at most the probability that i is first hit, times the probability that l is first hit after i; summing the latter over $l > i$ gives one. Consequently $EN_j^2 \le 1 + 2 = 3$. An h-configuration programmed by finitely many such adaptive moves can therefore control approximation errors without losing the low measure of each separate seam. This lemma assumes independent tests; arbitrary cocycles need not have this form.

R264. A signed-to-unsigned odd-support pair theorem

Scope and status. Result with proof.

Result. Unsigned odd supports on odd prime axes have pair-avoiding density $O((1+\sum 1/p)^{-1/3})$; divergent reciprocal mass gives pair recurrence in all actions.

Scope. Length two only, obtained through quantitative small expansion and the signed estimate.

Qualifications. Passing to a slope product does not preserve the required diagonal probability and invariance.

Let $G = \prod_{i\le n} C_{p_i}$ with odd moduli $p_i \ge 3$, and let V_+ consist of all sums of an odd number of distinct coordinate unit vectors. Write $S = \sum_i 1/p_i$. Every set having no pair of difference in V_+ has density $O((1+S)^{-1/3})$, with an absolute constant. Thus on the infinite squarefree group with divergent reciprocal-prime sum, the **unsigned** odd supports recur for every positive event at length two, on every tail. The proof uses the signed pair bound in odd-signed-primary.md [R169], whose stronger menu allows ± signs independently. The argument also works for any finite action by averaging its orbit masks. The theorem controls pairs only.

Small expansion. In a finite abelian group let C contain zero with $-C = C - v$ for some v. Suppose a subset L expands by relative-to-the-entire-group mass $\eta = m(L+C) - m(L)$, where m is uniform probability. Fix $K = 1+\beta$, $\beta > 0$. Minimize $m(Z+C) - Km(Z)$ over $Z \subset L$, including the empty subset. Comparison with L gives $m(L \setminus Z) \le \eta/\beta$ since $Z \subset Z + C$. Comparison with the empty set gives $m(Z+C) \le Km(Z)$. For any $Y \subset Z$, minimality gives

$$m(Z+C) - m(Y+C) \le Km(Z \setminus Y).$$

For any finite list of translations u, append them one by one. On adding u let Y be those z for which $z+u$ already belonged to the preceding union of translates of Z. At most $m(Z+C) - m(Y+C)$ new mass is added to the union of translates of $Z+C$, whereas the union of translates of Z gains exactly $m(Z \setminus Y)$. Summing, with the preceding union empty on the first step, gives $m(Z+C+C) \le Km(Z+C)$ on choosing the translations to be C. Consequently $Z \subset Z + C - C$ and $m(Z+C-C) \le K^2 m(Z)$. This is the marginal-growth argument in Petridis's proof of Plünnecke's inequality, here used after minimizing a penalized difference to retain nearly all of L. See Petridis.

Odd supports. Work in $G \times C_2$, write $w = (0,1)$, $s_i = (e_i, 1)$, and start with $B = A \times \{0\}$ of mass $a/2$, where $a = m_G(A)$. For a finite panel J let $C(J)$ be all subset sums of its s_i, including zero. Such a set is its own negative translated by its full sum. Select m disjoint consecutive panels each having reciprocal sum at least H. The sets

$$L_j = B + C(J_1) + \cdots + C(J_j)$$

are increasing and disjoint from $w + B$: an intersection would give a positive unsigned step of odd cardinality joining two points of A. Thus $B \subset L_j \setminus (w + L_j)$ for every j, with $L_0 = B$. At a stage of growth at most $1/m$, apply the small-expansion lemma to $L = L_{j-1}$,

$C = C(J_j)$ with $\beta = 1/\sqrt{m}$. Put $E = Z \setminus (w + Z + C - C)$. Since $Z \subset L$ and $Z \subset Z + C - C$, comparison with B gives

$$m(E) \geq a/2 - 1/\sqrt{m} - (2/\sqrt{m} + 1/m)m(Z) \geq a/2 - 3/\sqrt{m} - 1/m.$$

It has no pair with difference in $w + C - C$. Each signed odd-support direction on the chosen panel, with flip coordinate zero, is in $w + C - C$: write its plus and minus sets as disjoint subsets. Hence each slice of E obtained by fixing the other coordinates and the flip coordinate is independent for all signed odd supports on that panel. The signed box bound gives $m(E) \leq 1/H$. For large S take $H = S^{1/3}$ and greedily fill panels. Each completed panel has sum $< H + 1/3$, so there are m$\gg S^{2/3}$. This proves the asserted bound. The small-S cases are harmless.

In the infinite group an $h = 2$ counterexample restricted to any finite panel gives orbit masks avoiding the indicated finite steps. The density bound therefore applies without any ergodicity assumption. Together with the summable-character argument for $\Sigma 1/p_i < \infty$, this decides precisely when this unsigned family recurs at length two. It does not settle its all-length critical density. Passing to a product of copies with slopes 0,. . .,$h - 1$ does not preserve both the needed diagonal measure and invariance under the slope action.

R265. Local capacities on arbitrary unsigned odd finite sums

Scope and status. Result with proof.

Result. Any unsigned odd finite-sum family with a positive pair witness has arbitrarily high-capacity members in every ultrafilter, hence the interpolation property.

Scope. No lacunarity, divisibility or uniqueness of representations is assumed.

Qualifications. A capacity upper bound below one remains necessary to obtain intermediate or prescribed integer thresholds.

Suppose $D = \{\sum_{i\in F} n_i : Ffinite, |F|odd\}$ for any positive integers n_i, and D has a positive-density pair witness. Every ultrafilter containing D then contains sets of all-length critical value arbitrarily close to one. In particular if such a D has critical value strictly less than one, the partition interpolation theorem [R145] gives every prescribed integer threshold. This applies without divisibility, lacunarity or uniqueness of the representations of the sums. It does not establish $b(D) < 1$.

Choose an ergodic pair-avoiding system (X, μ, T) with A of positive mass. On $\Omega = X \times C_2$ write T for the transformation fixing the second coordinate, w for the sheet flip, $S_i = wT^{n_i}$ and $B = A \times \{0\}$. On a common conull set all odd finite sums avoid pairs. Let

$$L_m = \bigcup_{F\subset[m,\infty)finite} S_F B, L = \bigcap_m L_m,$$

where S_F is the product over F, including the empty one. L contains B and L_1 misses wB. For every $F \subset [m, \infty)$ finite, $S_F L \subset L_m$, by representing membership in L on a tail beyond F. Consequently the symmetric-difference distance of $S_F L$ to L is $\leq 2\eta_m$ where $\eta_m = \mu(L_m \setminus L)$ tends to zero. This holds uniformly in the tail set F.

Lift any ultrafilter containing D to the finite odd index sets by choosing one representation for each integer. Its choices on every finite prefix are fixed, compatibly, and produce a path. For its prefix F_m on indices $< m$ write $R_m = wS_{F_m}$. On the factor generated by all T- and w-translates of L these maps and their inverses converge strongly in the measure algebra: differences between any two prefixes are tail sums, first uniformly small on the named generators and then by approximation on the whole factor. Denote the resulting invertible limit by R. It is nonidentity, since $S_{F_m} L \subset L_1$ (use a representation with later support) misses wB whereas wL contains wB. Indeed $\mu(R_m L \triangle \mathrm{L}) \geq \mu(B)$ at every m. For each m take any odd word matching its frozen prefix; the difference between its actual power T^d and R_m acts by a tail sum, so these powers also converge strongly to R. Hence R fixes the invariant σ algebra of T pointwise. This factor has at most two T-ergodic components (X was ergodic and there are two sheets). On at least one of these components R is nonidentity; there it is either aperiodic or of finite period $q \geq 2$ with almost every orbit of exact size q, because its periodicity events are T-invariant.

Uniform two-sided tracking for finite packets. Given a finite list of odd words with the chosen prefix, let C be all subset products of their combined tail panel. In additive notation $C - C$ is a translate of $C + C$. We have $\mu(L + C) \leq \mu(L) + \eta_m$. The penalized-minimization proof in the signed-to-unsigned theorem [R264] applies to translations on Ω, with a ν-minimizer over measurable $Z \subset L$. In its marginal-growth telescoping allow an

error $|C|\nu$, and an error ν/β in the estimate for $\mu(L \setminus Z)$, where $K = 1+\beta$. Take ν as small as needed after C is fixed and use $\beta = \sqrt{\eta}_m + 1/m$. We obtain a defect set

$$(L \setminus Z) \cup [(Z + C - C)\mathbb{Z}]$$

of measure $o(1)$ as $m \to \infty$, uniformly over the packet. If two points outside it differ by any element of $C - C$, their L bits agree: a point of Z moves into $Z + C - C$ and at a safe endpoint is therefore in Z. Use the reverse move for the converse. The approximation Z may depend on the finite packet and need not belong to the factor. Its measure estimates on either original sheet, and on the pullback of the selected positive-mass factor component, lose at most a fixed constant. All tracking uses the actual commuting maps on Ω and their fixed translates, before restriction to the component.

Apply the local path argument [R256] with R_m in place of its time l_m. Explicitly, for any factor mask F depending on finitely many translates of L and any fixed H, delete from the initial mask all the corresponding tests of the defect set at R_m^j, $0 \le j \le H$. The cost is $o(1)$ uniformly over finite packets. Along a retained d-run, compare positions $(T^d)^t R_m^{j+1} x$ and $(T^d)^{t+1} R_m^j x$. They differ by the tail element associated with d, so F agrees at both whenever the named translated bits are safe. Telescoping along the diagonals gives the R_m-name for the required finite length. The same comparisons work from each relevant time residue in an integer-root tower. If using factor masks that are supported on the selected component, R_m fixes that component for all sufficiently large m by strong convergence on its indicator and the at-most-two-component permutation action.

If R is aperiodic on that component, a Rokhlin tower gives a mask of mass $> 1 - \epsilon$ with no H-run under R at some fixed H (delete the bottom level and the outside of a sufficiently tall tower). Approximate it by a finite-bit mask F, and choose m so large that the entire R_m-origin H-run cylinder of F has mass $< \epsilon$. Delete that cylinder and the tracking defects; an actual retained H-run would begin in the deleted cylinder. In the finite-period case, take a measurable single representative from each R-orbit. Its indicator has density $1/q$, period q and no adjacent ones. Approximate it by F. In r independent integer towers of heights 1,q,...,q^{r-1}, delete the tracking defects and the base points where the first HR_m-names of F differ from the corresponding R-names, with $H = 1 + \sum_{j<r} q^j$. Conditional on the base component each of these errors can be made arbitrarily small. Mark sites where at least one copy belongs to F and none has a defect. Along an all-marked progression adjust the time successively by zero or q^j to miss row j; future adjustments are multiples of its trace period on the selected residue. This produces an unmarked member, a contradiction. The mass tends to 1 as first r is chosen large and the subsequent approximation errors are made small.

Thus for each ϵ the frozen-prefix member has a common fixed-order avoider of mass $> 1-\epsilon$ on every finite packet, with the order and prefix independent of the packet. Compactness of stationary binary laws gives the same cap for the whole member. The principal-ultrafilter case is also immediate from the one-step periodic sieve. This proves the assertion.

R266. Odd weighted boxes of macroscopic width force all lengths

Scope and status. Result with proof.

Result. Odd weighted boxes of macroscopic relative width force all progression lengths at every positive density, uniformly on lifted free integer axes.

Scope. A positive lower bound on width/prime along infinitely many axes; primes may vary.

Qualifications. Unit-width directions with divergent reciprocal-prime sum are not covered.

Let infinitely many odd primes p_i tend to infinity. On axis i permit speeds whose representative modulo p_i is in $[0, L_i]$, with binary tag equal to the parity of that representative. Permit zero on unused axes, and require an odd sum of tags across the finite support. If L_i/p_i is bounded below by some $c > 0$ on infinitely many axes, then this family gives h-term returns at every positive measure for every h. This holds already on the free integer axes with the allowed residue classes fully lifted, and hence on their native cyclic quotient. The statement permits different primes at every site. It does not cover the unit-width divergent reciprocal-prime case.

Fix $h \geq 3$, put $l = \mathrm{lcm}(1, \ldots, h-1)$ and choose a fixed odd q with $1/q < c/2$. Only sufficiently large primes are used. Let Γ consist of the tuples $t \in (Z/q)^h$ whose coordinate sum is zero. Choose a vector $b \in (Z/q)^h$ of sum one, write $[t]$ for the usual representative, and put $\phi(t) = \lfloor p[t]/(ql) \rfloor$. Define two lists of integer AP queries parametrized by t:

$$d_+(t) = l(b_0 + \sum_i \phi(t_i)), x_+(t) = -l \sum_i i\phi(t_i),$$

$$d_-(t) = l(b_0 + \sum_i \phi(t_i + b_i)), x_-(t) = -l \sum_i i\phi(t_i + b_i),$$

where b_0 on the right of the two d expressions is a fixed buffer integer $\geq 2h$ (unrelated to the vector's coordinate at player zero). All query symbols are l-multiples of magnitude O_h,$c(p)$. Modulo p the plus representatives lie in $[lb_0 - hl, lb_0]$; the minus ones lie in $[lb_0 + p/q - hl, lb_0 + p/q]$. Both intervals are inside $[0, L_i]$ for large p. At player j the observable is ljb$_0$+$l \sum_i (j-i)\phi(t_i)$. Thus plus and minus player lists agree exactly after translating Γ by a vector whose ith coordinate is $-b_i$ for $i \neq j$, the remaining coordinate chosen to preserve its total sum.

Their aggregate tag difference is odd. In the plus representative, subtract exactly $p\Sigma[t_i]/q$ from the integer d_+. In the minus representative subtract $p(\Sigma[t_i + b_i] - 1)/q$ from d_-. Over Γ the sums of d_+ and d_- are equal by their equal one-coordinate marginals. The two indicated sums of the $[t_i]$ representatives before the -1 is inserted are also equal. The resulting difference in the total representatives is $-p|\Gamma|/q$, an odd integer. Therefore the signed sum of the binary tags is one.

Apply the translation symmetrization argument [R218] in the free class-$(h-2)$ quotients of the degree-zero player groups on the finitely many l-multiple symbols used here. Normalize each query by a fixed zero query. The word F_{h-1} applied to the two Γ-indexed lists has equal coordinates in these class-$(h-2)$ groups, by the preceding individual translations of

the indices. Each entry of either list occurs $|\Gamma|^{h-3}$ times for $h \geq 3$ and this multiplicity is odd. The difference word has tag one. The actual integer game supplies all requisite isolated γ_{h-1} subgroups in its commutator subgroup: for each different player i,j, every pair of the used j-side symbols lies in one pair-graph component since their difference is divisible by $j - i$. A path uses normal/allowed velocities between zero and L_i, and steps of $(j - i)$ times any integer of magnitude at most L_i are obtainable by a two-edge path. The length of a path connecting any two of the needed symbols is O_h,$c(1)$, since their magnitudes are O_h,$c(p)$ and $L_i \geq cp$. The tags of these auxiliary paths do not matter since they enter only as commutator lifts. Normalized letter differences between arbitrary needed symbols and the zero reference therefore have bounded lifts. This supplies a finite ordered contradiction at the original h-player game.

Its packet size can be bounded independently of p. The Γ-symmetrization has bounded length and uses boundedly many normalized symbols per player. For any word of that bounded length in γ_{h-1} take a fixed expression as a product of conjugates of length-$(h - 1)$ iterated commutators in those letters. There are finitely many word and identification patterns up to relabeling, so these expressions have some common finite bound. Each constituent player word has a bounded two-edge-path construction as in the preceding paragraph. Thus both auxiliary queries and their alphabets have bounded cardinality, and there are only finitely many possibilities for the resulting marked query support up to relabeling. An infinite subsequence shares one type. Give its finite queries uniform positive probabilities. Braverman–Minzer, Section 3 supplies exponential rectangular character cancellation for tensor powers of this fixed type. The finite density Hales–Jewett mixture and the integration over an independent stationary starting point described in the restricted-charge theorem [R218] turn this into a positive same-set return at odd total tag. The packet and its witness word use actual integer queries, hence no cover-period dependence enters. Projection also gives the two-slot conclusion.

Chapter 5

Transfer principles and nilpotent action classes

R003. Uniformly field-definable prime rows have only endpoint thresholds

Scope and status. Result with proof.

Result. Fresh-prime integer constructions with one fixed ring-language formula have critical value zero or one.

Scope. The formula complexity and parameter-tuple length must be bounded uniformly in the prime.

Consider the fresh-prime construction of suspension.md [R247], with 0 excluded from each row. Suppose that the subsets S_p of their prime cycles are described by a single fixed first-order formula in the language of rings, allowing a fixed tuple of parameters that can vary arbitrarily with the prime. Then the critical threshold of the lifted integer construction is either zero or one.

Here is the argument including the uniformity in the parameters. Kowalski, *Exponential sums over definable subsets of finite fields*, Corollaries 13 and 16 gives two facts for each fixed one-variable formula. Its fibers are either uniformly bounded in size or have density at least a fixed positive constant for large p. Every nontrivial normalized additive Fourier coefficient of each fiber indicator is $O(p^{-1/2})$, uniformly in the parameters and the nontrivial character. In the dense case apply Corollary 16 to the linear phase: no level set of the phase has more than one point, and the implied constant depends only on the formula and degrees. In the bounded case use the size bound directly.

In fact the centered functions $f_p = 1_{S_p} - |S_p|/p$ have $||f_p||_{U^s} = o(1)$ for every fixed s. For any fixed number of shifts, a multiplicative derivative of f is a bounded linear combination of indicators of intersections of translates of the original formula. The enlarged formula has bounded complexity independent of the shifts, so every nontrivial Fourier coefficient is again $O_s(p^{-1/2})$. For a derivative g of order $s-1$, use $\|g\|_{U^2}^4 = \sum_\chi |\widehat{g}(\chi)|^4$, Parseval, and the uniform bound to get $\|g\|_{U^2}^4 \le |\mathbb{E}g|^4 + O_s(p^{-1})$. Averaging the bound over the shifts yields

$$||f||_{U^{s+1}}^{2^{s+1}} \le ||f||_{U^s}^{2^s} + O_s(p^{-1}),$$

since $|\mathbb{E}g| \leq 1$ and the mean over shifts of $|\mathbb{E}g|^2$ is $||f||_{U^s}^{2^s}$. Start with $U^1 = |\mathbb{E}f| = 0$.

If there are infinitely many dense fibers, the generalized von Neumann inequality for the pairwise nonproportional forms $d, x, x+d, \ldots, x+(h-1)d$ (for $p > h$) makes their h-term weighted count in any positive-density mask agree with $(|S_p|/p)$ times the ordinary count, up to $o(1)$ uniformly over the mask. The ordinary count is uniformly positive at any prescribed density by finite Szemerédi and Varnavides. Thus arbitrarily late rows force every fixed length at every positive density. Restriction to a fresh prime avoiding M in the suspension criterion gives global $b = 0$.

If all but finitely many fibers have bounded size m, choose independently on each late prime a multiplier that maps every member of its row at least $1/(8m+8)$ away from zero on the circle. A random multiplier succeeds with positive probability by the union bound and uniformity on a nonzero prime class. The combined assignments define a character on the direct sum of the prime subgroups. Extend it first to Q/Z (the circle is divisible) and pull it back to Q. It uniformly separates all the late rows from zero, hence those lifted rows have arbitrarily high density avoiders via a root of this character. Each early row has a nonzero component in one of a fixed finite list of primes and is also repaired at arbitrarily small loss. Thus $b = 1$ in this case.

More generally a finite-color short obstruction on large prime cycles forces the normalized difference indicator to have a Gowers norm bounded away from zero at a fixed order: use the color class of density at least $1/c$, an ordinary Varnavides lower bound and the same generalized von Neumann inequality. This does not preclude a sequence of templates whose defining complexity increases with the primes. It rules out the bounded field complexity strategy for realizing intermediate thresholds.

R122. A converse test using fixed finite torsion

Scope and status. Result with proof.

Result. When all horizontal slice thresholds are zero or one, a finite-torsion extension has an exact finite-mask capacity formula.

Scope. A fixed finite abelian factor; for intermediate slices only a linear-programming upper bound is established.

Let K be a fixed finite abelian group, H a countable abelian group, and $W \subset H \times K$ an arbitrary family of steps. For $s \in K$ put $D_s = \{d : (d, s) \in W\}$. Allow zero as a difference in this discussion, with any family containing zero having capacity zero, and the empty family having capacity one. If each $b_H(D_s)$ is zero or one, then the critical value of W is exactly

the maximum of $|B|/|K|$ over all $B \subset K$ that contain no coset of $\langle s \rangle$ for any s with $b_H(D_s) = 0$.

In particular an irrational critical value on ANY fixed finite extension $H \times K$ of a torsion-free countable abelian group would already give a strictly interior integer critical value (by the torsion-free integer reduction). This necessity does not decide whether there is such an extension. The continuum construction uses an infinite Boolean tag, so it does not trigger this converse.

Proof. Let A be a stationary h-avoider for W. For each s whose slice has capacity zero, $C_s = \bigcap_{j<\mathrm{ord}(s)} U_s^{-j} A$ has measure zero: if positive, multiple recurrence of D_s under H applied to C_s gives an h-slot return at some $d \in D_s$. Since C_s is U_s-invariant and contained in A, (d, s) returns to A. This includes the trivial $s = 0$ case and a zero d. On a conull set, therefore, each vertical mask $\{k : U_k x \in A\}$ obeys all the indicated coset exclusions. Averaging its size proves the upper bound. Conversely use a maximizing finite mask for the zero slices, taking product with independent horizontal avoiding events of mass arbitrarily near one for the finitely many capacity-one slices. The chosen finite mask blocks each zero-slice step within $|K|$ slots; the horizontal factors block the other steps at the maximum of their finitely many horizons. If the displayed maximum is zero the upper bound already suffices. This proves the identity.

More generally, for intermediate slice values $p_s = b_H(D_s)$, the same proof gives finite linear inequalities on the distribution of a random vertical mask B: for every coset l of $< s >$, $P(l \subset B) \leq p_s$. Maximizing $E|B|/|K|$ subject to these inequalities gives an upper bound for $b(W)$. Exactness when all p_s are endpoints is the part proved above.

R128. Fresh prime cycles as test spaces

Scope and status. Proved subcases and unresolved extensions.

Result. Fresh-prime tests provide sufficient finite-model criteria, spectral restrictions, and finite exhaustive computations.

Scope. Uniform quantifiers across arbitrarily large primes are unproved; experiments must remain labeled computational evidence.

Qualifications. The results for primes 17, 19 and 23 are finite exhaustive computations, with scripts and data in the computational supplements. They do not establish a bound uniform over large primes.

Assume the suspension and transfer lemmas from the main research thread, which reduce a saturated rational step set through Q/MZ to integer recurrence. A prescribed threshold is not proved here.

Necessity for the version with a finite-color obstruction. Suppose $D \subset N$ has $0 < b(D) < \delta_0 < 1$ and admits a c-coloring avoiding monochromatic H_0-progressions. For each h, stationary compactness gives a finite nonempty $F \subset D$ with $a_h(F) < \delta_0$. For every prime $p > 2H_0 \max F + 2$ take $S = F \bmod p$. A subset of C_p avoiding its h-progressions lifts periodically to Z, so its density is smaller than δ_0. Extend the finite obstruction coloring to a two-sided word by taking a limit of large positive translates. Color the residues of C_p by its pair of colors on ordinary interval representatives with cuts at 0 and $\lfloor p/2 \rfloor$. A cyclic progression with step in S traces a positive run of spatial span at most $(H_0 - 1) \max F < p/2 - 1$. At least one cut is uncrossed, so one coordinate of the coloring is an honest integer progression of the original step and length. This gives a c^2-color obstruction on every such prime; no control on the cyclic lifts of this one prime is needed in this direction.

Size and structure of a viable prime row. For prime $p > h$, every cyclic h-progression of nonzero step has h distinct sites. Given $0 < t < 1$, retain vertices independently with probability t, then remove one vertex of every surviving forbidden edge. The surviving density can be at least $t - |S|t^h$. Thus $a_h, C_p(S) \le \delta$ requires $|S| \ge (t - \delta)t^{-h}$ for every $t > \delta$. For large h, optimizing with $t = \delta h/(h-1)$ gives the lower bound $(\delta/(h-1))(\delta h/(h-1))^{-h}$, of order $\delta^{1-h}/(eh)$ for fixed δ. This inequality requires no assumption on a low obstruction.

The count localizes in the actual prime cycle. Fix a desired total deletion budget $\eta > 0$ and m fixed monomials of bounded degree t_j. A Weyl recurrence criterion for the circle implies that if a polynomial in one variable of degree t_j avoids an arc of length η/m throughout a length-h segment, then some $q_j \le C$ has $||q_j\gamma|| \le Ch^{-t_j}$, where γ is its leading monomial coefficient (absorbing the factorial into q_j). Take L a multiple of all possible q_j and assume $p > L$. For prescribed integers u_j, choose $v_j \bmod p$ with $Lv_j = u_j \bmod p$. Delete from C_p those residues x for which any $v_j x^{t_j}/p$ falls in a respective short arc, choosing translates with total deletion at most η by averaging. On any entirely retained h-progression the Weyl test implies $||u_j s^{t_j}/p|| \le LCh^{-t_j}$ for every j. This is a direct periodic version of the polynomial obstruction in Alweiss–Farhangi; alternatively use the torus case of Green–Tao, Theorem 2.9.

Let S' be the subset satisfying all these bounds. Intersect the deterministic survivors with an independent random subset of density t, or first obtain a blocker for S' by the previous alteration and translate that blocker in C_p. The density of an S-avoider obtained by the latter method can be at least $(1 - \eta)(t - |S'|t^h)$. Thus $|S'| \ge (t - \delta/(1 - \eta))t^{-h}$ whenever

$t > \delta/(1-\eta)$ and the original row forces all masses $> \delta$. The Weyl constants may depend on the degree, m, η and are harmless when these are fixed before h; the exponential base in the cardinality bound may be taken arbitrarily close to $1/\delta$.

Spectral consequence for the short obstruction. Normalize counting measure on C_p and the Fourier transform accordingly. Suppose A of density $a > 0$ avoids H_0-term progressions at all $s \in S$. For each s, every occupied point of its cyclic ordering is at most $H_0 - 2$ steps before an exit from A to its complement, since all occupied runs have length at most $H_0 - 1$. Hence $\mu(A(A-s)) \geq a/(H_0-1)$ and $||T^s 1_A - 1_A||_2^2 \geq 2a/(H_0-1)$. For any fixed number m of frequencies Λ, the preceding linear test finds an s satisfying $||ks/p|| \leq C/h$ for every $k \in \Lambda$ if h is large enough (otherwise its complementary blocker alone would have mass $> \delta$). Parseval gives

$$\sum_{k \notin \Lambda} |\hat{f}(k)|^2 \geq a/(2(H_0-1)) - O_{m,\delta}(a/h^2), f = 1_A.$$

This applies in particular to a largest cell of a fixed-color obstruction. Thus a positive fraction of its Fourier mass must escape any fixed number of coefficients in a viable sequence of long tests. Uniformly bounded Fourier algebra norm would be impossible, since sorting the coefficients of an ℓ^1-bounded vector gives tail ℓ^2 mass $O(1/m)$ after its m largest entries.

Equivalently the translated 1_A cannot be covered by a bounded number of balls of some fixed small radius in $L^2(C_p)$. If m centers suffice at radius u, their linear span approximates every translate to squared error at most u^2. Average the errors over the translations. The resulting covariance operator is diagonal in the character basis, with eigenvalues $|\hat{f}(k)|^2$; the sum of all eigenvalues except its m largest is at most u^2 by the elementary maximum-trace principle. Taking $u^2 < a/(2(H_0-1))$ contradicts the preceding estimate for long tests. In particular the VC dimension of the family of translates of such an obstruction cell cannot be uniformly bounded, by the finite VC packing theorem (a symmetric-difference packing at fixed resolution in a family of bounded VC dimension has bounded cardinality independently of the underlying finite probability space).

For a *binary triple* obstruction a second restriction is available. Put $g = \pm 1$ for its two colors and $\rho(n) = Eg(x)g(x+n)$. The elementary identity $1 + 2\rho(s) + \rho(2s) = 0$ holds for every allowed residue. In the Bohr subset S' where $||ks/p|| \leq C/h$ for $k \in \Lambda$, either $\rho(s)$ or $\rho(2s)$ is at most $-1/3$ and all terms from Λ have nonnegative real parts once h is large. Applying Parseval to the remaining autocorrelation Fourier series shows $|S'| \leq 18p \sum_{k \notin \Lambda} |\widehat{g}(k)|^4$ (multiplication by two permutes the large prime cycle). Together with the exponential lower bound on $|S'|$, this forces residual Fourier peaks above random noise. Neither spectral restriction precludes the large periods permitted in the criterion.

Small exhaustive experiment. The script *prime_frontier.py* enumerates all binary words for $p = 17, 19, 23$ and all symmetric step families supporting a binary forbidden-length coloring. Its output is in the `prime-frontier-*.json` files. For forbidden length three and test length eight the minimum possible maximum avoiding cardinality among these prime families is 13, 14, 17 respectively. In those small cases a minimizing family is the symmetric extension of $[3,6]$, $[3,7]$, $[3,8]$ respectively. The least best single-character separation, over all binary triple-obstructible symmetric step families, is 5/17, 5/19, 6/23 respectively. These finite computations do not imply separation in arbitrarily large primes, and there are larger prime examples with separation below 1/4. For instance the $p = 53$ example in the preceding research notes [D21] has best gap 13/53.

R129. Polynomial recurrence and Lie nilrotation recurrence on the free saturated cover

Scope and status. Result with proof.

Result. Fully saturated free-coordinate circuit families pass ordinary polynomial tests and Lie nilrotation tests.

Scope. Kernel corrections in the free host need not survive a bounded-digit embedding into the integers.

Let $H = \bigoplus_i \mathbb{Z}^{n_i}$, let $G_i = (C_{p_i})^{n_i}$ for distinct primes, and in each G_i specify a finite menu S_i with zero included and a scalar charge zero at the origin. The full-lift words $d \in H$ have residues in the menus and charge the sum of their residue charges. A finite head may be kept at residue zero while its actual free coordinates range over all multiples of its primes. The circuit menu [R033] and its shifted variant [R034] each give all polynomial and Lie nilrotation conclusions below, with any prescribed charge.

All ordinary circle polynomials, including products across free blocks Suppose that in the native direct sum, even when each G_i is replaced by $(C_{p_i^{e_i}})^{n_i}$ for *any* sequence of finite depths and the menus/charges are pulled back, every finite list of ordinary circle-valued group polynomials vanishing at zero can be simultaneously arbitrarily small on a word of charge arbitrarily near a prescribed t, on every tail. Then the identical assertion holds on the free group H, for any finite list of genuine ordinary group polynomials on H.

Write the test as $P : H \to \mathbb{T}^k$ of degree D with $P(0) = 0$. On each finite block packet use a real-coordinate polynomial representative in ordinary monomials (the Newton expansion gives compatible representatives modulo one on integer points). The rationality of its nonconstant coefficients is intrinsic to each integer test row; switching representatives by integer-valued polynomials affects only rational coefficients. Let $\Lambda \leq \mathbb{Z}^k$ consist of rows with all nonconstant coefficients rational globally. This is a primitive lattice. There is a finite head already detecting this row space: take the descending rational linear subspaces of $\mathbb{Q}^k$ testing rational coefficients on increasing heads; their finite-dimensional chain stabilizes at the intersection. A unimodular coordinate change lets us treat the first r test components as rational polynomials. Any nonzero integer combination involving another component has some irrational nonconstant coefficient on this fixed head.

For the rational tests and every block i, take the p_i-primary parts of the circle values of its rational pure-block coefficients. They have some common depth e_i. These coefficients times ordinary integer monomials define a polynomial Q_i on $(C_{p_i^{e_i}})^{n_i}$; ordinary powers make reduction modulo $p_i^{e_i}$ harmless. Their sum over the direct sum is a genuine group polynomial of degree at most D. Apply the native hypothesis on a tail after the head to get a finite set of active block residues with $\sum_i Q_i(d_i)$ near zero and charge near t.

Take the finite packet consisting of these blocks and the initial head. Clear all denominators of the finitely many rational coefficients on this packet by a common modulus L. Use the Chinese remainder theorem to lift each selected i-th block agreeing with its prescribed p_i-primary coordinates and divisible by every prime-power factor of L for primes *other than* p_i. Use zero in the head. In any mixed-block monomial, for each prime dividing its denominator at least one of its factors comes from a different block and is divisible by that entire prime-power factor; hence this monomial vanishes modulo one. The pure-block contribu-

tions are exactly the chosen p_i-primary values. Thus the rational test coordinates are near zero. All these rational test values and residue charges are unchanged on the congruence coset of these lifts for a modulus divisible by L and all selected $p_i^{e_i}$.

On that entire coset vary the coordinates of the packet freely, including the residue-zero head. Every integer character on the remaining $k - r$ torus coordinates has an irrational nonconstant coefficient in its polynomial; nonsingular rational translation and dilation of a polynomial preserve the property that all nonconstant coefficients are rational, in both directions. The multivariate polynomial Weyl criterion therefore gives equidistribution on $\mathbb{T}^{k-r}$. In particular some actual word makes these remaining coordinates arbitrarily small as well. This proves the free-group assertion. It applies equally to a tail by choosing its stabilizing finite head internally.

For ordinary polynomial tests there is also a finite uniformization. Fix k, D, ϵ and a charge neighborhood. The space of zero-based polynomial maps $H \to \mathbb{T}^k$ of degree at most D is compact in pointwise topology: use the closed finite-difference equations inside the compact power $(\mathbb{T}^k)^H$. Each test map admits a word hitting the open ϵ-neighborhood and the prescribed charge interval, and these open sets cover the compact parameter space. A finite word list thus works for every test simultaneously. In any integer packet construction using group homomorphisms on the realized finite rank, images of this list still have the same polynomial-hitting property by pullback; select successive such lists for all accuracies, dimensions and degrees. The corresponding uniformization for nilsystems is not asserted here.

Uniform Lie-group recurrence on the full cover This step needs only the linear part of the polynomial assertion. Let a homomorphism of the free group into a connected simply connected nilpotent Lie group act by translations on a compact nilmanifold. The logarithms of its commuting elements lie in one finite-dimensional abelian real Lie algebra, and the logarithm of the homomorphism is additive. This works together for any fixed finite tuple of such actions by taking a product. Write L for the closure of the log image in its real vector space and M for the closure of the log image of the full kernel of residue reduction. The abelian Lie group L/M is compact. Indeed L, a closed subgroup of a finite-dimensional real vector space, is compactly generated, as is its quotient. The image in L/M of the free group is torsion since residue reduction has locally finite image. A compactly generated abelian Lie group with dense torsion has no Euclidean factor, and its finitely generated discrete component group is finite, proving compactness.

The induced compact Lie group L/M is thus detected by finitely many circle characters. Tail Bohr recurrence with prescribed charge chooses a word whose log projects arbitrarily close to zero in L/M. By openness of the quotient and density of the kernel image in M, add an actual finitely supported kernel word so that the resulting full log is arbitrarily close to zero in the real vector space. This addition preserves every local residue and the charge. Thus the true niltranslations approach the identity in the Lie topology, giving uniform convergence on the whole compact target. On any fixed positive Haar-measurable event, at any fixed number of slots the correlations can be taken arbitrarily close to the measure of that event by strong continuity of translations. Performing the construction using only axes on a given tail gives the same tail conclusion. Connected nilpotent target groups for which the closure of the abelian image is a compactly generated abelian Lie group admit the identical direct quotient argument in that closure.

This property for saturated words need not uniformize over integer packet embeddings. Under a homomorphism from the free group to the integers the kernel adjustment could send a nonzero free-group word to the integer zero; bounded-digit integer transfer prevents that collapse but then forbids arbitrary kernel additions.

R149. Finite-dimensional polynomial orbit tests in a locally compact Conze–Lesigne system

Scope and status. Result with proof.

Result. Locally compact Conze-Lesigne models admit finite-dimensional Lie orbit-map tests, and primary roots force finite-index horizontal ergodicity.

Scope. Orbit maps need not be factor actions; weighted characteristic estimates remain separate.

This is a structural reduction relevant to a possible four-term argument for the deep quadratic model. It does not prove the required weighted nilsequence estimate or arbitrary four-term recurrence.

Use a good translational model N/Λ from Jamneshan–Shalom–Tao, Theorem 1.7: N is locally compact, second countable and two-step nilpotent, C is a compact central Lie subgroup containing all commutators, and Λ is an abelian discrete cocompact subgroup meeting C trivially. The acting countable abelian group translates by a homomorphism ϕ. The approximating Lie models below are models of orbit *maps*. The full group action need not descend to an action on such a model.

Take a compact set of representatives $P \subset N$ with $N = P\Lambda$ (finite subcover by translates of relatively compact open neighborhoods). Generate L by P, C and a compact identity neighborhood. Then L is compactly generated and open, is normal because it contains all commutators, and $N = L\Lambda$. With $\Lambda_0 = \Lambda \cap L$, the natural map $L/\Lambda_0 \to N/\Lambda$ is a homeomorphism and identifies the invariant measures.

There are arbitrarily small compact subgroups J central in L such that L/J is Lie. Here a general use of Gleason–Yamabe for all of L without an open-subgroup qualification would not suffice. Indeed choose an open subgroup L_0 as in Gleason–Yamabe and a sufficiently small compact normal $B \triangleleft L_0$ with L_0/B Lie. Arrange $B \cap C = \{1\}$ using the no-small-subgroups property of the Lie group C. Then $[B, L_0] \subset B \cap C$, hence B is central in L_0; in particular it is abelian. The group $L/(CL_0)$ is discrete finitely generated abelian, since $\mathtt{CL_0}$ is open and normal. Lift a finite generating list g_i. Put $J = B \cap \bigcap_i \ker([g_i, \cdot])$. Commutators are homomorphisms on B. Thus B/J is isomorphic to a closed subgroup of a finite product of C, hence Lie, and J centralizes all of L. Now L_0/J is locally compact and has no small subgroups: use the no-small-subgroups neighborhoods in its Lie quotient L_0/B and its Lie kernel B/J simultaneously. It is Lie by the no-small-subgroups theorem. The larger group L/J has this Lie group as an open subgroup and is Lie by the same criterion.

The quotient $Y_J = (L/J)/(\Lambda_0 J/J)$ is a finite-dimensional compact nilmanifold in the sense allowing disconnected Lie groups. Its lattice is discrete since the projection with compact kernel is proper and Λ_0 is discrete; it is cocompact by compactness of L/Λ_0. Any continuous function on $N/\Lambda \cong L/\Lambda_0$ can be uniformly approximated by functions on Y_J, by averaging over the arbitrarily small compact left J-action. The same reasoning yields approximation in measure for measurable bounded tests.

For any finite list $d_1, \ldots, d_m$ of group elements choose $\lambda_i \in \Lambda$ projecting to the same coset of L as $\phi(d_i)$. At a basepoint $x = g\Lambda$, $g \in L$, and parameters $n \in Z^m$, the orbit admits the representative

$$w(n) = \phi(\sum_i n_i d_i) g \prod_i \lambda_i^{-n_i} \in L.$$

It is a degree-at-most-two polynomial map into the filtered two-step group L. Indeed it is the product of two homomorphisms out of the free abelian group and a constant into the ambient two-step group; collect the factors, obtaining products of fixed elements raised to linear integer forms and central commutators raised to polynomials of degree at most two. All resulting second derivatives lie in $C \subset L$. The map projects to the same orbit in N/Λ since the last factor is in Λ. After quotient by J one gets a polynomial orbit map in the *fixed* Lie nilmanifold Y_J, independent of the size and orders of the chosen generating packet. The same applies in products, to integer multiples of the parameters, and at different starting points. In particular on a finite subgroup of the deep prime sum one can choose a single cyclic generator for the full packet. The resulting nilmanifold-valued polynomial sequence is periodic, though its group representatives need not have that period.

Separately, for the native prime group $G = \oplus C_{p^{e_p}}$, let $H = \oplus p C_{p^{e_p}}$. If H acts ergodically in a G-system, then *all* its finite-index subgroups act ergodically. It suffices to prove this after deleting finitely many entire primary axes, since any finite quotient kills every axis whose prime does not divide the quotient order. For the subgroup H' omitting just one prime p, the finite quotient H/H' acts transitively on the H'-ergodic components, so these form a finite cycle of order p^k generated by pe_p for the unit e_p. The G-action of e_p commutes with this cycle, hence is a cyclic translation, and its p-th power is the generating translation. This is impossible if $k > 0$. Repeat inside the action after deleting that axis (the other unit roots still act). More generally, this proves the same assertion on a component stable under all the corresponding primary roots and on any tail. It implies that an ergodic horizontal Kronecker factor in this case is connected, without making a conclusion about its higher nilfactors.

R209. A single frequency controls carries for pure-coordinate steps

Scope and status. Result with proof.

Result. A single radix frequency yields equality of all-length capacities between pure-coordinate free-group directions and their integer realization.

Scope. Digits are bounded by one quarter of the radix; no summability or growth hypothesis is required.

Qualifications. The equality concerns all lengths, not necessarily the same fixed progression length.

Let $b_i \geq 2$, $\lambda_1 = 1$, $\lambda_{i+1} = b_i \lambda_i$, and F_i be finite sets of nonzero integers with $|f| \leq b_i/4$ for every $f \in F_i$. Empty rows are allowed. In $H = direct \sum_i Ze_i$ set $V = \cup_i F_i e_i$ and on the integers set $D = \{|\lambda_i f| : f \in F_i\}$. Then $b_Z(D) = b_H(V)$ without any growth or summability condition on the ratios. This is an equality of suprema over lengths, not of individual length caps.

The group homomorphism $e_i \to \lambda_i$ pulls an integer avoider back to H, giving $b_Z(D) \leq b_H(V)$. For the other direction take a stationary H-event A of mass a avoiding V at length h. Reversing progressions replaces F_i by their positive absolute values without changing avoidance. Write $z_i(n)$ for the standard mixed-radix digit of $n \geq 0$. For random x from the witness system mark n if $U_{\sum_i z_i(n) e_i} x \in A$. Each mark has probability a. All countably many translated forbidden patterns in the H-action can be excluded simultaneously.

Fix the integer threshold $K = 2h$. For any stage step $\lambda_i f$ with $f/b_i < 1/K$, a $2h$-term progression has fixed lower digits, and the unreduced ith digit traverses a distance less than b_i. There is at most one carry out of place i. On at least h consecutive slots the upper digits are constant and the ith digit advances by f. These slots would give a forbidden H-progression, so no entire such run is marked.

To handle the other steps use the circle frequency $\theta = \sum_{j \geq 1} 1/\lambda_{j+1}$. Since each earlier denominator divides λ_i, the phase of $\lambda_i f$ modulo one equals the real number

$$(f/b_i)(1 + 1/b_{i+1} + 1/(b_{i+1}b_{i+2}) + \ldots).$$

It lies between f/b_i and $2f/b_i \leq 1/2$. Thus all remaining steps have distance at least $1/K$ from zero under this one character. Such a family has $b = 1$. Explicitly for any small open arc I take M divisible by all positive integers through $2/|I|$; on the circle rotation by θ/M each such step has velocity in the compact set $||Mz|| \geq 1/K$. Rational velocities in this set have order greater than $2/|I|$ and irrational rotations are dense. A compactness argument gives one uniform finite hitting time for I from every start and every such velocity. Use an independent uniform clock phase and keep n only when its phase is outside I. Every n survives both tests with probability $a(1 - |I|)$, and the length can be raised to the larger of $2h$ and the hitting bound. Averaging, or taking a point with a sufficiently large limsup of prefix averages, gives an integer avoiding set of upper density at least $a(1 - |I|)$. Send $|I|$ to zero and take the supremum over h and A.

This prevents creating different limiting thresholds merely by tightening the radices of pure-coordinate packets whose digits stay within the stated fraction. For each desired avoid-

ing length, the single auxiliary frequency removes the relatively long digit strides at arbitrarily small loss of upper density.

R212. Rank-one rational linear actions and Bohr recurrence

Scope and status. Result with proof.

Result. Bohr recurrence suffices for rational rank-one linear Haar actions; bounded-degree polynomial tests suffice for the stated affine profinite actions.

Scope. Compact abelian Haar automorphisms, and the separate affine profinite extension.

Qualifications. The affine assertion applies to the specified profinite systems, not arbitrary connected affine systems.

For any additive subgroup $G \subseteq \mathbb{Q}$ acting by continuous group automorphisms of a compact metrizable abelian group with Haar probability, every positive measurable h-return set contains an ordinary Bohr zero neighborhood, with a uniformly positive correlation there. This allows $G = \mathbb{Q}$, where finite-index tests alone give no information. The group ring need not be Noetherian.

It suffices to verify the diagonal-containment step of the Haar graph argument [R002] for any ultrafilter p sending all circle characters on G to zero. Let m_j be a dual tuple for which $P(u^g) = \sum_{0 \le j < h} (u^g)^j m_j = 0$ on a member $E \in p$, with the usual units of $R = \mathbb{Z}[G]$. Suppose $s = P(1) \neq 0$, choose a maximal ideal $\mathfrak{q} \supseteq \operatorname{ann}_R(s)$, and let $M = \sum Rm_j$. Then $s \neq 0$ in $M_\mathfrak{q}$, and Nakayama gives a nonzero tuple in $M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q}$. Thus $u^g \bmod \mathfrak{q}$ takes only finitely many roots on E. Its fibers are cosets of the kernel H of this residual homomorphism. Every subgroup of an abelian group is Bohr closed: evaluate a nonzero element of the quotient by a circle character, extending a character of the cyclic subgroup using divisibility of the circle. Therefore one root fiber belonging to p forces root one, and $H \in p$.

For the principal filter at zero there is nothing to prove. Otherwise choose distinct nonzero $g_1, \dots, g_h \in E \cap H$. If the residue field has prime characteristic (ℓ), choose their (ℓ)-adic valuations pairwise distinct. Indeed the shell $\{g : \nu_\ell(g) = a\}$ is taken into the finitely many nonzero elements of order (ℓ) by the standard additive character $g \mapsto [\ell^{-a-1}g]_\ell \in \mathbb{Q}_\ell/\mathbb{Z}_\ell \hookrightarrow \mathbb{T}$. No finite union of shells belongs to p. Any finite generated subgroup of $G \subseteq \mathbb{Q}$ is cyclic, so write $g_i = b_i e$ for a generator e of their subgroup. Then $e \in H$ and $u^e \equiv 1 \bmod \mathfrak{q}$.

We may interpolate the polynomial P at (1) from these nodes in the localized module, despite possible zero divisors. In the rational function field in a formal variable X, the weights for evaluation at one are products of

$$(1 - X^{b_j})/(X^{b_i} - X^{b_j}), j \neq i.$$

Each is regular for substitution $X = u^e$ in $R_\mathfrak{q}$, after formal cancellation. In characteristic zero of the residue field cancel $(X-1)$; the remaining denominator at $X = 1$ is $b_i - b_j \neq 0$. In positive residue characteristic put $d = \nu_\ell(b_i - b_j) = \min(\nu_\ell(b_i), \nu_\ell(b_j))$. Both powers minus one are divisible, as Laurent polynomials, by $X^{\ell^d} - 1$. After cancellation the denominator, up to a Laurent monomial, takes the value $(b_i - b_j)/\ell^d \neq 0 \bmod \ell$ at $X = 1$, so it is a unit at u^e. Signed exponents are harmless for Laurent polynomials.

The scalar identities $\sum_i L_i(1)(X^{b_i})^j = 1$ for $0 \le j < h$, where $L_i(1)$ is the product of those regular ratios, hold in the ring obtained by inverting the remaining denominators, as that Laurent-polynomial ring embeds in its rational function field. Substitution is thus legitimate before applying the identities in $M_\mathfrak{q}$. Multiplying by the corresponding m_j and

summing gives $s = \sum_i L_i(1)P(u^{g_i}) = 0$, contradicting the localization choice. This proves diagonal containment. The Haar disintegration and compactness argument gives the claimed neighborhood. This proof concerns automorphisms, not arbitrary affine translations on a rank-one rational group.

For all notes using Haar on a subgroup $J \supseteq \Delta K$, its lower bound on B^h can be taken to depend only on $m_K(B) = a$ and h. Indeed the group $V = J/\Delta K$ parametrizes differences in K^{h-1}. For independent uniform $v_i \in V$ and independent uniform $x \in K$, the word map $w \mapsto x + \sum_i \pi_{w_i}(v_i)$, $\pi_0 = 0$ and π_j the difference projections, has expected B-density a. With probability at least $(a/2)$ it has density at least $(a/2)$. Take the finite dimension from density Hales–Jewett. Each fixed combinatorial line has the same all-B probability $m_J(B^h)$, since its nonempty wildcard sum is Haar in V and its starting point is Haar independently. Sum over the finite number of lines for the asserted positive constant. The neighborhood or particular difference need not be uniform over systems.

The affine case for a profinite abelian phase group Suppose now the action is affine and K is compact profinite abelian, still with Haar probability. Group-polynomial recurrence of degree $(h-1)$ suffices. Let p kill all zero-constant circle-valued group polynomials of degrees at most $(h-1)$; the principal case is immediate. Its linear graph-joining limit contains the diagonal by the previous result. For a character tuple satisfying (1) p-often, let E be its whole relation locus (including zero). Take the affine linearization $m'_j = (m_j, 0)$ as in the squarefree note [R239], so $\sum_j m'_j = 0$ and $V(g) = \sum_j (u^g)^j m'_j = (0, \phi(g))$ on E is globally fixed. Each character m_j has finite order, so every $(\phi(g))$ belongs to one fixed finite cyclic subgroup of the circle. There are only finitely many primary components to test.

Fix a prime (ℓ) among these and localize at the augmentation maximal ideal $(\ell, u^a - 1 : a \in G)$. Every finite phase in this fixed scalar subgroup localizes to its (ℓ)-primary component, and that component embeds injectively by the augmentation test of the squarefree note. Choose distinct nonzero nodes $a_1, \dots, a_h \in E$ of pairwise distinct (ℓ)-adic valuations. For any $g \in E$ with

$$\nu\ell(g - a_j) \geq \nu\ell(a_i - a_j)(\quad \text{all} \quad i \neq j),$$

the weighted interpolation above works also at u^g. Clear one cyclic generator e of the finitely many rationals involved. Each formal Lagrange ratio $(X^{g/e} - X^{a_j/e})/(X^{a_i/e} - X^{a_j/e})$ becomes regular at $X = u^e \equiv 1$ after canceling the matching $X^{\ell^b} - 1$, exactly as in the linear proof. Its augmentation is the rational number $(g - a_j)/(a_i - a_j) \in \mathbb{Z}_{(\ell)}$, as is seen by taking $X \to 1$ in the canceled expression. On a fixed (ℓ)-primary phase multiplication by this interpolating coefficient is just multiplication by that (ℓ)-integral rational number, since every unit in its denominator acts by its augmentation on the fixed submodule. Thus we have the genuine circle identity

$$\phi(g)_\ell = \sum_i [\prod_{j \neq i} (g - a_j)/(a_i - a_j)] \cdot \phi(a_i)_\ell.$$

The zero value at $g = 0$ also satisfies the valuation conditions for any pairwise distinct anchor valuations. For each i extend the additive map $x \mapsto x\phi(a_i)_\ell$ from $\mathbb{Z}_{(\ell)}$ to $\mathbb{Q}$ using

divisibility of (mathbb T). Applying the resulting circle character to the indicated ordinary rational polynomial in g, and summing, gives a circle-valued group polynomial of degree at most $(h-1)$ on G, zero at zero.

The valuations along p tend to $+\infty$ or to $-\infty$, since every finite set of valuation levels is excluded. In the first case fix any such anchors: all g with $\nu_\ell(g) > \max_i \nu_\ell(a_i)$ satisfy the valuation conditions, so the polynomial agrees with the phase on a member of p. In the second case, for each finite $F \subset E \cup \{0\}$ choose h anchors of distinct valuations below those of all its nonzero entries. The same polynomial construction fits $\phi(\cdot)_\ell$ at every point of F. The space of circle-valued group polynomials of degree at most $(h-1)$ and value zero at zero is compact in the topology of pointwise convergence, since its finite-difference identities define a closed subspace of $\mathbb{T}^G$. Hence a polynomial agreeing on all of E exists in the second case. Summing the finitely many resulting primary polynomials, and intersecting the finitely many applicable p-members, proves that the affine phase tends to zero. The affine graph limit is the linear Haar limit, and the same positive-return and neighborhood conclusions follow. No such finite-primary decomposition is assumed for characters of an arbitrary compact connected group.

R213. Saturation by the integers makes Bohr recurrence sufficient for rational niltranslation tests

Scope and status. Result with proof.

Result. Integer-saturated Bohr-recurrent rational sets return simultaneously in finite-dimensional niltranslation actions.

Scope. Genuine rational-group actions with integer saturation.

Qualifications. R222 supplies a second proof of substantially the same conclusion; avoid counting this as two independent new theorems.

Suppose $D \subseteq \mathbb{Q}$ satisfies $D + \mathbb{Z} = D$ and meets every simultaneous zero neighborhood of any finite tuple of circle characters on $\mathbb{Q}$. For every finite tuple of actions of $\mathbb{Q}$ by left niltranslations on compact finite-dimensional nilmanifolds, some differences in D act arbitrarily close to the identity uniformly on all these spaces. Thus they yield positive returns at any fixed length for arbitrary positive Haar events, with correlations arbitrarily close to their measures. This tests genuine homomorphisms of the rational group into transformation groups, rather than testing a prescribed polynomial sequence in a nilmanifold.

First let B be a connected nilpotent Lie group and $\phi : \mathbb{Q} \to B$ a homomorphism. Its closed abelian image closure A is connected. Indeed the component group of a closed subgroup in a connected nilpotent Lie group is finitely generated: take the preimage in the simply connected nilpotent universal cover, whose identity component L is closed and connected. The normalizer of L there is connected and simply connected, as follows by the nilpotent exponential map and invariance of its Lie algebra under $e^{\operatorname{ad} X}$ equivalently under $\operatorname{ad} X$. The component quotient in this cover is a discrete subgroup of the simply connected nilpotent Lie quotient of the normalizer by L, hence finitely generated (the standard lattice theorem for discrete subgroups of simply connected nilpotent Lie groups). Its image surjects onto the original component group. A finitely generated abelian group has no nonzero divisible subgroup, so the image of $\mathbb{Q}$ in A/A^0 is trivial; taking closure proves connectedness. Consequently $A \simeq \mathbb{R}^a \times \mathbb{T}^b$. The real coordinates of $(\phi(q))$ are linear in q. Subtract from each torus coordinate a real linear function of q with the same value at 1; the residual characters η_j vanish on (mathbb Z).

Apply Bohr recurrence jointly to all the residual characters for the finite list and $q \mapsto q \bmod 1$. From the resulting $d_0 \in D$ subtract its nearest integer. The new $d \in D$ is arbitrarily small in the usual real absolute value, while the residual character values are unchanged. Hence all $(\phi(d))$ are arbitrarily close to the identity. In a simply connected target there is just a continuous one-parameter flow (the roots of $(\phi(1))$ are unique).

For completeness the conclusion also holds for a presentation allowing disconnected nilpotent Lie groups. A compact manifold has finitely many components, and the divisible group $\mathbb{Q}$ cannot act nontrivially on this finite set. Fix a component, writing it as N/Λ for a connected simply connected nilpotent Lie group N and lattice Λ, by passing to the universal cover of the identity component and conjugating the stabilizer at the chosen point. Every left translation preserving the component acts there affinely: in coordinates at the identity coset its translating element can be expressed as $g_0\gamma$ with g_0 in the identity component and (γ) in the lattice, so conjugation by (γ) supplies the lattice-preserving linear part. The affine group maps to $\mathrm{Out}(\Lambda)$ by its action on the fundamental group. This group is residually finite, since it is arithmetic for a finitely generated torsion-free nilpotent group, by Baues–Grunewald,

Theorem 1.1 and its nilpotent predecessors. Thus the image of $\mathbb{Q}$ is trivial. An affine map inducing the identity outer automorphism is a left translation: its linear part acts on the lattice by conjugation by some $\lambda \in \Lambda$, and then agrees with this inner automorphism on N by uniqueness of extension from the lattice. In N/Λ the right factor λ^{-1} disappears. The effective group of these left translations is $N/(Z(N) \cap \Lambda)$, again connected nilpotent Lie. Apply the first argument simultaneously to all components. Continuous group action on a compact space turns convergence to the identity in this Lie group into uniform convergence of the transformations; continuity on L^1 gives the measurable assertion.

In particular the deep quadratic rational set [R049] meets these finite-dimensional nil-translation tests while failing a short test in an affine Haar action on an infinite-dimensional compact abelian group. No arbitrary-system density upper bound is asserted.

R220. Saturated words and affine actions on finite-dimensional nilmanifolds

Scope and status. Result with proof.

Result. Saturated Bohr-recurrent sets return under finite-dimensional affine nilmanifold actions; a separate affine Haar result follows from polynomial recurrence and finitely generated linear image.

Scope. Saturation by J with torsion quotient and genuine group homomorphisms are essential.

Let G be countable abelian, $J \leq G$ with torsion quotient G/J, and $D+J = D$. If D is Bohr recurrent in G, its image approaches the identity in every continuous action homomorphism from G into the affine automorphism group of a compact connected nilmanifold $X = N/\Gamma$ (N simply connected nilpotent, Γ a lattice). This gives all finite-length returns for every positive Haar-measurable event by strong continuity, indeed correlations arbitrarily near the original mass. It applies to the free dyadic pure menu when the usual Bohr recurrence holds: the saturation subgroup is $\oplus 2z_i Ze_i$. This enlarges the commuting simply connected translation case in full-lift-polynomial-nil.md [R129]; no analogous statement for arbitrary maps through bounded integer packet realizations is asserted.

The effective affine group is $(N \rtimes Aut(\Gamma))/\Delta\Gamma$, with $\Delta\Gamma = \{(\gamma, Ad(\gamma^{-1})) : \gamma \in \Gamma\}$: an affine map fixing every coset has this form by discreteness of Γ and connectedness of N. Automorphisms of Γ extend uniquely to N by the rational Mal'cev completion. The identity component is $N/(\Gamma \cap Z(N))$, and the component group is $Out(\Gamma)$. Any closed abelian subgroup in the identity component is compactly generated. Indeed quotient out the real span of the indicated central lattice to get a simply connected nilpotent group, with compact kernel on the already effective group. The projection of a closed subgroup is closed by properness. Its abelian image in the simply connected nilpotent group is, in logarithms, a closed additive subgroup of a finite-dimensional real vector space; this is compactly generated. Its lift has compact kernel.

Abelian subgroups of $Out(\Gamma)$ are finitely generated. Map to the integral linear group on the torsion-free part of the abelianization; an abelian subgroup of $GL_r(Z)$ is finitely generated by Mal'cev's theorem that solvable integral linear groups are polycyclic. The kernel before dividing out inner automorphisms is polycyclic as well. In fact its action on the lower-central graded Lie algebra is trivial, since the graded Lie algebra is generated in degree one. On the full lattice spanned by $\log \Gamma$ (a lattice by rational Mal'cev coordinates and bounded BCH denominators) it is an integral unipotent group preserving the rational central filtration. In a basis adapted to the saturated integral filtration this is a subgroup of the unitriangular integral matrices, hence polycyclic. Dividing out inner automorphisms gives the stated kernel in $Out(\Gamma)$. Thus the part of a closed abelian affine subgroup in the identity component is compactly generated and its discrete component image is finitely generated. Together they give compact generation of the whole closed abelian subgroup.

For the action homomorphism ρ, put $L = closure(\rho G)$ and $M = closure(\rho J)$. The compactly generated abelian Lie group L/M has dense torsion, so it is compact: in its standard decomposition there is no Euclidean factor, and its finitely generated discrete component group is finite. Pull back finitely many circle character tests for this compact quotient. Bohr recurrence gives an element of D whose image coset is arbitrarily near the identity. By openness of the quotient map and density of ρJ in M, correct it by an

actual element of J so its action on X is arbitrarily near the identity; saturation preserves membership in D.

The same argument includes a compact nilmanifold with finitely many connected components, with the affine action permuting them. The component-preserving subgroup has finite index. Its closed abelian image in the product of its componentwise image closures is compactly generated: it is a closed subgroup of a finite product of compactly generated abelian Lie groups, hence has the same property (project through the compact torus onto a Euclidean group times a finitely generated discrete group). The remaining finite permutation quotient does not affect compact generation. The quotient-and-correction argument then works on the whole space at once.

A separate affine Haar consequence without saturation. For any affine Haar action on a compact abelian group with finitely generated linear image, all-orders ordinary polynomial-Bohr recurrence of D already suffices, by the finite-rank affine Haar theorem [R002]. Split the linear image as $Z^l \oplus F$, choose a homomorphic lift $s : Z^l \to G$ of its free summand, and let $\ell : G \to Z^l$ be the free-coordinate projection. On the kernel of the entire linear image the commuting action is pure translation by a homomorphism τ. Require that the remaining finite automorphism component of $g - s\ell(g)$ be trivial by finitely many rational characters. To make its translation near identity, test any finite list of characters composed with τ: these extend from the zero-linear-image subgroup to $\ker \ell$ by divisibility of the circle, and compose with $g \mapsto g - s\ell(g)$ on G. Select the identity neighborhood so small that the resulting translates of the tested event at its finitely many slots change its correlation by less than half the lower bound from the finite-rank theorem. The latter theorem supplies finitely many ordinary polynomial tests composed with ℓ and a fixed positive lower bound for the $s\ell(g)$ correlation. Ordinary polynomial-Bohr recurrence satisfies all the tests simultaneously. The linear image is automatically finitely generated for toral affine actions by the integral linear group argument above. Inverse limits of action-preserved factors of this type follow by conditional density clipping on a sufficiently large factor.

R221. Saturation after two sums in one integer system does not force recurrence

Scope and status. Result with proof.

Result. One stationary integer witness can have near-full avoidance even though its multiple joining limits stabilize after two sums and its pair limits stabilize after one.

Scope. The identities concern a selected witness system and selected ultrafilters.

Qualifications. This is not a counterexample to stronger identities quantified over all actions.

Given $\epsilon > 0$, there is an integer probability-preserving system S, an event A of mass greater than $1 - \epsilon$, a progression length h, and an ultrafilter P on integer steps with the following properties. A has zero h-term return at every step of some P-member. For this same system the entire joint law at times `j d`, for any finite list of integer j, has the same ultralimit along all additive powers P^{*r} with $r \geq 2$. Even the two-slot laws at P itself agree with those at every subsequent power. These identities are properties in the one displayed system. P is not an additive finite-order ultrafilter in the universal Stone–Čech semigroup.

The finite-exponent action. Take $q = 2^m$ large, let $n = q - 1$ and choose a generator ζ of the multiplicative group of the finite field F_q. On the cyclic alphabet C_n the characters $t \mapsto \zeta^{st}$ $(s \in C_n)$ form an F_q-basis of all functions. Choose $S_0 \subset C_n$ containing 0 and 1, of size at most $2\lceil\sqrt{n}\rceil + 2$, whose two-fold sum is C_n. For example use the small nonnegative integers less than $\lceil\sqrt{n}\rceil$ together with their radix times each small integer through $\lfloor n/\lceil\sqrt{n}\rceil\rfloor$. Let V be the span of these character functions. It is translation-invariant and contains all constants.

For the discrete group $G = \oplus_{i\geq 1} C_n$, construct a random function F on G by making its restriction to any coordinate cube C_n^I, for finite I, uniform in $V^{\otimes I}$. This defines Haar on a compact subgroup of F_q^G: consistency of restriction to a smaller cube follows from surjectivity of coordinate evaluation at zero since constants lie in V. Its distribution is invariant under translations from G, using translation invariance of V in any larger cube. Take independent uniform $y_i \in C_n$. For a fixed deterministic $g : C_n \to F_q$, put

$$Z(x) = F(x) + \sum_i [g(y_i + x_i) - g(y_i)], x \in G.$$

The law of Z is translation-invariant. Indeed translation by a group element w replaces F by the shifted Haar noise, y by $y + w$, and adds a spatial constant determined by y,w. Conditional on y that constant is absorbed by Haar noise, whose subspace contains all constants. Thus use the shift on the function space with the law of Z.

Choose $H \subset F_q$ of density between $\epsilon/4$ and $\epsilon/2$ (for $\epsilon < 1$). Choose g initially iid uniform over its n values. For every y and $v \in V$ the probability that all $g(t + y) + v(t)$ miss H is $(1 - |H|/q)^n$. There are only $nq^{|S_0|}$ such choices, so a deterministic g avoiding that failure for every choice exists for large enough m. On any singleton-axis line the observable $t \mapsto Z(x + te_i)$ is exactly a translate of g in the argument plus some member of V (including its spatial constant). Thus $B = \{Z : Z(0) \notin H\}$ has mass $1 - |H|/q$ and exactly avoids every n-slot singleton-axis progression.

For r distinct axes outside I and their $\sum w$, consider the slices F restricted to $C_n^I + tw$, $t \in C_n$. Their joint law is uniform on

$$V^{\otimes I} \otimes \operatorname{span} t \mapsto \zeta^{st} : s \in rS_0.$$

This follows directly by restricting the independent Haar noise from the enlarged cube to that diagonal. When $r \geq 2$ the sumset is all of C_n. The n slices of the noise are then independent uniform copies on the old cube. The old forcing $\sum_{i \in I}[g(y_i + x_i) - g(y_i)]$ is shared across slices, and the extra forcing from the r moving axes only translates the independent constant coefficients in each separate slice. Thus the complete law of the Z-slices on any old cube is independent of $r \geq 2$ and of those late axes. Joint laws of any time list reduce modulo n. For $r = 1$, any two distinct time slots already have independent Haar noises: the evaluation of the functions 1 and ζ^t is invertible on those two slots. Equal time slots modulo n are identical. Hence all two-slot laws have the same values for $r \geq 1$.

An integer system. Choose fast-growing radices M_i divisible by n, with $\Sigma 1/M_i < \epsilon/4$. Let X be the mixed-radix compact adding machine with Haar product digit law, and let d_i be the place value $\prod_{k<i} M_k$. Read the i-th digit of x modulo n and denote the resulting infinite product sequence by $\psi(x)$. For every fixed integer d the componentwise difference $c_d(x) = \psi(x + d) - \psi(x)$ has finite support almost surely, since an integer addition produces only finitely many carries almost surely. The differences satisfy the cocycle law. On the product with the preceding field put

$$S^d(x, Z) = (x + d, T_{c_d(x)} Z),$$

where T is the G-shift. This is an integer measure-preserving action modulo the common invariant null set. Take as A the event $Z(0) \notin H$ and every radix digit different from zero. Its mass is $> 1 - \epsilon$ by the union bound. A full run of n retained slots with step d_i cannot have a carry across the i-th digit between two such slots, since that carry lands on digit zero. Along the entire run the successive c-increments are therefore precisely e_i, contradicting the finite-exponent avoidance. Thus A exactly avoids all d_i.

Let P be the image of any free ultrafilter on indices under $i \mapsto d_i$. Fix finitely many time slots j and cylinder observables of the integer system, so only finitely many old radix digits and finitely many field sites are tested. Along a sum of r distinct late d_i, up to all the prescribed times, the probability of a carry from a selected digit is bounded by $O_j(\sum_{i selected} 1/M_i)$; for negative times include lower radix boundaries too. On the remaining samples the old odometer digits are frozen and the field follows precisely the finite-exponent r-axis motion. In the nested index ultralimits defining P^{*r} the indices are distinct and all escape the old coordinates; the carry error tends to zero. The limiting joint cylinder law is consequently independent of $r \geq 2$, and for two slots independent of $r \geq 1$. Cylinder approximation in each fixed marginal extends this to arbitrary measurable tests. This applies to arbitrary positions of the action observed by the event A as well.

Finally P is not of finite additive order. On nested sums of distinct d_i the radix digits have exactly r ones (and zeros elsewhere), so the ultrafilters P^{*r} are all distinct. In fact choosing a frequency whose d_i-phases all approach any fixed nonzero circle value is possible by choosing its successive fast-radix digits. The example prevents an argument that uses

only the finite-additive-power identities **after projecting to the proposed witness or its finite powers**. The universal identities in βN, which test every integer subset, could be stronger.

R222. Saturated Bohr-recurrent rational sets return on classical niltranslations

Scope and status. Result with proof.

Result. Saturated rational Bohr recurrence implies recurrence for classical niltranslations, including the stated disconnected presentations.

Scope. A cleaner structural proof of the result recorded in R213.

Qualifications. No integer-set nilrecurrence follows merely by transferring arbitrary capacities.

Let $D \subset Q$ satisfy $D + Z = D$ and suppose it is Bohr recurrent, meaning it meets every zero neighborhood defined by finitely many circle characters of Q. For every finite list of actions of Q by homomorphisms into the translating groups of compact finite-dimensional nilmanifolds, there are elements of D returning arbitrarily close to any specified starting points, simultaneously also at any fixed finite list of integer multiples. This includes the deep quadratic example [R049]. The assertion is about actions of the rational group itself. It does not identify the generalized-polynomial return sets for a separate mixed-radix realization on the integers.

For a homomorphism $\phi : Q \to G$ into a **connected** nilpotent Lie group, the maximal compact subgroup K is a central torus and G/K is simply connected. One sees this by representing G as the simply connected nilpotent covering group modulo a discrete central subgroup; quotient also by the real span of that subgroup in the center. Powers in the simply connected nilpotent group have unique roots by exponential coordinates, so the image in G/K is $\exp(qX_0)$. Lift the generator to X in the Lie algebra of G. Then $\phi(q) = \exp(qX)\chi(q)$, with $\chi : Q \to K$ a homomorphism since the discrepancy is central. Split its value at one by choosing a logarithm Y in the torus Lie algebra. The homomorphism $\chi_0(q) = \chi(q)\exp(-qY)$ vanishes on the integers. Apply Bohr recurrence to the ordinary character $q \bmod 1$ and all coordinates of the finitely many such χ_0. After subtracting a nearest integer the resulting $q \in D$ has arbitrarily small usual absolute value and all the same small central-character values. Thus every $\phi(q)$, and each of its finitely many required powers, is near the identity.

For completeness one can reduce a disconnected nilpotent Lie presentation to this case. The finitely many connected components of the compact manifold are fixed by Q, since a finite group has no nontrivial divisible abelian subgroup. On a chosen component present the connected group action as N/Λ for a simply connected nilpotent Lie group N and a lattice Λ. A component-preserving left translation is affine in this presentation. Indeed, using a basepoint, its translating element is a product of an element of the identity component and of the ambient stabilizer lattice; conjugation by the latter preserves the lifted lattice Λ. Changing the decomposition changes its automorphism by an inner automorphism from Λ. Hence the outer lattice automorphisms induced by the Q action form a homomorphic image of Q.

The subgroup of lattice automorphisms arising by these conjugations is contained in a finitely generated nilpotent matrix group. To check this, their differentials preserve the full additive lattice spanned by $\log \Lambda$; the lattice sandwich sufficient for this assertion is given, for example, in Cornulier's account of Malcev lattices. The ambient nilpotency gives a common filtration with trivial action on its successive real vector-space quotients. More explicitly the spans obtained by iterating operators $Ad(g) - I$ vanish after at most the nilpotency class,

by differentiating successive group commutators at the identity. Their spans are rational for the lattice-preserving subgroup because its matrices are integral. Choose an integral lattice basis adapted to this rational flag after refining the trivial-action layers; all the matrices are upper unitriangular. Every subgroup of the upper unitriangular integral group is finitely generated nilpotent. The same is true in the quotient by its inner lattice automorphisms. Thus the image of Q, an abelian divisible subgroup of a finitely generated nilpotent group, is trivial. Every induced affine map can consequently be written as a pure left translation: its lattice-inner automorphism can be absorbed into the translating element since the remaining right factor belongs to Λ. Lie automorphisms on N are determined by their lattice restriction by rational spanning. The pure translation parameter is unique modulo the kernel of the action, which is a discrete central subgroup (a normal subgroup of a discrete lattice has singleton conjugacy orbits under the connected group). The pure translations therefore supply a homomorphism $Q \to N/\ker$, a connected nilpotent Lie group. Apply the preceding argument on each component under test.

R239. Affine Haar actions of the squarefree group and group-polynomial tests

Scope and status. Result with proof.

Result. Affine Haar actions of the squarefree torsion group have return sets containing bounded-degree group-polynomial neighborhoods; linear actions need only degree one.

Scope. Finite primary components and compact abelian Haar affine actions.

Qualifications. This affine Haar result is not a characteristic-factor theorem for arbitrary actions.

Let $G = \bigoplus_{\ell \in \mathcal{P}} C_\ell$ for any list of distinct primes. A group polynomial $P : G \to \mathbb{T}$ of degree at most s has all differences of order $(s+1)$ zero. Fix $h \geq 2$. For every Haar probability affine action $S_g x = A_g x + b_g$ on an arbitrary compact metrizable abelian group K, the positive h-return set of every positive measurable event contains a neighborhood cut out by finitely many zero-constant group polynomials of degrees at most $(h-1)$, with a uniform positive correlation on that neighborhood. For actions by group automorphisms, degree one suffices. In particular the polynomial recurrence already verified for the paired odd marks [R156] gives all-length recurrence for every such affine Haar action. An arbitrary skew product is outside this statement.

We prove this without finite generation of the acting group or Noetherianity of its group ring. The Haar graph-joining, diagonal-disintegration, and compactness deductions are as in the finite-rank theorem [R002].

The linear joining contains the diagonal Choose an ultrafilter p sending every circle character $G \to \mathbb{T}$ to zero. The linear graph joining has Haar limit on a subgroup $J \subset K^h$. A tuple of characters $m_j \in \widehat{K}$ annihilates J exactly when

$$\sum_j (A_g *)^j m_j = 0 (1)$$

on a member of p. Such a tuple has $s = \sum_j m_j = 0$. Otherwise let $R = \mathbb{Z}[G]$, $M = \sum_j R m_j$, and take a maximal ideal $\mathfrak{q} \supseteq \operatorname{ann}_R(s)$. Then $s \neq 0$ in $M_\mathfrak{q}$, and the generating tuple has a nonzero image in the vector space $M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q}$ by Nakayama. Under (1), $u^g \bmod \mathfrak{q}$ is a root of a nonzero polynomial of degree at most $(h-1)$ over the residue field (apply a scalar functional to the vector polynomial). Thus on a member of p, the homomorphism $g \mapsto u^g \bmod \mathfrak{q}$ is constant. Its image is a countable locally cyclic torsion subgroup of a field's multiplicative group and embeds in the circle: finite subgroups of a field's units are cyclic, and compatible embeddings into roots of unity are obtained along a nested exhaustion (an embedding of a cyclic subgroup extends to an embedding of the larger cyclic group, by choosing a suitable primitive root). Hence the constant value must be one by choice of p.

Let the residue characteristic be l, allowing zero. When $l > 0$ the subgroup of G with zero l-coordinate belongs to p, since the corresponding primary component is finite. Take a g satisfying (1), lying in this subgroup when needed, and with $u^g \equiv 1 \bmod \mathfrak{q}$. Its order n is prime to the residue characteristic, and $(u^g)^n = 1$. The geometric sum $1 + u^g + \cdots + (u^g)^{n-1}$ is a unit of $R_\mathfrak{q}$, since it reduces to $n \neq 0$ in the residue field. Therefore $u^g = 1$ in the localization.

Relation (1) now says $s = 0$ there, a contradiction. This establishes $\Delta K \subset J$. The argument works without changes for a torsion abelian group with finite primary components.

Separation of the primary affine phases For the affine conclusion let p send all zero-constant circle-valued group polynomials of degree at most $(h-1)$ to zero. In particular it satisfies the previous character condition. Fix a tuple satisfying (1) p-often, so $\sum m_j = 0$. Use the discrete G-module

$$(m, z) \in \hat{K} \oplus Tdiscrete, u^g(m, z) = (A_g * m, z + m(b_g)).$$

The affine cocycle identity and commutativity check the module law. Write $m'_j = (m_j, 0)$. Let E be the entire locus of (1), including zero. For $g \in E$ one has

$$v(g) = \sum_j (u^g)^j m'_j = (0, \phi(g)), \phi(g) = \sum_j m_j(b_{jg}).$$

This value is fixed by every group unit. Applying $1 + u^g + \cdots + (u^g)^{n-1}$, where $n = \mathrm{ord}(g)$, gives $nv(g) = 0$, since the group norm annihilates every difference $((u^g)^j - 1)m'_j$ and $\sum m'_j = 0$. Thus $\phi(g)$ is torsion, of squarefree order dividing n.

For a prime $\ell \in \mathcal{P}$ localize the module at the augmentation maximal ideal $\mathfrak{m}_\ell = (\ell, u^a - 1 : a \in G)$ of R. Every prime-to-(ℓ) group unit u^a is exactly one there, by the geometric-sum argument above. Inside the fixed summand $0 \oplus \mathbb{T}$, a finite-order phase localizes to its (ℓ)-primary component; the finite subgroup of order (ℓ) embeds injectively in the localized module. Indeed every element of $R \setminus \mathfrak{m}_\ell$ acts there on fixed phases by its integral augmentation, prime to (ℓ). Therefore, for $g \in E$, the (ℓ)-primary component of $(\phi(g))$ depends only on the residue $r = g_\ell \in C_\ell$. Write this value $f_\ell(r) \in \mathbb{T}[\ell]$ on the set of represented residues, with $f_\ell(0) = 0$. Specifically, using the unit u of the chosen generator of C_ℓ, one has in the localization

$$\sum_j u^{jr} m'_j = (0, f_\ell(r)) \quad (2)$$

on every represented residue. No assumption is made that re_ℓ alone belongs to E.

Uniform degree at each fresh prime Suppose $\ell > h$. If fewer than h residues are represented, interpolate f_ℓ on them over $\mathbb{F}_\ell$, identifying $\mathbb{T}[\ell]$ with the additive field. The interpolating polynomial has degree less than h and value zero at zero. Otherwise choose distinct represented $r_1, \ldots, r_h$ in $[0, \ell - 1]$. Put $X = u - 1$ in $R_{\mathfrak{m}_\ell}$. Since the right side of (2) is fixed, all h equations $\sum_j u^{jr_i} X m'_j = 0$ hold. They imply

$$X^h m'_j = 0 \quad \text{for all} \quad j.$$

Here is a cancellation-free verification over this ring. In the formal Laurent polynomial ring in U, let $x_i = U^{r_i}$ and $d_i = \prod_{k \neq i}(x_i - x_k)/(U-1)^{h-1}$, meaning the product of the polynomial divided differences. Invert the individual d_i's. Ordinary Lagrange interpolation gives, for each $j < h$, the polynomial identity

$$(U-1)^{h-1} Y^j = \sum_i x_i^j \prod_{k \neq i} (Y - x_k)/d_i.$$

It holds first in the rational function field and hence already in the indicated localized polynomial ring. At $U = u$ each d_i reduces mod $\mathfrak{m}_\ell$ to the product of the nonzero differences $r_i - r_k \bmod \ell$, so substitution in the actual ring is valid even when X is a zero divisor. Multiply the identities by $X m'_j$ and sum in j. All terms on the right vanish, and comparison of powers of Y proves the claim.

From $u^\ell = 1$, $X^h m'_j = 0$, and $\ell > h$, we get $\ell X m'_j = 0$. In fact the truncated binomial expansion of $(1+X)^\ell - 1$ on m'_j is (ℓX) times an element congruent to one modulo the maximal ideal, hence a unit. Thus all $X^k m'_j$ for $1 \leq k < h$ are in the localized module's (ℓ)-torsion vector space. Since the constant sum vanishes,

$$\sum_j u^{jr} m'_j = \sum_{j<h} \sum_{1 \leq k < h} \binom{jr}{k} X^k m'_j$$

is an (ℓ)-torsion-vector-valued polynomial in $r \in \mathbb{F}_\ell$ of degree at most $(h-1)$. The binomial polynomials are valid over the field because $k < \ell$. Extend the scalar identification of the embedded phase line $0 \oplus \mathbb{T}[\ell]$ to an (ℓ)-linear functional on this vector space. Equation (2) therefore supplies a scalar polynomial of degree at most $(h-1)$ agreeing with f_ℓ at all represented residues, again zero at zero.

Choose such an extension on every prime $\ell > h$. Summing its value $f_\ell(g_\ell)$ over the finite support of g gives a circle-valued group polynomial of degree at most $(h-1)$, by coordinatewise finite differences. On the relation locus with the finitely many small prime coordinates zero it equals $(\phi(g))$, by primary decomposition of the finite torsion phase. This locus belongs to p. The affine graph-joining coefficient for the tuple in (1) therefore tends to one. For tuples not satisfying (1) p-often, the coefficient tends to zero. All affine graph joinings have the same p-limit Haar on $J \supseteq \Delta K$ as the linear joinings. Disintegration and compactness as in the finite-rank note complete the positive-return and neighborhood assertions.

R240. Sparse direction sets on bounded-size prime supports have critical density one

Scope and status. Result with proof.

Result. Bounded-size prime supports with boundedly many vectors per support have tail capacity one and corresponding rational and integer endpoint one.

Scope. Both support size and menu multiplicity are uniformly bounded.

Let p_i be pairwise distinct primes and put $P_0 = \oplus C_{p_i}$ inside Q/Z. Suppose $V \subset P_0 \setminus \{0\}$ has the following properties, with fixed s, r.

1. Every $v \in V$ has at most s nonzero coordinates.
2. For each exact support I there are at most r such vectors with that support.

Then $\lim_J b_{P_0}(V_J) = 1$, where $V_J = V \cap \oplus_{i \geq J} C_{p_i}$. By squarefree-tail transfer [R241], the corresponding saturated rational family and its integer realization have critical avoidance density one. Coefficients may be arbitrary, including unrelated choices for every edge $\{i, j\}$. Genuine mixed bilinear characters on distinct primary factors vanish, so we use ordinary integer representatives on a suitable subprogression.

Polynomial hitting input. Given s and a desired arc length $\eta > 0$, there exist M and K such that, simultaneously for $1 \leq j \leq s$, any real polynomial in an integer variable t of degree j with leading coefficient γ and $||M\gamma|| \geq 1/(4r)$ hits any translate of an open arc of length η on every K consecutive inputs. In this statement K may also depend on r. Indeed the circle case of Green–Tao, Theorem 2.9 gives an integer $q \leq C(s, \eta)$ with $||q\gamma|| \leq C(s, \eta)/K^j$ when an arc is missed. Absorb the degree factorial in q when passing between binomial and ordinary coefficients. The constants and initial length depend only on s, η. Take $M = \mathrm{lcm}(1, \ldots, \lceil C \rceil)$, then choose K sufficiently large compared to `MrC`. This is also the fixed-degree polynomial sieve used in the preceding research notes.

Slow steps. Fix these M, K, set $N = 2(s+1)K$ and $Q = N^s$, and take J large enough that all p_i on this tail exceed Q. For each $v \in V_J$ with exact support I, simultaneous Dirichlet pigeonholing applied to its $|I|$ normalized residues gives one $1 \leq q_v \leq N^{|I|} \leq Q$ for which $||q_v v_i / p_i|| \leq 1/N$ on every active coordinate. Denote by $b_{v,i}$ the centered integer representative of $q_v v_i \bmod p_i$. It is nonzero since $p_i > Q$ and $v_i \neq 0$ there. We have $|b_{v,i}| \leq p_i/N$.

For each nonempty support I choose a real λ_I which separates all the at most r nonzero integers $m_v = \prod_{i \in I} b_{v,i}$ in the sense $||\lambda_I m_v|| \geq 1/(4r)$. Uniform random $\lambda_I \bmod 1$ fails for any one of these with probability $1/(2r)$, so such a choice exists. Supports containing no prescribed directions can have coefficient zero. This choice is made after fixing q_v for every vector. It does not need to separate every possible slowing multiplier for the same vector.

Write $[x_i]$ for the integer representative in $[0, p_i - 1]$, and set

$$F_j(x) = \sum_{|I|=j} (\lambda_I / M) \prod_{i \in I} [x_i], 1 \leq j \leq s.$$

Only supports contained in `supp(x)` contribute, so the functions are well-defined on the direct sum. Choose independent random translates of arcs of length η in each circle, and keep

those x for which every $F_j(x) \bmod 1$ avoids its designated arc. The probability of deletion at any fixed x is at most $s\eta$. The coefficients divided by M are actual real numbers: one first selects a real representative of each λ_I and then divides it; no division in a finite cyclic group is needed.

Fix $v \in V_J$ and observe $H = 1+Q((s+1)K-1)$ positions in its progression. Sample every q_v-th position for $(s+1)K$ slots, which fit in the stated length. In each active coordinate the integer representative moves by $b_{v,i}$ except when passing the end of its representative interval. Its total absolute unwrapped displacement along these slots is less than $p_i/2$, so it has at most one wrap transition. Cutting at the at most s wrap transitions leaves a subrun of K or more slots on which every coordinate is an affine function of the sample index (constant on the inactive axes). For $j = |I|$, the leading coefficient of F_j on that subrun is exactly $\lambda_I m_v/M$. Only the term on the exact support uses j varying coordinates. The other terms have strictly lower degree, irrespective of all the passive coordinate values. The stated hitting input applies, including arbitrary initial shift, and forces a deleted site. Thus the survivor set avoids every H-term progression with a step in V_J, for every choice of the arcs.

On each of the finite coordinate subgroup Følner sets take some choice of the arcs attaining retained density at least $1 - s\eta$. Translation-average these full survivor configurations on that Følner set and pass to a weak limit of the binary laws. The limit is invariant and still simultaneously avoids every specified H-term progression, since these are local cylinder conditions. This proves $b_{P_0}(V_J) \geq 1 - s\eta$; coordinates before J can be passive or handled by restriction/coinduction. Letting $\eta \to 0$ yields the limit one. The argument provides no such conclusion if there are unboundedly many unrelated vectors on each fixed support, or if the support sizes grow without bound.

R241. Arbitrary tails in the squarefree torsion subgroup

Scope and status. Result with proof.

Result. Rational saturation over the squarefree torsion group has capacity equal to the limit of tail capacities, and admits integer realization; bounded-exponent lift amplification is explicit.

Scope. The equality is a tail-limit statement, not equality with every native tail capacity.

Qualifications. A native obstruction alone may disappear after rational lifting.

This is a consequence of the suspension formula [R247]. Let p_i be pairwise distinct primes and $P_0 = \bigoplus_i C_{p_i}$ the subgroup of $\mathbb{Q}/\mathbb{Z} = P$ generated by its order-p_i points. For arbitrary $V \subseteq P_0 \setminus \{0\}$, including steps supported on several prime coordinates, set $V_J = V \cap \bigoplus_{i \geq J} C_{p_i}$. Then

$$b_{\mathbb{Q}}(\pi^{-1}V) = \lim_{J \to \infty} b_{P_0}(V_J).$$

In particular every such tail limit is an integer critical threshold. No coloring or uniform forbidden length across J is assumed.

For the upper bound, given M, take J so large that M is prime to every p_i, $i \geq J$. On the tail direct sum multiplication by M is an automorphism. Hence $[M]^{-1}V$ in P contains an automorphic copy of V_J in that tail subgroup. Restrict an avoiding action to this subgroup. Its capacity there equals $b_{P_0}(V_J)$, by restriction in one direction and coinduction from the subgroup on its identity-coset coordinate in the other. Since the tail capacities increase with J, the quotient cap is bounded by their limit, and the suspension formula gives the upper bound.

For the reverse bound, coinduce a V_J-avoiding P_0-system to P, and pull the resulting P-action back through $\pi : \mathbb{Q} \to P$. Each remaining direction in $\pi^{-1}(V \setminus V_J)$ has nonzero image in at least one of the first $J-1$ prime subgroups. For each such prime there is a character on $\mathbb{Q}$ obtained by composing π with projection of P to its Prüfer p_i-component, followed by its usual inclusion into the circle. On these steps a character has distance at least $1/p_i$ from zero. Its M-th root is the character $x \mapsto \chi(x/M)$. For an arbitrarily short open arc I, choose M divisible by all positive integers up to $2/|I|$. Every rotation velocity u with $\|Mu\| \geq 1/p_i$ hits I from every start in a common finite time, by the circle density/compactness argument of the suspension proof. The complement of I in the independent root-rotation system is an arbitrarily near-full avoiding event for this prime. Independent products over the finitely many primes give an arbitrarily near-full repair for all remaining steps. Thus $b_{\mathbb{Q}}(\pi^{-1}V) \geq b_{P_0}(V_J)$ for every J. The integer transfer for arbitrary countable torsion-free groups applies to the $\mathbb{Q}$-system.

For one-coordinate templates with a uniform finite-color obstruction, distinct-prime templates are necessary as well as sufficient for an integer separation with such an obstruction. Given a candidate D with $b(D) < \delta$ and a c-color obstruction at length H, extract for each h a finite $F \subset D$ with $a_h(F) < \delta$. In each prime cycle of sufficiently large size $p > 2H \max F$, its residue image still has cap $< \delta$, since any cyclic mask lifts to a periodic integer mask. To retain colorability, color representatives $0, \ldots, p-1$ by the integer blocking color together with a bit specifying a half of that interval. A progression of small positive step wrapping modulo p visits opposite halves immediately before and after the wrap, whereas one with no wrap is an integer progression. This gives a $2c$-color obstruction at the original H and arbitrary large distinct primes.

Bounded-order residue templates amplify under roots If $P \geq 2$ is an integer and $A \subseteq (\mathbb{Z}/P)^n$ of density $a > 0$ contains no whole cyclic line of any step $s \in S$, there is a P^k-term avoiding mask in $(\mathbb{Z}/P^k)^n$ of density $1 - (1-a)^k$ for **every** lift of every step in S. Mark a vector whenever at least one of its k base-P digit vectors lies in A. These digits are independent and uniform. For a start x and a lifted step d, inductively add jP^l to the time, with $0 \leq j < P$, to arrange that digit l is outside A. Lower digits stay fixed, and the l-th digit ranges along the arithmetic line of step $d \bmod P$. The final time is at most $P^k - 1$. Consequently a bounded list of base exponents P with a fixed lower bound on a cannot yield the desired uniform higher-density forcing through all lifts, regardless of the ranks n or the number of tagged directions. The remaining pairwise-prime possibility involves escaping base exponents.

R247. Periodic suspension and an alternative finite cyclic criterion

Scope and status. Result with proof.

Result. Periodic suspension identifies saturated rational capacity with a supremum over cyclic quotients and gives a finite template criterion for an intermediate integer threshold.

Scope. All-length capacities, with a common positive short witness and uniform long forcing over the required fresh primes and covers.

Qualifications. The compatible finite templates required by the criterion are not constructed.

For a countable abelian group G, a_h is the maximum stationary mass of an event avoiding every h-term progression of the indicated differences, and $b = \sup_h a_h$. Avoidance means that each corresponding intersection has measure zero. h may increase when approximating b.

Removing the integer action Suppose D is a subset of the additive rationals with $D + Z = D$, and let D_M be its image in Q/MZ for positive integral M. Then

$$b_Q(D) = \sup_M b_{Q/MZ}(D_M).$$

One can use only factorials, with the caps increasing as the factorial increases. Pullback proves the direction $\geq$ and monotonicity under divisibility. If `0 in D`, each side is zero. Otherwise take an h-avoiding event A of mass a under a probability-preserving action U of Q on X.

Fix a small open circle interval I of length ϵ and an integer $R > 1/\epsilon$. Take M divisible by each integer from 1 through R. On the product of X with $[0, M)$ with normalized Lebesgue coordinate t, define the measure-preserving suspension flow

$$S_s(x, t) = (U_{M\lfloor (t+s)/M \rfloor} x, (t + s) \bmod M).$$

It commutes with every U_d on the first coordinate. The action $V_d = U_d S_{-d}$ has $V_M = id$, so factors through Q/MZ. Choose $0 < \eta < 1/4$. Mark a point if $U_{floort} x \in A$, the fractional part of t lies in $[\eta, 1 - \eta]$, and $t/M \bmod 1$ is outside I. Its mass is at least $a(1 - 2\eta - \epsilon)$ by Fubini. At time `jd` the tested point of the original action is exactly

$$U_{jd+\lfloor t-jd \rfloor} x.$$

Indeed the clock reduction modulo M cancels its integer carry against the suspension cocycle.

Consider the circle velocity d/M. Remove open neighborhoods of radius η/M around every rational of denominator at most R. For every velocity in the compact remainder, every

forward orbit hits I. This follows from density for irrational rotations and mesh $< \epsilon$ for rational rotations in the remainder. Compactness of the velocity/initial-point pairs gives a common finite hitting time, also for rotations with the negative sign of the suspension flow. Thus a surviving sufficiently long progression must have $|d - n| < \eta$ for an integer n, using the divisibility of M. At every surviving slot its fractional part $\{t - jd\}$ is away from both endpoints by η, so $\lfloor t - (j+1)d \rfloor - \lfloor t - jd \rfloor = -n$. Through h such slots the tested points form an ordinary progression of step `d-n in D`. Its measure is zero by the original avoidance. The countable choices of d and starting floor present no exceptional-set problem. Enlarging the length to include the common hitting bound therefore gives a genuine avoider for D_M. Send ϵ and η to zero. No claim of preserving h is used.

Equivalently write $P = Q/Z$, $\pi : Q \to P$ the projection and $D = \pi^{-1}(E)$. Under the isomorphism $Q/MZ \to P$ given by division by M this reads

$$b_Q(\pi^{-1}E) = \sup_M b_P([M]^{-1}E).$$

This argument also works when the cyclic kernel is generated by an infinite-order element u in any countable abelian group G. Extend the homomorphism $\nu \to n$ to a real-valued homomorphism $\phi : G \to R$ (divisibility of the additive reals), use U_{Mu} in the suspension and use the flow time $-\phi(d)$. For $D + Zu = D$ the corrected difference is $d - \nu$. The rank-one case alone is needed below.

Sufficient finite templates using unlifted cycles Suppose there exist fixed c, H, $0 < \delta 0 < 1$, and distinct primes p_i with nonempty $S_i subset C_{p_i}$, such that each S_i has a c-coloring of its cycle avoiding monochromatic H-runs. Suppose also that for every h there are arbitrarily late i at which every set of density $> \delta 0$ in C_{p_i} contains an h-run with step in S_i. The finite steps are tested on the base prime cycle. Equivalently one can demand an unbounded sequence of forcing lengths at successively distinct primes.

Identify C_{p_i} with the subgroup of order p_i in the ith primary component of P. Use $E = \cup_i S_i$ with each S_i on its own pure component. For each M, arbitrarily late active primes are coprime to M. $[M]^{-1}E$ therefore contains, inside the order-p_i subgroup, the automorphic copy $M^{-1}S_i$. Restrict any P-action to this finite subgroup. Averaging an event of density $> \delta 0$ over its finite orbit and then over the ambient probability space gives a positive-measure family of starts with orbit masks of density $> \delta 0$. Such a mask has a progression, and the finite number of positions and steps supplies a positive-measure progression of the original event. This works for every h, so the caps in the suspension formula are at most $\delta 0$.

For a low witness, extend each cyclic coloring to its primary component by choosing representatives of the cosets of the subgroup of order p_i, taking the identity coset representative to be zero. Sum the colors of the finitely many nonzero primary coordinates modulo c after subtracting the color at zero. Along an E-step only one coordinate changes and the corresponding color varies on each H-run. Averaging translates of the partition over finite subgroups exhausting P yields a stationary positive-density color-class witness (density at least $1/c$), and pullback along π preserves it. Consequently $1/c \le b_Q(\pi^{-1}E) \le \delta 0$.

For integer realization enumerate finite rational packets F_j exhausting D, with a common denominator L_j on packet j. Place the bounded integer packet $L_j F_j$ on its own axis of a countably generated free abelian group. The map sending the jth generator to $1/L_j$ pulls

any Q-witness back to a witness there. Conversely an avoiding stationary law on the axes yields single-axis laws for each packet, which one coinduces from $L_j^{-1}Z$ to Q on the identity coset coordinate. Compactness of stationary binary laws gives a Q-witness imposing every packet; thus all fixed-length caps agree. The integer transfer [D06] now preserves b. Signs of integer differences are harmless by progression reversal.

This reduction leaves the construction of compatible prime templates open. It does not permit one to use the same small prime over and over. Any fixed list of primes eventually divides some M in the supremum. The earlier criterion that asks for all root depths at a single period remains sufficient and does not follow for a specified row from its base-cycle estimate. The present criterion uses different, coprime fresh periods to deal with each fixed M.

R261. A finite prime-product criterion for an exact prescribed density

Scope and status. Result with proof.

Result. Exact intermediate integer capacity is equivalent to one-axis long forcing together with compatible joint short masks on all finite prime products.

Scope. An equivalence of finite criteria, using a tail limit and the integer endpoint theorem.

Qualifications. Separate short masks on individual axes do not establish the joint test. No first intermediate row system is constructed.

The previous rational prime-host theorem [D17] already proves universality of a fixed list of rapidly growing distinct primes for every integer D, including the fixed-order tail formula and the representative construction for joint avoiders without a finite-color assumption. The following reformulates that theorem and the squarefree-tail result [R241] for prescribed exact endpoints.

A number $\delta \in (0,1)$ is an attained integer critical density if and only if there exist distinct primes, nonzero residue sets $S_i \subset C_{p_i}$ (empty rows can be ignored), and numbers $\epsilon_i \downarrow 0$ satisfying both of the following conditions. Here indices for the long tests start at $i \geq 2$.

- **One-axis upper tests.** The largest density avoiding i-progressions in C_{p_i} of step in S_i is at most $\delta + \epsilon_i$ for each i.
- **Joint short tests.** For every $r \geq 1$ there exist $k_r \geq 2$ and J_r such that for *every* $N \geq J_r$ the group $\prod_{J_r \leq i \leq N} C_{p_i}$ admits a mask of density at least $\delta - 1/r$ avoiding k_r-progressions in every pure-coordinate step $S_i e_i$.

For necessity take an integer D with $b_Z(D) = \delta$. At each long order i take a finite $F_i \subset D$ with $a_i, Z(F_i) \leq \delta + \epsilon_i$. Choose distinct $p_i > 2 \max F_i$ sufficiently large that $\sum_i \max F_i/p_i < \infty$. Reduction gives the one-axis upper tests by periodic lifting. For each r take a fixed-order original avoider of mass $> \delta - 1/(2r)$, at order k_r. Choose J_r late enough that the endpoint losses $k_r \sum_{i \geq J_r} \max F_i/p_i < 1/(2r)$. The representative construction yields all joint short tests.

For sufficiency, stationarity restricted to a single axis yields $a_h, P_0(V_J) \leq \delta$ for all h, J by passing along $i \to \infty$, since an h-avoider is also an i-avoider for $i \geq h$. The joint short tests give $b_{P_0}(V_{J_r}) \geq \delta - 1/r$ by the finite-mask compactness argument. Thus the tail limit is exactly δ, and squarefree-tail suspension and integer realization apply. The strict endpoint result for the integer definition then yields δ-density-largeness at equality.

For merely *some* intermediate value it suffices to fix one short order k, one density $a > 0$, and a single tail for the joint lower tests, together with an upper bound $\delta < 1$ on the one-axis long tests. Conversely such joint data are necessary from any intermediate integer example. Separate one-axis short masks do not establish joint compatibility. In graph language the order-two joint test is independence in the Cartesian product of the one-axis Cayley graphs. For order $k \geq 3$ use the hypergraph in which each edge is an entire k-progression on one axis. A common bounded coloring is one sufficient way to glue the tests, as in fresh prime tests [R128].

This reduces all lengths and the exact threshold to finite cyclic products with no common-color assumption. It does not produce prime rows satisfying both test directions at a first intermediate value.

R269. Finite-dimensional prime rows and the dependence on rank

Scope and status. Result with proof.

Result. Suspension and squarefree-tail formulas extend to fixed rational rank; a rank-dependent quantitative version gives a variable-rank finite-template criterion.

Scope. The suspension denominator and required length depend on rank, which must be fixed before choosing the prime in the criterion.

Qualifications. No satisfying variable-rank rows are constructed. Whole-line forcing does not bypass the denominator dependence.

The suspension argument extends from $\mathbb{Q}$ to each fixed rational rank m. Write $P = \mathbb{Q}/\mathbb{Z}$, let $\pi_m : \mathbb{Q}^m \to P^m$, and set $D = \pi_m^{-1}(E)$. Then

$$b_{\mathbb{Q}^m}(D) = \sup_{M \geq 1} b_{P^m}([M]^{-1}E).$$

One proof successively applies the cyclic-kernel lemma in the suspension note [R247] to the m standard basis generators. Later lattice generators still have infinite order in the partial quotients, and saturation by each of them is preserved. The result has independent denominators $M_1, \ldots, M_m$, which a common multiple refines by pullback. Equivalently, use one suspension clock per coordinate and take their joint product.

For fixed m and distinct primes p_i, put $P_0 = \bigoplus_i \mathbb{F}_{p_i}^m \subset P^m$. For $V \subset P_0 \setminus \{0\}$ let V_J denote the steps with all prime coordinates before J zero. Then

$$b_{\mathbb{Q}^m}(\pi_m^{-1}V) = \lim_{J \to \infty} b_{P_0}(V_J).$$

The upper proof of squarefree tails [R241] applies unchanged, since multiplication by M is an automorphism of each fresh vector group. For the lower bound, each of the finitely many head vector groups is detected by its finitely many coordinate characters. The corresponding roots on $\mathbb{Q}^m$ give arbitrarily near-full avoidance for steps with a nonzero head, while a coinduced and pulled-back tail witness handles the other steps. Thus finite-rank prime vector spaces can replace prime cycles in the same sufficient criterion, and the common finite-color low witnesses sum on their respective primary components. Transfer from a countable torsion-free group gives an integer realization.

There is a useful quantitative observation for variable rank. For any fixed m, h, ϵ there are integers $M = M(m, \epsilon)$ and $L = L(m, h, \epsilon)$ such that an h-avoider of mass a for D gives an L-avoider of mass at least $a - \epsilon$ in the quotient for M, independently of D and the starting system. In the simultaneous suspension use the same normalized Lebesgue clock on $[0, M)$ independently in each coordinate. Exclude an arc of length $< \epsilon/(3m)$ and a band of width $2\eta < \epsilon/(3m)$ near integer boundaries on each clock. Take M a multiple of all denominators through $R > 3m/\epsilon$ chosen larger than the reciprocal arc length. The corrected action is $U_d \prod_i S_{-d_i}^{(i)}$, where the i-th suspension wraps by U_{Me_i}, so it kills $M\mathbb{Z}^m$. At time td the old action sees $td + \lfloor x - td \rfloor$ componentwise, where x is the starting clock vector. Away from a fixed small neighborhood of the finitely many velocities of denominator at most R, a rotation hits the excluded arc within a common finite time. Hence every long surviving progression must have each d_i within η of an integer n_i; the boundary bands enforce the same rounded integer at all retained slots. The recorded old sites form an h-progression

of difference $d - n \in D$. This is the asserted length bound. It is not a dimension-free comparison.

For example, a sufficient variable-rank criterion is as follows. For every j choose m_j first, then find a prime $p_j \nmid M(m_j, 1/j)$, a subset $S_j \subset \mathbb{F}_{p_j}^{m_j} \setminus \{0\}$ forcing length $L(m_j, j, 1/j)$ above a fixed density $\delta_0 < 1$, and a coloring of that vector group with fixed c colors avoiding length h_0 at S_j. Regard S_j as the order-p_j vector subgroup of P^{m_j}. Saturate its preimage as a row on $\mathbb{Q}^{m_j}$, and put those pure rows on a direct sum of the rational vector spaces. Restricting any stationary h-avoider to row $j \geq h$ and using the quantitative comparison bounds its mass by $\delta_0 + 1/j$; use the automorphic copy of S_j in the M-quotient. The sum of the pulled-back row colors supplies a positive short obstruction. This would give a first intermediate integer threshold. A finite proper coloring of the order-p_j subgroup extends to P^{m_j} by coloring each coset by its translate from the subgroup. No condition that all rows use the same rank is needed. No such rows are established here.

There is a simple obstruction to a stronger demand in this criterion. In $\mathbb{F}_p^m$, deleting the m coordinate hyperplanes leaves density $(1 - 1/p)^m$ and misses every *whole* line. Thus whole-line forcing below a fixed density requires m at least a constant multiple of p. In that regime one cannot infer $p \nmid M(m, 1/j)$ from growth of p alone. Finite-length forcing, with the dependence on the suspension parameters respected, is the relevant requirement.

Chapter 6

Critical spectra, interpolation, and fixed-system thresholds

R004. Every exact threshold on one countable abelian group, with the required endpoint

Scope and status. Result with proof.

Result. One countable mixed group realizes every critical value in $(0, 1)$, with endpoint recurrence and infinite-order steps.

Scope. The horizontal endpoint argument uses the finite-color hierarchy; the newer distinct-prime mixed-group theorem is a stronger replacement for the main exposition.

Qualifications. R204 gives the prime-product mixed-group theorem. The present entry supplies a separate bounded-exponent construction.

Let $G = \mathbb{Z} \times \bigoplus_{p \text{ prime}} \mathbb{F}_p^{(\mathbb{N})}$, using countably infinite direct sums, and use the stationary definition of density largeness for its actions. For every $0 < t < 1$ there is a set of infinite-order steps with critical value t such that each fixed-length avoidance cap is strictly below t. Consequently it forces all lengths on events of measure at least t, and it fails at every smaller threshold. This is an analogue on a group with torsion, not a positive integer construction. The strictification below uses the finite-color hierarchy input [R139] that for every integer q there is a q-large integer set E of critical value one. The vertical spectrum construction is independent of that input.

Finite odd tag. Choose a prime p and $K = F_p^m$ with a nonempty set S of nonzero velocities. An allowed $B \subset K$ contains no full affine line in any direction of S; let α be the largest possible density. Such α are dense in $(0, 1)$: fix a target s in that interval, take p large, and take m so large that the maximum for all directions is below s by the density Hales–Jewett theorem on $[p]^m$. Start with no directions and add them one at a time. One new direction reduces maximum size by at most a factor $1 - 1/p$, by deleting a uniformly translated hyperplane meeting all its lines. At the first crossing the value lies between $(1 - 1/p)s$ and s. Choose odd p and such α with $\max(0, 2t - 1) < \alpha < t$.

Finite interpolation. Start with a Boolean line generated by e; attach the amalgamated Hamming blocks [R021], with their antipodes identified with e. At a finite stage let L be the group, $D_L = e + \Sigma(K_i - K_i)$ (with K_i the radius-one balls), and set $W = D_L \times S \subset L \times K$.

Every W-direction has order 2p. The cap is the maximum relative size of a mask avoiding full W-orbits. This follows by averaging over the regular action, including for nonfree actions. Regard a mask as sections $B_x \subset K$. For each s-line l the vertices whose sections contain l form a D_L-independent set. Indeed the orbit of (d, s) visits every point of l in each of its two x-sections, since p is odd. This characterization is exact. With no Hamming blocks, $B_0 \cap B_e$ must be line-free, so the maximum is $(1 + \alpha)/2$, attained by filling one section entirely and the other with a maximizer in K.

After adding a block of size $2k + 1$, the maximum b' satisfies $c(k)b \le b' \le b$, where b is the previous maximum and $c(k) = 1 - \binom{2k+1}{k}/2^{2k}$. The upper bound is restriction to the embedded old subgroup and coset averaging. On the new cube use the disjoint radius-$(k-1)$ balls about zero and the antipode. Their union has density $c(k)$. Give them labels 0 and 1; antipodal translation exchanges labels, and two safe points at Hamming distance at most two have the same label. For an old mask on $L \times K$, retain only safe points of the new cube and evaluate the old mask at $(x + label \cdot e, y)$. This prescription descends through the antipodal identification. Its mass is exactly $c(k)b$. Any full W'-orbit in it has identical label at its two safe new-block positions when the common central flip is represented in the old L, so it would project to a full forbidden old orbit. This proves the lower bound.

If infinitely many copies of a fixed size k are attached, the bound for each line-containing event is $(1/2)\prod c(k_i)$ by the Boolean theorem, tending to zero. The number of lines in K is finite; outside the union of these events, sections have density at most α. Thus finite caps decrease to α; the constant choice of a line-free section gives the lower bound α. Explicitly the upper is $\alpha + (numberoflines)(1/2)\prod c(k_i)$.

Starting above t, append copies of one size up to the last cap still above or equal to t before the next copy would cross below it. Such a crossing occurs after finitely many copies by the preceding bound. At that point the cap is at most $t/c(k)$ unless it already equals t. Choose successively larger k so $c(k) \to 1$, and repeat, stopping if equality occurs. The decreasing caps tend to t. In the infinite case any stationary avoiding mask on the growing locally finite subgroup satisfies every finite upper bound; conversely a stationary weak limit of uniformly translated maximizing masks coinduced from the finite subgroups gives a mask of mean t. Since each parent step has order 2p, avoiding all full orbits has the same meaning as avoiding at length 2p. Embedding into the specified countable direct sum of prime vector spaces does not change any capacity, by restriction and coinduction.

Strict endpoint by horizontal extension. Put $V = L \times K$, with W as above, and consider

$$W' = \{(n, w) : n \ge 1, w \in W\} \cup \{(d, 0) : d \in E\} \subset Z \times V,$$

where E is $|K|$-large and $b_Z(E) = 1$. Product of the maximizing vertical event with horizontal E-avoiders of means tending to one proves $b(W') \ge t$. For any fixed-length W'-avoider A let T be the horizontal action and U the vertical action. For $w \in W$, if the whole U_w-orbit intersection of A had positive mass, ordinary multiple recurrence under T on that intersection would produce a positive return along some (n, w). Therefore A is a vertical full-orbit avoider and $\mu(A) \le t$.

Suppose equality holds. Let F have the points whose entire K-orbit misses A. If F has positive mass, attach an independent regular K-clock to the vertical system and add one clock level on F (the original A is pulled back on all levels). This raises its mass by $\mu(F)/|K|$

without creating any full W-orbit. Indeed if an added point occurs in such an orbit, look at all positions with its same Boolean parity: moving two steps advances only by $2s$ in K, so stays on the same empty K-orbit of the original action. The new clock visits p distinct states there and only one is marked. A full orbit using no added point was already impossible. The resulting excess contradicts the vertical upper bound. Hence $\mu(F) = 0$.

Outside a common null set, at every horizontal time m there is therefore a $k_m \in K$ with $T^m U_{k_m} x \in A$. Color m by one such k_m. By the $|K|$-largeness of E a monochromatic progression of the forbidden length exists, giving an all-A horizontal progression of an E-difference, contradicting the simultaneous pointwise avoidance. Stationary compactness attains each fixed-length cap, so each is strictly smaller than t. Finally use the subgroup restriction/coinduction observation to place every example in the single group G specified above. All the steps in W' have positive horizontal coordinate, hence infinite order.

The usual bounded-exponent lift obstruction applies to this construction on attempts to port it to a torsion-free group: in the interpolation with $t > \alpha$ the Boolean projection has a positive pair-avoiding mask (the union bound over finitely many lines shows its capacity is at least $(t-\alpha)/(numberoflines)$). Taking binary roots can use this mask to block the vertical directions arbitrarily near density one. The additional horizontal-only family E already has the same near-full obstruction on its own. The realization on $Z \times V$, even with exclusively infinite-order steps, does not provide the requested example on Z.

R005. All exact thresholds in one Kronecker rotation

Scope and status. Result with proof.

Result. One fixed ergodic infinite-torus rotation realizes every prescribed relative threshold, with a strict gap at each fixed length.

Scope. The corresponding integer difference sets have unrestricted critical value one.

There is a single ergodic rotation T of the compact infinite torus such that for every $0 < t < 1$ there is a set $D \subset N$ which, in this one Haar probability system, forces positive returns at every length for every measurable event of mass at least t, and fails at every smaller threshold. It can have unrestricted integer critical value one. The construction below uses no weak mixing. In particular it includes the values between one half and two thirds.

Finite product step. In a finite abelian group K take a nonempty finite set S of nonzero velocities. Write $c(K, S)$ for the largest density of a subset containing no whole affine cyclic orbit in these velocities. Finite order identifies this number with the all-length avoidance cap. Denote the full list of these orbit sets by L and its cardinality by C.

Take a fresh prime p coprime to the exponent of K. Add a coordinate $V = F_p^m$ and replace the velocity list by $S \times R$, where R is a nonempty union of nonzero projective directions in V (all their nonzero scalar multiples). The orbit of (s, r) from (x, y) is exactly the Cartesian product of the old orbit and the fresh full line: the two periods are coprime. Let $c = c(K, S)$ and $\alpha = c(V, R)$. Then

$$c \le c(K \times V, S \times R) \le c + C\alpha.$$

The lower bound pulls back an old maximizing mask. For the upper bound, for each old orbit l mark the y whose entire l-section lies in the product mask. This marked set contains no full R-line, by the Cartesian-orbit observation, and has relative size at most α. Outside the union of these C sets the old section has relative size at most c.

Removing one projective direction from R increases the cap by at most $1/p$. Indeed from a maximizing mask for the relaxed family delete all points over any fresh hyperplane transverse to this one direction. This deletes at most $1/p$ of the full vertex set and breaks every reintroduced product orbit. At the other extreme, a single remaining fresh direction has product cap at least $1 - 1/p$, by retaining the complement of such a hyperplane.

Choosing the stages. Start with one odd prime p_0, all nonzero directions in $F_{p_0}^{m_0}$, and choose m_0 sufficiently large that the full-line cap is strictly below the target t, by finite density Hales–Jewett on the p_0 symbols. Given any finite product stage of cap $c < t$, take a new prime p larger than all earlier primes with $1 - 1/p > t$. For all nonzero directions in V, choose m so large that their full-line cap is smaller than $(t - c)/(2C)$, again by density Hales–Jewett. Thus the enlarged cap is still below t. Remove fresh projective directions successively until the cap first reaches or exceeds t; this must happen by the single-direction lower bound. Use the immediately preceding family. Its cap lies in $[t - 1/p, t)$ and is at least the old cap by projection. Repeating with primes tending to infinity gives finite projection caps increasing to t. Only the rank of a *new* coordinate is changed at each stage.

The compact product and strict endpoint. Let $V_\infty = \prod_{i \ge 0} V_i$ with its Haar probability, and $S_\infty = \prod_{i \ge 0} R_i$. All-length capacity in this regular compact translation action

equals t. Lower bounds lift from the finite quotients. For the upper bound, let A be an event avoiding all h-runs at every velocity in S_∞, for fixed $h \geq 2$. Let B_n be the set of first-n coordinate tuples at which the conditional mass of A exceeds $1 - 1/h$. It avoids every h-run at velocities in the finite product $\prod_{i<n} R_i$: if all slots of a projected run lay in B_n, lift its velocity to S_∞ and average over Haar tail starting points. Each slot misses A with conditional probability less than $1/h$, giving positive probability of all memberships. The masses of B_n tend to the mass of A by martingale convergence (indeed their indicators tend almost surely to 1_A). Every B_n is bounded by the all-length cap of that finite stage. This proves the claimed upper bound.

More strongly the fixed-h cap is strictly below t. Choose a coordinate with prime $p_i > 2h$. For each h-avoider A of mass a, some single slice $\{x_i = y\}$ contains at least $(1 - a)/|V_i|$ of the total complement mass. Add this whole slice to A. Every $2h$-block in an allowed velocity has at least two original misses, one in each h-half, apart from a null set for this velocity. These occur at time indices less than p_i apart. Since the ith velocity is nonzero, the two corresponding V_i values differ, and at most one is in the added slice. Thus the enlargement still avoids every $2h$-run. Its mass is at least $a + (1 - a)/|V_i|$; the already proved all-length bound makes this at most t. This supplies a strict gap independent of the choice of A at this h.

A single integer Kronecker system. Fix $X = T^N$ with Haar measure and a translation $T = T_\alpha$ whose positive powers are dense, for instance any α whose scalar coordinates together with one are rationally independent. This choice precedes t. For each constructed product identify the coordinates of each finite V_i with p_ith roots of unity in disjoint torus coordinates. This embeds V_∞ as a closed subgroup of X. The translation action of its elements on X is the regular Haar action on V_∞ times a trivial spectator: disintegrate Haar over the compact quotient X/V_∞ and choose a Borel representative on each coset. A countable dense subset $S_* \subset S_\infty$ suffices for every fixed measurable mask, because translations are strongly continuous in L^1 and progression correlations are continuous in the translation. Disintegration over the spectator can therefore be used simultaneously for the countably many velocities and followed by that continuity. The relative cap and its strict fixed-order bound are the same on X.

Write s_j for a schedule repeating each element of S_* infinitely often, and fix measurable masks A_i in X with masses tending to t, each avoiding every target at some fixed length h_i (finite quotient masks suffice). Inductively choose strictly increasing positive d_j so that $d_j\alpha$ approaches s_j in the compact-group metric with error tending to zero and, for $i \leq j$ and every $l < h_i$,

$$\mu((A_i - ld_j\alpha)\triangle(A_i - ls_j)) < 2^{-j}.$$

These are finitely many open-neighborhood requirements, possible by dense positive powers and translation continuity. Also require the first finite stage projection to be uniformly separated from zero in the coordinate torus; each s_j has a nonzero first-stage p_0 residue. Put $D = \{d_j\}$. Every actual D-avoiding mask avoids each limiting target by a constant-target subsequence. Thus the strict fixed-order upper bounds pass to this integer set.

Conversely, for a fixed i, the sum of the masses of actual h_i-run starting cylinders inside A_i for $j \geq J \geq i$ is at most $h_i \sum_{j \geq J} 2^{-j}$. Delete the union of those starts. To handle the finitely many earlier d_j, delete a narrow open strip in the first circle coordinate. For each

early step its first-coordinate rotation is irrational and every orbit meets the strip, uniformly in the start in some finite number of steps by compactness. Thus all early directions are blocked at one finite length and the additional Haar loss can be arbitrarily small. Taking J large and then i large proves all-length lower value t. For every lower density threshold $t' < t$, some such avoiding event has mass above t'. The first-projection separation gives unrestricted integer critical value one by the Bohr root sieve, so the proof concerns only the one stated rotation. The geometric construction may depend on t, whereas the dynamical system X, T is unchanged.

R006. Every critical value in the same weakly mixing integer transformation

Scope and status. Result with proof.

Result. One fixed weakly mixing Gaussian transformation realizes every prescribed relative threshold, including the endpoint.

Scope. The initial construction is Bohr-separated and has unrestricted critical value one; no all-system integer threshold is obtained.

There is a single weakly mixing probability-preserving integer transformation T such that for every $0 < \delta < 1$ there is a set $D \subset N$ with

$$\sup\{\mu(A) : \quad \text{some} \quad h \geq 2 has \mu(\bigcap_{j<h} T^{-jd} A) = 0 \quad \text{for every} \quad d \in D\} = \delta.$$

This quantifies over events in **this fixed system**. Every measurable event of mass **at least** δ has a return at every finite length, while every smaller threshold fails. The same D can be chosen Bohr-separated on the integers and has unrestricted integer critical value one. The stronger packet construction [R020] makes the sets nil-Bohr dense with the same relative statement and unrestricted endpoint. Neither construction proves the unrestricted integer conjecture.

A universal Gaussian system and a transfer lemma Choose a Cantor Kronecker set $E \subset T$ and a nonatomic probability σ on it. One elementary construction nests disjoint binary-tree intervals: choose rationally independent centers together with 1 for the finitely many intervals at each level; for every assignment from a finite $1/k$-net of phases on that level, choose an arbitrarily large positive exponent approximating the prescribed values at all its centers by Kronecker's theorem, then shrink each interval about its center to keep all these comparisons valid. Split and repeat, with shrinking diameters. Every continuous unimodular function on the Cantor limit is uniformly approximable by arbitrarily large positive powers of the coordinate character. The fair Cantor product measure is nonatomic.

On the realification of $L^2(\sigma; C)$, let the first-chaos unitary be multiplication by the coordinate $e(\theta)$. Take its isonormal Gaussian action T. Its first-chaos spectrum is nonatomic, hence its Gaussian action is weakly mixing (on each nonconstant chaos the spectrum is dominated by symmetric convolution powers of nonatomic measures and their reflections). Every measurable phase multiplier $f : E \to T$ is a strong limit of positive powers on the first chaos: approximate it in $L^2(\sigma)$ by cylinder-constant phases and use the Kronecker property; for bounded multipliers the same gives strong convergence on every vector. Strong convergence of the associated Gaussian transformations follows, for instance by testing their tensor-chaos polynomials and then using density. Partition E into countably many disjoint positive-measure spectral regions, giving independent infinite-dimensional Gaussian factors; unused regions may act trivially under any specified target multipliers. Fix one point $\theta_0 \in E$. Since σ has no atom there, all these approximations can simultaneously require $d\theta_0 \to 1/2 \bmod 1$, by modifying the continuous approximant on an arbitrarily small clopen neighborhood of θ_0 before choosing its positive power.

For a countable family L of these Gaussian transformations let $c(L)$ be the analogous supremum for the required h-orbits of each map in L. This is a capacity within the one fixed Gaussian probability space. It is realized as the displayed integer critical value for some D. To see this, enumerate a sequence of targets $V_n \in L$ repeating each infinitely often. Choose masks A_k of measures tending to $c(L)$ that avoid all targets at lengths h_k. (The zero case requires only the upper argument.) Choose successively larger positive d_n so T^{d_n} is strongly close to V_n on the first n members of a determining set in the measure algebra, with error tending to zero. Additionally require $\mu(T^{-jd_n}A_k \triangle V_n^{-j}A_k) < 2^{-n}$ for $k \le n, j < h_k$; these are only finitely many strong-neighborhood constraints and powers are continuous in the strong topology on transformations. Require $\|d_n\theta_0 - 1/2\| < 1/(10n)$. Set $D = \{d_n\}$.

Every integer avoiding mask avoids each target by strong convergence on its constant-target subsequence. In the other direction fix k. Starting sufficiently late, the union of all actual h_k-run cylinders inside A_k costs at most $h_k \sum_{n \ge J} 2^{-n}$. Delete it. The finitely many early differences can be avoided at a common larger length in the same system by intersecting a mask of measure arbitrarily close to one. Concretely, in a very tall Rokhlin tower for the aperiodic T delete the uncovered part, an upper strip of width at least $P \max_{n<J} d_n$, and every level with index $0 \bmod P$, where P is a large prime exceeding all early steps. In a retained run of P slots the level indices must hit the deleted class, and the extra strip prevents leaving the tower. First choose P and then the tower height to make the loss small. This proves equality of the capacities. The fixed condition at θ_0 makes every D so obtained Bohr-separated, hence globally of integer critical value one by the usual circle-root construction.

Compact order-two graphs For $0 < r \le 1/2$, choose finite Boolean groups G_i and nonempty direction sets S_i whose normalized independence ratios $r_i < r$ tend to r. They exist from the explicit finite antipodal Hamming products in boolean-antipodal-products.md [R021]: the factors $c(k)$ tend to one, so multiplying $1/2$ by just enough equal factors to go strictly below r approaches r from below, also when $r = 1/2$. Replace each graph by its connected component through zero, the subgroup generated by S_i; all components are translates with the same independence ratio. On $K = \prod_i G_i$ use the compact velocity set $S = \prod_i S_i$. Every velocity is an involution. The edge graph of the nth finite projection is the categorical product of the finite Cayley graphs. Its independence ratio is $\max_{i \le n} r_i$, since the reciprocal of the independence ratio is the fractional chromatic number for a finite vertex-transitive graph and fractional chromatic numbers take their minimum in a categorical product. See X. Zhu, The fractional version of Hedetniemi's conjecture is true, or H. Zhang's direct vertex-transitive result.

In the regular compact Haar action of K its all-order critical value for S is r. It suffices to use any countable dense subset $S_* \subset S$, by L^1-continuity of translations, and avoiding any length is exactly pair avoidance for involutions. If a measurable B avoids these pairs, its conditional density on the first n factors, thresholded at $> 1/2$, is independent for the finite projection. Indeed any two neighboring projected fibers of such densities would intersect after translation by a lift of that edge. These finite masks tend in measure to B by martingale convergence, giving the upper bound. Pulling back finite independent masks proves the lower bound.

No independent measurable mask has mass r. We give the measurable version of the strict-ratio part of Zhang's argument. Fix a single finite connected factor H of ratio $r_H < r$; let G denote the remaining product, whose supremal independent mass is still r by

the preceding calculation. In G, use the Cayley graph for a countable dense family of its tail velocities, so neighborhoods of measurable sets are measurable; all product edges with each finite H-velocity are legitimate by continuity. For an independent $P \subset G$, its closed graph-neighborhood $N_G[P]$ (itself and all its neighbors, not topological closure) satisfies $\mu P \leq r\mu N_G[P]$: insert P into a uniformly translated independent mask of mass near r, deleting that mask on $N_G[P]$, and average. The same inequality holds in finite H with r_H. The null exceptions from countably many edges may throughout be removed on an invariant conull set.

Let $B \subset G \times H$ be independent and $X_g = \{y : (g, y) \in B\}$. Let Y_g consist of the vertices of X_g having no H-neighbor in X_g. Put $P_y = \{g : y \in X_g \setminus Y_g\}$. Each P_y is independent in G: if neighboring g, g' both belonged, a neighbor of y in X_g would conflict with $y \in X'_g$. Moreover $N_G[P_y]$ is disjoint from $\{g : y \in N_H[Y_g]\}$. For the point $g \in P_y$ itself this is exactly the definition of the isolated part. For a neighbor g', if y neighbors a vertex of Y'_g, that conflicts with $y \in X_g$; and if $y \in Y'_g$, it conflicts with a neighbor of y in X_g. Hence

$$\mu B = (\int |Y_g| d\mu(g) + \sum_y \mu P_y)/|H|$$

$$\leq r + \int (|Y_g| - r|N_H[Y_g]|) d\mu(g)/|H| \leq r.$$

For equality the isolated part must vanish almost everywhere since $r_H < r$. Then $P_y = \{g : y \in X_g\}$ modulo null sets, and every P_y is independent of mass r since their average is r. For adjacent y, y', the union $P_y \cup P'_y$ is independent by cross-independence of their sections, so $P_y = P'_y$ almost everywhere. Connectivity shows that B is independent of the coordinate H. Applying this to every finite single factor makes B invariant modulo null sets under all finite-coordinate changes. It is therefore a tail event and has Haar mass zero or one, inconsistent with $0 < r < 1$. Disintegrating over any trivial spectator preserves this nonattainment.

This regular action is available inside the Gaussian host. Represent every binary coordinate by a sign multiplier on one spectral region. Almost surely a chosen nontrivial real Gaussian coordinate of that region is nonzero, so its sign is a uniform bit exchanged by the multiplier. Normalize the entire region by that sign. This identifies its space measurably with one independent uniform bit times the normalized data; the bit is independent by Gaussian symmetry. Doing this on all regions identifies the compact action with its regular Haar action times a trivial spectator. Arbitrary masks disintegrate along the spectator, preserving the cap because there are only countably many initially tested targets.

Odd tags and multiplication For any odd $q \geq 3$, include one extra cyclic uniform phase of order q by using a complex Gaussian pair on another region. Its angle modulo $2\pi/q$, together with compensating rotation of the whole region, gives a measurable product of a regular C_q-coordinate and invariant data, just as with the sign. Use target velocities $(s, 1) \in K \times C_q$, $s \in S_*$. Their order is $2q$. For a measurable mask let $B \subset K$ be its base positions whose vertical section contains every tag. A full $(s, 1)$ orbit is present exactly where two s-neighboring positions are in B, since the even and odd time indices each visit all q tags. This applies in measure by finite averaging. Thus its density is at most $1 - 1/q + r/q$.

Conversely fill $q-1$ tags at every base point and all tags on an independent base mask of mass approaching r. This proves equality. The endpoint $2/3$ separately comes from a pure order-three phase. The intervals $(1-1/q, 1-1/(2q)]$ for $q = 3, 5, 7, \ldots$ overlap and cover $(2/3, 1)$.

The tagged value with $r > 0$ cannot be attained by an avoiding mask, since equality would force its fully tagged sections to give an independent mask of mass r. Finally, attainable critical values multiply within this same fixed Gaussian host. Suppose countable target families L_1, L_2 on disjoint infinite Gaussian factors have values b_1, b_2 within their respective factors. On the product use all (V, W^m) for $V \in L_1, m \in N$, together with (I, U) for $U \in L_2$. Here W is any weakly mixing Gaussian multiplier on the second factor, such as the restricted original coordinate rotation. Every resulting transformation is again a single measurable phase multiplier on the host's first chaos; these actions need not be full integer powers until applying the transfer lemma.

If A avoids them all at length h, then almost every vertical section A_x avoids L_2, hence has mass at most b_2. Its positive-support set $B = \{x : \mu_2(A_x) > 0\}$ avoids every $V \in L_1$. In fact if a positive measure of first coordinates saw only positive sections at $x, Vx, \ldots, V^{h-1}x$, then for each such x average over m the correlations in the second system of the h sections at the powers W^{jm}. Weak mixing gives the product of their measures as the Cesàro limit; integrate by bounded convergence. A positive mean contradicts avoidance of every (V, W^m). Consequently $\mu(A) \le b_2 \mu_1(B) \le b_1 b_2$. Conversely products of masks approaching the two capacities avoid all the listed transformations at the maximum of their two lengths. Any remaining spectator factor acts trivially and changes neither bound. There are countably many disjoint nonatomic spectral pieces available to install the finitely many factors for each desired construction.

Given $0 < \delta < 1$, choose a positive integer k with $\delta^{1/k} > 2/3$. Use a tagged construction with $r > 0$ for that root and multiply k independent copies. Equality for a product mask in the upper argument would force an avoiding positive-support set of mass b_1 in the first factor. Nonattainment in the first tagged factor therefore gives nonattainment for the product. The transfer lemma applied on the **unchanged** weakly mixing Gaussian system gives the claimed integer set and system-relative cap, with nonattainment preserved by its strong-limit upper bound. The uniform finite-control property needed for an unrestricted upper bound does not follow from strong approximation of the target maps: the approximating directions are chosen for determining observables and countably many specific lower masks, while each different upper test event uses convergence along a whole subsequence.

R021. All avoidance thresholds up to one half on one countable Boolean group

Scope and status. Result with proof.

Result. Every value in $(0, 1/2]$ is an attained pair-avoidance maximum on one countable Boolean group.

Scope. The endpoint is attained by an avoider, opposite to the endpoint recurrence required on the integers.

Let L be a countably infinite-dimensional vector space over F_2, acting by measure-preserving transformations. For every $0 < \delta \leq 1/2$ one can choose $D \subset L \setminus \{0\}$ with **maximum** measure of a progression-avoiding event exactly δ. This even holds at the fixed length two, since every nonzero element of L is an involution. The endpoint is attained by an avoider: sets of measure strictly greater than δ force every length and some event of measure δ avoids all lengths ≥ 2. Thus this is a related exact spectrum, with a different group and different endpoint behavior from the integer question.

The blocks. Take $L_i = F_2^{2k_i+1}$, $k_i \geq 1$, e_i its all-ones element, and K_i the ball of radius one about zero. Identify all e_i to one involution e by quotienting the direct sum by the span of $e_i - e_j$. The resulting L is a countably infinite-dimensional binary vector space whenever the list is infinite. Each L_i embeds, and the subgroup on the first n blocks has size $\prod_{i \leq n} |L_i|/2^{n-1}$. For a finite list I write $K_I = \sum_{i \in I} K_i$ in this quotient. Let D be the union, over finite I, of $e + K_I - K_I$. It excludes zero: a vector from $K_i - K_i$ has weight at most two and cannot be e_i, of weight at least three. In a finite group on the chosen blocks a mask A avoids D-pairs if and only if $A + K_I$ does not contain an e-pair. This follows simply by writing out the difference of two elements of the sumset.

One-block expansion. Let $V_r = \sum_{j=0}^{r} \binom{2k+1}{j}$ and put $c(k) = V_{k-1}/V_k = 1 - \binom{2k+1}{k}/2^{2k}$. If $B \subset F_2^{2k+1}$ and $|B + K| \leq 2^{2k}$, then $|B| \leq c(k)|B + K|$. By Harper's vertex-isoperimetric theorem, one may minimize the neighborhood at a given size by filling the lowest Hamming levels first. The ball of radius k has exactly half the cube, and adding any point of weight k to the radius-$(k-1)$ ball gives a strictly larger neighborhood. Thus $|B| \leq V_{k-1}$. For a partly filled level r, where $|B| = V_{r-1} + u\binom{2k+1}{r}$, the neighborhood size is at least $V_r + u\binom{2k+1}{r+1}$. This follows by counting upper-shadow incidences, or directly for $r = 0$ at the only nonempty size one. The ratios V_{r+1}/V_r decrease with r by log-concavity of partial sums of binomial coefficients. Comparing the bounds at the two endpoints $u = 0{,}1$ gives the assertion. The radius-$(k-1)$ ball attains equality.

Finite capacity. Starting with a pair-avoiding mask A on the first n blocks, add K_i one at a time. All intermediate expansions are still free of e-pairs, since $0 \in K_i$. On every coset of L_i, the expanded fiber therefore has size at most $|L_i|/2$. The one-block bound shows that each expansion increases total size by a factor at least $1/c(k_i)$. The final expansion has at most half the group. Conversely, the image of the product of balls of radii $k_i - 1$ has density exactly $(1/2)\prod_{i \leq n} c(k_i)$, since each ball is disjoint from its antipodal translate and hence the quotient is injective there. Its expansion lies in the image of the product of balls of radii k_i, still disjoint from its e-translate, so it is independent. These prove equality.

Infinite capacity. In every action an avoiding event induces independent masks on almost every orbit of each finite subgroup; averaging gives the product upper bound. For the converse take a uniformly randomly translated maximizing mask on n blocks, extend the

action by coinduction from that finite subgroup, and use the origin indicator. Any weak limit of the resulting stationary binary laws on the whole countable group avoids every specified difference since pairs use finitely many blocks. Its density is $(1/2)\prod_i c(k_i)$. The infimum over finite products is consequently attained. Since $c(k) \to 1$ from below, one can make this product equal 2δ for any $\delta < 1/2$: choose successive positive terms $-\log c(k)$ no larger than the remaining required sum and whose sum tends to $-\log(2\delta)$ (e.g. use repeated terms of sizes tending to zero, greedily in batches). For $\delta = 1/2$ the single step e suffices.

The bounded-exponent digit argument in the squarefree-tail note [R241] explains why simply pulling this construction to the integers does not give the conjecture. In every finite binary quotient a positive independent mask lifts, through a fixed number r of base-two digit layers, to an avoiding mask of density $1 - (1 - \delta)^r$ at length 2^r. The length is independent of the dimensions and number of binary blocks.

R036. Exact reduction for a fixed torus rotation and arbitrary integer steps

Scope and status. Result with proof.

Result. For an arbitrary step set in a fixed m-torus rotation, finite torsion grids approximate the relative all-length capacity with error at most m/M.

Scope. The infinite-product formula takes the grid limit first, then the supremum over finite factors; positive relative capacity implies unrestricted integer value one.

Let T be the rotation by a topological generator α of $K = T^m$ with Haar probability, let D be any nonempty set of positive integers, and set $S = closure\{d\alpha : d \in D\}$. For $P = \mathrm{lcm}(1, \ldots, M)$ let $G_P = K[P]$ be the full finite P-torsion grid. Write c_M for the largest fraction of G_P occupied by a mask which contains no full coset of the cyclic group generated by any $v \in S \cap G_P$. An empty menu imposes no constraint. If it contains zero, the largest mask is empty. No openness or density condition on the integer menu is required.

The all-length avoidance threshold **in this rotation** is

$$b_T(D) = \lim_{M\to\infty} c_M, 0 \le c_M - b_T(D) \le m/M (M \ge 2).$$

For the upper bound an h-avoider has zero Haar overlap at slots $0, v, \ldots, (h-1)v$ for every $v \in S$, by continuity in measure of translations. Its sections on almost all cosets of G_P cannot contain any full orbit in the finite menu. Disintegration gives the upper bound. The c_M decrease by restriction and coset averaging.

For the lower bound tile the torus with the half-open cubes of side $1/P$ indexed by G_P. Keep the cubes of an extremizing finite mask and trim a strip of thickness $\tau > 0$ from each face. The retained fraction tends to c_M as $\tau \downarrow 0$, with P fixed. For a velocity v within $\tau/(2P)$ in the supremum metric of $w \in S \cap G_P$, no run of P sites can stay in the retained cubes. Indeed if its first site is retained, each unperturbed site at times $0, \ldots, P-1$ has the same internal cube coordinate as the first. Replacing w by v keeps the cube index of all these sites. The unperturbed orbit visits a missing cube.

Remove also the union of m coordinate bands where a circle coordinate lies in an open arc of length $\eta > 1/M$, of total cost at most $m\eta$. In the compact remainder of S outside the small open neighborhoods of $S \cap G_P$, every velocity has some coordinate of order strictly larger than M (or infinite), since otherwise every coordinate would be in the P-torsion grid. From every starting point that coordinate hits its open band at some nonnegative time: a finite rotation of order $q > M$ has spacing $1/q$, and an infinite one is dense. The tests that some site hits a band are open in the pair (velocity, starting point), so compactness makes the maximum necessary time finite. Together with P this gives a single avoiding length for *every* velocity in S. Let $\tau \downarrow 0$ and $\eta \downarrow 1/M$. This proves the error estimate, with the trivial bound if $m/M \ge 1$.

In particular, on a circle the limit is exactly $1 - 1/q$, where q is the least order of a torsion point in S, with $1/\infty = 0$, even for non-open D. On the grid, the consecutive missing segment of length P/q_P, where q_P is the least order in the finite menu, meets every allowed orbit, and a single orbit direction of least order gives the matching upper bound. An empty finite menu has cap one. In higher rank the full graph of allowed velocities matters; for

example all nonzero directions of C_2^m in the grid jointly permit at most one representative on each grid coset.

For the countable product torus with a dense cyclic rotation, the corresponding threshold is the supremum of the thresholds of the finite coordinate projections. Any h-avoider of mass a induces a cylinder avoider of mass arbitrarily close to a. On a sufficiently large finite coordinate factor take the set where the conditional density exceeds $1 - 1/(2h)$. If the projected slots were simultaneously in this set with positive probability, the conditional union bound in the original system would give a positive return there. The reverse inequality is pullback. This also distinguishes the two limits in the infinite-rank case: first take the torsion-grid limit in each fixed dimension, then take the supremum over dimensions.

These statements do not transfer a positive threshold on a compact rotation to the integer conjecture. A positive-measure h-avoider on any compact rotation already forces the allowed steps to miss a Bohr neighborhood of zero, by strong continuity on its indicator. Across all integer systems these steps have threshold one by the circle-root sieve (and finite intersections of such tests).

R040. A finite cyclic coordinate acts affinely on the critical density

Scope and status. Result with proof.

Result. A cyclic coordinate transforms a critical value t into $1-(1-t)/r$ on Z times C_r, and the note computes an elementary pair-capacity continuum on Z.

Scope. Free integer congruence lifting gives endpoint one, so the affine formula is a mixed-group result.

Write $a_h(D)$ for the stationary fixed-length avoidance cap and $b(D)=\sup_h a_h(D)$. The group is indicated when it is not the integers. For an abelian group action, steps are group elements and the successive times are their nonnegative multiples.

Let D be a set of positive integers, let $r\ge 2$, and let $V=\{(d,1 \bmod r) : d\in D\}$ inside $Z\times C_r$. Then

$$b_{Z\times C_r}(V)=1-(1-b_Z(D))/r.$$

For the upper bound, in any action write T for the infinite generator and R for the cyclic generator. For an h-avoiding event A, let B be the intersection of $R^{-j}A$ for $0\le j<r$. This event is R-invariant, is contained in A, and has measure at least $1-r(1-\mu(A))$. Every positive T-progression in B of step d gives a positive A-progression under T^dR, since the points of B stay in A under every power of R. Thus $\mu(B)\le a_h(D)$.

For the reverse bound start with a stationary k-avoider of `rD` with density a. Use its product with the uniform r-cycle, keeping every point on sheets other than zero, and keeping the original event on sheet zero. In any `rk` consecutive samples along $(d,1)$ there are k samples on sheet zero spaced by `rd`, so the event avoids V. Its measure is $1-(1-a)/r$. Here $b_Z(rD)=b_Z(D)$ (indeed equality holds at each length): restriction to the rth power gives one inequality, and the height-r tower over an arbitrary transformation gives the reverse.

In particular, setting $D=N$ gives exact thresholds $1-1/r$ in these groups. The formula also identifies precisely why a free integer replacement of this single cyclic coordinate fails. Replacing 1 mod r by every integer congruent to 1 mod r gives a set in Z^2 of cap one. On the vertical coordinate use a clock of period r^s. Each congruent step is a unit modulo r^s; hence deleting one clock state, of mass r^{-s}, meets every r^s-term progression. This is an exact obstruction even for $D=N$. A group homomorphism from $Z\times C_r$ to Z also kills the marked coordinate. Arbitrary integer realization of this affine operation would already give a first intermediate threshold from $D=N$.

For comparison, at pair length alone all thresholds in $(0,1/2)$ do occur on Z. Fix irrational θ, choose $0<t<1/2$, and take $D_t=\{d>0 : ||d\theta||>t\}$. An interval of phases of length t provides a pair-avoider of density t. Conversely the closure K of the phases of a pair-avoider has $K-K$ contained in the closed arc $[-t,t]$. Circle Kneser gives $m(K-K)\ge \min(2m(K),1)$; thus $m(K)\le t$. Uniform distribution and outer regularity bound the upper natural density by $m(K)$. This is equality of $a_2(D_t)$ with t. Each such D_t has $b(D_t)=1$, since it misses the zero Bohr window $||d\theta||<t$. Pair capacity and all-length capacity have different spectra in every construction in this note for which both were computed.

R121. Exact critical endpoints on groups with one finite factor

Scope and status. Result with proof.

Result. A finite-factor construction enforces recurrence at its exact finite avoidance cap and realizes a dense family of such caps.

Scope. Groups Z times a finite abelian group; the group may vary with the requested approximation and the construction uses a finite-color hierarchy input.

Qualifications. The finite-color hierarchy construction in D23 supplies the large set whose unrestricted capacity is one.

This research note concerns $Z \times K$ with finite abelian K, using the stationary measure-preserving definition of density largeness for group actions. For this group one can equivalently use upper densities on $[1, N] \times K$ by the usual correspondence; the directional event includes all group starts and positive steps in the horizontal coordinate. The construction does not transfer to Z by replacing the finite factor with an unrestricted free lift.

Let $S \subset K \setminus \{0\}$ be nonempty. Call $B \subset K$ allowed if it contains no full coset of $\langle s \rangle$ for any $s \in S$. Write α for the greatest possible fraction $|B|/|K|$. This number is strictly between zero and one. Let q be the number of maximizing subsets (using $2^{|K|}$ as an overestimate also works). Let E be a q-large integer difference set with $b(E) = 1$, for example a set from the finite-color hierarchy [R139]. Then

$$V = (N \times S) \cup (E \times \{0\}) \subset Z \times K$$

is α-density-large, including the endpoint, and is not α'-density-large for any $0 < \alpha' < \alpha$.

Upper bound and the endpoint. In any pmp action write T for the horizontal generator and R_k for the finite action. Let A avoid V at some length h. For each s, the R_s-invariant event $C_s = \bigcap_{j=0}^{\mathrm{ord}(s)-1} R_s^{-j} A$ is null. Otherwise ordinary multiple recurrence for the positive event C_s under T would supply a positive h-return at some $n \geq 1$; since C_s is R_s-invariant and contained in A, the original (n, s)-return would be positive. Therefore for almost every x, the subset $B_x = \{k : R_k x \in A\}$ is allowed (use the finitely many R_k-translates of the null exceptions). Averaging its cardinality gives $\mu(A) \leq \alpha$.

If $\mu(A) = \alpha$, then B_x is a maximizer almost surely. Fix a point x which is typical for this property at every horizontal translate and for the countably many forbidden horizontal progression cylinders, including all finite translates. Color $t \in N$ by $B_{T^t x}$, in at most q colors. There is a monochromatic h-term E-progression. Its common subset is nonempty. Choosing one element k in it produces an all-A progression starting at a translate of $R_k x$, a contradiction. Thus the measure of every fixed-length avoider is strictly smaller than α, and every measurable set of measure at least α has all lengths. For the correspondence direction, a stationary avoider obtained from an integer set has zero-measure forbidden cylinders; conversely a high-mean ergodic component gives a generic set of starts of the required upper density. No uniformity in the sizes of the return correlations is asserted.

Lower bound. For each $\epsilon > 0$, take a stationary E-avoider A' with mass $> 1 - \epsilon$ and some fixed forbidden length. On its product with the regular translation action on K, use $A' \times B$, with B maximizing and of mass α. Along (n, s) the finite coordinate visits a hole within $\mathrm{ord}(s)$ sites. Along $(d, 0)$ with $d \in E$ the horizontal coordinate blocks the run. Thus

this is a fixed-length V-avoider of mass $> \alpha(1-\epsilon)$. This proves the claimed exact critical value. Unlike the finite factor alone, it forces recurrence at its supremal avoidance threshold.

For $K = C_r$ and $S = \{1\}$, $\alpha = 1 - 1/r$. Here any binary-large E with $b(E) = 1$ suffices even though there are r maximal sets. At equality each B_x has one missing point. Partition the possible missing points into two nonempty classes. On a monochromatic run of that coarsened coloring, any point of the opposite class lies in every B_x, giving the same contradiction.

A dense set of exact critical values across finite factors. Given $0 < \delta < 1$ and $0 < \epsilon < \delta$, choose a prime p large enough that $\delta/p < \epsilon$, and set $K = F_p^m$. For sufficiently large m the largest density avoiding a whole line in every nonzero direction is smaller than δ: density Hales–Jewett applied to $[p]^m$ already forces a line with wildcard difference in $\{0,1\}^m \setminus \{0\}$. Start with no constraints (maximum fraction one) and add nonzero directions successively. Adding one direction decreases the maximum fraction by at most a factor $1 - 1/p$ in the sense $\alpha_{\text{new}} \geq (1-1/p)\alpha_{\text{old}}$. Choose a transversal of the cosets of the new one-dimensional subgroup (density $1/p$), take a random translate of it, and delete it from an old maximizer; the expected remaining fraction is exactly $(1-1/p)\alpha_{\text{old}}$. The first crossing below δ therefore gives some S with $\delta - \epsilon < \alpha < \delta$. Applying the previous result realizes this exact α and gives a δ-density-large set failing $(\delta-\epsilon)$-density-largeness on a group $Z \times F_p^m$. The group and the set may depend on ϵ. This does not yield a single set at every real δ, or a set with such a critical density on Z.

R123. Every extremal density at one prescribed length occurs on the integers

Scope and status. Result with proof.

Result. For each fixed h at least two, every value in $[0, 1 - 1/h]$ is an integer h-term avoidance cap.

Scope. The forbidden length is fixed; the nontrivial angular examples have unrestricted all-length cap one.

Qualifications. Main proof read in full. It uses zero weighted means, finite nilfactor approximation, and absolutely continuous motion on an AP orbit closure, not pointwise characteristicity at exceptional differences.

Fix $h \geq 2$. The set of integer fixed-length caps $a_h(D)$, for nonempty $D \subseteq \mathbb{N}$, is exactly $[0, 1 - 1/h]$. More explicitly, fix an irrational α, let U_s be the open circle arc of radius s centered at $1/h$, and take $D_s = \{n \geq 1 : n\alpha \in U_s\}$. The function $s \mapsto a_h(D_s)$ is continuous on $0 < s \leq 1/2$, has limit $1 - 1/h$ at zero, and takes value zero at $s = 1/2$. The right-continuity argument below is the main point. Each a_h is a maximum, so an avoider at its extremal mass exists. These are **one-length** thresholds, as distinct from the user's required supremum over all lengths.

A finite nilmodel for the mean. Take a maximizing ergodic integer system and a measurable event A avoiding all h-runs on D_s, of measure a. Let $g = \mathbb{E}(1_A \mid Z_{h-1})$. The normalized time mean of $1_{U_s}(n\alpha) \int \prod_{j<h} T^{jn} g$ is zero. Indeed the original weighted correlations vanish identically; the difference on replacing by g has zero mean absolute value by Bergelson, Host and Kra, Corollaries 4.5–4.6, and by Wiener's theorem for pairs. Equivalently one can use linear characteristic factors after inserting characters of $n\alpha$ via the product with a circle rotation, then approximate the interval indicator off its Haar-null boundary.

Fix $\xi > 0$. Choose $t > 0$ with $B = \{g \geq t\}$ of measure at least $a - \xi$. Its weighted intersection mean is zero by $1_B \leq g/t$. Approximate this pronil event by a finite nilfactor Y, taking $A' = \{\mathbb{E}(1_B \mid Y) > 1 - 1/(2h)\}$ of measure at least $a - 2\xi$; the existence follows by martingale convergence. For each fixed n, the B-correlation is at least half the A'-correlation. Indeed over initial basepoints whose h images all lie in A', the conditional union bound gives probability of all memberships in B at least one half. Thus the weighted A'-mean on Y is zero. This proof needs only zero mean in the finite model, not an unproved assertion about individual exceptional differences.

Averaging on the orbit closure. Present $Y = G/\Gamma$ with ergodic nilrotation S. The group G need not be connected. In $Y^h \times \mathbb{T}$, fix $x_0 \in Y$ and take the closure L of

$$((S^{m+jn} x_0)_{0 \leq j < h}, n\alpha), \qquad (m, n) \in \mathbb{Z}^2.$$

By Leibman's multiparameter orbit theorem, L is a finite disjoint union of compact connected translated subnilmanifolds $L_i = H_i z_i$, with H_i connected Lie subgroups in the ambient product group. The orbit is equidistributed for a measure Λ giving the components positive weights and their normalized Haar laws. This equals the Cesàro mean in n of the AP measures obtained by integrating in $x \in Y$ at phase $n\alpha$. For continuous tests first integrate m at each fixed n, using unique ergodicity; choose the first side length of a rectangle tending sufficiently rapidly to infinity with the second side to approximate each finite sum. These

rectangles are Følner. For the product of event indicators and the arc indicator, approximate them in their Haar marginals; all slot marginals equal Haar, and the boundary of the arc has angular Haar measure zero. We have proved

$$\Lambda((A')^h \times U_s) = 0.$$

Every tuple $((S^{jn}x)_j, n\alpha)$ is in L, by minimality of the original S-orbit. The angle map on each connected H_i has nonzero differential. Otherwise it is constant on L_i, which would give an atom in the Haar angular marginal of Λ.

Moving the angle. For each i, choose a Lie-algebra vector with angle derivative one and complete it by a basis of the kernel of the angle derivative. Use exponentials of random elements of the resulting small coordinate box, with all basis coordinates independent uniform on $[-\rho, \rho]$. Denote this law on H_i by $\sigma_{i,\rho}$. It is absolutely continuous with respect to Lie Haar in a neighborhood of the identity, and the resulting increment in the angle is uniform on $[-\rho, \rho]$. Choose $\rho < s$ and small enough for these charts. Retain in B_ρ the points x such that for each of the finitely many i and slots j,

$$\Pr_{w \sim \sigma_{i,\rho}} (w_j x \in A') > 1 - \frac{1}{\{} 10h\}.$$

Each averaged indicator tends in $L^1(Y)$ to $1_{A'}$, by strong continuity of translation on Haar space. Thus B_ρ tends in measure to A'.

If $n\alpha \in U_{s+\rho/4}$ and all $S^{jn}x$ lie in B_ρ, the corresponding tuple z belongs to some L_i. Under $w \sim \sigma_{i,\rho}$ the probability that the angle enters U_s is at least $3/8$, and the probability that any slot misses A' is at most $1/10$. Yet the law of wz is absolutely continuous relative to the component Haar measure, by the homogeneous-space quotient map. Its positive probability of landing in $(A')^h \times U_s$ contradicts the displayed zero. Thus B_ρ avoids every longer-window difference in $D_{s+\rho/4}$. This proves right continuity since ξ is arbitrary. Left continuity follows directly by compactness of stationary binary laws as $D_v \uparrow D_s$ for $v \uparrow s$.

Range and limitation. As $s \downarrow 0$, the h points with increment in U_s have largest gap at most $1/h + 2hs$; this follows by perturbing the regular h-gon. Delete a slightly longer arc in the α-rotation to get an h-avoiding event of measure approaching $1 - 1/h$. The matching upper bound holds for every nonempty direction set since each h-block at one fixed step must contain a miss. The bound is attained for $D = \{n : n \equiv 1 \bmod h\}$ by the cyclic clock. At $s = 1/2$, all positive integers belong to D_s, so $a_h = 0$. The intermediate value theorem gives every intermediate value. In fact D_s contains an IP set already at $s = 1/h$ by successive small positive phase choices. When $s < 1/h$ it stays separated from phase zero and has $b(D_s) = 1$ by the circle-root sieve. This pinpoints why the interpolation at fixed length does not yet interpolate the all-length cap.

R124. Fixed-order interpolation preserving nil-Bohr denseness

Scope and status. Result with proof.

Result. Anchored interpolation realizes all fixed-order values strictly between zero and $(1-1/h)^2$ on nil-Bohr-dense integer sets.

Scope. Every tail is nil-Bohr dense at every integer; the unrestricted cap remains one, and this interpolation has the stated square barrier.

Qualifications. The Gaussian packet construction supplies nil-Bohr density at every integer center; its proof is included among the supporting arguments.

The last proposition of Section 4 of the typeset notes gives, for any fixed integer menu E, all fixed-length caps below $(1-1/h)a_h(E)$ by adding an angular window to E. In particular, every $h \geq 2$ and every $0 < t < (1-1/h)^2$ admit a set whose every tail is dense in the topology of finite-order nil-Bohr tests, with $a_h(D) = t$ and $b(D) = 1$. The denseness holds at every integer, not just at zero.

Here is the seed for this consequence. In the nil-Bohr-dense Gaussian packet construction [R020] use one complex channel Z and choose the same primitive hth root ζ as target on every sufficiently late packet. The compact-parameter construction at stage n covers all prescribed finite families of nil-Bohr test shapes, coefficient boxes, and integer targets at that stage; discarding finitely many packets preserves every one of the density assertions. Outside a single exceptional set of summable probability per stage, the deviations between the observations at T^{jd} and $\zeta^j Z$ tend to zero uniformly for the packet times d and the finitely many $j < h$. The relevant joint error estimate is given in the linked construction and does not grow with the cardinality of a packet.

In the circularly symmetric Gaussian channel delete an open angular sector slightly longer than $1/h$ of a full turn, and delete a small disk about zero. The retained probability can be taken greater than $1 - 1/h - \epsilon/2$. Every retained starting value has one of its first h ideal rotates in the interior of the deleted sector with a uniform angular margin, by the regular h-gon. Its distance from zero controls the conversion from absolute complex error to angular error. Delete the exceptional events on every stage in a sufficiently late tail, of total probability $< \epsilon/2$. The remaining mask has no all-one h-run along any step in this tail. Thus the selected nil-dense seed E has $a_h(E) > 1-1/h-\epsilon$. It has $b(E) = 1$ by the Gaussian root amplification for finite nonidentity phase targets in the same linked construction (or by the approximate cyclic-label argument in the root-tower proof [D20]).

Choose ϵ so that $(1-1/h)a_h(E) > t$ and apply the proposition. Since $t > 0$ the selected radius is below $1/h$: at $1/h$ the window contains an IP set. Thus the angular family also has all-length cap one. Independent products give $b(E \cup D_s) = 1$ and the retained seed ensures every tail is nil-Bohr dense. At $t = 0$ the full set of integers already realizes the fixed-order zero value, with unrestricted all-length zero.

Upper limit of this particular interpolation. For a seed E of finite-order nil-Bohr recurrence and a nonempty open angular window D_s as above,

$$a_h(E \cup D_s) \leq (1-1/h)a_h(D_s).$$

Indeed take an ergodic avoiding event A, and let g be its conditional density on the characteristic pronilfactor. On each finite nilfactor the conditional density is at most $1-1/h$:

if its superlevel strictly above that value had positive measure, E supplies an h-return of that superlevel on the finite nilsystem. At the corresponding step a conditional union bound would give an A-return. Martingale convergence gives the same bound for g. The weighted-mean reduction and angular smoothing proof, applied to superlevels of g with threshold tending to zero, give a D_s-avoider of mass arbitrarily close to the measure of $\{g > 0\}$. Thus integrating g proves the inequality. In particular the supremum for this anchored construction cannot exceed $(1 - 1/h)^2$, no matter how nearly maximal the seed is.

R130. Multiplicative products of ultrafilters and local critical densities

Scope and status. Result with proof.

Result. Local ultrafilter capacities decrease under multiplicative products; quantitative strict loss yields a regular-point dichotomy and fixed-order decay.

Scope. Also proves full recurrence for a nil-Bohr-recurrent factor multiplied by a positive-upper-Banach-density factor; multiplication order matters for ultrafilters.

Qualifications. The product convention is $E \in qp$ if and only if $\{e : E/e \in p\} \in q$. The orientation is used consistently in the ultrafilter argument.

The multiplicative-product estimates below give exact fixed-order iteration bounds, a regular-point dichotomy, and a positive-Banach-density multiplier theorem. These restrict any possible intermediate critical density without assuming that one exists. The supporting amplification argument [D01] supplies the underlying multiplicative-semigroup method.

For a set of positive steps E, write $a_h(E)$ for the maximum density of an event in an invertible stationary system whose h-fold progression correlation vanishes at every $d \in E$. Write $b(E) = \sup_{h\geq 2} a_h(E)$. For an ultrafilter p on the positive integers set

$$u_h(p) = \sup_{E\in p} a_h(E), s(p) = \sup_{E\in p} b(E) = \sup_h u_h(p).$$

Both u_h and s are lower semicontinuous on the Stone space. A strict lower bound is witnessed on a single clopen neighborhood $\bar{E}$. Write $E/e = \{d : ed \in E\}$, $E_p = \{e : E/e \in p\}$. Our multiplication convention is $E \in qp$ iff $E_p \in q$.

One elementary extreme case is exact even before taking the supremum in h. If the Bohr image of p is nonzero, then $u_h(p) = 1 - 1/h$ for every $h \geq 2$. Indeed $\theta \mapsto p\text{-}\lim_n n\theta$ is a nonzero homomorphism from the divisible group $\mathbb{T}$ to the circle. Its image is divisible and nonzero, hence infinite and dense. For arbitrarily small $\eta > 0$ choose θ with this value within η of $1/h$. On a member of p, the phases $d\theta$ are within 2η of $1/h$. The points $0, d\theta, \ldots, (h-1)d\theta$ then have maximum circular gap at most $1/h + 4h\eta$. Deleting one open arc longer than that bound in the Haar rotation by θ obstructs every such h-progression. Let η tend to zero and use the elementary $1 - 1/h$ upper bound. In particular every nonzero additive integer translate of a finite-supremum interior ultrafilter attains this sharp upper bound at every order, since such an ultrafilter must have zero Bohr image.

Monotonicity For every q, p and h,

$$u_h(qp) \leq \min(u_h(q), u_h(p)), s(qp) \leq \min(s(q), s(p)).$$

In fact if $E \in qp$, pick any $e \in E_p$. Restricting a system to its e-th power gives $a_h(E) \leq a_h(E/e) \leq u_h(p)$. For finite $F \subseteq E_p$, there exists $d \in \bigcap_{e\in F} E/e$, giving $dF \subseteq E$. Restricting to the d-th power gives $a_h(E) \leq a_h(F)$. Take the infimum over finite F, equal to $a_h(E_p)$ by weak compactness of stationary binary laws with the prescribed cylinder exclusions. Since $E_p \in q$, this proves the other bound.

Strict loss Fix $k \geq 2$ and $0 < \gamma \leq 1$. Choose $W = W(k, \gamma) \geq k$ such that every subset of $[W]$ of density at least $\gamma/4$ contains a nonconstant k-term progression. The finite Szemerédi theorem supplies W. Whenever $s(q) \leq 1 - \gamma$,

$$1 - s(qp) \geq (1 - s(p))/(1 - u_k(p)^W). (1)$$

The denominator is positive since $u_k(p) \leq 1 - 1/k$. A nonempty step set imposes at least one miss in each k-block. If $u_k(p) = 0$ the assertion follows from monotonicity. In particular if $0 < s(p) < 1$ and $s(q) < 1$, then $s(qp) < s(p)$.

Proof. Let A_0 be a k-avoiding event for a member $D_0 \in p$ with mass $a > 0$ arbitrarily close to $u_k(p)$. Set $H_0 = A_0^c$. For any $E \in qp$, take an arbitrary h-avoiding event A for E, write $H = A^c$ and $\rho = \mu(H)$. All needed avoidance and hitting properties, including time translates, hold on invariant conull sets because there are only countably many positions and steps.

Since $E_p \in q$, $a_h(E_p) \leq 1 - \gamma$. Choose a finite nonempty $F \subset E_p$ with $a_h(F) < 1 - 3\gamma/4$, by the compactness assertion above. For all sufficiently large L, every binary word of length L internally avoiding h-runs at F has density at most $1 - \gamma/2$. Otherwise translate and average such words (extended by zeros), and take a stationary weak limit to contradict the cap for F. Choose a sufficiently large such L divisible by W. Put $P = D_0 \cap \bigcap_{e \in F} E/e \in p$.

For any $d \in P$, every length-L d-line in the original system sees an A-word that internally h-avoids F, because $ed \in E$ for all $e \in F$. Consequently some one of its consecutive length-W subblocks has H-density at least $\gamma/2$. That block contains k sites in H forming an arithmetic progression in the time indices of stride $1 \leq c \leq W$.

Use independent copies of the integer c-root towers of the system of H_0, for every $1 \leq c \leq W$, independent also of the system of H. A root has uniform level in $\{0, \ldots, c-1\}$, advances one level per unit time and applies the original transformation at a wrap. Use the mark $H_{0,c}$ on every level. At stride cd the c-root advances the original state by the original transformation to the power d, so the indicated k sites contain an $H_{0,c}$-mark because $d \in D_0$. Thus

$$H' = H \cap (\cup over 1 \leq c \leq W of H_{0,c})$$

hits every length-L progression at every $d \in P$, almost surely. Its mass is $\rho(1 - a^W)$. Its complement avoids $P \in p$, so $1 - s(p) \leq \rho(1 - a^W)$. Infimizing ρ over E, h, A, and taking $a \uparrow u_k(p)$, proves (1). The finite set F and the final length L may depend on E, h. The shrink factor $1 - a^W$ does not.

Regular points and long products If $p = qp$ for a q with $s(q) < 1$, then $s(p) \in \{0, 1\}$. More generally any multiplicatively regular p, meaning $p = ptp$ for some ultrafilter t, has $s(p) \in \{0, 1\}$. Indeed $s(pt) \leq s(p)$, and one applies strict loss to the left factor pt and right factor p. This includes multiplicative idempotents. The principal ultrafilters have $s = 1$.

The closed set $K_\gamma = \{p : s(p) \leq 1 - \gamma\}$ is a multiplicative subsemigroup (possibly empty). For $q_i \in K_\gamma$, $r = q_1 \cdots q_n$, $n \geq 2$, we have

$$u_k(r) \le [1-\gamma^{1/(n-1)}]^{1/W(k,\gamma)}.(2)$$

Every suffix has u_k at least $u_k(r)$ by monotonicity. Apply (1) to successive suffixes from right to left with the same W, starting with $1-s(q_n) \ge \gamma$. Since $1-s(r) \le 1$, rearrangement gives (2). As the right side tends to zero at each fixed k, lower semicontinuity yields

$$p \in \bigcap_{n\ge 1} closure(K_\gamma^n) impliess(p) = 0.$$

For $\gamma = 1$ this also follows directly from monotonicity. In particular, any local supremum strictly between 0 and $1-\gamma$ must have finite depth in this multiplicative sense. The order k is fixed when applying (2). It gives no uniform convergence in the forbidden length and does not exclude nonregular ultrafilters of finite depth. Multiplicative pattern thinness can persist even for fully density-large sets, so regularity cannot be imposed on an arbitrary witness without an additional argument.

Ordinary multiplicative products For nonempty ordinary step sets D, F, define $DF = \{df : d \in D,\ f \in F\}$. Monotonicity gives $a_h(DF) \le \min(a_h(D), a_h(F))$ by restriction to a nonzero power in each direction. If $b(D) \le 1-\gamma$, then

$$1 - b(DF) \ge (1-b(F))/(1-a_k(F)^W), W = W(k,\gamma).(3)$$

Take an arbitrary h-avoider of DF and use its complement H as in (1). Choose a finite test $J \subseteq D$ with $a_h(J) < 1 - 3\gamma/4$. On each f-line with $f \in F$, the allowed steps $ef \in DF$ for every $e \in J$ imply the same long-line density bound for H. Use a k-avoider for F itself as a seed. The new hitting event blocks every $f \in F$, giving (3). In particular a fixed-order bound just as in (2) holds for $a_k(D_1 \cdots D_n)$ whenever all $b(D_i) \le 1-\gamma$. Formulas (1) and (3) use only stationary compactness, finite Szemerédi, and the explicit integer towers.

No positive lower bound for $b(DF)$ follows from the two factor thresholds alone even at the upper endpoint. In fact $\{1, 2, 4, \ldots\}$ (use the frequency 1/3) and the odd numbers (use the frequency 1/2) each miss a neighborhood of zero on the circle, so both have threshold one. Their product is all of $\mathbb{N}$, with threshold zero.

A positive-density factor gives full recurrence If $D \subseteq \mathbb{N}$ meets every finite-order nil-Bohr zero neighborhood and $F \subseteq \mathbb{N}$ has positive upper Banach density, then DF is Szemerédi-large. More precisely, for each positive-measure event and each fixed return length there is $d \in D$ such that the successful $f \in F$ have positive upper Banach density.

Here is the elementary dilation calculation used in the proof. Let $\mathcal{N} \subset \beta\mathbb{N}$ denote the ultrafilters containing every finite-order nil-Bohr zero neighborhood. Then $\mathcal{N}$ is a two-sided ideal for multiplication. That $p \in \mathcal{N}$ implies $qp \in \mathcal{N}$ for arbitrary q follows by restricting to the e-th power of each nilrotation at each fixed e. We prove $q \in \mathcal{N} \Rightarrow qp \in \mathcal{N}$ as well, with any p.

Use the characterization by finitely many special generalized-polynomial expressions from Alweiss–Farhangi, Definition 1.6. Let $P(dn)$ be one fixed such expression. Starting from the

inside, represent the previous nearest-integer output, for q-many d and then p-many n, as a finite sum $m_\ell(n)F_\ell(d)$. The coefficients are integers, and each $F_\ell(d)$ is d^v or $d^v\langle Q(d)\rangle$ with Q special and $v \geq 0$. Start with the interior placeholder 1. The next layer multiplies by $\alpha_i n^{w_i} d^{w_i}$, $w_i \geq 1$. Split the real coefficients $\alpha_i n^{w_i} m_\ell(n)$ into integers $M_\ell(n)$ and centered fractional parts $c_\ell(n)$. Let c_ℓ be the real p-limit of the latter in the closed interval $[-1/2, 1/2]$. Each $Q'_\ell(d) = c_\ell d^{w_i} F_\ell(d)$ is a special expression (or identically zero). On a q-member every one of these has distance $< \eta$ from the nearest integer. For each such fixed d, restrict the p-member of n's further until

$$|c_\ell(n) - c_\ell| \cdot |d^{w_i} F_\ell(d)| < \eta$$

for all ℓ. Taking 2η times the number of terms below $1/2$, the rounded output at this layer is exactly

$$\sum_\ell M_\ell(n) d^{w_i} F_\ell(d) + \sum_\ell \langle Q'_\ell(d)\rangle.$$

This has the prescribed inductive form. At the last, unrounded layer the same bound puts $P(dn)$ within any prescribed tolerance of an integer. Intersect the finitely many q-members for a given nilwindow. This proves the claim, without interchanging ultrafilter limits or requiring continuity at a rounding boundary. The resulting tests have total degree at most the degree of the original expression, since powers added at each stage are the same input weights.

Choose $q \in \mathcal{N}$ with $D \in q$. Choose an ultrafilter p containing F and the complements of all sets of upper Banach density zero; the finite-intersection property follows from $d^*(F) > 0$. Every member of qp also has positive upper Banach density since it contains $e(E/e)$ for a fixed e with $E/e \in p$. Now qp contains every nil-Bohr zero neighborhood and omits every set of upper Banach density zero. For an ergodic system and a positive-measure event A, its positive h-term return set R contains a finite-order nil-Bohr zero neighborhood after a set of zero upper Banach density is deleted. This is Huang–Shao–Ye, Theorem 2.2.4, using the Bergelson–Host–Kra structure theorem and the positivity at zero of the nilsequence component. Hence $R \in qp$. Choose $d \in D$ with $R/d \in p$; then $F \cap (R/d) \in p$ has positive upper Banach density. Ergodic decomposition and stationary correspondence give the claim for Szemerédi largeness. A set with an interior avoidance threshold is necessarily nil-Bohr recurrent by Alweiss–Farhangi, Theorem 1.8, so in particular multiplying it by any positive-upper-Banach-density set gives threshold zero.

The same conclusion holds if F has positive upper density in the indices of the sparse sequence $r_j = Q(p_j)$, where p_j is the j-th prime and Q is a nonconstant integer polynomial eventually positive. It holds as well for $r_j = Q(j)$. In fact choose p containing F and the complements of all subsets of index density zero along r_j. By Le's theorem on null components along subsequences, the canonical error N of the single-transformation arithmetic progression multicorrelation is null along $dQ(p_j)$ for each fixed $d > 0$, and likewise along $dQ(j)$. Thus for each $c > 0$ and each $d > 0$, $\{f : |N(df)| \geq c\} \notin p$. Its undilated counterpart is consequently absent from qp. In the proof of Huang–Shao–Ye's theorem the exceptional set can be taken specifically as a superlevel of $|N|$, since the structured component has

strictly positive value at zero. The previous nilkernel argument therefore gives a $d \in D$ and a positive upper density of successful indices j with $r_j \in F$. This does not assert that arbitrary uniformly null sequences remain null on the primes.

The two orders can give opposite endpoints The Gaussian nil-Bohr packet construction [D08], gives finite packets F_j whose every tail union is nil-Bohr recurrent. We can choose its parameters so that the expanded set

$$G = \cup over jof\{fd : d \in F_j \quad \text{and} \quad fodd \quad \text{with} \quad 1 \le f \le j\}$$

has $b(G) = 1$ as well. Indeed in the packet construction the good spectral arcs have their d-phase within $\varepsilon_j = m_j^{-2}$ of the antipode, with at most t_j bad arcs. For each permitted odd f, the resulting good fd-phase is within $j\varepsilon_j$ of the antipode, with no new bad arcs. In the normalized arc-indicator Gaussian estimate multiply the good error bound by j and leave the bad coefficient bound 2 unchanged. The simultaneous bound is consequently at most a constant times

$$(jm_j\epsilon_j + t_j)m_j^{-1/2} \max_{i,\ell} |Z_{i,\ell}|.$$

The ranks m_j may be taken large enough that the probability this exceeds a prescribed $u_j \downarrow 0$ is summably small. At each of the at most $j|F_j|$ remaining tests the continuous spectral refinement multiplies the original arc approximation error by at most j. Choosing the child widths *after* fixing F_j again makes the residual Gaussian tail errors summable over all these tests. The rest of the cited argument now gives pair-avoiders on the expanded tails with masses tending to $1/2$, and consequently $b(G) = 1$ by the near-saturation estimate from the torsion note [D20] and the finite-prefix sieve.

Choose $q \in \mathcal{N}$ containing every tail $\bigcup_{j \ge J} F_j$, possible by the hitting construction and the finite-intersection property of nilwindows. Choose p essential (all its members have positive upper Banach density) containing the positive odds. For each fixed odd f, G/f contains a tail of the packets and hence belongs to q. Thus $G \in pq$, and $s(pq) = 1$. The result for the other order gives $s(qp) = 0$. In particular, requiring the *outer* ultrafilter to be essential and the *inner* one to be nil-Bohr zero does not yield the same conclusion. The growing finite schedules here can preserve exact obstructions of very large density despite exhausting the odd multipliers on every sufficiently late packet.

R145. Multiplication of critical densities

Scope and status. Proved subcases and unresolved extensions.

Result. Nested-set limits, local ultrafilter attainment and thinning criteria are proved; spectral multiplication and a density-step inequality use a substantial conditional nilfactor input.

Scope. The nested-set and finite-partition conclusions are unconditional. Multiplicative closure and the positive-density inequality depend on D11. The full prescribed spectrum still requires an intermediate seed and the stated finite-partition hypotheses.

Qualifications. Multiplicative closure of the spectrum and the positive-density inequality depend on the conditional nilfactor inequality in D11 and its stated localization and free-cover inputs. They do not follow from the Host–Kra factor theorem alone. Nested decreasing-set realization, local ultrafilter attainment and finite-partition interpolation above the starting capacity do not require that conditional inequality. Multiplication is used to reach smaller targets.

For a countable abelian group G, write $a_{h,G}(D)$ for the largest measure of an event avoiding every h-term progression of every difference in D in a probability-preserving G-action, and $b_G(D) = \sup_{h\geq 2} a_{h,G}(D)$. For $G = \mathbb{Z}$ this is the critical upper-natural-density threshold by stationary correspondence. I abbreviate it to $b(D)$.

Conditional Proposition (assuming the nilfactor-density inequality (1)**).** If $D_1, D_2 \subseteq \mathbb{N}$, the number $b(D_1)b(D_2)$ is the critical threshold of a subset of $\mathbb{N}$. More precisely, for

$$W = (D_1 \times \mathbb{Z}) \cup (\{0\} \times D_2) \subseteq \mathbb{Z}^2$$

one has $b_{\mathbb{Z}^2}(W) = b(D_1)b(D_2)$, and the group threshold can be transferred to $\mathbb{Z}$.

The upper bound uses the previous, substantial conditional nilfactor density inequality [D11] from these research notes:

$$\|\mathbb{E}(1_E \mid Z_s(S))\|_\infty \leq b(D) \tag{1}$$

for an event E avoiding D at some fixed length under an ergodic transformation S. This note is a deduction from that research result, which is distinct from the published characteristic-factor theorem. For nonergodic S, the Host–Kra factors disintegrate over the S-ergodic decomposition as their component factors, so (1) follows componentwise. For this application all spaces can be taken standard, by first passing to the binary $\mathbb{Z}^2$-shift factor of the event under test.

Proof of the upper bound. Let T, S generate a $\mathbb{Z}^2$-action, and let A avoid W at length h. Write $Z = Z_{h-1}(S)$, using a safe characteristic order for the h consecutive slopes. The transformation T preserves Z, by the functoriality of the Host–Kra factors under automorphisms commuting with S. Put $g = \mathbb{E}(1_A \mid Z)$ and $B = \{g > 0\}$. Since A avoids all $(0, e)$ with $e \in D_2$, (1) gives $g \leq b(D_2)$ almost everywhere.

For $d \in D_1$, the correlation of $T^{jd}S^{jn}1_A$, $0 \leq j < h$, vanishes for every integer n. Averaging in n and applying the Host–Kra characteristic-factor theorem for S gives zero mean of the same correlations with g in place of 1_A, since T commutes with conditional expectation onto Z. The statement for nonergodic S follows by ergodic decomposition and bounded convergence, with the functions $T^{jd}1_A$ treated as arbitrary tests on each component.

If B had an h-return at d under T, then for some $u > 0$ the set

$$C = \bigcap_{j=0}^{h-1} T^{-jd}\{g > u\}$$

would have positive measure. For every n the g-correlation above is at least $u^h\mu(\bigcap_{j=0}^{h-1} S^{-jn}C)$, whose Cesàro liminf is positive by the finite Szemerédi theorem and averaging. This contradicts the zero mean. Hence B itself h-avoids D_1 under T. In particular

$$\mu(A) = \int_B g\, d\mu \le b(D_2)\mu(B) \le b(D_2)a_h(D_1).$$

Taking the supremum over h proves the upper bound.

Lower bound. Take independent stationary D_i-avoiders A_i of masses approaching $b(D_i)$, each at its own fixed length h_i. Their product $A_1 \times A_2$ in the product action avoids W at length $\max(h_1, h_2)$, since along (d, n) the first projection has a forbidden d-run and along $(0, e)$ the second projection has a forbidden e-run.

Integer realization. In the group $H = \bigoplus_{m\ge1} \mathbb{Z}^2$, put $W_m = W \cap [-m, m]^2$ in the m-th summand and $V = \bigcup_m W_m^{(m)}$. Summing all coordinates $H \to \mathbb{Z}^2$ pulls any W-avoider back to a V-avoider. Conversely, an h-avoider for V gives, under each summand, a stationary binary $\mathbb{Z}^2$-law of the same mean avoiding W_m. A weak limit as $m \to \infty$ has that mean and avoids W, since the indicator of every forbidden progression cylinder is continuous. This proves equality at each fixed length between the rank-two problem and the replicated group problem. The coordinate bound in summand m is m, so the mixed-radix transfer [D06] now yields an integer set with the same all-length critical threshold.

Thus a single intermediate threshold t gives all t^n, $n \ge 1$, and in particular intermediate thresholds arbitrarily close to zero. More generally, any family of already achieved thresholds generates its finite multiplicative semigroup. This calculation does not produce the first intermediate threshold, multiplicative roots of t, or a limiting infinite product. Indeed an infinite lexicographic construction on active levels automatically contains the finite sums of a sequence of generators, and therefore has critical threshold zero.

A criterion yielding every real threshold Theorem. Suppose one integer step set D has $0 < b(D) < 1$, and, for every $\eta > 0$, admits a *finite* partition into cells C with $b(C) > 1 - \eta$. Then every $\delta \in (0, 1)$ is the critical threshold of an integer step set.

For subsets $F \subseteq D$, set $\phi(F) = -\log b(F)$. It is finite, monotone, and subadditive. Indeed an independent product of avoiders for F, G, with their lengths increased to the larger one, avoids $F \cup G$ and has the product mass. Also $\phi(\varnothing) = 0$.

Take nested finite partitions $\mathcal{P}_n$ of D for which every cell has $\phi \le 2^{-n}$, refining by intersections. Associate the cells with consecutive closed subintervals of $[0, 1]$, starting with D on $[0, 1]$. In each interval, divide equally among its children in any order, using an empty extra child if there would be only one child. The maximal interval length then tends to zero. Give each $d \in D$ the unique label $\ell(d) \in [0, 1]$ in all its successive closed intervals. Define $F_x = \{d : \ell(d) \le x\}$.

For a fixed n, every interval of parameter length less than the smallest $\mathcal{P}_n$-interval length meets at most three of the closed intervals (two suffice aside from harmless endpoint book-

keeping). Therefore for such $s < t$,

$$0 \le \phi(F_t) - \phi(F_s) \le \phi(F_t \setminus F_s) \le 3 \cdot 2^{-n}.$$

Thus $x \mapsto \phi(F_x)$ is continuous. The points labeled 0 lie in the first cell at every depth, so $\phi(F_0) = 0$, whereas $\phi(F_1) = \phi(D)$. The intermediate value theorem realizes every threshold in $[b(D), 1]$ by nested subsets of D. Given $0 < \delta < b(D)$, take an integer k such that $\delta^{1/k} \ge b(D)$ and use the product proposition k times. The strict finite-length lemma [D19] handles the endpoint at every positive threshold so obtained, as required for largeness at density δ itself.

The unresolved partition hypothesis has an exact alternative. Its failure means that for some $\eta > 0$ there is an ultrafilter p containing D such that

$$\sup_{C \in p} b(C) \le 1 - \eta.$$

Indeed if no finite partition into cells of $b > 1 - \eta$ exists, no finite family of such subsets can cover D, since taking differences turns a finite cover into a partition and cannot reduce b. The complements of all such sets together with D therefore extend to an ultrafilter. Conversely one cell of any finite partition belongs to p. Equivalently, every positive-measure h-term return set for an event of measure greater than $1-\eta$ belongs to p, for every h. To see the converse equivalence use stationary witnesses for every strict inequality in the supremum definition of b. The low-density witness for D fails along p at a single length. Thus if an intermediate example exists but the user's full assertion fails, every intermediate example carries one of these ultrafilters with a threshold uniformly below one.

Increasing values from decreasing step sets Let D_n decrease under inclusion with $b(D_n) \uparrow L$. Then L is itself attained by an integer step set, even when the intersection of all D_n has a larger capacity. For $n \ge 2$, take a finite $F_n \subseteq D_n$ with $a_n(F_n) \le L + 1/n$, using compactness of stationary binary laws at the fixed length. Put $V = \bigcup_{n \ge 2} F_n e_n \subseteq \bigoplus_{n \ge 2} \mathbb{Z} e_n$. Every h-avoider for V has mass bounded by $a_n(F_n)$ for all $n \ge h$, by restricting to the n-th axis (an h-avoider is also an n-avoider). This gives $b(V) \le L$.

For fixed j, pull a stationary D_j-avoider back via summation of the coordinates. It avoids all $F_n e_n,\ n \ge j$. The finitely many preceding directions admit an avoiding event of mass arbitrarily close to one: on the finite preceding axes take a common cyclic clock of large prime modulus, sending each generator to 1, and delete its zero residue. Every nonzero preceding coefficient is a unit if the prime exceeds their absolute values, so a progression of length equal to the prime hits the deleted residue. Taking an independent product and then approximating the D_j cap establishes $b(V) \ge b(D_j)$. The single-axis bounds are finite, so transfer gives an integer realization of L.

One source of such paths is independent thinning. Assign independent uniform $[0, 1]$ marks to the elements of D, and let D_s denote the subset with marks $\le s$. For each s, $b(D_s) = f(s)$ almost surely for a deterministic $f(s)$. Measurability follows by the finite-subset formula for a_h and the countable supremum in h. A finite change of differences leaves b intact by the cyclic-clock argument and the product inequality. Kolmogorov's zero-one law then gives the determinism. For $0 \le s < t \le 1$ the strip $D_t \setminus D_s$ has the law of D_{t-s}, giving

$$f(t) \ge f(s) f(t - s).$$

The decreasing-step-set principle realizes $\lim_{s\downarrow 0} f(s)$. If the limit equals 1, finite deterministic partitions of D with cell capacity arbitrarily near 1 exist: partition the mark interval into k equal intervals, and all k cells simultaneously have capacity $f(1/k)$ almost surely. This yields every threshold by the preceding theorem. The hypothesis on the thinning limit does not follow merely from $0 < b(D) < 1$. In the replicated single-axis realization $\bigcup_j \lambda_j(D \cap [1, j])$, independent thinning with any fixed positive retention almost surely preserves $b(D)$. Indeed infinitely many disjoint stages give independent trials of fixed positive success probability for keeping a scaled copy of each finite $D \cap [1, m]$. Every fixed-length avoider on the thinned steps therefore has mass at most $a_h(D \cap [1, m])$ for every m, by restriction to powers and compactness. The unthinned realization supplies the reverse all-length bound.

For any ultrafilter p, the number $s(p) = \sup_{C\in p} b(C)$ is an attained integer threshold. Choose $C_n \in p$ with $b(C_n) > s(p) - 1/n$, successively intersecting with all preceding C_i. The resulting decreasing sets remain in p, so their capacities are bounded by $s(p)$ and increase to it. Apply the decreasing-step-set principle. More generally

$$\inf_{p:\, D\in p} s(p) = \sup_{\mathcal{P}} \min_{C\in\mathcal{P}} b(C),$$

where $\mathcal{P}$ ranges over finite partitions of nonempty D. In one direction some cell belongs to each p. In the other, if the left-hand side is greater than v, the clopen neighborhoods supplied by the members of $b > v$ cover the compact clopen set of ultrafilters on D. A finite subcover can be disjointized to a partition, whose nonempty cells still have $b > v$.

A bound for positive-density families of steps If D has positive upper Banach density as a set of differences, then

$$a_h(D) \le (1 - 1/h)b(D) \qquad (h \ge 2).$$

When $b(D) > 0$, take an ergodic h-avoider A and write $g = \mathbb{E}(1_A \mid Z_\infty)$. Inequality (1) and passage through finite steps imply $0 \le g \le b(D)$. For every n, the integral of $\prod_{j<h}(g/b(D)) \circ T^{jn}$ is at least $1 - h(1 - \mu(A)/b(D))$ by the union bound for functions in $[0, 1]$. A strictly positive uniform lower bound would force the original correlations to be positive outside a set of differences of zero upper Banach density, by Glasscock et al., Theorem 5.8. This contradicts the positive upper Banach density of the allowed differences. Ergodic decomposition handles arbitrary witnesses. The bound with $b = 0$ is tautological. This identifies a necessary rate of approach to the critical threshold for a syndetic example, independent of the number of colors.

R146. Realization after adding a full IP-set

Scope and status. Result with proof.

Result. Any attained threshold has a realization $E + FS(q_i)$ with the same value and with every positive translate of capacity one.

Scope. Uses the earlier rigid-system realization with divisible heights; it supplies no first intermediate threshold.

Qualifications. The weakly mixing transfer and polynomial-translate constructions in D22 and D16 are prerequisites.

Proposition. Every attained critical threshold $t = b(D)$ has an integer realization $W = E + \mathrm{FS}(q_i)$, where $E \subseteq \mathbb{N}$, FS uses nonempty finite sets of indices, and $b(W) = b(E) = t$. All positive translates $W + a$, $a \geq 1$, have critical threshold one. This proposition uses the rigid-system realization [D22] with its divisible-height modification [D16] from the earlier research notes. It is a normalization of potential thresholds and does not supply an intermediate one.

Use that realization to choose E with threshold t, such that for every L all but finitely many members of E are multiples of L. In a single rigid transformation T, choose witnesses A_l of means tending to t, with fixed respective avoiding lengths h_l for E. The upper bound below works for $t = 0$ without using nonzero witnesses. The case of the empty difference set with threshold one may be realized instead by any nonempty finite step set.

Recursively choose $L_i \mid L_{i+1}$, $L_i > 2\sum_{j<i} q_j + i$, and positive $q_i \in L_i\mathbb{N}$, increasing strictly, such that

$$\mu(A_l \triangle T^{-jq_i} A_l) \leq 4^{-i} \qquad (l \leq i,\ |j| \leq h_l).$$

To achieve this for the finite set of tests, multiply a sufficiently late ordinary rigidity time by the fixed L_i. For each fixed jL_i, strong convergence of the corresponding powers follows by telescoping unitaries.

For $m \geq l$ let $B_m = \{0\} \cup \mathrm{FS}(q_i : i \geq m)$. There is a set $G_{l,m}$ of measure at least $1 - 2h_l \sum_{i \geq m} 2^{i-m} 4^{-i}$ such that

$$1_{A_l}(T^{js}x) = 1_{A_l}(x) \quad (x \in G_{l,m},\ s \in B_m,\ |j| \leq h_l).$$

Indeed for each $i \geq m$, each of the 2^{i-m} earlier subset sums u, and each such nonzero j, remove points where the entries at ju and $j(u + q_i)$ differ. All required identities follow by induction on the support. The cleaned event $A_l \cap G_{l,m}$ avoids all $E + B_m$ at length h_l. An all-marked line at $d + s$ would give membership in A_l at all $0, d, \ldots, (h_l - 1)d$, by shifting the point in slot j back by js, a contradiction. Countably many differences and starts are harmless.

For every nonzero early subset sum $v = \sum_{i \in I \subseteq [1,m)} q_i$, $0 < v < L_m$, and $b(E + v + B_m) = 1$. Indeed almost all $d \in E$ are divisible by L_m, so the corresponding differences $d + v + B_m$ are all congruent to $v \not\equiv 0 \bmod L_m$. A family missing all multiples of a fixed modulus has avoiding sets of density arbitrarily near one by the periodic sieve [D06]. For each remaining individual d, choose $k \geq m$ with $d + v + \sum_{m \leq i < k} q_i < L_k$. Reducing $d + v + B_m$ modulo L_k leaves precisely positive residues strictly below L_k. Thus each exceptional family also has threshold one. Their finite union with the first family has threshold one by independent products. Since all the early offsets form a finite set, independent product with a near-full avoiding event for each nonzero v yields $b(W) \geq \mu(A_l \cap G_{l,m})$. First let $m \to \infty$ and then $l \to \infty$.

Conversely, suppose A is an h-avoiding event for W. If it had a return at $d \in E$, put $C = \bigcap_{j<h} T^{-jd}A$ of positive measure. The IP multiple recurrence theorem furnishes a nonempty sum $s \in \mathrm{FS}(q_i)$ with $\mu(\bigcap_{j<h} T^{-js}C) > 0$. This gives a return for A at $d + s$, a contradiction. For this single-transformation application the usual finite density Hales–Jewett theorem suffices: sample words $w \in [h]^r$ at times $\sum_i w_i q_i$ from a random starting point, use a combinatorial line on the positive-density visits to C, and integrate over one of the finitely many line patterns of positive probability. Hence A also avoids E and $b(W) \le t$.

Finally take $a \ge 1$ and choose m with $a + \sum_{i<m} q_i < L_m$. All early offsets $a + v$ are nonzero and less than L_m, including $v = 0$. The same periodic sieve and finite exceptional-d argument now apply to every piece of $W + a$. Their finite union has threshold one.

R151. Exact thresholds with tests in uncountably many systems

Scope and status. Result with proof.

Result. Exact thresholds occur over one fixed torus rotation together with all systems missing at least one of finitely many prescribed eigenfrequencies.

Scope. The difference set can have any natural density in $(0,1)$, yet its unrestricted cap is one; infinitely many independent positive auxiliary windows force full recurrence.

Qualifications. Use the corrected Moreira-Richter theorem including its corrigendum; the tested class is defined by missing eigenfrequencies and is not all ergodic systems.

Fix an ergodic rotation T_0 on the countable product torus as in the compact product construction [R005]. Choose any $k \geq 1$ of its generating coordinate angles $\beta_1,\ldots,\beta_k$, independent over the rationals together with 1. Let the test class comprise T_0 and **all** ergodic probability-preserving integer systems for which, for at least one i, no nonzero multiple of β_i is an eigenfrequency mod one. For any $0 < t < 1$ the exact prescribed-threshold conjecture holds over this class, with one integer difference set D, and the smaller-threshold failures occur in T_0. In fact every positive event in each system satisfying the missing-eigenfrequency condition has returns of every length. The class includes all weakly mixing systems and all ergodic systems whose point eigenfrequency group modulo torsion has rational rank less than k. Its definition and the tested rotation precede the choice of t. The difference set can simultaneously have any prescribed natural density $0 < \rho < 1$. It is still Bohr-separated as an unrestricted integer set.

An auxiliary family for one irrational angle β. Fix $0 < c < 1$. Choose positive ϵ_j summing to less than $1-c$. For each rational x_j choose an arc J_j about it of length less than $\epsilon_j/2$ and an M_j so that its visit frequency under $n\beta$ for $n = 1,\ldots,$N is less than ϵ_j for every $N > M_j$, by equidistribution. Shrink to an arc I_j about x_j excluding every point $n\beta$ for $1 \leq n \leq M_j$. Thus I_j has visit frequency at most ϵ_j on every prefix. The closed complement of all I_j has measure greater than c and no rational points. Intersect it with an interval to obtain a closed set $C\infty$ of measure exactly c, using continuity of cumulative Lebesgue measure. Write C_j for that interval with just the first j arc exclusions. The visit indicators of C_j and $C\infty$ differ on at most a fraction of $\sum_{l>j} \epsilon_l$ in every prefix. Thus $F_\beta = \{n : n\beta \in C\infty\}$ has natural density exactly c.

If β is a rotation coordinate of T_0, then F_β has all-length avoidance cap one inside that rotation. For any arbitrarily short open circle arc every velocity in $C\infty$ hits that arc from every start, since every such velocity is irrational. Openness and compactness of the product of the circle with $C\infty$ give a common finite hitting time. The complement of the pullback of the arc to the rotation provides the claimed avoider.

Return averages in the other systems. For an ergodic integer transformation T let $c_A(n) = \mu(\bigcap_{j<h} T^{-jn}A)$ and m_A its ordinary mean, positive for $\mu A > 0$ by Szemerédi recurrence. If all nonzero multiples of β miss the eigenfrequency group of T, the corresponding twisted means of c_A vanish. Indeed Moreira–Richter, spectral refinement of the multiple-correlation decomposition, including the corrigendum approximates the sequence, up to a density-null term and an arbitrarily small uniform error, by a nilsequence on a system whose eigenfrequencies belong to that group. Its twisted means at any excluded frequency vanish, by considering the product orbit with the circle rotation and projecting the circle character

onto the nilsystem in a joining. This projection is an eigenfunction of the excluded frequency and hence zero. For finite unions C_j of intervals, trigonometric approximation and equidistribution of $n\beta$ give mean $c_j m_A$ with weight $1_{C_j}(n\beta)$.

The uniform prefix discrepancy just proved implies that the mean of $1_{F_\beta}(n)c_A(n)$ over all initial prefixes converges to $cm_A > 0$. No enumeration of the tested systems enters the construction.

Combine the sets. Add the finitely many F_{β_i} to the compact-product set constructed for T_0 at t. A finite union of families of relative cap one in T_0 again has cap one: take masks of masses arbitrarily close to one and intersect them. Intersect the resulting near-full mask also with any of the lower witnesses for the compact-product set, at the maximum of the forbidden lengths. Thus its threshold is still exactly t, and adding differences preserves all strict fixed-length upper bounds there. In every other system in the class the appropriate auxiliary family already gives all the stated positive returns. Each auxiliary family misses a circle neighborhood of zero, and the original set misses a neighborhood of zero in finitely many coordinates. Their union therefore misses the intersection of all those neighborhoods, so that the unrestricted integer cap remains one. This last obstruction explains why increasing k changes the example: one cannot remove the finite-rank cutoff in this construction.

To prescribe the natural density ρ, take $c = 1 - (1 - \rho)^{1/k}$ for every auxiliary family. The k-angle rotation is equidistributed, and the finite-union-of-intervals approximations have uniformly small natural-density errors. Their union therefore has density $1 - (1 - c)^k = \rho$. Take $d_j > 2^j$ in the original compact construction, so its set of steps has natural density zero.

Why countably many independent tests instead give full recurrence. If the β_i generate a group of unbounded rational rank modulo torsion and every auxiliary mass c_i is positive, their union is recurrent at every length in every system. A positive common h-avoider would give, by the BHK characteristic comparison and the two conditional superlevel replacements in the fixed-length theorem, a single positive event on a finite nilfactor whose weighted correlation means vanish for all the auxiliary indicators. All weights are bounded by one, so no simultaneous quantitative rate or uniform lower bound on c_i is needed. Some β_i has no nonzero multiple in the finite-rank eigenfrequency group of that nilfactor, contradicting the positive weighted mean $c_i m_A$ there.

R170. Exact odd-step threshold inside an ergodic integer system

Scope and status. Result with proof.

Result. The exact odd-difference cap in an ergodic integer system is $1 - 1/Q$, where Q is its maximal dyadic cyclic-factor order.

Scope. This is a fixed-system formula; arbitrary systems give capacity one, and subaction roots need not cohere globally.

For an ergodic probability-preserving transformation T, let Q be the largest power of two for which T has a cyclic factor of order Q, taking $Q = 1$ if there is no even-order cyclic factor, and $Q = \infty$ if these orders are unbounded. The supremum of masses of events in this particular system that avoid, for some fixed length, every progression of odd difference is exactly $1 - 1/Q$ (with $1/\infty = 0$). This is a fixed-system result. Allowing all integer systems gives the familiar endpoint one for the odd differences themselves.

Lower bound. On a cyclic factor of order 2^k omit one state. Every odd difference traverses the entire factor within 2^k slots. Take the largest order or let it increase if $Q = \infty$.

Residue averages. For an invertible transformation W for which W^q is ergodic, the mean of any fixed-length correlation on the class $d = qm + 1$ equals its ordinary Cesàro mean: apply Frantzikinakis, Theorem 6.4 to the slot functions $W^j f_j$ and shift the ordinary average by one. In particular the mean for h copies of an event C of mass $\geq c$ is bounded below by a positive constant $\kappa(h, c)$, uniform in the system, by finite Szemerédi and Varnavides averaging. This proves the upper bound if $Q = 1$ by applying it with $q = 2$.

Odd-depth root for a nilfactor. Suppose S is an ergodic nilrotation with S^q ergodic, q a power of two. It has an extension of the form Z, W^q with W^q ergodic. To see this first suppose its nilmanifold is connected. On its connected, simply connected cover group N with lattice Γ, present the transformation as $x\Gamma \mapsto a\phi(x)\Gamma$: write the translating element as a times a representative in the lattice of its component. Conjugation by the latter induces the unipotent rational automorphism ϕ of N preserving Γ. The rational automorphism $\phi^{1/q} = \exp((\log \phi)/q)$ takes Γ to a commensurable lattice. The intersection Γ' of $\phi^{j/q}\Gamma$ for $0 \leq j < q$ is a finite-index sublattice of Γ invariant under $\phi^{1/q}$. In the simply connected nilpotent group $N \rtimes R$ with real action $\exp(t \log \phi)$, take the qth root of $(a, 1)$. It acts on N/Γ' as an affine transformation W_0 with W_0^q lifting S. That lift is ergodic: a nonzero obstructing horizontal character on the connected finite cover would have a nonzero multiple descending to an obstructing character downstairs, by the horizontal criterion for ergodic affine nilrotations. The commensurability and horizontal criteria are the usual Malcev and nilrotation facts. If the manifold has l connected components, ergodicity cycles them and l is coprime to q. Apply the connected construction to S^l on a component. Build the l-level tower W over its root W_0. The map W^q cycles the levels transitively and its lth power on the initial level factors onto S^l. Explicitly set $\pi(W^{qj}y) = S^j\pi_0(y)$ for $0 \leq j < l$ and y in the initial level; at the wrap the definition agrees because W_0^q factors onto S^l. Thus Z, W^q is ergodic and π is the desired factor. This proof needs a finite cover; taking the usual q-level tower over S would leave its qth power nonergodic.

Upper bound for $2 \leq Q < \infty$. Fix A with $\mu(A) > 1 - 1/Q$ and h. Choose one of the Q ergodic components Y of T^Q, with normalized measure and $S = T^Q$ there. The components are cycled by T, and S^Q is ergodic on Y because the 2-primary order of T is exhausted. Put

$B_r = \{y \in Y : T^r y \in A\}$ for $0 \le r < Q$. Their normalized masses sum to $Q\mu(A) > Q - 1$. On the starting sheet Y, the desired AP correlations, with $d = Qm + 1$, test

$$S^{jm+\lfloor j/Q \rfloor} y \in B_{j \bmod Q}, 0 \le j < h.$$

The linear characteristic-factor theorem for the ergodic integer action S lets us replace the indicators by their expectations on its finite-order Host–Kra factor in this Cesàro mean (the fixed shifts commute with projection). This factor is an inverse limit of ergodic nilsystems. On each finite nilfactor let g_r be the corresponding conditional expectations. For fixed sufficiently small $\epsilon > 0$ the sets $D_r = \{g_r \ge \epsilon\}$ satisfy $\sum_r \mu(D_r) - Q + 1 \ge c > 0$ uniformly over these factors, since $\mu(g_r) \le \epsilon + (1 - \epsilon)\mu(D_r)$. Take the root extension W of the factor just constructed. The set

$$C = intersection_{r<Q} W^r \pi^{-1}(D_r)$$

has mass at least c. Whenever $W^{jd} x \in C$ for all j, projecting $W^{jd-(j \bmod Q)} x$ gives all the membership tests above on this finite factor. The finite-factor mean for the g_r is at least $\epsilon^h \kappa(h, c)$, by the residue-average identity. Martingale convergence and the h-term telescoping bound pass this uniform positive inequality to the Host–Kra expectations. The original all-zero avoidance is impossible.

Application to candidates with pure marks. If an integer subgroup generator R has every odd multiple among tested directions, the same proof on its ergodic components shows every h-avoider has mass at most $1-\text{integral}(1/Q_R(x))$, where $Q_R(x)$ is the componentwise 2-primary depth. If avoiding masses in one action tend to one, then $Q_R(x) = \infty$ almost surely for every such R. For countably many R these are simultaneous conull conditions. General $b = 1$ is witnessed in one ergodic action: take products of systems with deficits summable and select an ergodic component on which the sum of their deficits remains finite. The obstruction lengths may still vary. This forces unbounded rational spectral orders on each individual occupied direction. Such subaction information alone does not guarantee a global common eigenfunction.

R195. Exact threshold on a fixed ergodic system for unions of residue classes

Scope and status. Result with proof.

Result. For an open union of residue classes in a fixed ergodic system, the all-length threshold is $1 - 1/k$, with k determined by cyclic factors and allowed residues.

Scope. The formula is relative to one system. Unrestricted integer capacity for these open periodic families is only zero or one.

Let $D \neq \varnothing$ be a union, possibly infinite, of positive-integer residue classes. Let T be an invertible ergodic probability-preserving transformation. Write $\mathcal{Q}(T)$ for the positive integers q for which T has a cyclic factor of order q. Include 1. Then the supremum of the measures of events in this particular system that avoid all D-progressions at some finite length is

$$1 - \frac{1}{k}, \qquad k = \sup_{q \in \mathcal{Q}(T)} \min_{d \in D} \frac{q}{\gcd(q, d)}.$$

We use $1/\infty = 0$. Equivalently k is the least order of an element of the closure of D in the maximal rational Kronecker factor of T, with infinite order allowed. To see this, realize the procyclic factor as an inverse limit over a divisibility-increasing cofinal sequence of finite cyclic factors. Orders can only increase when passing to a higher factor. If the supremum is finite, the compact sets of points in the closure whose projections have order at most k have the finite intersection property. A point in their intersection has order at most k: its order in the increasing quotients eventually stabilizes because of divisibility. The reverse bound is immediate. For odd steps this recovers the dyadic-order formula [R170].

Lower bound. For a finite cyclic factor of order q, set $M = \max_{d \in D} \gcd(q, d)$. Delete the consecutive residues $0, \ldots, M-1$. Each d-orbit is a residue class modulo $\gcd(q, d)$, so all its orbits meet the deleted interval within q slots. The resulting event has measure $1 - M/q$. This argument works for completely arbitrary D.

Upper bound. Suppose $k < \infty$ and an event of measure $a > 1 - 1/k$ avoids every legal progression of some fixed length h. For each residue class $d = d_0 + \ell n$ contained in D, its normalized Cesàro correlation mean is zero. Replace every indicator by $g = \mathbb{E}(1_A \mid Z_{h-1}(T))$ in this mean. This is the ordinary linear characteristic-factor theorem applied at the specified slopes and shifts. One can also apply it to T^ℓ, since the positive-order Host–Kra factors agree for nonzero powers (see Frantzikinakis, Applications of ergodic theory to combinatorics, Appendix A). All these statements refer to normalized averages within the class.

Let f be the conditional expectation of 1_A on a finite-dimensional ergodic nilsystem factor approximating g in L^1. It has integral a and lies in $[0, 1]$. We show that on **some** residue class inside D the correlation mean for f is at least a constant depending only on a, h, k. Telescoping the product costs at most $h\|f - g\|_1$ uniformly for every choice of that class, so this claim suffices.

Let m be the number of connected components of the nilfactor, a cyclic factor order of T. Pick $d_0 \in D$ with $u = m/\gcd(d_0, m) \leq k$, write $t = \gcd(d_0, m)$, $b = d_0/t$, so $\gcd(b, u) = 1$. As D is open in the profinite topology we can take a class $d = d_0 + \ell n$ inside D with $m \mid \ell$. The action T^t partitions the components into cosets of size u. A positive fraction of cosets,

bounded below in terms of a, k, have normalized f-mean larger than $1 - 1/k + \eta$, with $\eta = (a - 1 + 1/k)/2 > 0$. This follows by integrating the coset means in $[0, 1]$.

Fix such a coset and a starting component Y. Use its normalized measure and write $S = T^m$ on Y. This is a totally ergodic affine nilrotation on a connected nilmanifold. Set $f_r(y) = f(T^{rt}y)$ for $0 \le r < u$. For $\epsilon > 0$ sufficiently small in terms of η, the superlevel sets $E_r = \{f_r \ge \epsilon\}$ obey $\sum_r (1 - \mu(E_r)) \le 1 - c$ for $c > 0$ depending only on η, k. Indeed the sum is at most $u(1/k - \eta)/(1 - \epsilon)$, with $u \le k$.

Use the connected nilrotation root construction [R170]: on a finite connected cover there is a totally ergodic affine map W with W^u factoring onto S by π. For clarity, on the simply connected nilpotent Lie group an affine map $a\phi$ has unipotent rational automorphism part ϕ; use the lattice $\bigcap_{0 \le i < u} \phi^{i/u}\Gamma$ and the u-th root in the real nilpotent semidirect product. Ergodicity of the lifted power follows by descending an integer multiple of any obstructing horizontal character. Connectedness then gives total ergodicity. Put $C = \bigcap_{r<u} W^r \pi^{-1} E_r$, of measure at least c.

On this sheet the physical displacements are $jt(d/t)$. If $W^{jd/t}x \in C$ for all $0 \le j < h$, choose r_j to be the representative of $jb \bmod u$. Subtracting r_j from the exponent and projecting through W^u gives

$$f_{r_j}\big(S^{(jd/t - r_j)/u}\pi x\big) \ge \epsilon.$$

These are exactly the original slot tests on the starting component. The means for W at the positive velocities $d/t = b + (\ell/t)n$ coincide with the ordinary means for the same event C, by total ergodicity and the arithmetic-progression mean identity (Frantzikinakis, Theorem 6.4), applied after fixed shifts to the slot functions. The ordinary mean has the system-independent positive lower bound from finite Szemerédi and Varnavides averaging at h, c. Multiplying by ϵ^h supplies the asserted lower bound on this sheet. Summing over qualifying cosets, even if we use only one starting sheet in each, loses a factor at most k and the fixed qualifying fraction. This proves the upper bound, including $k = 1$.

Scope and variants. In the full integer problem the unrestricted cyclic orders give only the two endpoints for profinite-open D. If 0 lies in its closure in $\widehat{\mathbb{Z}}$, then D contains an IP set: successively pick differences divisible by all moduli protecting the previously chosen sums, and a residue neighborhood inside D for each new sum. Then finite-sums recurrence applies at all positive densities. If 0 is outside that closure, some modulus separates all of D from zero, and arbitrarily deep cyclic lifts give cap one. For a fixed-system upper bound the proof only uses the following local condition: for every $m \in \mathcal{Q}(T)$ and $\xi > 0$ there is a positive residue class $d_0 + \ell\mathbb{N}$ with $m \mid \ell$, $m/\gcd(m, d_0) \le k$, and relative upper natural density of D on the class greater than $1 - \xi$. Choose ξ smaller than the uniform lower bound, and compare normalized averages along prefixes on which the deleted fraction is small. Entirely arbitrary integer step sets lack this local-density condition.

R211. Two ends of the independent-selector regime

Scope and status. Result with proof.

Result. Independent selectors of probability $O(1/n)$ have capacity one almost surely; subpolynomially dense selectors have capacity zero almost surely.

Scope. Two endpoint regimes for random selectors.

Qualifications. The intermediate polynomial-probability regime is not settled.

Let R select each positive integer independently with probability p_n. If $p_n \leq C/n$, its unrestricted all-length critical value is one almost surely. In fact no integer is a Bohr accumulation point of R. This uses Kahane and Katznelson, Entiers aléatoires, ensembles de Sidon, densité dans le groupe de Bohr et ensembles d'analyticité, which gives almost-sure nondensity in the Bohr group in this regime.

It is enough by monotone coupling to use a decreasing majorant q_n equal to C/n eventually, capped at one. Let E_a mean that some Bohr neighborhood of the integer a meets the selected set only finitely often. This is a measurable tail event: for any finite rank, fixed positive radius, and finite initial exception, simultaneous separation at all remaining selected n by a finite tuple of characters is a closed condition in the compact character-parameter torus. Existence is a compact projection; take the countable union of the resulting tests. Almost-sure nondensity implies $\bigcup_{a\in\mathbb{Z}} E_a$ has probability one, since ordinary integers are dense in the Bohr compactification. Each E_a has the same probability as E_0. Indeed for a fixed shift a the selector laws with probabilities q_n and q_{n+a} admit a common-uniform coupling differing at only finitely many positions almost surely: $\Sigma|q_n - q_{n+a}|$ is finite by monotonicity. Remove the harmless finite initial portion. The zero-one law and the countable union therefore give probability one for every E_a simultaneously. For $a = 0$ shrink the neighborhood to remove the finitely many nonzero exceptions, so the set misses a Bohr zero neighborhood. The polychromatic theorem or circle-root sieve gives its all-length critical value one.

In the opposite direction, if $p_n \geq n^{-o(1)}$ in the sense that $p_n \geq n^{-\epsilon}$ eventually for every $\epsilon > 0$, the threshold is zero almost surely by an elementary dilation argument. For a fixed m choose positive integers t in successive dyadic intervals, greedily taking a positive fraction in each interval so that all the sets $t[1, m]$ are disjoint (each conflicts with at most m^2 others; alternatively take prime $t > m$). The probabilities of retaining a full $t[1, m]$ then have divergent sum: use $\epsilon = 1/(2m)$. Borel–Cantelli supplies such a copy for every m simultaneously. Every event of positive density has a progression of any fixed length at a difference in $[1, m]$ for sufficiently large m, uniformly over stationary systems by finite Szemerédi; apply this to the t-th power to use the retained copy. The two endpoint arguments do not decide the all-length threshold in the intervening polynomial-sparsity regime.

R243. Dilution of every nearly minimizing family of hitters

Scope and status. Result with proof.

Result. If an intermediate capacity existed, near-minimal hitter families would have vanishing stationary stride-hit frequency by a root-thinning inequality.

Scope. A necessary consequence under the hypothesis $0 < b(D) < 1$.

Qualifications. This does not prove that an intermediate D exists.

Let D be a nonempty set of integer steps with $0 < b(D) < 1$, write $\kappa = 1 - b(D)$, and fix one positive k-avoiding event of mass a. Suppose H is a stationary binary hitting mask: for some finite h every h-slot run at every $d \in D$ has a hit, almost surely. Write p for the mass of H. In the closed support of the binary process let Y be the closed shift hull of all stride names read at $d \in D$, including arbitrary starting positions. For a positive integer M let R_M be the subshift of binary strings whose ones have no k-term progression of stride in $[1, M]$.

$$p(1 - a^M) < \kappa implies Y \cap R_M \neq \varnothing.$$

If the two subshifts were disjoint, some common finite length L would force a k-term progression of hits of stride at most M inside every length-L word of Y. Otherwise center arbitrarily long violating words and use compactness. Intersect H with the union of independent copies of the complement of the k-avoiding seed, one copy in each of the integer root towers at strides 1,. . .,M. On an original d-run its selected k hits at stride c become a d-run in the c-root tower, so one of the selected hits survives. The new mask hits every L-run and has mass $p(1 - a^M)$. This contradicts the definition of κ. The same proof holds if Y was formed from any shift-invariant closed full-measure set containing the actual support.

Consequently, for any sequence H_n of finite-length hitting masks with $p_n \to \kappa$, the least stationary frequency of hits on their respective stride hulls tends to zero. For each fixed M, eventually $p_n(1 - a^M) < \kappa$, so the corresponding hull has a stationary measure supported on R_M (average shifts of one point of the intersection). In each M-site window of R_M there are at most $r_k(M)$ hits, where $r_k(M)$ is the usual Szemerédi extremal number. Its stationary hit frequency is at most $r_k(M)/M$. Taking the limsup in n before letting $M \to \infty$ proves the assertion. The global densities p_n themselves stay bounded below by κ. The stride names here generally come from varying d; rescaling a configuration by d need not preserve its original forbidden difference menu. This is why the local low-hit-frequency limit alone does not contradict that bound.

Chapter 7

Higher digits, affine models, and finite-order obstructions

R007. Binary saturation for nonlocal relative eigenfields

Scope and status. Result with proof.

Result. Binary saturation gives all-order bounds $1 - 2^{-s}$ for specified relative eigenfield classes, and the fixed-h bound $1 - 1/\lceil h/2 \rceil$ in the stated integral-cocycle class.

Scope. The relevant rank is the observed modulation rank or primitive tag rank, not just the rank of one target character.

Use the angular one-clock family [R008]. Let $Y = \mathcal{I}(S)$, and suppose the event is measurable in Y and r relative circle eigenphases $\boldsymbol{\phi}$, with high eigenlabels $\boldsymbol{\lambda}(y) \in \mathbb{T}^r$. Assume that all their increments under the selected low group belong almost surely to a countable subgroup $\Gamma \leq \mathbb{T}$, that $\theta_i \in \Gamma$, and that their classes in $\Gamma/2\Gamma$ are linearly independent over $\mathbb{F}_2$. For example take $\Gamma = \bigoplus \mathbb{Z}\theta_i$ for independent irrational tags, or enlarge individual summands by arbitrary odd-primary rational division. No restrictions are placed on which of the infinitely many low coordinates affect a given increment.

Every such event of measure $> 1 - 2^{-r}$ has returns of all finite lengths. More precisely replace r in this bound by a uniform upper bound r_0 on the rank of the observed binary translations defined below. This is a system-relative bound. The rank of the one target character alone does not bound that translation rank.

On each component set $L_y = \Gamma + \sum_{l=1}^{r} \mathbb{Z}\lambda_l(y)$. This abstract countable subgroup is unchanged by every low action because the increments belong to Γ. The kernel of $\Gamma/2\Gamma \to L_y/2L_y$ is finite dimensional of dimension at most r. In fact it is the image of the map from the 2-torsion of L_y/Γ which sends the class of z to $2z \pmod{2\Gamma}$; a group generated by r elements has 2-torsion of rank at most r. Membership of each countable class in this kernel is measurable by enumerating integer combinations of the fixed labels and Γ. Thus in a jointly ergodic system the kernel is a fixed subspace almost everywhere. All finite sums of sufficiently late tags are nonzero in $L_y/2L_y$: intersect the kernel with the span of the independent tags, choose a basis for the finite intersection, and omit its finite coordinate support. Equivalently, of $r+1$ disjoint packets, at least one packet's entire nonzero tag span misses the kernel.

Let $E_y \le \mathbb{F}_2^r$ consist of the possible binary evaluations of a homomorphism $L_y \to \mathbb{F}_2$ on the observation labels $\lambda_l(y)$. This is a measurable subgroup: a bit assignment belongs precisely when it respects all binary linear relations among the $\lambda_l(y)$ in $L_y/2L_y$, a condition on finitely many masks, each checked by a countable search. Regard the binary values as zero or $1/2$ in the circle. Each vector in E_y acts by a measure-preserving translation on the conditional joint distribution of the observed phases. Indeed that distribution is invariant under the closure of all powers of their high rotation vector, and every such binary evaluation belongs to that closure by Kronecker's theorem. Identify the original event by its measurable fiber sections in these phase coordinates. Define B to require membership in A after *all* translations in E_y, including zero. Its mass satisfies

$$\mu(B) \ge 1 - 2^{r_0}(1 - \mu(A)), \qquad r_0 = \operatorname*{ess\,sup}_y \dim E_y \le r,$$

by the union bound on each component. In a non-jointly-ergodic system restrict to jointly ergodic components of positive mass satisfying the desired density and then use countability of actual directions. The high factor and relative eigenfeature identities may equivalently be interpreted on the original ergodic-component coding; an eigenlabel on a component of a jointly invariant restriction restricts the original fiber relation.

Low DHJ on any sufficiently long packet with this kernel property gives a nonempty sum of marked moves $v = \sum_{i \in J} a_i$ with a positive low B-line. At the base components where the corresponding conditional intersection has positive measure, choose a homomorphism $\zeta : L_y \to \mathbb{F}_2$ with $\zeta(\theta_J) = 1$, possible by the nonzero binary class. It gives translations from the allowed $E_{U^{jv}y}$ simultaneously in every slot. Hence the conditional correlation using high translations $j\zeta$ at slot j is positive. To see this without assuming that the original system admits a simultaneous high involution, form the finite rotation vector

$$(\theta_J, \boldsymbol{\lambda}(y), \boldsymbol{\lambda}(U^v y), \dots, \boldsymbol{\lambda}(U^{(h-1)v} y)).$$

The binary evaluations of ζ are a point in the compact closure of its integer powers, again by Kronecker's theorem. Starting low-translated observations at all slots have a fixed conditional joint law. Their product correlation varies continuously in the translations on this compact group: it suffices to approximate each measurable slot section in its invariant one-slot law by a bounded continuous function. Each slot marginal is invariant under its own projected group, so the approximation error is uniform. The angle at the chosen point is $1/2$. Thus a positive-measure open set of high group parameters has both angular distance greater than c and positive conditional return. Equidistribution of integer powers gives a positive average when the high angular parameter is multiplied by a nonnegative continuous bump that is positive at $1/2$ and supported in the allowed region. Integrating over the base proves that an actual integer high time works. This pointwise argument needs no jointly measurable selection of ζ.

The upper bound extends to arbitrary high extensions if the appropriate high Host–Kra conditional expectation is measurable in this factor and its integral mass exceeds the stated threshold: apply the preceding proof on a superlevel set of the conditional expectation of mass exceeding the threshold, which is possible since $0 \le g \le 1$ and $\int g$ exceeds it. The fixed low move and angular rotation are bounded linear weights for the ordinary characteristic comparison. The natural real six-slot local tables have coefficients in odd-primary divisions of their respective independent tags, because the inverse of each cyclic difference on the

zero-sum integer lattice has determinant a power of the odd prime. They are covered by this test when the finite high observation list has bounded binary rank. Saturation over all binary shifts can cost 2^{r_0} even when only one specified functional on the high group is queried: deleting one point in every coset of $\mathbb{F}_2^{r_0}$ destroys the full orbit at density $1-2^{-r_0}$. A width-independent bound needs more than this saturation step.

Using the rank of a single tag's low variations Suppose $\chi_i : \Gamma \to \mathbb{F}_2$ read the independent tag bits, with $\chi_i(\theta_j) = 1_{i=j}$ and $\chi_i(\gamma) = 0$ eventually for every $\gamma \in \Gamma$. For any finite list of r labels suppose along infinitely many indices that all observed low cocycle increments obey

$$\chi_i\big(\boldsymbol{\lambda}(U^g y) - \boldsymbol{\lambda}(y)\big) \in W_i \quad \text{a.e. for every selected low } g, \qquad W_i \le \mathbb{F}_2^r, \quad \dim W_i \le s.$$

The subspaces here are fixed for their respective sites, but all the coefficient functions may have arbitrary dependencies on all low coordinates. It suffices to assume this for the subgroup generated by an infinite admissible subsequence. Then the cap is at most $1 - 2^{-s}$, independently of r. In particular rank-one low variations give the half bound for an arbitrarily large finite list. If a countable high factor is used and the hypothesis holds for every finite observation list at the same s, cylinder approximation transfers the bound in the manner explained below.

For the given list keep an infinite subsequence with the same W, possible since r is finite. Saturate only by the available shifts in $W \cap E_y$. Let $H_y = \langle \lambda_1(y), \dots, \lambda_r(y) \rangle$. For each fixed y, its subgroup $H_y \cap \Gamma$ is finitely generated since H_y is finitely generated abelian. Consequently χ_i vanishes on this subgroup for all sufficiently large i, depending on y. The event of this vanishing is measurable by testing the countably many label combinations in Γ; its probability tends to one. On that event, χ_i extends to a homomorphism $\zeta_i : L_y \to \mathbb{F}_2$ vanishing on the original H_y, by the elementary sum of homomorphisms on subgroups agreeing on their intersection. Choose the finite low DHJ packet so late that this exception costs less than the positive uniform line mass for every one of its indices. For the chosen nonempty mask J fix any $i \in J$; on every good initial component ζ_i has zero bits on all original labels and its bits on every translated slot belong to W by the displayed cocycle condition. It still has $\zeta_i(\theta_J) = 1$. Thus exactly the preceding argument applies to this smaller saturation.

For an event on the full countably generated relative eigenfactor, choose a finite-feature event A' of mass at least a fixed amount above $1 - 2^{-s}$ approximating it in measure. The uniform low DHJ line mass κ for its saturation is determined by that margin and h, so first choose $\mu(A \triangle A') \ll \kappa/h$. At the chosen low row lift the binary high assignment on the finite labels to a (possibly nonbinary) circle-valued homomorphism on the countable joint eigenlabel group of all its slots and the target; such a lift exists by divisibility of the circle, or by surjectivity of the compact joint rotation closure onto its finite projection. Conditional measures are preserved under this full compact translation at each slot, so the corrected return mass for A at that point is bounded below by the finite B'-line mass minus $\sum_{j<h} \mu_{U^{jv}y}(A \triangle A')$, on good components. The latter error integrates to at most $h\mu(A \triangle A')$. Thus positivity holds on a positive set of components for any such pointwise lift, with no measurability choice required. As before the conditional correlation is continuous on the full compact joint rotation closure by its invariant one-slot marginals. A positive actual high

average follows. This compares the sets at the selected *point* of the rotation group before averaging, and does not require a uniform positive Haar size of its good neighborhood.

Two independent parity shifts without a rank assumption For the original integral cocycles in the independent tag lattice, at a fixed horizon h all events of measure greater than $1-1/\lceil h/2\rceil$ recur, with no bound on the number of observed features or the modulation ranks. Thus for $h=4$ the cap here is at most one half. The bound still approaches one as the horizon increases.

Begin with finitely many observed phases and write $m=\lceil h/2\rceil$. Let $\rho(x)$ be the density of the translated phase point in A as the sign runs uniformly over E_y; its mean is $\mu(A)$. For sufficiently small $\epsilon>0$ the event $B=\{\rho>1-1/m+\epsilon\}$ has uniformly positive mass in terms of the original mass margin. In $\Gamma/2\Gamma$ the subspace of tag classes falling in the span of the observed labels of a **single slot** in $L_y/2L_y$ is $(2\Gamma+(\Gamma\cap H_y))/2\Gamma$, since $2L_y+H_y=2\Gamma+H_y$. It is finite dimensional. Its intersection with the span of the independent tags has finite index support depending measurably on y. Consequently the probability that any given nonzero sum of sufficiently late distinct tags belongs to this single-slot span is uniformly small. By stationarity this is true at every one of the h slots of any fixed candidate low sum, without any independence of those slots.

Choose a late low DHJ packet with a uniformly positive B-line, and discard its small probability of any single-slot exception. Fix its mask J. At any good component let η_0,η_1 be independent Haar binary homomorphisms on $L_y/2L_y$, respectively conditioned on their values at θ_J being zero and one. The evaluated sign of either map at the labels of **each separate slot** is still uniformly distributed on the full $E_{Uj^v y}$, since the target class is outside that slot span. At a point on the B-line use η_0 in all even slots and η_1 in all odd slots. Within either parity class the sum of the individual miss probabilities is at most $m(1/m-\epsilon)$. Thus the probability of satisfying both classes is at least $(m\epsilon)^2$, by independence of the two homomorphisms. In each success write the applied shift as $\eta_0+j\zeta$, where $\zeta=\eta_1-\eta_0$ has angular value $1/2$. After integrating over starting points, the common translation η_0 does not change the conditional joint correlation: that joint law is invariant under the closure of high powers simultaneously applied to every slot. Averaging the binary choices therefore gives a positive integrated correlation at some angle-correcting point on the good components. The same compact-rotation continuity and high equidistribution finish. The quantitative lower at the selected points is at least a fixed multiple of the low line mass before averaging over the full high torus, so the preceding finite-feature approximation argument also applies to countably generated relative eigenfactors.

Removing the fixed cocycle group by primitive tag rank For arbitrary relative eigenlabels define the countable group L_y to be generated by all the tags and all the translates $\lambda_l(U^g y)$ under the countable low group. It is the same group at all low points in an orbit. Membership of any given finite tag sum in $2L_y$ is a measurable invariant condition by countable enumeration. On a jointly ergodic system the binary linear relations of the tag images are therefore almost surely fixed. If their images span an infinite-dimensional subgroup of $L_y/2L_y$ almost everywhere, choose an independent infinite tag subsequence once and for all. Every finite-slot binary saturation argument above using E_y now goes through without an increment assumption. In the two-parity proof, each single-slot label span in $L_y/2L_y$ is finite dimensional, so its intersection with the span of the independent subsequence has finite coordinate support almost surely. The probabilities of hitting it by any nonzero late

tag sum are uniformly small; the countable tests for a relation again ensure measurability. This gives the rank-based all-order bound for a finite observable system and the dimension-free fixed-order bound for countably generated relative eigenfields under the stated infinite primitive rank. The same saturation argument over any fixed prime q gives $1 - q^{-r}$ if the images have infinite rank modulo qL_y, using a target value j/q with distance greater than the angular cutoff. The two independent parity shifts specifically use exponent two. A joint spectrum containing all rational roots of all the tags has zero primitive tag rank, so this is not an unrestricted upper.

The marked orbit alone suffices for the rank test In the finite-feature modulation argument, it in fact suffices to have, for each fixed horizon, an infinite tag subsequence and subspaces W_i of dimension at most s for which $\Pr(\chi_i(\boldsymbol{\lambda}(U^{ja_i}y) - \boldsymbol{\lambda}(y)) \notin W_i) \to 0$ for each $0 \le j < h$. Keep one common subspace on a subsequence. Select the low packet inductively. For any fixed previous subset sum w, the increment for $j(w + a_i)$ is the sum of the increment for ja_i and the increment for the fixed move jw at $U^{ja_i}y$. The tag bit of this latter term vanishes in probability as $i \to \infty$, since each fixed group element has eventually zero bit and the shifted base point is stationary. Choose the next index so late that all such finitely many errors and all lift failures are arbitrarily small relative to the predetermined low-line mass. On the eventual DHJ line use its *last* selected index for χ_i. Its correction evaluations belong to the common subspace at all the slots except on this small error set. No conditions on its responses to other-coordinate shifts, or independence of local tables across the packet, are needed.

R008. Angular tags, individual resonances, and programmable high parameters

Scope and status. Proved subcases and unresolved extensions.

Result. The real angular-clock family retains a six-slot obstruction; Fourier coefficients impose exact high-spectrum resonances and yield recurrence in the stated finite-rank-spectrum classes.

Scope. General relative eigenfields can contain matched atoms; no arbitrary-system all-order upper bound follows.

On the one-clock model [R237] fix real numbers θ_i and $0 < c < 1/2$. The low residues are $0, 1, L_i$. If J is the nonempty set at the marked velocity L_i, admit (u, v) when $\|u\theta_J\| > c$, where $\theta_J = \sum_{i\in J} \theta_i$. All arguments apply to a family defined with $\geq c$ by using continuous weights supported in the strict region. The dyadic choice is $\theta_i = 1/(2z_i)$.

The six-slot obstruction persists for real tags. Solve the same finite low-table equation over $\mathbb{R}$ with right side $\theta_i h_i$, by invertibility of the differences on the real zero-sum subspace. For a finite packet let $B \subset \mathbb{R}$ be the countable additive group generated by the table entries and θ_i. Take a Haar character $x \in \widehat{B}$, where B is discrete. Advancing the clock adds the character $b \mapsto b \bmod 1$ in additive notation. Use the observations $x(b_i(y_i))$. Their tested filter after a step (u, v) gives exactly $u\theta_J$ on the same safe low residues. Small boxes in the finitely many spatial circle channels, chosen using c and the filter coefficients only, give a uniform positive six-avoider across finite packets. This does not require a common finite high period for all the θ_i.

Arbitrary stationary systems. Fix an event A and a low move v. Put $a_v(u) = \int \prod_{j<h} S^{ju} U^{jv} 1_A$. If a Fourier–Bohr coefficient of this high correlation at a fixed $\xi \in \mathbb{T}$ is nonzero, then $e(\xi)$ is an eigenvalue of the global high action S, without a high ergodicity assumption. On an ergodic component this follows from the Moreira–Richter spectral refinement, with their corrigendum included. A continuous nilsequence approximation whose spectrum is contained in that component's spectrum has zero coefficient outside it: take a joining of its nilorbit with the circle character and condition the circle character on the nilfactor. That conditional expectation must be an eigenfunction there. Cesàro-null and small L^1 errors do not change the conclusion. Integrating components gives the assertion. Indeed occurrence of a fixed eigenvalue on positive component mass gives a global eigenfunction, as seen from the twisted mean ergodic projections of a countable generating dense list of observables.

Choose $w \geq 0$ continuous, supported in $\{\|s\| > c\}$, with Haar mean $w_0 > 0$. Low density Hales–Jewett gives a uniform lower bound $\kappa > 0$ for the mass of $C = \bigcap U^{-jv} A$ for one nonempty marked sum v in every sufficiently long packet. Uniform Szemerédi applied to C makes the unweighted high average of a_v at least $\kappa' > 0$. Approximate w uniformly by a finite trigonometric polynomial with the same mean and error below $w_0\kappa'/4$. If the angular family avoids A, the weighted average of $a_v(u)w(u\theta_J)$ is zero, so at least one of the finite nonconstant frequencies $t\theta_J$ has a Fourier coefficient of absolute value bounded below by a positive constant fixed by h, A, w, uniformly over the low packet. For the dyadic choice take the axes late enough that no nonzero listed t annihilates θ_J. This replaces an entire high dyadic shell by finitely many exact atoms.

In particular, when the θ_i are rationally independent modulo torsion, every positive event

recurs as long as the global high eigengroup modulo torsion has finite rational rank. More generally it suffices that arbitrarily long late packets have their $\mathbb{Q}$-span disjoint from the rational span of that eigengroup modulo torsion. Finite rank guarantees this by extending the independent tag list to an algebraic basis and deleting the finitely many indices in the supports of a spanning set. The selected nonzero frequencies then have no global high eigenvalue. This conclusion applies to arbitrary low actions and arbitrary extensions, including systems whose high action is nonergodic.

Arbitrary relative eigenfields with countable low cocycles. With the notation of the relative clock test [R236], testing a temporal row on relative eigenphases produces $\beta(v, y) = \sum_j j c_j \cdot \boldsymbol{\lambda}(U^{jv} y)$. The nonconstant weight term of frequency t survives only if $\beta(v, y) = -t\theta_J$ exactly. Suppose the low increments of the tested labels take values almost surely in a countable subgroup $\Gamma \leq \mathbb{T}$. For every fixed finite prior sum v_0, the difference $\beta(v_0 + a_i, y) - \beta(a_i, y)$ belongs to a fixed finite part of Γ with probability as near one as prescribed, uniformly in the next axis: telescope the fixed jv_0 increments at the stationary points $U^{ja_i} y$. Thus a sufficient condition for positive recurrence on this factor is an infinite subsequence where, for all finitely tested rows, nonzero bounded t, and all fixed $g \in \Gamma + \langle \theta_i : i \geq 1 \rangle$,

$$\mathbb{P}\big(\beta(a_i, y) = -t\theta_i + g\big) \longrightarrow 0.$$

Indeed at each stage truncate all the previous increment sums, include the finitely many shifts $-t\theta_{J_0}$ from all previous submasks, and choose the next axis to evade those finitely many exact atom tests. For every subset the weighted correlation then differs from w_0 times its unweighted correlation by an error small uniformly in the chosen subset. Low DHJ and the unweighted high recurrence finish, using bounded Fourier approximation as in the relative note. In particular a uniformly vanishing maximum atom suffices. A matched atom can occur in the six-slot low witness; this criterion does not bound all stationary systems.

Programming the high parameters. For rationally independent tags together with one, any avoiding law on the original high integer and finitely many low axes yields an avoiding law of the same mass on arbitrary independently specified free high parameters. For integer generators $e'_1, \dots, e'_r$, give each a desired vector $\gamma_l \in \mathbb{T}^m$ on a finite packet of m low sites. Choose integers $n_l^{(k)}$ so that $n_l^{(k)}(\theta_1, \dots, \theta_m) \to \gamma_l$ by Kronecker. Pull the binary law back along $e'_l \mapsto n_l^{(k)} e_0$, retaining all low axes, and pass to a stationary weak limit. For any strict angular inequality with programmed high parameters the finite forbidden cylinder was zero for all sufficiently large stages. A countable diagonal deals with countably many programmed axes and sites. The same conclusion holds with a free rational vector group $\bigoplus \mathbb{Q}$ for high parameters and any homomorphism from it into $\mathbb{T}^m$: at each stage treat a finitely generated subgroup, choose images for a free basis by Kronecker, and coinduce its law to the full group to run the same compactness argument. This transfer does not impose finite order on the high transformations.

R009. Angular tags on integer word cubes

Scope and status. Result with proof.

Result. A k-knob angular family forces every length $h \le k+1$ in arbitrary actions but has a positive avoiding event at a later length at most 2^{k+1}.

Scope. k is fixed; the obstruction length grows with k.

For any fixed $k \ge 2$ choose at each low site i integer velocities $b_{i,0}, \dots, b_{i,k-1}$ whose subset sums are distinct modulo the site prime p_i. Assume $\sum 1/p_i < \infty$ and use independent irrational tags θ_i, jointly rationally independent with one. Permit all subset sums of the $b_{i,l}$, including zero, but mark only the sum of *every* one of the k choices. A high integer u yields a legal step when $\|u \sum_{i \in J} \theta_i\| > c$, with $0 < c < 1/2$ and mark mask J. As before, every lift of a permitted low residue is allowed on a free host.

Every positive event in an arbitrary pmp action recurs on this family at each length $h \le k+1$, even using only canonical low lifts. On the other hand some positive event avoids a common longer length at most 2^{k+1}. These statements concern a fixed k; no all-order upper threshold follows.

First, for $h \le k$, in one test pool sample independent fair Booleans $\omega_{i,l}$ and choose $r_i = \sum_l b_{i,l}\omega_{i,l}$. For fixed high u a nonconstant angular Fourier term of frequency $t \ne 0$ contributes the weight

$$W_u(\omega) = e\Big(tu \sum_i \theta_i \prod_{l=0}^{k-1} \omega_{i,l}\Big).$$

Shift the common starting point by $-\sum_{i,l} l\, b_{i,l}\omega_{i,l} e_i$. By stationarity this preserves the correlation. At time $j < h$ the shifted observation is independent of the entire Boolean column $(\omega_{i,j})_i$ for that fixed starting point. Dummy players equal to one cover the other columns. The complete multipartite box Cauchy–Schwarz inequality bounds the magnitude of each weighted averaged correlation by $\|W_u\|_{\square^k}$. In passing to a pool within a larger wildcard sum, hold the other pools fixed, their angular weights being a scalar of modulus one independent of this pool. Take the final average over starting points.

The 2^k-th power of this box norm is

$$\prod_i [1 - 2^{-k}(1 - \cos(2\pi t u \theta_i))].$$

Indeed each coordinate's mixed Boolean difference is the product of its k independent Bernoulli differences, and conditioned on being nonzero its sign is symmetric. By Kronecker, the high integer Cesàro mean of the product equals $(1-2^{-k})^{\#\mathrm{pool}}$. Jensen therefore bounds the average box norm itself by $(1-2^{-k})^{\#\mathrm{pool}/2^k}$.

There is one more independent input, the high integer u itself. For $h = k+1$ shift the common high start by $-ku$ as well, so the last slot is independent of this input. Use the $(k+1)$-column box norm on the product of the fair low columns and a uniform high Følner interval. The mixed derivative is the character at $t(u^0 - u^1)\sum_i \theta_i \prod_l (\omega_{i,l}^0 - \omega_{i,l}^1)$; contributions from any other fixed low pool are killed by a low difference. Taking the two independent high averages gives zero unless all the displayed integer coordinate products vanish, since the tag list is independent modulo torsion. The probability of that event is $(1-2^{-k})^{\#\mathrm{pool}}$,

yielding exponentially small box norm after its fixed root. This includes the extra time slot without any assumption on the system.

For completeness assign disjoint large test pools to every one of the finitely many low DHJ variables. For each realization the DHJ line bound makes the sum of low h-return correlations over all nonempty wildcard subsets uniformly positive. For at least one subset on a set of samples of uniformly positive measure, the low joint event thus has uniformly positive mass. Szemerédi on that event under the high action gives a positive uniform lower bound on the sum of the ordinary high averages of the original correlations. A nonnegative continuous angular weight supported strictly in the allowed arc and of positive mean retains this lower bound times its mean up to arbitrary error: approximate it uniformly by a fixed trigonometric polynomial and apply the preceding bound to each nonconstant term on each candidate subset. The pool size can be taken arbitrarily large before sampling. A positive weighted sum gives an actual legal return.

For the low obstruction use the real-valued clock tables [R008]. At each site take one squared factor $(T - S^{r})^{2}$ for each of the $2^{k} - 1$ unmarked velocities, and take $T - S^{r_*}$ at the marked velocity. The filter degree is $2(2^{k} - 1) + 1$, hence the displayed slot length. Each difference $S^{r_*} - S^{r}$ is invertible on the real zero-sum group-ring subspace of the prime cycle. Solve their squared product against the zero-sum forced label θ_i away from one exception and $(1 - p_i)\theta_i$ at that exception. The same finite-channel safe coloring has positive mass uniformly over all finite packets. Zero is unmarked, so the temporal sums at every channel vanish. This proves the later-order obstruction.

R010. A four-slot spectral obstruction to uniform angular damping

Scope and status. Result with proof.

Result. A four-slot relative eigenfield can have nearly maximal matched angular atoms; random lifting yields a per-direction soft obstruction.

Scope. An approximate or soft obstruction does not establish an exact all-step avoiding event.

This concerns the angular high clock [R008]. It exhibits actual relative eigenfields having large matched angular atoms for four slots, even on every low support in a finite packet. The result is a soft bound, with no all-step avoidance conclusion.

Use any finite set of distinct sites with low group $\prod C_{p_i}$. Put $D_i = L_i(L_i-1)$ (this notation here is a scalar denominator, not a clock depth). Take the dimension-free random lattice section [R125] on the low torus and write $Y_i(y) = p_i s_i((y_l/p_l)_l) \in \mathbb{Z}$ for the lift. For a given low direction with canonical residues $r_i \in \{0, 1, L_i\}$, a uniform initial y and random section, the event that **all** these lifts along slots $j = 0, 1, 2, 3$ equal $Y_i(y) + jr_i$ in every coordinate has probability at least

$$1 - 6\pi\Big(\sum_{i:r_i\neq 0} (r_i/p_i)^2\Big)^{1/2}.$$

Let $c = (-1, 3, -3, 1)$, $b = (1, -2, 1, 0)$. At each slot use the fixed real-valued label

$$f_j(y) = \sum_i \frac{\theta_i}{D_i}\left(\frac{c_j}{6}Y_i(y)^2 - \frac{b_j}{2}Y_i(y)\right).$$

On the common good event the two exact identities are

$$\sum_{j=0}^{3} f_j(y + jr) = 0, \qquad \sum_{j=0}^{3} j f_j(y + jr) = \sum_i \theta_i \frac{r_i^2 - r_i}{D_i} = \theta_J.$$

The cubic finite-difference row c annihilates every quadratic, and $\sum j^3 c_j = 6$. The second-difference row b annihilates affine polynomials and $\sum j^2 b_j = 2$. Both identities are real, without reduction modulo any prime. On a finite packet the section can be taken as part of the probability space. All possible lifts of the finitely many points belong to the countable integer lattice. Let $B \subset \mathbb{R}$ be the countable group generated by all possible labels, and give $\widehat{B}$ Haar law. The high clock adds the character $b \mapsto b \pmod 1$, the low group shifts y and fixes the random section. The phases $e(x(f_j(y)))$ are genuine relative eigenfunctions of this system. The angular Fourier coefficient of the four-slot product at θ_J is therefore at least the displayed success probability for every selected direction, by orthogonality in the initial character x and in the high time u. These phases use only two channels, with slot weights c_j, b_j.

The corresponding two torus observables, quadratic and linear in the lifted low variables and affine in the high clock, follow time polynomials of degrees at most three on any correctly lifted window. The four-slot filter has output $u\theta_J$. For a rational-rooted lift, apply the finite-window reasoning of the polynomial root gate [R027]: after choosing the density target, there

are finitely many rows, root strides and a fixed horizon. In each residue class of each finite row, the probability of an erroneous low lift on its tested window is at most the window length times the section bound. A union bound over these finitely many rows and slots gives a near-full soft obstruction with correlation $O_{h,\eta,c}(\sqrt{\sum_{i:r_i\neq 0}(r_i/p_i)^2})$ at every legal step, on the free pullback to the rational-rooted host. In particular when $p_i \asymp L_i^3$ and about L_i^2 sites are used at each level, this error tends to zero uniformly on tails, even though $\sum_i 1/L_i^2$ diverges. This computation explains why the unconditional three-slot Fourier attenuation alone cannot extend to the same uniform matched-atom estimate for four slots. A positive perhaps very small true correlation, and an exact simultaneous all-step obstruction, are different issues.

R030. Exact centered integer charges in repeated integer progression games

Scope and status. Result with proof.

Result. Replicated integer query games with an ordered charge-one relation force every prescribed exact signed total charge; several digit-statistic consequences follow.

Scope. Signed centered charges are allowed. Exactly one nonnegative mark is not covered by this argument.

This refines the finite cyclic tag argument. Fix a finite packet of tuples $e(x,v) = (x + jv)_{0\le j<h}$ and integer charges $\tau(x,v)$. Velocities and starts can be arbitrary integers. Suppose a query of charge $+1$ is a word in zero-charge queries, in the product of the free groups on individual player letters. Choose a local law positive on every query in this word as well as a negative-charge query, with mean charge zero. Its charge support contains $0, 1$, so it is aperiodic, and it has positive variance. No independence of start and velocity is required. As in the other notes, a conclusion on actual $\mathbb{Z}$ requires the ability to place the specified fixed local task in arbitrarily many disjoint dilated copies with inactive zero velocity and additive total charge.

For any fixed integer t, every positive event admits an h-term progression whose total charge is **exactly** t in such repeated copies. This allows *signed* contributions from arbitrarily many axes. It does not assert a return with precisely one exceptional axis when every other charge is nonnegative.

The integer Fourier estimate Cyclically rotate the ordered relation so the unique challenge of charge $+1$ occurs at the very last position; invert the entire relation if its sign is negative. Normal challenges in this word have charge zero. Apply the gluing proof of Braverman–Minzer, Lemmas 3.1–3.5. It applies with *integer* targets by using $e^{i\theta\tau}$ in place of a prime-order character; conjugation negates the exponent, and the Cauchy–Schwarz inequalities are unchanged. More explicitly, during a gluing step a new proof challenge at position j is made from the old challenge there and the challenge at the opening index of its cancellation pair, and its charge is their difference. The last position always closes a pair in the maximal reading sequence, since all player words reduce to the identity. Hence that position is never used as an opening query. Induction keeps a charge $+1$ at the last position and zeros at all other positions, through the no-player stage. Consequently the terminal law has atoms of charge zero and one of positive probabilities. For any $-\pi \le \theta \le \pi$, its characteristic function has modulus $\le 1 - c\theta^2$, by expanding its squared modulus with two independent copies and retaining the $(0,1),(1,0)$ pairs. Tensorization with possibly different angles in different coordinates, followed by the fixed number of Cauchy–Schwarz roots, gives

$$\left|\mathbb{E}\prod_{j<h} f_j(Y_j)\, e^{i\sum_i \theta_i\tau_i}\right| \le \exp\left(-c'\sum_i \theta_i^2\right), \qquad |f_j| \le 1.$$

Here Y_j is the vector of all player-j letters in independent packets and the constants are for the fixed proof and law. A useful biased variant has no negative query, ordinary proof probabilities bounded below, and the true query probability $p \to 0$. All terminal ordinary proof rows then still have probability bounded below independently of p, and the

distinguished one has probability at least cp. Indeed gluing two positive-probability rows along agreeing shared inputs gives probability equal to their product divided by a marginal probability, hence at least their product. The unique distinguished terminal row has only one original rare leaf. The resulting Fourier bound has exponent $-c'p\sum_i \theta_i^2$.

Matching a prescribed total Take the finite density Hales–Jewett dimension $m = m(a,h)$ for the given event of mass $a > 0$. Each trial block consists of n independent copies. Let K_l denote its integer charge sum. By the one-dimensional lattice local central limit theorem, $\mathbb{P}(K_l = 0) \asymp n^{-1/2}$. This follows directly by Fourier inversion: the characteristic function is strictly below one off the origin, with expansion $1 - \sigma^2\theta^2/2 + O(|\theta|^3)$ there.

First sample each block conditional on $K_l = 0$. Independently of the law of these trial velocities, averaging the ordinary density Hales–Jewett word cube in the stationary starting point gives a positive lower bound on the sum of the uncharged line-return expectations. Thus one line with nonempty wildcard set I has conditional expected return at least a constant $c_0(a,h) > 0$. The choice of line may depend on n, which poses no issue since the list is finite. The directions in fixed-letter blocks outside I change only the starting translate; invariance disposes of them after averaging. In the original unconditioned independent law define

$$b_I(\mathbf{k}) = \mathbb{E}[\text{line return } 1_{K_l = k_l \text{ for all } l \in I}].$$

It follows that $b_I(0) \geq c_0\mathbb{P}(K_1 = 0)^{|I|} \gg n^{-|I|/2}$. Its Fourier transform on $[-\pi,\pi]^{|I|}$ has modulus at most $\exp(-c'n\sum_{l\in I}\theta_l^2)$: freeze the independent stationary point in a player's rectangular test, include all common query starts in that test, and then integrate. For any fixed $u \in \mathbb{Z}^I$ Fourier inversion and $|e^{iu\cdot\theta} - 1| \leq \sum_l |u_l|\,|\theta_l|$ give

$$|b_I(u) - b_I(0)| = O_u(n^{-(|I|+1)/2}).$$

Choose u with one coordinate t and all other coordinates zero. For every sufficiently large n, $b_I(u) > 0$, giving an actual positive return at total charge t. The argument is uniform over the finitely many lines and uses the same finite word relation supplied for the whole local law.

There is a vector version for charges in $\mathbb{Z}^s$: assume zero-charge words realize one query of each standard basis charge and that a finite local law on the full packet can be centered. Ensure zero receives positive probability. For each basis coordinate use its gluing argument, retaining terminal atoms at zero and that basis vector. Among the s inequalities one yields exponential decay in the sum of the squared norms of all the angles divided by s. The full-rank aperiodic lattice local central limit bound gives $\mathbb{P}(K_l = 0) \asymp n^{-s/2}$. The inversion error is $O(n^{-(s|I|+1)/2})$, yielding any fixed exact vector charge in the same way.

For instance, in any integer base $b \geq 4$ allow digits $0,1,2,3$ and require exactly t more occurrences of 2 than of 3. The zero-charge local menu contains the IP of 0/1-digit sums. Local words with one 2 and otherwise only $0,1$ can be made divisible by every q: place all nonzero digits beyond the contribution of the part of q supported on primes of b, and add enough 1's at equal power residues modulo the coprime part to cancel the 2. The integer ordered lemma [R077] in its arbitrary-order IP-menu form (see the main research notes) consequently supplies the single $+1$ relation. The one-digit word 3 supplies negative charge;

inactive zero velocities have charge zero. Shift the finite packet to disjoint digit blocks by powers of b. Thus the prescribed integer digit-imbalance sets are measurable multiple recurrence sets on $\mathbb{Z}$ at every length.

In fact one ordered word along a prescribed primitive vector $u \in \mathbb{Z}^s$ suffices to get all fixed multiples of that vector, even if the other directions admit separated player representations. Assume the centered tag law and its projection perpendicular to u are full-rank aperiodic on their respective integer lattices. Use a unimodular coordinate change to make $u = e_1$. The distinguished gluing then bounds the full rectangle Fourier transform by $\exp(-cn \sum_{l \in I} \theta_{l,1}^2)$, irrespective of all transverse angles. Impose zero transverse charge in each wildcard block by integrating the transverse angles. The resulting scalar-per-block Fourier transform has both this same bound and the bound $O(n^{-(s-1)|I|/2})$, since its absolute value does not exceed the unconditional probability that all these transverse sums are zero. In Fourier inversion split the remaining angles at Euclidean norm $C\sqrt{\log n/n}$. In the small ball use the transverse-probability bound, its volume, and a factor $O_t(\sqrt{\log n/n})$ for the shift by tu in one block. Outside use the exponential bound with large C. The difference of the exact mass at all-zero block charges and at this shifted vector is $O(n^{-(s|I|+1)/2}(\log n)^{(|I|+1)/2})$. The ordinary all-zero DHJ lower still has order $n^{-s|I|/2}$. This proves the stated one-direction refinement. These hypotheses can also be applied in the actual lattice generated by the local tag support after removing any deterministic congruences.

The ordered-relation charge lattice For scalar centered tags, suppose the subgroup of total signed charges of all ordered relations in the local game is $q\mathbb{Z}$, $q > 0$. Choose finitely many relations with totals k_j of gcd q. Apply the same gluing procedure to each relation, without asking for a single distinguished leaf. In each no-player compound law the differences between terminal atom values generate a subgroup containing k_j: the signed sum remains k_j, and the signs sum to zero because the letters canceled. Consequently the squared modulus of its characteristic function is at most $1 - c_j \|k_j\theta/(2\pi)\|^2$. Indeed express k_j as the corresponding fixed integer combination of atom differences, then use the circle triangle inequality and the positive probabilities of those pairs of atoms. Taking the strongest of the finitely many tensor inequalities and using Bézout gives an exponential bound in $-c\sum_i \|q\theta_i/(2\pi)\|^2$. The Fourier difference by any fixed multiple of q vanishes at every one of its resonances. The preceding exact-neutral DHJ proof thus forces every total in $q\mathbb{Z}$.

On disjointly replicable digit packets this makes the fixed-length threshold for an exact charge t equal to that for charge t modulo q, in the case where the exact steps and modular steps use the same digit language and concatenation preserves all admissibility requirements. Given any positive modular return at a step d_0 of charge $t_0 \equiv t \pmod q$, use its positive h-fold intersection C as the event for a fresh, disjoint packet. A return of C at exact charge $t - t_0 \in q\mathbb{Z}$ yields a return of the original event at exact charge t on adding the two directions. The opposite implication is inclusion. This concerns each fixed h and the charges realized at that order. It does not give a uniform bound on the indices q across orders.

A dichotomy for balanced digit statistics on the integers Fix $b \geq 2$, an allowed digit alphabet $\mathcal{A} \subseteq \{0, \ldots, b-1\}$ containing zero and a nonzero digit, and a map $w : \mathcal{A} \to \mathbb{Z}^s$ with $w(0) = 0$. Suppose 0 is in the relative interior of the convex hull of the weights. For a given integer vector t let D_t consist of the positive integers with allowed digits whose weights sum to exactly t. If D_t meets $q\mathbb{Z}$ for every positive integer q, it is a set of measurable multiple

recurrence at every order and every positive density. If it misses such a subgroup, it has all-length avoiding threshold one by the finite-residue sieve.

Indeed relative interior and the integral data give a strictly positive rational convex combination of the digit weights equal to zero. Thus there is a nonzero balanced word, and its copies in disjoint digit positions supply an infinite IP in the zero-charge local menu. When $t = 0$, this already suffices by IP Szemerédi. Otherwise, for the given horizon h, use the arbitrary-order integer IP-menu lemma: for normal menu V_0 containing zero and an infinite IP, the free player tuple $e(0, r)$ is a word in zero queries for every sufficiently divisible integer r. Pick such $r \in D_t$. The resulting finite query relation contains a single charge t. Add queries at all single-digit velocities to its finite packet, and choose a strictly positive law of mean zero, possible by the relative-interior hypothesis even after assigning the finitely many additional proof letters small positive masses. Work in the lattice generated by the digit weights, where the law has positive-definite covariance and is aperiodic since zero is in its support. In integral coordinates write $t = ge_1$, $g > 0$; the one-direction proof applies with exponential damping at $g\theta_1$. The Fourier difference by t vanishes at all the resulting resonances, exactly as in the scalar lattice proof. Thus exact t can be forced. Place the finite query packet in disjoint digit intervals by multiplying by sufficiently separated powers of b. Zero inactive motions and the signed total are honest arithmetic ones. This proves the claim. In particular a fixed digit-statistic search for intermediate capacities must leave the balanced regime (or go beyond finitely many additive statistics).

The divisibility condition here is decidable from a single finite integer-lattice test. Let g be the gcd of the positive allowed digits and let m_* be the largest divisor of $(b-1)g$ coprime to b. It is enough and necessary that $(0, t)$ belong to the subgroup of $\mathbb{Z}/m_*\mathbb{Z} \times \mathbb{Z}^s$ generated by $(d \bmod m_*, w(d))$ for the allowed digits. Indeed for any q coprime to b, arbitrarily late positions realize every occurring exponent residue r modulo the period of $b \pmod q$. The semigroup of attainable pairs of residues and tags is generated by $(b^r d \bmod q, w(d))$. It is a group: use the positive balanced integer counts of the digits on every exponent residue, then repeat q times to make also the horizontal residue vanish. This is a positive zero relation using every generator, so each inverse belongs to the semigroup. Differences between two successive exponent types supply $((b-1)d, 0)$ and hence $((b-1)g, 0)$ by Bézout. Modulo $\gcd(q, (b-1)g)$ all the exponent types of the same digit coincide, reducing membership to the asserted finite test. The test at m_* gives all smaller moduli by projection. For general q, shift the word far enough to clear all prime powers supported on prime divisors of b, choosing a shift that leaves its residues at the coprime part unchanged. At $t = 0$ a nonempty word can always be retained by the nonzero balanced relation.

Sparse nonnegative count sets For an unsigned 0/1 charge using the original single-special relation, let $S \subset \mathbb{N}$ have unbounded counts of distinct elements in intervals $[\lambda - C\sqrt{\lambda}, \lambda + C\sqrt{\lambda}]$, for some fixed C and arbitrarily large λ. Then restricting the total count to S also preserves full positive recurrence. This condition covers many zero-density sets and needs no positive asymptotic density of the counts.

Keep a fixed local true probability $0 < p < 1$. For the m trial blocks choose favorable centers $\lambda_1, \dots, \lambda_m$ going to infinity in successive experiments, with each previous sum $o(\sqrt{\lambda_l})$ and each favorable window containing a number of points of S tending to infinity. Choose block sizes n_l with $n_l p = \lambda_l + O(1)$. Any wildcard set I therefore has total mean $\Lambda = \sum_{l \in I} n_l p$ within $o(\sqrt{\Lambda})$ of its largest favorable center. Along a subsequence the same line has unweighted expected success bounded below (there are only finitely many). Define $b(k)$

as its success mass at total mark count k; the Fourier bound is $|\widehat{b}(\theta)| \le e^{-c\Lambda\theta^2}$. Consider

$$F(z) = \frac{\sqrt{\Lambda}}{2\pi} \int_{-\pi}^{\pi} \widehat{b}(\theta) e^{-i(\Lambda + z\sqrt{\Lambda})\theta} \, d\theta.$$

These entire functions are bounded on every compact subset uniformly in the experiment, by Gaussian integration after substituting $u = \theta\sqrt{\Lambda}$. At $z = (k-\Lambda)/\sqrt{\Lambda}$ they equal $\sqrt{\Lambda}\, b(k)$. Binomial variance and the positive total success mass yield such a value bounded away from zero at some point in a fixed compact real interval. Passing to a normal-family subsequence gives a nonzero entire limit. If all counts in S were excluded, the functions would have arbitrarily many distinct zeros in a fixed compact interval, from the favorable window of the largest wildcard block. This contradicts local uniform convergence to a nonzero analytic function (choose a surrounding disk whose boundary avoids its zeros and apply Rouché). This proof does not cover exactly one unsigned mark or a bounded number of available counts.

R031. Charged tensor products of local arithmetic-progression queries

Scope and status. Result with proof.

Result. Charged tensor products have a cover-uniform Gowers estimate, yielding full recurrence whenever the local tag is not a low-degree polynomial.

Scope. The full local velocity family is required; diluting exceptional marks changes the game and is not justified by fixed-game repetition.

Let G be a finite abelian group, $h \geq 2$, $\tau : G \to F_\ell$ a local tag for a prime ℓ, and $C \subset F_\ell$ nonempty. In G^n retain the differences d for which $\sum_i \tau(d_i) \in C$. This first model permits **every** local velocity before taking the charge condition.

A uniform finite-cover estimate. Let K be any finite abelian group with a quotient map $\pi : K \to G^n$. For every $A \subset K$ of density a, with x,d sampled independently and uniformly in K, one has

$$E[1_C(\sum_i \tau(\pi(d)_i)) \prod_{j=0}^{h-1} 1_A(x + jd)]$$

$$\geq (|C|/\ell)c(a,h)$$

$$- \sum_{k \in F_\ell, k \neq 0} |\hat{1}_C(k)| \cdot ||e(k\tau/\ell)||^n_{U^h(G)}.$$

The constant $c(a,h) > 0$ is independent of the finite group K. To see this choose a density Hales–Jewett word dimension m at density $a/2$ and alphabet $\{0, \ldots, h-1\}$. Sample independent uniform $x, y_1, \ldots, y_m \in K$. The fraction of words u with $x + \Sigma u_i y_i \in A$ has expectation a and hence exceeds $a/2$ with probability bounded below in terms of a. On this outcome there is a word line entirely in A. Each fixed nonempty wildcard set gives a uniform difference $d = \sum_a ctive y_i$, and the independent offset x makes its starting point jointly uniform with d. Summing over the finitely many word lines gives the claim. See the density Hales–Jewett theorem.

For a bounded phase w, repeated Cauchy–Schwarz in the full $Haar(x,d)$ variables gives $|Ew(d) \prod f_j(x + jd)| \leq ||w||_{U^h(K)}$ when the f_j are bounded by one. At the j-th elimination use shifts $(x,d) \mapsto (x - jt, d + t)$ preserving the j-th slot. This requires no integer to be invertible in K. Uniform cubes push forward through π to uniform cubes in G^n, and the Gowers norm of a tensor product is the product of its norms. Fourier expansion of 1_C proves the estimate. In particular the same required tensor length n works simultaneously over *all finite covers* of G^n, with no dependence on the cover depth.

If τ is not a group polynomial of degree $< h$, all the displayed nonconstant norms are strictly below one: for a unit-modulus finite-field phase, equality in its Gowers norm is equivalent to vanishing of every h-fold additive difference of τ. Multiplication by $k \neq 0$ does not change the latter property. Thus all positive-density events have a charged return for all sufficiently high powers n, uniformly over their finite covers, at this fixed h. Consequently a

uniform positive-density obstruction at a fixed length h_0 for arbitrarily high powers of a **fixed** ordinary query template requires τ itself to be a genuine group polynomial of degree $< h_0$. If ℓ is prime to $|G|$ every such polynomial is constant: by induction each first difference is constant, and the resulting homomorphism $G \to F_\ell$ vanishes. In this coprime case all nonconstant full-velocity charges therefore force every positive density at every fixed length as n grows. With changing local templates, the n required may vary and such a conclusion is not uniform in G or τ.

What parallel repetition says for restricted velocities. A local game with queries $(x+jv)_j$, uniform x and a prescribed velocity law for v may be compared with recent results, but their conclusions are different from the scalar charge above. The theorem of Braverman–Minzer, Theorem 1.3 bounds the probability of correctly answering the XOR equation in *each* of n independent games. The charge condition here is a single sum of local targets. The distinction persists with an imperfect classical local game: for even-parity binary inputs x,y,z, the target $(x + y + z)/2 \bmod 2$ has no perfect classical bit-valued strategy (sum its four equations), but its sign is exactly $i^x i^y i^z$. Tensoring gives correlation one for separable complex tests with the product of all target signs. In the paper, the Cauchy–Schwarz gluing estimates for games admitting an ordered contradiction are applicable to individual character correlations; the BLR reduction for the general AND-of-equations theorem is not itself a scalar-XOR bound. See Sections 1.3 and 3.

The no-marginal-Abelian-embedding hypothesis in Bhangale et al. fails on genuine h-term progression queries already for $h \geq 4$ on any nontrivial odd-prime group: a marginal containing the first three views satisfies $x - 2(x + v) + (x + 2v) = 0$. Likewise the standard Hązła–Holenstein–Mossel same-set hitting theorem for correlation strictly below one does not apply to a genuine progression tuple at $h \geq 3$: the first view is determined by the next two. The restricted $\{0, 1\}$-velocity symmetric-set result and the general diagonal-support conjecture are distinguished in Hązła's paper. Even an uncharged same-set hitting bound does not by itself control a rectangle restricted by the global target.

If τ takes values in F_ℓ, $\ell \nmid |G|$, and at least two charges differ among the allowed velocities, no nonconstant charged character has a perfect scalar circle embedding on *any* such velocity support with all starts. Indeed the product over x of $\prod_j f_j(x + jv)$ is independent of v, whereas the product of a charged phase is its $|G|$-th power, which remains nonconstant. This rules out scalar embeddings, not commuting embeddings or all high-dimensional tensor correlations. In the deep local grid the charge order is p and the ambient group is p-primary. The nine-slot affine stencil supplies an explicit perfect scalar circle embedding on allowed velocities at $h \geq 9$: fix one point of the affine torsor, and view each of the nine character observations at its translate by the corresponding player input as that player's function. Their product equals the target phase on every local query. Thus the relevant tensor character can have separable-test norm identically one, consistent with the known low obstruction.

For the small-budget paired grids, repeating an identical finite pair bounds the *number* of charged coordinates by a constant depending on that pair. Any fixed i.i.d. local query law with a positive probability of a true mark assigns vanishing mass to those legal supports as the number of copies grows. Diluting marks with n changes the game with the tensor dimension. Neither cited parallel-repetition theorem provides the required uniform estimates for that operation. The full-velocity estimate above gives a concrete, transferable positive case, but it rules out its own fixed positive low obstruction. It does not construct an intermediate integer threshold.

R032. A quadratic test principle for primary-block menus on Conze–Lesigne systems

Scope and status. Result with proof.

Result. A quadratic polynomial/charge criterion yields all-positive recurrence on the stated primary Conze-Lesigne systems.

Scope. The native and finite-primary-cover conclusions do not supply arbitrary free-integer-cover bounds.

Let $G = \bigoplus_i G_i$, with each G_i a finite abelian p_i-group, where the distinct primes tend to infinity. In each block choose a menu $S_i \subset G_i$ containing zero and a scalar charge $\tau_i : S_i \to \mathbb{T}$ with $\tau_i(0) = 0$. An admissible word has a finite support, takes $d_i \in S_i$, and has charge $\tau(d) = \sum_i \tau_i(d_i)$. Suppose the following two properties hold on every tail.

1. For any finite list of ordinary circle-valued group polynomials P on G of degree at most two, vanishing at zero, the joint closure of $(P(d), \tau(d))$ over admissible words contains $(0, t)$ for every $t \in \mathbb{T}$.
2. If a character χ of G tends uniformly to zero on all words in tails ($\sup \|\chi(d)\| \to 0$ as the tail start tends to infinity), then $\chi|_{G_i} = 0$ for all sufficiently large i.

Then on every Conze–Lesigne G-system, any positive measurable event admits every finite progression length along admissible words with charge arbitrarily close to any prescribed t, on any tail. In particular both the local quadratic circuit menu [R033] and its four-slot shifted deformation [R034], on native blocks or arbitrarily deep primary covers, have this conclusion **despite their positive short obstructions in general systems**. For $t = 0$ the zero word is immaterial to the application; use small nonzero targets nearby if a nonzero word is required.

Proof Use the canonical open-orbit model, including all finite central components, from the primary Conze–Lesigne reduction [R168]. This applies to any abelian primary block of arbitrary rank or finite depth: its image preserves the finitely many orbits and has commutators in the central torus after discarding the finitely many primes dividing the relevant finite group orders. Fix $h \geq 2$, and form the slope envelope K of admissible words with a circle charge coordinate in place of the earlier parity. Specifically intersect the closures over all tails, without choosing an ultrafilter of axes. It is a compact group of tuples of bijections, by the disjoint-support semigroup and idempotent argument. Let A be the compact abelian horizontal data group from the reduction, $\rho : G \to A$ the induced homomorphism. Write $J = \bigcap_m \overline{\rho(\bigoplus_{i \geq m} G_i)}$.

The projection of K to $A \times \mathbb{T}$ is $J \times \mathbb{T}$. For any character (ψ, k) annihilating the projected envelope, its value tends uniformly to zero on all admissible tail words by compactness. Property 1, applied to $\psi \circ \rho$, forces $k = 0$. Property 2 then says ψ annihilates J. The other inclusion is immediate. This same reasoning works for any finitely many additional additive-across-block tags that are ordinary degree-at-most-two polynomials on G, to put $(0, t)$ in the joint horizontal-and-tags-plus-charge projection: test each character of the enlarged compact abelian horizontal and tag group as a polynomial in Property 1. No assertion of property 2 for the enlarged group is needed.

Let C_{sh} be the closed subgroup of central torus shifts occurring as scalar z in the commutator tuples $(j^2 z)_j$, C_{sh}^0 its connected identity torus, and $\Lambda_{\mathrm{sh}} = (C_{\mathrm{sh}}^0)^\perp$. Carry out the finite-commutator-row argument for a basis of Λ_{sh}, with m killing $C_{\mathrm{sh}}/C_{\mathrm{sh}}^0$, exactly as in the torus proof [R167]. The restricted image J' is compact in the Bohr and ordinary topologies of a countable-component abelian Lie group L, by Glicksberg.

The images of the *entire* blocks G_i in L/J' vanish for all sufficiently large i. Every continuous character of this quotient extends by the evaluation formula to the original compact horizontal data and kills J, and hence kills each G_i eventually by property 2 (or directly from the definition of J and the finite-character argument). For a precise last step, if infinitely many images in the discrete component group were nontrivial, thin their indices so that $\sum 1/p_i < \infty$. A Haar-random character of that discrete group fails to detect a given nonzero block image with probability at most $1/p_i$; almost surely it detects all but finitely many in the subsequence. This contradicts eventual triviality for that fixed character. Inside the identity component finitely many extended characters separate all torsion points, completing the proof. The eventual-triviality statement for a character killing J also follows from the following elementary observation: the descending closed generated subgroups in A have intersection J, so all sufficiently late subgroups project into any small circle arc about zero, and a subgroup contained in such an arc is trivial.

Thus on a tail *all* elements of all blocks project into the compact restricted torus and the common isotropic real subspace V for the forms in the Λ_{sh} directions. On the base-origin coset the quotient fiber by C_{sh}^0 splits as in the preceding proof. For $g \in G_i$ its action there is

$$(b, z) \mapsto (b + b_g, z + \theta_i(g) + \xi_g(b)),$$

where ξ_g is a continuous character on this compact base coset group. Composition shows $\xi_{g+h} = \xi_g + \xi_h$ and $\theta_i(g + h) = \theta_i(g) + \theta_i(h) + \xi_g(b_h)$, so both $\theta_i(g)$ and $\lambda_i(g) = \xi_g(b_g)$ are ordinary degree-at-most-two group polynomials vanishing at zero. For g, h on distinct blocks, $\xi_g(b_h) = 0$ by coprime primary orders. Hence the two initial scalar coefficients on every finite word are $\sum_i \theta_i(d_i)$ and $\sum_i \lambda_i(d_i)$, and these sums define ordinary degree-two polynomials on the direct sum. Encode the two local tags in the compact slope envelope: ordinary additivity on disjoint supports suffices both for the semigroup argument and for surjection onto K.

By the enhanced property-1 projection above, choose a kernel tuple over horizontal zero, both tag vectors zero, and charge t. Its central trajectory has form $ju + \binom{j}{2} v$ entirely in C_{sh}^0. The enlarged envelope surjects onto the original K; its commutator shifts have charge and extra tags zero, with the entire closure $(j^2 z)_j,\ z \in C_{\mathrm{sh}}^0$, available. Remove curvature and obtain a target jw, $w \in C_{\mathrm{sh}}^0$, with the prescribed charge intact. The Baire finite-shear escape argument, the countable base-point tests, and the independent Haar limit of the shears from the full locally compact proof [R168] apply to this target in the enlarged envelope. Its shifted Haar measures yield positive correlation for any positive measurable event. Select an actual nearby charged word. The conditional-density inverse-limit argument finishes the passage from good models to any Conze–Lesigne system; given a prescribed charge neighborhood, every step selected on a factor was already in that neighborhood.

For the two circuit menus property 1 was proved on all finite primary lifts, even for test polynomials of any fixed degree. Property 2 follows because circle concentration on a full local affine ray in a single primary block forces the character to annihilate its direction: an

entire nontrivial cyclic subgroup cannot fit into a short arc. The normal class rays span the whole U_i; in the shifted case the arbitrary second-type clock rays kill the full clock direction as well. Full primary lifts give the same result there. The proof supplies no upper critical density on arbitrary systems and no unsaturated integer-group transfer.

R033. A local quadratic charge without any bounded-degree global polynomial

Scope and status. Result with proof.

Result. A local quadratic charge can fail every bounded-degree global polynomial representation; the construction nevertheless has critical value one.

Scope. The local register remains uniformly quadratic along all allowed directions, enabling root amplification.

For every odd prime p there is a finite elementary abelian p-group U, a velocity family S closed under $\mathbb{F}_p$-scaling, and a scalar register $f : U \to \mathbb{F}_p$ such that

$$\Delta_v^2 f(x) = \tau(v), x \in U, v \in S,$$

while no quadratic polynomial on U equals τ on S. In fact no polynomial of degree less than $p-1$ does. This is a counterexample to the global interpolation speculation for arbitrary large primary rank, beyond the sparse-axis and symmetric-gauge theorems. It still gives endpoint one, not an intermediate density, because its own register is uniformly quadratic along every allowed line.

The normal slots and marked directions Choose a prime $q > 2p+1$ primitive modulo p^2; Dirichlet's theorem in a primitive invertible residue class modulo p^2 gives arbitrarily many choices. Let $M = 2p-3$, $n = q^M$, let $H = \mathbb{F}_q^{M+1}$, and take two disjoint sets of coordinate positions

$$A = \{(1,u) : u \in F_q^M\}, B = \{(-p,u) : u \in F_q^M\}.$$

Let $K/\mathbb{F}_p$ be a finite field containing a primitive q-th root ζ. The velocity space is the additive group $U = K^{A \cup B}$, viewed as an elementary abelian p-group. The normal directions consist of all vectors supported on one projective class of coordinate positions, that is, one one-dimensional $\mathbb{F}_q$-subspace in H. For each $r \in H$, the marked velocity is the Fourier vector $v_r(i) = \zeta^{r \cdot i}$ on the listed positions, and we include each $\mathbb{F}_p$-multiple of it.

Call J a circuit if it consists of p linearly independent normals $a_1, \ldots, a_p \in A$ together with the one special position $b = -\sum_{l=1}^{p} a_l \in B$. Every proper subset is linearly independent, and in particular each projective class occurs at most once. Let N be the number of circuits. Choose $\kappa \in K$ with field trace one, and put

$$f(x) = \mathrm{Tr}(K/F_p)[\kappa \sum_{J circuit} \prod_{i \in J} x_i].$$

The function is affine on lines in every normal direction, so $\tau = 0$ there.

Second differences on marked directions At a marked velocity the product of its entries around each circuit is one, since its positions sum to zero. In the expansion of the second difference of the sum before taking trace, the coefficient on the monomial x_W is therefore the number of circuits containing W, multiplied by $v_r(W)^{-1}(2^{p+1-|W|}-2)$ when $|W| < p+1$; the coefficient for $|W| = p+1$ is zero by direct cancellation. The coefficient for $|W| = p$ also vanishes, as does the one for $|W| = 1$ since $2^p = 2 \bmod p$.

For $2 \le j = |W| \le p-1$ the number of completions is divisible by p. In the all-normal case W must be independent, and the completion count is

$$\prod_{i=j}^{p-1}(n-q^{i-1})/(p-j)!.$$

Indeed $i \ge 1$ chosen independent normals have span meeting the affine slice A in exactly q^{i-1} points. If W contains one special and $s = j-1$ normals, these and the special must be independent, and the completion count is

$$\prod_{i=s}^{p-2}(n-q^{i})/(p-s)!.$$

Choose further normals independently of the special and those already selected; their new spans each meet the affine slice in exactly q^i positions because the special has nonzero first coordinate. The last normal is forced by the unique unit-coefficient relation and lies in A. It is independent of all the other normals and distinct from the previous sites. Every completed circuit is counted $(p-s)!$ times by the ordered procedure, including the choices for which element was last. Two or more specials cannot occur. Both products have a factor $n-q^{p-2}$, divisible by p, and neither denominator is divisible by p.

The total count, choosing all normal positions independently with no special prescribed, is

$$N = n\prod_{i=1}^{p-1}(n-q^{i-1})/p!.$$

Only the endpoint $i = p-1$ gives a factor divisible by p: writing $M = 2p-3$, the exponents $i-1$ range from 0 to $p-2$ and $q \bmod p$ has order $p-1$. This factor has exact p-valuation one because $M-(p-2) = p-1$ and q has order $p(p-1)$ modulo p^2. Thus $N \not\equiv 0 \bmod p$. The constant coefficient in the second difference is $(2^{p+1}-2)N = 2N \bmod p$. Taking the trace gives $\tau(v_r) = 2N \neq 0$; along tv_r, $t \in \mathbb{F}_p$, homogeneity (or iteration of the constant second-difference identity) gives $\tau(tv_r) = 2Nt^2$.

No global polynomial of uniformly bounded degree For a polynomial of degree $d < p-1$ agreeing with τ, use all scalar multiples in $\mathbb{F}_p^\times$ to isolate its homogeneous quadratic part, which would still agree (and first exclude degrees below two). The same quadratic reduction is immediate already when the original degree is two. Since the whole subspace

on each projective class is normal, the quadratic has no within-class monomials, using any $\mathbb{F}_p$-basis of K on each position.

Sum this quadratic over v_r, $r \in H$. Each coordinate functional on K evaluated at $\zeta^{r \cdot i}$ is a linear combination over K of its Frobenius powers, by the trace representation of $\mathbb{F}_p$-linear functionals. A cross-term for positions i, j therefore has Fourier frequencies $p^a i + p^b j$ in H. These are nonzero when i, j are in different projective classes, so their sum over r vanishes by finite-field Fourier orthogonality ($p \neq q$). Thus the predicted total is zero. The true total is $|H|\, 2N \neq 0 \bmod p$, a contradiction.

On finite products of such stages one may sum the scalar registers divided by p and allow only directions whose summed local charge lies a fixed distance away from zero in the circle. Partitioning this one register into fine arcs gives a fixed-color obstruction to three-term progressions. The same positive obstruction passes by compactness to stationary models of all stages. However restriction of the register at every allowed direction is itself polynomial of degree at most two in time, uniformly in p and its rank. Thus the usual higher-root periodic-row amplification on free integer coordinates applies to this charge construction, just as for the unmixed quadratic potential in the earlier analysis [R156]. Its global degree diverges with p, which defeats a bounded-degree global-polynomial obstruction classification, but the local degree suffices for endpoint one.

Full polynomial recurrence with any prescribed charge on native prime powers For each of distinct primes $p_i \to \infty$ use one copy of this finite vector space of rank n_i, or use $(C_{p_i^{e_i}})^{n_i}$ for arbitrary finite depths $e_i \geq 1$, allowing all lifts of the indicated velocities and pulling the tag back from U_i. In their direct sum, given any fixed finite list of ordinary torus-valued group polynomials P_ℓ with $P_\ell(0) = 0$, any target charge $t \in \mathbb{T}$, and any tolerance, there is a legal finite-support word with every P_ℓ arbitrarily close to zero and the total $\sum_i \tau_i / p_i$ arbitrarily close to t. The assertion holds on each tail and imposes no reciprocal-divergence condition.

Let D bound the degrees. On distinct primary blocks all mixed derivatives vanish, since a derivative on a p^e-block is annihilated by a power of p on fixed-degree polynomials (iterate the cyclic binomial identity); use coprimality as in the Abramov analysis [R171]. Consequently each P_ℓ is a sum of its individual-block restrictions. Form the intersection over tails of the compact closures of attainable tuples (the values of the P_ℓ and the charge) in a finite-dimensional torus, taking arbitrary finite disjoint products of local choices. This intersection is a compact subgroup by disjoint support approximation. An annihilating character gives, on all sufficiently late blocks, a scalar polynomial Q_i plus $c\tau_i / p_i$ uniformly very close to zero on every allowed input. Here c is its integer coefficient on the last coordinate.

For any allowed lift w, consider all multiples $zw,\ z \in C_{p^e}$. Its scalar test is a polynomial in z of degree at most $\max(D, 2)$, vanishing at zero, with values in p^e-torsion for $p > D + 2$ (use the Newton expansion and the periodicity relations successively from top degree down, or the cyclic binomial identity with unit nilpotent cofactor). If nonconstant, its complete exponential mean is $o_{p \to \infty}(1)$, uniformly even in the prime-power depth. Indeed let k be the highest degree with a nonzero coefficient mod one. After $k - 1$ Cauchy–Schwarz differences the remaining linear coefficient is nonzero unless at least one differencing variable is divisible by p; the bad probability is at most $(k-1)/p$, and $k!$ is a unit. Thus uniform concentration on a small arc forces the polynomial to vanish identically. It follows that

$$Q_i(w) = -c\tau_i(w \bmod p)/p$$

on all allowed lifts and all sufficiently late blocks.

If $c \neq 0$, take p also larger than $|c| + D + 2$. As $D < p$, the multivariate Newton expansion gives a polynomial with ordinary monomials and p^e-torsion coefficients. Scaling by the Teichmüller representatives of $\mathbb{F}_p^\times$ in $(\mathbb{Z}/p^e)^\times$ and applying character orthogonality isolates the homogeneous quadratic part with the same values on every allowed lift. (Use $p > D + 2$, so the exponents are distinct modulo $p - 1$.) On each pure normal coordinate e_j the quadratic is zero. Also every $e_j + pz$, for arbitrary z, is a legal lift of zero tag, as is pz. Subtracting the two diagonal evaluations shows that p times the bilinear cross-coefficient of e_j, z vanishes in the circle. This holds in every basis coordinate of the $\mathbb{F}_p$-expansion of K, since each such coordinate lies in one normal projective class. The quadratic thus descends to U_i, contradicting the proved impossibility. Therefore every annihilating character has $c = 0$. Duality puts $\{0\} \times \mathbb{T}$ in the compact attainable subgroup, proving the assertion. This argument concerns native primary groups and their arbitrary finite primary lifts, not arbitrary polynomials on a free integer cover.

R034. A four-slot positive obstruction with an unconstrained clock

Scope and status. Proved subcases and unresolved extensions.

Result. The sheared circuit family has a positive four-slot obstruction with an unconstrained clock and passes the indicated native polynomial tests.

Scope. Its general all-length threshold remains unknown; the scalar identity tests only specific gauge representations.

Use each elementary-abelian prime template U, S, f, τ in the circuit construction [R033], with f, τ valued in $\mathbb{F}_p$. Add one clock coordinate, choose $a \in U$, put $A = (a, 1)$ and $b(x, z) = zf(x)$. Use precisely the following local velocities:

- $d = (a + v, 1)$, $v \in S$, with tag $\tau(v)$.
- $d = (a, 1 + s)$, $s \in \mathbb{F}_p$, with tag zero.
- $d = 0$, with tag zero.

The two active modes agree at $(a, 1)$, where the first tag is zero. Take $a \neq 0$, so no marked active velocity equals zero. Along each word retain differences with the summed tag $\sum_i \tau_i/p_i$ a fixed distance away from zero on the circle; arbitrarily many distinct prime stages of all types may occur. The following four-slot obstruction is unconditional. If T denotes the temporal shift by d and S_A the predetermined spatial shift, then

$$(T - I)(T - S_A)^2 b = S_{2A} \Delta_d \Delta^2_{d-A} b.$$

For the first type, $\Delta^2_{(v,0)} b = z\tau(v)$, so the output is exactly $\tau(v)$. For the second type $\Delta^2_{(0,s)} b = 0$. For the inactive direction $\Delta_0 = 0$. These identities hold at every starting point. On a finite product, partition the three registers $\sum_i b_i(x_i + jA_i)/p_i$, $0 \leq j \leq 2$, into sufficiently fine fixed arcs. Each spatial offset in the expanded filter has temporal coefficient-sum zero because of the factor $T - I$. Thus a monochromatic four-term progression would make the full filter small, contrary to its charged value. The number of colors is fixed, independent of primes or dimensions. On the entire direct sum subtract the base-point constants from each spatial register so only finitely many summands differ from zero. A coloring class of positive upper Banach density, or stationary translation averaging, supplies a positive four-term avoiding event. Full integer lifts in countably many free coordinates preserve the same positive obstruction by composition with the finite-block quotients.

The proof of all ordinary polynomial tests returning at any prescribed summed charge on native prime-power covers in the original construction [R033] also applies here, on every tail. For an annihilating character on the compact attainable subgroup restrict to the allowed line $d_z = (a + zv, 1)$ at each stage, also varying over arbitrary lifts of its coordinates. For a sufficiently small fixed arc the scalar restriction on each allowed line must be constant, by the prime-power exponential bound proven there. Its common value at the translated anchor need not be zero. More explicitly for a point (x, z_0) lifting $(a, 1)$ vary the remaining coordinate lifts in its whole residue coset to obtain constancy there; for each allowed lifted ray $\tilde{v}$ vary the affine parameter t in $(x + t\tilde{v}, z_0)$. This covers all lifts of type I. Subtract the value

at one fixed lifted anchor, translate a to the origin, and fix one clock lift congruent to 1. The homogeneous quadratic extraction and the normal-coordinate cross-term test used previously show that $c\tau$ cannot equal any bounded-degree local polynomial when $c \neq 0$. Consequently every unshifted finite list of global ordinary polynomial tests on the primary direct sum can be arbitrarily small together with any prescribed tail charge. No corresponding argument for free integer covers or an upper capacity below one is asserted.

The simple earlier root proof relied on bounded time degree on *all* allowed directions. This no longer holds for the displayed register if the spatial vector a is chosen appropriately. Pick a circuit J and let a have value one on $p-1$ of its positions, zero elsewhere. Let x be one on the two remaining positions and zero elsewhere. All monomials other than J vanish on $x+ta$, and the restriction of f equals t^{p-1} since $\operatorname{Tr}\kappa = 1$. The allowed second-type direction $(a, 0)$, at clock $z = 1$, gives this same unbounded-degree time sequence in b. Likewise the gauge $f(x-za)$ rectifies first-type directions, but on second-type directions it tests arbitrary multiples of a, so it does not immediately give a bounded-time-degree root amplification. A different blocker may still yield endpoint one.

Finite lists of rectifying clocks force many active normal axes Choose the above $p-1$ nonzero entries of a among the p independent normal positions of a circuit. For each fixed m, on all sufficiently late stages there are m field-linear functionals $\lambda_1, \ldots, \lambda_m : U \to K$ with $\lambda_l(a) = 1$, such that each original allowed normal or Fourier velocity v gives $\lambda_l(v) = 0$ for all but at most one l.

Indeed the selected $p-1$ sites in the field-affine slice $A = \{(1, u)\}$ are affinely independent. Choose a scalar affine coordinate π on the position space, nonconstant and taking distinct values at these selected sites; independent sites allow arbitrary prescribed values and $q > p$. On m distinct selected π-fibers take functionals supported on the A-positions in that fiber with Fourier weights $\zeta^{-s_l \cdot u}$, choosing the frequencies with distinct restrictions to $\ker(\pi - \pi(0))$. Each functional has a nonzero value on a, because just one of the nonzero positions of a lies in its support. Rescale it. A normal projective class meets A in at most one position, so at most one support responds there. On a Fourier velocity v_r, the restricted character sum vanishes unless the variable frequency matches s_l on that kernel. Again at most one can respond. The same statement holds for all scalar multiples.

In gauge $x \mapsto f(x - \lambda_l(x)a)$, a first-type step $(a + v, 1)$ with $\lambda_l(v) = 0$ becomes just v, so its time trajectory is quadratic with second difference $\tau(v)$; every second-type step becomes zero. Therefore the version allowing at most b first-type active stages, and arbitrarily many second-type compensations, has endpoint one on its free saturated cover and standard integer realization: use $m = b + 1$ independent finite clock choices, one of which works simultaneously on every first-type stage of each given direction. Each individual clock family with its common quadratic tag has the usual bounded-time-degree root amplifier at arbitrary small loss; take their finite product. Finitely many early stages can be handled by separate finite-prime near-full repair for words with nonzero early residues. Thus a putative intermediate obstruction in this specific deformation requires arbitrarily many first-type activations, including neutral normal activations if the number of marks is bounded.

The exact three-player scalar identity For this paragraph identify $\mathbb{F}_p$ with the subgroup $\mathbb{T}[p]$ via division by p, writing the charge as τ. A circle identity $\sum_{j=0}^{2} g_j(y+jd) = \tau(d)$ for all starts in $U \times \mathbb{F}_p$ and all allowed velocities exists if and only if there is an original tag

gauge $f : U \to \mathbb{T}$ with $\Delta_v^2 f = \tau(v)$ for every original $v \in S$, and $\Delta_a^3 f = 0$.

Indeed type II gives $g_0(x,z) + g_1(x+a, z+t) + g_2(x+2a, z+2t) = 0$ for all z, t. By the elementary Pexider equation on $\mathbb{F}_p$ (the middle input is the average of the other two), these three functions of the clock are affine, with slopes $B(x), -2B(x), B(x)$, respectively, in $\mathbb{T}[p]$. Write their constant terms $F_0(x), F_1(x), F_2(x)$, so $F_0 + F_1 + F_2 = 0$. Inactivity yields $\Delta_{-a}^2 B = 0$ by comparing clock slopes. Type I gives $\Delta_v^2 B = 0$ in the same manner. The remaining constant equation for each $w = tv$ is

$$\Delta_w C + \Delta_w^2 F_2 = \tau(w), C = F_1 + 2F_2 + 2B.$$

Comparing v with $-v$ gives $(I + S_v)\Delta_v C = 0$. On a cycle of odd order this derivative must be constant of order at most two; its sum around the cycle vanishes, so it is zero. Since the normal rays generate U, C is constant. Inactivity on the constant terms now gives $\Delta_{-a}^2 F_2 = 2\Delta_{-a} B$, and hence $\Delta_a^3 F_2 = 0$.

Conversely, if such an f exists, $\Delta_{-a} f$ is p-torsion by summing the affine derivative around the p-cycle (use that p is odd). Set $B = \Delta_{-a} f/2$ in this p-torsion group. Then $\Delta_a^2 B = 0$, every $\Delta_v^2 B = 0$, and $2\Delta_{-a} B = \Delta_{-a}^2 f$. Take $F_2 = f$, $F_1 = -2f - 2B$, $F_0 = f + 2B$ and the clock slopes above. Reversing the calculations supplies the identity. The displayed high-degree restriction of the explicit circuit gauge excludes that particular gauge, not every gauge with the same tag.

R035. The actual three-channel circuit Haar system has threshold at most one sixth

Scope and status. Result with proof.

Result. The explicit three-channel sheared-circuit Haar action has all-length capacity at most one sixth and a positive short obstruction.

Scope. This is one specified action, with a replication schedule, not an upper bound for arbitrary actions.

Use the four-slot sheared circuit family [R034], with a equal to one on $p-1$ of the independent normals of a fixed circuit, zero elsewhere. For each prime p_i take a sufficiently large finite number of independent copies of $U_i \times \mathbb{F}_{p_i}$; the number is chosen below and can vary freely by stage. The direction family allows independent local choices in every copy subject to one common total charge bounded away from zero, say $\| \sum \tau/p_i \| \geq 1/8$. On the product Y of uniform local base states use three common Haar torus registers, indexed by $l = 0, 1, 2$. A finite direction translates the base and adds to register l the sum of the changes of $b(x + lA)/p_i$ in its active copies. This is a measure-preserving action on $Y \times \mathbb{T}^3$.

The four-slot filter gives an actual positive avoiding event in this particular Haar system: take the three registers in sufficiently short prescribed arcs. In every monochromatic run each of the three time-sum-zero offset contributions is small, contradicting the separated charge. The following complementary assertion is also true in this same system, uniformly over all its measurable events: **every event of mass greater than** $1/6$ **admits every length.** This is a bound for this one specified action, not for the strong conjecture over all systems.

The replicated uncharged controls Choose $h_i \to \infty$ with $h_i + 2 < p_i$, dropping finitely many stages. In an uncharged copy sample, independently of its uniform starting state, either no move with fixed positive probability or a second-type velocity $d = (a, 1 + s)$ with s uniform. Regard its vector of increments at times $t = 0, \ldots, h_i - 1$, offsets $l = 0, 1, 2$, as a random vector in the corresponding finite product over $\mathbb{F}_p$. Write Q_i for the subgroup generated by its support. Convolution of sufficiently many independent copies is arbitrarily close in total variation to uniform on Q_i: the positive atom at zero excludes periodicity, and every character nontrivial on Q_i has Fourier coefficient of modulus strictly less than one. Choose a finite number of copies for error $1/i$, with one more copy reserved for a true move.

For $x = x_c + qa$ in the witness from the sheared construction [R034], $f(x + ka) = (k+q)^{p-1}$ on the entire prime field. On the finite observed set of $k = t + l$, denote this profile by $F_q(k)$. Such profiles span the whole space of functions on this finite set: choose $-q$ outside the window to get the all-ones vector, and successively at each position to erase just that entry. Starts x as above have positive probability for each fixed stage, however small.

At start (x, z) the increment tuple for $d = (a, 1 + s)$ is exactly

$$(z + t + l + ts)F_q(t + l) - (z + l)F_q(l).$$

Subtracting two choices of s, then two choices of z, shows that Q_i contains the whole subgroup of independent controls

$$P_{t+l} - P_l + tS_{t+l},$$

for arbitrary scalar arrays P, S on the window. This does not require a generic starting profile, only the generated subgroup and the subsequent choice of enough copies. On the true copy choose a fixed marked velocity $(a+u, 1)$ with $\|\tau(u)/p_i\| \geq 1/8$; this is available for all large primes since the scaled quadratic values $2Nt^2/p_i$ are asymptotically equidistributed (the complete quadratic sum bound, $p_i \nmid 2N$). Nothing about its initial profile needs to be controlled. All sampled words are legal actual directions.

Limit and measurable events Fix h and project to its slots. Along a subsequence the total true-plus-normal increments in the torus registers have a limiting law. This law is invariant under every torus-valued free control $P_{t+l} - P_l + tS_{t+l}$: discretize P, S on the p_i-grids and use the approximation to uniform on Q_i, then take the weak limit. Equivalently write the limit as an arbitrary random offset array plus independent Haar free controls on these two streams. This uses only translation invariance on a compact group; convolving that law with Haar controls leaves it unchanged.

The whole limiting orbit law has one common initial base point y in all slots, independent of these increments. Indeed every fixed finite part of the base is unchanged and independent of the active copies at all sufficiently late stages, and the initial torus registers are independent Haar throughout. At each slot the limiting register values thus have the form

$$r_l + W_{t,l} + P_{t+l} + tS_{t+l}, l = 0, 1, 2,$$

where the initial independent Haar registers absorb the former $-P_l$ and remain independent of P, S. Conditional on y, W, r, the free streams are independent Haar. Writing $tS_{t+l} = (t+l)S_{t+l} - lS_{t+l}$ converts this to the exact form of the three-offset theorem [R108], by a Haar-preserving change of P.

If $E \subset Y \times \mathbb{T}^3$ has measure greater than $1/6$, a positive set of bases have vertical sections of Haar measure greater than $1/6$. For every such section the three-offset result gives positive probability that all h register slots are in that section, for every deterministic offset $W + r$. Integrating gives positive limiting correlation. Continuous cylinder tests suffice first, and then arbitrary indicators follow by L^1-approximation since the true and limiting one-slot marginals are all the original Haar law. Some actual sampled word therefore has a positive h-return. All choices can be made in an arbitrary tail. The same argument allows a fixed prescribed open neighborhood of any charge away from zero.

In fact the compact three-channel action has no other invariant probability measure. The marginal of any such measure ν on Y is Haar since the finite-support base rotations are dense. The conditional mean $F(y) = \mathbb{E}_\nu[e(k \cdot r) \mid Y = y]$ of any fixed register character intertwines that character's cocycle. Its modulus is constant by base ergodicity. If nonzero, the unit-modulus phase $F/|F|$ expresses the register increment as a multiplicative coboundary on Y. Every late motion of the base changes this fixed function by $o(1)$ in L^2, uniformly over all such motions, by approximation on a finite initial set of coordinates. But the normal uncharged trials alone at $h = 2$ have one-slot register increments tending to full Haar on $\mathbb{T}^3$: the values S_{1+l}, $l = 0, 1, 2$, already fill all coordinates independently in their free-control subgroup. Thus every nontrivial character has $F = 0$. Fourier uniqueness identifies the conditional

fiber laws as Haar, giving the original product measure. The action is continuous and the sole invariant probability has full support; hence it is minimal as well by compactness and amenability.

R042. The explicit deep affine witness has a system-relative upper bound of one half

Scope and status. Result with proof.

Result. The explicit deep quadratic affine Haar witness, and its indicated diagonal powers, have relative capacity at most one half.

Scope. The finite list of returning directions depends on the tested event; weak containment does not transfer this to all actions.

Qualifications. The diagonal-power extension applies to the stated diagonal action; it does not cover a general coinduced product.

In the particular affine Haar system of the deep mixed-characteristic construction [R049], every event of measure greater than $(1/2)$ has positive returns of every length along V in every tail. Its system-relative all-length avoidance capacity therefore lies between the positive short color bound and $(1/2)$. This does not bound the supremum over other systems, nor the capacity on the coinduced product with an infinite number of separate copies.

A convenient extension of the compact torsor In the local coefficient rings of that note put $v_p = u_p + u_p^{-1}$. Enlarge the original character module to the subgroup of $\prod_p B_p$ generated by the coordinate tuples $F_0 = 1, F_1 = v, F_2 = v^2$ and the full finite-support submodule $W = \bigoplus_p B_p$. It is a G-module since every group action changes only finitely many coordinates. It contains all the initial nine character tuples, whose generating polynomial is

$$P(T, v) = (T - 1)^4 (T^2 - vT + 1)^2.$$

As an additive group this module is the direct sum of the three free generators and W. Indeed any nonzero Laurent polynomial $L(Z) \in \mathbb{Z}[Z^{\pm 1}]$ has ϖ-valuation $O_L(1)$ at $Z = \zeta_p$ for all sufficiently large p: expand at $Z = 1$, absorb a power of Z to remove negative exponents, and use the first nonzero integral Taylor coefficient, a unit at sufficiently large p. Apply this to any nonzero polynomial in $Z + Z^{-1}$ of degree at most two. The resulting value does not vanish in $\mathbb{Z}[\zeta_p]/(\varpi^{2D_p(p-1)+7})$, even before adjoining (ϵ).

Prescribe the same E_p-character as before on the compact dual. Haar on the extension coset projects to the original affine Haar law, since restriction of characters from the enlarged module is surjective. The three free coordinates are jointly Haar on $\mathbb{T}^3$, independently of all finite-support character data. The latter are independent finite local coordinates with the prescribed E_p-values. Every event can consequently be approximated in measure by one on $\mathbb{T}^3$ and finitely many local coordinates, called its head.

Limiting profile from two fresh primes Fix ω strictly inside the open charge gap, $\|\omega\| > \gamma$. At a fresh marked p use $r = 1 + p^{D_p} t_p$ with $t_p^2/p \to \omega \bmod 1$, uniformly outside the forbidden near-zero arc eventually, and $p \nmid t_p$. Such choices follow from the quadratic Gauss sum estimate. Independently take a fresh normal prime $q \neq p$ and use speed q at q. Take any sequence of these pairs with both primes tending to infinity, zero elsewhere. Every sufficiently late direction lies in V.

Write $F_k(t)$ for the kth free coordinate at time t on that progression, $0 \le k \le 2$. A finite character of the array is encoded by $R(T, v) \in \mathbb{Z}[T^{\pm 1}, v]$ of degree at most two in v. Its Haar expectation conditional on any fixed head vanishes unless the local evaluations at all other primes are fixed-line multiples of their E_p's and the full module element has no remaining free-coordinate contribution. When these conditions hold, the coefficient is the corresponding prescribed phase, since the head of the start does not move. We claim the limiting coefficient is zero unless $R = B(T)P(T, v)$, in which case its phase is $B(1)\omega$.

At the fresh normal prime q, if $R = (T-1)^k R_0$, $k = 0, 1, 2$, where $R_0(1, v) \neq 0$, the coefficient of ϵ^k under substitution $T = u^q$, $v = u + u^{-1}$ is $q^k R_0(1, \zeta_q + \zeta_q^{-1})$. Its valuation is $k(q-1) + O_R(1)$ by the Taylor calculation. It is nonzero for all sufficiently large q and, even for $k = 2$, lies below the fixed scalar line at valuation $2D_q(q-1) + 6$. Thus a surviving R must be divisible by $(T-1)^3$.

At the fresh marked prime, if $(R(T, v))$ does not vanish identically under $v = T + T^{-1}$, its constant (ϵ)-coefficient $R(\zeta_p, \zeta_p + \zeta_p^{-1})$ is nonzero of bounded valuation. Thus $T^2 - vT + 1$ divides R in the Laurent polynomial ring (the variable T is invertible). If $R = (T^2 - vT + 1)R_1$ but the same factor does not divide R_1, its (ϵ)-coefficient at the mark is

$$\zeta_p p^{D_p} t_p (\zeta_p - \zeta_p^{-1}) R_1(\zeta_p, \zeta_p + \zeta_p^{-1}).$$

Its valuation is $D_p(p-1) + O_R(1)$, so it too prevents a surviving coefficient. The two distinct factors are relatively prime in this Laurent polynomial UFD. Since R has degree at most two in v, the remaining possibility is $R = Q(T)(T-1)^3(T^2 - vT + 1)^2$. On the mark its value is

$$p^{2D_p} t_p^2 \epsilon^2 \zeta_p^2 (\zeta_p - 1)^3 (\zeta_p - \zeta_p^{-1})^2 Q(\zeta_p).$$

If $Q(1) \neq 0$, its valuation before ϵ^2 is $2D_p(p-1) + 5$ eventually, one too low to belong to the scalar line. Otherwise $R = B(T)P(T, v)$. It vanishes on inactive sites and at the normal prime, while at the mark it evaluates to $4t_p^2 B(1)E_p$ by the exact low stencil and invariance of E_p. Its full free-coordinate contribution is zero. This proves the claimed Fourier limits independent of any fixed head.

Write S for temporal shift. By Fourier uniqueness the joint weak limit is Haar on the affine profile coset

$$(S-1)^4[(S^2+1)^2 F_0 - 2S(S^2+1)F_1 + S^2 F_2] = constant\omega.$$

Indeed the indicated operator is onto on the full shift (one may prescribe F_0, F_1, integrate the fourth difference, then solve for F_2); its dual annihilator is exactly the Laurent multiples of P. Evaluation of such a multiple gives $B(1)\omega$. Equivalently first sample the Haar-affine solutions W of $(S-1)^4 W = \omega$. Independently let F_0, F_1 be iid Haar circle streams and solve for F_2 by the displayed bracket equation with value W.

The uniform half-density argument in this action For a measurable set $B \subseteq \mathbb{T}^3$ of mass $> 1/2+\delta$, $0 < \delta < 1/2$, write $(b(z))$ for its third-coordinate section measure at $z \in \mathbb{T}^2$. Layer cake gives some $u \in [0,1]$ for which $P_0 = \{z : b(z) > u\}$ satisfies $\mu(P_0) > 1 - u + \delta$. At time t the required $F_2(t)$ is a prescribed translate of

$$-\mathrm{F_0(t+2)}-2\mathrm{F_0(t)}-\mathrm{F_0(t-2)}+2\mathrm{F_1(t+1)}+2\mathrm{F_1(t-1)}.$$

Conditioning on W is harmless. Start with the input pairs at $(-2,-1,0,1)$ in P_0. In chronological order choose each fresh pair at $(t+2)$ in P_0 while enforcing that $F_2(t)$ lies in the required section at the already chosen pair at t. The coefficient of its fresh first coordinate is (-1), so the latter condition has conditional mass $> u$. Their intersection has mass $> \delta$ at every step. This gives an all-B window with probability at least δ^{h+4}, independently of W and the set's complexity.

Now let A in the full Haar torsor have mass $a > 1/2$. Choose $\delta = (a - 1/2)/4$, and approximate A by an event B on a finite head and the three torus coordinates to arbitrary L^1-error $\eta < \delta$. A proportion bounded below in terms of a of its head sections have mass greater than $1/2 + \delta$, by averaging. On these sections the limit law just obtained gives a uniform positive h-return. Weak convergence gives convergence on the actual measurable product sets by continuous L^1 approximation on each of the finitely many head sections, since all one-time torus marginals are fixed Haar. The total error on replacing B by A is at most $h\eta$. Choose it smaller than half the uniform lower bound. Thus some two-prime permitted difference gives positive return. Projecting from this extension proves the assertion for the original affine system.

Diagonal powers, independent labels, and the finite-list obstruction Fix the pair-prime sequence above once and for all. For each fixed event A of mass $a > 1/2$, the argument gives $\liminf_n c_h(A; d_n) \geq c(a,h) > 0$. It holds on every finite diagonal product of copies: for a fixed d_n the profile laws in independent copies factor, so their limits are independent copies of the same affine coset law. Bundle the free inputs and outputs into three channels with values in $\mathbb{T}^k$, using the vector product Haar measure. The only latest coefficient is still $(-I)$, so the section argument and its lower bound are independent of k. Approximation by finite cylinders therefore gives the same conclusion on countably infinite diagonal products. Independent passive labels can be conditioned on, since a positive fraction of sections still has mass separated from one half. The same applies to independent labels indexed by primary axes whose every cylinder stays unchanged for all sufficiently late d_n. For an independent Bernoulli G-shift, approximate events by finite label cylinders. The h shifted finite coordinate sets are pairwise disjoint for all sufficiently late n, so the conditional acceptance probabilities multiply. With global mean $> 1/2$, a positive-level set of the resulting acceptance function on the affine factor has mass separated from $(1/2)$; apply the existing lower bound there. Cylinder approximation proves this assertion for general events. None of these product arguments makes the onset of the lower bound uniform over A.

Indeed the enlarged affine action is free. For a unit $r \bmod p$ the element $(u^r - 1)B_p$ contains E_p by multiplying a suitable ϵ^2-multiple of $\varpi^{2D_p(p-1)+5}$; the coefficient $(\zeta^r - 1)/\varpi$ is a unit. For $0 \neq r = p^k v \bmod p^{e_p}$, $1 \leq k \leq 2D_p$, v prime to p, one has $u^r - 1 = r\epsilon + \binom{r}{2}\epsilon^2$. Multiplication by $p^{2D_p-k}v^{-1}\varpi^6\epsilon$ gives E_p. Thus for every nonzero group element the augmentation-difference image contains some E_p, on which each point of the torsor has prescribed nonzero value. Such a group element fixes no point.

Consequently on every finite subgroup $H \subset G$ this very system can realize *every* stationary binary H-mask law. Its H-orbits are free; the quotient of the nonatomic standard probability space by these finite orbits is nonatomic. Choose a measurable transversal, partition its quotient to sample the desired distribution on translation-orbits of masks, and label each actual orbit by a representative of its chosen mask orbit. Uniform location on each free H-orbit gives exactly the desired law. Hence the finite-list fixed-length capacity of this one system is already the unrestricted finite-list capacity.

This prevents a finite sampler list uniform in all events. On a tail the two-prime sampler steps are all Bohr separated from zero in the primary group: the character assigning $\lfloor p/2 \rfloor / p$ to the unit of the pth cyclic axis vanishes on all normal components and sends the sole active $+1 \bmod p$ to a neighborhood of one half. For any $\eta > 0$, choose a common root order M divisible by all small arc denominators, and then discard the finitely many primes dividing M. The character has an M-th root on the tail primary group. The complement of a small circle arc under this root rotation has mass $> 1 - \eta$ and avoids some fixed finite horizon $h = h(\eta)$ for *every* one-active sampler difference, since its rotation velocities stay in the compact set $\|M\theta\| \geq 1/4$. Restrict that stationary law to the subgroup generated by any finite sampler list, and realize it inside the enlarged affine action as above. Thus each such finite list has an exact avoider of mass $> 1 - \eta$ inside the same system. The set varies with the list. A nonprincipal ultrapower contains an event of that mass obeying the entire countable sampler family at the one fixed length, by exhausting finite lists. By the same finite simulation and compactness of binary shift laws, the ultrapower's capacity on the *full* V equals its unrestricted capacity, which remains unknown. The head/cylinder needed to approximate the event, and the subsequent time at which its Fourier tests have converged, are explicitly where the individual-event proof loses uniformity.

R043. Boundedly many true marks in the mixed-characteristic deep quadratic model

Scope and status. Result with proof.

Result. The deep quadratic family restricted to boundedly many true marks has saturated integer critical value one.

Scope. A true mark is a nonzero high displacement, even when its quadratic charge happens to vanish.

Qualifications. Correct the typographical formula token to in the displayed charge condition.

Use the deep quadratic model [R049]. At prime p, put $e = 2D_p + 1$, $D_p \to \infty$, and choose moves which are passive multiples of p or active $r = s + p^{D_p} t$ with $s = \pm 1$. A **true mark** means $r \neq s \bmod p^e$. The other active moves $r = \pm 1$ may be arbitrarily numerous, as may the passive moves. If every allowed difference has at most a fixed number B of true marks and

$$\|Q\| \geq \gamma, \qquad Q = \sum_{\text{active}} t^2/p,$$

its tail capacity and saturated integer critical threshold are one, irrespective of the prime reciprocal sum. Additional high digits of the true marks are unrestricted. This does not prove the assertion for the original model with unbounded mark counts.

Graph and event. Use the dissociated expanding cube graph and the site test of the signed expander blocker [R232]. For a given loss this gives a fixed vertex set with distinct even integer locations n_x, label offsets $j_{xl} = (n_x - n_y)/2$ at an edge from x to $y = x \oplus D_l$, spatial test sets J_l, and a finite bandwidth such that the incident integers at each vertex admit no nonzero relation of that bandwidth. Its event keeps central values in a near-full closed circle set missing an arc and requires all symmetric arm empirical character means to be small. Independent spatial Haar translations make the site loss as small as prescribed. Its graph implication is the following: an all-kept run cannot contain a translate of this vertex pattern for which the symmetric mirror errors on edges have sufficiently small average character error. Indeed the edge averages would then be small while expansion compares them to squares of the empirical central Fourier coefficients, at least one forced non-small by the omitted arc. This implication only requires average accuracy and permits arbitrary complex labels at the endpoints.

On a finite prime packet define fields $F_j(x) = \xi_j + \sum_p f_p(x_p + j)$ for $j = 0, \pm J_l$. Use the full cyclic spatial unit $+j$. Passive moves with $r \in pC_{p^e}$ contribute spatially arbitrary temporal polynomials of degree at most two to the mirror errors for the fields below. Plain active moves $r = \pm 1$ cancel from these errors exactly, without any lift restriction and regardless of their count. The Haar constants add fixed terms. Give each incidence one half of the polynomial error on its unoriented edge, with the same half for both endpoints, by lifting the finitely many circle polynomial coefficients and dividing by two. This is an arbitrary polynomial in the time translation of degree at most two.

Quadratic field. Choose a vertical polynomial degree R as specified below and take

$$f_p(x) = \beta_p x^2 [x \bmod p]^R / p^e \pmod 1.$$

The multiplication in the numerator is modulo p^e, $[x \bmod p] \in \{0, \ldots, p-1\}$ is the fixed ordinary lift, and β_p will be the inverse of one common integer modulo p^e. Along a true

mark $r = s + \delta$, $\delta = p^{D_p} t$, the endpoint at temporal location $u = T + n_x$ is reached from the midpoint by temporal half-difference $j = j_{xl}$. The corresponding spatial sample changes the group argument by `s j` instead of `rj` and has exactly the same low residue as the endpoint. On any interval avoiding the low-digit wrap, the incidence contribution to the mirror error is therefore

$$\frac{\beta_p}{p^e}[-2j\delta(x + ru) + j^2\delta^2](v + su)^R.$$

Here v is an integer lift valid throughout the interval, and high-coordinate arithmetic is polynomial modulo p^e even across its wrap. Reversing the sign of the low speed merely exchanges the two spatial samples on the unoriented edge.

Use the same average incidence-control argument as in the expander proof. This time normalize each incidence phase to be the displayed contribution (summed over true marks) plus its half of the nuisance polynomial. Their sum on an edge is its mirror error; controlling their distance to zero suffices. For clarity, compare the phases to $j_{xl}U_x$ plus arbitrary fixed offsets, with iid Haar circle U_x. The normalized Fejér product has mean one in this comparison, and its mean times the average of $1 - \cos$ on incidences is the reciprocal of the Fejér order. This follows by the per-vertex bounded dissociation. Both trigonometric polynomials have one fixed finite Fourier index list λ, determined **before** the temporal polynomial degree R. It is enough to show that for every nonzero index in this list its actual exponential average on a sufficiently long clean interval is uniformly small. The same positivity-weighting and Cauchy–Schwarz as in the expander proof then give some translate with the required small average edge errors.

Put $A_x = \sum_l \lambda_{xl} j_{xl}$, $B_x = \sum_l \lambda_{xl} j_{xl}^2$. The first vector is nonzero by dissociation. Let a be the first nonnegative moment index with $A' = \sum_x A_x n_x^a \neq 0$; it is smaller than the vertex count by Vandermonde. Define b, B' similarly for the second vector, with $b = \infty$ if it vanishes. In the character polynomial the highest possible degree is $d = \max(R+1-a, R-b)$. Choose R even and exceeding the vertex count plus three. The only terms that can reach this degree are the leading triangles from the linear-in-u and the j^2 terms in the display. Indeed an arbitrary lower polynomial multiplying A_x loses the same a initial moments, and the corresponding B_x polynomial loses the first b. This removes all dependence of the top coefficient on the starting site, vertical lifts, or local translations.

Write

$$Q_p = t^2/p, \quad S_p = st/p^{D_p+1}.$$

The ordinary coefficient in degree d equals $\sum_p \beta_p(v_0 Q_p + w_0 S_p)$ for integers

$$v_0 = -2C_A + C_B, \qquad w_0 = -2C_A,$$

where $C_A = \binom{R+1}{a} A'$ when the first candidate reaches d, and zero otherwise; $C_B = \binom{R}{b} B'$ when the second reaches d, and zero otherwise. This uses $\delta r/p^e = Q_p + S_p$, $\delta^2/p^e = Q_p$, and $s^R = 1$. There is a single even R for which $v_0 \neq 0$ on the full finite list: if exactly one candidate reaches d, it is immediate; in a tie $a = b + 1$, and the vanishing condition after division by $\binom{R}{b}$ would be $B' - 2(R+1)A'/a = 0$, excluding at most one R. Multiply both coefficients by `d!` to obtain v, w for the binomial-basis top coefficient. These are fixed integers and every v is nonzero.

For each of the finitely many v define

$$S_v = \sum_p [v^{-1}]_{p^{D_p+1}} \, st/p^{D_p+1} \pmod 1.$$

Discard finite small primes. If all these have norm below $\gamma_0 = \gamma/(2(1 + \max|w|))$, the quadratic field works. In fact fix the tolerance needed in exponential averaging of the finite Fejér expansion. The one-dimensional polynomial Weyl criterion, equivalently the torus case of Green–Tao, Theorem 2.9, gives a bounded nonzero multiplier k and bounded binomial $C\infty[H]$ norm if that tolerance fails. Choose M divisible by all possible nonzero `kv` and take $\beta_p = M^{-1} \bmod p^e$. Multiplying the small alleged leading coefficient of the multiplied polynomial by $M/(kv)$ would yield a small value for

$$Q + wS_v,$$

whose norm is at least $\gamma/2$. This identity uses inverses in the actual prime-power denominator, rather than dividing an already reduced circle sum. For sufficiently large fixed H this is impossible; $d \geq 3$ also puts the field's leading degree above all nuisance polynomials. The estimates are uniform under arbitrary translations of the polynomial interval.

Linear fallback probes. Use a separate expander test and site-loss allocation for each preselected v, after fixing the entire quadratic event and its finite list. For that test take odd degree R' exceeding its vertex count plus three and

$$f_p(x) = \beta'_p[v^{-1}]_{p^e}\, x[x \bmod p]^{R'}/p^e.$$

The incidence contribution on a clean interval is now $-j\beta'_p v^{-1}\delta(v' + su)^{R'}/p^e$, where v' denotes the low-digit lift. Its top coefficient for every frequency in the new Fejér list is a fixed nonzero integer factor times the sum of the local $\beta'_p v^{-1} st/p^{D_p+1}$; the sign arises from odd degree. Choose β'_p inverse to an integer clearing the bounded Weyl multipliers and these factors, just as before. Thus the probe succeeds whenever $||S_v|| \geq \gamma_0$. All the finite shifted fields can use independent Haar constants. Their individual losses total less than any prescribed fixed loss. Combined with the quadratic probe they cover every allowed direction.

Wrap and tail uniformity. The number of true marks is at most B. After the finite graph spans and polynomial averaging lengths for all probes are determined, choose a long finite initial run large enough to fit one clean interval including those spans. In a sufficiently late tail all low primes exceed the entire run with margins; each marked low digit wraps at most once on the necessary enlargement. Removing the corresponding bounded neighborhoods of at most B cut positions leaves a clean interval by taking the original run sufficiently long. No plain active move introduces a cut in its wave identity, and the passive terms keep their bounded degree everywhere. This proves a uniform pointwise forbidden length on finite packets. Weak compactness of the stationary binary laws and the cyclic-prime tail suspension (the prefix repair works equally for prime powers of any fixed finite lengths) give the claimed threshold on the saturation. The limitation on true marks concerns exact residues $r = s$, not just whether the charge happens to vanish: a zero charge with higher nonzero high digits can still generate an uncontrolled shear.

R044. Necessary orbit depths on the base of the integer ergodic components

Scope and status. Result with proof.

Result. Any avoiding ergodic rational action for the deep family must move horizontal ergodic components with at least the stated p-power depths on every sufficiently late prime.

Scope. This necessary component-orbit condition gives no avoiding-mass bound.

For the saturated deep quadratic model [R049], every positive-measure finite-length avoidance witness in an ergodic rational action must move the ergodic components of its integer generator in cycles of length at least p^{D_p+1} at **every sufficiently large primary prime**. This concerns the permutation of components, distinct from a rational eigenvalue of the integer action on one component. The statement does not by itself bound the mass of a witness.

Work on $K = \pi^{-1}(\bigoplus C_{p^{e_p}}) \subset \mathbb{Q}$, where $e_p = 2D_p + 1$, or restrict a full rational action and then pass to its K-ergodic components. Write Y for the ergodic-component base of U^1. The action on Y factors through $G = K/\mathbb{Z}$, acts ergodically there if the original action is ergodic, and has an almost surely constant stabilizer $J \leq G$: each measurable fixed-point set of a group element is invariant under the whole abelian group. Every subgroup of the primary direct sum splits into primary subgroups, by applying the elementary CRT projections to each particular finite-support element. Thus the order p^{k_p} of the unit generator on Y is well defined almost surely, and the kernel at that prime consists of its multiples of p^{k_p}.

A single coset with componentwise trivial motion. If reduced a/q fixes Y, then $1/q$ fixes it too by Bézout, since the integer action fixes Y. On each fiber the transformation $S = U^{1/q}$ obeys that $S^q = U^1$ is ergodic. For any positive event C on the fiber, the multiple progression return average along S^{a+nq} equals the unrestricted average, by Moreira–Richter, Theorem 2.4; the unrestricted average is positive by multiple recurrence. This applies at every length and to any residue $a \bmod q$. On a set of fibers of positive measure where C has positive conditional measure, the corresponding unrestricted lower bounds integrate to a positive number. In particular any positive-measure C in the original space returns somewhere along $a/q + \mathbb{Z}$. This also applies on the positive-density fibers of any subgroup-fixed locus in a nonergodic ambient system. No upper bound on nilmanifold complexity or on the order of its nilfactor is involved.

Application to a perturbed actual return. Suppose $k_p \leq D_p$ for infinitely many p. At those primes the shift $z_p = p^{D_p} w_p = 1/p^{D_p+1}$ fixes Y, where $w_p = 1/p^{e_p}$. Given A of positive measure and h, choose sufficiently many distinct such unit directions. Finite density Hales–Jewett produces a nonempty subsum $v = \sum_{p \in F} w_p$ with $C = \bigcap_{j<h} U^{-jv} A$ of positive measure, by the word-cube argument in deep-triples.md [R058]. Fix one $p \in F$. Choose $t \in \mathbb{Z}$ with $\|t^2/p\| > \gamma$; this is possible at every sufficiently large prime by the quadratic Gauss estimate. The coset $tz_p + \mathbb{Z}$ gives an h-return for C, so for some $n \in \mathbb{Z}$, $v + tz_p + n$ gives a positive h-return for A. It is permitted with the single charged axis p, the other sites of F active with zero high digit. Whichever finite set of early primes is excluded, the argument still applies.

Consequently every joint-ergodic positive witness to the failure of any finite length has $k_p > D_p$ eventually. In particular every primary generator must eventually act on the U^1-component space with very large orbit order. A system pulled back from the free primary

group action can satisfy this necessary condition; it does not decide the conjectured all-system threshold. The earlier periodic root-splitting theorem [D15] serves a different purpose, giving periodic masks for fixed finite periodic difference tests from the rational spectra of individual components.

R045. Deep one-mark recurrence with countably many nonlinear horizontal symbols

Scope and status. Result with proof.

Result. The deep family has full recurrence for the specified countable collection of bounded-degree nonlinear horizontal symbols.

Scope. Diffuse nonlinear symbols are outside the proof.

Let $H = \oplus_p p C_{p^{2D_p+1}}$ in the deep quadratic model [R049]. Suppose the event under test belongs to the closed linear span of a family of bounded complex tests of the following type (or of their finite products, with the same condition). A test is a bounded H-invariant coefficient times a unit phase f. At almost every starting point x its horizontal orbit has the form

$$f(T_h x)/f(x) = e\left(L_x(h) + \sum_p Q_{p,\nu(x)}(h_p/p)\right).$$

Here L_x is an arbitrary measurable family of characters of H. The label ν has at most countably many possible values. For each fixed label the displayed nonlinear parts are predetermined polynomials of one cyclic coordinate $h_p/p \in C_{p^{2D_p}}$, with uniformly bounded degree and no constant or linear monomials, and circle coefficients in $p^{-2D_p}Z/Z$. Only finitely many terms occur for each individual argument. Classes with finitely many different degree bounds among a finite test are equally allowed. A genuine bounded-degree horizontal phase polynomial on these groups has the coordinatewise ordinary polynomial form on sufficiently late primes, by coprime mixed-difference cancellation and the binomial expansion on prime powers of characteristic exceeding its degree. The countability of its entire nonlinear coefficient array is an extra hypothesis. The characters L_x are unrestricted and need not have countably many possible values.

Under this assumption every positive event has returns at every length already with one true mark, arbitrarily far out. Arbitrary passive horizontal velocities and plain low $+1$ normal moves are used. This is an all-length sufficient condition for the subgroup model, not a claim about every relative Host–Kra factor.

Fix h, $a = \mu(A) > 0$. Density Hales–Jewett gives a constant $\kappa(h, a) > 0$ and finitely many trial low axes such that some nonempty pure sum $v = \sum_{i\in F} e_i$ among them satisfies $\mu(C) \geq \kappa$, $C = \bigcap_{j<h} T_{jv}^{-1} A$. The axes themselves can be selected after the approximations below. If $K \leq H$ is any finite subgroup, unrestricted uniform perturbation has the lower bound

$$\mathbb{E}_{w\in K}\mu\Big(\bigcap_{j<h} T_{j(v+w)}^{-1} A\Big) \;\geq\; \mathbb{E}_{w\in K}\mu\Big(\bigcap_{j<h} T_{jw}^{-1} C\Big) \;\geq c_0(\kappa, h) > 0.$$

Indeed use a random word cube with independent uniform K-directions and common stationary starting point; its expected occupied fraction is $\mu(C)$, while each nonempty wildcard sum is uniform on K independent of the common starting point after other directions are frozen. Finite density Hales–Jewett and a finite union bound give the last estimate uniformly in the subgroup, without needing any fixed amount of horizontal absolute compactness.

Choose a continuous nonnegative circle bump $g \leq 1$ supported where $||x|| > \gamma$, of positive integral. We show its weight on the active coordinate costs asymptotically only its mean. Approximate g uniformly by a finite trigonometric polynomial and approximate 1_A in L^1

by a finite linear combination of the indicated phase tests. Bounded approximants can be obtained from a generating algebra by clipping, or one can telescope with a mixed L^2 approximation by bounded combinations in the stated closed span. To avoid dependence of approximation errors on the approximant's sup norm, approximate the h slots successively: the accuracy for the next approximate slot is chosen after the finite sup norms of all earlier ones, while the still-original indicators have norm at most one. Telescoping and stationarity work for every fixed ordinary direction, so the total averaged error can be made arbitrarily small relative to $c_0 \int g$. Truncate the countable labels of these finitely many functions to finite sets, with the union of the exceptional sets at the h base slots of negligible measure uniformly over all v. This changes the estimates only by arbitrarily small bounded test errors.

In each term of the resulting finite expansion, conditional on the starting point x and the labels at $T_{jv}x$, the horizontal phase of the slot product equals an arbitrary linear character of w plus some member of a *finite list* of coordinatewise nonlinear polynomial arrays. The substitutions $w \to jw$ merely multiply degree-k coefficients by j^k. All constant factors have uniformly bounded modulus and do not depend on w.

For each array which has some nonintegral nonlinear coefficient at infinitely many primes, choose a large fresh passive prime among them, beyond the finitely many low trial axes, and include its entire horizontal cyclic subgroup in K. The normalized exponential average of the corresponding polynomial there is $o(1)$ uniformly in the additional linear character as this prime increases. In detail clear the largest denominator p^b of the nonconstant coefficients, so the resulting polynomial is not constant modulo p. If $b = 1$, use the finite-field polynomial exponential-sum bound (linear nonconstant phases sum to zero). If $b \geq 2$, sum on the last base-p digit: only zeros of the reduced derivative contribute, a fraction at most $(d-1)/p$ since $p > d$. A larger linear denominator only makes that derivative a nonzero constant. This passive-axis cancellation works both with and without the indicator-approximating weight on the marked axis.

For the remaining finite list of arrays all nonlinear terms are integral beyond a fixed head. Choose the low trial axes beyond this head and sufficiently large for the linear/quadratic Gauss estimate, take v by density Hales–Jewett, and select some fixed $i \in F$. Include the subgroup $p_i^{D_i} C_{p_i^{2D_i+1}}$ in K on this one coordinate, parametrized uniformly by $t \in C_{p_i^{D_i+1}}$. The dependence there for each remaining array is linear, however its character varies with x. Multiplying by any tested nonzero charge Fourier mode $e(kt^2/p_i)$ gives mean of modulus at most $p_i^{-1/2}$ by the estimate in the three-slot proof [R057]. The finite list of passive choices can be made beyond all the low axes even when the latter depend on the event; one may select the finite low axes first, then add the required fresh passive ones. All estimates were uniform in the choices of labels and linear characters at the actual starts.

Consequently the weighted average differs from $(\int g)$ times the unrestricted average by less than half the lower bound, once the fixed tolerances and prime sizes have been taken sufficiently far. Its positivity supplies actual w with $g(t^2/p_i) > 0$ and a positive ordinary return at $(v + w)$. This direction has exactly one charged axis and arbitrary legally passive velocities in the auxiliary coordinates. The argument leaves open situations in which the nonlinear symbols have diffuse variation and their combinations across the vertical slots cancel on precisely the passive axes.

R046. A three-dimensional affine Haar witness for the full rational action

Scope and status. Result with proof.

Result. The deep short obstruction extends to one rational affine action with underlying dimension three; connected finite-dimensional affine rational actions have full polynomial-test recurrence.

Scope. The disconnected torsion part of the three-dimensional example is essential to the comparison.

The short obstruction and system-relative upper bound in deep-affine-half.md [R042] can be realized on **one action of the full rational group** on a compact affine torsor whose underlying compact abelian group is the product of $\mathbb{T}^3$ and a zero-dimensional compact metrizable abelian group. In contrast, on a connected finite-dimensional compact abelian group every affine $\mathbb{Q}$-action has multiple returns at every positive density along every set passing all ordinary group-polynomial zero tests. The latter class includes finite-dimensional solenoids. Neither assertion gives an upper bound over arbitrary actions.

Extending each prime separately Write $H_p = C_{p^{e_p}} \subset P_p = C_{p^\infty}$. The local finite H_p-module B_p and invariant E_p of order p are those of the deep example [R049]. Induce the discrete module to P_p, obtaining, additively, the direct sum of copies of B_p indexed by its cosets modulo H_p, with action by coset permutations and H_p-twists. Identify all the copies of E_p: quotient by the subgroup generated by $u^t E_p - E_p$, $t \in P_p$. Denote the quotient by B_p^*. The original distinguished copy of B_p embeds. Indeed the subgroup of identified lines in the induced module is the augmentation-zero subgroup of their direct sum of order-p cyclic groups, so it intersects any single copy trivially. The identified E_p is nonzero, of order p, and fixed under the full P_p. The countable B_p^* is torsion of bounded p-power exponent, since all induced coset copies have the same finite exponent.

On $\prod_p B_p^*$ let $\mathbb{Q}/\mathbb{Z}$ act coordinatewise by its respective primary components (trivially on any prime outside the construction). Every element has finite prime support. Put $W = \bigoplus_p B_p^*$ and adjoin the three native tuples $F_0 = 1$, $F_1 = v$, $F_2 = v^2$, $v_p = u_p + u_p^{-1}$, each read inside the distinguished copy of B_p on every axis. The resulting additive module $M = \sum_{j=0}^{2} \mathbb{Z}F_j + W$ is invariant, since any acting group element changes a tuple only at finitely many prime coordinates. It is additively $\mathbb{Z}^3 \oplus W$: the nonzero integer polynomial of degree at most two on $\zeta_p + \zeta_p^{-1}$ remains nonzero in B_p for all sufficiently large p, by the bounded-order Taylor valuation argument in the native three-coordinate proof [R042]. The coefficient tuples of the nine-slot stencil belong to M.

Take the torsor of all characters agreeing with the prescribed E_p-values. It is a Haar affine $\mathbb{Q}/\mathbb{Z}$-space, pulled back to $\mathbb{Q}$; extension of the fixed-line character exists by divisibility of the circle. The group of differences of characters is the dual of $M/\bigoplus_p \langle E_p \rangle$, additively $\mathbb{T}^3 \times \prod_p \widehat{B_p^*/\langle E_p \rangle}$. The second factor is totally disconnected, by the torsion of each countable dual module, so this compact metrizable group has topological dimension three. Haar on the torsor gives three independent continuous Haar coordinates and an independent product of local fixed-line Haar laws.

The restriction of the local fixed-line Haar law to the native distinguished copy of B_p is exactly its original fixed-line Haar law: restriction is a surjection of compact torsors since any

character on B_p extends to B_p^*. Thus for the native two-prime sampler, conditioned on any finitely many early prime factors, the profile law on the three free circle coordinates is the one calculated in deep-affine-half.md [R042]. More explicitly, a finite temporal Fourier test of those coordinates changes the character tuple only inside the two native distinguished copies at the fresh primes. Nonvanishing conditional expectations there are exactly the old fixed-line tests. The finitely many early coordinates are unchanged by these steps, irrespective of their potentially infinite local dual factors. Approximation of arbitrary events by a finite set of early prime coordinates and the three torus coordinates therefore gives the same uniform-in-cylinder lower estimate for each fixed mass margin above $1/2$. Sections of arbitrary measurable events over a finite product of the extra zero-dimensional factors can, if needed, first be reduced to a finite measurable partition of that head and torus events. It follows that every event of mass $> 1/2$ in this full rational action returns at every length on the saturated directions. The finite nine-slot coloring obstruction still holds by the native character identities. Thus its system-relative all-length avoidance threshold belongs to $(0, 1/2]$.

Connected finite-dimensional groups Let X be a compact connected metrizable abelian group of dimension $m < \infty$, with a continuous-affine-map action $T^q x = A_q x + b(q)$ of the discrete group $\mathbb{Q}$. Its discrete dual M is torsionfree of rank m. The homomorphism $\mathbb{Q} \to \mathrm{GL}(M \otimes \mathbb{Q})$ from the dual linear parts has unipotent image. To see this, the finite-dimensional commutative rational algebra they generate has, modulo its nilpotent radical, a finite product of number fields. The multiplicative group of a number field contains no nontrivial divisible subgroup: all prime-ideal valuations of such elements vanish, and the unit group of the ring of integers is finitely generated by the unit theorem. Divisibility of $\mathbb{Q}$ sends its image to the identity in each residue field. The remaining commuting unipotent matrices admit a common strictly upper-triangular form after subtracting the identity. In particular all mixed products of m augmentation differences vanish. Moreover $A_q^* = \exp(qN)$ on the rational vector space with $N = \log A_1^*$, by the unique roots among commuting unipotent matrices.

For each character χ, $q \mapsto \chi(b(q))$ is a group polynomial of degree at most m vanishing at zero. Indeed the first difference with shift s is $\chi(A_q b(s))$, and subsequent differences apply commuting augmentation differences to χ. Now fix a horizon h. The Haar joining of $(T^{jq}x)_{j<h}$ has Fourier coefficient at $(m_j)_j$ equal to zero unless $\sum_j \exp(jqN)m_j = 0$; if this equality holds the coefficient is the exponential of the circle polynomial $\sum_j m_j(b(jq))$. The vector equality holds identically as a rational polynomial in q, or has only finitely many roots. On any nonprincipal ultrafilter along which all the indicated degree-m circle group polynomials go to zero, the limiting coefficient is one in the identically-zero case and zero otherwise. This gives Haar on the annihilator of the identically-zero tuples. It contains the diagonal since each identically-zero tuple has $\sum m_j = 0$, on evaluation at $q = 0$.

For any positive Haar-measurable event A, the limiting Haar subgroup gives positive mass to A^h. Average over translations by the diagonal; at its identity coset the translated intersection has mass $m_X(A) > 0$, and it is continuous as a function of the tuple of translations by L^1 continuity. Thus it remains positive on an open set of positive Haar measure. Approximation of A by continuous functions gives convergence of the correlations on the ultrafilter, using the common Haar marginal at each site. Consequently the h-return set contains a simultaneous zero neighborhood of finitely many degree-m group polynomials, with a positive lower bound there: otherwise the zero windows together with arbitrarily small-correlation

constraints would have the finite intersection property and yield a contradictory ultrafilter. Finite nonzero q's can be excluded by linear characters; the principal ultrafilter at zero has correlation $m_X(A)$. Applying this to the rational polynomial-recurrent deep family proves full recurrence in every such connected compact affine system.

R047. Higher-charge variants with unconditional finite recurrence orders

Scope and status. Proved subcases and unresolved extensions.

Result. Higher-charge models give unconditional recurrence through length r, ordinary group-polynomial recurrence, and a finite-color obstruction at length $2r + 2$.

Scope. The quadratic case has a separate stronger triple theorem; all-length intermediate capacity is unresolved.

Qualifications. Ordinary group-polynomial recurrence and generalized-polynomial recurrence are distinct hypotheses.

Fix $r \geq 2$. Take arbitrarily large primes p, integers $D_p \to \infty$, $e_p = rD_p + 1$, and primary axes $C_{p^{e_p}} \subset \mathbb{Q}/\mathbb{Z}$ with rational unit lift $w_p = 1/p^{e_p}$. Admit arbitrary passive multiples of p and positive active moves $1 + p^{D_p} t_p$. Require

$$|| \sum_{pactive} t_p^r/p|| \geq \gamma, 0 < \gamma < 1/2.$$

On both the primary group and its saturated rational cover every positive-measure event has all lengths $h \leq r$ in every tail. The same upper assertion transfers to the integer realizations at identical lengths. For all r the model still hits every simultaneous finite tuple of zero-constant circle-valued group-polynomial tests on $\mathbb{Q}$, at arbitrary fixed degree and tolerance. Nevertheless it has a finite palette forbidding every monochromatic $(2r+2)$-term progression on the group itself. Thus the all-length threshold is positive; its strict upper bound remains unknown. For $r = 2$ the relative Kronecker argument [R058] supplies the stronger unconditional $h = 3$ case. The short palette transfer to the integer realization need not preserve the stated horizon.

Unconditional weighted comparison For any measure-preserving transformation S, any 1-bounded $f_0, \ldots, f_{h-1}$, and any fixed integer $k \neq 0$, the weight $b_p(t) = e(kt^r/p)$ satisfies

$$\limsup_N |E_{1 \leq t \leq N} b_p(t) \int \prod_{j<h} S^{jt} f_j| = o_{r,h,k}(1)$$

as $p \to \infty$, uniformly over transformations and functions, provided $r \geq h$. This includes finite cyclic actions regardless of their orders. To prove it, shift the integral to anchor a chosen slot at slope zero and bound the square of the remaining L^2 norm by van der Corput. At a fixed lag s, invariance makes the resulting expression another scalar weighted average with one fewer slot, slopes translated by their first representative, bounded multiplicative derivatives of the previous functions, and weight $\Delta_s b_p$. For the last single slot shift the integral to remove its slope, leaving the scalar Cesàro mean of the remaining weight. After $h - 1$ differencings the coefficients and starting functions can depend on all lags, but the remaining weight is independent of those functions, whose integrals have modulus at most one. If needed apply scalar van der Corput up to a total of $r-1$ differencings. The surviving linear phase has slope $kr! \prod_{\ell=1}^{r-1} s_\ell/p$; its Cesàro mean is zero unless $p \mid \prod_\ell s_\ell$, for $p > r, |k|$. In

the iterated limiting uniform lag averages the latter event has probability at most $(r-1)/p$; taking finitely many square roots still gives $o(1)$. To justify the order of limits, first hold the averaging windows of the lags finite as $N \to \infty$, apply Cauchy–Schwarz or Jensen at every level to bring the resulting finite averages inside the finite power bound, then increase the windows to infinity successively. Difference-of-two-uniform lag windows or the usual triangular interval weights have the same limiting divisibility probabilities.

Given A of mass $a > 0$, use finite density Hales–Jewett on any sufficiently many fresh unit generators. There is a nonempty subsum v such that $C = \bigcap_{j<h} U^{-jv} A$ has mass at least $\kappa(a,h) > 0$. Choose any active prime p of v and $z = p^{D_p} w_p$. The unrestricted Cesàro average of the h-correlation of A at $v + tz$ is bounded from below by that of C at tz, hence by a positive constant depending only on κ, h, by ordinary finite Szemerédi–Varnavides for the transformation U^z. Choose a smooth nonnegative bump g of positive mean supported where $\|x\| > \gamma$. The comparison above, after a uniform trigonometric approximation to g, shows that inserting the weight $g(t^r/p)$ changes this lower bound by multiplication by its positive mean and an $o(1)$ error. In this application absorb the fixed U^{jv} shifts into the test functions f_j. Choose the fresh primes above that error threshold before the word argument. Some integer t therefore gives a positive h-return at the permitted $v + tz$. This proves the universal assertion even using only a single nonzero high mark and zero high digits at all other active low units.

There is a stronger special case for any h: it suffices that the tested event is generated, over the horizontal invariant factor, by horizontal phase polynomials of degrees at most $r-1$. Here a genuine phase observable has degree at most $r-1$ on almost every horizontal orbit, and products of such observables retain that degree. Approximate the h indicator slots by bounded finite combinations or telescope them successively with accuracies adjusted to the already known previous sup norms. For each starting point the phase test product on $v+tz$ is a circle polynomial in t of degree at most $r-1$, with coefficients allowed to vary measurably over the starts and the baseline slots. Multiplication by $e(kt^r/p)$ has Cesàro mean $o(1)$ uniformly in ALL its lower coefficients, by the same final $r-1$ differencings. This does not assume countably many possible nonlinear labels. In particular when arbitrary events can be reduced to such a horizontal factor the same recurrence follows. No such classification of arbitrary relative Host–Kra factors is asserted.

The short obstruction Use a primitive pth root ζ, $\varpi = \zeta - 1$, $p > r+3$, and let

$$B_p = (Z[\zeta]/(\varpi^{rD_p(p-1)+r+2}))[\epsilon]/(\epsilon^{r+1}), u = \zeta(1+\epsilon).$$

The coefficient ring has $p^{e_p} = 0$ and $u^{p^{e_p}} = 1$: for $1 \le j \le r < p$ the binomial coefficients at p^{e_p} are divisible by p^{e_p}. Define $E_p = p^{rD_p} \varpi^{r+1} \epsilon^r$; it is a nonzero fixed vector of order p. The scalar polynomial

$$P_p(Y) = (Y-1)^{r+1}(Y-u)^r$$

vanishes at u^b for every passive b, since $p \mid b$ implies $u^b - 1 \in (\epsilon)$. At $b = 1 + p^{D_p} t$, $(u^b - u)^r = \zeta^r p^{rD_p} t^r \epsilon^r$ in the truncated ring. The other factor can then be evaluated at $\epsilon = 0$, giving ϖ^{r+1}. Since $\zeta^r \equiv 1 \bmod \varpi$, one has exactly $P_p(u^b) = t^r E_p$.

In the countable module generated by the tuples of the $2r+2$ polynomial coefficients and the single-prime fixed vectors, prescribe the additive fixed-vector character $E_p \mapsto 1/p \bmod 1$ and take Haar measure on the torsor of all extensions, as in the quadratic construction [R049]. Each of the $2r+2$ coefficient observations has zero total at one point since $P_p(1) = 0$. Across the window their filtered sum is exactly the charge, so coloring the observations by arcs of diameter less than $\gamma/(2r+3)$ forbids a monochromatic progression. Coinduction and pullback give the saturated rational obstruction. The rational group-polynomial recurrence proof in that same note only requires that a sufficiently large active prime permit a digit with $\|t^r/p\| > \gamma$, supplied by the preceding scalar exponential estimate; its passive-prime analysis is unchanged.

If the two signed low speeds are both desired with charge $\sum s_p t_p^2/p$ in the quadratic case, one can instead take $P_p(Y) = (Y-1)^3(Y-u)^2(Y-u^{-1})^2$, $\epsilon^3 = 0$, and truncate at $\varpi^{2D_p(p-1)+6}$. The ϖ^5-factor at an active speed has residue $4s_p$ after normalization, yielding an eight-term obstruction; for the unsigned quadratic charge with both signs the earlier nine-term model remains the one analyzed.

R048. A pivot comparison for all bounded-degree polynomial phases

Scope and status. Result with proof.

Result. A pivot comparison gives all-order recurrence for deep-family correlations in the stated uniform phase-integral class, including Gaussian and Abramov cases.

Scope. Uniform approximation is used; arbitrary multicorrelations are not asserted to belong to the class.

Use the deep mixed-characteristic candidate V on $G = \bigoplus C_{p^{e_p}}$, and its rational saturation D, from deep-quadratic-primes.md [R049]. Both systems of directions force h-returns for every positive event whose h-return function is a uniform limit of finite linear combinations of integrals of circle polynomial phases, of finite degree at each approximation. In particular this holds for every ordinary pair test, all orders of phase-polynomial (Abramov) systems, and all orders of Gaussian systems. The lower bound for some return depends only on the event mass and h. These are strong positive subclasses; the finite-color obstruction proves that arbitrary multicorrelations cannot all belong to the asserted phase-integral class here.

The finite panel distributions on the primary group Fix m disjoint panels of R distinct sufficiently late primes each. In each panel choose a uniform pivot, setting its direction to (1). At every other index independently choose either (1) or a uniformly random element of $pC_{p^{e_p}}$, with probabilities one half each. This gives m independently sampled baseline directions $y_1, \dots, y_m$. For any nonempty $I \subseteq [m]$, modify the pivot of just one panel in I from (1) to $1 + p^{D_p} t_p$, using a preassigned t_p with $\|t_p^2/p\| \geq \gamma$. All other active sites have zero high digit, so the modified sum y'_I is always in V. Write $y_I = \sum_{i \in I} y_i$ for the unmodified sum.

For any degree-d circle polynomial Q on G, with R large and every panel prime and D_p sufficiently large in terms of d, one has

$$|E \exp(2\pi i Q(y'_I)) - E \exp(2\pi i Q(y_I))| \leq C/R$$

with an absolute constant once the prime threshold depending on d is met. Indeed after subtracting its constant Q is a sum of its one-axis restrictions by the coprime-difference calculation in the prime-square note [R205]. Since $p > d$, its restriction is an ordinary polynomial in r_p divided by p^{e_p} modulo one, by the same Newton argument. Call an index bad when the phase on (pz) is nonconstant; then its normalized exponential mean on the normal subgroup has absolute value at most $O_d(p^{-1/2})$ by the finite-field/last-digit estimate of the deep-prime note. Otherwise the original polynomial coefficients have p-adic valuations at least minus their corresponding degrees, and replacing (1) by the deep mark has no effect. Take the primes large enough that on each bad nonpivot index the independent mixed token has exponential multiplier of modulus at most (3/4). If the pivot is good, the two terms coincide. If there are $b > 0$ bad coordinates in its panel, the absolute difference is therefore at most $2(b/R)(3/4)^{b-1} \leq C/R$. All other panel factors have modulus at most one. The argument is uniform over the coefficients of Q.

For any positive event A of mass a, with return function c_h, take $m = m(a, h)$ from density Hales–Jewett. Deterministic word averaging, then averaging the baseline directions, gives

$$\sum_{\varnothing \neq I \subset [m]} Ec_h(y_I) \geq \kappa(a,h) > 0.$$

Indeed on a random start the word map $w \mapsto T_{\sum w_i y_i} x$ has expected occupied density a, so density at least $(a/2)$ occurs with probability at least $(a/2)$. Sum over the finite list of combinatorial lines, removing their base translates by invariance. For a uniform approximation to c_h by the stated phase integrals, the pivot comparison integrates and then sums (incurring the finite combination norms). First choose its uniform error much smaller than $\kappa/2^m$, then take R large and the primes sufficiently late for its degree. The modified sum of expectations is still at least $\kappa/2$, giving an actual positive return in any prescribed tail.

The rational saturation Apply the same word argument to independent rational lifts of the panel directions. Concretely first sample the residues as above, representing a direction by the sum of its chosen fractions r_p/p^{e_p}. Independently add to each of the m directions a uniform integer in $([1,N])$. On modifying one pivot use the same integer offsets. All resulting modified differences belong to D. For every fixed rational group polynomial $Q(g) = \sum_j \chi_j(g^j) + Q(0)$ the expectations have limits as $N \to \infty$, and the difference of those limits obeys the same (C/R) bound, uniformly among all polynomials of the fixed degree.

To verify this carefully write the characters in archimedean and finite-adic components as in the deep-prime note, absorbing rational archimedean coefficients into the finite-adic characters. On every fixed fractional residue choice and fixed nonempty I, the integer offset is a sum of finitely many independent long-uniform integers. If the archimedean polynomial has a nonzero residual irrational coefficient of highest nonzero degree, its exponential mean tends to zero, even after imposing any fixed modulus on the offset and adding the finite-adic phase. Indeed along the relevant residue classes all finite-adic contributions are constant (for the fixed polynomial and fractions), and Weyl's finite differencing estimate for the highest irrational coefficient is uniform over the lower coefficients and the translates, so one may average one of the independent offsets first. If all archimedean coefficients are zero, the offset tends to uniform measure on every finite congruence quotient. By CRT the limiting exponential factors into one-axis averages over p-adic integer offsets of the chosen rational fraction. There are only finitely many nontrivial factors, the finite list of chosen primes and the finite list where some adjusted coefficient is nonintegral. On an active chosen axis the normalized p-adic variable $r = p^{e_p} g$ runs through the $p^{e_p}\mathbb{Z}_p$-coset of the sampled residue; all contributions from the other fractions are integral shifts in g and hence do not affect this coset. On a randomly normal axis the mixed residue sampling and integer offset together give Haar on $p\mathbb{Z}_p$ in r-coordinates. Therefore the good/bad classification, invariance under the deep mark in the good case, and the uniform exponential bound in the bad case are exactly as before. Nonparticipating exceptional primes give fixed factors of modulus at most one. This proves the comparison, including for coset offsets with very deep conductors depending on the individual polynomial.

Integrate in the approximating phase measure by dominated convergence; approximation phases need not share any bound on their denominators. The deterministic density-Hales–Jewett sum is uniform in all choices of the baseline lifts and offsets before taking $N \to \infty$. The same argument as on G, using a liminf if needed for a residual uniform approximation error in c_h, gives the positive rational return and the bound depending only on (a,h).

Applicable functions For a system generated by measurable phase polynomials, approximate 1_A in L^h by finite sums/products of circle phase functions (conjugates and products retain finite degree). Hölder and telescoping give a uniform approximation to the correlation by an integral of finite sums of group-polynomial phases in the direction variable. The countability of the acting group lets all finite-difference identities hold on a common conull set of starts. This does not require uniform bounds on the degrees over all approximations. If $H = \bigoplus pC_{p^{e_p}}$ acts ergodically and the event is generated by H-phase polynomials, the same conclusion holds: the top H-derivatives of any such function are H-invariant constants, so applying any G-derivative lowers its H-degree by at least one. Iterating proves it was already a full G-phase polynomial of the same degree.

Pairs belong to the phase-integral class by the spectral theorem. For Gaussian systems approximate the event indicator in L^h by polynomials of finitely many first-chaos variables, and use Wick's formula. Products and sums of the resulting covariances are Fourier–Stieltjes transforms on the abelian group, and the approximation is uniform in the direction by Hölder. No assertion about arbitrary nonlinear factors, non-Haar measures on affine models, or the unknown general all-length threshold follows.

R049. A deep marked quadratic phase which survives rational polynomial tests

Scope and status. Proved subcases and unresolved extensions.

Result. The deep quadratic family has a nine-slot finite-color obstruction on its rational saturation and passes every fixed finite list of ordinary group-polynomial zero tests.

Scope. Generalized-polynomial or nil-Bohr recurrence does not follow from this particular argument alone; unrestricted upper capacity remains unknown.

Qualifications. Nil-Bohr recurrence requires the additional separately proved nilrotation tests; it is not established by the ordinary-polynomial paragraph itself.

Let $D_p \to \infty$ be positive integers over an infinite set of primes $p \geq 11$, and put $e_p = 2D_p+1$. In $G = \bigoplus C_{p^{e_p}} \subseteq \mathbb{Q}/\mathbb{Z}$, with residue r_p embedded as r_p/p^{e_p}, allow directions whose p-coordinate is one of

- a normal $r_p \equiv 0 \bmod p$ with no further restriction;
- an active $r_p = s_p + p^{D_p} t_p$, $s_p = 1$ or (-1).

Retain such finitely supported directions exactly when $\| \sum_{p \text{ active}} t_p^2/p \| \geq \gamma$, for a fixed $0 < \gamma < 1/2$. Call the result V, and set $D = \pi^{-1}V \subseteq \mathbb{Q}$, with $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$. The active digits $t_p \bmod p$ determine the charge. Higher digits are unrestricted on every permitted residue, and there is no size restriction on the active set.

Results. D has a fixed finite-color obstruction at length nine, already from a Haar affine action of the rational group on a compact metrizable abelian torsor. It is recurrent for every finite list of zero-constant circle-valued group-polynomial tests on the rationals, of every fixed finite degree and joint tolerance. Thus this example defeats the bounded-degree polynomial lift which ruled out the prime-square attempt [R205]. No bound $b(D) < 1$ is proved here. If such a bound can be obtained, the rational-to-integer realization gives a first intermediate integer threshold.

A mixed-characteristic nine-slot obstruction Let (ζ) be a primitive pth root of unity, $\varpi = \zeta - 1$, $L = 2D_p(p-1)+7$, and form the finite ring

$$B_p = (Z[\zeta]/(\varpi^L))[\epsilon]/(\epsilon^3).$$

In its coefficient ring p is a unit times ϖ^{p-1}, since $p = \prod_{1 \leq k < p}(1-\zeta^k)$ and the quotients by $1-\zeta$ are units at the prime above p. Thus $p^{2D_p+1} = 0$. The unit $u = \zeta(1+\epsilon)$ has order dividing p^{2D_p+1}, by binomial expansion with $\epsilon^3 = 0$. Take the additive module action by this unit and the polynomial

$$P_p(Y) = (Y-1)^4(Y-u)^2(Y-u^{-1})^2 = \sum_{j=0}^{8} m_{j,p} Y^j.$$

For $r \equiv 0 \bmod p$, $u^r - 1 \in (\epsilon)$ and $P_p(u^r) = 0$. For $r = s + p^{D_p} t$, $s = \pm 1$, squaring the factor $u^r - u^s$ gives $\zeta^{2s} p^{2D_p} t^2 \epsilon^2$; terms involving higher powers of (ϵ) vanish. The two other factors can therefore be evaluated at $\epsilon = 0$. Modulo the one surviving (ϖ)-layer, $(\zeta^s - 1)/\varpi \equiv s$, $(\zeta^s - \zeta^{-s})/\varpi \equiv 2s$ and $\zeta^{2s} \equiv 1$. Consequently

$$P_p(u^r) = 4t^2 E_p, E_p = p^{2D_p} \varpi^6 \epsilon^2.$$

This $E_p \neq 0$ is fixed by u and has additive order p. As in the prime-square note, let $M' \subset \prod_p B_p$ be the countable module generated by the nine coefficient tuples and the individual E_p. Prescribe the fixed-subgroup character $E_p \mapsto [4^{-1}]_p/p$, and use Haar probability on its extension coset in $\widehat{M'}$. Along any allowed direction the sum of the nine indicated character observations is exactly the charge, whereas their sum at one point is zero since $P_p(1) = 0$. Coloring the torsor by the cells of these nine observations gives a uniform finite palette avoiding monochromatic 9-runs; take cell diameters $< \gamma/10$. Extend this action from G to $\mathbb{Q}/\mathbb{Z}$ by coinduction (product torsor over the cosets, with coordinate permutations and the given commuting affine actions); on the identity-coset coordinate every step of V retains its obstruction. Pull back through π. This is a compact metrizable affine Haar system; in particular the positive-measure obstruction is exact. The underlying group has unbounded prime orders and is not assumed profinite; the combined coefficient tuples can have infinite additive order.

Rational group-polynomial recurrence Every degree-k zero-constant circle group polynomial on $\mathbb{Q}$ is a sum of expressions $\chi_j(g^j)$, $1 \leq j \leq k$, with additive characters $\chi_j : \mathbb{Q} \to \mathbb{T}$. Indeed its kth difference is symmetric multiadditive and factors through $\mathbb{Q}^{\otimes_{\mathbb{Z}} k} = \mathbb{Q}$ by multiplication. Subtract the matching kth-power term, dividing the rational argument by (k!), and induct. Write the character as

$$\chi(g) = \alpha g + \sum_p [a_p g]_p, a_p \in Z_p,$$

after choosing α so $\chi(n) = \alpha n \bmod 1$ on the integers. Its difference factors through $\mathbb{Q}/\mathbb{Z}$, giving its independent p-adic dual coefficients. For a finite tuple of tests, strip the rational real coefficients into the finite-adic side: in a rational basis including one for the finite span of the real coefficients and 1, move the coefficient of 1 using $y \bmod 1 = \sum_p [y]_p$ for rational y. The local coefficients are still integral outside finitely many primes. Now every integer linear combination of the tests has archimedean polynomial either zero identically or of irrational highest nonzero coefficient (the constant terms are zero).

Fix a small zero neighborhood in the test torus. Use a continuous nonnegative function $F = f * \widetilde{f}$ supported there with nonnegative absolutely summable Fourier coefficients and positive constant c_0, taking a small continuous nonnegative bump f. Retain a finite symmetric list of modes so the discarded coefficient sum is less than $c_0/4$.

For a retained mode k, let $Q_{p,k}(r)$ be its local p-adic polynomial with argument $g = r/p^{e_p}$, before applying $[\cdot]_p$. On an infinite set of large primes every mode has a constant label, good or bad, according as $Q_{p,k}(pz)$ is integral throughout $z \in \mathbb{Z}_p$ or not. In the good case all its coefficients in z are integral, by Newton interpolation modulo p when p exceeds the degrees.

The coefficient of r^j in $Q_{p,k}(r)$ thus has valuation at least $(-j)$. For D_p larger than the degrees, the phase at $1 + p^{D_p}t$ equals its phase at one for every $t \in \mathbb{Z}_p$.

In the bad case the additive exponential mean of $Q_{p,k}(pz)$ over Haar $z \in \mathbb{Z}_p$ tends uniformly to zero on these primes. Clear its largest coefficient denominator p^b, producing $p^{-b}H(z)$ with a nonconstant coefficient a unit; the constant term is zero. If $b = 1$, the usual finite-field additive Weil bound is $O_k(p^{-1/2})$. If $b \geq 2$, sum over the last base-p digit using $H(z + p^{b-1}t) \equiv H(z) + p^{b-1}tH'(z) \bmod p^b$. Only zeros of $H' \bmod p$ contribute; since $p > \deg H$, their proportion is at most $(\deg H - 1)/p$.

Pass to a subsequence where the finite vector of phases at $r = 1$ converges. For the good modes choose a fixed positive number of copies m so that m times their joint limiting phases are as close to zero as desired (pigeonhole on a torus). Choose m sufficiently late active primes from this subsequence, all with base $r = 1$; on one of them take $r = 1 + p^{D_p}t$ instead with $\|t^2/p\| \geq \gamma$, possible by the same quadratic estimate or equidistribution of quadratic residues. This modification does not change any good mode. Choose a further very large passive prime from the same subsequence, putting $r = pz$ there and averaging z uniformly modulo a sufficiently large power. Every bad mode now has mean of arbitrarily small absolute value; good modes have passive contribution exactly zero. The required charge condition holds for every one of these choices.

Every assigned collection of local residues (to all the finite precisions needed by the tests and by V) is realized by a rational g of denominator dividing the product of the chosen p^{e_p}, by the Chinese remainder theorem on its numerator. Require also that g be highly divisible at the finitely many other primes where test coefficients are not integral. At all remaining primes the local phases vanish. After each such choice add multiples of a common sufficiently divisible integer. This preserves all specified local phases and direction restrictions. Averaging along those integer multiples kills every mode whose archimedean polynomial has irrational highest nonzero coefficient, by elementary Weyl equidistribution; for the other modes the archimedean contribution is identically zero. In the resulting mean of F, good modes with zero archimedean part have real part as close to $(+1)$ as prescribed. The bad-mode contribution is arbitrarily small and the other retained modes with irrational real phase have zero mean. The constant c_0 and the Fourier tail bound give a positive mean, hence a realized $g \in D$ returning to the target neighborhood. All participating primes can be required to exceed any prescribed bound.

Scope There is no conclusion here for generalized-polynomial/nil-Bohr tests of arbitrary nilpotency order. The recurrence proof treats ordinary circle-valued group polynomials on the additive rationals. The main unresolved inequality is uniform multiple recurrence for arbitrary systems above a density bounded strictly below one.

R050. The deep model is recurrent on every affine Haar action of torus dimension at most two

Scope and status. Result with proof.

Result. Every affine Haar action of compact dimension at most two has full recurrence along the deep quadratic family.

Scope. Combined with the dimension-three obstruction this is sharp within the indicated affine category.

Let $G = \bigoplus_p C_{p^{e_p}}$, $e_p = 2D_p + 1$, $D_p \to \infty$, and use the deep quadratic directions [R049]. On **every** probability-preserving affine action of G on a compact metrizable abelian group X of dimension at most two, every positive Haar event has every length along these differences. The proof uses only positive low unit steps, freely sampled high marks and arbitrary passive multiples of local primes. It also applies to a full rational action factoring through $\mathbb{Q}/\mathbb{Z}$ by restriction to G, with saturated differences. Arbitrary torsion in the discrete dual is allowed. In combination with the three-dimensional affine counterexample [R046], dimension three is sharp for such affine Haar actions on this family. This is not an unrestricted measure-preserving recurrence assertion.

Isolating local fixed phases Write $M = \widehat{X}$. The linear part of the action is trivial on $M \otimes \mathbb{Q}$ at all sufficiently large primes: a matrix of finite p-power order in $\mathrm{GL}_2(\mathbb{Q})$ is the identity for $p > 3$, by the cyclotomic minimum polynomial. Only the rational rank of M, not a finite generation hypothesis, is used below. Use the group ring $\mathbb{Z}[G]$ and the affine linearization $M' = M \oplus \mathbb{T}_{\mathrm{discrete}}$ from the squarefree Haar argument [R239], writing u^g for its units. Fix a horizon h and a finite list of temporal Fourier tuples $m_0, \dots, m_{h-1}$. Its linear relation locus E is the set of g for which $\sum_j A_g^{*j} m_j = 0$. On E the group-ring word applied to $m'_j = (m_j, 0)$ equals a fixed scalar

$$\sum_j (u^g)^j m'_j = (0, \phi(g)).$$

If $\sum m_j \neq 0$, exclude E using a finite window of ordinary linear characters of G. Indeed if zero lay in the Bohr closure of E, the linear diagonal-containment lemma for torsion groups with finite primary components [R239] would force $\sum m_j = 0$; its proof in that note allows arbitrary finite primary components. Add the finitely many corresponding character angles to the test list to be made small. Hence work only with tuples of sum zero. Their phases at $g \in E$ have order dividing $\mathrm{ord}(g)$, by applying the geometric norm $1 + u^g + \cdots + (u^g)^{\mathrm{ord}(g)-1}$ to the displayed identity.

Localize at the ideal of augmentation modulo p. All units from the other prime components equal one in this localization (their geometric sums are units). A fixed finite-order scalar reduces precisely to its p-primary component, and the p-primary scalar subgroup embeds: all inverted elements act on it by integer augmentations prime to p. Consequently, on each represented primary residue $r = g_p$ of E, there is a uniquely determined scalar $\phi_p(r)$, of order dividing that of r, given in the localization by $\sum_j u^{jr} m'_j$, with u the p-axis generator. Also $\phi_p(0) = 0$ and $\phi(g) = \sum_p \phi_p(g_p)$.

Call a tuple bad if infinitely many primes have an allowed local residue (passive pz or positive active $1 + p^{D_p}t$) missing from the projection of E. In each finite panel of axes select one disjoint fresh disabling prime per bad tuple, and use such a missing residue there. Regardless of the other coordinates the resulting direction cannot belong to that tuple's E. All other tuples are called good. Outside a fixed finite head, each good tuple has defined local phases on **every** allowed residue. Bohr characters added above have their ordinary single-axis values everywhere. Thus on an allowed direction define the torus vector W as the sum of all local phase vectors for good tuples and all the additional character angles. Whenever a good tuple has a true linear relation, its coordinate of W is its actual affine phase. Proving W can be arbitrarily close to zero while the total charge is in the gap will suffice.

A single fresh prime Fix an integer linear combination of the good-tuple coordinates and the character coordinates, to be used as a Fourier test. Outside finitely many primes the combination of the affine tuple phases has a single localized representation

$$P(u^r)m + Q(u^r)n = \phi_p(r),$$

where m, n are two fixed localized affine module vectors and P, Q are polynomials of degrees at most $h - 1$ with rational coefficients regular at these primes; one or both may be absent. Indeed choose a rational basis of the span of the finite list in $M \otimes \mathbb{Q}$, using vectors from M; multiplying by a fixed nonzero integer clears denominators, and the finitely many resulting torsion remainders from M' have bounded orders. Their orders and denominators are invertible on fresh augmentations. Rational scalars in this argument mean the corresponding local scalars. Both polynomials vanish at one since the original tuples sum to zero. One extra genuine character adds the linear term $r\gamma_p$, with $\gamma_p \in \mathbb{T}[p^{e_p}]$. Write $f(r) = \phi_p(r) + r\gamma_p$.

If the polynomial span has dimension two, choose a rational change of basis so that P, Q have distinct positive orders $s < t$ of vanishing at one. Write $k = t-s$, $P(Y) = (Y-1)^s A(Y)$, $Q(Y) = (Y-1)^t B(Y)$, $a = A(1) \neq 0,\ b = B(1) \neq 0$. All the finitely many constants needed below are units at large p. Put $X = u - 1,\ Z_r = (u^r - 1)/(u - 1)$ as a Laurent polynomial, $M_0 = X^s m,\ N_0 = X^t n$. At $r = 1$ the displayed word is fixed, and $A(u)$ is a unit in the localization. Eliminating M_0 gives

$$\phi_p(r) = r^s \phi_p(1) + Z_r^s K_r N_0,$$

$$K_r = Z_r^k B(u^r) - A(u^r)B(u)/A(u).$$

Here action on the fixed scalar uses augmentation. At $r = p$, K_p has unit augmentation $b(p^k - 1)$, making $\eta_0 = Z_p^s N_0$ fixed. In fact it is a p-primary scalar, by solving the very same relation at p; the other primary actions are trivial in the localization. Thus on every passive residue the formula becomes

$$f(pz) = z^s(p^s \phi_p(1) - b\eta_0) + bp^k z^t \eta_0 + pz\gamma_p. \quad (1)$$

At $r = 1+p^{D_p}$, the factor K_r is divisible by Z_{r-1} with unit quotient. Indeed $Z_r-1 = uZ_{r-1}$ and $u^r-u = (u-1)uZ_{r-1}$. The augmentation of the quotient is $b(1+r+\cdots+r^{k-1})$, reducing to $bk \neq 0 \bmod p$. Also Z_r is a unit. Solving now gives a fixed p-primary scalar $\eta = Z_{p^{D_p}} N_0$, of order dividing $p^{e_p-D_p}$: multiply by the geometric norm of $u^{p^{D_p}}$, and use $Z_{p^{e_p}} X^t = 0$. On all active residues $r = 1 + p^{D_p} v$ the formula is therefore

$$f(r) = r^s \phi_p(1) + r\gamma_p$$

$$+bvr^s(1 + r + \cdots + r^{k-1})\eta. \quad (2)$$

If the polynomial span has dimension one, write it as $P(Y) = (Y-1)^s A(Y)$ with $A(1) \neq 0$. The relation at $r = 1$ gives $X^s m$ fixed, so $\phi_p(r) = r^s \phi_p(1)$ for every represented r. Dimension zero gives only the extra linear character.

Choose one local direction by an independent fair coin: either pz uniform in the passive subgroup, or $1+p^{D_p}v$ with v uniform modulo p^{D_p+1}. Its local charge Q_p is respectively zero or v^2/p. For every fixed nonzero integer j, all sufficiently large primes have

$$|E \exp(2\pi i(f(r) + jQ_p))| \leq 3/4. \quad (3)$$

To prove this, the exponential mean of the passive polynomial (1) is $o(1)$ uniformly unless all its nonconstant coefficients vanish. Its coefficients have p-power denominators, and the polynomial is periodic in $z \bmod p^{e_p-1}$. It can be averaged over a larger multiple of all its denominators with the same answer. Clear the largest denominator p^l to get an integer polynomial with a nonconstant coefficient a unit. For $l = 1$ the finite-field exponential sum is $o(1)$ by elementary successive differencing to a nonzero linear phase; for $l \geq 2$ averaging the last digit leaves only the boundedly many roots modulo p of its reduced derivative. If the passive mean is small, (3) follows at once.

Otherwise if $s \geq 2$, (1) gives $p\gamma_p = 0$, $p^k\eta_0 = 0$, $p^s\phi_p(1) = b\eta_0$, so $\phi_p(1)$ has order dividing p^t. If $s = 1$, merge the linear character into $\phi_p(1) + \gamma_p$; this sum has order dividing p^t. Thus in (2) the first two terms are constant throughout the active coset for $D_p > h$. Its remaining term is a scalar binomial expansion in v: since $p^{D_p+1}\eta = 0$, only the linear coefficient $bk\eta$ and a quadratic coefficient of order at most p survive. If the linear coefficient has order greater than p, averaging over the lifts of a fixed residue $v \bmod p$ gives zero even after inserting the quadratic weight. Otherwise the quadratic coefficient from η vanishes, so the inserted jv^2/p gives a complete Gauss sum of modulus $p^{-1/2}$. This proves (3). The one-polynomial and zero-polynomial cases follow by the same passive-zero test: for $s \geq 2$ one obtains $p^s\phi_p(1) = p\gamma_p = 0$, and for $s = 1$ the combined coefficient has order at most p. The active test is constant apart from the inserted quadratic weight.

Pooling the local laws and returning Take arbitrarily many disjoint panels on increasingly late prime axes. In each put the finitely many predetermined disabling coordinates, then a number tending to infinity of fresh independent local samples as in (3). The charge Q of a panel is the sum of its active squared charges, and W is its summed phase vector. For every fixed Fourier frequency (λ, j) with $j \neq 0$, the coefficient of the joint panel law tends to

zero: for that combination λ each sufficiently late sampled axis contracts its absolute value by at least the factor $3/4$. The finitely many fixed disabling coordinates only multiply by unit complex numbers. Every subsequential weak limit on the finite torus consequently has the form $\sigma \times m_{\mathbb{T}}$, with the second coordinate the charge.

Fix z in the support of σ and a desired zero window for W. Choose a positive integer l so that lz is close enough to zero to lie in its interior. Fix ω strictly inside the charge gap and $\omega' \in \mathbb{T}$ with $l\omega' = \omega$. In each of l sufficiently late distinct panels of the convergent subsequence there is an actual realization with W close to z and Q close to ω', by positivity of the limiting mass of any product of these open neighborhoods. Add those disjoint realizations. The resulting finite-support difference is legal, disables all bad tuples, and puts W in the zero window. In particular the ordinary characters exclude every relation for nonzero-sum tuples, as required.

Applying the finite intersection property to successively finer windows and arbitrary finite lists of tuples gives an ultrafilter of allowed differences on which every surviving Fourier relation is zero-sum and its affine phase tends to zero. Thus the limiting h-tuple Haar law is Haar on the annihilator of the subgroup of surviving tuples, which contains the diagonal. As in the connected-case argument [R046], averaging Haar over diagonal translations gives positive mass to A^h for every positive Haar event, and fixed Haar marginals permit continuous approximation. Hence some actual permitted step returns, at every horizon and on every prescribed tail.

R051. Arbitrary root strides: what follows from the two-mark Fourier calculation

Scope and status. Result with proof.

Result. The rooted two-mark Fourier calculation excludes a uniform bounded-state trajectory argument across unrestricted root orders.

Scope. An omitted-prime root restriction is material; this is a method obstruction, not a general density bound.

For the independent rooted copies of the full spatial mixed module, use the sampler of deep-spatial-periodic-profiles.md [R056]. The Fourier divisor at stride M is

$$B_M = (T-1)^3(T^M - S)^2(T^M - S^{-1})^2.$$

Indeed at the fresh passive prime the coefficients of ϵ^j, $j = 0, 1, 2$, successively require $(T-1)^3 \mid R$. At sign $s = \pm 1$ the constant term specializes $S = T^{sM}$, so $T^M - S^s$ divides R; a second copy is necessary since its ϵ-coefficient otherwise has valuation $D_p(p-1) + O_R(1)$ below the fixed line. The primitive p-root substitution $T = \zeta_p^{s/M}$ has first ramified digit s/M modulo p. For $R = B_M C$ the mark value at order $\varpi^5 p^{2D_p}\epsilon^2$ prevents survival unless $C(1,1) = 0$. On the next layer the normalized scalar coefficient is

$$t_p^2([M^{-4}]_p C_T(1,1) + s[M^{-3}]_p C_S(1,1))/p \bmod 1.$$

The square brackets denote inverses in the finite field; they cannot be replaced with division of a circle value without selecting the roots. A simultaneous choice across finitely many strides is provided in the cited spatial note. When a single prime ℓ is omitted from all the strides and allows an ℓ-torsion charge in the open gap, its bound $1 - 1/\ell$ extends to arbitrary finite cylinders and countably many independent copies.

Bounded dyadic exponents. More generally suppose all strides have $v_2(M) \le v$, while odd factors are unrestricted. For a finite list choose L divisible by their eighth powers with $v_2(L) \le 8v$, put $\theta = 1/2^{1+v_2(L)}$ and choose the two local marks with $[L^{-1}]_p t_p^2/p$ converging to the same ξ satisfying $2\xi = \theta$. Then $\Omega = L\theta = 1/2 \bmod 1$; the signed terms cancel. The spatially constant cubic-kernel test is realized by

$$z_M(t) = (L/M^8)\theta \binom{t}{8}.$$

Indeed on a surviving $B_M C$, $C(T,1) = (T-1)C_1(T)$ with $C_1(1) = C_T(1,1)$; $(T^M-1)^4 = (T-1)^4(1 + \cdots + T^{M-1})^4$. Thus the profile produces $M^4(L/M^8)\theta C_T = (L/M^4)\theta C_T$, the actual selected root-scaled charge. The joint profile across all these roots ranges over at most 2^{1+8v} states, since every coordinate is an integer multiple of the *same* 2^{1+8v}-torsion value. The conditional Haar-law argument then gives returns for every event of mass strictly greater than $1 - 2^{-(1+8v)}$, uniformly in the number of roots. For a countable list insist at every stage on $v_2(L) = 8v$ and odd part $L/2^{8v} \equiv 1 \bmod 2^{1+8v}$, multiplying by an auxiliary

odd number if necessary. Then every earlier z_M is unchanged modulo one under later choices, by uniqueness of odd division in the fixed dyadic group. Use the single resulting sampler, take its full compact product joining, and apply the diagonal-translation argument directly there. This does not require a quantitative cylinder-approximation estimate uniform over measurable sets.

At most four actual spatial offsets per copy. Here there is no root restriction or density threshold. For a nonzero Laurent Fourier character $R(T,S) = \sum_{a\in J} r_a(T)S^a$ with $|J| \le 4$, the double roots in S at T^M, T^{-M} are impossible. After adjoining additional distinct spatial exponents if necessary, its four equations on the coefficients $r_a(T)$ have a nonsingular matrix over $\mathbb{Q}(T)$. For four distinct $a_1 < \cdots < a_4$, the rows are $T^{Ma_i}, a_iT^{Ma_i}, T^{-Ma_i}, a_iT^{-Ma_i}$ read as four tests. The largest determinant exponent is obtained uniquely by putting the largest two a_i's in the positive-exponent rows. The coefficient there is, up to sign, $(a_4-a_3)(a_2-a_1) \neq 0$. Thus the two-mark law projects to independent Haar on all times at these actual spatial positions, even conditionally on any finite head. Independent copies still multiply for completely arbitrary root orders. An event measured on at most four spatial indices per copy and on arbitrary head/local data therefore has all positive returns; for general measurable dependence on the local data use finite-head approximation as in the affine half proof [R042]. This observation is for the individual spatial monomials and is not a bound for four arbitrary linear combinations of many such monomials.

An obstruction to a bounded-state proof for unrestricted roots. Every supported profile satisfies

$$(T-1)B_M(T,S)F = \Omega_M,\ M^4\Omega_M = \Omega.$$

Consequently at fixed s, $B_MF(t,s)$ ranges over a translate of the whole cyclic orbit generated by Ω_M as t runs over all integers. If the *whole spatial row* $F(t,\cdot)$ took at most b values, $B_MF(t,s)$ could take at most b^{20} values: its temporal inputs lie at $j+kM,\ 0\le j\le 3,\ 0\le k\le 4$, irrespective of spatial dependence of those row values. For $\Omega = 1/2$, the order of Ω_M is at least $2^{1+4v_2(M)}$. With factorial root strides, no uniformly bounded order is possible for any nonzero limiting charge (if all orders were at most N, M divisible by $1,\dots,N$ would give $\Omega = 0$). Thus even two profiles whose difference is a nontorsion spatial vector cannot supply a uniform bounded-state construction for all root orders. This excludes one proof method; it does not decide recurrence above a gap for the unrestricted spatial process.

R052. Arbitrary scalar circle arrays on the deep grids give full recurrence

Scope and status. Result with proof.

Result. Independent scalar or finite-vector torus arrays on deep grids satisfy full or dimension-dependent recurrence bounds, including $1 - 1/r!$ in the stated r-channel class.

Scope. Divergent reciprocal-prime mass and the indicated passive conditioning are required.

Qualifications. The vector extension uses the scalar construction and definitions supplied in the same entry.

Vector torus fibers and the dimension bound For any fixed r and arbitrary arrays $f_p : C_{p^{2D_p+1}} \to T^r$, the independent single-prime extension has all-length upper threshold at most $1-1/r!$, again on a prime list of divergent reciprocal mass. If the normal annihilator has full Laurent rank, every positive event returns. The same statement works conditionally on passive variables fixed across all chosen moves, provided they leave an independent sequence of active prime arrays with divergent reciprocal mass.

Use full-time profiles, include all constant even trajectories, and let L be the subgroup generated by those and abundant normal profiles. Its annihilator $R \subset Z[T^{\pm 1}]^r$ is shift invariant: shifting an anchored profile changes its uniform starting point, up to an additive constant. Construct K_f from the real span of the coefficients of the rational Laurent nullspace vectors of R, as in the toral-dimension argument [R253]. For every integral annihilator character ℓ of this connected subtorus, $Q_\ell(T)\ell \in R$ for some nonzero integer Laurent polynomial, by rational linear algebra and clearing denominators.

It remains to show an abundant mark projects into the projection of L on the quotient by K_f. Test a finite list of quotient profile characters annihilating L, representing each as a balanced temporal word of integral row coefficients $\ell_{j,n}$ in the quotient lattice. For each nonzero coefficient include its scalar relation $Q_\ell(T)\ell$ in the finite list of normal tests. Choose the good prime subsequence with uniform conditional errors exactly as before. Apply the passive difference and near-extremiser argument directly to the scalar phase $\ell_{j,n} f_p$ at its slot in the original word. This supplies a uniformly accurate horizontal-polynomial model $F_{j,n}$ for each required slot projection. Pure unit constancy of $Q_\ell(S)F_{j,n}$ gives the global polynomial plus low-only decomposition of every such slot model separately. No divisibility or primitivity of the quotient coefficients is needed.

The combined passive word $\sum_n F_{j,n}(x + npz)$ is a bounded-degree polynomial in high starting coordinate y and speed z, modulo a constant at fixed low residue. A small mean error forces that polynomial constant. The separately selected models need not be exactly balanced, so its constant need not vanish, but in particular its value does not depend on z. Thus for the genuine global polynomial pieces $P_{j,n}$, the scalar polynomial $Q_x(v) = \sum_n P_{j,n}(x + nv)$ satisfies $Q_x(pz) = Q_x(0)$. The identical coefficient-valuation argument gives $Q_x(1 + p^D t) = Q_x(1)$; the removed low-coordinate functions agree at these two speeds. Uniform slot approximation now establishes all the simultaneous quotient tests, and compactness gives the projected abundant mark claimed above.

Choose an odd representative in $K_f^m athbbZ$ by adding a normal profile sum. The full Haar streams given by Laurent nullspace vectors belong to L. The pivot-order compression proof in the dimension bound [R253] gives uniformly positive finite-window return in arbitrary

forcing above relative density $1-1/(\dim K_f)!$. If $K_f = 0$, use instead the flat profile directly. An independent common Haar translation in the whole torus supplies an adequate fraction of quotient cosets with this relative density whenever the full fiber event has a fixed margin above $1 - 1/r!$. As in that proof the lower bound depends only on h, r and the margin, not on the event's finite-cylinder complexity. Approximate the control law by finite supported profile mixtures, then by the tilted index pools; uniform marginal bounds and translation continuity on each finite head section pass the positive bound to arbitrary measurable events.

In particular, if a block-local cocycle can be restricted to independently moving individual primes with a divergent reciprocal sum after freezing the rest, apply the result separately to each passive environment having conditional event mass above the threshold. There are countably many ordinary legal directions, so positivity for some direction in each such environment yields positivity for one fixed direction on a positive-measure set of environments. General groupings into finite blocks can leave only a reciprocal-summable set of representatives, and this corollary does not silently assume otherwise.

Assume $\sum_p 1/p = \infty$ in the deep family [R049]. For each prime choose an **arbitrary** function $f_p : C_{p^{2D_p+1}} \to T$. On the Haar product of these cycles with one independent Haar circle coordinate take the compact extension whose cocycle for a finitely supported step d is

$$\sum_p (f_p(x_p + d_p) - f_p(x_p)).$$

Every positive-measure event in this system has a return of every length on every tail, already using exactly one true high-digit mark, arbitrary passive normal moves, and plain low $+1$ normal moves. This statement covers events depending on both the entire base and the circle. It concerns independent single-prime arrays; arbitrary measurable cocycles coupling many axes are a separate question.

Fix a length $h \geq 2$. A single-axis profile in T^{h-1} records increments from time zero at slots $1, \ldots, h-1$, at a uniform initial site. For each of two normal distributions use the step $+1$ or a stepuniform on $pC_{p^{2D+1}}$. A profile point is abundant for a given distribution if every neighborhood has divergent occurrence sum over primes. Let P be the compact subgroup generated by the abundant normal profiles. Fix a marked speed $1 + p^D t_p$ on each sufficiently large prime with $||t_p^2/p|| > \gamma$ (take t_p near $\sqrt{p/2}$). We prove that at least one abundant marked profile lies in P.

For any finite list of annihilating characters in $P^\perp$ write their temporal words as integer polynomials $A_j(T)$, inserting minus the sum of the positive-slot coefficients at zero so each is balanced. For each word, each fixed tolerance, and each normal distribution, the probabilities that its norm on that profile exceeds the tolerance have finite sum: otherwise a compact set of violating profiles would contain an abundant point, by finite covering. Discard identically zero words. Because the reciprocal primes diverge, for a fixed such list there is a sequence of distinct primes tending to infinity along which the number p times the sum of these failure probabilities tends to zero even at a tolerance tending to zero. For example choose increasingly far primes with the product below $1/n$ at tolerance $1/n$, using the finite sums just noted. This makes the normal errors small **at every low residue** $u = x \bmod p$, after averaging in the high coordinate $y = (x - [u])/p \in C_{p^{2D}}$. No uniformity in the character list is needed.

Automatic horizontal polynomial approximation. For a fixed low residue u, write

$f = f_p([u] + py)$. The passive normal relation

$$\sum_{n<h} a_n f(y + nz) \quad (A_j = \sum a_n T^n)$$

approaches zero in circle distance in mean over independent uniform y, z, uniformly in u on the selected sequence. Fix any nonzero coefficient a_l. Apply the $h-1$ commuting differences on (y, z) in directions $(-nb, b)$, one for each $n \neq l$, with independent uniform b. Every other summand is killed, whereas the lth summand gives the $h-1$st difference of $a_l f$ at uniform independent increments $(l-n)b$; all $l-n$ are units for sufficiently large p. The mean norm of this difference goes to zero by the triangle inequality over its 2^{h-1} translated tests. Hence $e(a_l f)$ has U^{h-1} norm tending to one uniformly in u. By Eisner–Tao, Theorem 1.1, it approaches in L^1 a genuine circle phase polynomial of degree at most $h-2$ on the horizontal cyclic group, uniformly in u.

Let $g > 0$ be the gcd of all the nonzero coefficients of all the words. An integral Bézout combination of the approximants to the coefficient-weighted functions gives a model F_p for gf_p, polynomial of that bounded degree on each horizontal fiber, with mean circle error there tending to zero uniformly in u. The model may have arbitrary circle constants on its separate fibers. On a cyclic group $C_{p^{2D}}$ each nonconstant coefficient of an ordinary polynomial representation can be taken in $p^{-2D}Z/Z$ for $p > h$, by Newton expansion of its periodicity. No attempt to divide an approximate circle phase by g is necessary: each original test on f_p approximates the word $B_j = A_j/g$ on F_p. This comparison is uniform in the step, since the full set of high starts at a fixed low residue translates to the full set of high starts on the new residue at any slot. It also holds for averages over speeds.

Two exact consequences on the selected primes. First, every $B_j(S)F_p$ is constant in the high coordinate on each low fiber, where S denotes unit shift. It is a bounded-degree polynomial in the high coordinate, even across a low-digit wrap since the high input is then translated by one. If it were nonconstant, the mean of its exponential would have modulus $o(1)$ as the large prime increases, uniformly in all the polynomial coefficients and in its constant term. This contradicts the uniform small mean circle distance of that test to zero on each low fiber. For the complete-sum estimate clear the largest nonconstant denominator p^b. If $b = 1$, use the finite-field polynomial exponential sum estimate (linear nonconstant phases have zero mean). If $b \geq 2$, sum on the top digits; only zeros of its nonzero reduced derivative contribute, a proportion at most $(h-1)/p$. The fixed degree is smaller than p. The same argument in two variables gives $o(1)$ for any nonconstant bounded-degree polynomial in the passive starting coordinate and speed: in the finite-field case slice along a variable with genuinely nonconstant reduction, using the one-variable bound off the zero set of its leading coefficient; in the higher-power case use the zero bound for one nonzero partial derivative. Therefore the small passive test $B_j(S^{pz})F_p([u] + py)$ is constant in y, z on each low fiber, and is identically zero by balance on putting $z = 0$.

Take any one nonzero B_j. Its pure-normal constancy gives $B_j(S)\Delta_p F_p = 0$, where $\Delta_p = S^p - I$. The circle array $\Delta_p F_p$ takes values in the finite group of p^{2D}th roots since low-fiber constants cancel. Factor $B_j(T) = (T-1)^a B_0(T)$ with $B_0(1) \neq 0$. On the module of these arrays $B_0(S)$ is invertible for sufficiently large p: modulo p the group-ring augmentation ideal is nilpotent, so a nonzero augmentation is a unit, and the inverse lifts modulo p^{2D}. Consequently $(S-I)^a \Delta_p F_p = 0$ exactly. The integer-index sequence $\Delta_p F_p$ is thus an ordinary circle polynomial of degree below a. Lift its coefficients to rationals and integrate the difference of stride p on ordinary polynomials to obtain P_p of degree at most a

with $\Delta_p P_p = \Delta_p F_p \bmod 1$. The circle function $F_p - P_p$ is p-periodic, hence depends only on the low residue, and P_p modulo one also descends to the full primary cycle by periodicity of F_p. We have obtained a single bounded-degree global polynomial plus an arbitrary low-coordinate function without bounding the prime-power depth.

Agreement at the mark. For each j and each starting site x, put $Q_x(r) = B_j(S^r)P_p(x)$ as a polynomial circle phase in the step r, of degree at most a. Its value is zero for every multiple $r = pz$, by the passive exact identity and cancellation of the low-coordinate function by balance. Its coefficients may be chosen rational with only p-primary denominators as a function on the original cyclic group, by Newton expansion (or discard p-integral parts in the local character). Integrality of $Q_x(pz)$ for all integer z forces each degree-k coefficient indicated locally to have valuation at least $-k$: interpolate in the ordinary monomial basis at $0, \ldots, a$ since $p > a$. Thus at $r = 1 + p^D t_p$ and at $r = 1$ these phases agree for all sufficiently large $D > a$. The low-coordinate part likewise agrees since both moves have the same low residue. All finitely many tested marked model profiles therefore have the same values as at the plain normal, pointwise in the start on these primes. Returning to f_p by the horizontal approximation shows that the probability of simultaneous smallness of the original finite character list at the mark tends to one along the selected infinite sequence.

This proves the desired compact-subgroup assertion: otherwise the compact set of abundant marked points could be separated from P by finitely many of its annihilating characters and fixed positive tolerances, contradicting the divergent occurrence probability of their common approximate zero condition. To justify this last use of abundance directly, a compact closed set having divergent total occurrence probability contains an abundant point; use slightly smaller closed neighborhoods when needed.

Finally approximate the inverse of the resulting abundant marked point by a positive sum of finitely many abundant normals (positive semigroup closure in a compact group is the generated subgroup). Realize each approximating increment profile from its own disjoint sufficiently large pool of primes, using one marked choice total. For a required neighborhood with per-prime probabilities s_i, choose the index with probability s_i/Λ, $\Lambda = \Sigma s_i$, and condition that prime's local start/speed, leaving the others independent Haar. At each time the base density on this pool is $\Lambda^{-1} \sum_i g_{i,t}(x_i)$ with $0 \le g_{i,t} \le 1$ and means s_i; its squared distance from one in L^2 is at most $1/\Lambda$. This remains true with random passive speed, since without conditioning the shifted start is uniform. Choose all pools sufficiently large for their joint marginal L^2 norms to be below two. Independent uniform common circle fiber gives the same marginal estimate on the extension. For an early-base/fiber cylinder approximating a positive event, the pools leave the early coordinates fixed and their near-zero increment tuple gives a return probability arbitrarily close to the cylinder mass, by translation continuity of the finitely many measurable fiber sections. The original-cylinder error costs at most $2h\sqrt{\mu(A \triangle A')}$, so the cylinder tolerance can be set beforehand using only the positive original mass. One actual chosen direction therefore returns, since choices have finite support in directions and absolutely continuous law of starting states. This proves the claim.

R053. Sharp affine-Haar recurrence order for the higher-charge family

Scope and status. Result with proof.

Result. For higher charge r, affine Haar recurrence holds through length $2r+1$ with a sharp obstruction at $2r+2$; at arbitrary horizons dimension at most r suffices and $r+1$ is sharp.

Scope. These sharp statements quantify over affine Haar systems, not all probability-preserving systems.

Fix $r \geq 2$ and the positive-speed directions of deep-order-hierarchy.md [R047], on $G = \bigoplus_p C_{p^{e_p}},\ e_p = rD_p + 1,\ D_p \to \infty$. **Every positive Haar event in every compact metrizable abelian affine G-action, without a dimension bound, returns at every length $h \leq 2r+1$. At length $2r+2$ one such action has a positive avoiding event.** For $r = 2$ this gives positive five-term recurrence also in the two-sign version, by containment of the positive-only steps. The theorem applies on every tail, and to full rational actions factoring through $\mathbb{Q}/\mathbb{Z}$ by restriction. Its upper assertion is for affine Haar systems, and its sharp lower example is from the higher-charge note.

Use the finite list of Fourier tuples, augmentation-local fixed phases, bad-tuple disabling, and compact-torus panel pooling of deep-rank-two-haar.md [R050]. Those pieces are independent of dimension. Fix a Fourier combination of the good tuples together with the finitely many linear Bohr controls, and a nonzero scalar charge frequency. Good tuples have zero sum and all individual passive and active residues are represented at all sufficiently large primes. In the localization at augmentation modulo a fresh p, the sum of their affine linearization words has the Newton form

$$F(c) = \sum_{l=1}^{n} (u^c - 1)^l m_l, n = h - 1 \leq 2r.$$

This follows just by expanding the original time indices as powers of $1 + (u^c - 1)$; there is no constant coefficient. Its value on every allowed local residue c is a fixed p-primary scalar. The only other phase is an ordinary character $c\gamma$, with $p^{e_p}\gamma = 0$.

Put $X = u - 1,\ Y = u^p - 1,\ a = p^{D_p - 1}$. At the passive inputs $c = pz,\ 1 \leq z \leq n$, the coefficients of the unknowns $Y^l m_l$ in $F(pz)$ are the l-th powers of $((u^p)^z - 1)/(u^p - 1)$, of augmentation z^l. For $p > n + 2$ this is an invertible Vandermonde matrix in the local ring. The right sides are fixed, so all $\eta_l = Y^l m_l$ are fixed p-primary scalars, and

$$F(pz) + pz\gamma = \sum_l z^l \eta_l + pz\gamma.$$

Either the exponential mean of this polynomial is $o_h(1)$, uniformly in the finite-order coefficient denominators, or all its coefficients vanish. This is the same prime-power exponential estimate as in the rank-two proof. In the first case the half-passive, half-active local law already contracts the combined charge Fourier coefficient to modulus $\leq 1/2 + o(1)$. In the second case $Y^l m_l = 0$ for $l \geq 2$ and $Y m_1$ is fixed, so $Y^2 m_1 = 0$.

We prove that in the second case $F(1+p^{D_p}t)$, on integer t, is a scalar polynomial of degree strictly less than r. The argument below is an identity in the localized module, so it allows zero divisors. Since Y acts nilpotently on each m_l, the binomial expansions of $(1+Y)^{at}$ on them are finite, give ordinary polynomials in t, and use only denominators prime to p. Fixedness at all these inputs implies $XF(1+p^{D_p}t)=0$ at all integers t, hence coefficientwise by the invertible Vandermonde at small consecutive integers. Its degree is at most $\max(1,n-1)$. The extra $c\gamma$ is scalar and adds only a linear term on the active coset.

For $n-1\geq L\geq r$, set $b_L=a^{n-L-1}$. Descend in L to show, simultaneously for all integers l with $L<l\leq n$, that

$$pb_LX^{l-L}Y^Lm_l=0. \tag{1}$$

If $n-1<r$ this assertion is vacuous and the active degree is already $<r$. In the induction at L, put $k=n-L$ and $w_l=b_LX^{l+1}Y^Lm_l$ for $l=L+1,\dots,n$. For $j=0,\dots,k-1$, multiply the t^j-coefficient of $XF(1+p^{D_p}t)=0$ by

$$a^{k-1-j}X^jY^{L-j}.$$

All these exponents are nonnegative since $n\leq 2r\leq 2L$. To compute the surviving coefficients write

$$u^{1+p^{D_p}t}-1=X+u((1+Y)^{at}-1).$$

The coefficient of t^jY^m in $((1+Y)^{at}-1)^v$ is zero unless $m\geq j$ and $m\geq v$, and is always a rational multiple of a^j with bounded denominator invertible at large p; only the finite degrees acting on m_l matter. After multiplication, $m>j$ produces a layer $K=L-j+m>L$ with common factor b_L and X-exponent at least $l-v+j+1\geq l-K$. When $K\geq l$ the term is zero directly by $Y^lm_l=0$ (the $l=1$ case also vanishes since $K>L\geq 2$). Otherwise induction (1) at K kills it because b_L/b_K is a positive power of a, divisible by p once $D_p\geq 2$.

The layer $m=j$ leaves an equation in the w_l. Its matrix coefficient at (j,l), reduced modulo the augmentation maximal ideal, equals $\binom{l}{j}$. Indeed expand the l-th power of $X+u((1+Y)^{at}-1)$; after factoring X^{l+1}, each $v<j$ has the extra factor X^{j-v}, while $v=j$ contributes precisely $\binom{l}{j}u^j$ (the lowest term inside this j-th power is $(atY)^j$). The k-by-k binomial matrix on the distinct consecutive indices $l=L+1,\dots,n$ is invertible modulo all large p, by Vandermonde in the polynomials $\binom{x}{j}$. Thus every $w_l=0$.

Now take $v_l=b_LX^{l-L-1}Y^Lm_l$. We have $X^{L+2}v_l=w_l=0$, and $Yv_l=0$. For the latter equality, if $l=L+1$ use $Y^lm_l=0$; otherwise use (1) at $L+1$ and $p\mid b_L/b_{L+1}$. Since $p>L+2$, the binomial identity

$$0=Yv_l=((1+X)^p-1)v_l=pXU(X)v_l$$

holds with $U(0) = 1$, a unit in the local ring: the X^p term vanishes, and the coefficients at $X^1, \dots, X^{p-1}$ are divisible by p. Thus $pXv_l = 0$, exactly (1). This completes the induction, with higher layers absent at its initial step.

Finally every t^m-coefficient with $m \geq r$ in the active Newton expansion is a sum of terms having factors $a^m X^{l-v} Y^K m_l$, where $K \geq m,\ K \geq v$, and $K < l$ or the term already vanishes. Since $n \leq 2r$ and $K \geq m \geq r$, one has $m > n - K - 1$. Hence a^m is divisible by pb_K and $l - v \geq l - K$. Every term vanishes by (1). The resulting ordinary polynomial in the localized fixed scalar subgroup can be read as a circle polynomial on the integers: its coefficients are determined by its bounded-degree values at small integers through denominators prime to p, or equivalently by integer finite differences in the binomial basis.

Inserting the charge frequency $e(jt^r/p)$, $j \neq 0$ fixed, into this active test now gives an $o_r(1)$ exponential mean uniformly over all the lower-degree scalar phases, by $r-1$ successive differencings to a nontrivial linear phase as in the higher-charge comparison [R047]. The finite cyclic average over t mod $p^{(r-1)D_p+1}$ is the same as the integer Cesàro average. Thus, in every case, the half-passive/half-active law contracts each Fourier coefficient with nonzero charge frequency to modulus at most $3/4$ on all sufficiently large sampled primes. The panel construction of deep-rank-two-haar.md [R050] then gives an allowed difference with any finite list of surviving tuple phases near zero and no surviving tuple with nonzero sum. The limiting Haar subgroup contains the diagonal, so every positive event returns. The polynomial module construction with $(Y-1)^{r+1}(Y-u)^r$ in the higher-charge note shows that the bound $2r+1$ cannot be increased within this category.

Sharp dimension threshold at unrestricted horizons For **every** horizon h, the same recurrence holds if the compact group's topological dimension is at most r, regardless of the size or primary decomposition of its disconnected part. Conversely the obstruction at $h = 2r+2$ lives on a compact affine torsor of dimension at most $r+1$. The two independent cutoff statements are therefore sharp together. Here dimension equals the rational rank of the discrete dual of the compact abelian group.

In fact the induction only requires

$$\min(\dim X, h - 1 - r) \leq r.$$

For finite dimension k, the action on the rationalized dual is trivial on all sufficiently large primes: a nontrivial finite p-power-order matrix over $\mathbb{Q}$ requires dimension at least $p-1$. For a fixed temporal Fourier combination choose a rational basis as in the rank-two proof [R050], clearing the fixed coefficient denominators and the orders of the finitely many torsion remainders. After invertible rational basis operations in the span of the coefficient polynomials, the fresh-primary local expression is

$$F(c) = \sum_i P_i(u^c) m_i, P_i(T) = (T-1)^{s_i} A_i(T),$$

with distinct positive $s_i \leq n = h - 1$, nonzero rational $A_i(1)$ and a number of terms at most k. The coefficient span can only decrease if some initial polynomials were dependent. For infinite dimension retain the Newton form above. For the passive Vandermonde use any

integer inputs z on which the vectors $(z^{s_i}A_i(1))_i$ span: such a finite list exists by polynomial linear independence, so its nonzero determinant is a unit for all sufficiently large p. It follows that $Y^{s_i}m_i$ are fixed and the passive polynomial-zero case kills them unless $s_i = 1$, for which $Y^2 m_i = 0$ as above.

At layer $L \geq r$ the number $k_L = \#\{i : s_i > L\}$ is bounded by both k and $n - L$. Thus $k_L \leq r$, so $j < k_L$ implies $j \leq L$. When $k_L > 0$ use instead $b_L = p^{n-L}a^{k_L-1}$. For $K > L$ with $k_K > 0$, b_L/b_K is divisible by p even if no order s_i is crossed. Multiply each t^j-coefficient of $XF(1 + p^{D_p}t)$ by $p^{n-L}a^{k_L-1-j}X^jY^{L-j}$, for $0 \leq j < k_L$. To compute the spatial v-th Taylor coefficient at $X = u-1$, use $u^v P_i^{(v)}(u)/v!$, divisible by X^{s_i-v} when $v \leq s_i$. For higher surviving layers $K > L$ with $K < s_i$ one always has $v \leq K - L + j < s_i$; all other higher layers vanish directly as before. Hence the preceding induction still removes every higher layer. On the target layer the matrix on $b_L X^{s_i+1}Y^L m_i$, $s_i > L$, now has augmentation $A_i(1)\binom{s_i}{j}$, again invertible modulo every sufficiently large prime. Its rows are the first k_L binomial polynomials evaluated on the distinct s_i. The binomial $Yv = 0$ step is unchanged and proves

$$pb_L X^{s_i-L}Y^L m_i = 0 (s_i > L).$$

The final active coefficients with time degree $m \geq r$ involve a^m, divisible by pb_K for every $K \geq m$ with $k_K > 0$, since $m > k_K - 1$ and $D_p - 1 > n + 1$ eventually. Thus the active phase again has degree strictly below r; the panel argument finishes the proof.

For the matching construction, the coefficients of $(T-1)^{r+1}(T-u_p)^r$ are integral linear combinations of the $r+1$ tuples $1, u, \ldots, u^r$ in the local modules of the higher-charge note [R047]. Including all the finite-support torsion modules makes the rational rank at most $r + 1$. To extend to the full rational group at the same dimension, induce each prime module separately to its Prüfer group and identify the coset copies of its order-p fixed line, exactly as in the three-dimensional extension [R046]. No additional free generator results. The constructed positive obstruction and the preceding recurrence theorem then force its dimension to equal $r + 1$.

R054. Approximate avoidance in every deep-prime direction

Scope and status. Result with proof.

Result. Deep directions admit near-full soft blockers with an explicit per-direction error controlled by the square-sum of reciprocal primes.

Scope. The estimate rules out a uniform supersaturation approach but is not exact simultaneous avoidance.

For every $\epsilon > 0$ there is a fixed H such that, after discarding finitely many primes, the full deep model [R049] admits stationary event laws of mass at least $1 - \epsilon$ satisfying, simultaneously for every permitted d,

$$\lambda_H(B; d) \le 2\pi(H-1)\Big(\sum_{p : r_p \bmod p \neq 0} p^{-2}\Big)^{1/2}.$$

Here λ_H is the mass of an all-kept H-window at that specific step. Thus no choice of a probability measure on permitted steps in increasingly late tails can give a uniform positive supersaturation bound at this horizon and density. The construction pulls back to the rational group. It does not provide an exact stationary avoider for the full family at this length.

On a finite prime packet, use the random lattice section [R125] only on the low residues $a_p = x_p \bmod p$. Write $\widetilde{a}_p = p\, s(a/p)_p \in \mathbb{Z}$. On a single step, the probability that some low lift fails to increase by its true signed speed $s_p \in \{0, \pm 1\}$ is at most $2\pi(\sum_{s_p \neq 0} p^{-2})^{1/2}$, at every spatial start. There is no lifting restriction on the high coordinates. Choose a short open arc of length $< \epsilon$. As in the degree-three gate proof of the same note, repeated elementary differencing or the torus quantitative Weyl criterion yields M, H with the property that every circle polynomial of degree at most four whose fourth finite difference c satisfies $\|Mc\| \ge \gamma$ hits that arc before H steps, uniformly in its other coefficients and initial value. Remove the primes dividing $24M$.

Choose c_p so $24Mc_p = 1 \bmod p$. Test the scalar

$$F(x) = \sum_p c_p (x_p - \tilde{a}_p)^2 \tilde{a}_p^2 / p^{e_p} \bmod 1,$$

with a uniform independent additive constant; its arc-deletion costs less than ϵ. The high x_p can be read using any integer representatives, since their modulus is p^{e_p}. On a trajectory with correctly increasing low lifts the summands are polynomials in the time parameter. At a marked coordinate the high difference $x_p - \widetilde{a}_p$ moves by $p^{D_p} t_p$, while the low lift moves by $s_p = \pm 1$. Hence its fourth difference is $24 c_p t_p^2 / p$. The other active axes with zero charge obey the same formula, and passive axes have degree at most two. Consequently $M\Delta^4 F = \sum_p t_p^2/p$ on the actual lifted quartic, so an all-kept window requires at least one of the $H - 1$ failed low-lift updates. The union bound proves the estimate. Translate to a stationary finite mask law, extend arbitrarily off the packet and take compactness limits along an exhaustion; each fixed difference eventually lies inside the packet. On late tails the right-hand side tends to zero uniformly in the full speed support and its size.

If a restricted family has uniformly bounded VC complexity in the positive and negative low supports (or any of the simultaneous small-width hypotheses of the lattice-section note), remove every simultaneous-disagreement site in addition to the arc. The resulting exact tail avoiders give endpoint one under those hypotheses. For unrestricted low supports $\sum 1/p^2 < \infty$ alone gives only the displayed per-direction estimate, even though the sums of $1/p$ may diverge. In particular a proof by uniform strict positivity for a single preassigned distribution of directions on all late tails would contradict this calculation; qualitative returns without that uniformity remain possible.

R055. The deep mixed-characteristic model on a reciprocal-summable prime list

Scope and status. Result with proof.

Result. On a reciprocal-summable prime list the full deep quadratic construction has saturated integer critical value one.

Scope. No bound on mark count is needed, but reciprocal summability is essential to this proof.

On any prime list with $\sum_p 1/p < \infty$, the full deep quadratic construction [R049] has tail capacity one, with unrestricted true mark counts and passive speeds. It continues to have the rational ordinary-polynomial recurrence proved in that note. The argument chooses a special extension of the finite-module fixed-line character and analyzes its orbit.

Write $D = D_p, e = 2D + 1, \zeta$ for the local primitive pth root, $\varpi = \zeta - 1$, and $L = 2D(p-1) + 7$. Construct a periodic circle sequence $w_p(v)$ with $v \in C_p$ as follows. Take an integer c with $4c = 1 \bmod p$, and for representatives $0 \le v < p$ put $n(v) = c\binom{v}{6}$. Its sum is divisible by p for $p > 7$ by the telescoping binomial identity. Change $n(0)$ by subtracting this entire sum. Put $w(v) = n(v)/p^{2D+1} \bmod 1$. The word has sum zero, so it defines a character $\ell : Z[\zeta] \to T$ by $\ell(\zeta^v) = w(v)$: the one integral relation among the p consecutive powers is their sum. On multiplying the word by p^{2D} it becomes $c\binom{v}{6}/p \bmod 1$ on all residues, even at the correction site, and this is a genuine polynomial on the prime cycle. Its seventh difference vanishes and its sixth difference equals c/p everywhere. Hence ℓ annihilates $p^{2D}\varpi^7 Z[\zeta] = \varpi^L Z[\zeta]$, since p is a unit times ϖ^{p-1} in this cyclotomic ring. It sends $p^{2D}\varpi^6$ to c/p. On the mixed module $B_p = (Z[\zeta]/\varpi^L)[\epsilon]/\epsilon^3$ take this character on the ϵ^2 coefficient alone. This gives exactly the prescribed value on the fixed line $E_p = p^{2D}\varpi^6\epsilon^2$.

For each of the fixed nine module coefficients m_j, the observable in this finite orbit is $\ell((u^x m_{j,p})_{\epsilon^2})$, where $u = \zeta(1+\epsilon)$. Expanding $u^x = \zeta^x[1 + x\epsilon + \binom{x}{2}\epsilon^2]$, this is a finite sum of circle values $w_p(x+a)$ with coefficients that are rational polynomials in the integer exponent x of degree at most two, integer-valued at integer inputs. The spatial shifts a range over one fixed finite interval independent of p, D, because the original filter coefficients are fixed combinations of u, u^{-1} and constants. Coefficients from expanding $(1+\epsilon)^{-1}$ cause no new spatial shifts. The circle polynomial formulas are valid with any fixed ordinary integer lift of the progression $x + tr$; the module action has order dividing p^{2D+1}.

Delete a band of a fixed number of low residues about zero on each prime, wide enough for this finite shift list and the adjacent low-digit boundaries. At a retained segment in one direction, an active axis advances by ± 1 on its low prime cycle. It cannot wrap on the segment without encountering a deleted site. Every $w_p(x + tr + a)$ along it therefore equals a degree-six ordinary polynomial of the time index divided by p^{2D+1}, using suitable integer lifts. On a passive axis it is constant. All nine summed observations on a wholly retained finite packet trajectory are thus circle polynomials of degree at most eight on that segment, uniformly over all starts, ranks and directions. Across the whole tail the deletion cost is at most a fixed constant times $\sum_{p \in tail} 1/p$ by Haar averaging over starting group elements. Add independent constant translations to the nine torus observations: their stencil sums over time compared to the starting values are unchanged, and no short orbit can be contained in any translate of a fixed small torus ball since the quadratic charge has norm at least γ.

For each desired density choose finitely many roots and bands by the bounded-polynomial

gate argument [R027] with fixed degree eight and nine torus coordinates. On a root row with stride M remove the same low-digit bad bands in the root state. All tail primes can be taken prime to the finitely many roots, so scaling by M^{-1} is an automorphism of each finite prime packet and an M-subsample along d is an unrooted d-trajectory. The union of the seam-band costs over these finitely many rows tends to zero on far tails; the other gate losses are those of the cited polynomial argument. Although polynomiality is asserted only on wholly retained segments, this is enough: an assumed arbitrarily long surviving sequence has polynomial observations on each time residue of every rooted row, so the same compact polynomial-coefficient extraction and periodic binary-label synchronization applies verbatim. Arbitrary spatial ball offsets and independent rooted packet starting points are allowed. This gives a fixed blocking length before exhaustion of the finite packets. Stationary compactness and the prime-power version of the squarefree-tail transfer conclude the proof.

R056. Full spatial rows and roots omitting one prime

Scope and status. Result with proof.

Result. The full spatial deep affine action has a relative bound $1 - 1/\mathrm{ell}$ when all root strides avoid a suitable prime ell, with additional small-window independence results.

Scope. Arbitrary root orders and arbitrary systems are outside this conclusion.

In the deep two-sign quadratic example, one may enlarge the explicit affine model to infinitely many spatial coordinates and take countably many independent roots. If their strides are all odd, every event of measure greater than one half has all lengths. More generally the bound is $1 - 1/ell$ for roots all coprime to a prime ell that admits a nonzero `k/ell` in the interior of the required charge gap. This is an action-relative statement. The denominator restriction and the passage from this action to arbitrary systems are material.

Use the ring B_p, $u_p = \zeta_p(1+\epsilon)$, fixed element $E_p = p^{2D_p}(\zeta_p - 1)^6\epsilon^2$ and the fixed character of the deep example [R049], assigning $[4^{-1}]_p/p$ to E_p. Inside the product of these rings take the countable additive module consisting of all finite local data $\oplus B_p$ and all integral Laurent combinations of the single tuple $U = (u_p)_p$. As an additive group it is the direct sum of these subgroups, with $Z[U, U^{-1}]$ freely generated by the Laurent monomials: a nonzero polynomial evaluated at a primitive pth root is nonzero of bounded $(\zeta_p - 1)$ valuation at every sufficiently large p (expand at one). Take Haar probability on the nonempty character coset agreeing with the prescribed independent E_p data. Its free spatial coordinates indexed by $s \in Z$ are iid Haar circle samples, independent of the local data. Multiplication by $\prod u_p^{d_p}$ means the coordinatewise operation in this product module. It is the identity except at finitely many primes and preserves the module and the coset. A root of stride M uses d_p/M interpreted modulo the group order on primes coprime to M. Motion on the finitely many other primes may be dropped. We may work on arbitrary tails.

Three fresh primes. Use a normal prime with velocity p, a positive active prime with velocity $1 + p^{D_p}t_p$, and a negative one with velocity $-1 + p^{D_p}t_p$. Choose all three fresh and tending to infinity. Temporarily fix finitely many strides M coprime to ell, let P be a common multiple of their fourth powers coprime to ell, and put $\Omega = k/ell$, $||\Omega|| > \gamma$. At both marked primes make $[P^{-1}]_p t_p^2/p$ converge to the same circle value ξ with $2\xi = [P^{-1}]_\ell k/ell$. The incomplete choice of an arbitrary phase can be made to any fixed accuracy at all sufficiently large primes by Weyl's criterion and the quadratic Gauss bound, uniform in the invertible numerator. We can take nonzero $t_p \bmod p$. The total charge therefore tends to Ω and eventually is legal. For countably many strides diagonalize these tests, making each fixed root and sign converge to the desired sum and difference; this only uses a finite common multiple at each stage and increasingly large primes chosen after the stage parameters.

Fourier calculation. Write T for time shift, S for spatial shift, and

$$B_M = (T-1)^3(T^M - S)^2(T^M - S^{-1})^2.$$

For the free row of the stride-M copy a Laurent character $R(T, S)$ of the entire space-time array is tested at selected primes by $T = u_p^{\lfloor} d_p/M]$, $S = u_p$. Unless $R = B_M Q$ with $Q(1, 1) = 0$, the limiting Fourier expectation vanishes, also conditional on any fixed finite head coordinates. Here are the relevant valuations. At the normal prime $u_p^{\lfloor} p/M] = 1 + c\epsilon + \ldots$

with $v_p(c) = 1$. If the power of $T-1$ dividing R is $j \le 2$, the coefficient at ϵ^j is $c^j R_0(1, \zeta_p)$ for a nonzero Laurent specialization. Its valuation is $j(p-1) + O_R(1)$, below the fixed scalar layer for large p. At a sign-s marked prime $s = \pm 1$, the constant-in-ϵ temporal specialization V is a primitive pth root with $V^M = \zeta_p^s$. If $T^M - S^s$ does not divide R, evaluation reduces to the nonzero polynomial $R(V, V^{Ms})$ and has bounded valuation. If exactly one factor divides, the leading ϵ coefficient contains $p^{D_p} t_p$ times such a nonzero specialization and a unit. It too cannot lie in the fixed line. The three factors are pairwise relatively prime in the Laurent polynomial UFD, giving divisibility by B_M. In the remaining case, the second ϵ coefficient at the active prime is

$$p^{2D_p} t_p^2 \zeta_p^{2s} (V-1)^3 (\zeta_p^s - \zeta_p^{-s})^2 Q(V, \zeta_p).$$

A nonzero $Q(1,1)$ leaves valuation one below the E_p line. Otherwise divide by E_p and use $(V-1)/(\zeta_p - 1) \equiv s/M (\mathrm{mod} \zeta_p - 1)$. After the fixed-character factor 4^{-1} this gives the scalar

$$(t_p^2/p) \cdot [M^{-4} Q_T(1,1) + s M^{-3} Q_S(1,1)]_p,$$

where coefficients in brackets are evaluated modulo p. The normal prime gives zero, and so do all inactive primes because of $T-1$. Thus the two signed choices give precisely $Q_T(1,1) \cdot [M^{-4}]_\ell k/ell$ in the limit. These calculations also show eventual zero of the non-retained Fourier modes even after adding a fixed character supported on finitely many old primes. In the infinite limiting joining the full old local data are constant in time and independent of the free profile. The free profile has Haar measure on the affine coset specified by these Fourier coefficients. Equivalently its direction subgroup is the common kernel of the characters $R = B_M Q$, $Q(1,1) = 0$. This follows directly from uniqueness of Haar Fourier coefficients. Multiple independent models have independent profiles for these same deterministic directions.

A periodic orbit inside the profile. In each copy any array constant in time, with arbitrary spatial row, belongs to the direction subgroup because all these characters have a factor $T-1$. A particular point of the affine coset, constant across space, is

$$z_s(t) = ([M^{-8}]_\ell k/ell) \binom{t}{8} (\mathrm{mod} 1).$$

Indeed on spatially constant profiles $B_M = (T-1)^3 (T^M - 1)^4$, and its application to the binomial of degree eight gives a degree-one polynomial with slope M^4. Thus $B_M Q$ evaluates to $Q_T(1,1)[M^{-4}]_\ell k/ell$, as required (write Q in the ideal generated by $T-1, S-1$). As t varies the joint spatial rows of *all* these profiles take at most ell distinct values, parametrized by $\binom{t}{8}$ mod ℓ. For odd roots choose ell=2, $k = 1$.

Now condition on the whole vector of stationary local data. For an event of total measure greater than $1 - 1/ell$, a positive-measure family of these sections B has spatial Haar mass strictly greater than that bound. Translate the particular profile by one arbitrary spatial row on each copy, kept constant in time and Haar distributed. Every finite window then

belongs to B^h with probability at least $1-ell(1-m(B)) > 0$, by the bound on the number of distinct rows. More generally the integral of its B^h indicator over these constant translations is a continuous function of the h rows (translation continuity in $HaarL^1$). It is positive on a neighborhood of the particular profile, hence has positive integral for the limiting Haar coset law. Averaging in the local data proves positive limiting correlation. Weak convergence of joinings with fixed marginals passes this assertion to measurable events by continuous L^1 approximation slot by slot. This argument applies to countably many spatial coordinates and copies by finite character approximation, without assuming a uniform rate of convergence over events.

For strides divisible by different powers of two, the binary periodic profile cannot synchronize in the same way: if an array constant across space solves $(T-1)^4(T^M-1)^4 w = 1/2$ using a single binary switch, then doubling M introduces additional factors of T^M-1 over F_2 and makes the corresponding eighth difference zero. This diagnoses a limitation of the particular profile argument. An unrestricted Haar profile has travelling-wave inputs and may still hit high-density sets with these strides.

Two quantitative consequences. If the 2-adic valuations of all strides are at most v, the finite-state construction still gives the common threshold $1-2^{-1-8v}$ throughout these products. Work in the circle subgroup of order 2^{1+8v}. For a finite set of roots write their lcm as $2^b u$, u odd, $b \le v$. In choosing charges use $P = (2^b u)^4$ and place both signed phase contributions at the same ξ with $2P\xi = 1/2$ and 2ξ of order 2^{1+4b} (the odd part is invertible). The limiting $\Omega_M = (P/M^4)2\xi$ then has order $2^{1+4v_2(M)}$. Solve $M^4\xi_M = \Omega_M$ in the 2-power circle with order at most $2^{1+8v_2(M)}$. The profile $z_s(t) = \xi_M \binom{t}{8}$ has the required Fourier integrals, and all rows together take at most 2^{1+8v} values. For infinitely many roots, at each finite stage any choices satisfying the finite constraints can be used, followed by passage to a subsequence stabilizing the induced phases for every fixed M. This bound is uniform in the number of factors.

Conversely arbitrary spatial dependence cannot give a bounded number of joint states by this method. Every supported profile satisfies $(T-1)B_M z = \Omega_M$. At each spatial index the observation $B_M z$ takes at least $\mathrm{ord}(\Omega_M)$ values, since it advances by that velocity. It is a function of at most twenty full time rows, indexed by $t+j+iM$ for $0 \le j \le 3$, $0 \le i \le 4$. If there are only b distinct full time rows then $\mathrm{ord}(\Omega_M) \le b^2 0$. For a common gap charge tending to 1/2 one has $M^4\Omega_M = 1/2$ and hence $\mathrm{ord}(\Omega_M) \ge 2^{1+4v_2(M)}$; even without fixing the gap value, bounded orders are impossible for a factorial sequence of roots. Thus no uniform finite-state profile can cover all such roots. This does not rule out high-density recurrence by other profiles.

For observables depending on at most four distinct spatial coordinates in each copy there is full recurrence at every positive density, with unrestricted strides. A nonzero Laurent character $R = \sum_{j=1}^{4} f_j(T) S^{a_j}$ (allow fewer terms) cannot be divisible by $(T^M-S)^2(T^M-S^{-1})^2$. For generic positive real $T > 1$ the Laurent polynomial in S would have double zeros at T^M and T^{-M}. A nonzero real combination of k distinct powers on the positive axis has at most $k-1$ zeros with multiplicity, by dividing out the lowest power and applying Rolle inductively. Choose T at which a nonzero coefficient does not vanish. The limiting projection therefore has iid Haar samples across all times and selected spatial coordinates. Static local data do not affect the assertion, by conditioning on a section of positive mass. This statement about independent copies permits simultaneous tests in arbitrarily many such small windows; it does not allow mixing several windows within a single copy.

R057. Full three-slot recurrence in the deep quadratic model

Scope and status. Result with proof.

Result. Three-term positive-density recurrence holds for the deep quadratic rational saturation and its length-preserving integer realization.

Scope. Use the direct characteristic-factor proof for fixed length; the soft all-length suspension identity alone is insufficient.

Qualifications. Reorder the note if needed: it begins with the rational extension before supplying the native setup. R058 provides a coherent full exposition.

The rational saturation and integer realization. Apply the same proof directly to the subgroup $G_Q = \pi^{-1}(G) \subset Q$ with horizontal subgroup $H_Q = \pi^{-1}(H)$. Its finite-index omission is $\pi^{-1}(H_{F^c})$. For the pure vertical generator on axis i take the rational number $1/p_i^{2D_i+1}$; the horizontal perturbation is now $t/p_i^{D_i+1}$ for an arbitrary integer t. It belongs to H_Q. The relative eigencharacters on this element can have arbitrary circle frequencies. Nevertheless the same finite Bohr/quadratic test works uniformly in those frequencies: the integer Cesàro mean of $e(kt^2/p + t\alpha)$ is zero unless $\alpha \in p^{-1}Z \bmod 1$, by averaging each p-residue progression; in the remaining case use the same quadratic Gauss bound. The convolution-square argument is unchanged. Countably many choices of t suffice for the fiber argument.

The Z_1 characteristic estimate remains valid for the rank-one rational subgroup: multiplication by two has index at most two, so the usual linear Cauchy–Schwarz average uses a finite-index difference subgroup with the same U^2 nullspace. All averages may be taken along Følner intervals in an increasing exhaustion by cyclic subgroups, with lengths tending sufficiently fast to ensure the Følner property. The ordinary positive triple-return lower bound is uniform on those intervals by finite Szemerédi–Varnavides applied to the one generator of each subgroup. Arbitrary actions of the ambient rationals restrict to G_Q, proving full rational three-slot recurrence. Finally finite replication of the rational packets on free axes preserves fixed-length capacities by restriction, coinduction and compactness, and each integer obstruction would pull back along the mixed-radix homomorphism to the replicated free group without changing its length. Thus the actual integer realization forces three-term progressions in every positive-upper-density set. This proof does not rely on preserving lengths through the suspension identity for the all-length capacity.

Let $G = \oplus_p C_{p^{2D_p+1}}$, with $D_p \to \infty$, and $H = \oplus_p pC_{p^{2D_p+1}}$. In every measure-preserving G-action, every positive-measure event has a three-term return in every tail of the deep model [R049]. It already suffices to use exactly one true high-digit mark, with arbitrary plain low $+1$ normals and arbitrary passive multiples of p. For every h, the same positive-mass assertion holds for any event in the relative Kronecker factor of the H-action, without requiring ergodicity under H or constant spectra of its ergodic components. This is distinct from recurrence only for Fourier–Stieltjes correlation functions.

Finite-field ingredient. Given any fixed finite number of characters of C_{p^m}, $m \geq 1$, a fixed small Bohr radius, and $0 < \gamma < 1/2$, for every sufficiently large prime p there is a $t \in C_{p^m}$ simultaneously in those character zero neighborhoods and with $||t^2/p|| \geq \gamma$. All bounds are uniform in the characters and in m. For every fixed nonzero quadratic frequency k and every linear character of frequency b, the absolute normalized exponential mean

$$\left|\mathbb{E}_t e(kt^2/p + bt/p^m)\right| \leq p^{-1/2}$$

when $p \nmid k$; first sum on the lifts of t mod p and then apply the elementary quadratic Gauss identity if that sum is nonzero. On the finite-dimensional test torus take a nonnegative convolution square `f*tilde f` of a nonnegative bump of small support near zero. Its support fits the Bohr window, its constant coefficient is positive, and its Fourier coefficients are nonnegative and absolutely summable. Hence its unrestricted mean along the character tuple is bounded below uniformly by its constant coefficient. Multiply by a nonnegative continuous bump supported in the open region $||\cdot|| > \gamma$. Finite Fourier approximation, the Gauss estimate, and positivity of the constant coefficient give a uniformly positive mean for the product.

Perturbing an actual vertical return. Write Y for the H-invariant factor and μ_y for the ergodic conditional measures. The ambient group preserves this factor and transports its conditional measures. Write B for a fixed relative-Kronecker event of mass $a > 0$. For any chosen h the finite density Hales–Jewett averaging lemma gives $m = m(h, a)$ and $\kappa > 0$ such that among nonempty sums of any m fresh coordinate units there exists $v = \sum_{i \in F} e_i$ with ordinary unmarked h-return measure at least κ. Indeed the expected density of B on translates of an h-letter cube is a, and a positive fraction of its translations have density at least $a/2$; sum over its finitely many combinatorial lines.

Approximate 1_B in L^2 by a bounded finite sum f of relative H-eigenfunctions with bounded coefficients. Eigenlabels are measurable functions of the base y and may vary arbitrarily from component to component. Such finite sums are dense in the relative Kronecker factor: on each ergodic component this is precisely the pure-point subspace, and its eigenspaces are one-dimensional; measurable enumeration of the eigencharacters can be obtained from the atoms of a measurable maximal spectral measure, then the orthogonal projections of a countable dense family select measurable normalized eigenfunctions. Truncate the invariant coefficients, or compute on a positive set where they are bounded, to obtain the stated bounded approximation.

Choose the approximation error sufficiently small in terms of κ, h, and then choose the fresh primes sufficiently large for the finite-field ingredient for the resulting finite number of labels and a small radius fixed by the coefficient bounds. Apply Hales–Jewett as above. On a positive set of y, the conditional v-return probability is at least $\kappa/2$ and all the approximation errors in $L^2(\mu_{T_{jv}y})$, $0 \le j < h$, are smaller than $\kappa/(16h)$. Indeed the conditional correlations are bounded by one and have mean at least κ, while a union bound and Markov's inequality control the errors at these translates uniformly in v.

Fix any $i \in F$. For each such y use the finite-field ingredient with $m = D_i + 1$ for the restrictions of the finitely many eigenlabels at **all** the base states $T_{jv}y$ to the cyclic subgroup generated by $p_i^{D_i} e_i$. Choose the radius sufficiently small to cover the factors $j < h$. The selected t makes translation by $jtp_i^{D_i} e_i$ nearly fix f in every corresponding fiber. Telescoping the product of the translated indicator functions, using Cauchy–Schwarz and invariance under H within each fiber, gives a positive conditional return at

$$d = v + tp_i^{D_i} e_i.$$

For detail, each slot changes by at most twice the conditional approximation error plus the norm change of the finite eigenfunction sum, which can be kept below $\kappa/(4h)$ by the radius choice. The choices of t may depend on y. This causes no interchange of ultralimits or uniform approximation across spectra: the finitely many candidate values of t cover the indicated positive-measure set by measurable positive-correlation subsets, so at least one

choice yields a genuine positive integral. Its charge is t^2/p_i. This proves all lengths on the stated factor.

Transfer to arbitrary events at three slots. Suppose an event A avoids all allowed three-term progressions and let $f = E(1_A|Z_1(H))$, $B = \{f > 0\}$. The latter has at least the mass of A. For a candidate d just constructed and its finite active index set F, every $d+u$ with $u \in H_{F^c}$ is still allowed. Average the zero correlations of A in u over this subgroup. Its index in H is finite. Its U^2 nullspace equals that of H, as in the common horizontal characteristic-factor proof [R014] (equivalently use the finitely many ergodic components under restriction and the equivalence of pure-point subspaces across a finite-index restriction). The ordinary three-slot progression average has Z_1 characteristic, by the linear Gowers–Cauchy–Schwarz estimate on the locally finite odd-prime tails. Thus the same averaged correlation with f in the three translated slots vanishes.

If B returned at d, some $C = \bigcap_{j=0}^{2} T_{jd}^{-1} f > \delta$ has positive mass for $\delta > 0$. The averaged f-correlation is bounded below by δ^3 times the average ordinary triple-return correlation of C under H_{F^c}, which is strictly positive by finite Szemerédi averaging over the finite subgroups (or by the finite density Hales–Jewett argument). This contradiction proves B would also avoid the constructed d, contrary to the preceding relative-Kronecker result. Finite early primes can simply be ignored by working in a tail subgroup. The same argument gives pair recurrence, alternatively obtainable by the positive spectral multiplier method.

For the saturated rational model, the same fixed-order recurrence holds in every quotient with a finite suspension period: restrict to fresh prime components, using the automorphic multiplication-by-period copy of the tail. The soft suspension identity for b itself increases the avoided length, so it does **not** alone transfer this fixed three-slot statement to every Q-action. The native/full relative-Kronecker assertion above is on the displayed torsion group.

R058. Every positive-density set has a triple along the deep quadratic directions

Scope and status. Result with proof.

Result. The deep quadratic family has unrestricted three-term recurrence, with all-length recurrence on its relative compact factor.

Scope. This is a fuller exposition of the triple argument, overlapping R057.

Use the directions of deep-quadratic-primes.md [R049]. The statement is valid both on the primary group $G = \bigoplus_p C_{p^{2D_p+1}}$ and on the torsion-free rational cover $\widetilde{G} = \pi^{-1}(G)$, with $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$. On the latter the step set is the full inverse image of the primary directions. Every positive-measure event in every probability-preserving action of either group has a positive three-term return at some legal difference. This still holds after discarding all coordinates in any prescribed finite set of primes (on the rational cover all integer lifts remain allowed).

For the rational-to-integer realization, the same three-term assertion holds on the integers. The separate finite-color obstruction at nine terms on the group shows that the all-length threshold is positive. The following result gives no strict upper bound for that threshold.

All lengths on a fiberwise compact system Let K denote the group in either case. Define H by requiring every coordinate to be divisible by its primary prime, taking the full inverse image on the rational cover. For finite S let H_S impose the additional condition that every coordinate in S vanish. Then H_S has finite index in H. Disintegrate an arbitrary standard probability-preserving K-action over the ergodic components Y of its H-action, with conditional measures μ_y. The K-action permutes these components. The relative Kronecker factor used here consists of the ordinary Kronecker factors within those ergodic components. Equivalently an event on this factor has pure-point spectral measure for its indicator on almost every component. All arguments can first be carried out in the separable factor generated by the countably many translates of the individual event under test.

Claim. Every event B of mass $a > 0$ in this factor has a positive h-term return at a legal difference, for each finite h. In fact only one active site needs a nonzero high-digit mark. The number of active sites is bounded in terms of a, h, and their primes can be taken arbitrarily large.

Choose $m = m(a, h)$ from density Hales–Jewett over the alphabet $\{0, \dots, h-1\}$ at density $a/2$. For any choice of m steps w_i, some nonempty subsum $s = \sum_{i \in F} w_i$ has return of mass at least $\kappa(a, h) > 0$. Indeed at a random point label a word v by the indicator of B at its translate by $\sum_i v_i w_i$. With probability at least $a/2$ the word set has density at least $a/2$. Each combinatorial line contributes the return along a subsum, after integrating in the point and using invariance. Divide $a/2$ by the finite number of lines.

On each fiber the spectral measure on $\widehat{H}$ of 1_B is atomic, of total mass at most one. Given small $\rho, \eta > 0$, there is L such that, outside a base set of measure $< \eta$, at most L atoms account for all but ρ of that measure. In particular for EVERY fixed $v \in H$ all but ρ of the spectral measure of T^v on that fiber is supported on at most L circle points (push the same atoms forward). This selection of a good base set is measurable: conditional spectral measures form a measurable kernel by their countably many Fourier moments, and the maximum mass on L atoms is a Borel function of the measure, e.g. by shrinking unions

of L closed balls in a compact metrizable group. No common eigenvalues across fibers are required.

Here is the numerical approximation used below. For any $l, \tau > 0$, all sufficiently large primes p have the following property, uniformly over every tuple $\alpha_1, \dots, \alpha_l \in \mathbb{T}$: there is $t \in \mathbb{Z}$ such that

$$\max_i ||t\alpha_i|| < \tau, ||t^2/p|| > \gamma.$$

To see this take a nonnegative smooth autocorrelation bump f supported in the τ-box in $\mathbb{T}^l$, with absolutely summable nonnegative Fourier coefficients and positive constant term. Multiply by a nonnegative smooth circle bump g supported where $\|z\| > \gamma$, of positive mean. Average $f(t\alpha)g(t^2/p)$ over $1 \le t \le N$, then let $N \to \infty$. The constant Fourier mode of g contributes at least $\widehat{f}(0)\widehat{g}(0)$. Each mode $k \neq 0 \bmod p$, multiplied by an arbitrary linear phase, has limiting mean bounded by $p^{-1/2}$, by splitting $t \bmod p$: the limit vanishes unless the linear frequency is in $p^{-1}\mathbb{Z}/\mathbb{Z}$, and otherwise is a complete quadratic Gauss sum. The total absolute contribution of the nonzero modes divisible by p is at most $\|\widehat{f}\|_1 \sum_{|k|\ge p} |\widehat{g}(k)| = o(1)$. This proves the stated uniformity.

Take $\eta \ll \kappa/h$, and then ρ, τ small enough that the spectral estimate below is $< \kappa/(4h)$. Choose m distinct tail primes exceeding the numerical bound for $l = hL$. Use their unit generators w_i (the rationals $1/p_i^{2D_{p_i}+1}$ on the cover). Apply the word argument and fix its s. The base set on which $\mu_y(\bigcap_{j<h} T^{-js}B) > \kappa/2$ has measure at least $\kappa/2$. Discard the y's for which any of the h components at $T^{js}y$ is bad; positive measure remains.

Choose any one active prime p in s and put $v = p^{D_p} w_p \in H$. For each remaining y, use the numerical approximation for the union of the h fiberwise lists of heavy eigenvalues of T^v, to choose t with $\|t^2/p\| > \gamma$. On each relevant fiber the L^2-distance of 1_B from $T^{jtv}1_B$ is at most $((2\pi h\tau)^2 + 4\rho)^{1/2}$. This bounds its L^1-distance as well. Hence $\mu_y(\bigcap_{j<h} T^{-j(s+tv)}B) > \kappa/4$. The permitted mark is at the chosen p, and the other active primes have zero mark. Since t runs through a countable set of integers, some one fixed t gives positive global return. No measurable eigenvalue or perturbation selection is needed.

Passing to arbitrary actions at length three Put $g = \mathbb{E}(1_A \mid Z_1(H))$. For a permitted d, write S for its finitely many active primes. All steps $d + u,\ u \in H_S$, are permitted; with a prescribed finite-prime tail include the excluded primes as additional zero conditions in H_S. The relative Kronecker factors of H_S and H coincide modulo null sets. On each ergodic H-component the finite-index subgroup H_S has only finitely many ergodic components; they are permuted transitively. Compactness of a vector's translation orbit in L^2 is equivalent for a group and a finite-index subgroup, by a finite coset decomposition. Applying this on the finitely many component summands gives the assertion. It also shows the factor is invariant under every translation of K.

On any Følner sequence of H_S, the three-correlation average along $d + u$ is unchanged in the limit if each indicator is replaced by g. For clarity, on an ergodic H_S-component this is the elementary three-term characteristic-factor argument. Reanchor the three factors if needed so that an orthogonal factor occurs at some nonzero slope l, with the other nonzero slope m; the three slopes have absolute pairwise differences at most two. The Hilbert-space van der Corput bound for $\mathbb{E}_u U_{lu} f\, U_\mu b$ reduces the fixed-lag inner average to

$$\int (overline f \cdot U_{lk} f) E(overline b \cdot U_{mk} b | I((m-l)H_S)).$$

This follows by shifting the integral by $-lu$ and using the mean ergodic theorem. The displayed conditional expectation is constant on each member of a finite partition: $(m-l)H_S$ is finite index. Indeed $[H_S : 2H_S] \le 2$ for a subgroup of $\mathbb{Q}$, and the odd-order primary group has index one. If f is orthogonal to the Kronecker factor, then for each atom C of the partition, the Følner average over k of $|\langle U_{lk} f, 1_C f\rangle|^2$ tends to zero by the spectral theorem. Here restricting to the finite-index subgroup lH_S preserves continuous spectrum, since the corresponding dual restriction map has finite fibers. Cauchy–Schwarz and van der Corput give zero for the term containing f. Integrate over the H_S-components and telescope the replacements. A standard van der Corput proof averages lags $k - k'$ over two Følner copies; these averages also tend to zero by the same spectral theorem. The fixed translations jd can be absorbed into each factor since the factor is K-invariant.

Suppose now that every permitted d has zero triple return for A. The event $B = \{g > 0\}$ has positive measure and is on $Z_1(H)$, so the preceding claim gives a permitted d and $c > 0$ such that $C = \bigcap_{j=0}^{2} T^{-jd}\{g > c\}$ has positive measure. For every $u \in H_S$ the g-triple correlation at $d+u$ is at least $c^3\mu(\bigcap_{j=0}^{2} T^{-ju} C)$. The latter has positive average on a suitable Følner sequence, as follows.

Choose R so that every subset of $[1, R]$ of density $\ge \mu(C)/2$ contains a three-term progression. For every u, averaging in x on $\{k \le R : T^{ku}x \in C\}$ shows that the sum over $1 \le j \le R$ of the triple return of C at ju is at least a fixed positive number (divide $\mu(C)/2$ by R, an upper bound for each number of starts). On the primary group average over an exhaustion by finite subgroups generated by primes $> R$, which may include a fixed finite number of smaller prime components as well. The multiplication maps $u \mapsto ju$ there have uniformly bounded kernels, since the orders at each of these finitely many small primes are fixed. On the rational group exhaust H_S by cyclic groups $a_i\mathbb{Z}$, and average u over the intervals $[-L_i, L_i]a_i$ with L_i increasing sufficiently rapidly for these to be Følner. All ju lie in $[-RL_i, RL_i]a_i$, another Følner sequence, and each fixed multiplication is injective. In both cases the positive lower average follows. This contradicts the zero average for 1_A and the characteristic-factor replacement.

Integer version Apply the rational-cover-to-integer realization in this workspace, keeping the horizon fixed for its upper direction. More explicitly exhaust the permitted differences by finite packets F_i, take the free direct sum of the cyclic groups $\langle F_i\rangle$ on separate axes, and put a copy of F_i on axis i. Its h-capacity is the original h-capacity: restrictions to the individual axes have capacity at most the infimum of the finite-packet capacities, which by compactness is the full capacity; conversely pull back the group action by the coordinate-summation map. Finite packet capacities are the same in their original subgroup and the ambient group by coinduction/restriction. Choose the mixed-radix integer homomorphism of the realization and take the positive absolute values of its packet steps, using the symmetry of progressions under reversal. Every integer avoider pulls back at the identical h; thus its three-term capacity is zero. The reverse direction needed for the positive *all-length* lower bound is the separate mixed-radix realization theorem and can change the avoidance horizon.

R108. One independently adjustable slope per endpoint

Scope and status. Result with proof.

Result. Independent endpoint slopes permit specified finite-window control, including square-root discrepancy rounding for finitely many statistics.

Scope. Bounds depend on the number of statistics and spatial range; the forcing model is a surrogate, not every arithmetic model.

On a finite spatial set of distinct integers J consider the surrogate

$$F_j(t) = X_{t+j} + jY_{t+j} + w_j(t) \bmod 1,$$

with two free Haar circle streams. Deterministic forcing $w_j(t) = j^2P(t + j) + C_j(t)$, including arbitrary quadratic C_j, is one subclass. These positivity results do not assume any arithmetic realization of the full stream law.

A central cut and concentrated row tests. Take $0 \in J$. Condition each central value $F_0(s)$ independently to lie in a positive-measure set A_s for a finite horizon. The $Y's$ remain independent Haar. Conditional on the central states, *all nonzero arms of any fixed row are independent Haar*, even for a completely arbitrary forcing array. This holds since the endpoints are distinct and multiplication by a nonzero integer is onto the circle. If further row constraints have conditional failure probability $\le u$ under these laws, the bad events have dependency degree at most $|J - J| - 1 \le |J|^2 - 1$. Thus $e|J|^2u < 1$ gives a positive probability of every required row being acceptable on any finite horizon by the local lemma. For this step it suffices that failures have probability $\le u$ under the conditioned product law on endpoint pairs; conditioning on fixed central values is even stronger.

In particular statistics with exponentially small tails for samples of exponentially large independent arm sets, and polynomially many such tests, do not give the signed statistical trap of the expander construction [R232] on this new model. Each forbidden event involves only the finite set of endpoints in its own row, independently of the size of the numeric offsets.

Two fresh projections. Let b,c be the largest and second-largest offsets. When earlier time rows are revealed, the endpoint $t + b$ has never been queried, and $t + c$ has had at most one query, at its offset b. Distinct slopes $X + jY$ of one Haar endpoint pair are independent Haar in pairs, since their integer determinant is nonzero and the torus map is onto. The two current arms at b,c are therefore jointly fresh Haar conditional on the previously observed labels and the other current arms. In particular an arbitrary event whose (b, c) sections have positive measure at almost all interior values admits every finite forced run: reveal current interior values, then the fresh pair. The interior prefix's unconditional law is Haar, so its null exceptional set still has probability zero under any earlier positive-probability restrictions.

Two compulsory cuts with Haar remainder. Even two offset conditions, at a,b with $b - a$ odd, can be imposed simultaneously while leaving every other arm of each row exactly Haar. At each of the two selected offsets and every time exclude an arbitrary translated interval of length at most 1/2. For each cut choose a circle probability density w supported in its complement with $w(z) + w(z + 1/2) = 2$ a.e.; allocate mass 2 to the admissible point of a pair if the other lies in the interval and use 1 on both points if both are admissible. Apart from the constant, all Fourier coefficients of w have odd index. At each endpoint use the

product density $w_a(X + aY)w_b(X + bY)$ on the original Haar pair, choosing the two phases to absorb the deterministic forcings at their respective rows. The factors are independent under the original Haar pair, so the density integrates to one. Endpoints are still sampled independently.

For some other offset j, a nonzero Fourier mode k of its marginal could survive only if integers $n_a + n_b = k$, $an_a + bn_b = jk$ both have nonzero w coefficients. Neither can be zero unless j equals a or b. Thus both must be odd, making k even, `jk` even and $an_a + bn_b$ odd, impossible. This Fourier test follows directly by integrating against Haar in X,Y; positive densities can alternatively be approximated in L^2 by Fejér means of the same parity support. Thus the remaining arms are Haar and mutually independent within the row, and statistically rare further tests on these arms can again be imposed over every finite horizon with the local lemma. Their values are also independent of the arms at a,b within that row as the endpoints differ. This extension uses parity rather than the numeric width of the offset set; it does not imply that every high-mass arbitrary row event is feasible.

Any fixed collection of compulsory individual cuts. Let a finite set I of offsets have events $E_a \subset T$ with $\sum_{a\in I} m(E_a^c) < 1$. At each endpoint s independently sample (X, Y) with density proportional to

$$\prod_{a\in I} 1_{E_a}(X + aY + w_a(s - a)).$$

The normalizing integral is bounded below by $1 - \Sigma m(E_a^c) > 0$, even under arbitrary deterministic forcing. As the shifts vary in a compact finite torus, these normalized densities form a compact subset of $L^1(T^2)$: each translated indicator is L^1-continuous in its translation, including after pullback by a Haar-preserving torus homomorphism, and finite products of $[0, 1]$-valued factors satisfy the sum-of-errors bound. On a compact L^1 family, Fourier coefficients tend uniformly to zero as their frequency leaves any fixed finite subset (use finite nets and the ordinary Riemann–Lebesgue lemma). Hence at any fixed nonzero harmonic k, the bias of $e(kF_j(t))$ tends to zero as $|j| \to \infty$, uniformly in t and all forcing. The arms from distinct endpoints are independent in each row. Their independent empirical averages on any sufficiently large set of distinct offsets concentrate exponentially near zero, and the same holds for products over disjoint pairs, e.g. symmetric-mirror averages. Consequently the same rare-failure local-lemma bound as above applies to such tests at any prescribed positive tolerance once their offset sets are sufficiently large compared to the fixed cuts. This does not give uniformity over growing compulsory collections or cover arbitrary row constraints.

Three offsets. For any three distinct offsets every measurable row event B of Haar mass strictly greater than 1/6 has a positive finite simultaneous-run probability, for every deterministic forcing. To prove it use the usual prefix rearrangement: for a tolerance tuple (p_1, p_2, p_3), work backwards through the row coordinates, retaining a prefix if its next-coordinate section of previously retained prefixes has mass greater than its tolerance. The three-dimensional volume of tolerance tuples for which the empty prefix survives is exactly $m(B)$, by layer cake in succession. Therefore some successful tuple has positive entries and $\Sigma p_i > 1$, since the lower simplex $\Sigma p_i \leq 1$ has volume 1/6.

Reveal the independent endpoint pairs in increasing order of their index. At any endpoint s, at most one row requests the next coordinate at each of the three offsets, in its own

prefix order. Their required subsets of the circle have masses greater than the selected p_i; fill unused requests by the full circle. The fresh three projections are Haar on a coset of the two-dimensional subtorus given by one primitive integer relation $\Sigma c_i z_i = u$ with all c_i nonzero. This follows directly from the rank-two matrix with columns 1 and j. Three measurable positive-mass subsets of masses totaling more than one have positive common probability on every such coset. Indeed push their restricted Haar densities forward under multiplication by the c_i. The positive supports have measures at least the respective masses. By circle Kneser, the positive support of the convolution of the first two has measure at least the lesser of one and the sum of their support measures: one can take inner compact subsets of their density points, whose sum lies in the positive convolution support. It therefore overlaps every translate of the reflected third support in positive measure. Disintegration under the primitive relation, or Haar integration over its kernel, gives the stated positivity. Iterating the positive conditional success probabilities over finitely many reveals proves the claim.

At any number r of offsets one sufficient condition for the success of a request tuple is that for some i of positive mass, $\sum_{j \neq i}(1 - p_j) < 1$: condition on request i, use independence in pairs, and take the union bound. The volume of this safe tolerance region is $(r - H_{r-1})/(r - 1)!$, where H is the harmonic number. To see this write deficits $u_j = 1 - p_j$ and select their largest. The volume is r times the integral of one minus the maximum over the standard $(r-1)$-simplex. On that simplex the mean maximum of its $r-1$ coordinates equals H_{r-1}/r, by inclusion-exclusion on the events that given coordinates exceed x and integrating in x. Thus any event of mass greater than one minus this volume supports all finite trajectories. The bound deteriorates with r.

The threshold 1/6 for three distinct offsets is sharp, even with zero forcing. For the primitive relation vector c above choose $\eta > 0$ and keep exactly the spatial vectors z with $\{c_i z_i\} \geq \eta$ for all i and $\sum_i \{c_i z_i\} \leq 1 - \eta$, using fractional parts in $[0, 1)$. Their Haar measure tends to the simplex volume 1/6 as η tends to zero. At each common endpoint s of a legal unforced run the sum $\sum_i \{c_i F_{j_i}(s - j_i)\}$ is an integer by the exact relation; positivity of every summand makes this at least one. In a putative H-slot run, the interior endpoints (all three indicated centers in the run) give total at least $H - O(\mathrm{span}(J))$. Summing instead by the time rows gives at most $(1 - \eta)H$, since unused boundary summands are nonnegative. A sufficiently long run is impossible. This gives extremal masses arbitrarily close to 1/6 from below.

Finite exponent and circular slopes. Over a bounded-exponent compact alphabet of exponent q the unforced problem admits near-full traps: at times t in one residue mod$q$ the row is a single spatially shifting stream, since the coefficient of Y_s is $s - t \bmod q$. Use the ordinary one-stream marker of the de Bruijn argument [R127] on the stride-q spatial coordinates. The nontrivial finite-alphabet case can use two cells of positive probability with arbitrarily rare marker words. This is not a circle result.

For any shift-invariant law of circle slopes (Y_s), any fixed m and $\eta > 0$, there exists a bounded positive q such that $\max_{|j| \leq m} ||qY_{jq}|| < \eta$ with positive probability. Adjoin independent Haar circle registers Z_s. The spatial shift and the transformation adding Y_s to Z_s commute. Multidimensional Szemerédi for the single short-arc event on Z_0 and points $(j, 0)$,$(j, 1)$, over all $|j| \leq m$, gives such q; finite-density averaging gives a bound depending only on m,η. Thus there are approximately shift-only subpatterns. A near-full mask checking the resulting, unknown q at the needed pattern length has not been obtained from this observation.

Bounded frequency complexity of the slope. There is a pointwise near-full obstruction if all slopes on the run belong to combinations of r fixed circle values with integer coefficients bounded by m^c, for fixed r,c and the marker parameter m below. The values themselves may depend on the trajectory. The case of at most r distinct slopes follows. Take a central arc I of small length s, and a concentric subarc I' of half the length. By simultaneous Dirichlet approximation there is $1 \le q \le Q = \lceil Crm^{c+1}/s \rceil^r$ such that $||qY_u|| < s/(16m)$ at every tested endpoint (increase the absolute constant C as needed). In the spatial row keep just the sites with $F_0 \notin I$ and, for *every* $1 \le q \le Q$, some $F_{jq} \in I', 1 \le j \le m$. Its Haar loss is at most $s + Q(1 - s/2)^m$, which can be made arbitrarily small because Q grows polynomially with m at fixed r,c,s.

For the unforced model, the arms at the good q agree to within $s/16$ with the corresponding later central values, contradicting this rule on a sufficiently long run. The same result holds with completely unknown quadratic nuisance polynomials C_j and one specified forcing $P(u) = \alpha\binom{u}{D}$ for irrational α and $D \ge m + 2$. For any of the finitely many $q's$, a common bounded translation t exists with

$$||(jq)^2 P(t + jq) + C_{jq}(t) - C_0(t + jq)|| < s/16 (1 \le j \le m).$$

Indeed for any nonzero bounded integer character of this vector, Vandermonde on the m distinct offsets jq makes some moment through $m - 1$ of the weighted character coefficients nonzero. The first such moment leaves an irrational leading coefficient of degree at least $D - m + 1 \ge 3$, above the nuisance. The quantitative polynomial Weyl criterion gives a common finite hitting time uniformly over the quadratic coefficients and lower polynomial terms; take the maximum over the finite q list. The arm values then agree within the central margin with $F_0(t + jq)$, again a contradiction. Include all tested temporal offsets in the run length. This excludes low-complexity slope witnesses, not the arbitrary independent circle streams.

See also bounded-coordinate forbidden predicates [R024], where the positivity threshold depends on the smallest forbidden certificate width rather than on the size of the spatial window.

Rounding finitely many row means at square-root scale. Suppose the spatial offsets lie in an interval of length r. At each row demand bounds on k statistics of the form $\sum_{j \in J} f_{i,t,j}(F_j(t))$, with arbitrary deterministic measurable real functions bounded by one. For every finite horizon there are controls achieving absolute discrepancy $\le C\sqrt{kr}$ in each statistic from its expected value under independent endpoint sampling, even with arbitrary coordinatewise restrictions already imposed on the independent endpoint pairs. For instance force all central $X's$ into large sets; all other arms then retain their uniform-circle expectations. The absolute constant C does not depend on the horizon or forcing. With strict slack in the inequalities the resulting finite run has positive probability under the original Haar controls, provided the sampling restrictions have boundedly absolutely continuous laws (or merely absolutely continuous marginals). There is no claim at arbitrary smaller multiples of the standard deviation.

Here are details. Split the endpoints into consecutive blocks of length r. Each block affects at most 2r time rows, or $m \le 2kr$ real tests. Its possible values at each single endpoint are vectors in $[-1, 1]^m$. Approximate the independent expectation at that endpoint by a uniform multiset of 2^a possible choices; the approximation may be as accurate as desired since there

are finitely many tests. To halve all multisets, pair their entries, keep one from every pair, and use a global signing of the vectors of pair differences. The discrepancy of a matrix with m rows and entries bounded by two is at most $C_0\sqrt{m}$ regardless of its number of columns. For at most m columns this is the matrix form of Spencer's theorem. For more columns, start at fractional signs zero and move in the kernel of the row map until at most m entries are nonintegral. Round these with the same bound up to a universal factor: ordinary successive binary-digit rounding of a fractional vector costs at most a geometrically weighted sum of the hereditary discrepancy bound on its column subsets. Halving from b choices down to $b/2$ changes the sum of the per-endpoint averages by at most $C_0\sqrt{m}/b$ up to the harmless choice of normalization. Summing over dyadic b leaves $O(\sqrt{m})$. Each time row meets at most two blocks, yielding the assertion. For positive measure, choose the preliminary sample points as support points of the finite-dimensional contribution-vector laws on each endpoint. Neighborhoods of their selected contributions then have positive probabilities; the strict simultaneous margins permit perturbing within their finite product. This establishes robustness without continuity assumptions on the $f's$.

R126. Four-point progression averages on ergodic finite-index subgroups

Scope and status. Result with proof.

Result. Four-term progression averages agree across ergodic finite-index subgroups under the stated finite-index doubling and tripling conditions.

Scope. All-length equality is asserted on the second Host-Kra factor; the deep-model application requires horizontal ergodicity.

Let Γ be a countable abelian group, $H \leq \Gamma$ a subgroup of finite index m, and let both act ergodically on the same standard probability space. Suppose $2\Gamma, 3\Gamma$ have finite index. For any four bounded functions the scalar progression average on each coset $w + H$ agrees with that on Γ. For functions on the second Host–Kra factor $Z_2(\Gamma)$ there is equality at every fixed length. All averages may be taken along arbitrary Følner sequences in the corresponding group. This extends the finite-index comparison at four points to this abelian setting. In particular it gives a new restricted four-term assertion for the deep quadratic example, including non-phase-polynomial systems.

Reduction to $Z_2(\Gamma)$. The U^1 seminorms for Γ and H agree by ergodicity. Inductively the 2^s-th powers of the $U^s(H)$ seminorms are at most m^{s-1} times the 2^s-th powers of the $U^s(\Gamma)$ seminorms. Apply the previous inequality to each multiplicative derivative, use its nonnegative seminorm power, and bound its mean over H by m times its mean over Γ by splitting the latter into the finitely many cosets. The recursive seminorm averages exist along any Følner sequence; the unions of the finitely many translates of an H-Følner sequence are a Γ-Følner sequence. Thus orthogonality to $Z_2(\Gamma)$ implies orthogonality to $Z_2(H)$. Since $2H, 3H$ also have finite index, the characteristic-factor estimate for the slopes $1, 2, 3$ applies to both groups. Each of the three shifted slots can be projected to $Z_2(\Gamma)$ in both averages. For the unshifted slot one can use invariance, recenter the progression at the opposite endpoint and reverse its direction, so that it too becomes a shifted slot. See Shalom, Proposition 2.3 and the norm-convergence theorem for amenable nilpotent averages of Zorin-Kranich; here and below it also suffices to use the van der Corput estimates for discarded terms and the explicit limit for retained terms.

The same Conze–Lesigne group. The two Kronecker factors coincide. Every H-eigenfunction generates a one-dimensional eigenspace since H is ergodic, and each Γ-transformation preserves that eigenspace. Denote this rotation by Z. By Jamneshan–Shalom–Tao, Theorems 1.10 and 1.11 and the argument of Section 4.2, $Z_2(\Gamma)$ is the inverse limit of factors $Y = Z \times_\rho U$ with U compact abelian Lie, Z their actual Kronecker factor, and ρ satisfying the Conze–Lesigne equation. More specifically, first choose Z to be the full Kronecker factor and approximate the compact vertical group by its Lie quotients. Their Kronecker factors still equal Z since they are intermediate factors of $Z_2(\Gamma)$ containing Z. For torsion Γ pull back the action to a countable torsion-free free abelian cover as in Section 4.2: if the equation for a lift of a base translation holds there with a homomorphism c', evaluating at a kernel element gives $c' = 0$ on the kernel since both the cocycle and the base action are then trivial. Thus these lifts and their homomorphisms satisfy the same equation for Γ itself.

On each such factor the full Conze–Lesigne group N_Γ is transitive and gives a homogeneous presentation of Y by the cited construction. Its elements are the pairs (z, F) covering

rotation by z on the base with $\Delta_z\rho = dF + c$ for some group homomorphism $c : \Gamma \to U$. Restricting to H gives lifts in N_H. Conversely fix any (z, F) for H and choose a Γ-lift (z, F_0) at this same z. Their difference $F - F_0$ has constant translation derivative along each H-element. For every character $\chi \in \widehat{U}$, $\chi(F - F_0)$ is an H-eigenfunction on Z. Its eigenspace is one-dimensional and Γ preserves it. Thus its derivative along each Γ-element is constant. Using a finite generating set for $\widehat{U}$ shows that the U-valued derivative is constant for each element and these constants form a homomorphism. Therefore (z, F) is a Γ-lift. This proves $N_H = N_\Gamma$ as groups acting on Y, with the same topology induced by $Z \times L^0(Z, U)$ and the same homogeneous coordinatization. No global replacement of Γ by the torsion-free cover is needed when applying the finite-index conditions on $2\Gamma, 3\Gamma$.

Comparison. By Shalom, Corollary 5.5 with $a = 1, b = 2$ the commutator subgroup of this transitive Conze–Lesigne group is 2-divisible. The hypotheses are precisely the finite indices of $\Gamma, 2\Gamma, 3\Gamma$ and likewise for H, and ergodicity holds. By Shalom, Theorem 6.1, the limit of every progression average on each good factor has the Hall–Petresco formula with Haar variables in N/Λ and its commutator subgroup. All the terms on the right depend only on this same presentation of Y and the input functions; it is identical for Γ and H. Approximate finitely many bounded functions in $Z_2(\Gamma)$ by these factors, with uniformly bounded sup norms. In the integrated average the sum of their L^1 approximation errors times the product of the remaining bounds suffices, uniformly over sampled directions. This proves the comparison for the subgroup. Apply it to inputs translated by `j w` and translate the full-group average to obtain the coset comparison. Nonnegative same-event unrestricted return averages at every finite length are strictly positive for a positive-measure event by ordinary multiple recurrence (or uniform density Hales–Jewett word sampling). The proof does not assume that the action of every finite-index subgroup of Γ is ergodic; only H itself was required.

Application to the deep model. Use $G = \oplus C_{p^{e_p}}$ and $H_0 = \oplus pC_{p^{e_p}}$ as in the deep quadratic note [R049]. Suppose H_0 acts ergodically. Fix any sufficiently large p and any admissible marked residue u on that single prime (including the quadratic charge condition). Put $K = \oplus_{q \neq p} qC_{q^{e_q}}$ and $\Gamma = K \oplus C_{p^{e_p}}$. The subgroup K acts ergodically by the finite-component root observation [R149]: its components are cyclically permuted by pe_p, which has the commuting pth root e_p. Applying the comparison to $K \leq \Gamma$ yields a positive four-term return on $u + K$ for every event of positive measure. Every direction in this coset is legal. On $Z_2(\Gamma)$ the conclusion holds at every length. This argument also applies on H_0-ergodic components whenever the root e_p preserves them; in particular if it induces the identity on the H_0-component space then every positive event returns at four terms with this particular prime. The unrestricted nonergodic-horizontal case is not resolved.

R152. A mixed-primary pure tag and its cyclic clock

Scope and status. Proved subcases and unresolved extensions.

Result. The mixed-primary pure-tag proposal has a fixed short obstruction and quantitative lower bounds on normal polynomial defect costs.

Scope. The unrestricted free-host threshold is undetermined; divergence of summed defect costs only excludes the stated deletion method.

Take low coordinates C_{p_i} at distinct odd primes and high coordinates $\mathbb{Z}$. Let $\sum_i 1/p_i < \infty$ and $p_i > 2^{k_i}$. Each finite-support direction has an arbitrary high speed t_i and low speed either $0, 1, 2^{k_i}$. Its tag is the parity of $\sum_{i:r_i=0} t_i$, and admissibility means odd tag. The free low lift permits **all** integer representatives of each of these residues, so every multiple of p_i is an unrestricted correction at that site under every mask. On a finite packet the whole low product is cyclic by CRT. The high ranks still grow. This is a free-host proposal; its all-action value is undetermined.

There is a positive fixed-length avoiding event in the native low product and the two-element high quotient. At site i over $\mathbb{F}_2$, choose h_i equal to one except at residue zero, where it is zero. Its sum vanishes. For the cyclic low shift S, choose

$$(S-I)^2(S^{2^{k_i}}-I)^2 b_i = h_i.$$

Indeed each shift difference is invertible on the zero-sum subspace, as its only kernel consists of constants and p_i is odd. At local state $(x, y) \in C_2 \times C_{p_i}$ evaluate $f_i = xb_i(y)$. In the sum of these observations use also the finitely many spatial channels needed for $(T-S)^2(T-S^{2^{k_i}})^2(T-I)$, where the coefficient $S^{2^{k_i}}$ is one concomitant shift across all sites. Along a normal low speed 1 or 2^{k_i}, the corresponding squared factor kills the local observation because it is affine in the high input and both factors move the same low amount. At a pure site the output is $t_i h_i(y_i)$, independently of the starting high bit. If all the initial low residues are nonzero, the total output is the legal odd bit. Each single spatial channel has on its temporal coefficients sum not necessarily zero, but the entire filter contains $T-I$, so every channel does have zero temporal sum. Equal channel vectors at all six times would give zero output instead.

In finite packets use independent uniform low residues and high bits. The probability of all lows nonzero is $\prod_i (1-1/p_i)$ over the packet, bounded below independently. One of the finitely many channel color cells intersected with this safe set has uniformly positive measure. It is six-avoiding. Coinduce to the whole group and take stationary compactness limits of the indicators (or color and indicator jointly); each particular finite direction eventually has its sitewise constraint. Thus the fixed lower bound passes to the full host. In the quotient with order-two high axes every *pure* singleton step already forces recurrence at density above $1/2$. The high orders on the free lifts are unrestricted.

There is a simple elimination for bounded or summably approximable low speeds. If the two normal speeds in centered integer representatives have absolute values bounded by a fixed B and positive 2-adic valuations at most a fixed v (valuation zero is allowed), the free lift has endpoint one whenever $\sum_i 1/p_i < \infty$. More generally it suffices that after independent changes of low generator the sum of the absolute normal speeds divided by p_i converges, and their 2-adic valuations on one tail have a common finite bound. Along an all-safe run of any prescribed fixed length L the centered low lifts have no wraps after

deleting boundary strips of cost $O(L)\sum_{\text{tail}}\max_j |r_{i,j}|/p_i$. On integer low coordinates choose $b_i(y) = \binom{y}{J_i} \bmod 2$, $J_i = 2\cdot 2^{v_2(r_{i,1})} + 2\cdot 2^{v_2(r_{i,2})}$. Over characteristic two the operator $(S^{r_{i,1}} - I)^2 (S^{r_{i,2}} - I)^2$ has multiplicity exactly J_i at $S = 1$, with remaining value one; it maps b_i identically to one. The b_i and the needed shifted channels have a uniformly bounded dyadic period (depending only on v). On safe runs the whole time profile of the sum of their binary observations therefore has a uniformly bounded period, and the same six-slot filter precludes a constant-color run. Use independent roots at hierarchically divisible temporal strides and keep the union of a fixed positive color class across the rows, requiring all rows safe. A sufficiently long all-retained run has some temporal position outside that class in every row: in each row at its stride the filter supplies a miss within six samples and the bounded period preserves all earlier misses under subsequent stride shifts. The mass of the constructed event tends to one with the number of rows and then with the starting tail. This is an exact root obstruction for the stated small-displacement regime.

For the full $1, 2^{k_i}$ menu this argument cannot simply use the original low generators with growing valuations, even when the primes themselves are arbitrarily large. If frequencies at all sites can simultaneously approximate bounded nonzero dyadic phases in both directions with summably small errors, there may be another bounded-period construction. Including infinitely many axes at each fixed, arbitrarily high k defeats this particular sufficient approximation uniformly: on a fixed such level the two phases satisfy the exact relation $\theta_2 = 2^k\theta_1$; a limit pair of bounded-order dyadic phases with both nonzero is impossible for sufficiently high k. An apposite upper would still have to hold in arbitrary stationary free high covers. On finite high covers the local arrays can instead be solved over $\mathbb{Z}/2^a$ for any a, with $h_i = 1$ off the exceptional residue and $1 - p_i$ there. This supplies an odd unit in the six-slot filter; it does not alone produce an avoiding event of density near one.

Stronger exclusion without a valuation bound. On free high lifts it suffices that after changing the low generators independently, the centered nonzero representatives a_i, b_i of the two normal speeds satisfy $\sum_i (|a_i| + |b_i|)/p_i < \infty$. In the original coordinates this applies whenever $\sum_i 2^{k_i}/p_i < \infty$. Set

$$u_i(y) = \frac{y^4}{48a_i^2 b_i^2} \pmod 1$$

on the centered integer chart, and place the high coordinate x on a cyclic cover of period $48a_i^2b_i^2$. Use $xu_i(y)$ and its finite spatial translates in the same filter, now in circle variables. The identity $\Delta_{a_i}^2\Delta_{b_i}^2(y^4) = 24a_i^2b_i^2$ makes its pure output exactly $t_i/2$ on safe points, while both normal outputs vanish. High wraps are harmless by the chosen cover period, including along every shifted channel. For any fixed finite horizon all spatial and temporal chart wraps can be excluded by site boundary strips whose summed measure is $O_h(\sum_{\text{tail}}(|a_i| + |b_i|)/p_i)$. On the remaining runs all observed traces are ordinary circle polynomials in time of degree at most five, uniformly in normal magnitudes, denominators, rank and high speeds. The bounded-polynomial root-gate argument [R027] therefore yields avoiding events of arbitrarily large density on the free lifts. To use its safe-run form, fix the desired density and its root tests and uniform horizon first, and then take the tail so that the total strip cost across every independent row is sufficiently small. Sum the basic phases on finite packets and use stationary weak limits; the common uniform horizon and the fixed forbidden cylinder pass to the limit. This eliminates mixed-clock variants with summable normalized radii, even when every bounded-order dyadic phase test just discussed would fail.

A quantitative lower bound on simultaneously polynomial normal traces Let p be an odd prime and $a, b \neq 0 \bmod p$. Define

$$\ell = \min\{|u| + |v| : (u, v) \in \mathbb{Z}^2 \setminus \{0\},\ ua + vb = 0 \bmod p\}.$$

For any $f : C_p \to \mathbb{F}_2$ with $\Delta_a^2 \Delta_b^2 f \neq 0$ at at least $p/2$ sites, and every fixed $s \geq 0$,

$$|\operatorname{supp}(\Delta_a^{s+1} f)| + |\operatorname{supp}(\Delta_b^{s+1} f)| \ \geq c_s \ell.$$

In other words, allowing exceptional sites with bounded-degree normal time traces cannot be done at arbitrarily small total cost on a well-rounded two-generator row.

Put $j = s + 1$ and write $E_a = \Delta_a^j f,\ E_b = \Delta_b^j f$, with combined support count n. Use the word metric from $\pm a, \pm b$, and join two sites of the supports if their distance is at most $2j$. It suffices to consider $10(j+1)(n+1) < \ell$, since otherwise $n \gtrsim_s \ell$ already (nonzero filter implies $n \geq 1$). Each component C of size n_C has its sites within word distance $2jn_C$ of a fixed root. Its sites and all their translates by at most j generators lift consistently to $\mathbb{Z}^2$ near a chosen lift of the root: two competing paths have total length at most $4jn_C + 2j < \ell$ and hence the same coordinate displacement. The translated neighborhoods of different components are disjoint, by the definition of the components.

Commuting differences give $\Delta_b^j E_a = \Delta_a^j E_b$. Restricted to one lifted component and the indicated neighborhoods this is the Laurent polynomial identity

$$(Y - 1)^j \widetilde{E}_a = (X - 1)^j \widetilde{E}_b \quad \text{in } \mathbb{F}_2[X^{\pm 1}, Y^{\pm 1}].$$

Use inverse monomial conventions if necessary so a shift of the array is multiplication by the corresponding variable. Coprimality of the two powers gives a Laurent polynomial H_C with $\widetilde{E}_a = (X - 1)^j H_C,\ \widetilde{E}_b = (Y - 1)^j H_C$. If one residual vanishes, the other vanishes as well on that component. The exponent span of H_C in each variable is no larger than that of the nonzero residual as can be seen from the extreme nonzero exponent terms. Thus H_C has $O_j(n_C^2)$ monomials. Descend their sum H to C_p; the residual differences of $f - H$ vanish there. The a-difference is invertible on the zero-sum subspace of $\mathbb{F}_2^{C_p}$, so $f - H$ is constant. It follows that $\Delta_a^2 \Delta_b^2 f$ has support at most $O_j(\sum_C n_C^2) \leq O_j(n^2)$, whence $n \gtrsim_j \sqrt{p}$. The elementary pigeonhole bound $\ell \leq 2\lceil \sqrt{p} \rceil$ finishes the proof.

Exactly the same estimate holds for tables with values in $\mathbb{Z}/2^d$ for any d, with constants independent of d. The difference along a is invertible on zero-sum arrays since p is odd. In the Laurent polynomial step, multiplication by $(Y - 1)^j$ is injective after quotienting by $(X - 1)^j$, by its unit leading coefficient when viewed as a polynomial in Y. This yields divisibility by $(X - 1)^j$, and cancellation is again legitimate by its unit extreme coefficients. The same extreme-coefficient observation bounds the exponent spans of H_C.

For a concrete surviving schedule let $L = 2^k,\ a = 1,\ b = L$, and take on this level about L different primes in $[L^2, 2L^2]$ (for all sufficiently large k these exist by the prime number theorem). The levels are disjoint. A relation shorter than cL for small absolute c has $|u + vL| < p$; equality to zero over the integers then forces length at least $L + 1$. The obvious relation $(-L, 1)$ gives $\ell \asymp L$. The summed densities of the individual low tag defects $1/p$ converge over all levels. In contrast, for every fixed s, the sum of the forced degree-s normal defect costs is at least a constant per level and therefore diverges. This excludes independently deleting all local bad degrees in these models; it is not a proof of any all-system upper bound for the mixed clocks.

R153. Relative clock registers without independence of the system

Scope and status. Result with proof.

Result. Relative mixed-clock eigenregisters force full recurrence for at most two channels and recurrence above one half for at most three.

Scope. Requires a weighted independent coordinate subsequence or the explicit bounded-dependency condition.

Use the well-rounded mixed clocks and depths [R154]. Let Y be any factor fixed by the whole high subgroup $H = \bigoplus_i \mathbb{Z}e_i$. A list of r circle-valued phases has relative eigenlabels $\rho_{l,y} \in \widehat{H}$, meaning $\phi_l(U^g x) = \phi_l(x)e(\rho_{l,y}(g))$ for $g \in H$. The labels at coordinate i are assumed periodic of period dividing p_i under its own low generator and invariant under every other low coordinate outside a set $S_i \ni i$ of uniformly bounded cardinality. This periodicity is automatic on native low prime coordinates, or on a coarse eigenlabel factor fixed by all their period multiples. Actual high actions and labels may be free and irrational. Neither independence of the local states nor a global coboundary presentation is imposed. For events in the σ algebra generated by $Y, \phi_1, \dots, \phi_r$, there is full positive recurrence for $r \le 2$, and all events of measure above $1/2$ recur for $r \le 3$.

Make an undirected edge i–j whenever $j \in S_i$ or $i \in S_j$, excepting loops. Each finite induced graph has average degree at most $2\sup|S_i|$, even on a subset of vertices, so the infinite graph has a proper coloring with $2\sup|S_i|+1$ colors by degeneracy and compactness. Keep a color class J with divergent $\sum_{i\in J} 1/L_i$. On this class each tested label at i is invariant under the low actions at all other tested sites. More generally existence of such an independent class suffices; the total number of low dependencies away from it then need not be bounded.

For $i \in J$ and a fixed starting y, put

$$\mathbf{b}_i^y(a) = (\rho_{l,U_i^a y}(e_i))_{l=1}^r \in \mathbb{T}^r, \qquad a \in C_{p_i}.$$

For a finite balanced Laurent row P as in the circle-register note, let $\delta_i(P; y)$ be the fraction of failures of $P(S^n)\cdot\mathbf{b}_i^y = Q_P(S^n)\cdot\mathbf{b}_i^y = 0$, averaging the argument a uniformly over C_{p_i} and the normal velocity n equally from $1, L_i$. These deficits are invariant under every selected low coordinate (their own one is averaged and the others leave all these labels alone), and are functions on the high-fixed Y. Disintegrate over the countably many deficits. Each conditional law is still stationary for all tested low and high moves, and all the sums of deficits and ensuing modules R, W can be treated as fixed there. For a global positive event use the conditional components where its own mass is positive, or above $1/2$ for the half bound. Countably many possible true directions let component-dependent choices give one globally positive return.

The sparse-potential argument works for these circle tables. Divide the common *integer gcd* out of the scalar polynomial D appearing there and multiply the table by this gcd. For every abelian coefficient group A a primitive integer Laurent polynomial $D(X)$ acts injectively on finitely supported A-valued Laurent arrays. Indeed if a nonzero kernel element of smallest exponent span exists, its multiple by the leading coefficient of D has strictly smaller span (the leading product coefficient was zero), so this multiple vanishes. Remove that coefficient from the relation and repeat with each remaining leading coefficient; their gcd one contradicts a

nonzero kernel. After quotienting the X-arrays by $D(X)$, the same statement applied to that abelian quotient proves injectivity of $D(Y)$ there. Hence $D(Y)E_x = D(X)E_y$ on a lifted cluster yields $E_x = D(X)H_C,\ E_y = D(Y)H_C$ also with circle coefficients. Its support remains in the y-rows of E_x times the x-columns of E_y, giving the uniform n_C^2 bound.

For large p_i, the matrix for $D(S)$ on the zero-sum integer lattice of the p_i-cycle has odd nonzero determinant $d_i(D)$; this is the same no-common-root argument modulo two used in the finite-cover note. If $D(S)b = 0$ for a circle array, then $p_i b - \sum_a S^a b$ is zero-sum and is annihilated by $D(S)$; the adjugate identity on the zero-sum coordinates shows $p_i d_i(D)b$ is constant. Thus after removing sparse potentials and multiplying by this allowed odd factor, any constant projection needed in the full/constant W cases is constant off the same $o(p_i)$ sites. All odd fixed denominators, gcds and determinants can be included in the odd multiplier of the one fresh pure site; all fixed 2-parts are absorbed by $z_i \to 0$ in the 2-adic topology. For each finite list choose a common product multiplier. The estimates are uniform in y once the deficits are fixed, regardless of the exceptional sets' positions. The normal Q_P tests make the resulting pure signatures vanish on typical own starting residues just as in the register proof [R154].

Here is the needed probability argument in lieu of product site laws. Sample far disjoint normal pools and independent candidate normals and high velocities as before. In each tested joint correlation, insert a uniform **initial low shift** on every used C_{p_i}, available by stationarity on the conditional law. The low generator need only have period p_i on the tested labels; the average may use ordinary representatives $0, \dots, p_i - 1$. Also average independently over initial high shifts and all normal high velocities along large finite integer Følner boxes. For a spectral term with integer coefficient rows c_j, high translation of the initial point contributes the $P(S_i^n)$ restriction on every normal site, because $\rho_{l,U^j v^y}(e_i)$ depends only on that site's own low movement among the selected ones. Averaging its normal high velocity gives the $Q_P(S_i^n)$ restriction there. For any finite pools the orthogonality limits are exact as the Følner sizes go to infinity. The remaining site and base amplitudes have modulus at most one, and Y-tests are unaffected by the high shifts. Conditional on the state before the uniform low shifts, the necessary normal conditions on the different sites involve **only their own independent low starts and normal choices**. Their joint success probability is precisely the product of $1-\delta_i(P)$. Hence all uncontrolled Fourier coefficients of the trajectory law tend to zero, even when multiplied by arbitrary bounded tests on the full base trajectory. This argument only uses boundedness of the remaining conditional amplitude; it does not assert independence of the phases at different sites.

For a balanced controlled row, a fresh pure mark at $i \notin \operatorname{supp} v$ contributes the factor $e(z_i t_i Q_P(1) \cdot \mathbf{b}_i^y(a))$ relative to the ordinary product, where its initial a can again be averaged uniformly. Other selected low moves leave this site's labels alone. Thus in the full/constant W cases, after the chosen odd multipliers, the discrepancy between the charged and uncharged correlations tends to zero for every controlled term. Unbalanced Fourier terms require a small qualification: their original initial-high-shift test $P(S_i^n)$ still applies with the actual, possibly unbalanced P. Define R in this argument to include *all* rows with summable normal deficits, without imposing $P(1) = 0$, since the original registers need not be independently Haar. It is still a Laurent submodule, the derivative-stability argument is unchanged, and the preceding tests apply verbatim. This also covers the constant Fourier row on the base. Approximation of a bounded joint observable of $(y, \boldsymbol{\phi})$ by finite Fourier sums with bounded base coefficients uses the actual one-site marginal measure, for instance by rectangle approximation and continuous trigonometric approximation on the compact torus. The

common-marginal error bounds are uniform over all candidate directions. DHJ now gives a true positive return for any positive event in the full/constant W cases, including $r \leq 2$.

In the remaining rank-one R case for $r = 3$ its primitive generator $v(T)$ has nonconstant rational direction. Every subsequential joint trajectory limit for a normal wildcard sum plus the independent pure site is invariant under Haar on $\ker v(T)$, holding its entire base trajectory fixed. Indeed all joint Fourier tests outside R vanish even against arbitrary bounded base tests, as just proved. Work on a compact metrizable model of the base, or on a countable compact coding that includes the needed measurable base tests; common one-slot marginals allow passage from continuous to measurable tests. Since $\ker v(T)$ projects onto the whole torus at any single time (there is no constant-row character in its annihilator), the one-site register marginal conditional on the base is necessarily Haar. For an event of mass above one half choose a positive-measure base set B on which the sections A_y have mass above $1/2$ and share one section threshold u with $m\{x : m((A_y)_x) > u\} > 1 - u + \epsilon$ for some $\epsilon > 0$; layer cake and countable perturbation to rational thresholds suffice. DHJ for B gives a fixed wildcard subset with positive limiting base return, since the high mark fixes the base. On those base trajectories the rank-one forward kernel recursion [R154] applies with a different A_{y_j} at each time but the *same* u, ϵ. It gives positive conditional probability on every residual forcing. This proves the half bound with these genuinely relative registers.

The finite-dependency or independent-subsequence condition is material. For example $S_i = \{1, \dots, i\}$ blocks every pair from being mutually independent, even though each individual dependency set is finite. Arbitrary relative eigenfields need not have any summably weighted independent subsequence. Conditional eigenfunctions with general unbounded-rank collections are not claimed to obey a uniform bound by this proof.

R154. Complete recurrence in scalar affine clock registers on arbitrary cyclic covers

Scope and status. Result with proof.

Result. Scalar and two-register mixed-clock affine Haar systems have full recurrence; three registers have the stated half bound.

Scope. Arbitrary finite cyclic covers and local coefficient functions are allowed, but the specified register representation remains essential.

Use the well-rounded levels of the mixed-clock proposal [R152]: $L = 2^k$, about L distinct primes $p_i \in [L^2, 2L^2]$ on each level, and low normal speeds $1, L$. A pure mark has low velocity zero, and positive high velocity $z_i t$ with t odd. Normal high velocities are unrestricted integers. At a site take independent Haar $x_i \in C_{N_i}, y_i \in C_{p_i}$, for **arbitrary** finite N_i with no uniform bound or relation of their 2-primary heights to the chosen z_i. Let $b_i : C_{p_i} \to \mathbb{Z}/N_i$ be arbitrary. Give a common independent Haar circle register w the coboundary increments of $f_i(x, y) = xb_i(y)/N_i$ at the finitely many shifted sites. Local states shift by their two velocities. This defines a stationary action even though an infinite sum of the f_i need not exist. Every positive measurable event in this entire system recurs along the allowed directions at every finite length, even with just one pure marked site. This does not assume the b_i have any degree, regularity or bounded range. The many-channel low coloring uses simultaneous observations that a single scalar register need not contain.

Fix a horizon h. Write $P_i(r, v) \in \mathbb{T}^{h-1}$ for the increments at $j(r, v)$, $1 \le j < h$. For a character $c = (c_j)_{1 \le j < h} \in \mathbb{Z}^{h-1}$ set $c_0 = -\sum_{j>0} c_j$, $P_c(T) = \sum_j c_j T^j$, $Q_c(T) = \sum_j j c_j T^j$. At a normal axis sample r equally from $1, L$ and v uniformly modulo N_i (using integer representatives). Averaging first the independent original x_i and v shows that the Fourier coefficient of $P_i(r, v)$ is a real number $1 - \delta_i(c) \in [0, 1]$, where

$$\delta_i(c) = \tfrac{1}{2} \sum_{r=1,L} \mathbb{P}_{y_i}\big[P_c(S^r)b_i(y_i) \neq 0 \text{ or } Q_c(S^r)b_i(y_i) \neq 0 \bmod N_i\big].$$

The subgroup $\Lambda = \{c : \sum_i \delta_i(c) < \infty\}$ is finitely generated (subgroup of $\mathbb{Z}^{h-1}$); the failure of a sum relation is contained in the union of the corresponding failures. Off Λ the sums diverge. For any prescribed number m of independent normal trials there are disjoint finite pools far out whose summed profiles each tend weakly to Haar on $\Lambda^\perp$. Indeed for each controlled c the tail sum of deficits tends to zero, whereas the products for every other c can be made to tend to zero; allocate disjoint subpools for finite initial lists of the countably many uncontrolled tests at successive stages. These weak limits follow from Fourier uniqueness. Independently omitting each sampled normal coordinate with probability $1/2$ is also permissible and merely replaces each deficit by its half.

Choose generators of Λ and let δ_i^* be the sum of their deficits. The schedule has $\sum_i 1/L_i = \infty$, so arbitrarily far out there are axes with $\delta_i^* L_i \to 0$. They can be kept disjoint from the normal pools at each stage. We show on any such axis sequence that its pure profile is invisible to every generator. The pure velocity used is $z_i t_i$, where t_i is the odd part of N_i, which removes all odd-order components of the measured phase.

Fix a nonzero generator c. In the 2-primary group $A_i = \mathbb{Z}/2^{v_2(N_i)}$, divide all coefficients of P_c, Q_c by the greatest common power 2^e in the coefficients of P_c, and replace b_i by its 2-primary projection multiplied by 2^e. Write the new polynomials P, Q and table b; $P \bmod 2 \neq$

0. The exact identities being tested have not changed in the 2-primary component. If that component is trivial there is nothing to show. Each of the supports

$$\operatorname{supp} P(S)b, \quad \operatorname{supp} P(S^L)b, \quad \operatorname{supp} Q(S)b$$

has size $o(p_i/L_i) = o(L_i)$, since failure of a primary-component identity implies failure of the original one. We need a uniform-in-height version of the sparse two-generator observation.

Put $E_a = P(S)b$, $E_b = P(S^L)b$. With $n = |\operatorname{supp} E_a| + |\operatorname{supp} E_b| = o(L)$, group their supports into components at bounded word-metric radius $2\deg P + 2$ for the steps $1, L$. The shortest nonzero lifted relation is comparable to L, so each component and its bounded neighborhood lift consistently to $\mathbb{Z}^2$. The characteristic identity reads $P(Y)\widetilde{E}_a = P(X)\widetilde{E}_b$ in the Laurent ring over A_i. For every such component C there is H_C with

$$\widetilde{E}_a = P(X)H_C, \qquad \widetilde{E}_b = P(Y)H_C, \qquad |\operatorname{supp} H_C| \le n_C^2.$$

Here is why neither degrees nor support bounds grow with $v_2(N_i)$. Modulo two, $P(X)$ and $P(Y)$ are nonzero, relatively prime as Laurent polynomials in independent variables, so the potential exists. Its y-rows lie in precisely the permitted set of rows meeting $\widetilde{E}_a$, since multiplication by $P(X)$ mod 2 is injective on every finite Laurent row. Likewise its x-columns must meet $\widetilde{E}_b$. Lift the coefficients of this potential to the current ring, subtract $P(X), P(Y)$ times that lift from the two residuals, and divide both residuals by two. They retain the relation at the next lower modulus. No new y-rows for the first residual or x-columns for the second are introduced. Repeat to reconstruct a potential on the same product of the original projected sets. The count of these sites is bounded by n_C^2, irrespective of the modulus. A constant polynomial P mod 2 causes no problem.

Descend the potentials to C_{p_i} and sum them into H, supported on at most $n^2 = o(p_i)$ sites. The array $b - H$ is annihilated by $P(S)$. For sufficiently large p_i this implies it is constant. In fact a nonzero root of the fixed polynomial P mod 2 belongs to some fixed finite field, hence has multiplicative order bounded independently of p_i. The only possible p_i-th root shared with P for late primes is 1. The prime is odd, so the cyclic shift diagonalizes over a splitting field of characteristic two; $P(S)$ mod 2 is invertible on the zero-sum part. Since p_i is a unit in A_i, this part splits off from constants there and is a free module of rank $p_i - 1$. The same operator is invertible modulo every power of two by elementary lifting (equivalently its determinant on that module is odd). Thus $b = H + c_i^*$ with c_i^* constant. Since $Q(S)b$ vanishes outside $o(p_i)$ sites and $Q(S)H$ is supported on $o(p_i)$ sites, we have $Q(1)c_i^* = 0$ identically. Consequently $Q(1)b(y_i) = 0$ with probability $1 - o(1)$.

Restoring the original notation, the nonconstant pure high profile at time j is $jz_it_ib_i(y_i)/N_i$. Its c-Fourier test is $e(z_it_iQ_c(1)b_i(y_i)/N_i)$. On the typical y_i's just found its 2-primary part is zero, and the chosen odd multiplier annihilates the odd part. In particular its expectation tends to one. This holds for all fixed generators simultaneously. Every cluster law of the pure profile is supported on $\Lambda^\perp$, so by independence convolving it with any of the nonempty sums of the limiting normal pools still gives the same Haar law.

Finally begin with an event depending on the global scalar and finitely many local head variables. All fresh directions fix the head. For each choice of m disjoint sampled normal trials, apply finite density Hales–Jewett averaging at a random initial point to the h-letter word grid; the sum of its uncharged line correlations has a positive lower bound depend-

ing only on $h, \mu(A)$. Fixed initial word shifts disappear by stationarity. In each of the finitely many possible nonempty wildcard subsets, adding one fresh pure mark at its prescribed odd velocity leaves the limiting increment-profile law unchanged, including jointly with the head and initial common Haar register. The same total lower bound survives for the corresponding true correlations. The use of measurable event indicators follows by finite translation-invariant-marginal L^1 approximation from continuous tests of the common circle; all one-slot global registers are Haar independent of the head. Arbitrary positive events can first be approximated in measure by the finite-head form with an error much smaller than that uniform DHJ lower bound divided by the finite slot and line counts. This yields a positive actual return at some true direction. No uniform conclusion for every stationary free action is asserted.

Two registers at positive density, and three above one half Allow r simultaneous registers with table vectors $\mathbf{b}_i : C_{p_i} \to (\mathbb{Z}/N_i)^r$, coupled through the same local high state x_i. For $r \leq 2$ the preceding full positive assertion persists. For $r \leq 3$ every event of mass greater than $1/2$ has all lengths of recurrence. Arbitrarily high cyclic covers and the full local σ algebra are permitted.

It is convenient to use all finite Laurent row tests $P(T) = \sum_j c_j T^j$ with $c_j \in \mathbb{Z}^r$, including negative times. Add the independent common Haar register to a normal profile, so all nonbalanced rows $P(1) \neq 0$ vanish. Let R consist of the balanced rows with summable deficits for the two scalar tests $P(S^a) \cdot \mathbf{b}_i$, $Q_P(S^a) \cdot \mathbf{b}_i$ on both normal speeds $a = 1, L_i$, where $Q_P = TP'$. This additive group is a Laurent submodule: shifting a row by T^m shifts the first test in its argument, and combines the second test with a fixed multiple of the first. Choose the finite normal pools diagonally for all these countably many tests. With the independent common Haar register the limiting normal trajectory law is Haar on $R^\perp \subset (\mathbb{T}^r)^{\mathbb{Z}}$. Convolving with the independent profile of the single fresh pure mark produces a mixture of its cosets. For arbitrarily many independent normal DHJ trials the sum on each nonempty wildcard subset enjoys the same conclusion. Every fixed local head stays fixed and independent of these limiting global laws; hence by finite-dimensional cylinder tests the same holds for the whole countable head. The common-marginal approximation gives the corresponding assertions for measurable events.

In the Laurent row space $\mathbb{Q}(T)^r$ put $W = \mathrm{span}_{\mathbb{Q}(T)}(R, \{Q_P : P \in R\})$. If W is full, or if W has a basis of constant rational rows, every controlled test is invisible to a suitable fresh pure mark. Indeed for each fixed integral constant row $l \in W$, clear denominators to express $D(T)l$ for a nonzero scalar integer D as a finite Laurent polynomial combination of rows P, Q_P with $P \in R$. Thus both $D(S)\, l \cdot \mathbf{b}_i$ and $D(S^{L_i})\, l \cdot \mathbf{b}_i$ have exceptional supports bounded by a constant times the sum of the finitely many corresponding deficits times p_i. On axes where those deficits are $o(1/L_i)$ the sparse lemma gives that $2^e l \cdot \mathbf{b}_i$ in the 2-primary group is constant off $o(p_i)$ sites, after dividing the common power 2^e out of D as in the scalar proof. Take a constant rational basis for W, with integer representatives. For any finite list of controlled rows, all of their individual coefficients belong to W; express their coefficients in this fixed basis and multiply by a common positive denominator m, enlarging m by all the required 2^e. On typical points and boundedly many translates, all vectors of these coefficient evaluations multiplied by m are independent of the point. For each fixed row P, the normal relation $Q_P(S) \cdot \mathbf{b}_i = 0$ holds outside $o(p_i)$ sites, so in the 2-primary group $mQ_P(1) \cdot \mathbf{b}_i$ is zero on typical points. Since z_i is divisible by the 2-part of m eventually, the actual pure mark annihilates these same signatures there; its odd multiplier kills all odd-primary parts.

Increasing the finite list and using the divergent weight $\sum_i 1/L_i$ gives a diagonal sequence working for all R. Consequently the pure-profile coset law is simply Haar on $R^\perp$, giving positive recurrence as in the scalar proof.

If $\operatorname{span}_{\mathbb{Q}(T)} R = W$, this subspace is closed under differentiation, hence has a basis of constant rows. For comparison, take a basis matrix A with $A' = MA$. Every maximal minor has derivative $(\operatorname{tr} M)$ times that minor by multilinearity, so the ratios of nonzero minors are constant. Row reduction at one fixed nonzero minor gives a constant matrix for the subspace. In dimension $r \le 2$, R of positive rank either has the full W or satisfies this stability (rank zero is automatic); hence the result for two. In dimension three the only additional case has $\operatorname{rank} R = 1$.

For completeness in that case take a primitive polynomial row $v(T)$ generating the rational line of R. Then every row in R is an integral Laurent polynomial multiple of v: cancel its polynomial gcd over $\mathbb{Q}[T]$, clear content to make the remaining coefficient list primitive, and use polynomial division/Gauss's lemma. Consequently every coset of $R^\perp$ is invariant under the Haar kernel $L = \ker v(T)$. If v has a constant direction, W is constant and the earlier proof applies. Otherwise choose an integral unimodular basis of outputs putting its top coefficient at degree d into $(a, 0, \dots, 0)$, $a \neq 0$. Let $e < d$ be the last degree where one of the other coefficients is nonzero, with vector $b \neq 0$, and put $s = d - e > 0$. In a mixture-of-cosets trajectory fix an arbitrary residual path; convolving with Haar on L preserves the law. The randomized states solve $v(T)(x, y) = w$ for some fixed forcing, with $x \in \mathbb{T}$ and $y \in \mathbb{T}^{r-1}$. At time t, the equation indexed $t - e$ takes the form

$$a x_{t+s} = \text{known earlier data} - b \cdot y_t.$$

This is the rank-one section recursion [R190]. Under Haar on the solution coset, each new y_t is freely Haar and x_{t+s} is uniform on the finite fiber of multiplication by a; these choices extend from any finite boundary data by forward and backward solving, using surjectivity of the leading and trailing nonzero coefficient maps. Equivalently, the resulting independent-parameter rule has the Haar coset law on the requested window by its translation equivariance under the corresponding kernel projections.

For a measurable fiber section $B \subset \mathbb{T}^r$ of mass $> 1/2$, choose u with $E = \{x : m(B_x) > u\}$ of measure $> 1 - u$ by layer cake. Initialize the needed x's in E; their joint Haar projection is unrestricted for the finite initial block of length at most the degree span, by the same extension in both directions. At $x_t \in E$ choose $y_t \in B_{x_t}$ with right side in aE. The second condition has Haar measure at least $m(E)$ because $b \neq 0$, so the choices have strictly positive measure; a $1/|a|$ fraction at least of possible x_{t+s} then lies in E. Repeating through a finite window gives strictly positive Haar probability, uniformly in residual forcing at fixed B, h, v. For an event of global mass $> 1/2$ its sections above a positive set of fixed local heads meet this criterion. Integrating the positive conditional probabilities and using the invariant marginal approximation gives a positive actual legal return. This completes the three-register assertion.

R203. Depth-uniform interpolation for characteristic-p affine filters on cyclic p-groups

Scope and status. Result with proof.

Result. Scalar characteristic-p filters on cyclic prime-power groups admit bounded-degree interpolation on the rational cover, uniformly in depth; many first-digit slopes force interpolation in arbitrary coefficient characteristic.

Scope. Fixed temporal degree; the last extension requires at least h represented first-digit slopes. Unrestricted higher primary rank is not covered.

Fix d. For primes $p > d$, let $e, N \geq 1$ with $N \leq p^e$, $u = 1 + X$ in $B = \mathbb{F}_p[X]/X^N$, and $P(Y) \in B[Y]$ of degree at most d. Set

$$S = \{r \in C_{p^e} : P(u^r) belongs to F_p X^{N-1}\}.$$

Interpret the scalar coefficient of $P(u^r)$ on S as a phase in $p^{-1}\mathbb{Z}/\mathbb{Z}$. There is $Q \in \mathbb{Q}[t]$ of degree bounded solely in d such that the p-adic additive character $[Q(t)]_p$ equals this phase for every $t \in \mathbb{Z}_p$ reducing to $r \in S$. In particular allowing unbounded depth e, arbitrary centers, and coefficients varying with p does not escape rational polynomial tests in this characteristic-p scalar model.

The allowed residues lie in a bounded number of balls at comparable depths Lift P to a polynomial over the power-series ring $\mathbb{F}_p[[X]]$. The zero and constant cases are immediate (if no admissible residues there is nothing to match). Use an extension of the X-adic valuation $v(X) = 1$ to an algebraic closure of $\mathbb{F}_p((X))$, and factor $P(Y) = a\prod_{j=1}^{m}(Y - \theta_j)$, with roots repeated, $1 \leq m \leq d$. For a root θ_i define

$$F_i(R) = v(a) + \sum_j \min(R, v(\theta_i - \theta_j)), F_i(R_i) = N - 1,$$

using $v(0) = \infty$. The real piecewise-linear function has slopes between 1 and m, so this level exists uniquely. The sublevel locus $v(P(z)) \geq N - 1$ is the union of the balls $v(z - \theta_i) \geq R_i$: at z use its closest root, so every root distance equals the minimum of the closest distance and the distance between the roots. Conversely membership in any listed ball is sufficient term by term. Two of the balls are either equal or disjoint. Indeed if the root-center distance has valuation at least the smaller threshold, the two functions have identical values up to that threshold, giving equality of the thresholds and the balls.

For distinct occupied balls (i, j), put $b = v(\theta_i - \theta_j) < \min(R_i, R_j)$. Then $F_i(b) = F_j(b)$, so their slopes give

$$R_i - b \leq m(R_j - b), R_j - b \leq m(R_i - b).$$

Here *occupied* means containing $z = (1 + X)^r$ for an integer r. For two different occupied balls $b \geq 1$, because it also equals the valuation of the difference of such two integer powers. Thus the ratios of the radii as numbers R_i are at most $m < p$. As

$$v((1+X)^r-(1+X)^{r'})=p^{\nu_p(r-r')}$$

for distinct integers (r,r'), the integer preimage of each occupied ball is a single congruence class $r\equiv s_i \bmod p^{D_i}$. Here $D_i=0$ if $R_i\le 1$, and otherwise $D_i=\lceil \log_p R_i\rceil$. All occupied depths differ by at most one. They are at most e: the sublevel condition is invariant under changing $r \bmod p^e$ since $N\le p^e$. If a class had depth greater than e, its entire mod-p^e parent would have to be covered by these disjoint classes; none can cover that whole parent because of disjointness, and fewer than p classes cannot cover its p children. Thus we obtain at most d disjoint balls $B_i=s_i+p^{D_i}\mathbb{Z}_p$.

Within each ball only the next digit is read Expand $P((1+X)^{s_i}(1+Z))=\sum_{k=0}^{m} b_k Z^k$ in the power-series ring, choosing nonnegative integer representatives s_i. On $r=s_i+p^{D_i}t$ substitute $Z=(1+X^{p^{D_i}})^t-1=tX^{p^{D_i}}+O(X^{2p^{D_i}})$. The smallest value $\min_k(v(b_k)+kp^{D_i})$ is at least $(N-1)$; otherwise its leading coefficient as a function of $t \bmod p$ would be a nonzero polynomial of degree at most $m<p$, nonzero somewhere on this permitted ball, contradicting the sublevel condition. At level $(N-1)$ the coefficient is consequently a polynomial in $t \bmod p$ of degree at most d, including the possibility of the zero polynomial. Lift its coefficients to ordinary integers. Dividing by p and replacing t by $(r-s_i)/p^{D_i}$ gives a rational polynomial $q_i(r)$ which gives the desired phase on B_i. At inputs outside this ball it can have deeper denominators.

Gluing with bounded degree Put $d_{ij}=\nu_p(s_i-s_j)<\min(D_i,D_j)$ for distinct balls. The depth comparison gives $D_i-d_{ij}\le 2(D_j-d_{ij})$. For each i, multiply $q_i(r)$ by

$$W_i(r)=\prod_{j\neq i}((r-s_j)/(s_i-s_j))^{a_{ij}}.$$

Choose the positive integers a_{ij} successively in decreasing order of d_{ij}, taking $a_{ij}=1+2(d+\sum_{k:d_{ik}>d_{ij}} a_{ik})$. This recursive list has at most $(d-1)$ entries, hence all exponents are bounded solely in d. On B_i the multiplier is one modulo p, preserving the original phase since $q_i\in p^{-1}\mathbb{Z}_p$ there. On B_j its j-factor gains at least $a_{ij}(D_j-d_{ij})$ in valuation. Negative contributions from other factors occur only when $d_{ik}>d_{ij}$, in which case the numerator on B_j has valuation d_{ij}. Their total deficit is at most $(D_i-d_{ij})\sum_{k:d_{ik}>d_{ij}} a_{ik}$. By the depth comparison and choice of a_{ij}, the net gain is at least $1+d(D_i-d_{ij})$. This dominates the denominator of q_i on B_j, so W_iq_i is integral there. It follows that $Q=\sum_i W_iq_i$ works on all the balls and has degree bounded only in d.

Rational-cover consequence For each appearing prime choose any of the displayed filters, with common bound d, possibly different depths e_p, N_p, and $P_p(1)=0$. For a direction $r\in\bigoplus C_{p^{e_p}}$ require its local values on all coordinates to lie in the indicated scalar lines, and suppose their total scalar phase has norm at least a fixed positive (γ). Arbitrary subsets of this family are allowed. If the pth polynomial is Q_p, the function on $\mathbb{Q}$

$$g\mapsto \sum_p [Q_p(p^{e_p}g)]_p$$

is well defined and has zero constant term: at every prime where $g \in \mathbb{Z}_p$ the argument reduces to the admissible residue zero and has zero phase. The sum is locally finite on every finite list of rationals, so it is a group polynomial of the same bounded degree. It equals the separating scalar phase on the rational preimage by exactly the congruence computation of the prime-square example [R205]. For the elementary root sieve one can separate its homogeneous components after discarding the finitely many primes up to the uniform degree bound: each polynomial $Q_p(p^{e_p}g)$ is integer-valued on $\mathbb{Z}_p$; when its degree is less than p, its ordinary coefficients are p-integral, by Newton interpolation on $0, 1, \ldots, D$. The discarded primes have cyclic direction components of fixed orders and are independently repaired by a finite-component sieve whenever their components are nonzero. Collect the remaining like powers as additive characters of $\mathbb{Q}$ at g^k. At least one term has a uniform gap. Taking its rational character roots and deleting small randomly translated arcs, as in the preceding note, gives capacity one. If several invariant scalar lines from a single primary component are summed, interpolate the summands separately and add their local polynomials; the number of summands does not increase the degree. This argument specifically uses finite characteristic on cyclic prime-power coordinate modules and a fixed temporal degree; it does not settle unrestricted mixed-characteristic or higher-primary-rank models.

Many first-digit slopes in arbitrary coefficient characteristic Suppose the affine graph relation from the squarefree note [R239] is tested instead on $\bigoplus C_{p^{e_p}}$. Its output $v(g) = \sum_{j<h} (u^g)^j m'_j$ is fixed on its relation locus, has order dividing the order of g, and $\sum_j m'_j = 0$. Localize at the augmentation maximal ideal above p. The geometric-sum argument again makes every prime-to-p group unit the identity, and the p-primary part of the fixed phase embeds there. If at least h distinct residues modulo p occur among the represented p-coordinates, choose such nodes r_i. With $X = u - 1$ and u the p^{e_p}-group generator, $X \sum_j u^{jr_i} m'_j = 0$. The divided differences $(u^{r_i} - u^{r_j})/X$ are units because they reduce to $r_i - r_j \neq 0 \bmod p$. The cancellation-free Lagrange calculation in the squarefree note gives $X^h m'_j = 0$ in the localization. For $p > h$, expansion of $u^{p^{e_p}} - 1$ then gives $p^{e_p} X m'_j = 0$: the low positive binomial coefficients through degree $h - 1$ are all divisible by p^{e_p}, and the remaining factor after extracting $p^{e_p} X$ is a local unit. The Newton expansion of $v(r)$ is consequently a polynomial in r of degree at most $h - 1$, with positive-degree coefficients killed by p^{e_p}. Extend the fixed-phase identification on the localized fixed torsion subgroup to an additive character into the divisible p-primary torsion circle group. The resulting polynomial has ordinary coefficients of denominator at most p^{e_p}, and gives the phase on every represented r. This conclusion imposes no finite-rank assumption on the compact affine fiber. Thus unboundedly many first-digit slopes detected by a single fixed affine check are insufficient on their own. The deep mixed-characteristic example uses just three first-digit slopes, with very unequal subsequent digit depths.

R205. A prime-square affine obstruction invisible to group polynomials on the base

Scope and status. Result with proof.

Result. A prime-square affine example defeats native group-polynomial tests, yet a degree-eight polynomial on its rational cover forces capacity one there.

Scope. Native primary action and rationally saturated action have different capacities.

Qualifications. Native polynomial invisibility does not persist on the rational cover.

Let $G = \bigoplus_{p\geq 11} C_{p^2}$ over distinct primes, with each group realized in $\mathbb{Q}/\mathbb{Z}$ by $r \mapsto r/p^2$. Fix $0 < \gamma < 1/2$. Allow the finitely supported steps r for which

$$r_p = s_p + pt_p (\mathrm{mod} p^2), s_p \in \{-1, 0, 1\}, t_p \in F_p,$$

$$\| \sum_{s_p \neq 0} t_p^2/p \|_T \geq \gamma.$$

Write V for this set. All high digits on inactive axes are unrestricted. There is no cost on the count of active axes. The family V is group-polynomial recurrent in G and has a uniform finite-color short obstruction in a compact abelian affine Haar action. Its rational saturation nevertheless has all-length avoidance capacity **one**.

The nine-slot filter For each p take the additive module $B_p = \mathbb{F}_p[X]/(X^{2p+7})$, put $u = 1 + X$, and let its cyclic group C_{p^2} act by multiplication by u. This works since $2p + 7 \leq p^2$ and $(1 + X)^{p^2} = 1 \bmod X^{2p+7}$. Put $E_p = X^{2p+6}$, a nonzero fixed element, and set

$$P_p(Y) = (Y - 1)^4 (Y - u)^2 (Y - u^{-1})^2 = \sum_{j=0}^{8} m_{j,p} Y^j.$$

For $r = pt$, $u^r - 1$ has X-order at least p, whereas the remaining two factors have order at least one; therefore $P_p(u^r) = 0$. For $r = s + pt$, $s = \pm 1$, $u^{s+pt} - u^s$ has leading term tX^p (and is zero when $t = 0$). The difference with the opposite u^{-s} has leading term $(2sX)$, and the difference with one has leading term (sX). Hence

$$P_p(u^{s+pt}) = 4t^2 E_p, P_p(1) = 0.$$

Take the countable G-submodule $M' \subset \prod B_p$ generated by the nine coefficient tuples $m_j = (m_{j,p})_p$ and by the individual fixed axes E_p. The latter span a direct sum $L \simeq \bigoplus_p C_p$. In the compact dual $\widehat{M'}$ prescribe the character on L by $E_p \mapsto [4^{-1}]_p/p$ (residue inverse modulo p). This character extends to (M') by divisibility of the circle. Its extension coset is a compact torsor for $\widehat{M'/L}$; give it its translated Haar probability. The G-action on it is affine because all E_p are fixed. For any starting point x on this coset,

$$\sum_{j=0}^{8} m_j(S_{jr}x) = \sum_{p:s_p\neq 0} t_p^2/p, \sum_{j=0}^{8} m_j(x) = 0.$$

Color by the cells of the nine circle values $m_j(x)$, each cell of diameter less than $\gamma/10$. A monochromatic 9-run would compare each sampled value to its value at the start, contradicting the charge gap. This fixed finite palette gives a positive-measure avoiding color. One can alternatively restrict the coloring to an orbit and take any invariant average, since the prohibition is pointwise.

All fixed-order group-polynomial tests return on the base For a circle-valued group polynomial Q of degree d and value zero at zero on G, its value on a finite-support tuple is the sum of the one-axis values. Indeed for any two group elements (a, b) of coprime orders, commuting finite differences satisfy $\Delta_a^{d+1} = \Delta_b^{d+1} = 0$ on translates of Q. The identities $(1 + \Delta_a)^{\mathrm{ord}(a)} = 1$ and its b-version show that $\Delta_a\Delta_b Q$ is killed by powers of both coprime orders, hence is zero. $Iterate(n\Delta_a = -\sum_{j\geq 2} \binom{n}{j}\Delta_a^j)$. On a p^2-cycle with $p > d$ its Newton coefficients of positive degree are p^2-torsion: induct downwards in the coefficients of $\Delta_{p^2}Q = 0$, since $\binom{p^2}{j}$ is divisible by p^2 for $1 \leq j \leq d$. The Newton polynomials can consequently be written as ordinary polynomials divided by p^2, using inverses of $j!$ mod p^2. At $(1 + pt)$ they are affine functions of $t \in \mathbb{F}_p$, with slope a multiple of $(1/p)$.

For a given finite tuple of tests choose a nonempty finite set F of large axes on which the sum of their values at (1) is as close as desired to zero. For example use successive partial sums of the tuple in the compact torus and the pigeonhole principle; a contiguous difference of two sums suffices. On any one selected axis change (1) to $(1 + pt)$, keeping all other high digits zero. For every sufficiently large prime, uniformly over the finitely many slopes, one can choose t so all the linear changes are as small as prescribed and $\|t^2/p\| > \gamma$. To verify this, the linear Bohr window has a nonnegative continuous cutoff with positive lower mean independent of the slopes: choose a convolution square of a small continuous bump supported near zero, so all Fourier coefficients are nonnegative and its zero coefficient positive. Average its product with a nonnegative continuous circle function supported where the quadratic phase has norm $> \gamma$. Finite Fourier approximation and the elementary quadratic Gauss bound $p^{-1/2}$ at every nonzero fixed quadratic frequency show that the mean differs by $(o(1))$ from the unrestricted linear cutoff mean times the positive integral of the circle function, uniformly in the linear slopes. All axes and the chosen return may be taken arbitrarily late.

The rational saturation is polynomially separated Write $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ for the standard quotient. Fix for each p an integer c_p with $4c_p \equiv 1 \bmod p^3$. Use the usual p-adic additive character $[\cdot]_p : \mathbb{Q}_p \to \mathbb{Q}/\mathbb{Z} \hookrightarrow \mathbb{T}$ with kernel $\mathbb{Z}_p$ and $[a/p^k]_p = a/p^k \bmod 1$. The group polynomial

$$Q(g) = \sum_p [c_p p^{-3}(p^2 g)^4((p^2 g)^2 - 1)^2]_p$$

has degree at most eight on the additive rationals. The sum is finite on each input, as its summands vanish at all primes where $g \in \mathbb{Z}_p$; the same finite union applies for every finite-

difference identity. If $g \in \pi^{-1}V$, then $p^2 g \in \mathbb{Z}_p$ and $p^2 g \equiv r_p \bmod p^2$. At an active residue $(s + pt)$ the numerator before c_p is $4p^2t^2 \bmod p^3$, and at (pt) it is zero. These statements hold for all $p^2\mathbb{Z}_p$-lifts by the same expansions. Thus $(Q(g))$ equals the charge and avoids a fixed zero neighborhood.

For detail on the capacity consequence write $Q = Q_4 + Q_6 + Q_8$, each $Q_j(g) = \chi_j(g^j)$ for an additive circle character of $\mathbb{Q}$, by collecting like terms of the finite p-adic expression. One of these three components has norm at least $\gamma/3$ on every allowed difference. For a small prescribed arc use the degree-j polynomial hitting test of the Euclidean-transfer note [R125], with M a common denominator for the three tests. On the rational group take the observable $g \mapsto \chi_j(g^j/(j!M))$; the jth difference on a line, multiplied by M, is exactly $Q_j(d)$. Delete a random translate of the small arc in each of the three observables. The surviving set avoids one uniformly bounded length and has arbitrarily high density in a stationary realization: sample the arc offsets and average the translated binary names over a Følner sequence in the countable group, then take a weak limit. The expected single-site mass exceeds one minus three arc lengths and the violations are identically zero. Hence $b_{\mathbb{Q}}(\pi^{-1}V) = 1$. The same capacity holds after the integer realization of this rational model. A bounded number of fixed vertical residue centers can sometimes admit similar rational Hermite lifts; the closure on the saturated group must be checked separately for every proposal.

R208. Pure dyadic marks in a finite register over independent axes

Scope and status. Result with proof.

Result. Pure dyadic marks return at every positive density in finite registers over independent local axes.

Scope. The register is a fixed finite abelian group and the base is an independent product.

Here is an all-positive recurrence result for a subclass of free actions in the pure-mark construction [R098]. In particular it permits arbitrarily complicated measurable local tables, with different laws on every axis. Only the normal residue 1 is used.

Fix a row with modulus $L = 2^K$, and let $z_i = 2^{D_i}$, where the depths go to infinity. Take independent stationary integer systems (Y_i, S_i), a fixed finite abelian group B, and arbitrary measurable $f_i : Y_i \to B$. Axis i acts on $\prod Y_i \times B$ by shifting its local state and adding $f_i(S_i y_i) - f_i(y_i)$ to the common Haar register. No infinite sum of the f_i is required. A legal direction may use finite support, arbitrary normals 1 mod L, and **exactly one** other coordinate with value $z_i t$, t odd. At every positive measure, such directions yield progressions of every finite length. The same proof works on any tail. Inverse limits follow by approximation if their directions and row indices agree.

Control of a marked local profile. Fix a length h. For a velocity v put $P_i(v) = (f_i(S_i^{jv} y_i) - f_i(y_i))_{1 \le j < h} \in B^{h-1} = K_0$. For a character $\chi = (\chi_j)$ of K_0, let

$$g_i(\chi) = \sup_{v \equiv 1 \pmod L} \big(1 - \Re\mathbb{E}\chi(P_i(v))\big).$$

There is one positive odd integer t depending only on h, B, L for which

$$1 - \Re\mathbb{E}\chi(P_i(z_i t)) \le C_{h,B} g_i(\chi)$$

on all sufficiently late axes. To see this, write the character exponent in the additive group of order $b = \exp(B)$, with functions $F_j = \chi_j f_i$ written additively, and $F_0 = -\sum_{j>0} F_j$. For any normal v, the probability that $\sum_{j=0}^{h-1} F_j(S_i^{x+jv} y_i) \neq 0$ is $O_b(g_i(\chi))$, uniformly in the integral starting shift x. The trivial register needs no argument. Fix $j > 0$, and choose w divisible by $L \operatorname{lcm}(1, \dots, h-1)$. Apply the $h-1$ commuting differences on the variables (x, v) with increments $(-sw/(j-s), w/(j-s))$, one for each $s \neq j$. Start at $v = 1$. Every tested velocity is normal. All slots except j cancel exactly, and we obtain $\Delta_w^{h-1} F_j(S_i^x y_i) = 0$ except on an event of measure $O_{h,b}(g_i)$, uniformly in x, by stationarity.

Choose C a positive multiple of $b(h-1)!$, at least h, so $b \mid \binom{C}{s}$ for $1 \le s < h-1$. Let t be the odd part of $CL \operatorname{lcm}(1, \dots, h-1)$. For sufficiently large D_i, take $w = z_i t / C$, which satisfies the required divisibilities. In $(1 + \Delta_w)^C - 1$ the terms of degree less than $h-1$ vanish modulo b. The others are fixed sums of translates of Δ_w^{h-1}. Thus $F_j(S_i^{z_i t} y_i) = F_j(y_i)$ except on a set of probability $O_{h,b}(g_i)$. Iterating at most $h-1$ times gives the same estimate for $j z_i t$, simultaneously over j, proving the bound.

Tail averaging. Let $\Lambda = \{\chi : \sum_i g_i(\chi) < \infty\}$. It is a subgroup by the elementary estimate on the squared distance from 1 of the product of two unit phases. Write $P = \Lambda^\perp \le K_0$. In a far finite pool select, independently on each used axis, its normal velocity or zero with probability 1/2 each. Pools and velocities may be chosen so that the distribution

of the sum of their profiles tends to Haar on P, in total variation. Indeed for each of the finitely many characters outside Λ, allocate pairwise disjoint finite subpools in succession with arbitrarily large sum of g_i, and at each allocated site select a velocity with loss at least $g_i/2$. The corresponding Fourier product tends to zero since $|(1+u)/2|^2 \le (1+\Re u)/2$ for $|u| \le 1$. For a character in Λ, the product tends to one since all individual deficits are summable in the tail: on the finite phase alphabet, $|1-\mathbb{E}\chi| \le C_b(1-\Re\mathbb{E}\chi)$. The same conclusion holds after adding a fresh, independent profile $P_i(z_i t)$, by the first estimate on Λ; outside Λ the previous vanishing is unaffected. Fresh disjoint pools can be used for each of any fixed number of independent trials. For any nonempty subset of trials their sum again has limiting profile Haar on P, with or without the one additional marked site.

Uniform return for measurable events. Start with an event E depending only on a finite head and on the register, with mass $a > 0$. Choose m so that any density $a/2$ subset of the h-letter m-dimensional word cube contains a combinatorial line. Give each word its site under the sums of m independent unmarked trial directions. Under the stationary initial law, its expected density in E is a. The sum over all lines of their probabilities of lying in E is therefore at least $a/2$. On a line with nonempty wildcard set J, translating away the fixed letters leaves the ordinary h-slot correlation at the sum of the J-directions; stationarity permits this for each choice of the velocities. In that correlation the head is unchanged, the initial register is independent uniform, and all register displacements have exactly the profile law calculated above. Every line therefore has the same limiting correlation, at least $a/(2N)$ for N total lines. Adding the one marked site to any such sum gives the same limiting law and hence the same lower bound. For a general event of mass at least $a > 0$, approximate it in measure by head-register events of mass at least $a/2$; fix the word dimension using this smaller mass first. Errors in all correlations are at most h times the approximation error, uniformly over every direction. Thus a genuine legal return follows.

The independence used here is between the local axes, conditional only on the initial Haar register. This does not treat arbitrary actions of the free abelian host; in particular its fixed finite register cannot model the growing cyclotomic obstruction.

R218. Restricted AP products with a coprime global charge

Scope and status. Result with proof.

Result. Pairwise-connected finite AP query products with a coprime nonconstant charge have an ordered contradiction and exponential charged cancellation at every fixed length.

Scope. The cancellation constant depends on the local query law; uniformity over covers requires R260.

Qualifications. Translation symmetrization includes binary charges at every finite length when the charge is coprime to the underlying group order. The separate BCH argument retains its own characteristic restriction.

Let G be a finite abelian group; local queries have the form $(x + jv)_{0\le j<h}$ for every x in G and v in a set V containing 0, with uniform x and positive probability for every velocity. The target is $\tau(v) \in F_\ell$, with $\tau(0) = 0$. Suppose every pair of player views gives a connected bipartite support graph. This holds, for example, on a cyclic prime group larger than h with two distinct permitted velocities, and on mixed prime pairs when the velocity differences generate the group and the small nonzero differences of slot indices act invertibly.

At $h = 3$, if ℓ is prime to $|G|$, every nonconstant τ admits an ordered contradiction for its XOR game. At $h = 4$, assume furthermore ℓ is odd and there are $a, b, c, d \in V$ with $a+b = c+d$ but $\tau(a)+\tau(b) \neq \tau(c)+\tau(d)$. Again the game admits an ordered contradiction. The target consequence of either assertion is exponential cancellation for the *single global charge* against all bounded rectangular tests on tensor powers. This is stronger than quoting the general AND-of-games theorem, which is distinguished in the full-velocity note [R031].

The main result, proved immediately below, covers every finite length: **for all $h \ge 3$ and all $\ell \nmid |G|$, any nonconstant target has exponential cancellation against arbitrary bounded rectangular tests on tensor powers.** In particular every positive-density set in sufficiently high identical powers admits a charged h-term progression, using the Hales–Jewett mixture below. The subsequent sections give complementary lower-class arguments and a separate characteristic-sensitive scalar-embedding extension.

Translation symmetrization at arbitrary class Use the group H, the free groups F_j and degree-zero kernels K_j defined in the elementary group argument below. Pairwise connectivity places every isolated $\gamma_{h-1}(K_i)$ in $[H, H]$. Write $c = h-2$; the quotient by these isolated subgroups embeds on the K-part in a product of groups nilpotent of class at most c. Fix a base query tuple e_0 and let $g_{x,v} = e_{x,v}e_0^{-1}$. These elements have all coordinates in K_i. For two velocities v,w their ordered lists of player-j projections, indexed by $x \in G$, differ by the fixed translation $j(v-w)$ of the index set. This is more than agreement as multisets.

For any G-indexed list z of elements of a group N and index translation $u \cdot z = (z_{x+u})_x$, define a positive word recursively by

$$F_1(z) = z_0, F_{j+1}(z) = \prod_{u\in G} F_j(u \cdot z),$$

using the same arbitrary fixed order on G at every application. By induction $F_j(z)$ is invariant under index translations modulo $\gamma_j(N)$ (the start $j = 1$ is tautological). Indeed all factors on the right then have the same image modulo $\gamma_j(N)$, so commute with one another

modulo $\gamma_{j+1}(N)$. Translating the original list only permutes those factors. Hence F_{c+1} is exactly invariant in every class-c group, and has length $|G|^c$.

It follows that $F_{c+1}((g_{x,v})_x)F_{c+1}((g_{x,w})_x)^{-1}$ has every coordinate in $\gamma_{h-1}(K_i)$ and so is in $[H,H]$. Were there no ordered contradiction, its target value would be $|G|^c(\tau(v)-\tau(w))$, nonzero for differently tagged velocities. This contradiction proves the all-length claim. No BCH denominators, coefficient-size restrictions or Freiman relation on V enter this proof. The tensor cancellation step uses Braverman–Minzer, Lemma 3.5. The constants are for a fixed local law. Applying the argument to successively deeper prime-group covers separately is not a cover-uniform bound.

Here are the elementary group arguments. Let F_j be free on the symbols $u_{j,x}$, let K_j be its zero-exponent subgroup, and let H in the product of the F_j be generated by the query tuples $e_{x,v}=(u_{j,x+jv})_j$. An ordered contradiction is a word in these tuples and their inverses whose product in every F_j is trivial but whose signed sum of targets is nonzero. In its absence τ extends to a homomorphism $H\to F_\ell$ sending $e_{x,v}$ to $\tau(v)$. For each pair $i\neq j$ one can prescribe any element of K_i and simultaneously the identity at j in a suitable element of H. Indeed alternating walks in the connected bipartite graph give all elementary differences between letters on the i side while the j letters cancel. The full fiber product of the two free groups with equal ordinary exponent is generated: differences $u_xu_z^{-1}$ and their conjugates by the surjective first projection generate the zero-exponent kernel. Thus iterated commutators of $h-1$ elements of K_i can be lifted one factor at a time while killing a different one of the other players. Their iterated commutator has only the i-coordinate nontrivial and lies in $[H,H]$. Conjugation by H supplies the normal generation in K_i. In particular each isolated $\gamma_{h-1}(K_i)$ lies in $[H,H]$.

For $h=3$ compare two velocities v,w with different targets. For arbitrary order on G form $P=\prod_x(e_{x,v}e_{x,w}^{-1})$. Its jth coordinate is in $[K_j,K_j]$: writing $k_y=u_{j,y}u_{j,0}^{-1}$, the sum in the abelianization of K_j is $\sum_x(k_{x+jv}-k_{x+jw})=0$. The isolated commutator result places P in $[H,H]$. But its prescribed tag is $|G|(\tau(v)-\tau(w))\neq 0$, contradiction.

For $h=4$, make a list of all the words $e_{x,a}e_{x,c}^{-1}$ and $e_{x,b}e_{x,d}^{-1}$ for $x\in G$. At each coordinate the multiset represented by this list is invariant under inversion, since the two ordered pairs of symbols differ by opposite shifts. Modulo $\gamma_3(K_j)$ form the product of this list followed by the identical list in reverse order. In a class-two group this palindrome is unchanged by reordering the original list: a swap introduces opposite central commutators in the respective halves. At a given j reorder to put inverse pairs next to one another, making its palindrome the identity. Different reorderings may be used to check the different projections of the same original word. Thus the original product lies in $[H,H]$. Its target is $2|G|(\tau(a)+\tau(b)-\tau(c)-\tau(d))$, nonzero under the hypotheses. This proves the four-slot claim. This particular four-slot argument does not treat characteristic two because of its factor 2; the translation-symmetrization proof above covers binary charges at every finite length in the coprime setting.

For each nontrivial charged character, an ordered contradiction implies a fixed finite Cauchy–Schwarz gluing proof reducing to a game without players with nonconstant target. The gluing is compatible with tensor powers, so for some $\rho<1$ independent of n the absolute integral of the product charge against arbitrary bounded player tests is $\le\rho^n$. This is the character-correlation conclusion of Braverman–Minzer, Section 3, in particular Lemma 3.1 and Lemma 3.3; no BLR reduction or inference from a parallel-AND value is needed.

Using cancellation without an uncharged same-set theorem. At either proven length, and at any other fixed h for which the same rectangle bound is available, take a

density Hales–Jewett dimension $m(a,h)$, partition $n = mL$ coordinates into m blocks, sample independent local velocities in each block and call their block vectors D_1,...,D_m. With independent uniform X, consider all words $X + \Sigma t_i D_i$ with $0 \le t_i < h$. A positive proportion in expectation belong to A, so with probability bounded below a full word line belongs. Each fixed line produces direction $D_I = \sum_{i \in I} D_i$ with nonempty I. Its starting translate is uniform independent of this direction. Therefore the sum of uncharged correlations over the fixed finite list of lines has a positive lower bound depending only on a,h. For the same test weighted by any nontrivial character of the global tag, freeze the coordinates outside I (their velocities are zero) and apply the rectangle bound on the $L|I|$ active-block coordinates, obtaining absolute value at most ρ^{L}. Fourier-expanding the indicator of any nonempty subset of F_ℓ gives a positive total charged return for L sufficiently large. Directions sampled here may have zero on entire blocks, as the uncharged law is an HJ mixture, not the single fixed product query law. The result does not apply when an upper bound on the total number or cost of true marks excludes these samples.

Arbitrary nonconstant targets through length ℓ+1 Write $c = h - 2 \ge 1$ and assume ℓ>c. Choose v,w with different target, and any one fixed base query tuple e_0. Make a new list of elements g_s consisting of $e_{x,v} e_0^{-1}$ for all x and $(e_{x,w} e_0^{-1})^{-1}$ for all x. Every g_s has zero exponent at each coordinate. Its coordinate list, counting multiplicities, is invariant under inversion: the letters observed at player j run over exactly the same alphabet at velocity v and at w, and the right base letter is fixed. The signed target sum of this list is $T_0 = |G|(\tau(v) - \tau(w))$, which is nonzero. In particular for $h = 4$ one can use this list directly in the palindrome argument; no Freiman relation is required.

Work in $\prod_i K_i/\gamma_{c+1}(K_i)$. The factors are free nilpotent groups of class at most c: K_i is a subgroup of a free group, hence free, and only finitely many of its free generators enter these words. They are torsion-free and embed into their rational BCH groups. See the rational BCH hull construction. Equivalently one can work from the outset in the BCH group of a free class-c rational Lie algebra on the finitely many free generators used here.

For integer k define $W_2(k) = \prod_s g_s^k$ in any fixed order. Its logarithm is a polynomial in k, each degree practically equal to the bracket length in the logs of the original letters g_s; hence it has degree $\le c$, and degree-one part zero by the coordinatewise inversion symmetry. In the last statement the letters for the bracket-length accounting are the whole fixed logs of the g_s, each given degree one; they need not be homogeneous in the original free generators. Let A represent a primitive unit modulo ℓ. For $j = 2, \dots, c$ recursively put

$$W_{j+1}(k) = W_j(Ak) \cdot W_j(k)^{-A^j}.$$

If the logarithm of $W_j(k)$ begins at degree $\ge j$, the degree-j term cancels in this new logarithm. The additional BCH commutators have degree at least 2j and the degree bound c still holds, since any term of degree r is a linear combination of length-r brackets of the original fixed logs. By induction $W_{c+1}(1)$ is the identity in every class-c factor. For $c = 1$ the initial word at $k = 1$ is already the identity in the abelian quotient. The original word is thus in the product of isolated $\gamma_{h-1}(K_i)$ subgroups and hence in $[H, H]$.

On the other hand its target is

$$T_0 \prod_{j=2}^{c} (A - A^j) \bmod \ell,$$

which is nonzero because $1 \le j - 1 < \ell - 1$. Thus no target homomorphism exists on H, and the ordered contradiction and exponential global-character bounds follow. The HJ mixture then gives a charged h-return at every positive density for all sufficiently high powers of this base query. Once h$\ge\ell$+2, the j=ℓ factor vanishes for every choice of A; this argument supplies no deduction at such lengths. Also its constants may depend on the entire fixed local alphabet, so applying it separately at finite cover depths does not supply a uniform cover comparison. The uniform-velocity Gowers argument is distinct in having cover-independent constants.

The BCH proof also applies outside coprime translation games. For *any pairwise-connected finite h-player query game*, absence of a scalar circle embedding is equivalent to a signed integer trade λ_e of its permitted queries: every individual player/letter signed multiplicity is zero, but $\sum_e \lambda_e \tau(e)$ is nonzero mod ℓ. This follows from dividing the free abelian group on player/letters by the query subgroup: every circle character of a subgroup extends to the free abelian ambient group, so failure of extension from the prescribed query targets means that one of the integer relations violates them. For a fixed base query e_0, each coordinate of ee_0^{-1} depends only on its player letter. Thus the linear term of $\log \prod_e (ee_0^{-1})^{k\lambda_e}$ vanishes identically by the integer trade. Its target at $k = 1$ is the nonzero original sum, since $\Sigma\lambda_e = 0$. Apply the same BCH cancellation procedure for $c = h - 2 < \ell$. It yields an ordered contradiction without any hypothesis on the size of the alphabets or on a translation symmetry. Consequently for such h, any non-scalar-embeddable tag has exponential tensor character cancellation even when ℓ divides the alphabet group's order. A scalar embedding conversely gives product tests with absolute correlation one. In the coprime AP setting the full translation symmetrization above bypasses this restriction on ℓ.

R224. Every positive density on totally disconnected horizontal Host–Kra fibers

Scope and status. Result with proof.

Result. Shallow primary directions recur in systems with totally disconnected horizontal Host–Kra fibers, with a stated penultimate-factor variant.

Scope. A restricted structural class, even though the top compact fiber may be arbitrary.

Qualifications. Connected higher horizontal factors remain outside this theorem.

On the shallow primary model [R226] write $G = \bigoplus C_{p^{D_p+1}}$, $H = \bigoplus pC_{p^{D_p+1}}$, $D_p \to \infty$. Suppose the order-h horizontal characteristic factor in an arbitrary G-system has totally disconnected structure groups on almost every ergodic H-component. Then every positive event has an h-return on every tail of the quadratic directions. The assertion already works with one true mark, plain positive low units, and arbitrary passive horizontal corrections. One may use order $h-2$ for the h-term characteristic factor when $h \geq 3$; any safely larger fixed order also suffices. This leaves the systems with connected horizontal structure groups unresolved.

More generally the argument for a positive event B on the factor needs only that its restriction on almost every horizontal component is approximable on a generalized Weyl extension by finite sums of bounded-degree polynomial phases. No simultaneous lift of the vertical moves or of the different component extensions is assumed.

Descent of a polynomial on an integer lift of one cyclic axis Let $p > s$, $N = p^D$, and $P(n)$ a circle polynomial of degree at most s on the integers. The kernel average

$$F(n) = \lim_{M\to\infty} E_{1\leq m\leq M} \exp(2\pi i P(n+Nm))$$

defines an N-periodic word. In $L^2(C_N)$ it is within $\sqrt{s/p}$ of a constant of modulus at most one times a polynomial phase with denominator N and degree at most s. All estimates are uniform in P and D.

If any nonconstant coefficient in the ordinary monomial representation of P is irrational, its highest such coefficient remains irrational at the same degree in every shifted polynomial in m, so the mean vanishes by Weyl's criterion. Otherwise split the rational coefficients into their p-primary parts and coprime parts; an arbitrary fixed circle constant is harmless. By CRT the coprime polynomial contributes a constant of modulus at most one, independent of n, since N is invertible modulo its period. If the denominators of the nonconstant p-primary coefficients divide N, its contribution is exactly the asserted polynomial. Otherwise let p^{D+b}, $b \geq 1$, be their largest denominator, and write the primary polynomial $A(n)/p^{D+b}$ with a nonconstant coefficient of A a p-unit. In the kernel average shift the uppermost digit of m by p^{b-1}. This changes the phase by $A'(n)/p \bmod 1$, since $Nm \equiv 0 \bmod p$. The mean is therefore zero unless $A'(n) = 0 \bmod p$. This nonzero polynomial modulo p has at most $s-1$ roots. The remaining word has L^2-norm at most $\sqrt{s/p}$, so use the zero approximant.

Componentwise approximation, stated on the original fibers For a K-action, a generalized extension means an extension for a covering group together with a surjection

of that group onto K, with the given projected dynamics. Pull the action back along any surjection from a free countable abelian group; this preserves ergodicity, the cube joinings and the structure groups since the image is the full acting group. Applying the torsion-free Jamneshan–Shalom–Tao, Corollary 1.7(i) there gives a generalized Weyl extension for the original totally disconnected system. Part (ii) of the corollary is stated for bounded torsion; that hypothesis is not being applied to H. A Weyl system has its L^2 space densely spanned by bounded-degree circle polynomial phases: induct on its polynomial-cocycle tower and expand in characters of each vertical compact abelian structure group.

Fix one horizontal ergodic component y of the stated factor and an event B. Pull 1_B to this extension and approximate it in L^2 by a finite sum of phases. For an axis generator $a_p = pe_p \in H$, of order p^{D_p}, use any single lift to the free group. Project the finite sum by averaging the powers of p^{D_p} times that lift. This is an L^2-contraction and leaves the pulled-back 1_B unchanged. On almost every orbit of the lift, each phase is an ordinary bounded-degree circle polynomial in the time variable; by the pointwise ergodic theorem its projected word is the periodic kernel average just studied. The descent lemma changes each such word to a scalar multiple of a degree-bounded ordinary phase on $C_{p^{D_p}}$, up to $O(\sqrt{s/p})$ in the normalized orbit norm. By stationarity and contraction, the expected orbit approximation error of the original event is consequently at most the chosen L^2-error plus $O(\sqrt{s/p})$ times the sum of absolute coefficients. The conclusion lives downstairs: the original p^{D_p}-orbit word at a point admits an approximation with these bounds, independent of which lift of the point gives it.

One can take the orbit approximants uniformly bounded by two for any prescribed small integrated error. First project the real part of the sum pointwise to $[0, 1]$, which only improves the error against the indicator. Approximate this clipping function uniformly on the bounded input interval by a fixed real polynomial. Products of a finite number of ordinary phase terms are again ordinary phase terms of the same maximal degree. This gives an approximant of modulus at most two and with some finite number, coefficient bound, and degree of terms, independent of the prime. The error still tends to the originally chosen accuracy. The degree bound on the ordinary input polynomials may of course depend on the particular component.

To work simultaneously across components, for a fixed accuracy define on the downstairs finite orbit word at x the least $L^2(C_{p^{D_p}})$ error from at most K ordinary polynomial phase terms with coefficients of modulus $\le K$, degree $\le K$, and a resulting sup norm at most two. This error is a measurable function of the word: polynomial coefficients in the cyclic phases take finitely many values, their external complex coefficients range over a compact set, and the sup bound is closed. Measurable minimizers exist by finite enumeration and compact coefficient minimization, or use a countable rational approximation with a slightly enlarged sup bound. Integrate this least error over each horizontal component. For almost every component and all sufficiently large primes the integrated least error is below the desired accuracy for some finite K, by the preceding construction. The sets of components where this holds for the same K and *all* $p > K$ are measurable, increase with K, and exhaust almost surely. Thus a common K works outside a base set of arbitrarily small measure. This avoids selecting the extensions on a measurable family or lifting any vertical transformations to them.

Perturbation by one true mark Fix h and a positive B on the factor. Density Hales–Jewett gives a bound $m = m(h, \mu(B))$ and $\kappa > 0$ such that some nonempty subsum v of m

arbitrary fresh low unit axes has $\mu(C) \geq \kappa$, $C = \bigcap_{j<h} U^{-jv} B$. For any chosen active p and $z = p^{D_p - 1} a_p$ of order p, averaging the h-correlation at $v + tz$ over uniform $t \bmod p$ is at least the average ordinary C-return at tz. This is bounded below uniformly by $c(\kappa, h) > 0$, for example by the density Hales–Jewett averaging argument on the finite cyclic group. Choose the approximation accuracies and exceptional base mass sufficiently small in terms of this constant and h, then choose the low trial primes beyond the resulting K's, large enough also for the cancellation below. This order of choices is independent of the actual subsum or its transported component spectra.

For each fixed nonzero frequency k of a cutoff for $\|t^2/p\| > \gamma$, estimate the correlation average with the extra factor $e(kt^2/p)$. Introduce an independent common starting displacement $n_0 a_p$, n_0 uniform in $C_{p^{D_p}}$, which does not change the integrated correlation. At a point x, approximate the h finite orbit words based at $U^{jv}x$ using the preceding downstairs condition. In their product one reads the j-th word at $n_0 + jp^{D_p - 1} t$. The independent uniform n_0 makes each integrated replacement error cost at most its mean orbit error, uniformly in t, j, v. The exceptional component mass at any of the h slots costs at most a constant depending on h times the chosen exceptional mass, by invariance and the bounded approximants. Approximants can be fixed as functions of x, independent of n_0, t.

Each ordinary denominator-p^{D_p} phase at $n_0 + jp^{D_p - 1} t$ is linear in t for $D_p \geq 2$. Terms of time degree two or above have coefficient divisible by $p^{2(D_p - 1)}$. For every fixed n_0, the product expansion times the charged frequency therefore has t-mean $O(p^{-1/2})$ times a constant depending only on the uniformly bounded phase list. The small exceptional and approximation errors were fixed first. A finite trigonometric approximation to a nonnegative bump supported where $\|t^2/p\| > \gamma$, of positive mean, now shows a weighted return is positive. It yields the allowed $v + tz$.

Finally if an arbitrary event avoids the full family, pass to the support of its projection on the common relative horizontal characteristic factor as in the horizontal reduction [R014] and the triple proof [R058]. In this locally finite horizontal group with p-power axes, multiplication by all relevant nonzero small slopes is bijective on late finite subgroups, and omitting the finitely many active axes is passage to a finite-index subgroup; the same seminorm comparisons work without requiring prime rather than prime-power orders. The positive return for the factor support lifts after averaging over the wholly passive coordinates, giving the contradiction.

Thus on the shallow host any genuine finite-horizon avoidance witness requires a positive-measure set of horizontal components whose corresponding finite-order factor is not totally disconnected. The argument proves recurrence already with one charged mark in the stated class, but controls neither all connected structure groups nor the unrestricted all-horizon threshold.

The top compact structure group need not be totally disconnected For an ergodic horizontal system of order $s \geq 2$ it actually suffices that its **penultimate factor** $Y = Z^{s-1}$ be totally disconnected. Over the free acting-group cover write the system as $Y \rtimes_\rho U$, with an arbitrary compact abelian U and a type-s cocycle. Take a totally disconnected, ergodic, ∞-divisible Weyl extension $\widetilde{Y} \to Y$ by the same Corollary 1.7(i), and pull the skew product back to $\widetilde{Y}$. The cocycle remains of type s: a witnessing cube coboundary pulls back under the factor map, and the Host–Kra cube measures project to the original ones. On this ergodic base apply Jamneshan–Shalom–Tao, Proposition 5.2 and the proof of Corollary

5.4 separately to each character $\chi \in \widehat{U}$. Its scalar cocycle is cohomologous to a cocycle whose γ-increments are bounded-degree polynomials on $\widetilde{Y}$. Indeed the genuine cocycle meets the quasi-cocycle hypothesis trivially, yielding the asserted scalar quasi-coboundary; no simultaneous character transfer is needed here.

Consequently the vertical character corrected by that base transfer is a phase polynomial on the pulled-back skew product, since its first derivatives are those base polynomial increments. Any finite vertical Fourier expansion is approximated in L^2 by finite sums of polynomial phases: absorb each transfer into its corresponding measurable base coefficient and approximate that coefficient on the Weyl base. No ergodicity of the pulled-back skew product is required by the cyclic-orbit argument. For $s = 1$ the original ergodic compact rotation already has characters spanning L^2. Thus the preceding componentwise proof applies under this weaker hypothesis. In particular at four slots it is enough that the horizontal Kronecker factor be totally disconnected; the final compact abelian fiber of the Conze–Lesigne system is unrestricted.

R225. A genuine horizontal Conze–Lesigne resonance on the high subgroup

Scope and status. Result with proof.

Result. A concrete Heisenberg model exhibits a nonzero horizontal Conze–Lesigne quadratic resonance on the high subgroup.

Scope. An explicit counterexample to a proposed uniform weighted-cancellation estimate.

Qualifications. The resonance is not itself a progression-avoiding event or an all-system nonrecurrence theorem.

An estimate obtained on totally disconnected penultimate factors [R224] fails on a single fixed connected-by-profinite two-step model. This concerns the proposed weighted-cancellation argument on a single horizontal axis. It gives neither an avoiding event nor a counterexample to four-term recurrence. The construction realizes the resonant motions simultaneously as commuting torsion translations.

Take odd primes p tending to infinity, $N_p = p^{D_p}$, $D_p \to \infty$. Set $H = \bigoplus_p C_{N_p}$, $J = \prod_p C_{N_p}$, with pairing $\langle a, j\rangle = \sum_p a_p j_p / N_p$ into the circle. On $\mathbb{R}^2 \times J \times H \times \mathbb{T}$ take the group operation

$$(r, j, a, z)(r', j', a', z') = (r + r', j + j', a + a', z + z' + r_1 r_2' + \langle a, j'\rangle).$$

It is locally compact, second countable and two-step nilpotent. The commutator pairing on its abelian base is $\omega(r, r') + \langle a, j'\rangle - \langle a', j\rangle$, where $\omega(r, r') = r_1 r_2' - r_2 r_1'$.

Put $v_p = (p, 2)/N_p$, $s_p(n) = N_p \omega(v_p, n) \bmod N_p$ for $n \in \mathbb{Z}^2$, and $s = (s_p)_p$. In the abelian base generate a subgroup Λ_b by $l(n) = (n, s(n), 0)$ and $l_p = (-v_p, j_p, e_p)$. Take $(j_p)_p = 0$ and at $q \neq p$ set

$$(j_p)_q = -[N_p^{-1}]_q s_q(N_p v_p) \in C_{N_q},$$

with inverse taken modulo N_q. We have $N_p l_p = l(-N_p v_p)$, also at the pth coordinate since $\omega(v_p, N_p v_p) = 0$. Thus the intersection with the open zero-H subgroup is precisely the discrete graph $l(\mathbb{Z}^2)$. The full subgroup is closed, discrete and cocompact by reducing the H-coordinate and then the two real coordinates.

The commutator of two lattice generators is zero in the circle. For $l_p, l(n)$ this is the definition of s_p. For $p \neq q$, decompose $\omega(v_p, v_q) \in \mathbb{T}[N_p N_q]$ into its p- and q-primary parts. Multiplication by N_q shows $\langle e_p, j_q\rangle = -\omega(v_p, v_q)_{(p)}$; similarly $\langle e_q, j_p\rangle = \omega(v_p, v_q)_{(q)}$. For $l(n), l(n')$ it follows by integrality. Hence the central extension over Λ_b is abelian. Injectivity of the divisible abelian group $\mathbb{T}$ splits it. We can use any fixed splitting on the subgroup $l(\mathbb{Z}^2)$: differences of splittings are characters and characters on a subgroup extend because $\mathbb{T}$ is injective. Use vertical coordinate zero there, which is a splitting because $n_1 n_2'$ is integral. Write Λ for the split lattice in the full group. It is still discrete and cocompact. The quotient has Haar probability, represented in coordinates by independent uniform $0 \leq r_1, r_2 < 1$, $j \in J$, $a = 0$, $z \in \mathbb{T}$.

The horizontal group acts by the elements $a_p = (0, 0, e_p, 0)$, genuinely of order N_p, all commuting. It acts ergodically. First the induced rotation on the compact abelian quotient

by the center is ergodic. A character there has the form $(r, j) \mapsto \alpha \cdot r + \langle b, j\rangle$, with $b \in H$ and α rational since it vanishes on $l(\mathbb{Z}^2)$. Evaluation at the p-generator is $\alpha \cdot v_p - \langle b, j_p\rangle$ mod one, a value of order dividing N_p. For p outside the finite support of b and the denominators of α, the second term has order coprime to p, so the value is exactly the p-primary part of $\alpha \cdot v_p$. Writing $m\alpha$ integral, annihilation of every such a_p implies $N_p \mid (m\alpha_1)p + 2m\alpha_2$. Since $D_p \to \infty$ this forces $\alpha = 0$. The lattice condition and coprime splitting now force $b_p s_p(n) = 0 \bmod N_p$ for every n, p, whence $b_p = 0$ by surjectivity of s_p.

In each given nonzero vertical frequency k, an H-invariant function has constant modulus by ergodicity on the base, and any two are proportional by taking their ratio there. The connected real Heisenberg subgroup commutes with H and preserves the vertical frequency. If an invariant function existed it would act on its one-dimensional complex span by scalars, and hence trivially through commutators. But the commutators of the real Heisenberg subgroup fill the whole central circle, acting by the nontrivial vertical frequency. Vertical Fourier expansion proves ergodicity. In particular this is a two-step translational model of Conze–Lesigne type; equivalently its central cocycle over the compact base rotation has the Conze–Lesigne translation equation, by lifting each base translation in the group and using central commutators with H.

Let $m = N_p/p$, $g = a_p^m$, and $F(x) = \exp(2\pi i z)$ in the above fundamental domain. To evaluate $a_p^n x$, multiply on the right by the inverse nth power of a lift of l_p. This returns to zero H-coordinate and gives real coordinates $r + nv_p$. The central coordinate changes by a term linear in n, depending on x, p, and $\frac{1}{2}n^2 v_{p1} v_{p2}$. Indeed the central part of the inverse power contributes $\binom{-n}{2} v_{p1} v_{p2}$, and the pairing with $-nj_p$ from the H-coordinate vanishes since $(j_p)_p = 0$. At $n = mt$ the real increments are $t, 2t/p$. Reduce the real first coordinate by the integer t using the chosen lattice splitting, and then the second by $\lfloor r_2 + 2t/p\rfloor$. Only the second reduction changes the central coordinate, by $-r_1\lfloor r_2 + 2t/p\rfloor$. Thus at all integer t,

$$F(g^t x) = e(z + t\theta_p(x) + t^2/p - r_1\lfloor r_2 + 2t/p\rfloor), e(u) = \exp(2\pi i u).$$

In the product $F(x)\overline{F(g^t x)}^2 F(g^{2t}x)$ the quadratic term is $2t^2/p$. After multiplying by $e(-2t^2/p)$, integration in r_1 leaves the indicator of

$$\lfloor r_2 + 4t/p\rfloor - 2\lfloor r_2 + 2t/p\rfloor = 0.$$

Its remaining integrals are nonnegative. For $0 \le t < p/8$ and $0 \le r_2 < 1/2$ it equals one. Hence the weighted mean over uniform $0 \le t < p$ has liminf at least $1/16$. A common extra random shift along the original p^{D_p}-orbit does not change the answer, by invariance. Three slots suffice to disprove cancellation for general bounded test functions. Adding an identically-one slot gives a four-slot failure of the same estimate. Multiple recurrence with independent additional passive coordinates remains possible.

Exact curvature simultaneously on every high axis With the modified observable $\widetilde{F} = e(z - r_1 r_2)$, every floor disappears. Along $g_p = a_p^{N_p/p}$ the first real coordinate is unchanged. In the preceding computation the linear cross term is $2r_1 t/p$. Subtracting r_1

times the updated second real coordinate cancels both it and the floor term. The result, with a constant c_p independent of the point, is

$$\tilde{F}(g_p^t x)/\tilde{F}(x) = e((j[p]/p + c_p)t + t^2/p).$$

Here $j[p]$ means the p-coordinate of j reduced modulo p. Under one high move g_q the update of $j[p]$ modulo p is 2 when $q = p$, since $(j_p)_p = 0$, and is zero when $q \neq p$. Indeed the correction from the two real lattice reductions is $-s_p(1, \lfloor r_2 + 2/q \rfloor) \equiv 2 \bmod p$, and the q-lattice correction at $q \neq p$ is

$$-(N_q/q)(j_q)_p \equiv (2p - 2q)/q \equiv -2 \bmod p.$$

Thus the spatial functions in the linear terms are individually constant along all other high axes. For any finite packet and any $g = \prod_p g_p^{t_p}$ we obtain the exact identity

$$\tilde{F}(x)\tilde{F}(g^2 x)/\tilde{F}(gx)^2 = e(2\sum_p t_p^2/p).$$

The positive weighted correlation is now one, for every packet and every choice of digits. Thus merely increasing the number of charged high coordinates does not yield cancellation, even on genuine commuting torsion translations in a single ergodic two-step model. The point is that the scalar is a quadratic polynomial phase on the **high subgroup**. The observable on the entire horizontal system need not be a polynomial phase, and the passive orbit data, low marked motions, and positive-event recurrence have not been addressed by the identity.

R226. Shallow primary presentation: identical rational capacities but no native affine Haar witness

Scope and status. Result with proof.

Result. Shallow and deep presentations have identical rational fixed-length capacities, while the shallow native family returns in every compact abelian affine Haar action.

Scope. Native stationary avoiders exist but lie outside the affine Haar class; rational and native capacities must be distinguished.

The integer threshold problem for the deep construction can be tested with local exponents only D_p+1. On that native primary group, every positive event in **every compact abelian affine Haar action** returns at every order, even though a positive avoiding event exists in some measure-preserving action. Nominal group exponents and the actual p-adic separation between the low speeds are distinct issues.

Rational invariance under changing unused residue digits For prime depths $e_p \geq f_p$, write $K_e = \pi^{-1}(\bigoplus_p C_{p^{e_p}}) \subset \mathbb{Q}$ and K_f similarly. The p-residue of $g \in K_e$ is $p^{e_p} g \bmod p^{e_p}$ in $\mathbb{Z}_p$. Suppose a difference family $D_e \subset K_e$ is saturated by the integers and its residue rule is the full coordinatewise pullback of the residue rule for $D_f \subset K_f$, using reduction modulo p^{f_p}. Then at each fixed h,

$$a_h^Q(D_e) = a_h^Q(D_f).$$

For a finite packet in D_e, collect all primes in its denominators in a finite set P. Choose an integer $L \neq 0$ with $L \equiv p^{e_p - f_p} \bmod p^{e_p}$ for $p \in P$. On multiplying every packet element by L, its denominator at p has order at most p^{f_p}, its f-residue is the desired reduction of its e-residue, and no other prime appears. Thus the scaled packet is in D_f. Conversely for a finite packet in D_f, choose $L = U/\prod_{p\in P} p^{e_p - f_p}$, with a nonzero integer U making $p^{e_p - f_p} L \equiv 1 \bmod p^{f_p}$ at each $p \in P$, possible by CRT. Multiplication maps the packet into D_e: its deep residue reduced to p^{f_p} is the original f-residue. Each such scalar gives a homomorphism $\mathbb{Q} \to \mathbb{Q}$, so pullback of arbitrary stationary avoidance witnesses and compactness on finite packets prove the equality, with no change of horizon. Restriction/coinduction identifies the same fixed-order capacities on the corresponding groups K_e, K_f.

Apply this with $e_p = 2D_p + 1$ and $f_p = D_p + 1$ to the deep signs and squared-charge rule. Both integer realizations have the same all-length critical density. A full finite-color obstruction over the shallow rational group can also be obtained directly by taking the local homomorphisms $g \mapsto p^{f_p} g \bmod p^{e_p}$ on K_f and forming their homomorphism to the **product** of the local cyclic groups. For any allowed g all but finitely many coordinates are passive (not necessarily zero); the original nine-slot module stencil vanishes on all of those and realizes the same finite sum of fixed lines. The product coefficient module is still countably generated by the finitely many tuples and countably many individual fixed lines under the countable acting group. Its rank need no longer be finite; integer shifts can change infinitely many coordinates. This separate direct argument is unnecessary for the equality of maximal stationary masses.

Native affine recurrence, in arbitrary dimension On $G_f = \bigoplus_p C_{p^{D_p+1}}$ let V_f be the reduced family, i.e. passive pz arbitrary and active $1 + p^{D_p}t$ with nonzero total quadratic gap (allowing the negative speed as well only enlarges the family). For every affine Haar G_f-system a positive event has all h-term returns on any tail. Use the same finite-character reduction and panel scheme as in the sharp-order argument [R053]. The zero-sum good tuples give the localized Newton word $F(c) = \sum_{l=1}^{h-1}(u^c - 1)^l m_l$, plus a scalar linear Bohr character if needed. Substantial passive exponential mean forces the passive polynomial to be identically zero, so $Y^l m_l = 0$ for $l \geq 2$, and $Y^2 m_1 = 0$, where $Y = u^p - 1$. The nonsubstantial passive case already contracts the charged Fourier mode.

For every l, Y thus acts nilpotently of index less than p on m_l at sufficiently large primes. Now $(1 + Y)^{p^{D_p}} = 1$. Truncating this binomial identity on m_l yields $p^{D_p} Y V(Y) m_l = 0$, with $V(0) = 1$ and coefficients in $\mathbb{Z}_{(p)}$: for $1 \leq j < p$, $\binom{p^{D_p}}{j}/p^{D_p}$ is p-integral. Hence $p^{D_p} Y m_l = 0$ in the augmentation localization. Along $c = 1 + p^{D_p}t$, expand $u^c = u(1 + Y)^{at}$ with $a = p^{D_p - 1}$. Every time coefficient of degree at least two in each Newton term has a factor $a^2 Y$ with p-integral remaining coefficients. Since $D_p \geq 2$ eventually, $p^{D_p} \mid a^2$. Thus the entire active scalar phase is linear in t, independently of the rank of the character module and without restricting h. Inserting a nonzero quadratic charge frequency gives the usual $o(1)$ Gauss bound. The independent panel laws send the charge to Haar independently of all the finitely many good phases, while disabling any bad tuples. The limiting graph argument therefore gives the required positive return. This reasoning works for the r-th charge for any fixed $r \geq 2$ with the same minimal residue depth $D_p + 1$, since the active phase is still linear.

The rational sharp obstruction implies $b_{\mathbb{Q}}(D_f) > 0$. By the suspension and tail formula [R241], whose proof uses only a finite cyclic component at each prime and applies to any prescribed finite exponent there, some native tail $V_{f,J}$ has a positive stationary avoider at a finite length. This gives a positive native avoider also for V_f: multiply by the event that the point in an independent regular action on the finitely many early components is zero. Its mass is positive, and every step with nonzero early component is blocked already as a pair. No horizon from the original rational affine witness is asserted here. In particular the native avoider just obtained cannot occur in any compact abelian affine Haar action.

There is also a simpler all-horizon result for the native shallow group on relative horizontal phase-polynomial factors. Here $H = \bigoplus_p pC_{p^{D_p+1}}$. On any ergodic H-component, a bounded-degree phase polynomial restricted to the p-axis of $H \cong C_{p^{D_p}}$ has the usual ordinary polynomial form in that axis at p exceeding its degree, with nonconstant coefficients of orders dividing p^{D_p}. This follows from the binomial expansion on $\mathbb{Z}$, periodicity, and invertibility of the small factorials modulo p. Translating by $p^{D_p - 1}t$ in that coordinate makes the phase a linear function of t because $2(D_p - 1) \geq D_p$. The finite horizontal phase-polynomial approximation and active quadratic cancellation in deep-order-hierarchy.md [R047] therefore apply even without a global bound of one on the original degrees or countability of their varying nonlinear labels. This assertion does not classify arbitrary horizontal Host–Kra factors.

R235. Concentration in a common high clock

Scope and status. Result with proof.

Result. Independent low axes with a common dyadic clock and finite toral register have dimension-dependent recurrence thresholds, including full recurrence in one dimension.

Scope. The special independent-axis affine-register model.

Qualifications. The threshold tends to one with dimension and does not bound arbitrary actions uniformly.

Use the single-clock mixed marks [R237] with distinct increasing depths and the well-rounded low-prime schedule. Write $D = \mathbb{Z}[1/2]/\mathbb{Z}$, and D_e for its subgroup of order 2^e. Consider independent uniform low sites $y_i \in C_{p_i}$, a common independent Haar odometer coordinate $x \in \mathbb{Z}_2$, and independent Haar circle vectors $\lambda, z \in \mathbb{T}^r$. Fix arbitrary tables $b_i : C_{p_i} \to D^r$, with no common range exponent. The high generator acts by

$$S : (y, x, \lambda, z) \longmapsto (y, x + 1, \lambda, z + \lambda).$$

At low site i, shift y_i by a, add $d_i = b_i(y_i + a) - b_i(y_i)$ to λ, and add $x\, d_i$ to z. Multiplication here is the standard $\mathbb{Z}_2$-module action on D. These are commuting measure-preserving actions, including commutation with S; at successive times the z-register becomes

$$z_t = z + tu\lambda + (x + tu) \sum_i [b_i(y_i + tr_i) - b_i(y_i)].$$

This model allows genuine coupling of the common rational clock and irrational fiber frequencies. For its whole σ algebra there is full positive recurrence when $r = 1$, threshold at most $1/2$ for $r = 2$, and threshold at most $1 - r^{-r}$ for every r. The bound for $r \geq 2$ is system-relative and uniform for that dimension; it is not a universal high-system estimate.

Dichotomy for the Prüfer sums. If ν_i are laws on D^k, take their lazy mixtures $\mu_i = (\delta_0 + \nu_i)/2$. Exactly the following implication is needed: if for every finite subgroup $F \subset D^k$ the sum $\sum_i \nu_i(D^k \setminus F)$ diverges, convolution over sufficiently many any-tail indices has maximum atom arbitrarily small. For otherwise the L^2-norms of growing tail convolutions do not tend to zero. By Plancherel on $\widehat{D^k} = \mathbb{Z}_2^k$, a positive-measure set of characters then has $\sum_i (1 - \Re\widehat{\nu_i}(t)) < \infty$, since $|\widehat{\mu_i}(t)|^2 \leq (1 + \Re\widehat{\nu_i}(t))/2$. On a positive-measure subset this sum is bounded by a single constant. The squared triangle inequality gives a constant bound on its difference set, a neighborhood of zero, by Steinhaus. Integrate over an open subgroup $2^e\mathbb{Z}_2^k$ therein. Orthogonality gives $\sum_i \nu_i(D^k \setminus D_e^k) < \infty$, a contradiction. Disjoint finite pools can be allocated for each of finitely many tests in as many independent trials as needed.

Fourier constraints and sampling. For a finite integral Laurent coefficient row $P(T) = \sum_j c_j T^j$ on the z-registers, put $Q(T) = TP'(T)$. Haar in z, λ forces $P(1) = Q(1) = 0$ in the limiting large-high-velocity averages. Indeed additional tests on λ_t impose a fixed integer constant on $uQ(1)$, holding for at most one integer u when $Q(1) \neq 0$. With the balance conditions imposed, Haar in the **single** x tests

$$\sum_i P(S^{r_i}) \cdot b_i(y_i) = -a$$

for a fixed dyadic hull a from any finite list of x_t-characters. High averaging tests the height of $\sum_i Q(S^{r_i}) \cdot b_i(y_i) + a'$, again with fixed $a' \in D$ if such characters are included. With u in

the exact class $v_2(u) = D_j$, a nonzero contribution requires this latter value in D_{D_j+1}; when it is already in D_{D_j}, the phase agrees with that for high multiples of any $2^{D'}$, $D' \geq D_j$. Other tests on the y, λ-registers contribute at most unit absolute value and may be dropped for upper bounds. All these formulae can be checked first for finite Fourier tests, then for bounded measurable tests by the stationary one-slot marginals. In particular finite-support initial low tests can be accommodated untouched on fresh axes.

A balanced row belongs to R if for some e the summed probabilities of falling outside D_e are finite for each of the three local variables

$$P(S)b_i(y_i), \qquad Q(S)b_i(y_i), \qquad P(S^{L_i})b_i(y_i).$$

This definition intentionally does not require the Q-test at L_i. The set R is a Laurent submodule since $Q_{T^m P} = T^m(Q_P + mP)$. Take the finite number of DHJ trials used at the tested mass and horizon. In each trial first place optional low marks, each independently using L_i or zero with equal probabilities, and then one deterministic lead mark at a fresh much later index. Build these trial blocks successively. After all leads have been chosen adjoin to each trial a disjoint far-future pool of optional normals, each independently using 1 or zero. In any nonempty union of trials the greatest marked depth is then its last included lead; optional lower marks cannot change it.

For a row whose first two normal tests are not jointly concentrated in any fixed D_e^2 with summable error, the future normal pools disperse the pair of sums by the concentration lemma. These pools may be chosen **after fixing all marked depths**. The probability of landing in the bounded finite target from both Fourier conditions is then arbitrarily small, uniformly in the contributions already present, even though the finite target size can depend on that stage's depths. If instead the mark $P(S^{L_i})$ has no such concentration, the optional earlier marks disperse the initial P-sum at any given point. This needs no bound on any later depth, so those pools can be chosen before their lead. In either case the row vanishes in both charged and uncharged limiting trajectory laws, including jointly with the entire base (y, x, λ) trajectory tested against bounded functions. Countably many rows and finite circle/cyclic test lists can be handled by a diagonal sequence. For base functions depending on infinitely many low coordinates take finite-head approximation at each fixed one-slot test; the upper bound on the spectral survival probability is in fact unaffected by additional bounded amplitudes.

Controlling the mobile quotient. Over $\mathbb{Q}(T)$ take polynomial basis columns v_l for the orthogonal complement of R, after clearing denominators. Let $K_f \leq \mathbb{T}^r$ have rational tangent space the span of their coefficient vectors. If l is a constant integral row annihilating K_f, a nonzero scalar polynomial $D_l(T)$ times l is an integral Laurent combination of rows in R. Both $D_l(S)l \cdot b_i$ and $D_l(S^{L_i})l \cdot b_i$ consequently have summably small probabilities of escaping a common fixed finite dyadic subgroup. For a finite set of such rows, multiply by one common fixed power of two to turn these defects into sparse supports. On arbitrarily late indices where the combined failure probability times L_i tends to zero (available since $\sum_i 1/L_i = \infty$), the sparse two-shear lemma [R153] implies that a fixed integer multiple of each $l \cdot b_i$ is constant off $o(p_i)$ sites. Any extra odd determinant is invertible on D and is immaterial here. These good indices are used for the successive leads.

For $P \in R$ whose individual coefficients annihilate K_f, use a basis of this constant lattice. The $Q(S)$-test is in the fixed finite dyadic subgroup off $o(p_i)$ sites, so after the mentioned multipliers, the constant projection values satisfy the same bounded-height condition for $Q(1)$. Thus $Q(S^{L_i})b_i$ on such a lead also has bounded height off $o(p_i)$ sites. Translations

of the boundedly many exceptional sets preserve their size. Choose every lead depth larger than the heights of all table entries at **all previous marked sites in the constructed packets**, which is possible by leaving gaps; also exceed the fixed test denominators and the chosen finite concentration subgroups. On its future normal pools, the probability of any Q outside the finite subgroup is small by summability and far-tail choice. Hence with probability tending to one the complete high test is already in D_{D_j} at the relevant last lead j. True and uncharged high averages for such a row are then identical up to $o(1)$; all other amplitude factors are unchanged. The uncharged comparison uses a common subgroup of high multiples deeper than every lead, and can be generated by independent Følner-sampled high trials in that subgroup. All sampling choices concern finite tests before their diagonal enlargement.

Consequently the projected base/quotient trajectories have identical charged and uncharged limiting laws, whereas the full charged limits, holding base data fixed, are invariant under the independent Haar input processes generated by the v_l. The dimension-only pivot bound [R120] applies verbatim: use a positive DHJ line through base/quotient data with mobile section mass $> 1-(\dim K_f)^{-\dim K_f}$; each corrected row has that same projected law, and adding the invisible moving noise solves every finite window with positive conditional probability. If $K_f = 0$ this gives recurrence at every positive mass. For $r = 1$, either R is zero (one completely free circle channel) or $K_f = 0$, giving the stated full conclusion. For $r = 2$ the only case not already full is a rational-rank-one R whose primitive polynomial row has nonconstant direction; apply the rank-one section argument [R154] to obtain the sharper half bound.

R236. Tight and escaping relative frequencies for the single high clock

Scope and status. Result with proof.

Result. Tightness or suitable shell decay of relative frequency increments gives positive recurrence for the single-clock family.

Scope. Explicit relative-eigenfield hypotheses.

Qualifications. The matching dyadic-shell mass required of an obstruction is a necessary condition, not a constructed obstruction.

Use the single-clock, two-low-shear model [R237], and write $a_i = L_i e_i$ for the marked low move. Let $Y = \mathcal{I}(S)$ and suppose ϕ_l are circle-valued relative eigenfunctions with $\phi_l(Sx) = \phi_l(x)e(\lambda_l(y))$. In additive notation their labels are $\lambda_l : Y \to \mathbb{T}$. Let O be the subgroup of odd-order rational points, D_e the dyadic subgroup of size 2^e, and $\pi : \mathbb{T} \to \mathbb{T}/O$ the algebraic quotient. All uses of membership in a union of countably many fibers of π are measurable. Projection is injective on the dyadic group. This quotient has no Hausdorff topology in the argument.

Tight increments. Suppose that for every finite set of labels and every finite horizon there is an infinite subsequence of marked axes on which all the random increments

$$\pi\big(\lambda_l(U^{ta_i}y) - \lambda_l(y)\big), \qquad 0 \le t < h,$$

are uniformly tight in the discrete sense: for every $\eta > 0$ some finite subset of the quotient catches every listed variable with probability $> 1 - \eta$, uniformly on the subsequence. Then every positive-measure event generated by Y and these relative eigenfunctions has returns of all finite lengths. There is no independence, finite-coordinate support or stationarity of the increment fields under other low generators in this assumption. In particular arbitrary odd-primary changes of the frequencies cost nothing, even if there is no measurable choice of representatives of their orbit classes.

For a Fourier monomial in the successive slots, its high-time character has frequency

$$\beta(v, y) = \sum_{j<h} j\, c_j \cdot \boldsymbol{\lambda}(U^{jv}y)$$

with integer coefficient vectors c_j. Its remaining amplitude can depend arbitrarily on the base and phase fibers but has modulus bounded by a constant independent of u. If the greatest chosen marked index is i, compare the AP high-time averages in $u = 2^{D_i}(2n+1)$ and $u = 2^{D_i+1}n$. **These two characters differ only on the single shell** $B_i = D_{D_i+1} \setminus D_{D_i}$. On that shell the discrepancy has magnitude two. One may instead compare any deeper subgroup, but the single-shell version below is sharper.

Approximate the tested event indicator by a bounded finite sum of such characters with bounded Y-coefficients. Finite Fourier approximation follows by simple rectangles in the joint law on $Y \times \mathbb{T}^r$ and bounded continuous approximation in the torus variable; for instance one can choose a bounded cylindrical continuous approximation and take Fejér means. One-slot marginal stationarity bounds both product-approximation errors uniformly over all low and high times. Preselect the positive uniform low DHJ line lower bound κ and the usual

uniform Szemerédi correlation lower bound for sets of mass $\geq \kappa$, then choose the approximation and probability tolerances.

For a low sum v of at most the fixed DHJ number m of marked axes, telescope each increment $\lambda_l(U^{jv}y)-\lambda_l(y)$. The summands are single-axis increments at measure-preserving translates of y. By a union bound, except on a uniformly small set, the quotient class of $\beta(v,y)-\beta(0,y)$ belongs to a fixed finite list. For each member c of this list, the probabilities $\Pr(\pi\beta(0,y)+c \in \bigcup_{D\geq M}\pi(D_{D+1}\setminus D_D))$ tend to zero as $M\to\infty$, by continuity from above and the disjointness of the shells. Thus the true and comparison correlations on every candidate differ by less than the preselected uniform recurrence margin. With sufficiently late marked axes, choose v by low DHJ so $C=\bigcap_{j<h}U^{-jv}A$ has mass $\geq\kappa$. Uniform Szemerédi on C for $S^{2^{D_i+1}}$ makes the comparison average positive by that margin, and hence the charged average is positive too. This applies also if only a sufficiently high S-Host–Kra projection of the indicator is on this factor: take the low DHJ line in a positive superlevel set of that projection and transfer by the usual absolute-mean characteristic bound for fixed low move.

Only the last marked coordinate needs to be tested. Suppose instead that all the increments of the finite list of labels under the subgroup generated by the tested marked axes are rational almost everywhere, with unrestricted dyadic heights and odd parts. Assume that along an infinite subsequence of distinct axes, simultaneously for each of the finitely many Fourier tests needed at each stage,

$$\mathbb{P}\big(\pi\beta(a_i,y)\in\pi(B_i)\big)\longrightarrow 0.$$

Then the same positive recurrence conclusion holds. This can handle frequencies escaping to dyadic orders strictly higher than the marked order, as well as tight ones. For a fixed previously selected low sum w, the difference $\beta(w+a_i,y)-\beta(a_i,y)$ is a finite sum of increments under the fixed translations jw evaluated at $U^{ja_i}y$. Each is rational and, modulo O, in a fixed finite dyadic subgroup with arbitrarily high probability, uniformly in the as-yet-unchosen i, by stationarity. Select the next axis with D_i beyond the dyadic height bounds for all finitely many previous subset sums and tests. Adding such an increment cannot change membership in $\pi(B_i)$. Thus every constructed nonempty subset has its true/comparison discrepancy bounded by the tested single-axis shell mass plus a preassigned arbitrarily small error. The same low DHJ and subgroup recurrence proof finishes. With countable nontorsion cocycle values one can instead require the single-axis shell condition after each fixed translation in a countable group containing the cocycle's quotient range. Truncate the prior increments to a finite list in that group before selecting the next axis.

An implication useful for auditing potential obstructions is that for any avoiding relative-eigenfield observable with rational low increments, some fixed finite list of bounded Fourier tests has a nonnegligible single-axis shell hit on every sufficiently large axis in any proposed subsequence. This is a necessary statement; the common six-slot low witness can exhibit such a matched shell, so shell exclusion is not automatic for arbitrary events.

R237. A single high clock and two low shears

Scope and status. Proved subcases and unresolved extensions.

Result. A one-clock, two-shear family has a common positive six-term avoiding event and returns in several substantial clock-spectrum classes.

Scope. The positive lower witness is unconditional for this family. Upper results require bounded dyadic spectrum, ergodic clock with native low primes, precompact low orbit, or global discrete projected spectrum.

Qualifications. The unrestricted upper threshold remains open.

Let $G = \mathbb{Z}e_0 \oplus \bigoplus_i \mathbb{Z}e_i$, or take the low summands modulo distinct odd primes p_i. Use the well-rounded choices [R152] $L_i = 2^{k_i}$, $p_i \asymp L_i^2$, about L_i primes per level, so that $\sum_i 1/p_i < \infty$. Assign strictly increasing distinct positive depths D_i tending to infinity and set $z_i = 2^{D_i}$. A difference $(u, (r_i))$ has finitely supported low part. At each site its low residue modulo p_i can be 0 or 1 without restriction on the common high integer u. The other permitted low residue, L_i, is marked and requires $z_i \mid u$. Require the total of u/z_i over the marked sites to be odd. Thus for nonempty mark set J with greatest index j the permitted high coordinates are the entire arithmetic coset $v_2(u) = D_j$. Low normals may occur at *any* sites, including later ones. The test uses one integer high action, not a different high group for each mark.

Positive fixed-horizon obstruction There is a common six-slot positive avoiding event on either host. On a finite packet take a single uniform high integer clock x modulo $2\max_i z_i$ and independent low residues y_i. Over $\mathbb{Z}/(2z_i)$ solve

$$(S^{L_i} - I)^2 (S^{L_i} - S)^2 b_i = h_i, \qquad h_i(y) = 1 \ (y \neq 0), \quad h_i(0) = 1 - p_i$$

on C_{p_i}. Each difference is invertible on its zero-sum part, since $1, L_i, L_i - 1$ are nonzero modulo the prime and p_i is odd. The right side has sum zero. Constant shifts of b_i or of the exceptional residue are harmless.

Use the local circle observations $xb_i(y_i)/(2z_i)$ and the finitely many global sums of their spatial translates that occur in $(T - I)^2 (T - S)^2 (T - S_{L_i})$. On low0 the time signal is affine in $x + tu$ and killed by the first squared factor. On low1 the own spatial shift coincides with its low motion, so the second squared factor kills its affine high dependence. On low L_i the last factor gives $ub_i(y_i + L_i)/(2z_i)$, and the two squares evaluate to $uh_i(y_i + L_i)/(2z_i)$. On states with all $y_i + L_i \neq 0$, the aggregate signal is exactly $1/2$ at an admissible direction. The safe set has mass $\prod_i (1 - 1/p_i)$ uniformly on all finite packets. The coefficients for each spatial channel sum to zero over time (factor $T - I$). Color the finitely many circle channels by sufficiently short fixed intervals. If six sites have the same color, their tested filter cannot equal $1/2$. Some color intersected with the safe set has uniformly positive mass and is a six-avoider. Passing to stationary weak limits (coinduce the low tails) gives the asserted common obstruction.

Using a single marked low center with only low0 normals instead would have endpoint one: on no-wrap integer representatives the bilinear direction signature $u \sum_{i \in J} r_i/(2z_i)$ reads the tag for marked speed one, at summed wrap cost $O_h(\sum_{\text{tail}} 1/p_i)$. The second normal shear and the nonsummable common chart cost matter. The two-shear family's unrestricted upper is open.

Every system with bounded dyadic spectrum in its clock components recurs Write $S = U^{e_0}$. Suppose that on almost every S-ergodic component the torsion part of the eigenvalue group has bounded 2-primary height, with the bound initially allowed to depend on the component. Then every positive-measure event has every finite length of recurrence in the proposed family, even if all selected low velocities are marked at distinct depths and no normals are used. There is no discrete-spectrum or low-component regularity assumption.

Reduce to a jointly G-ergodic system. For each fixed dyadic rational eigenvalue its occurrence on an S-component is measurable and invariant under G (compute twisted ergodic projections of a countable dense list and use commutation). Thus the dyadic spectrum is a single finite cyclic group of order at most 2^M for almost every component. On the S^{2^M}-ergodic components inside each of these components there is no eigenvalue of even order: the finite cyclic decomposition for powers of an ergodic transformation gives that $\mathcal{I}(S^{2^{M+1}}) = \mathcal{I}(S^{2^M})$. Equivalently, an extra two-cycle on such a component would give a cyclic factor of the original S of order greater than 2^M.

Fix any low translation V, and bounded functions in the successive slots. On an S^{2^M}-component apply Moreira–Richter, Theorem 2.4 to S^{2^M}. For every $D \geq M$ the Cesàro multiple progression average in the high times $u = 2^D(2n+1)$ equals that along all $u = 2^M n$; choose modulus $q = 2^{D+1-M}$ and residue 2^{D-M}. The slot functions may be $V^j f$ restricted to the component; V itself need not preserve it. Multiply by the unshifted slot and integrate. Disintegration yields equality of the total integrated correlations for that fixed low move and for every such D.

By finite density Hales–Jewett choose a nonempty sum $v = \sum_{i \in J} L_i e_i$ among finitely many distinct sufficiently late levels so that $C = \bigcap_{0 \leq j < h} U^{-jv} A$ has positive measure. Ordinary multiple recurrence gives a positive Cesàro h-return average for C along $S^{2^M n}$ (the usual Varnavides bound). Each such C-return contributes an A-return at $(2^M n, v)$. By the equality of progression averages with the absorbed low slot functions, the average at $(2^{D_{\max J}}(2n + 1), v)$ is positive too. These are admissible directions. Thus an actual avoiding action on this family must have *all dyadic rational eigenvalues* on almost every active clock-ergodic component of a jointly ergodic witness.

If S itself is ergodic, the odd-degree cyclic nil-joining lemma [R039] and the Host–Kra reduction also handle the infinite dyadic spectrum for native low prime-periodic actions: approximate an event on a finite S-nilfactor, take the finite cyclic self-joining under a low move V of odd order, and absorb the finite factor's 2-primary spectrum by choosing sufficiently late levels. The finite joining adds no dyadic prime to a return power that was dyadically ergodic. One way to see the needed exponent bound is that a dyadic cyclic quotient of order bigger than 2^M over a base with maximal dyadic factor C_{2^M} has an independent residual two-cycle over each base component under the 2^M-th power; the odd fiber size excludes it just as in the joining lemma. The surviving case allows an odometer in each clock component and varying coupled observations under the low group.

Two further positive-recurrence classes First suppose that on almost every clock-ergodic component the entire low orbit of the tested event's indicator, restricted to that component, is precompact in its component L^1. The component need not be preserved by the low group. For each fixed finite tuple of bounded slot functions on an ergodic component, the integrated AP averages along $2^D(2n+1)$ and $2^{D'}n$ differ by $o(1)$ as $D \to \infty$, uniformly for $D' \geq D$. Indeed project to a sufficiently high Host–Kra factor (the discarded error is

Cesàro-null in absolute value on all high times, and remains so on each fixed positive-density class), then approximate the resulting tuple in L^1 by functions on one finite nilfactor. For this finite factor the averages agree for all $D' \geq D$ beyond the finite exponent of its dyadic component group, by the power-component argument above. All progression averages are Lipschitz in the slot functions on the unit ball, with bound given by the sum of the L^1 errors. A finite net for the componentwise precompact low orbit makes this comparison uniform over all low moves there. Integrate the supremum over the countable low group by dominated convergence. Apply low DHJ on distinct late levels and uniform Szemerédi to the corresponding joint event along a common uncharged subgroup deeper than all selected levels. The true average differs by arbitrarily little uniformly over those choices, so the selected true progression has positive measure.

Second, suppose the conditional expectation $g = \mathbb{E}(1_A \mid \mathcal{Z}_s(S))$ for a sufficiently high order s lies in the closed span of global S-eigenfunctions. No precompactness for its low translates is needed. Take $a > 0$ such that $\{g > a\}$ has positive measure. Finite DHJ supplies a uniform $\kappa > 0$ for an h-term low return of this set along a nonempty sum of marked generators on any sufficiently long distinct block. Choose a finite global eigenfunction approximation to g with error far below $a^h\kappa/h$. For its finitely many eigenvalues $e(\alpha_l)$ the compact subgroup $\overline{\mathbb{Z}\boldsymbol{\alpha}}$ in a finite torus has finite component group. For every sufficiently large D, the integers u of exact valuation D for which all $u\alpha_l$ are close to zero have positive lower density: in the cyclic component group first enforce divisibility by its odd part, then use equidistribution of the remaining progression on the identity component (or on the finite union containing it). Thus at every such u, all $S^{ju}g$ are near g in L^1, also after any common commuting low translation. Choose a late low DHJ sum v. Its g-correlation is at least $a^h\kappa$ before the high displacement, so stays positive by a fixed margin on this positive-density set of exact high times. For fixed v, replacing the successive low translates of 1_A by their translates of g changes the integrated high correlation by a sequence of zero Cesàro absolute mean. Here one can use the ordinary Host–Kra generalized von Neumann bound on the square/product system with one additional seminorm order, or apply it on each clock-ergodic component and integrate. A commuting low translation preserves these characteristic projections. Therefore some charged high time has a positive original return. This conclusion includes arbitrary extensions for which this conditional expectation has global discrete spectrum. Host–Kra, Bergelson–Host–Kra.

R252. Three temporal slots in finite primary modules admit bounded-degree interpolation on the cover

Scope and status. Result with proof.

Result. Three-slot scalar filters in finite primary modules interpolate on covers with degree three, uniformly in depth and in the stated rank range.

Scope. Characteristic p exceeds rank plus three; the scalar output is a trivial primary line.

Qualifications. The degree-three refinement strengthens the auxiliary bound depending on rank. Both statements concern the specified scalar primary-module model.

Degree three independently of rank when $p > r + 3$ The apparent degree-$r + 1$ cost on the nonexceptional projection lines can be removed. Compute all ordinary coefficients of H_c through degree $r + 1 < p$. Its coefficient at t^{n+2} is a polynomial in $c \bmod p$ of degree at most $n + 2$, vanishing at $c = 0, 1$. Its quotient by $c(c - 1)$ is a polynomial $U_n(c)$ of degree at most n, with $U_0 = 1$. Let $G(t) = \sum_{n=0}^{r-1} U_n(0)t^n$. Divide the normalized series by this common series and write the resulting truncation as $1 + \sum_{n=1}^{r-1} V_n(c)t^n$. Each V_n has degree at most n and vanishes at zero, since the normalized quotient at $c = 0$ is one.

For a vector under test put $c_I = \sum_{i \in I} a_i$ as before. The function of I given by $V_n(c_I)$ is a polynomial in the Boolean variables $1_{i \in I}$ of degree at most n, without a constant term. Each nonempty Boolean monomial with support of size at most n is the restriction of an ordinary homogeneous polynomial of degree n on the lines after stripping the common factor t^n: pad one positive exponent. Thus there is an actual arrangement-ring unit V_a with constant term one giving the specified degree-$1, \dots, r - 1$ jets on every line. Replace h_a by $V_a^{-1}h_a$; this leaves its trivial-line output unchanged and does not change the generated ideal.

On each line with $n_I = 2$, the relevant coefficients of the new generator equal the fixed coefficients of $t^2G(t)$ multiplied by the quadratic $c_I(c_I - 1)$. On an exceptional line $n_I = p^k + 1$, the allowed $c_I \bmod p$ is zero or one, so the relevant jets of V_a^{-1} are fixed on each of those two branches. The resulting coefficients remain fixed branchwise multiples of the upper digit z, with the same cubic interpolation. The line-separation proof applies to the normalized generators as well. Consequently each signature is a sum of low-digit quadratics and univariate branchwise next-digit terms, realized by a cover polynomial of degree at most **three**, including when ranks grow subject to $p > r + 3$. The number of terms does not affect this degree.

Use the scalar signature definition in the slope-projection note [R157]. Fix the ambient rank bound r, allow arbitrary prime-power depths, and let $h = 3$. For all sufficiently large primes relative to r, the scalar signature is interpolated on the integer cover of its admissible directions by rational circle polynomials of degree bounded only in terms of r, with no slope or depth restriction. In the ordinary balanced situation $0 \in E, \phi(0) = 0$ the interpolating polynomial can be zero at the origin. The statement concerns signatures in a fixed trivial line over the prime field.

Reduction to generators. Choose an anchor $d_0 \in E$ and restrict to the subgroup $H = \langle E - d_0 \rangle$. Translating the stencil replaces its slot vectors by $U^{jd_0}m_j$. Select differences from the anchor generating H, with number $r' \le r$, by choosing a basis in H/pH. Use the integer group on these chosen generators as a cover, so the translated allowable coordinate vectors include zero and the individual standard generators. Two lifts of the same point have

the same scalar signature. For notational simplicity write r for the new generator count. Rank zero is immediate.

Let $x_i = U_i - 1$; they act as commuting nilpotents on every module, since a sufficiently large common p-power kills them individually. Thus the action extends from the polynomial ring to $R_0 = F_p[[x_1, \dots, x_r]]$. If $a = (a_1, \dots, a_r)$ is an admissible integer coordinate vector, put

$$z_a = \prod_i (1 + x_i)^{a_i} - 1 = \sum_i x_i P_i(a, x)$$

using any formal series decomposition; its constant coefficients satisfy $P_i(a, 0) = a_i \bmod p$. Formal binomial expansion is legitimate for negative integer exponents as well.

Writing the three-slot coefficient column in the binomial basis gives $v_a = (1, z_a, z_a^2)$. Subtract the anchor column and $\sum_i P_i(a, x)(v_{e_i} - v_0)$. Only the last component remains, equal to

$$h_a = z_a^2 - \sum_i x_i^2 P_i(a, x).$$

Its image under the original module map is

$$\big[\phi(d_0 + a) - \phi(d_0) - \sum_i a_i(\phi(d_0 + e_i) - \phi(d_0))\big]\eta.$$

In this formula $d_0 + a$ abbreviates its image in the finite group. Moreover $x_j(v_{e_i} - v_0) - x_i(v_{e_j} - v_0)$ gives the pure last component $x_i x_j(x_i - x_j)$ and has image zero because the output line is trivial.

The lines. Modulo the ideal $F = (x_i x_j(x_i - x_j) : i < j)$ the power-series ring embeds into the product of $F_p[[t]]$ over all nonempty subsets $I \subset [r]$, by sending $x_i = t$ for $i \in I$ and $x_i = 0$ otherwise. Indeed each two monomials of a fixed ordinary degree and the same support are equal modulo F: move surplus exponents between members of their common support using the cubic relations. Each monomial with support J restricts to $t^d 1_{J \subset I}$ on the lines I. For fixed degree the different supports give independent functions of I by Boolean Möbius inversion; the constant term is immediate. Taking homogeneous parts proves the assertion for series, since remainders can be written as convergent, degreewise multiples of the finitely many cubics. For every nonempty I there is a homogeneous degree-r polynomial e_I whose restriction is t^r on that line and zero elsewhere, by the same Boolean basis, since every nonempty support is available in degree r.

Write $A_0 = R_0/F$ and let J be the ideal generated by the images of all h_a. The residual scalar signature is a linear functional on $J/\mathfrak{m}J$, $\mathfrak{m} = (x_1, \dots, x_r)$. On line I one has

$$h_a|_I = H_c(t) = z_c(t)(z_c(t) - t), \qquad c = \sum_{i \in I} a_i, \quad z_c(t) = (1 + t)^c - 1.$$

This follows from $z_a = \sum_i x_i P_i$, irrespective of the chosen decomposition. Put $n_I = \min_a \operatorname{ord}_t H_c$ with infinity if all these series vanish. Every $g \in J$ whose line restrictions vanish through orders $n_I + r - 1$ is in $\mathfrak{m}J$. To see this, on each line with $n_I < \infty$ choose a generator h_{a_I} of exact line order n_I. The product $e_I h_{a_I}$ lies in $\mathfrak{m}J$, is zero on the other lines, and starts with a nonzero coefficient at order $n_I + r$ on the chosen one. Multiply by a suitable formal scalar series in any x_i with $i \in I$ to match the entire prescribed restriction of g. Add the finitely many results; injectivity on lines gives equality in A_0. Consequently

the functional is a fixed linear combination of coefficients of H_c at degrees $n_I, \ldots, n_I + r - 1$, over the finitely many lines.

Univariate digits. When some allowed projected residue $c \bmod p$ differs from both zero and one, $n_I = 2$. Its finitely many tested coefficients are polynomials in the low digit through degree at most $r + 1$ for $p > r + 2$. Otherwise all projected residues lie in $\{0, 1\}$. If there is a nonzero series on this line, $n_I = p^k + 1$ for some $k \geq 1$, and every allowed c lies in 0 or 1 mod p^k. Indeed the order of $(1 + t)^n - 1$ for nonzero integers is $p^{v_p(n)}$; among the two factors only one can have order greater than one. Write $c = b + p^k z$, with $b \in \{0, 1\}$. For $p > r + 2$, Frobenius expansion through degree $p^k + r$ gives

$$H_c(t) \equiv \begin{cases} -zt^{p^k+1} & b = 0, \\ zt^{p^k+1} + zt^{p^k+2} & b = 1, \end{cases} \qquad (\text{mod } t^{p^k+r+1}),$$

where unneeded terms above the truncation may be dropped. All these tests are therefore branchwise constant multiples of the next digit $z \bmod p$.

To realize any such branchwise slope as a circle polynomial on the integer cover, choose an integral polynomial $Q(c)$ of degree at most three with $Q(0) = Q(1) = 0$ and prescribed integer derivatives at $0, 1$. This is elementary Hermite interpolation at two nodes of difference one. At $c = b + p^k z$, the scalar $Q(c)/p^{k+1} \bmod 1$ equals $Q'(b)z/p \bmod 1$; the quadratic remainder is integral because $k \geq 1$. Low-digit polynomial terms and the anchor affine term are interpolated directly at denominator p. Summing over the at most $2^r - 1$ projections gives fixed degree $\max(r + 1, 3)$, with local coefficients allowed to vary over the prime.

If the original finite group was presented as the quotient of an ambient integer lattice, let Λ be the sublattice mapping to H. Lift the images of a basis of Λ to integer vectors in the chosen generator cover. The resulting linear map on Λ agrees with the finite quotient and extends rational-linearly to the ambient rational space. Subtract an ambient lift of the anchor and substitute this linear map into the interpolants. With anchor zero, the resulting polynomial vanishes at zero if the signature does so; each exceptional cubic and each projected function in that case has zero at zero. In the standard rational primary cover the same construction is performed on each local lattice. The resulting local polynomials have integral values on the kernel lattice, where the step is zero, when the balanced zero signature is imposed. Evaluating their p-adic fractional parts and summing on rational inputs is then locally finite, because an integral input belongs to the native zero kernel at all but its finitely many denominator primes after the cover-coordinate scaling. This yields a genuine bounded-degree group-polynomial detector for such simultaneous scalar signatures. The algebraic conclusion does not treat four or more temporal slots.

R257. A trivial central circle bundle has only compact torsion dynamics

Scope and status. Result with proof.

Result. A continuously trivial central circle bundle in a two-step translation model has compact dynamics for every abelian torsion subgroup; dual rank one ensures triviality.

Scope. Connected compact abelian base, continuous global section, and torsion acting subgroup.

Qualifications. No infinite-order analogue is asserted, and nontrivial rank-two bundles lie outside the theorem.

Let N be a second-countable locally compact two-step group acting transitively on a compact central principal circle bundle X over a connected compact metrizable abelian group B. Assume that elements of N cover base translations, act equivariantly for the central circle, and that their commutators act by constant vertical rotations. This is the relevant special case of the locally compact translational models of Jamneshan–Shalom–Tao. Suppose the bundle has a continuous section. Then the action of every abelian torsion subgroup G of N is simultaneously conjugate to an action by translations on $B \times T$. No infinite-order analogue is asserted.

In the section coordinates write $n(x,u) = (x+b_n, u+u_n(x))$, with $u_n : B \to T$ continuous. Every such circle map has a unique character χ_n for which $u_n = \chi_n + v_n \bmod 1$ with v_n real continuous. For existence uniformly approximate the circle map by a nonvanishing trigonometric polynomial and normalize. The approximant factors through a finite-dimensional connected torus, whose continuous circle maps have a character times a real exponential by lifting to the universal cover. The approximation error has a continuous logarithm. For uniqueness, a character with a real lift V would give an integer-valued continuous function $V(x+y) - V(x) - V(y) + V(0)$ on the connected space B^2; it is zero at the origin. Thus V differs by a constant from a real homomorphism on a compact group and is constant.

Composition shows $\chi_{nm} = \chi_n + \chi_m$, since translating an argument changes its character by a constant factor. The dual of B is torsion-free, so $\chi_g = 0$ for every $g \in G$. In the commutator identity for n and g, the difference of the two circle increments is constant. Removing the character χ_n leaves $\Delta_{b_n} v_g - \Delta_{b_g} v_n$ real and constant modulo integers. Connectedness makes it a genuine real constant, and integration in x gives zero. Here $\Delta_t v(x) = v(x+t) - v(x)$.

Choose a compact set of representatives in N mapping onto B (compactness of B and openness of the orbit quotient map). The assignments $n \mapsto u_n$ are continuous uniformly in x. On a finite cover of this compact set by sufficiently small neighborhoods, choose v_n from the lift at one center plus the small continuous logarithm of its ratio; the character stays constant on each neighborhood. A Borel choice among these charts gives uniformly bounded lifts. Take measurable representatives $n(t)$ over $t \in B$ there and put $V(x) = \int_B v_{n(t)}(x)\,dt$, a continuous function by bounded convergence. Averaging the exact difference identity gives

$$\int_B v_g - v_g(x) = \Delta_{b_g} V(x).$$

Thus one change of vertical coordinate by V replaces every u_g by the constant $\int v_g \bmod 1$. The constants form a homomorphism on G by the group law, completing the proof. The same proof applies to a torus fiber coordinate by coordinate.

In particular if B is a connected compact abelian group of dual rank at most one, its Čech H^2 vanishes: it is an inverse limit of circles (or a point), and Čech cohomology is continuous on these inverse limits. The standard classification of locally trivial principal circle bundles over compact metrizable spaces then gives a section, so this rank-one class is compact. For mixed prime-axis sums with divergent reciprocal sum, positive all-length recurrence follows by the native Bohr theorem [R169], even for originally nonergodic translations by ergodic decomposition. Nontrivial bundles in rank two do exist, as in Shalom's example [R171]; the displayed argument requires a single global trivialization on the connected base being considered.

R260. A cover-uniform charged-product theorem at all fixed lengths

Scope and status. Result with proof.

Result. A fixed periodic charged query product forces every fixed progression length at every positive density uniformly over covers; explicit integer trades include binary tags.

Scope. Fixed local modulus and charge, unrestricted counts of charged coordinates; local-law constants need not be uniform when the modulus varies.

Qualifications. The product argument allows unrestricted mark counts. Conditioning on bounded counts or costs requires a separate estimate. The same-primary deep quadratic charge is outside the coprime theorem.

Let p be prime, $p > h \geq 3$, let $V \subset F_p$ contain zero and at least one nonzero residue, and let $\tau : V \to F_\ell$ be nonconstant with $\tau(0) = 0$, $\ell \neq p$ prime. For $a > 0$ and any prescribed nonempty tag class (in particular any prescribed nonzero tag), some finite n suffices for the following: in **every** measure-preserving Z^n-action, an event of measure at least a has a positive h-return at an integer vector with all coordinate residues in V and total tag in the prescribed class. A fixed finite list of actual integer query coordinates suffices, hence the bound is independent of all covering periods. The theorem concerns unrestricted counts of true marks. It excludes low obstructions in this regime rather than providing an intermediate set.

An explicit translation trade on actual integer progressions Write $L = \mathrm{lcm}(1, \ldots, h-1)$. Choose residues w,v with $Lw, Lv \in V$ and $\tau(Lw) \neq \tau(Lv)$; L is invertible modulo p. Use ordinary integer representatives when necessary. Take

$$\Gamma = \{t = (t_0, \ldots, t_{h-1}) \in F_p^h : \sum_i t_i = 0\},$$

and choose $c_0, \ldots, c_{h-1} \in F_p$ with $\Sigma c_i = v - w$. Write $[u] \in \{0, \ldots, p-1\}$ for the usual integer representative. Build two Γ-indexed lists of actual integer queries, with respective starts and velocities

$$A_+(t) = -L \sum_i i[t_i], D_+(t) = L(w + \sum_i [t_i]),$$

$$A_-(t) = -L \sum_i i[t_i + c_i], D_-(t) = L(w + \sum_i [t_i + c_i]).$$

The velocities have fixed residues Lw and Lv, respectively. For each player j let g_j be the element of Γ with $(g_j)_i = -c_i$ for every $i \neq j$ (its j-coordinate is fixed by zero sum). Then

$$A_+(t) + jD_+(t) = A_-(t + g_j) + jD_-(t + g_j)$$

as integers. Indeed both sides are $Ljw + L\sum_{i\neq j}(j-i)[t_i]$. In particular the two player-j query lists differ by the Γ-index translation g_j. This is an exact integer identity, not an embedding of the cyclic group in Z.

Normalize each query tuple on the right by a fixed all-zero query inverse, producing a tuple of zero-exponent free words in the L-multiple symbols. The paragraph on pair walks below proves that the isolated $\gamma_{h-1}(K_i')$ lie in the full query commutator subgroup. In the product of their class-c quotients, $c = h-2$, use the explicit word F_{c+1} from the native translation argument [R218], now on Γ-indexed lists. Each player's outputs are equal under that word, by the exact translated-list identity. Thus the difference of the positive-list word and the negative-list word lies in the commutator subgroup. Its prescribed target is

$$|\Gamma|^c(\tau(Lw) - \tau(Lv)),$$

nonzero because $|\Gamma| = p^{h-1}$ and ℓ is prime to p. It follows that the full integer query game has a finite ordered contradiction. Choose a finite law containing all queries in a witnessing word and apply the Hales–Jewett / rectangle-cancellation argument below, which needs no uniform distribution on the local starts. This proves the main theorem. The direct integer construction works even for ℓ=2 with arbitrarily many slots $h < p$.

The same construction works for any finite abelian quotient $G = \Lambda/K$ of a finite-rank free lattice Λ whose order is coprime to ℓ and all the nonzero slot differences. Choose representatives $[u] \in \Lambda$ independently for $u \in G$ and replace F_p^h by G^h in the Γ formula. Require the allowed velocity differences to generate G so pair walks on Λ reach every difference in $L\Lambda$. The bracketed representative equality is unchanged. This includes repeated copies of a fixed odd prime pair with p,$q > h$ and its full periodic lifts.

For a tag τ with values in any finite abelian group C, write J for the subgroup generated by the tag differences and $M = |G|^{(h-1)(h-2)}$. Every character nontrivial on MJ has a choice of w,v for which the translated-list word has nontrivial character target. Applying the same Cauchy–Schwarz proof for that character produces exponential rectangle cancellation. One may impose the finitely many witnessing query lists in a single finite local law. In the Fourier expansion of the indicator of a target set $B \subset C$, the terms surviving this estimate give exactly the average of 1_B over MJ evaluated at the actual tag, a nonnegative function bounded below by $\min_u |B \cap (u + MJ)|/|MJ|$. Thus every positive-density set has a forced return if each such coset meets B. The full character/gluing argument works for finite abelian targets as indicated in Braverman–Minzer, Remark 1.4. The corollary applies also to deep tags whose orders exceed the relevant annihilation bound, but not to the order-p charge in the deep quadratic example.

Why bounded mark counts are a different regime. If a fixed low stencil approximates the phase $\tau(v)/2$ by a sum of player-symbol circle functions with zero error on normals and at most η error on individual true marks, every signed marginal trade with nonzero binary target must contain at least $1/(2\eta)$ marked occurrences with multiplicity: summation cancels all the player functions, while the target is a half-turn. An ordered contradiction is in particular such a marginal trade. Independently, a fixed regular product law with positive probability for a true mark puts exponentially vanishing mass on directions with any fixed bound on the number of true marks as the number of copies tends to infinity. Thus the unbudgeted conclusion cannot be transferred to the bounded-cost regime merely by conditioning this product law.

For comparison the following scalar-interpolation argument gives another proof in the narrower range h≤ℓ+1 and is not needed for the general theorem.

No scalar embedding on the integer game Suppose there were circle-valued functions f_j on the integer player symbols with

$$\sum_j f_j(x + jr) = \tau(r \bmod p)/\ell \bmod 1$$

for all integer x and all r with residue in V. Put $L = \mathrm{lcm}(1, \ldots, h-1)$ and $M = pL$. Holding r mod p fixed, take differences in each of the $h-1$ changes $(x, r) \mapsto (x - ipz, r + pz)$ with i running over slots other than j. Each other slot is killed by one such change, and the target is constant under all changes. The surviving equation says

$$\prod_{i \neq j} \Delta_{(j-i)pz_i} f_j = 0.$$

Choosing integer $z_i = L/(j-i)$ shows that on every M-residue each f_j is an integer-parameter circle polynomial of degree $\leq h-2$. This statement is exact and needs no continuity. In the Newton basis it follows by taking the initial finite differences.

Restrict x and r to multiples of L by putting $x = La + My$, $r = Ls + Mt$, where $0 \leq a, s < p$ and Ls mod$p$ is in V. At slot j write $b = (a + js) \bmod p$. The left side there is

$$P_{j,b}(y + jt + (a + js - b)/p), P_{j,b}(n) = f_j(Lb + Mn).$$

In this integer-parameter polynomial identity formally evaluate $(y, t) = (-a/p, -s/p)$, after multiplication by a fixed p-power $\mathrm{p^N}$. This use of rational arguments in circle polynomials has a precise integral meaning: expand each polynomial on Z^2 in binomial monomials. For every bounded degree k and integer m, $\binom{m/p}{k} \in Z[1/p]$. Indeed for every other prime q, approximate m/p by integers modulo arbitrarily high powers of q and use the integral binomial polynomial's q-adic continuity (or use the numerator product and count q-adic valuations). There are only finitely many degrees and residue pairs in use. Choose N large enough so that all evaluations, including the resulting univariate ones, become integers after multiplication by $\mathrm{p^N}$. Evaluation is then an integer linear combination of the circle coefficients. Substitution of the displayed integer-affine argument agrees with evaluation in the original Newton basis: the finite coefficient identities are rational identities, and after multiplication all weights are integers. No root choices in T occur.

Consequently there are well-defined circle values $z_{j,b}$ given by the $\mathrm{p^N}$-cleared evaluation of $P_{j,b}$ at $-b/p$ such that

$$\sum_j z_{j,(a+js) \bmod p} = p^N \tau(Ls \bmod p)/\ell.$$

Summing over $a \bmod p$ makes the left side independent of s. The right side remains nonconstant because $\ell \neq$p and L acts invertibly mod$p$. This contradiction proves the scalar nonembedding. The same proof applies if we insist beforehand that every query vertex and every step be divisible by L: after rescaling all integer symbols by L, the allowed residue set is $L^{-1}V$ with nonconstant reindexed tag.

A finite ordered contradiction on integer symbols Absence of the circle embedding on the L-multiple query subgame gives a finite integer signed trade among queries in this subgame, with zero signed multiplicity at every player/letter and nonzero total tag. Indeed if all integer relations among the query incidence vectors annihilated the target, their subgroup in the free abelian group on player/letters would admit the target character, and divisibility of the circle extends a character from every subgroup. Such a trade has finite support by definition of that free group.

Take the full integer query game for the free-group proof, so extra symbols and velocities away from the L-multiple subgame may be used. Fix a base tuple e_0 in the subgame, for example the all-zero query. Let K'_j be the subgroup of the free player group generated by normalized letters $u_{j,Ln}u_{j,0}^{-1}$ for $n \in Z$. In each pair of players i,j the permitted query graph has left components separated by $(j-i)Z$: a two-edge walk moves the left letter by $(j-i)(v-w)$, up to sign, for any actual permitted v,w, and all translations of these walks are available. The permitted velocity differences generate Z, since they include p and a residue generating F_p. In particular two left letters divisible by L are in the same pair component. An alternating walk therefore lifts each normalized L-multiple letter at player i to a word with prescribed i-coordinate and trivial j-coordinate. Products give the same for the whole K'_i.

As in the finite-game proof [R218], iterated commutators isolate each $\gamma_{h-1}(K'_i)$ inside the commutator subgroup of the full query group H. Conjugation by the necessary K'_i letters lifts as well. The finitely many normalized query tuples in the signed trade lie in $\prod K'_i$. Since their individual player/letter signed multiplicities vanish, the sum of their logarithms with the integer trade weights is zero in the BCH class-$(h-2)$ product of free nilpotent quotients. Apply that note's degree-elimination word with a primitive unit mod ℓ. Its total tag is nonzero because $\ell > h-2$. Its coordinates are all in the isolated lower-central subgroups and hence its H-image lies in $[H,H]$. If there were no ordered contradiction, the tag would define a homomorphism $H \to F_\ell$, giving a contradiction. All words witnessing this conclusion have finite length, including the auxiliary alternating walks. Thus there is a **finite** collection of integer queries supporting an ordered contradiction independently of any lift depth.

Choose a finite local query law giving every query needed by this contradiction positive probability. Its starting integer and velocity need not be independent or uniform. Braverman–Minzer, Lemma 3.5 gives exponential rectangle cancellation of the nontrivial character in tensor powers, with a constant for this one law. Its nonzero multiples have the same contradiction. Choose m from density Hales–Jewett at $a/2$,h, then split $n = mK$ coordinates into blocks, independently sampling the local starts x_i and velocities d_i. In any Z^n-action sample a stationary starting point z independently and consider words at times $(x_i)_i + \sum_b t_b D_b$, where D_b is the velocity vector of block b. Each word has probability at least a to belong to the given event. Thus the sum over the finitely many word-line patterns of uncharged acceptance probabilities is at least a fixed positive constant. For a fixed word line

condition on z and all inactive blocks' starts and velocities. Its slots are bounded functions, one per player, of that player's active integer query symbols $x_i + jd_i$. The global tag of this line is the sum over active coordinates since $\tau(0) = 0$. The exponential bound controls each nonconstant Fourier mode by ρ^K, independent of the stationary action and n. Fourier expansion as in the finite-game proof supplies a positive total acceptance with the prescribed tag when K is sufficiently large. Some one of the finitely many realized differences therefore has positive return. This proves the stated cover-uniform result.

The characteristic limit in this last scalar-interpolation/BCH proof is eliminated by the explicit translated lists at the beginning of the note. The periodic full-product theorem still does not address an upper bound on the total charged-coordinate cost or the same-primary characteristic of the deep quadratic example.

Chapter 8

Marked prime coordinates and extensions

R011. Arbitrary profinite abelian cocycles over the paired Haar base

Scope and status. Result with proof.

Result. Arbitrary profinite abelian cocycles over the specified paired Haar base have all-positive multiple recurrence along single-mark free-horizontal directions.

Scope. The full Haar product base and unrestricted horizontal corrections are used; this does not cover arbitrary base actions.

On the Haar product $X = \prod_i (C_{p_i} \times C_{q_i})$ let the paired direct sum act by translations. For the free-horizontal rule [R182], take only one marked positive vertical unit in a finite set of positive vertical units; the remaining vertical-active axes are pure normals, and the horizontal speed at the marked axis may be any one prescribed admissible residue from its row. Horizontal velocities are unrestricted on all vertically inactive axes. Suppose the primes escape to infinity. In **every** skew product over X by a compact metrizable profinite abelian group with an arbitrary measurable cocycle, every positive-measure event has a return of every length along these single-mark directions, in every tail. No reciprocal divergence, multiplicative-order condition, or independent-array assumption is needed here. The proof uses independent extensions only after cohomological replacement and a three-block spacing. The previously studied more restrictive paired directions lack the unrestricted horizontal motions used below.

Finite alphabet stability. Let G be a finite abelian group with no nontrivial homomorphisms to the finite abelian group J, and let $g : G \to J$. Write $\delta_s(g)$ for the probability that its s-fold additive difference at independent uniform base point and increments is nonzero. Then $\min_b \Pr(g \neq b) \leq C_s \delta_s(g)$. At $s = 1$ independent uniform values at two points give the inequality directly. Inductively $\Delta_v g$ differs from its most frequent value $c(v)$ with probability at most $C_{s-1}\delta_{s-1}(\Delta_v g)$; the mean of these errors is $e \leq C_{s-1}\delta_s(g)$. The derivative identity implies $c(v + w) = c(v) + c(w)$ except with probability $\epsilon \leq 3e$. If $6\epsilon < 1$, the elementary additivity test gives a true homomorphism within normalized Hamming distance $\epsilon/(1 - 2\epsilon)$ of c. Indeed, at each fixed v, compare $c(v + y) - c(y)$ at independent uniform $y, y + z$ using the two additivity tests at $(v + y, z), (y, z)$. Their disagreement probability is at most 2ϵ; the

most frequent value $C(v)$ of the increment therefore occurs with probability at least $1-2\epsilon$. Three simultaneous correct increments imply $C(v+w)=C(v)+C(w)$ for all v,w. Also where $c(v)\neq C(v)$, an additivity failure occurs with probability at least $1-2\epsilon$. Here $C=0$ by hypothesis, so $\delta_1(g)\leq e+\epsilon/(1-2\epsilon)$. When $6\epsilon\geq 1$ enlarge the constant and use the trivial probability bound. This proves the induction, for instance with $C_s=21C_{s-1}$.

Let $g_0,\ldots,g_{h-1}:G\to J$, allowing differing functions. Suppose $\sum_t g_t(u+ta)\neq 0$ with probability β for uniform independent u,a. If multiplication by all $1,\ldots,h-1$ permutes G, then each g_t is within $C_{h-1}2^{h-1}\beta$ of a constant. Apply the $h-1$ commuting differences in (u,a) by $(-t'b,b)$, one for each $t'\neq t$ with independent uniform b's. All summands except t are killed. The remaining expression is the $(h-1)$-difference of g_t with independent uniform increments $(t-t')b$ and uniform base $u+ta$. Any failure is among 2^{h-1} original translated failures. Apply the preceding bound.

The independent block calculation. In each of infinitely many disjoint finite blocks choose a finite-valued potential f_j in a fixed finite abelian group K, allowed to depend on all coordinates of that block together with independent neighbor data which do not move. In the block sample as normal a uniform horizontal velocity on the central block at vertical zero, or as another normal a fixed single vertical generator. The marked alternative is that same generator with its chosen horizontal residue. Set $W=K^{h-1}$ and form the temporal increments from each local potential relative to time zero. Let $P\leq W$ be generated by all normal profiles whose occurrence probabilities have divergent sum over the blocks. For every character χ annihilating P, the total probabilities β_j of a nonzero horizontal character word are summable, since there are finitely many profiles. Balance the word by adding its time-zero component. Conditional on the whole vertical position and passive neighbor data, the slot functions of the central horizontal variable take values in a fixed finite subgroup J of the circle. The central horizontal group is a product of escaping prime cycles, all coprime to $|J|$ and exceeding h after dropping finitely many blocks. The finite alphabet stability lemma applies. Upon averaging the conditioning data, each slot observable $g_t=\chi_t f_j$ differs from a function of the vertical/passive variables with probability $O_h(\beta_j)$. At the marked move and the vertical normal each slot has the *same* vertical/passive variables and two horizontal positions differing by a fixed translation; both are individually uniform on the central horizontal group. A union bound over slots thus shows that the two evaluations of χ differ with probability $O_h(\beta_j)$. The vertical normal itself has summable probability of nonzero χ. Consequently marks outside P have summable probability, using the finite dual annihilator. At least one mark profile $z\in P$ is abundant. Realize $-z$ by a finite positive sum of abundant normal generators, since W is finite. Exactly one abundant marked generator and these normals now give a zero total increment profile.

Use the tilted-pool argument [R186] on disjoint pools of these independent blocks. For a normal with random horizontal velocity include the independent speed in the conditioning experiment. The unconditioned translated starts remain uniform, so each one-block conditional density numerator is a function between zero and one of the full block with mean its matching probability. The normalized sum over a pool of total matching probability Λ therefore has squared L^2-distance from one at most $1/\Lambda$, at each time. Choose arbitrarily large pools, far beyond all coordinates in a cylinder approximant. The total fiber increments vanish with probability one, while every time marginal on the full base has L^2 norm below two. The upper bound on the number of normal slots depends on K,h, but the marginal bound can be chosen independently of that number and of the chosen profiles.

Block localization of the cocycle. For finite abelian K, write $\rho(g,x)$ for the additive

cocycle. On each finite prefix $G_{[1,n]}$ its action on the base is free. Choose the point with all first n coordinates zero in each finite prefix orbit, and define $F_n(x)$ by the cocycle displacement to x from that point. On an invariant conull set $\rho(g,x) = F_n(x+g) - F_n(x)$ for every g in that prefix, simultaneously over all n. Put $F_0 = 0$; the difference $r_n = F_n - F_{n-1}$ is invariant under the preceding prefix. It has a measurable representative depending only on coordinates with index at least n. Approximate it by f_n depending on $[n, N_n]$, with $\sum_n \Pr(f_n \neq r_n) < \infty$. Such finite-valued approximations follow from the cylinder σ-algebra on the tail. Almost surely only finitely many errors occur. Thus the sum of $r_n - f_n$ is a measurable transfer function, and ρ is cohomologous to $\sum_n \Delta_g f_n$, a finite sum at each fixed finitely supported move. This step permits arbitrary central-block sizes; no independence is imposed on the original cocycle.

Choose strictly growing boundaries m_j with $m_{j+1} \geq \max_{n \leq m_j} N_n$, and put $B_j = (m_j, m_{j+1}]$. The potentials summed over indices in B_j give one function b_j of $B_j \cup B_{j+1}$. Restrict moves to central blocks B_j with j in one fixed class modulo three, discarding a finite prefix if needed. For such a block the only moving potentials are $b_{j-1} + b_j$, depending on B_{j-1}, B_j, B_{j+1}. Distinct retained central blocks have disjoint three-block neighborhoods. They are independent Haar blocks with the neighbors passive, exactly as required above. Every retained block has at least one marked-axis candidate. There is no requirement to retain a divergent reciprocal mass of these representatives. All blockwise moves used in the proof are admissible in the full family, and the restrictions allow an arbitrary late starting tail.

For a general profinite abelian fiber first approximate the given event in measure by an event on the full base and a finite fiber quotient, to any tolerance fixed only from its positive mass and h. Carry out the quotient cohomological change, then approximate the changed finite-quotient event by an early/base cylinder at another arbitrarily small tolerance. The preceding choices make that cylinder constant through the tilted trajectory; its initial probability is its Haar mass because the early base and initial finite fiber are uniform. Both approximation errors cost at most twice the square root of their measures per time. This holds in the original full fiber as well: initial Haar is independent of the chosen moves, all fiber translations preserve it, and a base-dependent gauge is Haar-preserving on each fiber. Some finite ordinary direction consequently gives an actual positive return. This argument does not treat arbitrary base actions, or a nonabelian fiber by silently commuting group factors.

R012. Every fixed compact abelian Lie cocycle over the paired base has threshold below one

Scope and status. Result with proof.

Result. Every fixed compact abelian Lie cocycle over the paired Haar base has system-relative threshold strictly below one.

Scope. The gap depends on the cocycle and may require several marked axes; there is no uniform gap across systems.

Let a compact abelian Lie group K be the fiber of an arbitrary measurable cocycle over Haar $X = \prod_i (C_{p_i} \times C_{q_i})$. For the paired family with odd marked count and arbitrarily small positive harmonic-cost budget, the system-relative all-length avoidance capacity is strictly below one. No bound uniform in the cocycle is claimed. Unlike the profinite result [R161], the proof may require several marked axes on one progression; it does not cover the single-mark-only variant. It also works for the original paired unit moves [R156], not just the rule with unrestricted horizontal moves. These statements assume escaping primes so that a single marked move far out has cost tending to zero.

The prefix cohomological argument [R011] applies to compact abelian Lie fibers, approximating each tail function r_n by a finite cylinder f_n with summable L^1 metric errors for a translation-invariant metric. The sum of error norms is finite almost surely, so $\sum_n (r_n - f_n)$ converges in the complete group and is a measurable transfer. Group the cylinder potentials into successive overlapping pairs of blocks as before. Freeze alternating separator blocks. Conditional on this passive environment y, the cocycle on the moving cores is a sum of independent single-block potentials, with the neighbors fixed. In each active core use only one arbitrarily chosen pair and its legal horizontal, vertical and diagonal unit moves (plus no move). The core starts are independent Haar; any unselected coordinates of the core are passive local data for these particular trials.

Form in $K^{\mathbb{Z}} \times C_2$ the closed subgroup generated by all abundant increment profiles with their normal/odd tags and all even constants. Here abundance means divergent total probability in every neighborhood under the given type of single-core Haar trial in the fixed environment. A marked abundant point exists by compactness. Shifting time corresponds to shifting the starting site and subtracting a common constant, so the even subgroup L_y and its odd coset are shift-invariant. The odd coset defines a shift-fixed element of order dividing two modulo L_y.

By the periodic-profile argument (see the independent Artin–Rees note [R196]) the odd coset contains a periodic trajectory with values of 2-power torsion. Precisely the even annihilator R is a submodule of the finitely generated Noetherian module $A = \widehat{K}[T^{\pm 1}]$ over $\mathbb{Z}[T^{\pm 1}]$. Its odd character $\lambda : R \to C_2 \subset \mathbb{T}$ vanishes on $(T-1)R$. Artin–Rees gives $R \cap (T-1)^m A \subset (T-1)R$ for some m, so λ factors through the subgroup image of R in $A/(T-1)^m A$. Extend to a character of that group with values in the divisible group of 2-power roots of unity. The quotient is finitely generated as an abelian group, so the image exponent is finite. Its coordinate trajectory has m-th temporal difference zero, hence is given by its finitely many Newton coefficients; the finite 2-exponent gives a common 2-power period for this trajectory. In the case where the even and odd cosets coincide, a constant trajectory already works.

There are only countably many possible periodic trajectories of this kind, since K is a

compact abelian Lie group. For each fixed such trajectory its membership in the conditional odd coset is measurable in y. One way to see this is to test each basis neighborhood of the specified tagged trajectory for approximation by a finite sum of generators. Use a countable base of open balls in the profile space: an abundant occurrence sum for a ball ensures its closure meets the abundant-point set, by compactness and finite covering. One can therefore test by countably many finite lists of balls with divergent occurrence sums whose closed Minkowski sum (also allowing a constant from a countable dense subgroup) stays inside the target neighborhood. Conversely every genuine approximation admits such a list on shrinking the balls. These occurrence sums are measurable in the passive data by Fubini. Thus some fixed trajectory of period P is attainable on a measurable set E of passive environments of positive measure.

Set $\delta_f = 1 - \mu(E)/(2P) < 1$. If an event A in the cocycle system has mass greater than δ_f, a positive-measure subset of E has conditional fiber/base mass above $1 - 1/P$; otherwise its global missing mass would be at least $\mu(E)/P$. Work in each of those environments after the gauge. Approximate the conditional event by a cylinder on finitely many moving cores and the full torus/Lie fiber to an arbitrarily small tolerance. A positive fraction of its early-coordinate sections has fiber mass above $1 - 1/P$ by a margin depending only on the conditional original mass. For each such fiber set B, independent uniform translation of the fixed trajectory gives an all-B h-window with probability at least $1 - P(1 - \mu B) > 0$, since at most P distinct points occur. This lower bound is uniform over the sections with that margin.

The tilted pool construction approximates the trajectory by a finite number of abundant blocks, with odd mark count and arbitrarily small total marked cost, beyond all the cylinder coordinates. Select one index per pool conditioning its local profile into the required neighborhood, and otherwise leave the pool coordinates uniform, as in the profile pooling proof [R186]. The time marginal of the full base has L^2 norm at most two under the auxiliary law, independently of the number of pools. The finite-window fiber test after uniform common translation is continuous by L^1 continuity of translations, so the cylinder trajectory has a uniformly positive lower bound (arbitrarily close to the one above). The cylinder approximation can be chosen beforehand small enough that the actual event still has a positive return, since its total error is at most twice the square root of its measure per time. Countability of the possible ordinary directions gives a positive return under one fixed direction on a positive-measure collection of environments, by Fubini. This proves the fixed-system claim; the possibly very large period and the environment mass in δ_f can depend on every part of the original measurable cocycle.

R013. Vertical reciprocal-prime cost of a forcing packet

Scope and status. Result with proof.

Result. A finite paired forcing packet has near-full avoiding masks with loss bounded by a prescribed ϵ plus its vertical reciprocal-prime mass.

Scope. This forces macroscopic reciprocal cost in any candidate forcing packet; boundedly many late axes cannot suffice.

Consider a finite set I of pairs $C_{p_i} \times C_{q_i}$, or an integer-horizontal model with the first entries read modulo p_i. Write horizontal coordinates u_i and vertical v_i; ordinary vertical moves s_i are in $\{0,\pm 1\}$, and horizontal r_i is unrestricted where $s_i = 0$. For a fixed tolerance $\tau < 1/4$ consider either of the following rules:

1. On active vertical axes write centered integer representatives r_i. Their sum is odd and $\sum_{s_i \neq 0} |r_i|/p_i < \tau$.
2. Each active r_i/p_i is associated with center 0 or 1/2, an odd number of centers are 1/2, and their total circle distance from their centers is $< \tau$.

For every $\epsilon > 0$ there exist a length H and a bound P depending only on ϵ,τ such that if all $p_i > P$ there is an H-avoiding set for **every step of the indicated rule on this packet** of density at least

$$1 - \epsilon - \sum_{i \in I} 1/q_i.$$

The same holds for any restriction of these directions. In particular no fixed finite number of arbitrarily late axes, however freely their horizontal moves are allowed to vary, can provide the high-density forcing gadgets needed in the conjecture. A macroscopic aggregate reciprocal mass of the vertical axes is necessary. The number of marked moves in a single step need not be bounded.

Proof. For a circle arc of length $< \epsilon$, choose M clearing every small denominator in the quantitative degree-two Weyl test, then H so large that every quadratic circle trajectory of binomial second coefficient c with $||Mc|| \geq 1/4$ hits that arc from every start within H sites. Equivalently this follows by compactness and Weyl equidistribution after clearing the bounded rational denominators relevant to an arc of that length. Choose P larger than 4M. Set

$$F(u, v) = \sum_{i \in I} c_i u_i [v_i]_{q_i}/p_i \bmod 1,$$

where $[v_i]$ is the representative in $\{0, \ldots, q_i - 1\}$ and c_i is an integer. Delete all sites with any vertical coordinate zero, costing at most $\Sigma 1/q_i$, and the sites where F hits a randomly translated arc. An arc translation with loss $\leq \epsilon$ exists by averaging. Along any entirely retained legal line the vertical representatives never wrap, since unit vertical steps cannot cross the cut without meeting zero. Horizontal representatives can always be interpreted by

linear lifts since F uses them modulo p_i. Thus F is a quadratic along that line with second coefficient $c = \sum_{i:s_i \neq 0} 2c_i r_i s_i / p_i \bmod 1$.

For rule 1 choose $4Mc_i \equiv 1 \bmod p_i$. Then $2Mc_i/p_i$ is an odd half-integer plus $1/2p_i$, so Mc is within $\tau/2$ of an odd half-integer by parity of $\Sigma r_i s_i$. For rule 2 choose $2Mc_i \equiv 1 \bmod p_i$. Then $Mc = \Sigma s_i r_i / p_i$ on the circle and is within τ of a half-turn. Both violate the missed-arc requirement. The model with one integer horizontal coordinate inherits the construction by the CRT map, which is onto the whole finite horizontal product. This finite cost estimate also gives threshold one on tails whenever the total vertical reciprocal series converges.

R014. Full horizontal normal moves and a common characteristic factor

Scope and status. Proved subcases and unresolved extensions.

Result. The asymmetric family has a finite five-slot color obstruction, an exact directional characteristic-factor reduction, and unconditional fixed-order upper bounds including one third for triples.

Scope. The general h-bound tends to one; the all-system intermediate threshold remains open.

Let $G = H \oplus V$, with $H = \oplus C_{p_i}$ and $V = \oplus C_{q_i}$ on pairwise distinct odd primes, generated by a_i, b_i respectively. Take vertical speeds $s_i \in \{0, \pm 1\}$. Wherever $s_i = 0$, permit **any** horizontal speed $r_i \in C_{p_i}$. At the remaining finitely many coordinates one may take horizontal zero for normals and a prescribed speed for marks. The number of marks is odd. Two concrete rules with fixed short finite-color obstructions are given below. The ability to add an arbitrary element of $H_{S^c} = \oplus_{i \notin S} C_{p_i}$, where $S = supp s$, is what makes the characteristic reduction possible.

A fixed finite-color obstruction On each pair let S denote unit spatial shift in the *vertical* coordinate. Over F_p choose $k : C_q \to F_p$ and write $f(u, v) = uk(v)/p \bmod 1$, using the canonical additive identification of F_p with the pth roots in the circle. Use the filter $(T - I)^2(T - S)(T - S^{-1})$, where T is temporal shift along the unknown direction. At any pure horizontal speed—including every element of C_p—the sequence is affine in time, so the filter vanishes. At pure vertical ± 1 it also vanishes. At speed $(r, 1)$ it is $rH(v)/p$ where

$$H = (S - I)^2(S - S^{-1})Sk.$$

At speed $(r, -1)$ it is $-rH(v-4)/p$. Indeed use first the factor whose spatial displacement is the actual vertical speed and then $(S^{-1} - I)^2 = S^{-2}(S - I)^2$. The operator displayed for H is onto all zero-sum arrays: over F_p the cyclic shifts diagonalize over a splitting field since $p \neq q$; their nonconstant eigenvalues are qth roots different from both 1 and -1, and all displayed factors are invertible there.

For a unit horizontal mark $r = \pm 1$, prescribe the zero-sum array $H(v)$ to be integers nearest to a number $p/2 + O(p/q)$ modulo p: choose the target total to be a multiple of p nearest to `qp/2`, and distribute this integer total as equally as possible among all q entries. Each normalized entry is within $1/(2q) + 1/p$ of $1/2$. Thus any odd set of marks whose summed errors are below a fixed $\kappa < 1/2$ yields a filtered value within κ of a half-turn at every starting position.

Alternatively take $q \geq p$ and prescribe only the values $H(v) \in \{+1, -1\}$ with integer sum p, possible since q, p are odd. Then for a single mark *every* horizontal r with $||r/p - 1/2|| \leq \kappa$ gives the same half-turn conclusion; arbitrary odd numbers of marks and arbitrary near-zero horizontal moves at nonzero vertical speeds are also allowed if the sum of their distances to the assigned centers (half-turn or zero) is at most κ. This is a macroscopic permitted horizontal interval for a single mark.

On a finite packet, observe the circle-valued sum of $f_i(u_i, v_i + b)$ at each of the finitely many spatial offsets b of the filter. Color these observed sums by fixed small-arc partitions.

Coefficients at each spatial offset sum to zero because the temporal filter contains $T - I$. Hence a monochromatic five-slot line would put the filtered sum arbitrarily near zero; the true value is bounded away by $1/2 - \kappa$. The palette and length are uniform in the packet. Compactness (or translation limits of such finite colorings) yields a positive-mass stationary avoiding event on the infinite group. Thus the family has a positive all-length threshold. The question whether it has threshold strictly below one is unresolved.

Exact reduction at each length Here is the more general statement. Let $D \subset H \oplus V$ satisfy $d + H_{S^c} \subset D$ for each $d = (r, s) \in D$, $S = \mathit{supp} s$. For a fixed length $h \geq 3$, discard the finitely many horizontal generators of order $\leq h$, using their remaining tail action for all averages. Let $l = h - 1$; call the factor detected by the order-l Host–Kra seminorm $Z_{l-1}(H)$. The notation denotes the *directional* factor of the horizontal action, which need not be ergodic. It is invariant under all of G, by commutativity. Any h-avoiding event in a stationary G-system gives an h-avoiding event of at least the same measure in this one factor. Conversely any witness in such a factor is of course a witness in a system. Thus fixed-order capacities are **exactly** determined by these horizontal factors. The common factor works simultaneously for every omitted finite support S and every permitted mark size. For $h = 2$ one can use the order-two seminorm, giving a possibly larger factor. The factors still depend on h.

Finite-index comparison. For $l \geq 2$ the seminorms of H and $H' = H_{S^c}$ have identical nullspaces. The details below use rectangular box averages, allowing the subgroup sizes in different increments to be sent to infinity in the iterated order used in the Host–Kra definition. Exhaust H by finite subgroups $H_n = P \oplus K_n$, where $P = \oplus_{i \in S} C_{p_i}$ on the retained axes and $K_n \uparrow H'$. Put $m = |P|$. At fixed sizes $n_1, \ldots, n_l$, work on a finite H_M containing all these subgroups. The rectangular seminorm of a function on H_M averages the usual conjugated cube product over the base uniformly on H_M and increments independently uniform on H_{n_j}. These are box seminorms with the Gowers–Cauchy–Schwarz inequality: write the increments as differences of two independent points from each increment subgroup and supply an independent uniform base displacement. Its modulation by a character of H_M leaves it unchanged because $l \geq 2$. Multiplication by 1_{K_M} does not increase this seminorm: it is the average of the m quotient-character modulations, so use the triangle inequality. But this multiplication restricts the base and all l increments to their respective K-subgroups, with probability $m^{-(l+1)}$. Thus the restricted seminorm is bounded by $m^{(l+1)/2^l}$ times the full one. For the reverse inequality split each increment into its P-component and its K_{n_j}-component and the base into a P-coset. Each mixed box average is bounded by the product of the K-seminorms of the appropriate translated functions. Apply these arguments to orbit functions $g \mapsto f(T_g x)$ and integrate in x; invariance and Hölder replace every translated factor by the same integrated K-seminorm. Iterating the increment-size limits proves both comparisons for the action seminorms. The limit formulas are the usual mean-ergodic definition for countable abelian actions, and do not require a diagonal cubic averaging theorem.

Characteristic estimate. For bounded f_t and a finite subgroup $K_N \subset H'$ consider $E_{u \in K_N} \int \prod_{t<h} T_{tu} f_t$. Its limit superior in absolute value is bounded (for tests bounded by one) by the $U^{h-1}(H')$ seminorm of any one distinguished slot. To see the estimate using iterated limits, for each of the other slots successively duplicate the averaging variables by $(x, u) \mapsto (T_{-tb} x, u + b)$ with b uniform in any chosen earlier $K_{n_j} \leq K_N$, applying Cauchy–Schwarz to remove the unchanged slot. The resulting increments in the distinguished slot are $(t_0 - t)b$ and are uniform on that same subgroup since nonzero differences have size

$< h$ and all remaining primes exceed h. The final bound is the rectangular action box seminorm at the chosen n_j. Take the limit superior in N and then the iterated limits in n_j. Consequently a function with zero conditional expectation on $Z_{l-1}(H)$ contributes zero by the finite-index comparison. Telescoping gives characteristic replacement of all slots, including tuples $f_t = T_{td}f$ for any fixed d. For $h = 2$, direct use of the mean ergodic theorem gives the H' invariant projection instead, controlled by $U^2(H')$ by monotonicity.

From a positive reduced return to an allowed original return. Write $g = E(1_A|Z_{l-1}(H))$, $B = \{g > 0\}$. If B has an h-return along some $d \in D$, there is $c > 0$ so $E = \bigcap_{t<h} T_{td}g \geq c$ has mass $b > 0$. Uniformly over finite K_N, $E_{u\in K_N}\mu(\bigcap_{t<h} T_{tu}E) \geq \kappa(b, h) > 0$. Indeed sample $m(b, h)$ independent uniform increments in K_N, apply the elementary density Hales–Jewett word-cube inequality from the graph note [R016] to their subset sums, and note every nonempty subset sum is uniform. Thus the reduced average at $d + u$ is at least $c^h\kappa(b, h)$, since $T_{tu}E \subset T_{t(d+u)}g \geq c$. By characteristic replacement the original average is eventually positive too. Every one of its directions belongs to D. Contraposition gives exact avoidance for B, whose mass is at least that of A.

Unconditional fixed-length upper bounds For the full asymmetric family every event of mass **greater than one third** has a three-term return, in every stationary action. More generally at length $h \geq 3$ a mass strictly greater than $1 - 1/(h - 1)$ suffices. Only the horizontal closure and the existence of a marked legal lift for each nonempty positive vertical subset sum are required. In particular this is valid for the macroscopic interval of horizontal marks. The estimate for general h tends to one.

Set $\sigma = 1-1/(h-1)$, let $g = E(1_A|I(H))$ and take $\gamma > 0$ with $Y = \{g > \sigma+\gamma\}$ of positive measure. For each tail a nonempty sum s of a bounded number of distinct generators b_i has an h-return on Y of measure at least a fixed $c > 0$, by the same density Hales–Jewett cube bound. Fix a legal horizontal lift d of s (for example one marked speed and zero at its other vertical positions), and $H' = H_{S^c}$. Horizontal-invariant Y has the same return measure along d. Furthermore $g_S = E(1_A|I(H'))$ converges to g in L^2, uniformly over all finite omitted sets S in increasingly late tails. In fact $I(H) \subset I(H') \subset I(H_{firstJ})$ when S misses the first J axes, and reverse martingale convergence for the exhaustive finite subgroups gives the bound by $||E(1_A|I(H_{firstJ})) - g||_2$. Thus, on choosing the tail sufficiently late, the H'-invariant set on which all $T_{td}g_S$, $0 \leq t < h$, exceed $\sigma + \gamma/2$ has measure at least $c/2$; use the uniform error, a union bound, and the return set for Y.

Restrict to this invariant set E and average the original intersections along $d + u$ over larger finite subgroups of H'. Anchor the product at one slot j by translating the integration variable by $-ju$ and put $f_t = T_{td}1_A$. The product at the remaining $h-1$ positions is bounded below by $\sum_{t\neq j} T_{(t-j)u}f_t - (h - 2)$. Its u-mean tends in L^2 to $\sum_{t\neq j} T_{td}g_S - (h - 2)$, since all remaining prime orders exceed h and the nonzero slopes are automorphisms of each finite averaging group. This is bounded below on E by $(h - 1)\gamma/2$. The anchor $\int_E f_j$ is positive, again by its conditional density. Some direction therefore returns in the original system. For three slots the two endpoint terms after anchoring at the middle are just the translates by $\pm u$. For pairs the analogous argument works at every positive mass, since after anchoring there is only one other factor.

For the improvement at $h = 3$, first select a horizontal tail with every prime $> P$, where P is large in terms of the margin above $1/3$. Use this tail as H throughout the conditional-density argument (the vertical finite-support lifts still allow all horizontal corrections in this

tail outside S). Repeat the argument giving a positive H'-invariant set with all conditional densities above $1/3+\gamma$, now taking $\sigma = 1/3$. The margin γ can be fixed in advance in terms of $\mu(A)-1/3$, independently of P, by the mean of each invariant projection. For a large finite subgroup $K_N \subset H'$ the orbit sections $B_t(x) = \{u \in K_N : T_u x \in T_{td}A\}$ thus all have density exceeding $1/3 + \gamma/2$ for a positive-measure set of starts, by the mean ergodic theorem. In a finite abelian group whose prime divisors all exceed P, Kneser's sumset theorem implies $|B_0 + B_2| \geq \min(|K_N|, |B_0| + |B_2| - |K_N|/P)$: either the stabilizer is the whole group or its index exceeds P. Since doubling is bijective, $(B_0 + B_2) \cap 2B_1$ is nonempty for large P. A point of this intersection gives an affine three-term progression meeting the three respective sections. Finitely many choices of its initial point and difference suffice on this positive-measure set of starts, so one resulting $d + u$ is a positive original return. This proof needs no qualitative nilfactor approximation.

Using only these individual horizontal-component densities cannot supply a uniform threshold over h. For any $a < 1$ there are a length and circle arcs I_t, all of Haar measure a, with no pair $x, d \in T$ satisfying $x + td \in I_t$ at every slot. To see this choose independent uniform arc centers and a fixed small $e > 0$ with $a + 4e < 1$. Use an e/h grid for both variables. If the original system is feasible, a grid pair satisfies all arc constraints enlarged by at most $2e$ in radius. There are only $O_e(h^2)$ pairs, each passing all enlarged arcs with probability $(a + 4e)^h$, so for large h some choice has no such pair. A uniform-in-length argument for the candidate must also use the relationships between vertical translates, which this heterogeneous horizontal estimate discards.

An ergodic Abramov subcase Suppose on this factor the horizontal action is ergodic and its bounded-degree phase polynomials generate the σ algebra (the degree may vary among observables). Every vertical prime-order transformation V of order q then multiplies each such phase by a constant, i.e. the vertical action has a basis of eigenfunctions. Induct on polynomial degree. For a horizontal generator u of order p, with $D = T_u - I$ on circle-valued phases of degree $\leq d$, one has $D^{d+1} = 0$, $(I + D)^p = I$; successively applying the binomial relation shows $p^d D = 0$. The lower-degree phase $D_u P$ is p^d-torsion and by induction its difference under V is constant. This constant is also q-torsion, hence zero by coprimality. It follows that $D_V P$ is invariant under every horizontal generator, so constant by horizontal ergodicity. The induction starts with invariant constants. The one-sided compactness theorem [R016] now gives positive recurrence at every density for the paired template, including macroscopic horizontal marks: choose each DHJ horizontal generator to be a fixed allowed marked speed, and use two vertical generators of opposite signs; exactly one is paired to its active horizontal counterpart in the matching. Pure horizontal speeds remain unrestricted. A general directional factor can have moving horizontal ergodic components, with vertical transformations transporting the fibers. Even in the ergodic case arbitrary horizontal factors of all orders have no available general Abramov conclusion: Shalom, Theorems 1.19, 1.21, 1.26 and 1.31.

Local torus arrays with a horizontal annihilating word Let $f : C_p \times C_q \to T^k$ and let the length-h temporal word have coefficient vectors $a_0, \ldots, a_{h-1} \in Z^k$. Suppose uniformly for all horizontal speeds and starts that $||\sum_t a_t \cdot f(u + tr, v)|| \leq e$. Then for each index j the projected function $a_j \cdot f(\cdot, v)$ is within $2^{h-1}e$ in sup norm of an exact circle polynomial of degree $\leq h-2$, provided $p > h$ and $3 \cdot 2^{h-1}e < 1$. Apply differences in (u, r) with increments $(-tg_t, g_t)$ for every $t \neq j$. The tth summand vanishes under its own difference. The only

survivor is the $(h-1)$-fold difference of $a_j \cdot f$ with horizontal increments $(j-t)g_t$, which can be arbitrary. Its circle norm is bounded by $2^{h-1}e$.

For clarity, uniform near-polynomiality in the last step has an elementary exact correction on any finite abelian group. Suppose all order-b differences of F have circle norm $\le \delta < 1/3$. Lift each to its real representative $D(x; k_1, \dots, k_b) \in [-\delta, \delta]$. Its additive derivative cocycle identity in each increment is exact over the reals, since the proposed left side and right side differ by an integer of size at most $3\delta < 1$. Set $E(x) = (-1)^b E_{k_1,\dots,k_b} D(x; k_1, \dots, k_b)$. Taking a difference Δ_s and averaging the cocycle identity in one increment replaces that averaged increment by s and changes the sign: translate the increment average by s. Iterating proves $\Delta_{s_1} \cdots \Delta_{s_b} E(x) = D(x; s_1, \dots, s_b)$. Thus $F - E \bmod 1$ is an exact degree-$< b$ polynomial and $|E| \le \delta$.

For $p > h$ an exact degree-$d \le h-2$ circle polynomial on C_p is a constant plus an ordinary F_p polynomial divided by p: Newton expansion on Z, periodicity, and the divisibility of $\binom{p}{j}$ for $1 \le j < p$ give p-torsion of each nonconstant Newton coefficient, successively from highest downward. Consequently write $a_t \cdot f(u, v) = c_t(v) + P_{t,v}(u)/p + O_h(e)$ on the circle, with $\deg P_{t,v} \le h-2$ and zero constant coefficient. If the same word has residual $\le e$ along either vertical speed $s = \pm 1$, then for every v the nonconstant polynomial coefficients in $\sum_t P_{t,v+ts}(u)$ vanish exactly for all large p once e tends to zero. Otherwise a nonconstant polynomial of degree $\le h-2$ has at least $p/(h-2)$ distinct values, whereas the residual lives in a translated circle interval of length $O_h(e)$. Applying the same argument to the horizontal residuals implies that all nonconstant coefficients of the polynomial in (u, r) given by $\sum_t P_{t,v}(u + tr)$ vanish (first vary one variable, then the other). This is an array-level obstruction test, not a replacement of arbitrary systems by local array fibers.

Two-periodic profiles are not compelled. Multiply the earlier filter by $T + I$. The new mean-zero operator defining H acquires the invertible factor $S + I$ on the odd q-cycle. At a negative vertical mark it gives the same values up to an additional shift, still $\pm r/p$ in the macroscopic construction. Every exactly two-periodic temporal list of whole spatial observations is killed by $(T - I)^2(T + I)$ because its adjacent sum is time constant. Thus none of those lists can represent the forced mark profile even with r selected anywhere in the macroscopic interval. Nor can horizontal variation on a single vertical axis by itself give the desired upper bound: the union restricted to any *single supported* vertical motion $\pm b_i$ has threshold one by the common character $b_i \mapsto (q_i - 1)/(2q_i)$ and the root argument. The full asymmetric model permits arbitrary finite vertical supports.

Why a single active vertical coordinate cannot bootstrap the first example Let $S_i \subset V_i \setminus \{0\}$ be completely arbitrary vertical templates and $R_i \subset H_i$ nonempty horizontal marker sets. Form `D*` from exactly the steps (u, s_i) with a *single* active vertical coordinate $s_i \in S_i$, arbitrary horizontal coordinates away from i, and $u_i \in R_i$. Write $S = \bigcup S_i$. Pulling back vertical actions with trivial horizontal action gives $b(D*) \ge b(S)$. Conversely if the vertical union is fully density large on all tails (critical threshold zero), the same holds for `D*`.

Fix the length h, a mass $a > 0$, any tail, and take the usual horizontal word-cube constants $m(a, h), \kappa(a, h)$. Choose successively m disjoint finite blocks of vertical indices within the tail such that the union of their S_i in each particular block forces an h-return above mass $\kappa/2$. The blocks exist by finite compactness for the vertical union on each remaining tail, since its critical threshold is zero. For each block J form a horizontal vector u_J with one chosen

marker from R_i at **every** index $i \in J$, and zero elsewhere. An h-return of mass at least κ occurs at a nonempty subset sum u of these block vectors by DHJ. Write $B = \bigcap_{t<h} T_{tu} A$. Select any participating block. Its vertical templates supply some i in it and some $s_i \in S_i$ which is an h-return for B. Then $u + s_i \in D*$ is an h-return for A. The construction works on every tail. With the squarefree-tail transfer [R241], an intermediate saturated threshold for `D*` therefore requires that its vertical projection already have an intermediate saturated threshold. This does not apply when steps with several active vertical indices are also added.

In particular taking all nonzero vertical speeds at each marked individual coordinate forces all densities, by ordinary finite Szemerédi on large q_i (even allowing a growing initial interval of speeds, or its unit dilation, suffices). In this case each forcing block above can be a singleton. A positive short obstruction in the asymmetric model uses the ability to vary the active vertical support together with restrictions on the velocities at each active index.

R016. Bipartite prime graphs and the active-edge condition

Scope and status. Proved subcases and unresolved extensions.

Result. Bipartite prime graphs admit several endpoint-one exclusions, a matrix representation of parity, random blowups, and recurrence when one side of the tested orbit is compact.

Scope. Unbounded graph rank with two noncompact sides is not resolved.

Take disjoint prime axes indexed by two sets L, R, a bipartite graph E, and signed unit steps supported on finite sets $U \subset L, W \subset R$. Require the induced active edge count $|E \cap (U \times W)|$ to be odd and the sum over active edges of $1/p_u + 1/p_w$ to be at most $\kappa < 1/2$. Constants can be adjusted if the cost is written in half reciprocals as in the paired construction. All results are about this specific rule. No all-system upper bound below one has been proved for an infinite-rank graph.

Finite rank is excluded. If the F_2-incidence matrix has finite rank, there are finitely many left-neighborhood types C. For an admissible direction some type contributes an odd number of active edges. Thus $|C \cap U|$ is odd and its common neighborhood meets W. Every member of $C \cap U$ then belongs to an active edge, whence $\sum_{u \in C \cap U} 1/p_u \leq \kappa$. Give the left axes of C character values $(p_u - 1)/(2p_u)$ and give the other axes zero. The direction's phase is within $\kappa/2$ of one half, including independent signs of the entries. The finitely many such characters cover the family by Bohr-separated subclasses. The root-rotation repair and squarefree tail formula [R241] give saturated capacity one. This includes complete bipartite graphs and bounded-rank blowups of finite graphs. Exceptional finitely many prime axes can be discarded by the tail formula.

No ordinary $\mathtt{FS}_3$ forcing. In late tails all prime orders exceed 3. If signed unit steps v_j have all nonempty subset sums again signed unit steps, at each coordinate at most one $+1$ and at most one -1 can occur (apply the condition to the pair otherwise). Thus occupancy modulo two of each sum is the sum in F_2 of the occupancies of its summands. The parity of its induced active edges is a Boolean polynomial of degree at most two in the subset indicators, with zero constant term. It cannot equal one on every nonempty subset of three generators, as the unique multilinear polynomial for the latter function has a nonzero cubic term. No assumption of disjoint support is needed. The same degree obstruction applies to higher rank.

A summable vertex cover is also excluded Suppose some vertex cover C has $\sum_{u \in C} 1/p_u < \infty$. Then the saturated graph family has capacity one, irrespective of its incidence rank. This includes the case that either entire side has convergent prime reciprocals. Orient every edge (u, w) from an endpoint $u \in C$ and fix the orientations once. For each vertex w let $\chi_w(x_w)$ be the prime-cycle character with generator image nearest to $1/4$ in T; in particular $||2\chi_w(e_w) - 1/2|| \leq 1/p_w$.

On an arbitrary finite packet of tail axes define the scalar

$$\Phi(x) = \sum_{(u,w)} [x_u]\chi_w(x_w) \bmod 1$$

over oriented edges of its induced graph, where $[x_u]$ is the integer representative from 0 to $p_u - 1$. If along signed-unit increments d these representatives add as integers for the

covered coordinates, then on such a run the scalar is a circle polynomial of degree at most two. Its second difference is $2\sum_{(u,w)} d_u \chi_w(d_w)$, within κ of $1/2$: only the active edges occur, the signs of the half-turns are immaterial, their number is odd, and the summed errors are bounded by $\sum_a ctive 1/p_w \le \kappa$. In particular the second-difference distance to zero is at least $1/2 - \kappa > 0$. The coefficients of linear terms from inactive endpoints need not be controlled.

Here are details giving uniformity in the packet. Let $c_0 > 0$ satisfy $4c_0 < 1/2 - \kappa$. For each root stride R use $z_R(x) = \Phi(R^{-1}x) + \gamma_R$, where γ_R is an independent uniform scalar. Exclude positions where any coordinate $u \in C$ of $R^{-1}x$ equals 0 or $p_u - 1$. Their Haar cost for that row is at most $2\sum_t ail, C1/p_u$, uniformly in the packet. Thus along an allowed line, consecutive positions at stride R add d_u as ordinary integer representatives on every retained residue stretch. Some observation among each three consecutive slots at that stride must have circle norm $\ge c_0$, by the second-difference bound. Call those observations active.

Given a loss budget η, choose r with $(1 - 2c_0)^r < \eta/4$. Delete simultaneous activation of all r roots, and also delete a small fixed open arc and a thin fixed band around the activation boundary in every row; choose their total probability below $\eta/4$. These probabilities, for each group position, follow from the independent uniform offsets. The list of root strides can be constructed without any prior packet-size or prime cutoff, as follows. At one fixed root, limits of the scalar values on arbitrarily long centered legal stretches exist diagonally. On each residue modulo R their scalar sequence is a degree-at-most-two circle polynomial: extract the three Newton coefficients, with ordinary digit addition as established above. The missing open arc forces rationality of the nonconstant coefficients by the elementary Weyl criterion, hence that limiting sequence is periodic on the residue. Because of the band at the threshold, every limiting binary activation code is periodic. Its family on full bi-infinite stretches is a closed subshift; it has a common period. To see the last assertion generally, a compact subshift of periodic words cannot have unbounded periods: otherwise a convergent subsequence converges to a periodic word w with increasingly long exact central matches but eventual mismatches. Shift to the first mismatch on a matching segment boundary, passing to a subsequence of offsets in one phase of w and choosing a side with matching lengths tending to infinity. The resulting limit agrees with a periodic word on one half-line and differs at the next site, so is not periodic. Thus only bounded periods are possible.

Take the next root divisible by all preceding common periods. The uniform code family for a root can equivalently be defined by all degree-two residue polynomials missing the arc and band with the stated three-slot activation bound; this makes the common periods independent of any prospective axes. After fixing the finite list of roots, start sufficiently far on the axes that none of their primes divides any root and that the total endpoint losses for all roots are below $\eta/4$. Failure of a uniform line length would give joint limiting codes along ever longer legal stretches, across arbitrary finite packets. Starting at time zero activate each row successively by a shift of $0, R$, or $2R$ in its own stride. Later shifts preserve earlier codes. At the final time all rows would be active, contradicting retention. Thus all finite packets admit masks of density $> 1 - \eta$ avoiding one common length. Average over the offsets to fix them; translate-averaging and compactness over increasing packets give the stationary infinite-tail avoider. The squarefree-tail formula [R241] proves the assertion. The same conclusion holds if the edges partition into finitely many subgraphs each of finite incidence rank or possessing a summable cover, since an odd total has an odd part and each part's active-edge cost is at most the total cost.

A matrix model of the edge parity. For a finite packet use a qubit for each left vertex. Assign the Pauli matrix X_u to that qubit and $Z_w = \prod_{u\ w} Z_u$ to each right

vertex. The X_u commute with one another, as do the Z_w; a cross pair anticommutes exactly when it is an edge. Replace the eigenvalue -1 of each involution by a nearest root of the corresponding odd prime order, keeping the eigenvalue 1. Call the resulting unitaries A_u, B_w. Entries on either side still commute, all nonedges commute exactly, and on an edge $||A_uB_w + B_wA_u|| \le 2\pi(1/p_u + 1/p_w)$ by the triangle inequality and $||A_u - X_u|| \le \pi/p_u$, $||B_w - Z_w|| \le \pi/p_w$. Inverses obey the same estimate. With A, B the signed products over active vertices, commuting past each other one at a time therefore yields $||AB-(-1)^{|E(U,W)|}BA|| \le 2\pi\kappa$. This estimate charges only active edges. Along a legal line the normalized unitary observable is $A_xA^tB^tB_x$, within $2\pi\kappa\binom{t}{2}$ of $(-1)^{\binom{t}{2}}A_x(AB)^tB_x$. This supplies a finite-rank representation but no dimension-independent finite palette or density bound: the Clifford representation dimension grows with the graph rank. The scalar finite-color stencil in the paired construction is separate.

What a random graph can contain. Give every pair of the two countable prime lists independent Bernoulli(1/2) adjacency, assuming the harmonic weights on each list diverge. Almost surely, on either side, the common exact adjacency pattern to any fixed finite set on the opposite side has divergent harmonic weight on every tail. For each fixed pattern the membership indicators of vertices are independent with constant expectation 2^{-k} and variances bounded by one. The centered harmonic-weighted series converges almost surely, since its variance sum is bounded by $\sum_n n^{-2}$. Intersect the countably many resulting probability-one assertions.

Consequently such a graph contains a blowup of each prescribed countable bipartite graph, with all its new vertex classes of divergent harmonic weight. Recursively choose finite batches of new vertices for a class in the requested side, imposing exactly its prescribed adjacency to the finitely many previously placed opposite-side vertices. The preceding property allows a batch past all previous primes of harmonic mass at least one. Schedule each class infinitely often. This builds the desired pairwise disjoint classes and all their prescribed edges and nonedges. In particular the model includes induced matchings, independent rectangles, and any chosen finite incidence pattern with a separate large reservoir for every vertex. It does not itself transfer a finite-group density estimate across different prime orders, nor overcome the small cost on active edges.

A one-sided compactness theorem Assume the F_2-incidence matrix restricted to every prime tail has infinite rank. Allowing signed unit steps as above, in any measure-preserving action and at any fixed length $h \ge 2$, the following suffices for positive recurrence of an event A of mass $a > 0$: each of the $h-1$ families $\{T_{te_w}1_A : w \text{ on the right side}\}$ is relatively compact in L^1. No assumption is made on the left action, on the entire right orbit, or on unrelated observables. There is a positive lower bound depending only on a, h; every prime tail contains a returning direction with just two active right axes. The same statement holds after switching sides.

Choose $m = m(a, h)$ by density Hales–Jewett for the alphabet $\{0, \dots, h-1\}$ and density $a/2$. For any chosen m left generators there is a nonempty subset with sum s for which the h-return measure is at least $\kappa(a, h) > 0$: in expectation a translated word cube has occupied fraction a, hence with probability at least $a/2$ that fraction is $\ge a/2$; sum over the finite number of combinatorial lines in the cube and use invariance. Take a far tail on which $4m/\min p < \kappa_c ost$, with $\kappa_c ost$ the active-cost allowance. On its right coordinates finitely color the axes by the $h-1$ tuples of translations above, so any two in one cell have pairwise

distances $< \kappa(a,h)/(2h)$ in every component.

Let V be the finite-dimensional F_2 space of functions on these right vertices which are constant on each cell. Infinite tail rank allows choosing m left rows independent modulo V. Find a returning subset sum s of the corresponding generators by the word-cube bound. The adjacency parity function of its support is not in V; there are right vertices w_+, w_- in the same cell on which the parity values differ. Put $v = e_{w_+} - e_{w_-}$. Translation invariance gives $||T_{tv}1_A - 1_A||_1 < \kappa(a,h)/(2h)$ for every $1 \le t < h$. Telescoping the h bounded factors shows that $s+v$ has return measure at least $\kappa(a,h)/2$. Its induced edge count is odd by the two opposite parity values, and at most $2m$ edges are active, so the tail choice guarantees its required reciprocal cost. This proves the theorem for all positive a, including for arbitrarily high incidence-rank matchings. It says nothing about an action with neither side compact as specified.

R038. Audit of the cyclic nilsystem joining and its component restriction

Scope and status. Proved subcases and unresolved extensions.

Result. The cyclic niljoining proof is checked and several nonergodic extensions and countermodels delimit its scope.

Scope. An averaged cyclically symmetric joining need not have the required symmetry on every ergodic component.

Qualifications. The counterexamples to dropping ergodic-component symmetry are part of the result, not contradictions of the stated joining theorem.

The connected-flow, suspension and prime-degree arguments in the cyclic joining theorem [R039] admit the following detailed justification. For an ergodic joining under the flow choose an ergodic time and a generic point. Its full flow orbit has the same closed support as the discrete-time generic orbit and is connected. The Leibman nilorbit theorem gives a connected homogeneous support `Lz` with Haar measure. Discrete ambient stabilizers imply that evaluation at z has injective differential, hence that the connected setwise stabilizer is L. Its Lie projection to each ambient horizontal copy is onto. Cyclic permutation preserves L, sends its lattice to a conjugate, and fixes the diagonal one-parameter group. It therefore acts identically on the horizontal torus, because that projected one-parameter group is dense. The induced map on the Lie algebra is unipotent (all lower-central graded pieces are generated from the abelianization), so finite order forces it to be the identity. All of L is diagonal. The cyclic orbit of the individual base points is obtained by a single rational element h with h^q in the stabilizer lattice. In a Malcev basis adjoining this rational element yields a discrete overgroup of the lattice. The finite nilpotent quotient by the core of the original lattice has no Sylow p for $p \nmid q$, because the p-parts of both the original lattice and h lie inside the original subgroup. Thus the resulting finite covering degree has only primes dividing q.

An ergodic discrete nilsystem's roof-one suspension equals the roof-c suspension of its connected component return affine map, where c is the number of cyclically permuted components. On the simply connected lift of that component the affine unipotent automorphism extends to a real one-parameter automorphism group; its logarithm lowers the inherited central filtration. The semidirect product by the real line is nilpotent, and the affine return followed by quotienting the real coordinate by its integer lattice gives precisely the return affine map on the section. Cyclicity and ergodicity pass to the common-clock suspension of the joining, and the common cover degree holds for the original clock section by equivariance and disintegration. A cyclic factor of order p in the joining is independent of the base in the absence of that eigenvalue on the base. Equal fiber counts force p to divide the covering degree. Moreira–Richter, Theorem 2.4 has the required rational-spectral hypothesis for the congruence-class averages.

Why cyclic symmetry of an average is insufficient For any two distinct primes p, q, take the real Heisenberg group with integer lattice, and a translation u with rationally independent horizontal coordinates together with 1, so T_u and T_u^p are ergodic. Let ζ be the central element of height $1/p$. Choose a horizontal h satisfying $[u, h] = \zeta$; for instance in coordinates with group law $(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab')$ take $u = (\alpha, \beta, 0)$ and $h = (0, 1/(p\alpha), 0)$.

In Y^q take the uniform mixture, over $0 \le k < p$, of graph measures

$$(x, h\zeta^k x, x, \dots, x), x Haar \in Y.$$

The graphs are disjoint; the diagonal action takes graph k to graph $k+1$. On each graph its pth power is isomorphic to T_u^p. Hence this mixture is an ergodic horizontal self-joining with a cyclic p-factor. Its average over cyclic coordinate permutations is a cyclically invariant self-joining. It retains that p-factor in each horizontal ergodic component. For $q = 2$ the two cyclic images are distinct because the first graph and its reversal have different horizontal offsets ($2/(p\alpha)$ is not an integer); for $q > 2$ the images are distinguished by the location of the exceptional coordinate. The cyclic group permutes these ergodic components transitively, so the joint action with the order-q permutation is ergodic. Consequently horizontal ergodicity in the joining lemma cannot simply be replaced by joint ergodicity. This example treats one prescribed pair of primes, not all tails of an integer construction.

A nonergodic extension with jointly invariant approximants Use the horizontal residue property [R141] for $W \subset Z \times V$. Suppose an event A of positive measure can be approximated arbitrarily well in symmetric difference by events B on **jointly invariant** factors Y. Require that for almost every ergodic component of the horizontal transformation on each such Y, only finitely many of the indicated horizontal primes p_i occur in its discrete spectrum. No bound on this finite number, the component nilsystems or the cardinality of the component-orbits under V is needed. Then A returns at every length along W.

Fix h and a lower bound $a > 0$ on the mass of an approximant. Density Hales–Jewett gives constants $m, \kappa > 0$ depending on h, a so some nonempty subset F of *any* m of the vertical indices gives a positive vertical return of B of mass at least κ; write $C = \bigcap_{0 \le j < h} U^{-jv} B$, $v = \sum_{i \in F} b_i$. Before choosing the indices, pass to a tail in which the horizontal components of Y possessing any of the remaining primes have total measure below $\kappa/2$. This is possible by continuity from above using the componentwise finiteness assumption. The integral of the conditional mass of C over the other components is at least $\kappa/2$, so components where this conditional mass is at least $\kappa/4$ have measure at least $\kappa/4$. On each of them the Moreira–Richter congruence identity applies to C and the prescribed residue modulo $\prod_{i \in F} p_i$. The ordinary mean of the horizontal h-return of an event of mass at least $\kappa/4$ has a positive lower bound depending only on κ, h by finite Szemerédi averaging. Integrating gives a uniform positive lower bound on the return average of B for that residue and vertical v, since a horizontal C-return implies the corresponding mixed B-return. Take the approximating error for A smaller than this bound divided by $2h$ to conclude. Joint invariance ensures C lies on the very same factor used in choosing the exceptional components.

This covers inverse limits of joint factors with, for example, finite-dimensional ergodic nilsystems on their horizontal components, even when the full system has all the indicated rational eigenprimes. An arbitrary approximation by individually chosen horizontal finite nilfactors lacks joint invariance; cyclic averages alone cannot supply it, as the example illustrates.

Finite groups of transports. The cyclic finite-cover lemma extends to a finite group K permuting the coordinates of an ergodic nilsystem joining transitively and preserving its measure. In the connected flow each element of K induces a finite-order automorphism of L fixing the diagonal flow. Its differential is the identity by the argument above. Transitivity

then gives $L = diagG$. For each $g \in K$, the permutation of the basepoint tuple agrees with left translation by some element $h_g \in G$; coordinate permutations commute with diagonal translations. Thus $h_g^{\mathrm{ord}(g)}$ fixes every basepoint coordinate. Move the initial basepoint of one coordinate to the lattice point. The h_g are rational, and the lattice they and Γ generate is a discrete finite-index overgroup of Γ. In its finite nilpotent quotient by the core of Γ, the primary components of each generator at any prime $p \nmid |K|$ lie in the image of Γ, as before. The projection degree consequently has only primes dividing $|K|$. The roof suspension extends this discrete-time statement without a change to the group action.

In particular let the horizontal system be $(I \times X_0, id \times T_0)$ with product measure and ergodic T_0. Use **one fixed measurable identification** with this product throughout. For each late finite index set F, suppose the fiber transports over its finite vertical group in this fixed identification lie in a finite subgroup K_F of transformations commuting with T_0, with $|K_F|$ coprime to $m_F = \prod_{i \in F} p_i$. The action on component labels need not have finite orbits for the whole vertical group. Then every positive event has multiple returns on the asymmetric W. Changing the identification separately for each F would also change the finite horizontal approximant and does not satisfy this hypothesis.

Indeed the horizontal characteristic-factor comparison from the cyclic joining note reduces the observable to an event on $I \times Z_h(T_0)$; the commuting maps preserve this factor. Approximate the event in measure by B on $I \times Y$ with Y a single finite ergodic nilfactor of T_0. As usual preselect the approximation tolerance using only the two finite density-recurrence lower bounds. Take all vertical primes so late that their matching horizontal primes are absent from Y's rational spectrum, and obtain F, v from vertical density Hales–Jewett with the vertical intersection C of mass at least κ. On each base fiber, C is measurable on the join Y' of the K_F-translates of Y in X_0. This joining is T_0-ergodic and invariant under the regular permutation of its factor list by K_F. By the finite-group lemma its spectrum has none of the horizontal primes in m_F. The ordinary positive horizontal return average of C therefore also holds on the required residue on each fiber having positive C-mass, with a uniform lower bound globally as before. One can allow K_F to depend on the component orbit as long as the specified finite transports on almost every orbit are contained in such a group; the componentwise residue conclusion is unchanged. This result uses the prime divisors of the *fiber transports*, not just the order of the overall vertical map.

Sparse horizontal component movement. A different extension applies if a positive-measure collection of horizontal ergodic components give positive mass to A and are each stabilized by infinitely many whole vertical prime axes. On each such component use the ergodic-horizontal theorem with exactly those vertical axes. Its conclusion is a countable union over possible $(n, v) \in W$ of positive-return assertions, so a single pair has positive global return. The subset can depend on the component. If the $Z \times V$ action is jointly ergodic, each vertical generator's fixed-point set in the horizontal component space is V-invariant and thus null or conull. A counterexample must therefore move horizontal component labels essentially freely on an entire prime tail: a stabilizer in $\oplus C_{q_i}$ is the sum of the coordinate subgroups it contains, because multiplication by suitable integers isolates each coordinate of each of its elements.

Gauge for a fixed finite vertical subgroup. Suppose a finite horizontal eigenprime p occurs in almost every horizontal component, and a finite vertical subgroup H has order coprime to p. There is a global horizontal cyclic label of order p invariant under H. Start with a measurable componentwise normalized eigenfunction taking values in the pth roots,

obtainable by disintegration or by the spectral projection of a countable dense family of tests and measurable selection. On each finite H-orbit of components choose one representative (finite relations admit measurable selectors) and extend its eigenfunction by transport around the orbit. A stabilizer element multiplies the eigenfunction by a constant pth root on that ergodic component, since it commutes with T. The stabilizer order is coprime to p, so this constant is one; transport is well-defined. This finite gauge does not assert compatibility over all finite subgroups. Even after a common horizontal cyclic label has been fixed, correlations of several individually label-independent observables can reveal the label.

The short obstruction already occurs in a bundle of compact horizontal rotations For the specific asymmetric template and arrays $k_i : C_{q_i} \to F_{p_i}$ of the horizontal normal-move construction [R014], let L include every finite spatial offset used in its filter. Give independent Haar law to

$$y \in \prod C_{q_i}, c \in \prod C_{p_i}, \theta, s \in T^L.$$

The horizontal T fixes (y, θ), adds 1 to all c_i and sends s to $s+\theta$. The ith vertical generator U_i replaces y_i by $y_i + 1$, fixes c, and sends

$$\theta_b \mapsto \theta_b + d_{i,b}/p_i, s_b \mapsto s_b + c_i d_{i,b}/p_i,$$

$$d_{i,b} = k_i(y_i + b + 1) - k_i(y_i + b) \in F_{p_i}.$$

All these transformations preserve Haar by successive fiber translations. Distinct U_i commute; $U_i^{q_i} = id$ since the increments telescope. The increase of θ_b in TU_i exactly matches the contribution from $c_i + 1$ in U_iT, so these maps give a $Z \times V$ action. For Haar-almost every θ the rotation by $(1, \theta)$ on $\prod C_{p_i} \times T^L$ is ergodic by the character criterion and independence over the rationals; the horizontal components are therefore exactly the fibers at (y, θ), modulo null sets. They are compact group rotations with a uniformly bounded connected dimension, while their eigenvalue groups vary with θ.

At time $t(n, v)$ put $I = suppv$. The observed s_b equals, up to a term affine in t,

$$\sum_{i \in I} (c_i + tn) k_i(y_i + b + tv_i)/p_i \bmod 1.$$

Indeed before dropping the affine term it equals $s_b + tn\theta_b + \sum_{i \in I}(c_i + tn)[k_i(y_i + b + tv_i) - k_i(y_i + b)]/p_i$. This involves only finite sums. Applying the fixed spatiotemporal filter from the asymmetric construction kills the discarded affine terms, the pure vertical normals and every pure horizontal normal. The remaining marked term obeys the same half-turn estimate at every site. A fine fixed arc coloring of the finitely many observables s_b consequently forbids a monochromatic progression at the same fixed length. A positive-measure cell gives a genuine low-density avoidance witness in this explicit bundle. Thus even bundles of ergodic Kronecker components can support the nonergodic obstruction; this does not decide its all-system critical density.

Identical horizontal component systems and a profinite component action One can instead arrange that *all* horizontal components are the same ergodic system, with the vertical group acting by the regular profinite translations on the component labels. Let y, c be as above and now use independent Haar $x, s \in T^L$. Choose $\alpha \in T^L$ with its coordinates and 1 rationally independent. Set

$$T(y, c, x, s) = (y, c + 1, x + \alpha, s + x).$$

For U_i use the preceding formulas with x in place of θ: advance y_i, fix c, increase x_b by $d_{i,b}/p_i$ and s_b by $c_i d_{i,b}/p_i$. These preserve Haar, commute with each other and T, and have $U_i^{q_i} = id$, by the same direct computation. On every fixed y the horizontal transformation is ergodic. Indeed a Fourier mode with nonzero s-frequency l is sent through infinitely many distinct x-frequencies $k + tl$, so no such mode appears in an invariant square-integrable function. At $l = 0$, the only invariant mode is constant by the choice of α and the density of the diagonal integer generator in $\prod C_{p_i}$. Thus the component labels are exactly y. The horizontal system on each is the product of the same affine two-step torus transformation with the prime odometer. Regular translations of V are ergodic on the y-space, so the full action is jointly ergodic as well.

At $t(n, v)$, the observable $s_b(t)$ is the same finite sum $\sum_{i \in suppv}(c_i + tn)k_i(y_i + b + tv_i)/p_i$ as above plus a scalar polynomial of t of degree at most two. The new term is $\binom{tn}{2}\alpha_b$. Accordingly use the six-slot filter

$$(E - I)^3(E - S)(E - S^{-1}),$$

where E shifts the temporal slot and S shifts the spatial offset. This annihilates all quadratic terms at the individual spatial offsets and all pure normal contributions. On a positive marked vertical step at a site the output is $rH'(y_i)/p_i$, with horizontal residue $r = n \bmod p_i$ and $H' = S(S - I)^3(S - S^{-1})k_i$. On a negative marked step the same calculation gives $rS^{-5}H'(y_i)/p_i$; the shifts and sign conventions have no effect on nearness to a half turn. To verify the second formula, first apply $E - S^{-1}$ to $(c + tr)k(y + b - t)$ to obtain $rS^{-1}k$; apply $E - S$ and then use $(S^{-1} - I)^3 = -S^{-3}(S - I)^3$.

As in the original asymmetric construction, the operator defining H' is invertible on the zero-sum subspace of the q_i-cycle over F_{p_i}; its nontrivial Fourier frequencies have odd order $q_i \neq p_i$ and no listed factor vanishes there. Choose its prescribed zero-sum output near $p_i/2$ for the unit-mark version, or in $\{-1, +1\}$ with integer sum p_i when $q_i \geq p_i$ for the macroscopic horizontal mark. The original mark/error rules then give the same separation from zero uniformly over all starts. Because the coefficients at every spatial offset sum to zero, a sufficiently fine fixed coloring of the finite list of s_b by arc cells forbids monochromatic six-slot runs. This is a positive-measure short obstruction even with the same horizontal component system everywhere and with compact component action. The finite horizontal approximants here are not jointly invariant. No claim about nearly full avoiding sets follows.

R039. A cyclically symmetric self-joining of an ergodic nilsystem

Scope and status. Result with proof.

Result. An ergodic cyclically symmetric q-fold nilsystem joining is a finite cover whose degree uses only primes dividing q; this gives recurrence in the horizontal-ergodic asymmetric model.

Scope. Horizontal ergodicity or the explicitly stated component assumptions cannot be omitted.

This concerns the model with an integer horizontal generator in horizontal residue sections [R141]. It does not assert that a general joint action is ergodic under the horizontal generator.

Lemma. Let (Y,T) be an ergodic finite-dimensional nilsystem, and let λ be an ergodic joining of q copies of (Y,T), invariant under cyclic permutation of the q coordinates. The projection of this joining to its first coordinate is finite-to-one with uniform fibers of some degree D whose prime divisors all divide q. In particular if T^m is ergodic on Y and m is coprime to q, the diagonal T^m is ergodic on the joining.

Connected nilflow version. Start with a connected simply connected nilpotent Lie group G, a lattice Γ, and a one-parameter translation b_t on G/Γ. Assume an ergodic flow joining of q copies that is cyclically invariant. Its support is the closure of a single flow trajectory. It is a connected subnilmanifold Lz_0 for a connected closed subgroup L of G^q, with L$\cap$Stab(z_0) a cocompact lattice; its invariant measure is Haar. Indeed an ergodic flow on a standard probability space has an ergodic time τ (exclude the countable resonances of its eigenvalues). Apply the nilorbit closure and equidistribution theorem to the time-τ transformation and a generic point. The entire flow orbit has the same closure, which is connected. Projection onto each G/Γ is onto, so each Lie projection of L to G is onto: a transitive connected Lie subgroup on G/Γ must have full Lie dimension since stabilizers are discrete.

The identity component of the setwise stabilizer of Lz_0 in G^q is L itself: any infinitesimal stabilizer must be tangent to Lz_0 at z_0, and evaluation in $G^q/\mathrm{Stab}(z_0)$ has injective differential. Thus the diagonal flow lies in L, and cyclic permutation s on G^q induces a finite-order automorphism ψ of L. Write $sz_0 = hz_0$ for some $h \in L$. If Γ_0 is the stabilizer lattice of z_0 in L, then $\psi(\Gamma_0) = h\Gamma_0 h^{-1}$. The induced automorphism on the abelianization therefore preserves its projected lattice. It fixes the projection of the diagonal one-parameter group pointwise since that group is diagonal in G^q. This projected flow is dense on the horizontal torus of L/Γ_0 (it is a factor of the ergodic flow). It follows that ψ is the identity on the horizontal torus, hence on the abelianized Lie algebra. Its differential is consequently unipotent, since it is the identity on every successive lower-central quotient. A finite-order unipotent operator in characteristic zero is the identity. Connectedness now gives $\psi = Id$ on L. Hence L consists of diagonal elements, and surjectivity onto a coordinate shows $L = diagG$.

Write $z_0{=}y_0, \ldots, y_{q-1}$ with cyclic indices and take h in the *diagonal* G just identified. Possibly reversing the cyclic convention, $y_j = h^j y_0$ and $h^q y_0 = y_0$. Conjugating by a common left translation we may set $y_0 = \Gamma$. The orbit is G/Γ' where $\Gamma' = \bigcap_{j=0}^{q-1} h^j \Gamma h^{-j}$. Its projection onto G/Γ is a finite cover of degree $[\Gamma : \Gamma']$. Every divisor prime of this degree divides q. Indeed $h^q \in \Gamma$. By logarithms in a rational Malcev basis h is rational, and the finitely generated nilpotent group K=$\langle\Gamma$,h$\rangle$ is discrete and contains Γ with finite

index. For the elementary discreteness argument, the generators have rational coordinates in a Malcev basis, and collecting words through the fixed nilpotency class introduces only bounded denominators (equivalently use the lattice generated by their finite iterated brackets with fixed BCH denominator buffers). In the finite nilpotent quotient $K/\operatorname{core}_K(\Gamma)$, write H for the image of Γ. For any prime p not dividing q the p-primary component of h belongs to H: take the p-component of $h^q \in H$ and invert q on its cyclic p-subgroup. A subgroup of a finite nilpotent group contains the primary components of each of its elements. Since H and h generate the quotient, H then contains its entire Sylow-p subgroup, which is normal and hence contained in the core. Thus the finite quotient has only primes dividing q. All the displayed indices divide its order. Haar measures on the resulting finite cover give uniform conditional point masses.

Suspending an arbitrary ergodic finite-dimensional nilsystem. Its finitely many components are cyclically permuted by T. The suspension of roof one is equivalently the suspension of the return map T^c on one component with roof c. On this component present the connected simply connected nilmanifold as N/Δ and the return as $g\Delta \mapsto b\phi(g)\Delta$, where ϕ preserves Δ and acts unipotently. To obtain this, choose the component of the lattice point, write the c-th power of the translating element as $b\gamma$ with b in the identity component and γ in the lattice, and lift conjugation by γ to the simply connected cover. Its action is unipotent and its logarithm lowers the lower-central filtration inherited from the original nilpotent group. Consequently $\phi^t = \exp(t \log \phi)$ gives automorphisms for all real t (the bracket identities extend polynomially from integer t), and $N \rtimes_{\phi^t} R$ is a connected simply connected nilpotent Lie group. The subgroup $\Delta \rtimes_\phi Z$ is a cocompact lattice. The one-parameter subgroup through $(b, 1)$ in time c gives the required suspension flow by section at real coordinate 0. This includes finite discrete nilsystems, whose cyclic suspension is a circle flow.

Suspend the joining with a *common* roof-one clock. It gives a cyclically invariant ergodic flow joining on the product of these identical connected nilmanifolds; its marginal on each is Haar (equivalently the uniform suspension of the invariant measure). The previous calculation applies. The full support lies in the locus with all clocks equal; thus its finite cover has precisely the same finite degree on the section over any one clock. Disintegration in the clock first gives this assertion modulo null sets on almost every section; flow equivariance transports such a section back to clock 0 with the original joining and uniform fiber conditionals. Finally if a new prime p dividing m occurs in the joining spectrum but T^p is ergodic on Y, a p-cycle factor of the joining is independent of Y. Indeed Y has no eigenvalue of that order and the finite cyclic Fourier expansion gives the product joining. On each fiber of size D the p labels would therefore have equal counts, forcing p|D. This proves the ergodicity conclusion.

Standard nilorbit input: Leibman, *Pointwise convergence of ergodic averages for polynomial sequences of translations on a nilmanifold.* The argument uses cyclic invariance of the joining, not invariance of the chosen individual factor under the finite-order transformation.

Consequence for the asymmetric integer-horizontal model Every action of $Z \times V$ **ergodic under the horizontal integer generator T** has multiple returns on W for every positive-measure event A. First reduce to the horizontal pronilfactor for the h-term averages. For each fixed $(n_0, v) \in W$ with the residue property, averaging over $n = n_0 + km_F$ uses the ordinary Host–Kra characteristic-factor theorem for the distinct nonzero multiples of T. One may take $Z_h(T)$, with the slot shifts $T^{jn_0}U^{jv}$ absorbed into its test functions. All

the vertical maps preserve this factor as they commute with T. If $g = E(1_A|Z_h(T))$ and $B = \{g > 0\}$ has a positive return at (n_0, v), the joint return of $\{g > u\}$ has positive mass for some $u > 0$. The g-average in the same residue is bounded below by u^h times the positive multiple-recurrence average of that joint event under T^{m_F}. The original average is therefore positive. In this step it suffices to obtain a return of B at a residue whose *entire* congruence class at that vertical v is included in W.

In the pronilsystem approximate B in measure by an event B' on one finite ergodic T-nilfactor Y. The error can be chosen in advance from h and $\mu(B)$: vertical density Hales–Jewett supplies a positive $\kappa(h, \mu(B)/2)$ for a vertical return of B', and ordinary finite Szemerédi gives a positive uniform lower bound $\kappa'(\kappa, h)$ for the horizontal return average of any event of mass $\geq \kappa$. Choose primes p_i beyond the finite prime divisors of the number of components of Y, and apply vertical DHJ on their b_i. This supplies $F \neq \varnothing$ with $C = \bigcap_{t<h} U^{-tv} B'$ of measure $\geq \kappa$, $v = \sum_{i \in F} b_i$. Since U^v has finite order q dividing $\prod_{i \in F} q_i$, the factor generated by all q translates of Y is the joining in the lemma: the law of $\phi(x), \phi(U^v x), \ldots, \phi(U^{(q-1)v} x)$ is T-ergodic and cyclically invariant. All p_i in m_F are coprime to q and are absent from $Y's$ spectrum, since an ergodic nilrotation on a connected component under its return power is totally ergodic there. No such prime is in the joining spectrum. Moreira–Richter, Theorem 2.4 gives for C the ordinary positive return average even inside the prescribed residue $\bmod m_F$. Each such C-return yields a B'-return at (n, v). Its uniform lower bound κ' absorbs the original cylinder approximation to B. This proves the assertion. In joint actions with an atomic finite orbit of horizontal ergodic components, the far vertical primes also preserve each component (finite permutations of large prime order act trivially). The same proof works there by restriction. An arbitrary nonatomic space of horizontal components, permuted by the vertical group, remains outside this argument.

R060. The marked mass must have a part spread over many small coordinates

Scope and status. Result with proof.

Result. Tightness of marked reciprocal mass on finitely many physical axes forces endpoint one, including the corresponding ultrafilter statement.

Scope. The theorem is stated for the positive-vertical model; signed versions use the separate expander argument.

Work on the positive-unit vertical model with $q_i/p_i \geq c > 0$ in a tail. Use centered integer horizontal amplitudes a_i on the vertical-active sites and assume their total parity is odd and $\sum |a_i|/p_i \leq \tau < 1$. Write $w_j^{\downarrow}(d)$ for the decreasing rearrangement of the finite list $|a_i|/p_i$, padded by zeros.

Proposition. If a subfamily V satisfies

$$\lim_{B\to\infty} \limsup_{J\to\infty} \sup_{d\in V \text{ supported in } [J,\infty)} \sum_{j>B} w_j^{\downarrow}(d) = 0,$$

then its saturated integer critical threshold is one. More locally, if P is an ultrafilter containing all tails and $\lim_{B\to\infty} P - \lim_d \sum_{j>B} w_j^{\downarrow}(d) = 0$, then $\sup_{E\in P} b(E) = 1$ in the tail model. The total marked mass need not go to zero, and there is no requirement that q_i/p_i be irrational, converge, or obey prescribed congruences. The statement is specific to the positive vertical sign proof below.

Fix a final site-loss budget and the finite star test from the positive star proof [R200]. It supplies a multiplier M and a test window H, including offset and differencing margins, such that a trajectory polynomial W of the specified degree D, with constant top difference z and $\|Mz\| \geq (1-\tau)/2$, creates a forbidden center. The center still works under perturbations in its H-window smaller than a fixed positive tolerance. Choose $\xi > 0$ so that total curvature error $< \xi$ in a window permits such a perturbation by discrete summation. These choices are made before B, the number of heavy sites.

In all sufficiently late prime groups put $c_i = (2M)^{-1} \bmod p_i$. For any legal step, $z = \sum a_i c_i/p_i$ has $\|Mz\| \geq (1-\tau)/2$. Indeed $2Mc_i = 1 + \ell_i p_i$ with ℓ_i odd, so $Mz \equiv \frac{1}{2} + \frac{1}{2}\sum a_i/p_i \pmod 1$. For any integer $B \geq 1$ set $L = 2B+2$ and $N = LH$. The normalization correction e_i, the centered integer congruent to $-q_i c_i$ modulo p_i, satisfies $|e_i| \leq p_i/2$. On a tail $q_i \geq 2N$. Choose $\lfloor q_i/N \rfloor$ sites of the q_i-cycle with cyclic spacings at least N, for instance multiples of N starting at zero. Distribute the integer total e_i as evenly as possible, with a common sign, among these sites. Every correction has magnitude at most

$$\left\lceil \frac{|e_i|}{\lfloor q_i/N \rfloor} \right\rceil \leq 1 + Np_i/q_i \leq (1 + c^{-1})N.$$

Take $\Delta^D b_i$ equal to c_i plus these corrections in $\mathbb{F}_{p_i}$. Its sum is zero, so b_i exists on the q_i-cycle by invertibility of Δ on the zero-mean subspace. Use the resulting arrays in the star observable. Its active signal is $W(t) = \sum a_i b_i (y_i + t)/p_i$.

In any segment of N consecutive curvature slots, each site can contribute at most one correction. Split the segment into L disjoint H-windows. The largest B amplitudes spoil at most B windows. On all the remaining windows together, the sum of the circular distances

between $\Delta^{D}W$ and z is at most $(1+c^{-1})N\sum_{j>B} w_j^{\downarrow}(d)$. Some remaining window thus has total error at most

$$\frac{(1+c^{-1})LH}{L-B}\sum_{j>B} w_j^{\downarrow}(d) \le 2(1+c^{-1})H\sum_{j>B} w_j^{\downarrow}(d).$$

In particular the bound is independent of B apart from the tail mass itself. Now choose B so this right side is below ξ on a sufficiently late tail, or on a member of P, respectively. The long segment and its margin have fixed length before exhausting the finite groups. The star gives the high-density blocker on all those steps. The squarefree-tail suspension yields the claimed integer conclusion in the first case. In the ultrafilter case the blocked subfamily belongs to P, and the site-loss budget was arbitrary.

Thus a prospective obstruction with a uniform gap in the ultrafilter version needs positive total marked mass escaping every fixed number of the largest amplitudes. This conclusion is consistent with a charge concentrated on finitely many *free ultrafilters* of axes, as in the character classification [R136]; the new test is about physical coordinates in each individual step.

R140. Recurrence above a model-dependent threshold for measurable events in horizontal polynomial fibers

Scope and status. Result with proof.

Result. Specified independent horizontal-polynomial fiber models have all-length recurrence above a model-dependent threshold, with a half bound in the stated rank-one case.

Scope. The horizontal degree is bounded; arbitrary coupled fibers and all higher characteristic factors are excluded.

This is for the free-horizontal paired family [R182], using an additional freedom in choosing its prime pairs. Require $\sum_i 1/q_i = \infty$, with $p_i, q_i \to \infty$ and multiplicative orders $\mathrm{ord}_{q_i}(p_i) \to \infty$. One may simultaneously have $q_i/p_i \to \theta > 1$ irrational, and even $\mathrm{ord}_{p_i}(q_i) \to \infty$. To see this, use disjoint dyadic shells $X = 4^n$, take a small fixed-relative-width interval of possible p's around X, and the interval scaled by θ for possible q's (choose the width so no source/target intervals overlap). Subdivide into matched bins of relative width tending to zero slowly. The prime number theorem first for a fixed finite number of bins, followed by a diagonal choice increasing that number slowly, ensures that the target count exceeds the source count by a fixed positive fraction in every bin. For a fixed p, at most $O_\theta(d)$ primes $q \asymp p$ divide $p^d - 1$, by their size. Also at most $O_\theta(d)$ integers $q \asymp p$ satisfy $q^d = 1 \bmod p$, by the root bound in the prime field. Greedily pair within bins, excluding both possibilities for $1 \le d \le D_n$, where $D_n \to \infty$ is so slow the surplus exceeds all the exclusions. The source reciprocal mass on each shell is $\gg 1/n$. One can take $1 < \theta < 2$ to keep the shells disjoint.

Use the independent-grid compact extension of paired-profinite-fibers.md [R186] with fiber $K = \mathbb{T}^r$ and arbitrary arrays of the form

$$f_i(u, v) = b_i(v) + \frac{1}{\{p_i\}} \sum_{d=1}^{D} u^d k_{i,d}(v) \pmod 1,$$

where $D < \infty$ is fixed, $k_{i,d} : C_{q_i} \to \mathbb{F}_{p_i}^r$, and $b_i : C_{q_i} \to K$ is unrestricted. For **every such fixed system** there is $\delta_f < 1$ for which every measurable event of mass above δ_f has positive returns of every length in every tail at arbitrarily small marked cost. This includes arbitrary events depending on the entire base and the fiber together. If the even subgroup defined below has annihilator rank at most one, $\delta_f = 1/2$ works; in dimension $r = 2$ the rank-two case has actual constant attainable profiles, so the same half-density assertion holds in all two-dimensional examples of this form. If $r = 1$, every positive mass suffices. The toral coloring giving the new short obstruction uses precisely the case $r = 2, D = 1$. These are statements about specific families of actions, not all horizontally Host–Kra systems.

Pool generators. Use only three distributions of single-grid moves: a uniform random purely horizontal speed, the fixed vertical unit normal, and the fixed diagonal unit mark. Each is sampled independently of the uniform starting site. Write $Z_i \in K^{\mathbb{Z}}$ for the resulting increment profile $f_i(x_i + tw_i) - f_i(x_i)$. For a fixed type call z abundant if in every neighborhood $U \ni z$ the summed occurrence probabilities $\sum_i \Pr[Z_i \in U]$ diverge. Form the closed subgroup in $K^{\mathbb{Z}} \times C_2$ generated by the abundant profiles with their corresponding normal/odd tag and by all even constant profiles. There is at least one abundant mark, by compactness. Write L for the even subgroup and $z + L$ for the odd coset. Shift invariance holds: shifting time in a single-grid profile corresponds to shifting the uniform starting site, then adding its time-one profile value as a constant. More formally the anchored time-shift

map $w(t) \mapsto w(t+n) - w(n)$ preserves the law of each type, also for random horizontal speeds conditional on the speed, so it preserves abundant anchored profiles. The constants supply the difference from plain time shift. The annihilator $R = L^{\perp} \subset \mathbb{Z}[T^{\pm 1}]^r$ is therefore a Laurent submodule of balanced words.

Finite positive sums of generators can approximate the whole odd coset because the closure of the positive semigroup in a compact abelian group is the subgroup closure. Every one of the finitely many slots of an approximation can be realized arbitrarily far out, choosing exactly one index from its own disjoint pool. For its specified profile neighborhood let H_i be the matching event, now in uniform grid starts and a possible extra independent horizontal speed. Take the pool total $\Lambda = \sum \Pr(H_i)$ arbitrarily large. Choose index i with probability $\Pr(H_i)/\Lambda$, and condition its draw on H_i, leaving all other starts uniformly independent. As in the profinite proof [R186], the time-t marginal density on the pool is $\Lambda^{-1} \sum_i g_{i,t}(x_i)$, where $0 \le g_{i,t} \le 1$ has mean $\Pr(H_i)$. This holds because the unconstrained shifted start is uniform even for the auxiliary random speed. Thus the density has mean one and squared L^2-distance from one at most $1/\Lambda$. The finitely many pools can have product marginal norm below two, and their bounded number of actual unit marks can be made to have arbitrarily small total reciprocal cost. Finite probability mixtures of these approximations retain the same bound. Independent uniform initial fiber makes the marginal bound valid on the full space even though the chosen moves are correlated with the base. Common translates of a target profile do not affect the Haar-averaged fiber return test.

Exact relations from the horizontal coefficients. Fix $A \in R$ and a threshold $c > 0$. For either normal distribution the probabilities that $\|A(Z_i)\| \ge c$ have finite sum; otherwise the compact closed set of failed profiles would contain an abundant point by finite covering. Write $A = \sum_t a_t T^t$ with coefficient rows a_t, each evaluating K in the circle. At a vertical normal the part of this phase depending nontrivially on u is

$$p_i^{-1} \sum_{d=1}^{D} u^d \, (A(Y) k_{i,d})(v).$$

If any coefficient is nonzero for even one v, the conditional probability over uniform u of circle norm greater than $1/8$ is bounded below by an absolute positive constant for late i, independent of its constant term. One elementary proof avoids exponential-sum estimates: a nonconstant degree-at-most-D polynomial on $\mathbb{F}_p$ has at most D preimages per scalar value. Thus use instead threshold $c = 1/(16D + 16)$, so the probability in the shorter arc is at most $D(2c + 2/p) < 1/2$. Hence $A(Y)k_{i,d} = 0$ identically for all $d \ge 1$ except on indices of finite total $1/q_i$ mass. At a horizontal normal the filter is a polynomial in independent uniform u, a over $\mathbb{F}_{p_i}$, with the $b_i(v)$ terms cancelled by balance. A nonconstant polynomial of total degree bounded by D has at most Dp preimages of any specified value on $\mathbb{F}_p^2$, by the elementary polynomial zero bound. Its value at $a = 0$ is zero, so if it is not identically zero it is nonconstant. The same arc estimate shows the horizontal filter vanishes for every u, a, v outside another index set of finite reciprocal mass. In the case $D = 0$ there are only the arbitrary vertical-only offsets, which cancel automatically at horizontal moves.

Filling the quotient. Choose integral nullspace vectors $v_j(T)$ and the connected coefficient subtorus K_f from R exactly as in the toral control proof [R190]. For a basis l of the character lattice of K/K_f, some nonzero scalar Laurent word $Q_l(T)$ has $Q_l(T)l \in R$. Thus outside finitely many sets of finite total $1/q_i$ mass, $Q_l(Y)(lk_{i,d}) = 0$ for all $l, d \ge 1$. For large primes its reduction is nonzero. The cyclic vertical shift diagonalizes in a splitting

field over $\mathbb{F}_{p_i}$; every nontrivial eigenvalue is a primitive q_i-th root whose degree over $\mathbb{F}_{p_i}$ is $\mathrm{ord}_{q_i}(p_i)$. This eventually exceeds the degree of the fixed word, so every $lk_{i,d}$ is constant in v. If the projected odd coset missed zero, parity would extend to a balanced character word A in those quotient coordinates, annihilating the even subgroup and taking value $1/2$ on all abundant marks. The horizontal identity for this word holds for every u, a, v on the same good indices after one more finite-weight exception. There the projected array is a sum of a horizontal-only polynomial and a vertical-only offset. Thus its A-value at the diagonal mark equals its value at the vertical normal: the extra horizontal contribution is precisely the zero horizontal filter. On the infinitely many good indices the two distributions have identical character values. Compactness makes an abundant mark with value away from $1/2$, since normal failures of any fixed small arc about zero have summable probabilities. Contradiction. We may therefore choose the representative z with all coordinates in K_f. If R has full rank this says directly that the odd coset contains zero.

Transfer to arbitrary events. As in toral control [R190], the independent Haar streams $Y = \sum_j v_j(T)U_j$ lie in L, have all marginals Haar on K_f, and a fixed finite dependency degree Δ. Include an independent common Haar translation on K. For any fiber set of mass at least $\beta > 1 - 1/(e(\Delta + 1))$, a fraction bounded below in terms of the margin of the K_f-cosets has relative mass at least some β' strictly above that threshold. Conditional on the common translation in any such coset, the local lemma gives a uniform positive lower bound, depending on h, β, Δ, for the whole translated $z + Y$ window to be in the set. If $K_f = 0$, just use a constant Haar trajectory, which hits any positive-mass fiber set with at least its mass.

Approximate a given event A of mass strictly above this threshold by a cylinder A' depending on finitely many initial grid pairs and on the fiber. Its mass is still above the threshold by a fixed margin when the approximation error is small. A fraction bounded below of its early-coordinate sections has fiber mass above the threshold by a (possibly smaller) fixed margin. The uniformly positive bound in the preceding paragraph therefore works after integrating these sections. For each of the finitely many possible early-coordinate cells, the Haar-translated fiber return expression $\int_K \prod_{t<h} 1_{A'_x}(k + w_t)\, dk$ is continuous in the profile w. Approximate the law of the control profiles on this finite window by a finite probability mixture, then approximate its supported odd profiles by the tilted pool constructions beyond all cylinder coordinates, with arbitrarily small loss in these finitely many continuous tests. The resulting actual trajectory has A' probability bounded below solely in terms of the original margin, h, and the model. Each time marginal has L^2 density at most two relative to full Haar; the loss for replacing A' by A is thus at most $2h\sqrt{\mu(A \triangle A')}$. The approximation tolerance can be set before the cylinder and pools are chosen, giving a positive A-trajectory. The mixture has finite support in actual directions and absolutely continuous conditional starting laws, so one ordinary legal direction has positive Haar return.

Rank one. If R has rank one, use the primitive relation and unimodular output change in toral control [R190]. In the nonconstant-direction case abbreviate its fiber coordinate as (x, y). For B of mass $\geq 1/2 + b$, some u has $E = \{x : \mu B_x > u\}$ with $\mu E + u \geq 1 + b/2$, by integrating the deficiency (if necessary take a slightly smaller margin). For arbitrary affine forcing the Haar law on solutions of the primitive equation gives probability at least $\mu(E)^s (b/2)^h$ that a given h-window is inside B, where $s > 0$ is the delay in the earlier note. Indeed initialize the finitely many registers freely with Haar, with the first s middle x's in E. At time t draw uniform y_t and then a uniform root x_{t+s} of $a x_{t+s} = c_t - b_0 \cdot y_t$. Both new marginals are uniform conditional on the past since $b_0 \neq 0$; the probability of $y_t \in B_{x_t}$

and $x_{t+s} \in E$ is at least $b/2$ by the union bound. This is Haar law on the finite solution window: its finite boundary registers are unrestricted, each subsequent equation is solved by the uniform conditional Haar on its admissible pairs, and all finite solutions extend in both directions using a nonzero extreme coefficient. Equivalently this is the projection of Haar on the translate of the compact full solution subgroup. Adding an independent common Haar translation preserves the lower bound, since it simply translates B in the preceding calculation. The constant-direction case follows by applying the quotient-filling argument with that single constrained row; if all constraining rows lie on one constant direction, eliminating the odd anomaly on its projection leaves a constant in the odd coset. The same pool and cylinder transfer now gives the half-density assertion.

The finite polynomial degree bound is horizontal only. The vertical dependence $v \mapsto k_{i,d}(v)$ has no degree or smoothness restriction. Fibers with coefficients depending jointly on multiple grid pairs, and general high-order horizontally characteristic factors, are not included.

R141. An integer horizontal coordinate with periodic sections

Scope and status. Result with proof.

Result. Every positive avoiding event in the integer-horizontal model forces almost every horizontal component to carry all sufficiently late prescribed prime factors.

Scope. Also gives recurrence for the indicated pure-point or phase-generated observations; it does not control arbitrary nonuniform components.

Let $G = Z \times V$ where $V = \oplus C_{q_i}$ on distinct odd primes, with specified generators b_i. Choose further distinct odd primes p_i, disjoint from the q_i. Assume a step family W has the following property. Given a sufficiently far nonempty finite set F of vertical indices, there is a residue $r_F \bmod m_F = \prod_{i\in F} p_i$ such that $n, \sum_{i\in F} b_i$ belongs to W for **all** n in that residue. The asymmetric horizontal construction has this property: set one $n \bmod p_i$ to a marked residue, all other active horizontal residues to zero, and apply CRT. The finite-color short obstruction also transports to this group, by coloring any finite packet using the CRT projections of its horizontal integer and only the finitely many vertical axes it involves, followed by coloring compactness. The vertical lift to $Z \times Q$ is covered by the one-cyclic-subgroup suspension argument and the fresh q-prime tails; transfer from a torsion-free countable group then applies. This is an alternative model; equality with a model having a locally finite horizontal coordinate is not required or asserted.

Rational spectrum required by every obstruction. Let a jointly ergodic G-action have a positive h-avoiding event A for W. For all sufficiently late indices i, almost every ergodic component of the horizontal integer transformation T must have $1/p_i$ in its additive discrete spectrum. Here sufficiently late means that there are only finitely many exceptions. In particular, on almost every horizontal component there is an odometer factor projecting onto the full product of all the corresponding C_{p_i}. This conclusion applies to every positive mass and fixed forbidden length.

For each fixed p_i the set of T-ergodic components possessing that eigenvalue is measurable and is invariant for the vertical group because its transformations commute with T. Measurability follows, for instance, by taking ergodic projection onto the indicated frequency of a fixed countable dense family of test functions and examining its norms on the components. Joint ergodicity therefore makes the set conull or null. Suppose infinitely many p_i are absent. Choose arbitrarily far m of these indices, for m a density Hales–Jewett dimension at density $\mu(A)/2$ and alphabet h. Using the word-average argument on the m vertical generators, obtain a nonempty subset F such that

$$C = \bigcap_{t=0}^{h-1} U^{-tv} A, v = \sum_{i\in F} b_i,$$

has positive measure. The spectrum of almost every horizontal ergodic component is disjoint from the nonzero characters of order dividing m_F: any common eigenvalue would generate one of order p_i for some $i \in F$. On the positive-measure set of components where C has positive measure, Moreira–Richter, Theorem 2.4 gives equality of the average h-return correlation of C along $n = r_F \bmod m_F$ and along all n. The latter average is positive by ordinary multiple recurrence, equivalently finite Szemerédi averaging. The equality and positivity pass through ergodic decomposition by bounded convergence. Thus some n in that

congruence class gives a positive return of C under T, which gives a positive return of A at (n, v), a contradiction. For the odometer assertion, in each component choose normalized finite cyclic eigenfunctions for the countably many orders; the joint cyclic rotations are equidistributed on every finite product by CRT. This last claim is componentwise and does not claim the vertical group acts trivially on the odometer coordinates.

Pure point observable. Even in the presence of the whole odometer, no avoider can have 1_A of discrete spectrum for the global horizontal T. Approximate it in L^2 by a linear combination f of finitely many horizontal eigenfunctions, with error chosen in terms of h and the uniform positive vertical Hales–Jewett correlation $\kappa(h, \mu(A))$. For their tuple of eigenvalues λ in a torus, the closure K of its integer multiples is a compact abelian Lie group with some finite number M of components. Multiplication by an integer m coprime to M is onto K, so every residue class $\bmod m$ gives a dense subsequence $n\lambda$ in K. Use vertical indices beyond the primes dividing M, and take the uniform positive vertical return as above. Some n in its prescribed residue class makes $T^n f$ arbitrarily close to f in L^2. Thus $T^{tn}1_A$ is within t times the resulting error of 1_A for $0 \le t < h$. Telescoping against the vertical functions shows that the positive v-return survives at (n, v). This proof does not assume horizontal ergodicity or simple eigenvalues.

Remaining issue. The necessary odometer can exist. On a bundle over the T-invariant σ-algebra, vertical moves may change finite eigenfunctions by base-dependent rotations, and even a finite componentwise spectral approximation of A can acquire fresh rational frequencies after a vertical move. The spectral argument does not establish recurrence in this case. The horizontal characteristic-factor reduction likewise retains these ergodic components.

Ergodic horizontal actions with quasi-discrete spectrum Suppose T is ergodic and its circle-valued phase polynomials generate the σ-algebra, or just the event A under test. The step family W above then has h-returns at every positive mass. It is enough that the functions used to approximate the event are phase polynomials for T; no phase polynomial assumption is made on the vertical action. This conclusion holds also when the global T action has infinitely many of the specified prime cyclic factors.

Write multiplicative derivatives as $\Delta_T f = (\mathrm{f}\circ\mathrm{T})/\mathrm{f}$. For any commuting S of finite order q and any phase polynomial f of degree $\le s$ one has $(\Delta_S f)^{q^s} = 1$. At $s = 0$ the polynomial is constant by ergodicity. Induct using $\Delta_T \Delta_S f = \Delta_S \Delta_T f$, so $(\Delta_S f)^{q^{s-1}}$ is T-invariant and hence constant. Its product round an S-orbit is one, making this constant a qth root of unity. Further, $\Delta_S f$ belongs to the cyclic factor fixed by T^Q for an integer Q with prime divisors only from q. Indeed its T-trajectory is a polynomial of degree $\le s$ in n with binomial coefficients, valued in the q^s-th roots of unity and with Newton coefficients in that group. Choose Q a sufficiently large power of q that q^s divides $\binom{Q}{j}$ for $1 \le j \le s$; the binomial translation identity gives period Q. If $q = 1$ the assertion follows immediately. Thus a finite-order S acting on a finite list of T-phase polynomials only adds finite T-periodic data with prime divisors of q.

A factor Y generated by finitely many T-phase polynomials *and all their T-translates* has no rational spectrum at primes outside a finite list independent of S. For clarity, take the joint torus coordinates consisting of each phase and its finitely many iterated T-differences. These generate Y and evolve by an affine unipotent map. Along one generic orbit their coordinates are ordinary circle polynomials in n. Weyl's criterion splits their orbit into finitely

many classes modulo some M on each of which they are totally equidistributed on a connected subtorus coset: take a basis of the subgroup of integer characters whose resulting polynomial has only rational nonconstant coefficients and clear its denominators; the subgroup is primitive since an integer multiple of a real number is rational only if the number is rational. All other characters stay equidistributed on every subprogression. The same assertion on further progression samples shows that multiplication of time by m coprime to M leaves the limiting distribution unchanged, including from every residue mod$m$. This implies T^m is ergodic on Y: equivalently a period-m eigenfunction would give differing distributions on the m residue classes of a generic orbit. One may choose the original point generic for each of these countably many powers and a dense set of continuous tests, on its relevant ergodic component. Adjoining a finite cyclic factor of order dividing Q only adds primes dividing Q to this list. In detail, the ergodic joining with that finite factor is an ergodic component of the product (any joining is absolutely continuous with respect to the product); an eigenfunction on it extends by zero to the product. Product eigenvalues here are sums of eigenvalues of Y and multiples of $1/Q$, by finite Fourier expansion. If such a sum has prime order p coprime to Q, the Y eigenvalue already generates a nontrivial p-order eigenvalue.

To conclude, fix A of mass a and h. Finite DHJ gives a lower bound $\kappa = \kappa(a/2, h) > 0$ for the largest vertical return in a fixed sufficiently large finite sum family. Finite Szemerédi averaging gives a lower bound $\kappa' > 0$ for the ordinary h-return Cesàro average of every event of mass $\geq \kappa$ (valid also in a nonergodic action). Approximate A in measure by a set A' from a finite phase-polynomial factor Y as above, of error less than $\min(a/2, \kappa'/(2h))$. Pick all vertical indices from a tail avoiding the finite exceptional primes for Y, then use DHJ for A' to choose a nonempty F and its positive vertical return C of mass $\geq \kappa$. With $S = U^v$ of order q dividing $\prod_{i \in F} q_i$, C is measurable on Y joined with a finite cyclic factor of order dividing a power Q of q. This follows first for each generator and S^t by the derivative lemma, and then for the generated σ-algebra by measurability. Thus T^{m_F} is ergodic on the factor of C, because every prime of m_F avoids both the exceptional set and q. Moreira–Richter's theorem yields the same h-return average for C along the prescribed residue as for all n. Some allowed (n, v) therefore has return measure for A' at least arbitrarily close to κ'. Telescoping loses at most h times the approximation error on returning to A. This proves the conclusion.

R144. The two explicit integer-horizontal bundles have a high-density bound

Scope and status. Result with proof.

Result. Two explicit integer-horizontal torus bundles have all-length recurrence above $1 - 1/L!$, with a half bound for the two-channel case.

Scope. System-relative bounds, depending on the observed number of channels; no unrestricted integer bound.

Take the prime matching with $\sum 1/q_i = \infty$ and $\mathrm{ord}_{q_i}(p_i) \to \infty$ from the horizontal polynomial fiber note [R140]. In either explicit system from the cyclic nil audit [R038], let L denote the number of observed upper torus channels $s \in \mathbb{T}^L$. In that fixed $\mathbb{Z} \times V$-action, **every measurable event of mass greater than** $1-1/L!$ **returns at every length** along the associated asymmetric difference family, in every far tail, with arbitrarily small positive marked budget. This is a system-relative statement. For the four-slot unsigned choice with two upper channels, every event above one half returns, while the fixed coloring gives a strictly positive avoiding mass. Thus these particular explicit systems have a system-relative all-length avoidance threshold strictly between zero and one. The family allows arbitrary jointly stationary test systems in the original integer conjecture, and no such universal upper bound follows here.

Here are details of the comparison. Write the finite vector of local vertical words as $k_i(v) \in \mathbb{F}_{p_i}^L$, encompassing all the spatial offsets used. Use the augmented virtual paired-array model

$$(u, v) \longmapsto (k_i(v)/p_i,\ uk_i(v)/p_i) \in \mathbb{T}^L \times \mathbb{T}^L,$$

with the first torus called Θ, the second S. The three single-axis distributions in the polynomial-fiber argument [R140] are unchanged, except that the marked horizontal speed can be any prescribed a_i rather than one: take one for the centered-odd unit/small-cost rule, or the nearest integer to $p_i/2$ for the macroscopic marked rule. Each single-axis special cost tends to zero, there are finitely many per chosen odd step, and each pure normal is legal. Normal horizontal speeds are uniform in C_{p_i}.

Keeping the lower torus flat. In the tagged Palm profile group take the even subgroup L_0, and let L_S denote its slice with Θ-profile identically zero, viewed on S alone. Put $R_S = L_S^{\perp}$. Every $A \in R_S$ lifts to a word $(\eta, A) \in L_0^{\perp}$: on L_0 evaluation of A factors continuously through projection to Θ, and a continuous circle character on that closed subgroup extends to the full torus of Θ profiles. Thus the polynomial-fiber proof of exact normal identities applies to the horizontal u-coefficient of each lifted word, which ignores η.

Choose $K_f \subset S$ and the Laurent nullspace streams from R_S as in the dimension bound [R253]. For each lattice row l in the quotient S/K_f, some scalar nonzero $Q_l(T)$ has $Q_l(T)l \in R_S$. Lift it as above. Except for axes of finite total $1/q_i$-mass the vertical normal identity gives $Q_l(Y)(lk_i) = 0$ exactly. The choice of multiplicative orders forces lk_i to be constant in v there, simultaneously for a basis of the quotient. The projection of the odd profile coset to $\Theta \times (S/K_f)$ contains zero: otherwise parity extends to a character (η, ψ) on those output profiles, zero on every abundant normal and half on every abundant mark. Outside one more finite-reciprocal exceptional set its horizontal filter vanishes for all horizontal velocities and starts. On the good axes the Θ-increment of a mark is identical to its vertical normal increment, and its projected upper increment adds only the horizontal contribution of the constant lk_i,

killed by the word's horizontal identity. Thus for this character the two distributions are identical on infinitely many good axes, contradicting the asserted different values on their abundant profiles. We can choose an odd representative identically zero on Θ and valued in K_f in the upper channels. The polynomial control streams from R_S alter only S. Their dimension is at most L.

Integer CRT sampling. Fix a finite tilted-pool scheme and an early-coordinate cylinder, as in the polynomial-fiber argument. Impose all chosen horizontal residues on the finite pool axes by CRT and put zero residues on the early axes, using zero also on inactive pool axes. Take the actual integer n uniformly through an increasing finite segment of the resulting residue class, conditionally on the finitely many choices of moves. Outside these finite axes the horizontal velocity is unrestricted, and each vertical velocity is zero. All actual directions are legal. The full base marginal density at each time still has L^2 norm below two: within the finite pools the shifts agree with the prescribed velocities, on the early axes the shift is zero, and outside all pools and early axes the base is independent Haar. Even if the chosen n depends on the finite pool data, translation keeps this latter Haar product uniform conditional on all those data. For fixed n, the transformation on the two initial torus variables is affine Haar-preserving for each fixed base state, hence those variables at any one time are also uniform independent of that time's base. The same marginal bound consequently applies on the whole system, uniformly in the CRT averaging length.

For a finite pool move vector a with vertical component v, compare its virtual array trace with the actual formula in the cyclic audit. The upper torus gets one additional term $-t\sum_i a_i k_i(y_i)/p_i$ relative to the virtual trace, including the purely horizontal chosen moves. In the first bundle its other extra term is $tn\theta$. As $n \to \infty$, the initial Haar θ and $n\theta$ become independent Haar in distribution, by character orthogonality; adding the known finite base-dependent slope to $n\theta$ leaves this limit independent Haar. The lower trace has no additional term. Thus the limiting additional drift is a random linear upper-torus slope independent of the virtual grid profiles and initial torus variables.

In the second bundle the lower torus also gains $tn\alpha$, and the upper torus gains $tnx + \binom{tn}{2}\alpha$ in addition to the displayed base slope. For Haar x, the averaged tuple $(x, nx, n\alpha, \binom{n}{2}\alpha)$ tends to joint independent Haar along each fixed CRT class. Indeed integration in x kills every mixed Fourier term with nonzero nx coefficient for all large n; the remaining nonconstant terms in n vanish by Weyl's polynomial criterion and the rational independence of the coordinates of α together with one. Since $\binom{tn}{2}\alpha = t\binom{n}{2}\alpha + \binom{t}{2}(2\binom{n}{2}\alpha + n\alpha)$, the extra drift has independent Haar linear lower, linear upper, and quadratic upper coefficients. The base slope is again absorbed in the independent linear upper coefficient. These limit identities apply conditionally on the finite base data under each pool selection and hence also in finite mixtures. Weak convergence suffices for continuous observables; for the finite early-cylinder events with arbitrary measurable torus sections one first approximates their indicators in L^1 by continuous functions and uses the uniform marginal bound at all times on both sides.

Allowing the polynomial drift. In the quotient by K_f (including all lower torus coordinates), the limiting additional drift is a polynomial with jointly Haar coefficients in a compact abelian coefficient subgroup, independent of the Haar starting quotient and the pool profiles. The following elementary statement includes the case of correlated Haar coefficients on a subgroup. For a quotient cell E of mass $e > 0$,

$$\Pr\left[C + \textstyle\sum_d \binom{t}{d} A_d \in E \quad (0 \le t < h)\right] \ge c(h, e) > 0.$$

Use the finite density Hales–Jewett dimension for density $e/2$ and test the random word

cube consisting of $C + \sum_{j,d} \binom{n_j}{d} A_{j,d}$, with independent Haar coefficient tuples for the word coordinates and common uniform C. A nonempty wildcard sum has the original Haar coefficient-tuple law, and its base offset is uniform independent of that tuple because of C. The expected occupied proportion is e; the union bound over the finite word lines gives the positive lower estimate. This is uniform in the quotient cell and the coefficient subgroup. Degree zero coefficients may be included in C.

For each early-coordinate cylinder section of mass above the claimed threshold by a fixed margin, a fixed positive fraction of quotient points have K_f-fiber mass still above $1 - 1/k!$ by a margin, $k = \dim K_f \le L$. Partition those quotient points into finitely many measurable classes by a successful common vector of compression thresholds from the dimension argument [R253]. A finite rational grid suffices: decrease a successful threshold vector by a sufficiently small amount in each coordinate, preserving a fixed positive margin in the union bound. The largest class has mass bounded below in terms of the starting margin and L. By the word-cube estimate, with uniformly positive probability the quotient path is entirely in that class. Conditional on its drift, the fiber sets at the h times may differ but **share their compression thresholds**; shifts inside K_f preserve all rearranged thresholds by translation of prefixes. The pivot process therefore has conditional probability at least c^{kh} of meeting every such section. In the case $k = 0$ only the word-cube estimate is needed. All these constants may depend on h, L and the starting margin, but not on the chosen early cylinder. A positive fraction of its early sections has such a margin whenever its total mass does.

Finally approximate the odd control profile law by a finite mixture of actual tilted pools beyond the cylinder coordinates. The Haar-integrated finite-window tests, also after averaging over the independent polynomial drift, are continuous in the profile, by continuity of translations in L^1. Take the CRT averages just described so the actual cylinder trajectory has the same uniform positive lower bound up to an arbitrarily small loss. For a full measurable event choose its cylinder approximation error using this lower bound *before* choosing its axes or pools, then apply the uniform marginal estimate $2h\sqrt{\mu(A \triangle A')}$. A positive averaged return occurs at some fixed legal integer/vertical direction under an absolutely continuous start law, so its ordinary Haar return is positive.

Exactly one mark suffices. Both this integer-bundle result and the horizontal polynomial-fiber result [R140] actually permit only one marked horizontal residue on a vertical-active axis, all the other finite vertical moves being pure positive normals. The marked residue may be any prescribed choice $a_i \in R_i \neq \varnothing$ on each late axis. Replace the even subgroup L_0 above by P, the subgroup generated by abundant **normals alone** together with constants. Its $\Theta = 0$ slice, annihilator R_S, and free coefficient subtorus K_f are constructed exactly as before, with the same normal identities. Write π for the projection onto $\Theta \times (S/K_f)$ including all times. There exists a single abundant marked profile z with $\pi z \in \pi P$.

Indeed choose any finite list of characters of the quotient profiles annihilating πP. The previous exact normal identity and the high multiplicative orders imply that, outside a set of finite reciprocal mass, all $lk_{i,d}$ in the projected upper polynomial channels are constant in v, and the horizontal filter of each listed word vanishes for every value of the chosen horizontal velocity. Thus on those infinitely many good axes the whole character list at the mark has the same joint distribution as at the vertical normal, including all the vertical-only offsets and Θ observations. The total over all axes of the probability that even one of these characters is a fixed distance from zero at a vertical normal is finite by compactness and

absence of abundant normal failures. Consequently on any tail the summed probabilities that the marked profile satisfies the finite list within a given tolerance diverge. The compact set of abundant marked profiles therefore meets every such simultaneous closed approximate zero condition: otherwise a finite cover of the closed condition by neighborhoods of finite occurrence sum gives a contradiction (one may first use half the tolerance). The finite-intersection property gives one abundant marked z annihilated by the full annihilator of πP after projection, i.e. $\pi z \in \pi P$. Add an element of P to obtain a representative with flat lower projection and upper coordinates in K_f. Every target in this translated subgroup now uses exactly one abundant marked generator followed by a normal subgroup element. Its approximation uses one marked pool only, since normal positive sums are dense in P up to constants. The pivot and CRT arguments are unaffected. In the purely local model with no Θ-variables just omit that projection from this reasoning.

R161. Nonabelian profinite fibers also return with a single mark

Scope and status. Result with proof.

Result. Single-mark recurrence extends to nonabelian profinite Haar fibers over the full independent paired-grid base.

Scope. Unrestricted horizontal speeds off active vertical coordinates are essential; this is one extension class.

The arbitrary profinite abelian cocycle theorem [R011] holds for every compact metrizable **profinite group**, with no commutativity assumption on its fiber. The base must still be the full product Haar action of the paired grid coordinates, and the horizontal speeds in a vertical-inactive block must be unrestricted. The mark is exactly one chosen residue at one vertically active prime. The primes escape; no harmonic divergence is needed.

Localization and separators. Work in a finite fiber quotient K, with left multiplication by a cocycle ρ. On finite prefixes choose F_n so $\rho(g,x) = F_n(x+g)F_n(x)^{-1}$ there, as in the abelian note. Now $r_n = F_{n-1}^{-1}F_n$ is invariant under the earlier prefix, hence a tail function. Approximate it by f_n on $[n, N_n]$ with summable mismatch probabilities. The products $F_n^* = f_1 \cdots f_n$ in this indicated order give a coherent cocycle on prefixes: factors added beyond the support of g are invariant under g and cancel on the right in $F_n^*(x+g)F_n^*(x)^{-1}$. Almost surely $F_n(x)(F_n^*(x))^{-1}$ stabilizes after the last disagreement of the tail factors, because all equal later factors cancel. Its limit is the cohomological change of fiber coordinate.

As before choose blocks B_j with each grouped product b_j depending on B_j, B_{j+1}. Hold the odd-numbered blocks fixed as separators and allow movements only on the even cores. Conditional on the whole separator environment, the moving cores are independent uniform blocks. The consecutive products $f'_j = b_{2j-1}b_{2j}$ depend only on one moving core each and the fixed separators. A possible fixed initial prefix causes only common conjugation, which does not affect the equality of trajectories with their time-zero state. Below fix the passive environment, write f_j for these ordered one-core potentials, and drop primes on the notation. This conditioning is essential; adjacent two-block functions were not assumed independent.

A deterministic gauge and finite mixing. Let $W = K^h$ with componentwise product. Fix a modal value $c_j \in K$ of each f_j, and put $c_{<j} = c_1 \cdots c_{j-1}$. Use the new core function

$$g_j(x) = c_{<j} f_j(x) c_j^{-1} c_{<j}^{-1}.$$

It equals the identity with probability at least $1/|K|$. On every interval of cores the ordered product of these gauged functions is the ungauged product with diagonal endpoint factors on the two sides (at all trajectory times these factors are the same). Thus a time-profile product in $\Delta K \subset W$, the subgroup of constant tuples, is equivalent for the two gauges. Its value at time zero is the product with all moves inactive; consequently it gives exactly identity cocycle displacement throughout that interval.

For each core draw a normal move independently of all starts, using probabilities $1/2, 1/4, 1/4$ for no move, uniform horizontal motion on the entire core, or a fixed positive unit on one vertical axis. Write μ_j for the law of $(g_j(x+tw))_{0 \le t < h} \in W$. A letter is abundant if the sum over j of its μ_j-probabilities diverges. Let $P \le W$ be generated by the abundant letters. The probability mass outside P is summable; $\mu_j(e) \ge 1/(2|K|)$. For every sufficiently late starting core, an arbitrarily long product of these independent normal

profiles has law arbitrarily close to uniform on P, up to twice the outside-tail sum. To justify this, first condition each μ_j to P. The product-law change in total variation is bounded by the outside-tail sum by telescoping, and each identity atom remains uniformly positive. For fixed starting index and any initial probability on P, successively right-convolve with the conditioned laws. Its squared ℓ^2-norm decreases by at least

$$c \sum_{g \in P} \mu_j(g) \|R_g u - u\|_2^2$$

at step j, where u is the previous probability vector, R_g is right translation and $c > 0$ bounds the identity atom below. This follows from the Hilbert-space variance identity for the convex combination of the translated vectors. It also bounds the squared successive-vector increment by a constant times the energy drop, by Jensen's inequality. The sum of the k smallest vector entries is nondecreasing for every k, since these averaging matrices are doubly stochastic. Thus every entry in sorted order has a limit. If there were a gap in these limits, the set of low entries would eventually be constant as an index set, since the successive-vector increments tend to zero and the gap is fixed. For every right translation crossing that cut the squared translate distance stays bounded below. The energy drop would make that translation's cumulative probability finite. This contradicts generation of the transitive right regular group by the abundant letters. The sorted limits are all equal, proving convergence to uniform. This proof needs neither commuting consecutive laws nor a symmetry assumption.

One mark has the same subgroup constraint. Define $K' \le K$ by the abundant values of the one-time random variables $g_j(x)$. Every time marginal of each move law equals this one-time law. The no-move part implies $\Delta K' \subset P$, and the marginal observation gives $P \subset (K')^h$. Replacing g_j by the identity on its values outside K' changes any profile with total probability summable over the cores, because all its time marginals are uniform-start evaluations and h is fixed. We can work inside K' in the following argument and then restore the original arrays.

For each slot t let $N_t \le K'$ consist of the elements whose tuple supported only at that slot belongs to P. It is normal in K' since $\Delta K' \subset P$; the product of the coordinate-supported N_t is contained in P. Let β_j be the probability that the pure horizontal profile does not belong to P at a uniform start and uniform horizontal velocity. Its sum is finite. Condition on all vertical and passive data within a core. For the map $(u, a) \mapsto (g_j(u + ta, v))_{t<h} \in (K')^h$, successively apply multiplicative differences $G(x+d)G(x)^{-1}$ in the directions $(-t'b, b)$, with independent uniform b's, one $t' \ne t$ at a time. When a slot has been killed, it remains identity. Before its own killing, every entry at slot t' still depends on $u + t'a$ and the previous increments only, so the indicated difference kills it irrespective of noncommutativity. If every original evaluated tuple was in the subgroup P, the final coordinate-supported tuple is in P. Thus the $(h-1)$-difference test of g_j in the finite quotient K'/N_t fails with probability at most 2^{h-1} times the conditional horizontal failure.

The finite alphabets in the stability proof [R011] can be nonabelian. Indeed $\Delta_{v+w} g(u) = \Delta_v g(u+w) \Delta_w g(u)$, so the constants at the inductive step give an approximate homomorphism with the stated order. In the additivity test use the modal value of $c(v+y)c(y)^{-1}$, comparing it at $y, y+z$ by the two product tests at $(v+y, z), (y, z)$. The same concentration and product identity give a genuine homomorphism. No nontrivial one exists from a central horizontal group whose prime orders are all greater than $|K|$ into K'/N_t. The proof thus gives, for every slot after averaging the conditioning, that the coset of g_j modulo N_t is within

Hamming distance $O_h(\beta_j)$ of a function of the vertical/passive data only.

For the prescribed marked move versus that vertical normal, at each slot the vertical/passive data agree and both horizontal variables individually have their uniform marginal. Except on a set of probability $O_h(\beta_j)$, the tuples therefore differ by a member of $\prod_t N_t \subset P$ (normality makes either multiplication convention harmless). Vertical normal failures are summable, so the probability that a marked absolute profile is outside P has finite sum over j. In particular this probability tends to zero.

Trials with full marginals. On a far interval of cores run independent normal moves until their ordered-product law is within $1/(4|P|)$ of uniform on P, as permitted by the mixing result and small outside-tail sum. Append a fresh terminal marked move with probability of being in P greater than $3/4$. Independently multiplying a uniform law on P by any random element of P keeps it uniform. Thus the probability that the interval product equals the identity tuple, in particular lies in ΔK, is at least $1/(2|P|)$ with the displayed tolerances or smaller error. The equivalence of gauges implies zero displacement of this entire interval's raw cocycle for all times. This is a single-mark interval trial; the mark costs arbitrarily little in a late block for the original rules, or simply lies in its prescribed residue set for the single-mark template. There are infinitely many disjoint such trials conditional on the fixed environment.

Pool many whole independent trials and use only **one**. Give trial j weight proportional to its success probability, condition it on success, and leave all other blocks inactive with unconditioned uniform starts. As in the tilted-pool lemma, the base marginal at each time has density within $1/\sqrt{\Lambda}$ of one in L^2, where Λ is the sum of trial success probabilities. This works because the unconditioned trial's directions were independent of its uniform input, and different trials use disjoint cores. Any prefix and suffix outside the selected interval, as well as the passive separators, give identical factors at all times and do not affect the comparison: the selected interval has its original time-zero product at every time. In algebraic terms both full products are UVW with the middle product V identical, so the cocycle displacement is identity. An early/base cylinder event in the changed quotient is thus constant throughout the trajectory with its original positive conditional mass. Approximation costs at most twice the square root of the conditional error per time.

Finally, for an arbitrary profinite fiber first approximate the full event in measure by a measurable event on the base and a finite group quotient. After the finite quotient gauge, freeze the passive environment and approximate this changed event by a cylinder on finitely many moving cores. If the original event has positive mass, a positive-measure set of environments has positive conditional mass bounded below and conditional first-approximation error small, by choosing its global error sufficiently small in terms of the original mass and h. On each of these environments the preceding trial construction, beyond the cylinder cores, proves a positive full-event return with a finite ordinary direction: the lifts preserve Haar on every fiber and the conditional base marginal bound controls both approximation errors. One need not select directions measurably across environments; the possible directions form a countable set, hence some one of them gives positive unconditional return by Fubini. This proves the assertion, also when the profinite group is an inverse limit of increasingly large finite nonabelian quotients.

R167. Odd prime sums recur on class-two circle bundles over connected compact groups

Scope and status. Result with proof.

Result. Divergent-reciprocal odd prime sums give multiple recurrence on the specified class-two compact circle or torus bundles.

Scope. Haar systems and their stated locally compact translational structure; general characteristic-factor reduction is not supplied.

Let $G = \bigoplus_i C_{p_i}$, where the p_i are distinct odd primes with divergent reciprocal sum, and let D consist of sums of odd numbers of distinct coordinate generators. For every h, every tail of D is a set of h-term measurable recurrence on the following Haar systems. A second-countable locally compact group acts transitively on a compact central principal circle bundle $X \to B$ over a connected compact metrizable abelian group. The action covers translations of B, is circle-equivariant, and its commutators are constant central rotations. The action tested is that of an arbitrary homomorphism from G into this group. The invariant measure is Haar on the base and Haar on its circle fibers. This includes bundles with nonzero first Chern class and does not assert anything about all measurable systems.

A harmonic-ultrafilter lemma Choose a free ultrafilter $\mathcal{U}$ on the axes containing the complements of all sets with finite reciprocal sum. Given points a_i in any compact Hausdorff abelian group L, with $p_i a_i = 0$, let K be the intersection over $I \in \mathcal{U}$ of the closed sets of finite sums of $(a_i, 1) \in L \times C_2$, with no repetitions, drawn from I. Then

$K = J \times C_2$, where $J = \bigcap_{I \in \mathcal{U}} \overline{\langle a_i : i \in I \rangle}$.

Indeed K is a compact subsemigroup, by taking disjoint supports in successive approximations, and hence a subgroup. If a character (ψ, e) vanishes on K, compactness puts every finite subsum of $\psi(a_i) + e/2$, $i \in I$, in a fixed small neighborhood of zero for some $I \in \mathcal{U}$. Using separately positive and negative representatives, the sum of their distances to zero is finite. For $e = 1$ every summand costs at least $1/(2p_i)$. For $e = 0$ the nonzero summands cost at least $1/p_i$, so remove their indices from I. The character annihilates $J \times C_2$, as required by duality. In particular, every continuous character annihilating J vanishes on all individual a_i on some member of $\mathcal{U}$.

Canonical coordinates A connected compact abelian B is the inverse limit of its finite-dimensional torus quotients. Principal circle bundles on compact metrizable spaces are classified by integral Čech H^2. Continuity of this cohomology under compact inverse limits shows that the bundle is the pullback of one over a torus under a quotient $\nu : B \to \mathbb{T}^n$. Circle bundles over this torus are represented by the usual central Heisenberg nilmanifolds for an integral alternating form ω on $\mathbb{Z}^n$, including degenerate forms. To see the coordinate formulas, use the group law on $\mathbb{R}^n \times \mathbb{R}$ with central cocycle $\omega(a, a')/2$. The lattice is generated by lifts of the standard basis vectors and a central unit. Its projection is the ordinary integer lattice and its central intersection is the integers, because all basis commutators are integral. The commutator pairing is ω, giving the prescribed Chern class with a consistent orientation choice.

Write $(H(a))$ for the lift $((a, 0))$ on the torus bundle. When $k \in \mathbb{Z}^n$, $(H(k))$ acts on the fiber over y as $ell_k(y) + c_k$, where $ell_k(y) = \omega(k, y)$. The constant accounts for the choice of

lattice section. Write V_s for central rotation. Every element n of the acting group covers a translation b_n. Compare it to the reference map R_n which translates B by b_n and applies $H(a_n)$ to the torus-bundle coordinate, where a_n lifts νb_n. Their difference is a continuous circle map on B; it has the form $\chi_n + v_n \bmod 1$ for a unique continuous character χ_n and a real continuous function v_n, unique up to an integer. The elementary lifting argument is in the trivial-bundle note [R257]. The commutators of the references are constant, so the same averaging calculation applies to the real functions: $\Delta_{b_n} v_g - \Delta_{b_g} v_n$ is real constant by connectedness and is zero by Haar integration. Choose group representatives over $t \in B$ in a compact set; on finitely many local charts for the horizontal lifts and circle logarithms, choose uniformly bounded measurable real lifts $v_{n(t)}$. Changes in a horizontal lift add a character and a constant to the comparison, compatible with the asserted identity. Averaging gives a single vertical gauge which removes all real parts except constants. The reference exists on the comparison chart even though $(H(a))$ alone need not act on the full pullback.

Thus an individual map has data (b, a, η, c), acting by base shift b and $H(a)V_{\eta(x)+c}$ at $x \in B$. For genuine maps $\eta \in \widehat{B}$. Pointwise closure allows abstract characters $\eta \in \widehat{B_{\mathrm{disc}}}$. All these maps are bijections, multiplication is right-continuous, and the total space of the data is compact with its pointwise topology. This follows directly by using compact local charts for a modulo the lattice and the compact pointwise space of abstract characters; the constants and characters can be recovered by evaluating at base zero and then at arbitrary base points. Modulo constants this is a genuine compact Hausdorff abelian group A of triples with $a + \mathbb{Z}^n = \nu b$, with the identifications $(b, a + k, \eta - \ell_k \circ \nu) \sim (b, a, \eta)$. For two maps the central commutator is

$$\omega(\mathrm{a},\mathrm{a}\prime) + \eta(\mathrm{b}\prime) - \eta\prime(\mathrm{b}).$$

This expression is independent of the lifts. The central term of the j-th iterate, apart from $(H(ja))$, is $j(\eta(x) + c) + \binom{j}{2}\eta(b)$.

The slope envelope In the pointwise compactification of the tuple of powers $j = 0, \ldots, h-1$, adjoin a C_2 parity coordinate and intersect the closed finite-sums sets over $I \in \mathcal{U}$. The resulting K is a group. For the subsemigroup assertion approximate the left input by a genuine finite word, which is continuous, and the right input using a tail disjoint from this word; right multiplication by any fixed map is continuous. Any compact right-topological subsemigroup of bijections is a group: for $x \in K$, the compact left ideal (Kx) has an idempotent, necessarily the identity, so $x^{-1} \in K$. This uses the usual Ellis–Numakura idempotent lemma.

For $h \geq 2$ the horizontal projection $slot(j = 1)$ and parity has image $J \times C_2$ by the lemma. All slot horizontal triples are the j-th multiples of that first one. All elements of K have data in each horizontal chart; for commutators the slopes give exactly $(j^2 z)_j$. A tuple over the horizontal identity has central data $(ju + \binom{j}{2}v)_j$: take a subnet of genuine words whose horizontal lifts at the origin tend to zero, and also let the two scalar coefficients converge. For $h = 2$ the coefficient v is immaterial.

Infinite commutator subgroup The closure of the infinite commutator subgroup contains the entire compact circle $((j^2 z)_j, 0)$. Start with an odd element over the horizontal identity. Multiplying by a suitable point on this circle $solve(2z = -v)$ leaves a central tuple $(jt)_j$, odd.

In every neighborhood of this target and every $I \in \mathcal{U}$, genuine words in that neighborhood can be found with normalized shear outside an arbitrary finite set in $\widehat{B}$. Otherwise choose a smaller open chart near the target. Every member of K in that open subset is a limit of genuine words with shears in the fixed finite list, so is a tuple of continuous maps. Such elements commute with all of K: the original action is abelian, commutation with each genuine word passes to pointwise limits, and then to all limits by continuity of the given element. An open subset of the center makes the center open, and compactness by right translations makes its index finite. A group with center of finite index has finite commutator subgroup (Schur's lemma, or directly in class two take finitely many coset representatives and bound the orders of their pairwise central commutators). Contradiction.

Select odd words with $b \to 0$, $a \to 0$, $c \to t$, $\eta(b) \to 0$ $when(h \geq 3)$, and η escaping a countable exhaustion of $\widehat{B}$. The scalar tests use the action at any one point over base zero in slots 1 and 2, so they are encoded in neighborhoods of the target. For Haar $x \in X$, the joint laws of $x, \eta(\pi x)$ tend to Haar on X times independent uniform circle measure: for each nonzero integer r the characters $r\eta$ escape by torsion-freeness of $\widehat{B}$, and Fourier coefficients of $L^1(B)$ vanish at infinity. The small translations $(H(ja))$ and base shifts converge uniformly to the identity in the product realization. Hence the limit h-correlation of a continuous function is its correlation averaged over $(x, V_s x, \ldots, V_{(h-1)s} x)$, with s uniform (the added t is harmless). Approximation by continuous functions, using invariant Haar marginals, gives the same for arbitrary measurable indicators. It is positive for a positive-measure set by strong continuity of the circle rotations near zero.

Finite commutator subgroup Choose $m \geq 1$ annihilating this subgroup. Write $B_J \subset B$ for the projection of J; it is compact. For every fixed $(b', a', \eta') \in J$, the commutator formula identifies $m\eta'(b)$ with $m\omega(a, a') + m\eta(b')$ on J. The latter is continuous there in horizontal charts, including across their lattice changes. Its value depends only on b; the compact quotient onto B_J implies

$$m\eta' restrictedtoB_J isacontinuouscharacter.$$

Introduce the abelian Lie group, with countably many components,

$$L = (R^n \times (\widehat{B_J})_{\mathrm{disc}})/\{(k, -m\ell_k \circ \nu restrictedtoB_J) : k \in Z^n\}.$$

The horizontal restricted image $J' \subset L$ consists of $(a, m\eta|_{B_J})$. It is compact for the Bohr topology on L. In fact every continuous character of L tests a and evaluation at a fixed $y \in B_J$, and so is continuous on J. Glicksberg's compactness theorem shows (J') is compact in the Lie topology as well.

Let q_i be the restricted image of the actual axis in (L/J'). Every continuous character of (L/J') vanishes on q_i for i in some member of $\mathcal{U}$. Indeed its formula using evaluation at a fixed point of B_J extends to a continuous character of the original compact horizontal group A and annihilates J. We claim $q_i = 0$ on a member of $\mathcal{U}$. Project onto the discrete component group. On the compact dual of that discrete group choose a Haar-random character. A nonzero component image of q_i lies in its kernel with probability $1/p_i$. The expectation of the sum of $1/p_i$ over these undetected nonzero component images is at most $\sum 1/p_i^2 < \infty$.

Fix a choice where it is finite. Since this character vanishes on a member of $\mathcal{U}$, the axes with nonzero component cannot belong to $\mathcal{U}$. Inside the identity component of (L/J') finitely many continuous characters separate all torsion points (it is a Euclidean group times a finite-dimensional torus); they extend to (L/J') by Pontryagin duality. Intersect their finitely many ultrafilter members. This proves the claim.

For all i in this divergent subfamily, after removing finitely many primes, the restricted images of the axes belong to the identity torus of (J'): they are p_i-torsion and (J') is a compact abelian Lie group with finitely many components. The identity torus has the form $V/(V \cap \Lambda)$, where $V \subset \mathbb{R}^n$ is real-linear and its period lattice consists of integral k with $m\ell_k \circ \nu|_{B_J} = 0$. Its projection to the horizontal torus is contained in $\nu(B_J)$, since (J') is the image of J. For k in the period lattice and $v \in V$, $m\omega(k, v) = 0 \bmod 1$, and varying v over the real vector space shows $\omega(k, v) = 0$. The period lattice spans V, so V is isotropic. The projected subtorus is $T_V = V/(V \cap \mathbb{Z}^n)$, and every selected horizontal step projects into T_V.

On each coset of T_V in the horizontal torus, the circle bundle admits a trivialization in which $(H(v))$, $v \in V$, acts by constant vertical rotations and base translations. Indeed the $(H(v))$ form a true additive action since V is isotropic. Start with a point z_0 over a coset representative. The action of the lattice $V \cap \mathbb{Z}^n$ at z_0 is by a character of that lattice. Extend the character to a real-linear functional on V modulo one and compensate by the corresponding vertical rotation, giving a periodic section on the coset. Origin and real lifts of finitely many lattice values can be chosen measurably in the coset. Recenter the horizontal representatives of the selected axes inside V; this only changes their shears by continuous characters and their constants. Thus in these coordinates each acts as

$$(b, z) \mapsto (b + b_i, z + t_i(coset) + \chi_i(b)),$$

with χ_i a continuous character and the coset label invariant. Base characters and the fiber circle coordinate are then phase polynomials of degrees at most one and two, respectively, and generate the full fibered probability space (including the invariant coset factor). The Abramov recurrence theorem for distinct odd primes [R171] applies on this divergent subfamily; alternatively apply it on each ergodic component. This proves positive multiple recurrence.

Any finite-dimensional central torus The foregoing theorem holds also for central fiber $T = \mathbb{T}^r$. Choose a single finite torus quotient supporting all r Chern classes and use the vector-valued alternating form ω; the preceding canonical-coordinate construction applies componentwise. The closure of the slope commutator shifts is $\{(j^2 z)_j : z \in C\}$, where C is a closed subgroup of T. Write C^0 for its connected identity torus and let $\Lambda = (C^0)^\perp \leq \mathbb{Z}^r$. This is a primitive lattice. Pick a basis $\ell_1, \ldots, \ell_s$, and m annihilating C/C^0. In the finite-commutator argument impose the s finite commutator equations simultaneously. The group L becomes

$$(R^n \times ((\widehat{B_J})_{\mathrm{disc}})^s)/\{(k, -(m\ell_l\omega(k, \nu\cdot)|_{B_J})_{l \leq s}) : k \in Z^n\}.$$

The identical compactness and Lie-group argument supplies one $\mathcal{U}$-large subfamily I of axes in a torus T_V isotropic for every $\ell_l\omega$. If $s = 0$, no such selection or tags are necessary. Replacing all axes by $I \in \mathcal{U}$ does not alter their slope envelope.

Apply the coset trivialization just constructed to the quotient fiber T/C^0, identified through the basis characters of Λ. It suffices to work on the base-origin coset $\nu^{-1}(T_V)$ to find tags. There the action in each basis coordinate has the form $z \mapsto z + \theta_i + \xi_i(b)$, with ξ_i a continuous character. Set $\lambda_i = \xi_i(b_i)$. Both θ_i, λ_i are p_i-torsion: the second statement follows from $p_i b_i = 0$, the first by iterating p_i times at the origin and using oddness. For $i \neq k$ the same identity on bases $0, b_k$ implies $p_i \xi_i(b_k) = 0$, and $p_k \xi_i(b_k) = 0$ as well, so $\xi_i(b_k) = 0$. Thus at the origin the j-th fiber shift of a word F is exactly

$$j \sum_{i \in F} \theta_i + \binom{j}{2} \sum_{i \in F} \lambda_i$$

in these quotient gauges. Enlarge the slope envelope by carrying both finite-dimensional additive tags. Its projection onto the untagged envelope is surjective by nested compactness, and the commutator shifts in the enlarged envelope are the original ones with zero tags. Its compact abelian horizontal-and-tags projection still has odd parity over the identity by the ultrafilter lemma. Take an odd element over this identity. Since $b = 0$, the change between the quotient gauge and the original gauge cancels at the test origin by continuity of the trivialization on the origin coset. Its central tuple is therefore $(ju + \binom{j}{2} v)_j$ with $u, v \in C^0$ (only u matters if $h = 2$). Adding $(j^2 z)_j$ with $z \in C^0$ solving $2z = -v$ yields a target $(jt)_j$ with $t \in C^0$, horizontal and tags zero, parity odd.

In each small chart about the chosen target and for any prescribed finite list of characters $\ell \notin \Lambda$, their shear characters $\ell\eta$ can simultaneously avoid arbitrary finite lists. Otherwise a relatively open subset of the compact envelope is covered by finitely many closed finite-shear conditions. Normalize the horizontal lift near the chart origin. After shrinking the chart the possible shears in a finite list are closed by pointwise evaluation. By the finite-cover Baire argument one projected shear condition holds on a relatively open subset. All transformations in the ℓ-projected slope tuple there are continuous. The preimage of the center of the ℓ-projected group has a relatively open subset and hence has finite index; therefore the projected commutator subgroup is finite. This contradicts $\ell \notin \Lambda$, which entails an infinite projected commutator subgroup. If the list is empty the approximation follows from the definition of the envelope.

Choose one $b_* \in B$ at which every nonzero continuous character has nonzero value, possible since B is connected, metrizable, and each nontrivial character has Haar-null kernel. Select a sequence of words in the preceding charts with $b \to 0,\ a \to 0,\ c \to t,\ \eta(b) \to 0$ when $h \geq 3$, and with $\eta(b_*) \to 0$ (another pointwise test in the target). Exhaust the offending shear lists so every $\ell\eta$, $\ell \notin \Lambda$, escapes. For all remaining ℓ, pass to one subsequence to stabilize the dichotomy: $\ell\eta$ either escapes the countable discrete dual of B, or is constant along that subsequence, necessarily zero by the test b_*. The limiting Haar joint law of the base/full initial state and the shear vector is the product of the initial Haar measure and Haar on $S = \{\ell : \ell\eta = 0 \text{ eventually}\}^\perp \leq T$, as follows from the same Fourier argument as for one circle. Its annihilator is contained in Λ, so $t \in C^0 \subset S$. The limiting slot shifts are $j(s + t)$ with s Haar on S. This gives a positive multiple return for any positive Haar-measurable event by strong continuity. In the special case $C^0 = 0$ the earlier Abramov argument also suffices.

The compact translational models in Jamneshan–Shalom–Tao motivate the hypotheses. A base mixing connected and disconnected parts requires further arguments. The higher-

order characteristic-factor problem along odd finite sums in arbitrary measurable systems is separate.

R168. Odd finite sums are multiple recurrence on Conze–Lesigne systems

Scope and status. Result with proof.

Result. Odd prime sums are multiply recurrent on all Conze-Lesigne systems under reciprocal divergence.

Scope. An order-at-most-two system-class result; no characteristicity for sparse averages in arbitrary systems.

Let $G = \bigoplus_i C_{p_i}$ for distinct odd primes with $\sum 1/p_i = \infty$. Every tail of the odd binary finite sums of its coordinate generators is a set of multiple measurable recurrence on every Conze–Lesigne G-system (a system of Host–Kra order at most two). The model theorem used below concerns the ordinary Host–Kra factors. This result supplies no characteristic-factor theorem for sparse odd-sum averages in arbitrary systems.

We give the locally compact structural reduction that removes the connected-base restriction from the circle/torus proof [R167]. By Jamneshan–Shalom–Tao, Proposition 4.4 and the inverse-limit structure theorem it suffices to treat the good models $X = N/\Gamma$: the locally compact second-countable group N has a compact central abelian Lie subgroup $C \supset [N, N]$, the lattice Γ is abelian, and $\Gamma \cap C = 1$. The acting group has a homomorphism into N. The conditional-density inverse-limit argument [R164] requires no uniform quantitative lower bound. Use ergodic decomposition where necessary.

Canonical models on open orbits

1. The group $L = N/C$ is locally compact abelian, and Λ, the image of Γ, is discrete cocompact. Discreteness holds because projection by compact C is proper and Γ is discrete and closed. The classical structure theorem gives $L = \mathbb{R}^n \times L'$ where L' has a compact open subgroup K. The commutator map on L is a continuous alternating bicharacter with values in C. It vanishes on $\mathbb{R}^n \times K$ because the dual of K is discrete, applied coordinatewise to finitely many characters of C separating its points. On $K \times K$ it has finite image: its adjoint is a continuous map from compact K to a finite product of discrete duals. Shrink K to an open subgroup in the kernel of this adjoint. The preimage N_0 of $\mathbb{R}^n \times K$ now has commutator entirely inside the torus $T = C^0$, since the remaining bicharacter comes from the real factor.

2. The compact abelian preimage M of $\{0\} \times K$ in N_0 admits a decomposition $T \times M'$ as a compact abelian group with the indicated T: every basis character of T extends to M, giving a continuous retraction. This incorporates any finite component group of C into the compact horizontal coordinates. The preimage of $\mathbb{R}^n$ is a Lie group (compact Lie kernel, Lie quotient; equivalently no small subgroups). Its identity component is a central torus extension of $\mathbb{R}^n$, modeled by the lifts $H(a)$ with product $H(a)H(a') = V_{\omega(a,a')/2}H(a + a')$, for an alternating real bilinear form ω valued in the Lie algebra of T. This follows from the Lie algebra and the simply connected vector quotient; exponentials of a linear complement provide the lifts. The group M' is central in N_0. Thus $N_0/T = \mathbb{R}^n \times M'$; this uses lifts $H(a)$ only modulo T. Set $\Gamma_0 = \Gamma \cap N_0$, and let $\Lambda_0 \subset \mathbb{R}^n \times M'$ be its discrete cocompact image. Cocompactness follows since the open orbit $N_0\Gamma/\Gamma$ in N/Γ is also closed; N_0 is normal, $N_0\Gamma$ is open with finite index, and

multiplication identifies this orbit with N_0/Γ_0. Projection of Λ_0 to $\mathbb{R}^n$ is a full lattice P, since M' is compact and Λ_0 is discrete cocompact.

3. The tail generators preserve each of these finitely many open orbits: $N/N_0\Gamma$ is a finite abelian group and $p_i \to \infty$. Indeed $N_0\Gamma$ is normal since $C \subset N_0$. Conjugate the original generators by an orbit representative to identify a chosen orbit with N_0/Γ_0. The conjugated n_i still commute, satisfy $n_i^{p_i} = 1$, and can be written $m_i\gamma_i$, $m_i \in N_0$, $\gamma_i \in \Gamma$. On the orbit, at $y\Gamma_0$, this acts as

$$m_i y[\gamma_i, y]\Gamma_0.$$

The commutator $[n_i, y]$ is p_i-torsion in C and so belongs to T after excluding the finitely many prime divisors of $|C/T|$. Also $[m_i, y] \in T$, whence $[\gamma_i, y] \in T$. This is a continuous character in y, factoring through the compact abelian base $B = (\mathbb{R}^n \times M')/\Lambda_0$, because γ_i centralizes T and Γ_0. The same coordinates hold for all products without needing the entire ambient action to preserve the orbit.

4. We obtain the canonical data on this orbit with base not assumed connected:

$$(a, s, \eta, c) : y \mapsto H(a)S(s)V_{\eta(b)+c}y, a \in R^n, s \in M', b = base(y).$$

Here S is the central embedding of M', $\eta : B \to T$ is a continuous character tuple for genuine words, and $c \in T$ constant. A lattice element projecting to $(k, s') \in \Lambda_0$ acts on the orbit by a scalar constant plus the character ℓ_k given by $\omega(k, a)$, where a represents the real part of the test base point. The value descends to a character on B because Γ_0 is abelian. In particular $\omega(P, P)$ lies in the period lattice of T. Modulo constants the compact abelian horizontal group A consists of (a, s, η), with η any abstract character tuple on B, modulo the lattice shifts

$$(a, s, \eta)\ (a + k, s + s', \eta - \ell_k), (k, s') \in \Lambda_0.$$

Compactness follows from compactness of the base and of the characters in the pointwise topology; local base lifts exist as the lattice is discrete. When a horizontal triple approaches the identity we may choose $(a, s) \to (0, 0)$. Its central commutator formula with a second triple is $\omega(a, a') + \eta(b') - \eta'(b)$. The term $S(s)$ is central, so the slot-power formulas from the torus proof are unchanged.

The two changes in the torus proof Use the harmonic-ultrafilter slope envelope and its compact horizontal image J as in the torus proof [R167]. Let C_{sh} denote the closure of the scalar commutator shifts z in the torus T, write C_{sh}^0 for its identity component, and set $\Lambda_{\mathrm{sh}} = (C_{\mathrm{sh}}^0)^\perp$. For a basis of this lattice form the restricted Lie group as before, now with the horizontal period lattice P in place of $\mathbb{Z}^n$. The compact s-coordinate is simply forgotten:

$$(R^n \times ((\widehat{B_J})_{\mathrm{disc}})^r)/$$

$$\{(k, -(m\ell_l \ell_k|_{B_J})_l) : k \in P\}.$$

All characters used in compactness, random component detection, and the Lie-torus separation argument still extend continuously to A, since the forgotten compact coordinate never occurs in the lattice correction ℓ_k. This gives $I \in \mathcal{U}$ and $V \subset \mathbb{R}^n$ isotropic for all projected forms $\ell_l \omega$, with all selected axes admitting lifts (a_i, s_i), $a_i \in V$. Here $\nu : B \to \mathbb{R}^n/P$ is projection of the real part, and $\nu(B_J)$ contains the entire projection of V, exactly as in the preceding proof.

For the scalar tags at the origin, consider only the quotient torus T/C^0_{sh}. The orbit of the subgroup generated by $H(V), S(M')$, and all central torus elements over the subgroup $\nu^{-1}(T_V)$, $T_V = (V + P)/P$, is a compact abelian group in that quotient. More explicitly the subgroup over real coordinate V after quotient by C^0_{sh} is abelian by isotropy. Its quotient by the intersection of the image of Γ_0 is compact, identifies with the preimage of $\nu^{-1}(T_V)$ in the homogeneous space, and its central T/C^0_{sh} embeds since $\Gamma_0 \cap C = 1$. One may also see compactness directly: V has a full lattice of periods in P, all compact horizontal coordinates are available, and the central torus is compact. This compact abelian group extension splits over its central torus, by extending a basis of the torus characters. Consequently the reference horizontal translations act by constant vertical shifts in the splitting on this orbit. With the additional shears of the true axes the action becomes

$$(b, z) \mapsto (b + b_i, z + \theta_i + \xi_i(b)).$$

The torsion tags $\theta_i, \lambda_i = \xi_i(b_i)$, vanishing of mixed derivatives, and the parity-preserving enlarged envelope now follow word-for-word from the torus proof. They yield an odd target with shifts jt, $t \in C^0_{\mathrm{sh}}$, and approximants forcing $\ell\eta$ to escape for every $\ell \notin \Lambda_{\mathrm{sh}}$.

Finally, the original Haar argument invoked a single separating base point when B was connected. For any compact metrizable abelian B, instead choose a countable list of base points separating its countably many continuous characters from zero, and impose successively their pointwise tests in neighborhoods of the horizontal identity. Diagonal passage ensures that each $\ell\eta$ either escapes or is eventually constant. The latter constant must vanish by the pointwise tests. Fourier analysis again gives independent Haar shear on a closed subgroup containing C^0_{sh}, and thus positive recurrence of every positive Haar event at every prescribed number of slots. Normalized Haar on each open orbit is the restriction of the original Haar measure. This completes the reduction to the translational models.

R179. The paired odd-prime directions force multiple recurrence on Abramov systems

Scope and status. Result with proof.

Result. Paired odd-prime single-mark families force all positive-measure multiple returns on systems generated by global phase polynomials.

Scope. The class restriction is essential because a four-term obstruction exists in general actions.

Set G to be the direct sum of the distinct odd order-prime axes C_{p_i} and C_{q_i}, grouped in pairs, and a_i,b_i to be generators in the two summands. In each tail allow all steps v with one index i at which the move is $a_i + b_i$, and all other moves chosen independently among 0,a_j,b_j with finite support. No negative moves or second marked index are needed. In a measure-preserving G-system whose circle-valued phase polynomials generate the measurable σ-algebra, every positive-measure event A has positive h-returns along these steps, for every h. The supremum of the return measures in any tail is bounded below in terms of h and $\mu(A)$ alone. This is a theorem about this class of actions. On general actions a four-term obstruction exists; the unknown assertion only concerns events above one fixed threshold.

Splitting at coprime orders. A phase polynomial is written additively P with values in R/Z, with all $(s+1)$-fold additive differences zero for some s. For g of prime order p write $D = T_g - I$. On polynomials of degree $\le s$, $D^{s+1} = 0$ and $(I + D)^p = I$. Hence pD is an integer combination of D^2,D^3,... and $p^s D = 0$. For u,v of distinct prime orders, $D_u D_v P$ is killed by two coprime powers and is zero. All assertions hold simultaneously off one null set for our countable group. In particular for $g = \sum g_i$ in distinct axes, $P(T_g x) - P(x)$ is the sum of the individual displacements $P(T_{g_i} x) - P(x)$. The same applies at time t using tg_i. This argument includes all cross terms between the two axes of a marked pair.

Adding the mark. First replace 1_A by a bounded approximation made from a finite linear combination of circle phases. Such approximations taking values in $[0, 1]$ are available in L^1. Indeed condition on finitely many generating circle factors, approximate the bounded function on their joint compact torus by a continuous $[0, 1]$-valued function in L^1 of the image measure, and take positive Fejér means. A trigonometric monomial is again a single phase polynomial. Telescoping bounds the resulting error of the h-correlation uniformly in v by h times the L^1 error. In each term of the product expansion, the integrand has the form $C(x)$ times the product of separate unit-modulus factors for each active axis, by the splitting just proved. In particular write $z_j(x)$,$w_j(x)$ for the factors corresponding to a_j and b_j; $a_j + b_j$ contributes exactly $z_j(x)w_j(x)$. Treat the finitely many terms simultaneously.

On a background block choose moves independently, equally likely 0,a_j,b_j. Its expected multiplier on each term is the product of f_j=$1 + z_j + w_j/3$. For that term let R=$\{$x: $\sum_j$ $1 - |f_j(x)|<\infty\}$. On R both z_j and w_j tend to 1, since three unit numbers including 1 have average of modulus tending to 1 only if the other two tend to 1. Off R the product of $|f_j|$ from any starting index tends to zero as the end tends to infinity. For any precision, build m disjoint successive blocks each of whose multiplier has arbitrarily small L^1 norm off its R, simultaneously for all terms, by dominated convergence. Choose a fresh mark i after all blocks so $z_i w_i$ is arbitrarily close to 1 in L^1 on each R. For every nonempty selection of the blocks, multiplying their product by $z_i w_i$ changes its integrated term arbitrarily little. Off R use any one of the selected blocks and the bound 2; on R use the closeness of the mark. No convergence of the complex phases of the block products is required.

Choose m by density Hales–Jewett for h letters at density $\mu(A)/2$. As in the spectral argument [R181], the sum of the h-correlations over all nonempty subset sums of m arbitrary group elements is at least $\kappa(h, \mu(A)) > 0$. Sample those group elements independently as the background moves of the m blocks. For some nonempty selection its expected correlation is at least $\kappa/2^m - 1$. Choose the approximation error and each subsequent integrated error smaller than a fixed small fraction of this number. Then adjoining the fresh marked move preserves a positive expected correlation for the *original* indicator. Some actual legal step in the prescribed tail therefore returns, with a lower bound depending only on h and $\mu(A)$.

The same proof works for a circle phase polynomial test at any prescribed basepoint on which its identities hold, and extends to any finite system of test functions with the displayed multiplicative separation of the orbit profiles. It does not assert that Host–Kra factors in all actions of this group are Abramov, or that Følner characteristic factors control this sparse step family. For context see Shalom, *$\oplus F_p$-systems as Abramov systems* and Shalom, *Host–Kra theory for $\oplus F_p$-systems and multiple recurrence.*

R180. Adaptive normal choices for sums of grid profiles

Scope and status. Result with proof.

Result. Adaptive normal background controls generate Haar limits on the annihilator of absolutely summable profile characters.

Scope. A compact-fiber sum model; the marked-budget cosets reachable in arbitrary actions remain unresolved.

Let K be compact metrizable abelian and fix a trajectory length h. At stage i let $Z_i^x, Z_i^y \in K^h$ be the random horizontal and vertical profile **increments** of a local array, relative to its flat profile. The two random variables may be dependent within a stage, but the grid starts in different stages are independent. A normal controller at stage i can choose zero or either of Z_i^x, Z_i^y after inspecting that stage's start. It can also use coins shared within a finite block. Every such scheme is supported on allowed normal choices. In finding a positive return this adaptive sampling is legitimate: in a finite block, if its event has positive mass, at least one of the finitely many ordinary directions has positive mass when combined with the original unconditioned grid starts. No uniform lower bound on that last mass is asserted if the number of directions grows.

Put

$$R_1 = \left\{ \chi \in \widehat{K^h} : \sum_i \mathbb{E}\big(\|\chi(Z_i^x)\| + \|\chi(Z_i^y)\|\big) < \infty \right\}.$$

This is a subgroup by the triangle inequality, where characters take values in $\mathbb{R}/\mathbb{Z}$ and $\|\cdot\|$ is distance to zero. In arbitrarily far tails the laws of finite controlled sums approximate Haar measure on $R_1^\perp$. An extra independent uniform constant profile replaces R_1 by its intersection with the balanced characters. For each fixed frequency in R_1, the characteristic coefficient is close to one for all such sums starting sufficiently far out, because the expected norm of its total selected phase is bounded by the sum of both absolute-error tails.

For $\chi \notin R_1$, one of the two types has a divergent absolute expectation sum. Denote its principal phase by $Y_i \in [-1/2, 1/2]$. If $\sum_i \mathbb{E}Y_i^2 = \infty$, independently taking the given normal or zero with equal probability makes its block multiplier arbitrarily small: the inactive atom bounds the squared characteristic modulus by $1 - c\mathbb{E}Y_i^2$ for an absolute positive constant. Otherwise for some fixed sign the nonnegative part $W_i = (\pm Y_i)_+$ has divergent mean sum and summable second moments. Select exactly the stages where that sign occurs. Far out, choose a finite block on which $\sum \mathbb{E}W_i$ is within $o(1)$ of $1/2$, possible since the individual means tend to zero. Independence and the tail second-moment bound make $\sum W_i$ concentrate there, so the characteristic coefficient of the selected sum approaches -1. With equal probability use that full block or make the whole block inactive. The new coefficient approaches zero. For finitely many characters outside R_1, use disjoint successive blocks and multiply the independent block laws. Each targeted coefficient becomes small and remains so as later factors have modulus at most one. Exhaust the countable dual and use the character uniqueness theorem to obtain the claimed Haar limit.

In particular the annihilator surviving arbitrary normal background choices requires absolute summability, which is stronger than the square summability detected by independent unconditional normal choices. The row/column obstruction [R185] applies directly to these absolute norms. The remaining issue is how the allowed positive budget $\sum_{i \text{ special}} \eta_i < 1/8$ for diagonals constrains the cosets or neighboring laws one can reach while this larger normal

subgroup is freely available. The approximation is in the compact-fiber sum model, not in arbitrary actions.

R181. A spectral recurrence theorem for one diagonal move

Scope and status. Result with proof.

Result. One-mark paired directions recur for all pair correlations, and at arbitrary fixed length for correlations in the uniform Fourier–Stieltjes closure.

Scope. Signed paired backgrounds; the higher-order conclusion requires the Eberlein hypothesis. Pair return supremum is at least the square of the event mass.

Let G be countable abelian and let a_i,b_i be elements of G. In every tail consider the steps

$$a_i + b_i + \sum_{j \neq i} v_j, v_j \in \{0, +a_j, -a_j, +b_j, -b_j\},$$

with finite support. Assume that these are nonzero, as is automatic for the paired distinct-prime axes. For any measure-preserving G-action, let $c(d)$ be the h-term return measure of an event of measure $a > 0$. If c is a uniform limit of Fourier–Stieltjes transforms, some step in each tail has $c(d)$ bounded below by a positive constant depending only on a,h. In particular all positive-measure pair tests return, without any divergence condition on the reciprocals of the primes. This statement refers to the signed-normal version of the paired prime construction, with at least the one-mark moves allowed. It does not handle arbitrary higher correlations.

For a character χ of G write $z_i = \chi(a_i)$, $w_i = \chi(b_i)$, and take independent background moves at each site with probabilities 1/2 for zero and 1/8 for each of the other four options. Their real multipliers f_i=$2 + \operatorname{Re} z_i + \operatorname{Re} w_i/4$ lie in $[0, 1]$. Write R for the characters satisfying $\sum_i (1 - f_i) < \infty$. On R the tail products tend uniformly in their ending index to one, pointwise in the character. Also $z_i w_i$ tends to one there. Outside R, for each starting index, the products to a moving ending index tend to zero. Given finitely many finite complex measures ν, in any sufficiently late tail there is a finite block whose product of multipliers approximates 1_R in L^1 of the sum of their variation measures, to arbitrary prescribed precision. First discard the common initial segment using dominated convergence on R, then choose the block end large by dominated convergence on the complement. Repeat for finitely many disjoint successive blocks. Each nonempty subcollection of the block products approximates 1_R with error at most the sum of its block errors by telescoping. Multiplying any nonempty block product by $z_i w_i$ for a sufficiently late fresh diagonal axis gives the same approximation.

To see positivity when the approximating measures are complex, choose m by finite density Hales–Jewett for alphabet size h and density $a/2$. For *every* choice of g_1,...,g_m in G one has

$$\sum_{nonemptyFsubset[m]} c(\sum_{j \in F} g_j) \geq c_0(a, h) > 0.$$

Indeed the expected occupied fraction of the translates of the word cube $\{0, \ldots, h-1\}^m$ is a. With probability at least $a/2$ this fraction is at least $a/2$, so there is an occupied combinatorial line. Bound its probability by the sum over possible lines, each of which has return probability c at its wildcard step, and absorb the finite number of bases into the

constant. Choose a uniform Fourier–Stieltjes approximation c' with error much smaller than $c_0/2^m$. Give the g_j independent background moves in the m disjoint blocks above. Each nonempty subset correlation for c' has mean arbitrarily close to the same number $\nu(R)$, for a measure ν representing c'. The displayed inequality yields a positive lower bound on $\operatorname{Re}\nu(R)$, independent of the approximation. A background block with one added fresh mark has expectation arbitrarily close to this number for c', and hence positive, with the asserted bound, for c. At $h = 2$ one may instead use the exact nonnegative spectral measure, which gives the stronger supremum bound a^2 from its atom at the identity. Characters and spectral measures are on the full compact dual of G.

For the prime application the two axes of a pair have orders p_i,q_i. In the cyclic presentation $C_{p_iq_i}$ they may be represented by speeds p_i,q_i; the mark has speed $p_i + q_i$. In the filter for the four-term unsigned construction, allowing both signs of the normals adds the two factors $T - S^{-p_i}$, $T - S^{-q_i}$. At the marked speed the new increments $2p_i + q_i$ and $p_i + 2q_i$ are units modulo p_iq_i for distinct odd primes. Every unit-step difference is surjective on the real space of zero-mean arrays on the cycle, and on the real row/column zero subspace, where it has no kernel. Modulo one this also holds on the connected row/column zero torus by choosing real lifts with exactly zero sums. Consequently the signed-normal filter still allows the approximate half-turn at every marked cell, with six slots in place of four. The squarefree transfer still requires the high-density upper bound over all measurable actions; Eberlein correlations form only a subclass.

R182. A paired family with unrestricted horizontal moves

Scope and status. Proved subcases and unresolved extensions.

Result. Unrestricted horizontal normals and small odd diagonal marks admit a common finite coloring without monochromatic four-term progressions.

Scope. Positive lower capacity transfers from prime-pair packets; no upper capacity below one is proved.

Qualifications. The candidate's unrestricted all-system upper bound remains open.

On the squarefree direct sum of pairs $C_{p_i} \times C_{q_i}$ of distinct odd primes with $p_i < q_i$, write a single-coordinate step as (a, v), taking the centered representative $|a| \le (p_i - 1)/2$. Permit $v \in \{0, 1\}$ independently in every coordinate. When $v = 0$ there is no restriction on a. On the finitely many vertical-active coordinates require

$$\sum_{i:v_i=1} \frac{|a_i|}{p_i} < 1/4, \qquad \sum_{i:v_i=1} a_i \quad \text{odd}.$$

The step has finite support. This enlarges the unsigned paired directions from the earlier construction [R156]: all horizontally active but vertically inactive axes can now take *every* value in their prime group. At a vertical-active coordinate large integer horizontal velocities are also permitted within the displayed budget. The analogue with no horizontal movement at some vertical-active axes remains available.

There is a fixed finite coloring with no monochromatic four-term progression in these directions. Choose a function $h_i : C_{q_i} \to \mathbb{F}_{p_i}$ taking only the two values $(p_i - 1)/2, (p_i + 1)/2$ and having sum zero. The number of occurrences of the upper value can be chosen in $[0, q_i]$ to give the required congruence modulo p_i, since $q_i > p_i$. Writing Y for the vertical unit shift, choose $b_i : C_{q_i} \to \mathbb{F}_{p_i}$ with $(Y - I)^2 b_i = h_i$. Indeed $Y - I$ is injective on mean-zero arrays, as a constant with zero sum there is zero when $p_i \nmid q_i$. In finite dimension it is an isomorphism on this subspace.

The circle-valued potential $f_i(u, v) = ub_i(v)/p_i \bmod 1$ is well-defined on both cycles. Use two spatial channels, the sums of $f_i(x_i)$ and $f_i(Yx_i)$, with fixed basepoint values subtracted on the direct sum. The temporal/spatial filter $(T - I)^2(T - Y)$ uses only these channels at the four times $0, 1, 2, 3$. At a horizontal speed $(a, 0)$ it vanishes for every a, because the potential and its vertical translate are affine functions of u as circle functions. At a speed $(a, 1)$, the first operation $T - Y$ gives $ab_i(v + 1)/p_i$, independent of u; the remaining two temporal differences give $ah_i(v + 1)/p_i$. This is within $|a|/(2p_i)$ of $a/2$ on every cell. Summing over axes, the filter is within $1/8$ of a half-turn on every allowed line. Coloring the two channels by cells of width less than $1/64$ precludes a monochromatic four-term line, since each channel's coefficients have time-sum zero and the filter has total coefficient norm at most eight. The coloring is uniform in the primes and rank, so gives positive capacity in every tail, with the integer transfer from the squarefree tail formula [R241].

The unrestricted horizontal moves invite averaging over the horizontal subgroup. For a fixed finite set of vertical-active coordinates that averaging is unconstrained on every other horizontal axis; on the active axes one can also use small positive-length intervals of residues by assigning their harmonic cost in advance. This observation by itself does not prove a uniform upper bound for arbitrary events. When the original paired ratios tend to a positive irrational, all positive-measure profinite independent-grid tests return, and pure fiber events

above half return in every two-dimensional toral test, by the results for the subset family in the profinite [R186] and toral [R190] notes. The new low coloring itself uses two toral channels.

R183. Saturation onto a horizontal characteristic factor

Scope and status. Result with proof.

Result. A progression-avoiding event for the free-horizontal family can be enlarged to the support of its horizontal Host–Kra projection without creating a forbidden return.

Scope. A relative horizontal characteristic factor, not an identification of the full action with a nilsystem.

Qualifications. The arbitrary centralizer action on that factor remains a separate issue.

This is a reduction for the family with unrestricted horizontal moves [R182]. Fix the forbidden length h and work on a tail on which all horizontal primes exceed h. Let $H = \bigoplus_i C_{p_i}$ denote those horizontal axes, V the vertical axes, and U any measure-preserving $H \times V$-action. Write Z for the horizontal Host–Kra factor detecting zero $U^h(H)$ seminorm. If A avoids all specified directions at length h, then $B = \{\mathbb{E}(1_A \mid Z) > 0\}$ also avoids them at the same length, with $\mu(B) \geq \mu(A)$. Therefore in the tail capacity calculation it suffices to test systems whose horizontal action is a factor of this order. For example, if the horizontal subaction is weakly mixing, every positive-measure event returns at every length, since then its horizontal Host–Kra factors are trivial. The arbitrary nonergodic horizontal subactions are important; the two-torus low witness is horizontally a family of compact rotations with varying rotation data over the vertical coordinates.

For a legal step w with finite vertical-active index set I, every $w+u$ for $u \in H' = \bigoplus_{i \notin I} C_{p_i}$ remains legal. This is a finite-index subgroup of H. Its average of

$$\int \prod_{j=0}^{h-1} U_{ju} U_{jw} 1_A \, d\mu$$

is thus zero on every finite-subgroup Følner set.

The characteristic-factor replacement for this average uses the same Z for all choices of I. Here are the relevant facts. The horizontal group cubes define the seminorm and its factor for any countable abelian group; see Shalom, *Host–Kra theory for $\bigoplus \mathbb{F}_p$ systems*, §1. For h distinct integer slopes with differences invertible on H, an unweighted scalar correlation average vanishes if any one test function has zero $U^{h-1}(H)$ seminorm; the safe order h suffices. This follows by the usual van der Corput induction along the finite subgroup Følner sequence. Shift the slopes to put an index other than the target at zero, discard its bounded test by Cauchy–Schwarz and apply van der Corput on a fixed finite subgroup H_m. The translated products have one fewer function and the target is replaced by $\Delta_{cv} f$ at difference $v \in H_m$, where $c \neq 0$ is a slope difference. Use the inductive bound, followed by Hölder and the derivative recursion for the seminorm as $m \to \infty$; multiplication by c permutes all the H_m. The two-function case is the mean ergodic theorem. This argument works on nonergodic horizontal actions as well. The finite-index weight $1_{H'}$ on H is a finite sum of quotient characters. Each such weight is absorbed by taking the product with the finite rotation H/H', tensoring the tests at slopes zero and one with opposite characters. For every order $s \geq 2$, tensoring with a character preserves the U^s seminorm by the cube formula (every multiplicative cube derivative of that character is one). This proves the needed replacement, using order $h \geq 2$, without assuming the same first invariant factor for the subgroup and the full group. Horizontal factors are invariant under all transformations commuting with H, by the seminorm characterization. Thus the average with $g = \mathbb{E}(1_A \mid Z)$ in all slots is zero in the limit.

Suppose B had a positive return at w. Then for some $b > 0$ the set $C = \bigcap_{j=0}^{h-1} U_{-jw}\{g > b\}$ has positive measure. The g-correlation at u is at least $b^h \mu(\bigcap_{j<h} U_{-ju} C)$. The average of the latter over increasing finite coordinate subgroups of H' has positive liminf. To see this directly, finite Szemerédi applied to 1_C along scalar progressions in a single arbitrary direction u gives $\sum_{d=1}^{R} \mu(\bigcap_{j<h} U_{-jdu} C) \geq c > 0$ for fixed R, c depending only on $h, \mu(C)$, by averaging the occupancy of sufficiently long blocks. Averaging now in uniform u over finite coordinate subgroups, the term with du is at most its ordinary uniform average times the index of multiplication by d, bounded by the product of those distinct horizontal primes dividing this fixed d. Positivity contradicts the characteristic-factor replacement, proving the assertion.

This reduces the unrestricted-action question to horizontal Host–Kra systems. It does not identify their full vertical centralizer action, and in particular does not imply that the required systems are measured by any fixed finite-dimensional model.

R184. Arbitrary prime pairings with limited total marked weight have threshold one

Scope and status. Result with proof.

Result. Odd paired marks with bounded total logarithmic weight have critical density one for arbitrary prime pairings.

Scope. Weights are $(\log p)^{-1/40}$; this does not cover a bound on the reciprocal-prime cost.

Qualifications. The prime-pair counting step uses the cited uniform sieve estimate.

Let $p_i < q_i$ be any disjoint pairs of odd primes. Use the paired model with normal coordinate speeds $(0,0),(\pm1,0),(0,\pm1)$ and special speed $(1,1)$. For every fixed $K>0$, the squarefree rational saturation of the directions with odd special count and

$$\sum_{i \text{ special}} (\log p_i)^{-1/40} \le K$$

has avoidance capacity one. So does its integer realization. The result includes every bounded number of special axes, with any number of normal axes and arbitrary relative sizes of the two primes. The condition in the original candidate, on the sum of $1/p_i + 1/q_i$, allows substantially more directions; this result does not cover it.

It suffices to find, for every fixed integer root M, circle potentials on the paired grids whose worst-site speed errors as in the seam construction [R185] have summable means. Write $p<q$ for one pair and $L=(\log p)^{1/40}$. Choose $a \bmod p$ so

$$\left\| \frac{2Ma}{p} - \frac{1}{2} \right\| < \frac{1}{16(K+1)L}.$$

In particular one can restrict a/p to an interval about $1/(4M)$ of length comparable to $1/L$, with constants depending on K, M. For each sufficiently large p this contains an admissible numerator. If r is the centered residue $aq \bmod p$, the seam potential with strip width comparable to $\sqrt{|r|p}$ gives

$$\mathbb{E}e_i \ll \frac{\sqrt{|r|/p}}{q}.$$

The same computation works for all $q>p$: take the strip width between fixed constant multiples of $\sqrt{|r|p}$, sufficiently small to be at most $q/2$. The nonzero numerator and coprimality give $|r| \ge 1$. Without controlling r at all the bound is $O(1/q)$. This suffices on the pairs with $q \ge pL$, since $\sum_{p \text{ prime}} 1/(p(\log p)^{1/40}) < \infty$ by the prime number theorem (Chebyshev's upper bound and dyadic summation already suffice).

For the remaining pairs look for some integer r with $|r| \le p/L$ for which $a = q^{-1}r \bmod p$ lies in the prescribed interval of numerators. On all successful pairs the cost is bounded by $1/(q\sqrt{L})$, again summable. The exceptional primes p have summable reciprocals. Here are details of the latter assertion so that the needed sieve uniformity is explicit.

Put $\alpha = [q^{-1}]_p/p$, $N = \lfloor p/L \rfloor$, and denote the interval length by $\lambda \asymp_{M,K} 1/L$. Dirichlet approximation gives $1 \le j \le C/\lambda$ with $0 < \|j\alpha\| < \lambda/2$, for an absolute C, since $j<p$. If all $r\alpha$ for $0 \le r \le N$ miss the interval, then $\|j\alpha\| \le 2j/N$: otherwise multiples of $j\alpha$ with indices $tj \le N$ traverse the circle with gaps at most $\|j\alpha\|$, forcing a hit. Thus the nonzero centered residue $b = [jq^{-1}]_p$ satisfies $|b| \le C_{M,K}L^2$, with

$$bq - kp = j, \qquad 0 < |k| \le C_{M,K}L^3,$$

because $q < pL$. Changing all signs if necessary makes b, k positive, allowing j to have either sign but $1 \le |j| \ll_{M,K} L$.

On a dyadic interval $X \le p < 2X$ put $L_X = (\log(2X))^{1/40}$. There are $O_{M,K}(L_X^6)$ possible triples (b, k, j). For each triple admitting solutions let $g = (b, k) \mid j$, $b' = b/g, k' = k/g, j' = j/g$. Solve the coprime congruence $k'p + j' \equiv 0 \pmod{b'}$ by $0 \le s < b'$. Then $p = b'n + s$, $q = k'n + t$ with $t = (k's + j')/b'$, $1 \le n \le 3X$ for large X. The two forms have nonzero determinant $b't - k's = j'$. All coefficients have size less than $3X$.

The uniform two-linear-primes upper bound following from Green–Tao, *Restriction theory of the Selberg sieve*, Theorem 1.1 bounds their count by $O(\mathfrak{S}X/(\log X)^2)$, with an absolute implicit constant. Indeed their exponential sum is at least a constant times its zero-frequency value in real part on an arc of width a constant times $1/X$; apply their uniform L^3 estimate with $N = \lceil 3X \rceil$. If there is a local obstruction the count in our range is zero, as both primes are at least X and any obstructing prime divides $2b'k'j'$. At every prime away from $2b'k'j'$ the two excluded residue classes are distinct and the singular factor is $(1-2/s_0)/(1-1/s_0)^2 \le 1$, writing s_0 for the prime. At the primes dividing $2b'k'j'$ the factors are bounded by s_0^2. The crude estimate $\mathfrak{S} \le |2b'k'j'|^2 \ll_{M,K} L_X^{12}$ is sufficient. Consequently the reciprocal mass of all the exceptional p's in this dyadic interval is $O_{M,K}(L_X^{18}/(\log X)^2)$. Summing over $X = 2^n$ converges since $2 - 18/40 > 1$. Use any admissible numerator on those exceptional pairs and bound its cost by $O(1/q) \le O(1/p)$. This proves the asserted summability of expected worst-axis errors for the entire matching.

For completeness the rest of the proof is uniform in all allowed directions. Choose M and a quadratic hitting length H for a prescribed small arc as in the preceding note [R185]. On each finite sufficiently late block omit sites where the sum of local errors exceeds a sufficiently small constant $\tau \ll H^{-2}$ times the arc length, and omit when the shifted total potential $\sum_i f_i(x_i) + U$ lies in that arc. Markov's inequality and summability make the first loss arbitrarily small on a late tail; the second costs its arc length on average in U. On a retained allowed line the total potential is within $H^2\tau$ of a quadratic whose curvature c is $\sum_{\text{special}} 2a_i/p_i$. By the odd count and weight budget, $\|Mc\| \ge 1/2 - K/(16(K+1)) > 1/4$. Such a quadratic must visit the smaller concentric arc within H steps, a contradiction. Taking stationary orbit-law limits on increasing finite blocks and using squarefree tail transfer [R241] gives the stated result.

R185. Nearby matched primes permit near-full avoidance in the paired construction

Scope and status. Result with proof.

Result. Nearby congruent prime matchings have capacity one despite divergent reciprocal mass; irrational matchings obstruct a specified summable scalar-filter scheme.

Scope. The endpoint proof is for nearby matchings. The negative test concerns finitely many scalar filters, not all blockers.

Qualifications. R245 and R215 establish the broader matched-ratio endpoint theorem.

The paired-prime candidate in mixed-odd-marks.md [R156] depends essentially on how its primes are matched. There are pairings with divergent $\sum 1/\max(p_i, q_i)$ for which its squarefree tail threshold is nonetheless one. This includes the following congruence-matched family. In each dyadic interval $[X, 2X) = [2^n, 2^{n+1})$, partition the primes by residue modulo L_n, where L_n grows subexponentially in n and every fixed integer eventually divides L_n. For example use $\operatorname{lcm}(1, \dots, \lfloor\sqrt{n}\rfloor)$, bounded by $(\sqrt{n})^{\sqrt{n}}$. Within each class pair the sorted primes consecutively, discarding at most one. For each pair write $p < q$ and gap $g = q - p$. Discarded reciprocal mass is summable, since at most L_n primes are discarded in a dyadic interval. The reciprocal maxima of the retained pairs therefore have divergent sum by Euler's theorem. Also $\sum_{[X,2X)} g \le L_n X$, and Cauchy–Schwarz gives

$$\sum_{[X,2X)} \frac{\sqrt{g/p}}{q} \le X^{-3/2}\sqrt{2X \cdot L_n X} \ll \sqrt{L_n/X}.$$

Hence the sum of these quantities over all pairs converges. No estimate on prime gaps is used.

Fix an integer $M \ge 1$. Far enough out, $q - p$ is divisible by $4M$. Choose $0 \le a < p$ with $4Ma \equiv 1 \pmod p$, and write r for the centered residue $aq \bmod p$. Since $aq \equiv a(q-p) \equiv (q-p)/(4M) \pmod p$, here $|r| = g/(4M)$. The velocity $2Ma/p$ is within $1/(2p)$ of a half-integer with odd numerator: $4Ma = 1 + \ell p$ has odd ℓ.

Use CRT coordinates $u \in C_p, v \in C_q$ chosen so the normals are $(0,0), (\pm 1, 0), (0, \pm 1)$, and the special is $(1,1)$. Choose an integer $\ell \asymp \sqrt{rp}$, with $1 \le \ell \le q/2$ (a fixed lower multiplicative factor in the choice of ℓ suffices). There exists a real function k on $\mathbb{Z}$ satisfying $k(v+q) = k(v)+1$, integer-valued outside a strip of $O(\ell)$ residues, $\sum_{v \bmod q} |\Delta k(v)| \le 1$, and $\sum_{v \bmod q} |\Delta^2 k(v)| \ll 1/\ell$. For instance use constant slope $1/\ell$ for ℓ consecutive increments and zero on the other increments. Define a circle array using $0 \le u < p$ by

$$f(u,v) = \frac{auv}{p} - \frac{ru}{p}k(v) \pmod 1.$$

This expression is exactly q-periodic in v because $aq - r$ is a multiple of p. The chosen representative in u is essential.

For a site of this grid define $e(u,v)$ as the maximum over the six allowed directions w of the circle distance from $\Delta_w^2 f(u,v)$ to $\kappa(w)$, where $\kappa(w) = 0$ for a normal and $\kappa(1,1) = 2a/p$. Then

$$\mathbb{E}e \ll \frac{\ell+1}{pq} + \frac{|r|}{q\ell} + \frac{|r|}{pq} \ll \frac{\sqrt{g/p}}{q}.$$

For the horizontal second differences the unrestricted formula is affine in u. Its error from replacing u by its cyclic representative can occur only in a bounded number of seam columns and in the transition strip of k, of proportion $O((\ell+1)/(pq))$. For vertical second differences the unrestricted error is at most $|r||\Delta^2 k(v)|$, with the same mean bound for the negative step by translation. Away from the horizontal seam the mixed second difference, less $2a/p$, equals

$$-\frac{r}{p}\big(u\Delta^2 k(v) + 2\Delta k(v+1)\big).$$

At seam columns there is again no further error unless at least one of the relevant k-values fails to be integral, up to a bounded enlargement of its strip. These bounds use absolute errors, so every subset of axes is controlled at once. The means of all the e_i are summable.

For any requested loss $\varepsilon > 0$, use the ordinary quadratic circle equidistribution criterion: choose M divisible by all the integers bounded in that criterion for a missed arc of length $\varepsilon/4$. Then choose H so large that every quadratic $y_0 + tb + \binom{t}{2}c$ with $\|Mc\| \geq 1/4$ hits any arc of this length in $[0, H)$. This is the one-dimensional degree-two case of Green–Tao, Theorem 2.9; equivalently one can use compactness and the density of orbits of irrational-coefficient polynomials, choosing M to clear all small denominators. The quadratic coefficient in the binomial convention only changes these bounds by a factor two.

On any finite tail block take $F(x) = \sum_i f_i(x_i) + U$, with U a uniform independent shift. Omit sites for which F hits an assigned arc of length $\varepsilon/2$, or $\sum_i e_i(x_i) \geq \tau$, where $\tau > 0$ is chosen with $H^2\tau \ll \varepsilon$. The arc deletion costs on average $\varepsilon/2$, and Markov bounds the second deletion by $\tau^{-1}\sum_{i\geq J} \mathbb{E}e_i$, less than $\varepsilon/2$ for a far enough tail J, uniformly in the block end. Some choice of shift thus retains fraction at least $1 - \varepsilon$. On a retained H-line with allowed step w, the whole scalar trace is within $H^2\tau$ of a quadratic with curvature $c = \sum_i \kappa_i(w_i)$, by twice summing the errors. But Mc is within $\sum_{i \text{ special}} 1/(2p_i) < 1/8$ of a half-integer with odd numerator. The quadratic therefore hits the concentric smaller arc, contradicting retention. The argument is uniform in starts, backgrounds, and number of marked axes.

Exhaust the tail by its finite coordinate subgroups. Extend each finite-subgroup orbit law to the full group by coinduction, and take a weak stationary limit. Every fixed forbidden step is eventually inside the subgroup, so this gives the same H-avoiding law on the whole tail. Letting $\varepsilon \downarrow 0$, the squarefree tail formula [R241] shows the integer realization for these pairings has critical threshold one. The construction does not bound the threshold for pairings whose ratios approach a positive irrational or where modular mismatches keep $|r|/p$ bounded away from zero on a set of divergent reciprocal mass.

A converse for constant temporal filter schemes Fix a positive irrational $\theta \in (1, 2)$, for example $\sqrt{2}$. Divergent reciprocal-pair mass and $q_i/p_i \to \theta$ can be achieved simultaneously. Write P_j for the j-th prime. For j in the union of integer index intervals $[4^n, (1+c)4^n]$ with $0 < c < \min(\theta - 1, 4/\theta - 1)$, pair P_j with $P_{\lfloor \theta j \rfloor}$. Apart from finitely many initial indices the pairs are disjoint, and the second indices are injective since $\theta > 1$. The prime number theorem $P_j \sim j \log j$ gives both the stated ratio limit and a reciprocal contribution bounded below by a fixed multiple of $1/n$ on each interval.

Suppose now that one tries to use arbitrary arrays on such a grid, and the same integer-coefficient temporal filter on their horizontal, vertical, and diagonal profiles. The filter may use several channels $f^{(b)}$ and arbitrary finite Laurent words A_b. Write $F_w = \sum_b A_b(S_w) f^{(b)}$ for the resulting circle-valued array in direction w, and denote by X, Y the coordinate shifts. Summing F_{XY} over a complete horizontal row gives exactly the sum of F_Y on that row,

because every horizontal translation permutes the row; the analogue on a vertical column compares F_{XY} to F_X. Hence if the two normal filters have mean distance at most e from zero and the diagonal one has mean distance at most e from a scalar κ, then

$$\|p\kappa\| \le 2pe, \qquad \|q\kappa\| \le 2qe.$$

This identity holds for every fixed finite temporal filter, including every order of additive difference. It permits arbitrary spatial offsets, since those are just additional channels of the arrays.

If fixed M, C satisfy $\|M\kappa - 1/2\| \le C/p$, the maximum of $\|p\kappa\|, \|q\kappa\|$ is bounded below uniformly for all sufficiently late pairs of this irrational-ratio family. Otherwise pass to a subsequence and write $\kappa = (2k+1)/(2M) + c_i/p$ with k fixed modulo M and c_i bounded. Stabilize the residue classes of $p, q \bmod 2M$, and pass to $c_i \to c$. Then $p\kappa$ approaching an integer forces c to be rational and nonzero, since $p(2k+1)/(2M)$ has odd numerator and fixed fractional part. But $q\kappa$ approaching an integer would imply $\theta c \in \mathbb{Q}$. This contradicts irrationality. Consequently the largest of these three mean errors cannot have finite sum along the prime pairs. This excludes the summably accurate filter construction above for the irrational-ratio family. It is compatible both with the proved four-slot low-density obstruction, which spends errors of order $1/p + 1/q$ on its special axes, and with other possible methods of constructing near-full avoiders.

R186. All measurable events return in profinite independent-grid extensions

Scope and status. Result with proof.

Result. Every positive measurable event returns in independent-grid extensions with profinite abelian fiber and arbitrary local potentials.

Scope. All finite progression lengths, arbitrarily late odd marks of arbitrarily small reciprocal cost; this is a restricted action class.

Let $P_0 = \bigoplus_i (C_{p_i} \times C_{q_i})$ be the paired group, with p_i, q_i distinct odd primes escaping to infinity and $\sum_i 1/p_i = \infty$ (swap the two coordinate names if needed). On the compact product base $X = \prod_i (C_{p_i} \times C_{q_i})$ take Haar probability. Let K be any compact metrizable profinite abelian group and choose arbitrary arrays $f_i : C_{p_i} \times C_{q_i} \to K$. On $X \times K$ with product Haar use the action

$$U_g(x,k) = \left(x + g,\ k + \sum_i [f_i(x_i + g_i) - f_i(x_i)] \right).$$

Only finitely many increments are active for any $g \in P_0$, so this defines a measure-preserving action without assuming convergence of the sum of the arrays themselves.

For every positive-measure measurable $A \subset X \times K$, every h, every tail and every positive bound on the harmonic cost of marks, there is a positive h-term return with an odd number of diagonal unit moves, less than the prescribed total marked cost, and ordinary unit moves in the remaining active coordinates. In fact the proof uses only vertical positive moves as background when $\sum_i 1/p_i = \infty$, since it compares diagonal moves to vertical ones on a horizontal row. The same claim with a uniform bound on the number of marks as a function of h and a fixed finite fiber quotient holds for cylinder events. No bound on the return measure for a *fixed ordinary direction* uniform in the local arrays is asserted.

First take a finite quotient K_0 of K and finite trajectory space $W = K_0^{h-1}$. For a move $w \in \{X, Y, XY\}$ (horizontal, vertical, diagonal) form the increment profile

$$Z_i^w(x) = \big(f_i(x + tw) - f_i(x) \bmod \ker(K \to K_0) \big)_{1 \le t < h} \in W.$$

Call (z, w) abundant if $\sum_i \Pr(Z_i^w = z) = \infty$. In $W \times C_2$ take the subgroup L generated by $(z, 0)$ for abundant normal pairs and $(z, 1)$ for abundant diagonal pairs. Then $(0, 1) \in L$. Otherwise projection of L to W is injective and its parity character extends to a circle character χ on W. All outcomes with $\chi(Z_i^Y) \neq 0$ or $\chi(Z_i^{XY}) \neq 1/2$ are nonabundant, so their union probabilities have finite sum over i, since W is finite. But on **each horizontal row** the sum of $\chi(Z_i^{XY})$ equals the sum of $\chi(Z_i^Y)$: at time t, horizontal translation by t simply permutes the summation over the first coordinate, and the subtracted initial terms agree. In an odd row of length p_i, all diagonal values equal to a half-turn and all vertical values zero cannot have equal sums. At least one site on each row fails, so the combined failure probability is at least $1/p_i$, a contradiction.

Express $(0, 1)$ as a finite sum of these abundant generators. Positive repetitions suffice by finiteness of the group. For each slot in this fixed expression use a fresh finite pool of far-out axes of the required type (z, w), sufficiently far that the prescribed number of selected marks will have arbitrarily small total cost. For the matching event $H_i = \{x_i : Z_i^w(x_i) = z\}$,

the pool can have arbitrarily large $\Lambda = \sum_i \Pr(H_i)$, by abundance. Select index i with probability $\Pr(H_i)/\Lambda$, draw that start uniformly conditional on H_i, and draw all other starts independently uniformly. Choose exactly the move w on that index and no other move within this pool. Draw the pools independently. The quotient fiber increments sum to zero at every time with probability one under this law; the resulting direction has odd mark count.

At time t the marginal of the base on this pool has density $\Lambda^{-1} \sum_i 1_{H_i+tw}(x_i)$ relative to Haar product. The indicators in this sum depend on distinct independent coordinates under reference Haar. Its mean is one and the squared L^2 distance from one is at most $1/\Lambda$. Product over pools, whose finite number may depend on the quotient, and take Λ large in each pool. At every time $0 \le t < h$ the product marginal density then has L^2 norm at most two (indeed arbitrarily close to one). The full fiber starts from independent uniform Haar on K, so whatever increment is specified by the base and the choices, the full fiber at time t is uniform independent of that time's base. This bound therefore holds on $X \times K$ as well.

For the original A, approximate by a cylinder A' of symmetric-difference mass ϵ, depending on finitely many base axes and a finite fiber quotient. Such cylinders generate the σ-algebra because K is metrizable profinite. Take all the pools beyond the cylinder axes. Under the constructed law its membership remains constant throughout the trajectory and the initial membership has expectation $\mu(A')$, since the early coordinates and initial full fiber are Haar. The error probability at each time for replacing A' by A is at most $2\sqrt{\epsilon}$ by the marginal bound. Choose ϵ using only h and $\mu(A)$ so that $\mu(A') - 2h\sqrt{\epsilon} > 0$. Thus an entire A-trajectory occurs with positive mass under this auxiliary law. It selects among finitely many ordinary directions, and its conditional start law for each such direction is absolutely continuous relative to Haar. Some fixed ordinary allowed direction consequently has an actual positive-measure return.

The finite-group discreteness is used in classifying the finitely many possible failed profiles. The known toral coloring has diagonal filter values uniformly close to one half, with small nonzero errors across individual grids. The result above does not establish recurrence for arbitrary arrays in connected circle fibers or finite cocycles depending jointly on multiple grids.

R187. Single cyclic output on the paired prime grids

Scope and status. Result with proof.

Result. A fixed finite cyclic output on independent paired grids has positive returns with one diagonal mark and unit backgrounds.

Scope. The output alphabet is fixed and cyclic; vector outputs need not have the same period-two reduction.

Qualifications. The conclusion concerns a fixed cyclic output alphabet, not arbitrary finite-grid colorings.

Let the prime pairs of the paired construction [R156] satisfy

$$\sum_i \frac{1}{\{} \max(p_i, q_i)\} = \infty.$$

One can pair consecutive primes within each dyadic interval, discarding at most one in each interval. The discarded reciprocal sum is finite; each retained pair has comparable members, so Euler's divergence gives the displayed condition. Use CRT coordinates in which the two primitive motions are horizontal/vertical unit steps. Give every grid $C_{p_i} \times C_{q_i}$ an arbitrary function f_i with values in a **fixed finite cyclic group** K. A finite-block compact-fiber trajectory is formed by summing its independent grid contributions $f_i(x_i + tv_i)$ and adding an independent uniform constant in K; the x_i have independent uniform laws. At every length h, arbitrarily far along the axes, there are a finite block and a direction with exactly one diagonal mark and arbitrarily many horizontal/vertical normal choices such that all the trajectory values hit any prescribed nonempty $B \subset K$ simultaneously with positive probability. The lower bound can be chosen positive depending only on $|K|, h$, independently of the functions and the starting tail. One may include intermediate inactive axes: their contributions are constant along time and are absorbed by the independent uniform constant. This conclusion is restricted to compact-fiber sums with one cyclic output, and is not an assertion about arbitrary colorings of the finite grids.

Here is an elementary averaging proof. A frequency word $a = (a_0, \dots, a_{h-1})$ acts on trajectories by the temporal scalar filter $A(S) = \sum_{t=0}^{h-1} a_t S^t$, after identifying $K = C_m$ with $\mathbb{Z}/m$. Only balanced words $\sum a_t = 0$ survive the uniform constant. For each balanced word let β_i^x, β_i^y be the fractions of grid sites where the horizontal or vertical filter gives a nonzero value. Choose, independently in each normal coordinate, speed zero with probability $1/2$ and a unit horizontal or vertical speed each with probability $1/4$. Its characteristic multiplier $z_i(a)$ has $|1 - z_i(a)| \le 2(\beta_i^x + \beta_i^y)$ and $|z_i(a)| \le 1 - c_m(\beta_i^x + \beta_i^y)$ for some $c_m > 0$: compare its squared modulus to the expected character difference between independent draws, taking one draw inactive. The balanced words with summable $\sum_i(\beta_i^x + \beta_i^y)$ form a subgroup $R \subseteq \widehat{K^h}$ by the union bound for nonzero values.

For each fixed a there is a constant $P = P(a, m)$ with the following property for primes $p, q > P$. Except on a fraction at most $hp\beta_i^x + q\beta_i^y + \beta_i^x$ of grid sites, the diagonal filter also vanishes (with the first group of constants enlarged by a factor equal to the number of prime factors of m, if the argument is done componentwise). To see this, split into the prime power factors ℓ^r of m. On each nonzero projected filter, divide all coefficients by their common ℓ-power and work modulo the correspondingly reduced prime power, so at least one coefficient is a unit. All bi-infinite solutions of a primitive scalar recurrence $A(S)w = 0 \bmod \ell^r$ have

a common period. Modulo ℓ strip the vanishing coefficients at either end; the remaining relation has invertible extreme coefficients, yielding finitely many possible solutions with shift a permutation of them. Its common period P_1 exists, also in the monomial case. Then $(S^{P_1}-I)w=\ell w'$ with $A(S)w'=0 \bmod \ell^{r-1}$. Induction gives a period for w' dividing P_{r-1}. Use a common multiple L of P_{r-1},P_1 so $(S^L-I)w$, a sum of translates of $\ell w'$, is L-periodic. Thus $S^{mL}w=w$. This proves the claim, and any solution on a sufficiently large prime cycle has to be constant in the reduced group since its period divides both that prime and the common period.

A fraction at most $p_i\beta_i^x$ of horizontal rows contains even one horizontal defect, and a fraction at most $q_i\beta_i^y$ of columns contains a vertical defect. For the uniform base site (u,v), unless one of the rows $v+t$, $0\le t<h$, or the column at u is dirty, the preceding constancy in each reduced component gives $f_i(u+t,v+t)=f_i(u,v+t)=f_i(u,v)$ there. Balance kills the resulting constant in the original filter after restoring the coefficient power. Union bounds prove the asserted estimate (in fact one can use the defects of the original filter simultaneously to choose all the clean rows/columns). Since R is finite and its defects summable, the divergent reciprocal series gives arbitrarily large j such that $\max(p_j,q_j)\sum_{a\in R}(\beta_j^x(a)+\beta_j^y(a))$ is arbitrarily small. The diagonal multiplier on this single coordinate is consequently arbitrarily close to one simultaneously on R.

Now choose the normal block start sufficiently late that all summable defect tails on R are small. End sufficiently late that every balanced word outside R has arbitrarily small product of normal multiplier moduli, by divergence; there are only finitely many frequency words. Add one diagonal coordinate j past the block as above. Fourier inversion shows that the averaged trajectory law is arbitrarily close to Haar law on $R^{\perp}\le K^h$. This subgroup contains every constant trajectory because all its annihilators are balanced. Thus B^h has Haar mass at least $|K|^{-h}$ on it, as it contains a constant trajectory there. Taking the Fourier tolerance small enough gives probability at least $(1/2)|K|^{-h}$; some allowed direction in the finite mixture achieves at least that much. The unique special cost tends to zero with j.

One cannot infer a two-periodic representative for all vector-output odd cosets just from the paired origin. For example in $\mathbb{F}_2$ let a mixed stage target be $H(u,v)=1_{u\neq0}1_{v\neq0}$. Since the primes are odd, every row and column sums to zero. Each factor of the normal stencil at the special increment acts invertibly on the mixed subspace (diagonalize in a characteristic-two splitting field using that the group has odd order). Multiplying by $(T-I)^d$ still acts invertibly there. Thus an array can be chosen for which all normal filters vanish and the special filter with these additional differences equals H, of probability $(1-1/p)(1-1/q)$ near one. The resulting finitely many output channels permit an annihilator word divisible by an arbitrarily high power of $T-I$ with odd-coset value one. The uniform torus construction permits the same extra differences with the pointwise approximate-half template. This does not contradict the cyclic-output theorem, since the temporal word here acts on multiple coupled output coordinates.

R188. A sufficient mixing hypothesis for arbitrary-length paired returns

Scope and status. Result with proof.

Result. A spectral nonconcentration hypothesis gives arbitrary-length one-mark returns with supremum at least the corresponding power of the event mass.

Scope. The hypothesis excludes positive spectral mass on each relevant square-summability class.

Qualifications. Ordinary weak mixing alone is not asserted to imply this hypothesis.

Let G, a_i, b_i be as in the paired construction [R179], now using the signed ordinary moves $0,\pm a_i,\pm b_i$ and a single fresh mark $a_j + b_j$. The proof works verbatim for any abelian group with this move list. For a fixed positive h, suppose that for each centered f in L^2 of the action, with spectral measure σ_f on the dual group, and each $1 \le s \le h-1$,

$$(\sigma_f \times \sigma_f)\{(\chi,\psi) : \sum_i (|1-(\chi/\psi)(sa_i)|^2 + |1-(\chi/\psi)(sb_i)|^2) < \infty\} = 0. \tag{1}$$

Then for any event A the supremum of its h-return measures along the one-mark family in every tail is at least $\mu(A)^h$. No Fourier–Stieltjes approximation of the multiple correlation is assumed. This tests every centered observable, not just 1_A. Ordinary weak mixing alone does not give (1) along prescribed coordinates.

Uniformity in a translate. In fact for any finite list of correlations of bounded test functions with distinct slopes in $\{0,\dots,h-1\}$, any prescribed tail and any $\epsilon > 0$, one can find a finite distribution supported on unmarked steps in that tail whose translated average of *each* correlation approximates the product of the function integrals to within ϵ, uniformly over all common offsets o in G. Each sampled move has at most one active axis in each pair. One can add any later mark to this average, proving the assertion.

To prove this, in each chosen index sample zero with probability 1/2 and each of $\pm a_i,\pm b_i$ with probability 1/8. For two independent sums u,u' on a block, the expected squared magnitude of the autocorrelation of a centered f at $s(u-u')$ is

$$\int\int \prod_{i\in block} [(2 + \mathrm{Re}(\chi/\psi)(sa_i) + \mathrm{Re}(\chi/\psi)(sb_i))/4]^2 d\sigma_f(\chi) d\sigma_f(\psi).$$

By (1) and dominated convergence this tends to zero as the block end tends to infinity from *any* starting index. Negative s give the same bound. Finitely many such requests can use a common block.

Induct on the number of slots, allowing arbitrary distinct integer slopes with maximal difference $\le h-1$. For one slot invariance gives equality. Subtract the product of means by expanding into centered and constant factors. For each term with some centered factor, choose a different factor as baseline, translating its slope to 0 by invariance. The distinguished centered slot now has nonzero slope s. Constant slots may be kept, including the baseline. Take all tests bounded by one after scaling. First select an inner block u so that

the above mean squared autocorrelation of the distinguished slot is arbitrarily small, simultaneously for the finite list. For all values of u,u' in this finite support form test tuples with the baseline removed and other slots replaced by

$$g_l = (T^{s_l u} f_l) overline\{(T^{s_l u'} f_l)\}.$$

By induction choose a distribution of outer moves v on later disjoint pairs for which all those correlations are close to their products of means uniformly in o. In the original integral with direction $v{+}u{+}o$, use Cauchy–Schwarz on (x, v) to remove the bounded baseline factor, squaring the u-average of the product of the remaining slots. Expansion gives exactly the smaller correlations with g_l and direction $v + o$, averaged over u,u'. Their product-of-means contribution in absolute value is at most the average absolute autocorrelation of the distinguished centered function, which is small by the inner block estimate. The errors in both levels can be arbitrarily small independently of o. Taking square roots, adding the finitely many expansion terms, and using the sum of the independent inner and outer moves proves the induction. Their supports on disjoint pairs guarantee the result is an allowed normal move.

The necessity consequence is modest but explicit. If a stationary action avoids the signed paired family at h with any positive mass, some centered observable in that action fails (1) for some $s < h$. The quotient-square-summability relation in (1) is Borel and is an equivalence relation. Failure means that a single equivalence class has positive σ_f mass, by Fubini. The class and observable need not be common to different h. For root suspensions and weakly mixing rigidity constructions the distinction between true characters and these much larger equivalence classes matters.

R189. Temporal recurrences and integer row sums in the paired circle fibers

Scope and status. Result with proof.

Result. Full Laurent rank forces every positive pure-fiber toral event to return for irrational limiting prime ratio; one circle channel always has full rank.

Scope. Finite toral output, pure-fiber events, and the specified summably controlled temporal module.

Qualifications. Base-dependent events require additional arguments.

Use prime pairs with $q_i/p_i \to \theta$ positive irrational and $\sum_i 1/p_i = \infty$. On $C_{p_i} \times C_{q_i}$ take arbitrary arrays f_i valued in $\mathbb{T}^r$. Consider the compact-fiber extension in the profinite construction [R186], now with fiber $\mathbb{T}^r$. For an integer Laurent word $A = (A_1, \dots, A_r)$, write $A(S)f_i = \sum_j A_j(S)f_{ij}$, with $S = X, Y, XY$ denoting the horizontal, vertical, and diagonal shifts. Write

$$R = \{A : \sum_i \|A(X)f_i\|_{\infty,\mathbb{T}} + \|A(Y)f_i\|_{\infty,\mathbb{T}} < \infty\},$$

where the sum includes both terms, and the norm is supremum of circle distance to zero. This is a Laurent-polynomial submodule, since translating a normal word by a power of its own shift does not change either supremum.

If R has rank r over $\mathbb{Q}(T)$, then for every h the odd marked steps of arbitrarily small harmonic cost in every tail have fiber increment profiles arbitrarily close to the constant profile zero at some grid starts. In particular every positive-measure event depending only on the $\mathbb{T}^r$-fiber has a positive return for every h. When $r = 1$, the assertion about pure fiber events is unconditional. These are qualitative support statements. The grid starts giving the required profiles can have very small probabilities, so the proof does not transfer directly to arbitrary events depending also on the base coordinates.

Separating character. At this length a profile Z_i^w records $(f_i(x+tw) - f_i(x))_{1 \le t < h}$ with move $w \in \{X, Y, XY\}$, at any freely chosen start x. In the compact group $(\mathbb{T}^r)^{h-1} \times C_2$, permit sums of these profiles or zero, with normal moves tagged by zero and diagonal moves tagged by one and charged $1/p_i + 1/q_i$. Intersect the closures of the sets of sums with increasingly late supports and costs tending to zero. As in weighted finite sums [R270], the resulting compact set is a subgroup, by disjoint successive approximation and compact semigroup recurrence. If it does not contain $(0, 1)$, the parity character of its injective torus projection extends to the ambient torus. The extension is given by an integer temporal word A on the increments; equivalently each component is taken balanced by including its time-zero coefficient.

Uniform closeness of the nested compact sets to their intersection now forces $A \in R$ and

$$\|A(XY)f_i - 1/2\|_{\infty,\mathbb{T}} \le C/p_i$$

outside a set of indices of finite reciprocal mass. For detail, the norm of every centered sum in a sufficiently late tail at sufficiently small total cost is less than $1/8$. The individual centered errors tend to zero, uniformly over the available starts at each axis. For either normal type, select the maximizing absolute error at each axis. A divergent subseries of the same sign would give a finite subcollection first crossing $1/8$ with total less than $1/4$, at zero cost. For the special supremum, take the same argument only on axes where the error

exceeds C/p_i, with C so large that a subcollection of absolute errors below $1/4$ costs less than the permitted budget. Divergence would give the same contradiction; convergence gives finite reciprocal mass of this exceptional set. In each case the starts can be independently chosen across axes. In the scalar case the separating word is nonzero, so its membership in R already implies full rank.

Integer charges. Take r linearly independent words from R. The adjugate of their Laurent matrix gives one nonzero scalar Laurent polynomial D for which the normal residuals of $D(X)f_{ij}, D(Y)f_{ij}$ are summable in sup norm for all j. Multiply by $T-1$ if necessary so $D(1)=0$. Denote the sum of all these two-direction residuals and the normal residuals of the separating word A by e_i. Choose a subsequence outside the special exceptional set on which $p_i e_i \to 0$, possible by summability and divergence of the reciprocal mass. The comparable ratios give $q_i e_i \to 0$ as well.

Write $A(XY)f_i = 1/2 + v_i$ with a real array $|v_i| \le C/p_i$ on this subsequence. Since the shifts commute,

$$D(Y)A(XY)f_i = \sum_j A_j(XY)D(Y)f_{ij};$$

its circle supremum is at most a fixed constant times e_i. Since D is balanced and v_i uniformly tiny, this bounds the **real** absolute supremum of $D(Y)v_i$ by the same amount. The analogue with X also holds.

The horizontal row charge $b_i(v) = \sum_{u \in C_{p_i}} v_i(u,v)$ is bounded in absolute value independently of i. The sum of $A(XY)f_i$ on each horizontal row equals the sum of $A(Y)f_i$ there, term by term by permutation of the horizontal coordinate. As p_i is odd, $b_i(v)$ is uniformly within $p_i e_i = o(1)$ of a half-integer with odd numerator. Round it to such a bounded half-integer $m_i(v)$. We have $\|D(Y)b_i\|_\infty \ll p_i e_i \to 0$ as a real bound, so $D(Y)m_i = 0$ exactly for large i, since the values are discrete. A finite-alphabet bi-infinite sequence satisfying a fixed nonzero scalar recurrence with integer coefficients has a common period depending only on the alphabet and the recurrence: strip zero extremes, use the nonzero leading coefficient to determine the next symbol uniquely, and use the nonzero first coefficient to determine the preceding symbol. There are finitely many possible states and shift is a permutation on the possible two-sided solutions. Applied on the large prime cycle of length q_i, this forces $m_i(v) = m_i$ constant. Likewise all vertical column charges approach a constant bounded half-integer n_i.

Double summation of the real array v_i now gives $q_i(m_i + o(1)) = p_i(n_i + o(1))$. Pass to a subsequence with the bounded m_i, n_i fixed. Then $\theta m = n$ for two nonzero rational numbers, contradicting irrationality. The asserted approximate flat profiles follow. A positive-measure fiber set B has positive intersection with finitely many translates by all sufficiently small shifts, by continuity of translation in measure. The actual chosen starts on finitely many finite grids have positive Haar mass, giving the stated pure-fiber recurrence.

Thus any support obstruction for a vector of circle channels must leave at least one free Laurent rank in its normal trajectory constraints. The spatiotemporal stencil yielding the short obstruction does leave such freedom. Deciding what this forces for high-measure events in those remaining systems is open here.

R190. A high-density bound in each fixed toral model

Scope and status. Result with proof.

Result. Every fixed toral profile model has a recurrence threshold below one; rank-one residual controls give the half-density bound.

Scope. The original bound depends on the model. R253 gives a dimension/rank-only refinement within the same filling hypotheses.

Qualifications. The threshold depends on the model; no common threshold for all actions is established.

This concerns pure fiber events in a **fixed** independent-grid $\mathbb{T}^r$-extension as in the charge lemma [R189], on the same irrational-ratio paired prime set. There is a number $\delta_f < 1$, depending on its arrays $f = (f_i)$, such that every fiber event $B \subseteq \mathbb{T}^r$ of Haar mass greater than δ_f has positive returns of every length, on every tail and at arbitrarily small positive marked budget. If the even trace subgroup has annihilator rank at most one, one can take $\delta_f = 1/2$. Together with the full-rank charge lemma, this gives the half-density statement for all pure two-dimensional toral fiber events. No uniform bound below one over all choices of arrays is proved.

Work with two-sided profile increments $(f_i(x_i + tw_i) - f_i(x_i))_{t\in\mathbb{Z}}$ in $(\mathbb{T}^r)^{\mathbb{Z}}$. As in the finite-window charge argument [R189], take the compact zero-budget tail subgroup with tags C_2, and include all constant trajectories with even tag. Equivalently one permits one arbitrary common translation of the full output. Denote its even subgroup by L and its odd coset by $z + L$; odd profiles exist by single diagonal moves far out. These sets are shift-invariant, because translating time in any profile changes its starting grid point and adds a constant trajectory. The annihilator $R = L^{\perp} \subset \mathbb{Z}[T^{\pm 1}]^r$ is a Laurent submodule of finite words. Every word in R has summable normal sup residuals, by the normal finite-sums argument of the charge note, with arbitrary grid starts. The words are balanced because all constants belong to L.

Finite-memory controls. Choose integer polynomial column vectors $v_1(T), \ldots, v_d(T)$ spanning over $\mathbb{Q}(T)$ the nullspace of the rational span of R; clearing denominators permits integer coefficients. Let $K_f \leq \mathbb{T}^r$ be the connected subtorus generated by the coefficient vectors of the v_j, i.e. the image of the real coefficient span in the torus. It is closed since this span is rational. Let $\pi : \mathbb{T}^r \to \mathbb{T}^r/K_f$ be the quotient torus. One may choose z with $z_t \in K_f$ at every time. Indeed any integer row l vanishing on K_f annihilates all $v_j(T)$, so it lies in the $\mathbb{Q}(T)$-span of R. Clearing denominators gives a nonzero scalar polynomial Q with $Q(T)l \in R$. A basis of the character lattice of the quotient thus gives full-rank summably controlled normal words on the projected arrays. The charge lemma makes the projection of the odd coset contain zero at each finite window. At all windows simultaneously this follows by compactness; projection commutes with the nested intersections defining the compact subgroup. Hence the required representative exists. If $K_f = 0$ the entire odd coset already contains a constant.

For each j take an independent stream of independent Haar circle variables $U_{j,t}$, and form the vector sequence $Y = \sum_j v_j(T)U_j$. All R-words vanish on Y, so $Y \in L$. At each individual time Y_t is Haar on K_f, by surjectivity of the coefficient homomorphism onto that subtorus. If S is the finite union of all temporal coefficient supports of the v_j, the time-t inputs and all inputs of events with times outside $t + S - S$ are independent. Thus an event depending only on Y_t has a dependency graph of degree at most $\Delta = |S - S| - 1$, independent

of the requested progression length.

Replace B by its ambient Haar density points, which have the same measure. If $\mu(B) > 1-\varepsilon$, choose a coset $b+K_f$ on which its relative density is greater than $1-\varepsilon$. Constant translations are allowed, so $b+z+Y$ lies in the odd coset. At every time this trajectory is Haar on $b+K_f$. The unfavorable events of missing B each have probability less than ε and dependency degree at most Δ. For $e(\Delta+1)\varepsilon < 1$ the symmetric Lovász local lemma gives positive probability to avoiding them all in any finite window. Explicitly for $\Delta \geq 1$ put $x = 1/(\Delta+1)$, so $\varepsilon < x(1-x)^{\Delta}$; the usual conditional-probability induction bounds the probability of each unfavorable event given any already avoided subcollection by x. Their joint avoidance has probability at least $(1-x)^h$. The independent case is immediate. This proves the claimed fixed-model threshold.

From a profile to an actual return. Any tuple of finitely many ambient density points of B has a positive-measure set of simultaneous common translations which belong to B at every site: in a sufficiently small ball around zero each of the finitely many translated complements has relative mass below $1/h$. The measure of that intersection is continuous in the tuple of shifts by continuity of translation in L^1. Approximating the selected odd-coset profile by an actual finite sum and a common translate gives a positive return. Each of the selected grid starts on finitely many axes has positive Haar probability. This step uses pure fiber events, without an assumption of continuity on their boundaries.

One constrained row. If R has rank one, choose a primitive vector $P = (P_1, \ldots, P_r)$ spanning its rational line, with the polynomial gcd of its entries a unit in $\mathbb{Q}[T^{\pm 1}]$ and their total integer content one. Every element of R is an integer Laurent multiple of P. Indeed a rational multiplier has no nonconstant denominator by the polynomial gcd. A prime in the remaining integer denominator cannot cancel in every entry: at least one P_j is nonzero modulo that prime, and multiplication by a primitive numerator is nonzero over the Laurent polynomial ring over the prime field. Consequently all trajectories w satisfying $P(T)w = P(T)z$ belong to $z + L$. If P has constant direction, after removing its polynomial gcd it is a single primitive constant row. The scalar projection to that row has an odd profile arbitrarily close to constant by the scalar charge lemma [R189]. Compactness gives an odd-coset representative with constant projection. Subtract the appropriate constant and the whole remaining sequence lies in the kernel of the row, hence in L. In this case the odd coset contains a constant. Rank zero also gives all sequences.

In the other case make a unimodular constant change of output coordinates so the highest coefficient vector of P is $(a, 0, \ldots, 0)$, with $a \neq 0$. Write a symbol as $(x, y) \in \mathbb{T} \times \mathbb{T}^{r-1}$. The highest degree e where any y-coefficient is nonzero exists and is smaller than the top degree d. Write $s = d - e > 0$ and denote the nonzero vector of the y-coefficients there by b. Use the transformed set of density points, still denoted B. If $\mu B > 1/2$, there is u with $E = \{x : \mu(B_x) > u\}$ of measure greater than $1-u$. Otherwise integration of $\mu(E) \leq 1-u$ over $u \in [0,1]$ would give $\mu B \leq 1/2$.

Initialize the finitely many earlier boundary variables, with each initial x_t for a needed site in E. At time t the equation at time $t-e$ has the form $ax_{t+s} = c_t - b \cdot y_t$, with c_t known after initializing or previously selecting all x's through $t+s-1$ and y's through $t-1$. To maintain $x_{t+s} \in E$ choose $y_t \in B_{x_t}$ whose right side lies in aE. The latter condition has Haar mass $\mu(aE) \geq \mu(E) > 1-u$, because $y \mapsto b \cdot y$ is surjective and preserves Haar. It intersects B_{x_t}, of mass greater than u. If desired replace E by a compact subset still of measure $> 1-u$ so all images are measurably unambiguous. This recursion supplies every required

pair in B over the desired window, for the arbitrary forcing $P(T)z$. Extend forward solving with the nonzero top coefficient and backward solving using a nonzero bottom coefficient. Past initialization is needed only over the finite range in the equations for the window; each further backward equation can be solved in a new earlier variable by surjectivity of nonzero multiplication on the circle. This gives the full admissible profile and proves the half-density bound.

When $r = 2$ the only remaining rank is two, already handled by the full-rank charge lemma. For higher dimensions some fixed-model bounds from the finite-memory argument may approach one as the supports or the number of independent equations grow. The above results do not bound that complexity for the entire class of systems in the original density conjecture.

Several constraints with the same delay. The half-density estimate extends with no dependence on the number of equations whenever every row of R is an integral Laurent combination of a matrix of relations admitting the following form after a fixed unimodular output change, row operations and time shifts. Split the current alphabet into (x, y) and write the forcing equations as

$$Ax_{t+m} = B_0 y_t + c_t,$$

where $m > 0$ is common to all rows, c_t is determined by the fixed forcing and the variables x_j with $j < t + m$ and y_j with $j < t$, over a finite temporal span, and both integer matrices A, B_0 define surjections from their full respective tori to the common constraint torus. Apply the slice proof with vector-valued x and y. For a set of mass $\geq 1/2 + b$ some cutoff gives $\mu E + u \geq 1 + b/2$. Given past, draw Haar y_t, then draw x_{t+m} uniformly on the fiber over $B_0 y_t + c_t$. Both marginals are Haar; x_{t+m} is uniform independently of the prior state because B_0 is onto. Thus each step with $x_t \in E$ succeeds while supplying $x_{t+m} \in E$ with conditional probability at least $b/2$. The first m x's have probability $\mu(E)^m$ to be in E. This gives an h-window lower bound independent of the torus dimensions and the arbitrary forcing.

Backward solvability need not be separately assumed for this assertion. First restrict to a compact subset of the ambient density points of the test set, still with mass above one half with a margin. Start the unconstrained Haar recursion at time $-N$, with arbitrary uniform finite boundary registers, and run to positive times. By independence of each fresh x from the prior state, the first m x's to be tested at time zero are independent Haar. The same lower bound on the compact central h-event holds for every large N. A weak limit satisfies the full two-sided forcing equations at every fixed time, and the closed central event retains at least that probability. A common Haar translation may be added by using the same argument for every translate of the compact test. This yields the measurable-event transfer in the horizontal polynomial model [R140] as well. The same-delay and two-surjectivity assumptions are essential to this particular recursion; general Laurent modules need not display them.

R192. A partial bilinear prime tag with no bounded-degree polynomial extension

Scope and status. Proved subcases and unresolved extensions.

Result. A partial bilinear prime tag has no bounded-degree polynomial extension, admits exact noncommutative identities, and evades several scalar realization strategies.

Scope. The algebraic nonextension and word identities are proved; the desired uniform positive short avoiding event is not.

Qualifications. A bounded-order scalar identity does not itself provide a uniform event mass. No intermediate integer threshold is obtained.

This model gives an exact nonabelian three-slot identity and lies outside bounded-degree polynomial sieves. The known word identity does not furnish a uniform positive-density small mask, nor is there a universal high-density theorem for it. Thus it is not a solution to the density spectrum problem.

Fix an odd prime p, $U = F_p^r$ with $r \geq 3$, and $V = F_p^U$ the additive group of all scalar functions on U. Call (u, v) admissible when for all x the difference $v(x + u) - v(x)$ has a constant value $\tau(u, v) \in F_p$. This domain includes all $(u, 0)$ and $(0, v)$. The tag scales by $a \cdot b$ under independent nonzero scalings of u and v. In particular it is linear separately on the subspaces of inputs permitted for one fixed partner. It has no global bilinear extension on $U \times V$.

Dual line-sum obstruction A putative bilinear extension can be written $L(u, v) = \sum_x \sum_i u_i w_i(x) v(x)$ for coefficient functions w_i. For every u and every v constant along u-lines the tag is zero, so the sum of $\sum_i u_i w_i(x)$ on each such line is zero. Taking the coordinate functions $v(x) = x_j$ also gives $\sum_x w_i(x) x_j = \delta_i j$.

The equations are equivariant for coordinatewise dilations of the physical U by independently chosen elements of $\mathbb{F}_p^\times$. Average a solution over this group, whose order is prime to p. The resulting forms satisfy $w_i(Dx) = D_{ii}^{-1} w_i(x)$. They vanish if $x_i = 0$, and otherwise take the form $b_{i,Z}/x_i$, where Z is the set of zero coordinates of x and excludes i. The diagonal moment becomes

$$\sum_{Z \subset [r] \setminus \{i\}} (-1)^{r-|Z|} b_{i,Z} = 1.$$

For any disjoint nonempty A,B disjoint from a set Z, use a line with moving coordinates $A \cup B$, fixed zeros Z and other fixed coordinates nonzero. Arrange one common zero time a for A, a different common zero time b for B. At a generic time t the sum of $u_i w_i$ over moving indices is $\sum_{i \in A} b_{i,Z}/(t-a) + \sum_{i \in B} b_{i,Z}/(t-b)$. Each individual reciprocal sums to zero over the nonzero field residues. The deviations from this generic formula at the two zero times yield

$$\sum_{i \in A} [b_{i,Z \cup B} - b_{i,Z}] = \sum_{j \in B} [b_{j,Z \cup A} - b_{j,Z}].$$

Taking A,B singletons gives the discrete curl-free identities. Taking $A = \{i\}$,$B = \{j, k\}$ for distinct i,j,k, and using the two singleton identities on the right, gives $b_{i,Z\cup\{j,k\}} - b_{i,Z\cup\{j\}} - b_{i,Z\cup\{k\}} + b_{i,Z} = 0$. Hence each b_i is affine as a function of the membership indicators of the other indices (induction or Boolean Möbius inversion). Its alternating sum over $r - 1 \geq 2$ indices is zero, in contradiction with the moment above.

Any ordinary polynomial of fixed total degree D agreeing on the domain, for $p > D + 2$, could be projected by independent $\mathbb{F}_p^\times$ scaling averages onto the character of weight $(1, 1)$. Exactly terms of degree one in u and one in v survive, giving the impossible bilinear extension. The same works on all p-adic integer lifts, with τ read as τ/p in the circle: average over the independent $(p - 1)$st roots of unity in Z_p, using coefficients inverse to their residues (or taking the averaged polynomial with Teichmüller inverse factors, which act by their reductions on p-torsion phases). The average, divided by $(p - 1)^2$, preserves the target and leaves only a bilinear polynomial. Since the entire two axes are admissible with zero tag, testing any lift (pe_i, e_j) makes p times every coefficient integral in Z_p. Its restriction is therefore an honest F_p-bilinear form, a contradiction. Thus covers do not fix this failure of bounded-degree scalar extension.

A genuine three-slot group word, and the remaining gap On the finite monomial group of unitaries $A = X^x M_w$ on $\mathbb{C}^U$ (X is spatial translation, M_w multiplies coordinates by $e(w(y)/p)$), let (u, v) act by $A \mapsto X^u A M_v$. These are commuting measure-preserving actions for the uniform finite-group law. At a permissible pair, $X^u M_v X^{-u} M_v^{-1}$ is the scalar $e(\pm\tau/p)I$, the sign depending only on the shift convention. The same holds with M_v conjugated by the monomial state A, since its function is then a translate of v. For $A_t = X^{tu} A M_{tv}$, the successive quotient $A_{t+1} A_t^{-1}$ is $X^u (A M_v A^{-1})$ times that scalar to power t. It follows that

$$A_2 A_1^{-1} A_0 A_1^{-1} = e(\pm\tau/p)I$$

at every starting state. This gives short obstructions in sufficiently small balls in a bi-invariant metric, but the Haar masses of the balls so obtained need not stay positive as p grows. Evaluation of the simpler scalar $w(x)$ does not give the group-word certificate: its along-line second difference contains an uncontrolled second difference of the arbitrary starting table w.

In fact a fixed-h scalar F_p strategy $\sum_j f_j(x + ju, w + jv) = \tau(u, v)$ cannot furnish a certificate for this model at all sufficiently large primes. Along the pure horizontal and vertical factors all directions and their multiples are admissible normal moves. Along each such ray the elementary cyclic Lagrange/difference argument forces every varying-slot observation to have bounded degree in the single time variable on that factor; include the zero velocity equation to fold in the initial slot. The ordinary all-line criterion in prime characteristic then makes the observations polynomials of bounded degree separately on the two factors, and hence of bounded combined degree, using interpolation first in one and then the other. Evaluating at the origin gives the prohibited bounded-degree extension. This last observation is about additive finite-field strategies. It does not classify arbitrary stationary masks or show that nonabelian word obstructions admit such masks with uniform positive mass.

There is also a word with each of four players used just once: $A_2^{-1} A_3 A_1^{-1} A_0 = e(-2\tau/p)I$ for the convention $X^x M_w e_y = e(w(y)/p) e_{y+x}$. Evaluating right-to-left, the initial lamp

values cancel at y and $y-u$, leaving $2(v(y-u)-v(y))$. This is a word in one shared noncommutative group. It is **not** a commuting-player group embedding and does not disprove existence of an ordered contradiction in the query-group sense used in the tensor theorem [R218]. In fact the scalar argument extends to circle-valued observations: subtract a shifted pure-axis query with base shifted by minus one slot index times the velocity change to eliminate that slot. Successive eliminations give vanishing $(h-1)$-fold differences of each surviving observation on each pure factor, since all nonzero differences of slot indices are invertible $\bmod p$ on that factor. Differences along any generator then take values in p-torsion for $p > h$ (the binomial-nilpotence argument for phase polynomials). After removing constants, finite-field all-line interpolation applies as before. Thus for p sufficiently large the absence of a scalar circle embedding implies the ordered contradiction via the separately proved low-class trade lifting theorem; sufficiently large repeated tensor powers have positive h-returns. No uniformity over p or over numbers of tensor copies is claimed.

The central word alone cannot yield a positive uniform obstruction For any subset B of the finite monomial group with density a, every central scalar z *additively* $\in F_p$ occurs as such a three-slot twisted word within B when p is sufficiently large depending on a, with $r = 3$. A random sample of its solutions takes the form

$$(x, w), (x+u, w+v), (x+2u, w+(I+S^u)v+z),$$

with x,u,w,v independent uniform and S^u the translation of lamp coordinates (fix a sign convention). This larger word relation does not impose that the three lamp components be an ordinary abelian AP.

Fourier expansion of a bounded slot function on V at each fixed horizontal coordinate gives triples of frequencies with $k_2 = S^u k_0$, $k_1 = -k_0 - k_2$. The maps between any two of the frequencies are bijective at fixed u, since $I + S^u$ has nonzero augmentation 2 in the local group ring $F_p[U]$ and so is a unit. The difference in the count between z and zero only uses $\sum k_2 \neq 0$ (the sum of the entries of k_2 over U), precisely the unit elements of this ring. Cut off each slot's Fourier list at modulus η; Parseval and Cauchy–Schwarz against the other two bijective slots give total error $O(\eta)$, and at most η^{-2} frequencies per remaining list. For any fixed choices of list entries, put $h_l(b) = S^{-b/2}k_l(b)$; the endpoint constraint is $h_0(x) = h_2(x+2u) = h$. The two horizontal endpoints are independent Haar. Common labels of frequency at most θ in the first marginal contribute at most θ in total, upon summing their independent match probabilities. There are at most $1/\theta$ other labels. For a given middle point $y = x + u$ and such h, the equation $h_1(y) = -(S^{u/2} + S^{-u/2})h$ has at most two solutions u, since h is a unit and distinct unordered opposite pairs of group-basis elements give distinct ring elements (p is odd). Taking $\theta = |U|^{-1/2}$ bounds the central count discrepancy uniformly in the set and z by $O(\eta + \eta^{-6}|U|^{-1/2})$, which is $o(1)$ after optimizing η.

At $z = 0$ the count is the ordinary three-term geometric progression count on this finite p-group. It is bounded below by $\kappa(a) > 0$ uniformly for large p: apply the finite one-dimensional density theorem on a sufficiently long segment of each right-multiplication orbit, average the uniform starting point and multiplier, and note that every bounded positive power map on a p-group is bijective when p exceeds the exponent of that power map (the power index); its inverse is another integer power. The count lower bound and $o(1)$

discrepancy prove the assertion for all central z. The proposed short obstruction on the actual abelian direction family, if one exists, must use a condition beyond the central word test by itself.

R193. Dense sets of horizontal marks on the asymmetric template

Scope and status. Proved subcases and unresolved extensions.

Result. Bounded detectors permit dense sets of single-mark residues; simultaneous many-mark decoding imposes quantitative square-sum and first-power constraints.

Scope. Exact bounded filter constructions and necessary conditions for the stated decoder scheme.

Qualifications. No unrestricted upper threshold is proved.

This concerns the **single-mark** version of the asymmetric prime construction [R014]. Let p_i, q_i be arbitrary pairwise distinct odd primes tending to infinity. On the support F of the signed vertical unit move, prescribe zero horizontal residue on F except at a chosen mark i, where $r \in R_i \subset F_{p_i}$ is permitted. Horizontal coordinates off F remain unrestricted. The fixed six-slot filter and isotrivial horizontal model from the joining audit [R038] can be used; the five-slot version of the original construction works as well.

For the six-slot filter its output when there is a single mark is $rH_i(y_i + t)/p_i$ at a positive vertical mark, and a reflected/shifted version at a negative mark. Every prescribed zero-sum $H_i : C_{q_i} \to F_{p_i}$ is available, since $S(S - I)^3(S - S^{-1})$ is onto the zero-sum subspace. The same is true for the earlier filter with exponent two. To rule out long monochromatic blocks, it suffices that for every vertical starting site *at least one* of a bounded number of consecutive filter windows has output a fixed distance from zero; one does not need the separation at every site. Consecutive windows advance by ± 1 in the q_i-cycle.

Single coefficient. Choose any $a_i \in F_{p_i}$ and set $H_i = a_i$ except at one site, whose value is adjusted to make the sum zero. At least one of two successive windows observes a_i. Therefore all residues

$$R_i = \{r : ||ra_i/p_i|| \geq \gamma\}$$

can be marked, for a fixed $0 < \gamma < 1/2$. This works even when $q_i \ll p_i$. For $a_i = 1$ one gets the full macroscopic complement of a neighborhood of zero, without any lower bound on q_i/p_i. For $a_i = (p_i - 1)/2$, every odd standard representative $r \in [-(p_i - 1)/2, (p_i - 1)/2]$ has distance at least $1/4$ after multiplication by a_i/p_i. For the half-turn test used in the original proof, being a fixed distance from zero suffices since a monochromatic filter result lies as close to zero as desired.

Several coefficients. Fix L. At each site i choose $a_{i,1}, \ldots, a_{i,L} \in F_{p_i}$. Fill the q_i-cycle by copies of this block, with a final incomplete block and one corrected site at the seam. If $q_i \geq 4L + 2$, every cyclic interval of $3L + 2$ sites contains an unaltered complete copy of the block (a window cutting the seam loses at most one incomplete block and one regular block). Thus a fixed length, enlarged by at most $3L + 2$, works for all

$$R_i = \{r : \max_j ||ra_{i,j}/p_i|| \geq \gamma\}.$$

On a monochromatic run all filtered outputs on its consecutive windows must lie within γ of zero by choosing a sufficiently fine fixed coloring. Some window contradicts this. The

argument applies to both signs of the vertical move. It uses only one special mark, so no simultaneous good-window assertion across several marks is needed. For a random independent list of coefficients the expected excluded fraction is $1/p_i + O((2\gamma + O(1/p_i))^L)$ (the zero residue is always excluded). Suitable deterministic lists therefore give allowed mark density arbitrarily close to one at a fixed, larger forbidden length. The length and palette are independent of i. This gives no guarantee when the excluded fraction actually tends to zero at a fixed short-obstruction length: every finite-rank common character window of fixed tolerance contains a fixed positive fraction by finite Bohr covering.

These variants keep a fixed positive-density short obstruction even for very different growth rates of p_i and q_i. The uniform upper bound for all lengths remains unknown.

A decoder for unboundedly many horizontal marks Suppose the regular value at vertical axis i is a_i, with its single corrected value at 0 differing by $-q_i a_i$, and choose positive vertical unit speeds for the following formula. There may be arbitrarily many active marked axes. Write $z_i = r_i a_i / p_i \in T$, $w_i = q_i z_i$, and $\theta = \Sigma z_i$. Multiple horizontal prime axes above the same q_i can be included: when their patches are aligned, take z_i to be the sum of all their individual normalized horizontal charges. The correction is again $-q_i z_i$. Along L successive filter windows, $L < \min q_i$, the values are

$$\theta - \sum_{i \in M_t} w_i, t = 1, \ldots, L,$$

where the M_t are disjoint: each individual vertical unit path meets its patch at most once. Every such ordered collection of disjoint bins is realized by some choices of starting vertical positions, including all time shifts implicit in the filter. To leave a mark unassigned, choose its y_i not to encounter its seam during those L windows. Distinct vertical axes can be chosen independently. The same description holds after the appropriate fixed spatial shifts in the negative-speed version when the sixth-order filter is used.

Consequently the exact direction test for this particular normalized output is that **no** disjoint L-tuple of bins makes every bin sum within γ of θ (choose strict margins consistently). A sufficient condition, allowing any support size, is

$$\|\theta\| > \gamma \quad \text{and} \quad \sum_i \|w_i\| < L(\|\theta\| - \gamma).$$

At least one bin then has total distance smaller than $\|\theta\| - \gamma$, so its output is separated. This includes legal steps with arbitrarily many z_i near arbitrary q_i-th roots, not just zero or one half, when their summed errors satisfy the bound and θ is separated. The original fixed-color proof applies to all directions meeting the decoder, with a fixed run length enlarged by $L - 1$. One can also work in a fixed finite-dimensional torus of charges; the same bin argument bounds fluctuations in its maximum metric.

An inverse restriction on this decoder. If the bin sums in T^L miss the open box of radius γ about $(\theta, \ldots, \theta)$, then there exist constants K, C depending only on L, γ and an integer $1 \le k \le K$ such that

$$\sum_i \|kw_i\|^2 \le C.$$

Indeed assign each mark independently and uniformly to one of the $L+1$ bins (bin 0 means unassigned). The resulting vector-sum distribution has Fourier transform at $m \in Z^L$ equal to the product over i of $(1+\sum_{t=1}^{L} e(m_t w_i))/(L+1)$. Take a fixed nonnegative continuous bump of positive integral supported inside the box of radius γ around zero and translate it to the target. Its expectation is zero. Approximate uniformly, with error less than one quarter of that integral, by a finite trigonometric polynomial. The constant Fourier coefficient dominates the approximation error, forcing a nonzero bounded frequency m with product modulus at least a fixed $c = c(L,\gamma) > 0$. The identity $1 - |average(z_0,\dots,z_L)|^2 = (L+1)^{-2}\sum_{j<l}|z_j - z_l|^2$ on unit vectors bounds each factor by $\exp(-c_L\|m_t w_i\|^2)$ for each coordinate t (use $z_0 = 1$). Taking any nonzero m_t and logarithms proves the bound with $k = |m_t|$. Thus the allowed vectors cannot include unrestricted distributions of correction coefficients merely by permitting the decoder to skip a bounded number of windows. This necessary quadratic bound does not assert the converse.

In fact with the same k one has the stronger first-power bound $\sum_i \|kw_i\| \le C'(L,\gamma)$. Choose $0 < \eta < \min(\gamma/2, 1/4)$. At most C/η^2 terms have $\|kw_i\| > \eta$; these contribute at most $C/(2\eta^2)$ to the first-power sum. Write each remaining term as $w_i = a_i/k + \delta_i \bmod 1$, with $a_i \in \{0,\dots,k-1\}$ and $|\delta_i| \le \eta/k$. Group the indices into at most $2k$ classes according to a_i and the sign of δ_i. If the total first-power sum is unbounded compared to k, L, η, one class has $\Sigma|\delta_i| > L(1+\eta)+\eta$. Separate it into tuples of exactly k indices, dropping fewer than k terms and hence less than η from that mass. Every tuple sum, modulo one, is an increment with the same sign and magnitude at most η, since all its grid residues sum to an integer. A list of such increments of cumulative magnitude greater than $L(1+\eta)$ supplies L disjoint successive groups approximating the chosen target θ within η: in the selected sign direction use the representative in $[0,1)$, accumulate to it, and restart. Each group consumes at most $1+\eta$ total magnitude. Their original indices form the forbidden disjoint bins. This contradiction gives the asserted bound, with k uniformly bounded as above. The quantitative containment in an ℓ^1 grid tube is necessary; the target separation remains a separate condition.

R196. Periodic representatives in tagged trajectory groups

Scope and status. Proved subcases and unresolved extensions.

Result. Finite-generation of the fiber dual gives periodic representatives of shift-fixed torsion cosets and a model-dependent all-event recurrence bound; a solenoid gives a counterexample without that hypothesis.

Scope. Compact abelian Lie fibers, or the explicitly stated finite-generation condition after localization at the tag prime.

Qualifications. The all-event assertion assumes a compact abelian Lie fiber, or the stated finite-generation condition on its localized dual. Finite topological dimension alone is insufficient, as the solenoid counterexample shows.

Algebraic lemma. Let K be a compact abelian group with finitely generated character group (in particular a compact abelian Lie group). Let $L \leq K^Z$ be a closed subgroup invariant under the bilateral shift and its inverse. Every shift-fixed coset $\xi \in K^Z/L$ of order dividing two contains a trajectory of period dividing 2^k for some finite k depending only on L. All such cosets of this L can use the same period. In particular the bound is independent of any length subsequently tested. It need not be uniform in L.

Put $B = Z[T, T^{-1}]$, $M = (K^Z)^\wedge$, and $R = L^\perp$. Then M is a finitely generated module over the Noetherian ring B: it is the direct sum of copies of $\hat{K}$ at the integer times and finitely many generators at time zero suffice. Take the shift convention such that T is its dual. The coset defines a character $\gamma : R \to T$ that vanishes on IR, for $I = (2, T-1)$, by its order and shift invariance. The Artin–Rees lemma gives a fixed c such that $I^n M \cap R \subset I^{n-c}R$ for $n \geq c$. For $e_k = T^{2^k} - 1$ one has $e_k \in I^{k+1}$, inductively from $e_0 = T - 1$ and $e_{k+1} = 2e_k + e_k^2$. Therefore $R \cap e_k M \subset IR$ for $k \geq c$. The character γ descends to the image of R in $M/e_k M$ and extends to the entire discrete quotient because the circle is divisible. This extension is a point of K^Z annihilating $e_k M$, i.e. a 2^k-periodic trajectory, whose coset modulo L is exactly ξ.

The same proof works for a coset of prime order p, with $I = (p, T-1)$ and period p^k: the binomial theorem gives $T^{p^k} - 1 \in I^{k+1}$ inductively. This is a lifting of fixed torsion cosets to periodic points, not an assertion that the whole system has a common period.

More generally it suffices that the 2-localization $\hat{K} \otimes \mathbb{Z}_{(2)}$ be a finitely generated module over $\mathbb{Z}_{(2)}$, the integers localized at all odd denominators. Localize M, R and the Laurent ring at the odd integers. The order-two character γ extends to the localized submodule (multiplication by any odd integer is the identity on its range). The same Artin–Rees argument there kills $R \cap e_k M$ after localization, hence kills its image under the original γ. This permits arbitrary odd torsion or odd denominators in the dual, provided its remaining 2-local part is finitely generated. The analogous p-local statement holds for a prime-order tag. In particular if the dual has only odd-order torsion all such order-two characters already vanish.

Finite generation in this lemma cannot simply be replaced by finite topological dimension. Let K be the one-dimensional solenoid with dual $Z[1/2]$. In $M = Z[1/2][T, T^{-1}]$ take the $Z[T, T^{-1}]$-submodule generated by $g_k = (T^{2^k} - 1)/2^k$, $k \geq 0$. Every element P of it vanishes at 1 and has integral derivative $P'(1)$, because each generator has derivative one there. Then $\gamma(P) = P'(1)/2 \bmod 1$ is shift-fixed and has order two. For every positive integer $n = 2^k u$ with u odd, $(T^n - 1)/2^k$ belongs to this submodule and to $(T^n - 1)M$, and has $\gamma = u/2 \neq 0$.

The resulting odd coset therefore has no periodic representative of any finite period. This is an abstract profile-group example; no realization on the proposed paired coprime grids is asserted.

Period two test for a torus. Suppose the even group contains all constant trajectories, $K = T^r$. Its annihilator rows A are balanced, so $A = (T-1)C$. A sequence alternating between 0 and $e \in T^r$ evaluates this row at any shift to $\pm C(-1)\cdot e$. An order-two shift-fixed coset contains such a trajectory iff $\gamma(A) = 0$ on the kernel $C(-1) = 0$. Indeed this condition defines a character on the lattice of the $C(-1)$, which extends to Z^r to give the required e. In particular period two always works when R is saturated at $T+1$: if $C(-1) = 0$ then $A = (T+1)A'$ with $A' \in R$, and $\gamma(A) = 2\gamma(A') = 0$. Equivalently $T + I : L \to L$ is onto. This includes a group consisting exactly of constants plus the image of any integral finite Laurent matrix on arbitrary full torus streams: $T + I$ is onto on bilateral full streams by recursion, and doubling is onto on the constant torus. This sufficient condition does not require any bound on the number of streams or their delays.

An all-event bound in independent compact-fiber models Fiber hypothesis. In this subsection, assume that K is compact metrizable abelian and that $\widehat{K} \otimes \mathbb{Z}_{(2)}$ is finitely generated over $\mathbb{Z}_{(2)}$, the localization of the integers at the odd denominators. In particular, a compact abelian Lie fiber satisfies this hypothesis. This is the hypothesis used when applying the preceding periodic-coset lemma to obtain one fixed period. The nonabelian compact Lie extension at the end of the subsection is treated by its separate reduction.

Let $G = \oplus_i G_i$ with finite abelian coordinates. On $X = \prod G_i \times K$ take product Haar and the action

$$T_g(x,z) = (x+g, z + \sum_i [f_i(x_i+g_i) - f_i(x_i)]),$$

for arbitrary arrays $f_i : G_i \to K$. Choose one nonzero marked move $m_i \in G_i$ per axis and positive costs $w_i \to 0$. Let the allowed directions include all sums of an **odd** number of distinct chosen marks of sufficiently small total cost ($\le \kappa$ for some fixed $\kappa > 0$); additional allowed normal or marked moves are harmless. There is a number $a_f < 1$ for this fixed model such that every measurable event of mass $> a_f$ returns at every length, on any prescribed tail and even with arbitrarily small positive total marked cost. No size, rational-ratio, fiber polynomial, or normal-velocity hypothesis is needed. The number a_f depends on the arrays, the set of chosen marked moves, and the fiber group. This is not a bound over arbitrary stationary actions or over all arrays simultaneously.

For a uniform basepoint of coordinate i, let ν_i be the law of the anchored full trajectory $Z_i(t) = f_i(x_i + tm_i) - f_i(x_i)$, valued in compact metrizable K^Z. Call a trajectory abundant if every neighborhood U of it has $\sum_i \nu_i(U) = \infty$. At least one exists: otherwise a finite open cover by sets of finite total mass contradicts $\sum_i \nu_i(K^Z) = \infty$. The anchored shift $z(t) \mapsto z(t+1) - z(1)$ preserves ν_i and its inverse does also, by uniformity of the starting point. Take the closed subgroup $L^{\pm} \le K^Z \times C_2$ generated by the pairs $(z,1)$ of all abundant profiles and all `(constant,0)` trajectories. The preceding invariance implies that $L^{\pm}$ is invariant under ordinary shifts, using its constants. Its even subgroup L and nonempty odd coset are shift invariant and the latter has order dividing two modulo L. Apply the algebraic lemma, fixing a periodic trajectory `z*` in that coset with period N. Put $a_f = 1 - 1/N$ (zero if $N = 1$).

Every finite window of `z*` modulo a common constant can be approximated by a positive finite sum of abundant profiles of odd cardinality. In a compact group the closed semigroup generated by a set and zero is its closed generated subgroup (the inverses are limits of positive powers). Include constants in that assertion and combine them into a free common offset. The approximation remains possible arbitrarily far on the indices. Tolerances and the number of terms may depend on the window, but not the period N chosen above.

For each term of a chosen approximation take a disjoint finite pool of far-out coordinates. At each i in its pool let H_i be the basepoints giving profiles in a sufficiently small specified neighborhood of this abundant term. We may take $\Lambda = \sum_i \Pr(H_i)$ arbitrarily large by abundance; also take every possible mark cost at most κ'/s, where s is the fixed total number of terms and $\kappa' > 0$ the requested cost. Select one index with probability $\Pr(H_i)/\Lambda$ and condition its basepoint on H_i, sampling the other coordinates independently with their Haar measures. Independently repeat for all pools. Every chosen direction is an odd sum, total cost at most κ', and realizes the desired window tolerance. At time t the base marginal over that pool has density $\Lambda^{-1} \sum_i 1_{H_i + tm_i}(x_i)$ with respect to its Haar law. The mean is one and variance is at most $1/\Lambda$ by coordinate independence. Over the finitely many pools the full base marginal has L^2 density at most $\Pi(1 + 1/\Lambda)^{1/2} < 2$ if the pools are large enough. The fiber start is independent Haar, and remains conditionally Haar at each individual time. Therefore the same marginal bound holds on the full space.

Let an event have mass at least $1 - 1/N + g$, $g > 0$ (reduce g slightly as needed). Approximate it in measure by a cylinder A' depending on finitely many initial base coordinates and the full fiber, retaining mass greater than $1 - 1/N + 3g/4$. A proportion bounded below in terms of g of its early base cells have fiber sections B of mass $> 1 - 1/N + g/2$, by averaging. In each such cell the common-Haar-translation hitting probability for the fixed periodic trajectory is at least $1 - N(1 - \mu B) > Ng/2$, by the union bound on its at most N distinct states. This lower bound is uniform in the window length. The hitting probability as a function of the finite tuple of shifts is continuous by continuity of translations in L^1. Thus the profile neighborhoods and approximation tolerances may be chosen after these finitely many sections are known so the actual pool law retains a lower bound depending only on g, N. The probability lost in replacing A' by the original event at all h slots is at most $2h\sqrt{\mu(A \triangle A')}$ by the marginal bound. Choose the cylinder error accordingly before choosing the pools. The law uses finitely many actual directions and its starting law conditional on each is absolutely continuous relative to product Haar. Hence some ordinary direction has a positive-measure return.

One can also use a fixed nonabelian compact Lie fiber when each base-coordinate move acts on it by a left translation which depends only on that base coordinate. For different coordinates all attainable translation elements commute, by commutativity of the action and independence of the base coordinates. Let H_i be their closed generated local groups. The descending sequence of finite-prefix centralizers stabilizes: any descending chain of closed subgroups of a compact Lie group eventually has constant dimension and hence common identity component, after which only a descending chain in a finite component group remains. Write C for the stable centralizer. Every later H_i lies in C, and C centralizes every H_i, so the tail increments all lie in the compact abelian Lie group $Z(C)$. The preceding periodic-profile argument now applies to these tail increments. The initial common Haar translation may still be taken in the entire fiber; each individual translated state is Haar there and the same union bound works. No conclusion about general cocycles depending jointly on separate grids follows.

No uniform finite number of distinct states by this method Even for the asymmetric local models the number of different states needed by an attainable odd trajectory can grow without bound as the arrays vary. Given c, let W be the c-color van der Waerden number for the short filter length of the asymmetric model [R014]. Enlarge its torus output to record independent **observable coordinates** for each stride $r \le W$ (the base starts may be shared). On a prime pair beyond the divisors of these strides use the arrays from that construction after applying the coordinate automorphism r^{-1} and include the finitely many spatial offsets of its filter. At temporal stride r the usual filter sees exactly the original mark. For each legal odd move with negligible marked budget, the filter output stays bounded away from zero on every three/five-slot stencil in the appropriate row, whereas a constant row on that stencil gives zero. These are closed constraints with slack, so they also hold on the attainable odd coset from sufficiently late motions. If a length-W window took at most c values as full output vectors, van der Waerden would supply a constant arithmetic subwindow in time of the short length, with stride $r \le W$, contrary to that row's constraint. This only gives a restriction on proofs via a bounded number of representative states: some high-density sets can still hit all terms of a trajectory with many states.

R198. Quantitative bound for a fixed positive marked budget

Scope and status. Result with proof.

Result. Positive-vertical paired families with marked budget τ have lower capacity approaching one as τ tends to zero, with a square-root logarithmic bound.

Scope. A quantitative lower bound for a fixed positive budget, not an upper bound below one.

Use the unsigned vertical geometry of the star blocker [R200]. All horizontal moves at vertically inactive coordinates are allowed. Suppose $q_i \geq cp_i$ for fixed $c > 0$ and every tail step satisfies $\Sigma|a_i|/p_i \leq \tau < c$, with an odd sum of centered marked residues. For every integer $m \geq 1$ and $0 < s < 1$ satisfying $m\tau/2 < s/4$, the tail and integer-realized capacities satisfy

$$b(W_\tau) \geq 1 - s - (1 - s/2)^m.$$

As the fixed budget tends to zero this gives $b(W_\tau) \geq 1 - O(\sqrt{\tau \log(1/\tau)})$. Take, for example, $s = 5\sqrt{\tau \log(1/\tau)}$, $m = \lfloor s/(3\tau) \rfloor$ for sufficiently small τ.

Use the star lemma's hitting test at any radius strictly between 0 and $s/4 - m\tau/2$, with $D = m + 1$ and separation $\gamma = (1 - \tau/c)/2$. Let M be its clearing multiplier. On the tail primes $q_i > D$ coprime to $2M$, choose k_i with $2Mk_i \equiv 1 \bmod q_i$ and form

$$z_i(v) = (k_i/q_i)\binom{v}{D} \bmod 1, v \in C_{q_i}.$$

This polynomial is well-defined on the cyclic group since the binomial polynomial has denominator a unit modulo q_i and its numerator lies in $Z[v]$. At every site choose $b_i(v) \in F_{p_i}$ so that $||b_i(v)/p_i - z_i(v)|| \leq 1/(2p_i)$. The true marked profile W in the star formula lies uniformly within $\tau/2$ of the global circle polynomial $W_*(t) = \sum_i a_i z_i(v_i + t)$ at every time. This error controls the values themselves, irrespective of the time-window length. The top difference is $\kappa_* = \Sigma a_i k_i/q_i$. Since $2Mk_i = 1 + \ell_i q_i$ with ℓ_i odd, $M\kappa_* = (\Sigma a_i \ell_i)/2 + \Sigma a_i/(2q_i)$ modulo one and hence $||M\kappa_*|| \geq (1 - \tau/c)/2$. The box hitting test on W_* supplies a center where the test on W differs in arm j by at most $j\tau/2 \leq m\tau/2$. All arms miss the little arc as before. The inequality follows with the same single-site cut and the same finite-packet limiting argument. It applies even if arbitrarily many marked sites wrap simultaneously. The fixed-budget statement gives no upper bound below one.

R199. Sparse curvature corrections and a local mass parameter

Scope and status. Result with proof.

Result. Sparse corrections yield endpoint one for diverging prime ratios and local mass shells; two independent clocks remove a countable exceptional set.

Scope. Precise ratio, congruence and budget hypotheses remain essential.

Qualifications. For a single convergent prime ratio, R245 and R215 handle the exceptional shells left by this preliminary construction.

Work in the unsigned vertical paired setting of the positive vertical star [R200]. An active vertical coordinate travels by one, its horizontal speed is represented by a centered integer a_i, its total marked parity Σa_i is odd, and $\Sigma|a_i|/p_i \le \tau$ where $0 < \tau < 1$. Pure horizontal speeds at vertically inactive coordinates are arbitrary. Put $s(d) = \Sigma a_i/p_i$; this is a signed mass in $[-\tau, \tau]$. Threshold statements below concern tails, hence also the rational saturations and integer realizations of the tail formula [R241].

A uniformly sparse correction device. Fix a star with accuracy giving loss ϵ and with m arms, and take the toral hitting lemma in that note at a radius smaller than the available arc margin. Its separating multiplier M is determined by m and this radius before choosing the leading separation γ. Once a lower bound γ is given, take the finite hitting window H, extended to include all offsets and D-th differences. There is a finite number L, depending on H, τ and the remaining margin, for which the following works. On each prime pair choose a desired constant D-th difference $c_i \in F_{p_i}$. If the least magnitude e_i of a signed integer with $e_i \equiv -q_i c_i \bmod p_i$ satisfies $|e_i|/q_i < 1/(2LH)$, on a tail one can prescribe $\Delta^D b_i = c_i$ except at $|e_i|$ sites, using unit corrections of sign $sgn(e_i)$. The resulting array has sum zero $\bmod p_i$ and is therefore a D-th difference on the q_i-cycle. Place the sites with gaps at least $\lfloor q_i/|e_i| \rfloor > LH$ (when $e_i = 0$ there are no sites).

In any length LH segment of a direction each active coordinate has at most one correction. The sum over the entire segment of the absolute curvature-error magnitudes, with centered local increments, is at most $\Sigma|a_i|/p_i \le \tau$. Among L disjoint H-windows one therefore has total error $\le \tau/L$. On that window $W(t) = \Sigma a_i b_i (v_i + t)/p_i$ agrees with a circle polynomial of constant D-th difference $\kappa = \Sigma a_i c_i/p_i$ within $\binom{H+D}{D}\tau/L$, by matching the initial lower differences and summing errors. Choose L large enough that m times this bound is within the remaining margin. With $||M\kappa|| \ge \gamma$ the center of the star lies within the long run, proving avoidance. No independence of the starts or of the supports is required. Several probes with their own good windows can be used simultaneously; for each direction one detecting probe suffices, and one sums their site losses. The run length is uniform before exhausting the finite groups.

First consequence. If $q_i/p_i \to \infty$, this proves $b = 1$ at every $\tau < 1$. Choose $2Mc_i \equiv 1 \bmod p_i$. Then $|e_i|/q_i \le p_i/(2q_i) \to 0$ whereas $||M\kappa|| \ge (1 - \tau)/2$ by parity. There is no summability assumption on $\Sigma 1/q_i$.

Local shell theorem. Suppose $q_i/p_i \to \theta$ with θ positive irrational and that, for every K, eventually $p_i \equiv q_i \equiv 1 \bmod K$. For

$s \notin (2\mathbb{Z} + 1) + (\theta - 1)\mathbb{Q}$,

and for every $\epsilon > 0$, some neighborhood I of s in the real line satisfies

$$b(\{d : s(d) \in I\}) \geq 1 - \epsilon.$$

Here and below one can restrict the family by additional conditions on differences. For a proof choose the m-arm star and M giving the target loss. Write $\alpha = (\theta - 1)/(2M)$ and $\beta = s/2$. Among odd integers r one can arrange

$$||r\alpha|| arbitrarily small, ||1/2 + r\beta|| > 1/8.$$

Indeed the closure of odd multiples of (α, β) in the two-torus is a translate of the compact subgroup generated by $(2\alpha, 2\beta)$. The first coordinate projects that subgroup onto the circle. If the kernel is nontrivial, every kernel coset has a point of distance at least 1/4 from 1/2 in the second coordinate. If the kernel is trivial the subgroup is the graph of a continuous torus character $x \mapsto kx$. Then $\beta - k\alpha$ is either 0 or 1/2 modulo one, with the latter excluded by the hypothesis on s. In the former case the required fiber contains $(0, 0)$. This proves the assertion by density. In particular the separation is uniform before choosing r.

Use $\gamma = 1/16$ and fix H,L accordingly. Choose r odd with the first distance small enough compared with $\theta/(LH)$. Take $2Mc_i \equiv r \bmod p_i$. As $p_i \equiv q_i \equiv 1 \bmod 2M$, writing $2Mc_i = r + \ell_i p_i$ gives $\ell_i \equiv -r \bmod 2M$ and ℓ_i odd. Consequently

$$q_i c_i / p_i \bmod 1 \to r(\theta - 1)/(2M) \bmod 1,$$

$$M\kappa = (1 + rs(d))/2 \bmod 1.$$

The first identity ensures the needed sparsity after division by $q_i/p_i \to \theta$. The second gives separation γ uniformly for $s(d)$ in some neighborhood of s. Apply the correction device. Under these hypotheses such prime arrangements are compatible with divergent reciprocal masses. For example, in each stage fix a modulus and a relative bin width, use the prime number theorem in the corresponding reduced class on dyadic intervals and the intervals dilated by θ, and pair primes from the two bins with disjoint congruence classes or disjoint characteristic size ranges. For $\theta \neq 1$ one can keep bins disjoint, choose a fixed large geometric spacing between scales, and collect a divergent reciprocal sum at each fixed stage. Choose enough finite bins before proceeding to a smaller width and a larger modulus.

For the special rule restricting all $a_i \geq 0$, the budget parameter is exactly $s(d)$. Thus $t \mapsto b(\{d : s(d) < t\})$ is continuous at every positive t outside the displayed countable exceptional set: on a sufficiently small interval around t the shell has capacity $\geq 1 - \epsilon$, and independence of stationary avoiders gives $b(D \cup F) \geq b(D)b(F)$. This does not rule out jumps on the exceptional set. In particular this continuity statement does not yet construct an intermediate threshold.

Two synchronized vertical clocks. Suppose instead that each horizontal prime has two distinct vertical prime coordinates, both traveled together by the same unit or both inactive, and that $q_i^{(j)}/p_i \to \theta_j > 0$ where $1, \theta_1, \theta_2$ are rationally independent. Assume the same eventual congruences for all three moduli. Then $b = 1$ for the full signed-mass range, including arbitrary positive and negative centered horizontal marks subject to the

total budget. This uses just two probes, with site losses each below $\epsilon/2$. Fix their common M and $\gamma = (1-\tau)/6$, then H,L. Put $\alpha_j = (\theta_j - 1)/(2M)$. By Kronecker on odd integers take r with both $||r\alpha_1||$ and $||(2r-1)\alpha_2||$ as small as needed for the sparse-correction criterion. On the first clock use multiplier r and on the second use $2r-1$. Their tested curvatures are $(1 + rs(d))/2$ and $(1 + (2r-1)s(d))/2$ modulo one. Twice the first minus the second is $(1 + s(d))/2$, at distance at least $(1-\tau)/2$ from the integers. At least one probe therefore detects every allowed difference. Its mask gives the claimed endpoint after the tail limit. This is an exclusion for the augmented two-clock model. For the single-clock model with prime ratios converging to one fixed positive ratio, eventual congruence modulo every fixed integer, and marked budget below the smaller of one and that ratio, R245 [R245] and R215 [R215] establish capacity one. These results do not address supports mixing a continuum of limiting ratios.

R200. A star obstruction for positive vertical moves

Scope and status. Result with proof.

Result. Positive vertical stars with a single mark, or vanishing total marked budget, have near-full avoiding events.

Scope. Unsigned vertical movement and the specified prime-ratio lower bound.

Qualifications. Signed versions require $R231/R232$ and cannot be inferred just by changing signs.

In the asymmetric squarefree paired family suppose that vertical moves are **only zero and the positive unit**. Pure horizontal moves at vertically inactive coordinates remain completely unrestricted. At a vertical-active axis let $a_i \in C_{p_i}$ be its horizontal velocity. The following two subfamilies have squarefree tail capacity one, and so do their rational saturations and integer realizations.

1. There is only one active axis with nonzero horizontal residue (the mark). On each axis, every permitted marked residue $a \in R_i$ satisfies $\max_{1\le k\le r} \|ac_{i,k}/p_i\| \ge \gamma$ for some fixed finite list length r, fixed $\gamma > 0$, and arbitrary coefficients $c_{i,k} \in \mathbb{F}_{p_i}$. This includes single prescribed nonzero residues (choose one detecting coefficient), macroscopic half-turn intervals, and the bounded-list positive-vertical families in patching-mark-residues.md [R193]. There is no condition on the relative sizes of p_i, q_i.
2. In a tail the allowed differences have odd $\sum_{i:v_i=1} a_i$ in centered integer representatives, and the maximum of their cost $\sum_{i:v_i=1} |a_i|/p_i$ tends to zero on progressively later tails. The number of marks and the number of ordinary horizontal/vertical moves need not be bounded. Here suppose $q_i \ge p_i$. The fixed positive budget in the original asymmetric rule [R182] is not covered by the second assertion.

Uniform phase hitting. Fix a number $m \ge 1$ of spatial offsets and a desired radius $\eta > 0$. Put $D = m + 1$. There are an integer M and a finite time bound H for which the following holds. If a real-circle polynomial $W(t)$ has constant D-th difference κ with $\|M\kappa\| \ge \gamma$, then, for any affine functions $C_j + tL_j$, some $0 \le t < H$ has

$$\|jW(t+j) - C_j - tL_j\| < \eta \qquad (1 \le j \le m).$$

This also applies on any translated H-window, using a slightly longer stretch of values of W. Indeed missing a fixed box contradicts quantitative equidistribution unless a nonzero bounded integer character λ of the vector polynomial has bounded $C^\infty[H]$ norm, by Green–Tao, Theorem 2.9 applied to the torus, using a Lipschitz bump in the missed box. The character and norm bounds depend only on m, D, η. Expand the translation $W(t+j) = \sum_l \binom{j}{l} \Delta^l W(t)$. By the invertible evaluation matrix of the polynomials $j\binom{j}{l}$, $1 \le j \le m$, $0 \le l < m$, the first nonzero integer $s_l = \sum_j \lambda_j j\binom{j}{l}$ has $l \le m-1$. The leading binomial coefficient of the character polynomial is exactly $s_l\kappa$ in degree $D - l \ge 2$, unaffected by the arbitrary affine terms. Choose M divisible by all possible nonzero $|s_l|$ within these fixed bounds; then the C^∞ bound would give $\|M\kappa\| \ll MH^{-2}$. A sufficiently large H proves the claim. Carry out this test at a smaller radius if bounded small perturbations of the finite segment will follow.

The star rule. On each prime pair take an $\mathbb{F}_p$-valued vertical word $b_i(v)$. At offsets $0 \le j \le m$ observe on a finite packet

$$F_j(x) = \xi_j + \sum_i \frac{u_i b_i(v_i + j)}{p_i} \pmod 1,$$

with independent uniform circle shifts ξ_j. Let I be the open arc of radius $s/2$ about zero and I' its concentric subarc of radius $s/4$. Keep a site if $F_0 \notin I$ and at least one of $F_1, \dots, F_m$ is in I'. At each site the expected lost proportion is at most $s + (1 - s/2)^m$, so this is arbitrarily small by choosing s small and then m large. For several lists of words use independent shifts for each list and sum the losses.

Along a direction write $X(t) = F_0(x + td)$, $F_j(t) = F_j(x + td)$, and

$$W(t) = \sum_{i:\, d_i^{\mathrm{vert}} = 1} a_i b_i(v_i + t)/p_i.$$

Then for suitable arbitrary orbit-dependent constants C_j, L_j one has the exact formula

$$F_j(t) = X(t + j) - jW(t + j) + C_j + tL_j.$$

Indeed a vertical-active summand compares $(u_i + ta_i)b_i(v_i + t + j)$ to $(u_i + (t + j)a_i)b_i(v_i + t + j)$; the remaining discrepancy from a vertical-inactive summand is just affine in t. This works at every horizontal speed, including arbitrary residues at inactive sites. If a run is wholly kept, its later slot-zero values $X(t + j)$ all avoid I. A center satisfying the phase-hitting inequalities with radius less than $s/4$ therefore has all its arms outside I', contradicting its retention.

Only one mark. For list k choose $d_{i,k} \in \mathbb{F}_{p_i}$ with $Md_{i,k} = c_{i,k}$, on a tail prime to M. Choose $b_{i,k}$ so its D-th difference equals $d_{i,k}$ except at one corrected seam site. Give that site whatever value makes the sum over C_{q_i} zero. The operator $(Y - I)^D$ is an isomorphism on mean-zero $\mathbb{F}_{p_i}$-arrays, since $p_i \neq q_i$. In an allowed direction only its one mark contributes to W. On a long run there is a subrun of $H + m + D$ sites avoiding the marked word's seam, uniformly in every sufficiently large q_i: for instance take three times the desired length and then require q_i larger than this entire length. There W is a polynomial with $\kappa = a_i d_{i,k}/p_i$. Pick a list with $\|M\kappa\| \ge \gamma$, apply the hitting test inside the run, and use the star contradiction. Taking all lists only multiplies the arbitrarily small loss by the fixed r.

Vanishing total marked cost. In each grid prescribe $(Y - I)^D b_i$ to take the two nearest consecutive integer values bracketing $p_i/(2M)$, with total zero modulo p_i. A feasible number of upper values lies in $[0, q_i]$ since $q_i \ge p_i$. Now the D-th difference of W is uniformly within $\sum_{i:v_i=1} |a_i|/p_i$ of $\kappa = (\sum_{i:v_i=1} a_i)/(2M)$, at every slot and regardless of the positions. Here $M\kappa$ is a half-turn. On the fixed finite stretch needed in the hitting test, Newton interpolation with the initial differences through order $D - 1$ gives an exact polynomial of top difference κ approximating all values to at most $\binom{H+m+D}{D}$ times the error. The tail-cost hypothesis makes this as small as required. Hence the same center contradiction holds uniformly, with no restriction on how many vertical words change at once.

These are uniform finite-packet blockers. Exhaust each tail by its finite subgroups, coinduce the stationary uniform orbit laws of masks attaining the retained averages, and pass to a stationary weak limit; the forbidden length just proved is fixed before exhausting packets. Apply the squarefree tail formula [R241]. For negative vertical normal motions the expression for an arm involves a second independent temporal stream at $t - j$, so the positive-vertical argument above does not establish these conclusions for the signed family.

R201. Locality of exact prime-valued eigenvalue changes

Scope and status. Result with proof.

Result. Exact prime-valued eigenvalue increments force local finite factors; bounded recurrence order restricts the number of contributing axes, with quantitative stability for approximate increments.

Scope. The locality statements use exact identities; nonergodic fixed-spectrum conclusions are restricted to rank at most one.

Qualifications. No corresponding rank-two global spectrum conclusion is proved.

Let $V = \oplus C_{q_i}$ act ergodically on a standard probability space and write S_i for its generator on axis i. Let p_i, q_i be pairwise distinct primes, and let $\lambda : Y \to T$ be a measurable circle label, for example an eigenvalue selected in a field of integer-horizontal ergodic components. The results here require the displayed identities almost everywhere on the whole space; a resonance that holds only on a subset depending on the vertical step does not meet these assumptions.

One-axis differences. If $\Delta_i\lambda = (S_i - I)\lambda \in C_{p_i}$ almost surely for all i, then $\Delta_i\lambda$ is invariant under every S_j, $j \neq i$. In fact the commuting mixed differences lie in $C_{p_i} \cap C_{p_j} = 0$. The invariant factor for all axes except i is a transitive factor of the finite group C_{q_i}. To see this, ergodicity of the full action makes the residual finite-group action on that factor ergodic, hence the factor is a finite transitive probability space with uniform atoms (conditioning in a finite orbit gives the atomic assertion). It is either trivial or the q_i-cycle. Any finite collection of the nontrivial one-axis factors is independent: the generators translate the corresponding coordinates individually, so the joining is invariant under the full transitive product group. Consequently these differences are single-grid functions on independent axes; their sums along one cycle vanish. Each can be written $k_i(v_i + 1) - k_i(v_i)$ with k_i taking values in C_{p_i}. The factor generated by these axes and λ supports the resulting circle cocycle over their product. This does not require the infinite sum of the k_i to converge as a function on the product, and does not specify Haar conditional fibers if the extension is not relatively ergodic.

The same conclusion about independence holds if only $\Delta_i^b\lambda \in C_{p_i}$ with a fixed b. Indeed $p_i\lambda$ is then a degree-$< b$ polynomial along the individual q_i-cycle. When $q_i > b$, its first difference is q_i-torsion by Newton expansion and periodicity. Hence $\Delta_i\lambda \in C_{p_iq_i}$. Those orders are pairwise coprime, so all mixed differences vanish as before. Here the one-axis differences may include an eigencharacter of order q_i; a constant such difference may need to be kept as a vertical rotation rather than a primitive on the invariant factor.

In the exact first-difference case the spectral measure of $e(\lambda)$ for V is the product of the finite Fourier laws $|hat\{e(k_i)\}|^2$ on the dual coordinate groups (use constant k_i for a trivial axis). Its Fourier transform at a finite vector factors, by the one-axis formulas and independence of the corresponding finite grid factors. Uniqueness on the compact product dual identifies the measure. If the base itself is the profinite product of those axis factors, martingale convergence also gives $\sum_i(1 - |Ee(k_i)|) < \infty$ after ignoring a finite prefix: conditioning $e(\lambda)$ on the first n axes gives $c_n \prod_{i\le n} e(k_i)$, since the quotient is invariant under those axes. Its norm tends to one, and consistency gives $|c_m| = |c_n| \prod_{m<i\le n} |Ee(k_i)|$.

A pointwise parity signature thus forces continuous spectrum. More precisely let $P(T)$ be a fixed integer word with $P(1) = P'(1) = 0$. Suppose that on each axis of a sublist with

divergent $\Sigma 1/q_i$, all the values $P(S_i)k_i(v)$ on the entire q_i-cycle are represented modulo one by odd integers bounded uniformly in absolute value, divided by p_i. Then $e(\lambda)$ has atomless vertical spectral measure. To see this subtract any vertical linear character cv/q_i from k_i, which does not change the word value because the word kills constants and linear phases. If every value of that adjusted function were within $1/(8||P||_1)$ of some fixed center, use its real small lifts after subtracting the center. The word then has absolute real value $\leq 1/8$, so at each site it equals the given bounded odd numerator divided by p_i as an ordinary real number for all sufficiently large primes. Summing round the cycle yields zero, by balance, but an odd number of odd integer numerators has odd total. For every center there must therefore be at least one point outside this radius. Taking the center to be the phase of the Fourier mean (or any center for zero mean) bounds the absolute value of every Fourier coefficient of $e(k_i)$ by $1 - c_P/q_i$. Divergence makes the product law nonatomic. This excludes the signature in every purely discrete-spectrum vertical base, while allowing the compact skew extensions whose eigenvalue coordinate has continuous vertical spectrum.

Bounded-arity temporal recurrences. Let $A(T) \in Z[T]$ be a fixed nonzero primitive polynomial of degree d and suppose $A(S_i)\lambda \in C_{p_i}$ for each i. Discard the finitely many indices $q_i \leq d+1$. Set $M_i = |Res(A, \Phi_{q_i})|$, a nonzero integer since the cyclotomic polynomial has larger degree. The integer resultant identity expresses M_i as an integral-polynomial combination of A, Φ_{q_i} up to sign. Now $A(S_i)p_i\Delta_i\lambda = 0$, and $\Phi_{q_i}(S_i)\Delta_i\lambda = (S_i^{q_i} - I)\lambda = 0$, so $p_iM_i\Delta_i\lambda = 0$.

For any fixed prime ℓ put $I_\ell = \{i : \ell divides p_iM_i\}$. Outside the at most two exceptions $p_i = \ell$ or $q_i = \ell$, divisibility by ℓ requires a common root of A mod ℓ and Φ_{q_i} mod ℓ. The latter is exactly a primitive q_ith root over the algebraic closure of F_ℓ. The former is nonzero by primitivity, has at most d roots, and the distinct orders q_i require different roots. This proves $|I_\ell| \leq d+2$.

Decompose each finite cyclic difference $\Delta_i\lambda$ into its primary parts. Its ℓ-part is invariant under every S_j outside I_ℓ: $\Delta_j\Delta_i\lambda = \Delta_i\Delta_j\lambda$ is killed by the order of $\Delta_j\lambda$, prime to ℓ. The ℓ-component can thus depend only on the finite invariant factor of $\oplus_{j\notin I_\ell} C_{q_j}$. By ergodicity this is transitive under $\prod_{j\in I_\ell} C_{q_j}$ and is a quotient by a subgroup. Since these primes are distinct, any such quotient is the product of a subcollection of its coordinate cycles. Hence every primary piece is a local array of at most $d+2$ vertical coordinates; it need not live on the one axis i. For a fixed i there are only finitely many primary pieces, but their number need not be bounded across i. This establishes an exact local structure without assuming independence of the original observations.

A count-uniform error bound for first differences. Without assuming ergodicity or exact identities let $e_i = \Pr(\Delta_i\lambda \notin C_{p_i})$. For any finite vector $s = \sum_{j\in J} n_j b_j$ with $i \notin J$,

$$\Pr((\Delta_i\lambda) \circ S_s \neq \Delta_i\lambda) \leq 2e_i + 2\sum_{j\in J} |n_j| e_j.$$

Indeed $\Delta_s\Delta_i\lambda$ lies in C_{p_i} off two events of probability e_i. By telescoping in s, $\Delta_s\lambda$ belongs to the subgroup generated by the C_{p_j} for $j \in J$ off a set of probability at most $\Sigma|n_j|e_j$, also for negative coefficients by invariance. The commuting difference $\Delta_i\Delta_s\lambda$ lies in this latter subgroup except on two such sets. Their intersection is zero by coprimality. Notably e_i is charged only twice, however large the support of s. Thus if the errors are summable, any sufficiently late index has an almost single-coordinate increment uniformly against all

signed unit moves in the other tail coordinates. Available arbitrary group motions require a stronger weighted estimate, for example summability of $q_j e_j$ on the relevant tail.

A simple nonergodic pure-point situation. In a jointly ergodic commuting horizontal/vertical action, suppose the discrete spectrum Λ_y of almost every horizontal ergodic component is finitely generated of free rank at most one. Then these spectra are almost surely one fixed countable group. Indeed vertical transformations identify the components equivariantly and preserve their spectra as sets. Their torsion subgroups are finite sets; those sets and their elements can be measurably enumerated in a fixed Borel order. In the infinite case the set of atoms projecting to generators of the torsion-free quotient has exactly $2|Tor(\Lambda_y)|$ elements. Measurability follows for example by enumerating the atoms of a measurable maximal spectral type on each component: being a quotient generator says every enumerated atom is an integer multiple modulo the finite torsion group. Ordering this finite set canonically gives an invariant list of actual eigenvalues, constant by vertical ergodicity of the component base. Together with the torsion group it generates all of Λ_y. A function in the relative horizontal Kronecker factor therefore has **global** pure point spectrum: its componentwise spectral measures, hence their integral, are supported on the same fixed countable set. For three-slot recurrence this excludes such a component field by the common horizontal characteristic factor and one-sided compactness test. For rank at least two the possible bases need not form a finite set, and this argument cannot be applied as stated.

R231. A near-full blocker for boundedly many signed marks and completely free horizontal normals

Scope and status. Result with proof.

Result. Boundedly many signed marks with completely free horizontal normals admit near-full avoiding events; dense single-mark residue menus are also permitted.

Scope. The bound on the number of marks is fixed while the tail varies.

Qualifications. Unbounded mark counts with a reciprocal-cost constraint are not covered.

Theorem. Consider the squarefree direct sum of prime pairs $C_{p_i} \times C_{q_i}$, with all primes distinct and escaping. Fix an integer $B \geq 1$ and one prescribed nonzero horizontal mark $a_i \in F_{p_i}$ at each site. Allow arbitrary finitely many horizontal moves $(a, 0)$, with unrestricted a, and vertical normals $(0, \pm 1)$. Also allow an **odd** number at most B of marked moves $(a_i, \pm 1)$. The squarefree tail capacity is one, also with both independent sign choices $(\pm a_i, \pm 1)$ for each mark. The standard squarefree saturation and integer realization therefore have critical threshold one. The theorem holds on every subfamily as a tail-capacity conclusion. It places no restrictions on the ratios of the primes. For exactly one mark there is also a family of examples in which the fraction of permitted horizontal residues exceeds any prescribed fixed fraction less than one.

The preceding statistical surrogate [R258] supplied a spatial radius L, a set $B_0 \subset T^{[-L,L]}$ of Haar mass near one, and a uniform obstruction to a length-L unforced run. We need its following robust version. If all central values at a length-L block are $X_t = z_0(t)$ and

$$\|z_j(t) + z_{-j}(t) - X_{t+j} - X_{t-j}\| < \eta \quad (1 \leq j \leq L)$$

through that block, the rows cannot all belong to B_0. Here only centers and endpoints needed by the midpoint proof really matter. Choose $\eta > 0$ so that the induced phase discrepancy at characters $1, \ldots, K$ leaves its final strict inequality intact, or use compactness of these finite constraints and of the set of central values.

Choice of degrees. Set $M = 6L + 1$, and choose $D \geq 2M + 2$. In each prime pair use the scalar array

$$f_i(u, v) = p_i^{-1} \sum_{e=1}^{M} u^e \beta_{i,e} [v]^{D-e} \pmod 1,$$

where the numerator operations take place in the prime field and $[v] \in \{0, \ldots, q_i - 1\}$ is the ordinary vertical representative. On an arbitrary finite packet observe

$$F_j(x) = \xi_j + \sum_i f_i(u_i, v_i + j), \qquad -L \leq j \leq L,$$

with independent Haar circle constants ξ_j. Keep the sites whose spatial rows lie in B_0. Each fixed site has Haar row under this choice, so some constants give retained fraction at least the Haar mass of B_0.

Along a direction write $X_t = F_0(x + td)$. For each $s = 0, \ldots, L - 1$ and $j = 1, \ldots, L$ consider the symmetric error at center $t + s$,

$$E_{j,s}(t) = F_j(x + (t + s)d) + F_{-j}(x + (t + s)d) - X_{t+s+j} - X_{t+s-j}.$$

Every pure vertical normal, in either sign, cancels exactly. Every purely horizontal normal, even at an arbitrary speed, contributes a temporal circle polynomial of degree at most M. The constants contribute fixed errors too. These terms may depend on the direction and the initial site without any bound on their coefficients.

Moment computation. Take any nonzero integer character $\lambda = (\lambda_{j,s})$ on this finite error vector. For each $\lambda_{j,s}$ enter its value in positions $(s, s+j), (s, s-j)$ of an integer array c, and subtract its value in the two diagonal cells $(s+j, s+j), (s-j, s-j)$. The resulting array is nonzero: each off-diagonal entry is assigned by a unique pair (j, s). Every column sum is zero. Put

$$m = \min\{d + f : \mu_{d,f} = \sum_{k,l} c_{k,l} k^d l^f \neq 0\}.$$

All indices belong to an integer interval of length at most $3L$. Bivariate Lagrange interpolation on the two coordinate index sets shows that $1 \leq m \leq 6L$. Define

$$C_e = \sum_{d=0}^{m} \binom{e}{d} \binom{D-e}{m-d} \mu_{d,m-d}, \qquad 1 \leq e \leq M.$$

This is not the zero vector. As rational polynomials of e, the $m+1$ Bernstein-basis factors in the displayed sum are independent, by evaluation at $0, 1, \ldots, m$: the evaluation matrix is triangular with nonzero diagonal when $D \geq m$. Their nonzero combination has degree at most m and cannot vanish at every $e = 1, \ldots, M$.

On a marked site of horizontal velocity a and vertical sign σ, take a temporal interval on which all shifted vertical lifts in these expressions are ordinary affine functions without a wrap. Its contribution to the character error is

$$p_i^{-1} \sum_{e} \beta_{i,e} \sum_{k,l} c_{k,l} (u + a(t+k))^e (v' + \sigma(t+l))^{D-e}.$$

The integer v' is a lift consistent throughout this interval. The two signs exchange the two choices of $l = s \pm j$, so the same symmetric coefficient array works. All coefficients above degree $D - m$ are zero in the prime-field calculation, and the ordinary leading coefficient there, as a circle phase, is

$$p_i^{-1} \sum_{e} C_e \, a^e \beta_{i,e} \, \sigma^{D-e}.$$

Indeed each higher-degree coefficient is built from index moments of smaller total degree. At total degree m every term involving the initial shifts u, v' again contains a smaller index moment. This cancellation happens over the integers before reduction and is uniform in those shifts. Any choice of real lifts of the field coefficients gives the same binomial coefficients in the circle. Since $D - m > M$, the arbitrary horizontal polynomials do not affect this leading term.

Bounded characters and dyadic targets. Apply the torus polynomial quantitative equidistribution criterion, for example Green–Tao, Theorem 2.9, with fixed degree D, dimension L^2, and the small box of radius η about zero. Missing the box in a time interval of length H yields a nonzero character λ in a fixed finite list and a bounded $C\infty[H]$ norm for its temporal polynomial, with bounds independent of the coefficients and H. A Lipschitz bump supported in that box supplies the equidistribution discrepancy.

For the finite list, include the binomial-basis conversion in $C'_e = (D-m)! C_e$. Choose positive integers $s_1 < \ldots < s_M$ with all consecutive gaps and s_1 larger than the 2-valuations

of all nonzero C'_e occurring in this list. Set $\alpha_e = 2^{-s_e}$. For every character choose the largest e with $C'_e \neq 0$. The term $C'_e\alpha_e$ then has a uniquely smallest, strictly negative 2-valuation among the nonzero terms even if **every** term is separately multiplied by an odd integer. Thus the sum of the modified terms has distance at least 2^{-s_M} from an integer.

For each prime choose $a_i^e\beta_{i,e}$ to be a nearest prime-grid numerator for $p_i\alpha_e$, using invertibility of a_i. Across the odd marked collection, after accounting for both independent signs, the factor on $C'_e\alpha_e$ is a sum of an odd number of signs (the powers of the horizontal sign and vertical sign), so it is odd. The rounding error in the aggregate coefficient is at most $B\sum_e |C'_e|/(2\min p_i)$, uniformly small on a sufficiently late tail. The binomial coefficient at degree $D-m$ in every tested character is therefore separated from zero uniformly. The bounded quantitative equidistribution norm is impossible once H is chosen large enough. All character bounds and the selected H precede the choice of tail. The same argument works on a translated interval, since time translation leaves this leading coefficient unchanged.

On a fixed run of length large compared with $(B+1)(H+4L)$, assume the tail vertical primes exceed that entire run plus $4L$. In its enlargement needed for all offsets at most one cut per marked site occurs. Discard a radius at most $2L$ around these at most B cuts and discard endpoint margins. A clean interval of length H remains by making the initial run, for example, $(B+2)(H+8L+8)$ long. The moment calculation applies there, hence some starting time has all L^2 errors smaller than η. The robust midpoint trap then gives a forbidden row in the original run. This is pointwise, for every direction and start in every sufficiently late finite packet.

The usual finite-packet stationary compactness proof now gives tail capacity arbitrarily close to one as the desired retained fraction increases; see the squarefree tail formula [R241]. The conclusion concerns the supremum of tail capacities, hence also the torsion-free integer realization. Orders of the finitely many omitted early prime axes need not be varied.

Large marked-residue sets for a single mark. Fix $\rho > 0$. For stages $n = 1, 2, \ldots$ use traps with discarded Haar masses tending to zero. All dimensions, degrees, character lists, and integer vectors (C'_e) for a stage are fixed. At stage n, choose $\gamma_n > 0$ so small that the finite-list union of arcs of radius γ_n, over both signs, costs less than $\rho 2^{-n-2}$ in uniform target probability. For a large prime p_i, take the coefficients $\beta_{i,n,e}$ uniformly independently in the field. For each fixed $a \neq 0$ and each character/sign, the tested leading sum $\sum_e C'_e a^e \sigma^{D-e}\beta_{i,n,e}$ is uniform on the field, as some C'_e remains nonzero there. The fraction of allowed residues whose circle projection falls within γ_n of zero therefore has the stated expected bound up to the vanishing grid error. Choose the number of stages $J_i \to \infty$ slowly enough for all grid and finite-prime exceptions to fit. By averaging there are coefficient choices for which a proportion greater than $1-\rho$ of all horizontal residues are nonzero and pass every test at every stage $n \leq J_i$, on a tail. Take those as R_i. At each fixed n every allowed mark far out has leading binomial coefficient of circle norm at least γ_n, so exactly the same single-mark proof works, uniformly over its actual horizontal residue. If desired symmetrize by testing both signs of a as well. These examples are compatible with the distinct statement for relative densities tending to one [R193]: here the omitted fraction is fixed, not forced to vanish.

For a single mark chosen from any preassigned list of bounded cardinality per axis, a simpler union bound chooses the coefficients so every entry in the list passes at each fixed stage with a uniformly positive margin. The argument for unboundedly many marks would require a different way to control the simultaneous vertical wrap points, as the bounded-gap

choice of a seam-free common interval ceases to be uniform.

R232. A statistical blocker with signed vertical moves and arbitrary affine offsets

Scope and status. Result with proof.

Result. Small-bias controls turn a prescribed polynomial marked-forcing gap into a near-full signed statistical blocker, with arbitrary affine offsets.

Scope. A conditional construction with explicit local approximation and curvature hypotheses.

Qualifications. The curvature multiplier is chosen after the degree data; the lemma does not supply those hypotheses for every candidate.

Here is a signed replacement for the positive vertical star [R200]. It uses the symmetric mirror statistic from the two-stream statistical trap [R258]. The point is to cancel arbitrary spatial affine errors on enough endpoint pairs using just one scalar marked forcing stream. The pairs form a small-bias Cayley graph, and their integer half-differences admit average phase control with a fixed Fourier bandwidth.

Statement. Suppose the vertical speeds on the prime grids are $s_i \in \{0, 1, -1\}$, pure horizontal speeds at $s_i = 0$ are arbitrary, and configurations are tested using arbitrary scalar vertical words $b_i : C_{q_i} \to F_{p_i}$. Write a_i for the centered horizontal speed at a vertically active coordinate. Given $\epsilon > 0$ there exist an even degree D, integer M, finite spatial offsets, and an error tolerance $\eta > 0$ with the following property. For each $\gamma > 0$ there is a fixed finite required time length. Suppose that, in every sufficiently long run at a direction being blocked, some subinterval of that required length has

$$W(t) = \sum_{s_i \neq 0} a_i b_i (v_i + s_i t)/p_i$$

within η, uniformly, of a circle polynomial of degree at most D whose constant D-th difference κ satisfies $||M\kappa|| \geq \gamma$. The starting v_i may be arbitrary and the other directions and orbits impose no restrictions on the subinterval. Then there exists a common blocker of density $> 1 - \epsilon$ for all the directions under consideration, with the forbidden length uniform if the assumed run lengths are uniform. If there are finitely many word probes and at least one works for each run, sum their losses. The required polynomial approximation is local; the time at which the long run or the subinterval starts is immaterial.

The site test. Write $e(x) = \exp(2\pi i x)$. Choose a closed circle set A of mass $> 1 - \epsilon/3$ missing an open arc. By compactness and uniqueness of Fourier coefficients there exist K and $\kappa_0 > 0$ such that every probability measure supported on A has some character $1 \leq k \leq K$ with coefficient of modulus $\geq \kappa_0$. Put $\lambda = \kappa_0^2/10$. Fix an integer n large enough that a mean endpoint error $O(K/\sqrt{n})$ is $< \lambda$, with the absolute constants from the estimates below. Later bandwidths are at most $B = n + 2$. Choose C large depending on λ.

For all sufficiently large m there exist $d = \lceil Cm \rceil$ binary masks $D_1, \ldots, D_d \in F_2^m$ with these three properties:

1. The average of the translations by the D_l has operator norm at most λ on the orthocomplement of constants in normalized $L^2(F_2^m)$.

2. The m-by-d integer matrix with these zero-one columns has no nonzero integer kernel vector of coefficient magnitudes $\leq B$.
3. Each support has size at least $m/4$.

Take independent uniform columns. The Fourier eigenvalues away from the constant are means of d independent symmetric signs for each of the $< 2^m$ nontrivial cube characters; Hoeffding and a union bound prove (1). Another binomial bound proves (3). For a fixed nonzero integer coefficient vector with s nonzeros the probability that one row dot product vanishes is $\leq 1/2$ by conditioning on one sign, and $\leq O(1/\sqrt{s})$ by the elementary Littlewood–Offord bound. For completeness, change Bernoulli signs so the s coefficients are positive, at the cost of a constant shift of the sum. Its level sets are antichains of subsets, hence have size at most the middle binomial coefficient. The rows are independent. For s below any fixed s_0 there are polynomially many choices of vectors; use the 1/2 bound to sum their d-dimensional probabilities. Above s_0 the total is at most $(2B+1)^d(C'/\sqrt{s_0})^m$, tending to zero when the fixed s_0 is sufficiently large. This verifies simultaneous existence. The constant bandwidth B precedes m and d.

Choose positive integer radices R_i increasing so rapidly that a linear combination with coefficient magnitude $\leq$Bd *or* $\leq 4, whicheverislarger$ cannot vanish unless all its coefficients vanish. Put $n_x = 2\Sigma R_i x_i$ for $x \in F_2^m$. At the endpoint x the signed half-difference of the l-edge is

$$j_{xl} = (n_x - n_{x\oplus D_l})/2 = \sum_{i\in D_l} (2x_i - 1)R_i.$$

For fixed x the integers $j_{x1}, \ldots, j_{xd}$ have no relation with coefficients of magnitude $\leq B$: the radices reduce such a relation to the matrix relation in (2), after fixed row sign changes. For each l put $J_l = \{|\sum_{i\in D_l} \pm R_i|\}$. Its size is $2^{|D_l|-1}$; no sum is zero and only opposite sign patterns agree in magnitude. At any fixed midpoint t of l-edges the passive digits outside D_l are determined, whereas all the active signs vary freely. Thus the positive half-differences of its unordered edges run exactly over J_l, each once. This is why averages in the finite spatial row can test the edge graph. Rows may be taken at all integer times between the first and last embedded endpoints.

At any site use the channels

$$F_j = \xi_j + \sum_i u_i b_i (v_i + j)/p_i,$$

for $j = 0$ and all j in $\pm J_l$, using independent Haar shifts ξ_j on a finite packet. Retain a site if $F_0 \in A$ and, for each l and $1 \leq k \leq K$,

$$||J_l|^{-1} \sum_{j\in J_l} e(k(F_j + F_{-}j))| \leq \lambda.$$

For each fixed site the shifted coordinates are independent Haar. Within each J_l the symmetric sums are independent Haar, so Chebyshev and the union bound give extra loss

at most $dK/(\lambda^2 \min_l |J_l|)$, arbitrarily small as m grows. Choose m for loss $< \epsilon/3$ as well as the three graph properties. Some deterministic shifts give the required retained density.

Canceling the affine errors. Along any direction set $X(t) = F_0(t)$. For arbitrary signed vertical speeds the exact identity is

$$F_j(t) + F_{-}j(t) = X(t+j) + X(t-j) + C_j + tL_j$$

$$-j(W(t+j) - W(t-j)), j > 0.$$

On each vertically active grid the two shifted vertical words occur at $t+j$ and $t-j$ in either order, while the horizontal speed in the two central observations differs by $\pm j a_i$. On each vertically inactive grid the entire remainder is affine in t. The shifts ξ contribute a constant. In particular W has no factor s_i in front of a_i.

It suffices to prove that in a fixed sufficiently long polynomial window one can find a translation t of the full endpoint pattern with average distance error as small as needed on its edges. For a fixed oriented incidence x, l let $r_{xl} = (n_x + n_{x \oplus D_l})/2$. Choose an arbitrary real affine half

$$A_{xl}(t) = (C_{|}jxl| + (t + r_{xl})L_{|}jxl|)/2$$

of its affine edge error, taking identical lifts for the two endpoints of each edge. Put $Y_{xl}(t) = j_{xl} W_*(t + n_x) - A_{xl}(t)$, where W_* is the polynomial in the hypothesis. Take as G the normalized Fejér kernel of order n, so $G \geq 0$, has degree $< n$ and mean one, and $\int G(y)(1 - \cos(2\pi y))dy = 1/n$.

For comparison replace the values $W_* t + n_x$ by independent Haar scalars U_x, allowing any fixed affine-half values. Then the mean of $P = \prod_{x,l} G(Y_{xl})$ equals one exactly. Indeed each independent U_x kills every nonconstant bounded-degree monomial at that endpoint by the dissociation of its $j's$. Inserting $1 - \cos(2\pi Y_{xl})$ in one factor changes the mean to $1/n$, by the same argument at bandwidth $\leq n$. Therefore the comparison mean of $P \cdot ((2^m d)^{-1} \sum_{x,l}(1 - \cos(2\pi Y_{xl})))$ is $1/n$.

Choose the even integer D with $D \geq 2^m + 1$. The same two means along polynomial translations t can be made respectively within, for example, $1/4$ and $1/n$ of their Haar values uniformly over all the affine offsets and lower polynomial coefficients. To see this expand the two trigonometric polynomials; they have finite Fourier mass fixed before choosing M,H. For any nonzero multi-index λ' in this expansion the integer $S_x = \sum_l \lambda'_{xl} j_{xl}$ is nonzero for some x by the bounded dissociation. Among the moments $s_e = \sum_x S_x \binom{n_x}{e}$ for $0 \leq e < 2^m$, at least one is nonzero, by Vandermonde. For the first such e the combined scalar polynomial has degree $D - e \geq 2$ with binomial leading coefficient $s_e \kappa$; the arbitrary affine offsets do not affect it. The quantitative one-dimensional polynomial Weyl criterion, for instance the torus case of Green and Tao, Theorem 2.9, forces every relevant exponential mean to be arbitrarily small on a sufficiently long interval when $||M\kappa|| \geq \gamma$. In detail, for the chosen exponential tolerance a failure gives a bounded nonzero integer b with $||b s_e \kappa|| \ll H^{-(D-e)}$. Let M clear all bounded products $b s_e$ for the finite frequency list. Choose H large enough

after γ. The comparison constants do not depend on the values of the affine offsets or on the translate of the polynomial window. Include the endpoint span in its length.

Since $P \geq 0$ there exists t with $avg_{x,l}(1-\cos(2\pi Y_{xl}(t))) \leq 3/n$. The true W instead of W_* changes each phase by $\leq \max|j_{xl}|\eta$, so choose η after the spatial pattern to make the error as small as needed. By $|1-e(ky)| \leq k\sqrt{2(1-\cos(2\pi y))}$ and Cauchy–Schwarz, the averaged error for the k-th character at the two incidences of an edge is $\leq 2K\sqrt{6/n} + O(K \max|j|\eta)$, less than λ with the prior choices. The two incidence phases sum to the negative of the mirror error, because $j_{x\oplus D_l,l} = -j_{xl}$. This uses average phase accuracy; simultaneous sup control on d dependent frequencies was not needed.

If all slots were retained, summing the row tests over the l-edge midpoints (then over l) now bounds

$$|E_{x,l} e(kX(t+n_x)) e(kX(t+n_{x\oplus D_l}))| \leq 2\lambda$$

for every tested k. By the operator-norm estimate of the graph, this differs from $(E_x e(kX(t+n_x)))^2$ by at most λ: apply Cauchy–Schwarz to the zero-mean component in the complex bilinear average, using contraction on its orthocomplement. All endpoint-center values belong to A, so one of these squared averages has modulus $\geq \kappa_0^2 > 3\lambda$. This contradiction proves the statement. The finite packet bounds extend to stationary tail actions by the same compactness and orbit argument as in the positive-vertical note.

Consequences. The signed single-mark family with a bounded list of detecting residues on each axis has capacity one. Use $Mc_i \equiv$ the detector modulo p_i and a word with $\Delta^D b_i = c_i$ except at one seam. A sufficiently large q_i supplies a whole required polynomial subwindow along the sole mark in either orientation; D even makes its curvature $a_i c_i/p_i$ in both. For arbitrary numbers of marks, the vanishing total $\Sigma|a_i|/p_i$ case and the sparse-correction argument [R199] now apply for both vertical signs. In particular fixed cost < 1 at $q_i/p_i \to \infty$ gives endpoint one, and matched irrational ratios give the same local signed-mass shells and necessary continuous-character form [R136] for a hard tail ultrafilter. Negative vertical motions change correction locations but never their count because the step is still a cyclic unit. These are exclusions and restrictions, rather than an intermediate example for the integer conjecture.

R233. Signed stars with two freely controllable waves

Scope and status. Result with proof.

Result. Signed two-wave stars satisfy positive finite-window hitting under explicit local-lemma or convolution conditions.

Scope. A finite-window surrogate with freely chosen controls and prescribed forcing.

Qualifications. The surrogate does not automatically occur in an arithmetic construction.

This lemma concerns the surrogate two-wave Haar model. Passing from an actual direction family to this model requires a support argument. The lemma does not assert recurrence in the original action.

Take arcs (I, J) in the circle, of lengths $i < 1/2$ and $j > 0$. For integers t and $0 \le b \le m$, let

$$F_b(t) = U_{t+b} + V_{t-b} + w_{b,t},$$

where the entire deterministic forcing array w is arbitrary. Under independent Haar controls U_s, V_s, conditional on the conditions $F_0(s) \notin I$ on a finite interval, the pairs (U_s, V_s) remain independent across s. At each s, both individual circle marginals are Haar, including outside the conditioning interval. Hence at a fixed center t the m arm values $F_b(t)$, $1 \le b \le m$, are iid Haar: the arms use $(2m)$ distinct time indices and one individual marginal from each pair. Conditional on avoiding the central arc in the chosen interval, the bad-center event that all arms miss J has probability $(1 - j)^m$. Bad-center events more than $(2m)$ apart use disjoint time variables. The symmetric Lovász local lemma therefore gives a strictly positive probability of avoiding all such bad centers in any finite prescribed interval when

$$e(4m + 1)(1 - j)^m < 1.$$

Undoing the positive-probability conditioning, the original Haar law gives positive probability to the simultaneous central misses and at least one positive arm hit at every center in the interval. This bound is uniform in its length and the forcing, although the actual positive probability need not be. For J closed and I open, compactness gives a choice of circle controls satisfying these conditions for all integer centers.

The same reasoning applies to symmetric arms of the form $X_{t+b}+X_{t-b}+w_{b,t}$ after requiring $X_s \notin I_s$ at each time, where the arcs I_s may depend on s and all have length $i < 1/2$. The independent conditionally uniform controls have convolution density at least $(1-2i)/(1-i)^2$ at every point of the circle by the union bound on the two removed arcs. At each center the m disjoint pairs thus give bad probability at most $[1 - j(1 - 2i)/(1 - i)^2]^m$. Again choose m large enough for the symmetric local lemma.

An unsigned traveling wave locks the central time profile to its arms by translation. Allowing both signs can instead give the two independent waves above. In that case the algebraic star identity alone cannot exclude a long run bearing this marker, even with an arbitrary deterministic forcing term.

R270. Irrational prime ratios defeat simultaneous approximate polynomial curvature tests

Scope and status. Result with proof.

Result. Irrational matched prime ratios prevent finitely many summably approximate scalar curvature channels from separating all small-cost odd paired steps.

Scope. Arbitrary fixed difference orders, summable spatial mean errors, and normal constants are allowed.

Qualifications. This excludes only the specified approximate polynomial scheme; the Sturmian blockers $R245/R215$ use a different mechanism.

Take distinct odd prime pairs p_i,q_i with $q_i/p_i \to \theta$, where θ is positive irrational, and $\sum_i 1/p_i = \infty$. For instance, if l_n is the nth prime, use $p_i = l_{2i-1}$ and $q_i = l_{2\lfloor \theta i \rfloor}$ for any irrational $\theta > 1$. The prime number theorem gives the two assertions. Put $w_i = 1/p_i + 1/q_i$. On $C_{p_i} \times C_{q_i}$ let X,Y be the two coordinate unit translations. A paired step takes 0, X, Y, or XY independently on finitely many pairs. The XY moves are the marks. Require an odd number of marks and total marked cost less than any fixed $\tau > 0$. Here and throughout X,Y also denote the translation operators acting on circle functions.

Claim. Fix finitely many channels k, each of any fixed order $d_k \geq 1$. In channel k take arbitrary circle arrays f_{ik} on the pair grids, and target constants α_{ik}, β_{ik}, γ_{ik} for their d_k-th differences along X,Y,XY. Let e_{ik} be the uniform spatial mean of the maximum of the three circle-norm errors from those targets. Assume $\sum_i e_{ik} < \infty$ for each channel. Sum the target constants in each channel along a finitely supported step, according to the types of its active coordinates. The joint vector of these sums comes arbitrarily near zero on legal steps in every tail, even with arbitrarily small marked cost. The constants for the two normal moves need not vanish. Thus a *finite list* of scalar polynomial-root tests with these error budgets cannot separate all the steps, even if the detecting channel is allowed to vary with the step. Fixed integer root multipliers can be absorbed into the channels. The conclusion concerns this particular approximation scheme, not every possible blocker.

A weighted finite-sums observation. Write α_i,β_i,γ_i for the three vectors of target constants in T^m. In $T^m \times C_2$ form all tail sums by allowing 0,α_i, 0,β_i, 0,γ_i, 1 at each coordinate and charging w_i only for the last choice. Intersect over N the closures of the sums supported at indices $\geq N$ of cost $\leq 1/N$. The resulting compact set L is a subgroup. It contains 0, and closure under addition follows by approximating two elements on disjoint successive finite supports, giving each half the available budget. In a compact group any closed subsemigroup containing 0 is a group, by recurrence of positive powers. If $(0, 1)$ belongs to L the claim follows directly.

Otherwise L projects injectively onto a closed subgroup of the torus, on which the parity projection is a continuous character with values $\{0, 1/2\}$. Extend it to a torus character $x \mapsto k \cdot x$ using the character extension theorem for closed subgroups of compact abelian groups. The nested compact sets converge uniformly to L in the sense that they eventually lie in every open neighborhood of L. Thus for all sufficiently late supports of sufficiently small total marked cost, every finite sum of the centered errors $k \cdot \alpha_i$, $k \cdot \beta_i$, $k \cdot \gamma_i - 1/2$ has circle norm $< 1/8$. In particular the individual errors tend to zero, since $w_i \to 0$. The real absolute errors of the two uncharged types are summable: a divergent positive or negative subseries would have an arbitrarily late first finite subcollection crossing $1/8$, with total

$< 1/4$, a contradiction. For the charged type there is a constant C such that the indices with $|k \cdot \gamma_i - 1/2| > Cw_i$ have finite total weight. Indeed on such indices choose a single sign and, if its errors have infinite sum, take an arbitrarily late finite set first crossing $1/8$ but still below $1/4$. Its cost is $< 1/(4C)$, which we choose below the allowed budget. This proves absolute summability of the errors on the exceptional indices, hence summability of their weights. All real errors here are centered representatives of circle elements. Since the uncharged absolute errors are summable, the additional indices where either exceeds w_i likewise have finite total weight. Off the full exceptional set we obtain

$$k \cdot (\gamma_i - \alpha_i - \beta_i) = 1/2 + O(w_i) \bmod 1. \quad (1)$$

Row and column telescoping. Write $D_X = X - I$, $D_Y = Y - I$. For every integer $d \geq 1$, $D_{XY} = D_X + XD_Y$. Expanding its d-th power, the difference from $D_X^d + X^d D_Y^d$ consists entirely of mixed terms, each divisible by $D_X D_Y$. For $d = 1$ that difference is identically zero. Sum the mixed terms around a complete horizontal cycle, or a complete vertical cycle; each sum vanishes identically in the circle. The triangle inequality and translation invariance give, with $\lambda_{ik} = \gamma_{ik} - \alpha_{ik} - \beta_{ik}$,

$$||p_i \lambda_{ik}|| \leq 3p_i e_{ik}, ||q_i \lambda_{ik}|| \leq 3q_i e_{ik}.$$

For instance the mean pointwise error between the mixed expression and λ_{ik} is at most $3e_{ik}$; sum along a row and then average the rows. Apply these bounds to $k \cdot \lambda_i$ by the triangle inequality even though the channels can have different orders. Put $e_i^* = \sum_k |k_k| e_{ik}$, using k_k for the components of the character. Outside a set of finite total weight, (1) holds. There is a subsequence of that complement with $p_i e_i^* \to 0$: otherwise its divergent reciprocal mass would contradict summability of e_i^*. The ratio bound gives $q_i e_i^* \to 0$ too. Along this subsequence write $k \cdot \lambda_i = 1/2 + t_i/p_i \bmod 1$, with t_i bounded by (1), and extract $t_i \to t$. As both primes are odd, the two displayed norm inequalities imply t and θt are both half-integers of the form $n + 1/2$. This is impossible for irrational θ. The contradiction proves the claim.

The single-channel constant-curvature computation for order two and nearby-prime construction are in the matched-prime note [R185]. The subgroup argument above deals with multiple channels switching between steps, costs only on marked axes, and arbitrary ordinary-move constants and difference orders.

Chapter 9

Erdős–Taylor finite sums and translated recurrence

R018. Tail Bohr-dense finite sums are pointwise dense in pro-nil orbits

Scope and status. Result with proof.

Result. Tail Bohr-dense finite sums are pointwise dense on finite-step pro-nilsystems; the Erdos-Taylor family has the stronger property on all minimal compact metric distal systems.

Scope. The general first statement uses enveloping-group structure; the Erdos-Taylor strengthening additionally uses sweeping out.

Let a_i be integers. Suppose the nonempty finite sums from each tail are dense in the Bohr compactification of Z. Then for every finite-step minimal pro-nilsystem (X, T), the powers indexed by any tail are dense in its entire enveloping group in the topology of pointwise convergence. In particular every point has dense orbit along that tail, and simultaneous approximation at any finitely many points is possible. Translates have the same property. One can first pass to a minimal orbit closure or take products to test different powers of the same nilrotation. In particular, every translate of the Erdős–Taylor family passes these pointwise tests: its sweeping-out property [R254] applied to translations of a compact metrizable monothetic group ensures each tail Bohr-dense (a nonempty open set of rotations missed by a tail would give two small open sets with no allowed transition).

Argument. For a minimal pro-nilsystem of step s, its enveloping group E, the pointwise closure of T^Z in X^X, is a compact Hausdorff right-topological group. Its topological lower central series $C_1 = E$, C_{j+1}=closure of the subgroup generated by $[C_j, E]$ terminates at $C_{s+1} = \{1\}$. All the C_j are closed normal subgroups. We use Donoso, Theorem 1.1 and the compact Hausdorff filtered quotients in Qiu–Zhao, Lemma 4.10. In particular E/C_2 is a compact abelian topological group. The integer powers have dense image there.

Let K be the intersection of the closures in E of all the tail finite-sums powers. It is a subsemigroup. Given p,$q \in K$, a neighborhood U of pq and a prescribed tail, right continuity first gives a finite $\sum a$ in that tail with $T^a q \in U$. Continuity of T^a as a map on X gives a $\sum b$ with support later than a for which $T^{a+b} \in U$. The supports are disjoint. Moreover, Kp is itself a compact subsemigroup for every $p \in K$. Its idempotent, supplied by the Ellis–

Numakura lemma, must be 1 since E is a group. Thus p has its inverse in K, making K a closed subgroup. Its image in E/C_2 is the entire quotient by the hypothesis on every tail and compactness of the nested closed sets in each fiber.

For completeness, any closed subgroup K of a topologically nilpotent E with this surjectivity equals E. Induct on the length of the central series. In the abelian case the assertion is the hypothesis. In the higher-step case let C_s be the last potentially nonidentity group. It is central, and E/C_s with the image filtration has shorter length. The image of K is closed, and surjects onto its abelian quotient since the quotient's closed commutator subgroup is the image of C_2. The induction hypothesis gives $E = KC_s$. Centrality now yields $[E, E] \subset K$, hence $C_2 \subset K$ by closedness. Surjectivity modulo C_2 gives $K = E$. This proof avoids any assumption of joint continuity of multiplication in E itself.

Scope. Pointwise density on these distal systems does not yield positive multiple correlations in arbitrary measurable systems. Passing to such correlations along this particular sparse set would require additional control of the exceptional times in a characteristic-factor approximation.

A stronger conclusion for the Erdős–Taylor sequence. Its proof uses sweeping out, a stronger hypothesis than Bohr density. Let (X, T) be any minimal compact metric distal system and take an ergodic invariant probability; it has full support. For each nonempty open set from a countable base and each generator tail, sweeping out shows that almost every starting point reaches this open set using that tail. If the zero sum is excluded, fix one generator and apply sweeping to the remaining tail and a translate of the open set. A single point y thus has dense finite-sums orbit on every tail. In the compact Ellis group let K be the tail subgroup constructed above. For every $x \in X$, the nested compact tail closures have elements sending y to x, so some $k \in K$ sends y to x. This gives $Ky = X$. As K is a subgroup, its action is transitive from every point, including any prescribed starting point. The same holds for every translate and any minimal distal orbit closure in a finite product. This argument uses no nilpotency hypothesis and no assumption of metrizability of the Ellis group itself.

R109. A necessary arithmetic form of the bounded ET carry budget

Scope and status. Proved subcases and unresolved extensions.

Result. The Erdős-Taylor carry budget imposes arithmetic constraints and permits recurrence in a fixed measurable product-digit model.

Scope. A finite-chart high-density wrap obstruction prevents inferring arbitrary-system recurrence from the marginal construction.

Qualifications. Retain the finite-chart counterexample: marginal uniformity alone does not prove multiple recurrence for arbitrary observables.

Write $a_1 = 1, a_{i+1} = ia_i + 1$, and use $E = \exp(1)$ and $b_i = (i-1)!$. Thus

$$a_i = b_i \sum_{r=0}^{i-1} \frac{1}{\{r!\}} = Eb_i - w_i, \qquad w_i = b_i \sum_{r=i}^{\infty} \frac{1}{\{r!\}} = \frac{1}{i} + O(i^{-2}).$$

Let p be an ultrafilter on finite subsets $F \subset \mathbb{N}$ whose supports escape to infinity. Suppose the physical steps $t + \sum_{i\in F} a_i$ converge to zero in the Bohr topology along p, and suppose their harmonic masses are bounded on a member of p. The measures $\sum_{i\in F} i^{-1}\delta_i$ then converge weak-* along p on $\beta\mathbb{N}$ to a sum of precisely t unit point masses, with multiplicity, all on $\beta\mathbb{N} \setminus \mathbb{N}$. In particular t cannot be negative under bounded mass. When $t = 1$ all the limiting harmonic mass is at a single nonprincipal ultrafilter.

Indeed, let ν denote the measure limit and $\tau = \nu(1)$. For every integer $q \geq 1$ the character $1/(qE)$ sends the steps to $(t - \sum_F w_i)/(qE)$ mod 1 on the tail since $q \mid b_i$. The error replacing w_i by $1/i$ tends to zero. Consequently $(t-\tau)/(qE) \in \mathbb{Z}$ for all q, which forces $t = \tau$. For every subset J let $\eta = \sum_{r\geq 1} 1_J(r)/r!$. Modulo the integers, $b_i\eta = 1_J(i)/i + O(i^{-2})$. The character η/E now gives $\nu(J) \in \mathbb{Z}$. This holds on every clopen set. Such a positive finite measure of mass t is an integral point measure. For a direct proof, partition into as many disjoint positive clopens as possible. There are at most t, and each part's restriction charges exactly one side of each further clopen split, defining one ultrafilter with that part's integer mass.

The bare counts on a zero-harmonic-mass neighboring residue also have an exact constraint. Suppose $t = 1$, write U for its mass point, and choose a residue class J mod 3 in U. For *every* sequence $z_j \in \mathbb{T}$ on $J + 1$,

$$p\text{-}\lim_F \sum_{j\in F\cap(J+1)} z_j = -\lim_{i\to U} z_{i+1} \quad \text{in } \mathbb{T}.$$

Take representatives in $[0, 1)$. Set $d_j = \lfloor jz_j \rfloor$ on $J + 1$ and zero elsewhere; use $\eta = \sum_{j\geq 1} d_j/j!$. Then $b_i\eta \equiv z_i + O(1/i)$ on $J + 1$, $b_i\eta \equiv z_{i+1}/i + O(i^{-2})$ on J, and $b_i\eta \equiv O(i^{-2})$ on the remaining class. The first error sums to zero in the limit because $\nu(J + 1) = 0$. The second term tends to $\lim_U z_{i+1}$ modulo one, by the harmonic measure calculation (take the ultralimit of the representatives in $[0, 1]$, and finally reduce modulo one). Apply the character η/E and $\sum_F w_i \to t$ to obtain the assertion. For higher integer mass one obtains a sum of the successor evaluations for the mass points in J, provided $\nu(J + 1) = 0$. Arbitrarily many very distant dust coordinates can in principle supply these negative evaluations by Bohr approximation. This is a constraint, not a contradiction.

If the supports need not escape, there are still only locally finitely many possible bounded budgets along a Bohr-cancelling ultrafilter. Put $B = \{i : \{F : i \in F\} \in p\}$ and $\sigma = p\text{-}\lim_F \sum_F w_i$. The factorial sums have profinite limit $x = \sum_{i\in B} b_i$, since $q \mid b_i$ eventually for each q. For the character $1/(qE)$ one obtains $(\sigma - t)/E \equiv x \pmod q$. The first congruence, for $q = 1$, makes its left side an ordinary integer. The factorial expansion of an ordinary integer is eventually zero for nonnegative integers and eventually maximal for negative integers. For example, on truncating the digits just before $b_{n+1} = n!$, the partial sum is $o(n!)$, so the congruence to a fixed integer is an actual equality for sufficiently large n if that integer is nonnegative, and impossible otherwise. Consequently B is finite, $x = \sum_B b_i$, and $\sigma = t + E\sum_B b_i$. Remove this fixed finite head on a member of p; the assertions for escaping supports now apply with translate $t + \sum_B a_i$. For plain reciprocal weights the possible limits are $t + \sum_{i\in B}(a_i + 1/i)$, likewise a locally finite set. In particular a compact budget interval avoiding all the listed numbers gives a family separated from zero by a finite torus test. By the Bohr sieve its all-length threshold is one, and the independent-product bound shows that adjoining such a shell to a bounded-budget initial family does not change its threshold. This simple continuous clipping parameter cannot interpolate thresholds between successive resonances.

Sufficiency for Bohr convergence Conversely, let $U_1, \dots, U_s$ be any free ultrafilters each of whose members has divergent harmonic sum, and let $m_1, \dots, m_s$ be positive integers of sum t. There is an escaping ultrafilter of physical steps Bohr-converging to zero whose limiting weighted measure is $\sum_l m_l \delta_{U_l}$. The divergent-sum condition is also necessary for a mass point: on any set of finite harmonic sum the tail weights vanish uniformly. Thus the bounded-mass Bohr classification is exact. The zero-mass case $t = 0$ follows directly from the finite-dimensional dust subgroup below (nonempty dust sums approach the identity by taking a long multiple of a cluster value on arbitrarily late singleton indices).

For the positive-mass claim fix any finite torus test vector θ and put $g_i = b_i E\theta \in \mathbb{T}^d$. Define the closed group of negligible-cost corrections by

$$H = \bigcap_{M,\epsilon>0} \overline{\{\sum_{i\in F} g_i : \min F > M,\ \sum_{i\in F} 1/i < \epsilon\}}.$$

Include the empty sum. Disjoint successive tail choices show that H is closed under addition, so it is a compact subgroup. In the quotient torus $Q = \mathbb{T}^d/H$, all sufficiently late, sufficiently small-cost sums lie in any prescribed neighborhood of zero, by compactness and this definition. Identify Q with $\mathbb{T}^r$, and lift the images of the singleton g_i to real vectors $z_i \to 0$.

Choose $M, \delta > 0$ so every sum on that tail of weight $< \delta$ lies in the sup-norm $1/10$-neighborhood of zero in Q. If a finite set F on that tail has such weight, then $\sum_F \|z_i\|_\infty \le 2r/10$. Indeed all partial sums, including those within any subfamily, remain in the indicated arc in each coordinate. Individual summands are small, so the lifted partial sums cannot jump from the real interval around zero to one around a different integer. Apply this to the positive and negative summands for each coordinate. Choose R so large that $R\delta/2 > 2r/10$. Then $\{i : i\|z_i\|_\infty > R\}$ has finite harmonic sum, since otherwise it would contain a finite late subset of weight between $\delta/2$ and δ, contradicting the variation bound. Also $g_{i+1} = ig_i$, so $iz_i = u_i + z_{i+1}$ for some integer vector u_i. Outside the summable exceptional set, only finitely many u_i occur.

For each prescribed U_l-member J_l, pass to a harmonic-divergent subset of J_l on which u_i equals a single integer vector u. Arbitrarily far out choose a finite block F_l there with harmonic weight $m_l + o(1)$, choosing the blocks successively to keep them disjoint. Its image in Q has lift sum $u(m_l + o(1)) + o(1)$ because $z_{i+1} \to 0$. The image thus tends to zero. The total $\sum_l \sum_{F_l} g_i$ is correspondingly close to an element of H. By definition of H an additional disjoint dust block of arbitrarily small weight approximates that element's negative in the original torus. In the combined support, harmonic mass tends to t, and $(t - \sum_F w_i)\theta \to 0$ also. The physical vector $(t + \sum_F a_i)\theta$ tends to zero. Arbitrary finite lists of tests and of neighborhood restrictions on the U_l can be met simultaneously, by intersecting the respective requested members at each mass point. For a finite partition of the indices choose its cell in each U_l, ensuring the requested block masses assign the right weights to all cells up to arbitrarily small error. The ultrafilter lemma then supplies the asserted p. This conclusion concerns finite-dimensional toral tests.

A uniform window for finitely many characters For every positive integer t, fixed dimension d and tolerance $\epsilon > 0$, there is a finite $C = C(t, d, \epsilon) > 1$ such that, for every sufficiently large M and every $\theta \in \mathbb{T}^d$, a finite $F \subset [M, CM]$ satisfies

$$\left| \sum_F 1/i - t \right| < \epsilon, \qquad \left\| (t + \sum_F a_i)\theta \right\|_\infty < \epsilon.$$

The assertion does not prescribe an ultrafilter or a sparse sublist of indices. Here are details of a compactness proof. A contrary sequence has $M_n \to \infty$, $C_n \to \infty$, and vectors θ_n avoiding the displayed conditions throughout $[M_n, C_n M_n]$. Fix a free ultrafilter on the row indices n. Put $g_{ni} = b_i E\theta_n$. For $L \geq 1, \delta > 0$, let $K(L, \delta)$ be the closure of the union, over finite $R > L$, of all row ultralimits of sums $\sum_{F_n} g_{ni}$ with $F_n \subset [LM_n, RM_n]$ and weight less than δ. Include zero. Put $H = \bigcap_{L,\delta} K(L, \delta)$. This is a compact subgroup: approximate the first summand using some finite upper endpoint R, then approximate the second past $R+1$, each at half the required cost.

Put $Q = \mathbb{T}^d / H \cong \mathbb{T}^r$. For every sufficiently small $u > 0$, compactness gives L, δ so that the image of $K(L, \delta)$ is contained in the sup-norm u-ball about zero in Q. Consequently for **each** fixed finite $R > L$, on an ultrafilter member *all* supports of weight $< \delta$ in this entire row window have quotient sum in the $2u$-ball, and all singleton images there have real lifts z_{ni} of size at most $2u$. Otherwise bad supports could be selected on the contrary member. As in the preceding proof the true sums $\sum_F \|z_{ni}\|_\infty$ are at most $4ru$ for every such support, for $u < 1/20$. It follows that the set of indices in that fixed row window with $i\|z_{ni}\|_\infty > B$ has total harmonic weight at most δ when $B\delta/2 > 4ru$ (enlarge M_n so one weight is $< \delta/2$). Wherever the successor index is still in the window, $iz_{ni} = v_{ni} + z_{n,i+1}$ with $v_{ni} \in \mathbb{Z}^r$. Outside that exceptional set these integers take only $D = D(B, r)$ possible values.

Choose R now sufficiently large that $\log(R/L) > D(t+2) + \delta + 2$. On an ultrafilter member, one coefficient class in the window (discarding the final singleton) thus has harmonic weight above $t + 1$. Take F_n from that class with weight $t + o(1)$. Its sum has quotient distance $\leq 2ut + o(1)$ from zero. The resulting original torus ultralimit g is close to H, with its distance in the original torus tending uniformly to zero as $u \downarrow 0$ by compactness of the quotient. Choose u at the outset so this error is as small as required. Approximate an element of H close to $-g$ by a support after $(R+1)M_n$, with sufficiently small harmonic weight and some fixed finite endpoint $R'M_n$. On a row ultrafilter member the combined

support has both requested errors small, because the error in $\sum w_i - \sum 1/i$ is uniformly $o_n(1)$ in these late windows. Finally $R' < C_n$ on a member, a contradiction.

A continuous spectral measure for the same shell Fix $0 < c < 1$, take independent Bernoulli c-variables ξ_i for $i \geq 3$ and put $\theta = E^{-1} \sum_{i \geq 3} \xi_i / i!$. Its law on $\mathbb{T}$ is continuous, since the factorial expansion on these digits is injective and the product measure is nonatomic. Uniformly for $\min F \to \infty$ and $\sum_F 1/i \to t$, the characters of the physical steps tend strongly in L^2 of this law to the scalar $e^{2\pi i c t}$. Indeed modulo one their arguments are

$$(t - \sum_F w_i)\theta + \sum_{i \in F} \frac{\xi_i}{i} + O(\sum_{i \in F} i^{-2}).$$

The centered random term has variance $c(1-c) \sum_F i^{-2}$ and its expectation tends to ct. This holds for every such support without averaging in F. For $t = 1, c = 1/2$ the spectral limit is -1, despite the simultaneous Bohr recurrence just proved. On the complex Gaussian system of this atomless measure the powers converge strongly, along all these escaping supports, to the constant phase rotation on the first chaos, by the Gaussian functor (equivalently check polynomial cylinders and then use density). For $c = 1/2$ this is a genuine sign involution. A positive half-space is disjoint from its image, so its pair correlations tend to zero. Conversely any event of mass $a > 1/2$ meets its involution translate in mass at least $2a - 1$, giving the same lower bound on the limiting h-fold correlation in this Gaussian model for all h. Irrational c at $t = 1$ gives aperiodic Gaussian phase rotation instead. These are limits on supports with both conditions, not claims of exact avoidance. In particular arbitrarily close-to-zero pair correlations do not establish nonrecurrence of the step family. For the distinction between rigidity and exact recurrence see also Griesmer, *Separating measurable recurrence from strong recurrence via rigidity sequences*.

Finite panels give a fixed example with the indicated strong-recurrence failure. For $t \geq 1$, at stage r choose supports in a fresh tail with reciprocal mass within $1/r$ of t, demanding $\|d\theta\|_\infty < 1/r$ on $\mathbb{T}^r$. The sufficiency proof supplies a support at every vector of characters, and compactness gives a finite list covering all vectors. Choose the next tail beyond this finite list. The union of the finite panels is Bohr recurrent after every finite deletion, but with $c = 1/(2t)$ its Gaussian half-space pair correlations tend to zero along the whole union. This proves failure of strong recurrence, without deciding measurable recurrence for that union.

Bounded-degree polynomial IP maps are absent Fix an integer t with $-t$ not a nonnegative finite subsum of the a_i, in particular any $t > 0$. For each degree d there is a finite rank $r(d, t)$ such that there is no map $P : \mathcal{P}([r]) \to \mathbb{Z}$ with $P(\varnothing) = 0$, vanishing $(d+1)$-fold additive differences on disjoint unions, and $P(F) \in t + \mathrm{FS}(a_i)$ for every nonempty F. Choose M large enough that $a_m > 2^{d+1}(|t| + \sum_{i<m} a_i)$ for every $m \geq M$. Color each nonempty F by the first $M - 1$ digits in one chosen such representation of $P(F)$. The finite unions theorem gives $d + 1$ disjoint parameter blocks all of whose nonempty unions have the same color. On their Boolean cube the alternating sum of P vanishes. Any largest noncancelling digit $m \geq M$ would exceed the total possible error of the smaller terms. Thus all high contributions vanish. The low contribution of each nonempty vertex is now identically $t + u$ for a fixed finite subsum u, whereas the empty vertex has value zero. Its alternating sum is $\pm(t + u) \neq 0$, a contradiction. This includes the finite-sums case $d = 1$.

The window ratio in the finite torus result necessarily grows quickly with the dimension, even at a fixed small tolerance. Use $c = 1/(2t)$ and sample d independent copies of the factorial test. On any fixed ratio window $[M, CM]$, the normalized harmonic distribution of the d-bit columns tends almost surely to the d-fold Bernoulli c law P as $M \to \infty$, by the strong law for the finitely many column types and partial summation. If F of harmonic weight w near t is selectable and its physical phases are within small fixed ϵ of zero, write P_F for the column distribution weighted by $1/i$ on its chosen indices. Every mean $p_j = \mathbb{E}_{P_F}\xi_j$ satisfies wp_j within $O(\epsilon)+o_M(1)$ of an integer, by the displayed factorial formula above, while wc is close to $1/2$. Thus $|p_j - c| \geq 1/(4(t+1))$ for small enough fixed ϵ. Any weak limit P_* of such selected distributions satisfies $P_* \leq (\log C)/(t-\epsilon)\, P$ pointwise. On the one hand this bounds $D(P_*\|P)$ above by $\log(\log C/(t-\epsilon))$. On the other hand entropy subadditivity and the binary Pinsker inequality give $D(P_*\|P) \geq \sum_{j\leq d} D(\mathrm{Bern}(p_j)\|\mathrm{Bern}(c)) \geq d/(8(t+1)^2)$. It follows that any uniform ratio satisfies $\log C \geq (t-\epsilon)\exp(d/(8(t+1)^2))$. In particular any fixed-ratio family of bounded-error, positive-integer-budget shells on escaping windows is separated from zero by some finite torus test, and hence has all-length threshold one. The entire bounded-budget family permits supports extending over arbitrarily large ratios.

Choosing exactly one digit wrap per step Take disjoint physical pairs $(i, i+1)$ and use an independent low digit in $\mathbb{Z}/i$ and an upper digit in a very long uniform cyclic interval at each pair. Under a marked motion the low digit advances by one each time and the upper digit advances exactly when the low digit wraps. The physical displacement is a_i except on a wrap, where it is $a_i + 1$. Upper-digit wraps may be discarded with arbitrarily small probability over a fixed horizon by increasing their interval lengths.

Fix a horizon h and split lower indices into $h-1$ disjoint finite pools, with reciprocal sums λ_t as large as desired. For each $0 \leq t < h-1$, select an index from its pool with probability proportional to $1/i$, and plant its initial low digit at $i-1-t$. All nonplanted data start uniformly. Move just the planted pairs. Every edge of the resulting word has one wrap, so the whole physical word is an AP with step $1+\sum_{\text{planted}} a_i$. Each slot marginal of its low-digit vector is arbitrarily close to uniform: in pool t, at slot j the likelihood is N_{tj}/λ_t, where N_{tj} counts the digits equal to $i-1-t+j \bmod i$. Under the uniform law, this has variance at most $1/\lambda_t$. Upper coordinates preserve uniformity, apart from the discarded physical seams. All positions also agree on any prescribed finite head. Thus in a *fixed* measurable infinite product digit model every positive event admits such runs by finite-head approximation.

For arbitrary observables on the growing finite charts, that last assertion fails. Let all upper cyclic orders be multiples of h, and retain exactly those vertices where the sum of their upper digits is nonzero modulo h. This event has mass $1 - 1/h$ and misses every complete wrap-one path of the stated form, since its upper sum advances by one at each step. This obstruction also allows arbitrarily many additional **low-only** motions, provided there is just one total wrap per step. It ceases to be a certificate when independent ordinary upper-coordinate motions are also available. It is not a construction of an avoiding set on the physical integers. In a finite chart, the pullback of one measurable physical observable under the chart map can depend strongly on all late digits and changes with the chart. Unbiased marginal distributions alone do not give an all-system multiple return.

R110. Uniform, nearly exact harmonic cost in translated Erdős–Taylor panels

Scope and status. Result with proof.

Result. Translated Erdős-Taylor harmonic panels meet finite Bohr tests with nearly exact harmonic cost; thin panels have explicit Gaussian relative thresholds.

Scope. Uniform finite-dimensional Bohr tests and specified weakly mixing Gaussian actions; not an unrestricted all-length threshold.

Let $a_1 = 1$, $a_{i+1} = ia_i + 1$, $b_i = (i-1)!$, $E = \exp(1)$, and $w_i = Eb_i - a_i = 1/i + O(1/i^2)$. For integers $t \geq 1$, $d \geq 1$ and $0 < \epsilon < 1/4$ there are constants C, K, M_0 depending only on these parameters such that, for all $M \geq M_0$ and every $\theta \in (R/Z)^d$, there is a subset $F \subset [M, CM]$ satisfying

$$|\sum_{i\in F} 1/i - t| \leq K/M, \|(t + \sum_{i\in F} a_i)\theta\|\infty < \epsilon.$$

The harmonic error improves on the fixed-error window in et-carry-mass.md [R109]. In particular C need not grow when the target cost tolerance shrinks as K/M.

Proof. Choose real representatives for θ in a fixed bounded cube and put $g_i = Eb_i\theta \bmod 1$, with centered vector lifts G_i. Then

$$U_i = iG_i - G_{i+1} \in Z^d.$$

Dirichlet's approximation theorem supplies an integer $1 \leq N \leq L_0(d, t, \epsilon)$ for each $\beta \in T^d$ such that $\|N\beta\|\infty < \epsilon/[8(t+1)]$. Thus $l = (t+1)N - t$ lies between 1 and a uniform L, and $\|(t+l)\beta\|\infty < \epsilon/8$. Choose a positive mesh δ with $(t + d + 2 + L)\delta < \epsilon/8$. Pigeonhole the pairs (G_i, G_{i+1}) for indices of one fixed parity in $[M, CM - 1]$ into boxes of coordinate diameter δ. Choosing $\log C$ sufficiently large in terms of the number of boxes, one cell I has total harmonic weight exceeding $2^{d+1}(t+1)$, for all sufficiently large M. There are real vectors α, β with $\|G_i - \alpha\|\infty, \|G_{i+1} - \beta\|\infty \leq \delta$ throughout I.

On the integer points $P_i = (i, U_i) \in Z^{d+1}$, place weights $1/i$ for $i \in I$. The integer centerpoint theorem (or Doignon's lattice Helly theorem) gives an integer point P whose every closed halfspace has weight at least t if it contains P; see Basu and Oertel. For this finite measure, intersect the convex hulls of all subfamilies whose complements have weight less than the total divided by 2^{d+1}. Every collection of at most 2^{d+1} such hulls shares a support point, so all the hulls have a common lattice point. Any closed halfspace through that point has the requisite weight, since otherwise its strict complementary halfspace contains one of the hulls.

The convex compact set $\{\sum r_iP_i/i : 0 \leq r_i \leq 1, \sum r_i/i = t\}$ contains `tP`. In each linear-functional direction choose weight t supported in the upper halfspace about P, so its support function dominates evaluation at `tP`; apply separation. An extreme choice realizing `tP` has at most $d + 2$ fractional coordinates, by the number of linear equalities. Crucially the first component of P_i gives $\sum r_i = tP[1] \in Z$. Round the fractional coordinates to zero or one

with exactly their previous total. This rounding changes $\sum r_i/i$ by at most $(d+2)/M$. Its change in $\sum r_i G_i$ is bounded by $(d+2)\delta$, since the common contribution α cancels. Before rounding,

$$\sum r_i G_i = tP[2:d+1] + \sum r_i G_{i+1}/i = t\beta + O\infty(t\delta) \bmod Z^d.$$

Add l distinct successors of indices in I to the chosen support. They lie in the other parity class, each has G_{i+1} within δ of β, and the abundance in I makes these choices possible. The resulting g_i-sum has distance less than $\epsilon/3$ from zero. The added cost is at most L/M. Since $\sum_{i\ge M} |w_i - 1/i| = O(1/M)$, the bounded representative of θ gives the claimed physical estimate using

$$(t + \sum a_i)\theta = \sum g_i + (t - \sum w_i)\theta \bmod Z^d.$$

An integer consequence for thin shells. Choose increasing stages j with $d = j$, $\epsilon = 1/j$ (begin at $j \ge 5$), constants C_j, K_j as above, and then sufficiently large, disjoint starts M_j. Use all finite supports in $[M_j, C_j M_j]$ of harmonic weight within K_j/M_j of t, with physical differences $t + \sum_F a_i$. The union D meets every Bohr neighborhood of zero on every tail, by the theorem. Yet $b_Z(D) = 1$. The single-row Gaussian argument even gives a fixed positive event avoiding every three-term run on late stages.

Take independent Bernoulli $c = 1/(2t)$ variables ξ_i, and let $\theta_* = E^{-1} \sum_{i\ge 3} \xi_i/i!$ be the spectral variable, as in et-carry-mass.md [R109]. Let W be a real isonormal Gaussian process on the realification of its complex L^2 space and use the unitary multiplier $U = e(\theta_*)$ for the integer action. For $d = t + \sum_F a_i$ in stage j, put $V_F = e(d\theta_*)$. The uniform factorial expansion gives, modulo one,

$$d\theta_* - 1/2 = \sum_{i\in F} (\xi_i - c)/i + O((K_j + 1)/M_j).$$

Hölder or the fourth-moment bound for sums of independent bounded centered variables yields $\|1 + V_F\|_4 \le C(M_j^{-1/2} + (K_j+1)/M_j)$. Consequently the Gaussian variable $W((1 + V_F)^2)$ has standard deviation at most C'/M_j provided $M_j \ge (K_j+1)^2$. There are at most $2^{C_j M_j}$ supports. A direct Gaussian tail bound and a union bound show

$$\Pr[\max_F |W((1+V_F)^2)| > M_j^{-1/8}] \le 2\exp(C_j M_j \log 2 - c_0 M_j^{7/4}),$$

which we can make at most 2^{-j} by choosing M_j after C_j, K_j. This deliberately uses only a crude finite union estimate. If the point and its next two translates all have $W(1) > 1$, the second alternating-sign correction, i.e. $(I + T^d)^2 W(1)$, exceeds four. Thus after deleting exceptional tests on all sufficiently late stages, this halfspace has no such triple.

For the full cap, take independent Gaussian rows with root strides $R_l = 2^{l-1}$ and base multiplier $\exp(2\pi i \theta_*/R_l)$ on row l. The error tests for $(I + T^{R_l d})^2 W_l(1)$ have exactly the

preceding distributions. Use the argument of odd-block-gaussian.md [R162]: retain sites bounded in every row, at least one with value below -1, passing all increment tests in all sufficiently late stages. Its measure is arbitrarily close to one by first choosing the number of independent rows, then the bound, then the stage cutoff. Arbitrarily long centered retained runs with stage tending to infinity would give bounded bilateral limits on which each row, on a residue modulo R_l, solves $(I + S)^2 f = 0$. Hence each row changes sign at its stride and repeats with twice that stride. The successive binary stride choices produce one slot where all rows are nonnegative, impossible. The finitely many remaining stages have finitely many physical directions and admit near-full integer avoidance at a sufficiently large common length by deleting one point in a large prime cycle. This proves $b_Z(D) = 1$; the constructed set therefore does not have intermediate threshold.

For the *entire* family $t + FS(a_i)$, this same Bernoulli spectral action has all positive multiple returns. Indeed let the harmonic cost tend instead to $t + m$ for a fixed nonnegative integer m; uniform late-tail bounds show strong Gaussian convergence along such supports to multiplication by $e(c(t + m) - m\theta_*)$ on the first chaos. These operators are $R^t(RU^{-1})^m$, where R is the scalar complex phase $e(c)$ (all operators are real-linear on the realification). The Gaussian action of RU^{-1} is weakly mixing because its first-chaos spectral measure is nonatomic. Weak mixing gives the product of the means for the Cesàro averages of the shifted multiple correlations of these operators; some m gives a positive value for every positive event. This is a conclusion in this particular action, with its independent factorial digits, not for arbitrary integer actions.

For these **thin** panels the all-length cap in the particular weakly mixing Gaussian action is exactly $1/2$: the strong sign-flip limit gives a common intersection of mass at least $2a - 1$ at every length for an event of mass $a > 1/2$. For the lower bound take $W(1) > \gamma$ and delete all late increment exceptions. For any $\gamma > 0$ this gives triple-avoidance with mass arbitrarily close to its original Gaussian halfspace mass if the tail is taken sufficiently late. In this same aperiodic system a long Rokhlin tower with every Pth level and its top strip deleted repairs the finitely many remaining directions at a common large length, with arbitrarily small loss. Now let $\gamma \to 0$.

More generally use $c = 1/(qt)$, $q \geq 2$, and $\zeta = e(1/q)$. The strong scalar limit ζ has order q, giving the upper bound $1 - 1/q$ in this fixed action. The same fourth-moment and Gaussian union estimates apply to $(U^d - \zeta)^2$, using both real and imaginary observations of one complex Gaussian coordinate Z. To see the reverse bound, discard $|Z|$ above a large bound or below a small positive bound and discard an open angular sector of width strictly greater than $2\pi/q$. Also discard the exceptional late increment tests (for the complex row and its two real components). These losses approach $1/q$ since the angle is uniform. Arbitrarily long runs along stages tending to infinity would have a bounded two-sided limit $Z_n = \zeta^n P(n)$ for a polynomial of degree at most one; its modulus bound makes it constant after division by ζ^n, and the disk cutoff makes it nonzero. Some point of that cyclic q-orbit lies inside the discarded sector, contradicting the closed condition on retained angles. Repair finitely many earlier stages by the same Rokhlin argument. Thus the same thin Bohr-recurrent panels have exact fixed-system cap $1 - 1/q$ in the associated weakly mixing Gaussian model for that q.

R111. Multiple returns in measurable nilsystems along ET translates

Scope and status. Result with proof.

Result. Every translate of every Erdős-Taylor finite-sums tail has multiple measurable returns on pronilsystems, with a mass-and-length bound.

Scope. The additional product result concerns an independent mildly mixing factor, not an arbitrary extension.

Let S_m denote the nonempty finite sums in the tail of the Erdős–Taylor sequence. For any fixed $e \in \mathbb{Z}$, every ergodic inverse limit of nilsystems, any measurable A with $\mu(A) = a > 0$, and any $h \geq 2$, there is a constant $b(a, h) > 0$, independent of the system, with

$$\inf_m \sup_{n \in e+S_m} \mu \bigcap_{j=0}^{h-1} T^{-jn} A \ \geq b(a, h).$$

More generally, one can substitute any family of step sets having dense orbits on each minimal nilsystem, including on all finite product orbit closures. The tail sweeping and Ellis-group argument in unsigned-et-sweeping.md [R262] gives this property for the ET translates. Equivalently, one may apply the nil-orbit argument [R018]. The measurable step here uses Or Shalom, *A structure theorem for multiple correlation sequences on nilsystems*, Theorem 1.9.

For an ergodic nilsystem, write $c(n) = \mu \bigcap_{j=0}^{h-1} T^{-jn} A$. Its ordinary Cesàro lower mean is at least $\kappa(a, h) > 0$ by finite Szemerédi and Varnavides averaging (take κ nondecreasing in a). The cited theorem puts c in the sum of nilsequences and generalized nilsequences with Haar-a.e.-zero Riemann-integrable symbols. Given $\varepsilon < \kappa/10$, uniformly approximate these two parts by basic sequences within total error ε. Take the orbit closure Y of the joint basepoint in the finite product of the corresponding nilsystems. This is a minimal nilsystem, its orbit is Haar-equidistributed, and the projections to each ergodic null-symbol factor take its Haar measure to Haar measure. The joint real symbol F is continuous at Haar-almost every point on Y. Its orbit values differ from $c(n)$ by at most ε, so $\int_Y F \geq \kappa - \varepsilon$. Choose a continuity point with $F > 3\kappa/4$. A nonempty relatively open neighborhood has $F > 2\kappa/3$ throughout. Each prescribed nil-dense step set hits this neighborhood from the joint basepoint, giving $c(n) > \kappa/2$.

For an inverse limit choose a finite-factor event A' with $\mu(A \triangle A') < \min(a/2, \kappa(a/2, h)/(4h))$. Each h-term correlation of A differs from that of A' by at most $h\mu(A \triangle A')$, uniformly in the step. The conclusion therefore holds with $b(a, h) = \kappa(a/2, h)/4$. This uses Shalom's nilsystem theorem as stated. The ordinary null errors in the general ergodic decomposition need not have a nilsystem model with a Haar-almost-everywhere continuous symbol. Thus the argument does not yield the corresponding assertion for arbitrary stationary integer systems.

Independent mild mixing factors The class with measurable h-recurrence along the ET translates is preserved by adjoining an independent mildly mixing system. In fact this uses only that the difference family is a translate of the finite sums of a positive sequence, and that every positive event in the base already returns along that translate of each tail.

Write (Y,S) for the base and (W,R) for the mildly mixing factor. For a measurable event A in their product of positive mass put $f(y)=\mu_W(A_y)$. Some $\eta>0$ has $\{f\ge\eta\}$ of positive mass. In a prescribed tail find $d_0=t+s_0$ with $C=\bigcap_{j<h}S^{-jd_0}\{f\ge\eta\}$ of positive mass. Along *every* sub-IP of subsequent generators there is a v with $\mu_Y(\bigcap_{j<h}S^{-jv}C)\ge c(\mu_Y C,h)>0$. This is the uniform finite IP Szemerédi bound, obtained by density Hales–Jewett and averaging a finite translated word cube. Consequently for the selected v the f-product correlation at d_0+v is at least $\eta^h c$.

Now approximate 1_A in L^1 by a $[0,1]$-valued finite rectangular step function, and its base conditional by the corresponding base mean, with error sufficiently small compared with this fixed bound divided by h. Expand its products into finitely many rectangle terms. On a sub-IP of subsequent generators the correlations of the W-parts at d_0+v are simultaneously arbitrarily close to the products of their integrals. This follows from Bergelson–McCutcheon, *An ergodic IP polynomial Szemerédi theorem*, Theorem 6.10, with the fixed powers R^{jd_0} absorbed in the input functions. Equivalently the finitely many good sets belong to every nonprincipal idempotent carried by that tail IP and admit a common sub-IP. Choose the accuracy depending on the already fixed finite number of terms, then use the base IP return to obtain v. Base coefficients are bounded. Summing the terms and telescoping the uniform L^1 errors proves a positive return in the actual product. This applies in particular to inverse limits of ergodic nilsystems times a mildly mixing system. If desired, the bounds in the first part make the base return and its IP successor uniform in the tail in terms of a,h, by taking $\eta=a/2$ for a product event of mass a.

R112. All translates in the natural rigid rank-one action of ET tower heights

Scope and status. Result with proof.

Result. Translated Erdős-Taylor tails have all positive-measure multiple returns in the natural rigid weakly mixing rank-one action and stated variants.

Scope. The specified cutting-and-stacking model and its independent compact-rotation products.

Build an invertible rank-one transformation T by cutting the tower of height a_i into i equal subcolumns, stacking them in order, and adjoining one spacer over the whole stack. Its next height is $ia_i+1 = a_{i+1}$. The total measure is finite, since the tower areas are proportional to $a_i/(i-1)! \to \exp(1)$; normalize it to one. The refining tower-level partitions generate. This is an ergodic action by the rank-one construction. Its rigidity and weak mixing are verified below.

For every integer t, every $h \geq 2$ and every positive measurable event A, the family $t{+}\mathrm{FS}(a_i)$ has a positive-measure h-return on each tail in this action. When $t > 0$, the finite sum may be required to have exactly $t(h-1)$ summands, independently of the event and the tail. When $t \leq 0$, a finite number of summands depending on A, h, t suffices simultaneously for all tails. The recurrence conclusion continues to hold if the single spacer at each stage is placed over any chosen subcolumn, as detailed below.

Choose a level C of a fixed stage tower within which the conditional measure of A is greater than $1 - 1/(4h)$. This follows from tower generation, the martingale theorem, and the fact that the towers exhaust the space modulo null sets. Descendants of C are parametrized by independent uniform digits $x_i \in \{0, \ldots, i-1\}$ at the subsequent levels. Inside any common later tower, changing finitely many such digits changes the physical height exactly by $\sum_i (x'_i - x_i)a_i$. Finer digits remain unchanged throughout the construction.

If $t > 0$, put $P = t$. Otherwise fix one auxiliary digit j beyond this head so large that $P = t + a_j > 0$ and $h/j < 1/10$. Plan to move that digit by -1 per transition (temporarily take it modulo j); discard its underflow event, of conditional probability at most h/j. It will not be one of the summands in the legal direction.

Use $P(h-1)$ disjoint pools of large lower indices i, each paired with $i+1$, so that all pairs are disjoint, all lower indices exceed any prescribed tail, and each pool has arbitrarily large reciprocal mass $\lambda = \sum 1/i$. This is possible using, for instance, successive strings from one parity class. For each transition $0 \leq k < h-1$ give P pools the label k. In each pool independently select i with probability $1/(i\lambda)$, plant its initial lower digit at $i-1-k$, keep all other digits uniformly random, and move each selected lower digit forward one per step modulo i. Its paired upper digit moves forward only at its single wrap, initially taken modulo $i+1$. Discard any wrap of an upper digit; its total probability is at most $P(h-1)/(1+\min i)$. This can be arbitrarily small even for fixed large P.

At slot l, a given pool's low-digit density relative to conditional uniform digits is N/λ, where N counts the low digits equal to $i-1-k+l \pmod i$. Under the uniform law N has mean λ and variance at most λ. The total variation discrepancy is bounded by $1/(2\sqrt{\lambda})$. The paired upper digits remain uniform, as does the auxiliary cyclic digit before removing the seam. By the sum bound for product total variation each slot is arbitrarily close to the uniform descendant law on C. With positive probability all slots therefore belong to A, and

all seams are absent. All these are measurable events in the original digit space with a finite additional index choice.

For a selected pair with no wrap the physical increment is a_i; at its wrap it is $-(i-1)a_i + a_{i+1} = a_i + 1$. Exactly P wraps occur at every transition. Subtract a_j each time if the auxiliary digit was used. Thus the same physical increment at every position is

$$d = t + \sum_{i \text{ selected}} a_i.$$

These are actual iterates of the same point: use any common stage higher than all changed digits; all positions are levels within that tower, with unchanged later subcolumn coordinates. For each of the finitely many possible index selections the initial planted law is absolutely continuous relative to the conditional law on C. A positive probability of complete success consequently gives a strictly positive h-return for at least one actual direction. Once a single d_0 is obtained, ordinary IP multiple recurrence on $\bigcap_{l<h} T^{-ld_0}A$ gives infinitely many full-family returns of uniformly positive mass by adding later finite sums. This last strengthening does not retain the fixed number of summands.

For a spacer placed above subcolumn $r_i \in \{0, \dots, i-1\}$, the physical offset of descendant digit x_i is $x_i a_i + 1_{x_i > r_i}$. Advance the lower digit cyclically as above and increment its upper neighbor on the cyclic wrap. The net physical displacement is $a_i + 1_{x_i = r_i}$, except for the possible extra upper-spacer crossing or upper wrap. Plant $x_i = r_i - k \bmod i$ at the beginning to obtain its one prescribed spacer crossing. The probability of either bad upper occurrence is at most $2/(i+1)$. The auxiliary decreasing digit also discards both its own wrap and its own spacer, with cost at most $2h/j$; take j large enough. All digit likelihoods are unchanged, giving the stated extension. This is still a tower theorem, not a uniform finite recurrence certificate for every stationary integer action.

Finally $T^{a_i} \to I$ strongly on the measure algebra. On the first $i-1$ subcolumns of stage $i+1$, shifting by a_i preserves the position in the stage-i tower, hence any fixed earlier tower-level approximant. The remaining measure tends to zero. To check weak mixing let g be a normalized eigenfunction of eigenvalue $e^{2\pi i\alpha}$. For any $0 \le r \le i/2$ the same no-wrap comparison for the shift ra_i holds on a set of measure at least $1/2 - o(1)$. Approximate g by a fixed earlier tower-level function. On this safe set that function is identical before and after the shift, so $\sup_{r \le i/2} |e^{2\pi i r a_i \alpha} - 1| \to 0$, by taking the approximation error arbitrarily small. This implies $i\|a_i\alpha\| \to 0$. The relation $a_{i+1} - ia_i = 1$ then gives $\alpha = 0 \bmod 1$. Thus the nontrivial rank-one action is weakly mixing and rigid. The tower calculation uses the special descendant digit structure and does not assert recurrence in arbitrary rigid systems.

Product with independent compact rotations The full conclusion for positive-measure events and all finite lengths holds also on the product of this rank-one transformation with any compact metrizable abelian rotation with Haar law. Here the number of summands is unrestricted. For a nonpositive translate, freeze one sufficiently late generator to make the remaining translate positive and then work strictly after it. The argument thus reduces to an integer $t > 0$.

For a product event A, choose one tower level C and a positive-measure compact-group event B with A of conditional density greater than $1 - \epsilon$ on $C \times B$, for an arbitrarily small prescribed ϵ. Such rectangles follow from the martingale density theorem applied to refining product partitions. It is enough to make the rotation at the constructed direction

arbitrarily small on finitely many circle characters: by continuity of translations in measure and compact-group duality, this ensures B differs by arbitrarily little measure from all its translates at the first h times. Let θ be the resulting vector of circle frequencies, in a finite torus. Use the small-cost subgroup H_0 for $g_i = \mathrm{e}(i-1)!\theta$ from the harmonic calculation [R109], so all sufficiently late singletons project arbitrarily close to zero in the quotient by H_0.

Use the preceding planting procedure for $t(h-1)$ selected pairs, with all lower indices far out. All possible plants together have vanishingly small total harmonic weight and quotient sum. Choose a separate neutral filler set F_0 of weight arbitrarily close to t and whose g_i-sum is arbitrarily small in that quotient. This is supplied by the single integer-mass construction in the harmonic calculation, using a divergent class where the integral quotient digit is fixed. Place it after all planting pools. For each of the finitely many possible index selections, the total g_i-sum on plants and F_0 is close in the original torus to an element of H_0. Choose a later dust set F_1, of arbitrarily small harmonic weight, approximating that element's negative. The final support F has total harmonic weight close to t and $\sum_F g_i$ small. Since $a_i = \mathrm{e}(i-1)! - w_i$ with $w_i = 1/i + O(1/i^2)$, the physical direction $d = t + \sum_F a_i$ has $d\theta$ arbitrarily close to zero. The dust and filler depend only on finitely many selected indices, never on their digit values.

At every time advance the filler digits by one cyclically, also performing the earlier plant moves. Each filler digit stays away from both its wrap edge and its single-spacer edge over the full run with probability at least $1 - 2h/i$. Conditioned on the choice of indices these digits are independent and uniform, so the probability of avoiding every filler seam is bounded below by a constant $c(t,h) > 0$, since their harmonic weight is at most $t+1$. Plant upper-digit seams have arbitrarily small total probability. At each individual time, *before* discarding seams, all unplanted digits remain conditionally uniform under cyclic shifts even though the fillers depend on the selected indices. The slot marginal on the rank-one part therefore still differs arbitrarily little in total variation from uniform descendants of C. On seam-free paths every physical increment equals the same d.

Choose the initial compact coordinate independently with Haar law and restrict it to B. Conditionally on each choice of indices its law at time l is Haar on $B + ld\theta$, with the notation meaning the actual compact rotation translate controlled by those character coordinates. It is arbitrarily close in total variation to Haar conditioned on B, uniformly over all possible choices and slots. The mass lost from failure of membership in A over the run is at most $h\epsilon$ plus arbitrarily small errors, using the preceding marginal estimate and the chosen relative density on $C \times B$. Take $h\epsilon < c(t,h)/4$ and the individual errors small enough to retain positive mass. At least one finite support thus gives a genuine positive-measure return. This product statement requires independent compact Haar law and does not address a nontrivial compact cocycle over this rank-one base.

R254. Every translate of the Erdős–Taylor finite-sums set gives strong pair recurrence

Scope and status. Result with proof.

Result. Every translate of every unsigned Erdős–Taylor finite-sum tail has strong pair recurrence; its translates of any event sweep out its invariant saturation.

Scope. Pair recurrence in arbitrary invertible probability systems, with a more general divergent relation-panel criterion.

Qualifications. The proof does not upgrade to measurable multiple recurrence; R262 supplies a complementary proof and distal extension.

This integer result bears on the search for a short obstruction. It does not determine the all-length critical density. Griesmer asks about measurable, topological and strong recurrence for this example in his Warwick lecture, pp. 23–24 and about measurable and topological recurrence in his 2023 lecture, slide 20.

Let $a_1 = 1$ and $a_{i+1} = ia_i + 1$. Let S be the nonempty finite subsums of the a_i. Every translate $e + S$, for integer e, is a set of strong pair recurrence. This is true after passing to any tail. In fact, for every positive-measure event A, the union of its translates over the subsums of a tail, including zero, agrees modulo null sets with its full invariant saturation.

Proof. Fix a probability-preserving invertible T. Use successive disjoint pairs of indices $(i, i+1)$ starting at the first index of a chosen tail, so that $\sum 1/(i+1)$ diverges. Put $B_0 = A$, and on introducing a pair put

$$B_{j+1} = B_j \cup T^{a_i} B_j \cup T^{a_{i+1}} B_j \cup T^{a_i + a_{i+1}} B_j,$$

$$\delta_j = \mu(B_{j+1}) - \mu(B_j).$$

Each of the first two nonidentity translates of B_j has symmetric difference with B_j of measure at most $2\delta_j$. By translation invariance of symmetric-difference distance, the relation $a_{i+1} - ia_i = 1$ therefore gives

$$\mu(TB_j \triangle B_j) \le 2(i+1)\delta_j.$$

Since $\sum \delta_j \le 1$ and $\sum 1/(i+1) = \infty$, there is a subsequence with $(i+1)\delta_j \to 0$. As B_j increases to B, this proves $TB = B$ modulo null sets. By construction B contains A and is contained in its full invariant saturation, proving the saturation assertion. It follows that any fixed translate of this union meets A in measure at least $\mu(A)$. To require nonempty sums and arbitrarily large positive differences, first freeze one sufficiently late generator a_s and use all subsequent pairs with the translate $e + a_s$. Some $d_0 = e + a_s + u$ then satisfies $c = \mu(A \cap T^{-d_0} A) > 0$. The sign of the saturation shifts can equally be reversed, or one can use the symmetry of pair correlations.

Take $C = A \cap T^{-d_0} A$. Beyond any prescribed index after the support of $d_0 - e$, form M successive distinct partial sums $v_1, \ldots, v_M$ of tail generators. Cauchy–Schwarz gives

$\sum_{i,j} \mu(T^{-v_i}C \cap T^{-v_j}C) \geq M^2c^2$. For M sufficiently large an off-diagonal difference $t = v_j - v_i$, $j > i$, has $\mu(C \cap T^{-t}C) \geq c^2/2$. Then $d_0 + t \in e + S$ and $\mu(A \cap T^{-(d_0+t)}A) \geq c^2/2$. The tail can be arbitrarily late, giving positive limsup as $d \to \infty$. Measurable recurrence also implies topological recurrence: in a minimal compact system every invariant probability has full support.

Generality. The same saturation argument gives invariance under an integer c whenever infinitely many disjoint finite panels of generators express c as an integral combination with coefficient ℓ^1-norm w_j and $\sum 1/w_j$ diverges. For a relation using positive and negative generators alike, the relevant norm is still a sum of the absolute coefficients. In particular increments $a_{i+1} = q_i a_i + 1$ need only have one disjoint subsequence with divergent $\sum 1/q_i + 1$. For one-step recurrence, this direct proof bypasses the signed-box reduction.

Why the proof stops at pairs. The action $x_0, \ldots, x_{h-1} \mapsto x_0, Tx_1, \ldots, T^{h-1}x_{h-1}$ on a product space has the saturation property for any invariant probability. The AP test uses the distribution of $(x, \ldots, x)$, which is generally singular to such a probability. Invariance in the proof is essential to the estimate by $2\delta_j$. See also the pointwise nilsystem conclusion [R018].

R262. Unsigned Erdős–Taylor tails sweep out in measure

Scope and status. Result with proof.

Result. Unsigned Erdős–Taylor tails sweep out invariant saturation, have strong pair recurrence after every translation, and are pointwise recurrent for all finite tuples in minimal distal systems.

Scope. Measurable pair recurrence and topological distal multiple recurrence; the general spacer statement gives T^c invariance.

Qualifications. The diagonal measure for a slope action is not generally invariant, so this does not prove measurable multiple recurrence in arbitrary systems.

Let $n_{i+1} = q_i n_i + c$, with q_i positive integers and $\sum_i 1/q_i = \infty$. Write S_m for all nonempty positive finite sums of distinct n_i, $i \geq m$. Let T be an invertible probability-preserving transformation and A a measurable event. Introduce the empty sum temporarily, and set

$$B_m = \bigcup_{n \in S_m \cup \{0\}} T^{-n} A, B = \bigcap_m B_m.$$

Then B is T^c-invariant modulo null sets. In particular for the unsigned Erdős–Taylor sequence $q_i = i, c = 1, n_1 = 1$, every B_m agrees modulo null sets with the full invariant saturation of A. Every translate of every tail S_m is consequently a set of measurable pair recurrence. More strongly, for any two positive events A, C in the same ergodic system and any integer t, some $n \in S_m$ has $\mu(C \cap T^{-(t+n)} A) > 0$. The question about these unsigned translates, distinguished from the signed sumset, appears on Griesmer, slide 24.

Indeed $B_m = B_{m+1} \cup T^{-n_m} B_{m+1}$, by whether the m-th term is used. Because the sequence B_m decreases, the quantities

$$\delta_m = \mu(T^{-n_m} B_{m+1} \triangle B_{m+1})$$

$$= 2[\mu(B_m) - \mu(B_{m+1})]$$

have finite sum. Replacing B_{m+1} by B_{m+2} in its displacement at time n_m costs at most $2\mu(B_{m+1} \setminus B_{m+2}) = \delta_{m+1}$. Applying the triangle inequality for translation displacements and using $c = n_{m+1} - q_m n_m$ therefore gives

$$\mu(T^{-c} B_{m+2} \triangle B_{m+2})$$

$$\leq \delta_{m+1} + q_m(\delta_m + \delta_{m+1}).$$

Summability and divergence of $\sum 1/q_m$ give a subsequence on which $q_m(\delta_m + \delta_{m+1}) \to 0$, since otherwise the sum of these unweighted differences would diverge. Along it $B_{m+2} \to B$ in measure, proving the invariance.

The empty sum does not affect the union modulo null sets. Indeed $A \setminus \bigcup_{n \in S_m} T^{-n} A$ is disjoint from each of its S_m-translates. A positive event with that property cannot exist: the first k initial partial sums from the tail have all their pair differences in S_m, while k disjoint translates of a positive-measure event cannot fit in a probability space for arbitrarily large k. For $c = 1$, B contains A, is invariant, and by definition lies in its full invariant saturation; all intermediate B_m do also. This proves the sweeping conclusion. Applied to the diagonal action of T on X^r, the same argument gives simultaneous **pair** returns of finitely many positive events in the same system, by taking their product and its invariant saturation.

This proof gives pair recurrence, including its usual topological pair consequence. For the progression action $T \times T^2 \times \cdots$, a diagonal starting measure on X^h is generally not invariant under that action. The telescoping displacement identity cannot be applied with that measure in place of its invariant probability. Neither multiple recurrence in arbitrary systems nor a critical density below one follows from the argument.

Strong pair recurrence Every translate of every ET tail also gives infinitely many pair returns bounded below for any positive event. Choose $d_0 = t + n_{F_0}$ realizing $C = A \cap T^{-d_0} A$ of positive measure. In any tail beyond F_0 there is an n_F with $\mu(C \cap T^{-n_F} C) > \mu(C)^2/2$. Apply the second moment inequality to k translates of 1_C along successive prefix sums, with k sufficiently large in terms of $\mu(C)$; their distinct positive differences are allowed tail sums. Hence $\mu(A \cap T^{-(d_0+n_F)} A) > \mu(C)^2/2$. Taking successively later escaping supports gives the required infinite sequence of positive lower bounded returns.

Tail-orbit density beyond nilsystems The unsigned ET finite-sum tails in fact have dense orbits through every point of every compact metrizable minimal **distal** system. Let E be its Ellis group. The intersection $K = \bigcap_m \overline{\{T^n : n \in S_m\}}^E$ is a compact subgroup: given $p, q \in K$, first approximate p by a sum u with $T^u q$ in a prescribed neighborhood of pq, using right continuity, then approximate q using only indices later than the support of u, using left continuity of T^u. The sum stays in the desired tail. Each compact semigroup Kp contains an idempotent, necessarily the ambient identity, so $p^{-1} \in K$. An ergodic invariant probability on the minimal system has full support. Applied to every member of a countable open base, sweeping shows that almost every point x has every S_m-orbit dense. Evaluation commutes with intersections of nested compact sets, so $Kx = X$ for such an x. Group transitivity then gives $Ky = X$ for all y. Applying the argument on the minimal distal orbit closure of any finite tuple shows $K = E$ in the pointwise topology. It also applies to products of different powers of a distal system, yielding simultaneous multiple point returns along every translate. Uniform measurable recurrence in arbitrary systems does not follow from this topological group conclusion.

Chapter 10

Amplification and endpoint-one exclusions

R022. Every sufficiently high increasing Boolean gate below the row-density bound is schedulable

Scope and status. Result with proof.

Result. Every sufficiently high-measure Boolean gate below the row extremal bias is schedulable at arbitrary strides, with extensions to boundedly many patterns.

Scope. The equality bias is excluded from the general scheduling theorem; this is a trace constraint rather than an equivalence with all dynamical constructions.

Fix an integer $h \geq 2$ and a real $0 < a < 1 - 1/h$. There exists $\epsilon(a, h) > 0$ with the following property, uniform in the number of rows r, in their positive integer strides c_i, and in the choice of an increasing family $B \subset \{0,1\}^r$. If B has independent Bernoulli(a) measure greater than $1 - \epsilon(a, h)$, there are bits $b_i(t)$ for all integer times t such that every column belongs to B and row i contains no all-one h-progression at stride c_i. The bits can be produced by one common circle rotation, integer row multipliers, and translated row arcs. Thus an unschedulable increasing gate cannot have masses tending to one at any fixed strictly subextremal input bias. No disjointness or bounded-support hypothesis on the positive or negative certificates is required.

This is a combinatorial theorem about the trace constraint used to certify a basic root amplification, not an equivalence between that constraint and arbitrary measure-preserving constructions.

Change of bias. Use ones for holes in this proof and put $p = 1 - a$. The failure event C in the hole bits is increasing and has Bernoulli(p) probability ϵ. Fix constants

$$1/h < q < p_1 < p_2 < p.$$

Write $f = 1_C$, and let $I_i(b)$ denote the probability under bias b on all coordinates other than i that i is pivotal. Differentiating the multi-affine product expectation gives Russo's identity $(d/db)\mu_b(C) = \sum_i I_i(b)$. Therefore some $p' \in [p_1, p_2]$ has $\sum_i I_i(p') \leq \epsilon/(p_2 - p_1)$.

Abbreviate these influences by I_i. Coordinates with $I_i = 0$ do not affect f, since every assignment of the other finitely many bits has positive measure.

Smooth majorant. Choose a fixed smooth circle function $g_0 \in [0,1]$ equal to one on a closed arc I of length q, and of mean strictly less than p_1. It can be chosen zero on another open arc. Interpolate between g_0 and the constant one to obtain a smooth g of mean exactly p'. Write $u = (g-p')/\sqrt{p'(1-p')}$. It has mean zero and squared L^2 norm $\leq \rho < 1$, uniformly for the indicated p': equality in $\operatorname{Var} g \leq p'(1-p')$ would require a zero-one function, whereas the interpolation gives an open set with values uniformly inside $(0,1)$. All needed smooth norms of u are uniformly bounded as well.

For positive integer frequencies m_i to be chosen, take independent uniform circle phases θ_i. At any $z \in T$ set

$$F(z) = Ef(Z), \quad \text{with} \quad independentbitsofprobabilities Z_i = 1 equaltog(\theta_i + m_i z).$$

This is the multi-affine extension evaluated at those probabilities. If the hole pattern $H(z) = \{i : \theta_i + m_i z \in I\}$ belongs to C, then $F(z) = 1$, since its coordinates in $H(z)$ are forced to one and C is increasing. It suffices to arrange $||F||\infty < 1$.

A smoothed Fourier estimate. In the p'-biased orthonormal Boolean Fourier basis, write $\widehat{f}(S)$ for the coefficient on S. Then

$$F(z) = \sum_S \widehat{f}(S) \prod_{i \in S} u(\theta_i + m_i z).$$

Choose $\rho < \rho' < 1$. There is $\delta > 0$, depending only on the fixed parameters, for which

$$\sum_{S \ni i} |\widehat{f}(S)|^2 (\rho')^{|S|-1} \leq C I_i^{1+\delta}. (1)$$

Indeed the left side is $p'(1-p')||T_{\sqrt{\rho'}} \partial_i f||_2^2$ with T_λ the independent noise operator multiplying the Fourier coefficient of order d by λ^d. The p'-biased product cube has, for this $\lambda < 1$, an $\mathrm{L}^v \to L^2$ contraction at some $v < 2$, uniformly for $p' \in [p_1, p_2]$. Here is the elementary two-point argument if desired. Normalize a real two-point function to L^2 norm one. At $v = 2$ its squared output norm is $(Ef)^2 + \lambda^2 \operatorname{Var}(f)$, strictly less than one away from constants, so compactness preserves the inequality there for v sufficiently near 2. Near either normalized constant, Taylor expansion gives $||f||_v^2 = (Ef)^2 + (v - 1 + o(1)) \operatorname{Var}(f)$, uniformly on the compact bias interval and v near 2. Choose $v - 1 > \lambda^2$. The one-coordinate bound tensorizes by the mixed-norm Minkowski inequality since $v \leq 2$. Finally the pivotal derivative is zero or one, so its squared L^v norm equals $I_i^{2/v}$; take $\delta = 2/v - 1$.

Fix $0 < \eta < \min(\delta, 1)$, $s = (1+\eta)/2 > 1/2$. With circle Sobolev norm $||F||_{H^s}^2 = \sum_n (1 + |n|^{2s}) |Fhat(n)|^2$, independence and Fourier orthogonality of the θ_i give

$$E\theta ||F||_{H^s}^2 \leq C\epsilon + C \sum_i m_i^{1+\eta} I_i^{1+\delta}. (2)$$

For details, in a term indexed by S choose a nonzero circle frequency k_i at every $i \in S$. Distinct resulting multisite frequencies are orthogonal in the random phases. Their z frequency is $\sum_i m_i k_i$. The untwisted mass of the full sum is $\leq \sum_S |\widehat{f}(S)|^2 = \mu_{p'}(C) \leq \epsilon$. Bound $|\sum_i m_i k_i|^{1+\eta} \leq |S|^\eta \sum_{i \in S} m_i^{1+\eta} |k_i|^{1+\eta}$. For the selected i sum its circle Fourier weights against $|k_i|^{1+\eta}$ (bounded uniformly by smoothness); the remaining coordinates contribute $\rho^{|S|-1}$. Absorb $|S|^\eta$ into a constant times $(\rho'/\rho)^{|S|-1}$ (if needed take a larger positive ρ still below ρ'). Now use (1). This establishes (2).

Frequency choice. Select a short open interval J around $1/h$ mod 1 such that for every $\phi \in J$ the h points $0, \phi, \ldots, (h-1)\phi$ have maximum circular gap strictly below q. This follows by a small perturbation of the regular h-gon. For a uniform circle variable β let m_i be the first positive integer m with $m c_i \beta \in J$. There is a constant C_J with

$$P[m_i > L] \leq C_J / L, L \geq 1,$$

uniformly in the positive integer c_i. To check this take $\theta = c_i \beta$ uniform. Dirichlet's pigeonhole principle gives $1 \leq b \leq B_J$ with $||b\theta|| \leq |J|/2$. If $||b\theta|| \geq 2b/L$ and $L \geq 2B_J$, the initial multiples of $b\theta$ up to index $\lfloor L/b \rfloor$ traverse at least one full circle in mesh at most $|J|/2$, so one visits J within L original multiples. Thus failure implies $||b\theta|| < 2b/L$ for some bounded b, of total measure $O_J(1/L)$. Smaller L are harmless.

For $I_i > 0$ let $L_i = \lceil 1/I_i \rceil$. The probability that some m_i exceeds L_i is at most $C_J \Sigma I_i = O(\epsilon)$. Further, integrating the displayed tail gives $E[m_i^{1+\eta} 1_{m_i \leq L_i}] \leq C_{J,\eta} L_i^\eta$. Consequently

$$E \sum_i I_i^{1+\delta} m_i^{1+\eta} 1_{m_i \leq L_i}$$

$$\leq C \sum_i I_i^{1+\delta-\eta} \leq C \sum_i I_i \leq C\epsilon.$$

No independence between the frequencies is used. For sufficiently small ϵ, a choice of β avoids every cutoff failure and has the weighted sum $\leq C'\epsilon$ by Markov's inequality. Coordinates not affecting f can simply be assigned a constant zero success bit.

Conclusion. For that common β, (2) and the elementary circle Sobolev embedding $||F||\infty \leq (\sum_n (1+|n|^{2s})^{-1})^{1/2} ||F||_{H^s}$ give $E\theta ||F||\infty^2 \leq C\epsilon$. Make $\epsilon(a,h)$ small enough that this is strictly below one. There are phase shifts with F everywhere below one. The holes at integer t are precisely $\theta_i + m_i t\beta \in I$. Every stride-$c_i$ block of length h hits this arc by the definition of J, and its complement therefore gives the required success row. At every t the joint hole pattern avoids C, proving the assertion on both halves of the integer line.

For $a > 1 - 1/h$ an increasing density cutoff strictly between these two numbers has Bernoulli(a) mass tending to one but cannot be scheduled (every individual row has upper density $\leq 1 - 1/h$). No positive-measure h-avoider can supply such an input bias. The abstract equality case $a = 1 - 1/h$ is not covered by the proof. For the critical-density application, attaining this extremal fixed-length mass already forces critical threshold one by near-saturation amplification [D20].

Extensions The gate itself need not be increasing. Replace a general B by its upward closure and use its non-smaller Bernoulli(a) measure. At every time thin the attained support from the upward closure to a member of B. Deleting successes preserves all row exclusions.

One may impose up to k specified h-term strides on every row for fixed k if $a < (1-1/h)^k$. Pick a common covering arc length q just above $1/h$ with $1-(1-q)^k < 1-a$. Use the union of the k translated arc clocks as the required holes of that row, allowing repetitions where necessary. A smooth function on its k phase variables equal to one on this union, of mean strictly smaller than the target bias interval, is $1-\prod_\ell(1-g_0(x_\ell))$ for bumps with transitions sufficiently short. Interpolate toward the constant one to set the desired mean. The k-dimensional row-factor Fourier expansion is orthogonal between different rows as before; the frequency of the z-function at a multisite Fourier mode includes the k terms for each site. The inequality for its $(1+\eta)$-moment now has an additional factor bounded in terms of k, and the gradient weight for one row is bounded by a constant times $\sum_{\ell\le k} m_{i\ell}^{1+\eta}$. Apply the same individual first-hit estimates, union and expectation bounds to this finite number of clocks per row. The influence and hypercontractive estimates still concern the original one Boolean variable per physical row.

More generally fix $a > 0$ with $a < 1$ and k. There exists $H(a,k)$ such that the patterns on any physical row may be arbitrary sets of at least H **distinct integer offsets**, rather than consecutive multiples. Use k circle clocks per row as above. To cover a pattern F by an arc of hole length q from every translated starting point, it suffices that the largest gap of $\{s\alpha : s \in F\}$ is strictly below q. There are K,$\gamma > 0$ depending only on q such that a gap $\ge q$ implies $||F|^{-1}\sum_{s\in F} e(ls\alpha)| \ge \gamma$ for some $1 \le l \le K$. This follows for example by applying a continuous bump supported in an arc strictly shorter than q, of positive mean, and uniformly approximating it by a trigonometric polynomial; its translates have the same coefficient magnitudes. For definiteness use H large enough that the real trigonometric polynomial

$$G_F(\alpha) = 1 - \gamma^{-2}\sum_{l\le K} ||F|^{-1}\sum_{s\in F} e(ls\alpha)|^2$$

has mean at least 1/2. Its mean is exactly $1-\gamma^{-2}K/|F|$ by orthogonality; its Fourier ℓ^1 norm is bounded solely in terms of q. When no suitable multiplier m succeeds for β by time L, the average of $G_F(m\beta)$ for $1 \le m \le L$ is ≤ 0. The L^2 deviation of that average from its mean is at most $C_q/\sqrt{L}$, by the triangle inequality on its nonzero Fourier modes, each of which has squared L^2 average exactly $1/L$. Chebyshev gives the same first-hit tail C_q/L, independently of all the offsets and their sizes. This again supplies the frequency choices in the proof.

R027. Bounded periods in a horizontal-polynomial obstruction give endpoint one

Scope and status. Result with proof.

Result. A uniformly bounded-period polynomial trajectory obstruction on a torsion-free host amplifies to critical value one.

Scope. The common period, degree, number of channels, and short obstruction must be uniform over finite tests.

This restriction on a transfer from finite vertical tags applies even when the horizontal orders escape to infinity and the number of coordinates in a direction is unbounded. It concerns the torsion-free lift, which admits root actions.

Let H be a countable torsion-free abelian group and $V \subseteq H \setminus \{0\}$. Suppose that for every finite $F \subset V$ there is a stationary H-system and an observable f into the same torus $K = \mathbb{T}^r$, with the following uniform properties. The integers Q, s, L and a nonempty open ball $U \subset K$ are fixed for all the finite tests. For every $d \in F$, almost surely:

1. On each time class modulo Q, the trajectory $t \mapsto f(T^{td}x)$ is a K-valued polynomial of degree at most s in the quotient time variable. Its coefficients may vary with the start, residue, and direction.
2. No L consecutive values of the trajectory are contained in a common translate of U.

Then $b_H(V) = 1$. The same conclusion applies to an integer realization that preserves the critical threshold for H.

Why there is a common period. Fix a ball U' with closure in U, with boundary of Haar measure zero and Haar measure strictly between zero and one. Give the inside and outside of U' the respective binary labels zero and one. Disallow points in an open band of width $\tau > 0$ around the boundary. A trajectory with period classes Q' and polynomial degree s that avoids the band has a periodic label word. On each time class expand the polynomial in the binomial basis. Let Λ be the subgroup of integer characters for which all nonconstant coefficients of the resulting scalar polynomial are rational modulo one. This subgroup of $\mathbb{Z}^r$ is primitive. Clear the rational coefficients of a basis by passing to progressions modulo one integer (a further factor of $s!$ suffices for binomial coefficients). On each resulting class the polynomial has dense image in one coset of the connected annihilator of Λ, by Weyl's criterion. Indeed every character outside Λ still has an irrational nonconstant coefficient on every subprogression, by the invertible rational change of polynomial variables. A connected coset that misses the open band has a constant label. Interleaving the residue classes gives periodicity.

For fixed Q', s, r, τ, U', consider the full two-sided subshift of label words realized by the band-avoiding trajectories above. It is closed because the polynomial coefficients on the finitely many residue classes range over compact tori, and the two legal label sets separated from the boundary are closed. Every word is periodic by the preceding paragraph. Such a compact subshift is finite. Otherwise choose words from infinitely many distinct periodic orbits converging to a periodic word z. Translate each word by the position at which its long central agreement with z first fails to the right, passing to a subsequence with the same position modulo the period of z. A first failure exists because two periodic words agreeing on a half-line are equal. A limiting word agrees on the entire negative half-line with a translate

of z and disagrees at zero, so is aperiodic. This contradiction shows that one integer $P(Q')$ is a period of all the legal label words. The underlying torus trajectories need not have bounded period.

Root actions and synchronization. For $M \geq 1$ take an M-root of a witness by coinducing from $MH \leq H$, identified with H via $Md \leftrightarrow d$, and observing the original coordinate at the identity coset. On each time residue modulo M of an orbit under d, the observable on that residue is an original d-trajectory. This follows from the action of MH on the identity-coset coordinate after shifting the starting point. Hence the polynomial period-class parameter in the full time variable is $Q' = MQ$. Add to the observable an independent constant Haar shift, preserving both trajectory properties.

Fix $\eta > 0$. Take enough independent rooted rows r_0 that $(1 - m_K(U'))^{r_0} < \eta/3$. Choose the band so thin that the probability of a bad band in any row is less than $\eta/3$. First take $M_1 = 1$, then choose each subsequent M_j to be a multiple of $P(M_i Q)$ for all $i < j$. Keep a point if every row avoids the band and at least one row is inside U'. This event has measure at least $1 - 2\eta/3$. Some fixed length K_0, independent of the finite test F, its systems and directions, is forbidden for this event. Otherwise take longer and longer centered all-kept segments and pass to a diagonal limit in all row values. The limiting row words avoid the open bands, and at every time some row is inside, since band separation makes this a closed implication. Row j has period $P(M_j Q)$ in labels. From any time t, one of $t, t + M_j, \ldots, t + (L-1)M_j$ is outside U' in row j. This follows also in the limit since the original trajectories cannot be contained in U on this stencil, and $\overline{U'} \subset U$. Select such a time successively for rows $j = 1, \ldots, r_0$. All previously selected outside labels persist by divisibility of the strides. The resulting time has every row outside, a contradiction. The almost sure exclusion can be imposed at all starts simultaneously. Stationary compactness on the binary H-shift over increasing F now gives $a_{K_0,H}(V) \geq 1 - 2\eta/3$. Thus $b_H(V) = 1$.

Application. Pair horizontal prime cyclic factors with vertical cyclic factors whose orders divide a fixed Q. A typical observable is a finite sum of $u_i k_i(y_i)/p_i$ and vertical-only terms. On every time class modulo Q, the vertical sites are constant, and the horizontal variable contributes a linear circle polynomial even when it wraps modulo its prime. Sums of horizontal polynomials of uniformly bounded degree have the same property. Therefore a finite-channel coloring stencil which excludes every translated fixed ball on a uniformly short run via these observables proves endpoint one on the torsion-free lift. In particular, reusing arbitrarily many vertical involutions together with escaping horizontal primes in this way cannot yield an intermediate example. The argument permits unboundedly many horizontal factors per vertical axis. It does not apply when no common Q is available.

R029. A finite bracketing criterion for interpolation at any forbidden length

Scope and status. Proved subcases and unresolved extensions.

Result. Finite L1 bracketing of an ergodic avoiding orbit yields finite partitions into pieces with critical values arbitrarily near one.

Scope. The further full-spectrum implication invokes the separate interpolation result and its substantial conditional-factor input.

Qualifications. Separate the proved finite-partition conclusion from the downstream conditional interpolation dependency.

Let $D \subset N$ have an ergodic stationary witness A of positive mass avoiding h-term D-progressions at any fixed $h \geq 2$. Suppose the countable restricted class of sets

$$\{\{x : T^d x \in A\} : d \in D\}$$

has finite bracketing numbers at every positive L^1 width. In detail, at each width η there are finitely many pairs of measurable lower and upper sets with difference of mass $< \eta$, and every set in the class is almost surely between one of the pairs. Then for every $\eta' > 0$ there is a finite partition of D into sets of critical value greater than $1 - \eta'$. Consequently if $b(D) < 1$, the interpolation theorem [R145] gives the entire prescribed spectrum. Finite Vapnik–Chervonenkis dimension of this restricted class is sufficient for finite bracketing, by Adams and Nobel. Only the translates indexed by D are used. The proof does not show that every avoiding system or every possible intermediate integer threshold admits such a witness.

The local transformation. Fix an ultrafilter p containing D, and take the factor generated by all integer translates of A. Finite bracketing implies total boundedness of the D-translates in the symmetric-difference metric. Hence $1_A \circ T^d$ has a p-limit in L^2. Commutation gives limits on all its fixed shifts. Products and Boolean operations on indicators are continuous for this metric, and their algebra is dense in the factor. The powers T^{-d} have limits there as well since the distance of the inverse translates of each generator equals that of the forward translates for any pair d,d'. Passing products to the limit on vectors gives inverse measure-algebra automorphisms. Call the forward limit R, with observable convention $1_A \circ T^d \to 1_A \circ$R. R commutes with T. It is nonidentity: otherwise the h translates of A along d all converge to A in measure on p and have positive intersection on a member, contradicting avoidance. Since T is ergodic, R is either aperiodic or has a common period $q \geq 2$ with almost all orbits of size q. We may work with the standard factor coded by the integer translates.

Uniform name comparison. For arbitrary η choose a p-member $C \subset D$ on which the d-translates all use one bracket of width $< \eta$. Their limit $1_A \circ$R lies in that same bracket, since the bracket order interval is closed in L^2. Outside its difference E, one has $1_A(T^d x) = 1_A(Rx)$ simultaneously for all $d \in C$. The brackets may originally lie above the generated factor. If necessary project their bounds by conditional expectation onto that factor: where the conditional upper bound is < 1 all of the indicators in the bracket are zero, and where the conditional lower bound is > 0 all are one. Thus on the factor the region of possible

disagreement is where the conditional upper and lower bounds are respectively 1 and 0; it has measure at most η. This uses only binary indicators and countability.

For any finite Boolean combination F of fixed T-translates of A and any fixed H we can now track its first H names exactly on any retained d-run. Remove from F the union of the finitely many translates of E used by those bits and by R^j, $0 \le j \le H$. Indeed $F(T^d R^j y) = F(R^{j+1} y)$ at every safe actual site y. Repeating along a triangular array of positions $(T^d)^t R^j x$ gives $F((T^d)^t x) = F(R^t x)$ throughout a retained run. The number of deleted translates depends on F,H, while η is chosen afterward and C may shrink. Approximation of an arbitrary target mask by F contributes an additional error only in the finitely many R-name positions being compared.

If R is aperiodic take a high Rokhlin tower, delete its base and its uncovered part, and obtain a mask of arbitrarily large mass having no R-run at some fixed H. A finite-bit approximation F then has an R-origin H-run cylinder of arbitrarily small mass. Remove that cylinder and the tracking defects. On C the resulting stationary mask is H-avoiding and arbitrarily near full. For a finite period q, take a single representative per R-orbit and approximate it by a finite-bit F. On any fixed finite window its exact R-name has period q and cannot have adjacent ones. Use r independent integer root towers at heights 1,q,...,q^{r-1}. In addition to the tracking defects remove the base points where the R-name of F fails to agree with the representative's R-name on the window $H = 1 + \sum_{j<r} q^j$. Accept if some row has F and no row has a defect. During a hypothetical retained AP one may keep the current position or add q^j to obtain a miss in row j, while all later adjustments preserve it, by its q-period on that time residue. This gives a miss in all rows. Its mass approaches at least $1-(1-1/q)^r$ as approximation errors decrease. This is the same integer-root argument as in the path construction [R256], and uses R only to track finite names. Thus every p containing D has members with arbitrary near-one critical values. Compactness of the ultrafilter space, or the elementary finite cover criterion in the interpolation note, proves the assertion.

R037. The critical angular thickness in an antipodal Hamming construction

Scope and status. Result with proof.

Result. Critical near-antipodal Hamming thickness $O(n^{-1/2})$ with $O(\sqrt{n})$ exceptions still permits arbitrarily dense avoiders uniformly in dimension.

Scope. Character roots on integers or sufficiently large prime cycles are required.

This rules out the positive-thickness variant left undecided in the Hamming analysis [D04]. It concerns integer steps tested by n circle frequencies, or finite cyclic groups of sufficiently large prime order. One must have the indicated character roots. An action of the whole compact torus by all its group elements would not permit those roots.

Fix constants C,c. For each n take n arbitrary circle characters χ_i on the integers. A direction d is eligible if, except at at most $C\sqrt{n}$ indices, $||\chi_i(d) - 1/2|| \le c/\sqrt{n}$. For every $\epsilon > 0$ there are K and n0 depending only on ϵ,c,C such that for every $n \ge n0$ the eligible directions admit a K-term avoiding integer set of upper natural density greater than $1 - \epsilon$. Equivalently this can be proved with a stationary probability event. The conclusion also applies to masks on large prime cycles, with unrestricted n characters there. Here and below a circle phase is taken modulo one and $e(x) = \exp(2\pi i x)$.

Rows of observations. We use r independent copies of the product of n Haar circles, with the jth group action generated coordinatewise by roots χ_i/M_j. Choose any roots for these characters. For the integers this means division of real frequency representatives. On a prime cycle use modular inversion when the prime does not divide any of the finitely many M_j. At a position in this row observe

$$F = (1/\sqrt{n}) \sum_i \cos(2\pi x_i).$$

In each row the Haar coordinates are iid. The distribution of F converges to a centered Gaussian of variance 1/2. At progression indices t and $t + M_j$ the arguments differ by $\chi_i(d)$. The regular summands nearly flip sign, with cosine-sum at least $-2\pi c/\sqrt{n}$ per summand. The exceptional ones each cost at most 2. Thus one of the two normalized observations is at least $-\pi c - C$. Choose a fixed $L > \pi c + C + 1$, call $F \ge -L$ active, and choose $p0 < 1$ strictly above the limiting activation probability. For sufficiently large n the row is active with probability at most p0. Choose r with $p0^r < \epsilon/3$. Next choose B large and $\tau > 0$ small, with $\tau < 1$, so the sum of probabilities of $|F| > B$ or $|F + L| < \tau$ over the r rows is below $\epsilon/3$ for all sufficiently large n, by Chebyshev and Gaussian convergence. Retain sites where no row has either of these defects and the rows are not simultaneously active. Its product Haar measure is greater than $1 - \epsilon$.

Limit on one residue. Consider increasingly long centered stretches of otherwise legal observations with $n \to infinity$ and eligible differences and arbitrary starting phases. Fix a root M and a time residue u modulo 2M. On times $u + 2Mv$ the regular phases progress in v by $2(\chi_i(d) - 1/2) \bmod 1$, using centered representatives bounded by $2c/\sqrt{n}$. Taylor expansion of their normalized sum, with phases at u arbitrary, gives

$$b_n + v l_n + O_c(v^2/\sqrt{n}).$$

The slopes l_n are bounded in terms of c, and $|b_n| \le B + C$ on the legal stretch. The exceptional sum is the real part of a Fourier transform of a finite complex measure of norm at most C, absorbing starting phases and the normalization. Pass to subsequences so its measures converge weak-star and the constants and slopes converge. The whole limiting observation is bounded by B at every positive and negative v. The exceptional transform is bounded too, so the limiting slope is zero. Consequently on the residue the observation is real Fourier–Stieltjes of norm at most $B+2C$. Its values avoid the interval $(-L-\tau, -L+\tau)$.

Uniform approximate periods. A separated binary gate of a real Fourier–Stieltjes sequence of bounded norm is k-stable, where k depends only on the norm bound and the gap. Indeed approximate the gate on the two allowed compact ranges by a fixed polynomial to error $< 1/4$. Its sequence has bounded Fourier–Stieltjes norm. Such a sequence at sums $a_i + b_j$ has a Hilbert space factorization with products of vector norms at most that bound. A triangular half-graph forces a pairing of order $m \log m$ with the kernel $1/(j-i)$ off the diagonal, whereas the discrete Hilbert transform and the Hilbert factorization bound the pairing by $O(m)$. See the detailed argument [D18]. For any error $z > 0$ every k-stable binary word on Z is within upper Banach discrepancy z of a periodic word of period bounded in terms of k,z. This follows from Conant–Pillay–Terry, Theorems 2.3 and 1.3: first approximate modulo Banach-null discrepancy by a union of cosets of a finite-index subgroup. The periodic approximant retains k-stability since any finite half-graph could be translated by a multiple of its period away from all the null discrepancies. Apply their uniform finite theorem on its cyclic period group and lift back. Taking the least common multiple of bounded possible periods gives one period P. Interleaving the 2M residues provides a word of period `2M P` with upper Banach discrepancy at most z from the full limiting gate. All constants are uniform in the dimensions and differences. Here P can be taken independent of M as the residue estimate was uniform.

Synchronizing the rows. Use $r2^r z < 1$ and choose $M_1 = 1$, taking every later root divisible by $2M_jP$ for every earlier j. If no uniform K,n0 worked, there would be dimensions tending to infinity and arbitrarily long centered runs entirely retained. Along a subsequence all r binary gate words converge on the full integers. The preceding estimates apply to each residue after simultaneous subsequence extraction. Choose an integer t avoiding every discrepancy set shifted by every possible partial sum of the M_j with coefficients zero or one. Their total upper Banach density is at most $r2^r z < 1$. Starting at t, keep the current time or add M_j so that row j is active, which is possible by the two-position inequality. The periodic approximants agree at every tested time and later moves leave all earlier approximants unchanged. The final time has every row active, contrary to the retained limiting word. This proves the uniform finite length and dimension threshold.

The group action with Haar starting phases is already a stationary witness. A typical or high-mean-component sample, or averaging translated masks on a finite cyclic group, yields the actual density statement. For the prime-cycle claim, exclude the finitely many primes dividing the chosen M_j. No independence of the given characters is required: independent initial Haar phases, including across rows, supply the needed distributions. In particular a construction whose only forcing templates are of this critical near-antipodal type, in dimensions tending to infinity, cannot use them to force arbitrarily long runs at a single density below one.

R113. Unrestricted prime digits with a finite abelian acceptance rule have only endpoint critical densities

Scope and status. Result with proof.

Result. Finite-group acceptance of unrestricted prime digits has an endpoint dichotomy after rational saturation, including nonabelian acceptance groups.

Scope. The full digit alphabet is admitted; arbitrary local coordinate restrictions or finite monoid rules are excluded.

Qualifications. The endpoint dichotomy is for rational saturation and the resulting integer realization, not the native finite-primary group before saturation.

Research proof, November 2, 2026. Let $G = \oplus_{i\geq 1} C_{p_i}$ for pairwise distinct primes. Fix a finite abelian group K, arbitrary functions $f_i : C_{p_i} \to K$ with $f_i(0) = 0$, and an acceptance set $L \subset K$. Put

$$V = \{v \in G \setminus \{0\} : \sum_i f_i(v_i) \in L\}, H_i = \langle f_i(C_{p_i})\rangle \leq K.$$

Let H_∞ be the subgroup generated by all subgroups $H \leq K$ that equal H_i for infinitely many indices. If $L \cap H_\infty$ is empty, all sufficiently late tails of V are empty. If $L \cap H_\infty$ is nonempty, every tail contains all nonempty finite sums of arbitrarily many directions, with every one of these sums nonzero. Thus the saturated rational family $\pi^{-1}(V)$ with $\pi : Q \to Q/Z$ (using the usual embedding of these prime groups), and its integer realization via the transfer theorem, have critical density respectively one or zero. This result permits completely arbitrary local acceptance functions and nonconstant maps of sparse support. It requires all digit values to be tested by the one finite abelian sum rule; additional coordinate restrictions, such as restricting the digits to $0, \pm 1$, fall outside the statement.

A finite pattern lemma. Suppose p is prime to $|K|$, $f : C_p \to K$, $f(0) = 0$, and $H = \langle imf\rangle$. For any $m \geq 1$, write J for the $2^m - 1$ nonempty subsets of $[m]$. The pattern vectors

$$(f(\sum_{j\in F} x_j))_{F\in J}, x_1, \ldots, x_m \in C_p,$$

generate the full abelian group H^J. Otherwise a nonzero character of H^J vanishes on all patterns. Write it in additive circle notation as $\sum_F \chi_F(y_F)$ with characters on H, and take an inclusion-maximal F_0 with nonzero χ_{F_0}. Difference the identity on C_p^m once in every variable indexed by F_0, using arbitrary increments in those variables. Terms missing any of these variables vanish, and terms with strict supersets of F_0 already have zero character. Consequently all $|F_0|$th mixed additive differences of $\chi_{F_0} \circ f$, as a function on C_p, are zero.

Such a function is constant. Indeed its values are in the finite torsion subgroup of the circle annihilated by $|K|$. On functions with values in this subgroup, averaging over C_p makes sense because multiplication by p is invertible there. On the finite group of functions of average zero, the difference $T - 1$ at generator one has trivial kernel: a constant of average zero must be zero. Thus every power of this operator is injective on that group. Subtract

the constant average from $\chi_{F_0} \circ f$, and use the zero iterated difference at generator one to get constancy. Its value at zero is zero, so χ_{F_0} vanishes on `im f` and hence all of H, contradiction.

Combine different axes. Fix m and a tail starting after any prescribed cutoff and after all primes dividing $|K|$. For each subgroup type H occurring infinitely often, pass to an infinite subsequence of its axes on which the entire set of attainable patterns in H^J is the same (there are only finitely many such sets). By the lemma, this common finite set generates H^J; its semigroup of positive finite sums equals the group, since the group is finite. Thus every target in H^J can be written as a sum of patterns on finitely many *distinct* axes of this subsequence, placing one desired pattern on each axis. Taking all infinitely recurring subgroup types allows every target in $(H_\infty)^J$.

Choose $t \in L \cap H_\infty$ and demand the constant target $(t)_{F \in J}$. We may ensure all nonempty sums are nonzero, even when $t = 0$. First reserve m distinct tail coordinates of recurrent types and let the jth generator alone have value one on its jth private coordinate, with all other generators zero there. Their preliminary pattern values lie in $(H_\infty)^J$; correct them to the desired target using disjoint subsequent coordinates as above. (There is always at least one recurring type since K has finitely many subgroups, including the trivial possibility.) The resulting generators $w_1, \dots, w_m$ have every nonempty sum in V, and the private axes ensure none is zero. Choosing arbitrarily many such generators gives all positive-density multiple recurrence by the finite density Hales–Jewett theorem. Explicitly average the maps $a \mapsto T^{\Sigma a_i w_i} x$ on $[h]^m$; every word has the same event marginal, so some translate contains a combinatorial line, whose difference is a nonempty sum of the generators. Arbitrary lifts to Q of the generators give the same finite-sums conclusion for the saturated rational family.

If $L \cap H_\infty = \varnothing$, discard the finitely many indices of all transient subgroup types. Then the image sums on every remaining tail belong to H_∞, so no tail direction is accepted. The near-full root-rotation repair of finitely many head-prime components, or the squarefree-tail formula [R241], gives critical density one after saturation. No conclusion for intermediate thresholds follows; this rules out the entire stated class of constructions.

Nonabelian finite groups also work One may replace K by an arbitrary finite group, require $f_i(0) = 1_K$, and take the word product in increasing coordinate order, ignoring identity letters. The same dichotomy holds with H_∞ the subgroup generated by all recurring image-subgroup types. The proof of the finite pattern lemma extends as follows.

Let $L_p \le H^J$ be generated by all attainable patterns for an axis of prime order coprime to $|K|$. When there is just one slot this is H. Otherwise $k = |J| \ge 3$. The projection onto any pair of distinct slots is H^2: the two different subset-sum linear forms are linearly independent over F_p (their nonzero 0–1 coefficient vectors have different supports). One can prescribe the two input values independently, in particular obtain every pair of image values with the other entry equal to $f(0) = 1$.

This implies $\gamma_{k-1}(H)^J \subset L_p$, where γ_d is the dth lower-central group. Indeed fix a target slot, list the other $k-1$ slots, and use the pair projections to choose $k-1$ elements of L_p each with arbitrary component at the target slot and identity at the corresponding other slot. Their left-nested commutator vanishes off the target slot and realizes any simple commutator of weight $k-1$ there. Such commutators and their conjugates generate the indicated lower-central group, and conjugations in the chosen slot can be effected using the surjection of L_p onto that coordinate.

The abelian pattern lemma already proves surjectivity onto $(H_{\mathrm{ab}})^J$. Quotient by $\gamma_{k-1}(H)^J$; this leaves a finite nilpotent ambient group, still with full image on its abelianization. A proper subgroup of a finite nilpotent group lies in a maximal subgroup, and such a maximal subgroup is normal of prime index and contains the commutator subgroup. Hence this image is the entire nilpotent quotient. Since the kernel was shown to lie in L_p, we obtain $L_p = H^J$ as claimed.

For each recurring subgroup type, infinitely many axes have one fixed attainable pattern set. Their products in a finite group realize every element of the subgroup this set generates, because inverses can be expressed by positive powers. Place letters of any chosen word on fresh axes in the required order. The same applies to interleaving words from different recurring types, since each supplies arbitrarily late indices; their generated group on pattern tuples is $(H_\infty)^J$. Put private axes before the correcting active axes and use the multiplicative correction $P_{\mathrm{pre}}^{-1} P_{\mathrm{target}}$ to realize the constant accepting target. All subsequent recurrence and empty-tail conclusions are unchanged.

The invertibility in a finite group matters. For example the finite monoid tracking zero, one, or at least two marked coordinates, together with a dead state for disallowed local values, can encode arbitrary fresh-prime templates supported on exactly one coordinate. The group proof makes no assertion about that monoid.

One elementary exactly-one case. Suppose each axis has disjoint normal and marked sets $N_i, M_i \subset C_{p_i}$, with $0 \in N_i$, and the allowed directions have exactly one marked digit and all other digits normal. If infinitely many axes have nonempty marks and arbitrarily nearly full normal sets, in the sense that a subsequence has $|N_i|/p_i \to 1$, then every tail contains FS_m for all m. For each nonempty $F \subset [m]$ use a private fresh axis with normal deficit $< 1/2^m$. Fix a marked element s_i. Choose the local generating values uniformly subject to $\sum_{j \in F} x_j = s_i$. For every other nonempty F' the sum indexed by F' is uniform, by independence of the two different 0–1 coefficient vectors. A union bound shows there is a choice making every such other sum normal. Combining these $2^m - 1$ axes, each nonempty sum has its uniquely prescribed mark and all other digits normal. It is nonzero since zero is normal. Also, if a sequence of single-axis marked sets has density tending to one, sampling the m generating values on one such axis gives all nonempty sums marked by the same union bound and again full recurrence. These arguments do not require an unproved cyclic root estimate.

R114. Additive finite-palette colorings on squarefree rows

Scope and status. Result with proof.

Result. A fixed additive finite-palette coloring cannot obstruct a direction using too many nonconstant prime axes; binary support embeddings are likewise restricted.

Scope. Fixed palette and progression length, with sufficiently large primes; no general bound for arbitrary colorings.

Let q be a fixed prime, let $h \geq 2$, and on $G = \oplus C_{p_i}$ color by $c(x) = \sum_i [f_i(x_i) - f_i(0)]$ in F_q, with arbitrary local maps. Assume each active f_i is nonconstant and $p_i > \max(q, q^{h-1})$. Inactive coordinates can be removed. Any direction admitting no monochromatic h-run for this color has at most $M(q,h)$ active coordinates. The statement only limits this class of coloring certificates and gives no uniform bound on the number of bad prime steps on a single coordinate.

For any active coordinate and nonzero step s, let $P(s) \subset W = F_q^{h-1}$ consist of the possible profiles $(f(x+ts) - f(x))_{1 \leq t < h}$. Their affine span is all of W. Otherwise a nonzero linear functional is constant on $P(s)$. Its constant must be zero by averaging over x, since p_i is invertible in F_q. Write this linear recurrence as $\sum_{0 \leq t \leq j} a_t f((n+t)s) = 0$, $a_j \neq 0$, $1 \leq j \leq h-1$ (the $t = 0$ coefficient is minus the sum of the others). The length-j state determines the next symbol. If two such states match on a periodic word, the entire intervening difference is a period. A nonconstant word of prime period p_i therefore has p_i distinct states, contradicting $p_i > q^{h-1}$.

There are at most $2^{|W|}$ profile types. Let A be one profile and $a \in A$. The set $A - a$ contains zero and generates W. Its iterated sumsets increase strictly until equal to W: equality to the preceding sumset gives invariance under $A - a$, hence under all of W. Thus $nA = W$ for $n \geq |W| - 1$. If the number of active components of a direction exceeds $2^{|W|}(|W| - 2)$, some type occurs at least $|W| - 1$ times. Its sum is all of W, even after choices of the other profile values. Choose the components to sum to zero. Their chosen starting points provide a monochromatic h-run. This proves the bound, including $|W| = 2$ where every single profile already equals W.

There is also no nontrivial abelian-group embedding of the support of the h-word distribution at a nonzero step of a nonconstant *binary* prime word for $p > 2^{h-1}$. Explicitly, if $g : C_p \to \{0,1\}$ is nonconstant, $s \neq 0$, A is any abelian group, and maps $u_t : \{0,1\} \to A$ satisfy $\sum_{t<h} u_t(g(x+ts)) = const$ for every x, then all u_t are constant. If any is nonconstant, take the largest such index j. Given the preceding j bits the equality determines the value of the injective two-symbol map u_j, hence determines the next bit. When $j = 0$ it forces g constant directly; otherwise the same state-count contradiction applies. This is an assertion about support; it supplies no lower bound for the mass of individual words uniformly in p.

R115. Finite nonzero angular centers with critical Hamming errors

Scope and status. Result with proof.

Result. Finite nonzero angular centers with square-root Hamming errors admit arbitrarily dense avoiders at a uniform finite horizon.

Scope. Integer characters and sufficiently large prime cyclic groups admitting the required character roots.

This extends the stable-cosine argument for the single antipodal center. Fix a finite $K \subseteq \mathbb{T} \setminus \{0\}$ and constants c, C. For each growing dimension n, let allowed integer steps d be such that all but $C\sqrt{n}$ coordinates of $(d\alpha_j)_{j\le n}$ are within $c/\sqrt{n}$ of K. For every $\epsilon > 0$, there are N, L independent of the frequencies such that, for $n \ge N$, there is an integer set of upper natural density $> 1-\epsilon$ avoiding every L-term line with allowed difference. The same conclusion holds on prime cyclic groups whose orders tend to infinity, with the frequency coordinates being characters there. This is a statement about domains admitting the finite character roots used below. The whole torus with its translations as group elements need not admit those roots.

For the proof choose a nonnegative real polynomial $P(X) = \sum_{m=0}^{m_0} w_m X^m$, $\sum w_m = 1$, annihilating all $e(\beta), \beta \in K$. For a root of unity use its finite orbit average. For an irrational frequency finitely many orbit points have zero in their convex hull, because the orbit is dense and can approximate three vertices whose triangle contains zero in its interior. Multiply the single-center polynomials.

At root scale R, consider $F(t) = n^{-1/2} \sum_{j\le n} \mathrm{Re}(z_j e(t\theta_j))$, where z_j are unit initial phases and $R\theta_j = d\alpha_j$ modulo one. Then $\sum_{m=0}^{m_0} w_m F(t+mR)$ is at least minus a fixed constant depending on K, C, c. The regular terms cost at most $2\pi c \sum_m m w_m$ after normalization by Lipschitz continuity of P on the circle. The exceptional terms cost at most C. Thus some slot $t + mR$ has value $> -L_0$ for a fixed L_0 chosen with slack.

Keep only values satisfying $|F| \le B$ and a fixed boundary gap around $-L_0$; use one independent Haar phase vector per root row. At a fixed sampling point F has the distribution of $n^{-1/2}$ times a sum of independent Haar cosines. The central limit theorem gives a nontrivial limiting lower tail below $-L_0$, so the number r of independent root rows can first be chosen such that simultaneous activation costs $< \epsilon/3$. Next select B large and the boundary gap small so the total additional cuts cost $< \epsilon/3$ for all large n.

On one residue $t = b + Ru$, write the regular angles as $\beta + a_j$, $\beta \in K$, $|a_j| \le c/\sqrt{n}$. At fixed integer u their sum equals

$$\mathrm{Re} \sum_{\beta\in K} e(u\beta)(A_{\beta,n} + uB_{\beta,n}) + O_c(u^2/\sqrt{n}).$$

The $B_{\beta,n}$ are bounded by $2\pi c$. The exceptional coordinates give the real part of a Fourier–Stieltjes transform of norm at most C. Combine conjugate frequencies before taking limits, so the displays become linear combinations of the pairwise distinct $e(\pm\beta)$. On an expanding, centered legal segment the values $|F| \le B$ at the first as many integer indices as the number of distinct frequencies. The Vandermonde matrix there is invertible, so the combined coefficients $A_{\beta,n}$ are bounded in terms of B, K, C, c. Extract them, the slope coefficients, and the exceptional measures weak-star. The limit is bounded for all integer u. Therefore the

trigonometric polynomial $b(u)$ made of the combined slope coefficients must vanish identically: $ub(u)$ is bounded, forcing $b(u) \to 0$ as $u \to +\infty$, whereas its Cesàro mean square equals the sum of squared coefficients. The entire limiting profile is now uniformly Fourier–Stieltjes, with a bound independent of the word, including the exceptional part. These arguments only need $n \to \infty$; failure of any uniform eventual pair N, L would supply legal runs with both n and their lengths tending to infinity.

The margin implies uniform stability of the limit binary gate on each residue by the Hilbert-transform estimate in the stable witness note [D18]. Its stable regularity argument approximates each residue code by a binary word of uniformly bounded period up to arbitrarily small upper Banach density error. Take the next root a multiple of all preceding approximating periods after interleaving the finitely many residues. Off a union of at most $r(m_0 + 1)^r$ translates of these discrepancy sets, use the activation stencil above successively at each root, keeping earlier activated approximants fixed. There is a site where all gates activate. The conclusion for integer powers follows by independent phase randomization and reverse Fatou. In a prime group, take roots in its actual character group when the chosen primes exceed all root orders, so that the constructed masks really are periodic on that group; phase randomization then supplies a deterministic mask by averaging.

Allowing an unconstrained proportion of zero-centered coordinates invalidates the positive stencil. This extension therefore leaves the single-marked construction with backgrounds $0, \beta, -\beta$ unresolved.

R117. A finite set of moving angular centers

Scope and status. Result with proof.

Result. Moving finite angular centers and the specified controlled zero-mode extensions still force unrestricted endpoint one.

Scope. A uniform finite-row result under explicit separation, error, and Gaussian-width bounds.

The endpoint-one exclusions in the critical Hamming note [R037] and the fixed finite-center extension [R115] extend to the following direction families on the integers, or on prime cycles large enough to supply the finitely many character roots. Fix positive a, γ, C and an integer $s \geq 1$. Given n characters, suppose every eligible d admits at most s angular centers, all at distance $\geq a$ from zero, such that all but $C\sqrt{n}$ of its character phases are within $Cn^{-\gamma}$ of one of those centers. Centers may depend on d and n and may collide in the limit. For every $\eta > 0$ there exist h, n_0, uniform in the lists and directions, such that all these steps for $n \geq n_0$ have an h-avoider of density $> 1 - \eta$.

Use the same independent normalized cosine rows with character roots, upper and lower cuts at a fixed B, and a gap around $-L$ as in the critical Hamming argument. Take $L > C+2$ first, enough rows to make joint activation negligible, then B and the gap width from the Gaussian one-slot marginals. If arbitrarily long all-retained stretches existed for $n \to \infty$, fix a residue of a given root stride and split its raw normalized sum into regular and exceptional parts. The latter is the real part of a Fourier–Stieltjes sequence of norm $\leq C$, uniformly over all its frequencies and coefficients. Therefore on the observed stretches the regular part is bounded by $B + C$. Extract pointwise limits on the integer stride indices and subsequential limits of the centers.

Let T now denote the shift on the stride index. For each n form the monic shift polynomial P_n whose roots are both signs of the center frequencies (on the unit circle), each repeated k times, where $k\gamma > 1/2$. Use a subsequence on which the number of centers is constant. Each regular complex exponential is killed by this polynomial up to $O_{k,s,C}(n^{-k\gamma})$, since one factor is within $2\pi Cn^{-\gamma}$ and all others are bounded by two. Consequently $\|P_n(T)F_{\mathrm{reg}}\|_\infty = O(\sqrt{n}\, n^{-k\gamma}) \to 0$. Coefficient convergence of P_n gives a bilateral bounded limiting solution of $P(T)F = 0$. The ordinary constant-coefficient recurrence formula expresses it as a sum of polynomial multiples of the limiting exponentials. Boundedness removes every positive-degree coefficient: the highest-order coefficients would give a nonzero finite exponential sum tending to zero at infinity after division by that power, impossible by averaging against each distinct root. There are at most $2s$ frequencies, each at distance at least a from zero, and each coefficient is bounded by $B + C$ via its Fourier mean. In particular the full row limit on this residue has Fourier–Stieltjes norm at most $2s(B + C) + C$.

By the geometric-series bound the absolute average of the regular part on every K consecutive stride positions is at most $2s(B + C)/(K \sin(\pi a))$. Choose a fixed K making this less than one. Each such window therefore contains a full-row value $\geq -C - 1 > -L$. The fixed gate gap and Fourier–Stieltjes norm bound yield the same stable periodic approximation on each residue as in the critical Hamming note. Choose its discrepancy ζ so $rK^r\zeta < 1$, and successive roots divisible by the earlier full periods. In the limiting words avoid the discrepancy sets shifted by all possible K^r partial sums of roots with coefficients from $0, \ldots, K-1$. From that starting position choose one of the K offsets for each row to activate it, while preserving the earlier activations. This gives simultaneous activation inside the assumed

all-retained limiting run, a contradiction. All constants above depend only on the stated fixed parameters and η.

The original near-half case is the single center $1/2$, now at any polynomially vanishing width. For fixed finite center sets this argument also permits arbitrary sequence widths if there are no exceptions and one can subsample a slowly growing number of characters. The statement above with unrestricted exceptional locations uses the displayed rate. These exclusions by themselves do not give an interior threshold.

A large near-zero group with bounded width May additionally allow a set I_d of regular phases within $Cn^{-\gamma}$ of zero itself, where $\max_d |I_d|/n \to 0$ and

$$W_0 = n^{-1/2}\mathbb{E}\sup_d \left|\sum_{i\in I_d} g_i\right| \le C_0$$

for independent standard real Gaussians. These sets can be substantially larger than $\sqrt{n}$. At a site with independent Haar phases x_i truncate the supremum of the corresponding normalized cosine sums at $C_0' > \sqrt{\pi/2}C_0$. Its expected value is at most $\sqrt{\pi/2}C_0$: conditional on their magnitudes the cosines are symmetric signs, the independent contraction principle applies, and Jensen on the absolute Gaussian magnitudes compares signs to Gaussians. The supremum is $2\pi\sqrt{\max|I_d|/n}$-Lipschitz in product Euclidean distance, so the product torus Poincaré inequality implies the discarded mass tends to zero. This also holds for every rooted row with its independent Haar starting phases. Along each long retained trajectory the near-zero group's normalized stream is bounded by C_0', and its k-th difference vanishes in the bilateral limit by the same $\sqrt{n}\,n^{-k\gamma}$ estimate. It therefore tends to a constant of absolute value at most C_0'. The full limiting stream has bounded Fourier–Stieltjes norm and some value $\ge -C - C_0' - 1$ in every fixed long root window. Choose $L > C + C_0' + 2$, then perform the same synchronization. This covers moving zero-mode holes of small measure and bounded Gaussian width, even when they can contain any specified finite sample.

An unbounded number of distinct nonzero centers More generally partition the nonzero regular sites for each d into any number of clusters $I_{d,l}$, each within $Cn^{-\gamma}$ of its own center $\xi_{d,l}$, still a uniform positive distance a from zero. Replace the fixed bound on the number of centers by

$$n^{-1/2}\mathbb{E}\sup_d \sum_l \left|\sum_{i\in I_{d,l}} g_i\right| \le C_1.$$

At each retained position truncate the analogous sum for the complex phases $e(x_i)$ at C_1'. Its mean is at most $2\sqrt{\pi/2}C_1$, by applying the preceding contraction bound to the real and imaginary parts separately. Here C_1' may be chosen *after* the activation threshold and the number of rows, since it affects the length of the finite root window but not its required peak. Markov's inequality gives the needed small total cost with no concentration assumption.

On a stride residue place its complex normalized cluster coefficient at its center $\xi_{d,l}$, and demodulate that coefficient by $e(-t\xi_{d,l})$ at stride time t. This defines a signed complex measure $\nu_{n,t}$ on the circle of total variation bounded by C_1' along the whole retained stretch. Uniformly in t, the total variation of its k-th time difference is at most $O_{k,C}(\sqrt{n}\,n^{-k\gamma})$: each

input exponential now has frequency equal to its individual deviation from the assigned center. By diagonal weak-star compactness take a bilateral limit ν_t, bounded in total variation and satisfying $\Delta^k \nu_t = 0$. Pairing against every continuous test gives a bounded scalar polynomial and hence a constant, so $\nu_t = \nu$ is independent of t. The nonzero regular limiting stream is $\Re \int e(t\xi)\, d\nu(\xi)$. Its support stays outside the fixed zero neighborhood. The geometric-series bound now gives window average at most $C_1'/(K \sin(\pi a))$ in absolute value. The bounded near-zero and arbitrary exceptional pieces may be included as above, and the same stable periodic argument concludes. This criterion is a uniform finite-row exclusion, not only a per-direction correlation estimate.

R120. A dimension-only high-density bound for moving circle registers

Scope and status. Result with proof.

Result. Full-rank moving circle noise gives a dimension-only high-density bound independent of temporal degree.

Scope. For r observable channels the sufficient deficit is below r^{-r}; the system classes satisfying the noise reduction are specified.

Let $G(T)$ be any r-by-m matrix of integral Laurent polynomials. At time t its noise contributes

$$Z_t = \sum_{a,k} G_{a,k}\, Y_{t+a,k} \quad \text{in } \mathbb{T}^r,$$

for independent Haar circle inputs $Y_{s,k}$; $G_{a,k}$ denotes the column vector of coefficients at that lag and input. Suppose these vectors span $\mathbb{Q}^r$. Given arbitrary deterministic offsets c_t, and measurable sets $B_t \subset \mathbb{T}^r$ all of measure at least $1-\epsilon$ with $\epsilon < r^{-r}$, every finite window has

$$\mathbb{P}[Z_t + c_t \in B_t \text{ at every required } t] > 0.$$

The positive bound for a fixed window, matrix and density margin is uniform in the offsets and the individual sets. In particular no bound on the number or magnitude of temporal lags occurs in the density cap. For $r = 1$ the statement is valid whenever the sets have a common positive measure bound.

Order the finitely many pairs (a, k) lexicographically. By an integral unimodular output transformation give the r rows distinct *last* nonzero coefficient positions. Indeed at the largest remaining nonzero position integer gcd operations on the remaining output rows clear all but one entry. Put that row aside and continue with lower positions. The assumption on the coefficient span gives full row rank of this finite constant matrix, so every row acquires a distinct pivot. This is an automorphism of the output torus, preserving Haar and measurability. Order the resulting coordinates by increasing pivot.

Pick $\epsilon^{1/r} < \gamma < 1/r$. For each B_t, starting with the full good r-tuples B_t, call a shorter prefix good if its next-coordinate good continuations have Haar measure $> 1-\gamma$. At each truncation the bad-prefix measure is at most $1/\gamma$ times the previous bad measure. The empty prefix is good since $\epsilon/\gamma^r < 1$. Null choices of sections are handled by using Borel representatives and the actual coordinate integrations.

Expose the finitely many inputs needed for the window in increasing (s, k). Whenever one is a pivot for a row coordinate, that coordinate has the form $nY_{s,k}$ plus already known data and offset, with $n \neq 0$. Its earlier coordinates have already been processed because their pivots are earlier. If earlier prefixes were good, its good-continuation set for this Haar variable has measure $> 1-\gamma$, by surjectivity and Haar preservation of multiplication by n. For each of the r output coordinates there is at most one requesting time at a fixed input: the time must equal $s-a$ for that coordinate's unique pivot (a, k). Hence at most r requests meet here. Their simultaneous conditional probability is at least $1 - r\gamma > 0$, even if those requests are otherwise completely dependent. Multiplication through the finite exposures gives the theorem. It also works if the offsets are jointly random but frozen independently of the added fresh Haar noise.

Application to the mixed affine-register bounds. In the scalar and vector table models [R154], or the bounded-dependency relative eigenregister models [R153], let R be the normal controlled Laurent module on an invariant environment. Over $\mathbb{Q}(T)$ take a basis for its orthogonal complement, clearing denominators to make Laurent polynomial columns $v_l(T)$. Let $K_f \le \mathbb{T}^r$ be the connected subtorus with rational tangent space spanned by their coefficient vectors. The integer process noise obtained from those columns annihilates every row of R. All of the limiting charged normal trajectory laws are invariant under adding it even with the entire Y-trajectory held fixed, by Fourier vanishing outside R. This includes trials with true pure corrections: the normal-site survival upper bound was uniform in the added pure multipliers.

A constant integral character l vanishing on K_f belongs to the rational span of R: it annihilates every basis column as a polynomial. Clearing denominators gives $D_l(T)l$ as an integral Laurent polynomial combination of rows of R, with a nonzero scalar polynomial D_l. Thus on good pure axes the sparse clock argument [R153] makes a suitable fixed integer multiple of $l \cdot \mathbf{b}_i$ constant off $o(p_i)$ sites, after also multiplying by allowed odd site-specific determinants. For every controlled row all of whose coefficients vanish on K_f, every separately shifted Q_P coefficient lies in that same constant lattice. The normal Q_P equation therefore gives zero pure signature on typical sites after the multiplier. Taking a basis on the quotient and increasing the finite character list as before shows that the limiting joint law on Y and the quotient registers $\mathbb{T}^r/K_f$ is identical for true and uncharged samples, even for every nonempty subset of the DHJ trials. The relative setting permits an arbitrary bounded test of the base since all the corrections are high.

If $K_f = 0$, this already gives full positive recurrence. Otherwise use a unimodular choice of coordinates putting $K_f = \mathbb{T}^{r'} \times 0$ for $1 \le r' \le r$. One-slot Haar-noise invariance implies that the initial mobile registers are Haar conditionally on all the base and quotient data in the true limiting law; stationarity gives the same original one-slot law. For an event of mass $> 1 - (r')^{-r'}$, let B be a positive-measure base/quotient subset on which its mobile sections have mass exceeding the same threshold by a fixed margin. The uncharged word gives a positive DHJ return to B. By the equality of projected trajectory laws choose a true limiting word with positive all-B mass. Conditionally on such a trajectory (and any other residual offsets), add the independent noise $v_l(T)$. Its mobile coefficient span is full by definition, so the preceding pivot argument gives positive simultaneous A-membership. The fixed one-slot marginals extend the limiting conclusion to measurable events and give an actual admissible step.

In particular $1 - r^{-r}$ is an upper bound uniform over these entire r-channel classes and cover heights. The sharper three-channel conclusion [R154] still gives $1/2$ when $r = 3$. The same pivot estimate can replace a temporal-degree-dependent local-lemma bound in the paired toral control argument [R190] whenever its corresponding projected-odd-coset conclusion applies. The bound depends on the number of independent observable channels; it does not alone give a uniform upper below one over all stationary systems.

R125. Euclidean transfer of weighted finite progressions, with exact repair for intervals

Scope and status. Proved subcases and unresolved extensions.

Result. Euclidean transfer gives per-direction approximate avoidance, with exact repair for the stated bounded-complexity and Gaussian-width classes.

Scope. Uniform averaged forcing and arbitrary unrestricted support repair remain unresolved.

Qualifications. Retain the distinction between approximate avoidance per direction and exact simultaneous avoidance; arbitrary support repair is not proved.

This note distinguishes exact positivity for a particular direction and a positive lower bound for an average with respect to a direction law. On the mixed-prime graph examples, the latter fails uniformly even far above the mass of the known short avoiding events. The full exact question remains open.

A random lattice section There exists a random section $s : \mathbb{T}^m \to \mathbb{R}^m$ of the quotient by the integer lattice such that for every $x, u \in \mathbb{R}^m$

$$P[s(x + u \bmod Z^m) - s(x \bmod Z^m) \neq u] \leq 2\pi\|u\|_2.$$

The constant is independent of the dimension. At rational torus points the section is rational. The construction is the periodic tessellation/rejection procedure of Kindler, O'Donnell, Rao and Wigderson. Here is a proof with the particular kernel needed below.

Let $g_1(z) = \max(0, \cos(\pi z))1_{[-1/2,1/2]}(z)$, $g(z) = \prod_{i=1}^m g_1(z_i)$, and $f = g^2$. Sample iid proposals z uniform in $([0,1)^m)$ and b uniform in $([0,1])$. For an input x, take the first proposal with $b < f(x - z - k)$ for some $k \in \mathbb{Z}^m$, and set $R(x) = k$. That k is unique, and the probability of accepting a trial is $c = \int f = 2^{-m} > 0$. We only use this rule on countably many points at a time. It is lattice equivariant, and $s(x \bmod \mathbb{Z}^m) = x - R(x)$.

For two points, the probability that a single proposal accepts one but not the other with the same integer label, counted at most twice when the labels differ, is bounded by $\|f(\cdot+u)-f\|_1$, by integrating first in b, summing over the possible k, and unfolding the fundamental cube. The first proposal accepted by either point therefore disagrees with probability at most $\|f(\cdot + u) - f\|_1/c$. Now $\|f(\cdot + u) - f\|_1 \leq 2\|g\|_2\|g(\cdot + u) - g\|_2 \leq 2\pi c\|u\|_2$. For this last inequality integrate the directional derivative of g on the segment: the coordinate partials are mutually orthogonal in L^2, with squared norms $\pi^2 c$. There is no boundary jump in g, which vanishes on the faces.

Supersaturation with no increase of the forbidden length Take a finite product $K = \prod_i C_{p_i} \subseteq \mathbb{T}^m$ by dividing coordinate i by p_i. Equal p_i's are permitted. Let $V \subseteq K \setminus \{0\}$, and let $D \subset \mathbb{Q}^m$ contain the rational preimages of V under quotient by the integer lattice. If a probability-preserving action U of $mathbbQ^m$ has an event A of mass a exactly avoiding h-progressions along D, define a random finite mask by

$B(y) = 1_A(U_{s(y)}z)$, with z sampled from the system independently.

Its expected density is a. Write $\lambda_h(B;v) = |K|^{-1}\sum_x \prod_{j=0}^{h-1} B(x+jv)$. For every $v \in V$ and any lift $u \in \mathbb{Q}^m$ of v,

$$E\lambda_h(B;v) \le 2\pi(h-1)\|u\|_2.$$

Indeed except for probability at most the displayed quantity every successive rounded difference is exactly u, giving an actual rational progression. For every realized rational starting point the product is almost surely zero by exact avoidance and invariance. Averaging over starts proves the claim. In particular if every finite mask of density at least $a_0 < a$ has (ν)-weighted correlation at least $c_0 > 0$, the expected weighted correlation is at least $c_0(a-a_0)/(1-a_0)$. Consequently fixed-h supersaturation can transfer upwards when the (ν)-mean of the Euclidean lift length is small, in arbitrary rank. The estimate does not require any independent behavior of progresses for different velocities.

Soft obstructions on the signed prime graphs For the paired construction and its bipartite graph variant take finite horizontal and vertical vertex sets with primely periodic coordinates p_i (all distinct), and signed unit velocities $v_i \in \{0,1,-1\}$. Let $Q(x) = \sum_{i\sim j} x_i x_j$, summing each edge once, and require an odd number of edges with both endpoints active. Then $(Q(v))$ is odd for every allowed signed step, independently of the active counts or directions. Let $M = 2^b \ge 16$. For integers a odd and (b', c'), the polynomial $at^2 + b't + c' \pmod M$ attains all points in some residue class modulo (8) as t ranges over $([0, M-1])$. When (b') is even, complete the square, using that every 1 mod 8 is an odd square modulo every 2^b. This last assertion follows by induction, adjusting an odd approximate root by 2^{k-1} to toggle its square by $2^k \bmod 2^{k+1}$. When (b') is odd, the polynomial minus its constant takes both lifts of each attained even value on passing to the next power of two, by changing t by the previous modulus; it therefore takes all even values. Hence deleting eight consecutive residues of $Q(x) - w \pmod M$ blocks every M-term lifted progression with an allowed step. Random uniform w mod M gives mean density $(1 - 8/M)$ on any finite sampling set.

For $y \in \prod_i C_{p_i}$ define the random integer lift $x_i = p_i s_i(y)$. Apply that quadratic mask to x, using independent w. In the resulting random group mask the expected density is $(1 - 8/M)$ and simultaneously every signed allowed step has

$$E\lambda_M(B;v) \le 2\pi(M-1)\Big(\sum_{i\,active\in v} p_i^{-2}\Big)^{1/2}.$$

Translate the mask by a uniform group element to make its law stationary. On an infinite direct sum use the masks on increasing finite vertex sets (the other generators can act trivially at this stage), and take a weak limit of stationary binary laws. Every fixed word eventually lies in the finite set, so the same bound holds for all steps, including ones with arbitrarily many active normal or marked axes. The quadratic proof needs only the odd edge count and even works for arbitrary bounded signed integer velocities satisfying $(Q(v))$ odd, with v_i^2/p_i^2 in the sum. The unsigned case is included. Since $\sum_{i\ge J} 1/p_i^2 \to 0$, every fixed M of this form has stationary witnesses of mean $(1 - 8/M)$ whose M-correlations tend to zero uniformly over all direction laws on increasingly late tails. This precludes a tail-uniform positive weighted lower bound at that length and density. For fixed J, the upper bound in general does not tend to zero as we add more vertices. In particular the construction does not give exact avoidance on the full family.

An exact repair for one ordered interval per panel For each coordinate prescribe a signed motion v_i, possibly zero, and let $u_i = v_i/p_i$ with $|u_i| \le 1/4$. In a finite ordered panel of coordinates let $u_F = (u_i 1_{i\in F})_i$, where F ranges over contiguous intervals of indices, including the empty interval. Put $\rho = (\sum_i |u_i|^2)^{1/2}$. For $\rho \le 1/10$, at each fixed input y, the random section preserves **every one-step interval move simultaneously** with probability at least $1 - C\rho$. The statement is dimension free and has no count of the possible intervals.

To prove this view the acceptance thresholds in a fundamental cube as periodic functions. Under reference probability density proportional to f, the coordinates z_i are independent. Write $g_*(t) = |\cos(\pi t)|$ on the circle and let $r_i = g_*(z_i + u_i)/g_*(z_i)$, defining it arbitrarily on the reference-null set where the denominator vanishes. We have $Er_i^2 = 1$, and $a_i = Er_i \ge 1 - \pi^2 u_i^2/2$, by the L^2 translation estimate for g_*. Thus for $W = \prod_i r_i/a_i$ one has $EW = 1$ and $E|W-1|^2 = \prod_i a_i^{-2} - 1 = O(\rho^2)$.

Let P_l be conditional expectation on coordinates up to l and S_k conditional expectation on coordinates from k on. By independence $P_l S_k W$ is exactly the product of the normalized ratios on $([k,l])$ *onewhen*$(k > l)$. Applying Doob's L^2 maximal inequality twice gives

$$\|\sup_{k,l} |P_l S_k (W-1)|\|_2 \le 4\|W-1\|_2.$$

In detail take the absolute value after the outer conditional expectation, bound its integrand by the supremum in k, and apply the forward and the reverse maximal inequalities. Since all interval products of the deterministic a_i's are within $O(\rho^2)$ of one, it follows that $E\sup_F |\prod_{i\in F} r_i - 1|^2 = O(\rho^2)$. Cauchy–Schwarz then shows $E\sup_F |\prod_{i\in F} r_i^2 - 1| = O(\rho)$. In particular the one-trial probability of acceptance by at least one input is at most $c(1+C\rho)$, where $c = \int f$. The one-trial probability that every input accepts with the same integer label is at least $c(1 - C\rho)$: use the minimum relative threshold, and exclude those reference z's at which some centered coordinate $z_i + u_i$ crosses a fundamental face. Their probability is $O(\sum_i |u_i|^3) = O(\rho^3)$ because the one-dimensional reference density is proportional to cosine squared. Taking the first trial accepted by any input proves simultaneous agreement.

For a fixed number d of disjoint panels, each with its own interval choice, independently sample a rounding for each panel. The joint disagreement probability at any start is $O(\sum_{k=1}^{d} \rho_k) \le O(\sqrt{d}\rho)$. Mark a finite torus site if it survives the lifted quadratic mask **and** all the prescribed one-step moves are preserved there. The expected loss from this second cut is $O(\sqrt{d}\rho)$. An accepted trajectory of length M in any allowed direction would have a common integer increment at each step, hence would contradict the quadratic mask. Translate to stationarity and pass to the finite-volume limit as above. Thus for families with an odd edge detection and direction supports given by one interval in each of finitely many panels with prescribed signs, the tail capacities have limit one, even when the full reciprocal sum diverges and arbitrarily many coordinates may be active. The proof works as well for any uniform lifted blocker on the integer coordinate steps, in place of this particular quadratic. No bound is asserted for unboundedly many panels or arbitrary independent sign choices within an interval.

A Gaussian-width exact repair for arbitrary direction sets Let U be a finite set of velocity vectors in $\mathbb{R}^m$, containing zero, with $\max_{u,i} |u_i| \le 1/4$. Choose weights $a_i > 0$, write $D_a u = (a_i u_i)_i$, let G be a standard Gaussian vector, and put

$$w = E \sup_{u \in U} |\langle G, D_a u \rangle|, \rho = \max_{u \in U} \|D_a u\|_2, \tau = 2 \sum_i \exp(-a_i^2/32).$$

There is a random lattice section for which the one-step moves of every $u \in U$ are simultaneously preserved at any fixed point with probability at least

$$(1-\mathrm{w}-\rho^2/2-\tau)\ \exp(-\mathrm{w}-\rho^2/2).$$

Use the same proposals as before but the acceptance kernel $f(x) = f_0(x) 1_{[-1/2,1/2]^m}(x)$, where $f_0(x) = \exp(-\|D_a x\|_2^2/2)$. Let $Z = \int f_0$. The one-trial probability of any acceptance, after unfolding a sum over labels which dominates their maximum, is bounded by $\int \sup_{u \in U} f_0(x+u)\, dx$. Divide by Z to get an upper bound $E \exp(\sup_u |\langle G, D_a u \rangle|) \leq \exp(w + \rho^2/2)$, by the exponential-moment inequality for a ρ-Lipschitz Gaussian function. Conversely the probability that all inputs accept the same label is at least

$$Z E[1_{\forall i |G_i| \leq a_i/4} \exp(-\sup_u |\langle G, D_a u \rangle| - \rho^2/2)].$$

Indeed for such centers every point moves inside the truncation cube, and centers for distinct labels are disjoint. The expectation is at least $1 - w - \rho^2/2 - \tau$ by the elementary Gaussian tails and $e^{-x} \geq 1 - x$ for nonnegative x. Taking the ratio at the first accepted proposal proves the bound.

For distinct prime axes choose $a_i = \sqrt{128 \log p_i}$, so $\tau \leq 2 \sum p_i^{-4}$ tends to zero on tails. Whenever the weighted Gaussian width of a family of signed velocities tends to zero on successive tails (uniformly over its finite-axis restrictions), so does its maximum radius, since $E|\langle G, v \rangle| = \sqrt{2/\pi} \|v\|_2$. Deleting sites that fail simultaneous agreement therefore costs vanishing density. Every uniform lifted avoiding mask transfers exactly with arbitrarily small tail loss and with its own fixed forbidden length unchanged.

In particular this holds when motions belong to a fixed finite alphabet of signed integer speeds and each family of level sets of the velocity words has uniformly bounded VC dimension. For one level with coefficients $b_i = a_i/p_i$, the centered Gaussian increment metric between two support sets is $L(\sum_{F \triangle F'} b_i^2/L^2)^{1/2}$, $L^2 = \sum b_i^2$. By Haussler's packing theorem, its covering numbers at radius $L\eta$ are at most $C_d \eta^{-2d}$. Arbitrary coordinate probabilities in this statement follow from the uniform-coordinate version by replication for rational weights, then approximation on the finite set. Chaining nets of radii $L2^{-j}$ and using the elementary Gaussian maximum bound on every successive net difference gives an expected absolute supremum $O_d(L)$: the sum is bounded by a constant times $L \sum_j 2^{-j} \sqrt{1+j}$. Sum over the finitely many levels, with their bounded coefficients. Since $\sum_{p \geq P} (\log p)/p^2 \to 0$, the width hypothesis holds. Thus in a mixed-prime graph any genuinely intermediate subfamily must evade this exact repair in arbitrarily late tails. Allowing an arbitrary subset of every finite collection of normal axes has unbounded VC dimension and need not have small Gaussian width.

Universal exact agreement cannot persist for full signed horizontal backgrounds
Suppose the finite direction menu contains $v_0 + c$ for every signed unit word c on a panel of horizontal prime axes disjoint from a fixed marked step v_0, including $c = 0$. Let A be the

set of sites where a fixed normalized lattice section preserves every such move exactly, and write its lifted coordinates in integer units. If $x, y = x + c$ both belong to A, the common endpoint $x + v_0 + c = y + v_0$ shows that the integer lifts of (y, x) differ by exactly c. Color A by parity of the sum of its lifted horizontal coordinates. Each color class is independent for all odd signed horizontal steps on the panel. Slicing over the other coordinates and applying the odd-box pair bound [R169] gives

$\text{density}(A) \le 2/(\sum_{i \text{ in horizontal panel}} 1/p_i)$.

It follows that arbitrary random sections also have vanishing expected good density along divergent harmonic panels. This is a limitation of the simultaneous *exact-section* strategy for the full signed family, even when it has only one marked pair. A quadratic mask need only detect its polynomial trajectory, so exact preservation of all coordinates is a stronger demand.

Asymmetric horizontal alternatives and signed vertical activation Take vertical cyclic primes q_i with velocities $s_i \in \{0, 1, -1\}$, and a finite set of horizontal cyclic primes $p \in P_i$ associated to each vertical coordinate (the horizontal sets are disjoint). Allow arbitrary horizontal residues $r_p \in C_p$, including unrestricted inactive-axis velocities. Suppose every permitted step obeys at least one of a fixed number L of inequalities

$$\| \sum_{i : s_i \neq 0} \sum_{p \in P_i} a_{i,p,l} r_p / p \|_T \ge \gamma, 1 \le l \le L,$$

with $\gamma > 0$ fixed and arbitrary integral coefficient arrays. This includes one-mark composite-axis constructions and every subset of the corresponding multi-mark gap family, including windows with near-grid jitters. There is no harmonic cost condition on any one direction. If A is the set of its active vertical indices, then for each $\eta > 0$ there are H, P_0, depending only on η, L, γ, and a stationary mask law of mean $> 1 - \eta$ for which all such directions with horizontal primes in the panel larger than P_0 have

$$E\lambda_H(B; v) \le 2\pi(H - 1)(\sum_{i \in A} q_i^{-2})^{1/2}.$$

For clarity, make masks on panels with all horizontal primes larger than P_0, and pass to the weak limit on the tail direct sum. The estimate also applies to the integer-horizontal model by pulling back its finite quotient action through CRT.

Choose a short open arc of length $< \eta/L$. There are (M, H) so that any circle polynomial of degree at most three with third finite difference c satisfying $\|Mc\| \ge \gamma$ hits the arc during its first H sites. For detail, a polynomial trajectory missing the arc forever has a nonconstant Fourier mode of bounded frequency with positive-sized infinite mean, by trigonometric approximation of a bump inside the arc. The magnitude of each such mean tends to zero uniformly in the lower coefficients when the order of the third coefficient tends to infinity. Indeed finite van der Corput differencing twice at lags through a fixed large R bounds the fourth power by $(O(1/R))$ plus the mean of linear exponential sums whose frequency is the original frequency times the product of the two lags and the third coefficient; these have zero limiting mean whenever it is nonintegral. For third-coefficient

order larger than the bounded frequency times R^2 all positive lag products are nonintegral. A common multiple M therefore kills the third coefficient of every forever-avoiding polynomial. Compactness of all starting coefficients satisfying $\|Mc\| \geq \gamma$ yields a uniform finite hitting bound.

Assume now all the horizontal primes exceed $(6M)$. Solve $6Mc_{i,p,l} = a_{i,p,l} \bmod p$. Round only the vertical coordinates by a lattice section and use their integer-unit lifts $\widetilde{y}_i = q_i s(y)_i$ (here $(s(y))$ denotes the section). For each test form use

$$F_l(x,y) = \sum_{i,p \in P_i} c_{i,p,l} x_p \cdot (\tilde{y}_i)^2 / p \bmod 1.$$

Keep the sites where none of these L values lies in its randomly translated arc, incurring mean loss less than η. When rounding preserves the vertical increments along H sites, F_l follows a cubic whose third difference is $6 \sum c_{i,p,l} r_p s_i^2 / p$. Horizontal coordinates require no section since their values modulo p give exactly the same circle element after multiplication by the squared integer lift. The square of each vertical speed detects its activation independently of its sign. The guaranteed gap for one form makes an all-surviving correctly lifted trajectory impossible. Applying the one-step Euclidean bound only to the vertical axes proves the displayed estimate.

When the vertical velocities have vanishing weighted Gaussian width on tails, instead remove all sites of simultaneous vertical disagreement under the Gaussian section. The loss tends to zero and the same proof now gives exact H-avoidance. Thus the corresponding tail capacity is one, with no restriction on the number or pattern of horizontal alternatives. Bounded VC dimension for the positive and negative vertical support classes suffices. No claim of exact repair is made for completely unrestricted vertical supports.

R132. Bounded-width finite-order programming cannot give an interior threshold

Scope and status. Result with proof.

Result. Summably accurate finite-order Gaussian programming with bounded fixed-part width gives unrestricted integer capacity one.

Scope. The variance gap and bounded expected Gaussian supremum are substantive hypotheses.

Let T be the Gaussian action determined by a unitary multiplication operator U on a Hermitian first-chaos spectral space, and let f be a real unit vector. For positive steps d_n suppose that unitary multiplication targets V_n commute with U, satisfy $V_n^q = 1$ for the same finite $q \geq 2$, and preserve the Hermitian real form. Write $P_n = q^{-1} \sum_{j<q} V_n^j$ for their fixed-space projections. Assume

$$\sum_n \|(U^{d_n} - V_n)f\|^2 < \infty, \qquad \sup_n \|P_n f\|^2 \leq \rho < 1, \qquad C = \mathbb{E} \sup_n |X(P_n f)| < \infty.$$

Then the unrestricted integer avoidance capacity of $D = \{d_n : n \geq 1\}$ is one. Unlike a finite list of fixed spectral points globally protecting all these steps, this allows the fixed-space parts to vary without any finite hitting rule and to have nonvanishing measure. The prior summable period-repair lemma [D20] applies directly when the approximations are controlled on an actual short avoiding indicator; the hypotheses above concern only the Gaussian vector.

To prove the assertion fix $\eta > 0$. Use independent Gaussian roots at strides $M_i = q^{i-1}$; a spectral root gives the action, or one may use the integer root tower at each stride. In the latter case each residue subsequence of a row is the original action on its corresponding coordinate. Observe its real unit-variance coordinate $F = X(f)$. Write $p_L = \Pr(F < -L)$. By Gaussian concentration for a countable supremum (or its finite truncations and monotone convergence),

$$\Pr\{\sup_n |X(P_n f)| > L - 2\} \leq \exp(-(L - 2 - C)^2/(2\rho)) = o(p_L).$$

For $\rho = 0$ the supremum vanishes. Choose L large enough, and then r of order $\log(1/\eta)/p_L$, so that the probability of simultaneous $F \geq -L$ in all rows is below $\eta/4$, and the union cost of the displayed bad event is below $\eta/4$. Exclude also a narrow band about $-L$ in each row with combined cost $< \eta/4$. Finally exclude all tail errors, in every row, on which one of the coordinates

$$X((U^{jd_n} - V_n^j)f), \qquad 1 \leq j \leq q,$$

has magnitude greater than a fixed small fraction of the band width. Their variances are at most $j^2\|(U^{d_n} - V_n)f\|^2$ by commutation and telescoping, so Chebyshev and summability make the union cost arbitrarily small after dropping finitely many initial steps. The retained event also requires at least one inactive row $F < -L$; its mass is $> 1 - \eta$.

On any entirely retained progression, on each time residue modulo M_i, labels in the i-th row repeat after q stride steps, since the corresponding actual shift is within the band

width of the identity on that observable at every retained initial site. In any window of q consecutive stride positions a row is active at least once: the average of the ideal values is $X(P_n f)$, greater than $-L + 1$ on the retained point, and each actual value is within the chosen tolerance. These statements hold for all starts simultaneously on an invariant conull set. Successively add an offset $0, M_i, \dots, (q-1)M_i$ to hit an active value for each row, preserving earlier labels by divisibility. A long enough retained run would then have a site active in every row, a contradiction. A finite initial collection of step sizes costs arbitrarily little extra mass by the finite prime-modulus sieve at a longer horizon.

For signs, $P_n f$ is precisely the part of f on the unflipped spectral region. A low construction by a truncated Gaussian antipodal coordinate that bounds all such residuals with positive probability necessarily gives a bounded Gaussian supremum (Gaussian zero-one law and integrability). Any actual uniformly avoiding nonzero mass for these ideal signs on the factor generated by f also excludes a subsequence with $\|P_n f\| \to 1$, which would converge strongly to the identity on that cyclic Gaussian factor. Thus many positive Gaussian pair-obstruction programs formally furnish the last two hypotheses already. If the high program was chosen accurately on the first-chaos vector as above, an interior unrestricted capacity cannot follow from them.

R135. Audit of the amalgamated Hamming cubes and a barrier to lifting their tags

Scope and status. Result with proof.

Result. Hamming expansion is quantified, bounded-label root lifts are excluded, and a positive-density affine-compatibility counterexample is supplied.

Scope. The p-ary expansion argument fails without compatible choices along an entire affine line.

Finite expansion. Write $n = 2k+1$, $V_j = \sum_{a \le j} \binom{n}{a}$, $e = (1, \dots, 1)$, and $K = B_1$ in $\mathbb{F}_2^n$. Every set A with $|A+K| \le 2^{n-1} = V_k$ satisfies $|A| \le (V_{k-1}/V_k)|A+K|$. Indeed Harper's vertex-isoperimetric theorem reduces to an initial simplicial segment. Beyond size V_{k-1} its closed neighborhood strictly exceeds V_k (the first k-set included has supersets of size $k+1$). At size $V_{l-1} + \theta\binom{n}{l}$, $1 \le l \le k-1$, its closed neighborhood has size at least $V_l + \theta\binom{n}{l}l + 1\}$, since the upper shadow of a proportion θ of the l-level occupies at least that proportion of the next level by counting inclusion edges. The ratios V_{j+1}/V_j are nonincreasing (partial sums of a positive log-concave sequence are log-concave; alternatively compare successive ratios in $\binom{n}{j}/V_{j-1}$). The ratios at both ends of this linear interpolation therefore exceed or equal V_k/V_{k-1}. The singleton has neighborhood V_1, covering the omitted boundary case; empty fibers cause no difficulty.

For finitely many cubes take the quotient of their direct product by the even central flips. The common image e is a nonzero involution. A set S avoids pairs at $e + \sum_i (K_i - K_i)$ if and only if $S + \sum_i K_i$ misses its translate by e. Every partial expansion retains this property since all unexpanded K_i contain zero. In every coset of each local cube an e-pair-free set has at most half the points, so the local expansion bound applies at each stage, proving

$$|S|/|L| \le (1/2) \prod_i c_i, c_i = V_{k_i - 1}/V_{k_i}.$$

Equality holds. On the direct-product cover take the union over even binary words σ of $\prod_i (B_{k_i - 1} + \sigma_i e_i)$. These boxes are disjoint and their union is invariant under all even central flips. Their expansions are the parity-even boxes of the two complementary half-balls B_{k_i} and $e_i + B_{k_i}$, so the expansion is e-pair-free. The normalized density is exactly the displayed right side. Pass from finite packets to the countable group by restricting upper witnesses and taking stationary weak limits of the finite extremizers coinduced to the whole group (one may also independently place translated copies on the cosets). An invariant maximizing measure exists at horizon two. Since every difference has order two, this also gives the all-length critical value. Thus the native system fails recurrence at equality under the paper's inclusive density convention.

Here $c_i = 1 - \binom{2k_i+1}{k_i}/2^{2k_i} \uparrow 1$. Every number in $(0, 1/2)$ occurs. Write twice that number as e^{-u}. The positive numbers $-\log c_k$ tend to zero. Choose each term not exceeding the outstanding sum but at least half the supremal eligible term (or take the largest eligible term). As long as a positive residual persists, this rule eventually uses a fixed positive lower bound, which is impossible, so the series sums to u. If a choice exhausts the sum one can instead require a strict undershoot at each selection. The endpoint $1/2$ is readily realized by a single central difference without expansions.

Bounded labels on growing-prime models. A common way of imitating this construction cannot give an intermediate threshold on a group allowing roots, even if the physical axis orders are unbounded. Here is the precise barrier. Suppose the lifted tail differences have finite support, and at each site i there is a stationary factor with a label $\phi_i : X_i \to K$ in a fixed finite abelian group, and a safe event H_i of mass $1 - \epsilon_i$, with $\sum_i \epsilon_i < \infty$. Whenever the two sites at a local step are both safe, the increment of ϕ_i is the predetermined $\tau_i(d)$; inactive sites have no increment. Assume $\sum_i \tau_i(d) \neq 0$ for every allowed direction under consideration. The separate axis actions may live on a quotient of a free or divisible abelian host; only simultaneous stationary realization of each finite packet is required. In this assertion the root actions are available for the **host**: use $d \mapsto d/q$ on a rational vector space, or coinduce the action of the free abelian host's subgroup $qG \cong G$.

In a finite tail packet take the sum of its labels plus an independent constant uniform in K. Its marginal is uniform. Declare a site bad if some included local state is unsafe, at cost at most $\sum_{\text{tail}} \epsilon_i$. On each pair of retained adjacent sites in a legal direction, the summed label increases by its total nonzero tag, independently of the initial point. One can obtain a single stationary law on the full tail by taking weak limits of the finite alphabet label-and-bad processes. Stationarity and the deterministic implications survive, since they concern cylinder events; an exhaustion containing each particular step's full support eventually realizes its exact total tag.

Put $b = \exp(K)$ and choose independent root rows at strides $1, b, \dots, b^{m-1}$. Retain a point when no row is bad and at least one row's label is zero. Its mass is at least $1-(1-1/|K|)^m - m\sum_{\text{tail}} \epsilon_i$. In an all-retained segment along a legal step, the labels sampled at stride b^j in row j advance by the same nonzero tag, hence repeat after b such slots and cannot vanish at two consecutive slots. Choose a shift of zero or b^j at each stage to leave row j nonzero, retaining all previously secured nonzero rows by divisibility of later shifts by their periods. A segment of length b^m then contains a position with no vanishing row, a contradiction. Send the tail and then m to infinity. Where the usual head-removal equality holds, the full lifted critical value is one.

For the original exponent-two directions one can bypass the safe-site discussion entirely: coinduce the positive avoiding binary mask to the containing torsion quotient and pull back. Each indicated element acts on that system as an actual involution. Its entire sampled mask word is zero or alternates, so exactly the same root argument with $b = 2$ works without any bad sites. This applies to countably many free coordinates just as in finite rank.

The binary extremizers also fit the tag hypothesis directly: B_{k_i-1} and its antipodal translate are at Hamming distance at least three. An endpoint-safe move in $K_i - K_i$ cannot change their labels. Adding the antipode flips the label. Thus growing cyclic physical orders will require a change beyond preserving this bounded-label, summably safe mechanism. For $p > 2$ there is a separate algebraic obstruction to copying the expansion upper bound: two arbitrary corrections $k_0, k_1 \in K_i$ determine a slope, whereas a whole affine line needs the remaining lifts to satisfy $k_j = k_0 + j(k_1 - k_0)$; membership of all p translates in the ordinary enlarged set need not give compatible choices.

This last failure persists at positive density uniformly for large primes. In $L = \mathbb{F}_p^2$ take $e = (1, 0)$, $K = \{0\} \times \mathbb{F}_p$, b nearest to $p/4$, and $S = \{(x, y) : y - bx^2 \in I\}$, where I is an interval of length $\lfloor p/10 \rfloor$. Then $S + K = L$, but S contains no three-term progression at any $e + K - K = (1, z)$. The second difference of $y - bx^2$ along that progression would be $-2b$, of circle norm approaching $1/2$ after division by p, while any triple in I has second

difference of norm at most 1/5 after division. For an explicit finite neighborhood just take the entire indicated vertical subgroup; the obstruction is to the proposed implication even with an expansion as large as possible.

R139. The finite-color hierarchy can fail density largeness at every positive threshold below one

Scope and status. Result with proof.

Result. The existing finite-color hierarchy, including its least admissible radices, has unrestricted avoidance capacity one.

Scope. Topological forcing and the finite-color obstruction are imported from the earlier hierarchy construction.

Research proof, November 2, 2026. This is a strengthening of the existing finite-color construction [D23], taking its topological forcing and its finite-color obstruction as inputs. It applies also to the original choice of radices, including the least admissible ones. It does not settle the prescribed intermediate-density conjecture.

Write $b(D)$ for the supremum of upper natural densities of sets avoiding all length-K progressions with step in D, for some finite K. Equivalently use the supremum over stationary binary laws of the corresponding avoiding-event means. For every $s \geq 1$ the recursive construction can be made so that D_s is s-large, not $(s+1)$-large, and $b(D_s) = 1$. Thus for every fixed s no fixed density $\delta < 1$ is forced by s-largeness, even if the nilwindow strengthening from the color construction is assumed.

Radices and induction At stage i the finite packet F_i has been chosen before the radix b_i. The original construction requires $b_i > 2L \max F_i$ for the fixed color-forbidden length $L \geq 2$. Write $\lambda_1 = 1, \lambda_{i+1} = b_i \lambda_i$ and let $z_i(n)$ be the ith digit of $n \geq 0$ with weights λ_i. The empty base family has $b = 1$.

For any eventual required block length K call an original packet entry f heavy if $|f|/b_i \geq 1/(2K)$. These heavy differences jointly have $b = 1$, without any additional growth condition. Indeed put $\theta = \sum_{j \geq 1} 1/\lambda_{j+1}$. The terms before $j = i$ contribute integers to $\theta \lambda_i f$, and the remaining real sum is

$$(f/b_i)(1 + 1/b_{i+1} + 1/(b_{i+1}b_{i+2}) + \ldots).$$

The factor in parentheses belongs to $[1, 2]$, and $2|f|/b_i < 1/L \leq 1/2$. Consequently $||\theta \lambda_i f|| \geq |f|/b_i \geq 1/(2K)$ for every heavy entry, including signed ones. The class of all d with $||\theta d|| \geq 1/(2K)$ has fixed-length avoiders of density arbitrarily close to one. For a deletion budget $\eta > 0$, take $Q_0 > 2/\eta$ and $M = \mathrm{lcm}(1, \ldots, Q_0)$. With $\beta = d\theta/M$, choose by pigeonholing $1 \leq q \leq Q_0$ with $a = ||q\beta|| \leq 1/Q_0$. Divisibility `q|M` gives $a \geq q/(2MK) > 0$. At indices $0, q, \ldots, \lfloor 1/a \rfloor q$ the phases form a circle net of largest gap at most a, and the last index is at most `2MK`. Thus deleting one arc of length η under the rotation θ/M hits every such $2MK + 1$-term progression. In the stationary rotation model the deletion costs exactly η. This repair may be combined with other avoiding events by independence and the maximum of their lengths. If one chooses the optional growth $b_i / \max F_i \to \infty$, only finitely many stages are heavy for each fixed K, allowing the simpler prime repair below instead.

Suppose the inner family B has $b(B) = 1$. The color construction gives packets inside $G_{M_i} \cup B_i$ where $B_i \subset B$ and the G_{M_i} are the finite torus gates at the fixed palette size of this induction step. Split $D_{s+1} = \bigcup \lambda_i F_i$ into the union of

$$E = \bigcup \lambda_i(F_i \cap G_{M_i}), I = \bigcup \lambda_i(F_i \cap B_i).$$

Showing $b(E) = b(I) = 1$ suffices. Indeed independent stationary avoiding events for these two families have an intersection of product mass avoiding the union at the larger forbidden length. A finite step family can always be handled at arbitrarily small additional loss. Choose a prime larger than all its steps and sufficiently large that $1/p$ is below the desired loss, and delete the multiples of that prime. Every p consecutive samples of any such difference hit the deletion. This repair works after passing to the largest of the required lengths.

The inner packets Choose a stationary binary word w of mean $> 1 - \epsilon$ avoiding all length-K runs along steps in B, simultaneously at all starts (a full-probability property). Define

$$A_w = \{n \geq 1 : w(\sum_j z_j(n)) = 1\}.$$

At every n its membership probability equals the same stationary mean. Reverse Fatou on prefix densities supplies a realization with upper density at least $1 - \epsilon$ and the avoidance property above. For any nonheavy entry $d \in F_i \cap B_i$, the change in the integer quotient above digit i on a run of $2K$ sites of step $\lambda_i d$ is at most one, since $(2K - 1)d < b_i$. On some K consecutive sites every other digit is fixed and digit i grows by d at each move. The digit sums give a forbidden word in w if all these sites belong to A_w. Use the single rotation repair for all heavy entries. Letting $\epsilon \to 0$ proves $b(I) = 1$.

A uniform statement for the torus gates We include the details to clarify why growing dimension and arbitrary static offsets do not affect the density argument. The proof uses the mechanism of the original gate analysis [D05], applied to all stages at once.

Fix $Q = 100q$, $a = 1/Q$, $\eta = (10^6 Q^2)^{-1}$, $m = Q/2$ and $C = 3/Q + \eta$. For $d \in G_{M_i}$ write θ_j for the centered representative of $d\alpha_{ij}$. There is one index with $|\theta_j + a| < \eta$, and the remaining indices satisfy $\max |\theta_j| < \eta$, $\Sigma|\theta_j| < 2/Q$. In particular the sum over all indices is at most C. For an integer root $R \geq 1$ put

$$W_{i,R}(x) = \sum_{j \leq M_i} (e(x\alpha_{ij}/R) - 1), e(t) = \exp(2\pi i t).$$

Test $W_{i,R}$ with an arbitrary additive offset $z \in C$, including at starts $x = 0$. Its second difference on the three sites $x, x + mRd, x + 2mRd$ has modulus between 3.9 and 4.1. The marked term has modulus $|e(m\theta_j) - 1|^2 \geq (2 - 2\pi m\eta)^2$. The sum of the absolute values of the other terms is at most $(2\pi m)^2\eta(2/Q)$. All starting phases have modulus one, and the constant offset cancels exactly. Thus of the three values at least one has distance greater than $c = 1/4$ from `10 Z[i]`. Indeed otherwise their second difference would lie within $4c = 1$ of that lattice. A number of modulus between 3.9 and 4.1 cannot do so. We call a value with distance greater than c active.

Choose r independent root tests. Given ϵ, first pick r so $(1-\pi c^2/100)^r < \epsilon/3$; choose P a large integer multiple of 10 and $\tau \in (0, c/2)$ small. In the torus $C/PZ[i]$ declare a value illegal if a real or imaginary coordinate lies in a periodic gap of width $H > 2\pi C$, or its distance modulo $10Z[i]$ is within τ of c. Choose P, τ so the sum of the probabilities of any of these illegalities across r Haar-uniform values is less than $\epsilon/3$. These parameters are fixed before choosing the root denominators.

For a fixed R, form a compact subshift X_R consisting of two-sided binary limits of arbitrarily long centered legal gate words

$$1\{dist(z + W_{i,R}(x + td), 10Z[i]) > c\},$$

where stage, start, allowed step and offset can all vary. Equivalently require every centered block realizable at a single stage, with its sampled arguments nonnegative. Every sequence in X_R is periodic. Here are the relevant uniform estimates.

On a residue $t = b + Ru$ the change over one unit of u has modulus at most $2\pi C$, as follows by summing $|e(\theta_j) - 1|$. Legal real and imaginary values therefore stay within a fixed pair of adjacent-gap components after choosing their unwrapped lifts: jumping between distinct components requires a jump of at least H. Thus the increments relative to $u = 0$ are uniformly bounded by $\sqrt{2}P$ on the entire available segment. Arbitrary offsets cancel in these increments, while their values at $u = 0$ may be reduced modulo 10 and compactly extracted.

For a sequence of growing centered realizations reorder the θ_j by decreasing modulus, padding finite lists with zeros. Diagonally extract $\theta_j \to v_j$ and phases $e((x+bd)\alpha_{ij}/R) \to u_j$ for each j, together with the limit of their bounded first-moment sums $\Sigma e((x+bd)\alpha_{ij}/R)\theta_j$. Phases at padded zeros may be arbitrary. The bounds $\Sigma|\theta_j| \le C$ imply $\sum_{j>J} \theta_j^2 \le C^2/(J+1)$ and $\Sigma|v_j| \le C$. By Taylor subtraction of the linear term, the limiting unwrapped increment function on this residue is

$$g(u) = b_0 u + \sum_j u_j(e(uv_j) - 1)$$

for a constant b_0; the convergence here is pointwise in integer u, and g is bounded by the trapping estimate. The formula implies that its first difference is a constant plus an absolutely convergent trigonometric series on the compact rotation generated by the v_j (the coefficients include $e(v_j) - 1$). A bounded primitive of this form is Bohr almost periodic. For completeness, take the compact orbit closure of $(uv, g(u))$ under the continuous additive skew cocycle given by that first difference and a minimal subsystem L there. The projection to the compact base rotation is onto. If two points of L over one basepoint differ vertically by $z \neq 0$, then L and its vertical translate by z intersect, hence coincide by minimality, contradicting compactness under repeated translations. Thus L is a continuous graph over the rotation, and its values on the original orbit agree with g up to a constant. This proves almost periodicity. The fine gate boundary band separates its code, so the binary code is a continuous function of the almost periodic torus samples, hence itself almost periodic. A finite-alphabet almost periodic sequence has an exact period by taking a sufficiently close positive almost period. Interleaving the R residue classes proves the entire code periodic.

A compact subshift consisting entirely of periodic points is finite. Otherwise distinct periodic words could approach a fixed periodic word in this subshift. Shift to their first rightward mismatch and pass to a convergent subsequence preserving the reference period phase. The limit agrees with the reference word on the negative half-line and differs at the origin, contradicting periodicity. Distinct periodic words have a right mismatch even when the closest original mismatch lay to the left. Thus there is a common period P_R for all legal limit codes (take 1 if there are none).

Choose $R_1 = 1$ and each R_j divisible by all earlier P_{R_l}. There is a uniform K such that no single gate stage with any $d \in G_{M_i}$ and arbitrary offsets for its r rows has K consecutive legal positions with no simultaneous activation. Otherwise center longer counterexamples and extract limits in every X_{R_j}. Each limit row always has an active member among its times $t, t + mR_j, t + 2mR_j$, by the second-difference estimate and finite block realization. Starting at zero, choose at each row one of those updates to activate it. Previously activated rows remain active because the update is a multiple of their common periods. This gives simultaneous activation, a contradiction. All offsets were part of each compact subshift from the outset.

Assemble the gate packets For R in the finite list set $W_R(n) = \sum_i W_{i,R}(z_i(n))$, a finite sum since every summand vanishes at zero. Add independent uniform offsets on $C/PZ[i]$. Keep a site if all rows are legal and not all are active. Each site is deleted with probability less than $2\epsilon/3$ by the Haar estimates. Reverse Fatou gives one choice of offsets for which the upper natural density of retained sites is at least $1 - 2\epsilon/3$.

Whenever $d \in F_i \cap G_{M_i}$ is nonheavy we have $(2K - 1)d < b_i$. A retained $2K$-term gate progression with difference $\lambda_i d$ would contain K consecutive sites on which all other digits are fixed, as above. On each row their contributions together with its random offset are a static complex offset, and the active digit runs along the single-stage progression. This contradicts the uniform result for K. The single rotation repair for heavy entries costs arbitrarily little additional density. It follows that $b(E) = 1$ and completes the induction. If signed packets have the same norm/curvature conditions on the unsigned steps, reversal of arithmetic progressions makes the same result immediate for their absolute values; the digit-ratio separation calculation itself already handles either sign.

The proof gives some finite forbidden length for each density strictly below one. It places no bound on that length as the density tends to one. It uses no independence or equidistribution of the stage frequencies for these upper-density conclusions.

R159. Scheduling arbitrary high-probability rules for finitely many colors

Scope and status. Result with proof.

Result. For fixed color weights below $1 - 1/h$, sufficiently high-probability column rules admit simultaneous stride-wise nonconstant-row schedules.

Scope. A finite-color combinatorial feasibility theorem; it does not construct an intermediate integer density threshold.

Fix $h \geq 2$ and positive color weights π_j on a finite alphabet $[c]$, with $\Sigma\pi_j = 1$ and $\max \pi_j < 1 - 1/h$. There is an $\epsilon(h, \pi) > 0$ such that for every finite list of positive strides $v_1,\ldots,v_r$ and every $B \subset [c]^r$ of product π-measure $> 1-\epsilon$, there are words $w_i{:}Z \to [c]$ whose columns always belong to B, with no constant h-term progression of stride v_i in row i. Unlike the binary mark proof, there is no order on B, and the row constraint holds for *every* color. The words need not have independent or prescribed empirical column distributions. This is a combinatorial constraint result; it does not assert the existence of an integer step set with an intermediate threshold.

Allowed symbols and overlapping lists. Partition the circle into c successive arcs of lengths π_j. Enlarge each slightly in both directions to open I_j of length strictly below $1 - 1/h$. At all velocities in a fixed neighborhood J of $1/h$, every h consecutive steps on the circle visit the complement of each I_j. Convolve the original partition indicators with a smooth nonnegative mollifier of sufficiently small support to obtain a smooth partition of unity ψ_j subordinate to I_j, with means π_j and positive neighbor overlaps. Use one pair label $\{j, k\}$ for each distinct neighboring pair on the cycle (for $c = 2$ use only one). In each overlap choose a nonnegative smooth bump χ_{jk} of positive integral supported where both corresponding ψ are strictly positive; choose its amplitude small enough for the following construction. The list alphabet Ω consists of the singletons and these pairs. For $0 \leq u \leq 1$ take the smooth kernel

$$g_{jk}(z) = u\chi_{jk}(z), g_j(z) = \psi_j(z) - (u/2)\sum_{k\ j} \chi_{jk}(z).$$

It is a probability distribution on nonempty lists at each z, and a positive-probability list contains only permitted colors. Its mean distribution ν_u on Ω starts at singleton law π. For u in $[1/3, 2/3]$ all list atoms have uniformly positive mass. The associated expectation operator $K_u{:}L^2(\Omega, \nu_u) \to L^2(T)$, $K_u v = \sum_L g_L v(L)$, preserves constants. On the orthocomplement of constants its norm is uniformly < 1 on the parameter interval. Indeed equality in Jensen would force a function to agree wherever two list probabilities overlap; the positive singleton-pair-singleton connections connect the entire finite list graph. Compactness of the interval makes the strict bound uniform. Smooth norms are uniformly bounded.

Choice of a bias. Let f be the indicator that the product of r input lists contains no column in B. It decreases when lists grow. Write I_i for the probability of changing f upon redrawing only list i independently with the current law. For fixed other lists, if f is nonconstant in coordinate i, some singleton and an incident pair have different values, by connectedness; a singleton there has value one and the pair has value zero. Hence I_i is at most a constant times the drop in its conditional expectation per unit u, where that

derivative affects only coordinate i. The constant is uniform since there are finitely many positive integrals $\int \chi_{jk}$. Integrating the derivative of $E_{\nu_u^r} f$ over $[1/3, 2/3]$, and using $E_{\nu_0^r} f = 1 - \pi^r(B) < \epsilon$, gives some u with $\sum_i I_i \le C\epsilon$. Fix this u.

Smooth interpolation. At phases z_i form $F = E[f(L_1, \dots, L_r)]$ by independently sampling the lists with probabilities $g_L(z_i)$. Evaluate $z_i = \theta_i + m_i z$ with independent uniform θ_i. If $F(z) < 1$ then some permitted choice of lists admits a column in B. It is therefore enough to obtain $||F||_\infty < 1$ and $m_i v_i \beta \in J$ for a single β.

Here are details of the Boolean Sobolev argument [R022] with this alphabet. Write $f = \sum_S f_S$ for its orthogonal product decomposition into constants and mean-zero factors on the coordinates in S. Different supports stay orthogonal under average over the independent translations. On the Fourier side of the torus product, the z-frequency of a mode k_i is $\sum_i m_i k_i$. For $0 < \eta < 1$, its $(1+\eta)$-moment is at most $|S|^\eta \sum_i |m_i k_i|^{1+\eta}$. In the i-term, the map on coordinate i is K_u followed by the multiplier $|k_i|^{(1+\eta)/2}$; it is bounded from the fixed finite-dimensional list space by smoothness. On every other active coordinate K_u has norm $\le \sqrt{\rho} < 1$. Thus the contribution of this support is at most $C|S|^\eta \rho^{|S|-1} ||f_S||^2 \sum_{i \in S} m_i^{1+\eta}$. The ordinary L^2 contribution sums to at most E $f \le \epsilon$. Absorb $|S|^\eta$ into a larger $\rho' < 1$.

For fixed i the sum $\sum_{S \ni i} (\rho')^{|S|-1} \|f_S\|^2$ is the squared norm of the centered coordinate difference $f - E_i f$, after applying the $\sqrt{\rho'}$ noise operator in all other coordinates. The finite-alphabet biased hypercontractive estimate (the same two-point compactness and tensorization proof, or its finite-space version) bounds this by $C I_i^{1+\delta}$ for some $\delta > 0$ independent of r. Explicitly, on each i-slice use an $L^p \to L^2$ noise bound with $p < 2$. The slice L^1 norm of $|f - E_i f|$ is at most $I_i / \min_L \nu_u(L)$, since the full L^1 norm equals the resampling change probability for zero-one f. Also $|f - E_i f| \le 1$. Hence the slice squared L^p norm is at most $C I_i^{2/p}$. The constants are uniform because all atom masses have a common lower bound. For completeness the one-coordinate $L^p \to L^2$ bound near $p = 2$ follows by compactness of the L^2 unit sphere away from constants (noise contracts strictly there), and by Taylor expansion near a constant where the variance coefficient $p - 1$ can be taken larger than ρ'. The usual Minkowski tensorization gives the product estimate.

Choose $\eta < \delta$. A uniform random β has $P[m_i > L] \le C/L$ for the first positive integer m_i with $m_i v_i \beta \in J$. Indeed failure forces a geometric sum of a bounded nonzero multiple of $v_i \beta$ to be large, by a trigonometric bump minorant on J. Ignore colors independent of f in the truncation, choosing any hitting multipliers for them. For positive I_i truncate at $\lceil 1/I_i \rceil$: all truncations hold off a set of probability at most $C\Sigma\epsilon_i$ where $\epsilon_i = I_i$, and the expected sum of $I_i^{1+\delta} m_i^{1+\eta}$ on the respective truncations is $O(\Sigma I_i)$, as in the binary argument. Choose β satisfying all truncations with that $\sum O(\epsilon)$. The one-dimensional $H^{(1+\eta)/2}$ Sobolev inequality now gives $E_\theta ||F||_\infty^2 \le C\epsilon < 1$, so a choice of phases exists. At $z = t\beta$, choose an admitted column in B. Its colors live in the stated arcs and every row therefore obeys the specified prohibition. This completes the proof.

R160. Depth- and rank-uniform interpolation for scalar characteristic-p filters

Scope and status. Result with proof.

Result. Scalar characteristic-p filters over a single power-series valuation admit depth- and rank-uniform rational-polynomial interpolation.

Scope. The theorem applies to the scalar valuation class, not arbitrary multivariate Artin rings.

This extends cyclic scalar-filter interpolation [R203] to arbitrary finite primary rank. Fix d, $p > d$, and units $u_1, \dots, u_r \in 1 + XF_p[[X]]$. Set $\theta(x) = \prod_j u_j^{x_j}$ for $x \in Z_p^r$, with powers defined continuously. Let P be a polynomial of degree $\le d$ with coefficients in $F_p[[X]]$, viewed modulo X^N. On all x with $P(\theta(x)) \in F_p X^{N-1} \bmod X^N$ assign the additive phase of its top coefficient divided by p. A polynomial over $\mathbf{Q}$, of total degree bounded only by d, agrees with this phase under the p-adic fractional-part map on every such x. All integer lifts are included. In particular neither the number of generators r nor their possibly very unequal filtration weights enter the degree bound.

Local filtered lattices The zero/constant and singleton-image cases are immediate. Factor P over a valued algebraic closure as in the cyclic note. Its allowed unit arguments are a union of at most d disjoint valuation balls around roots, with thresholds R_i. For separate occupied balls their center distance valuation b satisfies $1 \le b < \min(R_i, R_j)$ and $R_i - b \le d(R_j - b)$ and vice versa. In particular $\max R_i / \min R_i \le d < p$. With one occupied ball no radius comparison is needed. Define $L_t = \{x : v_X(\theta(x) - 1) \ge t\}$. These are nested open finite-index subgroups of Z_p^r (all of Z_p^r for small t). The exact Frobenius identity is $v_X(\theta(px) - 1) = p v_X(\theta(x) - 1)$, interpreting infinity in the usual way. Each occupied ball therefore has preimage $B_i = s_i + L_{R_i}$ for an integer vector s_i. Use density of Z^r in Z_p^r for this choice.

Write $P(\theta(s_i)(1 + Z)) = \sum_{k \le d} b_k Z^k$ and let t_i be the least realized finite valuation of $\theta(x) - 1$ on L_{R_i}. If no finite valuation occurs there the local phase is constant. Otherwise take a vector attaining t_i with leading coefficient $c \ne 0$. Its multiples a, $0 \le a < p$, have leading coefficient `a c` at level t_i. The least of $v(b_k) + kt_i$ cannot be below $N - 1$: the coefficient there on these multiples would be a nonzero polynomial of degree $\le d < p$ in a, contrary to ball membership. Thus the output on B_i is an F_p-polynomial of degree $\le d$ in the additive homomorphism

$$\ell_i(x - s_i) = [X^{t_i}](\theta(x - s_i) - 1).$$

Here $t_i \ge 1$, so the leading coefficient is additive on its lattice. Lift ℓ_i to a $\mathbb{Q}$-linear form λ_i taking integral values on L_{R_i}, with its reduction modulo p there equal to ℓ_i. One way is to view ℓ_i/p as a finite character of the lattice, extend to Z_p^r modulo a finite-index sublattice by divisibility of Q_p/Z_p, and represent the extension by a rational linear form with a p-power denominator *timepproduces*λ_i. Lifting the scalar coefficient polynomial to integers and dividing by p gives $q_i(x) = p^{-1} A_i(\lambda_i(x - s_i))$, a rational polynomial of degree at most d representing the correct phase on B_i.

Rank-free gluing For different i,j write $b_{ij} = v_X(\theta(s_i - s_j) - 1)$, precisely the root-ball separation. Let $R* = \min(R_i, R_j)$, $n'_{ij} = ceil \log_p(R*/b_{ij})$, and $n_{ij} = ceil \log_p(R_i/b_{ij})$. Thus $1 \le n' \le n \le n' + 1 \le 2n'$. In the finite abelian group Z_p^r/L_{R*}, the class of $s_i - s_j$ has order exactly $p^{n'}$ by Frobenius. There is a rational linear form λ_{ij} integral on L_{R*} taking value of valuation $-n'$ there: prescribe on the cyclic class its faithful p-power character, extend to the finite quotient with values in Q_p/Z_p, and lift to a rational linear form on Z_p^r. Put

$$D_{ij}(x) = \lambda_{ij}(x - s_j)/\lambda_{ij}(s_i - s_j).$$

This is one mod p on B_i and has valuation at least n' on B_j. For a weight D_{il} on B_j with $l \neq i$,j, its valuation is nonnegative if $b_{il} \le b_{ij}$. In fact $b_{jl} \ge \min(b_{ij}, b_{il})$ by the ultrametric inequality, while $pL_{R_j} \subset L_{\min(R_i,R_l)}$ by radius comparison. The variation on B_j therefore contributes at worst -1 to the numerator, handled by the denominator order $n'_{il} \ge 1$. The center difference in this case has order at most $p^{n'_{il}}$ modulo the relevant lattice. If $b_{il} > b_{ij}$, the possible valuation deficit of D_{il} on B_j is at most n_{ij}: now $b_{jl} = b_{ij}$, and Frobenius clears the displacement into $L_{\min(R_i,R_l)}$ using $\le n_{ij}$ powers. Finally $p^{n_{ij}}(B_j - s_i) \subset L_{R_i}$, because $n_{ij} \ge 1$ and $pR_j \ge R_i$. Hence the valuation of q_i on B_j is at least $-1 - dn_{ij}$.

Choose $a_{ij} = 1 + 2(d + \sum_{l:b_{il}>b_{ij}} a_{il})$ in decreasing order of proximity b_{ij}. The complete weight W_i is the product of $D_{ij}^{a_{ij}}$ for $j \neq i$. On B_i it preserves q_i modulo Z_p. On B_j its gain is at least $a_{ij}n'_{ij} - n_{ij} \sum_{l:b_{il}>b_{ij}} a_{il} \ge 1 + dn_{ij}$, enough to make $W_i q_i$ integral. Therefore $Q = \sum_i W_i q_i$ agrees with all assigned phases. At most d centers occur, so all exponents and the total degree depend on d alone.

For prime-by-prime finite modules in this scalar DVR class with $P(1)$ having zero top phase, the rational-cover consequence follows exactly as in the cyclic argument: use coordinates $p^{e_{pj}} g_j$ for the p-primary finite cyclic factors, evaluate the newly obtained multivariate polynomial under the p-adic fractional-part map, and sum over primes. On finitely supported rational vectors all but finitely many local values vanish when the coordinates are p-integral. If a local filter is trivial outside a known subset one may restrict the represented directions arbitrarily. A uniform fixed-degree gap of these phases is the same polynomial-type root blocker as before. The proof is specific to a **single scalar power-series valuation** (or boundedly many tests combined additively); it does not replace a general multivariate Artin ring by a DVR.

R162. Signed odd directions in disjoint prime blocks

Scope and status. Result with proof.

Result. Signed odd prime blocks satisfying a quantitative harmonic/square-sum condition have saturated capacity one.

Scope. The full divergent-reciprocal signed family is pair recurrent but fails the stated van der Corput spectral criterion; arbitrary mixed all-length recurrence is unresolved.

Qualifications. The positive-atom spectral obstruction concerns the standard van der Corput criterion on the stated group and is compatible with pair recurrence.

Let $P_0 = \oplus C_{p_i}$ have distinct odd primes. Partition a subsequence of the axes into disjoint finite blocks B_j, with $\min_{i\in B_j} p_i \to \infty$. Write $u_j = (\sum_{i\in B_j} p_i^{-2})^{1/2}$ and $w_j = \sum_{i\in B_j} 1/p_i$. Let V be the union over blocks of any subsets of the vectors supported in that block with coordinates $0, \pm 1$ and odd support size. If $u_j^{m-1} w_j \to 0$ for some fixed integer $m \geq 2$, the saturated rational family has $b = 1$ and so does its integer realization. This includes uniformly bounded block harmonic mass. It even permits $w_j \to \infty$, in which case the pair-recurrence theorem [R169] applies to the full signed-odd boxes. No conclusion in this note covers arbitrary, extremely fast block growth, or mixed directions spanning different blocks.

A Gaussian field on the squarefree sum Sample independent signs $\epsilon_i \in \{\pm 1\}$ on a product probability space and put $\chi_\epsilon(e_i) = (p_i + \epsilon_i)/(2p_i) \bmod 1$. In this proof a character value is exponentiated using $e(t) = \exp(2\pi i t)$. The complex unit functions $e(\chi_\epsilon(x))$ belong to the real Hilbert space underlying $L^2(\epsilon; C)$, with real inner product. Let W be a real isonormal Gaussian process on this Hilbert space, so $X(x) = W(e(\chi_\epsilon(x)))$ is a stationary standard real Gaussian field on P_0. Its shift action on its path space is probability preserving. For a signed vector v in a block, as a function of the signs put

$$Z_v = \exp(i\pi \sum_i \epsilon_i v_i/p_i).$$

Thus for odd-support vectors $e(\chi_\epsilon(v)) = -Z_v$. Use all $v \in \{-1, 0, 1\}^{B_j}$, including zero, as indices for the following Gaussian supremum. Put $D_v = W((Z_v - 1)^m)$ and $S_j = \max_v |D_v|$. With constants depending only on m,

$$ES_j \leq C_m u_j^{m-1} w_j, \mathrm{Var}(S_j) \leq C_m u_j^{2m}.$$

Indeed $|Z_v - 1| \leq \pi |\Sigma \epsilon_i v_i/p_i|$. The fixed-moment Rademacher estimate gives $||Z_v - 1||_{2m} \leq C_m u_j$ and $||Z_v - Z_z||_{2m} \leq C_m (\Sigma (v_i - z_i)^2/p_i^2)^{1/2}$. Factor a difference of mth powers and apply Hölder to get canonical distance $||D_v - D_z||_2 \leq C_m u_j^{m-1} (\Sigma (v_i - z_i)^2/p_i^2)^{1/2}$. Comparison with $C_m u_j^{m-1} \sum_i v_i g_i/p_i$, for independent standard Gaussians g_i, bounds $E \max_v D_v$. Add the identical bound for $-D_v$ to control the absolute maximum since $D_0 = 0$. Finally all representing vectors have norm at most $C_m u_j^m$, so the maximum of absolute linear forms is $C_m u_j^m$-Lipschitz on its finite Gaussian span. The Gaussian Poincaré inequality bounds its variance by the squared Lipschitz constant. See Tropp's high-dimensional probability

notes, Gaussian comparison and Gaussian concentration for Lipschitz functions. The latter concentration estimate also gives the same variance with an immaterial universal factor.

Since $\sum_j u_j^2 \le \sum_{n\ge 2} n^{-2} < \infty$, Chebyshev shows that, outside events of total probability tending to zero on late tails, simultaneously for all $j \ge J$,

$$S_j \le \rho_j := C_m u_j^{m-1} w_j + u_j^{m-1} \to 0.$$

Indeed the exceptional sum is at most $C_m \sum_{j\ge J} u_j^2$. Our masks only require the bound on odd directions, where the mth power is up to sign $(e(\chi_\epsilon(v)) + 1)^m$. It thus corresponds directly to the path-space observable $|(T_v + I)^m X(0)|$, with T_v shifting by v. Multiplying by a unit character at an arbitrary site preserves the Gaussian law of all the increments. It suffices for the mask to test precisely these odd-direction increments.

Uniform line length after taking roots Fix $\eta > 0$. Take r independent Gaussian rows with root strides $R_l = 2^{l-1}$; all are invertible in every odd prime group. In row l use the stationary field $X_l(x) = X^{(l)}(R_l^{-1}x)$ so $(T_{R_l v} + I)^m X_l(0)$ has the law of the tested observable. Choose r with $P(N(0,1) \ge -1)^r < \eta/3$, then B with $rP(|N(0,1)| > B) < \eta/3$. For a sufficiently late J, the simultaneous second or higher difference tests just established fail with total probability $< \eta/3$ over all r rows. Keep a site when all row absolute values are at most B, all odd-direction tests for $j \ge J$ pass, and at least one row has value < -1. This is a stationary event of mass $> 1 - \eta$.

There are $J' \ge J$ and a finite length H such that the event avoids every H-run with difference in blocks $j \ge J'$. Otherwise take centered stretches of lengths tending to infinity along odd differences in blocks $j \to \infty$. Pass diagonally to limits of the finitely many bounded real observations at every integer time. On a residue class modulo R_l, the sequence f of limiting values satisfies $(T + I)^m f = 0$ since the available error ρ_j tends to zero. Thus $f(t) = (-1)^t P(t)$ for a real polynomial P of degree $< m$, which is constant by boundedness. Hence globally row l is negated by shift R_l and fixed by shift $2R_l$. Starting at time zero choose either zero or a shift of R_l to make the lth row nonnegative, successively in increasing l. Later shifts leave earlier rows unchanged. At that time all rows are nonnegative, contradicting the retained-site condition, whose closed implication in the limit is that some row has value ≤ -1.

Because blocks are finite and have escaping minimal primes, discarding any finite initial set of blocks is permitted in the squarefree-tail formula [R241]. This proves the claimed saturated cap. In particular signed unit boxes with moderately growing harmonic masses can have recurrence for **pairs at every positive density** and still have all-length capacity one.

A spectral distinction even for all mixed supports For the full mixed-support odd family, with no hypothesis on the reciprocal series besides distinct odd primes, there is a probability measure on the compact dual of P_0 with a positive identity atom and Fourier transform zero at every such direction. Let $c_i = \cos(\pi/p_i)$. In the minus product law take the independent two near-half characters above. The plus product has marginal $c_i\delta_0 + (1 - c_i)Haar$. At each local digit ± 1 the minus and plus coefficients are respectively $-c_i$ and c_i, and at zero both are one. The equally weighted mixture is therefore zero on every odd

support. The plus product gives the identity atom mass at least $\Pi c_i > 0$, since $\Sigma(1-c_i) < \infty$. Thus in the divergent reciprocal case the full signed family is a pair-recurrence set but fails the standard Fourier criterion for a van der Corput set on this group. This does not decide its all-length capacity.

R163. Odd-block symmetrization on a homogeneous row

Scope and status. Result with proof.

Result. Odd-block symmetrization preserves avoiding density and forces order-two residual compact motion in the resulting spectral limits.

Scope. It does not control all temporal joinings or provide a common involution for the full process.

Let B be a countable abelian group, $G = B^{(\mathbb{N})}$, and $D \subset G$ a direction family invariant under permutations of the coordinates and under copying each input coordinate into disjoint nonempty blocks of odd size (only images of individual directions need remain admissible). A stationary h-avoiding binary law can then be replaced by a shift-invariant h-avoiding law of the same origin density which is invariant in distribution under ordered disjoint odd-block copying. The conclusion extends to simultaneously observed finitely many stationary marks. It applies on each infinitely replicated homogeneous dyadic row, including its broader normal menu and parity-of-marks condition. It does not assert full affine exchangeability or allow even-size copying.

Construction. First average over the permutations of the first n axes and take a weak cluster limit; it is stationary, has the same h-avoidance and mean, and is invariant under all finite permutations. Given a cylinder specification involving l axes, copy its jth axis onto m_j fresh consecutive axes, and compute its probability in this symmetric law. This is a bounded function $F(m_1, \dots, m_l)$ of the positive odd lengths. Integrate successively in reverse order of the indices, using on each odd length the same positive normalized mean invariant under even translations (for example a cluster mean of the uniform measures on the first N odd integers). The resulting cylinder laws are consistent; ignored variables contribute nothing and changes of the locations of the finite blocks do not affect F. They define an ordinary countably additive law by the finite-dimensional extension theorem. It is shift-invariant, origin density is preserved, and each fixed forbidden cylinder still has probability zero. For any ordered substitution of l disjoint odd blocks at the output, the effective input lengths are sums of odd numbers of odd lengths. Within each block integrate the last variable after fixing all preceding lengths. Its even translate gives the same mean. This collapses the block to one variable. Disjoint consecutive blocks, with arbitrary unused gaps, therefore preserve the new law. Successive means need not commute; no permutation invariance of the new full law is claimed.

Weak limits and factors. Denote translation in axis i by $U_i(r)$, $r \in B$. On the separable space L^2 of the symmetrized law, $U_i(r)$ converges weakly as i tends to infinity to an operator Q_r. Indeed matrix coefficients of two cylinder functions stabilize as soon as the axis is later than all the coordinates they involve, by increasing singleton substitution. All operators belong to the abelian von Neumann algebra generated by the translations. Replacing one late axis by the sum of three later axes preserves its cylinder coefficient. Taking three weak limits successively (with increasing indices, last index sent to infinity first) therefore proves $Q_r^3 = Q_r$. Each Q_r is a positive unital contraction preserving the integral. Normality and the polynomial identity show it is self-adjoint with spectrum contained in $\{-1, 0, 1\}$. Hence Q_r^2 is a positive orthogonal Markov projection, the conditional expectation onto a sub-σ-algebra B_r. Indeed Jensen and equality of integrals make the range closed under squaring bounded real elements, and thus a factor. On B_r, Q_r is a measure-algebra involution, since it and its inverse are positive and unital. On this factor $U_i(r)$ converges strongly to Q_r. Also

$B_r \subset B_{2r}$ and $Q_{2r} f = f$ for f in B_r, by two successive applications of strong convergence and commutativity. If $U_i(r)$ has odd finite order q throughout the row, the involution on B_r is the identity: $U_i(r)^q f$ converges strongly to $Q_r^q f$ there.

For a cylinder vector f one may equivalently sample the positive normalized spectral measure of the tail coordinate characters. It is spreadable and hence exchangeable (the classical theorem on spreadable sequences). Its directing law conditional on the de Finetti parameter is almost surely Haar on a coset of a closed subgroup of the compact character group of B, the coset having order at most two in the quotient. In fact replacing the first sampled coordinate by the product of three others leaves its joint law with arbitrarily many later samples unchanged. Thus the conditional measure ν satisfies $\nu^{*3} = \nu$ almost surely. On the countable character group this forces each Fourier coefficient to be 0, 1 or -1; the characters of modulus one form a subgroup, define a single sign character there, and vanish in Fourier coefficient off it. Haar measure on its annihilator coset has exactly these coefficients. This is only a description of spectral measures; the operator proof of the involutions does not require de Finetti.

Limitation for the threshold. This shows that a symmetrized witness has only two-point residual compact motion at an individual coefficient on its own limiting factor. It does not show that the joining of h different temporal slots is controlled by these single-observation factors. Nor does it give a common involution for correlated low and high coefficients before an appropriate characteristic-factor argument. The $h \geq 6$ obstruction still satisfies the symmetrization hypotheses.

R210. A dilate threshold for logarithmic random prime templates

Scope and status. Result with proof.

Result. Logarithmically many random prime residues have a sharp noncritical dilation-separator threshold, yielding endpoint one for suitable random rows.

Scope. The strict inequalities $C\log(1/b) < 1$ and > 1 are treated; the equality case is not.

Qualifications. Typical random rows under the prescribed rapidly escaping schedule do not classify arbitrary prime rows.

Let $p \to \infty$ be prime, let B_p be the complement of a centered interval in C_p with $|B_p|/p \to b \in (0,1)$, and draw m independent uniform directions in C_p. Equivalently one can condition on all being distinct and nonzero when $m = O(\log p)$, since this succeeds with probability $1-o(1)$. If $m/\log p \to C$, with $C\log(1/b) < 1$, with probability tending to one some nonzero multiplier carries **all** the directions into B_p. If $C\log(1/b) > 1$, the probability tends to zero. The equality case is not asserted.

Write $b_p = |B_p|/p$ and let X count the successful multipliers. Then $EX = (p-1)b_p^m$, proving the second assertion by Markov. In the first range $EX \to \infty$. For ratios $r \neq 0$ let $c_r = E_x 1_B(x)1_B(rx)$. Normalized additive Fourier expansion and averaging over nonzero multipliers give

$$E_{r\neq 0}|c_r - b_p^2| \leq (\sum_{k\neq 0} |1_B - hat(k)|)^2/(p-1) = O(\log^2 p/p).$$

The interval Fourier l^1 estimate follows from the geometric sum and $|1_B - hat(k)| \leq \min(1, C/\min(k, p-k))$ for $1 \leq k < p$. Consequently for any $a > 0$, the exceptional ratio fraction where $|c_r - b_p^2| > p^{-a}$ is $O(p^{a-1}\log^2 p)$. The normalized second moment $EX^2/(EX)^2$ is the ratio average of $(c_r/b_p^2)^m$. On the good ratios this is at most $1 + O(mp^{-a})$. On the exceptional ratios it is at most $b_p^{-m} = p^{C\log(1/b)+o(1)}$. Choose $0 < a < 1 - C\log(1/b)$. The normalized second moment tends to one with a polynomially decaying error. Chebyshev or the second-moment inequality proves that $X > 0$ with probability tending to one.

For any bounded logarithmic sampling rate, choose b close enough to one and each typical row is Bohr-separated with a fixed gap. Along sufficiently rapidly growing primes, e.g. at least exponential in the row index, the polynomial failure bounds are summable. Almost surely only finitely many of those independent random rows then fail the common separation test. Choosing its multiplier on each remaining primary coordinate gives one global character, and the root-rotation repair of the finitely many initial primes shows that this saturated family has critical threshold one. This assertion for a specified rapidly growing sequence does not eliminate atypical rows when all primes are sampled. For the related infinite integer Bernoulli results see Kahane and Katznelson, random integers and Bohr density.

R223. Rigidity of exact scalar progression identities

Scope and status. Result with proof.

Result. Exact scalar progression identities on prime or rough cyclic groups force bounded-degree polynomial tags; the integer analogue forces finite or nil-Bohr separation.

Scope. The identity holds at every start for all permitted directions.

Qualifications. A general high-density avoiding mask need not furnish such an exact scalar identity.

Fix $h \geq 2$ and $M = \binom{h}{2}$. Let V be a menu of velocities containing zero. An exact scalar identity is

$$\sum_{j=0}^{h-1} f_j(x + jr) = \tau(r) \quad \text{for all} \quad x \quad \text{and} \quad r \in V$$

in the circle group. The functions may be arbitrary; on the integers no regularity assumption on them is needed.

Prime cyclic group. If the ambient group is C_p with $p > \max(M, h)$, then $\tau - \tau(0)$ on V is the restriction of $P(r)/p$ with $P \in F_p[r]$, $P(0) = 0$ and degree at most $h-1$. Conversely every such polynomial has an exact scalar identity there, so this characterizes the possible restrictions.

Sum over all x to see that $p(\tau(r) - \tau(0)) = 0$. If $|V| < h$, interpolate at its $|V|$ distinct field points. Otherwise choose distinct $v_1,\ldots,v_h$ in V. Write S for unit shift of a function and $g_j = (S-1)f_j$. The h equations obtained by differencing in x have coefficient matrix $((S^{v_i})^j)_{i,j}$. Its commutative adjugate gives

$$[\prod_{a<b} (S^{v_b} - S^{v_a})]g_j = 0.$$

Each g_j is killed by $\Phi_p(S) = 1 + S + \cdots + S^{p-1}$. For $1 \leq k < p$ the geometric $\sum 1 + S + \cdots + S^{k-1}$ is a unit modulo $\Phi_p(S)$ over the integers: multiply by $1 + S^k + \cdots + S^{(l-1)k}$ with $kl \equiv 1 \bmod p$ to get the geometric sum of length kl, congruent to 1. Units S and signs can be dropped too. Thus $(S-1)^{M+1} f_j = 0$ on the whole cycle. Newton's formula writes the lifted array as a binomial polynomial of degree $\leq M$. Periodicity by p makes each nonconstant coefficient p-torsion, by downward induction using $p \mid \binom{p}{i}$ for $1 \leq i \leq M < p$. Up to an added constant we have genuine F_p-polynomials P_j of degree $\leq M$ divided by p.

If d=max deg $P_j \geq h$, for each $0 \leq s < h$ the coefficient of x^{d-s} in $\sum_j P_j(x + jr)$ is a polynomial in r of degree $\leq s$. It vanishes at the h selected velocities, since $d - s \geq 1$ and the polynomial in x has degree $< p$. Thus its coefficient of r^s vanishes, giving

$$\binom{d}{s} \sum_j j^s [x^d] P_j = 0.$$

All binomial factors are units and the Vandermonde matrix on $j = 0,\ldots,h-1$ is invertible. All top coefficients vanish, a contradiction. Setting $x = 0$ proves the assertion with the sharper degree $h-1$. Conversely each monomial r^k with $k \le h-1$ is the kth finite difference of $x^k/k!$ at spacing r, a sum of permitted slot functions. A separate constant accounts for $\tau(0)$.

The same direct large-menu bound holds on C_N if every prime divisor of N exceeds $\max(M, h)$ and V has h representatives whose pair differences are units modulo N: use $1+\cdots+S^{N-1}$ on the zero-sum arrays and the same integral inverse when $(k, N) = 1$. Binomial coefficients $\binom{N}{j}$ are multiples of N for $1 \le j \le M$, and the interpolation and top-degree Vandermonde determinants are units modulo N. The small-menu interpolation without unit differences was asserted only for prime p.

Consequences for many cyclic axes. On squarefree prime tails any sum of exact scalar charges with such menus and a uniform circle gap reduces to a bounded-degree polynomial test, regardless of the local menus or functions. Write each P_i with zero constant, separate into at most $h-1$ homogeneous power sums Q_j, and suppose the norm of the total charge exceeds $\gamma > 0$ on permitted directions. One Q_j then has norm $\ge \gamma/(h-1)$. These are degree-j polynomials on the direct sum of the C_{p_i}. For any fixed integer multiplier L, on tails past all primes dividing L their coefficients have Lth roots modulo p_i. The circle-polynomial hitting argument of the sparse-support note [R240] produces simultaneous near-full avoiding masks: choose a short omitted arc per j; choose L divisible by the bounded denominators of leading coefficients for polynomials missing the arc, and apply the test to rooted $Q_j(x+tv)$. The length is uniform over p_i and the number of coordinates. Finite head directions can be removed on rational saturation by squarefree tail transfer [R241]. In particular these scalar identities cannot produce an intermediate integer threshold by combining distinct prime-cycle axes. The same argument works with vector-valued charges in a fixed finite product of circles and a gap in some coordinate. It does not apply to arbitrary finite color obstructions, which need not be scalar identities.

Exact scalar identities on the actual integers. For any $V \subset Z$ as above, if V contains h distinct velocities the same adjugate argument works before reducing modulo any period. Choose $L \ge 1$ divisible by all pair differences of the h selected velocities (using absolute values). Each $S^{v_b} - S^{v_a}$ divides $S^L - 1$ as a Laurent polynomial over Z, and $S-1$ does also. Consequently $(S^L-1)^{M+1} f_j = 0$. On each residue modulo L every f_j is an ordinary circle-valued binomial polynomial in the quotient variable, of degree at most M. On multiples of L we may lift it to a real polynomial $F_j(n)$ at Ln, converting binomial powers to ordinary powers. On $V \cap LZ$, $\tau(r)$ is the sum of these polynomial evaluations at jr/L.

If infinitely many nonzero velocities occur in $V \cap LZ$, the degree on a smaller congruence subgroup may be sharpened to $h-1$. At each of h distinct tested r/L, the polynomial $\sum_j F_j(x + jr/L) - \sum_j F_j(jr/L)$ takes integer values at every integer x. Its coefficients are rational by binomial interpolation. Take coefficients modulo Q, a vector space over Q, and apply the top-degree Vandermonde argument above. If any coefficient of largest degree $d \ge h$ survived in R/Q, the h moments would force all coefficients of that degree to vanish. Therefore every F_j coefficient in degree $\ge h$ is rational. Pass to multiples of $L' = L$ times a common denominator; all those higher powers then take integer values. There is a real polynomial Q of degree $\le h-1$ with $Q(0) = 0$ such that $\tau(r) - \tau(0) = Q(r)$ mod one for every $r \in V \cap L'Z$. If the original $V \cap LZ$ is finite, a further modulus removes all its nonzero velocities instead.

In particular, if $D = V \setminus \{0\}$ is positive and all its charges have $||\tau(r) - \tau(0)|| \geq \gamma > 0$, D misses a nil-Bohr zero neighborhood (impose the congruence and $||Q(r)|| < \gamma$). Thus $b(D) = 1$ by Alweiss and Farhangi, Theorems 1.8 and 1.9. A finite V is immediate. The integer result applies to identities holding at every start, a stronger condition than an h-slot obstruction on some positive-density event after removing carry boundaries.

R249. Hamming clocks covered by a bounded-entropy family of angle templates

Scope and status. Result with proof.

Result. Hamming clocks controlled by a bounded-entropy family of angle templates have arbitrarily high-density avoiders.

Scope. The template covering entropy, error scale, and required root synchronization are explicit.

Qualifications. High-entropy wandering angle families are not excluded.

This extends finite nonzero angular centers [R115]. Let $K \subset T \setminus \{0\}$ be finite, and fix c,C. At dimension n let $\beta^u \in T^n$, $1 \le u \le M_n$, be predetermined templates, all coordinates within δ_n of K, where $\delta_n \to 0$ and $\log(2M_n) = O(1/\delta_n^2)$. The compared characters may be on the integers or on a finite cyclic group of prime order tending to infinity, so that the finitely many needed character roots exist. An eligible difference d has its phases within $c/\sqrt{n}$ of the corresponding entries of one β^u, except at most $C\sqrt{n}$ coordinates. For every $\epsilon > 0$ there is a common finite length H such that these families, on every sufficiently large n, have H-avoiding stationary masks of density $> 1 - \epsilon$. The statement allows δ_n to tend to zero arbitrarily slowly, unlike a single common interval of width $c/\sqrt{n}$.

Take the fixed probability polynomial P with nonnegative coefficients that annihilates the roots corresponding to K, as in the finite-center note. On a root row, write $Z = n^{-1/2} \sum_i e(x_i)$, with independent Haar initial phases. At every retained site impose two extra conditions simultaneously for every u: the absolute value of $n^{-1/2} \sum_i e(x_i) P(e(\beta_i^u))$ is at most a sufficiently large fixed B_1, and that of $n^{-1/2} \sum_i e(x_i) P(e(\beta_i^u))^2$ is at most $\tau_n \to 0$. Each summand is centered and bounded by constants times δ_n or δ_n^2. Hoeffding on real and imaginary parts and a union bound over M_n make the discarded probability $o(1)$. Indeed take B_1 fixed large enough for the stated entropy constant, and take $\tau_n = C_1 \delta_n$ with C_1 sufficiently large. The conditions are stationary site tests and require no actual group element whose phases exactly realize a template.

Now follow an arbitrarily long entire retained d-run, letting R be the root stride and splitting the regular and exceptional coordinates. At any given site, the regular sum in the nonnegative P-stencil is bounded in absolute value by a constant independent of dimension: compare to the accepted B_1 template bound, remove $\le C\sqrt{n}$ template terms and use the $O(1/\sqrt{n})$ Lipschitz error per regular angle. The exceptional sum always has Fourier–Stieltjes norm $\le C$. This gives the same fixed-threshold activation stencil as the finite-center proof, with B_1 incorporated into the gate threshold.

For the P^2-stencil on the regular trace the error is now $o(1)$. Template terms removed for exceptional coordinates cost $O(C\delta_n^2)$, while a regular term's squared-polynomial discrepancy is $O(\delta_n/\sqrt{n} + 1/n)$. The tested template sum itself is $\le \tau_n$. Pass to bounded pointwise limits on each R-residue of arbitrarily long retained stretches (the base complex sum may be bounded at sites by an arbitrarily large fixed B, at negligible prescribed cost). The exceptional part is a Fourier transform of a measure of norm $\le C$, by compactness. The remaining bounded bi-infinite regular sequence is killed by $P(S)^2$, where S is the one-slot shift on that residue. Elementary constant-coefficient recurrence theory makes it a sum of exponentials for the unit roots of P; all non-unit roots and all positive-degree polynomial factors are excluded by boundedness in both time directions. Its Fourier–Stieltjes norm is

uniformly bounded because there are a fixed number of unit roots and each coefficient is bounded by the sequence bound (Cesàro extraction). The stable gate approximation, integer root synchronization, boundary-gap and density choices of the finite-center argument [R115] then apply unchanged.

In particular if the nonexceptional coordinate velocities all equal a single prescribed block direction, whose corresponding coordinate phases approach nonzero centers at any rate, there is one template. Large equal-prime Hamming blocks with arbitrary $O(\sqrt{n})$ exceptional coordinates around an all-ones direction therefore still admit near-full obstruction in the root-compatible model, including when n is much larger than p^2. This extension does not handle arbitrary high-entropy choices independently wandering around the centers.

R268. Exact repair of vanishing progression errors in a VC family

Scope and status. Result with proof.

Result. Vanishing progression-error probabilities in a uniformly finite-VC family can be repaired exactly in a fixed product power, retaining mass close to a^r for r exceeding the VC dimension.

Scope. Countable errors on a common conull set, fixed progression length, and a uniform VC bound.

Qualifications. Pointwise vanishing alone is insufficient; this constructs a positive lower witness but no upper threshold.

Let B be an event of mass a in a stationary integer system. For fixed h and d in a countable family D let

$$E_d = \{x : T^{jd}x \in B \quad \text{for every} \quad 0 \le j < h\}.$$

Assume $\mu(E_d) \to 0$ as d tends to infinity through D, and that the countable family of these error events has finite VC dimension v (on one common conull set suffices). For every integer $r > v$ there are h-avoiding events for sufficiently late D-tails of mass arbitrarily close to a^r. In particular $b(D) \ge a^r$, because finitely many omitted directions can then be blocked at arbitrarily small cost by allowing a longer length. This statement is useful even when $\sum_d \mu(E_d)$ diverges. It shows only positivity if $a > 0$; it does not put b strictly below one.

Here is the probability fact behind the repair. For a countable VC family C of dimension v on a probability space, with $r > v$ fixed,

$\Pr[\exists C \in \mathcal{C} : \mu(C) \le \eta,\ X_1, \ldots, X_r \in C] \to 0 \quad \text{as } \eta \downarrow 0.$

for iid observations. Given such a witness choose the first in a fixed enumeration. Conditional on the first r observations, independently sampling n more avoids this particular set with probability $\ge (1-\eta)^n$. Thus, up to this divisor, the probability in question is bounded by the probability that a set in the class selects exactly the first r sample indices from all $n + r$ observations. Conditional on the unordered sample, permuting the indices uniformly makes this at most $\sum_{j \le v} \binom{n+r}{j} / \binom{n+r}{r}$ by the Sauer–Shelah bound. This applies also with coincident samples, by counting traces on indices. Let $\eta \downarrow 0$ for fixed n and then $n \to \infty$.

In the product of r copies of the stationary system, mark B in every coordinate and remove all points of $\bigcup_{d \in D_{\text{tail}}} E_d^r$ from that mark. The removed mass tends to zero by the probability fact. At the beginning of any surviving h-progression of step d, all coordinates would have been in E_d; this contradicts the deletion. The remaining mass is at least $a^r - o(1)$.

The same proof with independent stationary realizations applies on finite packets whenever the individual error bound tends to zero and the VC dimension is uniformly bounded. Pointwise convergence of the probabilities alone supplies neither the VC bound nor the exact repair. The lemma is a possible way of obtaining a common positive lower witness in a template construction; the high-density forcing hypotheses still have to be proved.

Chapter 11

Finite combinatorial estimates and obstructions

R024. A dimension-free bound for row events with small forbidden certificates

Scope and status. Result with proof.

Result. Events having bounded-width forbidden certificates admit dimension-free cylinder-cover bounds and local-lemma applications.

Scope. Certificate width must be uniformly bounded; general predicates with unbounded certificates are outside the result.

Let a finite row have independent coordinates in arbitrary standard probability spaces. Suppose its forbidden event C agrees modulo null sets with the union of measurable cylinders, each depending on at most k coordinates. There is a cover of C modulo null sets by cylinders of the same width whose probabilities sum to at most $K_k P(C)$, with K_k depending only on k. A cylinder cover here may include points outside C. This bound does not require any independence of the original certificates, nor any bound on their number. Because the number of possible supports is finite, merge multiple cylinders on a common support first (for an arbitrary uncountable union assume the merged predicates are measurable and give the stated event).

Proof. For C of probability $\epsilon \geq 1/2$ take the whole space as one cylinder. For $k = 1$, independence shows $\sum_i p_i \leq -\log(1-\epsilon) \leq 2\epsilon$ for $\epsilon < 1/2$. Suppose the assertion is known through $k-1$. For each nonempty S with $|S| \leq k-1$, define

$$D_S = \{x_S : P(C|x_S) > c\}, c = 1/(2^{k+1}+1).$$

Let D be the union of these cylinders. We claim $P(D) \leq 2^k \epsilon/c$. Take two independent full samples X,X'. On $X \in D$ choose a witnessing S. Keep the coordinates X_S and replace the rest by X'; conditional on X this gives a bad point with probability $> c$. On the other hand, if *any* point mixing coordinates of X and X' is bad, at least one width-k certificate is witnessed. A fair independent choice between X and X' at every coordinate then gives a bad point with conditional probability $\geq 2^{-k}$, since it suffices to match that one certificate.

The random mixed point has the original product distribution, proving the claim. Use the induction hypothesis to cover D at total cost $\le K_{k-1}2^k\epsilon/c$.

Write the original merged cylinders as E_L. In each E_L keep only assignments having **no** heavy D_S for $S \subseteq L, 1 \le |S| \le k-1$; call the result R_L. Points of C not covered by D are still covered by these residual cylinders. For each S let

$$d_S(x_S) = \sum_{L \supseteq S} P(R_L | x_S).$$

For $|S| = k$ this is at most one. Induct downwards to show it remains at most one for $1 \le |S| < k$. Fix x_S and let N_S count the residual witnesses whose supports contain S, with all other coordinates independently sampled. Then

$$E(N_S^2 | x_S) \le d_S^2 + 2^k d_S.$$

Indeed ordered pairs L,L' with intersection exactly S contribute products of the conditional marginal probabilities and hence total at most d_S^2. For each fixed L and nonempty further intersection $U = L \cap L' \supsetneq S$, the product formula conditional on x_U and the bound $d_U \le 1$ show that the sum over such L' is at most $P(R_L|x_S)$. There are at most 2^k choices of U per L, including the diagonal case. This proves the estimate. Whenever x_S belongs to D_S the residual count is zero. Otherwise the conditional probability of a positive count is at most c (every residual witness is contained in C), so conditional Cauchy–Schwarz gives $d_S \le c \cdot 2^k/(1-c) \le 1$. Finally apply the same second moment estimate at the empty S, using ϵ instead of c, to obtain $\sum_L P(R_L) \le 2^k\epsilon/(1-\epsilon) \le 2^{k+1}\epsilon$. Adding the inductive cover of D proves the lemma, for example with $K_k = K_{k-1}2^k/c + 2^{k+1}$ and $K_1 = 2$. All conditional assertions are understood almost everywhere, which suffices because only finitely many coordinate sets occur.

Application to the endpoint-shear surrogate. Let

$$F_j(t) = X_{t+j} + jY_{t+j} + w_j(t)$$

with independent Haar pairs X_s,Y_s and arbitrary deterministic forcing. For each row the joint law is product Haar. If $P(C) < 1/(4k^2K_k)$ and the forbidden event C has width-k certificates as above, all rows of any finite horizon can simultaneously avoid C with positive probability, uniformly in the number and spacing of the spatial offsets and in the forcing (the positive probability itself may depend on the horizon). Use the new cover, of masses p_L. At time t a bad certificate is an event in the independent endpoint pairs indexed by $t + L$. A particular type L' at other times can share an endpoint with at most $|L||L'|$ of its translates (equivalently for fixed t,L there are at most this many time indices for L'). Under the asymmetric local lemma use parameters $2p_L$. The sum of parameters over neighbors is at most $2k^2\Sigma p_L < 1/2$, hence their product of complements is $\ge 1/2$. This verifies the criterion. A whole-space cylinder is excluded by the small-cost bound. Null cover defects do not affect any of the finitely many Haar-row marginals.

The same proof works for any model with independent input blocks, product one-row marginals, and this bounded intersection count for small coordinate certificates, including

the single moving stream. It places no restriction on how complicated each allowed k-coordinate predicate may be. It does not establish a dimension-free bound for general row predicates whose smallest forbidden certificates have unbounded spatial support.

R025. Bounded sums of powers always have integer critical value one

Scope and status. Result with proof.

Result. Bounded sums of signed powers, and automatic sets of polynomial word-count growth, have integer critical value one.

Scope. Polynomial growth in digit length is substantially stronger than zero density.

Fix $k \geq 2$, $m \geq 1$ and a finite set of signed integer coefficients. Let D consist of the positive absolute values of the nonzero sums of at most m terms ck^a, $a \geq 0$, with these coefficients. Then $b_Z(D) = 1$. In particular this applies to all differences of two powers, even allowing both signs; and it applies to any base-k automatic set whose language has polynomial word-count growth in the number of digits. This theorem gives arbitrarily dense avoiding sets at a finite length depending on the desired density. It gives no obstruction for a direction family with an unbounded number of power terms.

Digit marker for separated powers. Ignore zero coefficients and orient the whole sum positively, adjoining the negatives of all coefficients to the list if necessary. Let C be a bound on their absolute values. Choose s very large relative to m and C, and put $B = k^s$. Among the $\leq m$ exponent residues modulo s there is an empty arc of length at least $\lfloor s/m \rfloor - 1$. Some shift by $u \in \{0, \ldots, s-1\}$ puts all remainders $r = (a+u) \bmod s$ at most $s - \lfloor s/m \rfloor + 1$. After multiplying both the start n and the difference d by k^u, all terms have the form $\gamma_i B^{b_i}$, where $\gamma_i = c_i k^{r_i}$ are nonzero integers with $\max |\gamma_i| = G$ and $m(G+1)$ arbitrarily small relative to B by our choice of s. We test the finitely many markers $M_u(n)$ equal to the sum of the squares of all base-B digits of $k^u n$, reduced modulo a prime P. Choose P larger than all possible positive sums $\Sigma\gamma_i^2$, and with s/P smaller than the allotted site loss. Put $L = P$. Require for this paragraph that each gap between successive b_i is at least $L+2$, so the bands $[b_i, b_i + L]$ are disjoint with at least one unused position between them.

For every start $n \geq 0$ we can choose a time t with base-B digits $t_0, \ldots, t_L$ and all higher digits zero, so that $M_u(n + td)$ is any prescribed residue modulo P. In band i the signed contribution to the ordinary addition is $\gamma_i t$. Its incoming carries at variable positions and the guard position lie between $-G-2$ and $G+2$: carries entering from a separated lower band have absolute value at most one after one empty position, and elementary digit division gives the bound successively in the band. First choose the same guard digit t_L for all bands such that the outgoing guard carry $\lfloor (x_i + \gamma_i t_L + c)/B \rfloor$ is independent of all incoming carries $|c| \leq G+2$, where x_i is the original guard digit. There are $O(G + |\gamma_i| + 1)$ bad values for band i. Indeed the linear expression crosses $O(|\gamma_i| + 1)$ multiples of B; each crossing excludes $O(1 + (G+2)/|\gamma_i|)$ consecutive input digits. The absolute bound is $O(G+1)$ even for negative or noninvertible coefficients. Thus a guard digit survives for s sufficiently large. All digits between bands and the initial carries into the bands are now fixed, independent of the later branch choices. The guard digit's *output digit* may depend on its incoming carry; the procedure below makes that incoming carry identical on all branches as well.

Choose the L lower time digits sequentially. At each position their common incoming carries are fixed. The B possible values of the next digit split into at most $1 + \sum_i (|\gamma_i| + 1)$ intervals on each of which all outgoing carries are constant. Some interval has at least three consecutive values. On it the contribution of all newly determined output digits to M_u is a quadratic polynomial in the time digit with leading coefficient $\sum_i \gamma_i^2$, nonzero modulo P.

Thus two choices on this interval give different residues and have identical outgoing carries. Keep those two choices and proceed. At the end, all contributions outside the L variable levels, including every guard output, are the same on all branches. The sum of the L independent two-point residue menus is all of C_P: adding a two-point set with distinct elements increases the size of any nonempty proper set by at least one in a prime cycle. The chosen t is less than B^{L+1}. The digit calculation remains valid for signed coefficients since $n + td \geq 0$ and signed carries are simply Euclidean division in the usual base expansion.

For each of the s markers omit one chosen residue. Independent uniform choices of the omitted residues retain any n with probability at least $1-s/P$. Some choice therefore retains at least this upper natural density by reverse Fatou, and no progression at the separated steps survives B^{L+1} consecutive times, by the preceding deterministic property for its appropriate shift u.

Other gaps and induction. If clustering occurs for the chosen u, two successive original exponents differ by at most $s(L + 3)$. The corresponding terms can be combined into one power with a coefficient from the finite set $\{c+c'k^j : 0 \leq j \leq s(L+3)\}$, enlarging this list by the original coefficients and allowing zero terms to disappear. This case has at most $m - 1$ nonzero terms. Induct on m, with the one-term case handled by the separated construction. The threshold-one property is closed under finite unions by products of stationary avoiders, so allocate an arbitrary small loss to each of the two cases. The order of choices matters: s and the coefficient menu are fixed before choosing P, then the new finite coefficient menu for the induction is determined.

Automatic corollary. A regular language with polynomial growth is a finite union of languages of the form $w_0v_1 * w_1 \cdots v_l * w_l$. One proof decomposes accepting paths in a trimmed deterministic automaton into strongly connected components: every cyclic component must be a simple directed cycle, since a state with two different internal exits admits two distinguishable closed words of the same length (use powers to equalize lengths), hence exponentially many words of some lengths. There are only finitely many types of paths through the component graph and entry/exit vertices. On a bounded-language piece, the value of each word, after clearing a fixed denominator, is a sum of boundedly many fixed coefficients times powers of k. This follows from concatenation and the geometric formula for a repeated digit word; it includes constant terms as exponent zero. The common denominator can be chosen uniformly on the finitely many pieces. Dilation preserves b_Z (restriction to a power and the usual root tower), which proves the corollary. For background on this sparsity convention see Sparse Automatic Sets. Polynomial growth in digit length is stronger than merely zero natural or Banach density.

R028. A bounded number of word maps cannot by itself certify an intermediate threshold

Scope and status. Result with proof.

Result. A bounded number of maps on arbitrarily large word cubes cannot provide high-density-only forcing if every line contains a sufficiently long allowed progression.

Scope. The conclusion is full positive-density recurrence at the specified forbidden length.

Let G be a countable torsion-free abelian group and $D \subset G$. Fix $k, h_0 \geq 2$, and let $H = W(k, h_0)$ be an ordinary van der Waerden number. Suppose that for some fixed finite alphabet of size l, arbitrarily large word dimensions m admit maps $f_1, \dots, f_k : [l]^m \to G$ such that for every combinatorial line L the union $\bigcup_j f_j(L)$ contains an H-term progression with difference in D. Then **every positive-measure event in every G-action has an h_0-return along D**. The maps may be nonlinear, and the elements of the contained progression may come from different word positions or branches.

The motivation for considering this condition is the elementary union bound. For an event of measure $> 1 - 1/k$ and a random common translate, the expected density of words for which every branch is in the event is bounded below by $k\mu(A) - (k-1)$. One might hope to obtain high-density-only recurrence by building two or more tracks through a word cube even though individual tracks do not give legal differences. The stronger conclusion above rules out an arbitrary-dimensional construction of this type as the source of a genuine intermediate threshold.

Suppose A were a positive h_0-avoider for D. For every $q \leq H$, its q-root action gives a positive h_0-avoider for qD of the same mass. For a torsion-free G, identify G with qG by multiplication, transport the original action to qG, coinduce to G, and take the mask at the identity-coset coordinate. On a divisible host the explicit q-division action suffices. Intersect these masks in independent products and take an ergodic component or ergodic binary shift law with positive mass; denote this new event by B, of mass $b > 0$. It avoids $\bigcup_{q \leq H} qD$ at the same horizon h_0.

Choose m above the density Hales–Jewett threshold for alphabet l and density $b^k/2$. For a fixed word w, average over x and independently over group translates $t_1, \dots, t_k$ in large Følner sets. The probability that $U^{f_j(w)+t_j}x \in B$ for all j tends to b^k, by the mean ergodic theorem and bounded-function telescoping. The cube is finite. Consequently some common x and one fixed list of translates mark at least a $b^k/2$ fraction of its words. The point x can be taken outside all countably many exceptions for the forbidden runs and their G-translates. A combinatorial line is fully marked. Every site of its extracted length-H progression then belongs, in the orbit name of x, to one of the k translates $U^{-t_j}B$. Color each time by one such branch. A monochromatic h_0-term subprogression has step qd for $1 \leq q \leq H$, contradicting the defining avoidance.

This does not prohibit a template available only up to an explicitly bounded dimension depending on the large-density margin, or branches whose number grows without a suitable fixed-horizon restriction. A synchronous template available for arbitrarily long horizons, with unbounded dimension, does induce a fixed-horizon test if restricting the word alphabet to its first H letters preserves the first H extracted time slots. Bounded-torsion groups do not satisfy the root step: the known two-track involution construction there is consistent with this argument.

R041. Daisy-based rank probes for a single exceptional digit

Scope and status. Result with proof.

Result. Static daisy feature models and independent-column rank probes obey algebraic and asymptotic obstructions to implementing a one-mark exception.

Scope. Changing dictionaries, exceptional deterministic coefficients, and nonlinear rank fields are not covered.

Ellis–Ivan–Leader construct positive-density daisy-free families on middle layers. On a fixed set of four petals with a common root of size $r-2$, the six choices of two petals cannot all give bases over $\mathbb{F}_2$: after quotienting by the root span there are only three nonzero vectors in dimension two. Thus a general principle that every dense middle-layer family contains all bounded local rearrangements is false. Two features of a one-mark integer progression packet substantially limit a direct use of that construction.

First, if a single coordinate's normal query always gives exactly the same set of root features to two players, every free word in normals has that equality too. Indeed map the free player letters to the free group on feature-set labels, by the individual labeling maps. In a local packet where the special tuple e_* is a word in normal tuples, any two different feature-sets on the special must therefore differ on some normal row. An exact static-root daisy cannot be implanted locally by independently assigning features to the player letters.

Second, here is a quantitative asymptotic fact about the usual independent-column rank construction. Let $S_1, \dots, S_h$ be sets of r column labels each. To each distinct label assign an independent uniform vector of $\mathbb{F}_p^r$, p a fixed prime. If $\min_{j \neq l} |S_j \setminus S_l| \to \infty$, then the probability of simultaneous bases tends to

$$c_p^h, \qquad c_p = \prod_{l \geq 1} (1 - p^{-l}) > 0.$$

This statement is uniform over set incidence patterns under the stated divergence.

Let N_j be the space of row vectors annihilating all columns labeled by S_j. For fixed $t_1, \dots, t_h$, the joint moment of the numbers of injections $\mathbb{F}_p^{t_j} \hookrightarrow N_j$ tends to one. Sum over choices of linearly independent vectors *within each slot*. If all the vectors together are independent, their number is asymptotic to $p^{r \sum t_j}$; the joint probability is exactly $p^{-r \sum t_j}$ by summing, over labels, the dimension of the tester span required there. If the combined vectors have rank $s < \sum t_j$, pick a global basis from the testers, each basis vector tagged by its owning slot. There are $O_{p,h,t}(p^{rs})$ possible assignments of that rank. The probability exponent is at least rs, by counting every owned basis vector on the r labels of its owning set. Some nonbasis tester at slot j has a nonzero basis coefficient from a different slot l, since its own slot's original testers are independent. At every label in $S_j \setminus S_l$, that tester lies outside the span counted there, giving at least one more dimension. Thus this part is $O_{p,h,t}(p^{-\min_{j \neq l} |S_j \setminus S_l|})$. The case with one effective player has the familiar ordinary random matrix limit.

For completeness the moment implication needs no abstract determinacy assertion. For nullity d, the even partial sum

$$U_M(d) = \sum_{s=0}^{M} (-1)^s p^{\binom{s}{2}} \begin{bmatrix} d \\ s \end{bmatrix}_p$$

with even M lies above $1_{d=0}$; the next partial sum lies below it. For $d > 0$ the partial sum equals $(-1)^M p^{\binom{M+1}{2}} \begin{bmatrix} d-1 \\ M \end{bmatrix}_p$, interpreted as zero if $M \geq d$, by the Gaussian binomial recurrence. Thus $U_M \geq 0$, and its error over the indicator is at most $V_M(d) = p^{\binom{M+1}{2}} \begin{bmatrix} d \\ M+1 \end{bmatrix}_p$. Bound the desired product from above by the product of the U_M's and from below by subtracting $\sum_j V_M(d_j) \prod_{l \neq j} U_M(d_l)$ from this product. For fixed M, all these joint expectations tend to the products of their one-slot limiting expectations by the injection moments just proved. The one-slot expectation of V_M tends to $p^{\binom{M+1}{2}} / |\mathrm{GL}_{M+1}(\mathbb{F}_p)|$, which tends to zero as $M \to \infty$. The one-slot nonsingularity probability tends to c_p directly by sequentially exposing columns. This proves the claim, including its uniform consequence by contradiction.

More generally all the S_j may be replaced by arbitrary r-dimensional subspaces U_j in one common $\mathbb{F}_p$-space V, and the column test by invertibility of $M|_{U_j}$ for a single uniform random linear map $M : V \to \mathbb{F}_p^r$. If $\min_{j \neq l}(r - \dim(U_j \cap U_l)) \to \infty$, the same joint limit holds. The probability exponent for fixed tester spaces $V_j \leq (\mathbb{F}_p^r)^*$ is now $\dim(\sum_j U_j \otimes V_j)$. The global tester basis with owner slots gives the direct subsum of $U_{\mathrm{owner}(v)} \otimes v$, of dimension rs. If a dependent slot-j tester has nonzero coefficient at a basis vector owned by slot $l \neq j$, the image of its U_j-tensor space in the quotient by that direct subsum has dimension at least $\dim U_j - \dim(U_j \cap U_l)$, by inspecting this one coefficient modulo U_l. This replaces the extra exponent in the preceding proof. The uniform random codimension-r kernel version follows too: surjectivity of M has probability tending to one, since $\dim V - r$ diverges. This accommodates arbitrary local linear dependencies determined in advance.

Now let each symbol in a fixed replicated finite query alphabet select exactly d features from that coordinate's finite dictionary, or any d-dimensional subspace of its local linear ambient. At a vertex take all its selected features over independent coordinates (respectively the direct sum of their subspaces), and test the random basis/restriction as above. Choose one special query. For any pair of players that differs anywhere on the chosen special or normal rows, choose a normal proof row distinguishing that pair (if it differed only on the special this follows from the free-word argument). Repeat all such normal rows many times, using just one special among all the coordinates. Players with completely identical rows impose the same test and may be identified. Every remaining pair now has arbitrarily large feature-set differences or pairwise intersection codimensions, so the simultaneous rank probabilities are bounded away from zero independently of the ambient dimension. This rules out a deterministic daisy contradiction imposed solely by this feature-incidence pattern with a generic linear map. It does not rule out specifically chosen global nonrandom maps or other nonlinear rank mechanisms.

There is also an average-case obstruction to finding a global counterexample by this independent random coefficient model on cyclic digit lifts. In the homogeneous digit construction, partition positions into fixed-length blocks of a base b, or group individual digit features into the block feature dictionary. The finite word relation may be translated so all its local symbols fit in one block without a seam. Prescribe the finite true and repeated normal rows at finitely many fixed blocks, hence giving a *fixed* actual integer direction d_m with just one special and at least m feature differences for each distinct pair. Condition on the corresponding starting digits in these blocks, an event of constant positive probability independent of how many additional inactive digit blocks are used. All other blocks have zero motion. This is valid in arbitrarily high cyclic powers of the integer base, and also in any larger cyclic multiple using uniform starts on the digit part.

For random vector labels let Z be the fraction of these conditioned starts that give simul-

taneous bases. With h' distinct rows, $\mathbb{E}Z$ differs from $c_p^{h'}$ by at most $\epsilon_m \to 0$, uniformly once the dimension is large. For the second moment take two independent conditioned starts. If the replicated dictionary is nonconstant, every pair from different starts has at least m feature differences with probability tending to one, by independent inactive blocks. Applying the same moment limit with $2h'$ rows yields $\mathbb{E}Z^2 \le c_p^{2h'} + \epsilon'_m + o(1)$, $\epsilon'_m \to 0$. Hence the probability that $Z < c_p^{h'}/2$ tends uniformly to zero as $m \to \infty$, taking ambient size to infinity first. A limit of the random stationary laws consequently almost surely has a positive return on at least one such d_m. In the constant-dictionary case all rows coincide, so there is already no rank-based distinction. These conclusions concern independent uniform coefficient vectors with fixed local feature rules; an exceptional deterministic choice, a changing feature dictionary, or a nonlinear rank field requires separate analysis.

R116. A bounded number of choices at word vertices forces full density recurrence

Scope and status. Result with proof.

Result. A bounded number of vertex choices realizing every sufficiently large word line forces full positive-density recurrence.

Scope. Free abelian groups; the root/coinduction step is required to remove the stride introduced by van der Waerden.

Let G be a free abelian group, and $D \subset G$. Suppose some fixed q has the following property for each $H \geq 2$ and arbitrarily large m. One may assign to each word $w \in [H]^m$ a list of q group positions $x^1(w), \ldots, x^q(w)$ such that every combinatorial line $w(t)$, $0 \leq t < H$, has an admissible progression $a + td$, $d \in D$, with

$$a + td \in \{x^1(w(t)), \ldots, x^q(w(t))\} \quad \text{for every} \quad t.$$

The lists may vary arbitrarily with the vertex, and the selection rule with the full line. Then D has full multiple density recurrence. Thus deterministic bounded choices of this kind cannot by themselves prove a nonzero intermediate threshold. This is distinct from the fixed-offset polynomial IP barrier [R118]; it permits arbitrary vertex maps.

Given target length h and mass $a_0 > 0$, choose $H = W(q, h)$ from finite van der Waerden. At sufficiently large ambient word dimension, make one witness and one vector $(\epsilon_t)_{t<H} \in [q]^H$ of choices for each line. Regard that vector as a finite coloring of all one-parameter words. By the Graham–Rothschild parameter sets theorem, there is an M-parameter subword on which all one-parameter subwords have the same vector, for M as large as the density Hales–Jewett dimension for alphabet size h and mass $a_0/2$. The common vector has an h-term monochromatic subprogression of times $s, s + r, \ldots, s + (h - 1)r$, with common choice ϵ. Restrict the parameter letters to those h symbols. On each resulting h-line, the *single map* x^ϵ supplies an h-progression with difference rd, since that line extends to its full H-line.

Now in any measure-preserving G-action sample a random initial point. The expected fraction of restricted-word vertices whose $x^\epsilon(w)$ shift is in an event of mass a_0 equals a_0. With positive probability a density Hales–Jewett line exists. Summing the finitely many line probabilities gives an actual h-return of positive measure at an rd. Apply this argument to an r-root extension of the given action: realize its action on rG via $rg \mapsto T^g$, coinduce to G, and observe A in the identity-coordinate copy. Restriction to the rG-parameters rd recovers the required d-return in the original space. The root is taken after the finite array and its stride r have been selected. The claim follows at every length and density.

R118. A finite-offset IP proof would yield all-positive recurrence in a torsion-free group

Scope and status. Result with proof.

Result. A fixed finite-offset polynomial-IP realization in a torsion-free group yields recurrence at every positive density.

Scope. Requires the stated joint polynomial-IP recurrence input and a fixed offset count smaller than the progression length.

Qualifications. The polynomial IP recurrence theorem is an explicit hypothesis of this obstruction lemma.

Here is a restriction on a possible high-density proof, not a resolution of the prescribed-density problem. Let G be torsion-free abelian. Fix $q < h$, an offset set $B \subset G$ of size q, and a finite family of additive IP polynomials $P_0,\ldots,P_{h-1}$ of bounded degree with $P_j(\varnothing) = 0$ on finite subsets of N. Concretely these maps have vanishing alternating difference over $s+1$ disjoint sets together with the empty set, and the same is true on every disjoint-block IP subring. Suppose for every nonempty F there are a_F, $d_F \in G$ with d_F an allowed difference and

$$a_F + jd_F - P_j(F) \in B (0 \le j < h).$$

Intersecting q shifts of an event of measure $> 1 - 1/q$ leaves positive mass, so these data suggest a high-density argument by polynomial IP recurrence. In fact they imply h-recurrence at **every positive density** whenever the joint polynomial IP recurrence theorem applies to the listed P_j on each block subring.

Color F by its h-tuple of offsets in B. The finite-unions form of Hindman's theorem gives disjoint nonempty blocks whose nonempty finite unions have one fixed tuple $\beta_0,\ldots,\beta_{h-1}$. For each $j \le h-3$ the polynomial $P_{j+2} - 2P_{j+1} + P_j$ is constant on all nonempty unions in that subring, with value $-\beta_{j+2} - 2\beta_{j+1} + \beta_j$. Any such constant must be zero: evaluate the vanishing $(s+1)$st alternating difference of the polynomial on disjoint blocks; its empty-set value is zero, while every other value is the constant. Consequently the β_j form an h-term AP in B. Since G is torsion-free and $h > |B|$, all the β_j coincide (the case $h = 2$, $q = 1$ is immediate). On the subring the $P_j(F)$ themselves therefore form allowed progressions with step d_F. Joint IP recurrence on an arbitrary positive event supplies an allowed h-return. Absolute values of the integer steps can be used by reversal.

The point is the uniform finite list of offsets and the polynomial identities in F. Arbitrary measurable joinings, finite-valued programs that depend on the starting point, and a number of offsets growing with h are outside the claim.

R127. Arbitrary forcing does not give a dimension-independent density bound

Scope and status. Result with proof.

Result. Arbitrary forcing of moving input windows can block events of mass arbitrarily close to one when the output dimension grows.

Scope. An explicit surrogate forcing model; not an arithmetic counterexample to the conjecture.

Fix any $\epsilon > 0$. There exist a dimension r, a measurable $B \subset T^r$ with Haar mass $> 1 - \epsilon$, and a finite forcing list $w_0, \ldots, w_{h-1} \in T^r$ such that for every common offset $c \in T^r$ and every input sequence of circle values U_t, at least one of the vectors

$$(U_t, U_{t+1}, \ldots, U_{t+r-1}) + w_t + c$$

lies outside B. Thus a density threshold independent of dimension cannot be deduced from unrestricted finite linear Haar-stream controls for **arbitrary forcing**, even when all output coordinates have distinct leading degrees and independent Haar inputs are allowed. The forcing here is hand-picked; an even-profile subgroup with a shift-invariant odd coset permits choosing its representative, which this argument does not address.

Choose a binary marker word m of length l with fair-bit probability $p = 2^{-l} < \epsilon/8$. For a long window of $r = L + l + 1$ bits let its allowed mask require that m does not start at site zero but does start at some site $1, \ldots, L$. Since disjoint length-l test words are independent, choose L large enough that the probability of finding no marker among them is $< \epsilon/8$. The allowed probability is then $> 1 - \epsilon/4$. No bit stream has every window at times $0, \ldots, L$ allowed: an allowed window at zero requires another marker start at one of those later times, making that later window forbidden.

Lift this mask by the two half-circle cells on each coordinate. Retain only points at least $\eta > 0$ from each cell boundary. For small positive η the resulting compact set B still has Haar mass $> 1 - \epsilon$, by finite dimension. Select a finite $\eta/2$-net of the common-offset torus. Concatenate forcing segments, each of length $L + 1$ and identically minus one net point. For any common c one segment cancels it to within $\eta/2$. If the observations there all belonged to B, their half-circle bits would coincide with those of the unforced scalar input windows, by the boundary margin. This contradicts the binary marker property. Finite Haar input or any other input law therefore also fails on this specific forcing.

Even a four-bit example already exceeds one half. Take the allowed binary words of length four with Hamming weight at most two, of density $11/16$. Use XOR forcing words encoded by integers $0, 8, 8, 8, 14, 10, 12, 14$, with successive input bits read from most to least significant. The initially reachable binary triples $0, \ldots, 7$ are reduced after each slot to `0:6`, `0:5`, `0:4`, $\{0, 1, 2, 4\}$, $\{0, 2, 4\}$, $\{0, 1\}$, $\{0\}$, and the empty set (`0:k` denotes all integers from zero to k). The update appends a bit, tests the forced four-bit word and keeps its low three bits. The same margin and common-offset net convert this finite bit example to the torus.

R157. Short scalar signatures in finite modules force thin slope projections

Scope and status. Result with proof.

Result. Scalar finite-module signatures yield thin-projection restrictions and bounded-degree polynomial interpolation on bounded-rank elementary abelian groups.

Scope. Arbitrary depth inside boundedly many low-residue strips is not classified.

Complete consequence for elementary abelian prime groups of bounded rank If $G_p = F_p^r$ with fixed r, then every scalar signature on an arbitrary set of directions is a polynomial of degree bounded only in terms of r, h for large p. Induct on the rank. When every projection has at least h values, use the preceding theorem. Otherwise the set lies in $m \le h-1$ distinct parallel affine hyperplanes. On each hyperplane translate its argument by a representative a, multiplying each stencil vector m_j by U^{ja}. Restrict to the common kernel subgroup of rank $r-1$, still with a trivial output line. Induction supplies a bounded-degree polynomial on that hyperplane. Extend its linear coordinates to the full vector space, and sum the branch polynomials multiplied by the ordinary univariate Lagrange interpolation selectors of the transverse projection (degree at most $m-1$). The rank-zero case is immediate. When the original output at zero is zero, the resulting polynomial also takes that value. The rank-dependent bound grows at most linearly in r, for fixed h, by the explicit value of N and the cost of the Lagrange factors. The separate three-slot calculation [R252] also treats arbitrary p-power depth and gives degree three when $p > r+3$.

Fix a rank bound r and a progression length $h \ge 2$. Let

$$G_p = \prod_{i=1}^{r} C_{p^{e_i}}, \quad R = \mathbb{F}_p[G_p], \quad I = \ker(\mathrm{aug} : R \to \mathbb{F}_p)$$

with arbitrary depths $e_i \ge 1$. For group elements write U^d for their canonical units. We call $\phi : E \to F_p$ a scalar signature if there is an R-module with a nonzero trivial line generated by η and a tuple m_j, $0 \le j < h$, for which

$$\sum_{j<h} (U^d)^j m_j = \phi(d)\eta \qquad (d \in E).$$

All module elements, hence all spatial stencil coefficients, may depend on the prime. Several trivial output components can be tested one at a time. This is a statement about a particular source of short obstructions, not about arbitrary colorings or arbitrary actions.

Theorem. There is $C = C(r,h)$ such that for $p > C$, either the reduction $S = E \bmod p \subset F_p^r$ lies in a union of at most $h-1$ mutually parallel affine hyperplanes, or every scalar signature on E is the restriction of a polynomial in the coordinates of $d \bmod p$ of total degree at most C. Correlated slopes lying on nonlinear algebraic curves therefore do not by themselves evade scalar polynomial tests.

Put $z_d = U^d - 1$ and $v_d = (1, z_d, \dots, z_d^{h-1})$. The binomial transform between powers of U^d and powers of z_d is triangular and invertible. Let

$$W = \sum_{d \in E} R v_d \ \subset \ V = \bigoplus_{j<h} I^j.$$

A signature induces a scalar linear functional on W/IW with value $\phi(d)$ at v_d: use the transformed tuple to map R^h into the given module and note that its restriction to W has image in the trivial line. Conversely any such functional can be realized by quotienting $R^h \oplus F_p\eta$ by its graph on W. Only the forward direction is needed.

Use the presentation $R = F_p[x_1, \ldots, x_r]/(x_i^{p^{e_i}})$ via $x_i = U_i - 1$. In the grading by total degree the initial linear form of z_d is $L_d = \sum_i \bar{d}_i x_i$. With the shifts by j in the coordinates of V, the degree-zero leading tuple is

$$l_d = (1, L_d, \ldots, L_d^{h-1}).$$

This also applies when $L_d = 0$, with the corresponding positive-degree entries zero. First work in the untruncated polynomial ring $S_0 = F_p[x_1, \ldots, x_r]$. Select finitely many l_d spanning all of them as vectors over the **constant field** in $\oplus_{j<h}(S_0)_j$. This requires at most its dimension, depending only on r, h.

At a geometric point $x \neq 0$ the column rank is h, by Vandermonde, unless the linear projection $\bar{d} \mapsto \bar{d} \cdot x$ has fewer than h distinct values on S. A deficient projection over the algebraic closure already gives a deficient projection over F_p: partition S into its at most $h-1$ fibers, whose within-fiber differences span a proper F_p-subspace since they admit a nonzero common orthogonal vector over the extension field. Choose a nonzero base-field vector in the same orthogonal complement. Hence outside the stated hyperplane case every nonzero geometric point gives full rank.

Every nonzero maximal minor of the chosen matrix is homogeneous of degree $H = h(h-1)/2$. Their ideal contains $(x_1, \ldots, x_r)^N$ with $N = r(H-1)+1$. To verify this, base change to the algebraic closure. Choose r linear combinations of the minors of degree H whose only common zero is the origin: recursively avoid each positive-dimensional irreducible component of the preceding intersection, which is possible since no such component is in the common zero set of the full ideal. This is a homogeneous system of parameters in a polynomial ring, hence a regular sequence. The complete intersection quotient has Hilbert series $(1 + \ldots + t^{H-1})^r$, by successively using the multiplication exact sequences. All degrees at least N vanish. The containment descends by field extension and linear algebra in each homogeneous degree.

Each maximal minor times every coordinate vector lies in the column module by the adjugate identity. Consequently for every $n \geq N$ the entire homogeneous shifted layer $\oplus_{j<h}(S_0)_{n+j}$ lies in the span of these leading columns with coefficient polynomials of degree exactly n. Indeed all its coordinate monomials have degree at least N, so are in the minor ideal, and homogeneous components of the coefficients can be taken in the shifted grading. The same assertion holds after truncation to the associated graded of R. Therefore every element of $I^N V$ can be reduced modulo W successively from filtration layer N to $N+1, N+2, \ldots$, by using lifts of the indicated homogeneous coefficients times the actual v_d. The filtration is nilpotent, giving $I^N V \subset W$ and $I^{N+1}V \subset IW$.

The scalar functional on W extends linearly over F_p to $V/I^{N+1}V$. Its evaluation on v_d uses only degrees through $N + j$ in coordinate j. In the group-ring expansion of $U^d = \prod_i (1+x_i)^{d_i}$ each such coefficient is a polynomial of the same total degree in the low residues $d_i \bmod p$ when $p > N+h$, by the binomial formula and invertibility of the bounded factorials. Products forming z_d^j keep this bound. This proves the assertion with, for example, $C = N+h$. If low residues are already unisolvent for all polynomials of degree through $h-1$, the leading tuples span V/IV itself, by the invertible multinomial coefficients in powers of linear forms. Nakayama then gives the stronger $W = V$ and degree bound $h-1$.

For the homogeneous law $(U^d-1)^h m \in F_p\eta$ an alternative argument applies even with thin projections. In the subgroup $H_0 = \langle E \rangle$ choose at most r directions from E generating it, by taking a basis of H_0/pH_0 and using finite abelian p-group generation. Each corresponding augmentation increment has $(h+1)$st power killing the cyclic $F_p[H_0]$-module of m. Hence the $(rh+1)$st power of its augmentation ideal kills this module, by the pigeonhole principle and commutativity. For $p > rh+1$ all evaluations on H_0 are bounded-degree polynomials in coordinates of its minimal generators modulo p. This argument concerns the subgroup coordinates, which may lie at higher valuation in the original ambient presentation.

These results do not classify arbitrary short signatures when a nonzero linear projection takes only a bounded number of low residue values. Different lifts inside those parallel strips can occur at very different p-adic depths. A proof about torsion-free covers must handle those finer coordinates as well.

R172. A uniform obstruction to finite-field constructions based on one affine constraint

Scope and status. Result with proof.

Result. A uniform finite-character positivity neighborhood for one affine-constraint correlation excludes the associated high-rank polynomial gate constructions.

Scope. All lower coefficients must be freely averaged; arbitrary multiple correlations need not have this form.

Let $k \geq 3$, $K = \mathbb{F}_p^r$ with r arbitrary and p sufficiently large in terms of $k, a > 0$. If $B \subset K$ has density at least a, define

$$R_B(z) = \Pr_{x_0,\dots,x_{k-2} \in K} \left[\sum_{\ell=0}^{k-2} \binom{t}{\ell} x_\ell + \binom{t}{\{} k-1\} z \in B \quad (0 \leq t < k) \right].$$

There are constants $m, \sigma, \gamma > 0$ depending only on a, k and at most m characters $\chi_j \in \widehat{K}$ such that $R_B(z) > \gamma$ whenever $\max_j \|\chi_j(z)\| < \sigma$. The ranks of the output groups and the primes do not affect these bounds.

Lower bound at zero. Put $d = k - 2$ and $P(t) = (\binom{t}{1}, \dots, \binom{t}{d})$. Choose N so every subset of $[N]^d$ of density $a/2$ contains a translate of a positive integer dilate of the finite configuration $\{P(t) : 0 \leq t < k\}$, by the multidimensional Szemerédi theorem. Apply this to the points sent into B by $u \mapsto X_0 + \sum_j u_j X_j$, with independent uniform $X_j \in K$. The expected occupied fraction is at least a, hence the chance of density at least $a/2$ is at least $a/2$. For each copy with starting vector w and scale l, its success probability is precisely $R_B(0)$: $X_0 + \sum w_j X_j, lX_1, \dots, lX_d$ are jointly uniform, since $0 < l \leq N < p$. There are a bounded number of copies. The union bound gives $R_B(0) \geq c(a, k) > 0$.

Uniform continuity on a finite-character neighborhood. Let $f = 1_B$, let $c_t = (-1)^{k-1-t}\binom{k-1}{t}$, and evaluate characters additively modulo one. The first $k-1$ trajectory positions are independent uniform variables by the triangular binomial matrix. The only constraint on all positions is $\sum_t c_t y_t = z$. Thus the Fourier coefficients of R_B, up to harmless conjugations, are $\prod_{t<k} \widehat{f}(c_t \chi)$. Every fixed nonzero c_t acts bijectively on the dual when $p > k$ (or enlarge p beyond all these fixed coefficients). Use slot 2 as a cut: the sum of the absolute terms where $|\widehat{f}(c_2\chi)| < u$ is at most u, by Cauchy–Schwarz and Parseval on slots 0 and 1 and the bound of one at all other slots. The other terms lie on at most u^{-2} frequencies and their total absolute mass is at most one. Choose $u < c/8$, then a sufficiently narrow fixed window where all retained character values are close to one. The change from $R_B(0)$ is bounded by $2u + 2\pi\sigma < c/2$. This proves the assertion.

Application to prime rows. Suppose the row group is $\mathbb{F}_p^n$ and $Q : \mathbb{F}_p^n \to K$ is a homogeneous polynomial map of fixed degree $s \geq 1$. Assume every allowed direction v satisfies $R_B(Q(v)) = 0$ for some common row set B of density at least a. Then the row admits an arbitrarily near-full avoiding mask, with forbidden length independent of p, n, r, B, Q once p is sufficiently large. Indeed some detecting character has $\|\chi_j(Q(v))\| \geq \sigma$. Fix a short open arc of length less than $\eta/(2m)$. For a degree-s circle polynomial whose leading binomial coefficient is κ, there exist M, H depending only on the tolerance and degree so that $\|M\kappa\| \geq \sigma$ forces a visit to the arc in the first H terms, uniformly in all lower coefficients and the arc position. This follows from the torus case of the quantitative Leibman theorem (or the usual quantitative one-dimensional polynomial Weyl criterion), clearing the bounded

possible obstructions by M. Write $\chi_j(Q(x)) = u_j(x)/p \bmod 1$ with a homogeneous $\mathbb{F}_p$-polynomial $u_j(x)$. Evaluate $u_j(x)/(Ms!)$ in $\mathbb{F}_p$, followed by the standard circle embedding of its additive group. Its degree-s binomial coefficient along v, multiplied by M, is $u_j(v)/p$. Delete the sites mapping into an arc for each of the at most m channels. Choosing each arc location uniformly and averaging gives density loss at most $\eta/2$, and the retained mask excludes all length-H progressions at the allowed directions. Across distinct pure rows, sum each channel over the coordinate groups with an arbitrary common offset: along any row its other summands are constant and do not change the hitting implication. The same bound then holds for every finite packet. A compactness limit gives the corresponding tail conclusion whenever the squarefree realization applies.

This excludes even high-rank polynomial output gates under the **free lower coefficients** hypothesis in the definition of R_B. The zero of an ordinary multiple correlation in an arbitrary action does not imply this stronger hypothesis. In particular the argument does not exclude a prescribed subvariety of lower coefficients, or the signed paired candidate.

R202. Sparse directions in high-rank elementary primary filters

Scope and status. Result with proof.

Result. Sparse-direction scalar filters extract homogeneous tags below the characteristic, with rank bounds and an explicit Frobenius obstruction outside that range.

Scope. The sparsity condition is s times support size less than p; external variety-extension corollaries require their own high-rank hypotheses.

Qualifications. The Frobenius example disproves one extraction formula, not every possible low-degree extension.

Let V be a vector space over F_p, let S be a set of allowed directions, and choose a basis of span S from S. Below all x lie in this span; one coset suffices if the ambient V is larger. Suppose f:$V \to F_p$ and a fixed nonzero temporal polynomial $P(Y)$ of degree $< h$ satisfy $P(T_u)f = \tau(u)$ as a constant function of x, for every $u \in S$. Here T_u is translation by u. If $P(1) = 0$, put $s = \operatorname{ord}_{Y=1} P \geq 1$ and $P = (Y-1)^s Q$ with $Q(1) \neq 0$. Since $(T_u - 1)^p = 0$, $Q(T_u)$ is invertible on functions and commutes with all the differences. Applying $T_u - 1$ to the relation gives $\Delta_u^{s+1} f = 0$, and in the original equation this gives $\tau(u) = Q(1)\Delta_u^s f$.

Assume $p > s$. In particular, on each basis axis f has univariate degree $\leq s$ (binomial interpolation of its finite differences on the p-cycle). In the canonical polynomial representing f in these coordinates, all individual degrees are $\leq s$, by one-dimensional interpolation in each coordinate. Let f_s denote its homogeneous part of total degree s. If u has w nonzero basis entries with $sw < p$, the ordinary polynomial $f(tu)$ has degree at most $sw < p$ before reduction modulo $t^p - t$. On its p values it is a univariate polynomial of degree $\leq s$, so its coefficient of t^s is exactly $f_s(u)$. Consequently

$$\tau(u) = Q(1)s! f_s(u).$$

When all allowed directions are this sparse (in particular when the entire rank is $< p/s$) this is a polynomial extension of uniformly bounded degree. It covers arbitrary f, without a scalar DVR or cyclic-module assumption. Since it is a polynomial over F_p it gives the same phase on integer lifts by lifting coefficients, then dividing by p. If $P(1) \neq 0$, the same invertibility gives f constant from any single direction equation, and there is no nonconstant tag.

There is a multi-player variant when enough slopes occur on every relevant ray. Suppose $\sum_t (T_{au}^t - 1) f_t$ is a constant for at least h distinct native scalars a on each basis ray and each tested ray, with $0 \leq t < h$. Multiplying by $T_u - 1$, the observations give h values of the temporal polynomial (fold its constant term into the $t = 0$ coefficient) at $Y = T_u^a$. By Lagrange interpolation, $\prod_{b \neq a}(T_u^a - T_u^b) = (T_u - 1)^{h-1}$ times a unit, since each divided factor has constant term $a - b \neq 0$ in $F_p[T_u]/(T_u - 1)^p$. Without division by $T_u - 1$, summing the Lagrange basis identities yields $(T_u - 1)^h f_t = 0$ for every noninitial coefficient t (the folded constant is irrelevant). Thus each f_t has degree at most $h - 1$ on the individual axes and along the tested ray. For $w(h-1) < p$, the polynomial in k obtained by evaluating the registration at `k u` has ordinary degree $\leq w(h-1) < p$, and on its p inputs its degree is $\leq h - 1$. Summing its homogeneous terms only through degree $h - 1$ therefore gives a

global polynomial of degree $\leq h-1$ agreeing with the marked phase on S. This variant needs distinct sampled slopes on the specified rays; the one-player case does not.

A Hilbert-function obstruction independent of ambient rank For the single-player case put $N_s = \dim\{homogeneous degree - s forms identically zero on S\}$, calculated inside its linear span. Assume $s \geq 2$. If the total degree D of the canonical f is larger than s, then

$$N_s \geq \binom{\lceil D/(s-1)\rceil}{s}.$$

Indeed the coefficient of degree s in any individual basis variable is constant in the others by its constant s-th difference. Thus every top monomial x^α of degree $D > s$ has $\alpha_j \leq s-1$. Write F_D for the top homogeneous part. The degree $D-s$ part in x of $\Delta_u^s f(x)$ is the formal directional derivative $D_u^s F_D(x)$. It vanishes for all $u \in S$, since this difference is constant. Translation creates no alias in x: individual exponents in x stay $< p$. Choose the lexicographically leading monomial x^α of F_D. For each $\delta \leq \alpha$ with $|\delta| = s$, the coefficient at $x^{\alpha-\delta}$ in the formal derivative is a vanishing form with distinct leading monomial u^δ and nonzero coefficient equal to the original leading coefficient times $s!\binom{\alpha}{\delta}$. Every lower α' capable of contributing gives a lexicographically lower u-monomial. Thus all these forms are independent. The support of α has size at least $\lceil D/(s-1)\rceil$; choosing s distinct entries produces the displayed count. In particular, for $N_s < \binom{\lceil p/(s-1)\rceil}{s}$, one has $D < p$ and the global homogeneous-part extraction works for **every** direction, without a sparsity hypothesis. At $s = 2$ the bound $N_2 \geq \binom{D}{2}$ is exact as a count for a squarefree leading monomial.

There is a separate geometric consequence. On any affine line contained in S choose a basis for its linear span from two (or fewer) points of that line. The restriction of f to the resulting plane has individual degrees $\leq s$ and total degree $\leq 2s$. For $p > 2s$, extraction on the plane shows that τ restricts to a polynomial of degree $\leq s$ there. Thus on each affine subspace contained in S the line-test criterion makes τ a polynomial of degree $\leq s$: in the canonical function polynomial on such a subspace, any monomial α with total degree $> s$ would admit $\beta \leq \alpha$ of total degree $s+1 < p$. Expanding on arbitrary lines $x + tv$, the monomial $x^{\alpha-\beta}v^\beta$ in the prohibited t^{s+1} coefficient has nonzero binomial multiplier, and no other monomial contributes to the same x and v exponents, even after reducing t. It follows that τ is weakly polynomial of degree $\leq s$ on S. When S meets the hypotheses of the Kazhdan–Ziegler high-rank extension theorem, their global polynomial extension rules out a new bounded-degree obstruction. An arbitrary sparse S need not satisfy any such extension theorem.

Why the support-size argument stops For odd $p \geq 5$ let $n = 2p-1$ and let $f = e_{p+1}$ be the elementary symmetric polynomial in n variables. Every coordinate restriction is affine. On the full direction **1** its restriction is quadratic with leading coefficient -1 although its global degree-two homogeneous part is zero. Indeed

$$e_{p+1}(x + t1) = \sum_{k=0}^{p+1} \binom{p-2+k}{k} t^k e_{p+1-k}(x).$$

For $2 \le k \le p-1$ the coefficients vanish modulo p, while those at $k = p, p+1$ are respectively 1 and -1 by Lucas's congruence. On the prime field $t^p = t$, $t^{p+1} = t^2$. Thus the constant second-difference tag is -2 on the dense direction and zero on each basis direction. This example *does* have another quadratic extension on this small direction set, for example $-2u_1u_2$; it demonstrates failure of extracting f_s, not an intermediate threshold or failure of all bounded-degree extensions. Any true high-rank escape from the displayed criteria needs dense directions or a genuinely different multi-player pattern.

R207. Choosing one short-prohibited step on each of many projective lines

Scope and status. Result with proof.

Result. Fourier density increment bounds how many projective directions can carry pair-prohibited steps for a positive-density set.

Scope. Pair witnesses, with explicit rank and full-line restrictions; not arbitrary higher-order masks.

On the vector group F_p^m, forcing complete affine lines at high density would force arbitrarily long progressions, independently of which nonzero representative one chooses for each projective direction. The choice of representative could in principle create a uniform short obstruction. In the pair case there is the following dimension obstruction.

Proposition. Let $A \subset F_p^m$ have density at least $a > 0$ and let S consist of nonzero differences with $A \cap (A + d) = \varnothing$. Given $0 < \lambda \leq 1$, there is a subspace H of codimension at most $2\sqrt{p}/a^{3/2}\sqrt{\lambda}+1$ and a translate of H on which A has density at least a. Either H is zero, or the fraction of projective directions in H represented in S is less than λ. Every line at a direction outside H is hit by the union of codim H linear hyperplanes in the whole space, of total density at most codim(H)/p. In particular if S represents every projective direction, the ambient rank is at most $2\sqrt{p}/a^{3/2} + 1$. All the complete lines can then be hit at cost $O_a(p^{-1/2})$. Thus this particular complete-line argument cannot give fixed high-density forcing together with a pair witness of fixed positive density.

Proof. On any translate of an n-dimensional subspace, identify the slice with F_p^n. Write f for the indicator of the restricted set, of relative density $b \geq a$. If $n > 0$ and at least a λ fraction of its projective directions are represented in S, then a fraction at least λ/p of the p^n vectors have zero pair correlation $R(d) = E_x f(x)f(x + d)$. Indeed each represented direction contributes at least one of its $p - 1$ nonzero vectors, and $p^n - 1/p^n(p - 1) \geq 1/p$. Correlation zero is inherited by restriction to the slice. With normalized Fourier transforms, Parseval gives

$$\sum_{\xi \neq 0} |\hat{f}(\xi)|^4 = E_d |R(d) - b^2|^2 \geq \lambda b^4/p.$$

Since $\sum_\xi |\hat{f}(\xi)|^2$=b, there is a nonzero ξ with $|\hat{f}(\xi)| \geq b^{3/2}\sqrt{\lambda/p}$. The conditional densities b_t on the p cosets of $\ker \xi$ have average b and $|\hat{f}(\xi)| \leq E_t|b_t - b| = 2E_t(b_t - b)_+$. Restrict to a coset with $b_t \geq b + a^{3/2}\sqrt{\lambda/p}/2$. The restriction loses one dimension. The density never exceeds one, so it stops within the stated bound. Choosing a basis of the annihilator of H gives the hyperplanes: any direction outside H is nonzero on one basis form and a full line in that direction visits its kernel. This proves all assertions.

A useful version takes $\lambda = p^{-1/3}$. One can then hit every complete admissible line outside a subspace at cost $O_a(p^{-1/3})$; within that subspace admissible projective directions occupy at most $p^{-1/3}$ of all directions (or none when the subspace is zero). A sparse set of remaining directions in a very high rank could still force many lines. The proposition gives no corresponding collapse for three-term or longer short obstructions, and a blocker for complete p-point lines need not hit a shorter progression uniformly as p increases.

R227. Short-direction line covers using symbol-count vectors

Scope and status. Result with proof.

Result. Histogram masks hit full finite-field lines of short direction with bounds governed by profile counts and stabilizer ranks.

Scope. Finite profiles and full p-point lines.

Qualifications. The general rank estimate in R228 closes the logarithmic gap in this preliminary histogram argument.

This supplements short-vector-full-lines.md [R229]. It concerns finite full-line blockers only.

Let a profile consist of nonnegative integer counts $m(s,u)$ of moving coordinates with slope $s \neq 0$ and intercept u in F_p. Write $c(m) = \sum |s| m(s,u) \leq p/H$, using centered integer slopes. At time t its symbol-count vector is

$$A_t(y) = \sum_{s \neq 0} m(s, y - st) (y \in F_p).$$

Set $E_m = \{A_t : t \in F_p\}$, $L(m) = |E_m|$. The other coordinates contribute a constant translate. All counts in these moving profiles are at most p/H, irrespective of the original dimension.

Counting distinct rows. Fix any finite family of profiles F and any $0 < \rho < 1$. There is a blocker meeting all their full lines in every dimension with density at most

$$\rho + \sum_{m \in F} (1 - \rho)^{L(m)}.$$

Indeed take a Bernoulli-ρ subset of the finite histogram group $(Z/qZ)^p$, with $q > p$. For each profile and each translate of its unabbreviated vector set E_m missed by this subset, insert one point of that translate. Distinct rows remain distinct modulo q. After dividing by group size the expected final cost gives the displayed bound. In the original cube color by the histogram modulo q of every coordinate, including the constant ones. A uniform extra common histogram shift gives the same expected density as on the finite histogram group; choose one such shift. Hence no assumption on the law of a random vertex's histogram is needed.

In particular, given the rank-based enumeration in the companion note, a rank-R class of size `exp(C(R/H)log(eHp/R))` costs exponentially little whenever $\rho L(m) \geq 2C(R/H)\log(eHp/R)$ throughout the class (sum the resulting geometric errors over dyadic ranks, also retaining any polynomial counting factors). This result by itself does not remove the logarithm uniformly.

Genuine coincidences. Given any integers $d, H \geq 2$, take $p \equiv 1 \bmod d$ large enough that $p > (4Hd)^d$, and the multiplicative subgroup K of order d. Partition the unit cube into at most $(4Hd)^d$ boxes of side at most $1/(2Hd)$ (e.g. use exactly `2Hd` subdivisions). Pigeonholing the points $(zk/p \bmod 1)_{k \in K}$, $z \in F_p$, yields a nonzero w with $\sum_{s \in wK} |s| < p/H$. Take all these d slopes once, each with intercept zero. The nonzero-time rows are the indicator

vectors of the distinct multiplicative cosets of K. The zero-time row is d times the symbol-zero basis vector. Consequently both the number of distinct rows and their rational rank are $1 + (p-1)/d$. In particular distinct-row counting cannot universally improve the rank exponent when p/R tends to infinity.

Enumeration with an affine time symmetry. The global symmetry group of a non-constant moving profile is the group of $t \mapsto at + b$ which permute its multiset of affine-linear forms identically. This is stronger than equality at some times. No nontrivial translation can preserve the profile: a nonzero slope would run through at least p intercepts. Therefore projection to the multiplicative group is injective. This group is cyclic, of order $d|p-1$, and for $d > 1$ conjugate to a group of dilations fixing a unique point b. Normalize this point to zero. The profile is now specified by multiplicities of pairs (O, u), O a slope coset of the order-d subgroup and u the common intercept of its orbit of linear forms. Each such atom has cost $c(O) = \sum_{s \in O} |s| \geq d^2/4$. For $U \geq 1$, the number of cosets of cost at most U is at most $2U/d$, because every representative lies in the interval of radius U and the cosets are disjoint.

The number of invariant profiles with this d and $c(m) \leq p/H$ therefore has logarithm at most

$$\log p + C\, p \log(eHd)/(H\, d^2).$$

To verify, allow all pairs with repetition, and dominate their counting function by $\exp(\theta p/H) \prod_O (1 - \exp(-\theta c(O)))^{-p}$. Take $\theta = 64 \log(eHd)/d^2$, splitting costs into the bands $[2^j d^2/4, 2^{j+1} d^2/4)$. The sum of $-\log(1 - \exp(-\theta c(O)))$ is at most $Cd \sum_j 2^j (eHd)^{-16 \cdot 2^j} \leq C/(Hd^2)$. The prefactor p counts possible centers. This estimate remains true when the stabilizer is larger than d.

Thus the histogram blocker deals, for instance, with classes where the average row multiplicity is at most a fixed constant times the true affine stabilizer size, for fixed target blocker density and large enough H. More explicitly combine the two preceding displays only when $\rho L(m)$ dominates both the displayed logarithm and the desired error. For bounded d the rank-based bound handles high rank while the histogram estimate using the full profile count at degree d handles families with $L(m) \gg p$; in this last notation $\gg$ means a fixed positive fraction. Each genuine multiplicative example above has `L(m)~p/d`.

The estimate in this entry does not control configurations with many pointwise coincidences and no comparably large global stabilizer. The general rank estimate in R228 [R228] supplies the required short-direction line-hitting bound.

R228. A sparse-neighborhood rank bound for all short directions

Scope and status. Result with proof.

Result. Dimension-free masks of arbitrarily small density hit every full line whose direction has sufficiently small centered l1 norm relative to the prime.

Scope. Both the shortness parameter and the prime are sufficiently large, depending on the desired density.

Qualifications. The result is about whole p-point lines, not fixed-length arithmetic progressions.

For every $\eta > 0$ there are $H_0(\eta)$ and $p_0(\eta)$ such that, for all $H \geq H_0$, all primes $p \geq p_0$, and every dimension n, there is a subset of F_p^n of density at most η meeting each full affine line having a nonzero centered representative v with $\sum |v_i| \leq p/H$. In particular the dimension-free short-vector suspension comparison cannot supply a uniform positive forcing bound strictly below one using these full lines. This is a finite obstruction, not an integer capacity formula.

Fix a profile $m(s,u)$ of moving coordinates with slope $s \neq 0$ and intercept u as in short-line-histogram.md [R227], and let $A_m(t,y) = \sum_s m(s, y - st)$. For a complex exponential $e_k(y) = \exp(2\pi iky/p)$ its matrix maps e_k into a combination of e_{sk} over the slopes. The coefficient corresponding to s is $\sum_u m(s,u)e_k(u)$. For an arbitrary integer **difference** of two such profiles, if its coefficient profile at slope s is nonconstant as a function of u, this coefficient is nonzero for every $k \neq 0$, since the minimal polynomial of any primitive pth root is $1 + \cdots + X^{p-1}$. A nonzero slope coefficient that is constant as a vector is impossible for a difference of arrays of total weight at most p/H each, as a nonzero integer constant would force one of the positive arrays to have at least p entries counted with multiplicity. All matrix ranks below are over Q, equivalently over C.

Lemma. Suppose a nonzero difference is supported on $0 < |s| \leq M$, with $1 \leq M \leq p^{11/20}$. Its rank is at least `c p log(eM)/M`, for an absolute $c > 0$.

Choose any nonconstant slope s_0. On F_p^* form the simple Cayley graph with all generators s/s_0 and s_0/s for $0 < |s| \leq M$, deleting 1. Its maximum degree is at most $4M$. For an independent set I, restrict the Fourier matrix to columns I and rows `s₀ I`. The diagonal coefficients are nonzero and the off-diagonal coefficients all vanish, so the rank is at least $|I|$.

Every edge of the neighborhood of 1 gives a product of three generators equal to 1, each chosen slope or its reciprocal after normalization by s_0. The number of such products is bounded by a constant times the combined number of solutions

$$\text{ab} \equiv \text{c } s_0 \pmod{\text{p}}, \quad \text{abc} \equiv s_0^3 \pmod{\text{p}},$$

with $0 < |a|, |b|, |c| \leq M$. In the first case fix c, then enumerate the nonzero integers `ab` of absolute size at most M^2 in that residue. In the second enumerate the nonzero integers `abc` of size at most M^3 in the indicated residue. The usual elementary divisor bounds $d_2(n), d_3(n) = O_\epsilon(n^\epsilon)$ give counts

$$O_\epsilon(M(1 + M^2/p)M^\epsilon), O_\epsilon((1 + M^3/p)M^\epsilon),$$

where one can choose the exponent in the divisor bound a fixed factor smaller than ϵ. As $p \geq M^{20/11}$, both are $O(M^{3/2})$. All neighborhoods have the same bound by vertex transitivity. By the sparse-neighborhood coloring theorem of Alon, Krivelevich and Sudakov, a graph of degree at most D and with at most D^2/f edges in every neighborhood has chromatic number $O(D/\log f)$. Here use $D = 4M$ and f an absolute constant times $M^{1/2}$; for bounded M use the greedy bound instead. Equivalently one can separate graphs whose actual degree is less than $M/\log(eM)$ and use greedy on these. Thus $|I| \geq c(p-1)\log(eM)/M$ as required.

Profile counts. If a profile has k distinct slopes then $k \leq C\sqrt{p/H}$ by the cost and distinctness of centered integers. The preceding diagonal independent-set argument with a graph on only these k generators and its greedy bound gives rank $A_m \geq (p-1)/(2k+1) \geq c\sqrt{pH}$ for nonempty profiles and $p \geq H$. In particular rank grows uniformly as p grows.

For a dyadic rank bound R at least this minimum and at most $2p$, set $S = p/R$. For sufficiently large S choose $M = \lfloor c_1 S \log(eS) \rfloor$ with a sufficiently small absolute positive c_1. Then $M \leq p^{11/20}$ for all sufficiently large p, since $R \gtrsim \sqrt{p}$; by increasing the fixed lower cutoff for S, $cp\log(eM)/M > 2R$. It follows that a profile of rank at most R is uniquely determined by its counts on slopes $|s| > M$. Indeed two with the same such counts give a difference of rank at most $2R$, contradictory to the lemma unless zero. For bounded S we instead retain all slope counts.

The number of possible arrays supported on $|s| > M$ has logarithm at most `C p log(e H M)/(H M)`, ignoring integer rounding constants for $M \geq 1$. To see this, allow unlimited counts at each of $2p$ species for every positive magnitude $l > M$. The log of the generating-function bound is

$$\theta p/H + 2p \sum_{l>M} (-\log(1 - \exp(-\theta l))).$$

Take $\theta = 4\log(eHM)/M$. The sum is bounded by $C\exp(-\theta M)/(1-\exp(-\theta)) \leq C/(HM)$. For bounded S use the same count with all $l \geq 1$, taking $\theta = 4\log(eH)$ and absorbing an absolute constant in $R \gtrsim p$. It follows uniformly that the rank-$[R/2, R]$ stratum has at most

exp(C R log(eH)/H)

profiles. Indeed $M \asymp S\log(eS)$ for large S, and $\log(eHM) \leq C\log(eH)\log(eS)$. The constants, including the fixed cutoff for bounded S, are absolute.

Haar marker. Choose independent Haar circle elements $\phi(y)$, $y \in F_p$. For a rank-r profile, some r rows are independent over Q, so the corresponding elements $(A_m\phi)(t)$ are independent Haar by surjectivity of the associated torus homomorphism. The probability that their set misses some interval of length η is at most $C_\eta \exp(-\eta r/2)$, by a finite mesh of intervals of length $\eta/2$. Sum this over the dyadic rank strata, using the displayed profile bound. For $H \geq H_0(\eta)$ the sum is less than one for $p \geq p_0(\eta)$ because the minimum rank tends to infinity. Fix such a table ϕ. Along a full line the marker $\sum_i \phi(x_i + tv_i)$ is $A_m\phi$ plus a fixed constant from stationary coordinates. Consequently its values meet every interval of length η even after any common translation. Taking the inverse image of an interval of length η, and averaging its translate, gives the desired blocker of density at most η.

The multiplicative subgroup examples in short-line-histogram.md [R227] remain compatible with the proof: their Fourier rank can equal their number of distinct rows, but a small *difference* supported on small signed representatives gains a logarithm from the sparse local multiplicative graph. If a formulation requires all primes, increase H_0 to exceed p_0 so that there are no eligible positive-cost profiles at the remaining primes.

R229. A dimension-free suspension comparison for short vector representatives

Scope and status. Result with proof.

Result. Short vector representatives give a dimension-free rational-to-full-line comparison; bounded magnitude types admit sparse line hitters.

Scope. The fixed-length representative condition and full-line conclusion are explicit.

Qualifications. R228 establishes the general short-direction line-hitting bound.

Let p be prime, m arbitrary, and select projective directions in F_p^m, each with an integer representative $v = (v_i)$ of nonzero residue. Regard the chosen residues as the points v/p mod Z^m in the order-p subgroup of $(Q/Z)^m$, and saturate their preimage $D \subset Q^m$ by Z^m. Fix $h \geq 2$ and suppose each chosen representative satisfies

$$(h-1)\sum_i |v_i| < p.$$

Then $a_{h,Q^m}(D)$ is at most the extremal density in F_p^m of a set containing no full affine p-point line in any chosen projective direction. This comparison has no rank dependence, uses no denominator coprime to p, and loses no density. Only an upper bound on the fixed-h capacity is asserted.

Given a stationary h-avoider A, perform the rank-m suspension of the vector transfer [R269] with $M = 1$ and with no boundary cuts in the clock. Thus the $(Q/Z)^m$-action on $X \times [0,1)^m$ is $V_g = U_g S_{-g}$; mark the full preimage of A under the projection to X. Along the p-periodic restriction at $d = v/p$, the tested sites are $U_{td+\lfloor z-td \rfloor}x$, with floors taken coordinatewise. Over one full period $t = 0, \ldots, p-1$, each coordinate floor changes on exactly $|v_i|$ edges (the strict hypothesis makes $|v_i| < p$); the final wrap of the clock cancels the corresponding integer in `pd`. Hence at most $\Sigma|v_i|$ edges have any floor change, counted circularly. In the p cyclic starting positions fewer than p can see such an edge within their next $h-1$ edges. If the full line were marked, there would therefore be h consecutive old sites with common step $v/p \in D$, contradicting avoidance. This argument is pointwise outside the common null set for all old h-avoiding cylinders (finite slopes and countable integer starts). Restrict to the native finite subgroup action: each orbit mask, averaged over uniform group location, is a deterministic subset of F_p^m with no such line. The density is bounded by the finite extremal density.

An obstruction to exploiting this comparison. If all chosen representatives additionally have $\max_i |v_i| \leq C < h-1$, then $0 < \Sigma v_i^2 < p$; every chosen line has nonzero leading coefficient for the quadratic $Q(x) = \sum_i x_i^2 \bmod p$. Every nonconstant quadratic along a p-line takes $\geq p/2$ values. There is a set of $O(\log p)$ residues meeting the range of every quadratic with nonzero leading coefficient: choose that many independent uniform residues and apply a union bound over fewer than p^3 scalar quadratics. For every translate of this hitting set, the corresponding inverse-image is a full-line hitting set in F_p^m; averaging the translate gives some such set of density $O(\log p/p)$. Thus bounded amplitudes cannot force complete p-lines at any fixed density below one in this regime.

More generally, let K bound the number of distinct nonzero magnitudes among the coordinates of each chosen representative. Since the total count of nonzero entries is less than p,

some even power sum $\sum_i v_i^{2j}$ with $1 \le j \le K$ is nonzero modulo p whenever $K < p/2$. Otherwise the Vandermonde matrix on the K or fewer distinct nonzero squared residues would annihilate their multiplicities modulo p. Center each coordinate representative to avoid identifying two distinct magnitudes as negatives. Consequently some polynomial $Q_j(x) = \sum_i x_i^{2j}$ is nonconstant along every chosen line. Its image size is $\ge p/(2j)$. For each j use $O(j^2 \log p)$ random residues to hit, simultaneously, all nonconstant scalar polynomials of degree at most 2j (at most p^{2j+1} choices). Independently translate these hitting sets before taking the union of their Q_j-inverse images. The resulting full-line hitting set has density $O(K^3 \log p/p)$. This bound is nontrivial for $K = o((p/\log p)^{1/3})$. Finding a finite template with the required uniform positive short-color witness and full-line forcing in genuinely more complicated short directions remains open.

R230. A common random table for full lines with short representatives

Scope and status. Result with proof.

Result. A common random-table sieve bounds full-line hitters using direction rank and weighted shortness.

Scope. A quantitative rank sieve for full-line hitting; R228 supplies the complementary uniform bound.

Qualifications. The complementary rank estimate in R228 completes the low-rank regime not covered by the estimate in this entry.

Let p be an odd prime and let $H > 1$. In arbitrary dimension n, consider directions in $\mathbb{F}_p^n$ with nonzero centered vector v satisfying $\sum_i |v_i| \le p/H$. The question here concerns hitting **complete** lines, as in the short-vector comparison [R229]. We give a dimension-independent sparse hitter whenever $H/\log p$ is sufficiently large. More generally the same argument hits all the high-rank line patterns even when p is much larger. This is a restriction on one prospective finite template, not an upper bound on the critical density of the original integer conjecture.

Choose independent Haar phases $\phi(y)$ on the circle for $y \in \mathbb{F}_p$ and record $F(x) = \sum_i \phi(x_i)$. Apart from an arbitrary constant coming from the inactive coordinates, its values along a line are $A\phi$, where

$$(A\phi)(t) = \sum_{s \ne 0,\, u} m_{s,u}\phi(u + ts), \qquad m_{s,u} \ge 0, \qquad \sum_{s,u} |s| m_{s,u} \le p/H.$$

Write $R(A)$ for the real matrix rank. Every set of $R(A)$ independent rows gives independent Haar output phases: the corresponding integer matrix maps the input torus onto the full output torus, by its character criterion. For an arc of length η, a net of $O(1/\eta)$ translated arcs of length $\eta/2$ suffices to guarantee that $A\phi$ hits every arc of length η. A fixed matrix fails with probability at most $C\eta^{-1}\exp(-\eta R(A)/2)$. If all desired matrices pass at once, delete the inverse image under F of one translate of the length-η arc. Averaging over the translate gives deletion cost at most η, regardless of the ambient dimension.

Rank and enumeration. For a nonzero signed difference of two allowed profiles, supported on k slopes, the rank is at least $(p-1)/k$. Indeed the character at frequency $a \ne 0$ maps to a sum of at most k characters, at the distinct frequencies as. The coefficient at as is $\sum_u m_{s,u}\exp(2\pi i a u/p)$. For any slope where the signed profile differs, these coefficients are nonzero for every $a \ne 0$, by the minimal polynomial of a primitive p-th root: a vanishing integer coefficient list would be constant, and a nonzero constant difference on a slope would cost at least p in one of the individual nonnegative profiles. Thus the union of all Fourier column supports contains every nonzero frequency. The supports of any basis of columns must cover that union, proving the inequality. The same proof with the zero matrix gives $R(A) \gtrsim \sqrt{pH}$, since k distinct nonzero slopes have total centered cost at least $k^2/4$.

Group the allowed matrices by $R \le R(A) < 2R$, with R dyadic. For this group erase all slopes of magnitude at most $L = \lfloor (p-1)/(8R) \rfloor$. The erased part is uniquely determined by the rest: a nonzero difference there would have rank at least $(p-1)/(2L) \ge 4R$, whereas the sum of the two ranks is strictly smaller than $4R$. With $L = 0$ no uniqueness assertion is

needed. At most

$$\exp\left\{C\frac{R}{H}\big(\log(eH)+\log(1+p/R)\big)\right\}$$

profiles are possible in that rank range. For details, the count by cost on the remaining slopes is at most

$$\exp(\alpha p/H)\prod_{s>L}(1-e^{-\alpha s})^{-2p}.$$

Take $\alpha=[\log(4H)+2\log(1+L)]/(1+L)$; the logarithm of the product is bounded by $O(pe^{-\alpha(L+1)}/(1-e^{-\alpha}))=O(p/(H(1+L)))$. Also $1+L$ is comparable to $1+p/R$. These give the asserted bound. Specifying just an interval of ranks rather than one precise rank ensures that we sum only over $O(\log p)$ possible sizes.

For fixed $\eta>0$, choose H large in terms of η and retain only rank ranges with $\log(1+p/R)\le c\eta H$ for a sufficiently small absolute c. The failure probabilities summed over profiles in each retained range are $O_\eta(e^{-\eta R/4})$. The sum over all those ranges is less than one for sufficiently large p, since $R\gtrsim\sqrt{pH}$. Consequently all full lines in the retained ranks are hit with cost at most η, simultaneously and for every n. If $\log p\le c\eta H$ this covers every short direction. The argument allows arbitrary nonzero magnitudes, starting offsets, and inactive coordinates.

The preceding estimate leaves the range $R/p<\exp(-c\eta H)$ at large moduli; the rank bound in R228 [R228] completes this range. Algebraically possible sources of small rank are multisets of affine maps invariant under a large multiplicative subgroup of $\mathbb{F}_p^\times$. For generators v consisting of that subgroup and zero offsets, the matrix averages over its orbits and has rank $1+(p-1)/|\mathrm{Stab}|$. The centered cost restriction still has to be checked for any proposed subgroup, and the rank example alone does not force an intermediate density.

R234. A sharp density bound for a common lattice lift

Scope and status. Result with proof.

Result. A common exact lattice section for all short box edges has sharp product density at most the product of one minus the coordinate boundary proportions.

Scope. Simultaneous componentwise lifts on finite tori.

Qualifications. This is a limitation of a lifting method, not an upper bound on recurrence capacity.

Let $G = \prod_{i \le n} Z/m_i Z$, with all $m_i \ge 3$. Give an edge from x to $x + v$ for every $v \in \{-1, 0, 1\}^n$. On a set $E \subset G$ suppose the vertices of each component of the induced graph have lifts to Z^n that increase by exactly v along every such edge with endpoints in E. Then

$$|E|/|G| \le \prod_i (1 - 1/m_i).$$

Equality holds for the product of coordinate intervals $0, \dots, m_i - 2$. Thus the full-box variant of a common seam deletion in the lattice-section approach fails on a schedule with divergent reciprocal side lengths, even with a fresh choice of section on each component and no condition imposed at unretained sites. The statement applies to the low torus in the deep model. It is a restriction on this method of blocking progressions, not a proof of recurrence.

To prove the bound, identify the vertices of each component C with its chosen lifts. Put $B = \{0, 1\}^n$. All the $C + B$ project injectively and their images for different components are disjoint. Indeed, if $x + b$ and $y + c$ agree modulo the side lengths, the original vertices are equal or adjacent with signed displacement $b - c$. The prescribed rule gives $y - x = b - c$ as an integer vector whenever they are distinct. In the equal case the lift is unique and $b = c$. This proves the assertion.

For one component add the n unit segments of B successively. Just before adding segment i, denote the partial enlargement by D. Each nonempty lattice line section of D parallel to coordinate i contains at most $m_i - 1$ sites. Its enlargement by $\{0, e_i\}$ adds at least one site, and still injects into the torus; the enlarged section has at most m_i sites by this injectivity. Hence the new cardinality is at least $m_i/(m_i - 1)$ times the old cardinality. Multiplication over the coordinates and summation over the components yield the estimate. The product example has the lifts in the displayed integer intervals; in each coordinate any retained pair at displacement $0, 1, -1$ already has that same integer displacement (also for $m_i = 3$).

The same argument permits other integer ranges. If $m_i > 2k_i$ and the good-edge condition holds for velocities $|v_i| \le k_i$, enlarge by $\prod\{0, \dots, k_i\}$. A nonempty section gains at least k_i sites and the enlarged section has at most m_i, giving the sharp upper bound $\prod(1 - k_i/m_i)$ by the same interval example. Only the velocities actually required by the recurrence model may be used in this variant.

R246. A restriction on high-density proofs via several representatives of a word

Scope and status. Result with proof.

Result. Fixed finitely many sheets that cover all sufficiently large parameter subspaces must contain finite-sums patterns of every rank, forcing all-length recurrence.

Scope. The sheet count, alphabet and subspace data are fixed while the ambient parameter dimension grows.

Qualifications. Allowing these parameters to grow is not covered.

The earlier one-line sheet obstruction [D03] extends to unordered extractions from subspaces of any fixed finite dimension. Let G be torsion-free abelian and D a family of nonzero steps. Fix a sheet bound c, a finite alphabet $[L]$, a parameter dimension $d \geq 1$ and a target length $h \geq W(c, 3)$. Suppose for arbitrarily large n there are maps ϕ_s:$[L]^n \to G$, $1 \leq s \leq c$, such that in *each d-dimensional combinatorial subspace* U, the union of the $\phi_s(U)$ contains some h-term D-progression. The progression points may come from arbitrary words of U, in arbitrary order and from different sheets. Then D contains finite sums of every finite rank, so it forces progressions at every positive density and every length. The unbounded ambient n for the fixed data is important; this does not exclude an isolated finite gadget at each length.

Fix m. Color each ordered d-parameter subspace by a selected witness, recording for each of its h time slots just the sheet and the position tuple in $[L]^d$. There are finitely many colors. By the Graham–Rothschild parameter sets theorem a sufficiently large n supplies a dm-dimensional subspace on which all d-parameter subspaces have the same selected code. By ordinary van der Waerden choose three equally spaced time indices j,$j + r$,$j + 2r$ with the same sheet s. Denote their fixed parameter positions by u^0,u^1,$u^2 \in [L]^d$. The positions on this sheet are distinct since the progression is nonconstant.

In the large subspace order its dm parameters in d successive groups of m. Embed ternary words $z \in \{0, 1, 2\}^m$ by assigning the k-th parameter in group l the symbol $u_l^{z_k}$. Write $\Psi(z)$ for ϕ_s of the resulting word. Every ternary combinatorial line, with nonempty wildcard set E, comes from a d-subspace: in each of the d successive groups put its corresponding variable on indices E, and keep all other indices fixed at their assigned symbols. The same selected code therefore ensures that the three Ψ-values along the ternary line form an AP of step r v for some $v \in D$. This does not require the three parameter tuples themselves to be collinear or ordered in $[L]^d$.

The singleton lines make Ψ separately affine in all ternary coordinates. The lines with two simultaneous variables then show twice each mixed first difference vanishes when the other coordinates are fixed, hence it vanishes by torsion-freeness. Equivalently, expand a separately affine map on the cube in its multiaffine coefficients, and use the pair lines to kill all coefficients of order ≥ 2. Thus $\Psi(z) = g + \Sigma z_i w_i$. At the zero passive base the singleton lines give $w_i = rv_i$ with $v_i \in D$, and the lines for all nonempty E give $\sum_{i \in E} w_i = rv_E$ with $v_E \in D$. Cancel r to obtain $\mathrm{FS}(v_1, \ldots, v_m) \subset D$. The finite density Hales–Jewett word-averaging argument on these directions gives full multiple recurrence. The same calculation works in an odd squarefree group if all witness steps are supported on primes above h: there is no 2-torsion in the group and one can cancel r on the subgroup containing the differences in the last relation.

R248. Symmetric multiaffine scalar filters have a rank-uniform polynomial tag

Scope and status. Result with proof.

Result. Symmetric multiaffine scalar filters have a rank-uniform homogeneous polynomial tag; finitely many exchangeable types give a bounded list.

Scope. Characteristic and degree bounds, with a fixed number of exchangeability types.

Qualifications. Choosing different list elements at different primes does not automatically assemble a global separating phase.

Let $f(x) = \sum_D a_D e_D(x)$ be any symmetric multiaffine F_p-valued polynomial on F_p^n, with no upper bound on n or on its total degree. Suppose a direction u is admissible when $\Delta_u^s f$ is constant on the whole space, with fixed $2 \le s < p-1$. Equivalently one may use a scalar temporal filter of multiplicity s as in the sparse-direction lemma [R202]. There is one homogeneous degree-s polynomial Q on F_p^n whose value at **every** admissible u is that constant divided by s!. Consequently arbitrary signed combinations of elementary symmetric polynomials, including the Frobenius alias in the earlier example, do not give new unbounded-degree phase tests.

Write $f_u(t) = f(x + tu)$ on a line. On an admissible direction it has temporal degree $\le s$ with constant coefficient of t^s equal to the tag divided by s!, independent of x. Average $a^{-s} f(ax)$ over $a \in F_p^\times$; this preserves the tag at every original admissible direction and retains precisely the e_D terms with $D \equiv s \bmod (p-1)$. Thus assume this congruence on all degrees. If f now has degree at most s, the usual homogeneous part is the desired Q. Otherwise its largest degree D is at least $p + s - 1 \ge 2s$ and $n \ge D$.

An admissible vector has a common coordinate outside at most $s-1$ entries The top degree-$(D-s)$ spatial piece of $\Delta_u^s f$ vanishes. At each squarefree coefficient x_J, $|J| = D - s$, this says $e_s(u_{J^c}) = 0$. So every subset of size $n - D + s$ has vanishing s-th elementary symmetric sum in the entries of u. If u had s disjoint unequal coordinate pairs, choose an extra disjoint set of $n - D$ coordinates, possible since $D \ge 2s$. Select one member per pair together with this extra set and take all s alternating differences of the vanishings. The result is the product of the s nonzero pairwise differences, a contradiction. In the complete multipartite graph on the equal-value classes a maximum matching has size $\min(\lfloor n/2 \rfloor, n - largest_class_size)$ (pair two largest remaining classes greedily). Hence all but at most $s-1$ entries have a common value c. Write $u = c1 + d$ with this sparse exception vector d. If $c = 0$, the temporal degree is at most $s-1$ by multiaffinity in the active coordinates, and the tag vanishes.

If there is no admissible direction with $c \neq 0$ we are done. Otherwise put t equal to the largest possible number of exceptional coordinates in one with $c \neq 0$. For $l \ge 0$ put $T_l = \sum_{D \ge l} a_D \binom{n-l}{D-l}$, extending by zero above n. Expansion on the support of d gives for every allowed $c \neq 0$

$$tag(u)/s! = f(u) = \sum_{l \le t} c^{s-l} e_l(d) T_l,$$

since $D \equiv s \bmod (p-1)$ and $l \le t < s$. The first equality also uses $f(0) = 0$ and homogeneity in nonzero time scalars.

The full-line constraint ties the coefficients together For $q = m(p-1) > 0$ and every $0 \le l \le t$ we have $T_{q+l} = 0$. To see this use one admissible vector with exactly t nonzero exceptions. If $q > n - t$ then no contributing degree $D \ge q + l$ exists: the next residue-s degree past q is $q + s > n$. Otherwise $q \ge p - 1 \ge t$ and $q \le n - t$. Choose the monomial x_J with $|J| = q$, varying freely which exceptional coordinates occur outside J. Its coefficient in $f(x + zu)$ is a sum of scalar powers z^{D-q}, with all nonzero contributions of temporal residue s. The coefficient must be zero since the coefficient of z^s in the line polynomial is constant in x. Inserting any chosen subset E of the exceptions outside J, it equals

$$\sum_{l \le |E|} c^{s-l} e_l(d|E) T_{q+l}.$$

Successive differences over inclusion of a fixed l-subset E give $c^{s-l}(\prod_{i \in E} d_i) T_{q+l} = 0$. This proves the asserted vanishing.

For $j < t$, the reduced interpolation in z of $(1+z)^{D-j}$ on F_p is $(1+z)^{s-j}$ since both exponents are positive and congruent modulo $p-1$. Its linear coefficient is therefore $s - j = \sum_{q \equiv 1 \bmod (p-1), q \ge 1} \binom{D-j}{q}$ in F_p. Multiply by $a_D \binom{n-j}{D-j}$ and sum over D. Using $\binom{n-j}{D-j}\binom{D-j}{q} = \binom{n-j}{q}\binom{n-j-q}{D-j-q}$, all terms $q = 1 + m(p-1)$ with $m \ge 1$ vanish by the preceding result at $l = j + 1$. The remaining $q = 1$ term proves

$$(n-j) T_{j+1} = (s-j) T_j \quad \text{for all} \quad j < t. (*)$$

A squarefree degree-s extension Find coefficients w_I on s-subsets so that $\sum_{I \supseteq J} w_I = T_t$ for every t-subset J. The t-to-s inclusion matrix has full row rank over F_p because $p > s$ and $n \ge s + t$. For an explicit proof realize each t-subset coordinate vector using only columns in an $(s+t)$-set containing that subset. Inside this set complement identifies the matrix with the disjointness matrix A on t-subsets. For $|V| \le t$ write $v_V(J) = 1_{V \subset J}$. Then $A v_V = \binom{s-|V|}{t-|V|} \sum_{R \subset V} (-1)^{|}R| v_R$. The functions with $|V| = t$ span all functions and the nonzero leading multiplier on their cardinality filtration gives surjectivity (or prove it inductively on $|V|$). This also treats $t = 0$. Extending such a local construction by zero creates no nonzero row outside the chosen $(s+t)$-set.

For $j < t$ double-count supersets to see the j-subset sums of w_I are $\binom{n-j}{t-j} T_t / \binom{s-j}{t-j} = T_j$, the last equality by (*); denominators are units since $s < p$. Now $Q(u) = \sum_{|I|=s} w_I \prod_{i \in I} u_i$. Its expansion on every $u = c1 + d$ with at most t exceptions is exactly the formula for `tag(u)/s!`; it vanishes on every $c = 0$ admissible vector as that support has size at most $s - 1$. This proves the claim independently of the rank and high symmetric degree.

A finite covering by polynomials for boundedly many exchangeable types More generally assume f is multiaffine and invariant under all coordinate permutations within each of at most b types, with b fixed. For sufficiently large p in terms of b,s, the scalar tag on its

full admissible locus agrees pointwise with at least one of a list of at most s^b homogeneous polynomials, each of degree s. The choice in the list can depend on u. For packets living on *single* primary components with a separated local tag this suffices for the familiar finite union of polynomial-root sieves. For mixed supports with a sum of phases over many primes, independent choices of the correct list index at each prime are not one common polynomial test.

First consider a type on which the type-degree of f is at least 2s. If an admissible velocity had s disjoint unequal pairs of entries in this type, take all 2^s swaps; symmetry makes all swapped velocities admissible with the same tag. On each line polynomial in the temporal variable z the degree-s coefficient is this same constant. Hence the alternating sum of all these line polynomials has degree $\le s-1$. In the generating product for the elementary symmetric functions of this type, swapping a pair i,k changes its pair factor by $z(u_i-u_k)(x_k-x_i)y^2$, where y is the elementary-symmetric generating variable. Thus the full alternating sum equals z^s times the nonzero pair-factor product and the polynomial formed by lowering all the type degrees by 2s, evaluated on the remaining shifted variables. This lowered polynomial is nonzero: different elementary-symmetric type indices remain linearly independent, and some term has sufficiently high type degree. Its top-degree spatial coefficient is independent of z and nonzero; after adjoining the pair-factor product this gives a coefficient proportional to z^s, a contradiction. This also handles p-term aliasing, since the temporal exponent at this top spatial coefficient is exactly s. Hence each high-degree type has at most $s-1$ exceptions from a common value. Each remaining type has total type-degree $\le 2s-1$ in f.

For every high type fix $L=s-1$ prototype positions for its exceptions and represent its entries by a common center c and L deviations d. Let W be the linear space of these parameters together with all the original low-type coordinates. In $S\cap W$ choose $\le$bs admissible directions whose projections span the high-coordinate projection of this set (the projection dimension is at most s per high type). Use their scalar coefficients and arbitrary low-coordinate translations to parameterize the span of those directions plus the low-coordinate space. The pullback f in these coordinates has individual degree $\le s$ along each of the selected directions. It has total degree $\le b(2s-1)$ in low coordinates at any fixed values of those scalars, by its original type degrees; interpolation in the remaining scalars preserves this bound. The full polynomial in the new parameters has total degree at most $bs^2+b(2s-1)<p$ on a sufficiently large prime. Its s-homogeneous part therefore gives the tag divided by s! on $S\cap W$. Extend that homogeneous polynomial linearly in the ambient W coordinates if its parameterized subspace is proper, and average it over permutations among the L prototype positions of each high type; these small factorials are invertible. It still matches the tag, by the symmetry of f and of the full admissible locus.

Write this symmetrized polynomial with monomial symmetric functions in each prototype deviation list. Replace such a function by the matching sum over distinct indices of the entire corresponding type: on a list with at most L nonzero deviations this has exactly its old value, after a permutation and zero padding. Low coordinates and named centers remain as variables, and the degree is still s. Finally try one of the first s actual coordinates in each high type as its center, with all type deviations given by subtraction of this choice. At least one of those s entries is a true center, since the type has $\le s-1$ exceptions. There are at most s^b choices; on every admissible u the correct choice in all its high types gives the desired tagged polynomial. Each high type has at least 2s positions by the definition, so all the indicated choices exist.

R250. Exponential size is necessary for a cyclic forcing template

Scope and status. Result with proof.

Result. Cyclic long-forcing templates must have exponentially many directions, even after specified bounded polynomial localization.

Scope. A necessary cardinality lower bound, with explicit dependence on forcing density and length.

Qualifications. Exponential size alone is not sufficient for a compatible template.

This is a restriction on possible finite constructions, not an example of an intermediate density.

For an integer modulus $p \geq 2$ (primality is unnecessary), let $S \subset Z/p$ be a set of nonzero residues and let $D(p, S) = \{d \geq 1 : d \bmod p \in S\}$. Write $k = |S| \geq 1$. Fix a desired deletion budget $0 < \eta < 1$, take $Q = \lceil 4/\eta \rceil$ and $L = \mathrm{lcm}(1, \ldots, Q)$. For every $h \geq 2Q + 2$ there exists a periodic subset of Z containing no h-term $D(p, S)$-progression and having density at least

$$1 - \eta - (2\mathrm{L}/\mathrm{h}) \log(\mathrm{hk}) - 1/\mathrm{h}.$$

In particular, for any fixed $0 < \delta < 1$, a finite template whose full residue lift forces h-term progressions at density $> \delta$ must have at least $\exp(c_\delta h)/h$ residues, for all sufficiently large h (unless it contains zero). Thus the periods in any prospective nontrivial construction must grow at least exponentially in the requested progression lengths. An elementary split according to the additive orders of the residues makes this independent of the prime factorization of p.

Small orders. Write $t = \mathrm{ord}_p(s)$. If $t \leq (h-1)/(L)$ then for any integer $d \equiv s \bmod p$, the phase $\beta = d/(pL)$ has $||L\beta|| = ||s/p|| \geq 1/t$. Dirichlet pigeonholing gives $1 \leq q \leq Q$ and $a = ||q\beta|| \leq 1/Q$. Since $\mathtt{q|L}$, we also have $a \geq q/(Lt) > 0$. For any initial phase, the points sampled at indices $0, q, 2q, \ldots, \lfloor 1/a \rfloor q$ have largest circular gap at most a: in the positive orientation they are the regular a-mesh up to $\lfloor 1/a \rfloor a$, and in the negative orientation its reflection. The final index is at most $Lt \leq h - 1$. Thus every such progression hits any open arc of length η. The integer set deleted by one translate of this fixed arc under $n \mapsto n/(pL)$ has density at most η, on averaging its translation. It hits all the small-order progressions simultaneously.

Larger orders. For each remaining residue form the p hyperedges $x + [0, h-1]s$ in the base group. Their sizes are $\min(t, h) \geq h/(2L)$. Choose vertices independently for deletion with probability $v = \min(1, (2L/h)\log(hk))$. If $v = 1$ the asserted bound is trivial. Otherwise the expected number of uncovered edges is at most $pk \exp(-vh/(2L)) = p/h$. Add one vertex of every uncovered edge. There is a transversal of relative size at most $v + 1/h$. Its lift to the integers hits all progressions with these residues, regardless of higher digits. The union of the two deleted sets has period dividing $\mathtt{pL}$ and costs no more than the sum of the two density bounds.

The example also supplies a cyclic mask on some Z/p^m, as required by the root-uniform formulation. Factor $pL = PC$, where P has only prime factors of p and $(C, p) = 1$. Then $\mathtt{p|P}$ and $P|p^m$ for sufficiently large m. In the CRT product choose a densest slice of the surviving set at a fixed C-coordinate. Every residue step on Z/P lifts to an integer in $D(p, S)$ with zero C-coordinate, so the slice remains avoiding; now pull it back to Z/p^m.

For example, choose $\eta = (1-\delta)/3$ and take h large. If $\log(hk) < (1-\delta)h/(6L)$, the displayed density exceeds δ. This is incompatible with the forcing requirement for any such row. Fixed H-term colorability plays only the role of excluding the zero residue. The growth condition is necessary and gives no converse for exponentially many residues.

Localization to polynomial return neighborhoods. A forcing row actually needs exponentially many residues even after imposing any fixed number of polynomial congruence tests of bounded degree at their natural shrinking scales. Fix m, t and $\delta < 1$; choose r with $1/r < (1-\delta)/4$, and use Alweiss–Farhangi, Theorem 1.9. Fix integers u_j, degrees $1 \le t_j \le t$, and set

$$S' = \{s \in S : ||u_j s^{t_j}/p|| \le C h^{-t_j} \quad \text{for all} \quad j\}.$$

Here the initial threshold for h and the constant C only depend on r, m, t. For such h there is an r-coloring of the integers in which every h-term progression with difference reducing into $S \setminus S'$ sees all r colors. Deleting one randomly selected color has expected cost $1/r$; it hits these progressions for every choice. Combine with the periodic blocker above for S'. On long prefixes its second blocker has the stated limiting density; on a subsequence one fixed color costs at most $1/r$ in upper density loss (use the least frequent color on each prefix and pigeonhole). Thus the resulting avoiding set has upper density at least $1 - 1/r - \eta - (2L/h)\log(h\max(1, |S'|)) - 1/h$. If S' is empty the periodic blocker can be omitted entirely. The finite cyclic model principle transfers this integer lower bound to the supremum over all cyclic lifts. One can also work directly with the forcing statement for integer $D(p, S)$ without invoking that principle. Choosing $\eta < (1-\delta)/4$ proves the exponential bound on $|S'|$, uniformly in the polynomial numerators and p. This does not give a contradiction if the overall period is very large.

Chapter 12

Dyadic constructions, ordered words, and parity

R061. Half-density recurrence in the explicit dyadic affine witness

Scope and status. Result with proof.

Result. The explicit dyadic affine witness and its countable diagonal powers have relative all-length capacity at most one half.

Scope. This conclusion is tied to the displayed cyclotomic module action.

Work with the two odd-center candidate [R107], and assume $k_i \to \infty$, $K_i - k_i \to \infty$ and $D_i - K_i \to \infty$ (for example $K_i = 3k_i$, $D_i/K_i \to \infty$). Its explicit cyclotomic Haar witness has system-relative critical capacity at most $1/2$. The same is true on any countable diagonal power of that particular action. It gives a positive-density six-slot obstruction as already shown. These bounds are not uniform over arbitrary stationary actions; the finite-template argument below depends on the event through its head approximation.

Free characters and valuation. Use the local ring, notation u, E, ζ, ϖ and $\phi = 2^{K-1}$ of the candidate. Write $w = u^c$, $c = 1+2^k$, dropping coordinate subscripts when unambiguous. Enlarge the character module to the abelian group generated by the six global tuples $u^a w^b$ with $a = 0, 1, 2$, $b = 0, 1$, and all the finite-support local ring coordinates. The enlarged module is invariant since each group element changes only finitely many coordinates. The six global tuples freely generate modulo the finite-support part. For a fixed nonzero integer Laurent polynomial $Q(U, W)$, first divide its coefficients by their common maximal two-power and write Q in powers of $(W - U)$ with coefficients Laurent in U (clear negative W exponents first if necessary). Let m be the first such coefficient nonzero mod2, and j its order at $U = 1 \bmod 2$. At (ζ, ζ^c) the term selected this way has ϖ-valuation $m2^k + j$ for large i, since $v_\varpi(\zeta - 1) = 1$, $v_\varpi(\zeta^c - \zeta) = 2^k$. All higher m terms then have larger valuation as $k \to \infty$, and coefficients divisible by 2 have valuation $\geq \phi$, larger since $K - k \to \infty$. Restoring the common two-power adds its exponent times ϕ. In particular the valuation is $O_Q(\phi+2^k)$ and is less than the ring truncation depth $D\phi+2+2^{k+1}+1$ eventually. This proves the assertion on free generators. Haar on the character-extension torsor with $\chi(E_i) = 1/2$ projects onto the original Haar torsor; its free six circle coordinates are independent Haar and independent of all the local data.

One marked and two normal coordinates. Sample directions with speed 2^K at one fresh axis, speed $1+2^K$ at another, speed $c+2^D$ at a third, and zero elsewhere. All are allowed, with true tag only at the third axis. Take any sequence in which the three indices tend to infinity. For finitely many times, an integer character of the six-channel profile is represented by a polynomial $R(T,U,W)$ Laurent in T and of degrees ≤ 2 in U and ≤ 1 in W. Its conditional Haar integral given any fixed local head vanishes unless $R(1,U,W)=0$ as a polynomial (to remove free-coordinate dependence) and each fresh-axis evaluation $R(u^r,u,w)$ lies on the fixed line generated by E. Constant and ϵ components are checked separately.

- At the first axis, $u^{2^K}=1+2^K\epsilon$. If R is divisible by $(T-1)$ exactly once, its ϵ term is $2^K R_0(1,\zeta,\zeta^c)$ for $R_0=R/(T-1)$. Its valuation is $K\phi+O_R(\phi+2^k)$, strictly below that of the E-line by $D-K\to\infty$. Thus $(T-1)^2$ is necessary.
- At the second axis, $u^{1+2^K}=u+\zeta 2^K\epsilon$. If $(T-U)$ does not divide R, the nonzero constant component $R(\zeta,\zeta,\zeta^c)$ has valuation below truncation by the preceding estimate. If it divides only once, the ϵ term is $\zeta 2^K(R/(T-U))(\zeta,\zeta,\zeta^c)$, again too shallow for E. Thus $(T-U)^2$ is necessary.
- At the third axis, if $(T-W)$ does not divide R, the nonzero constant component $R(\zeta^c,\zeta,\zeta^c)$ similarly prevents survival.

These distinct factors are relatively prime in the relevant Laurent polynomial UFD. Since the U and W degrees are bounded, the remaining tests are exactly

$$R(T,U,W)=B(T)P(T,U,W),P=(T-1)^2(T-U)^2(T-W)$$

for integer Laurent B. Their limiting phase is $B(1)/2$: the two normal substitutions vanish and P gives E on the true mark, which is fixed by u. Thus every Fourier coefficient of the profile has its claimed limit, also upon fixing the starting local head (none of those axes moves).

Limiting law and half bound. Let $F_{ab}(t)$ denote the six free-coordinate circle traces and S the forward time shift. The limit law is Haar on the affine set determined by

$$(S-1)^2[S^3F_{00}-S^2(2F_{10}+F_{01})+S(F_{20}+2F_{11})-F_{21}]=1/2.$$

Indeed sample a Haar primitive W of $(S-1)^2W=1/2$, independently sample the five bivariate-indexed input streams other than F_{21} as independent Haar full shifts, and solve for F_{21} from the bracket. This construction has exactly the annihilator and the phases just proved; the scalar second difference on the full shift is onto and the F_{21} coefficient in the bracket is a unit.

Suppose $B\subset T^6$ has measure $>1/2+\delta$. For $u\in[0,1]$ let P_u be the set of five-input points whose F_{21}-section of B has measure $>u$. For some u one has $m(P_u)>1-u+\delta$, by integrating over u. Condition on W. In time order maintain five-input vectors in P_u, starting with the first three vectors. At time t the missing output $F_{21}(t)$ is a translate of the first coordinate $F_{00}(t+3)$ of the next input, with coefficient $+1$; all the other contributing inputs have already been chosen. The new input has conditional probability at least δ of belonging

to P_u and simultaneously giving an acceptable output in the current section. Thus every fixed h-window lies in B with probability at least δ^{h+3}, uniformly in B and W. The argument works for vector circle coordinates in any diagonal power.

For a general event of mass $> 1/2$ on the enlarged torsor, approximate by its finite head and the six free coordinates in L^1. A positive fraction of head fibers are separated from $1/2$ in density. On those fibers the previous bound applies to the limiting law. One-time Haar marginals make approximation by continuous tests valid for each finite head section, and then the original-L^1 error costs at most h times its mass. As in the deep affine half proof [R042], choose the approximation accuracy in advance using the fixed density gap and h. Hence some actual sampled difference has a positive return for that event. The conclusion descends to the original affine witness and to the action pulled back along the natural map from the free group. This provides a system with an intermediate system-relative threshold, not a bound on the supremum over all integer systems.

R062. Simultaneous affine returns and nilspace tests

Scope and status. Result with proof.

Result. The dyadic module half bound can be synchronized with finitely many compact finite-rank nilspace morphism return tests.

Scope. These are tests on directions; arbitrary measure-preserving nilspace actions are not included.

There is a choice of sufficiently replicated dyadic depth blocks on the free integer-coordinate host with the following simultaneous property. In **any** Haar ring-module action from the maximal affine theorem [R085], if an event has measure $> 1/2$, then for every finite h it has an h-return along legal directions meeting any fixed finite list of compact finite-rank nilspace morphism return neighborhoods. The morphisms may be defined on the free group or be compatible tests on its finite-coordinate packets. In particular multiplying by an independent compact abelian rotation in the measure-theoretic model preserves the half bound. The statement about nilspaces concerns pointwise tests on the directions, not arbitrary measure-preserving nilspace actions.

Even normal signatures Retain the word-signature setup of the polynomial word lemma [R095] at degree s. In the nilpotent integral signature group H generated by the velocity letters $g(r)$, replace the old normal subgroup-generating list by the **squares** $g(1 + 2^K j)^2$, $0 \le j \le s$. This is just the subgroup J generated by the displayed elements, without any assertion that J is normal. Its index remains a power of two with $\log_2[H : J] = O_s(K + 1)$: in the abelianized Vandermonde bound the old inclusion of M times the generating lattice is replaced by $2M$, and the successive commutator quotients obtain $(2M)^j$ in place of M^j. The finite coset action and its power-of-two period bound are unchanged in form. Thus for sufficiently large D/K there is an odd positive t with

$$g(c + 2^D t)g(c)^{-1} \in J.$$

This supplies a signed identity word with one $c+2^D t$ of exponent 1, one c of exponent -1, and all normal letters having **even total** exponent individually. Introduce cancelling pairs $g(r)^2 g(r)^{-2}$ at each of the speeds $2^K, 1, 1+2^K$, even when they were already present. Change each signed repetition count by the same even positive integer $2v$ as in the nilspace word proof. The resulting words have positive counts, every required normal speed is present, every normal speed has even total count, and both active speeds have odd counts. Their signatures at $v = 0$ remain exactly the identity. The topology, compact-uniform return bounds on v, and coherence of multivariable Taylor colors in the compact-nilspace proof [R069] apply unchanged. Thus on any sufficiently large homogeneous coordinate set at this fixed level, one of these positive words passes the prescribed nilspace return-neighborhood tests.

Keep the affine profile simultaneously Take an increasing sequence of equal-parameter stages, each with finite multiplicity. The stage sizes can be chosen in advance for a diagonal list of nilspace targets, tolerances, degrees and numbers of test observables; the countability of finite-rank target types and the uniform bounds over their morphisms are as in the nilspace

note. Also allow a list of at most j affine Fourier tests by stage j, with no constraint on their actual coefficient tuples. Color each coordinate singleton at the stage by the values $0, +1, -1$ of each of those Fourier tests at each speed in its fixed signed word, as well as the values of the socle signs on the finitely many individual coefficients under consideration. There are uniformly finitely many colors at that stage, with a bound depending on its finite word and the number of tests, independent of modules. Choose enough coordinates that one singleton type has at least the number needed by the finite nilspace Ramsey bound. Perform the nilspace selection within this type, and set all unused velocities to zero. The added singleton coloring can equivalently be included in the Ramsey setup from the beginning.

For any fixed countable affine model, along an escaping diagonal sequence take the first j tests of a countable exhaustive list, and require the chosen nilspace neighborhoods (shrinking them if desired). The compact increment laws have a convergent subsequence, and by further extraction the socle bits converge for each coefficient to a single order-at-most-two character π. On every surviving affine Fourier test the normal speed phases cancel by their individually even counts. The phases of the two active speeds have the same parity contribution as taking their difference. The interpolation identity in the maximal affine theorem [R085] still holds with the odd t multiplying its right side, which is order at most two. Thus the Fourier coefficient is exactly its value on $s_j = \binom{j}{5}\pi$; nonsurviving tests have value zero. The increment coset again contains s, initial Haar coordinates are independently distributed in the limit by tail support, and each event of measure $> 1/2$ therefore has a positive return. All the selected directions already pass the nilspace tests. Unlike the eight-axis template, these words can have arbitrarily many active axes. The latter matters since the fixed-two-active subfamily by itself is quadratically separated on tails by the active-mask calculation [R071].

For a compact abelian rotation one may request that each of the first j characters of the rotating element be within $1/j$ of identity, using degree-one simultaneous tests and diagonal enumeration. For an arbitrary fixed standard compact-spectrum action the same gives strong convergence to the identity on L^2 on that factor by density of its eigenfunctions. On an independent product with the Haar affine model the limit trajectories thus freeze the compact factor. The original event can depend on both factors. If its total mass is greater than $1/2$, a positive measure set of compact-factor fibers has affine mass greater than $1/2$. On each such fiber the affine intersection at the same common support point s is positive. Integrating, and then using continuity in L^1 of translation on the affine factor and the support of s, gives a positive actual return. No such diagonal-freezing assertion is made for general distal actions.

R063. Arbitrary invariant measures in the native cyclotomic module category

Scope and status. Result with proof.

Result. The maximal dyadic half bound extends to arbitrary invariant measures on the stated native single-axis coefficient-module actions.

Scope. The ring, depth, and replication assumptions remain essential; wider ring categories are separate.

The maximal affine half theorem [R085] extends to all invariant probability measures on the same native cyclotomic coefficient-group character actions if the depths and the replication are increased. In particular no relative-Haar or local-Haar hypothesis on the measure is needed. This remains a theorem about a special class of actions: the change of any global character under one generator depends only on that single local coordinate. The wider native rings in the distinct-window note [R070] are not the rings in this statement.

At level k, K, D use the exact rings $R_i = S_i[\epsilon]/\epsilon^2$ of the original model [R107], with truncation $D\phi + J + 1$ and $E = z\varpi^J\epsilon$. In particular $z\epsilon w(w-1)^2(w-u)^2 = E$ and $uE = E$, for $w = u^c$, $z = 2^D$. Allow arbitrary local modules M_i and countable additive subgroup $A \leq \prod M_i$ containing their direct sum (or obtain a factor from this case). The action on the dual of A is the dual of coordinatewise multiplication by $u_i^{d_i}$. Assume a sequence of equal-parameter stages, $D - K \geq 2(J+4) + 2$, and sufficiently many identical axes in each stage. We specify below a bound for replication uniform over modules, measures and characters at that stage. Then **every invariant probability measure and every event of measure** $> 1/2$ **has all finite lengths of recurrence along legal directions arbitrarily far out**.

The local base and arbitrary ergodic components For now assume standard probability spaces with countably many coefficient characters, and reduce by ergodic decomposition. At each axis, the induced action on the dual of M_i is cyclic of period dividing $q = 2^{D+1}$. Its orbit class is invariant under the whole group, hence constant in an ergodic measure. The local base Y, generated by all these coordinates, has product measure of the uniform measures on the selected finite cyclic orbits: the subgroup acting on any finitely many coordinates acts transitively on their product. Choose a representative local character χ_i for each orbit.

For a coefficient m, let $f_m = E[e(\chi(m))|Y]$. Its modulus is invariant and hence constant: under a group element the corresponding character is multiplied by an observable phase on finitely many local coordinates. If the constant is positive, $e(\chi(m))/f_m$ is globally invariant, hence constant, and conditional expectation shows $f_m = e(\chi(m))$ almost surely. Otherwise $f_m = 0$. Thus for the subgroup Λ of base-measurable coefficients the conditional Fourier coefficients vanish outside Λ, and the measure μ is invariant under the group $H = \Lambda^\perp$ of fiber translations. This last assertion also follows directly from the unconditional Fourier coefficients of μ. It is the elementary compact-extension description here; no assertion of Haar on the entire coefficient torsor is used.

Replicated local orbit tests For each finite trajectory character use the polynomial $R_i(Y) = \sum_{t<h}(Y^t - 1)(m_t)_i$ as in the Haar proof. The local integral at speed s is now the mean of the q values $e(\chi_i(u_i^a R_i(u_i^s)))$ on the cyclic orbit. All are qth roots of unity, as

$qR_i = 0$ in the native coefficient ring. A nonconstant such list has mean of modulus at most $1 - \eta_q$ for a number $\eta_q > 0$ uniform in the module: there are only finitely many lists of qth roots of length q. For a constant list its phase is a fourth root of unity, as verified below. A coordinate singleton can therefore be colored for any finite number of tests and the five speeds $L = 2^K, 1, 1 + L, c, c + z$ by finitely many categories: either nonconstant or one of the four constant phases. Also record the first finitely many bits $\chi_i(E_i m_i)$ for individual coefficients. The color count is bounded in terms of the number of tests alone.

At stage j request enough axes that one singleton color class can supply $4N_j$ copies each of the three normals, $4N_j - 1$ copies at c, and $4N_j + 1$ at $c + z$, for the first j tests and j bits. Choose N_j so $(1 - \eta_q)^{4N_j - 1} \le 1/j$ before selecting the stage size; the color count times $20N_j$ suffices. The resulting direction is legal and stage sizes can grow independently of the particular action. If a monitored test has any nonconstant category, its full product integral has modulus at most $1/j$. Otherwise the fourfold normal phases cancel, and the residue is the phase of the true mark minus the uncharged active speed, evaluated at any one of the mutually homogeneous axes.

Why orbit invariance suffices for Hermite interpolation Fix such an axis, suppress its index, and take the quotient N of M_i by the elements annihilated by all $u^a \chi_i$. The kernel is invariant under $Z[u]$, but need not be invariant under the entire ring R; the distinction is essential. The local integral is constant precisely when the image of the tested element belongs to $K_0 = \ker(u - 1)$ on N. As $(u^\phi + 1)^2 = 0$ in R, it follows on K_0 that $4K_0 = 0$. This proves the earlier statement about fourth roots of unity. All the following manipulations on N use only $Z[u]$ scalars.

Put $V = u^L - 1 = L\epsilon$, so $V^2 = 0$, and divide the module-valued R polynomial by $P(Y) = (Y - 1)^2 (Y - u)^2 (Y - w)$ with $w = u^c$, writing $R = PQ + B$, $\deg B < 5$. From survival of the normal and uncharged-active tests,

$$B(1) = 0, B(u), B(w) \in K_0, VB'(1), VB'(u) \in K_0.$$

Indeed the two derivatives are obtained at the speeds L and $1 + L$ by subtracting the corresponding baseline; u is a unit and fixes K_0. The two values and derivatives match those of R at the indicated double roots. By the confluent adjugate at $1, u, w$ we have

$$det(C) \cdot VB_l \in K_0 \quad \text{for every} \quad coefficient B_l,$$

$$det(C) = \pm (u - 1)^4 (w - 1)^2 (w - u)^2.$$

Here $VB(u), VB(w)$ even vanish since V kills K_0. We need an integer divisibility bound **inside** $Z[u]$, rather than in the full ring. In the variable X put $d(X) = (X - 1)^4 (X^c - 1)^2 (X^c - X)^2$ and $\Phi(X) = X^\phi + 1$. We have $\Phi(u)^2 = 0$. The resultant $|Res(\Phi, d)| = 2^{J+4}$: at the primitive root ζ the three factors have valuations $1, 1, 2^k$, respectively, and their norms are powers of two with precisely these exponents. The identities $Norm(1 - \zeta^m) = 2^{2^{v_2(m)}}$ for the indicated nontrivial roots follow by taking the norm down the cyclotomic tower. Consequently $2^{2(J+4)}$ is an integer-polynomial combination of Φ^2 and d (the Sylvester

determinant identity); in particular it belongs to $(detC)$ in $Z[u]$. Since $4K_0 = 0$ and $D-K \geq 2(J+4)+2$, we have $(z/L)VB'(w) = 0$ on N.

The exact difference of true and baseline R evaluations is $z\epsilon wR'(w)$, with $z\epsilon = (z/L)V$ an allowed scalar. The preceding calculation discards B. In the original module, hence also on N,

$$z\epsilon wP'(w)Q(w) = EQ(1) = E\sum_t \binom{t}{5}(m_t)_i.$$

For the first identity use $Ew = E$. For the second use $Eu = E = Ew$, so after multiplying by E the monic factor is $(Y-1)^5$; B has degree less than five. Thus the difference phase on a surviving test is exactly the fifth-binomial combination of the recorded bits. No division of elements of the orbit quotient by $\zeta - 1$, ϵ, or any integer was used.

The limiting joint law Diagonalize the stages over all trajectory characters and all individual coefficient bits, passing to a subsequence of stages for which the observed bits stabilize and the surviving/nonconstant categories stabilize for each test. The eventual bits are one additive order-at-most-two character π of A, zero on the local direct sum. The increment law tends to Haar on a coset containing $s_t = \binom{t}{5}\pi$, since every Fourier coefficient is either zero or the phase at s, as in the Haar proof [R085].

For $m \in \Lambda$ the original character is base-measurable, so all its increments at finitely many times tend to zero in measure as the selected axes escape: approximate the base observable in L^2 by a function of finitely many local coordinates, which all late directions fix. Therefore the increment law ν is supported on H^h, and s is a point of H^h. The initial character χ is independent of the increments in the joint limit. For an initial test outside Λ its conditional expectation on the base is zero and all increments are base-measurable. For an initial test in Λ, finite-coordinate approximation makes it independent of the chosen orbit coordinates on the tail by the product base law. Hence the joint limit is $\mu \times \nu$.

Translation by every element of H preserves μ. For any measurable event B, its return integral under an h-tuple of H-translations is continuous on H^h: approximate by continuous functions in $L^1(\mu)$ and use invariance. At the two-valued tuple s this integral is at least $2\mu(B)-1$, strictly positive in the claimed range. As s is in the support of ν, the limiting actual return integral is positive. Continuous approximation with the fixed one-slot marginals works for bounded measurable indicators as before. Finally, in any nonergodic measure a positive measure set of components has event mass greater than $1/2$, and there are only countably many candidate directions, giving a common direction with positive integrated return. For nonmetrizable character systems one may first pass to a countable invariant factor determining the event modulo null sets for a regular invariant measure, and include all its axis-local coefficient terms.

R064. A uniform threshold for arbitrary finite-rank compact abelian cocycles on the dyadic product base

Scope and status. Result with proof.

Result. Arbitrary finite-dimensional compact abelian cocycles on the dyadic product base have an all-order bound $1 - m^{-d}$ at bounded extra dyadic depth.

Scope. The bound depends on dimension and extra depth; it extends to the stated disconnected and solenoidal fibers.

In the setting of the profinite odometer result [R096], let the fiber be the fixed torus $K = \mathbb{T}^d$, with an arbitrary measurable cocycle and Haar fiber law. Write $Q_i = 2^{D_i+1+e_i}p_i$ with p_i odd. On a cofinal infinite sequence of eligible axes assume $e_i = e$ is fixed; this includes all minimal native cyclic axes and their arbitrary odd covers with $e = 0$. Put $m = 2^{e+1}$. At every length every event of measure

$$a > 1 - m^{-d}$$

has a positive return using only one true axis. The threshold depends on the torus dimension and the extra 2-depth, and is independent of the cocycle and the horizon. In particular the native bound for a circle is one half. If the extra 2-depths are bounded one can pass to a fixed-depth subsequence and use the corresponding bound. The lower ring example can itself be encoded by finitely many circle-valued local potentials, so a claim of full positive recurrence for all such connected fibers would be false. No arbitrary-action upper bound on the integer problem follows here.

Use the abelian instance of the head–separator localization [R176] on the full product base. Condition on a typical separator environment. The remaining moving cores have independent uniform data, each with one designated trial cyclic axis of the above cofinal type. Fix a metric on K and use additive notation. The ordered-interval displacement of any finite motion is now simply the sum of differences of the independent core potentials. For convenience use their *absolute* profiles; subtract a constant from each potential without changing any displacement. Pass to a subsequence so the single-time potential laws converge, and subtract any fixed point in the limiting support. This ensures zero belongs to the support of the limiting constant (no-move) profiles. Unselected intervening potentials need never be included in a displacement sum, since the group is abelian.

At core i write g_i for its normalized potential. Sample normal motion by mixing with fixed positive probabilities no move, a uniform $r \in LC_{Q_i}$ on the designated axis, and speed c on that axis. Take a common further weak limit of their whole two-sided profile laws $(g_i(X_i + jr))_{j\in\mathbb{Z}}$, including each of the component laws separately, and let $P \le K^{\mathbb{Z}}$ be the closed subgroup generated by the support of the limiting normal mixture. It is shift-invariant by the uniform starts. Define

$$N = \{n \in K : (\ldots, 0, n, 0, \ldots) \in P\}.$$

Then $N^{\mathbb{Z}} \subseteq P$, by shifting, adding finitely many isolated tuples, and taking closures. In the window $[-H, H]$ let P_H denote the projection of P, and N_H its subgroup of possible isolated entries at zero. These are decreasing closed subgroups with intersection N, by compactness.

Twice a native top displacement is in the isolated subgroup First take the native or odd-cover case $e = 0$; the proof below treats general fixed e at the same time. Fix H, $s = 2H$, and $M = L\,\mathrm{lcm}(1, \dots, s)$. In K/N_H, uniformly for every integer $v \in M\mathbb{Z}$,

$$\Delta_v^s g_i(X_i) \to 0 \quad \text{in probability.}$$

Indeed use the deterministic parameter cubes of the finite-fiber proof [R096] on slots $-H, \dots, H$, killing every slot except zero, with the individual parameter increments chosen to give stride v at zero. At every one of the 2^s vertices the start is uniform and the velocity uniform on LC_{Q_i}. Their profile laws concentrate in every fixed neighborhood of P_H. Their alternating sum is close to P_H with probability tending to one, uniformly in v. All but its zeroth entry are exactly zero; compactness shows that an isolated tuple approaching P_H has entry approaching N_H.

The required cyclic ring calculation uses just a *fixed* order 2^R, where R can depend on s, e but never on the core. In

$$\mathbb{Z}[Y]/(Y^{2^R} - 1, (Y-1)^s)$$

put $X = Y - 1$ and $u = Y^{2^{R-e-1}} - 1$. Then $u^2 = 0$ for large enough R, so $mu = 0$ by $(1+u)^m = 1$. For details take $B \geq \lfloor \log_2 s \rfloor$, R so large that $2^{R-e-1} \geq s$, and use $v_2\binom{2^A}{j} = A - v_2(j)$ for $1 \leq j < 2^A$. The cyclic relation reads $2^{R-B}X(2^B + XC(X)) = 0$ modulo X^s with C integral. Multiplying by the finite geometric sum for $2^B + XC$ gives $2^{R+B(s-2)}X = 0$ (the last error term contains X^s). The coefficients of u below degree s are all divisible by $2^{R-e-1-B}$, and its constant term is zero. Hence $u^2 = 0$ when $2(R-e-1-B) \geq R+B(s-2)$. This is an integral polynomial identity with coefficients depending only on R, s, e, permitting stable application to the previous convergence in probability.

Let M_{odd} be the odd part of M and set $t_i = p_i t$, where t is an odd multiple of M_{odd}. The same t_i must work at all windows, which needs care. Choose odd $t = t(i) \to \infty$ in divisibility, for example the product of the odd positive integers up to i; this is divisible by each fixed M_{odd} eventually. Write $v_i = (Q_i/2^R)t(i)$, so $v_i \in M\mathbb{Z}$ eventually and its translation Y has order 2^R. Its 2^{R-e-1}-th power translates by $z_i t_i$, $z_i = 2^{D_i}$. Apply the fixed polynomial identity to g_i in K/N_H for each fixed H. The term involving $Y^{2^R} - 1$ vanishes exactly, while any fixed integral combination of translates of $(Y-1)^s g_i$ vanishes in probability. Therefore

$$m\,(g_i(X_i + z_i t_i) - g_i(X_i)) \to 0 \quad \text{modulo } N_H$$

for every H. The same holds with any fixed multiple $jz_i t_i$, by telescoping and uniform starts.

A palette independent of the horizon Pass to a further subsequence on which the jointly sampled whole profiles at c and $r_i^* = c + z_i t_i$, with the *same* uniform start, converge in law. The first limit belongs to P. The difference at any fixed slot j has its m-multiple in every N_H, hence in N almost surely, by the preceding calculation, using the uniform start shifted by jc. The difference at slot zero is zero. The compact quotient K/N is a torus of dimension at most d; in particular its subgroup of m-torsion has size at most m^d, even when N has disconnected components. Choose a set $R_0 \subset K$ of representatives of these torsion classes, including zero as the representative of zero. It is one palette common to all slots.

For each fixed h, write $P^{(h)}$ for the projection on slots $0, \dots, h-1$. Almost every true profile's coset modulo $P^{(h)}$ contains a tuple in R_0^h with zeroth slot zero: subtract the coupled ordinary c-profile in $P^{(h)}$ and then adjust the individual entries by $N^h \subset P^{(h)}$. Independent normal profiles with the limiting mixture law convolve weakly to Haar on $P^{(h)}$, by Fourier characters (the mixture has zero in its support and its support generates the group). Adding an independent true profile therefore hits any neighborhood of these palette tuples with positive probability once enough normal profiles are used. To see positivity use a continuous bump supported in the neighborhood and positive at every one of its finitely many target tuples; on each of the above cosets Haar integrates it positively. Returning to the subsequence, there are arbitrarily many disjoint trials, each with some fixed number of normal cores and exactly one true core, with success probability bounded below for this environment and this tolerance. The induced *displacement* differs from each absolute sum by its zeroth entry; requiring that entry near zero leaves the required displacement within twice the chosen tolerance of the palette.

On a positive set of separator environments the conditional event mass exceeds $1-m^{-d}$. In such an environment approximate the event indicator by a finite-head function continuous in the fiber, in L^1 as accurately as required; the superlevel $\{f > 3/4\}$ can have mass arbitrarily close to the event mass. The set where every R_0-translated fiber belongs to this superlevel has positive measure by the union bound, since $|R_0| \le m^d$. Its lower bound can be chosen from the conditional mass gap alone, before the subsequence or palette is chosen. Uniform continuity gives a tolerance ensuring all the approximant values along a successful trial exceed $1/2$ from that starting set. Pool many whole independent trials as in the one-mark proof [R176], moving only the selected one: each slot's conditional marginal tends in total variation to the product law, independently of the possibly small individual success probabilities. This controls the finite-head approximation error and the starting mass. There are only countably many true differences, so positive return in the original event follows by Fubini over the qualifying environments. This completes the fixed-rank torus-fiber bound.

Disconnected abelian fibers and solenoids More generally let K be compact metrizable abelian and suppose its discrete dual has finite rational rank d. The same upper bound $1 - m^{-d}$ holds, without a bound on the torsion subgroup of the dual. Everything through the construction of N and the m-torsion assertion in $Q = K/N$ proceeds unchanged. For any continuous finite quotient $\psi : Q \to F$, eventually N_H is in the kernel of the composite $K \to Q \to F$: otherwise the decreasing compact subgroups would have an element of their intersection outside that open-and-closed kernel. At such a fixed window, the deterministic mixed difference conclusion holds *exactly with probability tending to one* in the finite group F. The finite-state period proof of dyadic-profinite-odometer.md [R096] shows that the top displacement, at any fixed slot, tends in probability to zero in this quotient. Indeed $z_i t_i$ is eventually divisible by the fixed period times the required mesh M, by the growing 2-depth and the cofinal odd divisibility of $t(i)$. This applies to every finite quotient of Q; their kernels intersect in its identity component Q^0. Thus the common palette only needs representatives of $Q^0[m]$.

The dual A of Q^0 is torsion-free of rank at most d, by exactness of Pontryagin duality. For a prime p the dimension of A/pA over $\mathbb{F}_p$ is at most d: a longer list would be rationally dependent, with an integral relation that can be chosen primitive, contradicting independence modulo p. Induction gives $|A/p^u A| \le p^{ud}$; in particular $|Q^0[m]| = |A/mA| \le m^d$. The same palette proof now works. At $d = 0$ this gives all-positive recurrence for arbitrary profinite

abelian fibers, and the separate argument already includes nonabelian profinite fibers.

R065. A support agreement lemma for the two coarse parity states

Scope and status. Result with proof.

Result. A bipartite support-agreement lemma localizes labels outside logarithmically many coordinates, with explicit density losses.

Scope. Local label agreement alone still allows encoding the binary tag and does not prove a uniform density threshold.

Let s and $\eta > 0$ be fixed, and let $E \subset \{0,1\}^n \times \{0,1\}^n$ have uniform density $\geq \eta$. Suppose each left vertex a and right vertex b carries labels $F_a(S), G_b(S)$ for every coordinate set S of size at most s, with values in arbitrary alphabets (or in a tuple of such alphabets). An edge of E is assumed to enforce

$$a|S = b|S \Rightarrow F_a(S) = G_b(S).$$

Then there exist an exceptional coordinate set of size $O_{s,\eta}(\log n)$, label functions $L_S : \{0,1\}^S \to \Sigma_S$ for supports avoiding those coordinates, a left vertex set of density at least $n^{-O_{s,\eta}(1)}$, and a right vertex set of density at least $\exp(-O_{s,\eta}((\log n)^2))$, such that

$$F_a(S) = L_S(a|S) \quad \text{for every} \quad \mathit{selecteda},$$

$$G_b(S) = L_S(b|S) \quad \text{for every} \quad \mathit{selectedb}.$$

The selected sets still span a graph with edge density at least $\eta/4$ relative to their product. Constants concern sufficiently large n. The labels need not be measurable outside their finite domains, and multiplicities for repeated coordinate directions can be included in a single tuple label.

First discard left vertices of degree below $\eta/2$, leaving at least $\eta/2$ of all left vertices and only using that side thereafter. Sample $m = \lceil C \log n \rceil$ independent uniform right anchors. Their common left neighborhood A' has expected density at least $r = (\eta/2)^{m+1}$. If there are t pairwise disjoint supports of size $\leq s$ each missing a pattern from among the anchors, its probability is at most

$$(O_s(n^s) \cdot 2^s)^t (1 - 2^{-s})^{mt}.$$

Indeed fix the supports and missing patterns; all anchors must independently avoid all the indicated patterns on disjoint coordinate sets. First take $C = C(s)$ large, then fixed $t = t(s,\eta)$ large so that this probability is below $r/2$. Some anchor realization therefore has common-neighbor density at least $r/2$ and missing-pattern hypergraph of matching number $< t$. Delete the vertices in a maximal such matching, of size $O_{s,\eta}(1)$. All local patterns on every remaining support occur among the anchors.

For any set of vertices of density $\beta > 0$, the hypergraph of its missing-pattern supports has matching number at most $\log(\beta)/\log(1-2^{-s})$, by the same disjoint-support probability estimate. Thus deleting $O_{s,\eta}(\log n)$ more coordinates ensures that all local patterns occur also in A'. For a support S outside the deleted coordinates and pattern u, choose an anchor with that pattern. Its label is independent of the choice: take any $a \in A'$ realizing u, and use acceptance with every anchor. This defines $L_S(u)$ and gives the assertion for A'.

Every $a \in A'$ retained right degree at least $\eta/2$. Hence at least $\eta/4$ of all right vertices b have a neighborhood $A_b \subset A'$ of density at least $\eta/4$ relative to A'. For each such b, its A_b has density polynomially large in n in the full binary cube. Its missing-pattern supports therefore have an $O_{s,\eta}(\log n)$-vertex cover. Off that cover and the earlier exceptions, find a neighbor $a \in A_b$ realizing $b|S$; the agreement gives $G_b(S) = L_S(b|S)$. There are at most $\exp(O_{s,\eta}((\log n)^2))$ choices for this last exceptional set. Select a right subfamily with a common choice, and keep A' on the left. Every selected right vertex still has relative degree at least $\eta/4$ to A', proving the stated edge density.

In the $K = k+1$ free two-center model, averaging a start in a finite block of R/P with $R = \bigoplus 2^{k_i}\mathbb{Z}e_i$, $P = 2R$, gives independent coarse endpoints $a, b = a + \sigma$. On an agreement coordinate the velocity is odd-normal, hence its passive variation is unrestricted. On a discrepant coordinate it is active, with only deep high-bit variation. For fixed componentwise polynomial terms in passive increments, a biased pair boundedly torsionizes its top derivatives on every support of normal coordinates. After multiplying by that order the even-endpoint and opposite odd-endpoint top coefficients are labels satisfying exactly the premise above. This localizes their surviving values to functions of the bits on their own supports off logarithmically many coordinates. The lemma does not give a uniform threshold by itself: even local functions of the two coarse bits can encode the binary tag.

R066. The Haar half bound with bounded-order interactions between axes

Scope and status. Proved subcases and unresolved extensions.

Result. Bounded-order Haar interaction modules satisfy the dyadic half bound; a sparse-support extension identifies a relative Haar structure for some invariant measures.

Scope. The last structural extension alone does not establish simultaneous high returns for arbitrary relative star cocycles.

The single-axis module theorem [R085] has an all-length threshold at most 1/2 also for the following **Haar interaction modules**. Finite order is allowed to vary from one observable to the next. The conclusion does not cover arbitrary measures on such modules or arbitrary stationary systems.

Use the original local rings R_i, D_i, K_i, k_i, u_i, E_i with the regimes of the ring theorem [R085]. Select axes on a tail with depths D_i strictly increasing; arbitrarily many such axes suffice and all arguments below use distinct depths. For each finite nonempty subset e of axes let M_e be an additive group carrying compatible commuting unital R_i-module structures for $i \in e$ (equivalently a module over their integral tensor product). Coordinates from axes outside e act trivially. Let A be an additive group of coefficient tuples $(m_e)_e$, closed under the axis actions, containing the local direct sum $L_0 = \oplus_e M_e$. For each $m \in A$ assume that its nonzero coordinates have bounded size $|e| \le s(m)$. The collection of finite hyperedges may be restricted arbitrarily. Let C_e be any subgroup fixed by all the incident u_i; prescribe an additive character θ on $C = \oplus C_e$, and give its character-extension torsor Haar measure. Each C_e has exponent at most two: on $\ker(u_i - 1)$, the identity $u_i - 1 = \varpi_i + \zeta_i \epsilon_i$ implies $\varpi_i^2 = 0$ as an operator, hence $2 = 0$ there. The earlier proof for one-axis models prescribed the smaller socle-image subgroup; the present formulation allows any locally fixed subgroup.

Claim. Every measurable event of mass $> 1/2$ recurs at every finite length along legal directions in any tail. In particular the pair model is covered with one threshold independent of h, and three or more finite interaction orders cause no extra loss at the Haar level. We work on a countable coefficient factor for the event; all arguments may be run on a countable invariant set of test characters inside the stated ambient product. Local Haar Fourier integration still refers to the original group C, so no extra saturation assumption on reduced local modules is needed.

Isolating singleton and mixed tests At a given stage take finitely many Fourier tests on h-slot increments with coefficients m_t, $0 \le t < h$. Let s bound all hyperedge sizes in these coefficients. Use four ordinary speed types $L_i = 2^{K_i}, 1, 1 + L_i, c_i$ and additionally the true type $c_i + 2^{D_i}$. For a finite set J of chosen axes, each assigned one of the five types, put

$$F_J = \sum_t [\prod_{i \in J} (U_i^{tr_i} - 1)] m_t.$$

It vanishes outside edges containing J and hence is zero for $|J| > s$. For a packet I, the total increment test is $\sum_{\varnothing \ne J \subset I} F_J$. Haar evaluation is zero unless this sum is in C. We select an infinite homogeneous set such that, if a full packet test survives, every one of its constituent F_J belongs to C.

Here are the exact free-set details. For every $1 \le |J| \le s$ and every tested assignment on J with $F_J \notin C$, preselect a witness of bounded size. If F has finite support but an entry is not in its C_e, use that edge e. If F has infinite support choose an infinite sunflower of its nonzero edges and use its finite kernel e; every edge contains J, so its kernel does also. The elementary infinite sunflower fact for bounded finite sets follows by induction on their size: either there are infinitely many disjoint sets, or a finite maximal disjoint family meets all sets and some element of its union occurs infinitely often. Remove this element and continue the induction. In the finite-support case unbounded edge sizes never occur, since the Fourier tests have arity $\le s$.

Select an infinite subset S of axes so that none of these specified e for $J \subset S$ meets $S \backslash J$. For each size r, test and assignment, finitely color increasing $(r+1)$-tuples by whether deleting each one of their entries leaves a J whose witness contains the deleted entry; also allow a value for good tests with no witness. Infinite Ramsey gives homogeneity for the finitely many such colorings. None of the containment flags can be true: within the resulting infinite ordered set one can choose fixed J with more than s elements at the relevant position (before it, after it, or in a chosen gap), contradicting $|e| \le s$. Refine by Ramsey on r-tuples to make the symbols `bad` or their actual signs $\theta(F_J) \in \{0, 1/2\}$ constant for each prescribed type order, test, and arity. All choices can start on an arbitrary tail of strictly increasing depths.

If a packet $I \subset S$ has any bad term, take one J minimal by inclusion. On an edge of its witness whose intersection with I is exactly J, no term with an axis outside J can contribute, and the proper-subset terms all belong to C. This prevents a single non-C_e finite witness from being repaired. In the infinite-support case the sunflower kernel misses I outside J and only finitely many petals can meet the finite I outside J, giving infinitely many unaffected nonzero entries even after adding the proper-subset terms of finite support. Thus a bad term indeed precludes total survival. This uses no independence assumption among overlapping edges.

Canceling all mixed signs Choose a power of two $M > s$, $M \ge 2$. In increasing axis order take **four blocks of M** axes, one per ordinary type, with the c block last. On its final axis replace c by the true speed $c + 2^D$; hence the full packet has odd tag. Every mixed term with that last axis is *identical* to its all-ordinary counterpart: a contributing incident edge contains another selected axis, of smaller D, and that earlier ring has additive exponent dividing $2^{D_{\text{earlier}}+1}$. Thus $2^{D_l} ast$ kills the entire edge module. Since $u_{\text{last}}^{2^{D_l} ast} = 1 + 2^{D_l} ast \epsilon_{\text{last}}$, the change from baseline to true disappears identically on this mixed term. This is why distinct depths and the order of the unique true axis matter.

On a surviving packet, replace the last singleton also by its c counterpart temporarily. The latter is good because some other c singleton is present and its category is homogeneous on the selected set. For each arity and ordered list of types, all signs now agree. A nonempty count vector $(j_1, \dots, j_4)$ of total size at most s contributes its common sign with multiplicity $\prod \binom{M}{j_i}$, which is even: some $0 < j_i < M$ and M is a power of two. Every term cancels, including all mixed terms. The sole remaining phase is the difference between the true and the c singleton on the final axis.

Interpolating the one residual star At the final axis i, regard the tuple of all its incident components as a module over R_i. The four ordinary singleton tests and the true test are all in C on this axis by the preceding homogeneity; sum over all incident edges, not just

those inside the packet. Write $R(Y) = \sum_t (Y^t - 1)(m_t|_i)$ with star-restricted coefficients. The ordinary surviving observations belong to $\ker(u_i - 1)$ in this module, whose elements are killed by ϖ_i^2. The integral Hermite calculation from the maximal theorem is still valid: the evaluations $\epsilon R(1), \epsilon R(\zeta), \epsilon R(\zeta^c)$ and $L\epsilon R'(1), L\epsilon R'(\zeta)$ are killed by ϖ^2. Divide by $P_0 = (Y-1)^2(Y-\zeta)^2(Y-\zeta^c)$ with remainder B of degree < 5. The confluent adjugate gives $\varpi^2 Ldet(C_H)\epsilon B_j = 0$, with determinant valuation $J + 4$. The ratio $z/(\varpi^2 Ldet(C_H))$ is integral on the tail where $(D-K)\phi \geq J + 6$. Therefore $z\epsilon B'(\zeta^c) = 0$. As in the maximal theorem,

$$R(u^{c+z}) - R(u^c) = E_i \cdot \sum_t \binom{t}{5} (m_t|_i).$$

The expression on the right is in C for this surviving test. For each i extend θ from the intersection of C with the order-two group $E_i \prod_{e \ni i} M_e$ (include this star group in the ambient product by zero off the star) to an additive binary character of the whole star group. This is an extension between elementary abelian two-groups. Pulling back along $m \mapsto E_i(m|_i)$ gives an order-at-most-two character π_i on A. It computes the residual phase above even though individual $E_i(m_t|_i)$ need not have finite support or belong to C. The π_i vanish on each fixed local direct-sum element on all sufficiently fresh axes.

The limiting law At stage n include the first n trajectory Fourier tests of a countable determining list and take a packet with $M_n > s_n$, where s_n bounds their arities. The support can be chosen escaping. Pass to a subsequence where all individual bit values at the final axis converge to one character π and all monitored survival/vanishing categories stabilize. For each tuple of coefficients the increment Fourier limit is zero or its phase at $s_t = \binom{t}{5}\pi$. Thus the limiting increment law is Haar on a coset containing s, exactly as in the single-axis theorem.

The initial state is independent of the increments in the joint limit. If the initial test coefficient has infinite support, an infinite sunflower in it has a fixed kernel. Every sufficiently fresh packet misses that kernel, and its finite axes can affect only finitely many petals. The infinite remainder survives unchanged, preventing the joint character's total coefficient from belonging to C. Its initial marginal has expectation zero likewise. If the initial coefficient has finite support, eventually the packet misses all its vertices, giving independence of its finitely many Haar-local coordinates from every increment. These tests prove the product joint limit. The character π annihilates the local direct sum, so it is an admissible Haar translation. At s any event of mass $a > 1/2$ has conditional intersection at least $2a - 1$ by the two-value property; continuity of its return integral and fixed Haar marginals give positive actual returns. The packet may depend on the event, which is enough. This direct mixed-sign argument is specific to Haar with local prescribed subgroups. Extending it to arbitrary invariant measures on the interaction modules would require further control of the star-base law.

An arbitrary-invariant sparse-support case Suppose the edges carrying the observed coefficients admit a vertex subset I with sufficiently many replicated deep levels for the one-axis arbitrary-invariant theorem [R063], and no such edge contains two vertices of I. Restrict the action to the axes I. Regroup every edge meeting I by its unique vertex i there: its whole

tuple now lives in a single R_i-module $\prod_{e \cap I = \{i\}} M_e$. The remaining edges are invariant under this subgroup. One may adjoin the entire direct sum of these regrouped modules and lift the given measure by Haar on the fibers of the restriction map of character groups; this lift is invariant since all the transformations preserve fiber Haar. The static part can be conditioned on in the ergodic decomposition. The one-axis theorem gives the same half upper for **arbitrary invariant laws** under this support hypothesis. For example uniformly bounded degree in the two-section graph gives such independent sets if the replicated blocks grow sufficiently quickly, by ordinary greedy selection, avoiding previously chosen finitely many neighborhoods at each later stage.

For dense pair supports and arbitrary ergodic laws there is still a structural filtration. Finite-edge observations form a compact rotation factor: any finite block lies on one finite orbit of its incident vertex group, and the inverse-limit measure is the corresponding compact Haar-orbit law. Characters supported on finitely many stars change under each tail generator by local coordinates. Global pair characters change under finite directions by star-supported coordinates. Projecting such a character relative to the star factor, the modulus of its conditional expectation is invariant; if nonzero its normalized quotient is invariant and it is already measurable on the star factor. Otherwise its conditional expectation vanishes. Thus the full ergodic law is a relatively Haar compact abelian extension over that factor. These statements do not establish simultaneous high returns for the relative star cocycles.

R067. Carry cocycles with ergodic parity kernel and exact thresholds in one action

Scope and status. Result with proof.

Result. Explicit carry cocycles have ergodic parity kernel and exact system-relative values $1 - 1/q$.

Scope. The free-host parity coset admits further root obstructions; no intermediate integer capacity follows.

Let $q = 2^a$, $a \geq 1$, $H = \bigoplus_i C_{2q}$, and $\chi(g) = (-1)^{\sum_i g_i}$. On $X = \mathbb{F}_2^{\mathbb{N}} \times C_q$ with Haar product probability put

$$U^g(y, c) = \left(y + \bar{g},\ c + \sum_i \left\lfloor \frac{y_i + g_i}{2} \right\rfloor \pmod{q} \right).$$

Here each y_i is read as 0 or 1, g_i can use any integer representative, and only its finite support contributes. Replacing a representative by $2q$ changes the carry by q; successive binary carries telescope, so this is a continuous measure-preserving abelian action. Every $2e_i$ is the same unit translation of C_q. Thus an H-invariant function is independent of c, and the Boolean rotations of y then make it constant. The same independence applies to an eigenfunction with character χ, since $\chi(2e_i) = 1$. The Boolean product rotation has only finite-support binary eigencharacters, excluding this infinite-support χ. It follows that $\ker \chi$ is itself ergodic: a function invariant under this index-two subgroup splits into its invariant and χ-eigenparts.

Yet for every $\chi(g) = -1$, its square U^{2g} advances c by the odd quantity $\sum_i g_i$, a unit modulo q. The event $c \neq 0$ of density $1 - 1/q$ avoids all $2q$-slot runs in those directions. Conversely any event A of density above $1 - 1/q$ has a positive-measure set $B \subset \mathbb{F}_2^{\mathbb{N}}$ of base points with their **whole** C_q-fiber in A. Some finite odd mask s returns B to itself with positive mass, since odd masks are dense even arbitrarily near identity in the compact Boolean product; use continuity of translation in L^1. Any lift g of that s modulo two has its entire orbit above the two points $y, y + s$. On this positive base set it gives all finite lengths of return simultaneously.

Thus in this particular ergodic action the all-length avoiding supremum for the odd coset is exactly $1 - 1/q$, even though the original high H-action has no eigencharacter χ. The action by $g \mapsto U^{2g}$ does detect that parity. This rules out an argument based only on ergodicity of the index-two kernel. The difference coset considered as a free-host or integer family admits arbitrarily high cyclic root obstructions; no integer critical threshold follows from the system-relative result. In the proposed transposed dyadic examples the normal-coordinate menu adds constraints not treated by this carry system.

R068. Full recurrence with an arbitrary common scalar component factor

Scope and status. Result with proof.

Result. A common scalar passive invariant algebra satisfying $I(mP) = I(R)$ for every m gives full recurrence for the replicated dyadic family.

Scope. The common-factor hypothesis is substantive, including when that factor is nonatomic.

Write $R = \bigoplus_i 2^{k_i}\mathbb{Z}e_i$ and $P = \bigoplus_i 2^{K_i}\mathbb{Z}e_i$, with $K_i > k_i$. Use sufficiently replicated dyadic stages for fixed CFR target tensor cancellation [R088]. Let a probability-preserving free-coordinate action satisfy

$$I(mP) = I(R) \quad \text{for every} \quad m \geq 1.$$

The common invariant algebra may be nonatomic, and the ambient group may move its components arbitrarily. Every positive-measure event has all finite lengths of recurrence along the legal directions on arbitrarily late stages. This only treats systems with the stated common invariant algebras.

Characteristic residual Fix the length h and the event indicator f. On almost every R-ergodic component every mP is ergodic. The ergodic subgroup cube comparison [R106] identifies the Host–Kra cubes of mP and R there; globally disintegrate over their common invariant algebra. Take $s = s(h)$ large enough for the usual generalized von Neumann bound for h slots, and $S = S(s)$ from Kuca's frozen-complement estimate [R073]. Put $g = \mathbb{E}(f \mid Z_S(R))$ and $r = f - g$, using an order at least S so that $\|r\|_{U^S(R)} = 0$. The common cubes give the same zero norm on $h!P$. Standard ergodic decomposition of the Host–Kra factors identifies g on almost every common component with a function on its order-S factor; this follows also by disintegrating cube dual functions and their invariant-algebra multiples.

At a fixed identical-parameter block choose a random envelope J of size b, from a block of $n \gg b^2$ axes. Let $F_J \leq P$ be the full passive subgroup with its coordinates in J set to zero (including all the other axes throughout the free host). For disjoint envelopes $h!F_J + h!F_{J'} = h!P$; the finite-action proof of Kuca's proposition [R073] by invariant translations, Cauchy–Schwarz and subgroup-mean convolution applies also to these countable amenable actions, with successive invariant means in the corresponding cube coordinates. Hence, for some envelope J, $\|r\|_{U^s(h!F_J)}$ is arbitrarily small for n/b^2 sufficiently large. Quantitative constants depend only on the slot count and desired error. The usual successive Cauchy–Schwarz/van der Corput estimate for translates on a progression with direction $p + d$, $p \in F_J$, bounds any residual-slot contribution by a quantity tending to zero with this norm, uniformly in fixed d. Indeed every nonzero difference subgroup jF_J, $|j| < h$, contains $h!F_J$, so monotonicity of box seminorms suffices; no equality of its cubes to the full P-cubes was assumed. The translations by d commute and disappear from the distinguished function's derivative norms. The estimate may be enforced on a sufficiently large finite Følner average uniformly over d and all other 1-bounded factors. One first fixes finite van der Corput increment sets approximating the bounded cube seminorm bound, and then takes

the averaging set sufficiently invariant under their differences. All other factors were removed by Cauchy–Schwarz, and the finite derivative expressions of the distinguished function are unchanged by its common translate. This avoids exchanging an arbitrary-residual mean with the later word-variable average.

Approximating the structured orbit process across components For a fixed continuous $\psi : Y \to [0,1]$ on a compact finite-rank nilspace, consider the compact shift-invariant set

$$M(Y,\psi) = \{(\psi(p(u)))_{u\in R} : p : D_1(R) \to Y amorphism\}.$$

Here R is abstractly free abelian on the countable axes. Morphism constraints are closed in Y^R, and the code is continuous, so the image is compact. On every ergodic component with the compact order-S nilspace model for g, approximate g in L^1 by a continuous observable of a finite-rank fibration, using the nilspace inverse limit theorem. Its joint orbit under R is supported in $M(Y,\psi)$ by the translation cube property, regardless of consistency of the fibration with R. There are countably many finite-rank isomorphism types and a countable dense list of continuous observables on each; use the same lists for all components and include constant codes.

Here are details avoiding any choice of a common fibre-coordinate presentation. Map the original system to its g-name $u \mapsto g(U^u x)$. Its conditional R-stationary law ν_y on $[0,1]^R$ is a measurable function of the R-component y. For each fixed code type, the space of invariant probability measures on $[0,1]^R \times M(Y,\psi)$ with first marginal ν_y is compact and nonempty. Its graph as a function of the marginal is closed in compact metrizable spaces, and the cost $\int |z(0) - w(0)|$ is continuous. Its minimum and a minimizing selection are therefore Borel measurable (e.g. compact-fiber measurable selection). For almost every component the infimum over the listed types is arbitrarily small by the preceding approximation. Select finitely many types which achieve cost $< \epsilon$ except on a base set of measure $< \epsilon$, choosing a best type there and a constant code elsewhere. The resulting mixed joining is stationary over the base. Relatively independently join it to the original system over the pair (component, g-name), obtaining a probability kernel assigning a random code W_x to each original site. For every $u \in R$,

$$E_x E_{W_x} |g(U^u x) - W_x(u)| \le 2\epsilon.$$

This is all that will be used; no action of low G-directions on these selected code models is needed.

The sampled directions Use m disjoint trial groups of b/m axes in the envelope, with m from finite density Hales–Jewett for $\mu(A), h$. Independently in each envelope coordinate sample the speed $1 + 2^{K_i} w_i$ in a normal branch or $1 + 2^{k_i} + 2^{D_i} w_i$ in an active branch, with a fair branch choice. To each trial group add an independent uniform passive increment p_t from the sufficiently large finite Følner set in F_J. A DHJ line uses a nonempty subset of the trial groups, with their added increments summed. In a residual-slot estimate, freeze all external variables except any one included p_t; the preceding finite uniform bound applies.

Telescoping therefore replaces the indicator by g with small error in each signed line mean; the tag is independent of every p_t, as they all vanish on the envelope and are normal zero off it.

For any one fixed line and realization of the finite passive increments, let v_0 be the sum of e_i over the included envelope axes and let p be the sum of its passive increments. The sampled direction is $p + v_0 + u$ with $u \in R$. At a random initial x, use the kernel above at the h points $x_j = U^{j(p+v_0)}x$, independently conditional on the original points. The error of replacing $g(U^{ju}x_j)$ by the code value $W_{x_j}(ju)$ has integrated expectation at most 2ϵ in each slot, uniformly in u, since each x_j has the original marginal. Crucially the codes are sampled before the mask and integer variables of u. Conditional on them the product is a fixed-CFR-morphism test in the included coordinate variables (scaling inputs by j preserves the morphism property). There are only finitely many products of the finite target types. Normal steps in R-units are $2^{K_i-k_i}$; active high steps $2^{D_i-k_i}$. The CFR product cancellation [R088] makes each signed average small. Pointwise polynomial nilsequence means exist, hence dominated convergence justifies integrating these code estimates. The fixed finite passive averages introduce no limit exchange.

Unweighted, the sum over the finite DHJ lines has a lower bound depending only on the event mass and h, by sampling the word cube at a random initial point and summing line probabilities. This holds in every finite box of word variables. The signed sums are smaller after the three errors (residual, code approximation, tensor-cancellation tolerance) are taken suitably small. Fourier expansion of the odd-tag indicator then produces positive actual correlation on some admissible trial. A single schedule suffices for all such events and systems: at each level select D then b/m for a growing initial segment of the countable list of targets, observables, and tolerances; select the full replication n still larger so that the Kuca error vanishes for the prescribed envelopes. The finite passive averaging size and actual witnesses may depend on the system; the original group permits all passive corrections.

R069. Word recurrence for morphisms into arbitrary compact finite-rank nilspaces

Scope and status. Result with proof.

Result. The ordered dyadic word identity gives pointwise recurrence for morphisms into arbitrary fixed compact finite-rank nilspaces, uniformly in the morphism.

Scope. This is not an assertion about arbitrary invariant-measure joinings on nilspaces.

The dyadic signed word [R095] gives positive pointwise recurrence not only for ordinary polynomial nilorbits but also for morphisms into any fixed compact finite-rank nilspace. Explicitly, given such a target Y of step s and $\eta > 0$, choose $D \geq C_s K + O_s(1)$ as in the word theorem. In every sufficiently large block of identical parameters, every nilspace morphism $p : \mathcal{D}_1(\mathbb{Z}^n) \to Y$ takes some legal direction d within η of $p(0)$. The bound on the size of the block depends on the fixed target and parameters but not on the morphism. Products cover simultaneous observations at prescribed integer multiples. Arbitrary compact nilspaces follow by passing through a finite-rank factor for a prescribed neighborhood. This is a pointwise assertion about morphisms.

By Candela–González-Sánchez–Szegedy, Theorem 1.7, Lemma 5.13 and Corollary 5.37, write $Y = F/\Gamma$, where $F = \prod_{l=1}^{s} \mathcal{D}_l(L_l)$, $L_l = \mathbb{R}^{a_l} \times \mathbb{Z}^{b_l}$, and Γ is a fiber-transitive higher-order lattice of translations. In particular the map is a fibration, the l-th structure group is the quotient of L_l by a discrete cocompact subgroup, and each element of this subgroup is induced by some $\gamma \in \Gamma \cap \Theta_l(F)$. This does not assert that Γ is cocompact inside the full translation Lie group of F.

Any morphism p from $\mathcal{D}_1(\mathbb{Z}^n)$ lifts through this quotient. Induct on the layers. Given a lift on the lower free factors, extend its l-th coordinate by zero, and compare its projection with p on the l-th factor. Their lower projections agree, so their pointwise difference in the top structure group is a morphism to the degree-l compact abelian nilspace: on cubes the two lifts have common lower cube and differ by a degree-l abelian cube, by the structure group axiom. This difference is an ordinary abelian polynomial of degree $\leq l$ on $\mathbb{Z}^n$. Its finitely many Newton coefficients can be lifted individually through the surjection from L_l. Adding the resulting free top coordinate gives the lift through layer l. Thus we have ordinary L_l-valued polynomial coordinates $g_l(x)$ of degrees $\leq l$.

All the Taylor coefficients of such a lift can be reduced into fixed compact sets, independently of n. Process $l = 1, \ldots, s$ in order. For each multi-index ν with $|\nu| \leq l$, apply the pointwise deck correction

$$g(x) \mapsto \gamma^{-\binom{x}{\nu}} g(x), \gamma \in \Gamma \cap \Theta_l(F).$$

It changes neither the projection nor the lower layers. In layer l the correction subtracts an arbitrary member of the structure lattice times $\binom{x}{\nu}$; choose that member to put this particular coefficient in fixed compact representatives for the quotient. Other coefficients of layer l are untouched. Changes in higher layers are allowed at this stage. These are legitimate morphisms after correction. Indeed a degree-l translation in a free nilspace adds to the j-th coordinate a polynomial of weighted degree $\leq j - l$ in earlier coordinates (the triangular description of translations in Section 3 of the same paper). Its integer power

m, with m assigned weight l, still gives weighted degree $\le j$ there: induct on j, take the one-step difference in m, substitute the lower-coordinate polynomials, and sum using integer binomials, which increases weight by l. Substitution of $m = -\binom{x}{\nu}$, $|\nu| \le l$, preserves the ordinary degree bound. This also applies to the constant coefficient.

With these compact coefficients the usual homogeneous Ramsey reduction [R095] applies separately in each free layer. For homogeneous ordered coefficients the value in L_l is its constant plus a linear combination of word-signature moments of total weight $\le l$, by the scalar multivariate Newton expansion. The fixed signed identity word at order s gives the initial point when its integer repetitions have their signed values. Increasing each count by $2t$ gives a curve $g(t)$ in F, polynomial of degree $\le l$ in its l-th coordinate, defined also before the counts become nonnegative. Its projection $a(t)$ is a morphism $\mathcal{D}_1(\mathbb{Z}) \to Y$ with $a(0) = p(0)$ for the homogeneous coefficient data.

Every such one-variable morphism returns near its initial point at arbitrarily large times. In fact the shift on the compact space $\mathrm{Mor}(\mathcal{D}_1(\mathbb{Z}), Y) \subset Y^{\mathbb{Z}}$ is distal. If two morphisms have shifts along the same net converging to the same point, their projections to the previous nilspace factor already coincide by induction. Their pointwise top-fiber difference is an ordinary polynomial $\alpha : \mathbb{Z} \to Z_s(Y)$ of degree $\le s$. Its translated values tend to zero simultaneously on any finite window, by continuity of the torsor difference map. Its top constant derivative must be zero, and repeating this argument on decreasing degrees gives $\alpha = 0$. The initial morphisms therefore coincide. Compact distal systems are pointwise recurrent, giving the required arbitrarily late times.

In the finite homogeneous parameter family, a fixed positive t giving a strict return remains successful on a neighborhood by continuity of the quotient and the finite word evaluation. Compactness bounds a successful t uniformly, taken sufficiently large to make all repetitions nonnegative. This fixes the required finite slot count before Ramsey. A sufficiently fine finite cover of the compact coefficient sets makes replacing the many true coefficients by their homogeneous representatives harmless for that finite list of bounded words. The odd high-bit tag and the permitted velocities are unchanged. Finally CFR nilspaces have only countably many isomorphism classes (see Candela and Szegedy), so the stage sizes may be chosen to cover every fixed target and tolerance along a single choice of the family. This does not address Haar joinings of general CFR nilspaces, arbitrary measure actions, or the all-system critical threshold.

R070. More distinct states on genuinely higher covers

Scope and status. Result with proof.

Result. Wider native rings produce short observation windows excluding every proposed bounded-state supported trajectory.

Scope. Failure of a bounded-state proof does not establish high-density avoiding events.

Qualifications. Bounded-state nonexistence is a statement about supports of selected observed trajectory laws, not an avoidance-density theorem.

The Haar-module half proof [R085] uses a supported trajectory taking only two values. A different ring, still acting on the **minimal native group**, has finitely many Haar observables for which no trajectory with at most q distinct observed states is possible on any sufficiently deep tail, for each fixed q. The same one native ring action serves every q (the observed channels and horizon vary). We first give this stronger fact, then separate variants for particular period lifts. Failure of this support argument does not establish a high-density avoiding set.

Distinct six-windows on the native group Keep $k \to \infty, K - k \to \infty$ and choose $B_i \to \infty$ with $J_i + B_i + 1 \le \phi_i$. Replace the truncation length by $D_i\phi_i + J_i + B_i + 1$; still $2^{D_i+1} = 0$, so multiplication by u factors through the minimal cyclic axis. Prescribe the half-turn on $E_{\mathrm{top}} = z\varpi^{J+B_i}\epsilon$, of order two and fixed by u. The original filter P at a mark is $tE'V$ with $E' = z\varpi^J\epsilon$ and V a unit of residue 1. At normals it vanishes identically.

For a nonnegative integer s put $m_{s,i} = \varpi_i^{B_i-2^s}$ on indices where the exponent is nonnegative, and zero on the exceptional head. Observe the six coefficient characters of $m_sP(Y)$ at one site. For every fixed positive lag d with $v_2(d) = s$, on a sufficiently deep tail the filter

$$m_s(Y^d - 1)P(Y)$$

evaluates to the fixed top half-turn along every legal direction. Indeed on a marked coordinate its residual factor on E' is $\zeta^{cd} - 1$, which has valuation exactly 2^s. Every unit reduces to 1 after normalization against the top element. Normals contribute zero, the sum of t is odd, and an arbitrary further starting time does not affect the top element. Consequently two length-six windows of those observed snapshots at starts differing by d cannot agree. The conclusion also excludes sufficiently close six-windows (six differences contribute with coefficients ± 1 after absorbing the coefficients into the chosen channels). The same row of six channels works for every lag with a given 2-adic valuation, at any fixed finite horizon on a sufficiently deep tail.

For $H = q^6 + 6$, any q-valued H-word has more than q^6 six-windows and two are identical. Use rows $0 \le s \le \lfloor \log_2 H \rfloor$ simultaneously. Every legal H-run must therefore have more than q distinct joint observed states, even after clustering into q sufficiently small sets. This much stronger recurrence-exclusion certificate comes from comparing equal windows and does not incur the factor a from taking a temporal stride. There is no contradiction with the half theorem: on these wider rings E' need not be a fixed order-two element. A high-density event can contain many distinct observed configurations.

Keep two-center notation, with $k \to \infty$, $K - k \to \infty$, and write $\phi = 2^{K-1}$, $J = 2 + 2^{k+1}$. Enlarge the ring truncation from $D\phi + J + 1$ to $(D+e)\phi + J + 1$, so the action factors through

$C_{2^{D+e+1}}$. Prescribe the character on the top invariant element $E_{\mathrm{top}} = z\varpi^{J+e\phi}\epsilon$ to be $1/2$ on each axis. Work with the group of all coefficient sequences if desired; each argument below only uses finitely many global character observations.

A q-state exclusion by temporal stride Fix $q \geq 2$ and take $H = W(q, 6)$ from van der Waerden's theorem. For each $a = 1, \ldots, H$ use the six-term filter

$$P_a(Y) = (Y^a - 1)^2(Y^a - u^a)^2(Y^a - w^a), w = u^c.$$

Its coefficients are at the times $0, a, \ldots, 5a$, and their sum is zero. For every normal integer velocity r, $P_a(u^r) = 0$ by the squared normal factor, since the difference from the normal root has zero constant term in ϵ. At an active $r = c + zt$, one obtains

$$P_a(u^r) = tz\epsilon a\zeta^{ac}(\zeta^{ac} - 1)^2(\zeta^{ac} - \zeta^a)^2.$$

For fixed a and large enough $K - k$ the scalar after $tz\epsilon$ has ϖ-valuation

$$v_2(a)\phi + J \cdot 2^{v_2}(a).$$

Indeed c is odd and $c - 1 = 2^k$. If $e > \max_{a \leq H} v_2(a)$, then on sufficiently deep tails this valuation is at most $J + e\phi$. Choose an unchanging global coefficient m_a with local entries normalizing the last expression to tE_{top}; units have inverses in the finite chain ring. Therefore for each of these strides, every allowed progression has a half-turn in its six-observation filter at that stride, including when the six positions start later along the same orbit, since $uE_{\mathrm{top}} = E_{\mathrm{top}}$. Observe all six coefficient characters of m_aP_a for each of the finitely many a.

If an H-term run had at most q distinct observed states, color its times by their states. A monochromatic six-term subprogression exists with some stride a. At its six positions the corresponding filter then gives the zero phase: for any fixed state the six coefficient characters sum to zero. The forced half-turn contradicts that. The same proof applies when the observations lie in the union of q clusters of sufficiently small diameter, by using actual samples as cluster representatives and requiring the six errors together to be less than $1/2$. The conclusion covers all legal directions supported sufficiently far out, not just the directions sampled in the native-module half proof.

Seven slots and two clusters on four extra levels A smaller horizon is available for $q = 2$. Take $e = 4$ and any nonconstant binary word $\sigma_0, \ldots, \sigma_6$. With $P(Y) = (Y - 1)^2(Y - u)^2(Y - w)$, define scalar coefficients $A = (P(S)\sigma)(0)$ and $B = (P(S)\sigma)(1)$ in the local ring, where S shifts the binary word in time. Use the seven-slot filter

$$R(Y) = mP(Y)(B - AY).$$

Its coefficient sum is zero, and its σ-weighted coefficient sum is identically $m(BA-AB)=0$. For an active velocity the scalar $B-Aw$ evaluated at $\epsilon=0$ is the value at $(U,W)=(\zeta,\zeta^{c})$ of the formal polynomial

$$C(U,W)=[(S-1)^{2}(S-U)^{2}(S-W)^{2}\sigma](0).$$

This is a nonzero polynomial: its coefficients at $U^{i}W^{j}$ are nonzero signed binomial multipliers of the second difference of σ at $4-i-j$, for $i,j\in\{0,1,2\}$. A length-seven binary word with all these second differences zero is constant. All coefficients have absolute value at most 8. By the two-variable valuation estimate [R061], $v_{\varpi}C(\zeta,\zeta^{c})\le 3\phi+O(2^{k})<4\phi$ on the tail (the O constant is uniform over the finitely many words). Hence m may be chosen to make the active value of R equal tE_{top}, absorbing the normalized unit from P. Normal values vanish. For a constant binary word one instead uses P with a scalar normalizer sending its active value to tE_{top}; only the ordinary zero coefficient sum is then needed.

Take the finite set of coefficient observations of these tests over the binary words. If the times in a seven-slot trajectory could be partitioned into two equal-state classes, the corresponding filter would give zero by both coefficient-sum identities, contrary to its actual value $1/2$. The same strict separation in finitely many circle-coordinate neighborhoods again rules out limit trajectories approximated by two states. The argument does not imply that a measurable subset of density near one meets no complete trajectory: an arbitrary high-density subset can contain many configurations, and the necessary number of observed channels grows with the requested number of states.

R071. A summable-period exclusion for the two-normal dyadic model

Scope and status. Result with proof.

Result. Summable low dyadic seam rates force endpoint one; complementary bounded-depth ratio conditions give ordinary polynomial separation.

Scope. The high replication schedules of the surviving candidate can violate these summability assumptions.

Use the coordinates of the high-subgroup note [R079]. Normal velocities are $0, 1 \bmod 2^{K_i}$ and marked velocities $1 + 2^{k_i} + 2^{D_i} t_i$, with odd total t_i; $k_i < K_i < D_i$. If $\sum_i 2^{-k_i} < \infty$, the critical density on the torsion-free coordinate host is one. The number of active coordinates is unrestricted. This includes strictly increasing integer k_i. The argument uses the full near-one obstruction proved for the single-normal parity model [R080], not just its low four-slot witness.

Fix a desired loss $\epsilon > 0$. For every finite coordinate packet the parity model with depths $D_i - k_i$ has a stationary event A of mass $> 1 - \epsilon$ avoiding some common length $h = h(\epsilon)$; use its uniform bounded-period/root-gate proof, or restrict the global high avoiding event to the packet. Introduce the finite clock $\prod_i \mathbb{Z}/2^{k_i}\mathbb{Z}$ with uniform independent coordinates. On the product with the system of A, let an integer vector d update the clocks by addition and the first system by the integer vector with coordinates $\lfloor (x_i + d_i)/2^{k_i} \rfloor$, using centered or ordinary integral representatives $x_i \in [0, 2^{k_i})$ in the latter convention. The ordinary carry identity makes this a measure-preserving group action. Mark A only when each clock is outside the last h residues. The cost on a tail starting at N is at most $h \sum_{i \geq N} 2^{-k_i}$.

Every allowed d_i has least residue $a_i = 0$ or $1 \bmod 2^{k_i}$. Starting at the retained point there is no extra clock wrap in the next $h - 1$ slots; the first coordinates therefore run with constant difference $b_i = (d_i - a_i)/2^{k_i}$. For either normal type b_i is even, and for a marked type $b_i = 1 + 2^{D_i - k_i} t_i$. Thus b is legal in the parity model. The h-fold retained cylinder is null by avoidance for A. The finite packet bound passes to the full tail by stationary compactness, without summing errors more than once.

To handle a finite head on the torsion-free host, any direction with a marked head has a fixed nonzero residue in one of finitely many head coordinates modulo its 2^{D_i}. Their finite union has near-one avoiders at some larger length by ordinary circle-root separation, and its product with the tail avoider still has near-full measure. Directions with no marked head have legal odd tag entirely in the tail, even if some head coordinates are normal and nonzero. Sending tail loss and parity-model loss to zero gives the assertion. If $\sum_i 2^{-k_i} = \infty$, the finite clock union estimate does not give an arbitrarily cheap tail cut; the argument gives no result for unrestricted such schedules.

The active mask has ordinary degree two to arbitrary fixed dyadic precision At one coordinate put $q_i(x) = x(x-1)/2^{k_i}$. At a normal r, $q_i(r)$ is divisible by $2^{K_i - k_i}$. At a marked $r = 1 + 2^{k_i} + 2^{D_i} t$,

$$q_i(r) \equiv 1 \bmod 2^{\min}(k_i, D_i - k_i).$$

All these values are integers. Consequently, when each of $k_i, K_i - k_i, D_i - k_i$ tends to infinity, on every sufficiently late tail and for every fixed s one has

$$\sum_i w_i q_i(d_i)/2^s = \sum_{iactive} w_i/2^s \bmod one$$

for arbitrary integer weights w_i. This is a single ordinary group polynomial of degree at most two on the direct sum, irrespective of the number of summands. The same holds with coordinatewise denominators 2^{s_i} for $s_i \le \min(k_i, K_i - k_i, D_i - k_i)$. In particular any subfamily on which the weighted active-mask sums are separated from zero jointly in a finite list of these circles has critical density one, by ordinary polynomial root coloring. Fixed nonzero bounded active counts and counts outside $2^s\mathbb{Z}$ are examples. A surviving ultrafilter with local threshold below one must send its active masks to zero along all these phase tests. This statement recovers the **mask of active coordinates**, not their odd-tag condition, which is still coupled to the individual high digits.

Ordinary interpolation on bounded lists of comparable residue balls On a single coordinate take finitely many disjoint 2-adic balls $a_j + 2^{b_j}\mathbb{Z}$, labelled normal or marked, the latter carrying the bit $(x-a_j)/2^{b_j} \bmod 2$. Suppose the number of balls is uniformly bounded and that, for each mark i and each other ball j,

$$(b_i - s_{ij})/(b_j - s_{ij}) \le R, s_{ij} = v_2(a_i - a_j).$$

The denominator is positive since the balls are disjoint. Equal-radius balls at arbitrary depth automatically qualify. Then the marked bit on the whole support is the reduction of a rational polynomial of uniformly bounded degree and 2-integral values on that support.

For a single mark take

$$Q_i(x) = ((x - a_i)/2^{b_i}) \prod_{j \ne i} ((x - a_j)/(a_i - a_j))^{e_j}.$$

On ball i all the factors in the product are units congruent to one modulo two. On another ball j its designated factor gains $e_j(b_j - s_{ij})$ in valuation. The linear term loses at most $b_i - s_{ij}$, and a factor for $l \ne i, j$ can lose valuation only if $s_{il} > s_{ij}$, in which case its loss is $e_l(s_{il} - s_{ij})$. Order j's with decreasing s_{ij} and choose each e_j exceeding $(b_i - s_{ij} + \sum_{l:s_{il}>s_{ij}} e_l(s_{il} - s_{ij}))/(b_j - s_{ij})$. This kills the result modulo two there. Every ratio in this expression is at most R, so the largest exponent is bounded in terms of R and the number of balls only. Sum the Q_i over the marked balls.

The reduction statement is 2-adic. To turn it into an ordinary circle polynomial, write each coordinate polynomial over a common integer denominator $m_i = 2^{d_i} u_i$ with u_i odd. Replace u_i^{-1} in every coefficient by an integer inverse modulo 2^{d_i+1}. All coefficients now have power-of-two denominator, and the polynomial's value at each integer differs from the old 2-adic value by an element of $2\mathbb{Z}_2$. Since its values on the permitted integers are 2-integral and have power-of-two denominator, these values are ordinary integers with the advertised parity.

If zero is a normal residue, subtract the even value at zero. The half-sum of the resulting coordinate polynomials is a bounded-degree group-polynomial test on the direct sum of integers, and equals a half-turn on every direction of odd total tag. Its ordinary polynomial Bohr separation gives critical density one by the finite-degree root-coloring theorem. The strengthened example with a broad normal ball at 1 mod 2^{K_i} and much finer mark at $1+2^{k_i}$ need not obey the ratio bound.

R072. Simultaneous polynomial-phase and Eberlein autocorrelations on the dyadic pure menu

Scope and status. Result with proof.

Result. One-mark words synchronize Eberlein tests with ordinary polynomial phases via a common ultrafilter/idempotent evaluation.

Scope. Uniform-algebra membership is required; general integrated polynomial phases cannot be interchanged with ultrafilter limits automatically.

The single-mark unitary argument [R174] can be synchronized with an actual bounded-degree polynomial relation, rather than applying two separate recurrence tests. For every fixed degree d and sufficiently large D/K in the truncated integer word construction [R095], the following holds on *infinitely many replicas of that one parameter type*. There is an ultrafilter on genuine one-mark words and an additive idempotent with the same evaluations on the entire uniform algebra generated by Eberlein functions and phases of ordinary group polynomials of degree at most d. The argument below also makes a diagonal consequence for rows with finitely many copies of each type.

Input and sampling. On free axes e_i take the normal velocities $1 + 2^K j$, $0 \le j \le d$, and one pure marked velocity $r_* = 2^D t$, with t odd and chosen as in the truncated signature calculation. Put $g(r) = 1 + \sum_{j=1}^{d} \binom{r}{j} X_j$ in the associative integral word algebra truncated at weight d, assigning weight j to X_j. The index argument there puts $g(r_*)g(0)^{-1} = g(r_*)$ in the subgroup generated by the normal $g(r)$. Thus choose a word in powers of these signed normal letters and with exactly one marked exponent $+1$, whose group product equals 1. Denote its ordered bases by r_l and signed exponents by b_l. Weight one gives $r_* + \sum_{normall} b_l r_l = 0$ exactly.

For large positive u change each **normal** exponent to $b_l + 2u$; keep the single mark unchanged. Use each positive repetition on its own fresh axis in word order. Fix a free index ultrafilter and take the corresponding successive index limits; the resulting ultrafilter is a sum of convolution powers of the individual letter limits, in the stated order. Finally let u tend to a free additive idempotent ν on N, giving an ultrafilter p on the allowed steps, on every tail. Its image in every cyclic unitary compactification is a projection, by the proof of the integer-menu variant of one-mark-eberlein.md. For clarity, in the region where all normal multipliers have modulus one, their convergence is strong and the weight-one relation identifies the marked multiplier. The resulting multiplier there is $(\prod_{normall} f_l^2)^u$; elsewhere the diverging normal counts force convergence to zero. Choose an additive idempotent e with the same images in all those compactifications.

Polynomial values. For each ordinary circle group polynomial P of degree at most d write the multivariate Newton expansion on finitely supported input vectors. For each ordered tuple of positive exponents $j_1, \dots, j_a$ of sum at most d, take the iterated index-ultrafilter limit of its torus coefficient on increasing axes. These finitely many limits specify an additive map Θ on the word augmentation ideal assigning those coefficients to $X_{j_1} \dots X_{j_a}$. At any fixed u the iterated limit of $P(v) - P(0)$ is precisely $\Theta(W(u) - 1)$, where $W(u)$ is the product of the positive-repetition signatures. Each term follows by expanding the product: the distinct ascending axes and their binomial values correspond exactly to the mixed Newton terms. The iterated limit on any sublist of ordered indices is its same prescribed coefficient. In particular no interchange with an integral over polynomial coefficients is involved.

The coordinates of $W(u) - 1$ are integer-valued polynomials in u, and vanish at zero by the initial free word. The ν-limit of any circle polynomial is its value at zero. More generally this holds on any abelian group at any additive idempotent: apply induction in degree to $P(s+t) - P(t)$ at fixed s, and use $e + e = e$ to obtain $\lim_e P = 2 \lim_e P - P(0)$. Thus p and e agree on all the specified polynomial phases, and also all Eberlein functions. Ultrafilter evaluations preserve products and uniform limits, proving the assertion about their joint algebra. A same-event multiple correlation in that algebra has positive p-limit by finite density IP recurrence at e.

Finite scheduled rows. Quantifier order matters when K, D must increase. Fix d, h, a lower event-mass bound $a > 0$, a bound on the number of terms, and a bound on the total variations of their Fourier–Stieltjes measures. Suppose a system has its h-correlation uniformly approximated within $\epsilon < \kappa(a, h)/3$ by a sum of products of such coefficients and polynomial phases of degree at most d, where κ is the uniform finite IP recurrence bound. For each valid type (K, D) there is a finite number of replicas of that type sufficient to force a genuine return in all these systems. Otherwise take counterexamples on arbitrarily large initial coordinate sets, pull the binary process laws back to the infinite free group by dropping later coordinates, extend polynomial coefficients by zero, and extend the spectral measures by trivial tail coordinates in $\widehat{G}$. Pass to subsequential weak limits of the process laws, all finitely many variation-bounded spectral measures and all of the torus-valued Newton coefficients. The resulting process has mass at least a, avoids every allowed finite marked word, and still has its triple or higher correlation within ϵ of the limiting algebraic approximation at every group time. But its p- and e-limits differ by at most 2ϵ; the first is zero and the second at least κ. This contradiction supplies the uniform size. Restriction to finite coordinate sets does not enlarge any of the approximation parameters.

Scheduling types with D/K increasing and sufficiently many replicas at successively increasing integer complexity bounds therefore gives positive returns in any system whose tested correlations belong to the uniform algebra generated by Eberlein functions and ordinary polynomial phases of arbitrary finite degree, without a universal bound on that degree from one system to the next. This does not assert that correlations of general stationary processes belong to this algebra, nor that arbitrary integrals of polynomial phases can be interchanged with the ultrafilter limits.

R073. What a dyadic energy plateau controls

Scope and status. Result with proof.

Result. Dyadic energy plateaus bound changes of invariant projections and give the stated frozen-complement estimates.

Scope. The estimates do not apply automatically to a return-product observable chosen after the envelope.

Qualifications. The family of sections is fixed before the envelope is chosen; the energy estimate is not uniform over sections selected afterward.

Write $P_l = 2^l P$ and $E_s(l; f) = \|f\|^{2^s}_{U^s(P_l)}$ for a bounded observable in a finite stationary block, with $|f| \le 1$. All formulas below also hold for fixed countable amenable subgroups using iterated Følner averages. The sequences are nondecreasing in l and lie in $[0, 1]$. In fact for any positive integer a,

$$E_s(l + a; f) - E_s(l; f) \ge$$

$$E_{w_1,\dots,w_{s-1} \in P_l}$$

$$\|E(\Delta_{w_1} \dots \Delta_{w_{s-1}} f | I(P_{l+a}))$$

$$-E(\Delta_{w_1} \dots \Delta_{w_{s-1}} f | I(P_l))\|_2^2.$$

Here $I(H)$ denotes the H-invariant factor. To see this change only the last subgroup in the box norm from P_l to P_{l+a}. The box norm after the other increments are fixed is the squared norm of the invariant-factor projection of the displayed derivative; subtracting the nested projections gives their squared distance. Shrinking other subgroups can only increase the box power. This last monotonicity follows by cube-coordinate symmetry and the same Hilbert projection argument one subgroup at a time. Consequently on disjoint intervals of length a across the gap $D - K$, all but $O(s_0 b/\epsilon)$ intervals have total energy change $\le \epsilon$ for b fixed observables and all $s \le s_0$. The same choice works for translates of those observables, by invariance.

There is also a uniform relative-error statement in an identical finite block. Put $T = 2^K C_{2^e}$ at each coordinate and choose $H \le T$ contained in jT for all $1 \le j < h$, e.g. $H = h!T$ or any deeper digit subgroup. Suppose a bounded remainder r has small cube norm for increments in H^n, measured with Haar in the entire ambient group, or equivalently by averaging over its H^n-cosets. For a uniformly random fixed-size b-set J of coordinate indices, the frozen-axis comparison [R100] gives

$$E_J \|r\|^{2^s}_{U^s(H^{J^c})} \le \|r\|^{2^s}_{U^s(H^n)} + o_{n\to\infty}(1)$$

at each fixed e, K, h, b. For that same J, the generalized von Neumann box bound on progression averages with difference $v+p$, p uniform in T^{J^c}, controls an r-slot by $\|r\|_{U^s(H^{J^c})}$ at a safe fixed order $s = s(h)$: all nonzero slot-difference subgroups contain H^{J^c}. The estimate is uniform over **every** fixed v supported on J. In particular there is no union bound over later candidate active subsets of J. This avoids an artificial loss exponential in $b(D-K)$ when first comparing a regularity remainder. It does not control varying e without a bound on the local second moment in the frozen-axis lemma.

The plateau compares nested **conditional** factors. A feature already determined by $I(P_l)$ can be arbitrary and is unchanged by an exact plateau. In particular the two odd low centers remain distinguished on every $I(P_l)$-fiber label, for all $l \geq 0$. There is even an exact three-even-slot binary decoder of their active mask. On C_{2^K} let S be unit translation, $c = 1 + 2^k$, $d_0 = 2 + 2^{k+1} < 2^K$, $b(y) = \binom{y}{d_0} \bmod 2$, well-defined and periodic. Use $R(T) = (T^2-1)(T^2-S^2)$. Then $R(S^r)b = 0$ for $r = 0, 1 \bmod 2^K$. At $r = c$, $S^{2c}-1$ is a unit times $(S-1)^2$ in the local group ring over $\mathbb{F}_2$, and $S^{2c}-S^2$ is a unit times $(S-1)^{2^{k+1}}$. Their product applied to b is one, since the leading unit coefficients are one and the d_0-th difference of b is the constant one. Multiplication by T gives the statement using just three odd slots. Spatial translates in the coefficients are allowed here. Thus bounded coarse features can recover the active mask separately on both temporal parities even at an exact plateau of their own P_l-energies. The missing step is a high-density recurrence theorem for the resulting, potentially different, structured conditional functions on these coarse slots, not the martingale or the random-axis estimate.

Uniformity over arbitrary per-axis group sizes Allow higher global degree in the uniformity hypothesis on the remainder. Then the local group-size loss of the frozen-axis total-variation estimate disappears. Namely write $H = \bigoplus_{i=1}^n H_i$, a subgroup of a finite abelian ambient group, and let $F_J = \bigoplus_{i \notin J} H_i$ for a random uniform b-subset J of $[n]$. For a 1-bounded r,

$$(E_J \|r\|^{2^s}_{U^s(F_J)})^C(s) \leq \|r\|_{U^S(H)} + O_s(b^2/n)$$

for fixed S, C depending only on s. Apply Kuca, Proposition 2.3, with each of its s input subgroups at J equal to F_J. Its right side averages a box seminorm with $S = O_s(1)$ pair-sum subgroups over 2^s independent index sets. Each pair always uses two *different sample labels*. With probability $1 - O_s(b^2/n)$ all these sampled sets are disjoint, hence every pair sum is H. All exceptional box norms are bounded by one. This proves the stated weaker bound, regardless of whether the output box power is retained. The same statement on a finite action follows either by running the invariance/Cauchy–Schwarz proof there, or applying the finite-group result to each orbit and Hölder (raising or lowering constants as necessary). For $s = 1$ one has an even sharper additive bound b/n on the excess squared invariant projection: decompose the finite-group representation spectrally, and any frequency nontrivial on H becomes trivial on F_J only when some one of its nontrivial axes belongs to J.

Thus for the h-slot progression one can discard any common residual small in $U^{S(s(h))}(H)$ on most frozen envelopes, with constants free of the orders of the cyclic lifts. This estimate concerns the original function and its translates. It does **not** imply the same for $C_v = \prod_{j<h} U^{jv} 1_A$ chosen after observing the envelope J. The six-slot example has exactly the capacity for such a product to depend on an active transverse high character.

R074. Recurrence with an ergodic finite-rank passive subgroup

Scope and status. Result with proof.

Result. An ergodic finite-rank passive subgroup gives full recurrence for the replicated dyadic family.

Scope. An infinite-rank ergodic passive group need not contain such a subgroup.

Use the sufficiently replicated free-coordinate dyadic family [R087]. Suppose some finitely generated subgroup P of the head coordinates acts ergodically, and adding any $p \in P$ to any legal tail step is still legal. The latter holds, for example, when all coordinates of P are multiples of their normal periods 2^{K_i}. Then every positive event in this action has every finite length of recurrence on legal directions. No individual generator of P is required to act ergodically. The hypothesis about the finite-rank passive action is substantial.

Commuting automorphisms and nilorbit words First let S be a totally ergodic invertible transformation commuting with the action U of the free coordinate axes, on a standard probability space. For a positive event B in a finite-order Host–Kra factor $Z_s(S)$, the uncorrected legal tail steps have an h-return to B (for any given h). The action on no individual finite-stage nilfactor is presumed. Here is the argument.

Take a finite-stage nilsystem factor $\pi : X \to Y$ of S. Since S is totally ergodic, the finite-stage nilsystem is totally ergodic and its space connected. If necessary use the connected nilflow suspension [R039] to view Y as a section in a connected simply connected filtered nilmanifold $\widetilde{Y}$, with its transformation given by the restriction of a translation there. Its measure-theoretic Host–Kra cubes are supported on the closed cubes of $\widetilde{Y}$: the dynamical parallelepipeds of the nilrotation lie in its ordinary Host–Kra cubes, and the Host–Kra cube measure is supported on their closure.

For almost every x, the map $w \mapsto \pi(U^w x)$ from any finite-rank free coordinate group into $\widetilde{Y}$ takes every ordinary additive cube to a nilmanifold cube. By Host–Kra, Lemma 5.5 and Corollary 5.4, every S-commuting automorphism preserves the cube measures when applied to any codimension-one face. Fix finitely many additive cube increments. If their based orbit cube fell outside the closed target cube set with positive probability, separability would supply open coordinate sets O_ω whose product is disjoint from the cube set and a positive event E on which every corresponding translated π-value lies in its O_ω. The product event E^{2^l} has positive l-cube measure, by the uniformity seminorm's lower bound by the mean. Apply each required increment to its face; projection of the preserved measure onto the factor still has to be supported on target cubes, whereas on that positive product event all its outputs belong to the impossible open product. The initial cube translate is absorbed by invariance of the starting measure. Countability gives the simultaneous conclusion for almost every x.

A cube morphism $w : \mathbb{Z}^n \to N/\Gamma$ into a connected rationally filtered nilmanifold lifts to an ordinary filtered polynomial in N. For completeness quotient by the top central group N_s and induct on the filtration degree. Lift the projected morphism by induction, then lift its polynomial logarithm through a filtration-respecting linear section of Lie algebras. This gives a polynomial orbit with the desired projection into the quotient by N_s. Its difference from the initial morphism is a function into the central torus $A = N_s/(N_s \cap \Gamma)$. On each

cube the two maps are cubes with the same projection, so their pointwise fiber differences form a cube in the degree-s abelian nilspace $\mathcal{D}_s(A)$, by the top structure group axiom for the filtered nilmanifold. Thus all its $(s+1)$-fold differences vanish. Its multivariate Newton coefficients can be lifted to the Lie algebra of A; multiplying the first lift by this central polynomial completes the induction. The step-one case is the same elementary torus lifting. See Candela, *Notes on nilspaces: algebraic aspects* for the structure group axiom.

Choose a countable set of typical x_j with $\pi(x_j)$ dense in Y, insisting also that $(\pi(U^{\ell d}x_j))_{\ell<h}$ belong to the support of their pushforward measure for every d under test; these are only countably many full-measure conditions. The nil-word theorem [R087] applied jointly to finitely many orbit maps and finitely many multiples produces a sequence of legal tail directions whose graphs approach the diagonal at increasingly fine dense samples. The graph probability λ_d, obtained by pushing forward the starting point to $(\pi(U^{\ell d}x))_{\ell<h}$, is an ergodic nilsystem joining since it is an S-factor in Y^h. Leibman's linear orbit theorem gives Haar measure on a subnilmanifold; this subnilmanifold is connected because the joining is totally ergodic (components otherwise form a finite cyclic factor).

By the connected homogeneous-measure compactness lemma of measurable-nil-returns.md [D10], every subsequential limit along those directions is homogeneous Haar on a connected subnilmanifold with all marginals Haar and with the full diagonal in its support. The same note's homogeneous joining lemma implies that for a measurable finite-stage event C of mass at least a, the limits charge C^h at least a fixed $c(a,h)>0$, independent of the finite-stage target. Its ambient connected homogeneous manifold may be taken to be Y itself: after the connected suspension its connected section is a subnilmanifold and the supports are inside its h-fold product. Passing from continuous tensor functions to indicators uses the fixed Haar marginals. Approximate any positive event B in $Z_s(S)$ by a finite-stage event with error below $c(\mu(B)/2,h)/(2h)$ and mass at least $\mu(B)/2$. The telescoping error bound $h\mu(B\triangle C)$ is independent of the directions. This proves the stated factor recurrence.

Passing back through a finite real suspension Choose an integer basis for P of rank $r\ge 1$. On $X\times[0,1)^r$, suspend its action to $\mathbb{R}^r$ by adding a real vector to the clock modulo one and applying the corresponding vector of integer carries to the old system. This $\mathbb{R}^r$-action is ergodic if P is ergodic: an invariant function is constant in the clock by the local translations and invariant under every old passive generator. Almost every $w\in\mathbb{R}^r$ has a totally ergodic time transformation $S=R^w$. Indeed in the spectral representation of the $\mathbb{R}^r$-action on the zero-mean space, the projection to invariants of R^{mw} is supported where $m\langle w,\xi\rangle$ is integral; every $\xi\neq 0$ imposes a Lebesgue-null set of w's. Apply Fubini and a countable dense list of test functions and nonzero integer m. The old group actions acting just on X commute with this suspension.

Mark $A'=A\times J^r$, where J is an open interval of length strictly less than $1/(4h)$ away from the clock seams, e.g. centered at $1/2$. Write $g=\mathbb{E}(1_{A'}\mid Z_{h-1}(S))$. The preceding result gives a legal tail v and $b>0$ for which $C=\bigcap_{\ell<h}U^{-\ell v}\{g>b\}$ has positive mass. The ordinary S-averaged g-correlations of $U^{\ell v}S^{\ell n}$ are bounded below by b^h times the ordinary S-progression average of 1_C, with positive liminf. The Host–Kra characteristic factor for the consecutive integer slopes, its invariance under commuting maps, and invariance to center the unshifted input if needed show the corresponding averaged $1_{A'}$-correlation is positive too. Hence for some integer n a positive set of x,θ has all the required slots marked.

For such returns, nw is within $|J|$ of an integer vector p because the first two clock slots are

both in J^r. On all h slots the residual displacement is less than $1/4$ and the initial phase is near $1/2$. The passive carry at slot ℓ is therefore exactly ℓp. The original X-coordinates form an arithmetic progression with single difference $v + p$ (using the chosen basis identification); it is legal by the head assumption. If w has no ties, p is determined by n; alternatively countability suffices to fix it on a positive subset. This completes the finite-rank passive result. A merely ergodic action of the entire infinite-rank passive group need not contain an ergodic finite-rank subaction, and passing to the finite-rank ergodic components alone would allow the low directions to permute those components. No such extension is asserted here.

R075. Invisible parity flips on components with finite binary spectral rank

Scope and status. Result with proof.

Result. Finite binary spectral rank on compact high components supplies invisible parity flips and full recurrence.

Scope. Finite detection is assumed; it is not established for all infinite-rank nilspace fibers or extensions.

Use the free group $G = \bigoplus_i \mathbb{Z}e_i$ or any of its indicated covers and put $H = \langle z_i e_i : i \in I\rangle$ on one infinite homogeneous row. The directions may use either the old nonzero marked center or the newer pure-zero marked center. In either case an arbitrary $g \in H$ can be added to an uncharged difference with the same low types, and the parity of its tag is the value of a binary character χ of H. The ordinary DHJ trial uses nonempty unions J of some arbitrarily fresh trial axes. In the old case the possible characters are χ_J, with each chosen marked center on the axes of J. In the pure case use normal center p on those axes and zero elsewhere, so the characters are $\chi_{\mathrm{all}}\chi_J$. The statements require these characters to be well-defined and the distinct singleton bits linearly independent; they hold on the free host and the usual coordinate cyclic covers. All arguments may be run on any tail where the hypotheses hold.

Finite-rank conditional spectra. Suppose on almost every ergodic component for the H-action the action has discrete spectrum, and the eigencharacter subgroup $\Lambda_\omega \subset \widehat{H}$ has only finitely many characters of order dividing two. No smooth variation of the full spectral subgroup across the components, nor global discrete spectrum for H, is assumed. Every positive-measure event has every finite length of recurrence along the legal directions.

It is enough to work on a G-ergodic component. Only countably many binary characters are needed, namely the span W of all the singleton bits and (where needed) χ_{all}. For each such fixed χ, whether it occurs as an H-eigencharacter on ω is a measurable property of the component. Indeed apply the χ-twisted Følner averages to the members of a countable measure-determining bounded continuous family on a compact model; on every ergodic component their norm limits calculate its spectral projection. Existence of a nonzero projection on some member detects occurrence. Since every G-map commutes with H, this property is G-invariant and therefore constant a.e. Consequently $\Lambda_\omega \cap W = V$ a.e. for a **single** subgroup V. Choosing a typical component shows V is finite. It is permissible for the other, nontorsion spectra of distinct components to be very different.

Choose m fresh singleton bits independent modulo $V + \langle\chi_{\mathrm{all}}\rangle$, with m the DHJ dimension at the desired density and horizon. The standard sampling of a low word gives a nonempty subset J and a baseline direction v for which $C = \bigcap_{j<h} U^{-jv}A$ has positive measure. Its given character χ is outside V in either version. On almost every H-component with $\mu_\omega(C) > 0$, for any finite list of its eigencharacters the closure of the joint character map on the $\chi = -1$ coset contains the identity in all other coordinates. Otherwise the graph would separate χ as a continuous character of the compact image of the others, making it their integer combination. This contradicts $\chi \notin \Lambda_\omega$. Approximate 1_C in that component's discrete spectrum by finitely many eigenfunctions. Pick g in the odd coset so its first h powers move that approximant arbitrarily little in $L^2(\mu_\omega)$. Telescoping gives a positive h-return for C there. The correcting g may depend on ω; by countability some genuine g yields a positive global return. Membership in C at the successive jg entails membership in A at

all $j(v+g)$. This works in particular for arbitrary finite-dimensional compact abelian group rotation fibers, and more generally for relative component spectra of any rank with finite 2-torsion.

Individually torsion high actions. Here is a stronger version on higher finite covers when the high translations have finite yet unbounded individual orders. Suppose on almost every component of an invariant subgroup containing H (for instance P) the H-action extends to translations on a connected finite-rank compact nilspace with its Haar law. Assume every element acts as a finite-order transformation there. No common bound on orders, nor common nilspace coordinates across the fibers, is required. Then every positive event again has all lengths of recurrence.

For each $g \in H$, acting as the identity modulo null sets on a component is measurable and invariant under G, by the same commutation. In a G-ergodic system the fiber kernel is consequently a single group L almost everywhere. Fix one typical fiber Y of the specified type. Its Haar law has full support, so a continuous translation acting as the identity a.e. is the identity everywhere. The abelian image H/L is torsion, and $(H/L)[2]$ is finite. For the last assertion, as in the connected nilspace argument [R026], the finite-order translations commute with the connected transitive group of first translations: finite-order conjugation on its Lie algebra is unipotent, hence trivial. They are determined by their value at a fixed base point. An order-two translation gives there a two-periodic morphism from $\mathcal{D}_1(\mathbb{Z})$ with that fixed starting point. A connected finite-rank nilspace Y admits only finitely many possible second sites of such morphisms. At structure layer s, two lifts of the same projected sequence with the same start differ by a degree-s structure-group polynomial alternating between 0 and b; its $(s+1)$-difference implies $2^s b = 0$. Each compact abelian Lie structure group has finite 2^s-torsion. Induct over the structure layers, starting with the one-point factor. The argument bounds the number of values independently of the orders of all other group elements.

For any torsion abelian group D with $|D[2]| = 2^r < \infty$, one has $\dim_{\mathbb{F}_2} D/2D \le r$. Indeed any finite list independent modulo $2D$ generates a finite subgroup F; its quotient by $2F$ has that many independent generators, and $|F/2F| = |F[2]| \le 2^r$. Thus the subgroup of binary characters of H vanishing on L is finite. Choose the trial singleton characters and successful baseline as before with its resulting character outside this finite subgroup. Now an actual $g \in L$ changes the tag to odd and fixes all the original fibers. The baseline return is preserved exactly. The same argument applies whenever the high image on one typical component is torsion abelian with finite 2-rank, regardless of nilspace models.

Strong operator detectability. A more general mechanism allows completely nondiscrete spectra and high elements of infinite order. Disintegrate over $\mathcal{I}(P)$, or any G-invariant factor fixed by H, and give the image of H on each conditional L^2-space the strong operator topology of its unitaries. Say that a binary character is detected when the identity is not in the strong closure of the images of its odd coset. If only finitely many binary characters on each typical fiber are detected, every positive event recurs at every length. In fact one may restrict the finiteness to the test group W from the first proof.

For a fixed character, detection is measurable: use countably many bounded continuous functions dense in the conditional L^2-spaces, and finite neighborhoods expressed by the integrals of their squared displacement. Commutation with G makes detection invariant. In a G-ergodic system the detectable members of the countable W are consequently a single finite set a.e. Choose trial bits independent modulo their span and $\langle \chi_{\mathrm{all}} \rangle$. For the successful

baseline its χ is undetected. On every pertinent fiber an odd correction can approximate the identity on 1_C in L^2 as closely as desired. Its first h powers then do the same by telescoping unitaries; hence C returns, and countability gives the result globally. Neither convergence between fibers nor global strong rigidity along that coset was needed.

One sufficient hypothesis is a continuous action on each fiber by a Lie group containing the image of H, with the closure J of this abelian image compactly generated as an abelian Lie group. Each detectable character must also be detectable in the Lie topology by continuity of the action into the strong operator topology. The closure of the image of its kernel is then a proper closed subgroup of J of index two, so the character extends continuously to J. There are only finitely many such characters on a compactly generated abelian Lie group, as follows from its decomposition into real vector, finitely generated discrete, and compact abelian Lie factors. In particular this holds when the H-action lifts to a homomorphism into a connected simply connected nilpotent Lie group acting continuously on the fiber, including genuine niltranslations of this form with their Haar measure. The logarithms of the commuting image elements commute in the Lie algebra (conjugation is unipotent and $\exp(\operatorname{ad} X)Y = Y$ implies $[X, Y] = 0$). Their logarithmic span is an abelian vector subspace and its closed subgroup given by the image closure is isomorphic to $\mathbb{R}^a \times \mathbb{Z}^b$. No ergodicity on the fiber is required. This also covers any compact Lie group action: its image closure is a compact abelian Lie group. The method does not assert finite binary detection on infinite-rank nilspace fibers or general measurable stationary extensions.

R076. A finite-period high derivative gives an all-length upper bound

Scope and status. Result with proof.

Result. A finite-period high derivative yields an all-length upper bound $1 - 1/(1 + L/2)$ in the specified action class.

Scope. The fixed derivative alphabet, lower-axis invariance, and period are required.

Here is a version of the parameter-word programming argument [R091] for wider native modules, without its nilpotence identity on the high derivative. Use cofinally many axes with the same fixed k, K, strictly increasing depths D_i, and put $L = 2^K, c = 1 + 2^k$. Suppose a countable family of generating circle observables has the following property on all sufficiently late such axes. Their multiplicative derivatives under the high involutions $z_i e_i = 2^{D_i} e_i$ take values in a fixed finite alphabet (binary signs suffice), are invariant under all lower-depth axes, and have period dividing L under their own axis. The full action is abelian and preserves the measure. No bound on interaction arity or on algebraic nilpotence of the high table is needed. Also no invariance under *other* high moves is needed in this version: every possible correction toggles just one actual axis.

In this category the threshold of the dyadic two-center family, at every length, is at most

$$1 - \frac{1}{\{} 1 + L/2\}.$$

Indeed use the Graham–Rothschild construction for the finite list of generating observables needed to approximate the event. In each of the m trial blocks use one parameter with fixed initial exponent zero and low motion c. In the terminal reference group use one parameter of initial exponent r and zero low motion for each *odd* residue r mod L. Color one-parameter words by the finite lists of high-derivative values at the last occurrence of their variable, at all own exponents modulo L, with the fixed letters of the word acting at their specified exponents. As in the linked proof the other occurrences of this variable are earlier axes and hence invisible, and all substitutions of other parameters give the same table on the homogeneity event. Denote the resulting table by $F(r)$, a vector of phases. All reference pivots have genuine fixed high moves w_r, whose effects are always $F(r)$ on the word vertices.

For a combinatorial line of wildcard trial blocks, designate its first wildcard pivot. All occurrences of all wildcard parameters move by the allowed uncorrected active speed c; toggle the high bit only on the designated pivot, so the true tag is odd. At odd slot j this has the same effect on every tested observation as applying the reference $w_{cj \bmod L}$ to the ordinary word vertex; at even slots its effect is zero. Here c is odd. Each corrected slot therefore agrees on the tests with one of just $1 + L/2$ genuine translations of that ordinary vertex.

Accept an ordinary vertex if it and all the $L/2$ reference translates belong to A. If $a = \mu(A)$ exceeds the stated bound, $1 - (1 + L/2)(1 - a) > 0$. The head-conditioning argument applies to all these translates simultaneously: the homogeneous-class probability lower bound is fixed before choosing the ergodic head, and all qualifying tables are invariant under that head. The positive conditional expected word density gives a line by finite density Hales–Jewett. As before, the approximation of A by finite observables is chosen with error small for all ordinary and corrected positions on the finite family of lines, prior to choosing the Ramsey pool. This yields a true positive return for A. Ergodic decomposition handles arbitrary

invariant laws. The proof and the numerical threshold are uniform in h for each fixed L; the bound tends to one as L tends to infinity.

In particular it covers native wider square-zero cyclotomic rings on a cofinal fixed-K family, whenever the high layer has exponent two, is invariant under all earlier axes, and U_i^L fixes it. For $u_i = \zeta_L(1+\epsilon_i)$, $\epsilon_i^2 = 0$, one has $u_i^{z_i} - 1 = z_i\epsilon_i$ and $(u_i^L - 1)(u_i^{z_i} - 1) = 0$. Native truncation of additive exponent at most 2^{D_i+1} guarantees its order-two claim; the earlier-depth exponent bound gives lower-axis invariance on shared modules just as in the original proof. This covers all such truncated layers at that fixed period, not arbitrary actions on the native group.

R077. Five-slot ordered recurrence on true integer lifts

Scope and status. Result with proof.

Result. An exact ordered query identity yields unrestricted five-slot recurrence on sufficiently deep replicated true integer dyadic lifts.

Scope. The proof concerns five slots and provides no uniform estimate for all longer lengths.

Qualifications. External finite-game ordered-contradiction input should be cited with its exact version and hypotheses.

Fix k, K as in the two-center dyadic construction [R107], and put $c = 1 + 2^k$. For all sufficiently large D depending on k, K, the five-player integer game has an ordered contradiction. More strongly, a query of velocity $c + 2^D t$ with **odd** t is a word in queries of normal velocities $r \equiv 0, 1 \bmod 2^K$ and queries at the single uncharged velocity c. This is an actual free-group word equality. Hence high enough identical powers force five-term progressions at every positive density, uniformly over arbitrary finite period lifts and all measure-preserving actions of the free abelian coordinate group. In conjunction with the six-slot low-density obstruction, the first missing length can genuinely be six in this integer construction. The argument does not give an avoiding-density bound uniform in longer progression lengths.

The isolated factors Write $h = 5$, $L = \mathrm{lcm}(1, \ldots, h-1) = 12$, and $M = L \cdot 2^K$. Let F_j be the free group on integer symbols $u_{j,y}$. In their direct product let H_0 be generated by $e(x, r) = (u_{j,x+jr})_j$ for all normal velocities and for $r = c$, with arbitrary integer starts. The group H_0 does not depend on D. For each integer y let $[y]$ be its representative modulo L in $[0, L)$, and in factor j set $k_{j,y} = u_{j,y} u_{j,[y]}^{-1}$. Let K_j be the subgroup generated by these normalized letters and J_j the subgroup of elements of K_j which, supported just at j, lie in H_0. The subgroup J_j is normal in K_j, because H_0 projects onto the whole F_j.

For each other player i, every normalized letter at j has a lift to H_0 with trivial projection at i. Indeed the two-player support graph contains the edges for velocities 0 and 1, and alternating between them connects any two j-symbols differing by a multiple of $j - i$. A path gives the corresponding difference of free letters, and L is divisible by every $|j - i|$. Thus we may independently lift the $h - 1$ inputs to a commutator in K_j, killing each of its other coordinates with a different input. All iterated commutators of length $h - 1$ and their conjugates are isolated in H_0. Consequently $G_j = K_j / J_j$ is nilpotent of class at most $h - 2$.

In addition, for every integer multiple w of M, the sequence of normalized symbols at j satisfies the exact identity $\Delta_w^{h-1} k_{j,y} = 1$ in G_j, where $\Delta_w p(y) = p(y + w) p(y)^{-1}$. Start with the tuple $e(x, r)$ for a normal $r \equiv 0 \bmod 2^K$. For each $i \neq j$ take its multiplicative difference under $(x, r) \mapsto (x - i t_i, r + t_i)$, with $t_i = w/(j - i)$ a multiple of 2^K. These differences are taken successively in any chosen order. At slot i each accumulated expression is a function only of $x + ir$, so the ith operation makes that slot identically trivial, retaining previous cancellations. At j the successive differences are exactly $\Delta_w^{h-1} u_{j,y}$. Since w is divisible by L, the first difference removes the fixed reference letter on the right, so this also equals $\Delta_w^{h-1} k_{j,y}$. It is an isolated element of H_0.

Each G_j is finitely generated. On each residue modulo M the recurrence just obtained generates the whole sequence from finitely many initial symbols and their initial successive differences. This follows inductively using $p(n + 1) = (\Delta p)(n) p(n)$ in both time directions. We need finitely many residues and finitely many players.

Set $\Gamma = \prod G_j$. The product quotient maps $H_0 \cap \prod K_j$ onto a subgroup $B \leq \Gamma$. The kernel of the product quotient on $\prod K_j$ is contained in H_0 by isolation of each J_j. Therefore a normalized difference tuple in $\prod K_j$ lies in H_0 whenever its image lies in B.

Polynomial paths and their right logarithmic derivatives Quotient the finite torsion subgroup of the finitely generated nilpotent Γ and embed the torsion-free quotient as a lattice in its simply connected real Malcev group $G = \prod G_j^R$. Let `B*` be the connected rational Malcev closure of B in G, and let $T \leq \Gamma$ be its inverse image. Then B has finite index in T. These are the standard finitely generated nilpotent lattice facts: the image of B is a lattice in its connected rational closure, and the ambient lattice intersected with that closure is discrete, so the former has finite index in the latter. The finite torsion kernel only adds a finite index. See Segal, Polycyclic Groups, Chapter 6.

On each residue modulo M, every image $k_j(y)$ in G_j^R has polynomial matrix entries in y in a unipotent triangular realization. One can use Leibman's polynomial sequences or argue directly. If $\Delta^s p = 1$, induction first makes the entries of $d(n) = p(n+1)p(n)^{-1}$ polynomial. In a unipotent upper triangular matrix the difference of p_{ab} is $d_{ab} + \sum_{a<i<b} d_{ai} p_{ib}$; induction on the diagonal gap makes this an ordinary polynomial, hence p_{ab} is polynomial. The polynomial curves at real arguments remain in the connected unipotent group: in matrix logarithms membership in its Lie algebra is given by linear equations, which hold on every integer of the residue class. We write $g_{j,s}(Y)$ for the real polynomial path on the residue s mod M, using the physical symbol variable Y.

For fixed residue classes $x = a, r = b \bmod M$ put

$$P(X, R) = (g_{j,a+jb}(X + jR))_j.$$

If b is normal, $P(X, R)$ belongs to the single right coset $B * P(a, b)$ for all real X,R. Indeed, at integers $x \equiv a, r \equiv b \bmod M$ the coordinates of $e(x, r)e(a, b)^{-1}$ are the products of the corresponding normalized letters and their fixed reference inverses, so its image is $P(x, r)P(a, b)^{-1} \in B*$. The coset identity extends from the full integer grid by polynomiality in X,R and the linear logarithm equations defining `B*`. For the uncharged velocity $r = c$ we get the same statement on the entire real X-line at $R = c$, one residue class a at a time. In particular the reference can depend on the whole residue pattern, yet **right** logarithmic derivatives along a right translate always lie in the same tangent Lie algebra $b* = Lie(B*)$.

Let λ on `Lie(G)` be any real linear functional annihilating `b*`. For each j and residue s, take the scalar polynomial

$$f_{j,s}(Y) = \lambda_j(g'_{j,s}(Y) g_{j,s}(Y)^{-1}).$$

For $b = 0, 1$ the two tangent equations give identities in independent real variables X,R,

$$\sum_j f_{j,a+jb}(X + jR) = 0, \sum_j j f_{j,a+jb}(X + jR) = 0.$$

For $b = c$ the first identity holds at $R = c$ for all X. Notice that the functions f are the same ones for all the residue patterns. Using the logarithm of PP_{ref}^{-1} without taking its right logarithmic derivative would generally lose this property.

Fourier interpolation at the five slots Write the finite Fourier and ordinary polynomial expansion

$$f_{j,s}(Y) = \sum_{\omega^M=1} \omega^s \sum_{d\geq 0} A_{j,d,\omega} Y^d,$$

$$L_{d,\omega}(U) = \sum_{j=0}^{4} A_{j,d,\omega} U^j.$$

Take Fourier coefficients in a in the two normal identities, and compare $X^{d-p}R^p$ for $0 \leq p \leq d$. The resulting moment equations say that each $L_{d,\omega}$ vanishes with multiplicity at least $d+2$ at both 1 and ω. Indeed the moments with powers j^p come from the X tangent and with powers j^{p+1} from the R tangent. For $\omega = 1$ this already kills every needed moment through $d+1$ at the active residue as well. For $\omega \neq 1$, degree ≤ 4 gives $L_{d,\omega} = 0$ for every $d \geq 1$. For $d = 0$, the X tangent at $R = c$ now says $L_{0,\omega}(\omega^c) = 0$. If this is a third root, the two double roots and the third root exceed degree four. If it equals one of the double roots, its derivative is already zero. In all cases the active R tangent vanishes after applying λ:

$$\sum_j j f_{j,a+jc}(X + jR) = 0 \quad \text{for every} \quad realX, R.$$

Since all annihilating λ give zero, on the active-residue surface the R tangent vector $\partial_R P \cdot P^{-1}$ belongs to `b*` everywhere. Holding X fixed, the path consequently stays in its initial right translate $B * P(X, c)$ (equivalently project the path to the left quotient $B * \backslash G$ and its derivative vanishes). In particular, at $x = 0$ and each integer n,

$$d(n) = (k_{j,j(c+Mn)} k_{j,jc}^{-1})_j \in T.$$

This conclusion is in the original finitely generated nilpotent group, via inverse image from its torsion-free quotient.

Removing the finite index without a cover-depth dependence Act by T on the finite set of cosets of B and quotient by the kernel of that permutation action. In that finite quotient $d(n)$ is periodic. Indeed each j-coordinate of d, being the normalized letter along an arithmetic progression of stride `jM` with a fixed right reference, has $(h-1)$st unit derivative identically one. The same holds in T and in its finite quotient. In any finite group such a sequence is periodic: its state consisting of its value and finitely many successive derivative values evolves deterministically and invertibly by $p(n+1) = (\Delta p)(n)p(n)$, starting the inversion at the highest derivative, which is constant. Let P be a positive period. It depends only on the normal queries and on c, since H_0, Γ, B, and the trajectory used to choose P are independent of D.

For $D \geq v_2(MP)$, there exists an odd positive t with $MP | 2^D t$. Set $n = 2^D t/M$. Periodicity gives $d(n) \in B$ since $d(0) = 1$. By the product-kernel observation above, the actual difference of the query tuples $e(0, c + 2^D t)e(0, c)^{-1}$ belongs to H_0. Thus the charged tuple

equals a product of uncharged tuples and their inverses. Only finitely many queries occur in this word. Projecting the identity to fewer players gives the corresponding shorter-length assertion at the same depth.

Finally apply the Braverman–Minzer ordered contradiction estimate with the Hales–Jewett subblock argument [R218] to a finite law assigning positive mass to every query in this word. This gives five-slot recurrence at sufficiently many copies for each fixed density, with the number of copies depending on the level, but independent of the ambient period or action. The proof used actual integer queries throughout. Finite rank-one transfers must retain the full finite signed packet whose existence it proves.

R078. An ordered four-slot contradiction on the true dyadic integer lifts

Scope and status. Result with proof.

Result. A four-slot ordered contradiction is established on true dyadic integer lifts with explicit depth requirements.

Scope. Its fixed-order conclusion is strengthened by R077, but the detailed class-two calculation remains a separate result.

Let $k \geq 2$, $K \geq k+3$, $c = 1+2^k$, $z = 2^D$, and assume $D \geq 9K+20$. The allowed integer velocities include every $r \equiv 0$ or 1 modulo 2^K (normal, tag zero) and every $r \equiv c$ modulo 2^D (active, tag the upper bit $(r-c)/z$ modulo 2). On the four-player AP game observing $x+jr$, j = 0,1,2,3, these tags have an ordered contradiction. This is stronger than integer scalar nonembedding [R103]. At a fixed level it gives four-term positive-density returns on sufficiently many copies, uniformly over period lifts. No copy count uniform over the levels is obtained.

Class-two reduction in the correct pair components Let F_i be free on letters $u_{i,y}$, $y \in Z$, and $H \leq \prod F_i$ generated by all legal query tuples $e_{x,r}$. If there is no ordered contradiction, the assignment $e_{x,r} \mapsto \tau(r)/2$ gives a homomorphism $\chi{:}H \to T$. Put K_i equal to the free subgroup on $k_{i,y} = u_{i,3y}u_{i,0}^{-1}$ for $y \neq 0$, and put $k_{i,0} = 1$. Every tuple $g_{x,r} = e_{3x,3r}e_{0,0}^{-1}$ with 3r legal has these normalized generators as its coordinates. The freeness follows, for example, by the Nielsen change of basis fixing $u_{i,0}$ and replacing $u_{i,3y}$ by its product with $u_{i,0}^{-1}$.

Each isolated lower central subgroup $\gamma_3(K_i)$ belongs to $[H,H]$. To check pair components, at any two players i,j the connectivity classes on the i side of their query support graph are the residues $\bmod |j-i|$, because velocities include 0 and 1 and all shifts of the starting point are allowed. An alternating path supplies any difference of two i-letters in the same class with identity at j. For $|j-i| = 1$ or 3 these lifts and full surjectivity onto F_i give lifts of every element of K_i (work with the original integer game, including intermediate symbols). For the unique j with $|j-i| = 2$ they give lifts of the subgroup $A_i = \ker(K_i \to Z)$ whose normalized even letters go to zero and odd ones go to one. Indeed A_i is normally generated by differences of normalized letters with equal parity, including each normalized even letter individually; conjugations in K_i can be lifted using surjectivity, preserving the zero at j.

Modulo $[A_i, K_i]$, the image of A_i is central with cyclic quotient, so that quotient group is abelian. Thus $[A_i, K_i] = [K_i, K_i]$. It suffices for $\gamma_3(K_i)$ to isolate iterated commutators $[[a,b],d]$ with $a \in A_i$ and b,$d \in K_i$ and their conjugates. Lift these three inputs to H so that each kills one different player other than i, assigning the distance-two player to a. The resulting triple commutator is isolated at i and is in $[H,H]$. Products and normal generation prove the assertion. Hence χ induces a character on the image of the normalized query subgroup in $\prod N_i$ with $N_i = K_i/\gamma_3(K_i)$.

Central phases and finite-dimensional symbol vectors Write $L_i = K_i^{ab}$, free on the nonzero normalized letters. The commutator center of N_i is naturally $\Lambda^2 L_i$. On the intersection of the normalized image with $C = \oplus_i \Lambda^2 L_i$ extend its character to all of C

using divisibility of the circle. This gives alternating circle pairings on each player alphabet, written $B_i(y, y')$, with $B_i(0, y') = 0$. Taking the two query commutators gives

$$\sum_i B_i(x + ir, x' + ir') = 0 \bmod 1$$

for all normalized integer x,x' with 3r,3r' allowed. Set $r' = 0$ and fix $x' = v$. On the normal class $3r \equiv 0 \bmod 2^K$, apply three differences which keep all but the designated ith slot, one by one, fixed under their respective differences. More explicitly, for each $j \neq i$ change (x, r) by $-jt_j, t_j$ with $t_j = 6 \cdot 2^K/(i - j)$; these are signed multiples of 2^K. Each such difference kills the term at j, and in the ith slot each shifts by $M = 6 \cdot 2^K$. Thus $\Delta^3_M B_i(\cdot,\text{v})=0$. Skew symmetry gives the corresponding assertion separately in its second variable.

Put $m = 2^{K+1}$, so $M = 3m$, and further restrict normalized x,r to 3Z (the original queries are now multiples of nine). For a normalized symbol $y = 3b + Mn$ with $0 \leq b < m$ define

$$E(y) = e_{b,0} + ne_{b,1} + \binom{n}{2} e_{b,2} - e_{0,0}.$$

In particular $E(0) = 0$. The pairings $B_i(y, y')$ factor through a biadditive circle form on the finite-rank free lattice generated by the $E(y)$ at each player: any relation among the $E(y)$ annihilates $B_i(\cdot,y')$ by the degree-two polynomial formula on each residue and the value zero at symbol 0, and the same holds in the other variable. The descended form is alternating. Lift each form on a free basis to a real alternating bilinear form b_i, and let b be their direct sum on the direct sum of the finite-rank lattices.

Let Λ denote the *original* exponent incidence lattice in $\oplus L_i$ spanned by these further restricted query tuples, and L its image in the finite-rank lattice, generated by

$$V(x, r) = (E(x + jr))_{0 \leq j \leq 3}, x, r \in 3Z, 3rlegal.$$

The restriction of b to $L \times L$ is integer-valued by the query commutator equation. We claim there are scalar circle functions f_j on 3Z and a function q on $L/2L$ such that

$$\tau(3r)/2 = \sum_j f_j(x + jr) + q(V(x, r) \bmod 2L)/2. (1)$$

Here q is a quadratic refinement over F_2 of the integral alternating form b reduced modulo two. To prove the claim, form the class-two group on the finite-rank lattice times T with product $(v, t)(w, u) = (v+w, t+u+b(v, w)/2)$. Send each normalized player letter to $(E(y), 0)$ in its own summand. Different summands commute, and this extends to a homomorphism from the product of the restricted free class-two groups (only symbols with $y \in 3Z$ are needed). On the commutator centers its central coordinate equals the extended character by construction of b. On the restricted query subgroup call its central coordinate $c(h)$, with exponent data $A(h) \in \Lambda$ and $V(h) \in L$. Then $\chi(h) - c(h)$ depends only on $A(h)$: two elements with the same A differ by an element of the product of the commutator centers,

on which χ agrees with c. Its additive defect at A,A' is $-b(V,V')/2 \bmod 1$, a half-integer. Any alternating form on the F_2-vector space $L/2L$ has a quadratic refinement q with that polarization (use the off-diagonal upper triangle in a basis). Subtracting $q(V)/2$ kills the defect, giving a homomorphism on Λ. Extend it to the original free exponent lattice using divisibility; its values on the individual letters give f_j. Every query generator has $c(g) = 0$, so (1) follows. Keeping the original exponent lattice at this step avoids assuming χ descends through the smaller polynomial symbol lattice.

The period of the residual sign We prove that $q(V(x,r))$ on normal queries is periodic in both normalized x and r with period $T_0 = M \cdot 4m^2$. More generally whenever two admissible pairs differ by a multiple of this period in one variable, their V difference lies in 2L, even if intervening velocities would not be admissible. This latter point is needed for the active step comparison.

Test an arbitrary character of $L/2L$, with values in $\{0, 1/2\}$, and extend it to the ambient finite-rank lattices with circle values. Write $g_j(y)$ for its evaluation on $E(y)$ at player j, so

$$g_j(3b + Mn) = A_j(b)\binom{n}{2} + B_j(b)n + C_j(b).$$

For all legal queries, $\sum_j 2g_j(x + jr) = 0$. Take $x = 3b + Mn$, $r = 3u + Mt$ for $u = 0$ or for an odd u with $9u \equiv 1 \bmod 2^K$. Write $b_j = b + ju \bmod m$ and l_j=$b + ju - b_j$/m, so the jth lifted parameter is $n + jt + l_j$. Comparing the coefficients of $\binom{n}{2}$, nt and $\binom{t}{2}$ gives $\sum_j 2A_j(b_j) = \sum_j 2jA_j(b_j) = \sum_j 2j^2A_j(b_j) = 0$. For four circle coefficients satisfying these three equations, doubling identifies them with $\lambda_j = (-1, 3, -3, 1)$ times the last doubled coefficient: subtract the second from the third to obtain twice the third-slot coefficient (index 2) as minus six times the last. Thus at $u = 0$ one has $4A_j(b) = \lambda_j U(b)$. At the other u, comparison of slots 0 and 3 gives $U(b + 3u) = U(b)$, so U is constant as m is dyadic and u odd. Choose $\beta \in T$ with $4M^2\beta = U$. Subtract $\lambda_j\beta\binom{y}{2}$ from g_j; their sum on every AP vanishes and their fiber quadratic coefficients change by $\lambda_j M^2\beta$. The residual A_j now have order dividing four.

Multiplying the normal equations by four and taking their n,t coefficients yields $\Sigma W_j(b_j) = \Sigma jW_j(b_j) = 0$ with $W_j = 4B_j$. Here and in the next section the constant terms of those equations may depend arbitrarily on the low residues. For completeness let S denote translation by u on the b-arrays. The $u = 0$ equations give the formal polynomial identity $\sum_j W_j X^j = (X - 1)^2(a + b'X)$. Evaluation at S and differentiation followed by evaluation at S give $(S - I)^2 v = 0$ and $(S - I)^2 b' = -2(S - I)v$, where $v = a + Sb'$. Hence $(S - I)^3 W_j = 0$ for all j. A degree-two binomial polynomial on the m-cycle has m times its quadratic coefficient zero and m times its linear coefficient equal to minus $\binom{m}{2}$ times the quadratic coefficient. Therefore

$$m^2(W_j(b) - W_j(0)) = 0 \quad \text{for every} \quad b, j.$$

Now shift x or r by $M\omega$ with $4m^2|\omega$. The resulting high-coordinate increments are ω, respectively $j\omega$. The residual quadratic contributions are zero since their coefficients have order dividing four and ω is divisible by eight. In the remaining sum the $B_j(b_j)$ can be

replaced by $B_j(0)$; their unweighted and weighted sums are killed by four by the $u = 0$ equations. Thus the whole change in Σg_j is zero, independently of intermediate velocities. The previously subtracted common polynomial had zero AP sum identically. Every $L/2L$-character consequently kills the V difference for admissible endpoints. Since characters separate $L/2L$, this proves the period assertion.

The scalar contradiction at the enlarged period Rescale $x = 3X$,$r = 3R$, so X,$R \in Z$ and the original velocities are 9R. In these variables the period just proved is $T_0/3 = 4m^3 = 2^{K'}$, with $K' = 3K+5$. On normal velocities equation (1) says the sum of the scalar functions $F_j(X+jR) = f_j(3(X+jR))$ has target periodic in both variables modulo $2^{K'}$. At two active velocities differing by a multiple of $2^{K'}$ the q-sign also agrees by the endpoint version of the period assertion.

Take triple differences eliminating all but one slot as before, now with increments $t_j = 6 \cdot 2^{K'}/(i-j)$. The periodic normal target is unchanged under each shift. Thus $\Delta^3_{M'} F_j = 0$ with $m' = 2^{K'+1}$, $M' = 3m'$. Restrict X,R to multiples of three, write out the same quadratic fiber polynomials, and take the normal choices $R = 3u+M't$ with $u = 0$ or an odd u satisfying $27u \equiv 1 \bmod 2^K$. The normal target is constant in the high variables n,t. Therefore the same three coefficient equations now hold for A_j themselves, giving $2A_j(b) = \lambda_j U$ with U independent of b. Subtract a cancelling common quadratic as above so all residual A_j have order at most two. The linear coefficient calculation gives $\Sigma 2B_j = \Sigma j \cdot 2B_j = 0$ at $u = 0$, and m'^2 kills the variation of each $2B_j$ array over the residues. This is exactly the interpolation in the scalar note [R103], with an arbitrary periodic normal target in place of zero.

Choose R divisible by three with $9R \equiv c \bmod 2^{D+1}$, and let R change to $R + 3z$. Both are active, and the original velocity changes by 27z so the binary tag flips. The q-sign is unchanged because $2^{K'}|3z$. The jth parameter in the M'-fiber changes by $j\omega'$ where $\omega' = 3z/M' = 2^{D-K'-1}$. Since $D \geq 3K'+5$, ω' is divisible by $2m'^2$ and by four. All residual quadratic changes vanish, and all $j\omega' B_j(b_j)$ may be evaluated at $b_j = 0$, where their sum vanishes. The cancelling polynomial itself has zero AP sum. Thus the scalar sum in (1) does not change either, a contradiction.

The ordered contradiction consists of a finite word even though the proof used the infinite integer query group. Choose a finite packet of query probabilities supporting that word. Braverman–Minzer, Section 3 and the finite Hales–Jewett mixture [R218] give the asserted single-charge rectangular cancellation and four-term recurrence in sufficiently many identical copies. For measure-preserving actions, integrate an independent invariant starting point; the integer query's starting symbol may be nonuniform and dependent on the velocity without affecting the unweighted correlations. The packet does not depend on any chosen cover modulus.

R079. High-coordinate spectra in the dyadic marked model

Scope and status. Result with proof.

Result. Compact high-subgroup actions have full recurrence for the dyadic family; the stated weak-mixing conditions on every jK give another positive class.

Scope. Weak mixing of K alone does not imply weak mixing of all its multiples when K has infinite rank.

On $G = \bigoplus_i \mathbb{Z}$, suppose a direction has, on its active set I, coordinates $d_i = a_i + 2^{D_i} t_i$, and is accepted if $\sum_{i \in I} t_i$ is odd. Assume that whole translates of the high subgroup $K = \bigoplus_i 2^{D_i}\mathbb{Z}$ preserve all coordinate types and shift these digits by the assigned high parameters. Assume also that all sums of distinct fresh active base choices $a_i e_i$, with other coordinates zero, are available before the parity test. This applies to the two-normal, one-mark model with normals $0, 1 \bmod 2^{K_i}$, active $a_i = 1 + 2^{k_i} \bmod 2^{D_i}$, $k_i < K_i < D_i$, $k_i \to \infty$. It also applies in the native minimal quotient $\bigoplus C_{2^{D_i+1}}$, where the image of K has exponent two, and to intermediate higher cyclic quotients in which the digit parity is well defined.

Complete recurrence for compact high actions on all these hosts Assume the restriction of the action to K has pure point spectrum, allowing measurable nonergodicity. It is enough that the indicator f of the event under consideration lies in the span closure of the K-eigenfunctions. Then every positive-mass such event has every finite length of recurrence, also arbitrarily far out on the axes.

For a finite index set I, write $\chi_I(\sum_i 2^{D_i} t_i e_i) = (-1)^{\sum_{i \in I} t_i}$. Fix length h and event mass $\alpha > 0$. Finite density Hales–Jewett and its averaging form supply $m = m(h, \alpha)$ and $c = c(h, \alpha) > 0$ such that among the nonempty sums of any m base directions there is a v with $\int \prod_{j<h} U^{jv} f \geq c$. Indeed on random translates of the word cube some fixed word line has acceptance probability bounded below in terms of the finite word count and α.

Approximate f in L^2 to accuracy $\eta \ll c/h$ by a finite linear combination of K-eigenfunctions, with eigencharacters $\lambda_1, \ldots, \lambda_s$. The finitely generated subgroup $\Lambda = \langle \lambda_1, \ldots, \lambda_s \rangle$ has only finitely many elements of order two. Choose the trial axes beyond the union of the supports of those members of Λ which are some finite χ_J. If working entirely on a tail, restrict the approximants and characters to that tail subgroup before doing so. Consequently the nonempty active set I of the chosen v has $\chi_I \notin \Lambda$. The closure of the joint character map $K \to \mathbb{T}^s \times \{1, -1\}$ contains $(0, \ldots, 0, -1)$. Otherwise its projection onto the torus would be injective, hence a compact-group isomorphism onto a closed subgroup, and χ_I would be induced by a continuous character of that subgroup. The character extension theorem for a closed subgroup of a compact abelian group gives an integer combination of the λ_j, contrary to the choice. Thus there is an actual $z \in K$ with $\chi_I(z) = -1$ and all $\lambda_j(z)$ sufficiently near one. On the finite approximant its first h powers are correspondingly close to identity, and $\|U^{jz} f - f\|_2 \leq 3\eta$ for $j < h$. The product telescoping bound on indicators changes the v-correlation by at most $3h\eta$. This gives the legal $v + z$. In particular this needs no order-two identity for the actual high generators on a lift.

The exponent-two native high subgroup For a baseline v and active set I, put $F_v = \prod_{0 \leq j < h,\ 2 \mid j} U^{jv} f$, $G_v = \prod_{0 \leq j < h,\ 2 \nmid j} U^{jv} f$. Its parity-conditioned high average is exactly

$$\langle F_v, P_1 G_v\rangle - \langle F_v, P_{\chi_I} G_v\rangle,$$

where P_λ is the orthogonal projection onto the λ-eigenspace of the entire discrete K-action, and limits can use exhausting finite high subgroups containing the support of I. This follows by character orthogonality and the mean ergodic theorem since all even slots are independent of the high toggle z. If the v-correlation is positive then $C = F_v G_v$ has positive mass, and $\langle F_v, P_1 G_v\rangle \geq \|P_1 C\|_2^2 \geq \mu(C)^2$. Hence only the transverse projection can prevent return.

If the span of the occurring K-eigencharacters meets the space of finite-support χ_I's in infinite codimension, choose m fresh singleton characters linearly independent in that quotient. Every nonempty DHJ support then gives zero transverse projection and positive return. In particular this holds if K acts weakly mixingly.

In fact if K acts ergodically here it is automatically weakly mixing. A K-eigenfunction can be normalized so that $f^2 = 1$, by ergodicity and exponent two. Each full-group element preserves its one-dimensional eigenspace, and therefore acts there by a scalar in $\{1, -1\}$. Every high generator $2^{D_i} e_i$ is itself an even power, so it gives scalar $+1$. The eigenfunction is thus constant. This also gives the tail conclusion, since a finite-index subgroup of a weakly mixing action remains weakly mixing (apply ergodicity of the diagonal square and the finite quotient).

There is a componentwise version of this argument. Let Y denote the K-ergodic component space, and $Y_i \subset Y$ the components preserved by $2^{D_i - 1} e_i$. On Y_i, any occurring χ_I with $i \in I$ would have a normalized $\{\pm 1\}$-valued eigenfunction on that component. Its eigenline is preserved by $2^{D_i - 1} e_i$, which consequently acts on it by a sign. Its square $2^{D_i} e_i$ cannot act there by -1. In particular every such global eigenspace vanishes on the inverse image of Y_i. Its projection contribution for bounded indicator products is at most $\mu(Y_i^c)$ by Cauchy–Schwarz and contraction, since the inverse image is K-invariant. If $\mu(Y_i) \to 1$ along any infinite list, take m such trial indices with $\mu(Y_i^c) < c^2/2$. The resulting transverse term cannot cancel the positive c^2 ordinary term. This proves all positive recurrence on the tail of that list. The case of high component cycles nearly always having full order is left open.

For arbitrary native actions and fixed h, a mass above $1 - 1/(2\lfloor h/2 \rfloor)$ suffices. The set of high components with conditional density above this threshold has positive mass. Restrict A to these components and use DHJ to find a positive baseline run there. On a positive set of components its even and odd slot products both have positive mass and every individual translated slot has conditional mass above the threshold (the full group permutes the high components). The union bound then gives $\mu_y(G_v) > 1/2$. On an ergodic component, averaging Haar over the odd parity coset cannot have zero correlation between a positive F_v and this G_v. Indeed if the component has no χ_I eigenfunction the transverse term vanishes. Otherwise the index-two kernel has two equal-measure ergodic components, interchanged by an odd move; G_v meets both, so its conditional mean on either such component is positive. Integrate on the stated positive base set. This improves the elementary fixed-step bound at odd lengths, but tends to one with h.

At $h = 3$ there is full positive recurrence on the native group without any spectral assumption. Indeed $G_v = U^v f$ has only one slot. By commutation, $\|P_{\chi_I} G_v\|_2 = \|P_{\chi_I} f\|_2$, and by Bessel's inequality this norm is uniformly small for all nonempty I on sufficiently fresh axes. Choose the tail so the transverse term is less than $c^2/2$, before finding the DHJ baseline with ordinary high term at least c^2. The same proof at $h = 2$ works in every higher quotient and in the free lift, since the two-slot parity-conditioned average always uses exactly one

translate by z, even if z has infinite order.

A weak-mixing subcase for the higher covers There is a separate three-slot result on arbitrary higher covers. If $2K$ acts ergodically, every positive event has a three-slot return. More generally this works if on arbitrarily many fresh trial axes the doubled low directions fix the $2K$-ergodic component factor Y pointwise. To see this, choose a DHJ baseline v with $C = \prod_{j=0}^{2} U^{jv} f$ of mass at least $c > 0$. The ordinary K-average of its perturbed correlation is bounded below uniformly by ordinary three-term recurrence on C, since $C, U^z C, U^{2z} C$ imply the corresponding perturbed slots. Uniformity in the mass follows by density Hales–Jewett averaging on this abelian group.

Recenter the signed correlation at its middle slot, putting $g = U^{-v} f,\ h = U^v f$. Apply Hilbert-space van der Corput to $\chi_I(z) U^{-z} g\, U^z h$. At a difference u, the integrated inner product averaged over z is $\chi_I(u)$ times the integral of $(U^{-u} g)\bar{g}$ against the conditional expectation of $(U^u h)\bar{h}$ on Y. Indeed changing starting point reduces the remaining average to one translate by $2z$ and the mean ergodic theorem for $2K$ applies. Averaging over u on Følner sets bounds the squared limsup norm by

$$\|P^R_{\chi_I}(g \otimes h)\|^2, R^u = (U^{-u}, U^u) on X \times_Y X.$$

This is a measure-preserving action there because U^{-u} and U^u agree on Y for $u \in K$. If $2v$ fixes Y, the transformation (U^{-v}, U^v) also preserves this relative product and commutes with R. The displayed norm then equals $\|P^R_{\chi_I}(f \otimes f)\|_2^2$. Its tail supremum tends to zero by Bessel. Fix the tail for the uniform unweighted lower bound before choosing v, and the signed term cannot cancel it. If $2K$ is ergodic, Y is a point and any v meets this condition. The argument does not require that K or $2K$ be weakly mixing.

If every jK for $1 \le j < h$ acts weakly mixingly, then every parity-constrained high coset works at length h for arbitrary bounded nonnegative tests of positive mean. In particular one marked axis suffices. For $K_I = \ker \chi_I$, each jK_I is of index at most two in jK and remains weakly mixing. Averaging $z \in K_I$ after one fixed high correction and low shift uses the progression slopes $0, \dots, h-1$. Their multilinear averages converge to the product of the means (in the integrated sense). For clarity, one standard van der Corput induction removes the last shifted function after translating the product to center it; remaining slopes are differences of the old ones. A multiplicative derivative of the distinguished zero-mean slot at increment w acquires mean $\langle f, U^{jw} f \rangle$; the mean absolute value over $w \in K_I$ vanishes by weak mixing of jK_I. The same property for pair differences supplies the inductions, giving the assertion by expanding into constants and zero-mean functions. Plain weak mixing of K need not imply weak mixing of jK when K has infinite rank. No arbitrary-action bound on either the native or free host is claimed.

R080. The dyadic marked high-digit construction has endpoint one

Scope and status. Result with proof.

Result. The single-normal dyadic marked-high-digit family has torsion-free and integer critical value one.

Scope. Its minimal native quotient can conceal the deeper root obstruction.

Let H be a countable direct sum of integer axes, indexed by i, and let $D_i \geq 2$ be arbitrary depths. At each active coordinate permit $r_i = 1 + 2^{D_i} t_i$, and permit every even speed as a normal coordinate. The finitely supported direction is legal if the sum of t_i over its active (*odd*) coordinates is odd. One may also allow both active low speeds, $r_i = s_i + 2^{D_i} t_i$, $s_i = \pm 1$. Then the all-length capacity of either family on H is one, and so is the capacity of an integer realization preserving this threshold. The same proof works for any subfamily of the legal directions.

In the *minimal native* group with cyclic coordinates $C_{2^{D_i+1}}$, adding a high-bit displacement 2^{D_i} at each member of a chosen subset genuinely changes only odd time positions of the trajectory; the displacement has order two. For a fixed unmarked direction with odd-coordinate set I, a Boolean high-bit vector with odd inner product against I flips the total tag. The actual integer family allows every lift of these native velocities. A four-slot obstruction is already visible in a cover with two more bits, and one must test arbitrarily deep lifts before inferring an integer upper bound from a calculation on the minimal native quotient. The argument below is instead a construction of high-density avoiders directly on the torsion-free group.

One active low speed. On a finite packet of axes let $M_i = 2^{D_i+3}$ and define on C_{M_i} the circle function

$$f_i(x) = (-1)^x x/M_i \bmod 1.$$

It is well-defined since M_i is even. Denote the full coordinatewise spatial unit shift by S and the time shift by T along a legal direction. At each axis apply the operator $(T-I)^2(T-S)$ to f_i. If r_i is even, f_i along any spatially shifted trajectory is affine in the temporal variable, so the second difference gives zero. At $r_i = 1 + 2^{D_i} t_i$ the first factor applied from the right gives

$$(T-S) f_i(x) = -t_i(-1)^x/8.$$

The second temporal difference multiplies this alternating sequence by 4. The resulting value is $t_i/2 \bmod 1$, independent of starts. Sum across all axes: the output is exactly a half-turn on every allowed direction.

Observe the two fields $\sum_i f_i(x_i + j)$, $j = 0,1$. Four consecutive observations contained in any translate of a sufficiently small fixed ball in T^2 would make the filter output close to zero, since the temporal coefficients on each spatial arm sum to zero and their absolute sum is at most eight. This contradicts the half-turn. At every start along any allowed direction

the two observed channels are affine on each of the two parity classes in time, since in each summand the sign is then fixed. All these assertions are uniform in packet size, depths and the allowed number of active marks.

Both active low speeds. Here take $M_i = 2^{D_i+6}$, $f_i(x) = (-1)^x x^2/M_i$, and the operator $(T-I)^3(T-S)(T-S^{-1})$. An even speed is killed by the third temporal difference. For any odd speed the two spatial-temporal factors yield $2(r_i^2-1)(-1)^x/M_i$, by expanding the four shifted squares. The third time difference multiplies by -8. At $r_i = s_i + 2^{D_i}t_i$,

$$-16(r_i^2-1)(-1)^x/M_i = t_i/2 \bmod 1,$$

since $r_i^2 - 1 = 2^{D_i+1}s_it_i + 2^{2D_i}t_i^2$ and $D_i \geq 2$. The three spatial arms -1,0,1 now give a six-slot common-ball obstruction; their trajectories are polynomials of degree at most two on time parity classes.

In either case every finite list of directions uses only finitely many axes. Pull back Haar on the corresponding finite packet to an H-action. The two preceding calculations verify exactly the uniform hypotheses of the bounded-period root lemma [R027], respectively with $(Q, s, L) = (2, 1, 4)$ and $(2, 2, 6)$. Thus there are stationary avoiders of density arbitrarily near one at lengths uniform over all directions of H, proving the endpoint. For the saturation into a divisible group containing H, when the tested steps still lie in H one can coinduce the stationary witness to the larger group. The note does not claim every possible rational lift through a differently specified quotient has these same residues; the actual directions and their subgroup must be identified before applying that transfer.

R081. Integer five-slot recurrence with a six-slot obstruction

Scope and status. Result with proof.

Result. An integer difference set can force five-term progressions at every positive density while admitting a positive-density six-term avoider.

Scope. The mixed-radix construction preserves the six-slot witness directly; its all-length critical value is unknown.

Qualifications. The final sentence about the missing bound beginning at six is specific to this constructed family.

There is a positive integer difference set for which every positive-density set contains a five-term progression, but some positive-density set contains no six-term progression. This is a fixed-length subsidiary result of the dyadic construction, consistent with the established recurrence hierarchy of Frantzikinakis, Lesigne and Wierdl. It does not determine the all-length critical density of this new example.

For stage $m \geq 2$ choose parameters k,K tending to infinity and D sufficiently deep as in the ordered five-slot proof [R077]. Work with the actual integer queries in its finite ordered contradiction. With sufficiently many replicated axes the five-slot result gives positive returns in every free abelian action for events of mass $\geq 1/m$. Only finitely many directions are required. For instance, positivity on the compact set of invariant shift laws of mean $\geq 1/m$ gives a finite subcover of the open positive-cylinder tests, or use the finite packet in the ordered proof directly. Take such a packet in its own finite coordinate block, choosing only legal steps. The direct sum of all these finite-rank blocks has a countable, coordinate-wise bounded family W of packet directions. Every event of positive mass in a W-avoiding action would avoid its stage-m packet on restriction to that block, which is impossible at length five for m sufficiently large.

There is one common six-slot avoiding action on all blocks, with a fixed positive mass a, by the cyclotomic character-torsor construction in the two-center note [R107]. For the actual integer embedding, enumerate the axes, let M_i bound all absolute coordinates at position i, and choose odd mixed bases $s_i > 12M_i + 3$ so large that $\Sigma 10M_i/s_i < 1/2$. Define $\lambda_1 = 1$ and $\lambda_{i+1} = s_i\lambda_i$; send v to $\phi(v) = \Sigma\lambda_i v_i$ and take positive absolute values. Coordinate bounds make this nonzero. The integer action pulled back along this homomorphism inherits the five-slot positive-return property. For the converse **only the single six-slot obstruction** is required: apply the balanced-digit argument of the free group transfer [D06] with the cut on every digit, including the first. The proportion surviving on the symmetric radix intervals, and therefore their positive halves, is at least $1 - \Sigma 10M_i/s_i > 1/2$. Each surviving start sees exactly six unreduced digit vectors $z + nv$, $0 \leq n \leq 5$. Mark it if its digit translation of a random point belongs to the six-avoider. Reverse Fatou gives upper density at least $a/2$ for some sample, and countably many translated tests exclude every forbidden six-term progression. This preserves the obstruction length, even though the unrestricted critical-value transfer alone takes a supremum over lengths.

The recurrence statement with positive upper natural density follows from correspondence, and shorter lengths follow by taking subprogressions. The missing uniform upper bound in the prescribed-density conjecture begins at length six.

R082. Half-density recurrence for arbitrary invariant measures with bounded interactions

Scope and status. Result with proof.

Result. Arbitrary invariant measures on the specified bounded-interaction dyadic modules retain the half threshold.

Scope. The high-on-two-axes vanishing and minimum-depth invariance are algebraic assumptions, not properties of general stationary systems.

Use the original cyclotomic rings and coefficient interaction modules of the Haar interaction theorem [R066]. The same half threshold holds for **every invariant probability measure** on the resulting compact character action if the schedule has infinitely many axes at one fixed $k \geq 2, K \geq k+3$ with strictly increasing, arbitrarily large D, beyond every finite head. Require $J+1 \leq \phi$, $J = 2+2^{k+1}$, $\phi = 2^{K-1}$. Additional axes, or other cofinal parameter regimes needed for other tests, may be included. The direction uses only the specified fixed-k,K subfamily. This conclusion concerns the interaction-character actions, including arbitrarily large but finite arity per observed coefficient. It is not a theorem for arbitrary actions of the native group or its integer covers.

Write $z_i = 2^{D_i}$, $L = 2^K$, $p = 2^k$, $c = 1+p$, and $J = 2+2p$. Fix an ergodic invariant measure μ, an event A of mass $a > 1/2$, and horizon h. We will first program only a finite list of coefficient characters $m_1, \ldots, m_b$. They have some common interaction arity bound s. The number of trial blocks $m = m(h,a)$ will depend on the original mass and h, not on this list. Working on a countable invariant factor determining A is sufficient. If local coefficient groups outside the factor are needed to describe an action, lift μ by Haar over fibers; the construction below tests only characters from the chosen list and their genuine group translates.

Homogeneous high tables without changing the starting marginals For an ordered list $i_0 < i_1 < \cdots < i_v$ of prospective axes, ordered by strictly increasing D, with $0 \leq v \leq s-1$, record at a starting state χ the evaluations of the following elements, for the finitely many m and all exponents b_j mod L:

$$(U_{i_0}^{z_{i_0}} - 1) U_{i_0}^{b_0} \prod_{l=1}^{v} (U_{i_l}^{b_l} - 1) m. \quad (1)$$

Here U denotes the coordinate action on coefficient tuples. The first operator is zero off the incident edges and multiplication by $z_i \epsilon_i$ on them. All entries have additive order at most two. On this high layer every U_j^L acts as identity, since $u_j^L = 1 + L\epsilon_j$ and L is even. Thus the tables require only a finite color set independent of D. A high entry with minimum distinguished axis i_0 is unchanged by **every** lower-depth axis: an edge containing one such vertex has additive exponent dividing its smaller 2^{D+1}, which kills the entire high layer. This applies to head axes even if they are outside the finite packet. The recorded values are therefore invariant under any fixed finite head once the packet is sufficiently deep.

Specify a finite slot count N as in the next section, before selecting a finite Ramsey pool. Color each increasing tuple of at most s indices by its displayed table. For each χ, finite

Ramsey supplies an N-element subpacket homogeneous for every one of these finitely many tuple arities. Thus some fixed subpacket has a homogeneity event $\mathcal{E}$ of measure at least $\gamma = 1/\binom{R}{N} > 0$, where R is a finite Ramsey pool size. This lower bound depends only on k, K, h, m, s and the finite test-list size, never on how far beyond the head the pool is taken. Since the qualifying fixed-k,K depths are cofinal, we may choose the head **after** determining γ. The event $\mathcal{E}$ says only that all equal-order tables agree, not what their values are. It is invariant under that head. Every translate of it under the abelian group has the same head invariance and the same positive measure.

For any fixed bounded test f, averages over the increasing finite head subgroups converge in $L^2(\mu)$ to its mean, by ergodicity. Hence by going far enough initially, every head-invariant event of mass at least γ has conditional f-mean arbitrarily close to its original mean (apply Cauchy–Schwarz to the head average, dividing by $\sqrt{\gamma}$). We use this for 1_A and one finite-list approximation error. It applies uniformly to *all translates* of $\mathcal{E}$, including all word points and prospective corrected runs. This prevents conditioning on a rare but unfavorable class of starting states.

A binary ring identity On the high layer of the distinguished axis put $X = \zeta - 1$. The coefficient algebra here has characteristic two, $X^{J+1} = 0$, $u = 1 + X$, and $u^c = (1 + X)(1 + X^p)$. Write polynomials

$$C(X) = 1,$$

$$B(X) = 1 + X^p + X^{2p-2} + X^{2p},$$

$$A_0(X) = (1 + X)(X^{2p-2} + X^{2p}).$$

For every odd positive t one has, in $F_2[X]/X^{J+1}$,

$$A_0 + B(1 + X)^t + C[(1 + X)(1 + X^p)]^t = \binom{t}{4} X^J. \quad (2)$$

Indeed set $\beta = \binom{t}{2} \bmod 2$, $\xi = \binom{t}{4} \bmod 2$. The third term minus $(1 + X^p)(1 + X)^t$ is $\beta X^{2p}(1 + X)^t$. In the remaining terms use $(1 + X)^t - 1 - X = \beta X^2 + \beta X^3 + \xi X^4 + O(X^5)$, valid because t is odd. The coefficients through $2p + 1$ cancel and the coefficient at $2p + 2$ is ξ. The higher term X^{3p} is beyond truncation since $p \geq 4$. On odd t this ξ equals $\binom{t}{5} \bmod 2$. On even t a genuine high toggle does not move the tth point at all. We will use (2) on a collection of axes, not divide any coefficient or impose any condition on the local orbit measure.

Expand each polynomial A_0, B, C in powers T^a where $T = 1 + X$, reducing a mod L if needed. These are binary expansion coefficients, denoted $b_{a,r}$ with r respectively $0, 1, c$. In particular the sum of all coefficients in the c type is odd (it is the constant 1). Also expand X^J in the same T basis with coefficients $b_a^r ef$.

Blocks, cancellation of later moves, and the reference involution Take a power of two $M > s$. Each of m ordered trial blocks contains all subtypes (a, r) for $0 \le a < L$, $r \in \{0, 1, c\}$, in a fixed order, with exactly M consecutive vertices of each subtype. A final reference block has M copies of every $(a, 0)$ subtype. Their total size is the N used above. The last vertex of each subtype group is its representative. On all these vertices the initial group move is its assigned a; denote the resulting starting shift by g. For a word $w \in \{0, \dots, h-1\}^m$, its additional group move adds $w_l r$ per vertex in trial block l, zero in the reference block; write g_w for the full initial and word move. The uncharged candidate difference for a wildcard collection of trial blocks has its prescribed r in those blocks and zero elsewhere.

On $\mathcal{E}$ all representatives have the same singleton table, written $F_m(a)$ for a coefficient m. At the word vertex $U^{g_w}\chi$, a high toggle on a representative in subtype (a, r) of block l contributes exactly $F_m(a + w_l r)$ to the m observation. To verify this, ignore earlier vertices since their actions are trivial on the high layer. Expand the effects of all later vertices as products of $1 + (U_j^b - 1)$ at their corresponding exponents. No more than $s - 1$ other vertices can matter in any mixed term. Every later subtype group is whole, since the representative sits at the end of its group. Within each group all exponents b are equal. The high tables in (1) are homogeneous by ordered tuple type. For any nonempty pattern of later-vertex counts the multiplicity $\prod \binom{M}{n_j}$ is even. All such contributions therefore cancel, leaving just the singleton. This proof includes all initial shifts a, all future inactive blocks with arbitrary fixed word letters, and the reference block. Distinct high toggles add their effects: their product's mixed high term contains two factors $2^{D_i} 2^{D_j}$, which kills every shared edge module. Thus a sum of toggles has the sum of their displayed binary effects at every word point.

Define one fixed reference high move w_0 on the final block by toggling the $(a, 0)$ representative iff $b_a^r e f = 1$. Its effect at *every* word vertex is $\sum_a b_a^r e f F_m(a)$, for every tested m. Given a combinatorial line, choose any one wildcard block l. Put an extra high move w on its subtype representatives with $b_{a,r} = 1$. Along the progression with baseline wildcard direction, adding w to the difference flips these bits at the odd times and not at the even times. Its legal tag is odd since the toggles among active c types have odd count; all other toggles are allowed at normal coordinates. At odd slot t, (2) says its effect on m equals the reference effect multiplied by $\binom{t}{5} \bmod 2$; this follows by linearity of the singleton table in the ring action, since homogeneous tables equal the evaluations on any one actual representative axis. Thus on the measured characters every corrected slot is **either** the ordinary word vertex **or** that same word vertex translated by w_0. Both w and w_0 are genuine fixed native group elements for their chosen blocks. No invariance under a merely formal ring endomorphism is being assumed. The baseline starting vertex of the line is the word with wildcard letters zero, so slot t has exactly the required low move `t r` in block l.

Density and the measurable event Choose the DHJ dimension m so that a positive proportion depending only on $2a-1$ of accepted words guarantees a line. On the conditioned event $\mathcal{E}$ take as accepted the word vertices for which **both** $U^{g_w}\chi$ and $U^{g_w + w_0}\chi$ belong to A. Every word has conditional acceptance probability at least $2a - 1 - o(1)$ by the head-average bound applied to the two translates of $\mathcal{E}$. In particular with conditional probability at least a fixed $\kappa(a) > 0$ the word set has density large enough for DHJ, and hence contains a line. The number of possible lines is bounded in terms of h and m alone.

To apply the programmed equalities to an arbitrary measurable A, choose a finite-coordinate approximation A' in the listed characters with error $\mu(\mathrm{A} \triangle A')$ small compared to $\kappa(a)$ divided by the number of tested word points and prospective true slots for **all** word lines (choose, for each line, its first wildcard block for the correction). These combinatorial counts were fixed before approximating A. Then compute the Ramsey number and γ for that character list. Only afterwards choose the ergodic head so that $1_{A \triangle A'}$ has conditional mean at most twice the chosen error on every translated $\mathcal{E}$; one may use any small positive bound for an error that is zero. By a union bound the probability of an approximate disagreement anywhere among these genuine translated snapshots is less than $\kappa(a)$. On a positive conditional set an accepted line therefore exists and all the relevant A/A' memberships agree. The character identities show every corrected slot is in A' and hence in A. A finite choice of lines gives one actual legal direction with strictly positive return after removing its fixed starting shift. Each such argument works on any tail. Ergodic decomposition and countability of the directions give the statement for arbitrary invariant measures.

The algebraic hypotheses here matter. In particular high-on-two-axes products vanish and the minimum-depth high correction is fixed by every earlier vertex. These need not hold for general stationary sets or for arbitrary compact actions whose high subgroup has the same abstract exponent. The proof does not settle the prescribed integer threshold conjecture.

R083. A complementary digit cut in the dyadic two-center construction

Scope and status. Result with proof.

Result. A complementary low-digit summability condition, or the summability of the coordinatewise minimum of two seam costs, forces endpoint one.

Scope. Arbitrarily replicated fixed levels can violate both conditions.

On the free-coordinate lift of the two-center model [R107], the critical value is one if

$$\sum_i (1 + 2^{k_i})/2^{K_i} < \infty.$$

Consequently, combining with the other digit cutoff [R071], it is enough that the sum of the coordinatewise minima of this quantity and 2^{-k_i} is finite. The result is for the unrestricted integer coordinates and passes to the rational or integer realizations when those realizations preserve the critical value. It does not bound a schedule with arbitrarily many copies at each fixed level.

Write $B_i = 2^{K_i}$, $L_i = 2^{D_i+1-K_i}$, $c_i = 1 + 2^{k_i}$. On a finite set of coordinates let $y_i(x) \in [0, B_i)$ and $z_i(x) \in Z$ be the residue and integer quotient of x_i by B_i. For a legal velocity v write $g_i \in \{0, 1, c_i\}$ for its low residue and $s_i{=}v_i - g_i/B_i$. At an active axis $g_i = c_i$ and $s_i = L_i t_i/2$ with integer t_i. At normals s_i is unrestricted. Hence

$$\sum_i [g_i^2 s_i - g_i s_i]/(2^{k_i} L_i) = \sum_a ctivec_i t_i/2 = 1/2moduloone.$$

To test the two terms separately choose a small open circle arc I. There are integers M,H depending only on I such that a real polynomial in t of degree $j = 2$ or 3 with leading coefficient ξ and $||M\xi|| \geq 1/4$ hits I in its first H values, for every choice of lower coefficients. This is the usual bounded-order obstruction from the circle polynomial hitting lemma used here [R240]. For each j mark the complement of a randomly translated I under the observable

$$F_j(x) = \sum_i y_i(x)^{j-1} z_i(x)/(M 2^{k_i} L_i).$$

These are exact observables for finite translation clocks of periods $B_i M 2^{k_i} L_i$; any independent uniform circle offsets are invariant. Cut away points whose low digit is in the last Hc_i residues, only on a sufficiently late tail where $Hc_i < B_i$. At a retained start, through H slots every low digit moves by g_i without wrapping and the quotient moves by s_i. Consequently F_j follows a real circle polynomial whose leading coefficient is $\xi_j = \sum_i g_i^{j-1} s_i/M 2^{k_i} L_i$. Since $M(\xi_3 - \xi_2) = 1/2$ mod one, at least one of the two polynomials must hit its omitted arc. Thus the simultaneous event excludes every odd-tag direction of the tail exactly, with measure at least $1 - 2|I| - H \sum_t$ail c_i/B_i. This proof costs no derivative errors and does not require any sign condition on s_i.

For any given desired error choose I and then H, followed by the tail. Finite packets suffice by stationary compactness. If no active coordinate occurs in the head, the tail has odd tag even if head normals move; simply use the tail-only observable. If some head coordinate is active, that velocity is in a fixed nonzero residue class there, and the finitely many such families admit independent near-full uniform-length sieves. This proves the first statement. For the minimum-weight assertion divide axes according to which summand is smaller. An odd total tag forces an odd tag in at least one part, so the product of the two near-full avoiding events blocks all such directions.

R084. A two-bit potential for the pure dyadic mark model

Scope and status. Result with proof.

Result. A two-bit potential gives a pure-mark short obstruction and endpoint one under summable reciprocal low-bit periods.

Scope. Divergent seam rates remain untreated by this construction.

On integer axes let $p_i = 2^{k_i} \geq 4$, $L_i = 2^{K_i} \geq 2p_i$, and $z_i = 2^{D_i}$ divisible by L_i. Use the pure-mark menu dyadic-pure-mark-transpose.md [R098]: each normal velocity is 1 or p_i modulo L_i; each other coordinate is $z_i t_i$, and the sum of the latter t_i is odd. The following elementary templates give its fixed short obstruction and an unrestricted endpoint-one conclusion when $\sum_i 1/p_i < \infty$.

Potential. Write $bit_j(x) = \lfloor x/2^j \rfloor \bmod 2$, $b_i(x) = bit_1(x) bit_{k_i}(x)$ and

$$f_i(x) = b_i(x)x/(4z_i) \bmod 1 on C_{4z_i}.$$

The bit b_i has period $2p_i$. On a normal axis with residue $a = 1$ or p_i, $(T_v - T_a)^2 f_i = 0$, because the step $v - a$ is a multiple of $2p_i$ and the function is affine on each such residue. For a pure velocity $v = z_i t$, $\Delta_v f_i = t b_i/4$. On this last periodic array T_v is the identity. The square of the difference at $-p_i$ applied to $b_i/4$ yields $bit_1/2$ modulo one: the high bit flips, the low bit does not, and the two choices of sign of $1/2$ agree. Finally the square of the difference at -1 applied to $bit_1/2$ is the constant $1/2$, as follows from the repeating array $(0, 0, 1, 1)$ modulo two. Including the spatial shifts, therefore,

$$(T_v - I)(T_v - T_1)^2 (T_v - T_{p_i})^2 f_i = t/2 on pure axes,$$

and the same is zero on normal axes. For finitely many axes take the sum of their potentials on the product of the indicated cyclic covers. The filter is a degree-five temporal polynomial, with the nine fixed spatial observations obtained from shifts by $a \cdot 1 + b \cdot p$, $0 \leq a, b \leq 2$ componentwise. The temporal coefficients at each of these spatial offsets sum to zero because of the factor $T_v - I$. Since the combined output on a true direction is $1/2$, six consecutive spatial vectors cannot belong to a common translate of a fixed small torus box. This gives the same finite palette independent of the axes, with one extra dyadic cover factor permitted on the free group. The statement applies to all finite restrictions, hence gives a stationary positive fixed-length avoiding mask by coloring and compactness (or one may realize the registers in a compact skew product over the infinite digit product).

When $4p_i | L_i$ one can use the period $2z_i$ instead: take $b_i = bit_1 \cdot bit_{k_i+1} = \binom{x}{2p_i+2} \bmod 2$ and $f_i = b_i x/(2z_i)$. On a pure high step the fifth-order filter on $b_i/2$ gives $1/2$ again, since over F_2 the two second differences combine to $\Delta_1^{2p_i+2}$ up to a shift. The basic $4z_i$ version suffices for the following free-group assertion.

Summable high-bit seam rates. If $\sum_i 1/p_i < \infty$, then the full torsion-free critical value is one. Here is the uniform control behind this conclusion. For a normal velocity congruent to 1, the high bit of $x_i + a + bp_i$ with $0 \leq a, b \leq 2$ advances by one modulo $2p_i$ at each step; remove starting coordinate residues where that advance would cross a high-bit boundary, for each of the fixed spatial offsets. Their Haar cost is $O(1/p_i)$. Fractional

notation for the potential is periodic on its raw domain since its x denominator is $4z_i$. Along any retained segment the relevant high bit of a residue-1 normal stays constant. Along a residue-p_i normal it alternates, while bit_1 is constant. Along a pure velocity the bits are both constant. All observed bit products on the segment are consequently constant on each time class modulo four. In that time class $x/(4z_i)$ evolves linearly in time, even when the raw cyclic coordinate wraps. The sum of every register is thus linear modulo one on each of the four time classes of the segment. There is no bound on the number of active coordinates and none is needed in this argument.

To apply this to a fixed target loss ϵ, use the rooted-row synchronization argument of bounded-vertical-period.md [R027] with degree one, period classes four and the fixed short box furnished by the filter. It is enough for the polynomial law to hold **on retained segments**, because the proof takes limits of arbitrarily long such segments, and the finite vectors of linear coefficients lie in compact tori. In a root row of stride M, each time residue modulo M samples an original trajectory at the original velocity; seam tests at the retained sites of that residue apply exactly as above. First choose the finite number of independent shifted rows, their boundary margins and their strides as in that proof, with the class period four fixed. Only then choose a cutoff J so the total cost of the bit tests on all tail coordinates $i \geq J$, across all these rows, is below the prescribed error. This order of choice keeps the argument uniform over every finite directional packet. For directions with all initial coordinates zero modulo $4z_i$, the initial registers are constant. If any of the finitely many initial coordinates is nonzero modulo $4z_i$, a finite union of ordinary single-coordinate character tests separates that class of directions from zero and the usual circle-root sieve gives near-full avoidance there. Intersect the independent near-full masks and use stationary compactness of all finite tests at the common resulting lengths. This proves the claim, including in the regime with unbounded k_i. The argument does not cover divergent high-bit seam rates.

R085. The half bound on arbitrary dyadic ring modules

Scope and status. Result with proof.

Result. Arbitrary native dyadic ring-module Haar actions satisfy an all-length half bound, with no fixed bound on observed channels.

Scope. This is the original-ring theorem; broader rings and arbitrary non-Haar measures need their separate statements.

Qualifications. R063 and R082 extend the conclusion to their stated classes of non-Haar measures. An extension to arbitrary non-Haar measures is not established.

The six-slot dyadic obstruction in the two-center construction [R107] cannot be amplified to near-full avoiding measure anywhere in the following Haar module class. In particular, an event can observe arbitrary coefficient sequences on the axes, and need not be generated by the six original global channels, their Laurent-polynomial translates, or any uniformly bounded list of channels. The conclusion is specific to these module actions.

Use the parameters of that construction on a tail, with $D - K \to \infty$ and $J + 4 < \phi$ eventually (for example $K = 3k$). Write $L = 2^K$, $z = 2^D$, $S = Z[\zeta]/(\varpi^T)$, $T = D\phi + J + 1$, $R = S[\epsilon]/\epsilon^2$, $u = \zeta(1 + \epsilon)$, $c = 1 + 2^k$, $E = z\varpi^J\epsilon$ at one axis. Each M_i can be **any module** over its local ring R_i. Let $A \leq \prod M_i$ be an additive coefficient group containing the local direct sum $L_0 = \oplus M_i$. Each finitely supported group element d acts on A by coordinatewise multiplication by $u_i^{d_i}$: its difference from the identity is locally supported. On the compact character group of the discrete A, prescribe arbitrary additive signs $\theta_i : E_iM_i \to T$ and take Haar on the torsor of characters restricting to those signs on $\oplus E_iM_i$. The signs take values 0 and 1/2 since $2E_i = 0$. The torsor is nonempty by divisibility and invariant since $u_iE_i = E_i$.

Claim. Every measurable event of measure greater than 1/2 in this action has positive returns at every finite progression length along legal directions supported arbitrarily far out. This includes the native, all period lifts through the native action, and any factor. Product torsors with separate signs are included by using module direct sums. One may also prescribe signs on a subgroup of $\oplus E_iM_i$: disintegrate over extensions to the whole socle-image subgroup, aggregate them into the coordinate signs, and use the countability of legal directions on the positive set of fibers where the conditional event measure exceeds 1/2.

It suffices to treat countable groups A and countable M_i. Every Haar-measurable event on the general compact torsor agrees modulo null sets with one observable through countably many characters, by density of continuous functions and of finite trigonometric polynomials in L^1. Restrict to the group they generate and adjoin the axis-local submodules generated by their entries. Close each smaller local submodule under socle preimages: iteratively for each of its elements lying in the ambient E_iM_i adjoin one E_i-preimage and generate again. The resulting countable union M_i' has $M_i' \cap E_iM_i = E_iM_i'$. Countability holds since the rings are finite, and projection of Haar to the required smaller torsor is onto by extending characters over the sum with the full prescribed socle subgroup. This reduction also treats arbitrary coefficient groups not initially containing the local direct sum, by adjoining the locals and projecting.

Homogeneous fresh coordinates Fix length h and a finite Fourier test of an increment trajectory, with coefficients $m_t \in A$, $0 \leq t < h$. At axis i put the module-valued polynomial

$$R_i(Y) = \sum_t (Y^t - 1)(m_t)_i.$$

It vanishes identically at $Y = 1$. For each of the speeds $L, 1, 1 + L, c, c + z$, the local Haar integral of $e(\chi_i(R_i(u_i^s)))$ is zero if the argument is outside $E_i M_i$, and otherwise is $e(\theta_i(R_i(u_i^s))) \in \{1, -1\}$. Along a subsequence of axes the value for every such speed and test is eventually constant, by countable diagonal extraction. Also arrange that each $\theta_i(E_i m_i)$ for $m \in A$ stabilizes. The latter limits give a single character $\pi : A \to \{0, 1/2\}$, vanishing on L_0. All selections can start on an arbitrary tail.

For each stage use eight distinct indices of this subsequence, escaping to infinity. On two indices each put normal speed L, 1, $1 + L$, on one put uncharged active speed c, on one put the true speed $c + z$; unused axes have speed zero. This direction is legal. The characteristic function of its increments $Z_t = U^{td}\chi - \chi$ at $(m_t)_t$ is the product of the eight local integrals. If this product survives, the duplicate normal signs cancel. Homogeneity allows the remaining two signs to be evaluated at the **same** representative index sufficiently far along the extracted subsequence; all five arguments there belong to $E_i M_i$. Their two signs add as the sign of the difference, since each has order at most two.

Integral Hermite interpolation, including modules with torsion At such an index omit subscripts. Because the true and the baseline powers differ by $\zeta^c z\epsilon$, the difference of their R evaluations is exactly $\zeta^c z\epsilon R'(\zeta^c)$. We show that this equals

$$E \sum_t \binom{t}{5} (m_t)_i.(1)$$

Work with coefficients in the given module, allowing zero divisors. The normal and baseline tests give

$$\epsilon R(1) = \epsilon R(\zeta) = \epsilon R(\zeta^c) = 0,$$

$$L\epsilon R'(1), L\epsilon R'(\zeta) \in EM.$$

The values at ζ and ζ^c follow by multiplying $R(u), R(u^c) \in EM$ by ϵ. The first derivative constraint follows from $R(u^L) = L\epsilon R'(1)$ and $R(1) = 0$. For the second, subtract $R(u)$ from $R(u^{1+L})$ to obtain $L\zeta\epsilon R'(\zeta)$; ζ is a unit and preserves EM.

Divide the module-coefficient polynomial by the monic scalar polynomial

$$P_0(Y) = (Y - 1)^2(Y - \zeta)^2(Y - \zeta^c), R = P_0 Q + B, \deg B < 5.$$

The values and derivatives in the two preceding lines agree for B and R. The confluent five-by-five evaluation matrix for B has determinant (up to sign)

$$\Delta = (\zeta - 1)^4(\zeta^c - 1)^2(\zeta^c - \zeta)^2,$$

of ϖ-valuation $4 + 2 + 2 \cdot 2^k = J + 4$. The adjugate matrix proves $L\Delta\epsilon B_j \in EM$ for every coefficient B_j. In S, $z = L\Delta b$ for some b divisible by ϖ: the relevant valuation difference is $(D - K)\phi - J - 4 > 0$ on the tail. This uses that S is a finite local chain ring, with an element of given valuation a unit times the corresponding power of ϖ. As $\varpi EM = 0$, it follows that $z\epsilon B'(\zeta^c) = 0$. No division of a module element has been made.

The derivative of P_0Q at ζ^c gives $z\epsilon\zeta^c(\zeta^c - 1)^2(\zeta^c - \zeta)^2Q(\zeta^c)$. The scalar in front of Q is E: its ϖ-valuation without $z\epsilon$ is J, and its normalized unit reduces to 1 in the residue field F_2, while $\varpi E = 0$. Thus this expression is $EQ(1)$. Modulo ϖ, $P_0 = (Y - 1)^5$, so multiplying by E and taking the fifth binomial Taylor coefficient at 1 gives $EQ(1) = E\sum_t \binom{t}{5}(m_t)_i$. The remainder B has degree less than five. This proves (1) for every local module and every surviving test.

A two-valued point in the increment law Put $s_t = \binom{t}{5}\pi$ in the compact group dual to A/L_0. The joint laws of the increments converge: every Fourier coefficient converges by the extraction. Its limit is either zero or the character evaluated at $s = (s_t)_t$, by (1) and the matching cancellation of baseline signs. Translating the limit law by $-s$, all its Fourier coefficients are zero or one. The coefficients equal to one are a subgroup (their characters equal one almost surely), and uniqueness of Fourier coefficients identifies the law as Haar on their annihilator. In particular s lies in the support of the limiting increment law ν. Each local character vanishes on all the increments in the limit, so ν is supported in the h-fold power of the dual of A/L_0.

The joint limit of the initial point χ and the increments is $\mu \times \nu$, with μ the Haar torsor measure. To verify this on characters, if the initial character is tested at $m \notin L_0$, averaging over characters annihilating L_0 gives zero for each direction. All increments depend only on the local coordinates. If $m \in L_0$, its finitely supported test is independent of the eight fresh local coordinates eventually, by the product Haar marginal on the local torsors. These two cases determine the joint limit. This also shows explicitly why dependence on infinitely many local head coordinates in the original event poses no problem.

For a measurable event B of measure $a > 1/2$, let $p(v) = \int \prod_{t<h} 1_B(\chi + v_t)d\mu(\chi)$ on the h-fold compact increment group. It is continuous, by continuity in L^1 of translation of each indicator. At the particular point s, the h translates are only B and $B - \pi$, since $2\pi = 0$; consequently $p(s) \geq 2a - 1 > 0$. The support statement gives $\int p d\nu > 0$. This is the limit of the actual progression-return measures. For measurable B one just approximates its indicator by continuous bounded functions in $L^1(\mu)$; both the actual and limiting slots have marginal μ, so the telescoping error is at most h times the approximation error. Some legal direction therefore returns positively, as claimed.

The uniform bound here is on the threshold over this entire Haar ring-module class. The positive return probability and the choice of directions can depend on h, the module and the measured event. Arbitrary stationary systems, Haar measures on unrelated modules, and general non-Haar invariant measures are outside the argument.

R086. A native half bound without any bounded-state profile

Scope and status. Result with proof.

Result. A wider native Haar module has a half upper bound despite excluding every bounded-state trajectory method.

Scope. Only the specified edge/slope registers are covered, not arbitrary added coefficient channels.

There is one native Haar module system with a positive six-slot avoiding event and all-length system-relative threshold at most $1/2$, for which **no bounded number of distinct observed states suffices in a supported arbitrarily long trajectory**. For each proposed number of states a finite collection of observables and a horizon already exclude this on every sufficiently deep legal tail. The upper bound comes from independent overlapping edges. This system uses the wider-ring truncation of the distinct-window note [R070], not the ring of the maximal half theorem.

Take native wide rings with $T_i = D_i\phi_i + J_i + B_i + 1$, $B_i \to \infty$, $J_i + B_i + 1 \le \phi_i$, and $K_i - k_i \to \infty$. Prescribe the half-turn on the invariant top element. Besides the entire local direct sum, use exactly the following global coefficient generators, and the resulting Haar measure:

$$m_s u^a w^b, 0 \le a \le 2, 0 \le b \le 1, s = 4, 5, \dots or s = *,$$

$$(m_s)_i = \varpi_i^{B_i - 2^s}, (m_*)_i = \varpi_i^{B_i}.$$

At any early index with a negative exponent set the entry to zero. Denote $l_s = 2^s$ and $l_* = 0$. These generators are independent over the integers modulo the locals. Indeed in a finite test factor out $\varpi_i^{B_i - l_{\max}}$; the constant-in-$\epsilon$ evaluation of the rest is the specialization at $(U, W) = (\zeta, \zeta^c)$ of

$$\sum_s (U-1)^{l_{\max} - l_s} R_s(U, W), \deg_U R_s \le 2, \deg_W R_s \le 1.$$

The supports in powers of $U - 1$ are disjoint between s, so this polynomial is nonzero for a nontrivial test. Its valuation at the specialization is at most $O_R(\phi + 2^k)$, by the Laurent specialization estimate [R061]; even adding $B_i - l_{\max}$ leaves it far below T_i. The full single-site global state is consequently a product of independent Haar circles, independent of the local head.

The $*$ row gives the ordinary low six-slot obstruction: $m_* P(u^r)$ is zero on normals and tE_{top} on a marked residue, with top normalization units of residue 1. All required coefficient characters belong to the six displayed monomials. The other rows rule out bounded-state trajectories. As in the native distinct-window argument [R070], along any fixed lag d with $v_2(d) = s \ge 4$ two equal six-windows are impossible on an escaping legal tail. For a q-state trajectory choose more than q^6 window starts that are multiples of 16. Two windows would agree and the valuation of their start difference is at least 4. The finitely many pertinent rows (one per possible dyadic valuation) exclude this, including the small-neighborhood version. All q use the same action.

Independent noise on successive edges Fix a finite h. Use the single legal velocity $c_i + z_i$ on one escaping axis, zero elsewhere, and take a subsequential limit of the h-trajectory law. For finite joint Fourier tests on the global states, combine their coefficients in row s into $R_s(T, U, W)$, with spatial degrees $\leq 2, \leq 1$ and temporal degrees $< h$. A surviving test must have $R_s(1, U, W) = 0$ for all rows, by independence of the single-site global coordinates. At the moving axis a surviving test must have its local argument in the constrained top line, which has zero constant term in ϵ. Using the substitution $T = \zeta^c$ on the true-mark constant part therefore requires

$$\sum_s (U-1)^{l_{\max} - l_s} R_s(W, U, W) = 0$$

as a formal polynomial. Otherwise the same valuation estimate prevents survival. Substitution $T = W$ does not raise the degree in U. The disjoint bands in $U - 1$ therefore give $R_s(W, U, W) = 0$ separately for each s. Conditioning on any fixed finite local head changes none of these necessary vanishings: the moving coordinate is independent and the head is fixed along its trajectory. The same conclusion holds jointly with an arbitrary character test on the entire local group by passing to finite support.

This annihilator condition means the limiting law is invariant under adding, independently at every (s, a), a moving full Haar sequence whose contribution at channel (s, a, b) and time t is $Z_{s,a,t+b}$. Indeed $R_s(W, U, W) = 0$ is exactly the condition for a character to vanish on these moving sequences, coefficient by coefficient in U and W. Haar convolution on that compact subgroup leaves the law unchanged by Fourier uniqueness. The local coordinates are constant through the window and the noise does not move them.

Let a measurable event have mass $> 1/2$. There is a positive measure set of local-head fibers on which its global Haar mass exceeds $1/2 + 2\delta$ for some fixed $\delta > 0$. In the preceding convolution, fix any other trajectory data over one such head. Collect the independent sequences into iid vector vertices $Z_t = (Z_{s,a,t})_{s,a}$. At time t the state after randomization consists of Z_t in all $b = 0$ channels and Z_{t+1} in all $b = 1$ channels, plus fixed offsets. Thus every constraint is a translate of the same two-vertex relation B determined by the head fiber.

Write $P_u = \{v : m(B_v) > u\}$ for sections in the first vertex. Layer cake and $m(B) > 1/2 + 2\delta$ give a threshold u with $m(P_u) > 1 - u + \delta$. Start in the translated first-vertex P_u at the initial slot. At each slot the eligible second vertices have measure greater than u and the translated good first vertices for the next slot have measure greater than $1 - u + \delta$; their intersection has measure greater than δ. Consequently every h-slot window succeeds with probability at least δ^{h+1}, uniformly in the other trajectory offsets. Integration proves positivity in the limit. Fixed marginal Haar measures allow continuous approximation of every indicator in L^1, so an actual legal tail direction has positive return. This argument even works on the *single-mark* directions at these Haar events.

The claim concerns the entire σ algebra of the displayed system, including all its local variables and every listed row simultaneously. Wider W-windows give a finite-range dependence argument whose allowable density loss deteriorates with the window size; it does not settle general actions.

Adding all slope registers preserves the half bound Adjoin any family of global coefficient tuples with zero constant term in ϵ, including $L_i\epsilon$ times arbitrary spatial registers. Let the base factor consist of them and all original finite-support local registers. Single-site global $m_s u^a w^b$ observations remain independent Haar given that whole base: in every putative dependence, the nonzero constant-in-ϵ part of a nontrivial combination of m rows persists by the disjoint-band calculation. Highbit toggles act trivially on each new global slope register because $\epsilon^2 = 0$.

For an event with total measure greater than $1/2$ take a positive-measure base set where its original edge-coordinate sections have Haar measure greater than $1/2 + 2\delta$. A single rational threshold u and a positive-measure base subset C have $m(P_{y,u}) > 1 - u + \delta/2$ for every y in C: for each qualifying y the layer-cake argument gives a threshold with a strict positive gap and it can be perturbed slightly downwards to a rational one. On arbitrarily fresh axes, density Hales–Jewett averaging gives a nonempty subset I with a baseline return through C of mass at least a fixed positive $\kappa(h, \mu(C))$, using the velocity c_i on I and zero elsewhere. Toggle the high bit on one coordinate of I. On the slope generators this changes nothing. Approximate C in measure by slope data and finitely many local coordinates, with error much less than κ/h. All sufficiently fresh toggles fix the finitely many local coordinates from the approximation. Thus we can choose legal directions with an actual h-return to C bounded below independently of the stage. Pass to a weak limit of their full trajectory laws; fixed one-slot marginals justify using the measurable C.

For a joint Fourier test on the edge, slope, and local data, a nonzero formal polynomial $\sum_s (U-1)^{l_{\max}-l_s} R_s(W, U, W)$ cannot be canceled by any added slope term, since all those terms have zero constant in ϵ. Every direction used has at least one escaping active coordinate, and its ϵ-free evaluation at a mark is precisely W. Thus the same independent W-moving noise exists on the edge coordinates **with every base trajectory held fixed**. On the positive mass set of C-trajectories the positive-gap section rule holds at the same u at every slot, giving probability at least $(\delta/2)^{h+1}$ conditionally on all the other data and hence a positive return in the enlarged system. Uncountably many registers cause no new issue after reduction to the countably many determining the event. This does not include arbitrary extra registers with nonzero ϵ-free parts or arbitrary additional W-translates of the original rows.

R087. Polynomial nilorbit recurrence from the ordered word identity

Scope and status. Result with proof.

Result. Ordered dyadic words give uniform pointwise polynomial-nilorbit recurrence on connected filtered nilmanifolds.

Scope. The realization must preserve the finite word tests explicitly; critical-value transfer alone is insufficient.

Take the two-normal dyadic family [R107], with sufficiently large identical blocks and D/K tending to infinity. The word identity [R095] gives recurrence on every fixed connected finite-step filtered nilmanifold, including simultaneous pointwise returns for arbitrarily long nilrotation progressions. This excludes a nil-Bohr obstruction of these orders. It does not bound the avoiding density for general stationary systems.

Precisely, fix a connected simply connected nilpotent Lie group N, a lattice Γ, and a connected rational filtration $N = N_0 = N_1 \geq N_2 \geq \cdots \geq N_{s+1} = \{1\}$, with $[N_i, N_j] \leq N_{i+j}$. Use a metric for the compact manifold $X = N/\Gamma$. Given $\eta > 0$, for an equal-parameter depth block of sufficiently many axes, every polynomial nilorbit $P : \mathbb{Z}^n \to X$ lifted by a filtered polynomial $g : \mathbb{Z}^n \to N$ has a legal direction v with $P(v)$ within η of $P(0)$. The number of coordinates depends on the fixed target, tolerance, and local depth, not on g. Products of fixed targets and the maps $x \mapsto g(jx)$, $0 \leq j < h$, give the simultaneous version. The usual connected nilrotations with commuting translating elements, at any starting point, are a subcase using the lower central filtration.

We give the proof for maps whose logarithms in a basis adapted to the filtration have ordinary coordinate degrees at most the corresponding basis weights. This is the standard first-kind description of filtered polynomial maps on these groups. It includes every multivariate Taylor product of filtered binomial powers, by the finite BCH formula. In particular when the translating elements commute, one can start from the linear Lie algebra combination of their logarithms and then multiply by the constant starting representative. The elementary argument below can equivalently be read using just this coordinate hypothesis. For the group and polynomial-orbit background see Green and Tao and Leibman's polynomial orbit closure theorem.

Compact Taylor representatives in arbitrary source rank Order the multi-indices $\nu \in \mathbb{N}^n$ of total weight $|\nu| \leq s$ by a linear extension of the coordinatewise partial order that is unchanged by inserting zero coordinates. For example compare at the largest coordinate where two multi-indices differ, using increasing exponent there. Write $p_\nu(x) = \prod_i \binom{x_i}{\nu_i}$. Each polynomial map has a Taylor form in this order,

$$g(x) = \prod_\nu b_\nu^{p_\nu(x)}, b_\nu \in N_{|\nu|}.$$

Here is the elementary verification. Products and inverses preserve the degree condition on logarithms: in the BCH formula a bracket with inputs in filtration weights $j_1, \ldots, j_l$ lies in weight at least $\sum j_i$, and the multiplied polynomial coefficients have degree at most that sum. Make the Taylor product agree with g successively at $x = \nu$. The missing coefficient is

the residual there, since the exponent is one and later monomials vanish there. For $|\nu| > 0$ the residual's lower-weight log coordinates vanish: they are ordinary polynomials of degree $< |\nu|$, already zero at all coordinatewise lesser nodes, and their ν-th difference is zero. Thus $b_\nu \in N_{|\nu|}$. Each binomial correction remains in the coordinate class by its weight. After all nodes are matched the residual logarithm is identically zero, by the scalar multivariate Newton formula.

Every coefficient can moreover be put in a compact set $K_j \subset N_j$ depending only on the target. Choose compact representatives for $N_j/(N_j \cap \Gamma)$, possible by rationality. Process the same nodes in order. Right-multiply the current $g(x)$ by $\lambda_\nu^{-p_\nu(x)}$, with $\lambda_\nu \in N_{|\nu|} \cap \Gamma$ chosen so that $b_\nu \lambda_\nu^{-1} \in K_{|\nu|}$. This preserves its orbit coset at every integer x. Earlier node values remain unchanged, since the exponent vanishes there; at ν the coefficient becomes exactly $b_\nu \lambda_\nu^{-1}$. The map stays a polynomial of the same class, so the recursion applies to each subsequent node. Reduce also the constant coefficient and remove it on the left temporarily. On reinsertion it ranges over the fixed compact set K_0. This use of compact coefficient representatives, rather than just reducing endpoint coordinates modulo a lattice, controls arbitrary cross-coordinate motions and unbounded original lattice parts.

Homogeneous coefficients give precisely the required word moments Imagine coefficients already agree for equal ordered patterns of positive exponents. Write $u_{(e_1,\dots,e_a)}$ for the common element, ranging over $K_{e_1+\cdots+e_a}$. For a finite coordinate word $r = (r_1, \dots, r_b)$, take the resulting Taylor product without its constant coefficient, and denote its logarithm by $L_u(r)$. Inserting zeros anywhere in the word leaves the product unchanged, since the vanished factors are exactly those involving the inserted coordinates, and the chosen total order retains the relative order of the other factors. The map L_u has total ordinary degree at most s, with each log coordinate respecting its filtration weight.

The Newton coefficient at a multi-index supported on $i_1 < \cdots < i_a$ depends only on the restriction to those coordinate axes: it is the corresponding mixed difference at zero. By zero-insertion consistency it therefore depends only on the ordered exponent pattern $e = (e_1, \dots, e_a)$. Call this coefficient $C_e(u) \in \mathrm{Lie}(N)$. It is continuous in the finite compact parameter u. It lies in $\mathrm{Lie}(N_{|e|})$ by the coordinate degree bound. With $g_{\mathrm{w}}(r_i) = 1 + \sum_{j=1}^{s} \binom{r_i}{j} X_j$ denoting the word-signature letter,

$$L_u(r) = \sum_e C_e(u)[X_{e_1} \cdots X_{e_a}](\prod_i g_w(r_i)).$$

The bracket here denotes extraction of the indicated coefficient in the truncated word ring, not a Lie bracket. This identity holds for every number of slots. In particular it does not treat the nilmanifold as an abelian quotient or neglect ordering of the Taylor factors.

Use the fixed signed identity word of the preceding note [R095], at order s, and replace its counts b_j by $b_j + 2t$. The right side of the displayed formula, evaluated by integer group powers in the word ring even at negative repetition counts, is a Lie algebra polynomial $Q_u(t)$ with $Q_u(0) = 0$. Each signature coefficient of weight w has degree at most w in t, because expanding a power of $1+$ a positive-weight term introduces at most w factors in that coefficient. Hence $\exp Q_u(t)$ is an ordinary filtered polynomial sequence on N. At all large positive t the counts are nonnegative and its coset is exactly the homogeneous Taylor orbit of the corresponding literal velocity word.

By Leibman's one-parameter orbit closure theorem this polynomial nilsequence returns arbitrarily close to the identity coset at arbitrarily large positive t. One uses the residue class of zero in the finite periodic decomposition; the polynomial orbit is well distributed on the subnilmanifold containing the initial point there. Reinserting the compact K_0 factor preserves the desired closeness to its initial coset by uniform continuity. The strict return inequality is open in u and this constant factor, since $Q_u(t)$ at any fixed t is continuous in them. A finite subcover therefore bounds a successful t uniformly over these compact parameter spaces, with the error first taken smaller than $\eta/2$. This supplies a common finite bound B on the number of needed word slots.

Finally color the increasing a-tuples of input coordinate indices by sufficiently fine finite compact cells for all their Taylor coefficients with that support. Iterated finite Ramsey gives B homogeneous indices if the block size is large enough. On these indices each coefficient is close to a common representative for its exponent pattern. The cells can be fixed before Ramsey: there are finitely many candidate words, all with B or fewer slots and entries in the fixed finite velocity alphabet, and the corresponding products are uniformly continuous functions of their compact coefficients. The error from replacing them by their homogeneous representatives is less than $\eta/2$. The positive-repetition word found above, with zeros elsewhere, has odd marked tag and completes the proof.

One family for all connected finite-step targets The isomorphism classes of connected simply connected nilpotent Lie groups equipped with a lattice and rational connected filtration are countable. For example use a lattice-adapted Malcev basis, rational structure constants and rational subspaces for the filtration; choices of an integral lattice inside a rational hull are also countable. Choose one compatible metric per class and enumerate finite products, reciprocal integer tolerances and degrees. Increasing equal-depth block sizes and depth ratios to serve the first finitely many tests at each successive stage gives a family meeting every finite list on every tail. These block sizes can again be enlarged to make the reciprocal depth sums diverge. Finite products cover the full set of nearby starting points, generators and polynomials within each enumerated target, since the block bound was uniform over the Taylor coefficients. A rank-one integer mixed-radix image of each entire finite word packet retains this nilorbit property, as a scalar polynomial orbit pulled back along its coordinate-linear projection is still a multivariate polynomial orbit. To realize both the capacity and these nilorbit tests one must include the finitely bounded velocity words guaranteed by each stage in the transferred packets.

R088. Tensor tag cancellation against fixed nilmanifold polynomial tests

Scope and status. Result with proof.

Result. Dimension-uniform tensor tag cancellation holds for fixed nilmanifold polynomial tests and then compact finite-rank nilspace morphisms.

Scope. The target and tolerance are fixed before depth and replication; there is no uniform bound across arbitrary host actions.

Here is a dimension-uniform analogue of the scalar product lemma [R102] for a fixed ordinary nilmanifold target. The 2-adic depth requirement can now be an unspecified function of K, rather than linear in K. There is no dependence on the input dimension in that requirement.

Let $Y = N/\Gamma$ be a compact nilmanifold with N connected, simply connected and nilpotent, with a rational Mal'cev basis. Fix $s, \eta > 0$ and a continuous $F : Y \to \mathbb{C}$ bounded by one. Given an integer $L \geq 1$, there are finite d_*, n_*, depending on these data and L, such that

$$|E_\sigma b(\sigma) a v_w F(g(x_\sigma(w))\Gamma)(-1)^{\Sigma \sigma_i w_i}| < \eta$$

whenever $n \geq n_*$, $|b| \leq 1$, and the following hold. The logarithm of $g : \mathbb{Z}^n \to N$ has ordinary real-valued polynomial coordinates of degree $\leq s$, with arbitrary coefficients. The bits σ_i are independent fair bits. At a normal coordinate $x_i = b_i + l_i w_i$ with $l_i > 0$ dividing L. At a marked coordinate $x_i = c_i + z_i w_i$, with $2^{d_*} \mid z_i$. The fixed offsets are arbitrary. The constants work uniformly over any compact family of the tests F in the uniform norm. The average exists along Følner boxes; the estimates concern the completed average rather than a convergence rate.

A horizontal inverse independent of the number of input variables For any fixed target, degree, bounded Lipschitz test and positive discrepancy tolerance, there is a finite list $\mathcal{E}$ of nonzero horizontal characters and an integer M with the following property. If an orbit $g : \mathbb{Z}^n \to N$ has invariant average of the test differing by at least the tolerance from its Haar integral, then some $\ell \in \mathcal{E}$ has all increments of ℓg mod 1 killed by M. The same statement holds for compact families of continuous tests by approximation. A polynomial in logarithmic coordinates restricts to a group polynomial of bounded filtration degree on every line. For example repeat successive terms of the lower central filtration s times; BCH shows that each iterated derivative lowers the remaining weighted polynomial degree, or use the ordinary Taylor expansion in coordinates of the second kind, whose degrees are bounded in terms of the target and s. Enlarge the stretched filtration by a further fixed factor if needed. Only a degree bound independent of the input dimension matters.

For a detailed proof choose the finite list of bounded horizontal characters in Green and Tao, Theorem 2.9, with line tolerance small compared with the alleged discrepancy divided by the Lipschitz norm. Suppose a sequence of counterexamples has no common increment exponent for any of these characters, choosing successive forbidden exponents divisible by all integers up to the sequence index. Pass to subsequences so that for each ℓ the largest order attained among all mixed derivatives of each positive order is either bounded uniformly

or tends to infinity (infinite order included). Write $Q_\ell = \ell g \bmod 1$ and let $j_\ell \geq 1$ be the highest derivative order for which this bound is unbounded. After multiplication by one fixed integer S, all higher derivatives vanish for every ℓ. The derivative $D^{j_\ell}(SQ_\ell)$ is then constant in the starting point and multi-additive in the directions, with values of unbounded order. By polarization its diagonal polynomial

$$H_\ell(a) = D^{j_\ell}_{a,\dots,a}(SQ_\ell)$$

also has unbounded orders. It vanishes at zero. The scalar high-bias lemma [R102] implies that for each fixed nonzero integer v, the invariant averages of $e(vH_\ell(a))$ tend to zero across the counterexamples, uniformly over choices of invariant mean. Thus the distributions of these diagonal values are asymptotically Haar on the circle for each member of the fixed finite list. By a finite union bound, in each sufficiently late example there is one direction a with all $\|H_\ell(a)\| \geq \delta > 0$, for a common δ depending only on the list. For every start x, the j_ℓ-th binomial coefficient of $Q_\ell(x + ta)$ has circle norm at least δ/S. The corresponding $C^\infty[H]$ norm therefore exceeds the fixed Green–Tao obstruction bound for large enough H, uniformly over starting points. The line is equidistributed up to the required tolerance in the specified testing norm. Averaging its length-H expression over x in the allegedly discrepant invariant mean gives a contradiction. This proves the horizontal inverse. Replace a recovered character by its primitive factor if needed, increasing M by a bounded factor.

We use the standard fact that a continuous polynomial nilsequence on $\mathbb{Z}^n$ has the same mean under every invariant mean, including coordinatewise iterated means. One direct justification is the multiparameter polynomial orbit equidistribution theorem of Leibman, with finite progression cosets if the closed orbit has several connected components. Equidistribution over translated large boxes yields a unique invariant mean: bound its deviation by replacing the test with its sufficiently long box average pointwise. Products with a periodic scalar character are still nilsequences, by taking a product with a finite rotation. The input dimension is fixed within each invocation of this fact. Consequently all frozen-coordinate selections below may use iterated means in whichever order puts the chosen coordinates first.

Removing large horizontal coefficients Argue by induction on $\dim N$; for a point target any mask with an active coordinate has zero conditional tagged average. Suppose the absolute mixture mean is at least η. A positive proportion of masks, bounded below in terms of η, have conditional tagged mean of modulus bounded below. Such a good mask with at least one marked coordinate gives a positive discrepancy for F on at least one of the two parity cosets in w. This follows by subtracting the two coset means; a constant Haar expectation contributes zero to their difference. The all-normal mask has negligible probability. Apply the horizontal inverse on that coset, identified with a full-rank integer lattice coset. For its ℓ, the increments of $\ell g(x_\sigma(w))$ have bounded order there and consequently throughout $w \in \mathbb{Z}^n$, with new uniform order. Indeed along $w_0 + t(w - w_0)$, every even t stays in the chosen coset; scalar polynomial interpolation at $0, 2, \dots, 2s$ costs at most $s!2^s$. Use a fixed common exponent M after this interpolation, regardless of the mask. At subsequent deletions lower the bias by a total of less than $\eta/2$, so all lists and exponents may be fixed in advance at that smaller tolerance.

For every $\ell \in \mathcal{E}$ maintain a multiple U_ℓ of $Q_\ell = \ell g \bmod 1$ and its current upper degree j_ℓ, initially $U_\ell = 1$ and $j_\ell \leq s$. If some current polynomial is constant we have the desired

horizontal reduction. Otherwise build its bad-support hypergraph as in the scalar matching proof [R102], using the derivative $ML^{j_\ell}D^{j_\ell}(U_\ell Q_\ell)$ on coordinate directions and all supports of size at most j_ℓ. For a mask to work with the given ℓ, no such support can be entirely normal, by multi-additivity and $l_i \mid L$. If every ℓ had a matching of size t, a union bound over the list would put the total proportion of good masks at most $|\mathcal{E}|(1-2^{-s})^t$, apart from the all-normal mask. Thus some ℓ has a maximal matching of bounded size depending only on the list, s, η. Fix its bounded vertex cover to actual legal values while preserving the mixture bias up to a preassigned arbitrarily small loss, using iterated means and bounded mask amplitudes. On the remaining variables $ML^{j_\ell}U_\ell Q_\ell$ has degree at most $j_\ell - 1$. Update its multiplier and degree. The degrees of other polynomials do not increase on restriction. After at most $s|\mathcal{E}|$ rounds some nonzero primitive horizontal character has UQ_ℓ constant, with bounded U independent of n, d_*. The number of deleted coordinates is bounded. These bounds may depend on L, Y, s, η, which is allowed.

Induction in the target dimension Let $\ell : N \to \mathbb{R}$ denote the real lift of that primitive horizontal character, and $H = \ker \ell$, a simply connected closed rational nilpotent subgroup of strictly smaller dimension. Put $\Gamma_H = H \cap \Gamma$. Choose $X \in \mathrm{Lie}(N)$ with $\exp X \in \Gamma$, $\ell(\exp X) = 1$; this is possible by primitivity. Define

$$V(r) = g(r)\exp(-\ell(g(r))X) \in H.$$

Its logarithmic coordinates are ordinary polynomials of a bounded degree by BCH, independent of the input rank and coefficients. If $\ell(g(r)) \bmod 1 = t$ with $0 \le t < 1$, then

$$g(r)\Gamma = \exp(tX)[Ad_{\exp(-tX)}V(r)]\Gamma.$$

Indeed the remaining integer power of $\exp X$ can be dropped on the right. The t-section test $F_t(h\Gamma_H) = F(\exp(tX)h\Gamma)$ belongs to one compact family in the uniform norm: the map on $[0,1] \times H/\Gamma_H$ is well-defined and continuous. The subsidiary polynomial $g_t(r) = \mathrm{Ad}_{\exp(-tX)} V(r)$ takes values in H and has the same uniform finite degree bound.

Since UQ_ℓ is constant, there are at most U phases t, and $Q_\ell(r)$ depends only on the coordinates $r \bmod B$ with $B = Us!$ by binomial periodicity. Assume 2^{d_*} is divisible by the 2-part of B. Before sampling the mask, independently pre-sample in every coordinate a residue modulo B for its possible normal branch and a residue modulo the odd part B_{odd} for its possible active branch. In the normal branch integrate over the remaining $w = w^{(0)} + Bw'$. In the active branch use $w = w^{(1)} + B_{\mathrm{odd}}w'$. Both substitutions keep the correct uniform (invariant mean) distribution by ordinary finite-index splitting. In the latter branch $z_i B_{\mathrm{odd}}$ is divisible by B, while tag parity still depends on w' with coefficient one. The extra known sign arising from the active residue can be absorbed into $b(\sigma)$. The normal divisors now divide LB, and all active 2-adic depths remain unchanged.

For fixed pre-sampled residues, t is a function only of σ. Select a residue realization and one of the at most U phases whose contribution still has absolute size bounded below, at cost a factor at most U and an arbitrarily small tolerance. Absorb its mask selector in b. The resulting expression is exactly of the original kind for H/Γ_H, a target of smaller dimension, a bounded (possibly increased) polynomial degree, and one of the compact family F_t. By the

induction hypothesis its absolute size is smaller for sufficiently large n, d_*. Only finitely many horizontal characters, bounded multipliers and subgroup targets can arise at this stage. The corresponding constants depend on L and the original fixed data but not on n or the high depths. Deletions reduce n by a bounded quantity. This proves the claimed cancellation.

Fixed finite-rank nilspace morphisms The same assertion holds for $F(p(r))$, where $p : \mathcal{D}_1(\mathbb{Z}^n) \to Y$ is any morphism into a fixed compact finite-rank nilspace Y. Here Y need not be connected. We detail the presentation reduction rather than identify its cube structure with the lower-central filtration of a Lie group.

Use Candela–González-Sánchez–Szegedy, Theorem 6.2 and Definitions 1.3–1.4, and the lift from the compact-nilspace word proof [R069]. Thus $Y = F_0/\Gamma_0$ for a free nilspace whose layers are $\mathbb{R}^{a_j} \times \mathbb{Z}^{b_j}$, by a finitely generated fiber-transitive, fiber-discrete, fiber-cocompact lattice of triangular translations. The lift $g_j(r)$ has ordinary polynomial coordinates of degree at most j. Global freeness of this lattice is not assumed.

There exists an integer Q_0 such that the discrete layer labels of any inputs congruent modulo Q_0 can be matched by Γ_0, by the triangular procedure in the component period lemma [R106]. For clarity, at layer j choose lattice elements of degree j effecting m_j times each discrete unit vector. They leave earlier coordinates unchanged, and their powers act polynomially on all higher coordinates, with bounded degree and rational coefficients on the discrete labels (use integer binomial expansion). Discrete outputs do not depend on real inputs. Choose the divisibility thresholds backward so that the integer powers needed for exact matching at this layer preserve all required congruences at later layers, uniformly in the inputs. For each of the finitely many residue vectors ρ mod Q_0, fix a literal representative b. On the inputs congruent to this representative, the entire matching procedure uses powers that are rational polynomials in the original polynomial discrete coordinates, integer-valued there. All resulting real coordinates are ordinary real polynomials of uniformly bounded degree on that branch; the formulas also have polynomial extensions to all real inputs. Since g_j's discrete coordinates are integer-valued polynomials of bounded degree, the branch depends only on r mod C for $C = Q_0 s_0!$, with s_0 the fixed degree bound. Express the original test as a finite sum of these branch selectors times their corresponding real-component tests.

Fix a discrete component $F_b \simeq \mathbb{R}^a$. Denote by Γ_b the **faithful restricted image** of its stabilizer acting on F_b, quotienting out the elements that act trivially there. Its quotient is one connected component of Y: the saturated union of F_b and its Γ_0-translates is open and closed in the free space, and its image is connected. Compactness gives compactness of this component.

The faithful action on F_b is free and proper. To see the essential local point, consider a translation in the stabilizer which acts trivially on all earlier real layers of F_b. Then for every $y \in F_b$, y and its image agree on all earlier layers of the free nilspace (including their discrete coordinates). By fiber-transitivity, their displacement in the current j-th layer belongs to $\eta_j(\Gamma_0) \cap \Theta_j((F_0)_j)$, a discrete subgroup of the full structure group. The discrete part of the displacement is zero. As F_b is connected, this displacement is constant on F_b. For $j = 1$ the first-layer translation is constant directly. Consequently any first nontrivial layer of an element of the faithful image moves by a nonzero constant. This proves freeness. Induct on layers for properness of the faithful actions on their real truncations. For a compact set meeting its translate only finitely many lower-layer transformations can occur, by induction. Divide out chosen representatives. The remaining transformations have constant top-layer

displacement in a discrete subgroup, so only finitely many distinct ones can meet the given enlarged compact set. Thus the restricted group embeds discretely in the connected simply connected nilpotent group $G_\triangle$ of real triangular polynomial translations of these real layers. Those translations add bounded weighted-degree polynomials in preceding coordinates, so composition and inversion are triangular polynomial operations with a nilpotent weight filtration; the coefficient parameters give ordinary Euclidean coordinates. The restricted image Γ_b is finitely generated since it is a quotient of a subgroup of the finitely generated nilpotent Γ_0.

Use the real Mal'cev hull N_b of the discrete subgroup Γ_b in $G_\triangle$. It is a closed connected simply connected nilpotent Lie subgroup containing Γ_b as a cocompact lattice (the usual discrete subgroup theorem for simply connected nilpotent Lie groups). Its action on F_b is proper: write elements in a fixed compact set times Γ_b, and apply properness for Γ_b on an enlarged compact set. The stabilizers are compact and therefore trivial. The free proper Lie action has a smooth quotient F_b/N_b. This quotient is compact since the lattice quotient is compact. It is contractible: the principal bundle has contractible total space and contractible group N_b, so its manifold base is weakly contractible and has CW type. A compact connected boundaryless manifold of positive dimension has nontrivial top mod-two homology. Thus the quotient is a point, and N_b acts simply transitively on F_b.

The inverse orbit coordinate $F_b \to N_b$ is polynomial in real coordinates, in a fixed system of Lie exponential coordinates. Indeed use successive triangular layers. When the earlier coordinates have already been brought to zero, the subgroup fixing the earlier zero coordinates acts on the current layer by constant translations. The displacement homomorphism onto that layer is surjective by transitivity: any two points with earlier coordinates zero are related by an element of this subgroup. For the connected subgroup at the first stage, this map has a surjective Lie differential; its kernel is connected, as can be seen in exponential coordinates for a homomorphism from a simply connected nilpotent group into a real vector group. Induction keeps the subsequent stabilizing subgroup connected. Choose a linear right inverse of the differential, exponentiate the negative required displacement, and apply that group element. The group action and the exponential-coordinate multiplication are polynomial (triangular unipotence and BCH). Continuing through finitely many layers leaves zero, and inverting the product gives the desired polynomial orbit parameter. Consequently the fixed branch of the lifted real polynomial determines an ordinary bounded-degree polynomial map to $\Gamma_b\backslash N_b$, or to N_b/Γ_b by inversion. The branch test becomes a fixed continuous nilmanifold observable.

Finally absorb the branch r mod C in the arbitrary mask amplitude by pre-sampling both potential per-axis residues exactly as in the subgroup induction (full C on a normal branch and only C_{odd} on an active branch). Each resulting test has one of finitely many fixed ordinary targets and a degree bound determined by that finite-rank nilspace alone. Increase the depth and replication thresholds to handle each target and sum the finite list. This proves the asserted extension to CFR morphisms. It does not give a universal bound for all host actions.

R089. Nilpotent residual in the ordered native query group

Scope and status. Result with proof.

Result. The ordered native query problem has a nilpotent residual on its pair-component kernels, including odd-slot toggle variants.

Scope. This is not a reduction of the whole ambient free-group product, nor a six-slot ordered contradiction.

The commutator lifting argument for pair-connected query alphabets already appears in restricted-charge-trades.md [R218]. Here it is applied to the dyadic pair-component kernels and then to the antisymmetric odd-slot toggles, whose coarse quotient is a free product of involutions.

Fix the native alphabet $A = \mathbb{Z}/q\mathbb{Z}$, $q = 2^{D+1}$, with $z = q/2$, and h slots. Write $F_j = F(a_{j,x} : x \in A)$. For start x and velocity v put $E_{x,v} = (a_{j,x+jv})_{0\le j<h} \in \prod_j F_j$. Write P for the subgroup generated merely by the uncharged normal queries at velocities $0, 1$ for all starts. Thus P is contained in the zero-tag subgroup for any consistent ordered binary tag.

Let M be the largest power of two dividing any nonzero slot difference, and assume $M \mid z$. Define $L_j = \ker(F_j \to F(C_M))$ by sending each generator to the generator indexed by $x \bmod M$. Then

$$\prod_{j<h} \gamma_{h-1}(L_j) \subset P,$$

where $\gamma_1 L = L$, $\gamma_{a+1} L = [\gamma_a L, L]$. In particular the ordered query problem on the direct product of coarse-label kernels can be tested in a product of nilpotent groups of class at most $h-2$. The kernels may have infinite rank. This is not a reduction of the full ambient product of free groups to a nilpotent group.

Indeed in the j, p projection of P one gets all pair edge words for a bipartite graph whose components are the residue classes modulo $\gcd(q, j-p)$. The correspondence at velocity zero identifies the two index sets, and following a velocity-one edge and a reversed velocity-zero edge moves through a step $j-p$. More explicitly, two pair words with a common p-coordinate give $(a_{j,x'} a_{j,x}^{-1}, 1)$ on division. Multiplying along a path gives all such differences in the same graph component. Their F_j-normal closure can also be obtained in the projection because the projection of P onto F_j is onto. Since $\gcd(q, j-p) \mid M$, this includes $L_j \times \{1\}$. Thus each element of L_j lifts to a member of P with identity in any one chosen other slot p.

For a simple $(h-1)$-fold commutator of elements of L_j, lift its entries choosing all the other slots once. The commutator of the lifts then reduces to the chosen commutator in slot j and identity elsewhere, because an iterated multilinear commutator vanishes whenever any one input is the identity. Conjugates by any L_j-element are handled using surjectivity of the single projection. These generate $\gamma_{h-1}(L_j)$. Multiply the single-slot subgroups to prove the claim. The high corrections $E_{x,v+z} E_{x,v}^{-1}$ are in $\prod L_j$, since the index increment jz vanishes modulo M.

Odd-slot high-toggle version There is a smaller residual degree for normalized velocity toggles alone. Use

$$B_{x,v} = E_{x,v+z}E_{x,v}^{-1}.$$

Every even player cancels, and the odd player j sees $k_{j,x+jv} = a_{j,x+jv+z}a_{j,x+jv}^{-1}$. In a fixed slot these $k_{j,x}$ generate a free group D_j on the z representatives $0 \le x < z$, with $k_{j,x+z} = k_{j,x}^{-1}$. For instance a free-basis change from $(a_{j,x}, a_{j,x+z})$ to $(a_{j,x}, k_{j,x})$ proves freeness. Let $\nu \ge 2$ be the number of odd players, let M be the maximum 2-part of their pair differences, and define L'_j to be the kernel of the map from D_j to the free product of M copies of C_2, sending $k_{j,x}$ to the involution of its residue class modulo M.

For P' generated by the normal toggles $B_{x,0}, B_{x,1}$, one has

$$\prod_{odd j} \gamma_{\nu-1}(L'_j) \subset P'.$$

The pair edge argument again identifies the generators in each residue modulo $\gcd(q, j-p)$ when quotienting by the one-player normal kernel; since the inverse of each generator occurs at $x+z$ in the same class, these common generators also have square one in that quotient. Thus their pair projection contains L'_j in the isolated j-slot, and the commutator argument applies verbatim. At late levels $c = 1 + 2^k \equiv 1 \bmod M$. Hence the charged toggle divided by its normal analogue,

$$B_{x,c}B_{x,1}^{-1},$$

lies in $\prod L'_j$. In this restricted comparison any residual obstruction lives on a kernel of nilpotency class at most $\nu - 2$. This is complementary to the native abelian incidence bound [R105]; the abelianization of a subgroup need not inject into the abelianization of the ambient group. The conclusion neither asserts an ordered contradiction at six slots (one is impossible there) nor forces a high-density return.

R090. Paired word representatives of a finite IP system modulo passive components

Scope and status. Result with proof.

Result. Paired word representatives realize finite IP systems modulo passive components while meeting nilspace tests, with recurrence consequences for stated rational-component classes.

Scope. The component and relative-spectrum hypotheses must be retained.

Use the sufficiently replicated free-coordinate two-odd-center family [R107], with late parameters k, K, D, $c = 1 + 2^k$, and $P = \bigoplus 2^{K_i}\mathbb{Z}e_i$. Fix a scalar m, an IP rank R, a compact finite-rank nilspace Y, a point-nearness tolerance, and finitely many morphism tests from the block to Y. On a sufficiently late block one can find pairs v_i^0, v_i^1, $1 \le i \le R$, supported on successive disjoint groups of axes, such that

- $v_i^0 = v_i^1 \bmod mP$. Each obeys the normal/active velocity list. Its total tag is respectively zero or one.
- For all nonempty subsets $I \subset [R]$, and **every** choice $\sigma_i \in \{0, 1\}$ for $i \in I$, the vector $\sum_{i \in I} v_i^{\sigma_i}$ returns near the initial point in all the tests.

In particular the coset FS system generated by the $v_i^0 \bmod mP$ has a legal, returning representative in each nonempty sum: set one of its chosen bits to one. Include tests at all required temporal multiples by composition. Terminal zero slots do not participate in the word.

Use the integer word construction [R095] at truncated weight s, adding a weight-one variable X_* recording the **number** of letters. Thus

$$g(r) = 1 + X_* + \sum_{j=1}^{s} \binom{r}{j} X_j.$$

The triangular Newton expansion starts with the constant $b_0 = g(0)$; the correction at $j \ge 1$ has no term of weight below j by the same interpolation as before. The subgroup H_0 generated by $g(1 + m2^K j)$, $0 \le j \le s$, has finite index n in $H = \langle g(r) : r \in \mathbb{Z} \rangle$. The binomial determinant is $(m2^K)^{s(s+1)/2}$ and the lower-central commutator argument gives $n \le (m2^K)^{C_s}$, with $s + 1$ generators. Without restricting prime factors, the permutation image on the cosets has exponent dividing $\operatorname{lcm}(1, \dots, n)$, whose 2-part is at most n. Take $D \ge C_s(K + \log_2 m) + O_s(1)$, also $D \ge K + v_2(m)$. Choose an odd t making $2^D t$ divisible by both $m2^K$ and $s! \operatorname{lcm}(1, \dots, n)$. Then $g(c + 2^D t)g(c)^{-1} \in H_0$, as the two velocities have the same Taylor image in the coset action.

Write the signed marked identity as the two **adjacent** letters $g(c + 2^D t)g(c)^{-1}$ followed by a word in integer powers of the normal generators. The signed exponents of this normal word sum to zero, by the coefficient of X_*. In the unmarked version of that very same list replace the marked velocity by c, and all its normal velocities by 1. Leave all repetitions and their order fixed. The adjacent letters now cancel, and the remaining signed product is $g(1)^0 = 1$. The corresponding velocities in the two lists are congruent modulo $m2^K$ at each slot.

Concatenate R copies of this pair, zeroing the omitted groups of slots for each subset, meaning that no letters from omitted groups enter the signature. At signed repetition counts every concatenation has the identity signature, for every choice of the two versions. Add the same $2u$ to all repetition counts. For each version of each concatenation, the signature coefficients in weight j are polynomials in u of degree at most j. The homogeneous free lifts of the fixed morphism tests consequently produce finitely many morphisms of $\mathcal{D}_1(\mathbb{Z})$ into the fixed product nilspace, simultaneously equal to their initial values at $u = 0$. Positive recurrence gives one common u, large enough for all counts to be positive. Compactness on the homogeneous coefficient parameters bounds u before choosing the Ramsey packet. Apply the compact nilspace word proof [R069] with the enlarged finite concatenation list to obtain actual positive words within tolerance. Removing unused letters corresponds exactly to zeroing the associated input coordinates in the original ordinary Newton expansions, which use only positive exponents. The auxiliary X_* terms are needed in the group identity and are assigned zero coefficient in these evaluations. The parity of each marked tag remains unchanged by the positive repetitions. This proves both assertions.

Rational-component events In an arbitrary measure-preserving G-action every positive event in $\mathcal{Y} = \bigvee_{m\ge1} \mathcal{I}_{mP}$ has popular returns at all lengths. More precisely the lower bound depends only on its mass $a > 0$ and the length h. First take an event in $\mathcal{I}_{mP}$. For the unmarked generators ce_i in an equal-parameter block, the finite density Hales–Jewett theorem gives a nonempty subset sum with all-slot intersection $\ge c_0(a, h) > 0$. Indeed on the word space $\{0, \dots, h-1\}^R$ translate a random base point by the indicated generator sum, average the mass of good words, and then sum their line probabilities for finite fixed $R = R(a, h)$. Add $2^D te_i$ to one active coordinate, t the odd part of m, choosing $D - K \ge v_2(m)$. The change lies in mP, fixes the event and each translate, and gives an admissible odd tag. For the join $\mathcal{Y}$, use the increasing factorial subgroups and approximate in measure at error below $c_0(a/2, h)/(2h)$.

The same argument gives full recurrence on any compact profinite finite-step nilspace with its Haar probability and any abelian translation action. Every finite-rank image in a profinite nilspace presentation by finite nilspaces can be taken finite. For a finite target fibration π, the component periodicity lemma in the passive theorem [R106] gives a single M such that $\pi(U^{Mg}x) = \pi(x)$ for every g and x. Thus its events belong to $\mathcal{I}_{MP}$, whether or not π is action-consistent.

Combining an arbitrary rational-component factor with a connected nilspace Let X_0 be an arbitrary probability-preserving G-system generated by $\bigvee_m \mathcal{I}_{mP}$, and Z a compact finite-step nilspace with Haar probability and an arbitrary commuting G-translation action. Suppose Z is connected, or is an inverse limit of G-consistent compact finite-rank nilspace stages (the latter may be disconnected). On the **independent product** $X_0 \times Z$ all positive events have popular legal h-returns, with a bound depending only on their mass and h.

First, for any fixed finite-rank fibration $\pi : Z \to Y$ in the connected case, and finite collection of measurable sets $C \subset Y$ of Haar mass $\ge b > 0$, there are finite point-nearness tests at all temporal multiples forcing, simultaneously for these sets,

$$Haar_Z\{z : \pi(U^{jd}z) \in C \quad \text{for} \quad 0 \le j < h\} \ge c_1(b, h) > 0.$$

This is the connected-graph argument in the passive theorem [R106], with no ergodicity assumption now needed: every finite-rank refinement of Z is connected by continuous surjection. For G-consistent finite-rank stages fix one such stage as Y; its finitely many connected components are all positive. The identity component of its first-translation Lie group acts transitively on each component by Candela, González-Sánchez and Szegedy, Remark 5.3. Require point returns of a dense sample in each component. Eventually the translations fix all the components, and the graph on each component is connected homogeneous Haar by equivariance. The uniform diagonal-joining lemma in the measurable nilrotation proof [D10] applies to every component on which C has mass $\geq b/2$; their total mass is at least $b/2$. Finiteness of the list allows one common test family, retaining one $c_1(b,h)$ regardless of the number of listed sets. This assertion concerns graph laws for the fixed Haar nilspace, not arbitrary joinings with X_0.

Approximate an event of mass $\geq a$ on the product by an event A' of mass $\geq a/2$ which takes finitely many sections on $\pi(Z)$ as the base x varies, with the section index measurable in $\mathcal{I}_{mP}(X_0)$. The approximation error will be chosen using only a, h. For every list of h possible sections form their intersection C, and include in the finite point tests every such C of mass at least b, with $b > 0$ specified below. Choose the paired IP words above with $R = R(a/2, h)$. On $X_0 \times Z$ now acting on X_0 only, the same finite word averaging gives a nonempty subset I for which

$$E_{x \in X_0} Haar_Z(\bigcap_{j<h} A'_{U_0^{jw} x}) \geq c_0(a/2, h), w = \sum_{i \in I} v_i^0.$$

Take $b = c_0(a/2, h)/2$. The measure of x for which this common intersection has mass $\geq b$ is at least b. Choose its legal representative d by marking one of the summands. Since $d = w \bmod mP$, all corresponding base section indices coincide almost everywhere. For every such good x, the finite point tests give the lower bound $c_1(b, h)$ for the Z-return to the common intersection. Independence and Fubini yield full correlation at least $bc_1(b, h)$. Choose the approximation error below $bc_1(b, h)/(2h)$ and apply the common-marginal telescoping inequality to obtain a positive return for the original event.

No claim of independence of arbitrary ergodic components follows from the word identity. A passive nilspace bundle varying over its component factor is a remaining case.

The same positive-return assertion passes to every invariant probability absolutely continuous with respect to this product. Indeed restrict the event to a positive-measure part on which the invariant density is bounded below by a positive constant, apply the product result there, and transfer the resulting positive correlation. Its threshold constant in the new measure need not be uniform in this argument.

Relative discrete spectrum over a single scalar component algebra Suppose in an ergodic G-system X that mP fixes a factor Y pointwise, and that X/Y has relative discrete spectrum: finite-rank G-invariant $L^\infty(Y)$-modules span a dense subspace of $L^2(X)$ after closure. Every positive event has full-length popular recurrence, whether or not the extension group or its cocycle is abelian.

Let $H = \bigoplus_{\text{tail}} 2^{D_i} m_{\text{odd}} \mathbb{Z} e_i$, where the tail has $D_i - K_i \geq v_2(m)$. Then $H \subset mP$. On a rank-r module choose a measurable orthonormal frame in its conditional Hilbert fibers; the rank function is G-invariant and constant almost everywhere. On each fiber the countable

abelian H therefore acts by a commuting family of unitary r-by-r matrices. This family is simultaneously diagonalizable; its unordered multiset of characters in $\widehat{H}$ is measurable in the base. One justification is that the normalized traces at the countably many H-elements are its measurable Fourier coefficients and determine the empirical measure of its r characters. Commutation with G makes the empirical measure invariant under the base G-action, since the representations at two translated points are unitarily conjugate. Ergodicity makes it constant almost everywhere. The entire module is therefore contained in the span of at most r global eigenspaces of H, for example by the orthogonal character projections. Density of the modules shows the whole H-action has pure point spectrum. The same reasoning applies if only the indicator under test is approximable by these modules.

Apply the high pure-point theorem [R079] using H in place of the full high subgroup: the extra factor m_{odd} on every normal high generator is odd and leaves the tag character and its independent fresh singleton restrictions unchanged. Thus the legal parity flip can be made while moving the finite spectral approximation arbitrarily little, with positive baseline mass uniformly from density Hales–Jewett. Ergodic decomposition also applies when the relative finite-rank hypothesis is assumed on almost every ergodic component. The conclusion holds even on the higher cyclic covers and the free host.

R091. The high-bit program with no bound on interaction arity

Scope and status. Result with proof.

Result. The binary high-coordinate programming argument extends to unbounded interaction arity and arbitrary invariant measures satisfying its stated derivative identities.

Scope. Native truncated-ring actions or the explicit representation-free high-derivative class; not all stationary actions.

The antidiagonal binary programming proof for bounded interactions [R082] extends to arbitrary interaction order. Keep the same fixed-k,K cofinal sequence of strictly increasing D, and the same native $J = 2 + 2^{k+1}$ ring truncation. The supplied measurable generators may now have nonzero entries on edges of **unbounded** size. Even infinite coordinate edges are allowed when all incident ring actions commute and make sense on each module; these force the appropriate exponent restrictions through every one of their vertices. The ambient coefficient group is any invariant additive subgroup on which each finite-direction action is defined. No Haar assumption is made.

The only place that needed an arity bound was the construction and cancellation of the later-coordinate tables. Replace that step by the Graham–Rothschild theorem on parameter sets, with trivial alphabet group. For any finite alphabet Σ, finite colors and number d of parameters, sufficiently many coordinates ensure a d-parameter word W for which substituting all possible one-parameter words into W gives words of the same color. A parameter is allowed to occur several times; W may also have fixed alphabet letters. We use ordinary substitutions in which exactly one of d parameters is left variable and all others are fixed, which the full theorem includes.

For the finite tested coefficient list, use the finite alphabet of all exponents needed in the previous program: every $a + jr$ with $0 \le a < L$, $0 \le j < h$, $r \in \{0, 1, c\}$, including every fixed reference exponent. Color a one-parameter word at starting state χ as follows. Let i be the **last occurrence of the sole variable**. For each tested coefficient m record the binary high observation $(U_i^z - 1)m$ after assigning each fixed letter its coordinate exponent, and record its evaluation at all own exponents b modulo L. All the other variable occurrences are earlier than i, so however one substitutes for them they act trivially on its high observation. The own period divides L on the high layer. This is a bounded finite coloring independent of D and independent of arity. The color and the homogeneity event are invariant under every sufficiently early head: any edge seeing both i and a lower-depth coordinate is killed by 2^{D_i}.

Take one parameter for each subtype (a, r) in each of the m DHJ blocks from the preceding proof, and one for each reference subtype $(a, 0)$. Apply the theorem in a finite pool. For every χ some parameter word W works; therefore one fixed W has its homogeneity event of probability at least the reciprocal of the number of possible W, a constant determined before the ergodic head is chosen. In this W designate as each parameter's high representative its last occurrence. When all parameters receive prescribed exponents, leave any one parameter temporarily variable. By homogeneity its representative high table agrees with the common one. On substituting b in that parameter, the earlier occurrences have no effect on its representative, so its resulting high evaluation is precisely the common table value $F_m(b)$. This conclusion includes arbitrary simultaneous substitutions into all other parameters, without expanding a later-axis interaction into finite-arity pieces. Fixed alphabet letters in W simply

contribute a fixed initial group move.

For clarity, the common table as a function b satisfies the distinguished-axis polynomial identity: take any one-parameter subword, hold every *fixed* coordinate at its displayed exponent, and vary the last variable's exponent b. Its other variable occurrences are all earlier than that pivot. Thus the table is obtained by applying the powers u_i^b to the high layer at one genuine axis and measuring at a single fixed shifted state. The proof of (2) in the bounded-arity note [R082] applies to it.

Give every occurrence of the parameter for (a, r) initial exponent a and low motion r; give fixed letters in W their initial exponents and zero low motion. At each accelerated word point substitute $a + jr$. A true high flip is placed on just the representative of each requested subtype, **not on its entire variable block**. Thus the tag is exactly the odd sum of the c-subtype flip coefficients, irrespective of variable multiplicities. All repeated low motions on that same parameter besides its representative occur strictly earlier and are invisible to its high observation. The terminal reference high move once again realizes the table combination X^J at every word point. The other representative high moves realize either this same reference effect or zero, with the temporal switch $\binom{t}{5} \bmod 2$. The linearity of high toggles uses the zero product of two high operators on a common edge; this follows from the smaller incident additive exponent and does not depend on edge size.

The conditional-head argument, density Hales–Jewett, finite-observable approximation with its error chosen before the Ramsey number, and the final positive-return conclusion are exactly those of the invariant proof [R082]. Thus **all invariant measures in this unbounded-arity native ring class still have all-length threshold at most** $1/2$. The method did not construct the requisite algebra in general stationary systems; the top high layer of a wider native ring need not satisfy the displayed binary identity.

A representation-free formulation On a standard stationary action of the native direct sum, it suffices to have a generating collection of circle observables f such that the following conditions hold almost everywhere on the cofinal fixed levels, on a common conull invariant set. For each high generator $z_i e_i$, the multiplicative derivative of f takes values in $\{1, -1\}$, is invariant under all high toggles and under every lower-depth axis, and its orbit under the ith unit generator admits the binary relation $(U_i - I)^{J+1} = 0$ (write the derivative as a function to F_2). Require these properties for all the finitely many f used at a time; a common tail suffices. On that high derivative $U_i = 1 + X$ and $U_i^c = U_i(1 + X^p)$, so the same native identity (2) holds. Invariance under the full high subgroup gives the binary additivity of simultaneous flips. Invariance under earlier axes makes the marked-word coloring and its head conditioning valid. The proof above then applies to arbitrary invariant measures on such actions, irrespective of the unmarked dynamics. The ring modules verify these hypotheses, but the hypotheses are stronger than the abstract group relation that the native high generators are involutions.

R092. Rank loss for translated-pattern entropy arguments

Scope and status. Result with proof.

Result. Independent parity masks recoverable from long lines have rank at most a constant times the number of vertices times h to the power minus one third.

Scope. A bound on this particular parity-pattern certificate; it does not construct high-density avoiders.

This concerns the native direct sum of cyclic groups in the two-center construction [R107]. Write K_B for the group of top-bit shifts on a finite set B of axes, a copy of F_2^B. On an uncharged admissible difference r write $\eta(r)_i = 1$ when the ith low residue is active, and zero on the two normal classes. Pair its restriction to B with the bits of a shift w in K_B.

Suppose A in a measure-preserving action avoids all charged progressions of a fixed length h. Let C be a finite arbitrary pattern of integer or native positions, and select any family of uncharged h-lines lying in C. If the full translate $x + C$ lies in A, then its other highbit translates $x + w + C$ which also lie in A have $\eta(r) \cdot w = 0$ for **each** of the selected lines. Indeed on one such line, starting on its even slots from the first translate and its odd slots from the second gives a charged h-line when the pairing is one. Adding w preserves each allowed velocity: all top-bit choices are freely permitted, and only its pairing with the active mask changes the total tag. Thus on every finite highbit orbit the set of complete starts of C lies in one coset of the common kernel, or is empty. This is an almost-everywhere claim for a measurable avoiding action, by finitely many null-set exclusions. If q independent mask restrictions are present, the occupancy has measure at most 2^{-q} even conditional on the K_B-invariant factor.

However the best rank per vertex obtainable this way goes to zero uniformly as h increases, including arbitrary overlapping lines and arbitrary choices of the pattern. Set $N = |C|$, $T = \lfloor h/3 \rfloor$ for $h \geq 3$. Then

$$q \leq 2NT^{-1/3}/(1 - e^{-1}).$$

Indeed give each position y the two vector labels in the product of as many circles as there are axes in B,

$$Q_i(y) = y_i^2/2^{k_i+2} \bmod 1, L_i(y) = y_i/2^{k_i+1} \bmod 1.$$

They are well-defined on the native groups since the periods 2^{D_i+1} are multiples of their denominators. On ALL permitted velocities,

$$\eta(r)_i/2 = (r_i^2 - r_i)/2^{k_i+1} \bmod 1.$$

It is zero on the two normals because $K_i \geq k_i + 1$. At the active residue $1 + 2^{k_i}$ it is a half-turn and adding any high bit preserves this value. On any three consecutive line positions the mask divided by two is therefore recovered as $\Delta_r^2 Q - \Delta_r L$, using only the two labels at each of those three positions. Each selected line supplies T disjoint recovering triples when

its positions are distinct. For a native direction with a nonzero mask, its active coordinate is odd and has order 2^{D_i+1}. The disjointness applies provided that order is at least h; this holds on every sufficiently deep tail for fixed h. For integer patterns there is no restriction.

Select each position independently with probability $p = T^{-1/3}$. The chance of recovering a given independent mask from at least one full triple is $1 - (1 - p^3)^T \geq 1 - e^{-1}$. In each outcome all recovered order-two characters belong to the subgroup generated by the labels at the selected positions. An abelian group generated by m elements has at most m independent order-two elements, by the structure of finitely generated abelian groups. The expected recovered rank is therefore at most `2pN`, proving the claim.

In particular, even an independent-pattern occupancy lower bound a^N compared against 2^{-q} from this parity argument could not force a fixed cutoff $a < 1$ for all h. Real pattern probabilities need not have such a lower bound; the statement identifies a limitation of that specific certificate model. It does not show the existence of near-full avoiding sets.

R093. Two-center dyadic steps in phase-polynomial systems

Scope and status. Result with proof.

Result. Sufficiently replicated two-center dyadic directions force multiple recurrence on systems generated by global phase polynomials.

Scope. All finite lengths on the phase-polynomial factor, including free integer-coordinate lifts; no characteristicity assertion for arbitrary systems.

Every event of positive measure in the σ algebra generated by global circle phase polynomials has recurrent progressions of every finite length, on the **free integer-coordinate lift** of the dyadic construction as well as on its minimal native group. No ergodicity hypothesis is needed. A circle phase polynomial of degree $\leq s$ means that all its $(s+1)$-fold group differences vanish in the circle almost surely, simultaneously on the countable group. The degrees available in the generated factor need not be uniformly bounded. This is a class of systems and does not prove a universal bound on the critical density.

A bias lemma If a degree-s polynomial Q on a countable discrete abelian group has a nonzero limsup of the absolute averages of $e(Q)$ along any Følner sequence, the differences $Q(g) - Q(0)$ all lie in one finite subgroup of the circle. The order of a sufficient subgroup depends only on s and a positive lower bound δ on that limsup.

We use the following elementary univariate Weyl fact. If real circle polynomials of degree $\leq j$ on Z have a common constant j-th difference θ, and the order of θ is infinite or sufficiently large depending on j,δ, then their long mean exponentials have limsup modulus $< \delta$, uniformly in lower coefficients. Indeed apply $j-1$ finite van der Corput steps with positive lags at most a sufficiently large fixed R. The last linear coefficients are the products of those lags and θ, nonzero when the order exceeds R^{j-1}. Their geometric sums tend uniformly to zero in the constant term; all averaging errors tend to zero as R increases. At $j = 1$ this is the geometric sum itself. Averaging a Følner mean over a fixed initial segment of any cyclic direction does not change its limit. Thus if $Q(x + nt)$ has degree $\leq j$ and its j-th difference $\theta(t)$ is independent of x, δ-bias bounds the order of $\theta(t)$ uniformly in t.

Descend from $j = s$ to $j = 1$. Suppose all mixed derivatives of orders $> j$ are killed by a common integer q, initially $q = 1$. Put $m = s!$ q. The expansion of Δ_{mt} in powers of Δ_t has coefficients $\binom{m}{l}$, divisible by q for $1 \leq l \leq s$. Thus $\Delta_{mt}^{j+1} Q = 0$, $\Delta_{mt}^{j} Q$ is independent of x, and $\Delta_{mt}^{j} Q = m^j \Delta_t^j Q$ (all higher derivatives have a coefficient divisible by q). Use the Weyl fact to bound the orders of all pure j-fold derivatives. The polynomial qQ has degree $\leq j$, so its j-fold mixed derivative is a symmetric additive form. Polarizing its diagonal, with the possible factor j! retained, bounds the orders of all mixed derivatives of order j. This completes the descending induction. At $j = 1$ the claimed finite subgroup follows. In particular if no integer kills all $Q(g) - Q(0)$, the mean is zero along every Følner sequence. The lemma does not claim that a polynomial with values in a finite subgroup has nonzero mean.

The free group Let $G = \oplus Z e_i$, $P = \oplus 2^{K_i} Z e_i$. Use levels with $D/K \to \infty$ and arbitrarily many equal-depth axes as required by finite-density Hales–Jewett. At a chosen level normal velocities include all 0 mod 2^K, and a marked velocity is $c + 2^D t$, $c = 1 + 2^k$. These ingredients alone suffice for this proof. Write T^g for the action. Approximate 1_A in L^2 by

a uniformly bounded finite sum of global phase polynomials of some common maximal degree s, viewed multiplicatively by exponentiating their circle values. Indeed approximate by bounded continuous functions of finitely many phase observations on a finite torus, then by bounded trigonometric polynomials. For a fixed h, all terms in the approximating progression correlation have the form $e(Q_x(g))$ where $Q_x(g) = \sum_{j<h} \phi_j(T^{jg}x)$ is a polynomial on G. There are finitely many such tests and a common conull set of x. Errors in the integrated progression correlation are uniform in g by telescoping and invariance.

For each test and each x classify Q_x as rational on P if some integer kills all $Q_x(w)-Q_x(0)$ for $w \in P$. These are countable measurable conditions. On a set of measure arbitrarily close to one there is a fixed common integer q killing those increments for every test classified as rational. This q is chosen before the level. Choose now a sufficiently deep level K,D. Finite density Hales–Jewett supplies a nonempty set I from a fixed finite number of its axes for which the baseline $v = \sum_{i\in I} ce_i$ has unperturbed h-return mass at least $\kappa(a,h) > 0$ for A. Put

$$P_I = (\oplus_{i\in I} 2^{D+1} Z e_i) \oplus (\oplus_{i\notin I} 2^{K_i} Z e_i).$$

It has finite index in P. Fix $i_0 \in I$ and let z be a suitable odd multiple of $2^D e_{i_0}$. All $v+z+w$, $w \in P_I$, are legal. We compare the correlation averages on $v+P_I$ and $v+z+P_I$.

If Q_x is not rational on P, both coset averages are zero. Otherwise a positive-bias subsequence on $c'+P_I$ for $c' = v$ or $v+z$ would force finite-order increments on that coset by the bias lemma. Pick m with mP$\subset P_I$ and $mc' \in P_I$. For each $g \in P$ the polynomial $Q_x(c'+n(g-c'))$ at $n = 0$,m,. . .,sm has increments in the fixed finite subgroup; circle-valued polynomial interpolation gives the same at $n = 1$ up to a denominator s!m^s. In detail, choose a real univariate polynomial lift from its Newton coefficients and use the Lagrange formula; after multiplying by s!m^s all interpolation coefficients are integral. This works also for $g = 0$, so $Q_x(g) - Q_x(0)$ would have bounded order on P, contradiction.

If Q_x is rational on P with the selected common bound q, let G_l denote the subgroup of all axes of the chosen level. For any $g \in P + G_l$ we have $2^K g \in P$. The same interpolation on $Q_x(ng)$ gives $q' = qs!\ \ 2^{sK}$ as a common annihilator of $Q_x(g) - Q_x(0)$ on $P + G_l$. Every progression $n \mapsto Q_x(g + ne_{i_0})$ then has period s!q', since all its nonconstant Newton coefficients are killed by q' and binom(s!q',j) is divisible by q' for $1 \le j \le s$. As $D/K \to \infty$ we can ensure that the 2-part of s!q' divides 2^D. Choose z to be 2^D times the odd part of s!q', in coordinate i_0. Thus $Q_x(v+w+z) = Q_x(v+w)$ for all $w \in P_I$.

Consequently the two averages of the approximating finite sum differ by as little as desired: the irrational terms tend to zero for almost every x where that case holds, the rational terms agree on the chosen large-measure set, and dominated convergence applies to the finitely many terms. The L^2-approximation accuracy for A can be fixed before all this relative to κ and the uniform uncharged lower bound. The exception-set accuracy for q can in turn be chosen after the finite coefficients of the approximation are fixed.

To see the uncharged lower bound, take $C = \bigcap_{j<h} T^{-jv} A$ of mass $\ge \kappa$. Its ordinary h-term correlation under $w \in P_I$ is dominated by the correlation of A at $v+w$. Every Følner liminf for the ordinary correlation of C is $\ge \kappa'(\kappa,h) > 0$. Indeed for any fixed list of sufficiently many increments u_i, density Hales–Jewett on the random translate of their h-letter cube gives a sum of the ordinary correlations at nonempty subset sums at least a positive constant, with only boundedly many terms. Integrate these inequalities in the u_i

with an iterated invariant mean on P_I; every nonempty subset sum then has that same mean law. This proves the bound for every invariant mean and hence for the Følner liminf. The positive lower bound and the coset comparison give a legal return. This includes the actual free host and its deeper finite cyclic quotients by pullback.

A second, direct proof on the minimal native group On the native group the high subgroup K is generated by the involutions $z_i = 2^{D_i} e_i$. For a degree-s phase ϕ and $z \in K$, $\Delta_z \phi$ is killed by 2^s since $\Delta_z^j \phi = (-2)^{j-1} \Delta_z \phi$. For $B = s + \lfloor \log_2(s+1) \rfloor + 2$, the coefficients $\binom{2^B}{l}$ for $1 \le l \le s$ are divisible by 2^s. Hence $\Delta_{2^B g} \Delta_z \phi = 0$. On a sufficiently late tail $K \subset 2^B G$, so $z \mapsto \Delta_z \phi(x)$ is a K-character constant along the K-orbit of x. Its values belong to $\{0, 1/2\}$.

Use a finite bounded trigonometric approximation to 1_A as above. Choose m DHJ trial blocks each of L fresh axes with $k_i \ge B$. Independently sample bits σ_i on all these axes and give each axis the baseline speed $1 + 2^{k_i} \sigma_i$. On any nonempty union J of the trial blocks the total displacement v has the same value modulo $2^B G$ for all σ, and the active mask $I = \{i \in J : \sigma_i = 1\}$ is uniform on subsets of J. In the odd-slot approximating product at v, each phase term has a K-character at x depending on J and x but independent of σ, by the derivative identity. The projection onto χ_I has that term only where its K-character equals χ_I. For each x and each term this has probability at most $2^{-|J|}$, even though the phase value itself varies. Summing the finite coefficients, taking squared norms and using the finite number $\le 2^m$ of trial unions, choose L and then σ so that for every J its odd-slot transverse projection is as small as required and I is nonempty. The approximation transfers this bound to the true indicator product. DHJ gives one of those J with positive baseline mass $\ge \kappa$. As in the high subgroup argument [R079], the odd-tag high average is the invariant even/odd product term, of size at least κ^2, minus a term bounded by that transverse projection. This yields a legal return. This native argument also applies if the phase is polynomial for a subgroup $\oplus 2^{u_i} Z e_i$ and both $k_i - u_i$ and $D_i - u_i$ tend to infinity, since the same differencing only uses translations in this subgroup.

R094. Polynomial phases modulo arbitrary passive-component data

Scope and status. Result with proof.

Result. The polynomial-phase recurrence theorem extends to phases whose full-group highest derivatives are invariant under a scalar passive subgroup.

Scope. The mixed derivatives must involve the full acting group; a polynomial condition only along the passive subgroup is insufficient.

There is another full positive-recurrence conclusion for the free two-center dyadic host, including $K = k + 1$. Suppose a circle phase ϕ on a stationary $G = \bigoplus_i \mathbb{Z}e_i$-system has the following property for some s, m: all s-fold additive derivatives along **arbitrary full-group directions** are invariant under mP, where $P = \bigoplus_i 2^{K_i}\mathbb{Z}e_i$. There is no condition on the action on this invariant base. Then every positive event measurable in the σ algebra generated by phases of this kind (allowing s, m to vary) has all finite lengths of recurrence along the dyadic family, on arbitrarily late stages. Unlike a polynomial hypothesis along P alone, the mixed derivatives include arbitrary low directions. This includes global phases of bounded degree modulo a possibly infinite single scalar-component base.

Here is the stronger product estimate used in the proof. Fix $s, \eta > 0, m \geq 1$. At axes $1, \ldots, n$ use fair independent masks, normal speeds $r_i = 1 + Lw_i$ or active speeds $r_i = c + zw_i$, with $L = 2^K$, $z = 2^D$, $L \mid z$. The starting offsets can also be arbitrary integers on the two branches. Suppose $Q : \mathbb{Z}^n \to \mathbb{T}$ need not be a global polynomial, but every normal-grid difference

$$D_i(r) = Q(r + mLe_i) - Q(r)$$

is an ordinary circle polynomial of degree at most $s - 1$ on the full $\mathbb{Z}^n$. For $D \geq C_s K + C(s, \eta, m)$ and n sufficiently large in terms of s, η, m, one has

$$|E_\sigma b(\sigma) a v_w e(Q(r))(-1)^{\Sigma \sigma_i w_i}| < \eta$$

for every $|b| \leq 1$. Conditional averages exist and have a unique value: split w first into its residue classes modulo m. There each basis difference in the remaining variables is a finite sum of translates of an ordinary degree-$\leq s - 1$ polynomial, so the restricted function is an ordinary polynomial of degree $\leq s$. Apply the scalar high-bias lemma and uniqueness of means [R102]. To retain the parity test if m is even, one must not condition the **active** variable on its full m-residue in the argument comparing masks.

In detail put $m_i' = m$ on a normal branch and $m_i' = m_{\mathrm{odd}}$ on an active branch, assuming 2^{D-K} is divisible by the 2-part of m. Before selecting the mask, pre-sample one residue for each possible branch of each variable, and write $w_i = w_i^0 + m_i' w_i'$. At normals the input step is now $L' = mL$; at actives it is $z' = m_{\mathrm{odd}} z$, divisible by L'. Every active w_i' still has odd coefficient in the tag, with its fixed residue giving a mask amplitude. If the claimed mixture had large modulus one could fix the pre-samples retaining its bias up to an arbitrarily small loss. For any fixed mask, Q restricted to its new w' variables has degree $\leq s$, as just noted. Use the coordinate deletion method [R102], always keeping the original biased mixture on

the remaining axes. Fixing deleted coordinates does not change the global polynomial nature of their other grid differences.

On a positive proportion of good masks the scalar high-bias lemma gives a uniform even exponent $M = M(s, \eta/4)$ killing all increments of the untwisted Q on that slice. At every point of a good slice, if i is normal then in particular $MD_i = 0$. Initially take $U = 1$ and a common degree bound $j = s - 1$ for all the currently restricted UD_i. For each listed i and ordered j-tuple of remaining coordinate directions (repeats allowed), put an edge on their support **together with** i if

$$M(L')^j \Delta_{e_{i_1}} \cdots \Delta_{e_{i_j}} (UD_i) \neq 0.$$

The displayed top derivative is constant. For $j = 0$ the test is MUD_i itself, now constant. No good mask makes all the vertices of an edge normal: D_i at such a slice is a bounded-torsion normal grid increment, also at all normal translations in the listed directions; multi-additivity transfers their top increments from $L'e_i$ to e_i. There are at most $j+1 \leq s$ vertices in each edge, even after combining the tests over **all** possible i. As before, the matching number is $O_{s,\eta}(1)$, since t disjoint edges avoid an all-normal edge with probability at most $(1 - 2^{-s})^t$. Delete a bounded maximal-matching cover, fixing those coordinates to retain mixture bias up to a small preset loss. On all remaining variables, $M(L')^j UD_i$ has degree at most $j - 1$ simultaneously for all remaining i. At $j = 0$ it is identically zero. After s rounds we therefore have $UD_i = 0$ pointwise for all remaining i, where U divides $M^s (L')^{s(s-1)/2}$.

Along a line of grid increment $L'e_i$, D_i is a polynomial killed by U. Its values have period dividing $U(s - 1)!$ in that line parameter by binomial interpolation. Summing over U full periods gives zero in the circle. Consequently Q itself has period $L'U^2(s - 1)!$ in each remaining coordinate. With D as stated, some odd positive t makes $z't$ a multiple of this period. On a mask containing a mark, shifting its w'_i by t preserves the phase and negates the tag, giving conditional mean zero. The all-normal mask has probability at most $2^{-n+O_{s,\eta}(1)}$, contradicting the retained absolute bias for large n. This proves the product estimate.

To apply it, approximate the indicator by a bounded trigonometric combination of finitely many of the phases, with a common scalar multiple m and common derivative order s. For any fixed temporal product and initial point x,

$$Q_x(r) = \sum_{j<h} \phi_j(U^{jr} x).$$

Its mLe_i-difference is a sum of $\Delta_{jmLe_i} \phi_j(U^{jr} x)$. Each such difference has all its s-fold full-group derivatives zero, by commutation and the mP-invariance assumption. It is therefore an ordinary polynomial of degree at most $s - 1$ in r, on a common conull set by countability. The fixed sampled trial groups of the density Hales–Jewett mixture now have signed means arbitrarily small by the estimate, uniformly in x. The ordinary sum of line means has the same positive lower bound as in the relative scalar proof [R102], and bounded slotwise approximation completes the proof. This does not allow arbitrary P-polynomials whose low-direction derivatives have no such control.

R095. Polynomial recurrence for sufficiently replicated dyadic depth blocks

Scope and status. Result with proof.

Result. A nilpotent integral word identity gives simultaneous ordinary-polynomial recurrence for sufficiently replicated dyadic depth blocks.

Scope. Bounded-degree phase tests, and a stated bounded-arity residue-polynomial extension; not measurable recurrence on every system.

For every d, m and $\eta > 0$, all sufficiently large ratios D/K (depending only on d) have the following property. With sufficiently many axes having the same parameters (k, K, D) in the two-normal dyadic model [R107], every list of m zero-constant ordinary circle group polynomials of degree at most d on the free host has an allowed direction v with all m phases within η of zero. The number of axes can depend on the parameters, degree, tolerance and list length, but not on the polynomial coefficients. This also applies to polynomials on any finite cyclic lifts by pullback. Taking successively large equal-parameter blocks by diagonalization thus gives an actual family admitting no finite ordinary polynomial obstruction of any degree, even when cross-coordinate terms are allowed. It still has the fixed six-slot avoiding action. This establishes a recurrence test, not an all-action critical-density bound.

An integer word identity Use letters X_j, $1 \le j \le d$, of weights j, in the free associative algebra over the integers modulo all words of weight greater than d. Denote its augmentation ideal by J. The group $1 + J$ is nilpotent of class at most d. Put

$$g(r) = 1 + \sum_{j=1}^{d} \binom{r}{j} X_j, H = \langle g(r) : r \in Z \rangle.$$

There are fixed $b_l \in H$, $1 \le l \le d$, with $g(r) = \prod_{l=1}^{d} b_l^{\binom{r}{l}}$, in that order. To see this, successively make the product agree at $r = l$, which determines b_l since $\binom{l}{l} = 1$. At each stage $b_l - 1$ has no component of weight less than l: the existing product has coefficients polynomial in r of degree at most their weights, agreeing with the target at the l preceding nodes. Multiplying by the new binomial power preserves that degree property by expansion of the nilpotent powers. Finally agreement at $0, \dots, d$ proves equality componentwise for every integer, also at negative integers. In particular the b_l's generate H.

Write $L = 2^K$. Let H_0 be generated by the $d + 1$ normal letters $g(1 + Lj)$, $0 \le j \le d$. In the abelianization the binomial evaluation matrix $(\binom{1+Lj}{l})_{0 \le j,l \le d}$ has determinant $M = L^{d(d+1)/2}$, by the ordinary Vandermonde determinant and the leading coefficient $1/l!$. Its adjugate shows that the image of every b_l^M belongs to the image of H_0. If $\gamma_j(H)$ is the lower central series, the simple commutators of j generators generate $\gamma_j(H)/\gamma_{j+1}(H)$. Pick representatives in H_0 of b_l^M mod $\gamma_2(H)$. Their j-fold commutators belong to $H_0 \cap \gamma_j(H)$ and give the M^j-multiple of the corresponding generator commutators modulo γ_{j+1}, by multilinearity of the bracket on these graded quotients. Each quotient has at most d^j such generators. Consequently the index $n = [H : H_0]$ is a power of two and

$$n \le M^{\sum_{j=1}^{d} j d^j}.$$

Indeed multiply the successive abelian quotient indices bounded by $(M^j)^{d^j}$, using the finite lower-central series. The action of H on the n cosets gives a finite nilpotent image F. It is a 2-group: in a finite nilpotent group odd Sylow factors are normal, and a subgroup of power-of-two index contains them all, hence they lie in the core of the stabilizer. Every permutation in this 2-group has order at most n, dividing n, by its power-of-two cycle lengths. By the Taylor formula for g, its image in F is periodic with period $nd!$; $\binom{r+nd!}{j} - \binom{r}{j}$ is divisible by n for every $j \le d$.

Choose $D \ge \log_2 n + v_2(d!)$, guaranteed by $D \ge C_d K + O_d(1)$. Choose an odd positive t making $2^D t$ a multiple of $nd!$, and set $c = 1 + 2^k$. The element $g(c + 2^D t)g(c)^{-1}$ lies in the kernel of the coset action, hence in H_0. Therefore there is a finite word in integer powers of $g(c + 2^D t), g(c)$, and the $d+1$ normal $g(1 + Lj)$'s whose product is exactly the identity and whose first marked letter occurs once with exponent $+1$. The other marked letter has high digit zero. The sum of marked high digits counted with these integer exponents is odd.

Positive repetitions, homogeneous coefficients and allowed integer steps Write this fixed word as $W(r_1^{b_1} \cdots r_s^{b_s}) = \prod_{\ell=1}^{s} g(r_\ell)^{b_\ell}$, here using b_ℓ for the word's integer repetition counts. For a nonnegative count n_ℓ, this is precisely the ordered signature of a word with n_ℓ coordinate slots holding velocity r_ℓ. Its coefficient at word $X_{j_1} \cdots X_{j_a}$ is the sum over all increasing a-tuples of these coordinate slots of the products $\prod_\nu \binom{r_{i_\nu}}{j_\nu}$.

Let Θ be any additive map from the finite-rank augmentation ideal J into $\mathbb{T}^m$. Put all counts $n_\ell = b_\ell + 2u$. The signature minus identity is an integral-coordinate polynomial in u, vanishing at zero. Thus $\Theta(W - 1)$ returns to within $\eta/2$ of zero at arbitrarily large positive integers u. This is ordinary polynomial recurrence on a finite-dimensional torus. For example write the finitely many real coefficients in a rational basis with 1, clear all rational denominators by taking a multiple, and apply multivariate Weyl equidistribution to the remaining rationally independent coefficients and ordinary positive powers of that multiple. The parameter space of all maps Θ is a compact finite torus. The displayed strict return inequality supplies an open neighborhood of each map with the same u, so some finite upper bound for u suffices uniformly in Θ. Require u initially large enough to make every count nonnegative. Choose a uniform B bounding the total number of slots used.

On a finite free coordinate packet an ordinary degree-d polynomial P into $\mathbb{T}^m$, with $P(0) = 0$, has the multivariate Newton expansion

$$\sum_{a \le d} \sum_{j_1 + \cdots + j_a \le d, j_\nu \ge 1} \sum_{i_1 < \cdots < i_a} \theta_{j_1,\ldots,j_a}(i_1, \ldots, i_a) \prod_\nu \binom{x_{i_\nu}}{j_\nu}.$$

Here $a \ge 1$, and the identity follows by successive finite differences, also at negative integers. Color the increasing coordinate a-tuples by sufficiently fine finite torus cells for all their coefficients, for each $a \le d$. The finite Ramsey theorem iterated over those arities gives a homogeneous ordered subset of B coordinates, uniformly over the coefficients. Pick

representative values $\Theta(X_{j_1}\cdots X_{j_a})$ in their common cells. The cells may be fixed in advance so small that for every word using at most B slots from the fixed velocity list r_ℓ, P differs from $\Theta(W-1)$ by less than $\eta/2$. Indeed there are only finitely many monomials for this fixed B, with uniformly bounded integer binomial factors. Put the chosen positive repetitions in order on the homogeneous slots and assign velocity zero elsewhere. The resulting direction has odd total marked tag, because every signed repetition was changed by an even number, and all test phases are within the required tolerance.

For instance take a sequence of stages with $k\to\infty$, $K=3k$, choose $D/K\to\infty$ large enough for the degrees at each stage, and repeat the identical parameters sufficiently many times for the stage's degree, list size and tolerance; increase those complexities and decrease the tolerance. One may simultaneously require the sum of 2^{-k} over the block to exceed one. Applying the result on any sufficiently late such block shows arbitrary-tail polynomial recurrence. The proof concerns bounded ordinary degree, and does not replace an integral over continuously varying polynomial phases by finitely many phases uniformly over an unbounded family of steps.

Bounded coordinate arity and bounded residue period The same result holds for fixed a,d,M when each test on a finite packet is a sum over coordinate subsets I, $1\le|I|\le a$, of circle functions $f_I(x_I)$ which vanish if any input coordinate is zero. On each coordinatewise residue class modulo M, assume f_I is an ordinary polynomial of degree at most d in the integer parameters within that class. The coefficients, and the number of terms in the global sum, are arbitrary. This includes coordinatewise sums of functions which are piecewise polynomial on any uniformly bounded common period. It does not allow arbitrary unbounded-arity functions of the entire residue mask.

Use only nonzero velocities $r=c+My$ with the same residue as a mark. Normal velocities $1 \bmod 2^K$ in this progression are given by $y_0+L'j$, $L'=2^K/\gcd(M,2^K)$, once $k\ge v_2(M)$. As letters now use X_e of weight $e+1$ for $0\le e\le d$, truncate at $b=a+d$, and take

$$g(y)=1+\sum_{e=0}^{d}\binom{y}{e}X_e.$$

This time $g(0)$ is not the identity: it represents the actual marked velocity c, whereas **unused** axes contribute the empty word. The Taylor representation starts with $b_0=g(0)$; the same correction at $y=1,\dots,b$ works. The determinant at $y_0+L'j$, $0\le j\le b$, is $(L')^{b(b+1)/2}$, irrespective of the translate y_0. Thus the index proof and period bound go through with $b+1$ generators and nilpotency class b. Take odd t making $2^Dt/M$ integral and divisible by the Taylor period, which is possible for $D\ge C_{a,d}K+v_2(M)+O_{a,d}(1)$. An odd-tag identity word is obtained, and normal $y_0+L'j$ can be chosen with the corresponding r nonzero. Homogenize the Newton coefficients of the restrictions on the single nonzero residue pattern. Their exponents can now be zero in any participating coordinate: an arity-s monomial of total usual degree $\le d$ has weight at most $a+d$. Terms involving unused coordinates vanish by hypothesis. Positive repetitions and the finite uniform bound work as before.

Consequently even finite-channel open-cell obstructions coming from coordinatewise sums of uniformly residue-periodic polynomial observations fail in these sufficiently replicated

blocks. Indeed if $F(x + jd) - F(x)$, for each observed coordinate and each fixed j, has the stated form as a function of the axis velocities d_i, apply the simultaneous recurrence theorem with a fixed starting point interior to a color cell. All slots through the desired finite length can be put into that same cell. This observation about open quantization cells is weaker than a measurable recurrence theorem for arbitrary sets in an action.

R096. Profinite odometer cocycles cannot obstruct the dyadic two-center family

Scope and status. Result with proof.

Result. The two-center dyadic family has all positive-measure multiple returns in profinite-group Haar extensions of the independent cyclic-product base.

Scope. One compact profinite extension of the specified product base, including higher period covers; not arbitrary actions.

Consider the independent product Haar base $Y = \prod_i C_{Q_i}$, with its direct sum acting by coordinate translations. On infinitely many cofinally available axes let $L = 2^K$, $c = 1 + 2^k$ be fixed, $D_i \to \infty$, and $2^{D_i+1} \mid Q_i$. Permit normal velocities zero modulo L, the uncharged speed c, and the true speed $c + 2^{D_i}t$ for every odd t. Inactive axes have zero velocity. This is a subfamily of the two-center family when just one true axis is used and the others move at the normal velocities. The statement also applies on higher cyclic period covers. Every positive event returns to every fixed finite length along this family, using exactly one true axis, in every measurable compact profinite-group Haar extension of this base.

Use the finite-fiber tail-cocycle localization and separator argument of nonabelian-profinite-cocycles.md [R161]. Work first in a finite fiber group K_f. Each independent moving core has a designated one of the above cyclic axes; the partition can ensure this by picking successive sufficiently late block endpoints. All other coordinates in that core are passive. As in the cited argument, after fixing the separators one has independent core potentials and deterministic modal gauges g_i; multiplication in increasing core order gives the actual cocycle displacement after cancelling the fixed prefix and suffix. At each core choose, with fixed positive probabilities, no motion, a uniform speed $r \in LC_{Q_i}$ on the trial axis, or its speed c, always with independent uniform start. Let $P \le K_f^h$ be generated by the abundant absolute normal profiles, those with divergent sum of probabilities over the cores. The sum of normal failures outside P is finite. Use the finite subgroup K' generated by the abundant time-zero modal-gauge values; as before $\Delta K' \subset P \subset (K')^h$, and the negligible values outside K' can be changed to identity with summable error at every slot for *any* of the motions. The gauge makes an identity atom uniformly positive. Normal product laws on sufficiently many successive far cores become arbitrarily close to Haar on P, up to a summable tail error, by the finite-group mixing argument in that note.

For each slot j let $N_j \triangleleft K'$ consist of those elements whose isolated j-coordinate tuple belongs to P; normality follows from $\Delta K' \subset P$. Set $M = L\,\mathrm{lcm}(1, \dots, h-1)$, and write β_i for the uniform-normal failure probability on core i. For **any** integer v divisible by M, in the quotient K'/N_j,

$$\mathbb{P}[\Delta_v^{h-1} g_i(y) \neq 1] \le 2^{h-1}\beta_i, \qquad \Delta_v g(y) = g(y+v)g(y)^{-1}.$$

Only the trial-axis input is translated. Indeed for each other slot l use the parameter increment $(x, r) \mapsto (x - lt_l, r + t_l)$, $t_l = v/(j-l) \in L\mathbb{Z}$. Start with r uniform on LC_{Q_i} and x uniform independent; every vertex of this parameter cube has the same law, including the passive core data. Successively taking multiplicative differences kills each unwanted slot and leaves the asserted isolated derivative at slot j. If all the initial cube profiles are in the subgroup P, so is the isolated tuple. Its argument $y = x + jr$ is uniform given the passive data. A union bound and averaging prove the inequality uniformly in v.

Put $P_0 = (|K_f|^{h-1})!$. Along the trajectory at stride v, if the $(h-1)$st derivative is trivial at the first P_0 positions, its state of consecutive multiplicative derivatives through order $h-2$ follows a fixed permutation of at most $|K_f|^{h-1}$ states. Indeed $a(y+v) = \Delta_v a(y)\, a(y)$ gives the triangular update, reversible from its highest constant derivative downward. After P_0 updates the state returns. Consequently

$$\mathbb{P}[g_i(y + P_0 v) \neq g_i(y) \bmod N_j] \leq P_0 2^{h-1} \beta_i.$$

Choose a fixed odd integer t divisible by the odd part of MP_0. For all late cores $MP_0 \mid 2^{D_i} t$. Apply the last bound at each slot to compare the true profile at $r_* = c + 2^{D_i} t$ with the uncharged profile at speed c, using $v = j 2^{D_i} t / P_0$. Except on a set of probability tending to zero their entries coincide modulo every N_j. The uncharged c-profile lies in P with probability tending to one, since it is one of the sampled normals and failures are summable. Also $\prod_j N_j \subset P$. Thus the true profile belongs to P at most starts, meaning with probability tending to one.

Append one independent true core to as many normal cores as needed to obtain a whole interval with the same constant profile, indeed identity profile, at all slots with a fixed positive probability. On every sufficiently late tail we can take arbitrarily many such disjoint intervals. The tilted-trial argument restores each slot's base marginal arbitrarily close to the conditional product law, while only the chosen interval moves. It gives a positive return in the actual measurable event by cylinder approximation exactly as in the cited proof. Finite quotient approximation handles all profinite fibers: one first chooses a finite quotient approximating the indicator, then works with its gauged approximation, with tilted marginals controlling errors independently of the success probability. This is a result for a single isometric extension of the full independent product base, not for arbitrary actions of its translation group. No bound on the higher period multipliers $Q_i / 2^{D_i+1}$ enters the derivative argument.

R097. Relative high corrections when the pure center carries the mark

Scope and status. Result with proof.

Result. Relative high increments of bounded even exponent q give an all-length upper bound $1 - 1/(1 + q/2)$; odd exponent gives full recurrence.

Scope. The event must be generated by observables whose high increments are measurable over the passive invariant factor.

Let $M = 2p$, $p = 2^k$, $M \mid z_i$, and take directions with each coordinate either normal 1 or $p \bmod M$, or pure $z_i t$. The condition is that the sum of the t's over pure coordinates is odd. Let P be the subgroup of M-multiples on the homogeneous row. Suppose a finite or countable family of circle observables generates A, with the differences $\Delta_{z_i e_i}\phi$ measurable on $\mathcal{Y} = \mathcal{I}(P)$ and taking values of exponent dividing q, for one fixed even q; sufficiently late axes suffice separately for each observable. There is then an **all-h upper bound** $1 - 1/(1 + q/2)$ for this class, even with freely varying component-dependent high differences. For odd q, use a pure high multiple qz_i which is admissible and invisible on all these generators, giving every positive event recurrence.

For even q, use the conditional density-Hales–Jewett and Graham–Rothschild argument in the binary program [R101]. Take the good $\mathcal{Y}$-set B on which the conditional density of A exceeds $1 - 1/(1 + q/2)$ by a fixed positive amount, and the conditional error of a finite-observable approximation is as small as needed relative to the final DHJ grid size. The two Ramsey steps find a positive-measure base event where every low word vertex lies above B and every last-axis representative has the same finite q-valued table $F(a)$, $a \bmod M$, on the tested observables.

Each trial block has a pure velocity-0 parameter and a normal velocity-p parameter, both at initial low phase 0. The reference uses two other parameters of velocity 0 with fixed initial phases 0 and p. Along a candidate word line, correct one wildcard block by a positive high flip on its pure representative and a negative one on its normal-p representative. The true direction is admissible and has odd tag. At time t, its effect in the finite observations relative to the uncharged word is

$$t\,\big(F(0) - F(pt \bmod M)\big).$$

This vanishes at even t, and is tV at odd t, with $V = F(0) - F(p)$. At every word vertex the reference high translation (positive on the initial-0 reference representative, negative on the initial-p one) produces precisely V. Thus the true observations agree with one of the fixed $1 + q/2$ original group translates of that word vertex: zero, or an odd residue times the reference high translation. Their joint probability of lying in A, conditional on the homogeneous base event, is bounded below uniformly by the union bound, since each stays over B. DHJ on accepted vertices, followed by the conditional approximation error estimate over the fixed slots, proves the upper bound. For the special case $q = 2$ it is $1/2$. On a whole action with every high element of order at most q one already has the cruder universal $1 - 1/q$ upper just from running a single pure direction. The refinement here uses only relative observables, includes higher covers, and is independent of k. It does depend on q.

Merely postulating that all corrected profiles agree with one fixed reference at odd slots and zero at even slots on **arbitrary** higher-order coefficients faces an algebraic obstruction. In the pure-mark cyclotomic filter on a higher cover, multiply the normal-killing factor $(T-u)^2(T-u^p)^2$ by T^2-1 instead of $T-1$. The resulting operator kills every two-periodic temporal term, whereas its pure high output is twice the former top response since the slope on a pure axis is linear in time. On an adequately deep cyclotomic cover this has the same detectable parity after placing the distinguished scalar one 2-adic slice deeper. This observes a limitation of parity-periodic interpolation, not an avoiding event near density one.

R098. Pure high marks and two unmarked low centers

Scope and status. Proved subcases and unresolved extensions.

Result. Pure high marks admit a common six-term obstruction, while native involutions and specified compact limits give upper bounds and spectral restrictions.

Scope. The free integer-coordinate family's all-system critical value remains undetermined.

On integer axes write $L_i = 2^{K_i}$, $p_i = 2^{k_i}$, $z_i = 2^{D_i}$, where $k_i < K_i < D_i$. A coordinate is normal at residues 1 or p_i modulo L_i, with all lifts allowed. Other coordinates are pure high multiples $z_i t_i$, including zero. Require finite support and an odd sum of t_i over pure coordinates. The mask of high parity is now the complement of the finite normal support. Every odd multiple of $z_i e_i$ is itself an admissible direction.

Fixed short obstruction. In the finite cyclotomic dual-number rings of the two-center construction [R107], use

$$P(Y) = (Y-u)^2(Y-u^p)^2(Y-1), u = \zeta(1+\epsilon), \epsilon^2 = 0.$$

It vanishes at every normal coordinate since the deviation from the corresponding root is a multiple of ϵ. At a pure high velocity zt its value is $tz\epsilon(1-\zeta)^2(1-\zeta^p)^2$, whose cyclotomic valuation in the coefficient of $z\epsilon$ is $J = 2 + 2p$. Take the same truncation at $D\phi + J + 1$, $\phi = 2^{K-1}$. The remaining top element E_i is fixed under u and has order two; prescribe its global character to be a half-turn on each axis, and use the six coefficient observations. Their filter sums to a half-turn while six coinciding observations would give zero, since $P(1) = 0$. This gives a common six-slot short obstruction of positive stationary mass on the free axes as in the existing ring proof. The minimal native period 2^{D+1} is available when $J + 1 \le \phi$; for $K = k+1$ use the finite deeper ring cover when needed. The actual free family is well-defined for either choice of K.

Native upper and necessary spectral conditions. In the abstract product of the minimal native cycles each individual $z_i e_i$ is an involution, so any event of mass above one half has all-length recurrence along this very step. The free integer axes instead admit arbitrarily deep roots. The fixed-system odd-step formula [R170] proves that a putative endpoint-one action on the free group must have unbounded 2-primary rational eigenvalue orders on almost every ergodic component of every pure odd direction (including every combination of pure $z_i e_i$ with odd total coefficients). This is a restriction on subactions, and by itself does not identify a common global eigenfactor.

Partial return under compact high motion. If in an odd-block-symmetrized homogeneous row the fresh high operators $U_i(z)$ converge strongly to an involution R on the entire observed factor, then events of mass $a > 1/2$ return. Choose $B = A \cap R^{-1}A$, of mass at least $2a - 1$. In the pure case even each single high step gives a return, since its finitely many powers approximate R^t on A. In the original nearby-center model one instead chooses an unmarked DHJ baseline $c \sum_{i \in F} e_i$ giving an h-return for B of uniformly positive mass on far indices, and toggles any one such axis by ze_i. The finitely many affected slots change only by arbitrarily small errors on the approximated event. The odd block spectral theorem [R163] gives this strong conclusion automatically on the range of Q_z^2, but only weak convergence on its complement. The all-system cap below one for the free family remains unproved.

R099. The pure mask and weakly mixing coarse components

Scope and status. Result with proof.

Result. The pure-mark family has all positive-measure returns for relative high-eigenfunction events when the coarse Boolean component action is weakly mixing.

Scope. The specified relative eigenfunction class and coarse-component hypothesis are essential.

In the pure-center model on an infinite row, let $M = 2p$, $P \supset M\mathbb{Z}^{(\mathbb{N})}$ and $H = \bigoplus z_i \mathbb{Z} e_i \subset P$, with binary digit parity defined on every high coordinate (including on the free host). At every coordinate permit either a normal velocity $p \bmod M$ or a pure high velocity. The additional normal $1 \bmod M$ in the proposed full model is not needed for this theorem. Let $Y = \mathcal{I}(P)$. The low shifts U^{pe_i} give an action of $V = \mathbb{F}_2^{(\mathbb{N})}$ on Y, and suppose this action is weakly mixing. Consider relative eigenfunctions ϕ, with $|\phi| \le 1$, satisfying

$$U^g \phi(x) = \lambda(y)(g)\phi(x) \qquad (g \in H),$$

where $\lambda(y) \in \widehat{H}$ is a measurable function of $y \in Y$. No restriction is imposed on the orders of these characters. Every positive-measure event measurable over a family of these eigenfunctions has all lengths of recurrence on the legal directions.

Partition the row into m disjoint infinite sets, so that the $V_I = \mathbb{F}_2^{(I)}$-action on Y is weakly mixing on each part. Such a partition exists for any fixed m under the given hypothesis. Use an iid uniform coordinate coloring with m labels. Every nonatomic spectral measure ν on $\widehat{V} = \mathbb{F}_2^{\mathbb{N}}$ restricts atomlessly to the dual of each color class almost surely: two independent samples from ν differ in infinitely many coordinates almost surely, as finite-difference equivalence classes are countable. Each color meets their difference infinitely often with probability one. Fubini identifies the sum of squared atoms of the restricted measure with zero. Apply this to a countable dense list of mean-zero spectral tests. The constant subspace remains constant. Equivalently this establishes weak mixing in the usual product sense as well.

On each block I_l sample η_l uniformly from all binary masks on an increasingly long finite prefix of that block, yielding low direction $v_l = p \sum_{i \in I_l} \eta_{l,i} e_i$. Also choose an independent high correction g_l uniformly from a large Følner set in H. The ordinary DHJ sum over the lines of the word grid generated by $v_l + g_l$ has a positive lower bound $\kappa(a, h)$, uniform in all the sampling distributions, for $m = m(a, h)$. This follows by sampling the initial point and summing line probabilities. In a fixed line with wildcard set $J \neq \varnothing$, its actual step is $v_\eta + g$, where the low masks on distinct blocks combine without overlap, and $g = \sum_{l \in J} g_l$. The tag of this candidate is the character $\chi_{\text{all}} + \chi_\eta$ on H, with the additive notation for the binary characters. Averaging the initial point removes all the fixed shifts of that word line.

In a signed mean of orbital products, approximate the event indicator in L^1 by a finite combination of products of the relative eigenfunctions and their conjugates, uniformly bounded to a fixed constant, by Stone–Weierstrass and simple measurable approximation on the finite observation spaces. The individual monomials are relative eigenfunctions of the same kind. For one term $\prod_{j<h} \phi_j(U^{j(v_\eta + g)} x)$, averaging signed over the high variables makes the result zero unless

$$L(y) + R(U^{\upsilon_\eta} y) = \chi_{\mathrm{all}} + \chi_\eta, \qquad L(y) = \sum_{2|j} j\lambda_j(y), \quad R(y') = \sum_{2\nmid j} j\lambda_j(y').$$

Indeed the even low velocities fix Y, and all the odd ones have the same action there. For any fixed finite low sampling domain the high Følner averages converge after integration by bounded convergence and ordinary orthogonality of characters. Only domination of the summands by the probability of the displayed equality is required.

That probability tends to zero. Freeze all low masks except one included block I. Its other-block shifts merely replace R by a translate. For r selected coordinates in I, the uniform V_I-averages of $(y, U^{\upsilon_\eta} y, \eta|_{[r]})$ converge, on every product of bounded one-site tests and a bit-pattern test, to independent copies of the Y law and a uniform independent r-bit vector. This is the ordinary twisted mean ergodic theorem: the zero twist gives the constant projection, every nontrivial twist has zero projection. To make the upper bound uniform in the frozen other-block shift, either take the block averages successively (all bounds are uniform after the limit), or take all included boxes jointly. For joint averaging, pass the outside-block unitary means to the first argument; each is a contraction and commutes with the I-mean, whose finite-test error tends to zero in L^2. Under the one-site weak limit apply continuous bump majorants at the 2^r possible values of the right-hand side on these r coordinates of $\widehat{H}$; take disjoint-support bumps so their sum is at most one. Continuous functions of the two vectors of phase values are uniformly approximable by sums of products of one-site tests. Thus the limiting upper probability is at most 2^{-r}, and r is arbitrary. This proof uses neither finite phase order nor uniform smoothness of the eigenlabel fields.

The finite DHJ sum of signed terms is smaller than its positive unweighted lower bound with the approximation error chosen first and then sufficiently long boxes and high Følner samples. Expanding the odd-tag indicator gives positive correlation at an actual admissible direction. This establishes the assertion. The same proof permits a finite-dimensional subgroup of the global finite-support V-eigencharacters on an ergodic Y: a typical partition as above preserves atomlessness of the continuous spectrum and restricts all infinite-support eigencharacters to infinite-support characters a.s.; then select arbitrary finite lists of bits admitting no nontrivial eigen-twist supported on them. The tuple lemma also extends relative to a V-invariant base by disintegration if this finite-rank condition holds on its ergodic components and the necessary infinite block partition works on almost every component; one must retain the relative, not independent, endpoint joining in that case.

R100. Freezing a bounded number of randomly chosen high axes

Scope and status. Result with proof.

Result. Freezing a bounded random coordinate set changes a product law by $O(n^{-1/2})$ in L2 and total variation.

Scope. Constants depend on the fixed local group and cube order; no uniform cover-depth or recurrence consequence is asserted.

This observation is about a fixed finite local group, so it has no uniformity in cover depth. In a large equal-depth native packet, restricting high directions on a small randomly chosen active set does not incur an exponential loss in a global cube-uniformity estimate.

Let μ be any probability law, $\nu \ll \mu$ with $M = \int (d\nu/d\mu)^2 \, d\mu < \infty$, and fix b. Mix the n-fold products with law ν on each coordinate of a uniform b-subset I and μ elsewhere. The $L^2(\mu^n)$-distance of the likelihood from one is

$$(E_{I,J} M^{|I \cap J|} - 1)^{1/2} = O_{M,b}(n^{-1/2}).$$

Indeed expand the square using independent uniform I, J. At a coordinate occurring once the likelihood factor has mean one; at a common coordinate its second moment is M. The chance of any overlap is $O(b^2/n)$, and its contribution is bounded by M^b. The same conclusion holds in total variation. No exchangeability of the function tested against this mixture is needed.

In particular, take μ to sample a uniform base point and s independent uniform increments in a fixed finite abelian group H_0, and ν to sample the uniform base point with all increments zero. Then $M = |H_0|^s$. For every bounded $f : H_0^n \to \mathbb{C}$, the mean over I of the fiber-averaged 2^s-power of its U^s-norm computed with increments in the complementary coordinate subgroup equals its ordinary U^s-norm power plus $O_{H_0,s,b}(n^{-1/2})$, for $|f| \le 1$. The test is just the signed cube product. The same estimate holds after averaging a family f_y over independent coarse data y. Uniform independence is still available if y is shifted by any function of I in a finite Haar rotation. Markov and union bounds allow one to choose a uniformly random fixed-size search set and impose estimates on each bounded subset later selected from it. The finite size and number of such tests must be fixed before taking $n \to \infty$.

This provides one possible route to high-coordinate arithmetic regularity: regularize separately on the cosets of a passive subgroup, control a uniform remainder there, and freeze a bounded set of prospective active coordinates. It does not align different structured functions on different coarse cosets, give a common bound through all cover depths, or establish recurrence.

R101. Full recurrence with arbitrary component-dependent high bits at the short shadow

Scope and status. Result with proof.

Result. Component-dependent binary high increments at the short shadow admit full positive recurrence; bounded higher order gives an order-dependent threshold.

Scope. The high derivatives are passive-factor measurable; arbitrary higher-cover obstructions need not satisfy this.

Use an infinite homogeneous row with $M = 2p$, $p = 2^k$, normal velocities $0, 1 \bmod M$, active velocities $c + zt$ with $c = 1 + p$, $z = 2^D$, and require odd total active t. Each use of a nonzero residue-1 normal allows its arbitrary M-multiple correction. Let P contain M times every row direction and suppose the high elements ze_i belong to it. Work in a countable probability-preserving abelian action, including actions on cyclic quotients. Assume there is a collection of circle observables ϕ generating the event of interest such that on every sufficiently late row axis

- $\phi(U^{ze_i}x) - \phi(x)$ takes values in $\{0, 1/2\}$, and
- this difference is measurable over $\mathcal{Y} = \mathcal{I}(P)$.

The tail requirement may vary with the observable. Relative degree-one phases for the P-action have this property on the native group where every ze_i has order two. The subgroup P may have a nonatomic invariant factor, and the binary functions may depend on the coarse coordinates in any fashion. Then **every positive event in this generated σ algebra has legal recurrence at every finite length.**

Write $f_i(x) = 2[\phi(U^{ze_i}x) - \phi(x)]$ in bit notation, i.e. the two possible circle increments are recorded as bits 0 and 1 (the displayed factor is taken on their representatives 0 and 1/2). Multiple high flips add their bits, since these bits are invariant under the whole P. It suffices to use an arbitrary finite list of observables and corresponding tuples of bits.

Here is a density argument that permits conditioning on an extremely rare homogeneous word event, with no invariance under any low head required. For the positive event A, choose a fixed $a > 0$ so that $\{\mathbb{E}(1_A \mid \mathcal{Y}) > a\}$ has positive measure. Choose m by density Hales–Jewett for the alphabet $[0, h-1]$ and density $a/2$. If each of its h^m words has acceptance probability $\geq a$, the probability of having a combinatorial line of accepted vertices is bounded below by a positive constant depending only on a. Fix $\epsilon > 0$ small relative to this bound divided by the number of all tested word and true-line slots, a quantity depending only on h, m. Approximate A in measure by a finite-observable event A' so accurately that the following set has positive measure:

$$B = \{\mathbb{E}(1_A \mid \mathcal{Y}) > a,\ \mathbb{E}(1_{A \triangle A'} \mid \mathcal{Y}) < \epsilon\}.$$

Consider an arbitrarily late but sufficiently large pool of row coordinates and the full word grid $\{0, \dots, M-1\}^N$, with a letter indicating the low shift on its coordinate. Every leaf has marginal probability $\mu(B)$ of lying above B. With positive probability in the initial $\mathcal{Y}$-state, the density of these leaves exceeds $\mu(B)/2$. Multidimensional density Hales–Jewett gives a large-dimensional parameter subword whose **every leaf lies above** B. Choose its dimension to support the next application of Graham–Rothschild. For a one-parameter subword, with

fixed low letters at its fixed coordinates, designate the last original axis i occupied by its variable. Its color is the finite tuple of its f_i-values for all the tested observables and all $b \bmod M$, when **all** occurrences of the variable take b. The M-period here is exact because all bits are $\mathcal{I}(P)$-measurable. Apply the ordinary Graham–Rothschild parameter-sets theorem to this vector coloring to obtain a subword with $2m$ parameters homogeneous on all one-parameter subwords. A parameter of the original subword can have repeated occurrences. The designated representative is always just its last actual original axis and the table varies all equal letters at once, so repetitions cause no change in this argument. Density Hales–Jewett and Graham–Rothschild are being used in this order. Some fixed final word achieves all of these properties on a success event $E \in \mathcal{Y}$ of strictly positive measure, by the finite number of choices.

Assign two parameters to each trial block. On the first use initial letter 0, velocity c; on the second use initial p, velocity 1. Low letters may be reduced modulo M for the **component tests and tables**, since the residue difference is in P. For each h-letter word on the m blocks use its actual integer shifts with these initial letters and velocities; fixed letters remain fixed. Every resulting low word vertex is above B on E, so after conditioning on E each has probability at least a of lying in A. The positive lower bound for finding an accepted word line therefore applies. Choose the first wildcard block of each prospective line. To that line's baseline difference add ze_i on both representatives of this particular block, one active and one normal. This is an admissible true difference and its tag is odd. At even times the high correction gives zero bit increment for every observable. At each odd time t, $ct \equiv p + t \bmod M$. The two representative high tables at that word vertex coincide by homogeneity, using the one-parameter subword in which the chosen parameter is variable and all others are fixed at their actual letters. Thus the two high bits cancel, and the true slot and the uncharged word slot have identical values on every tested observable. The fact that the other original axes for a parameter also move is included in the definition of its table. The high correction itself uses only one actual representative per parameter.

Every true slot above this same base event also lies above B, since its difference from the word slot belongs to P. Conditional on E, every potentially tested slot consequently has probability less than ϵ of an A/A' error. A union bound over the fixed final word grid and its candidate true lines shows positive conditional probability of a valid A-line with no errors at the relevant sites. The equality of the finite observables transfers membership at all the true slots. Finitely many possible returned directions give an actual direction of positive measure. This proves the claim without any quantitative lower bound for $\mu(E)$. The proof also works at varying depths D_i, as long as k, K are constant throughout each arbitrarily large pool.

In the original higher-cover short-shadow witness, the criterion need not hold: an increment at ze_i can have order greater than two even if it is constant on P-fibers. The proof asserts neither recurrence for all covers nor an all-system upper bound there. For higher bounded order q, the same conditional method can compare to at most $\operatorname{lcm}(M, q)$ reference states: introduce reference parameters with fixed initial letters $b \bmod M$ and zero speed. A true single active high flip has response $tF(ct)$, agreeing at time t with the reference translate by tze_{i_b}, $b = ct \bmod M$, on all the finite tested q-valued $\mathcal{Y}$-measurable increments. Each reference translation may depend on $t \bmod \operatorname{lcm}(M, q)$, but is fixed across word vertices. If $\mu(A) > 1 - 1/\operatorname{lcm}(M, q)$, choose B with conditional density above the same threshold by a fixed positive amount, and require all the reference translates simultaneously in A at every accepted vertex. Their probability is uniformly positive by the union bound. The rest of the

proof is identical, using the same low word pattern and a single chosen active high flip. This gives an order-dependent all-h upper for this relative class, including actions with nonatomic P-components.

R102. Product tag cancellation against ordinary polynomial phases, with relative fibre coefficients

Scope and status. Result with proof.

Result. Product parity tags cancel uniformly against bounded-degree polynomial phases, including the stated relative-fiber version.

Scope. The test subgroup must contain all low switches; unrestricted passive-relative phases are excluded.

Fix a degree s and tolerance $\eta > 0$. At n coordinates choose independent fair bits σ_i. At a normal coordinate ($\sigma_i = 0$) use integer parameter $x_i = b_i + l_i w_i$, and at a marked coordinate use $x_i = c_i + z_i w_i$. Here b_i, c_i are arbitrary integers, every positive l_i divides $L = 2^K$, and each z_i is a power of two divisible by 2^d. Independently average w_i over long integer intervals. The tag character is $(-1)^{\sum_i \sigma_i w_i}$.

If $d \geq [s(s+1)/2]K + C(s,\eta)$ and $n \geq N(s,\eta)$, then every ordinary polynomial $Q : \mathbb{Z}^n \to \mathbb{T}$ of degree at most s satisfies

$$|E_\sigma a v_w e(Q(x))(-1)^{\sum_i \sigma_i w_i}| < \eta.$$

The constants are independent of the dimension, the shifts, and the polynomial coefficients. Here the average is the limit along growing Følner boxes; no rate uniform in Q is asserted.

Use the following high-bias lemma: for each $s, \theta > 0$ there is $M = M(s,\theta)$ such that if the absolute invariant mean of a degree-s circle polynomial on a finite-rank free abelian group exceeds θ, then M kills all its increments. One proof bounds the leading directional coefficient by averaging on long integer lines and applying univariate Weyl differencing $s-1$ times. The bound on a coefficient of order q tends to zero with q, uniformly in all lower coefficients: the relevant van der Corput bound after taking the 2^{s-1} power is at most the density of tuples of $s-1$ integers with product divisible by q, at most $(s-1)\tau(q)q^{-1/(s-1)}$. An irrational coefficient gives zero; degree one follows from character orthogonality. Polarization gives uniformly bounded torsion of all mixed top derivatives. Descend one degree at a time: if all higher mixed derivatives are killed by a common integer M', take in each direction g a sufficiently divisible fixed multiple mg, e.g. m divisible by $M's!$. The expansion of $\Delta_{mg} = (1 + \Delta_g)^m - 1$ shows its jth power is $m^j \Delta_g^j$ up to terms of higher order with coefficients divisible by M'; its jth derivative is independent of starting point after the higher mixed derivatives are killed. The higher along-line derivatives vanish. The same univariate argument bounds the order of $\Delta_{mg}^j Q$ uniformly in g. Recover bounded torsion of $\Delta_g^j Q$ and polarize in the quotient by the bounded subgroup containing the higher derivatives, where the polynomial has degree $\leq j$. This induction reaches the first derivatives. For any invariant mean the comparison with a length-N line average is valid by invariance and positivity. On $\mathbb{Z}^n$ polynomial phases have a unique invariant mean: the lemma gives zero for every invariant mean unless all increments have bounded order, and in that case the polynomial is periodic (use its binomial expansion with a period divisible by $Ms!$). Consequently one may use iterated means in any order or Følner-box limits for these polynomial phases.

Assume the displayed product mean has absolute value at least η. We delete a bounded collection of coordinates at each of at most s rounds, fixing their legal values to preserve an absolute mean at least $\eta/2$ at every round. This is possible by using iterated means

with the deleted coordinates outermost: the retained-axes inner average exists and the full value cannot exceed the supremum of the conditional absolute values. Reduce by less than $\eta/(3s+3)$ per round if necessary. Fixed-coordinate tag signs do not alter an absolute value.

At any stage a proportion at least $\eta/4$ of masks on the unfixed axes have conditional absolute mean $> \eta/4$ (adjust the harmless constants if equality occurs). For each such good mask, apply the high-bias lemma to its polynomial in w, including the linear tag $\sum_i \sigma_i w_i/2$. Take an even multiple $M = M(s, \eta/4)$ of the lemma's output, independent of the mask and dimensions. It kills all increments of the *untwisted* $Q(x_\sigma(w))$ as well.

Suppose the currently remaining polynomial UQ has degree $\le j$, initially $U = 1,\ j = s$. For an ordered j-tuple of coordinate directions, repetitions allowed, put an edge on the underlying set of coordinates if

$$ML^j \Delta_{e_{i_1}} \cdots \Delta_{e_{i_j}} (UQ) \neq 0.$$

The derivative on the left is constant in the input, since it is the top derivative of UQ. No good mask can have every coordinate of an edge normal. Indeed on that slice the indicated mixed derivative with increments $l_{i_1} e_{i_1}, \ldots, l_{i_j} e_{i_j}$ is M-torsion by the increment bound on the good slice. Multi-additivity of the top derivative permits replacing each l_i with L. If the hypergraph had t disjoint edges, independent fair bits would give probability at most $(1 - 2^{-j})^t$ of avoiding an all-normal edge. Since good masks have probability $\ge \eta/4$, the matching number is $O_{s,\eta}(1)$. Delete all the vertices of any maximal matching, a vertex cover of size $O_{s,\eta}(1)$, fixing their values as above. On the remaining coordinates all the indicated mixed j-derivatives vanish identically, so ML^jUQ now has degree at most $j-1$. This reduces the degree of the multiple, while the biased correlation and further tests always concern the original restricted Q.

After all rounds UQ is constant on the remaining coordinates, with $U = M^s L^{s(s+1)/2}$. Fewer than $O_{s,\eta}(1)$ coordinates were deleted. For each remaining i a period of Q in the e_i-direction is $Us!$, by the one-variable binomial expansion on each line: the nonconstant coefficients are killed by U, and $\binom{n+Us!}{j} - \binom{n}{j}$ is divisible by U. Thus some **odd** t makes $z_i t$ a period for all i, by the lower bound on d. (Take an odd multiple of the odd part of $Us!$.) If a mask has any marked coordinate, shifting its w_i by t preserves Q and negates the tag, so its limiting mean is zero. Only the all-normal mask can contribute; its probability is at most $2^{-n+O_{s,\eta}(1)}$. This contradicts the retained $\eta/2$ bound for large n. The same proof works when z_i merely have 2-adic valuation at least d.

Relative phase-polynomial recurrence on the free dyadic host Let $G = \bigoplus_i \mathbb{Z} e_i$, with sufficiently replicated two-center dyadic stages and $D/K \to \infty$. Suppose A lies in the σ algebra generated by circle phases that, for each individual phase ϕ, are ordinary phase-polynomials of some finite degree along a subgroup

$$R_\phi = \oplus_i 2^{u_i} Z e_i, 0 \le u_i \le k_i on \quad \text{all} \quad sufficientlylateaxes.$$

These are measure-theoretic phases: the additive $(s+1)$-fold derivative along the subgroup vanishes almost everywhere. Values and degrees of coefficients can vary measurably across

its ergodic components, and no global degree on G is required. Then every positive-measure A has positive returns of every finite length along legal directions on arbitrarily late stages.

First approximate 1_A in L^1 by a trigonometric polynomial of finitely many such phases, bounded by $1+\epsilon$. This follows by continuous approximation under their joint torus law and uniform trigonometric approximation. The common subgroup has the maximum of their exponents u_i, still $\le k_i$ eventually. Fix h, take m trial groups by finite density Hales–Jewett, and use N distinct identical-level axes in each. On every axis independently choose an unmasked trial speed

$$v_i = 1 + 2^K w_i \quad \text{if} \quad \sigma_i = 0; v_i = 1 + 2^k + 2^D w_i \quad \text{if} \quad \sigma_i = 1.$$

The sum of the trial groups over a nonempty subset $J \subset [m]$ has global parity tag $\sum_{\sigma_i=1} w_i \bmod 2$ on its included axes. On unused groups put zero. For one fixed J, set $v_0 = \sum_{i \in J_{\text{axes}}} e_i$; **every** sampled velocity differs from this same base vector by an element of the common R, regardless of the mask. Expand a product of h translated approximants into finitely many terms. At a fixed initial point x, each term along these velocities is a constant times $e(Q_x(t))$ for one ordinary multivariate degree-$\le s$ circle polynomial, in $t_i = (v_i - 1)/2^{u_i}$. Indeed the shift by jv_0 is fixed before the R-polynomial is evaluated, and multiplication of the R-argument by the slot index preserves its degree. Simultaneous versions hold on one conull set by countability. Here $l_i = 2^{K-u_i} \mid 2^K$, the normal offset is zero, the active offset is 2^{k-u_i}, and $z_i = 2^{D-u_i}$ has valuation at least $D - K$.

The product lemma therefore makes every tilted intersection mean arbitrarily small, uniformly in x and the coefficients, by taking D/K large and then N large at the stage, with η small relative to the finite trigonometric coefficient sum. Actual box means converge pointwise for each term by the preceding uniqueness, hence dominated convergence allows integration in x. The unweighted sum over a fixed finite list of DHJ lines has a positive uniform lower bound depending only on $\mu(A), h$: start a word cube at a random base point, use the trial directions sampled independently, and absorb the fixed letters by invariance after integrating in the base. The same finite-box bound holds before taking limits. Each line's signed term is controlled as above, and replacing the approximants by the indicators costs at most $O_h(\epsilon)$ in the integrated products by the boundedness and common marginals. The indicator of an odd tag is $(1 - \text{tag})/2$; choose ϵ, η below the finite-line lower bound to conclude some actual retained trial with positive intersection. This proves the claim also for actions obtained through native and higher cyclic covers, by pullback to the free host.

The argument needs a subgroup for the phase tests that contains every switch $2^{k_i} e_i$; the passive subgroup with coordinate mesh 2^{K_i}, $K_i > k_i$, does not. In particular it cannot be applied to arbitrary passive-relative phase functions, a class already containing short obstructions.

R103. Short charged products for the two-center dyadic example

Scope and status. Proved subcases and unresolved extensions.

Result. Ordered trades prove native recurrence through five terms and integer recurrence through three terms; scalar four-slot nonembedding is also proved.

Scope. Fixed dyadic level with sufficiently many copies; the final four-slot open statement is superseded by the later ordered-four-slot note.

Qualifications. The four-slot integer ordered contradiction is established in R078.

Fix one dyadic level [R107], with $k \geq 2$, $K \geq k+3$, $c = 1+2^k$, and $D \gg K$. Write $z = 2^D$. The two normal residue classes are 0 and 1 modulo 2^K, and the active classes are c and $c+z$ modulo 2^{D+1}, with respective tag 0 and 1. Nonzero global binary tag is required. The following positive results concern sufficiently many identical axes at this fixed level; no copy count uniform over levels is asserted.

Exact ordered trades Write $e_{x,r} = (a_{j,x+jr})_j$ for the word of query symbols in the direct product of free player groups, with target $\tau(r)$. Both queries in $W_{x,r} = e_{x,r+z}e_{x,r}^{-1}$ are legal for $r = 0,1,c$. Their target difference is one at $r = c$ and zero at $r = 0,1$.

On the minimal native group $C_{2^{D+1}}$, the even-player coordinates of every W are the identity. For $h = 4$ or 5 only the two odd players at times 1,3 remain. Reindex by $y = x+r$. Their two coordinates are

$$W(y,r) = (k_y^1, k_{y+2r}^3),$$

where each k is the difference of two corresponding free letters separated by z. The alternating product $W(y,1)W(y+2,0)^{-1}W(y+2,1)W(y+4,0)^{-1}\dots W(y+2(c-1),1)$ telescopes to (k_y^1, k_{y+2c}^3), which is $W(y,c)$. All even coordinates remain trivial. Thus one true-mark difference equals a product of normal differences whose total tag is zero, an ordered contradiction. At $h = 3$ or 2 the same word may simply be projected to the relevant players. Repeating the fixed query law and using the Hales–Jewett rectangle argument [R218] forces every positive density through length five at large identical native powers.

For $h = 3$ **on actual integer coordinates**, player zero still vanishes but the two remaining slots 1,2 now have coordinates (k_y^1, k_{y+r}^2), where the shifts in their free letter differences are z and 2z respectively. The same unit-step telescoping word with y,$y+1,\dots,y+c-1$ gives $W(y,c)$ exactly. This finite integer query packet is independent of all covering periods. Therefore repeated copies force triples at every positive measure in all measure-preserving Z^n-actions when n is sufficiently large at that fixed level. In particular the same n works for every finite period lift of that level.

At $h = 4$ on integer lifts the W procedure retains three players, and trying to derive a relation only among those three-player normalized W reduces the target to the low active selector. That selector divided by two equals $r(r-1)/2^{k+1}$ mod 1 on permitted velocities and is an ordinary degree-two phase. It admits a three-slot scalar decomposition by second differences and a linear correction. Thus the preceding two-player telescoping argument does not extend in this form.

No four-slot scalar circle embedding on the integer lift Assume $D \geq 3K+5$. We claim that there are no circle functions $f_0, \ldots, f_3 : Z \to T$ satisfying $\sum_j f_j(x+jr) = \tau(r)/2$ on every integer start and allowed velocity. This result does not assert absence of a noncommuting embedding.

Set $m = 2^{K+1}$, $M = 3m$. Within either normal velocity class, take differences in the three changes $(x, r) \mapsto (x - i \cdot 2^K t, r + 2^K t)$ for the slots $i \neq j$. Each unwanted slot is killed. Choosing the three t to make all remaining shifts equal M proves $(T^M - I)^3 f_j = 0$. Restrict starts and velocities to actual multiples of three. For $0 \leq b < m$ write

$$f_j(3b + Mn) = A_j(b)\binom{n}{2} + B_j(b)n + C_j(b).$$

Select $x = 3b + Mn$, $r = 3u + Mt$ for $u = 0$ and for one odd u with $3u \equiv 1 \bmod 2^K$. Write $b_j = (b + ju) \bmod m$ and $l_j = (b + ju - b_j)/m$. Substituting $n + jt + l_j$ into the four displayed polynomials and using the normal zero-target equations, the coefficients of `binom(n,2),nt,binom(t,2)` are zero. Thus $\Sigma A_j(b_j) = \Sigma j A_j(b_j) = \Sigma j^2 A_j(b_j) = 0$. Twice this vector of coefficients is $(-1, 3, -3, 1)$ times its last doubled component, by subtraction of the three equations over T. At $u = 0$ this writes $2A_j(b) = \lambda_j U(b)$, $\lambda = (-1, 3, -3, 1)$. At the chosen odd u, comparing the zeroth and third slots gives $U(b + 3u) = U(b)$. Translation by 3u generates C_m, so U is constant.

Choose $A_* \in T$ with $2A_* = U$ and β with $M^2\beta = A_*$. Subtract $\lambda_j \beta \binom{x}{2}$ from f_j. The sum of these four subtracted values on every progression is zero, and afterwards every $A_j(b)$ has order at most two. The coefficients of n and t in the same polynomial substitution now yield, after doubling to kill all the A_j terms, the relations $\Sigma B'_j(b_j) = \Sigma j B'_j(b_j) = 0$ for both u choices, where $B'_j = 2B_j$.

Here are the period bounds implied by those relations. Regard the B'_j as arrays on C_m and let S be translation by the odd u just fixed. The $u = 0$ identities write $\Sigma B'_j T^j = (T-1)^2(a + b'T)$ for arrays a,b' (the indeterminate T in this formula denotes coefficient algebra only). The identities at u then give

$$(S-I)^2 v = 0, (S-I)^2 b' = -2(S-I)v, v = a + Sb'.$$

In the second equation differentiate the polynomial in T, multiply by T, set $T = S$ and cancel the common invertible S. Thus a and b', and hence each B'_j, are polynomial arrays of degree at most two along the cyclic generator S. For $P(n) = b_0 + b_1 n + b_2\binom{n}{2}$ of period m, $mb_2 = 0$ and $mb_1 + \binom{m}{2}b_2 = 0$. Hence $m^2(P(n) - P(0)) = 0$ for all n. In particular $2m^2(B_j(b) - B_j(0)) = 0$.

Take any active r divisible by three and congruent to $c \bmod 2^{D+1}$, and an x divisible by three. Changing r to $r + 3z$ flips the target. Its slot increments in their M-fiber parameters are $j\omega$, where $\omega = 3z/M = 2^{D-K-1}$. These increments kill every residual binomial quadratic contribution, since A_j has order at most two and ω is divisible by four. The phase change reduces to $\sum_j j\omega B_j(b_j)$. But $2m^2 | \omega$ under the depth assumption, so all $B_j(b_j)$ may be replaced by $B_j(0)$ in this formula. Their weighted sum is then zero by the $u = 0$ relation on B'_j and the evenness of ω. This contradicts the half-turn target change and proves scalar nonembedding on integer lifts. The argument uses extra high digits as actual integer

variables; it is not a finite native averaging argument. Recovering quantitative four-slot rectangle cancellation from this nonembedding in characteristic two is unresolved here.

R104. Four and five native slots with singular high-bit spectrum

Scope and status. Result with proof.

Result. Singularity of the event's high-bit spectral measure forces four- and five-term recurrence in the native two-center model.

Scope. Exponent-two high subgroup only, with no all-length or higher-cover conclusion.

In the native minimal two-center model [R107], let $K = \oplus_i C_2$ be its high-bit subgroup, and let ρ be the K-spectral measure of the indicator f of a positive-measure event A. If $\rho \perp m$ for Haar probability m on $\hat{K} = \prod C_2$, then A has a tagged return of length four and of length five in every tail. This concerns singular continuous measures as well as atomic ones, and places no restrictions on the spectral types of other observables. Conversely an avoidance at either of these lengths requires a nonzero absolutely continuous component of this particular spectral measure.

A spectral overlap bound. Write $\mathrm{Aff}(\rho, \sigma) = \int \sqrt{d\rho d\sigma}$ using any common dominating measure. For any translation U by an element of the ambient group, any K-character χ, and $M \geq 1$,

$$||P_\chi(f \cdot Uf)||_2 \leq M\,\mathrm{Aff}(\rho, \chi\rho) + 1/M.$$

Here P_χ denotes orthogonal projection onto the χ-eigenspace of the full K-action. To verify the bound, pair the product with any unit L^2 vector g in that eigenspace. Truncating g on $|g| \leq M$ retains the eigencharacter and incurs error at most $\int_{\{|g|>M\}} |g| \leq 1/M$, as the product indicators are bounded by one. Multiplication by the truncated eigenvector (or its conjugate) has L^2 operator norm $\leq M$ and shifts every K-spectral projection by χ. Thus the spectral measure of its product with Uf is dominated by $M^2\chi\rho$; U commutes with K, so Uf has the original ρ. Pairing with f is bounded by the spectral-measure affinity using orthogonality on every finite Borel partition and passing to the affinity integral. Characters are real here and conjugations do not change χ. This proves the inequality uniformly in U and also after a common translate of the two factors.

Many disjoint low-overlap blocks. Deleting finitely many coordinates from a measure singular to product Haar on the Boolean dual preserves singularity: a Haar-null Borel carrier has null projection since the deleted alphabet is finite. On any tail take a sufficiently long finite block B such that the affinity between its ρ marginal and uniform Haar on C_2^B is arbitrarily small; the affinities of exhausting cylinder marginals decrease to the affinity of the full tail measures, which is zero. Write p_u for the marginal masses. Averaging the finite marginal overlap with its shift by a random binary mask w gives

$$2^{-|B|} \sum_w \sum_u \sqrt{p_u p_{u+w}} = \mathrm{Aff}(\rho_B, m_B)^2.$$

Choose a nonempty $I \subset B$ with this shifted overlap as small as desired (zero shift gives the fixed mass $\rho(\hat{K})$). Any global flip whose restriction to B equals I has full overlap at most the marginal overlap, since projection increases affinity. Thus for any fixed m one can

choose m disjoint such trial blocks and nonempty masks $I_1,\ldots,I_m$ so that *every nonempty union* I of masks has $\mathrm{Aff}(\rho,\chi_I\rho)$ arbitrarily small.

Tagged return. Use the m standard density Hales–Jewett inputs for density $\mu(A)$ and the desired h, with base generator $\sum_{i\in I_s} c_i e_i$ for block s. Some nonempty union I gives an unmarked baseline v_I whose h-return probability is at least $\kappa(\mu(A),h)>0$. Write F,G for the even-slot and odd-slot products along this v, as in the high spectral note [R079]. The parity-conditioned high correction average is $\langle P_1F,P_1G\rangle-\langle P_{\chi_I}F,P_{\chi_I}G\rangle$. Its first term is at least κ^2, since the product FG is itself a positive subevent of both F and G and conditional expectation onto the K-invariants preserves order. When $h=4$ or 5 the smaller side of the parity split is a product of just two translates of f. By the overlap estimate its χ_I projection has norm $<\kappa^2$ after choosing M large, then the block overlaps small. The other side has norm ≤ 1. Positivity of the conditioned average now produces a legal high correction with positive return. This does not give full-length recurrence or any assertion for deeper high-bit covers where the high subgroup no longer has exponent two.

R105. Size of the native abelian high-bit obstruction at every fixed horizon

Scope and status. Result with proof.

Result. The native incidence module yields an explicit finite-rank exponent-two bound for the high-bit obstruction at a fixed horizon.

Scope. An algebraic obstruction module can still support complex temporal behavior; it is not a two-state factor theorem.

At a fixed dyadic level let $q = 2z = 2^{D+1}$, $L = 2^K$, and fix h. Use the integral abelian incidence module of the h-player native game,

$$A = Z[X]/(X^q - 1), b_r = (X^{jr})_{j=0,\dots,h-1} \in A^h.$$

Start translation multiplies the entire tuple by X. Let H_0 be the A-submodule generated by all normal-velocity tuples and the uncharged tuple at $r = c$. Then the whole high-bit difference module $(X^z - 1)A^h$, modulo what is already generated by H_0, is a quotient of

$$(F_2[\varpi]/\varpi^v)^h, v = L \sum_{0 \le j < k < h} 2^{v_2(k-j)},$$

provided $z \ge v$ and the selected velocities are distinct. In particular its rank as a binary vector space is bounded in terms of K,h, independently of the much larger depth D. This concerns abelian query incidences, not the ordered-word group or general actions.

Indeed the ideal $(X^z - 1)A$ is isomorphic as module to $O = Z[X]/(X^z + 1) = Z[\zeta_q]$, via multiplication by $X^z - 1$; its kernel in the polynomial ring is precisely $(X^z + 1)$. Use the normal velocities $r = Lk$, $0 \le k < h$, and multiply their translated tuples by elements of this ideal. In O the h-column evaluation matrix has Vandermonde determinant

$$\delta = \prod_{0 \le j < k < h} (\zeta_q^{Lk} - \zeta_q^{Lj}).$$

Each factor is a unit times $\varpi^{L \cdot 2^{v_2(k-j)}}$, $\varpi = \zeta_q - 1$. This is an assertion in the global integer ring: ϖ is the unique prime above two, and $1 - \zeta_q^m$ has absolute norm $2^{2^{v_2(m)}}$ as long as $2^{D+1} \nmid m$. Other primes do not occur. Thus δ generates (ϖ^v) and the adjugate supplies all of this ideal in every player coordinate inside H_0. Since 2 generates (ϖ^z), the residual quotient has exponent at most two; the ring $O/(\varpi^v)$ in this range is $F_2[\varpi]/\varpi^v$, as follows from the cyclotomic presentation or the successive ideals of index two. This gives the stated bound.

The difference between b_{c+z} and b_c belongs to this high-bit module: its even coordinates vanish and its odd ones are $(X^z - 1)X^{jc}$. This calculation helps isolate the native finite rank obstruction but does not imply a two-state trajectory, as the distinct-window construction [R070] demonstrates. In particular an elementary abelian obstruction group can support substantially more complicated temporal behavior than a single order-two rotation.

R106. Full recurrence for a totally ergodic infinite-rank passive action

Scope and status. Result with proof.

Result. A totally ergodic infinite-rank passive action gives full positive recurrence for the replicated two-center family.

Scope. Every scalar multiple of the passive subgroup must act ergodically; extensions to finite component data have the stated retwist conditions.

Let $G = \bigoplus_i \mathbb{Z}e_i$ and $P = \bigoplus_i 2^{K_i}\mathbb{Z}e_i$, the common normal-zero periods of the sufficiently replicated two-center candidate [R107]. If mP acts ergodically for every positive integer m, every positive-measure event has an h-return for every h on this family. There is no finite generation requirement on P, and no ergodic finite-rank subaction is assumed. Use the compact-nilspace word theorem [R069] when choosing the block sizes. The unrestricted density problem includes nonergodic scalar passive subgroups.

Ergodic subgroup cube comparison For ergodic abelian subgroup actions $H \subset G$, the Host–Kra cube measures and diagonal invariant algebras agree for H and G. Indeed an automorphism V commuting with H preserves the H-cube measure when applied on any codimension-one face. In dimension one this uses $\mu_H^{[1]} = \mu \times \mu$, so applying V to one endpoint preserves the measure. This map commutes with the diagonal H, thus taking its double on the relative independent square still preserves the next cube measure. Iterate, and use symmetry of the abelian cube measures to choose any face. Since acting by $V^{[l]}$ on just one half preserves the $(l+1)$-cube measure, which is a relative independent square over the H-diagonal invariant algebra, $V^{[l]}$ fixes that algebra pointwise: two copies of an invariant function agree on the square, and still agree on applying $V^{[l]}$ to one copy. Apply this to each $V = U^g$. On every H-cube measure the H-invariants therefore equal the G-invariants, and induction gives equality with the G-cube measures. This is the usual face argument of Host and Kra, Section 5; it uses group symmetry and ergodicity only.

For a legal v with finite active set I use the subgroup P_v that retains $2^{K_i}\mathbb{Z}e_i$ on inactive coordinates and uses $2^{D_i+1}\mathbb{Z}e_i$ on I. Every $v + p$, $p \in P_v$, remains legal. This subgroup has finite index in P, and every nonzero jP_v acts ergodically because it contains some mP. Consequently the h-slot averages over P_v have the single characteristic factor $Z_{h-1}(G)$, a safe choice of order. To detail the seminorm estimate, center the progression at any slot other than a distinguished one and use Cauchy–Schwarz to bound the integrated average by the norm of its other slots' average. Van der Corput at increment $u \in P_v$ changes each remaining function to its derivative at the corresponding slope ju. Recenter at another slot and apply the estimate for the shorter list to the derivative of the distinguished function. Hölder followed by the cube recursion bounds the derivative mean by the next Host–Kra seminorm, since the jP_v-cube measures are precisely the G-cube measures by the preceding paragraph. The base, two slots, is the mean ergodic theorem for the nonzero difference subgroup. This proves that the contribution of a slot perpendicular to the stated factor vanishes; all translations U^{jv} commute with projection to it. This argument avoids any finite-index assumption on jP in P.

By Candela and Szegedy, Theorem 5.11, the factor has a compact finite-step nilspace model Z with Haar probability and the full G acting by translations. In particular its mP-

actions remain ergodic. We next prove full positive recurrence for measurable events on this factor.

Connected finite-rank images under scalar ergodicity For any fixed compact finite-rank nilspace Y there is an integer $M(Y) \geq 1$ such that for every morphism $b : \mathcal{D}_1(\mathbb{Z}) \to Y$, $b(n)$ and $b(n + M(Y))$ belong to the same connected component for every n.

Use the free cover F/Γ of the compact-nilspace proof [R069], and lift b to an ordinary polynomial in each layer's real and discrete coordinates. We only need the discrete coordinates $\check{F} = \prod_l \mathbb{Z}^{b_l}$; matching them puts the lifts in the same connected component of F. Every translation in the higher-order lattice induces a triangular polynomial transformation on these coordinates: outputs in $\mathbb{Z}^{b_l}$ are independent of the real inputs by continuity and connectedness. There is an integer Q such that any two discrete label vectors congruent modulo Q coordinatewise can be brought to equality by lattice translations. Here are details. At each layer pick an integer $m_l \geq 1$ and finitely many lattice translations of degree l whose layer-l moves project to m_l times each coordinate basis vector of $\mathbb{Z}^{b_l}$. This is possible since the layer structure lattice projects onto a finite-index subgroup of the discrete coordinate group, by compactness of its quotient. Higher-coordinate changes on raising any of these fixed translations to a power u are integer-valued polynomials in u and the discrete coordinates; at $u = 0$ all changes vanish. For each prescribed modulus R, these changes are divisible by R on all inputs when u is divisible by some fixed C_R: use the integer binomial expansion or clear the fixed rational denominators. Working backward through the finitely many layers choose Q_l, terminating at $Q_{s+1} = 1$, so that if coordinate differences are all multiples of Q_l, the powers needed to match layer l exactly are multiples of $C_{Q_{l+1}}$. For example take Q_l a multiple of Q_{l+1} and $m_l C_{Q_{l+1}}$. The degree-l translations do not disturb earlier layers and the remaining differences stay divisible by Q_{l+1}. Take $Q = Q_1$. Since discrete orbit coordinates are integer-valued polynomials of degrees at most s, they are periodic modulo Q with common period $Qs!$. This proves the component assertion, including for copies of F whose real coordinates are not zero.

Let $\pi : Z \to Y$ be any finite-rank fibration. For $p \in P$, the map $n \mapsto \pi(U^{np}x)$ is a morphism by the translation property, for each x in the compact nilspace model. Its component is unchanged by $M(Y)p$. Hence the inverse image of each component is $M(Y)P$-invariant. There are only finitely many components, each open with positive Haar mass, since Y is a compact finite-dimensional manifold with full-support Haar measure. Ergodicity forces Y to be connected. The same reasoning applies to every compact finite-rank refinement, even when the original P-action does not descend as transformations of that refinement.

Graph measures are Haar on connected subnilmanifolds Fix such a fibration $\pi : Z \to Y$, and a direction $d \in G$. By Candela–González-Sánchez–Szegedy, Theorem 1.6, there is a finite-rank refinement $Z \to Y_d \xrightarrow{\{} p\}Y$, consistent with the single translation U^d. It induces a translation a of Y_d. By the preceding argument Y_d is connected. The graph distribution of the slots $\pi(U^{jd}x)$, $0 \leq j < h$, is the pushforward of Haar on Y_d by the maps $p_j(y) = p(a^j y)$, each a nilspace fibration.

For a connected finite-rank nilspace W, its connected first-translation Lie group $N_W = \Theta(W)^0$ is transitive, as in the proof of Lemma 5.2 here. It is nilpotent and preserves Haar, which is therefore its transitive invariant probability. The space is an ordinary homogeneous nilmanifold: pass to the simply connected cover $\widetilde{N}_W$ and the closed cocompact stabilizer H.

Its identity component H^0 is normal. Indeed the normalizer of the connected subgroup H^0 in a simply connected nilpotent Lie group is connected, by the exponential map and the Lie algebra normalizer (the logarithm of a unipotent adjoint operator is a polynomial in that operator). It contains H and hence is cocompact, while the quotient by a connected closed subgroup of a simply connected nilpotent group is a Euclidean space. The normalizer must be the entire group. Factoring out H^0 leaves a discrete cocompact stabilizer. This concerns the underlying homogeneous manifold; the full nilspace cube structure is not being identified with the lower-central filtration.

By Gutman–Manners–Varjú III, Theorem 1.28, each p_j intertwines N_{Y_d} with a continuous homomorphism into N_Y. Thus the joint graph is a transitive connected group orbit in the fixed compact nilmanifold Y^h, and its probability is its invariant Haar measure. It is closed because it is the image of the compact space Y_d. In the effective simply connected covers its acting connected subgroup is closed (connected Lie subgroups of simply connected nilpotent groups are closed). Hence this is Haar on a connected subnilmanifold.

Choose countably many $x \in Z$ whose images are dense in Y. The compact-nilspace word theorem applied jointly to finitely many such points and j-multiples gives a sequence of legal tail directions whose graphs approach the diagonal at increasingly many dense samples. All these graph points belong to the supports, by continuity and full support on Z. By the connected homogeneous-measure compactness and diagonal joining lemma [D10], any weak limit of those graph probabilities is connected homogeneous Haar with full diagonal in its support. For a finite-stage event C of mass $a > 0$, the limit therefore charges C^h at least $c(a, h) > 0$, independently of that target; measurable tensor approximation uses the fixed marginals. Any positive event B of Z can be approximated in measure by an event on one finite-rank fibration using the inverse limit theorem for compact nilspaces. Choose its error below $c(\mu(B)/2, h)/(2h)$; the finite-stage lower bound survives by telescoping, giving a legal d with a positive B-return.

Finally take $B = \{\mathbb{E}(1_A \mid Z_{h-1}(G)) > b\}$ with $b > 0$ and positive mass. At a legal d giving a B-return, let $C = \bigcap_{j<h} U^{-jd}B$. The averaged factor correlation on $d + P_d$ is at least b^h times the ordinary multiple-recurrence mean on C for P_d, strictly positive by finite density Hales–Jewett averaging. Characteristic transfer gives a positive original correlation at some allowed correction $d + p$. This proves the assertion. It does not discard the case where P is ergodic but some mP has a nontrivial component space, nor the case where P itself is nonergodic.

Finite total rational component factor The result also holds on any jointly G-ergodic system for which $Y = \bigvee_{m\geq 1} \mathcal{I}_{mP}$ is finite, whether or not P is itself ergodic. Indeed a single m_0 then satisfies $\mathcal{I}_{m_0 mP} = Y$ for every $m \geq 1$, because the factorial-multiple sequence increases to a finite algebra. The finite transitive G-action on Y is a regular action of an abelian quotient. Some fixed finite head realizes all its translations. For every i choose a head vector a_i whose action on Y equals that of e_i, and define $R(e_i) = e_i - a_i$. Change the head a_i's by multiples of a common kernel integer times their basis vectors to make the head matrix of R nonsingular over the rationals: adding a sufficiently large scalar diagonal avoids its finitely many determinant roots. The images $R(G)$ preserve each Y-component. Moreover $R(P)$ contains rP for some common r. On the head the image of its period lattice under a nonsingular integral matrix contains an integer multiple of the entire head group. On a tail period vector cancel its head correction after multiplying by that same integer, using a head period vector. Thus on every chosen component each $mR(P)$ is ergodic, since

it contains $m_0 mrP$ whose component algebra is exactly Y.

Apply the theorem there to the retwisted group action by R, restricting the selected coordinates to a tail past the correcting head. In the word selection impose also the finite linear congruence $\sum_i d_i a_i = 0$ modulo the normal periods on the head; it is an extra finite abelian nilspace target. In the last averaging step one restricts passive corrections by the kernel of the same congruence, still finite index in P_d. This ensures $R(d)$ changes the head only by normal-zero multiples; other coordinates are unchanged and the retained odd tag is legal. A component carrying positive conditional mass of the original event supplies the desired return. An infinite rational component factor need not admit this finite retwist.

If merely P is ergodic, its point spectrum has no nontrivial dyadic finite-order character. For an eigencharacter of order 2^a, normalize its eigenfunction so that $f^{2^a} = 1$. All U^{e_i} act on the one-dimensional eigenline by 2^a-th roots. Thus each $P_i = 2^{K_i} e_i$, $K_i \geq 1$, acts by a root of order at most 2^{a-1}, contradicting the character's order. Every dyadic finite-index subgroup of P is accordingly ergodic. The factor $\mathcal{I}_{2^a P}$ may still be nontrivial, but is weakly mixing for the full P-action: it factors through a bounded dyadic group and has no nonconstant eigenfunctions. This restricts the remaining obstruction without replacing total ergodicity in the main argument.

R107. A dyadic candidate with two odd centers

Scope and status. Proved subcases and unresolved extensions.

Result. The two-odd-center dyadic construction has a six-term positive-density obstruction and several positive spectral subcases.

Scope. This is a candidate family; no all-system upper bound below one is proved.

This is a candidate, not an upper bound for the conjecture. It avoids the immediate bounded-period certificate of the first dyadic family [R080]. Choose integers k_i,K_i,D_i tending to infinity with $K_i > k_i$, $2 + 2^{k_i+1} + 1 \le 2^{K_i-1}$, and D_i much larger than K_i. The nontrivial scheduling regime includes arbitrarily many axes at a given depth and $\sum_i 2^{-k_i} = \infty$; one can for example use finite but sufficiently large successive blocks with $K_i = 3k_i$ for $k_i \ge 3$. Put $c_i = 1 + 2^{k_i}$, and let the native group have axes $C_{2^{D_i+1}}$. Normal velocities are either zero or one modulo 2^{K_i}, with arbitrary higher digits. Active ones are $c_i + 2^{D_i} t_i$; take the finitely supported directions for which the sum of the active t_i is odd. On the free integer group take the same residue rules with unrestricted lifts.

A fixed six-slot obstruction already on the native group. At one axis write $\zeta = \zeta_{2^K}$, $\varpi = \zeta - 1$, $\phi = 2^{K-1}$, $J = 2 + 2^{k+1}$. Use the ring

$$(Z[\zeta]/(\varpi^{D\phi+J+1}))[\epsilon]/\epsilon^2, u = \zeta(1+\epsilon).$$

Here 2 is a unit times ϖ^ϕ and $v_\varpi(\zeta^{2^k} - 1) = 2^k$, by the cyclotomic factorization (or by taking the norm down the 2-power tower). Thus $2^{D+1} = 0$, and u has order dividing 2^{D+1}. Put $E = 2^D \varpi^J \epsilon$. It is nonzero, fixed under multiplication by u, and of additive order two. The degree-five polynomial

$$P(Y) = (Y-1)^2(Y-u)^2(Y-u^c)$$

vanishes at $Y = u^r$ on both normal residue classes: the appropriate difference has zero constant term in ϵ. On a mark, $u^r - u^c = \zeta^c 2^D t\epsilon$. In the other factors only ϵ-free terms survive. The valuations of $\zeta^c - 1$ and $\zeta^c - \zeta$ are respectively 1 and 2^k. Their normalized unit factors reduce to one in the residue field F_2. Hence $P(u^r) = tE$ exactly. Across all axes form the countable module generated by the six global coefficient tuples and the finite local coordinates, stable under the component actions. On its character torsor prescribe $E_i \mapsto 1/2$; Haar on the extension coset is invariant. The sum of the six corresponding observations at consecutive times along a legal direction equals $1/2$, whereas at any one site the sum equals zero because $P(1) = 0$. Quantizing each circle coordinate into cells of diameter $< 1/12$ gives a fixed finite coloring with no monochromatic six-slot run. The free lift also has this positive obstruction. Using the coordinatewise bounded packets in separate copies of the free group gives an integer realization preserving the all-length cap. The six-slot length need not be preserved in that last realization.

Where high-density recurrence would come from. At minimal depth the high subgroup is $V = \oplus_i \{0, z_i\}$, $z_i = 2^{D_i} e_i$. For any nonempty finite I put $v_I = \sum_{i \in I} c_i e_i$. Then all $v_I + z$ with $z \in V$ and $\chi_I(z) = -1$ are legal steps, where χ_I reads the high bits on I. In an arbitrary native action an h-run on this coset separates into the even-slot intersection F for the baseline v_I, and the odd-slot intersection G. The z-average is

$$\langle P_1 F, P_1 G\rangle - \langle P_{\chi_I} F, P_{\chi_I} G\rangle,$$

where the two projections are on the invariant and χ_I eigenspaces of V (averages along its increasing finite subgroups). Other choices of ordinary normal velocity may be added to the baseline. The eigenvectors that matter are those of the products F and G, not merely those of the original indicator.

For completeness suppose the tested event has pure point spectrum under the high subgroup; on the free lift use $V' = \oplus 2^{D_i} Z e_i$ instead. Approximate its indicator in L^2 by finitely many eigencharacters of this subgroup. The finitely generated group they span contains only finitely many order-two characters (including ones whose original eigenvectors have infinite spectral coordinate support). The Walsh characters χ_I of nonempty subsets of a sufficiently fresh block avoid that group. For a fixed such I there is z with $\chi_I(z) = -1$ returning all the finite eigencharacter values arbitrarily close to one: in the compact closure of the graph of the joint character map the contrary would make χ_I a continuous character of its projection into the finite torus, hence an integral combination of the tested characters. This uses the usual extension of characters from a closed subgroup of a torus. On the native Boolean subgroup one can take the tested eigenvalues exactly one. On a sufficiently long fresh block, finite density Hales–Jewett provides v_I with a positive h-return bounded below in terms only of the original density and h. Fix the approximation in advance relative to this lower bound, then choose z as above. The L^1 error of replacing the original indicator by its jz-translate is uniformly small for all slots. This proves a positive return at the legal step $v_I + z$. No conclusion on the capacity over all actions follows from this pure-point calculation.

In particular the exact high-bit involutions are relevant on the native group but their higher lifts can have arbitrary order. Also the candidate with only multiples of 2^{K_i} as normal velocities would be a subfamily of the excluded original dyadic example. The odd normal center and an unbounded quotient $D_i/K_i - k_i$ are essential features of this proposal.

R131. Exact unrestricted-length parity heights for a power-closed Gaussian system

Scope and status. Result with proof.

Result. A power-closed Gaussian action has an exact parity-coset capacity computed from atomic spectral images under dilations.

Scope. This one specified Gaussian action may attain its finite cap at finite length; no integer endpoint-forcing conclusion.

Let H be a countable abelian group, $\chi : H \to \{\pm 1\}$ nontrivial, and $D = \{v : \chi(v) = -1\}$. Use additive notation for scalar coefficients of elements of the discrete character subgroups below ($m\gamma$ is γ^m). Let σ be an arbitrary nonzero finite spectral probability on $\widehat{H}$, symmetrizing under conjugation for the realification when necessary. Consider the single Gaussian H-action made of countably infinitely many independent copies of the realified complex first-chaos representation for **each** dilation $v \mapsto nv$, $n \geq 1$, of $L^2(\sigma)$. Its first-chaos maximal spectral type is equivalent to the positive mixture $\tilde{\sigma} = \sum_{n \geq 1} 2^{-n}(n)_*\sigma$. The independent copies are part of the system and supply independent signs in the lower construction.

Write P_m for the atoms of $(m)_*\sigma$. For a positive dyadic integer k put

$$B_k = \langle (k/\gcd(k,m))\lambda : m \geq 1, \lambda \in P_m \rangle \subset \widehat{H}.$$

These are countable groups decreasing along powers of two. The all-length fixed-system avoidance cap for D is zero if $\chi \notin B_1$. If k is the largest dyadic integer with $\chi \in B_k$, the cap is $1 - 1/(2k)$ and is actually attained at a finite forbidden length. If there is no largest such k and the set of them is nonempty, the cap is one. This is a result for this specified Gaussian action, not a general integer threshold or a claim of endpoint forcing for these finite heights.

Limit along the odd coset. Let L be the discrete countable subgroup of $\widehat{H}$ generated by χ and all P_m, with $Q = \hat{L}$. Evaluation maps H densely to Q, since its image has trivial annihilator in L. If $\chi \notin B_k$, choose $w \in Q$ annihilating B_k with $w(\chi) = -1$: distinguish the nonzero order-two element in L/B_k by a circle character (extend its character using divisibility of the circle). For the enlarged spectral type $\tilde{\sigma}$ the atoms at index m are the union of all P_{mn}. For $\lambda \in P_{mn}$, $(k/\gcd(k,m))\lambda \in B_k$, since $k/\gcd(k,mn)$ divides $k/\gcd(k,m)$.

There is a sequence of odd $v \in H$ along which, simultaneously for all integral j, the scalar multipliers $\gamma(v)^j$ tend weak-star in $L\infty(\tilde{\sigma})$ to

$$\phi_j(\gamma) = w(j\gamma) \quad \text{if} \quad j\gamma \in L, \quad \text{and} \quad 0 otherwise.$$

Indeed, for each integrable test f, the Fourier coefficient at v of the pushforward $(j)_*(f\tilde{\sigma})$ splits into an absolutely summable atomic Fourier series with frequencies in L and a continuous-measure term. The latter is small outside a set of Følner density zero, by applying the spectral theorem to the squared modulus (Wiener's argument works on any countable abelian group). The atomic series is continuous on Q and at w equals the integral prescribed by the display: the countably many non-atomic single fibers with $j\gamma \in L$ have zero total measure. Test finitely many of these functions on a small neighborhood of w, whose visiting

times have positive lower Følner density, shrinking neighborhoods and diagonalizing over a countable L^1-dense family and over j. The parity on that neighborhood is odd. The same argument applies to finite H, with its uniform distribution, without any need to make the times distinct.

For each spectral γ outside a single null set let $m = m(\gamma)$ be the generator of $\{j \in Z : j\gamma \in L\}$, taking $m = \infty$ if there is no positive member. When $m < \infty$, the limiting conditional phase law is uniform on the m roots of $w(m\gamma)$, by the displayed moments; when $m = \infty$ it is circle Haar. In the first case $\lambda = m\gamma$ is an atom of $(m)_*\tilde{\sigma}$ on almost every point with such an m. As $w((k/\gcd(k,m))\lambda) = 1$, the set of roots contains a kth root of unity: the image of μ_k under the mth power map is exactly $\mu_{k/\gcd(k,m)}$. Select such roots measurably and with conjugation symmetry, using reflected choices on paired spectral points. At self-conjugate spectral points the phase law is supported on the two real signs and there is a permissible real choice (also when $k = 1$ the choice is 1).

The Gaussian selector result [R133] applies unchanged: it is a statement on the resulting Gaussian covariance and deterministic multiplier selectors, independent of the parameter group giving the limiting multipliers. A null h-rectangle for the limiting joining remains null for the graph joining of a selected deterministic multiplier. Here that multiplier has order dividing k, hence an event of mass greater than $1 - 1/k$ has positive simultaneous intersection on all its iterates by the finite union bound. Thus no such event can exactly h-avoid the entire odd coset. This proves the upper bound for every excluded k, including $k = 1$.

Building signs at the detected heights. For $\lambda \in P_m$ the spectral fiber $m\gamma = \lambda$ has positive σ-mass. In the $n = m/\gcd(k,m)$ dilation take a nondegenerate complex Gaussian coordinate supported on this fiber, using the complex realification to read its two components. Under the `k v` shift this coordinate is multiplied by $\lambda(v)^{k/\gcd(k,m)}$, simultaneously for every $v \in H$. Normalize by its absolute value, almost surely nonzero, to obtain a unit function with that character for `kH` (the assertion refers to $v \mapsto T^{kv}$, so does not require injectivity of multiplication by k).

If $\chi \in B_k$, multiply finitely many such functions and their conjugates to get a unit function Z with $T^{kv}Z = \chi(v)Z$. Every smaller dyadic `q|k` also has $\chi \in B_q$. Use independent copies for all $q = 1, 2, \ldots, k$ in powers-of-two order. In each row take a measurable sign of Z exchanged by multiplication by -1; it is balanced because an odd group element performs that exchange at its qth multiple. The OR of positive signs has mass $1 - 1/(2k)$ by independence. Along any odd v the sign in row q flips at `qv` and is invariant at `2qv`. Successively moving by zero or the current q makes all signs negative within $\sum_{q \le k} q$ steps, preserving previous signs. This proves the lower and completes the formula.

R134. Positive Eberlein returns for distinct parity corrections and the mixed dyadic model

Scope and status. Result with proof.

Result. Distinct parity corrections preserve some positive multiple return when baseline correlations have a common positive lower bound in Eberlein classes.

Scope. Includes Gaussian actions; the mixed-model application does not automatically transfer through arbitrary integer embeddings.

Let G be a countable discrete abelian group, $H \leq G$, and $\eta_n \in \widehat{H}$ distinct characters of order two. Fix a probability-preserving Gaussian G-action, a measurable event A, and $h \geq 2$. Suppose offsets $g_n \in G$ satisfy $\mu(\bigcap_{j<h} T^{-jg_n} A) \geq \kappa > 0$. Then the union of the cosets

$$g_n + \{u \in H : \eta_n(u) = -1\}$$

contains a positive h-return for A; the same holds on every tail in n. More generally it suffices that this tested multiple correlation be Eberlein (a uniform limit of Fourier–Stieltjes functions). There is a positive lower return bound depending only on κ, h, uniformly for all sufficiently late n, although the stage cutoff may depend on the system. This is not asserted for every stationary action.

Direct proof. Write $c(g) = \mu(\bigcap_{j<h} T^{-jg} A)$ and let m be any invariant mean on H. With $C_n = \bigcap_{j<h} T^{-jg_n} A$, the correlation $c(g_n + u)$ is at least $\mu(\bigcap_{j<h} T^{-ju} C_n)$. Therefore $m_u c(g_n + u) \geq \kappa'(\kappa, h) > 0$ uniformly. Indeed finite density Hales–Jewett gives a single finite generator count M and a positive uniform lower bound on the sum of these correlations over the word lines for arbitrary $y_1, \ldots, y_M \in H$; apply successive independent invariant means in the y_i. Invariance in an active variable makes every summand the same mean. Alternatively choose any one Følner accumulation mean on bounded functions.

If $c(g) = \int_{\widehat{G}} \theta(g) d\sigma(\theta)$ for a finite complex measure σ, along any Følner sequence in H one has

$$|mean_u \eta_n(u) c(g+u)| \leq |\sigma|\{\theta : \theta|_H = \eta_n\}, uniformly \in g.$$

Dominated convergence on the Fourier representation gives the formula and indeed the exact integral over this fiber; arbitrary invariant means agree here by the same Følner limit, or directly by the weak mean-ergodic projection. The fibers are disjoint, so the right side tends to zero. Uniform approximation extends the same limiting statement to Eberlein c. As $1[\eta_n(u) = -1] = (1 - \eta_n(u))/2$, the conditional odd-class mean is bounded below for all late n, proving the claim. In a Gaussian action Wick's formula for polynomial Gaussian cylinders expresses each multiple correlation as a finite sum of products of ordinary unitary matrix coefficients and their integer-dilation pullbacks; it is then Fourier–Stieltjes. Approximate bounded slot tests in L^h by polynomial cylinders and telescope using Hölder, giving uniform approximation on G. This is the observation in one-mark-eberlein.md [R174]. Thus all Gaussian actions are covered by this short proof.

The following longer Gaussian calculation additionally identifies a correcting sequence whose limiting Gaussian joining enlarges the original phase supports, which is stronger than simply finding a positive mean.

A separate strong-topology criterion for arbitrary systems. Without a Gaussian or Eberlein hypothesis, let $O = \{u \in H : \mu(T^{-u}A \triangle A) < \kappa/(2h^2)\}$. If all the correcting cosets avoided A, then every η_n would vanish additively on the same neighborhood O (equivalently $\eta_n(u) = 1$ there). Indeed telescoping the powers gives $|c(g_n + u) - c(g_n)| \leq \sum_{j<h} j\mu(T^{-u}A \triangle A) < \kappa/2$. Thus all η_n descend to the common discrete quotient by the subgroup generated by O. This is impossible if the closure of the image of H in the strong topology of the measure algebra is compactly generated, or more generally has finitely many order-two characters on every discrete quotient. To check the compact-generation statement, the subgroup generated by an identity neighborhood is open, and the discrete abelian quotient of a compactly generated group is finitely generated. The same condition holds for compact closures without any finite-dimensional restriction. In particular the correction method works for arbitrary events if these H-actions extend continuously from a dense homomorphic image into a compactly generated abelian topological group. This is a separate condition and is not automatic for H of infinite rank.

Spectral choice of the correcting directions. On the complexification of the real first chaos, let σ be a finite maximal spectral type on $\widehat{G}$, chosen symmetric; the argument works with arbitrary first-chaos multiplicity. For each nonzero integer l define $p_l(\theta) = \theta|_H^l$ in $\widehat{H}$, and let B_l be the countable set of atoms of $(p_l)_*\sigma$. For $l = 0$ the map is trivial. Fix a countable $L^1(\sigma)$-dense collection of tests. Passing to a subsequence, the uncorrected phase multipliers $\theta(g_n)^l$ have weak-star limits $\phi^{g_l}(\theta)$ for every l. Their fiberwise moments determine measurable probability laws ν_θ^g on the circle, and the limiting Gaussian h-joining has the corresponding moment covariance, as in gaussian-ultrafilter.md [R133]. Its self-rectangle for A has probability at least κ: finite-dimensional Gaussian cylinder approximation transfers convergence to arbitrary measurable rectangles because all one-slot marginals are unchanged.

Enumerate the countable union of all atom characters across l. Choose a further subsequence, relabeling the indices by n, so η_n is outside the subgroup generated by the first n atom characters. Indeed a finitely generated abelian subgroup has only finitely many elements of order two. There is a positive-Følner-density Bohr region of $u \in H$ where $\eta_n(u) = -1$ and those first n characters are within $1/n$ of one. To see this take the compact closure of their joint map along with η_n; absence of the order-two character from their discrete span puts $(1, \ldots, 1, -1)$ in that closure by duality (characters of a closed torus subgroup extend to integral torus characters). Haar convergence of the Følner averages gives positive lower density for a smaller open neighborhood there.

For fixed n, l and any integrable test f, the finite complex measure pushed to $\widehat{H}$ by p_l from

$$f(\theta)\theta(g_n)^l 1_{p_l(\theta) \notin B_l} d\sigma(\theta)$$

is nonatomic, by absolute continuity to the nonatomic part of $(p_l)_*\sigma$. Its Fourier coefficients have squared mean zero along every Følner sequence of H: expand the square over two copies, average their quotient character and use that the product measure gives zero to the diagonal. Therefore within the above Bohr region we may choose u_n making all these coefficients smaller than $1/n$ for the first n tests and $0 < |l| \leq n$. On the atom part, char-

acters in the finite list are near one on u_n; the remaining atom mass for each fixed l tends to zero by countable additivity. Thus the corrected phase laws have moments

$$\phi^{c_l}(\theta) = \phi^{g_l}(\theta) 1_{p_l(\theta) \in B_l}$$

for every integer l in weak-star limits. For σ-almost every θ, the indices whose $p_l(\theta)$ is an atom form a subgroup $J_\theta \leq Z$. Indeed countably many pairs of atomic fibers either have zero intersection (discard it) or have positive intersection, in which case the product or quotient of their values is an atom at the summed or subtracted index. Include the trivial zeroth index. Consequently

$$\nu_\theta^c = \nu_\theta^g * Haar(J_\theta^\perp), supp\nu_\theta^g \subset supp\nu_\theta^c.$$

Zero rectangles transfer from the corrected to the baseline law. This follows from the rectangle-nullity selector theorem in gaussian-ultrafilter.md [R133]: the full-rank fibers allow every measurable selector, and on the remaining fibers the corrected selector list contains every genuine baseline selector. The same proof applies over a countable abelian group, since only the circle of *one-time phases* at each first-chaos spectral point enters the covariance and the transfer of Gaussian shifts. On the self-conjugate part of $\widehat{G}$ all phases are real (± 1), including when this part is nonatomic; choose real branches there. The other spectral points can be paired by conjugation via a measurable choice of half of each pair.

Here there is also a direct simplification of the selector argument. On $|supp\nu_\theta^c| \geq h$, the corrected covariance is full rank, so a null rectangle projects to a null rectangle of positive conditional-support events on the remaining fibers (the independent Cameron–Martin argument in the cited internal note). On those remaining fibers $J_\theta^\perp$ is finite of order $q \leq h - 1$; split into its finitely many possible orders. The corrected process is the average in covariance of q rotated copies of each baseline phase branch $w(\theta)$, with rotations by the constant qth roots of unity on the chosen conjugation half and their conjugates on the other half. One can transfer all weights to the original branch w using the finite-rank Gaussian covariance transfer in the cited proof. In this case the target/source phase ratio is a fixed root on each half, so real finite-rank projections tending to the identity may be chosen to commute with it exactly (use conjugation-compatible complex-linear projections; on self-conjugate fibers $q \leq 2$). For each finite transfer of fraction < 1, the modified noises remain absolutely continuous with respect to the original finite-modification law, and the one-slot Gaussian covariance is *exactly* unchanged because the ratio commutes with the transfer projection. The intensities being transferred are equal for the copies of each branch, so standardization is bounded relative to the target intensity, even if this intensity itself tends to zero. Passing first through increasing projection ranks and then to total transfer preserves rectangle nullity by fixed-marginal cylinder approximation. This leaves precisely the baseline covariance on the finite-support part. As any nonnull baseline rectangle projects to a nonnull rectangle of its positive conditional-support events there, corrected nullity would contradict the baseline lower bound. This proves existence of an actual correcting $g_n + u_n$ with positive return.

Application. For $G = \oplus Ze_i$ and $H = \oplus z_i Ze_i$ in dyadic-pure-mark-transpose.md [R098], choose infinitely many disjoint finite packets of fresh generators. In each packet the finite density-Hales–Jewett word bound supplies a nonempty I_n with $g_n = \sum_{i \in I_n} e_i$ having h-correlation at least a fixed $\kappa(\mu A, h) > 0$. The I_n are distinct. The order-two character η_n on

H recording odd sum **only outside** I_n is therefore distinct at each stage. Every correction u with $\eta_n(u) = -1$ makes $g_n + u$ legal: inside I_n its coordinates are 1 mod L_i because $L_i | z_i$, while outside they are pure high multiples with odd total coefficient. Hence this free-group direction family has all positive returns at all finite lengths in every Gaussian model. The second normal residue was not needed. No assertion that weak limits of arbitrary Gaussian binary-event models are again Gaussian, or that this restricted conclusion automatically transfers through an arbitrary mixed-radix integer realization, is required here. The full free-group and integer capacities remain unresolved.

R142. Finite-subgroup roots need not yield any full-group root spectral mass

Scope and status. Result with proof.

Result. Infinite-rank weakly mixing parity avoidance can have mass arbitrarily close to one without full-group root spectral mass, while a parity character appears on an AP joining.

Scope. This separates joining spectrum and finite-subgroup spectral roots from the original full-group spectrum.

Qualifications. The construction concerns full-group root spectral mass, not merely absence of eigencharacters. Preserve the stronger stated exclusion and its comparison with AP-joining spectrum.

Let $H = \oplus_{i\geq 1} Ze_i$, and write its total parity as $\chi = \chi_1\chi_2$, where χ_1 records the parity sum on the even index set and χ_2 on the odd index set. They are independent nontrivial characters. For every $\epsilon > 0$ there is a weakly mixing H action and an event of measure $> 1-\epsilon$ avoiding progressions of one fixed length at **every** v with $\chi(v) = -1$. Nevertheless its full Koopman spectral type gives zero mass to

$$\{\lambda \in \widehat{H} : \lambda^k = \chi\}$$

for every positive integer k. Thus even a continuous spectral measure supported on a root coset of the specific full parity is not forced by these avoiders. This concerns the pure odd high subgroup only, and does not supply an avoider for the more restrictive mixed family which also allows normal coordinates.

Choose an integer n large. For each $j = 1, 2$ choose a list of powers of two $Q_{j,l} = M_j \cdot 2^{l-1}$ for $1 \leq l \leq n$, the two lists having no common terms. Use one independent complex Gaussian channel at each stride. Its first chaos uses multiplication by the coordinate characters of $\widehat{H}$ on the realification of $L^2(\rho_{j,l}; C)$, where $\rho_{j,l}$ is the uniform measure on the coset $\{\lambda : \lambda^{Q_{j,l}} = \chi_j\}$. Each is a translate of Haar measure on $K_Q = (C_Q)^N$, a compact infinite nonatomic group ($Q = Q_{j,l} \geq 2$). Thus the H-Gaussian action, also on the finite product of all channels, is weakly mixing: every nonconstant chaos spectrum is dominated by finite convolutions of continuous first-chaos spectral measures, including their reflections.

For a fixed j retain an origin when at least one of the n real Gaussian coordinates is positive. This has mass $1 - 2^{-n}$. For v with $\chi_j(v) = -1$, the lth row negates at time stride $Q_{j,l}$ and repeats at stride $2Q_{j,l}$. Over a long run, choose zero or one copy of the current stride to make its sign nonpositive. Later choices preserve all earlier signs. The run therefore has a missing mark by time $\sum_l Q_{j,l}$. The intersection of the two independent retained events has mass $(1 - 2^{-n})^2$, and every total-odd v is handled by at least one channel group.

For the spectral claim every nontrivial finite convolution is Haar measure on a translate of the largest one of its nested groups K_Q. If Q does not divide k, the equation $\lambda^k = \chi$ specifies either no point or a single coset of the kernel of the kth-power map on K_Q. The image $(C_{Q/\gcd(Q,k)})^N$ is infinite, so the kernel has Haar measure zero. If $\mathtt{Q|k}$, each strictly smaller participating stride divides $k/2$; consequently its character contributes the identity on taking kth powers. All terms at the maximal stride belong to a **single** list j, so their contribution is either the identity or χ_j, neither of which equals $\chi_1\chi_2$. This proves the zero-mass assertion

for each chaos and their direct sum; the invariant constants also avoid the coset. Finite-rank restrictions still have the eigencharacters required by joint-finite-factors.md [R147] and joint-parity-roots.md [R148].

Here the product of two **differently dilated** observed characters recovers χ. This suggests testing spectra of AP joinings or products of distinct powers instead of demanding a root at a single dilation on the full original group. No such necessity is proved here.

All independently dilated point spectra can also miss parity There is a different ergodic near-full avoidance example in which, for every $a \geq 1$, **all** eigencharacters of the action $v \mapsto T^{av}$ have finite coordinate support on H. Consequently no finite product of differently dilated independent copies has parity as an eigencharacter either. The extra eigenvalue on the singular AP joining below is a genuinely different condition.

Fix an integer n and put $M_l = 4^{l-1}$ for $1 \leq l \leq n$, $P = 4^n$. Over Haar $x \in (C_P)^N$ take n independent uniform bits y_l. Let $f_l(x_i)$ be the indicator that the corresponding base-four digit $\lfloor x_i/M_l \rfloor$ mod 4 is 3. Define the H-action by $x \mapsto x + v$ and

$$y_l \mapsto y_l + \sum_i [f_l(x_i + v_i) - f_l(x_i)] \bmod 2.$$

Every sum is finite, the cocycle telescopes, and product Haar is preserved. At stride $M_l v$, the local digit test is $\binom{\lfloor x_i/M_l \rfloor + j v_i}{3} \bmod 2$ at slot j. This uses the period-four Lucas identity for the third binomial polynomial. Its third difference in j is $v_i^3 \equiv v_i \bmod 2$. Thus the lth bit trace at this stride has period four, and its third difference is identically one when $\chi(v) = -1$. In particular it takes both bit values in a four-slot period. The OR of the bits, of measure $1 - 2^{-n}$, misses some slot of each such odd orbit by time $3 \sum_l M_l$, successively choosing one of four multiples of the current stride to reach zero; later strides preserve every earlier row's value.

To verify the spectral assertion, decompose functions by vertical bit characters $J \subset \{1, \ldots, n\}$. On a nonempty vertical character subspace simple cylinder vectors of the form $(-1)^{\sum_{l \in J} y_l} product_{i \in I} h_i(x_i)$, with I finite, are dense in linear span. For a fixed vector of this form, autocorrelations factor across axes. Outside I all local spectral marginals for the action $v \mapsto T^{av}$ are identical: they are the spectral law of

$$g_J(x) = product_{l \in J} (-1)^{f_l(x)} on C_P$$

under translation by a. The full spectral measure is the product over axes of these univariate spectral marginals, up to normalization of the vector and the finitely many changed factors on I (this follows directly by multiplying Fourier coefficients for each finitely supported v). Since the specified base-four digits are independent under uniform x, $Eg_J = (1/2)^{|J|} > 0$. If the local spectral law is singleton it is therefore the trivial character. Otherwise the infinite product of the identical non-point-mass laws is nonatomic. Neither case can give an eigencharacter with infinite support. The empty vertical character is the compact base and has only finitely supported Fourier characters. At $a = 1$ every nonempty g_J is nonconstant and the translation generates C_P, hence all nonempty vertical sectors have continuous spectrum; the base action is ergodic because finite-support translations are dense. The whole action

is ergodic. Finally, an atom in a finite convolution of spectral measures of independent dilations requires an atom in each participating measure, so finite products produce only finite-support eigencharacters. This example concerns the pure parity coset and does not include the normal cosets of the mixed model.

A coherent eigencharacter really is present on an AP joining For any countable abelian H-action with a positive event A avoiding an h-run at every v with $\chi(v) = -1$, the parity character occurs on an invariant arithmetic-progression joining, with nonzero projection of the single test 1_{A^h}. Specifically pass to the stationary binary H-shift factor generated by A, so the coordinate event is clopen. On X^h use the slope action $V^v = (T^{jv})_{0\le j<h}$ and diagonal starting law μ_Δ. Along a Følner sequence average the translates $(V^v)_*\mu_\Delta$ together with the deterministic label $\chi(v)$, and take a weak limit λ on $X^h \times \{\pm 1\}$. Its X^h marginal ν has all one-slot marginals μ; the joined action $V^v \times \chi(v)$ preserves λ. The conditional expectation of the sign given X^h is therefore a χ-eigenfunction for V on ν.

For any invariant mean on H, the average of $c(v) = \mu(\bigcap_{j<h} T^{-jv}A)$ is bounded below by a positive $\kappa(\mu A, h)$. Indeed the finite density-Hales–Jewett argument gives, for any finite list of arbitrary increments $y_1, \dots, y_M$ with $M = M(\mu A, h)$, a uniform positive lower bound on the sum over word lines of $c(\sum_{i\in I} y_i)$, with each variable set I nonempty. Apply successive independent invariant means in the y_i, obtaining the same mean of c on every summand by invariance in any one active variable. Because the odd terms of c vanish and the event is clopen, the inner product of the conditional sign against 1_{A^h} is at least $\kappa(\mu A, h)$. Thus this eigenfunction is nonzero on a single countable joining; no extremal choice of a different finite-rank eigenfunction for each group of generators is necessary. This elementary argument does not give an original `kH` eigencharacter and the preceding example shows why the distinction matters.

R143. Whole parity cosets in infinite rank

Scope and status. Result with proof.

Result. Whole parity cosets in infinite rank admit weakly mixing relative capacity one, and their three-slot capacities have the stated zero-or-half classification.

Scope. The finite-rank joint eigencharacter conclusion does not extend to infinite rank.

There is a weakly mixing action of $H = \oplus_n Z$ with all-length relative threshold one for the entire odd coset of a fixed nontrivial character $\chi : H \to \{\pm 1\}$. In particular the joint character conclusion for a finite-rank parity coset [R148] fails in infinite rank, even for ergodic actions and arbitrarily high avoiding masses. The same threshold on the integers for an actual odd residue class has a rational eigenvalue explanation; here the group action has no nonconstant joint eigenfunctions.

Put $q_i = 2^i$, $i \geq 1$. In $\widehat{H} = T^N$ take Haar probability on the closed coset $C_i = \{\lambda : \lambda^{q_i} = \chi\}$, a translate of $(\mu_{q_i})^N$; the product Haar probability there is nonatomic. It is conjugation invariant since $\chi = \chi^{-1}$. On the realification of the complex $L^2(C_i)$ use multiplication by $\lambda(v)$ for $v \in H$, and form the Gaussian action, taking the independent product for all i. Every nonconstant chaos has continuous joint spectral measure: it is dominated by a sum of convolutions of at least one nonatomic spectral probability (reflections cause no change). Thus the full countable product is weakly mixing. Choose a nonzero real first-chaos test in each row and retain if at least one of its first r independent Gaussian signs is positive. The mass is $1 - 2^{-r}$. For every v with $\chi(v) = -1$, the whole ith first chaos is multiplied by -1 at $q_i v$ and by 1 at $2q_i v$. Beginning at time zero, move by 0 or q_i to make each successive row nonpositive, preserving previous signs by divisibility. All signs are nonpositive by time $\sum_{i \leq r} q_i$. Thus each stated mask has a fixed forbidden length, proving the threshold. For $r = 1$ its three-slot obstruction at mass one half is exact. A spectral projection onto a whole root coset, possibly with continuous rather than atomic spectrum, is the natural distinction from finite rank.

For triples the half threshold is sharp when the parity itself is absent. Let a countable abelian group H act ergodically, let χ be an order-two character absent from its joint point spectrum, and set $K = \ker \chi$, $D = H \setminus K$. Then every event A of mass greater than one half admits a positive three-term return with a step in D. Every positive event admits a pair return. For the pair statement, K acts ergodically: its invariant σ algebra is an ergodic factor on which a group of order two acts, so it is trivial unless χ is an eigencharacter. Average the pair correlation over $v_0 + K$. This gives the product of the two means.

For triples let Z be the Kronecker factor for K. Each of its eigenspaces is one-dimensional and is preserved by all of H; consequently the factor is also the full H Kronecker factor. The K-action is dense on its compact rotation model. The middle slot may be projected to $g = E(1_A|Z)$ when taking the mean over $v = v_0 + u_1 + u_2$, where the two u_i run over independent Følner sets in K with nested limits. This is the two Cauchy–Schwarz cube estimate retaining the middle slot, whose surviving directional differences are $+1$ and -1; both generate all of K, even when $2K$ has infinite index. The retained function orthogonal to Z has zero $U^2(K)$ seminorm. The same conclusion holds with a continuous weight in the translation of v on Z: approximate by finite sums of characters and absorb each character into the endpoint test using a corresponding rotation eigenfunction and its conjugate at the middle. Multiplication by such an eigenfunction preserves Kronecker orthogonality. The fixed offset is absorbed into the endpoint functions.

For some $\eta > 0$ the base event $C = \{g > 1/2 + \eta\}$ has positive measure. All translations in a sufficiently small neighborhood of zero in the rotation group satisfy $\mu(C \cap (C + v) \cap (C - v)) > \mu(C)/2$, by translation continuity. Choose a nonnegative continuous weight supported there and with positive Haar integral. Conditional on the initial Z-coordinate in this triple intersection, the outer memberships at $-v, +v$ have individual probabilities exceeding $1/2 + \eta$, so simultaneously hold with conditional probability at least 2η. With the projected middle test the triple integral is at least $\eta\mu(C)/2$ for every such v, with room in the constants. The K-rotation is ergodic, so the nested Følner means give positive average weight also at the fixed translate v_0. The zero orthogonal error therefore proves the claimed positive triple return. This proof gives no uniform upper bound in the progression length, as the first construction demonstrates.

A direct middle-slot estimate. Single Følner averages also suffice. Shift the initial coordinate by $-v$, leaving functions at $0, v, 2v$, with the middle function u orthogonal to Z. Fixed offsets can be absorbed into the outside functions. By Cauchy–Schwarz and Hilbert-space van der Corput, square the mean in v of $T^v u \cdot T^{2v} w$ and bound by the mean of pairwise inner products at two progressively large auxiliary shifts s, s'. For fixed s, s', the ergodic theorem for K shows that the inner Følner limit equals the product of scalar correlations of u at $s - s'$ and w at $2(s - s')$: change the initial point by $-v$ to leave relative sampling slope one. The first correlation has mean square tending to zero in s, s' by the spectral theorem, since u has continuous K spectral measure. This proves the middle-slot replacement along every Følner sequence both on H and on every translate of K.

A sharper spectral test. Suppose again χ is absent from the joint point spectrum Λ of H. Write σ for the maximal Koopman spectral type on $\widehat{H}$. If

$$\sigma(\bigcup_{\lambda \in \Lambda} \gamma \in \widehat{H} : \gamma^2 = \chi\lambda^{-1}) = 0,$$

then the odd coset gives three-term returns at **every positive measure**. With $f = 1_A$, replace the middle by $g = E(f|Z)$ in means on both groups. Approximate g in L^1 by finite linear combinations of eigenfunctions; the trilinear test changes by at most the approximation error at every time. For a unit eigenfunction e_λ the correlation is

$$\lambda(v) \int f(x) e_\lambda(x) f(T^{2v} x) d\mu(x).$$

It is Fourier–Stieltjes. Its measure at χ vanishes under the hypothesis since the spectral projection of f onto the stated inverse-image fiber is zero. The mean of any Fourier–Stieltjes coefficient on $v_0 + K$ differs from its mean on H only by the sign times its Fourier mass at χ. The two means of the original triple test consequently agree and exist along every Følner sequence, by the uniform approximations and middle-slot estimate.

The full-group mean is positive. One elementary way to see this without any finite-index hypothesis on $2H$ is to use the finite density Hales–Jewett theorem: for some $M = M(\mu A)$ and $c > 0$, the sum of the triple correlations at all nonempty subset sums of any $v_1, \ldots, v_M$ is at least c. Indeed the expected occupied proportion of the parameter words in $[3]^M$ after a common random translate is μA, so with uniformly positive probability its occupied words contain a line; sum over the finitely many lines and translate away the fixed letters. Take

successive Følner means over $v_1, \dots, v_M$; the mean of each nonempty summand is the above full-group mean. The odd-coset mean is therefore positive.

Exact three-slot formula in an arbitrary weakly mixing action. Here $\Lambda = \{1\}$ and $g = \mu A$. The odd-coset mean of the triple correlation is exactly

$$(\mu A)\langle E(1_A|I(2K)), T^{2v_0}E(1_A|I(2K))\rangle.$$

Indeed this is the mean ergodic theorem for $v \mapsto T^{2v}$ restricted to K. On the factor $I(2K)$ the transformation T^{2v_0} is an involution, and its fixed-point set is invariant under the whole ergodic action. Thus either it acts trivially almost everywhere or it has no fixed points almost everywhere. In the trivial case every positive event has a triple return. In the nontrivial case a measurable selector of the two-point orbits has measure $1/2$. It is exchanged with its complement by every T^{2d}, $d \in D$, so it exactly avoids all triples at these differences. The preceding half bound is sharp. Equivalently the fixed-system three-slot cap is precisely zero or one half, according as the maximal spectral type gives zero or positive mass to $\{\gamma : \gamma^2 = \chi\}$. This equivalence follows because the annihilator of $2K$ splits into $\{\gamma : \gamma^2 = 1\}$ and $\{\gamma : \gamma^2 = \chi\}$, and only the second is negated by T^{2v_0}. The example with $q_1 = 2$ realizes the second case without joint eigencharacters.

R147. Finite joint rational factors for profinitely open directions on Z^m

Scope and status. Result with proof.

Result. Profinitely open directions in finite-rank ergodic actions force a joint finite rational factor separating all directions by order.

Scope. The parity formula is exact; an arbitrary finite-rank profinitely open family has only the stated upper bound.

Let $H = Z^m$ act ergodically on a standard probability space X, and let $D \subset H$ be a union of cosets of finite-index subgroups. For $k \geq 1$, if an event of mass strictly greater than $1 - 1/k$ avoids all h-runs in D at some fixed finite h, there exists a finite **joint rational factor** H/L of this action such that every $v \in D$ has order strictly greater than k in H/L. Equivalently, the all-length cap in this system is at most $1 - 1/s$, where s is the supremum over finite rational factors of the smallest order of any allowed element in that factor. This upper bound is exact for parity cosets. In several dimensions a lower bound for general D does not follow by deleting an interval in a cyclic factor, since the finite groups need not be cyclic.

This strengthens joint-parity-roots.md [R148] by using the original H-suspension for the marker and only an **external** finite-sheet clock for a tested difference class.

Full suspension. Form the R^m suspension over **all** of H with fundamental cube $[0,1)^m$. It is ergodic, and its real joint eigenfrequencies are

$$E = \{\xi \in R^m : h \mapsto e(\xi \cdot h) is a joint eigencharacter on X\}.$$

They include $H* = Z^m$. Choose a generic real time α as in the preceding note; $S = S_\alpha$ is totally ergodic, its point spectrum is exactly the injective image $\xi \mapsto e(\alpha \cdot \xi)$ of E, and the same map is injective on the countable group generated by $E \cup Q^m$. The natural clock is $P_H = R^m/H$, rotating by α. Mark $T^{floor z}x \in A$, excluding a fixed small fractional-coordinate boundary band to keep the mass above $1 - 1/k$. Whenever $v + K \subset D$ with K finite index and $n\alpha \bmod K$ sufficiently close to $v \bmod K$, every run in the marked event would yield an actual h-run on X at the neighboring integer lift in $v + K$, by the same floor calculation as in joint-parity-roots.md [R148].

For each such v, K the mean weighted by the indicator of a small open neighborhood of $v \bmod K$ in the external clock $P_K = R^m/K$ is zero. Use BHK uniform-null comparison before this bounded weighting to replace the marker by its $Z_{h-1}(S)$ conditional expectation. Take a positive superlevel of mass above $1 - 1/k$, approximate it by a finite nilfactor and use the finite-factor superlevel with conditional density $> 1 - 1/(2h)$ as in the previous note. These choices **do not depend on the tested coset**: pointwise conditional union bounds at fixed n transfer zero weighted means for every single clock test. Include P_H in this finite nilfactor and call the resulting high-measure event B on it. It remains to contradict any allowed velocity of order at most k in an actual joint finite factor of X derived from this one nilmodel.

Horizontal period lattice. Write the connected nilfactor of the totally ergodic S as $Y = N/\Gamma$, with N simply connected nilpotent and $S(x) = a\phi(x)$ an affine niltranslation,

obtained by presenting the original group as generated by the translating element and its identity component. Here ϕ is unipotent and preserves Γ. Its horizontal Kronecker torus Q has dual a finitely generated subgroup $E' \subset E$ containing `H*`. The factor onto P_H, after adjusting its additive constant, is induced by a real homomorphism $F : N \to R^m$ modulo H factoring through the horizontal projection $f : N \to Lie Q$; in dual coordinates this last projection onto R^m is restriction from E' to `H*`. The horizontal period lattice of Q is the image of Γ under f: the fibers of the map to Q are connected because $[G, G]$ is connected in the presentation generated by G^0 and the single translating element. See the rational-lift details and sources in the preceding note. In particular $L = F(\Gamma)$ is a finite-index sublattice of H.

We may represent the affine translation with $F(a) = \alpha$ exactly, not merely modulo H. Its horizontal displacement in Q is evaluation at α on E'. The real vector $\xi \mapsto \alpha \cdot \xi$ on E' is a lift of this displacement to `Lie Q`. Change the chosen a by right multiplication by an element $\gamma \in \Gamma$ so its horizontal projection is precisely that vector, using surjectivity onto the horizontal period lattice. Replace ϕ by $Ad(\gamma^{-1})\phi$, which gives the identical affine map on N/Γ. Also $F\phi = F$ identically, since the clock action is a translation, the possible discrepancy is in the discrete lattice H, and N is connected. This last identity persists under the inner change.

For each $\ell \in L*$, the function $e(\ell \cdot F(x))$ on Y is well defined on N/Γ and is an S-eigenfunction of eigenvalue $e(\ell \cdot \alpha)$. Since ℓ is rational and the generic-time map is injective on $E + Q^m$, it follows that $\ell \in E$. By the full-suspension eigenvalue identification, each such $\ell|H$ is a joint eigencharacter of X. Therefore H/L is indeed an original rational factor. In more detail choose cyclic generators of the finite group `L*/H*` and normalize the associated eigenfunctions to have values in their respective finite root groups (the appropriate powers are invariant constants on the ergodic system). Their joint law is invariant under the full finite dual translation group and hence uniform, giving the stated factor.

Periodic right translation and the clock test. Suppose $v \in D$ has order $d \le k$ in this factor. Choose K with $v + K \subset D$. The image of

$$N/\Gamma_K \to Y \times P_K, x \mapsto (x\Gamma, F(x) \bmod K), \Gamma_K = \Gamma \cap F^{-1}(K),$$

is the joint nilorbit closure at the identity. Indeed Γ_K has finite index and is ϕ-invariant. The lift of $a\phi$ to this connected finite cover is totally ergodic: any horizontal character obstructing an ergodic power there has a nonzero multiple descending to the corresponding horizontal character on Y, contradicting total ergodicity. This uses the usual horizontal ergodicity criterion in the same connected affine nilsystem presentation; the commutator group is independent of the lattice. The lifted dynamics maps to (S, R_α) by $F(a) = \alpha$. Thus this minimal cover gives the claimed joint orbit (indeed the map is injective). Its Haar measure supplies the joint means for fixed initial point by unique equidistribution.

Choose $\gamma \in \Gamma$ with $F(\gamma) = dv$, possible by $dv \in L$, and take its rational root $t = \gamma^{1/d}$. Then $F(t) = v$ and right translation by t has order dividing d on the finite cover of $\mathbf{Y}$ given by $\Gamma' = \bigcap_{i<d} t^i \Gamma t^{-i}$. It need not have order d on the external clock cover. The event B is on Y and ignores that extra sheet.

By Frantzikinakis–Host, Lemma 4.6 and Appendix B, every tuple $(xt^j\Gamma_K)_{j<h}$ belongs to the arithmetic-progression joining support of the minimal lift; the corresponding Hall–Petresco element is the product of a constant and a linear sequence. Its angular slope in

P_K is v mod K. The weighted finite nilfactor mean of B^h is exactly its Haar joining mass with angular slope in the tested neighborhood: integrate the diagonal start and the time parameter by the usual two-parameter nilorbit formula, noting that each slot's base marginal is Haar. One can also use rectangles whose start average is taken long enough for each finite time sum, as in fixed-length-all-thresholds.md [R123].

Take the small-motion core of B on Y: for each of the finitely many connected joining components and slots, require probability $> 1 - 1/(2h)$ of landing in B after a small Haar absolutely continuous component motion projected to the slot on Y. The core still has measure above $1 - 1/k$ by continuity of translation in L^1, and motions are small enough to keep every angle initially equal to v mod K inside its tested neighborhood. A tuple of core points of that angle would contradict zero weighted Haar mass by simultaneous union bound and absolute continuity of the orbit motion on its component. Yet pull back the core to N/Γ': by the union bound its first d right translates meet in positive measure. Any such x has all $(xt^j\Gamma)$ in the core, and the representative x in N gives exactly the tuple on N/Γ_K with the required slope. This contradiction proves the theorem.

Exact parity formula. For $D = \{\chi = -1\}$, if χ is not in the joint point spectrum then every finite rational factor has an allowed vector in its kernel, so the cap is zero. Otherwise write s for the largest $s \geq 0$ such that some joint eigencharacter λ satisfies $\lambda^{2^s} = \chi$, with value infinity if these heights are unbounded. Then the cap is $1 - 1/2^{s+1}$, with value one at infinity. For the upper bound, on a finite abelian rational factor Q_f, either χ does not factor through and the kernel contains an odd vector, or χ descends. In the latter case write the 2-primary part as a direct sum of cyclic $C_{2^{n_i}}$. The parity is nontrivial on specified cyclic factors, and the minimum order of a group element on which parity is nontrivial is $2^{\min n_i}$ over those factors. The divisibility height of χ in Q_f's dual is exactly $\min n_i - 1$, hence at most s. Apply the upper theorem. For the lower bound use a joint root λ of height s': it has order $2^{s'+1}$ and every odd vector is an odd generator on its cyclic factor. Deleting its zero point gives an avoider of measure $1 - 2^{-(s'+1)}$ at that length. Take all available heights. This includes the no-parity case without the extra two-sheet cost incurred in the earlier direct suspension over K.

R148. Joint rational roots forced by avoiding a whole parity coset

Scope and status. Result with proof.

Result. High-mass avoidance of a whole finite-rank parity coset forces a joint rational root of parity.

Scope. A finite-rank necessary condition; later finite-factor analysis sharpens it, and infinite-rank coherence is false.

Qualifications. The finite-rank necessity is sharpened by joint-finite-factors.md and explicitly fails to imply infinite-rank coherence, as shown in infinite-parity-spectrum-counter.md.

Let $H = Z^m$ act ergodically on a standard probability space, let $\chi : H \to \{\pm 1\}$ be nontrivial, and let $D = \{v : \chi(v) = -1\}$. Fix $k = 2^r$, $r \geq 1$. Suppose an event of mass strictly greater than $1 - 1/k$ avoids all h-term orbits in D at some finite h. Then there is a **joint** eigencharacter $\lambda : H \to T$ of the H-action satisfying $\lambda^k = \chi$. This is stronger than requiring deep eigenvalues separately in each cyclic subaction. The theorem gives a necessary condition on finite pure high subgroups of the dyadic construction, not an unrestricted capacity upper bound for that construction.

Suspension. Set $K = \ker \chi$ and choose $v_0 \in D$; the nonzero sheet point $w = v_0 \bmod K$ has order two in $P = R^m/K$. The K-ergodic components of an ergodic H-action are one or two equal-mass sheets permuted transitively by H: the factor of K-invariant functions has an ergodic action of the group of order two, hence is its single orbit. On any one component X_0 form the suspension $Y = (X_0 \times R^m)/K$ with $(x, z + k') \sim (T^{k'}x, z)$. Its R^m-translation action is ergodic and it projects to P. Its real joint eigenfrequencies form the countable additive subgroup

$$L = \{\xi \in R^m : k' \mapsto e(\xi \cdot k') is an eigencharacter on X_0 \quad \text{for} \quad K\}.$$

Indeed an eigenfunction on the suspension lifts to $e(\xi \cdot z)f(x)$ in product coordinates; the quotient condition gives exactly the eigenfunction equation on the component. This equality can equivalently be derived by Fubini and translation invariance on bounded clock boxes. L contains the full dual lattice `K*`.

For Lebesgue-almost every $\alpha \in R^m$, simultaneous time translation $S = S_\alpha$ is totally ergodic, and its eigenfrequencies, identified abstractly, are precisely L embedded by $\xi \mapsto e(\alpha \cdot \xi)$. To check this, take a maximal spectral measure for the strongly continuous R^m Koopman action. The pushforward of its continuous part under $\xi \mapsto e(\alpha \cdot \xi)$ has no atoms for almost every α, by Fubini on pairs of distinct frequencies (the equality of their projected values describes countably many hyperplanes). Exclude the countable collection of hyperplanes where distinct joint eigenfrequencies agree or a nonzero frequency gives a root of unity. This also makes the clock rotation by α mod K totally ergodic and identifies its dual with $K* \subset L$.

Let A be the hypothetical avoider on X. In Y use the marker $T^{floor z}x \in A$, with coordinatewise floor, and delete the band where any fractional coordinate is within η of an integer. The marker is well defined on the suspension. Using a domain comprising $[0, 1)^m$ and $v_0 + [0, 1)^m$, its mass before the band is μA: average over both H/K sheets. Choose

η so its new mass remains strictly above $1 - 1/k$. If $n\alpha \bmod K$ lies sufficiently close to w, there is $v \in v_0 + K$ with $\|n\alpha - v\|\infty < \eta/(2h)$. Every marked h-run under S^n would then lift, starting from a safe clock, to $T^{floor z+jv}x \in A$ for every $j < h$. A countable common null set suffices. Hence the lifted marker has zero correlation on an open clock neighborhood U of w.

Passage to a finite connected nilfactor. The ordinary Z Host–Kra characteristic factor, or the Bergelson–Host–Kra uniform-null comparison, allows replacing the marker by its conditional expectation g on $Z_{h-1}(S)$ in the normalized mean weighted by $1_U(n\alpha \bmod K)$; choose a small ball or box U with Haar-null boundary. Use an arbitrarily small positive superlevel of g of mass strictly above $1 - 1/k$. Its weighted mean remains zero by pointwise domination of g. Approximate this superlevel C by a finite-dimensional nilfactor; on that factor use $B = \{E(1_C|Y_0) > 1-1/(2h)\}$. Its mass too is strictly above $1-1/k$ if the factor is fine enough, and its weighted mean is zero. For the last assertion, at each fixed n condition on the initial factor point. If all projected slots lie in B, the probability all lifted slots lie in C is at least one half by the conditional union bound. Include the torus clock factor by taking the join, an ergodic nilsystem realized as the orbit closure in the product nilmanifold. Denote this final nilsystem by $Y_0 = G/\Gamma$; it is connected since S is totally ergodic. The clock map $\pi : Y_0 \to P$ factors through its horizontal Kronecker torus Q, and may be translated to send the identity basepoint to zero. Spectrally, Q is dual to a finitely generated subgroup $L' \subset L$ containing `K*`.

Horizontal torsion lift. If $K* \cap kL'$ is annihilated by evaluation at w, then w lifts to a point of order dividing k in Q: extend its character from the image of `K*` in L'/kL' to that whole group, using divisibility of the circle. Every value of the extension is k-torsion. Conversely such a lift requires the stated annihilation.

If the annihilation fails, some $\ell = k\xi \in K*$, $\xi \in L'$, has $e(\ell \cdot v_0) = -1$. The K eigenfunction with character $e(\xi \cdot k')$ on X_0 gives a joint H eigencharacter extending that K-character. When K is ergodic on X, its eigenspace is one-dimensional and preserved by T^{v_0}; when there are two components, induce either extension of the character across the two sheets by translating the eigenfunction. Any such extension is $e(\xi \cdot h)$ times either the trivial character or χ. Because k is even, its kth power is exactly $e(\ell \cdot h) = \chi(h)$. This is the desired conclusion, so under its negation take a k-torsion lift in Q.

A periodic progression in the actual joining. Use the usual presentation generated by G^0 and the nilrotation. Here $[G, G]$ is connected and lies in G^0; this is the elementary disconnected-group fact for a nilgroup generated by one element and its identity component (see Ackelsberg–Richter–Shalom, Proposition 2.2). The connected nilmanifold G/Γ is also N/Γ_0, using the simply connected cover of G^0 and the lifted lattice. The map to the horizontal Kronecker torus has connected fibers (they are the orbits of the connected commutator group). Consequently the horizontal projection of Γ_0 surjects onto the full period lattice of Q: equivalently use the fundamental-group exact sequence of this connected homogeneous fibration, or the description of $Q = G/[G, G]\Gamma$. Choose $\gamma \in \Gamma_0$ whose horizontal coordinate divided by k equals the chosen torsion class in Q, and put $t = \gamma^{1/k} \in N$. This rational element has clock image w. Its conjugates of Γ_0 are commensurable, and $\Gamma' = \bigcap_{j<k} t^j \Gamma_0 t^{-j}$ is a finite-index lattice normalized by t. Also $t^k \in \Gamma'$. On N/Γ' right multiplication by t is therefore a well-defined Haar-preserving map of order dividing k. For every measurable B' of normalized mass above $1 - 1/k$ its first k translates have positive common intersection.

By Frantzikinakis–Host, Lemma 4.6 and Appendix B, the closure of the diagonal/slope

orbit in a minimal singly generated nilsystem is its Hall–Petresco nilmanifold. In particular every tuple $(xt^j\Gamma_0)_{j<h}$ belongs to the true arithmetic-progression joining support: it is the projection of the product of a constant sequence and the linear power sequence in the Hall–Petresco group. The slope clock coordinate there is the difference of the first two clock coordinates, so these tuples have slope angle w. The joining is Haar on its finitely many components, by unique orbit equidistribution.

For completeness the zero weighted mean rules out even such a lower-dimensional tuple if its slots lie in a suitable core of B. On every connected component of the finite-slot joining, average by a small absolutely continuous identity-neighborhood law on its connected transitive Lie subgroup. Require a core point of Y_0 to enter B with probability $> 1-1/(2h)$ under each of these projected slot motions. The finitely many convolutions tend in L^1 to 1_B as the neighborhoods shrink, so this core still has mass above $1-1/k$. Make all motions small enough that slopes at w stay in U. A tuple of core points at angle w would give positive probability of landing in B^h with slope in U, although the perturbed tuple law is absolutely continuous on its Haar component and that event has zero Haar measure. Pull back the core to N/Γ' and apply the periodic right translation: the positive intersection just proved supplies exactly such a tuple, the contradiction. This finishes the joint-character theorem.

Use on the free high axes. For each finite family of pure high generators $2^{D_i}e_i$ in dyadic-pure-mark-transpose.md [R098], their subgroup is $H = Z^m$ on which every odd-parity coset element is legal. In a not necessarily ergodic H-action, every ergodic component having event mass $> 1-1/k$ therefore has a joint root of parity. A global event of mass a leaves at most $k(1-a)$ of the component measure outside that sufficient high-mass class. In a larger ergodic abelian group action, the event of H-components supporting any particular H eigencharacter is invariant under the whole commuting action and thus has probability zero or one. There are just finitely many possible kth roots of the fixed parity on a finite-rank H. Whenever $a > 1-1/k$, at least one of them consequently occurs on almost every H-component. This necessary condition does not guarantee an eigenfunction on the full infinite-rank high subgroup; eigenfunctions for larger and larger finite subgroups can vary without a nonzero common limit.

R158. Thue–Morse short-obstruction sets have integer endpoint one

Scope and status. Result with proof.

Result. Thue-Morse missing-pattern difference sets have integer endpoint one, with extensions to the specified permutation-automatic settings.

Scope. This does not assert a dichotomy for arbitrary automatic languages with unrestricted transient behavior.

Let $t(x)$ be the binary digit sum of the nonnegative integer x modulo two. For a fixed $H \geq 2$ and a prescribed H-bit pattern w, let D_w consist of the positive d for which no $x \geq 0$ gives

$$(t(x), t(x+d), \ldots, t(x+(H-1)d)) = w.$$

Then D_w belongs, after one fixed integer scaling, to bounded signed sums of binary powers. In particular its all-length critical density is one by the bounded-power sieve [D02] (only its separate bounded-power input is used here, **not** its characteristic-factor argument). This applies to one avoided color, two avoided colors, or any finite union of forbidden patterns. It does not assert that every bounded-complexity difference avoids a pattern. There is an extensive literature on the maximum monochromatic length at individual Thue–Morse differences, for example Aedo–Grimm–Nagai–Staynova; the short-pattern argument here concerns a uniform class of differences.

Local programming Fix M sufficiently large in terms of H, and put $L = M+4+\lceil \log_2(H+1) \rceil$. Call a bit position $b \geq 0$ good for d if, with $d_b = \lfloor d/2^b \rfloor$,

$$dist(jd_b, 2^M Z) > H \quad \text{for every} \quad 1 \leq j < H.$$

At the start of a length-L bit block of $x + jd$ there is an incoming carry c_j between 0 and H, determined by all lower bits of x and d. Write $u_j = jd_b + c_j$. For any incoming vector the u_j are distinct mod2^M, by the goodness assumption. Locally vary the L-bit block of x, an integer X ranging between 0 and $2^L - 1$. The bits in the corresponding result block are $X + u_j \bmod 2^L$. As X varies the variable part of the outgoing carry in each of the H sums changes at most once; the contribution of higher bits of d to u_j is fixed and irrelevant. Thus there is a common consecutive interval of X values of length at least $2^L/(H+1)$, on which every outgoing carry is fixed. Its length exceeds $2^{M+2} + 1$.

Over this safe interval let $p(X) \in F_2^H$ be the vector of digit parities of the H local result blocks. Its affine span is the whole F_2^H. Indeed its differences include all vectors $p(X+1) - p(X)$ with X,$X+1$ in the interval. The jth coordinate of this difference is $1 + v_2(X + u_j + 1)$ modulo two, by the carry rule, with no wrap because the outgoing carry is fixed. For any slot i find X,X' in the interval (each with its successor) with $v_2(X + u_i + 1) = M$ and $v_2(X' + u_i + 1) = M + 1$; congruences modulo 2^{M+2} suffice. For $j \neq i$ the two valuations both equal $v_2(u_j - u_i) < M$. Hence the two difference vectors differ by precisely the ith standard basis vector. This proves affine fullness.

Whenever there are at least 2^H disjoint such good blocks, start from the least significant one and fix all intervening bits of x, for instance at zero. Use one of the safe intervals at every selected block; all its possible choices give exactly the same outgoing carries, so later intervals and all data away from the variable blocks are independent of the choices. The possible final parity vectors are a translate of the sum of these local sets $p(I)$ in F_2^H. Adding any set of full affine span to a nonempty proper set strictly increases its size unless the sum is already the whole group: equality in cardinality would imply invariance of the original set under all differences from the added set, which generate F_2^H. Consequently the complete sum is F_2^H. A difference in D_w therefore has at most $O_H(1)$ good positions, since a list of starting positions always has disjoint length-L subblocks of cardinality at least its cardinality divided by L.

Reading the remaining binary language Set r_b=d_b mod $2^M/2^M$ and let R be the finite set of rational circle points of denominators at most $H-1$. At each nongood b there is a unique $u_b \in R$ with circle distance at most $H/2^M$ from r_b. Uniqueness holds when M is large by separation of distinct elements of R. If b,$b+1$ are both nongood then

$$r_b = 2r_{b+1} + bit_b(d)/2^M \bmod 1, u_b = 2u_{b+1} \bmod 1.$$

For the second equality note that $2R \subset R$ and the error is less than the minimal separation of R. Along each consecutive nongood interval, reading from its high end down, the rational centers thus have a uniformly bounded preperiod and period under doubling. On a nonzero odd-denominator cycle their distance from 1/2 and both circle endpoints is uniformly positive, so the bit at $b+M-1$, i.e. $\lfloor 2r_b \rfloor$, is exactly $\lfloor 2u_b \rfloor$ and forms a periodic word. On the zero cycle the r_b are either close to zero from above or close to one from below, and cannot change sides between consecutive zero centers. Indeed if r_{b+1} is near zero then $2r_{b+1} + \mathrm{bit}_b/2^M$ is still near zero without wrapping. If it is near one then this sum is strictly less than 2 (the grid spacing gives $r_{b+1} \le 1 - 2^{-M}$) and is near one after reduction. Hence in this case the bits are constant. The finitely many transient centers, the boundedly many good positions, and the first $M-1$ low bits are all a bounded number of exceptional digits.

It follows that the binary expansion of every $d \in D_w$ is a concatenation of a bounded number of periodic substrings with bounded periods, and boundedly many single digits; all bounds depend only on H. Choose P a common multiple of the possible periods. A string of repeated blocks of length l|P represents, up to its place shift, an integer of the form $B(2^{lm}-1)/2^l - 1$, with B bounded in terms of H. Multiplication by $2^P - 1$ turns this into a bounded signed sum of powers of two; a partial terminal block has bounded length and contributes a bounded coefficient. This proves the assertion.

Fixed finite abelian digit colors More generally let $q \ge 2$ and let $c(x) = \sum_i \psi(d_i(x))$ in a finite abelian group G, where $d_i(x)$ are the base-q digits and $\psi(0) = 0$. Replace G by the subgroup generated by the letter values. For any one color $g \in G$, every set of differences avoiding H consecutive g values has integer critical density one. The conclusion covers periodic position-dependent digit maps by grouping one period into a single larger-base digit. It also applies to any positive color class from merging these individual colors.

To prove it write $\delta_a = \psi(a+1) - \psi(a)$ for $0 \le a < q-1$ and let B be the subgroup generated by $\psi(q-1)$ and all $\delta_a - \delta_b$. In G/B write u for the common δ_a; then $\psi(a) \equiv au$

and $(q-1)u = 0$. Thus $c(x) \equiv xu$ in G/B. In particular d with d $u \neq 0$ lie in a fixed complement of a zero congruence class, which has endpoint one. For all d with d $u = 0$ it suffices to show that many good local blocks can program a constant g-word.

Use the same good-block definition and local incoming/outgoing carries with q^M replacing 2^M; enlarge the fixed block length so that some safe interval carries an entire set of residues modulo q^{M+2} along with their successors. The vector derivative when X advances by one has jth coordinate

$$\delta_a - r\psi(q-1),$$

where r is the number of trailing $q-1$ digits of $X + u_j$ and a is its next digit. At any chosen i, fix $X + u_i$ to end with at least M digits $q-1$. Since the shifts are pairwise distinct modulo q^M, the other coordinates' derivative values are determined by those prescribed M low digits alone, including their first non-$(q-1)$ digit. For the chosen coordinate we can specify $r = M$ or $M+1$ and any $a < q-1$ with the additional two digits. Differences between two such derivative vectors generate the ith coordinate copy of B. All derivative vectors project modulo B to the common diagonal u. Since G/B is generated by u, the differences among the local profiles generate the full subgroup

$$V = \{(v_j) \in G^H : v_j + Bisindependentofj\}.$$

Conversely each local profile difference lies in V, because the outgoing carries are fixed and a local digit $\sum \mathrm{mod} B$ equals the represented block's integer value times u. Thus each local alphabet occupies one coset of V and its differences generate V. After at least $|V|$ disjoint good blocks the cardinality-growth argument fills a whole translate of V; choices in all other blocks and the full high tail are fixed. Globally $c(x+jd) \equiv xu$ in G/B across every slot when d $u = 0$, so this entire translate is V itself. It contains the prescribed constant g-vector. This proves the claim for d having many good positions.

With boundedly many good positions the same finite rational-center proof works with multiplication by q. Nonzero centers on a cycle have denominator coprime to q and are separated from every digit boundary k/q; the base-q digits at $b + M - 1$ therefore agree with the periodic center digits. A run on the zero cycle has all high digits zero or all $q-1$, since $qr_{b+1} + d_b^{(0)}/q^M$ is strictly less than q even at the last grid point. Finite transients and periodic arcs yield bounded signed sums of q-powers after multiplying by a fixed $q^P - 1$. The bounded-power sieve and the product with the fixed-congruence complement finish the endpoint-one assertion.

Nonabelian groups and permutation automata The same endpoint conclusion holds for every coloring obtained from a finite permutation automaton, in particular any constant-length bijective substitution. On grouping a fixed number of base-q digits we may assume its all-zero digit acts as the identity permutation: group one full order of the zero permutation and take the resulting larger base. After inverting group elements if necessary, the group value is now $c(x) = \psi(d_0)\psi(d_1)\cdots\psi(d_r)$ in least-significant-digit first product order, with $\psi(0) = 1$. The original automaton color is a function of this value. The subgroup K

generated by the $\psi(a)$ is finite, and any particular attainable target color has an element $g \in K$ mapping to it. It suffices to reach the constant g-vector on H slots.

Put $B = \psi(q-1)$ and $D_a = \psi(a+1)\psi(a)^{-1}$. The derivative $c(X+1)c(X)^{-1}$ is $D_a B^{-r}$, as the high digit tail cancels from this left quotient. Let N be the normal closure in K of B and all $D_a D_b^{-1}$. The quotient K/N is cyclic on a generator u of order dividing $q-1$, since every D_a has the same quotient u, $\psi(a)$ has quotient u^a, and B becomes trivial.

On one safe good block let $P \subset K^H$ be its set of local profiles and let J be the subgroup generated by PP^{-1}. Its projection on each slot is K: suitable consecutive X in the long interval have $r = 0$ and next digit a for each $a < q-1$, so the projection contains all D_a, which generate K. Keeping the other slots' derivatives fixed as before, two values with the same $r = M$ and next digits a,b isolate $D_a D_b^{-1}$ at the target slot. Comparing $r = M$ and $r = M+1$ with next digit a isolates $D_a B D_a^{-1}$ there. Conjugate these isolated elements by members of J which have chosen values at the target coordinate, using surjectivity. It follows that J contains a full coordinate copy of N for each coordinate. Every block derivative projects to the common diagonal u, while all ratios of profiles project into that diagonal subgroup since the outgoing carries are fixed. Hence J is exactly the normal subgroup V of K^H consisting of tuples in the same coset modulo N.

Each P lies in a V-coset. In successive multiplication by such profile sets (also allowing fixed digit factors between them), cardinality strictly increases until a whole V-coset is filled. Indeed V is normal in K^H, and $|AP| = |A|$ would imply that A is right-invariant under PP^{-1}, which generates V. Fixed intervening factors preserve the coset property and its cardinality. For d with $u^d = 1$ the final coset is V itself by $c(x+jd)N = u^{x+jd} = u^x$. Thus $|V|$ disjoint good blocks suffice. All remaining differences are in the fixed congruence complement or the bounded-power family proved above.

Finite abelian counts of bounded digit patterns The same endpoint assertion holds when the finite abelian coloring sums contributions depending on boundedly many consecutive digits, for example binary Rudin–Shapiro. After grouping enough base digits into a digit of a fixed larger base Q, the color takes the form

$$f(n) = \sum_{i \geq 0} \phi(b_{i-1}, b_i), \phi(0,0) = 0,$$

including any upper boundary contribution from the zero-padded expansion and a fixed low boundary digit. Grouping represents each original finite-range contribution by a pair of adjacent enlarged letters. The resulting function can also have a separate function of the first enlarged digit; this makes only a fixed low-residue correction below and causes no change. Write G for its finite abelian target.

For the increment $g(n) = f(n+1) - f(n)$, a trailing run of r maximal digits $Q-1$ followed by a digit $a < Q-1$ and then b determines the increment. In particular for every $r \geq 2$ it has the form

$$g(n) = r\gamma + \delta_{a,b},$$

for a fixed element γ of G and a finite list δ, since every interior adjacent maximal pair becomes $(0,0)$, with just the lower boundary and the two upper adjacent pairs contributing

additional terms. Let B be generated by the differences of all the values $r\gamma + \delta_{a,b}$ with r ranging over any full period $\exp(G)$ and all a,b. Modulo B the increment is periodic: on the tail-max cylinder $r \geq 2$ all its values coincide, and elsewhere a fixed bounded number of low digits determines it. Therefore f modulo B is periodic too, since summing increments through one period adds a fixed element of the finite quotient. Let P be its period.

For d divisible by P all slots $f(x+jd)$ have the same coset modulo B. Program a long line by the preceding good-window method, now in base Q. Choose good positions sufficiently far above the bottom digits needed to fix x modulo the Q-supported part of P. In each local variable block fix the preceding digit vector from lower blocks. Attribute to this block the adjacent pairs ending inside it. Choose a long common consecutive interval of X so that for every slot both its outgoing carry and its top result digit are fixed. Only $O(QH)$ boundaries need be removed, and the block length depends just on Q,H,$\exp(G)$. This keeps the next interfaces fixed, hence the achievable profiles add independently in G^H. When a selected slot has a trailing maximal run well past the separating M low digits, variations through a full $\exp(G)$ possibilities and all choices of the next two digits produce all the deep increment differences generating B on that slot. The other slots' first nonmaximal digits occur strictly below M, so prescribing $M + 2$ common low digits fixes their derivative values, including the extra adjacent digit. Use runs at least $M + 2$. The lower boundary contribution at the selected slot depends on its predecessor digit, but is independent of r,a,b over these deep runs and cancels from the differences. Upper boundary terms are unaffected. Thus the local difference subgroup contains the coordinate copy of B for every slot.

After at least $|G|^H$ local blocks the sum of their profile sets is invariant under B^H. Indeed if adding any block fails to increase cardinality, its differences already preserve the preceding sum, hence B^H preserves it and all its subsequent translates and sums. Strict increases at every step would exceed $|G|^H$. For an attainable original color t choose the sufficiently low digits of x to agree modulo the Q-supported part of P with some x_0 giving $f(x_0) = t$. After arbitrarily fixing one viable selection in each variable block, select digits far above all of d and these blocks to make the resulting x also agree with x_0 modulo the coprime part of P, by invertibility of the power of Q there. Fix this common high tail. Its contribution is fixed for all allowed selections, since the last output interfaces were fixed; use fixed padding in between. The reachable profile set is still B^H-invariant and now contains one tuple whose cosets all equal $t + B$. It therefore contains $(t, \ldots, t)$. Nongood-window stretches and the complement of PZ have endpoint one by the same bounded-power and congruence arguments as before.

In particular, a translated sum box D meeting all finite-index subgroups and with infinitely many positive available increments supplies all lengths for each attainable permutation-automatic color, without any restriction on the spectral subgroup of these increments. Freeze a finite representation of $z \in D$ divisible by $\mathrm{ord}(u)$, enlarging it to be positive by appending sufficiently many positive disjoint zero-residue packets if necessary. Such packets exist on any prescribed tail by pigeonholing the partial sums of positive increments modulo the desired modulus. Choose further positive packets b_j on disjoint supports, divisible by $\mathrm{ord}(u)$ and by q^{m_j}, where m_j exceeds the digit lengths of every earlier contribution by an arbitrarily large buffer. For t fixed with $q^t > H$, each packet gives a good position t digits below its least nonzero digit. In fact there the relevant truncated number is q^t times an integer v not divisible by q. For each $1 \leq j < H$ the q-divisibility of jv is uniformly bounded in terms of H and q: select a prime factor of q for which v has valuation strictly below the valuation in q. Thus M can be fixed sufficiently large so that $jq^tv \neq 0$ modulo q^M; its distance to $q^M Z$

is at least $q^t > H$. Earlier and later packets do not affect this window by the chosen buffers. The positive step $z + \Sigma b_j$ has the required congruence and arbitrarily many disjoint good blocks, so has a full run of the prescribed color.

R173. Exactly one exceptional digit gives all finite-color recurrence

Scope and status. Result with proof.

Result. The integer language with exactly one digit 2 and all other digits 0 or 1 is multiply recurrent in every finite coloring and every piecewise syndetic set.

Scope. The resulting witness-size criterion would imply a density bound, but the required quantitative growth estimate is not proved.

Qualifications. Retain the warning that qualitative finite-color recurrence gives no all-length density threshold without the exponential witness-size bound.

Fix $h \geq 2$. Suppose a finite tuple of actual integer queries with one tagged query $e(x_*, r_*)$ and finitely many normal queries has an ordered identity expressing the tagged tuple in the subgroup generated by the normal tuples. Here $e(x, v) = (u_{j,x+jv})_{0 \leq j < h}$ uses a separate free group of symbol letters for each player. Suppose disjoint scaled copies at positions b^{B_i}, with $B_i \to \infty$, realize all normal choices, the zero choice, and precisely one tagged query as allowed positive integer differences. Then every finite coloring contains a monochromatic h-term progression with such a difference, using arbitrarily late copies. The same holds inside every piecewise syndetic integer set. In particular this applies to digits $0, 1$ and exactly one digit 2 in every base $b \geq 3$, by the integer word relation [R217]: the normal $0, 1$ subsums contain an IP set and the eligible marked values meet every modulus. For the latter, place one 2 in a sufficiently high position for the part of the modulus supported on primes of b, and add enough distinct later 1's in positions of the same power residue modulo the coprime part to cancel that mark.

Semigroup proof. Use addition in the compact right-topological semigroup $\beta\mathbb{N}_0$, with the usual convention that right translation by each ultrafilter is continuous and addition of an ordinary integer is continuous on both sides. Translate all query starts by one common sufficiently large integer first, to make the finitely many required starts and letters nonnegative. This harmless alphabet relabeling preserves the free word identity. Put

$$P = \bigcap_i \overline{b^{B_i}\mathbb{N}_0} \subseteq \beta\mathbb{N}_0.$$

Take $K \subseteq P^h$ to be the intersection of the tail closures of tuples of exponents obtained by adding finitely many disjoint unmarked block tuples $((x + jv)b^{B_i})_j$, permitting arbitrary nonnegative starts at the zero move. These tuples give arithmetic progressions with a common ordinary starting offset. K is a compact subsemigroup: approximate the left summand on the right summand, then approximate that right summand by blocks beyond the finitely many just used, using continuity of each ordinary left translation. Also $\Delta P \subset K$, since one arbitrary start at the zero move suffices in every tail.

Let p be a minimal idempotent of P, and put $e = (p, \ldots, p)$. The idempotent e is minimal in K. Indeed an idempotent below it has, at every coordinate, an idempotent of P below p, which must be p. In particular eKe is a group with identity e, and its coordinates lie in the group pPp. These are the standard minimal-idempotent properties for compact right-topological semigroups. One may choose a free minimal idempotent, for example in a minimal left ideal inside the free ultrafilters of P.

By compactness, along one subnet of escaping indices the ordinary integers yb^{B_i} tend to an element $p_y \in P$, simultaneously for every needed symbol y. Each normal tuple of such

limits belongs to K. Evaluate the free word identity in the product of the h groups pPp, replacing each letter by $p + p_y + p$ in the appropriate coordinate. Each compressed normal tuple is in eKe, so the compressed tagged tuple $e + p_* + e$ belongs to eKe as well. Let $k \in eKe$ be its group inverse. Then

$$e = k + e + p_* + e.$$

The right side is in every tail closure of actual tuples with precisely one tagged location. To see this in any neighborhood, first approximate $k + e$ by ordinary normal blocks on the fixed right argument $p_* + e$. Next choose the tagged location later than these blocks on the argument e, and finally approximate e by later normal blocks. This uses only right continuity and continuity of left translation by the fixed ordinary tuples. There is no inversion or multiplication of jointly varying ultrafilter limits.

If a color class belongs to p, its product neighborhood of e contains an actual such progression. Finite colorings therefore have a finite witnessing set by compactness of the coloring space, with as late a block tail as desired. If A is piecewise syndetic, a finite union of translates of A is thick. Color an appropriate translate of the finite witness by a successful translating offset. A monochromatic run shifts into A, proving the additional claim.

A quantitative route to density. For a step set D, let $N_{r,h}(D)$ be the least cardinality of a finite set of integer positions on which every r-coloring has a monochromatic h-term D-progression; allow infinity. Any stationary h-avoider of mass a satisfies

$$N_{r,h}(D)(1 - a)^r \ \geq \ 1$$

when the witness size is finite. Indeed use r independent samples of that system and color each covered position by one of its covering samples. If the left side were less than one, the expected number of uncovered positions would be less than one and an entirely covered realization would violate the finite coloring property. Consequently if for every h there exist witnesses with $\liminf_{r\to\infty} r^{-1} \log N_{r,h}(D) \leq L$ for a single finite L, then $b(D) \leq 1 - e^{-L}$. No such bound has been proved here for the one-mark set, and even a positive short avoiding event for that actual integer set is unknown. Qualitative finite-color recurrence does not supply an upper bound below one in the density question.

R174. A common additive-idempotent image for a single difference of two normal moves

Scope and status. Result with proof.

Result. Single marked linear combinations of normal moves share an additive-idempotent image on all Eberlein functions, yielding Gaussian multiple recurrence.

Scope. A correlation-class theorem; no universal all-system threshold follows.

Let G be a countable abelian group and let $A_i, B_i \in G$ be sequences. Form the finite words which use exactly one marked move $B_i - A_i$, and at other distinct indices any choice of $0, A_j, B_j$. All conclusions apply on every index tail. If zero is not accepted as a step, assume the genuine marked words used below are nonzero, as they are for the free-axis and fixed-base digital examples. The set of these words meets the positive-mass multiple-return sets of every system whose tested autocorrelation is an Eberlein function. In particular it forces all lengths at every positive mass in every Gaussian G-action. The dyadic pure-mark family allows $A_i = e_i$, $B_i = (1 + z_i)e_i$; the single mark is $z_i e_i$. This conclusion excludes every Gaussian obstruction to that family despite its known nongeneric six-slot obstruction in arbitrary actions. It is a statement about correlation functions and does not supply an all-system threshold.

Here Eberlein means a uniform limit of Fourier–Stieltjes functions (matrix coefficients of unitary representations). The following evidence is stronger than checking one fixed spectral measure at a time: there is a *single* ultrafilter p supported on the marked words and a genuine additive idempotent $e \in \beta G$ whose evaluations coincide on **all** Eberlein functions.

The construction. Fix a free ultrafilter on the indices. In βG denote its limits of $A_i, B_i, B_i - A_i$ by a, b, c respectively. Fix also a free additive idempotent $\nu \in \beta N$. Write `n a` for the *additive convolution power in the semigroup*, not the pointwise dilation of an ultrafilter. Put

$$p = \lim_{m \to \nu} [(m+1)a + (m-1)b + c], m \geq 2.$$

The iterated limits defining each summand allow all indices to be chosen successively larger than the previous ones and beyond any fixed tail. Thus the sets of genuine marked words on every tail belong to p.

For any unitary representation U of G, write $U(q)$ for the weak operator q-limit. Its values belong to the abelian von Neumann algebra of the representation. Separate weak continuity of multiplication there shows this is a semigroup homomorphism from βG. On a cyclic spectral model let $U(a), U(b), U(c)$ be multiplication by f, g, t, each of modulus at most one. On the measurable spectral region $E = \{|f| = |g| = 1\}$ the operators $U(A_i)$ and $U(B_i)$ converge **strongly** along the index ultrafilter to the two unitary multipliers on that region: expand the squared norm of the difference on each vector. Their inverses also converge strongly there. Since $U(B_i - A_i) = U(B_i)U(A_i)^{-1}$, it follows that $t = g\bar{f}$ almost everywhere on E.

On the complement of E, the scalar multiplier $f^{m+1}g^{m-1}t$ tends pointwise to zero as $m \to \infty$, hence strongly to zero on that spectral subspace. On E it equals $(fg)^m$. Evaluating the powers of this unitary along the additive idempotent ν gives an idempotent normal

contraction, necessarily an orthogonal projection. Thus $U(p)$ is an orthogonal projection in every cyclic spectral model. Map βG into the product of all cyclic unitary weak-operator compactifications, one for each spectral probability on $\widehat{G}$. This product has a compact image; the evaluation map is continuous and is a semigroup homomorphism. The fiber over the image of p is a nonempty compact right-topological subsemigroup because that image is idempotent. By the idempotent theorem the fiber contains an additive idempotent e of βG. Equality on cyclic matrix coefficients gives equality on all Eberlein functions by uniform approximation.

Recurrence. If $C(v) = \mu(\bigcap_{j<h} T^{-jv} A)$ with $\mu A > 0$, then the e-limit of C is bounded below by a positive constant depending only on $\mu A, h$. Indeed finite density Hales–Jewett gives a fixed finite-sums rank M and $\kappa > 0$ such that for any M generators some nonempty subset sum has correlation at least κ. Apply density Hales–Jewett to the word cube in a randomly translated copy of the action and sum over the finitely many lines. Were $\{v : C(v) < \kappa\}$ in e, the elementary finite-sums construction for an additive idempotent would place all nonempty subset sums of M generators in that set, a contradiction. Consequently every Eberlein such C is positive at a genuine marked step on every tail.

For a Gaussian action the correlation is Eberlein at every fixed order. For polynomial Gaussian cylinder tests Wick's formula writes the correlation as a polynomial in matrix coefficients at the integral differences of the slots, hence as a Fourier–Stieltjes function (products are tensor-product coefficients). Approximate the original bounded tests in L^h by polynomial cylinders and telescope using Hölder and stationary marginals to get uniform approximation in the group time. The same proof works with different bounded tests whenever their unmarked idempotent limit is known to be positive. It also works for arbitrary multiplicities and nonergodic Gaussian actions.

Integer-menu variant. There can be any fixed finite number of normal choices $A_{1,i}, \dots, A_{q,i}$, provided the mark is $M_i = \sum_j z_j A_{j,i}$ for fixed integer coefficients z_j. Use $(m - z_j)$ copies of normal choice j, followed by one mark, in the index-ultrafilter formula for large m. Off the common unit-modulus region some factor tends to zero; on that region strong convergence makes the mark multiplier $\prod f_j^{z_j}$, so the total multiplier is $(\prod f_j)^m$. Every subsequent step is identical. If a finite local integer packet has an ordered player-word expressing the marked tuple in the subgroup generated by normal tuples, abelianizing a player's free word and applying the integer position map, then subtracting two slots, gives just such an integral relation for the marked velocity. Thus the Eberlein conclusion applies also to replicated ordered integer menus, including the single-2 base-b digit language; for that example the simpler choices $A_i = b^i$, $M_i = 2A_i$ already suffice.

R175. Exactly one exceptional coefficient

Scope and status. Proved subcases and unresolved extensions.

Result. Exactly-one bounded exceptional coefficients give pair recurrence, while suitably chosen factorial packets admit polynomial endpoint-one obstructions.

Scope. The fully mixed and single-packet families must not be conflated.

Let a_i be positive integers and let k_i be bounded integers. The differences

$$k_i a_i + \sum_{j \in F} a_j, \qquad i \notin F,$$

give pair recurrence, including on every tail, when the steps are nonzero (in particular for positive k_i). To see this, let B_m be the union of all translates of A by positive subset sums, including zero, on the first m indices of the tail, and put $B = \bigcup_m B_m$. If $i > m$ then $T^{a_i} B_m \subset B$, hence $\mu(T^{a_i} B \triangle B) \le 2\mu(B \setminus B_m)$. The same conclusion with the error multiplied by $\sup |k_i|$ holds at $k_i a_i$. The analogous full union $B^{(-i)}$ excluding the index i contains B_m, so $T^{k_i a_i} B^{(-i)}$ tends in symmetric difference to $B \supset A$. In particular every sufficiently late i is the exceptional index of some positive pair return. The same proof allows any fixed uniformly bounded number of distinguished indices and bounded signed coefficients, on excluding vanishing steps or allowing zero in the formulation.

For higher-order recurrence a flexible sampling of even highly structured factorial scales can reintroduce a polynomial obstruction. Put $b_i = (i-1)!$. One can select successive finite packets of n factorial scales, for arbitrarily large n, for which the set of steps supported on *one* packet with one coefficient 2 and the others $0, 1$ has all-length threshold one. The omitted gaps between allowed positions in a packet are important. This does not settle the family using all factorial indices.

Put $c_0 = 0, c_1 = 1, c_{d+1} = dc_d + c_{d-1}$ for $d \ge 1$. For large d, $c_{d+1}/c_d \ge d$. There is $\lambda \in [0, 1)$ with $c_d \lambda \equiv 1/4 + O(1/d) \bmod 1$. Indeed successively choose the closest numbers $(m_d + 1/4)/c_d$ to the preceding approximants; the tail error after stage d is bounded by $\sum_{s>d} 1/(2c_s) = O(1/c_{d+1})$. Reduce the limit modulo one. Choose α by its b_i^2-radix expansion with digit $\lfloor i\lambda \rfloor$ at denominator b_{i+1}^2 for $i \ge 2$. Then $x_i = \{\alpha b_i^2\} \to 0$ and $ix_i \to \lambda$. For each fixed gap $d \ge 0$,

$$\lim_{i \to \infty} \alpha b_i b_{i+d} = c_d \lambda \quad \text{in } \mathbb{T}.$$

This follows inductively from $b_i b_{i+d+1} - b_{i+1} b_{i+d} = d b_i b_{i+d}$. The fixed-gap identity at $d = 1$ also describes the simple obstruction to treating the diagonals as wholly independent: $b_{i+1}^2 = b_i b_{i+2} - b_i b_{i+1}$. Choose β by ordinary factorial digits $\lfloor i/4 \rfloor$ at denominator b_{i+1}, so $\beta b_i \to 1/4 \bmod 1$.

For a packet of n scales first set their gaps to L_n with $L_n/n^2 \to \infty$, then take their common starting index M_n sufficiently large. The finitely many fixed-gap limits ensure all off-diagonal $\alpha b_i b_j$ in the packet are within $o(n^{-2})$ of $1/4$, and we can require diagonals αb_i^2 within $o(n^{-2})$ of zero and βb_i within $o(n^{-1})$ of $1/4$. For a desired step $d = \sum_i v_i b_i$, put $L = \sum_i v_i$. Then $\sum_i v_i^2 = L + 2$, while the two test phases are

$$\beta d = L/4 + o(1), \qquad \alpha d^2 = (L^2 - L - 2)/4 + o(1) \mod 1.$$

If $4 \nmid L$ the first phase is separated from zero; otherwise the second is close to a half-turn. The packet sizes and errors can be chosen large and small from the outset. The polynomial sieve thus gives threshold one for the union. The pair lemma for infinitely many distinguished positions and arbitrary neutral supports does not itself concern this one-packet restriction, but its finite analogue also gives positive pair returns here. Indeed if an A of mass a avoided the family on arbitrarily large packets, form B_j from earlier axes within a packet. The successive growth increments total at most one, and using one additional trial axis i, $\mu(T^{a_i}B_j \triangle B_j) \le 2[\mu(B_{j+1}) - \mu(B_j)]$. Since $T^{2a_i}B_j$ is disjoint from $A \subseteq B_j$, this forces each increment to be at least $a/2$, contradicting large packet size. Thus these examples distinguish positive pair recurrence from the all-length threshold even in factorial packets.

R176. One marked digit in arbitrary compact group extensions of the odometer

Scope and status. Result with proof.

Result. One-mark digit families have all positive-measure multiple returns in arbitrary compact metrizable group Haar extensions of the odometer.

Scope. The compact fiber and its jointly continuous multiplication are essential; this is not arbitrary measurable-fiber structure.

Fix a base $b \geq 2$ and a length h. Suppose finite integer query packets of normal velocities and a special velocity r_* have a free-word identity: the player tuple $e(x_*, r_*) = (u_{j,x_*+jr_*})_{j<h}$ is a word in normal tuples $e(x_l, r_l)$ and their inverses. Include zero among the normal velocities used in the law. Suppose placing finitely many such velocities on separated digit blocks by powers of b, with **one** special and all others normal or zero, gives legal physical differences. A standard example is a base-b word with exactly one digit 2 and all other digits $0, 1$, for $b \geq 3$: the unrestricted normal digit-word menu contains an infinite IP, and the words with exactly one 2 can meet every congruence subgroup by placing their digits sufficiently high and using enough additional 1's in repeated coprime residue classes. The arbitrary-order integer IP-menu lemma gives the requisite finite word for each h.

Every such one-mark family gives positive h-returns for all positive events in any measurable compact metrizable group Haar extension of the base-b odometer. The fiber need not be abelian, profinite, or a Lie group. In particular the general measurable binary Morse cocycles on this odometer cannot furnish a short obstruction to the one-mark family. The proof below treats an independent Haar fiber; finite-fiber invariant laws absolutely continuous to that model also follow by restricting to a positive density level of their invariant Radon–Nikodym derivative.

Localizing the cocycle Write $Y = \prod_{n\geq 1} \mathbb{Z}/b\mathbb{Z}$ for the digit space with product Haar law. Almost surely two points of one odometer orbit differ at finitely many digits, and conversely every finite digit change corresponds to their actual integer height difference $\sum_i (y'_i - y_i)b^{i-1}$, using digit representatives $0, \ldots, b-1$. Work on one conull invariant set. An odometer cocycle into the compact group K thus gives a cocycle over the coordinatewise action of $\bigoplus C_b$. Equip K with a compatible bi-invariant metric. The localization of the profinite-cocycle argument [R161] works in this metric. In detail, on each finite head of n digits a measurable function F_n satisfies $\rho(g, y) = F_n(y + g)F_n(y)^{-1}$, using the section with those digits zero. The functions $F_{n-1}^{-1}F_n$ are invariant under earlier digits. Approximate each by f_n depending on a finite interval beginning at digit n, with the sum of the expected metric errors finite. Write $F'_n = f_1 \cdots f_n$. The change from $F_{n-1}(F'_{n-1})^{-1}$ to $F_n(F'_n)^{-1}$ is bounded by that stage's error, by bi-invariance. These changes are summable almost surely, so their limit gives a measurable gauge in which F'_n represents the cocycle on every finite digit change. For a digitwise motion supported on a finite head, all subsequent factors are unchanged and cancel on the right.

Group digits in successive finite blocks so that factors starting in each block depend only on this and the next block. This is achieved inductively by taking the next endpoint beyond all the finitely many dependency endpoints already required. The block lengths can also tend to infinity as rapidly as desired. Freeze the odd blocks as separators. On fixing a

typical separator environment, the pair of consecutive products from an odd block and its following even block depends only on that even moving core and the fixed environment. Denote the resulting potential on core i by f_i, reusing the symbol. Its input X_i is uniform on $C_{b^{L_i}}$, viewing the L_i digits as an ordinary integer modulo b^{L_i}; the inputs are independent. Potential factors are multiplied in increasing core order. For the fixed packet of small local velocities normal or special it suffices to use shifts on these cyclic cores. A true physical addition changes no outside digits whenever the modular core addition by jv does not cross either endpoint of its core, at all $0 \le j < h$. The probability of any such seam at a selected core is at most $O(h \max |v|/b^{L_i})$, and the sum on all late cores is arbitrarily small. The same reasoning works on arbitrarily late tails of the physical blocks.

The exchangeable limit of local displacement tables Let $C_{<i} = \prod_{s<i} f_s(X_s)$ in the indicated order, and put

$$H_i(y) = C_{<i}\, f_i(X_i + y) f_i(X_i)^{-1}\, C_{<i}^{-1}, \qquad y \in \mathbb{Z}.$$

This random table lies in the compact metric space $K^{\mathbb{Z}}$, with $H_i(0) = 1$. For an increasing finite collection of active cores and their velocities v_i, the total gauge displacement at slot j is exactly the ordered product of $H_i(jv_i)$. This follows by inserting the unperturbed inverse factors, or by induction on the last active core. The statement applies to the actual odometer orbit when there is no seam.

Ramsey selection supplies a subsequence whose ordered finite table marginals converge as the least selected index goes to infinity. Indeed use a countable determining set of continuous functions on every finite power and successive finite color approximations. The limiting consistent sequence is spreadable (invariant under increasing subsequences). By the Ryll–Nardzewski/de Finetti theorem it is a mixture of iid table sequences; call its directing law Θ. Use the two-sided exchangeable realization, which has the same ordered finite marginals. Its infinite past determines Θ, for instance from the empirical averages on a countable determining family. Kallenberg, *Probabilistic Symmetries and Invariance Principles*, Chapter 1.

At one actual core i, replace X_i by $X_i + a$. This preserves the whole law, fixes all earlier tables, changes H_i into

$$(R_a H_i)(y) = H_i(y + a) H_i(a)^{-1},$$

and conjugates every later table by the single element $H_i(a)$. All these maps are continuous on the compact table space, so the same identity holds in the limiting two-sided sequence. It follows for almost every directing law that R_a preserves Θ, and that $\mathrm{Ad}_{H(a)}$ preserves Θ for Θ-almost every table H, simultaneously for all integers a. For the second assertion condition on the unchanged infinite past and the current table; its recentering R_a is invertible with inverse R_{-a}. The transformed future conditional law is the product of conjugated copies of Θ, whereas invariance of the entire sequence and the unchanged past identifies its law as the product of Θ itself.

Normal products and one corrected core Fix such a directing law. Draw $H \sim \Theta$ and a normal velocity v from a fixed finite law positive on zero and each normal velocity in the word. Let $P \le K^h$ be the closed subgroup generated by the support of $N_v(H) = (H(jv))_{0 \le j < h}$;

denote this mixture law by σ. At slot zero every member of P is the identity. The law σ has a positive atom at the identity tuple from $v = 0$.

For Θ-almost every H, the diagonal tuples $\Delta H(a)$ normalize P for every a, by the conjugation invariance just obtained. By recentering invariance every shifted normal tuple $(H(x + jv))_j$ equals an element of P times $\Delta H(x)$. Evaluate the actual free-word relation on this table by assigning to each player letter $u_{j,y}$ the group element $H(y)$. Moving the diagonal factors to the right using normalization shows that the resulting true tuple is $N\Delta k$ with $N \in P$. Its zeroth coordinate is $H(x_*)$, so $k = H(x_*)$. Recenter once more to conclude $N_{r_*}(H) \in P$ almost surely. Write σ_* for this last law.

Independent convolution gives $\sigma^{*m} * \sigma_* \to m_P$ weakly as $m \to \infty$, where m_P is Haar on P. To check this apply Peter–Weyl on P: on a nontrivial irreducible unitary representation the average of σ has no eigenvalue of modulus one, because the identity atom forces that eigenvalue to be one, and equality in the Hilbert-space triangle inequality would then fix a vector under the generating support. Its powers tend to zero. Continuous tests follow by approximation. Thus for every prescribed neighborhood of the identity, the normal-then-true products hit it with a positive probability in the exchangeable mixture. More precisely use a nonnegative continuous bump supported inside that neighborhood and positive at identity. For almost every Θ its Haar integral on P is positive, so its unconditional limiting integral is positive by dominated convergence.

Fixing m accordingly and returning to the original subsequence gives arbitrarily far finite trials of m normal active cores and one actual true core, with positive probability bounded below uniformly for this fixed environment and tolerance. All unselected intervening cores remain fixed. The limiting use of the bump only involves a bounded continuous function of these finitely many tables and a finite mixture of velocities. By choosing the tail sufficiently far we may require no seams at all in each successful trial and retain a uniform positive success bound.

Pooling and returning to the measured event Choose arbitrarily many such trials on disjoint finite intervals of moving cores. Although H_i uses the full earlier prefix, the event that every slot displacement has distance less than a given tolerance from identity depends only on the data of the trial's interval. Indeed its common pre-interval prefix conjugates the entire product by the same element; the distance is bi-invariant. Seams and all local velocities also use only this interval. Distinct trial events are therefore independent under the conditional product law on moving cores.

If their probabilities are $p_l \geq c > 0$, choose one trial with probability p_l/Λ, $\Lambda = \sum p_l$, and condition it on success; leave all other cores unconditioned and inactive. At every slot the density of this pooled moving-base marginal relative to the uniform moving data differs from one in L^2 by at most $1/\sqrt{\Lambda}$. In fact each unconditioned trial's cyclic translations, with velocities sampled independently of uniform starts, have uniform slot marginals. The likelihood is $\Lambda^{-1} \sum_l q_l$, where q_l is the conditional probability of the interval's success given its slot's data. The q_l are independent under the uniform reference law, are in $[0, 1]$, and have means p_l. The sampled starting fiber remains independent Haar; conditional on all base data each translated fiber is still Haar, so the same marginal estimate holds jointly with the fiber. On the selected interval every slot stays within the chosen distance of its starting fiber in the gauge coordinates.

For a positive event, condition on typical separator environments having positive condi-

tional event mass a. In the gauged coordinates approximate its indicator under that conditional product law by a $[0,1]$-valued function f depending on finitely many early moving cores and continuous in the compact fiber, with L^1 error arbitrarily small compared with a/h. It can be chosen so $\{f > 3/4\}$ has mass at least $a/2$. Choose the trial tolerance for uniform continuity, and run all trials beyond the early cores. On this set at time zero every slot then has $f > 1/2$. Its starting probability tends to at least $a/2$ as $\Lambda \to \infty$, while the probability that at least one such slot lies outside the actual event is at most $2h$ times the approximation error plus $o_\Lambda(1)$. This gives a positive-measure success conditional on the environment for some actual direction: each trial start law is absolutely continuous, and only finitely many choices were used. There is no need to select any of the constructions measurably across environments; there are only countably many integer directions and each conditional success measure for a fixed direction is measurable (the safe trials have unchanged separator coordinates). Fubini gives a single unconditional positive return. The same proof works on compact homogeneous Haar fibers by pulling an event back to the compact group extension.

The theorem uses compactness of the local displacement tables in the fiber group. The weakly compact operator closure of an arbitrary group of measure-preserving fiber transformations does not supply the joint continuity of the recentering and multiplication maps used above.

R177. Exactly one exceptional digit on an arbitrary sparse geometric subsequence

Scope and status. Result with proof.

Result. A sufficiently sparse geometric subsequence makes the exactly-one exceptional-digit family fail density largeness at every threshold below one.

Scope. The construction uses a fixed invertible residue class modulo a modulus at least three and does not cover factorial scales automatically.

For every integer base $b \geq 2$, there is a single infinite set of digit positions m_i such that the fully mixed family

$$D = \{2b^{m_i} + \sum_{j \in F} b^{m_j} : F finite, i \notin F\}$$

has integer all-length capacity one. Supports may mix freely across *all* positions of the subsequence. In contrast the family is pair recurrent by the one-mark pair lemma [R175]. This does not settle the full-digit one-mark language with no positions removed.

Fix $q \geq 3$ relatively prime to b, for example an odd prime not dividing b. Choose m_i among the multiples of $\mathrm{ord}_q b$, so $a_i = b^{m_i} \equiv 1 \bmod q$. Choose them sufficiently rapidly increasing and choose a common α such that

$$\alpha a_j = 1/q - 1/(qa_j) + e_j \bmod 1,$$

$$|e_j| \leq \eta 2^{-j}/(ja_j), \sum_{i<j} a_i/a_j \leq \eta.$$

To construct them, take successively nested closed intervals centered at $(k_j + 1/q - 1/(qa_j))/a_j$ of radius $\eta 2^{-j}/(ja_j^2)$. Inside the preceding nonempty interval, centers with any sufficiently huge allowed a_j are spaced by $1/a_j$, so a new interval fits strictly inside. Impose the ratio bound by making each new stage arbitrarily large. Their common limit gives α.

The diagonal entry αa_j^2 is within $a_j|e_j|$ of an integer. For $i < j$, the off-diagonal $\alpha a_i a_j$ is within $a_i/(qa_j) + a_i|e_j|$ of $1/q$ modulo one, since $a_i \equiv 1 \bmod q$. These error bounds sum to $O(\eta)$ across *all* diagonal and unordered off-diagonal pairs. For a word $d = \sum v_i a_i$ put $L = \sum v_i$; since precisely one coefficient is 2 and the others 0 or 1, $\sum v_i^2 = L + 2$. Uniformly over all words,

$$d/q = L/q \bmod 1,$$

$$\alpha d^2 = (L^2 - L - 2)/q + O(\eta) \bmod 1.$$

If $q \nmid L$ the linear phase is separated from zero by $1/q$. Otherwise the second phase is separated from zero by at least $\|2/q\| - O(\eta) > 0$ for a sufficiently small initial η. The fixed two-polynomial sieve/root blocker gives endpoint one. This applies as well to choosing a sufficiently sparse subsequence of any unbounded sequence containing infinitely many values in a common invertible residue class modulo some $q \geq 3$, with the linear phase and target appropriately rescaled by that class. The congruence hypothesis is not automatic for factorial scales.

R178. Unrestricted three-term recurrence for exactly one exceptional digit

Scope and status. Result with proof.

Result. The full one-digit-2 integer language is a set of measurable three-term recurrence in every integer system.

Scope. The all-length measurable conclusion remains unproved; the proof avoids the specifically identified false rational-cube lemma.

Qualifications. Explicitly excludes use of the disputed rational-cube Lemma 4.10 of arXiv:2506.17771v1; the proof depends instead on the compact-odometer extension and relative-cube estimate.

For each integer $b \geq 3$, the positive integers with digits $0, 1$ and exactly one digit 2 are a set of measurable **three-term** recurrence in every integer system. This holds on every digit tail. The argument uses the compact extension of the odometer theorem [R176] and the two-block estimate from the relative-cube calculation [R216]. We do not use the disputed rational-cube Lemma 4.10 of arXiv:2506.17771v1. The extension to arbitrary lengths on all systems remains unproved.

Fix an ergodic system and a positive event A. All transformations are invertible. Let Y_* be the factor spanned by the eigenfunctions whose eigenvalues, in additive circle notation, have orders dividing $2b^n$ for some n. Let W be the maximal relatively compact factor over Y_*, in the Furstenberg–Zimmer sense. We first explain why W suffices for the two smoothing blocks used below.

The two cube factors. At each block $I = [m, N]$ independently select binary digits with equal weights and set $u_I = \sum_{i \in I} \epsilon_i b^i$. An independent difference $u_I - u'_I$ has Fourier multiplier $\prod_{i \in I} \cos^2(\pi b^i \theta)$. The successive limits $N \to \infty$, $m \to \infty$ give the indicator of $\mathbb{Z}[1/b]/\mathbb{Z}$. Indeed square summability of $\|b^i\theta\|$ implies convergence to zero; sufficiently small centered representatives must then advance exactly by multiplication by b and so eventually vanish. By spectral dominated convergence, averaging a diagonal Koopman action at $k(u_I - u'_I)$ gives the orthogonal projection onto its eigenspaces of orders dividing $|k|b^n$, for some n. This remains true on nonergodic spaces by the spectral theorem.

In the generalized von Neumann bound for a three-slot correlation, retain slot j and eliminate the other two slots successively. Use two independent blocks of ordinary **unsigned** increments in the common difference, with their differences appearing only after Cauchy–Schwarz. At each elimination shift the initial point so the slot being removed is independent of that block, and apply conditional Cauchy–Schwarz. The fourth power of the absolute averaged correlation for inputs of modulus at most one is bounded by the standard two-dimensional cube correlation of the retained input. Its two cube increments are $k_1(u_1 - u'_1)$, $k_2(u_2 - u'_2)$, with k_i one of the nonzero differences of slot indices. This bound is uniform in any fixed additional common offset, including additional block sums; see the calculation in the linked cube note. Integrate first the innermost increment and take its two tail limits on the original system. The resulting joining is $\lambda_1 = \mu \times_{Y_{k_1}} \mu$, with $Y_{k_1} \subseteq Y_*$ the rational factor specified by this increment. On λ_1 the second increment then tests the rational eigenfunctions of the diagonal action $T \times T$. One can reverse the naming of the blocks to match the nested limits, placing the innermost-averaged digit block later than all previously fixed finite blocks.

Every eigenfunction of the diagonal action on λ_1 depends on $W \times W$. Here are details relevant when the joining is nonergodic. The two copies in λ_1 are conditionally independent over their two specified Y_*-coordinates; the coupling of these coordinates itself need not be independent. For a bounded tensor with at least one coordinate f orthogonal to W, its diagonal autocorrelation is bounded using the usual relative weak-mixing estimate for this input,

$$\lim_N N^{-1} \sum_{n<N} \|\mathbb{E}(f\,\overline{T^n f} \mid Y_*)\|_2^2 = 0.$$

This is the invariant-kernel characterization of the maximal relatively compact factor (see Frantzikinakis, Lemma 2.2). Conditional independence makes the tensor autocorrelation an integral of a product of the two conditional autocorrelations over the coupled base; the other bounded input costs only a fixed factor. Wiener's criterion excludes any atom in its spectral measure. Such tensors span the orthocomplement of $W \times W$, by conditional independence over the finer separate factors as well. In particular, if the retained input f has $\mathbb{E}(f \mid W) = 0$, then $f \otimes \bar{f}$ has zero projection on $W \times W$ under λ_1. The iterated two-block cube bound is zero. Telescoping therefore replaces 1_A by $g = \mathbb{E}(1_A \mid W)$ in these iterated normalized means.

The base of the compact extension. The ergodic rotation Y_* is a factor of the full b-adic odometer, possibly times the two-cycle when b is odd. Pull the relatively compact factor W back to that full base. An ergodic compact extension is an isometric homogeneous-space extension, or an inverse limit of such compact extensions, and can be lifted to a principal compact-group cocycle for the purpose of testing events. A finite intermediate cycle can also be absorbed into the compact fiber: on two labeled copies of K/H, the compact group $K^2 \rtimes C_2$ implements the permutation of the sheets and their individual rotations. Pullback or homogeneous-space lifting need not be ergodic; product Haar on the lifted fibers still projects to the required measure. The compact-cocycle theorem now gives one actual direction d_0 with exactly one exceptional digit such that $\bigcap_{j=0}^{2} T^{-jd_0}\{g > \epsilon\} = C$ has positive measure, where $\epsilon > 0$ was first chosen with a positive superlevel set. This use of the compact theorem can alternatively be made on a finite list of compact approximants and events of conditional density close to one.

Putting back the other factors. Take m large enough for the density Hales–Jewett theorem at alphabet size 3 and density $\mu(C)/2$. In each of m disjoint trials past the digits of d_0, take the sum of two unsigned random digit blocks as its velocity. On the word cube of the m trials, the expected proportion of visits to C is $\mu(C)$, by stationarity, for every choice of these velocities. The sum over the finitely many combinatorial lines of their success probabilities is at least $\mu(C)/2$. After translating away the fixed letters of a line, a success at its additional step v gives g-correlation at $d_0 + v$ at least ϵ^3 times the C-correlation at v. Average all block choices and take the nested tail limits (or iterated limsups with lower-order choices sufficiently accurate). For any fixed line there are at least two included blocks; its last included pair gives the zero cube bound for each orthogonal error term, uniformly in the earlier choices. Unused blocks integrate out. The finite sum of original A-correlations consequently cannot vanish throughout. Every actual step used has the digits of d_0 and only new binary digits in disjoint positions. This proves the claim.

R267. A necessary shattering property for varying-parity avoidance

Scope and status. Result with proof.

Result. Varying-parity avoidance forces quantitative shattering and an l1 sequence of correlation translates; tame correlations therefore recur. In the pure-mark model it also forces event-specific positive sequence entropy.

Scope. The entropy conclusion uses independent parity characters and specified unmarked direction lists.

Qualifications. Arbitrary multiple correlations need not be tame, and positive entropy elsewhere does not satisfy the stated necessary condition.

Suppose $H \le G$ are countable abelian groups acting on a probability space, A is measurable, and $c(g) = \mu(\bigcap_{j<h} T^{-jg}A)$. Fix $\kappa > 0$, distinct order-two characters η_n of H, and $g_n \in G$ with $c(g_n) \ge \kappa$. For every invariant mean m on H, write $f_n(u) = c(g_n + u)$. If every correction $g_n + u$ with $\eta_n(u) = -1$ avoids, then there are a constant $s = s(h,\kappa) > 0$, a subsequence n_i, and a constant $a > 0$ such that for any disjoint finite I, J the set

$$\{u \in H : c(g_{n_i} + u) > s \quad \text{for} \quad i \in I, \eta_{n_j}(u) = -1 \quad \text{for} \quad j \in J\}$$

has m-mass at least $a \cdot s^{|I|+|J|}$ (decreasing s if necessary). In particular values on the last indices vanish exactly. The mean and the subsequence may be chosen separately. It suffices throughout to have the odd-coset avoidance along an infinite subsequence only.

Indeed the varying-parity lemma [R134] gives $m(f_n) \ge c_0(h,\kappa) > 0$, via a uniform finite density Hales–Jewett bound on the subevent $\bigcap_{j<h} T^{-jg_n}A$. Regard m as a probability on βH, with subsets of H identified with their clopen closures. Put $B_n = \{f_n > c_0/2\}$, $C_n = \{\eta_n = -1\}$. Their respective indicators take the same values after extension to βH as those of their clopen closures. We have $m(B_n) \ge c_0/2$; the indicators of C_n converge weakly in $L^2(m)$ to $1/2$, by orthogonality of the nontrivial distinct characters under the invariant mean. Passing to a subsequence gives weak convergence of 1_{B_n} to $b \in [0,1]$ with integral at least $c_0/2$. The event $Y = \{b \ge c_0/4\}$ has positive mass. Start with Y. After choosing finitely many indices, for each surviving finite intersection D of choices restricted to Y the limit of $m(D \cap B_n)$ is at least $(c_0/4)m(D)$, and that of $m(D \cap C_n)$ is $m(D)/2$. Choose the next index large enough simultaneously for all such intersections, retaining a fraction $s = \min(c_0/8, 1/4)$ in either branch. This gives the asserted bound for full patterns. For noninitial patterns use the union of all possible prior unspecified letters, or repeat the same induction including the no-condition branch at each index. In the former argument direct unions can lose the advertised bound $s^{|I|+|J|}$, so we use the no-condition branch explicitly when choosing the common next index. The original mean then has the same value on its requested finite Boolean combinations as the Borel measure, by clopen realization. The constant $a = m(Y)$ suffices. There is no need for the η_n to be linearly independent, only pairwise distinct.

Consequently the functions $u \mapsto c(g_{n_i} + u)$ span an ℓ^1 sequence: for any finite real coefficients, realize the sign pattern favoring the positive coefficients and forcing the negative ones to zero, or its negative, whichever has the larger total absolute coefficient. The sup

norm of the linear combination is at least $s/2$ times its ℓ^1 coefficient norm. Distinct signs in an upper threshold can be chosen with an actual $u \in H$, since a positive-mean subset is nonempty. The converse lower implication is not asserted.

Thus the correcting cosets always give arbitrarily late positive returns if c is tame in the precise sense that its translates in $\ell\infty(G)$ contain no ℓ^1 sequence. In particular weak almost periodicity suffices: a weakly relatively compact sequence of translates has a weakly convergent subsequence, whereas an ℓ^1-equivalent sequence has no weakly Cauchy subsequence (test alternating signs on its span and apply Hahn–Banach). This extension does not require a Fourier–Stieltjes representation. In the pure-mark free family [R098], the disjoint unmarked normal supports given by finite density Hales–Jewett produce distinct η_n and all the corrected cosets are legal. The conclusion applies there for every fixed h. Arbitrary measure-preserving multiple correlations need not satisfy the tameness assumption.

A necessary measure entropy bound along the unmarked directions In the pure-mark application the η_i are actually linearly independent over the two-element field. Indeed for disjoint nonempty normal masks I_i, their parity characters count coefficients outside I_i. The product of any even nonempty number counts exactly the union of its masks, and a product of an odd number is nontrivial on an axis outside their union.

For any linearly independent finite list of these characters, write $P = \{A, X \setminus A\}$ and let H_μ denote Shannon entropy in bits. Avoidance forces

$$H_\mu(P) + \sum_{j=1}^{h-1} H_\mu(\bigvee_{i \le n} T^{-j g_i} P) \ge n c_0(h, \kappa).$$

The estimate needs no subsequence selection and holds on any finite sublist of the given baselines. To prove it sample a uniform u in increasingly invariant finite Følner sets of H, and sample an independent x μ. Pass for the fixed n to a limit of the joint laws of the finitely many binary variables

$$E_i = 1[\eta_i(u) = 1], Z_{ij} = 1_A(T^{j(g_i+u)}x), Y_i = \prod_{j<h} Z_{ij}.$$

The character vectors E tend to the uniform cube by independence and character orthogonality. For each i we have $Y_i \le E_i$ almost surely by avoidance on the negative coset. Its probability of being one is at least c_0 in the limit, by the uniform positive invariant-mean bound. Thus the mutual information $I(E_i; Y_i) \ge P(Y_i = 1) \ge c_0$: conditioning on a one discards the entire one bit of initial uncertainty, and conditioning on zero leaves at most one bit. Since the E_i are independent, the chain rule and monotonicity of conditional entropy give

$$n c_0 \le \sum_i I(E_i; Y_i) \le I(E; Y) \le H(Y) \le H((Z_{ij})_{i,j}).$$

Each column $(Z_{ij})_{i\le n}$ at fixed j has exactly the law of the unmarked μ-name at the directions jg_i, for every choice of u, by stationarity. At $j=0$ this is a single bit. Subadditivity across columns proves the claimed estimate. In particular avoidance demands a positive sequence entropy for P along some one of the normal lists jg_i and cannot occur if all of those specific name entropies are sublinear. The full group or transformation may have other positive-entropy sequences. This is an event-specific necessary condition in a genuine arbitrary measure-preserving action; it makes no claim of an unrestricted upper bound independent of h.

Part II

Supporting arguments and dependency record

Chapter 13

Supporting proofs

D01. Congruence localization and multiplicative amplification

Scope and status. Supporting proof and dependency material.

This supporting entry reproduces the prerequisite sections on congruence localization and multiplicative amplification. The later refinements are proved in R130 and the principal exposition. The unrelated historical automatic-set arguments from the same earlier notebook are not invoked; their corrected scope is explained in R015 and R214.

Removing nonmultiples costs no critical density For each integer $Q \geq 1$,

$$b(D) = b(D \cap QN) = b((D \cap QN)/Q).$$

For the first equality note that the complement of QN can be blocked by an arbitrarily sparse periodic deletion, uniformly in its steps. For each prime power p^e exactly dividing Q, delete residues 0,...,$p^{e-1} - 1$ modulo p^{m+e-1}. This costs p^{-m}; if p^e does not divide d, each orbit of adding d modulo this modulus visits a deleted residue in at most p^{m+e-1} steps. Take the union of these deletions. Apply to any D intersect QN witness and let m grow. For the second equality, lift a witness by filling every Q-block whose quotient lies in the witness, and in the other direction restrict to a densest residue along a subsequence and divide indices by Q. Same-difference indexing is preserved in both directions. This is a special case of the earlier Bohr-localization principle. It does not amplify a witness uniformly distributed across residue classes.

Question 6.1 for multiplicative semigroups and uniformly controlled multiplicative limits Write $\rho(D) = 1 - b(D)$. Equivalently $\rho(D)$ is the infimum of the measures of stationary hitting events H for which every D-progression of some finite length h hits H, at almost every starting point. Null exceptions for all integer times and all d may be removed simultaneously. Suppose for contradiction $0 < \rho(D) < 1$. If D is closed under multiplication, every h-hitting event H, whatever its mass, has the following property: for all sufficiently large K, every K-term D-progression at an admissible starting point has relative H-frequency

$\geq \rho(D)/2$. The bound on K may depend on H,h. Otherwise take longer and longer individual d_j-progressions with frequency $< \rho(D)/2$. On each, form the empirical measure of the binary H-name under the rescaled shift T^{d_j} over that segment. A weak-star subsequential limit is stationary and has mark density $\leq \rho(D)/2$. It hits every h-term progression at each $e \in D$ almost surely, since $ed_j \in D$ and countable original hitting held at all integer times. This contradicts the definition of $\rho(D)$. Extraction only uses the compact binary shift.

Fix a single D-hitter H_0 of mass $p_0 < 1$ and finite gap h_0; there is one since $\rho(D) < 1$. By finite Szemerédi, any interval of some length W containing a subset of relative density $\geq \rho(D)/4$ has an h_0-progression of step $1 \leq c \leq W$. W depends on $\rho(D)$,h_0 and NOT on the near-optimal hitter H used below. Partition a sufficiently long D-progression into consecutive W-blocks; by the preceding frequency bound one block has H-density $\geq \rho(D)/4$, accounting for its final incomplete block.

For each $c = 1,\ldots,$W take a separate cth root system of the stationary H_0 system, and use their independent product with H. The root on $X_0 \times \{0,\ldots,c-1\}$ advances the clock by one each time and applies T_0 when the clock wraps, with uniform clock law, so its cth power applies T_0 on the base at every point. Set H' equal to H intersected with the union of the lifted H_0 events in all these independent rows. On any sufficiently long D-line of step d, the progression of h_0 selected H-positions in the W-block has index stride some $c \leq W$. Along those positions, the cth row advances by T_0^d, and therefore hits its mark at least once. Hence H' is a D-hitter of finite uniform gap, of mass exactly $\mu(H)[1-(1-p_0)^W]$. Taking infimum over H yields $\rho(D) \leq \rho(D)[1-(1-p_0)^W]$, impossible. This proves that every multiplicative semigroup of positive integers obeys the zero-one alternative of Question 6.1.

More generally form the compact collection $T(D)$ of pointwise limits in $\{0,1\}^N$ of the quotients $D/d = \{e : de \in D\}$, $d \in D$, also including all the quotients themselves. If $\inf_{E \in T(D)} \rho(E) > 0$, exactly the same proof works with half that infimum as the uniform line-frequency bound: in the empirical extraction pass also to a convergent subsequence of D/d_j. For each fixed e in the limit, $ed_j \in D$ eventually, as required. In particular the criterion holds when there is a set B with $\rho(B) > 0$ such that $dB \subset D$ for every $d \in D$. A genuine intermediate D necessarily satisfies $\inf_{E \in T(D)} \rho(E) = 0$ despite the identity $\rho(D/q) = \rho(D)$ at every individual positive integer q. The infimum conclusion alone does not assert attainment by a single limit set, since the allowed hitting length can vary.

In fact some $E \in T(D)$ attains $\rho(E) = 0$ for every genuine intermediate D, by the following further argument. Write $b_h(F)$ for the fixed-length h maximum avoiding density. It is upper semicontinuous in the step indicator $F \in \{0,1\}^N$: take extremizing stationary processes on F_n and a subsequential weak limit, observing that $e \in F$ entails $e \in F_n$ eventually. For every e, $b_h(F/e) \geq b_h(F)$ since an h-avoider for F under T remains one for F/e under T^e. Consequently $b_h(E) \geq b_h(F)$ and $b(E) \geq b(F)$ throughout $T(F)$. Moreover $T(E) \subset T(F)$ whenever $E \in T(F)$: if $F/d_n \to E$ and $e \in E$, then $ed_n \in F$ eventually and E/e is the limit of F/ed_n. Take closure. Starting with intermediate $D_0 = D$ apply the uniform criterion to get $D_{j+1} \in T(D_j)$ with $b(D_{j+1}) > 1 + b(D_j)/2$, or with $b = 1$ immediately. The nested nonempty compact $T(D_j)$ have a common point E; by the just-proved monotonicity $b(E) \geq \sup_j b(D_j) = 1$. Thus absence of any $b = 1$ limit is already sufficient, even without a uniform assumed positive deficit on the hull. In particular any D whose pointed quotient hull is minimal (every E in it has the same hull, containing D) satisfies the dichotomy because both directions of monotonicity give the same value on the whole hull.

Strict gain under multiplicative products. For arbitrary nonempty D,F, if $0 <$

$\rho(D) < 1$ and $\rho(F) > 0$, then $\rho(DF) > \rho(D)$, where $DF = \{de : d \in D, e \in F\}$. Indeed on any stationary DF-hitter, every sufficiently long d-line, $d \in D$, has density $\geq \rho(F)/2$, by the same empirical rescaling with the fixed multiplier set F. Take a single D-hitter H0 of mass $p0 < 1$ and gap k, choose W by finite Szemerédi at density $\rho(F)/4$, and apply the root construction to any DF-hitter H. It produces a D-hitter of mass $\mu(H)[1-(1-p0)^W]$. Thus

$$\rho(DF) \geq \rho(D)/[1-(1-p0)^W] > \rho(D).$$

The hitter H need not itself hit D. By symmetry the product strictly beats either intermediate positive factor; if either factor already has $\rho = 1$ the product has $\rho = 1$ by containing its fixed dilation. In particular the multiplicative semigroup generated by any positive-deficit D is fully Szemerédi-large (it has at least that deficit and satisfies the semigroup dichotomy). For each fixed h and $a > 0$, finitely many product lengths therefore suffice to exclude a common h-avoider of density a on their union, by compactness. This last assertion has no uniformity in h.

Uniform capacity remains on bad directions. Fix $0 < \gamma < \rho(D)$. Choose W by finite Szemerédi at density $\gamma/2$ for the above fixed H0. Choose near-optimal H of mass p so that $\rho(D) - p[1-(1-p0)^W] > c > 0$. For large N let E_N contain $d \in D$ for which there exists an admissible N-step d-line of H-density $< \gamma$, i.e. some such word has positive measure in the stationary process. For d outside E_N every N-line has density $\geq \gamma$ almost surely. Partition into W-blocks as above, so $\rho(D \setminus E_N) \leq p[1-(1-p0)^W]$. The union bound for independent hitting events gives $\rho(E_N) \geq c$ for all sufficiently large N. Yet each fixed d is eventually absent because every stationary fixed-gap hitter on the d-axis has mean $\geq \rho(D/d) = \rho(D)$. If desired use the decreasing sets $\bigcup_{n \geq N} E_n$. Thus bad d-lines cannot merely be rare in the sense of the deficit set function, though their probability at a fixed d can be arbitrarily small.

Ultrafilter form. For an ultrafilter p on N put $g(p) = \inf_{A \in p} \rho(A)$. Use the multiplication convention $A \in q \cdot p$ iff $E = \{e : A/e \in p\} \in q$. If $0 < g(p) < 1$ and $g(q) > 0$, then $g(q \cdot p) > g(p)$. To see this, fix a p-set D0 with a hitting event H0 of mass $p0 < 1$ and finite gap k. For $A \in q \cdot p$ its multiplicative p-limit E as just defined has $\rho(E) \geq g(q)$. For a stationary h-hitter H on A, there is a finite $E0 \subset E$ with every h-hitter on E0 of mass $\geq g(q)/2$: this follows by compactness on the binary shift for that fixed h (finite test E0 is allowed to depend on A,h). The set of d with $E0 \subset A/d$ belongs to p. On that set, every sufficiently long d-line has H-density $\geq g(q)/4$, uniformly, by the usual fixed-E0 empirical extraction. Intersect with D0, still a p-set. The root construction with W chosen only from $g(q)/8$ and k gives a hitting event on a p-set of mass $\mu(H)[1-(1-p0)^W]$. Taking infima first over hitting events H, then over $A \in q \cdot p$ proves the strict gain. In particular a multiplicative idempotent p has $g(p) \in \{0, 1\}$. The argument does not show positive $g(p)$ for an ultrafilter containing an arbitrary positive-deficit D.

Separation by colors. If an intermediate D exists, for every fixed K it has an intermediate subset containing no two distinct members with ratio among $\{2, \ldots, K\}$: color by the total number of prime factors modulo an integer $> \log_2 K$ and use finite partition regularity of positive deficit. More generally color by the valuations modulo a sufficiently large integer at each of the finitely many primes $\leq K$, ruling out all nontrivial reduced ratios with numerator and denominator both $\leq K$ within a single cell. Some cell still has positive deficit, and its positive obstruction is inherited from D. Thus a positive solution cannot simply assume

self-multiplicative structure of all candidate step sets. No construction preserving a common positive witness under the full semigroup closure, or converse to the strict product bound, has been obtained here.

Uniform weighted counting and chromatic product bounds. For every positive integer r,

$$b(D) \geq P[Binomial(r, b(DF)) > rb(F)].$$

Given a DF-avoider A with mean a, every sufficiently long d-line for $d \in D$ has row occupancy $\leq b(F) + \epsilon$ uniformly, by the rescaled empirical argument. Mark a site in r independent copies exactly if its number of A-rows exceeds $rb(F)$. The threshold is integer-valued with a positive gap from $rb(F)$, so a sufficiently long marked D-line would violate the aggregate of the r frequency bounds. This gives a D-avoider of the displayed binomial mass at a; take a increasing to $b(DF)$. The same argument works with arbitrary nonnegative weights, using strict excess above $b(F)$ times their total.

If F is r-large in the finite-color sense, a stronger inequality is $b(D) \geq 1 - (1 - b(DF))^r$. Take the union (OR) of the r independent row indicators. On an all-marked D-line color each position by an active row. A sufficiently long line must contain an h-term monochrome subprogression with step $e \in F$, giving a forbidden $h - AP$ of step de in that row. If F is large for every number of colors and $\rho(D) > 0$, then $b(DF) = 0$. Also if D is r-large and F is s-large then DF is rs-large: group the rs colors into r groups, take a long $D - AP$ in one group, and apply the s-color statement for F along that arithmetic progression. Thus if D is 2-large and $\rho(D) > 0$,

$$b(D^n) \leq 1 - (1 - b(D))^{1/2^{n-1}} \to 0.$$

In particular this applies whenever $b(D) < 1/2$, since non-2-largeness supplies a binary avoiding coloring and one cell has upper density $\geq 1/2$. Even the binomial inequality proves the limit qualitatively in this latter range: the numbers $b(D^n)$ decrease by scaling and inclusion; if their limit s were positive, take r large with rs noninteger and P[Binomial(r,s)>rs]> $b(D)$, possible by the central limit theorem. Apply the inequality with $F = D^n$ and let n tend to infinity.

D02. A conditional automatic-set dichotomy and its obstruction

Scope and status. Superseded formulation; see corrections.

Correction. The proposed theorem for all automatic sets uses a rational-spectrum assertion that is not valid in the required generality. The unconditional conclusion is withdrawn. The corrected terminal-state theorem appears in R015; the counterexample to the problematic input is discussed in R214. The endpoint-one argument under the stated modular obstruction remains a separate valid result.

This conditional argument concerns automatic sets in the setting of Alweiss–Farhangi Question 6.1. It combines the rational-spectrum IP assertions discussed below with bounded-signed-power colorings.

Conditional claim (the unconditional assertion is withdrawn). Fix a base $k \geq 2$ and a k-automatic $D \subset N$. In a finite automaton for the language of its accepted base-k expansions (take the most-significant-digit first convention), call a state branching if it has two noncommuting loop words returning to that state. Let D_{br} be the numbers represented by words visiting such a state. If D_{br} contains a positive multiple of every integer, then D is fully Szemerédi-large, $b(D) = 0$. If D_{br} misses the multiples of some integer, then $b(D) = 1$.

The easy case and the variable cubes Use Byszewski–Konieczny, *Automatic sequences and generalised polynomials*, Proposition 3.4 and proof of Theorem 3.13. The accepted language of paths avoiding all branching states is arid: delete these states from the automaton (or redirect paths to a rejecting sink); no accessible coaccessible state remaining has two noncommuting loops. Its accepted numbers lie in a finite union of basic arid sets. After clearing a common denominator, arid numbers are bounded signed sums of powers of k: expand each repeated fixed word by a geometric-series formula, as in bounded-power-steps.md [D24]. That proof with r uniform colors is polychromatic and gives $b = 1$ for these sets, as explained in chacon-witness.md [D25]. If D_{br} misses all multiples of q, it too has $b = 1$ by the finite congruence complement. Intersections of independent high-density avoiding events show finite unions of such sets still have $b = 1$. This proves the second case.

Suppose now that D_{br} contains multiples of every integer. Given an arbitrary integer $L \geq 1$, choose an accepted positive d_0 in D_{br} divisible by Lk^M, where M is much larger than the automaton size and than $\log_k L$. The accepted word for d_0 ends in $\geq M$ zeros. One can choose a branching state on its path with a suffix long enough that $k^{suffixlength}$ is divisible by the factor $L_{\|}$ of L supported on primes dividing k. Indeed, if an occurrence is near the end of the long zero run, the zero-transition map on a finite set is eventually periodic after at most the number of states in the run. The same branching state then occurs before the desired terminal zeros too (take M larger than the required length plus twice the number of states). Earlier occurrences are already suitable.

Write the word as prefix U and suffix V at this cut. The chosen state has two distinct equal-length loops v_0, v_1, by taking powers of the two noncommuting words as in the cited proof. Upon interchange $c = k^{|V|}([v_1]_k - [v_0]_k) > 0$; put $Q = k^{|v_0|}$. Inserting m loops of either type gives, simultaneously for every $F \subset \{0, \ldots, m-1\}$,

$$b_m + c\sum_{i\in F} Q^i \in D,$$

$$b_m = [V]_k + k^{|V|}(Q^m[U]_k + [v_0]_k(Q^m - 1)/(Q-1)).$$

There are arbitrarily large m with $b_m \equiv 0 \bmod L$. Modulo $L_\parallel$ this follows from the choice of suffix and $d_0 \equiv 0 \bmod L$. For the complementary factor $L_\perp$ coprime to k, take m through multiples of the multiplicative order making $Q^m \equiv 1 \bmod (Q-1)L_\perp$; then $b_m \equiv d_0 \bmod L_\perp$. The numerical values are positive because the initial represented word d_0 is nonzero.

It is important to make *every* variable step divisible by L, without any loss in the eventual density lower bound. Choose B with $L_\parallel | Q^B$ and t with $Q^t \equiv 1 \bmod (Q-1)L_\perp$. Group the slots $B + jt, \ldots, B + jt + t - 1$ to be selected all together or none. Their weighted steps are

$$La_j = cQ^B((Q^t - 1)/(Q-1))(Q^t)^j, j \geq 0,$$

where every a_j is a positive integer and the sequence is a fixed integer multiple of a geometric progression of integer ratio Q^t. For any fixed number N of variable groups, all $b_m + L\sum_{j\in F} a_j$ with $F \subset \{0, \ldots, N-1\}$ belong to D whenever m exceeds the largest used slot. All choices (the automaton state, L, B,t, the common variable family) are made before we take the finite test averages below.

Uniformizing the characteristic-factor estimate over offsets Consider any ergodic binary witness system (X, μ, T) and positive event A of mass a, and fix required length $h = s + 1$. Write V for the coordinatewise slope action on h-tuples: its jth coordinate uses T^j, $0 \leq j \leq s$. Let J be the compact convex space of h-fold self-joinings of T, invariant under the *diagonal* action $T_\Delta = T \times \ldots \times T$. The topology is convergence on measurable rectangles; using a compact shift model, measurable rectangle tests are continuous on the fixed-marginal joining space by marginal L^1 approximation. In particular the graph joining λ_b, the law of $(x, T^b x, \ldots, T^{sb} x)$, belongs to J for every integer b.

Let ν_N denote the uniform subset-sum law of La_j, $0 \leq j < N$. For arbitrary bounded $f_0, \ldots, f_s$ define

$$E_N(\lambda) = \int E_{u\sim\nu_N} \prod_{j=0}^{s} f_j(T^{ju} y_j) d\lambda(y_0, \ldots, y_s).$$

When one f_j is orthogonal to a sufficiently high fixed finite Host–Kra factor $Z_s(X)$ (a harmless overestimate), $E_N(\lambda) \to 0$ for **every** $\lambda \in J$. To see this, the displayed expression is an integrated multiple average on (X^h, λ, T_Δ) with tests $f_j \circ \pi_j$. A pullback of a single-coordinate function has exactly its original Gowers–Host–Kra seminorms, by the iterated cube formula and equality of its marginal process; this remains true for nonergodic λ. Shift all slopes by one of the *other* slopes if necessary to arrange that the orthogonal function is at a nonzero slope. For $s \geq 2$ apply Kra–Shalom, Theorem 4.2. The sequences La_j and a_j have rational spectrum in their Definition 3.7: nonvanishing of any tail cosine multiplier requires

$||(La_j)\theta|| \to 0$, forcing eventual integrality since the fixed successive ratio is bounded. For $s = 1$ the same conclusion follows from the ordinary spectral theorem on the mean-zero continuous spectral component, or by integrating against a graph joining and the uniform single-unitary norm estimate. Thus the conclusion for all J holds; for $s = 1$, the projected single-coordinate function has the same spectral measure as the original even in the joining.

The same argument applies to all the finitely many error terms when telescoping the original correlation from 1_A to $g = E(1_A|Z_s)$. Each E_N is continuous in λ and bounded uniformly in N,λ. Pointwise convergence on all J is enough; we will take a **finite convex mixture of digit lengths**, which can be used in one single larger automatic cube. This avoids asserting uniform convergence of a raw sequence of IP averages.

A nilfactor lower bound independent of the modulus Choose b continuous on some finite-dimensional nilfactor, $0 \le b \le 1$, with $||g - b||_1$ as small as desired and $\int b \ge a/2$. Use inverse-limit approximation of the ergodic Host–Kra factor and then continuous approximation. Let (Y, ν, S) be this nilsystem. On Y^h take $V = (S^j)_{0 \le j \le s}$ and let $Dg(x) = (x, \dots, x)$ denote the diagonal point. Put $F(y) = \prod b(y_j)$. For any L, the ordinary Cesàro mean of $\int F(V^{Ln} Dg(x))d\nu(x)$ has a uniform lower bound $c(a, h) > 0$ independent of L and of this nilsystem. This is the finite Szemerédi/Varnavides bound applied to S^L and the set $\{b \ge a/4\}$ of measure $\ge a/4$: use a fixed ordinary length-R density-Szemerédi test at every dilation and average the finitely many possible small strides, as in amplification.md [D01]. Fix the L^1 approximation so $h||g - b||_1 < c(a, h)/8$ before selecting L.

The nilorbit closure of `Dg(x)` is a compact subnilmanifold with finitely many components cyclically permuted by V. Write $r(x)$ for their number. This is measurable; for example the compact-valued orbit-closure map is measurable, and the condition that its intended component is connected/clopen cyclic can be checked via the orbit closures of powers (equivalently count the distinct power-orbit closures). Choose L divisible by $r(x)$ on a set of measure $> 1 - c(a, h)/4$ (take a factorial of a sufficiently large integer). For those good x, $O_L(x) = closure\{V^{Ln} Dg(x) : n \in Z\}$ is the identity-index connected component of that orbit, and V^L is minimal and uniquely ergodic on $O_L(x)$. Indeed a connected minimal nilrotation is totally minimal; a power of a minimal map has finitely many cyclically permuted clopen minimal pieces, which connectedness forbids except the whole component. The Haar integral over $O_L(x)$ is the ordinary mean for that x. Consequently its integral over *good* x is $\ge 3c(a, h)/4$, the integrand being at most one.

Let K be the closed convex hull of the laws $\rho_b = law_\nu(x, V^b Dg(x))$ on $Y \times Y^h$ for $b \equiv 0 \bmod L$. Every $\rho \in K$ has first marginal ν and is supported on pairs with $y \in O_L(x)$. One way to check closedness despite varying orbit dimensions: $x \mapsto O_L(x)$ is measurable into the compact hyperspace, being the closure of countably many continuous orbit maps. By Lusin it is continuous off an x-set of arbitrarily small fixed ν-mass. Equivalently the nonnegative function $(x, y) \mapsto dist(y, O_L(x))$ is approximable in integral uniformly over all such laws by continuous functions (extend its restriction to the good compact x-set). Its integral zero is preserved under weak limits and convex mixtures.

Define the continuous affine test

$$P_N(\rho) = \int E_{u \sim \nu_N} F(V^u y) d\rho(x, y).$$

For good x and every $y \in O_L(x)$, Holikov–Shalom, Proposition 5.3, applied to V^L and the geometric spectrum of a_j, makes the inner average converge to the full Haar integral of F on $O_L(x)$. The orbit closure here is connected, so there is **no** $1/L$ factor in the lower bound. Elsewhere F is nonnegative. Fatou therefore gives $\liminf_N P_N(\rho) \geq 3c(a,h)/4$ for every $\rho \in K$.

A finite mixture, then an actual allowed difference On the compact product $K \times J$ take the bounded continuous real functions

$$H_N(\rho, \lambda) = P_N(\rho) - \sum_e |E_{N,e}(\lambda)|,$$

where e ranges over the finitely many telescoping errors. Pointwise $\liminf_N H_N \geq 3c(a,h)/4$. There is a finite convex combination of these functions with infimum $> c(a,h)/2$ on the whole compact product. For detail, otherwise finite minimax (allowing probability measures on the compact parameter space as one player) gives for every N a probability integrating all of $H_1, \ldots, H_N$ to at most $c(a,h)/2$. A weak limit would have this bound for every N, contradicting Fatou and the pointwise liminf. This is the needed uniformity for moving offsets.

Use the automaton construction with this L and choose a sufficiently late default-loop length m with $b_m \equiv 0 \bmod L$, long enough to contain all variable blocks used in the finite convex mixture. Test that mixture at ρ_{b_m} and λ_{b_m}. Telescoping from A to g and then the L^1 approximation of g by b show that the corresponding convex average of $c_A(b_m + u)$ is positive, with error at most $h||g-b||_1$ for the last replacement. Every one of its steps belongs to D. Thus some $d \in D$ gives an h-return. This holds for every positive event and every h, proving full Szemerédi-largeness in the branching-divisible case.

The theorem classifies one natural infinite-state collection of possible examples using finite automata. It does not reduce an arbitrary weakly mixing obstruction in Question 6.1 to an automatic one.

A finite decision test for the arithmetic alternative The condition on all moduli has an effective single-modulus version. For each branching state s fix two distinct equal-length loop words and let $c_s \neq 0$ be the difference between their numerical values. Let $c_s^{\perp}$ be the positive part of $|c_s|$ supported on primes not dividing k (use 1 when there are none). If the automaton has u states, put

$$M = k^{u+1} \cdot \prod_{branchs} c_s^{\perp}.$$

Then D_{br} meets every positive integer modulus if and only if it meets `M N`. Only sufficiency needs proof. Given such a word, choose any branching state s it visits. Its terminal state is on a zero-transition cycle, since the expansion ends with at least $u+1$ zeros. Appending arbitrarily many full periods of zeros keeps it accepted, retains s, and preserves divisibility by $c_s^{\perp}$. Given an arbitrary q, first append enough zero cycles that the suffix following the chosen s has k-power divisible by $q_{\|}$ and the new word value d' is divisible by $q_{\|}$. Use the

loop cube above and take m arbitrarily large with $Q^m \equiv 1 \bmod (Q-1)q_\perp$. Its default offset is 0 mod $q_\parallel$ and d' mod $q_\perp$. All variable increments are zero modulo $q_\parallel$. In the other modulus repeated slots $i \equiv 0 \bmod \mathrm{ord}_{q_\perp}(Q)$ all have weight $k^{|V|}c_s$ modulo $q_\perp$. That weight generates the subgroup $\gcd(c_s, q_\perp)Z/q_\perp Z$, which contains d'. With sufficiently many such slots choose a nonnegative number that cancels d'. The resulting accepted value is a multiple of q. If $q_\perp = 1$, no adjustment is needed. The language test for a branching-visited accepted word with value 0 mod M is finite-state reachability. Branching states can be detected by product-automaton reachability for two distinct equal-length return words (equivalent to admitting two noncommuting loops; different equal-length words cannot commute).

D03. A finite-sheet Hales–Jewett lifting would itself force all densities

Scope and status. Supporting proof and dependency material.

One conceivable way of proving a high-density threshold for directions D in a torsion-free abelian group is to assign to each word $x \in [h]^n$ a list $\Phi_1(x),\ldots,\Phi_c(x)$ of group points, for fixed c, such that every combinatorial line $x(t)$ can be lifted, with a sheet chosen separately at each t, to an h-term AP of D-step. A common random translate of all representatives of every word, followed by the union bound and density Hales–Jewett, would give the threshold $1 - 1/c$. There is an intrinsic obstruction to this approach, even permitting arbitrary nonlinear maps and time-varying sheet choices.

Fix c and a desired finite-sums length m. There exist fixed L,N depending on c,m such that for ANY $h \geq L$, $n \geq N$, these maps force the full nonempty finite-sums set $\mathrm{FS}(d_1,\ldots,d_m)$ to lie in D. Here the steps have the orientation in which the h-term lines are lifted. Thus existence for unbounded h,n would give $b(D) = 0$. In particular this approach cannot certify a genuine intermediate example.

Take L a van der Waerden length for a three-term progression in c colors. For every combinatorial L-line in the first L alphabet symbols, extend it to its h-line (keeping all passive symbols). Fix a successful sheet assignment there and color the L-line by its restricted L-symbol sheet pattern, with at most c^L colors. The Graham–Rothschild parameter-words theorem gives an m-dimensional combinatorial subspace of $[L]^n$ on which every combinatorial line has exactly the same sheet pattern. There are three equally spaced time values t_0,$t_0 + s$,t_0+2s in that pattern assigned the same sheet, say ϕ. Restrict the subspace's m parameters to these three letters and identify them with 0,1,2 in increasing order.

On this ternary cube ϕ has zero second difference along every combinatorial line. The singleton-variable lines imply separate affinity in all coordinates, giving a multiaffine expansion over the ambient group's rational vector space. On two-variable lines the second difference equals twice their mixed coefficient with the other coordinates fixed. Thus every two-variable mixed coefficient vanishes identically; in the multiaffine expansion every term of order$\geq$2 vanishes. Therefore $\phi(x) = u + \sum_j x_j w_j$. For a nonempty set S of wildcard parameters, the first two chosen same-sheet positions differ by sd_S in the lifted $h - AP$, hence $sd_S = \sum_{j\in S} w_j$. The singleton case gives $w_j = sd_j$, with $d_j \in D$. Torsion-freeness yields $d_S = \sum_{j\in S} d_j \in D$ for all S. Finite density Hales–Jewett (apply its word-average proof to these directions) then gives arbitrary positive-density multiple recurrence when the maps are available at unbounded dimensions. The proof genuinely uses torsion-freeness to cancel 2 in the mixed coefficient and s in the final slope. Torsion models with order-two markers do not automatically inherit this obstruction.

D04. Finite constructions and obstructions for intermediate critical densities

Scope and status. Supporting proof and dependency material.

Write $b(D)$ for the supremum over upper densities of sets avoiding all length-h D-progressions for some fixed h. A counterexample needs $0 < b(D) < 1$ for the same D.

Near-antipodal Hamming test In $(Z/2)^n$, a Hamming ball of radius $n/2 - c\sqrt{n}$ has positive density (depending on c) and avoids pairs whose differences have Hamming weight $> n - 2c\sqrt{n}$. In this group, every set of density $> 1/2$ contains a pair at any fixed difference, and that pair extends to a progression of every length because the difference has order two. This genuinely gives intermediate thresholds in the exponent-two model, not in Z.

A root lift changes the assertion. In $(Z/4)^n$ let w be the parity mask required of a direction v. There is a full four-term coset in A with $v \bmod 2 = w$ if and only if for some parity word u both fibers u and $u + w$ contain a pair separated by 2w in A. For necessity look at times 0,2 and 1,3. For sufficiency, choose first points x,y of the two pairs and take $v = y - x$. Then $2v = 2w$, and the four points are x,y,$x + 2w$,$y + 2w$. This is an exact reduction even when v is otherwise unrestricted. It does not extend mechanically to 2^r: even and odd suborbits share the same direction, unlike the freedom to choose their initial points.

Fixing t parity coordinates forever cannot suffice: if every w has all but at most t coordinates odd, delete the coordinate hyperplanes $x_i = 0 \bmod 2^s$, $i = 1,\ldots,t+1$. The loss in density is at most $(t+1)/2^s$, and every 2^s-term line in an eligible direction visits a deleted plane. Thus the possible number of exceptional coordinates must become unbounded. Taking it much bigger than $\sqrt{n}$ loses the low-density Hamming-ball pair obstruction. The interesting range is an unbounded radius no bigger than a fixed multiple of $\sqrt{n}$.

The dyadic obstruction even at critical radius This specific forcing strategy still fails. Uniformly for all sufficiently large n there is, for each $\epsilon > 0$, a set of density $> 1 - \epsilon$ containing no K-term line with direction v having at most $C\sqrt{n}$ even coordinates. K depends on ϵ,C but not on n.

Fix r, $K = 2^r$, and define $F_s(x) = \sum_{j \le n} \cos(2\pi x_j / 2^s)$ for $1 \le s \le r$ on $(Z/2^r)^n$. Delete the points H where simultaneously $F_s(x) \ge -C\sqrt{n}$ for all s. Every eligible line meets H. Starting at $t = 0$, choose successively whether to add 2^{s-1} to t. This does not change any lower F_j. In every odd coordinate the two cosine values in $F_s(x + tv)$ and $F_s(x + (t + 2^{s-1})v)$ cancel. Each of the remaining $\le C\sqrt{n}$ coordinates contributes at least -2 to their sum. Thus at least one of the two choices satisfies the new threshold. The final t is less than K.

For fixed r the vectors of cosine statistics divided by $\sqrt{n}$, under uniform x, converge by the iid multivariate CLT to independent centered Gaussians (one-coordinate covariance is diagonal by character orthogonality, variances 1 for $s = 1$ and $1/2$ for $s \ge 2$). The density of H therefore tends to $\Phi(C)\Phi(\sqrt{2}C)^{r-1}$; for fixed C that tends to zero as r grows. This is a dimension-uniform barrier, not merely the fact that finite D has $b(D) = 1$ with an uncontrolled length.

There is a torus-frequency variant. If all but $C\sqrt{n}$ coordinates of $d\alpha$ are within ϵ_n of $1/2$ and $\sqrt{n}\epsilon_n$ is bounded, use the statistics $\sum_i \cos(2\pi x \alpha_i / 2^{s-1})$. The same t-bit choice incurs errors $O(n\epsilon_n)$ at the choice and $O(2^r n\epsilon_n)$ from remaining changes of higher bits. If

$\sqrt{n}\epsilon_n \to 0$, for every fixed r these latter errors are $o(\sqrt{n})$. The exceptional $C\sqrt{n}$ coordinates may change the earlier statistic by at most $2\,C\sqrt{n}$ in total, comparing its chosen time directly to the final time. For finite torus frequencies rationally independent with 1, averaging over x in the integers makes the root phases Haar. For the real-root version, orthogonality of the dyadically related frequencies still yields independent centered Gaussians under the many-coordinate CLT, now all with variance 1/2. Again one obtains arbitrarily thin hitting sets for large dimensions. If $\sqrt{n}\epsilon_n$ is merely bounded away from zero, the error $O(2^r)$ in the normalized threshold must not be discarded and this calculation alone does not establish vanishing density as r grows.

A positive partial thickening lemma for pair obstructions Suppose B avoids h-term D-lines and C avoids pairs of every difference in $[h]D = \{td : 1 \le t \le h, d \in D\}$. Then B union C avoids $(2h)$-term D-lines. Indeed any such line would contain an element of C in each successive h-block; consecutive C-elements on the line are separated by at most h time steps (otherwise h consecutive terms are in B alone). This would give a forbidden pair of C-elements. For C built as a joining of scaled positive-density pair-obstruction sets, the available density may shrink exponentially with h. The lemma yields no density-uniform amplification.

Pointwise encoders without arithmetic compatibility Consider m binary rows in which no row contains h consecutive ones. Identify each column with the subset of active coordinates. A pointwise encoder accepting a family B of subsets excludes arbitrarily long all-accepted runs on the basis of that row constraint alone if and only if B is h-wise intersecting, with repetitions allowed. If h sets have empty intersection, repeat them cyclically: every consecutive h columns contains a zero in every row, but all columns are accepted. Conversely if every h have a common element then no h valid columns can all be accepted.

At Bernoulli(a) one-site density with $a \le (h-1)/h$, no h-wise intersecting family has measure $> 1-1/h$. In each coordinate sample h coupled bits by placing a uniform distinguished zero among h sites and retaining each other bit with probability a $h/(h-1)$, independently for distinct coordinates. Each site's column then has Bernoulli(a) product law and the h-fold intersection is empty. If each of the h sites were accepted with probability $> 1 - 1/h$, the union bound would give positive probability of simultaneous acceptance, impossible. This is an obstruction only to the black-box independent-product step, not to arithmetic encoders using more than the row constraints.

Finite-order caution for the original Alweiss topology The least-order obstruction applies only to actions strongly continuous for the original coordinates. For any number L, choose q divisible by all 1,..,L, replace $P(n)$ by $P_q(n)$ with coordinates $n\alpha_j/q$, so $qP_q(n) = P(n)$, and lift the allowed ambient set to the inverse image under multiplication by q. This pullback has no element of order $\le L$, because every such element is killed by q whereas zero is not allowed. Thus the fixed ambient torsion order is not an invariant of all actions of the integers presenting the same step set. This does not give the required sparse hitting construction on the root group.

Odd finite sums: a spectral constraint Suppose every positive odd-cardinality finite sum of an increasing sequence a_i is in D, and a set of positive density avoids *pairs* at all these

steps. Correspondence gives an indicator f of positive measure in an invertible stationary system with $\langle f, U^d f\rangle = 0$ for all such d. Restrict to the cyclic Hilbert space generated by f. Along a subsequence U^{a_i} has a weak operator limit V in the abelian spectral algebra of U. For every odd m, first taking iterated limits over m distinct indices gives $\langle f, V^m f\rangle = 0$, since every underlying correlation has an odd finite sum as its step. With σ the spectral measure of f and V multiplication by g, $|g| \le 1$, the Cesaro averages of the odd powers of g converge to 1 on $g = 1$, -1 on $g = -1$, and 0 elsewhere. Dominated convergence therefore gives $\sigma(\{g = -1\}) = \sigma(\{g = 1\}) > 0$; positivity follows from the constant component of f. On the former subspace U^{a_i} converges strongly to minus the identity, as the weak limit is already unitary there. This does not imply d-uniform antipodality on all D.

Virtual finite models: one-way transference Write $F = \{d_1, \dots, d_m\}$. From any density-a h-avoider on Z, choose a translate on which the map $x_1, \dots, x_m$ -> offset+$\sum x_i d_i$ from $[0, p)^m$ samples density $\ge a$ (invariant-measure averaging), $p \ge h$. On identifying the domain with the residue cube F_p^m this pullback avoids every full coordinate line, since in an e_i-line the represented integers run over p consecutive d_i-terms. A digit-product construction can amplify on the free domain. But the map is not a homomorphism from the residue cube to Z, and an equivariant section from Z taking d_i to e_i is impossible when additive relations do not hold among the axes. Nor is a density-one-minus-ϵ lift easily descendible by taking dense conditional fibers: finding a whole length-K lifted line using the union bound costs $K\epsilon$. For the elementary digit scheme $K = p^r$ and $\epsilon = (1 - a)^r$, so $K\epsilon = (p(1 - a))^r \ge 1$. This is a genuine limit on the shortcut, not a nonexistence theorem for lifts.

Odd-prime loophole in dyadic pointwise encoders for triples Let a proposed black-box construction take m binary rows and suppose row i has neither a run of three zeros nor a run of three ones on its own stride s_i. If some odd prime p divides none of the strides, *any* hole H in the m-cube of unbiased measure strictly below $1/p$ can be missed by the entire column sequence. In row i use a periodic coloring of Z/p whose colors along stride s_i follow an alternating p-letter cyclic word; the single seam gives a double letter but not a triple. Randomly complement each row independently. Each of the p column vectors has the uniform cube law. By a union bound, some choice misses H at all times. For strides that are all powers of two take $p = 3$. This is a barrier to marginal-only encoders, not a counterexample to integer density amplification.

Literature constraints Griesmer, Separating Bohr denseness from measurable recurrence, https://arxiv.org/html/2002.06994 , couples successive finite pair obstructions by Bohr-Hamming extensions; independent witnesses in each digit do not automatically glue with positive density (the direct intersection can have product density tending to zero). This paper proves Bohr denseness with pair nonrecurrence, not high-density multiple recurrence of the allowed steps.

D05. The original Alweiss example fails density largeness at every density below one

Scope and status. Supporting proof and dependency material.

Write $e(x) = \exp(2\pi i x)$. The frequencies α_j are positive and summable, and 1,α_1,...,α_J are linearly independent over the rationals for each J. If d belongs to the fixed Alweiss example E, choose the centered representatives σ_j of $d\alpha_j$. With $a = 1/100$, $b = 10^{-10}$, there is an i such that

$$|\sigma_i + a| < 2b, \max_{j \neq i} |\sigma_j| < 2b, \sum_{j \neq i} |\sigma_j| < a + 2b.$$

In particular $\Sigma|\sigma_j| \leq C = 2a + 4b$, and $\sum_{j \neq i} \sigma_j^2 \leq 2b(a + 2b)$. Only these necessary conditions on d will be used. Define the following absolutely convergent complex observables, indexed by positive integers M,

$$F_M(n) = \sum_j (e(n\alpha_j/M) - 1).$$

For every $\epsilon > 0$ we will exhibit integers K and a set of uniform density $> 1 - \epsilon$ that contains no K-term AP with difference in E.

Two elementary analytic facts Joint uniform distribution. For distinct indices M_1,...,M_r and any positive real period P, the vectors $F_{M_s}(n)$ are well distributed on $(C/PZ[i])^r$, also uniformly over the left endpoint of the averaging interval. Indeed take L to be a common multiple of the M_s. Each nontrivial real or imaginary character gives a real potential $\psi(x)$, a linear combination of $\mathrm{Re}(e((L/M_s)x) - 1)$ and $\mathrm{Im}(e((L/M_s)x) - 1)$, multiplied by $2\pi/P$. Distinct frequencies and a nonzero vector of character coefficients make ψ a nonconstant real Lipschitz trigonometric polynomial vanishing at zero. Set $\theta_j = \alpha_j/L$ and $z_n = \exp(i\Sigma\psi(n\theta_j))$. The summation converges for each n. For a fixed integer $h \geq 1$, the correlation z_{n+h} conjugate(z_n) is the product over j of $\exp(i(\psi((n+h)\theta_j) - \psi(n\theta_j)))$; the fixed-h tail error is bounded uniformly in n by $\mathrm{Lip}(\psi)h\sum_{j>J}\theta_j$. Consequently its uniform mean is $k(h) = \prod_j K(h\theta_j)$, where $K(t) = \int_{\mathbb{T}} \exp(i(\psi(x+t) - \psi(x)))\,dx$. Since ψ is nonconstant, $\int_{\mathbb{T}} |K(t)|\,dt < 1$. In fact equality $|K(t)| = 1$ requires $\psi(x+t) - \psi(x)$ to be constant by continuity, and this constant must be zero by integration; not every t is a period. For each J, $|k(h)| \leq \prod_{j \leq J} |K(h\theta_j)|$, whose Cesàro mean in h tends to $\left(\int |K|\right)^J$ by finite-dimensional equidistribution. Thus the Cesàro mean of $|k(h)|$ tends to zero. Van der Corput applied uniformly over starting points gives zero mean for z_n, proving the assertion by Weyl's criterion.

Trapped primitives. Suppose g:$Z \to C$ is bounded and its first difference has the representation

$$g(u+1) - g(u) = c_0 + \sum_j a_j e(uv_j), \Sigma|a_j| < \infty.$$

Then g is a Bohr almost periodic sequence. To see this let Y be the closure of $\{uv : u \in Z\}$ in T^N, with the minimal rotation by v. The difference is the restriction to this orbit of a continuous function f on Y, defined by the absolutely convergent series. The orbit of $(0, g(0))$ under $(y, z) \mapsto (y + v, z + f(y))$ has bounded second coordinate and hence compact closure. Choose a nonempty minimal closed invariant subsystem B in this closure. Its projection is Y. If a fiber contains two points separated by $s \neq 0$, the minimal sets B and $B + s$ intersect, and hence coincide; this contradicts compactness since translation by s is unbounded under iteration. Thus B is the graph of a continuous function on Y, and this function has first difference f. Along the orbit, g differs from the graph by a constant. This proves the claimed almost periodicity.

Every limiting legal gate code has a common finite period Fix once and for all a fine period $L_0 = 10$, a gate-center circle of radius c $takec = 1/4below$, and a coarse period P that is a large integer multiple of L_0. Remove from consideration values with real or imaginary part in $[0, H]$ mod P, where $H = 1 > 2\pi C$. Also remove values where distance to $L_0 Z[i]$ is within τ of c, where $0 < \tau < c$ will be small. Values outside both deletions are called legal; their code is 1 if the fine-torus distance is $> c$ and 0 otherwise.

Fix M. A centered finite word is realized if it is the gate word of the values $F_M(n + td)$, $|t| \leq T$, for a single $d \in E$, with every value legal and all the sampled integers positive. Let X_M consist of the two-sided binary sequences all of whose centered finite blocks are realized. This is a closed, shift-invariant subshift: a centered block at another position is a subblock of a larger centered realized block. Arbitrarily long legal finite segments yield an element by centering and diagonal extraction.

Fix any $x \in X_M$ and choose such a realization of x$|[-N, N]$ for each N, with starting argument n_N at $t = 0$ and difference d_N. For each residue $0 \leq r < M$ and $t = r + Mu$ the increment along one residue step has complex modulus at most $2\pi C$, by the triangle inequality for the defining exponential series (the increment uses $e(d_N \alpha_j)$). The surviving real segments $[Pq+H, P(q+1)]$ for $q \in Z$ are separated by a gap of length H. Each real track thus lies in one such component throughout its available u range, and the same holds for the imaginary track. These boxes can be translated by arbitrary P-lattice vectors depending on N and r; we use only that

$$g_{N,r}(u) = F_M(n_N + (r + Mu)d_N) - F_M(n_N + rd_N)$$

has modulus $\leq \sqrt{2}P$ for each available u, for $N \geq M$. Extract a subsequence with $F_M(n_N + rd_N)$ mod L_0 converging for all r. For each N order $\sigma_{j,N}$, the centered representatives of $d_N \alpha_j$, by decreasing absolute value, using the same order on the associated phases $z_{j,N,r} = e((n_N + rd_N)\alpha_j / M)$. A positive summable decreasing sequence has $\sum_{j>J} \sigma_{j,N}^2 \leq C^2/(J + 1)$. A diagonal subsequence gives $\sigma_{j,N} \to v_j$, $z_{j,N,r} \to z_{j,r}$ for each fixed j,r, and $w_{N,r} = \sum_j z_{j,N,r} \sigma_{j,N} \to w_r$. We have $\Sigma |v_j| \leq C$.

For each fixed u in Z, the increments satisfy

$$g_{N,r}(u) = \sum_j z_{j,N,r}(e(u\sigma_{j,N}) - 1)$$

$$\to g_r(u) = 2\pi i u(w_r - \sum_j z_{j,r} v_j) + \sum_j z_{j,r}(e(uv_j) - 1).$$

Indeed subtract the linear term $2\pi i u\sigma$ in each summand; the uniform tail of the resulting series is $\leq 2\pi^2 u^2 C^2/(J+1)$. This formula is simultaneous for all integer u and all the finitely many residues, with each g_r bounded on Z by $\sqrt{2}P$. Its first difference has the form of the trapped-primitive lemma with a constant term and absolutely summable coefficients $z_{j,r}(e(v_j) - 1)$, showing that g_r is Bohr almost periodic.

The fine-torus values along residue r are the sum modulo L_0 of the limiting base phase at $u = 0$ and $g_r(u)$. The limiting trajectory stays outside the open τ-band around the gate boundary, and the code $x(r + Mu)$ is recovered by the gate test. Equivalently, recover the code by evaluating a continuous function on the fine torus that equals zero at distances $\leq c - \tau$ and one at distances $\geq c + \tau$. This binary subsequence is thus Bohr almost periodic. A binary almost-periodic sequence is periodic, by taking an almost-period for tolerance 1/2. Interleaving the M periodic subsequences makes x periodic.

Finally, a compact subshift over a finite alphabet containing only periodic sequences is finite. If it were infinite, take distinct elements x_j converging to a periodic x. Their maximal intervals agreeing with x about the origin expand, and since the x_j are periodic but distinct from x, each has a first mismatch on the right. Shift to the mismatch, take a subsequence where the phase of x modulo its period is fixed and the shifted x_j converge. The resulting sequence agrees with a fixed translate of x on the entire negative half-line but differs at zero, so it cannot be periodic. This contradiction gives finiteness. In particular there is a positive integer T_M that is a period of every sequence in X_M (including the empty case, for which any T_M works). T_M may depend on M,P,c,τ, all fixed before the multiscale recursion.

Every allowed difference can activate the gate For all n,$d \in E$ as appropriate, compare F_M at n, n+50 Md, n+100 Md. Its second difference is $\sum_j e(n\alpha_j/M)(e(50\sigma_j)-1)^2$. The distinguished term has modulus at least $(2 - 2\pi \cdot 100b)^2$, since $50\sigma_i$ is within 100b of $-1/2$. The sum of the moduli for the other terms is at most $4\pi^2 \cdot 50^2 \cdot 2b(a+2b) < 0.000001$. Consequently the complex modulus of the total is > 3.9 and < 4.1. If the three positions were all inactive then the second difference would be within $4c = 1$ of the lattice $L_0 Z[i]$. A complex number with modulus between 3.9 and 4.1 cannot be that close to a lattice of spacing 10. Thus for every $x \in X_M$ and $t \in Z$,

$$\max\{x(t), x(t+50M), x(t+100M)\} = 1.$$

This statement follows by realizing one sufficiently large finite block and is independent of limits.

A single high-density avoiding set Given ϵ, the fraction of fine-torus area in the active gate is $p = 1 - \pi c^2/L_0^2 < 1$. Choose r with $p^r < \epsilon/3$. Choose P a sufficiently large multiple of L_0 to make $2rH/P < \epsilon/3$. Choose $\tau > 0$ so small that r times the fine-torus area fraction of the closed τ-band is $< \epsilon/3$. All of these choices precede selection of M. Set $M_1 = 1$; recursively choose M_s distinct, increasing, divisible by every T_{M_j}, $j < s$. This is possible since each X_{M_j} is a finite subshift constructed solely from the single corresponding row with

the already fixed gates. Delete every n where any of the r rows is illegal, and every n where all r rows are active. Joint uniform distribution at period P, a multiple of L_0, proves that the retained set B has uniform density $> 1 - \epsilon$. All sets are Jordan measurable in the finite product torus.

Suppose that B contains arbitrarily long E-progressions. Center such progressions and extract a joint limit x_1,. . .,x_r of the r binary codes. Then $x_s \in X_{M_s}$. At each $t \in Z$ not all $x_s(t)$ are 1. Start at $t = 0$; for $s = 1$,. . .,r choose an update from $\{0, 50M_s, 100M_s\}$ which makes x_s equal 1. All earlier codes retain their value, because the update is divisible by their common row periods. At the final t all codes are 1, a contradiction. Thus there is a single finite K bounding all E-progression lengths in B.

This proof gives no effective bound for K. Its compact-subshift argument replaces the shrinking-tail hypothesis used in a separate diagonal construction. It establishes $b(E) = 1$ for this specific $D = E$, not a general zero-one law for $b(D)$.

D06. Exact integer realization of a coordinatewise bounded difference family on a free abelian group

Scope and status. Supporting proof and dependency material.

Let $H = \oplus_{i\geq 1} Z$ and $V \subset H \setminus \{0\}$. Assume finite integers $M_i \geq 0$ satisfy $|v_i| \leq M_i$ for all $v \in V$ and all i, with no uniform bound in i required. Write $b_H(V)$ for the supremum of the measures of events A in probability-preserving H-actions such that, for some fixed h, every $v \in V$ has zero h-term intersection $\mu(\bigcap_{0\leq j<h} T^{-jv} A) = 0$. The ordinary integer quantity has the same dynamical formulation by correspondence.

Choose odd bases $s_i > 2M_i + 1$, $s_i \geq 3$, with $\Sigma M_i/s_i < \infty$. Put $\lambda_1 = 1$, $\lambda_{i+1} = s_i\lambda_i$, $\phi(v) = \sum_i v_i\lambda_i$, and $D = \{|\phi(v)| : v \in V\}$. Then

$$b_Z(D) = b_H(V).$$

Indeed no nonzero $v \in V$ maps to zero. All its coordinates are allowed balanced digits of the mixed bases ($-(s_i - 1)/2$ through $(s_i - 1)/2$), and such expansions are unique. The same fact gives $\lambda_m|\phi(v)$ iff $v_i = 0$ for all $i < m$. Pulling back any integer action by the homomorphism ϕ immediately gives $b_H(V) \geq b_Z(D)$; changing signs of steps just translates an intersection.

For the converse fix an H-action with an h-witness of measure a. For arbitrarily small ϵ choose m so that $\sum_{i\geq m} 2hM_i/s_i < \epsilon$ and $hM_i < (s_i - 1)/2$ for all $i \geq m$. Write $z(n) \in H$ for the finite balanced digit vector of $n \in Z$. Keep only n for which each digit $i \geq m$ has distance at least $(h - 1)M_i$ from both endpoints of its digit range. This condition only needs inspection of finitely many nonzero digits at each n, since zero is strictly interior to all relevant bands. Call the kept set G.

Its proportion is at least $1 - \epsilon$ along the nonnegative intervals $[0, (\lambda_{j+1} - 1)/2]$, $j \geq m$: on the symmetric interval the first j digits are independent uniform digits and all remaining ones vanish, so the union bound gives the estimate. The balanced sieve is symmetric about zero; on the nonnegative half there is at most an endpoint error tending to zero. If $n \in G$ and $v_i = 0$ below m, then for $0 \leq k < h$,

$$z(n + k\phi(v)) = z(n) + kv$$

by the boundary margins and balanced uniqueness; all signed digit sequences on the right stay legal.

For a random point x in the H-system consider $A_x = \{n \geq 0 : n \in G, T^{z(n)}x \in A\}$. For each v with zero early coordinates, almost surely there is no h-run of signed step $\phi(v)$ among the retained integers. The preceding identity reduces the event to an H-run starting at $T^{z(n)}x$; use countability over n,v. For negative $\phi(v)$, reverse the progression, using its other endpoint as the starting n. The expected retained proportion in the displayed intervals is a times the proportion of G by invariance, so reverse Fatou supplies an x with upper density at least $(1-\epsilon)a$ and all the stated prohibitions. We obtain $b_Z(D \cap \lambda_m Z) \geq (1-\epsilon)a$.

For clarity the elementary localization identity $b_Z(D) = b_Z(D \cap qZ)$ uses a periodic sieve of arbitrary small measure for the steps not divisible by q. For each prime power $p^e||q$, work

mod p^R, R≫e, and delete the p^{e-1} residues $0,\dots,p^{e-1}-1$, uniformly randomly shifted, using independent finite clocks if desired. If $v_p(d) < e$, a full orbit of step d contains one of these residues (they represent every class modulo $p^{v_p(d)}$). Thus any retained progression longer than the largest chosen period has step divisible by q. The density cost of all the sieves tends to zero. Apply localization with $q = \lambda_m$ and let $\epsilon\downarrow 0$. This proves equality.

In particular a counterexample on such a free abelian group would give one on Z without an additional gluing argument or a requirement that the original action be finite. This requires a single H-witness for the entire family; witnesses on separate summands with no common H-model do not by themselves meet the hypothesis. A bounded-exponent torsion-group model also does not qualify unless its high-density forcing survives lifting to every H-action.

One arbitrarily sparse host contains an exact copy of the whole threshold problem Given any increasing $f : N \to R$ with $f(N) \to \infty$, choose the odd radices in the theorem above once for all D, taking $M_i = i$, ensuring $q_i > 2i+1$, $\Sigma i/q_i < \infty$, and $f(\lambda_j) \geq 1 + j(j+1)/2$ for every $j \geq 2$ by taking each preceding radix large. Define

$$\Psi(D) = \bigcup_{i\geq 1} \lambda_i(D \cap [1,i]), U = \Psi(N).$$

Then $b(\Psi(D)) = b(D)$ for every D, and $\sup_{x\in Z} |U \cap [x, x+N)| \leq f(N)$ for all sufficiently large N. To see equality at the group level, put $V_D = \bigcup_{i\geq 1}(D\cap[1,i])e_i$. The homomorphism $H \to Z$ summing the coordinates pulls any h-witness for D back to one for V_D of equal mass. Conversely restricting an H-witness to the ith axis gives an h-witness for $D\cap[1,i]$. For fixed h, stationary compactness shows $b_h(D) = \inf_i b_h(D\cap[1,i])$: the finite-prefix subshifts decrease to the full subshift and the coordinate mean is continuous on their compact spaces of invariant measures. So the two fixed-h group values coincide. Apply the preceding theorem to pass to the integer images and the supremum over h.

The numerical stages are disjoint since $i\lambda_i < \lambda_{i+1}$. Thus Ψ preserves unions, intersections and relative complements in U, and is injective. If $\lambda_j \leq N < \lambda_{j+1}$, all stages later than j lie in $\lambda_{j+1}Z$, so at most one of their points can lie in the half-open interval of length N. Earlier stages total at most $j(j+1)/2$. This proves the stated local counting bound. In particular an intermediate threshold, if it exists, can be realized inside one fixed as-thin-as-prescribed host, including one with reciprocal sum finite and local counts considerably slower than logarithmic. This does not rule out intermediate b, since the same host already contains all-density examples, for instance U itself. The final integer coding claims preservation of b, not of every fixed-h integer threshold.

D07. A weakly mixing Gaussian pair obstruction at Bohr-recurrent differences

Scope and status. Supporting proof and dependency material.

This is a stress test for a possible proof of Question 6.1. It gives a Bohr-recurrent set D with an honest positive-measure pair-avoider in a weakly mixing Gaussian system. Its avoidance supremum is nevertheless one. Thus Bohr recurrence alone does not eliminate the Gaussian partially rigid case. No claim of novelty is made for this refinement of the Bohr-Hamming construction strategy.

A sparse list of Hamming increments tests every bounded-rank Bohr neighborhood Fix dimension r, tolerance δ and an integer m sufficiently large in terms of r,δ. Let $L = \lceil 2m/\delta \rceil$ and $t = \lceil \log_2((2mL+1)^r(m+1)) \rceil + 1$. For any $\beta_1, \ldots, \beta_m \in T^r$, there are nonnegative integers c_i with $\sum m$ and support of size $\leq t$ such that

$$|| \sum_i (1 - c_i)\beta_i ||_{T^r} < \delta.$$

To prove this choose centered real representatives and quantize $L\beta_i$ to nearest integer vectors w_i. The target $\sum_i (w_i, 1)$ initially has nonnegative integer representation coefficients all one. If a representation has support larger than t, two distinct subsets of its support have the same sum of the augmented vectors $(w_i, 1)$: there are at most $(2mL+1)^r(m+1)$ possible sums. Subtract their difference repeatedly until some positive coefficient becomes zero; both sign classes of this difference are nonempty because cardinalities agree. All coefficients stay nonnegative and their sum stays m. Repeat until support $\leq t$. The real quantization error in $\Sigma(1 - c_i)\beta_i$ is bounded by $m/L < \delta$. The universal list of vectors $v_i = 1 - c_i$ with these constraints has at most $(m+1)^{O(t)}$ members, all with at least $m - t$ entries equal to one and $\Sigma|v_i| \leq 2m$.

Building a single continuous spectral measure For stages $j \to \infty$ take e.g. $r_j = j$, $\delta_j = 1/(j+10)$. Choose m_j sufficiently large, a multiple of m_{j-1}, with $m_j > t_j$, where t_j has the preceding value. In particular ensure $m_j/(t_j^2 \log(m_j + 1)) \to \infty$ as rapidly as desired. At stage j choose m_j distinct spectral centers $\theta_1, \ldots, \theta_{m_j}$ rationally independent with 1, distributed equally into the small disjoint intervals from the previous stage. Kronecker's theorem lets us choose positive superincreasing n_i such that

$$||n_i\theta_k - 1_{i=k}/2|| < m_j^{-2}$$

simultaneously in k; require each $n_i > 2m_j \sum_{l<i} n_l$. Put $F_j = \{|\sum_i v_i n_i| : v \text{ is on the universal list}\}$. Every entry is nonzero by superincreasingness and $t_j < m_j$. For an arbitrary $\beta \in T^{r_j}$, apply the first lemma to $\beta_i = n_i\beta$ to obtain some $d \in F_j$ satisfying $||d\beta|| < \delta_j$. Hence $D = \bigcup F_j$ is Bohr recurrent (the same is true after removing finitely many stages).

For each $d \in F_j$, all but t_j centers have $||d\theta_k - 1/2|| \le 2/m_j$ (the sign of $\Sigma v_i n_i$ does not matter at the antipode). Shrink the disjoint child intervals around the centers until these inequalities persist with bound $3/m_j$ for every d in the finite list. Make diameters tend to zero. Assign equal mass $1/m_j$ to each interval, refining equally at later stages. The limit σ_0 is a nonatomic probability since $m_j \to \infty$. Use its symmetrization $\sigma = (\sigma_0 + reflection(\sigma_0))/2$ for a real Gaussian system. For each $d \in F_j$ this gives

$$\int |\exp(2\pi i d\theta) + 1|^2 d\sigma(\theta) \le C(t_j + 1)/m_j.$$

The weakly mixing Gaussian action is the one generated by the unitary multiplication by $\exp(2\pi i\theta)$ on the real symmetric subspace of $L^2(\sigma)$. Its weak mixing follows from nonatomic first-chaos spectrum (all positive tensor-chaos spectral types are nonatomic as well). With the constant unit vector f in its first chaos, write $X(f)$ for the associated standard real Gaussian coordinate. For $d \in F_j$, the residual $X((U^d + I)f)$ is a centered Gaussian with the above variance.

Exact pair avoidance and the threshold For a fixed $u > 0$, the probability of any residual exceeding u in absolute value on F_j is at most

$$2(m_j + 1)^{O(t_j)} \exp(-cu^2 m_j/(t_j + 1)).$$

Choose m_j recursively so that, for every fixed $u > 0$, this bound is summable in j with tails tending to zero (for instance enforce the bound $< 2^{-j-10}$ at $u = 1/(j + 2)$). Also arrange that the sum for $u = 1$ is less than $\Pr(N(0, 1) > 1)/2$, by further enlarging m_j. The event

$$A_{J,u} = \{X(f) > u \quad \text{and} \quad |X((U^d + I)f)| \le u \quad \text{for every} \quad d \in \bigcup_{j \ge J} F_j\}$$

avoids all pairs at these tail differences: if both $X(f)$ and its d-translate exceed u, their sum exceeds 2u, contradicting the displayed residual condition at the starting point. The probability of $A_{1,1}$ is positive, giving the claimed pair-avoider for the whole D. For each fixed u, $\liminf_{J\to\infty} \mu(A_{J,u}) \ge \Pr(N(0, 1) > u)$. First letting J grow and then u decrease to zero yields tail pair witnesses arbitrarily close to density one half.

The near-extremal $h = 2$ proposition in torsion.md [D20] now gives $b(D) = 1$: for each target $\eta < 1$, choose r with $1 - 2^{-r} > \eta$, choose J,u so the tail pair witness is sufficiently close to $1/2$ to force a 2^r-avoider denser than η, and impose the finitely many earlier differences using a sparse periodic sieve. This rules out these Gaussian examples as counterexamples to Question 6.1.

Strengthening to chromatic pair recurrence One can add to every stage F_j all absolute differences between the integer vertices $\sum_{i\in S} n_i$ for disjoint subsets S,S' of $[m_j]$ of size $k = \lfloor (m_j - j)/2 \rfloor$. Put $s = m_j - 2k$, so $j \le s \le j + 1$. By Lovász, *Kneser's conjecture, chromatic number, and homotopy*, these vertices contain a copy of the Kneser graph of chromatic number $s + 2 > j$. Every j-coloring of the integers therefore yields a monochromatic

pair with one of these differences. This asserts chromatic recurrence for pairs, not multiple topological recurrence.

The Gaussian estimate remains summable despite the much larger number of signed patterns. For a difference vector with ± 1 entries on the two disjoint supports and zero entries precisely on the remaining set J, $|J| = s$, the spectral multiplier $\exp(2\pi i d\theta) + 1$ is within $O(1/m_j)$ pointwise on the entire spectral support of $2 \cdot 1_{I_J}$, where I_J is the union of the stage intervals having indices in J and their reflected partners. This uses the same original m_j^{-2} approximation for each n_i and shrinking the stage intervals also for the finite added list. The leading residuals thus comprise at most $\binom{m_j}{s}$ vectors, each of squared norm $4s/m_j$. There are at most 3^{m_j} different remainder vectors, each of squared norm $O(m_j^{-2})$. By the Gaussian tail bound and splitting at $u/2$, the probability of any added residual exceeding u in absolute value is bounded by

$$2m_j^s \exp(-cu^2 m_j/s) + 2 \cdot 3^{m_j} \exp(-cu^2 m_j^2).$$

The choices of m_j can therefore achieve exactly the same summability targets for the enlarged stage sets. All previous positive-pair and density-one conclusions go through for this chromatically recurrent D. The spectral construction is weakly mixing, and its nonatomic subspace is partially rigid; its approaching the parity isometry on the first-chaos vector is essential to this argument.

D08. A nil-Bohr-dense set with a weakly mixing Gaussian pair obstruction

Scope and status. Supporting proof and dependency material.

There is $D \subset \mathbb{N}$ dense in the integer topology generated by nil-Bohr neighborhoods of every finite order (every tail is dense). It is not a set of measurable pair recurrence, even for a weakly mixing Gaussian system, and $b(D) = 1$. It can simultaneously be chromatically recurrent for pairs. In conjunction with measurable-nil-returns.md [D10], this gives pair nonrecurrence in a weakly mixing system despite multiple polynomial measurable recurrence on every pronilsystem. Compare Griesmer, *Separating Bohr denseness from measurable recurrence* for the ordinary Bohr case and Green, Propositions 3.2–3.3 for the joint nilorbit and compact-parameter argument. This does not decide whether either of the original Alweiss/Green sets is pair recurrent. No priority claim is made.

Joint nil hitting by Hamming shells. Fix $1, \theta_1, \dots, \theta_m$ linearly independent over $\mathbb{Q}$. Let B be the condition that all but t of the $d\theta_i$ lie within ϵ of $1/2$. For a polynomial test $p : \mathbb{Z} \to G$, $p(0) = I$, on a connected simply connected presentation G/Γ, and an integer target b, if $t \geq \dim G$ then arbitrarily late $d \in B$ have $p(d-b)\Gamma$ in any specified identity neighborhood. Indeed for the polynomial tuple $(n\theta, p(n)\Gamma)$, **a subsequence** $n = qk$ is well distributed in a connected subnilmanifold H/Λ through identity by Leibman's orbit theorem, or Green Proposition 3.2. Its projection to $\mathbb{T}^m$ is onto. Hence $\dim H \geq m$, and the kernel of projection of $\mathrm{Lie}(H)$ to $\mathfrak{g}$ contains an at least $m - \dim G$-dimensional subspace of $\mathbb{R}^m \times \{0\}$. Choose independent coordinate projections on this subspace. Some point of the kernel has on each selected coordinate the desired real representative of $1/2 - b\theta_i$. Exponentiate this vector to obtain a point of the subnilmanifold with group coordinate identity. Approach it along $n = qk$ and set $d = b + n$.

For stage j, test all products of at most j special generalized polynomials of degree at most j, shapes and coefficients bounded in absolute value by j, against all integer centers $|b| \leq j$. Equivalently use their AF polynomial upper-unitriangular block curves p with $p(0) = I$ and request canonical coordinates smaller than $1/j$ (take $j \geq 3$). This is an open condition through the local identity chart. Its satisfaction implies small wrapped values of the corresponding special polynomials, by AF Lemma 3.8. For each list of shapes, its coefficient domain is a compact box and the evaluations into the nilmanifold at fixed $d - b$ vary continuously. The previous hitting lemma with t_j the largest block-product dimension covers that box by hits at integers beyond any previously chosen cutoff. A finite subcover and the finitely many shape lists and centers give a finite set F_j of common Hamming-shell differences. By Huang–Shao–Ye's special generalized polynomial characterization, all stages together give the stated nil-Bohr density at every integer. There is no cardinality bound on F_j.

A uniform Gaussian estimate insensitive to that cardinality. At stage j choose $m = m_j$ mutually disjoint small child arcs centered on θ_i, refining the previous generation in equal numbers per parent, the centers rationally independent with 1. Take m_j a sufficiently large multiple of the preceding count, in particular $t_j\sqrt{\log(2m_j)/m_j} \to 0$ as fast as desired. Use shell tolerance $\epsilon_j = m_j^{-2}$. After the finite F_j has been chosen, shrink all child arcs around their centers so every phase $e(d\theta)$, $d \in F_j$, differs from its center value everywhere on a child arc by at most a prescribed η_j. Here η_j may be made arbitrarily small in terms of $|F_j|$. Make diameters tend to zero. The limiting equal-mass probability on these arcs is nonatomic. Use

its symmetrization σ, choosing original arcs in $(0, 1/2)$, for a real weakly mixing Gaussian action associated to multiplication by $e(\theta)$ on $L^2(\sigma)$, exactly as in gaussian-bohr-example.md [D07].

Let $X(f)$ be the isonormal Gaussian coordinate of the real (symmetric-below-conjugation) vector $f = 1$, and let U be the rotation. Replace the multiplier $e(d\theta) + 1$ by its step function on child arcs and their reflections. In each pair of arcs use the real and imaginary Hermitian indicators as real first-chaos vectors, normalized to unit variance. Their $2m$ Gaussian variables $Z_{i,\ell}$ are independent standard normals. Up to absolute constants, for **every** $d \in F_j$ the absolute Gaussian coordinate of its step function is bounded by

$$\frac{(\epsilon_j m + t_j)}{\sqrt{m}}\, C \max_{i,\ell} |Z_{i,\ell}|.$$

Indeed each good coefficient has modulus $O(\epsilon_j)$, the rest have modulus at most two, and every pair of arcs has mass $1/m$. No union over differences is needed here, even though their exceptional coordinate sets can change. The residual beyond this step function has variance at most η_j^2; a Gaussian tail and union over the finite F_j control its supremum. Thus one may ensure

$$\Pr\big(\max_{d \in F_j} |X((U^d + I)f)| > u_j\big) < \xi_j$$

for any preassigned $u_j \downarrow 0$ and sufficiently small summable $\xi_j > 0$, by first choosing m_j large enough and then η_j after F_j. For optional chromatic recurrence add the finite Kneser differences in gaussian-bohr-example.md [D07] before shrinking arcs; all but at most $j + 1$ center values are within $O(m_j^{-1})$ of the antipode, which suffices for the displayed bound. Increase t_j to at least $j + 1$.

For fixed $u > 0$, the event where $X(f) > u$ and $|X((U^d + I)f)| \le u$ for all tail differences avoids all those pairs, since two $X(f) > u$ marks would have sum greater than $2u$. As the tail cutoff increases its measure approaches at least $\Pr(Z > u)$. The whole D can be made pair-avoiding on a positive event in the Gaussian system already with $u = 1$, choosing $u_j \le 1$ and $\sum_j \xi_j < \Pr(Z > 1)$. Sending $u \downarrow 0$, the tail pair capacities approach $1/2$; torsion.md [D20] then gives long-run avoiders of densities arbitrarily near one, and a finite-prefix sieve preserves the limiting cap. Hence $b(D) = 1$.

D09. Restrictions on Gaussian weakly mixing witnesses

Scope and status. Supporting proof and dependency material.

These are partial statements about Gaussian systems; no reduction of an arbitrary weakly mixing witness to a Gaussian one is claimed. In particular, replacing an indicator by a Gaussian variable with its covariance need not preserve the original hard disjointness of events.

Pairs: mild mixing excludes even asymptotically vanishing returns Let T be a Gaussian automorphism, with real first-chaos orthogonal operator U, and let A, B have positive measure. If $d_j \to \infty$ and $\mu(A \cap T^{-d_j} B) \to 0$, some subsequence of d_j has $U^{d_j} \rightharpoonup P$, where $|P|$ has a nonzero subspace with eigenvalue 1. On that subspace the first-chaos operators converge strongly to the corresponding isometry. Taking differences of two divergent subsequences supplies a rigidity sequence on a nonzero real first-chaos vector. In particular a **mildly mixing Gaussian system is lightly mixing**: the liminf of every such correlation on positive events is strictly positive. A pair-avoiding D in this system is therefore finite.

Here is the proof, including the positivity point needed at the spectral endpoint. By WOT compactness on a separable Hilbert space pass to a subsequence with $U^{d_j} \rightharpoonup P$. All these operators belong to the abelian von Neumann algebra generated by U; P is a normal contraction. The full Gaussian Koopman powers converge weakly to the Gaussian Markov operator $\Gamma(P)$. Indeed on degree-k homogeneous Wiener chaos the action is the k-fold symmetric tensor power, and its matrix coefficients on pure tensors are finite sums of products of first-chaos coefficients. Finite sums of the chaoses are dense. For a contraction C, the Markov property of $\Gamma(C)$ can also be read directly from a jointly Gaussian pair with identity marginal covariances and cross-covariance C, using the conditional-expectation operator.

Write $P = VC$ with $C = |P|$, polar isometry extended to an orthogonal V on the zero part (equivalently multiply by a phase of modulus one in the spectral realization, compatibly with the real structure). Suppose C has no fixed vector. The semigroup $M_t = \Gamma(C^t)$, $t > 0$, consists of selfadjoint nonnegative Markov operators. In the spectral representation it equals $\Gamma(C)^t$ for positive t (use $0^t = 0$), hence its scalar matrix coefficients are real-analytic for $t > 0$. Since $C^t \to 0$ strongly by the spectral theorem, $M_t \to \Theta$, projection onto the constants, strongly on Gaussian space.

This semigroup improves positivity. For $g \geq 0$, $supp(g) \subset supp(M_t g)$ modulo null sets. Otherwise take $g_E = g 1_E$ on a positive-weight part E where $M_t g$ vanishes. Positivity gives $\langle M_t g_E, g_E \rangle = 0$, while positivity in the operator-order sense and the fixed constants give $\langle M_t g_E, g_E \rangle \geq (\int g_E)^2 > 0$. By the semigroup law this support containment shows that if $\langle f, M_1 g \rangle = 0$ for nonzero nonnegative f,g, then $\langle f, M_t g \rangle = 0$ for $0 < t \leq 1$. Analyticity extends the vanishing to every $t > 0$, contradicting its positive limit $(\int f)(\int g)$. Since $\Gamma(V)$ is a measure-preserving Gaussian automorphism, this also gives $\langle f, \Gamma(P) g \rangle > 0$ for any nonzero nonnegative f,g. The alleged zero-return limit contradicts this if C has no fixed vector.

Finally for any nonzero real f in the fixed subspace of C, $||Pf|| = ||f||$ and weak convergence gives $||U^{d_j} f - Pf|| \to 0$. Select pairs of indices with positive difference tending to infinity; their differences n_j satisfy $U^{n_j} f \to f$ in L^2. Such a nonconstant first-chaos vector is impossible for a mildly mixing transformation (equivalently, no nonconstant rigid functions).

Higher orders: a finite conditional phase law is necessary Let σ be a symmetric maximal spectral probability for the first-chaos U. From any $d_j \to \infty$ take a subsequence on which the circle-valued phases $\exp(2\pi i d_j \theta)$ have Young laws κ_θ over σ. Concretely, these are the conditional laws in a weak limit of the measures $(\theta, d_j\theta \bmod 1)_*\sigma$ on T^2. For every integer l the weak-* limit of $\exp(2\pi i l d_j\theta)$ in $L\infty(\sigma)$ is then $\phi_l(\theta) = \int z^l d\kappa_\theta(z)$ (identify the circle with unit complex numbers). Convergence against all L^1 tests follows from convergence against continuous functions and approximation. The symmetry can be respected throughout.

If $\#supp\kappa_\theta \geq h$ for σ-almost every θ, every subsequential Gaussian h-joining along $0, d_j, \ldots, (h-1)d_j$ gives positive probability to **every** product of h positive Gaussian events. In particular an h-term avoiding sequence must have $\#supp\kappa_\theta < h$ on a set of positive σ-measure for every such subsequential law.

In fact the joint Gaussian covariance on h copies of the first chaos is fiberwise the positive semidefinite matrix $\Sigma(\theta) = (\phi_{v-u}(\theta))_{0\leq u,v<h}$, times the identity on spectral multiplicity fibers. It is positive definite whenever κ has at least h support points, because a polynomial of degree $< h$ cannot vanish on all those points unless it is zero. Take $c(\theta) = \min(1/2, \lambda_{\min}(\Sigma(\theta))/2) > 0$ and regard c as a real-compatible scalar multiplication operator C. The Gaussian vector with covariance Σ may be represented by a jointly Gaussian residual with covariance $\Sigma - CI_h$, plus **independent** Gaussian noises on the h coordinates, each with covariance C. Normalize each residual coordinate by $(I-C)^{-1/2}$ to have standard Gaussian marginal law; this operator is bounded. Conditional on the joint residual the h events are independent, and each event's conditional probability is its indicator smoothed by $\Gamma((I-C)^{1/2})$ at the corresponding normalized residual. Since $c(\theta) > 0$ almost everywhere, this smoothing contraction has no fixed vector and the semigroup argument above makes the conditional probability strictly positive almost everywhere. Their product has strictly positive expectation. Passage from the first-chaos covariances to joint measures on all measurable rectangles uses finite-dimensional Gaussian cylinder approximation in each marginal (the error bound is uniform in the joining).

For $h = 2$ the support obstruction says precisely that κ_θ is a point mass on a positive spectral component, giving the partially rigid first-chaos result. At $h > 2$ a varying finite phase support does not in general provide an isometry in any of the individual pair marginals; the proof makes no such assertion.

D10. Positive Haar measure gives an actual nil-Bohr neighborhood of popular returns

Scope and status. Supporting proof and dependency material.

Theorem (internal deduction). Let $X = G/\Gamma$ be a compact finite-dimensional nilmanifold, T a nilrotation, μ Haar probability, and $A \subset X$ Haar-measurable with $\mu(A) > 0$. Given $h \geq 2$ and any Lebesgue density point x of A, there are a neighborhood U of $(x, \dots, x)$ in X^h and a number $c > 0$ such that

$$(x, T^d x, \dots, T^{(h-1)d} x) \in U$$

$$\Rightarrow \mu(A \cap T^{-d} A \cap \dots \cap T^{-(h-1)d} A) \geq c.$$

In particular the popular return set contains a fixed nil-Bohr-zero set of the same nilpotency step as X, without deleting a null set of differences. This is a statement about systems which are themselves finite-dimensional nilsystems. It does not assert a nil-Bohr neighborhood for all measurable systems. Combined with pointwise nil-recurrence at every order, it upgrades the 2-large conclusion in the Alweiss–Farhangi preprint to *measurable* multiple recurrence for Haar events of finite nilsystems.

Geometric compactness input. Given a sequence of connected translated subnilmanifolds S_i of a fixed compact connected nilmanifold and points $z_i \in S_i$ converging to z, their Haar measures have a subsequence converging to Haar measure on a connected subnilmanifold S containing z. Translate z to the identity coset by a fixed left translation. Using a local smooth section, translate each z_i there by elements tending to the identity. These small translations do not affect weak limits. In the resulting presentation all subnilmanifolds pass through the identity. Choose a minimum-dimensional connected subnilmanifold through the identity which contains infinitely many of them, and pass to those indices. In that smaller ambient nilmanifold no proper connected codimension-one subnilmanifold through the identity contains infinitely many of the sequence. Thus the Haar measures converge to its Haar measure by Leibman, Proposition 1.1 and its consequence immediately below. Translate back. This is also a special case of Mozes–Shah's homogeneous-measure compactness, but the nilmanifold proposition alone suffices.

Proof. For a sequence of simultaneous returns d_i with all the displayed positions tending to x, let C be the connected component of x. For all sufficiently large i, T^{jd_i} preserves C for each $0 \leq j < h$. Work with its normalized Haar probability μ_C. If G is connected the graph image of C under $y \mapsto (T^{jd_i} y)_j$ is a connected translated diagonal subnilmanifold of C^h. This remains true when G is disconnected: identify C after translation with $G^0/(G^0 \cap \Gamma')$ using the conjugated lattice, and write the power preserving the component as $a_j = u_j \gamma_j$ with $u_j \in G^0$, $\gamma_j \in \Gamma'$. The corresponding map on the connected presentation is translation by u_j following the lattice-preserving automorphism Ad_{γ_j}. The graph is the translate by the u_j of a closed connected subgroup image of the product of these automorphisms restricted to the diagonal. It is compact because the source quotient C is compact, and it has Haar measure the graph pushforward λ_i of μ_C. Every coordinate marginal is μ_C.

The graph contains $z_i = (x, T^{d_i} x, \dots, T^{(h-1)d_i} x)$. By the geometric input there is a subsequence converging to $Haar\lambda$ on a connected subnilmanifold S containing $z = (x, \dots, x)$.

Every coordinate marginal of λ is still μ_C. In particular every coordinate projection $S \to C$ is a smooth submersion. One way to check it is to use equivariance of the orbit map from the connected transitive subgroup of S: its differential rank is constant. Since the marginal has full support, the projection is onto, so its rank must equal $\dim C$.

The point z is a density point of $(A \cap C)^h$ relative to S. Indeed let $m = \dim S$, $n = \dim C$. In a smooth chart at z, the λ-mass of an S-ball of radius r is asymptotic up to constants to r^m. By the submersion theorem for coordinate j and smooth positivity of Haar volume, the mass in this ball whose jth projection is outside A is at most $O(r^{m-n})\mu_C(B_C(x, O(r))\backslash A) = o(r^m)$. The constants may depend on the *one limit* S, sufficient here. Sum over the h coordinates. In particular $\lambda(A^h) > 0$. Also $\lambda_i(A^h) \to \lambda(A^h)$: approximate the indicator of A in $L^1(\mu_C)$ by continuous $[0, 1]$-valued functions; their tensor powers approximate the events uniformly in all λ_i and λ by the common marginals. This does not require a Haar-null topological boundary of A.

If uniform c and U failed, choose d_i making the simultaneous return distance and the global intersection masses both less than $1/i$. The μ_C graph masses are bounded by the global intersection masses divided by $\mu(C) > 0$. The preceding positive limit contradicts this. Taking $T \times T^2 \times \ldots$ on a product nilmanifold gives the nil-Bohr formulation; orbit closures are minimal nilsystems, or use Host–Kra–Maass, Corollary 5.3.

The proof applies just as stated to graphs of arbitrary finitely many Haar-preserving affine automorphisms of a connected nilmanifold when their simultaneous images of x tend to x. The return-time description as a nil-Bohr set uses the nilrotation setting. No uniform c depending only on $\mu(A)$ is asserted. This argument also does not by itself replace the conditional-density inequality for arbitrary systems with nilfactors in nilfactor-density.md [D11].

Uniform constants and bounded-step inverse limits There is a stronger version if the nil-Bohr neighborhood can test finitely many starting points. Given $a > 0$, $h \geq 2$, $s \geq 1$, there is $c(a, h, s) > 0$ with the following property. In every nilsystem of step at most s, every event A of Haar measure $\geq a$ has its h-fold correlation at least $c(a, h, s)$ on an actual nil-Bohr neighborhood of order s. The same holds for every measurable inverse limit of such systems, with a possible decrease of c depending only on a,h,s. No dimension bound or density-topology regularity of A is needed.

Uniform homogeneous joining lemma. Let C be a compact connected nilmanifold of step $\leq s$ and $S \subset C^h$ a connected translated subnilmanifold containing the *entire* diagonal. With λ its Haar law, and A of Haar measure $\geq a$ in C, one has $\lambda(A^h) \geq c_0(a, h, s) > 0$. Work with the identity in C and its diagonal in S as basepoints, so S has a connected subgroup presentation L with a cocompact lattice. Choose $u = (u_0, \ldots, u_{h-1}) \in L$ so rotation by u on S is totally uniquely ergodic. For example choose its horizontal torus projection totally irrational and use the usual horizontal-character ergodicity criterion on a connected nilmanifold; when S is a point the assertion is immediate. For every $t \geq 1$ the h transformations on C given by u_j^t generate a group of nilpotency class $\leq s$. By Leibman's nilpotent multiple recurrence theorem there exist $N = N(a, h, s)$ and $\gamma = \gamma(a, h, s) > 0$ such that for some $1 \leq n \leq N$, the Haar measure of $\cap_j (u_j^{tn})^{-1} A$ exceeds γ. For uniformity, use the free nilpotent group on h generators of class s, and represent the event in the invariant binary shift. The compact set of shift measures of mean $\geq a$ is covered by recurrence-open sets for separate n; compactness and continuity give a uniform strictly positive maximum

over a finite set of n. The same argument also holds if the recurrence theorem includes A as an extra unshifted slot.

Sum these h-fold correlations over $n \le N$ and average over t. For any fixed n the average tends to $\lambda(A^h)$: integrate the orbit of $x^h \in S$ under the totally uniquely ergodic rotation u^n against $Haarx \in C$. This works for continuous test functions by unique ergodicity and then for measurable tensor powers by L^1 approximation, since each coordinate map preserves Haar; λ also has Haar marginals (each subgroup projection contains C and pushes its Haar probability to Haar). It follows that $N\lambda(A^h) \ge \gamma$. This proves the lemma.

Finite nilsystems. Suppose no neighborhood defined by simultaneous point returns has a given popular correlated lower bound. Choose increasingly fine returns for an increasing finite dense sample of points of X with correlations below that bound. In each component C, the graph measures of restricted Haar for those differences have a subsequence converging to connected homogeneous $Haar\lambda_C$, by the geometric compactness already proved. Its support contains $(x, \dots, x)$ for every x in the dense sample in C: for each such point, apply geometric compactness on a further subsequence, with the weak limit already fixed. The full diagonal lies in the support by closure. With finitely many components choose a single subsequence for all of them, and ensure the powers fix every component by testing a point in each component. Components in which A has conditional density $\ge a/2$ carry total Haar mass at least $a/2$. Apply the homogeneous joining lemma on these components and sum; approximate the measurable tensor by continuous tensors as before. Thus the limiting correlation is at least $(a/2)c_0(a/2, h, s)$. This contradicts a strictly smaller common bound. Taking the product system over the finite point sample makes the neighborhood a single order-s nil-Bohr neighborhood.

Inverse limits. On the inverse limit, approximate 1_A in L^1 by an indicator B on a finite stage with error less than both $a/2$ and $c(a/2, h, s)/(2h)$, using factor martingale convergence. Correlations change by at most h times this error uniformly in the difference. Apply the finite-stage theorem at density $a/2$. The resulting neighborhood works on the inverse limit and is still defined on a single finite product of finite-dimensional nilsystems of step $\le s$. This argument deliberately uses a fixed nilpotency step; the indicated uniform constants do not follow here if the inverse-limit steps grow without bound.

Step-independent bound by ordered words There is a second proof of the uniform homogeneous joining lemma with $c_0(a, h)$ independent even of the nilpotency class. It follows that for any inverse limit of finite-dimensional nilsystems, including one with unbounded steps, the popular return set of every positive Haar event contains a *finite-order* nil-Bohr neighborhood. The order of this neighborhood can depend on the stage needed to approximate that event. If $a \le \mu(A)$ the popular threshold depends only on a and h. In particular a set of pointwise nil-recurrence at every finite order provides multiple *measurable* recurrence, with a uniform positive lower bound along some of its differences, in every such inverse limit.

Use the G/Γ presentation of connected C with G connected and identity chosen, and let L be the connected subgroup in G^h presenting S through the identity in C^h. In this paragraph $\Delta G \subset L$ as groups: the full diagonal in C^h lies in S, so $\Delta G \subset L\Gamma^h$; at the identity, the tangent space of the embedded S is Lie(L) because Γ is discrete. Differentiating along ΔG gives $Lie(\Delta G) \subset Lie(L)$ and connectedness gives the group inclusion. Thus in a tuple product in G^h, both any element of L and the diagonal lift of any one of its coordinates belong to L.

Choose N by the density Hales–Jewett theorem for alphabet $\{0,\dots,h-1\}$ and density $a/2$. Let J be the finite number of combinatorial lines in this word space. Fix a uniquely ergodic translation u on S as above (total ergodicity is unnecessary). Independently sample $s_i = u^{n_i} = (s_{i,0},\dots,s_{i,h-1}) \in L$, where n_i is uniform on $[1, M_i]$; the M_i will be chosen in increasing index order. Also sample x Haar in C. Color a word w good when $s_{1,w_1}\cdots s_{N,w_N}x \in A$, with this multiplication order fixed. Every word has probability $\mu_C(A) \geq a$. With probability $\geq a/2$ the good words therefore have density at least $a/2$; on that event they contain a line. Hence the sum over lines of their all-good probabilities is at least $a/2$.

For any line let i be its rightmost wildcard. All factors to its right are the *same in all h slots*. Conditional on those factors their product transforms $Haarx$ to another Haar starting point and is independent of the prefix. The earlier factors form a tuple $P \in L$, since variable letters use an s_j and fixed letters use the diagonal of one $s_{j,b}$. The all-good probability for this line is therefore an average over finitely many possible prefix values P of

$$(1/M_i)\sum_{n\leq M_i}\int_C 1_{A^h}(Pu^n(x,\dots,x))d\mu_C(x).$$

For each fixed prefix this tends to $\lambda(A^h)$ as $M_i \to \infty$. For continuous tests this follows by unique ergodicity on S and Haar invariance under left multiplication by P. For measurable tensor powers approximate 1_A in $L^1(\mu_C)$ by continuous $[0,1]$-valued functions. All the finite averaging measures have marginals μ_C, since at fixed n and P each coordinate acts by a Haar-preserving translation on x, as does λ. Thus the approximation error is uniform in M_i. Choose M_i successively to make this difference $< a/(4J)$ for *every* earlier prefix possible at this i. The sizes of the later sampling ranges do not enter, since all suffixes were absorbed into $Haarx$. It follows that $J\lambda(A^h) \geq a/4$. This proves the step-independent assertion, without commuting the word factors.

Use this uniform version in the preceding graph compactness argument and sum over high-density components. The resulting popular threshold can be chosen as $(a/4)c_0(a/2,h)$ in finite nilsystems. For an arbitrary inverse limit approximate by a single finite-stage indicator B of mass $\geq a/2$ with symmetric difference smaller than $c(a/2,h)/(2h)$. The fixed finite-stage nil-Bohr neighborhood persists with at least half its uniform lower bound in the full system. This is exactly where independence from the step of the approximating stage is needed.

Polynomial parameters. The graph lemma applies word for word to $T^{p_0(d)},\dots,T^{p_{h-1}(d)}$ for integer polynomials p_j with $p_j(0) = 0$ $onep_jmaybezero$. In particular the finite list of point-nearness tests for these maps forces the same uniform positive correlation. Such simultaneous tests contain an actual nil-Bohr neighborhood: the product orbit of all tested points is polynomial of naive degree $\leq t$=max deg p_j; choose a continuous test of the distance to its time-zero value and apply Leibman's Proposition 2.3, which represents every polynomial basic nilsequence from an s-step space and naive degree $\leq t$ as an st-step basic nilsequence. A return near the identity point of the representing nilsequence therefore forces all required nearness tests. For fixed-step inverse limits the resulting neighborhood has order at most st; for general pronil inverse limits it still has some finite order. The method also permits products of distinct group powers with zero constant polynomial exponents in a finite nilsystem, provided the relevant nilrotation maps preserve the factor presentations in a stated inverse-limit version.

Dual characterization. For $s \geq 1$, a set R is pointwise recurrent for all minimal s-step nilsystems iff it meets the positive $(s + 1)$-term *measurable* AP return set of every positive event in every s-step nilsystem. The forward implication is the result above, with a popular lower bound and even with inverse limits allowed. Conversely measurable recurrence applies to every nonempty open set in a minimal nilsystem (it has positive Haar measure), hence gives s-fold topological recurrence and then pointwise recurrence by Host–Kra–Maass, Corollary 5.3, using Huang–Shao–Ye's nil-Bohr realization theorem. For such R, measurable AP recurrence at every length on the same class follows at once. This is an exact fixed-order equivalence, not a claim that an s-step set tests only s-step factors of an arbitrary higher-step nilsystem. Shalom's counterexample on pointwise decay of triple correlation errors illustrates why eliminating higher factors by pointwise decay would be a different assertion.

D11. A conditional-density inequality over connected nilfactors

Scope and status. Conditional result.

Dependency convention. Uses of this proposed conditional-density inequality elsewhere in the report are conditional on its nilmanifold localization and free-cover input. They are not deductions from the Host–Kra factor theorem alone.

This proposition is a partial result toward Question 6.1. It is derived from a polynomial-orbit argument. Theorem 1.9 of the Alweiss–Farhangi preprint controls special generalized polynomials of the *difference* and does not, on its own, imply the uniform flattening of orbit segments used here.

Let $b(D)$ be the supremum of the upper natural densities of integer sets avoiding every h-term AP with difference in D, for at least one finite h. If A in an ergodic probability-preserving system (Z, T, μ) avoids h-term D-progressions in measure, and has a finite-dimensional nilrotation factor (Y, S, m), $Y = G/\Gamma$ with G connected simply connected nilpotent, Γ cocompact discrete and m Haar, then

$$\|\mathbb{E}(1_A \mid Y)\|_\infty \le b(D). \tag{1}$$

An increasing inverse limit of such factors also satisfies (1), by martingale convergence. Every positive-measure avoiding event measurable with respect to such an inverse limit therefore implies $b(D) = 1$.

A small geometric transversal On any fixed connected nilmanifold $X = F/\Lambda$, there is a positive c_X such that every positive-dimensional translated connected rational subnilmanifold has diameter $\ge c_X$. Indeed $xH\Lambda/\Lambda$ contains a noncontractible loop in X: lift the path from x to $x\lambda$ for a nonidentity $\lambda \in H \cap \Lambda$. Compactness gives a uniform ball size below which balls lie in contractible coordinate charts. If the translated subgroup is also conjugated, apply this to its presentation as a translated rational subgroup.

For any $\eta > 0$, choose a finite partition of X into cells of diameter $< c_X/2$ with Haar-null boundary (e.g. successive differences of finitely many small metric balls of continuity radii). Let Σ be the union of their boundaries and let V be an open w-neighborhood of Σ having Haar mass $< \eta$, with $w > 0$. Every positive-dimensional connected translated rational subnilmanifold W meets Σ: otherwise it is contained in one cell, since its portions in the cell interiors form a disjoint relatively open cover. Thus there is a point in each such W whose entire ambient $w/2$-ball is in V. There is no restriction on the dimension, rationality complexity, or translation of W here.

Uniform flattening with rational factor For fixed filtered X as above, V forces the following standard consequence of Green–Tao factorization: there are R, N_0 such that *every* polynomial $p : Z \to F$ whose orbit segment $p(n)\Lambda$, $1 \le n \le N$, misses V, $N \ge N_0$, factors as $p(n) = \epsilon(n)\gamma(n)$ with ϵ (R, N)-smooth and every $\gamma(n)\Lambda$ an R-rational coset. Here R is independent of all coefficients, starting point and N.

More explicitly, use Green–Tao Theorem 1.19 on X, obtaining $\epsilon g'\gamma$ of parameter M between a prescribed sufficiently large M_0 and a fixed R. The rational orbit has period $q \le M$; on the progression $n \equiv 1 \bmod q$ in the initial segment $n \le N/M^3$, its

reduced representative is a single $M^{Ox(1)}$-rational u. If the equidistributed factor subgroup H is positive-dimensional, freezing ϵ at 1 makes displacement $\leq M^{-2}$ by right invariance. The comparison target is $\epsilon(1)u(u^{-1}Hu)\Lambda/\Lambda$. It contains a center of a $w/2$-ball in V. One must transfer total equidistribution from $H/(H \cap \Lambda)$ to the conjugated group with its target lattice; reducing γ modulo Λ and merely asserting the lattices are identical would be incorrect. The requisite transfer and ambient density estimates are given in this workspace's `outputs/2LargeSetsNilrecurrence.tex`, Lemmas `lem:rational-conjugation` and `lem:ambient-density`. The argument behind these auxiliary lemmas is as follows. The rational height of u makes $u^{-1}(H \cap \Lambda)u$ and $(u^{-1}Hu) \cap \Lambda$ commensurable with polynomial-in-M index. A bounded horizontal-character obstruction to equidistribution on the target, supplied by Green–Tao Theorem 2.9 on a long subprogression, pulls back after multiplication by that index to an obstruction with polynomial-in-M complexity on the source. Its small C^∞-norm gives an arc obstruction on a further subprogression of proportional length $M^{-O(1)}$. This contradicts total M^{-A}-equidistribution for a prescribed sufficiently large A. Charts in a bounded-complexity Mal'cev basis give Lipschitz bounds for the translate map and lower Haar bounds for pullbacks of ambient balls, with polynomial losses in M. Thus on the chosen progression of length $\gg N/M^4$, some conjugated orbit sample comes within $w/4$ of the indicated center. Choose M_0 large so freezing costs $< w/4$; choose the equidistribution exponent large in terms of the fixed manifold, the required length and w, before applying the factorization theorem. The resulting sample belongs to V, a contradiction. When H is trivial, g' is trivial and the claimed factorization follows. This argument invokes quantitative Leibman and factorization, not Theorem 1.9 of the posted paper.

A fixed free cover allows the rational factors to be cleared after R is known Write s for the nilpotency step of G. Choose a rational Lie algebra basis of `Lie(G)` for its lattice rational structure. Take F to be the simply connected free step-s nilpotent Lie group on that many generators. Map the generators to the chosen basis and integrate to a surjective homomorphism $f : F \to G$. Choose a homogeneous rational basis of `Lie(F)`, e.g. a Hall basis, and let U be its integer span. The graded dilations Δ_L multiply the weight-j subspace by L^j. After choosing a sufficiently divisible integer B, $\Lambda = \exp(BU)$ is a lattice and $f(\Lambda) \subset \Gamma$. Indeed BCH has rational coefficients and finite degree, so BU is closed under BCH for suitable B; its image is discrete and cocompact, by induction on the grading through central quotients. In Mal'cev coordinates for G/Γ, $\psi_G(\exp f_*(u))$ is a rational-polynomial function of rational linear coordinates of u with zero constant, so choosing B sufficiently divisible makes all such coordinates integral for $u \in BU$. Thus $X = F/\Lambda$ is fixed independently of the later dilation. Haar measure projects to Haar measure through every surjective homogeneous-space factor below.

Construct V, R, N_0 on X as above. Take an integer L divisible by every $1 \leq q \leq R$. The dilation Δ_L maps Λ into itself. If a coset $v\Lambda$ has an element w with $w^q \in \Lambda$, $q \leq R$, then $q \log(w) \in BU$. As all graded weights are positive, $\Delta_L \log(w)$ has each homogeneous coordinate in BU, hence $\Delta_L w \in \Lambda$. This also kills $\Delta_L v$ modulo Λ. Define

$$\pi : X \to Y, v\Lambda \mapsto f(\Delta_L v)\Gamma.$$

Lift the translation $Sy = ay$ to $S'x = bx$ by choosing $b \in F$ such that $f(\Delta_L b) = a$. For *every* $v \in F$, $d \in Z$, $n \mapsto b^{dn}v$ is a polynomial sequence in the lower-central-series filtration

on F. Therefore if its length-N orbit in X misses V, factorization followed by π leaves just $f\Delta_L\epsilon(n)\Gamma$. The ambient F-sequence ϵ stays in the R-ball and has successive distance $\leq R/N$. On this compact ball the fixed smooth homomorphism $f\Delta_L$ is Lipschitz with some fixed constant (equivalently use right-invariance and bound near the identity). Consequently all successive *projected* positions have distance $\leq C/N$, uniformly in starting point and difference. The rational part has been removed without passing to a subprogression; passing to stride q would generally change the allowed difference. This proof accounts for the shear from arbitrary starting points.

Conditional-density increment Write $\psi : Z \to Y$ for the original factor and $g = E(1_A|Y)$. Fix $p < ess\sup g$. By ergodicity finitely many translates of $\{g > p\}$ cover all but η of Y. Select on that portion one suitable translation index $j(y)$ so $g(S^{j(y)}y) > p$. By inner regularity there are pairwise disjoint compact cores K_j of its selection pieces covering all but 2η. Distinct nonempty cores have positive mutual separation ρ (if only one core is used, take any $\rho > 0$). In the system with relatively independent joining of (Z, T) and (X, S') over Y, accept a position (z, x) precisely when $x \notin V$, $\psi(z) = \pi(x) \in K_j$ and $T^j z \in A$ for the indicated j. Its probability is at least $p(1 - 2\eta) - \eta$. The joining exists by disintegration of the two invariant measures over the common Haar factor. Its ergodicity is unnecessary.

Choose $N \geq h, N_0$ so $C/N < \rho$. If every sample along an N-term orbit at any $d \in D$ is accepted, its lifted nilmanifold coordinates miss V. Consecutive projected samples are within C/N, hence within the same compact core. The selector index is fixed throughout and the first h translated original points belong to A, an event of probability zero. This is simultaneous for countably many starts and differences. Stationary correspondence via the individual ergodic theorem produces an integer avoiding set of upper density at least the mean (or arbitrarily close). Let $\eta \downarrow 0$ and $p \uparrow ess\sup g$ to obtain (1).

Disconnected-group and skew presentations Here is a reduction that also covers ergodic rotations on G/Γ for nilpotent Lie groups G not assumed connected. Two details matter: a nilmanifold may have finitely many connected components, and an ergodic skew action on a connected component need not itself be a rotation in a connected nilpotent Lie group.

Choose a component meeting $\{g > p\}$ in positive Haar measure. Translate its presentation to assume that it is the identity component $Y_0 = G^0/(G^0 \cap \Gamma)$. The ergodic rotation permutes the q components transitively, and S^q is ergodic for Haar measure conditioned on Y_0 (an invariant set there, and its q iterates, would produce an S-invariant set). Write $a^q = u\gamma$ with $u \in G^0$ and $\gamma \in \Gamma$. On Y_0, S^q is the affine map $h\Gamma_0 \mapsto u\phi(h)\Gamma_0$, where ϕ is conjugation by γ and $\Gamma_0 = G^0 \cap \Gamma$. Pass to the simply connected cover H of G^0, with Λ_0 the preimage of Γ_0, and lift u, ϕ. The unique lift of the automorphism preserves Λ_0.

On `Lie(H)`, ϕ_* is unipotent. Indeed $\phi_* - I$ lowers the filtration supplied by the Lie algebras of the closures of the successive lower central subgroups of the original nilpotent G. So does $D_0 = \log(\phi_*)$, a finite polynomial with no constant term in $\phi_* - I$. All real $\exp(tD_0)$ are Lie algebra automorphisms: the bracket identity is polynomial in t and holds for every integer t. They integrate to automorphisms ϕ_t of simply connected H. The connected simply connected semidirect product $\tilde{G} = H \rtimes_{\phi_t} R$ is nilpotent because the bracket with the extra direction, D_0, lowers the same central filtration as brackets with `Lie(H)`. The group $\tilde{\Gamma} = \Lambda_0 \rtimes_\phi Z$ is a cocompact lattice (fundamental representatives use $[0, 1)$ in the time coordinate and a

compact set for H/Λ_0, allowing the bounded family of automorphisms ϕ_t, $0 \le t \le 1$). The fiber over time 0 mod 1 in $\tilde{G}/\tilde{\Gamma}$ is H/Λ_0, and the translation by $(u, 1)$ on it equals the stated affine transformation. Thus Y_0 embeds as an invariant slice, with its own stationary Haar measure m_0, which generally is singular for Haar measure on the larger nilmanifold.

To accommodate the singular measure and the order in which the covering map is selected, use a fixed finite polynomial-orbit partition into r colors, **each** of which can serve as V, with a common structural factor bound R. Its construction on the free covering nilmanifold is exactly the random-ball argument in `outputs/2LargeSetsNilrecurrence.tex`, Lemma `lem:model-family`, Theorem `thm:log-coloring` and Theorem `thm:structural`, with $r \ge 2$ equiprobable colors replacing two: each candidate ball involves at most a fixed J independently colored cells, hence has probability at least r^{-J} of being any prescribed color. At scale $1/K$ there are $\gg K$ disjoint trials on each translated subnilmanifold and polynomially many elements of a Hausdorff net for the family of complexity $K^{O(1)}$. Taking large K works simultaneously for every color by a union bound. All constants may depend on fixed r, but **not** on the least-mass color chosen later. The factorization and conjugation proof then contradicts avoidance of even one color if its subgroup H is positive-dimensional. This is a statement about all polynomial orbits of the fixed nilmanifold, not an application of coefficient control for special polynomials.

Apply the free-cover argument to $\tilde{G}/\tilde{\Gamma}$ and obtain $\pi = f\Delta_L$ onto the larger nilmanifold. Lift the invariant slice law m_0 to an invariant probability ν in the free cover: take any lift under the continuous surjection (a measurable section exists), then average its translates and take a weak limit by compactness, preserving the projected law. Choose the color V of least ν-mass, at most $1/r$. The previously determined structural constant is independent of this choice. Successive projected positions on surviving orbit segments differ by $O(1/N)$ for the ambient $\tilde{G}/\tilde{\Gamma}$ metric. The compact selector cores of Y_0 have positive separation in that same metric.

On the original system condition μ on the preimage of the selected component, use T^q, and put $D/q = \{e : qe \in D\}$. The given event avoids h-term D/q-progressions in this system, and its conditional mean on the normalized Y_0 is $g|_{Y_0}$. Ergodicity of S^q on this factor suffices for selecting finitely many translates; no ergodicity of the ambient T^q system is required. Apply the compact-core/relative-joining proof. The resulting bound is $p \le b(D/q) = b(D)$, using dilation and congruence localization as proved in torsion.md [D20] ($b(D) = b(D \cap qN) = b(D/q)$). This extends (1) to conventional inverse limits of ergodic nilsystems of any order.

Scope Even (1) for all pronilfactors does not imply a positive witness has conditional probability exceeding $b(D)$: the function may be constant and well below one in every nilfactor. No implication making all weakly mixing exceptional directions recurrent has been supplied.

D12. Elementary quantitative amplification of a pair obstruction

Scope and status. Supporting proof and dependency material.

For any integer step set D, let $a = a_2(D)$, where a_h is the maximal stationary measure of an h-term avoiding event. Then

$$a_5(D) \geq 1 - (1-a)^2,$$

$$a_{2h}(D) \geq a_h(D) + (1 - a_h(D))a^h \quad \text{for every} \quad h \geq 2.$$

Here are direct proofs. Given an invertible system T with a pair-avoiding event A of mass a, its integer c-root is the uniform height-c suspension. One step moves up a level, applying T at the wrap. Use the mark A on every level. On a line with step $d \in D$, that root mark cannot occupy two indices c slots apart, since the cth power of the root advances the original system by T.

For the first bound take independent roots of orders 1 and 2 and accept their union. If five successive positions were accepted, color each by an accepting root. Writing 1 for the first color, the resulting five-bit word has no adjacent 11 and has no two zeros at distance two. The only four-bit word satisfying both conditions is 1001: if its first bit is 0 then its third must be 1, forcing its second and fourth to be 0, a contradiction. If its first bit is 1, the second is 0 and the fourth 1, forcing the third to be 0. Two copies of 1001 cannot overlap on three bits, so five positions are impossible.

For the second bound take an h-avoider B of optimal mean and a product of independent pair-avoiding roots of orders 1,. . .,h, independent also of B. Let C be the simultaneous mark of all h roots. Distinct C-marks on a given d-line must be separated by at least $h + 1$ time slots: use the root corresponding to the separation if it is at most h. If a segment of length 2h lay in $B \cup C$, both its first and last h-block would require C. Any consecutive C-marks are too far apart to avoid an intervening h-run in B. This proves the bound by independence.

Both arguments work without multiplicative closure of D. The estimates give concrete finite-length gains, but iterating the second one only yields a product with factors $1 - a^{2^j h}$. For $0 < a \leq 1/2$ that product is strictly positive, so the estimates do not establish $b(D) = 1$.

D13. Dense palettes for simultaneous proper path rows

Scope and status. Supporting proof and dependency material.

Given positive integer strides $q_1,\ldots,q_r$ and $B \subset [c]^r$, the schedule problem asks for b:$Z \to B$ with $b_i(t+q_i) \neq b_i(t)$ for all i,t. A threshold depending only on $c \geq 4$, uniformly over r and the strides, has not been proved here. Nor has a counterexample been found at $c \geq 4$. The reductions below distinguish this finite-alphabet problem from the further integer-difference realization needed for Question 6.1.

Algebraic reduction to downward closed palettes of degree vectors Work over the reals with the c symbols represented by distinct numbers. Fix lex order on the r variables, and let $S(B) \subset \{0,\ldots,c-1\}^r$ be the standard monomial exponent set of the ideal of B. It is downward closed and $|S(B)| = |B|$. The c-coordinate bounds follow from the univariate vanishing polynomials.

On any finite interval V, draw the edges of each q_i-path and orient them arbitrarily. In each row these orientations are acyclic, since each row graph is a union of finite paths. Suppose the vector of row indegrees at every t lies in $S(B)$. Form

$$P = \prod_{i,t,t+q_i \in V} (x_{i,t} - x_{i,t+q_i}).$$

Use a monomial order first comparing total exponents in each row lexicographically (row totals over all times), then refining by a variable order that induces a topological ordering of the oriented edges within each row. At each time t the restricted monomial order is precisely the prescribed lex order. The leading monomial of P has the chosen indegrees as its exponents. The initial ideal of B^V is the sum of the sitewise initial ideals in their disjoint variable sets: the union of the respective Gröbner bases is a Gröbner basis (relatively prime leading monomials across parts), and the quotient dimension is $|B|^{|V|}$. Thus the leading monomial is standard for B^V. Hence P does not vanish on B^V, giving a proper B-schedule on V. Compactness yields a two-sided schedule if the orientation test succeeds on every finite interval.

More generally one may prescribe capacities $v(t) \in S(B)$ at each t, and orient separately each row below those capacities. This row orientation exists iff the sum of capacities on any consecutive subinterval of its q_i-path is at least the number of internal edges (one less than its number of vertices). This follows from Hall's theorem matching each edge to a slot at one of its endpoints. Dense $S(B)$ admits a probability distribution on capacity vectors in $S(B) \cap \{0,1,2\}^r$ with strict average slack in every coordinate when $|B|/c^r$ is sufficiently near one and $c \geq 4$. Indeed for U uniform on $[0, c-1]^r$ the coordinate averages of $\min(U_i, 2)$ are $2 - 3/c \geq 5/4$; conditioning on $U \in S$ loses at most $2 \cdot P(U \notin S)$ in each numerator, and truncating coordinates preserves membership in S. These fractional inequalities do not in themselves meet all the path interval constraints.

Partial pinning certificates A certificate is a partially assigned word whose full cylinder lies in B. Suppose a distribution on certificates has pin marginals $p_{i,a}$ and

$$\sum_{i,a} p_{i,a}^2 < 1/16.$$

Then a schedule exists. Independently choose a certificate at every integer. The undesired event that the certificates at t and $t + q_i$ pin the same symbol in row i has probability $\gamma_i = \sum_a p_{i,a}^2$. Two events depend only if they share an endpoint. In the asymmetric local lemma take $x_i = 2\gamma_i$; the sum of x over all events meeting the two endpoints is at most $8\Sigma\gamma_i < 1/2$. Thus every finite interval admits a choice with no pinned conflict. In each row fill unpinned sites between the pins: paths with at least three colors extend any compatible pinned assignment, as follows either by choosing interior colors in order while reserving the terminal requirement or by noting any two prescribed endpoint colors can be connected by a walk of every length at least two in K_c. Compactness gives the full schedule. Sparse unions of independent pinning certificates fall within this test. Dense parity-type palettes can have only full certificates, so it is not general.

Two dimension-free estimates Independently for each row and each residue class modulo q_i, run the stationary uniform proper-color Markov chain with transition $K(a, b) = 1/(c - 1)$ for $b \neq a$, zero otherwise. This gives a stationary full-palette schedule. For a centered function f on $[c]^r$ with Hoeffding/ANOVA components f_S one has for $n \neq 0$

$$\mathrm{Cov}(f(b_0), f(b_n)) =$$

$$\sum_{S \neq \varnothing} ||f_S||^2 \prod_{i \in S} [1_{q_i | n} (-1/(c-1))^{|n|/q_i}].$$

Indeed coordinates along different residue chains are independent, and the transition acts by $-1/(c-1)$ on centered one-site functions. Summing in n and bounding each product via any one $i \in S$ yields $\sum_{n \neq 0} |\mathrm{Cov}| \leq 2\,\mathrm{Var}(f)/(c-2)$, independent of r.

The same Markov construction has a shift-register transition P on all last q_i symbols of each row. The next current-slot tuple is the product of the one-row kernels. Finite-state hypercontractivity for K (there is a $p < 2$ with $||K||_{p \to 2} \leq 1$ since all nonconstant singular values are < 1, and tensorization) gives

$$||PM_H||_{2 \to 2} \leq \mu(H)^{1/p - 1/2}$$

when M_H is multiplication by the forbidden event on the newly emitted tuple, with counting-product stationary measure. Condition on the untouched registers and apply the product-kernel estimate in the emitted coordinates. A nonzero spectral radius for $P(1 - M_H)$ would give a schedule. This small-norm perturbation by itself does not give a uniform spectral argument, since the full-register spectral gap can shrink to zero with the strides.

The simple product-pair palette surrogate fails. For $c = 4$, $H = \prod_i ([4] \text{ minus } \{a_i\})$ has density $(3/4)^r$, yet contains a full product of pairs for every choice of a partition in each coordinate into two complementary pairs. This does not obstruct the true schedule; avoiding this H only requires that at least one row take its distinguished color at every site, a cooperative path-coloring problem.

Exact answer in two rows For $c \geq 3$ the largest unschedulable two-row palette, maximizing also over the strides, has size $2c-1$. Regard B as a bipartite graph on the two copies of $[c]$. If it has a matching of size three, those three words are pairwise disjoint in both coordinates. The union graph of two strides is 3-colorable: on every finite time interval color in increasing order, avoiding at most two earlier neighbors, then use compactness. Mapping its three colors to the matching gives a schedule. Otherwise König's theorem gives a vertex cover of size at most two. Two vertices on opposite sides cover at most $2c-1$ edges. For a cover on the same side, if both fibers have size at least two, alternate the two corresponding colors along the first stride, then properly fill the other row from its resulting lists of size at least two (finite paths are greedily 2-choosable). If some fiber has size at most one the palette has at most $c+1$ words.

The sharp example uses B=$\{(x,y)$: $x = a$ or $y = b\}$, $q_1 = 1$ and $q_2 = 2$. An infinite schedule would induce a shift-invariant measure by taking empirical translates. At each time its first distinguished symbol and its second distinguished symbol cover the whole space. Both events have measure at most $1/2$ by their respective independence along the prescribed stride. Hence they partition the space modulo null sets and both have measure $1/2$. Each then alternates exactly along its stride. The first alternating event is invariant under shift by two, while its complement, the second event, alternates under shift by two, a contradiction.

Every central histogram gives a schedule Fix $r \geq 6$, $c \geq 4$ and a histogram $n_1, \ldots, n_c$ with $\Sigma n_a = r$ and $\max n_a \leq \lfloor r/3 \rfloor$. Then for any choices of strides there is a proper schedule in which this histogram occurs at every time. Let m=$\lfloor r/3 \rfloor$. List the r symbols grouped by symbol value and dispatch them in round-robin order into m blocks. Each symbol occurs at most m times, so each block receives distinct symbols, and each block has size three or four (for remainder two use $m \geq 2$). Partition the rows into blocks of the resulting sizes and give each block its prescribed distinct palette.

On any finite time interval, for a block of three rows the graph consisting of their stride edges is a disjoint union of paths. Strong 3-colorability of disjoint paths (the cycle-plus-triangles theorem of Fleischner and Stiebitz, packaged for paths at https://www.lars-daniel.se/papers/3or4characterisation.pdf §5) colors it properly with the three prescribed symbols so that the three sites at each time have all three symbols. For a block of four, use the independent-transversal theorem for maximum degree two and parts of size four to choose one independent representative of each time block and assign it the fourth symbol. Removing these leaves a disjoint union of paths with triples at every time, to which the three-color theorem applies. Combining row blocks and taking compactness proves the claim. Consequently every coordinate-permutation-invariant palette of any fixed positive density has a schedule once r is large depending on that density and c: a uniformly random histogram has maximum frequency tending to $1/c < 1/3$. The arbitrary nonsymmetric palette remains open in this investigation.

Arbitrary holes for a uniformly Bohr-separated family of strides Suppose $c \geq 4$ and an angle α satisfies $||q_i\alpha|| \geq \kappa > 0$ for every row. Then there exists $\epsilon(c,\kappa) > 0$ independent of r and of the strides such that density(B)$> 1 - \epsilon(c,\kappa)$ implies a schedule. More generally, it suffices that D fixed angles cover the rows in this sense, with ϵ depending also on D. The independence from r is important; merely taking a growing list of Bohr tests does not prove an unrestricted answer.

Use the algebraic reduction and write H for the upward complement of $S(B)$, viewed on $\{0,\dots,c-1\}^r$. Work first with a single angle. An edge $t \to t+q_i$ corresponds to a circle step $u_i = q_i\alpha$ at the base point $\theta = t\alpha$. Independently for each row build a random orientation profile on the circle, stationary under translating θ and with $O(1/\kappa)$ breakpoints. Take a partition into equal arcs of length $< \kappa/2$, shifted by a uniform random phase. Toss a fair coin for the construction method. In the first method, label arcs with independent continuous ranks, and orient an edge towards its endpoint of larger rank. At each θ its two neighbors $\theta \pm u_i$ occupy arcs different from the central arc. If they occupy distinct arcs, the probability of indegree zero or two is 1/3 each. If they occupy the same arc these probabilities are 1/2 each. In the second method use independent fair arc signs to orient the edge beginning at θ: use $\theta \to \theta + u_i$ for positive sign and the reverse otherwise. The indegree at θ then involves the signs in the two distinct arcs at θ and $\theta - u_i$, so the probabilities of zero and two are 1/4 each. The fair mixture consequently has $P(0) = P(2)$ between 7/24 and 3/8 at every θ, uniformly over the row. The indegree profile has at most a constant times the number of arcs as breakpoints, with a half-open convention at discontinuities. The construction refers to the two distinct time edges on the row path even when their two endpoints project to the same circle point.

For $0 \le s \le 1/48$ let D_s be the law on $\{0,1,2\}$ with tails $D_s([1,2]) = 17/24 + s$ and $D_s(\{2\}) = 3/8 + s$. This stochastically dominates each actual degree law, and is itself stochastically dominated by the uniform distribution on $\{0,\dots,c-1\}$. Independently in each row replace its degrees by larger virtual capacities of law D_s using a monotone coupling. To do so sample for each degree level a fixed random value from the conditional output kernel and use that same value every time that level occurs; this creates no additional breakpoints. At any fixed θ the virtual capacities of different rows are independent.

For an upward family H and x sampled in all coordinates except i from D_s, put $\Delta_i(s) = P(1_H(\text{x with } i=2) - 1_H(\text{x with } i=0) = 1)$. Then

$$d/ds P_{D_s^r}(H) = \sum_i \Delta_i(s),$$

since increasing s moves mass directly from 0 to 2. Every probability on the left is $\le \epsilon$ by stochastic domination, so some s has $\sum_i \Delta_i(s) \le 48\epsilon$. At each breakpoint of row i, the other rows are independent D_s coordinates independently of the random breakpoint location, by phase stationarity and row independence. Thus the expected number of changes of H around the whole circle is at most $O(\epsilon/\kappa)$. Simultaneous breakpoints from distinct rows have probability zero. Also P(H at $\theta = 0$)$\le \epsilon$. Taking ϵ small gives a realization in which no virtual capacity vector anywhere belongs to H. All actual indegrees are smaller, so their restrictions to every finite time interval lie in the staircase. The algebraic argument finishes.

For D fixed separating angles, partition the rows into D groups, one per angle. Eliminate one group at a time as follows. Keep uniform c-state degree variables in the uneliminated coordinates. For the current group use the circle construction and smooth only its coordinate laws through D_s. The same derivative estimate, averaged over the independent uniform coordinates outside the group, shows that one choice of the group profiles makes the probability of the upper family

$$H_{\text{new}} = \{remainingvectors x : \quad \text{some} \quad groupphase \theta has (x, virtual_group(\theta)) \in H\}$$

at most $C(\kappa)\epsilon$. Indeed the probability of H at $\theta = 0$ plus the number of its boundary crossings bounds existential occurrence pointwise, and the preceding derivative bound applies after integrating in x. The new family is upward. Iterate over the D groups, paying a factor $C(\kappa)$ each time. If $C(\kappa)^D\epsilon < 1$, the final empty-coordinate event is false, so H is avoided for every combination of the groups' clock phases. In particular it is avoided at the simultaneous integer phases $t\alpha_j$. This proves the stated extension.

Arbitrary strides at inverse-logarithmic deficit There is an absolute $\eta > 0$ such that for every $c \geq 4$ and $r \geq 1$, every palette of density $> 1 - \eta/(1 + \log r)$ supports a schedule for all r strides. In fact, the previous one-clock argument only needs a bound on the sum of influences weighted by the reciprocal angular displacements. Choose first s with $W = \sum_i \Delta_i(s) \leq 48\epsilon$. For an angle α with $||q_i\alpha|| > 0$ use an individual mesh finer than $||q_i\alpha||/2$ in row i. The stationary degree/bias bounds are unchanged, but the expected number of H transitions is now at most

$$C \sum_i \Delta_i(s)/||q_i\alpha||$$

for an absolute C. For $W > 0$ write $p_i = \Delta_i/W$. For uniform α the probability of some $||q_i\alpha|| < p_i/8$ with $p_i > 0$ is at most $1/4$ by the union bound. Moreover

$$E \sum_{p_i>0} p_i 1_{||q_i\alpha|| \geq p_i/8}/||q_i\alpha||$$

$$\leq 2 \sum_i p_i \log(4/p_i).$$

Thus one can select α outside all the excluded arcs, with all displacements nonzero and with the displayed weighted sum at most a fixed multiple of $W(1 + \Sigma p_i \log(1/p_i))$. The $W = 0$ case poses no boundary contribution at any nonresonant angle. This proves the stated estimate by the bound $\log r$ for entropy. In particular a much smaller effective entropy of the influence weights may certify a schedule even without a fixed Bohr gap. This theorem still loses a logarithm in the number of rows and does not supply the sought uniform threshold.

For coordinate-transitive upward staircases this logarithmic dependence can be removed via sharp thresholds, another result distinct from the exact histogram construction for the palette itself. Write $\phi(s) = P_{D_s^r}(H)$. On the upper half of the s-interval, the product-space influence theorem of Bourgain, Kahn, Kalai, Katznelson and Linial (https://research.ibm.com/publications/the-influence-of-variables-in-product-spaces) implies a coordinate influence at least $c\phi(s)(1 - \phi(s)) \log(r)/r$. The influence is precisely the chance the indicator is nonconstant upon changing that coordinate. Transitivity makes all influences equal, so $\phi' s{=}\Sigma\Delta_i(s) \geq c'\phi(s) \log r$ when $\phi \leq \epsilon < 1/2$. Integrating downward over a fixed interval shows $\phi(s_0) \leq \epsilon r^{-\kappa}$ at an interior point s_0, for an absolute $\kappa > 0$. Integrating the derivative over a further lower interval then gives a point with $W \leq C\epsilon r^{-\kappa}$, so $W(1 + \log r)$ is uniformly $O(\epsilon)$. More generally the same reasoning applies to product-space KKL/Talagrand's total-influence bound whenever the maximum influence is at most $r^{-\kappa'}$ throughout the upper interval, with constants depending on κ'. No transitivity is assumed in the logarithmic result itself.

Power improvement of the logarithmic bound For a probability U on totally ordered coordinate values denote by $U^{\lfloor\gamma\rfloor}$ the distribution whose kth upper tail is the kth upper tail of U raised to γ. For every upward event H and $\gamma \geq 1$,

$$P_{(U^{\lfloor\gamma\rfloor})^r}(H) \leq P_{U^r}(H)^\gamma.$$

For rational $\gamma = a/b$, take a independent vectors with coordinate law $U^{\lfloor 1/b\rfloor}$. The coordinatewise minimum over all of them has law $U^{\lfloor\gamma\rfloor}$. If this minimum belongs to H then for every b-subset J the corresponding minimum also belongs to H; each J has individual event probability $P_{U^r}(H)$. Finner's fractional Hölder inequality for these events of independent vectors (each sample index has total exponent weight one) gives the claimed power a/b. Extend to real γ by continuity.

The D_s upper tails are uniformly strictly below both relevant U tails, even for $c = 4$. Thus there is $\gamma > 1$ such that every D_s is dominated by $U^{\lfloor\gamma\rfloor}$. It follows that $W \leq 48\epsilon^\gamma$ at some s and the previous all-stride theorem holds already for deficits $\epsilon \leq C_c(1 + \log r)^{-1/\gamma}$.

More precisely one can attain any γ below the solution of $(1 - 1/c)^\gamma + (1 - 2/c)^\gamma = 1$. Choose a target $p < 1/2$ with $1 - p$ and p strictly below these two upper-tail powers, respectively; a small common upward perturbation s retains the inequalities. It suffices to provide a symmetric indegree law $P(0) = P(2) = p$ stochastically dominated by these perturbed laws, with the same entropy-controlled interface cost. For $u = q_i\alpha$ choose m so that $||mu - 1/2|| < \eta$ for fixed η with $1/2 - \eta > p$. One can take $m \leq C_\eta / \min_{1\leq h\leq H_\eta} ||hu||$ outside the null rational exceptions. Indeed choose a fixed trigonometric polynomial of positive mean that is nonpositive outside the stated target interval, by smooth approximation from below. If the interval is missed through M, one of its finitely many nonconstant Fourier geometric series has normalized modulus bounded below, implying $||hu|| \leq C_\eta/M$. Give the outgoing edge at circle position θ the sign indicated by which half-circle contains $m\theta + \phi_i$, with ϕ_i uniform. Its indegree-zero and indegree-two probabilities each equal $||mu||$. Mix with a constant-direction orientation so the two probabilities each equal the prescribed p exactly. The mesh has $O(m)$ endpoints. The entropy truncation applies to the finitely many hq_i as well, with constants depending only on η, and the area/bias interval now depends only on c and γ. This improves the inverse-logarithmic exponent, without making the deficit independent of r.

Two clocks and a second-pivotal sufficient condition At a state x sampled from product D_s, let $N(x)$ count the coordinates i for which $x \in H$ but replacing x_i by zero takes the state outside H. There is an absolute constant C such that a schedule exists whenever, for some $0 \leq s \leq 1/48$,

$$\epsilon + C\sum_i \Delta_i(s) + CE[N(x)(N(x) - 1)] < 1.$$

Here the bound is schematic up to an absolute C, and only asserts a sufficient condition. General small-measure monotone events are not known to meet this score.

Pick independent Haar angles α,β. Let $u_q = \max(||q\alpha||, ||q\beta||)$ and assign the row to the maximizing clock, using mesh of order $1/u_q$. We have $E[1/u_q] = O(1)$. Also for any positive a,b,

$$E[1_{a,bassigneddifferentclocks}/(u_a u_b)] = O(1).$$

For the second assertion reduce by gcd to coprime A,B. At a single uniform angle γ the measure of $||A\gamma|| \le x$, $||B\gamma|| \le y$ is $O(xy + \min(x/A, y/B))$: count the allowable integer determinants of the two approximating numerators, each with one solution modulo the coprime denominators, and multiply by the shorter interval length. For the two independent angles square this bound. If Bx+$Ay < 1$ then, at either angle, the approximating fractions coincide (at zero modulo one). Their two displacement vectors are proportional, so there is no contribution with different maximizing clocks. The dyadic reciprocal expectation is therefore bounded by a constant plus a sum of $\min(x/A^2y, y/B^2x)$ over dyadic x,y with Bx+$Ay \ge 1$. Fixing x, or fixing y, the sum over the other variable is $O(1/(AB))$ by a two-sided geometric series. Since $x \ge 1/(2B)$ or $y \ge 1/(2A)$, this part costs $O((\log(2A) + \log(2B))/(AB))$.

On the resulting clock torus, test first the two coordinate axes. The expected crossings there, summed over rows, are $O(\Sigma\Delta_i)$, as in the one-circle proof. If neither axis contains H but a bad rectangle exists, take the first horizontal strip containing any bad rectangle, then the first bad rectangle left-to-right in that strip. Crossing into it from either direction separately turns a nonbad state into a bad state. Two different row boundaries are involved. Conditional on a given pair of positions, all other arbitrary row phases are independent with the D_s law. For the pair i,j let Γ_{ij} be the sum, over a_i,$a_j \in \{1, 2\}$, of the probability in the other variables that H holds at those levels but fails when either of these coordinates is individually reduced to zero. Any two-axis activation contributes to this sum (one may lower to a minimal successful pair of levels in the activated rectangle), because the two predecessor values were nonbad and H is monotone. The probabilities of each nonzero D_s level are bounded below, so $\sum_{i \ne j} \Gamma_{ij} \le CE[N(N-1)]$. The number of candidate intersections for a pair of rows is $O(1/(u_{q_i} u_{q_j}))$; the harmonic estimate just proved applies because only different clocks intersect. Taking expectations over the two angles proves the criterion. This is different from controlling a fixed finite-rank Bohr gap; arbitrarily small pairwise angular separations are included.

Simultaneous permutation-of-degrees blocks and shifted holes An arbitrary collection of row triples admits simultaneous degree capacities which are a permutation of $(0, 1, 2)$ on each triple at each time. On a finite time interval consider the three disjoint rows of stride paths of one triple. Join their path components into one Hamiltonian cycle by dummy edges. Use intervals $[1, N]$ with $N > 3 \max_i q_i$. Traverse every path in increasing time, joining its terminal time $> N - \max q_i$ to the initial time $\le \max q_i$ of the next path. All dummy edges thus avoid the time triangles and the genuine stride edges, and the resulting Hamilton cycle is simple. The cycle-plus-triangles graph-polynomial coefficient of the monomial with degree two at every vertex is nonzero by Fleischner and Stiebitz, Discrete Mathematics **101** (1992), pp. 39–48 (the coefficient is nonzero modulo four). For a direct source discussing the graph polynomial identification see Tarsi and for the coefficient result see Petrov. In the Vandermonde product of each time triangle, every monomial uses a permutation of exponents $(0, 1, 2)$. Therefore some term of the cycle polynomial has complementary degrees which at every time also form such a permutation. The term comes from an orientation of its edges. Removing dummy edges leaves indegrees bounded by these permuted capacities. This proof works independently for arbitrarily many triples; leftover ungrouped rows can be oriented forward, with indegrees at most one. Consequently a downward S containing every vector

of this block type certifies a schedule.

Here is one dimension-free consequence, valid for all strides. Suppose that the upward H is closed under shifting any nonzero rank at position j to an earlier vacant position $i < j$ (leave zero at j). There is a universal $\kappa > 0$ such that $\mu(H) < \kappa$ implies schedulability for $c \geq 4$. It also suffices if this shifting property holds within each of a bounded number K of coordinate chains, with threshold κ^K.

Partition the ordered coordinates into successive triples, with at most two leftovers. Fix any candidate bad profile x which is a permutation of 0,1,2 on each triple and has ones on the leftovers. We show $\mu(H) \geq \kappa$ if $x \in H$. For a uniform raw vector Y, mark its coordinates independently with two disjoint labels H2,L1 such that each label has probability 3/8. An H2 coordinate is required to have $Y_i \geq 2$, and an L1 coordinate $Y_i \geq 1$. These labels can be sampled by using part of the mass at ranks ≥ 2 for H2, then extracting mass 3/8 from the remaining positive ranks: $P(Y_i \geq 2) \geq 1/2$ and $P(Y_i \geq 1) \geq 3/4$. On the first K_0 complete triple positions, with K_0 a large multiple of six, require that every $Y_i \geq 2$. Match all the nonzero tokens of x there in place, leaving $K_0/3$ vacant slots. Put half of these in reserve for each token type. For each type, the number of its tokens among the next n indices is at most $\lceil n/3 \rceil$, because this tail starts at a triple boundary. The subsequent corresponding pool-label counts are binomial $(n, 3/8)$. With probability at least 1/2, both these counts plus the reserve $K_0/6$ dominate $\lceil n/3 \rceil$ at all n: Hoeffding's inequality bounds the failure probabilities by $C \exp(-c(K_0 + n))$ after the trivial range of small n. Impose $Y_i \geq 1$ separately on the at most two leftover indices. Shorter instances may simply have $Y_i \geq 2$ on all complete triples. All these probabilities are bounded below by a constant independent of r and c.

Match each remaining token of each type injectively to a pool slot at or before its original position, possible by the prefix conditions. Process tokens in increasing original position; when a token is to move left, the target is vacant since any original token occupying it has already been assigned and the matching is injective. All moves preserve H by hypothesis. The relocated profile lies below Y, hence $Y \in H$. This proves the claim for every candidate x. If $\mu(H) < \kappa$ the full block template belongs to S, so the orientation argument applies. Coordinate independence gives the product probability κ^K for K chains. General upward families need not have this shifting property.

D14. Exact cyclic models for allowed residue classes at fixed length

Scope and status. Supporting proof and dependency material.

Let $p \geq 2$ (possibly composite), $S \subset Z/p$, $D = \{d > 0 : d \bmod p \in S\}$, and $K \geq 2$. Denote by a_r the largest density of a subset of $Z/p^r Z$ containing no whole K-term progression with step having image in S. Repetitions around a short cyclic orbit are allowed. Then

$$b_K^Z(D) = \sup_{r \geq 1} a_r = \lim_{r \to \infty} a_r.$$

If $0 \in S$ both sides are zero (finite Szemerédi for the left; the zero step on the right). For every other S the easy lower bound and monotonicity follow by pulling back a cyclic pattern. The hard direction does not state a bound on the required r, even for p fixed. This is a new internal deduction, not a claim of literature priority.

Approximate a nilrotation's AP statistics on cycles Suppose Y is an ergodic nilsystem with rotation a and continuous function f:$Y \to [0,1]$. Realize its orbit as a closed subnilmanifold in a compact nilmanifold G/Γ with G connected simply connected, using one time-one rotation $a = \exp A$. For clarity, Y has a finite tower of m_0 components over the connected nilsystem C with return map a^{m_0}. A connected nilsystem embeds in a nilflow on a connected simply connected nilmanifold; see Richter, connected nilflow embedding proof. If that time-one return is b, embed the tower at heights $r = 0,\ldots,m_0 - 1$ by $(x, r) \mapsto b^{r/m_0} x, r$ and use the time-one element $b^{1/m_0}, 1$ in the product with $R/m_0 Z$. Take the basepoint to be the identity coset by a fixed left translation and conjugation of a. Extend f continuously with range $[0,1]$ to the compact ambient manifold.

On $(G/\Gamma)^K$ write $\Delta(g) = (g, \ldots, g)$, $W(g) = (g^j)_{0 \leq j < K}$. For one g these two elements commute; W need not be a group homomorphism. Their powers acting on the identity give precisely $(a^{n+jd}\Gamma)_j$. By the nilrotation orbit-closure and equidistribution theorem, choose a positive multiple m of p such that the orbit for indices n,d both in mZ is dense uniquely ergodically on a connected subnilmanifold through identity $Y_0 = L\Gamma^K/\Gamma^K$ with L connected rational. Indeed the full Z^2-orbit closure is a finite union of connected pieces. A finite-index subaction fixing the pieces has an orbit closure on each piece equal to it: in this distal action the finitely many translates of that orbit closure are disjoint or equal and a connected component cannot split among them. The resulting dense niltranslation action is uniquely ergodic. The data for all pairs of residues n,$d \bmod m$ are the corresponding translates of Haar measure on Y_0.

The stabilizer of Y_0 is $\mathrm{L} \cdot N_{G^K}(L) \cap \Gamma^K$. Every stabilizer element sends the identity coset into Y_0, so is $l\gamma$; comparing tangent spaces at identity after conjugation by γ gives $Ad(\gamma)$Lie L=Lie L. Conversely every indicated $l\gamma$ preserves Y_0. Each coset $L\gamma$ of this stabilizer is a rational affine subspace in exponential coordinates:

$$\log(L\gamma) = LieL + \log \gamma.$$

In fact $\mathrm{ad}(\log\gamma)$ preserves Lie L (take the finite logarithm of the unipotent $Ad\gamma$ preserving it), so Lie L is an ideal in the algebra generated by it and $\log\gamma$. The equality follows by BCH or passage to that one-generator quotient. Both Lie L and $\log\gamma$ are rational for the lattice.

Now $\Delta(a)^m$ and $W(a)^m$ stabilize Y_0. Requiring that $\Delta(\exp A')^m$ and $W(\exp A')^m$ lie respectively in the SAME two stabilizer cosets amounts to rational affine-linear equations on A' in Lie G: their logs are $m\Delta(A')$ and $m(jA')_{j<K}$. The solution space contains A and has a dense set of rational vectors. Choose A_i rational there tending to A. Then $a_i = \exp A_i$ has some positive power in Γ by rational Malcev coordinates. Take periods N_i of its orbit at identity which are multiples of m and tend to infinity, padding actual periods if needed.

The joint orbit at multiples of m for $\Delta(a_i)$,$W(a_i)$ stays inside Y_0. Its uniform cyclic (n,d)-averaged law is invariant under both mth powers. Each subsequential weak limit is therefore invariant under $\Delta(a)^m$ and $W(a)^m$ on Y_0, hence is Haar there. For each pair of residues $\mathrm{mod}\, m$ the cyclic law is the translated one, converging to the same translate as for a. Thus with $f_i(n) = f(a_i^n\Gamma)$ on Z/N_i one has both

$$E_n f_i(n) \to \int_Y f,$$

$$E_{n,d:d\equiv v(p)} \prod_j f_i(n+jd)$$

$$\to \lim_M E_{d\le M:d\equiv v(p)} \int_Y \prod_j f(a^{jd}x)dx \quad \text{for every} \quad v \bmod p.$$

For the first display use the first marginal; its true Cesàro law is Haar on the original orbit Y by ergodicity. For the right side of the second, take a long independent mean in n first by unique ergodicity of Y at each fixed d. The iterated mean for each fixed pair of n,d residue classes $\bmod m$ is still the corresponding translated Haar measure on Y_0, by invariance under both mth powers. This verifies that the statistical limit just calculated agrees with the measure-theoretic class average. Individual rational approximant orbits need not lie in the original embedded Y; this is why we extended f to the ambient space.

Characteristic reduction and exact elimination Take an ergodic stationary K-avoider A of mass a for D, or an ergodic component attaining any lower target mass. Choose a global Host–Kra characteristic factor of finite order for the $K-AP$ residue averages of T. For each v shift the test slots by jv and take the ordinary average under T^p. Orthogonality to a fixed sufficiently high Host–Kra factor of T suffices: the seminorm of order s under T^p vanishes if that of order $s+1$ under T does, by the multi-average of squared correlations restricted to finite-index sublattices. This argument also handles nonergodicity of T^p. Applying the usual generalized von Neumann inequality and telescoping therefore preserves every one of the p class averages after conditioning 1_A on the chosen global factor. Write its conditional function as g, $0 \le g \le 1$, $\int g = a$. Its class averages for $v \in S$ vanish.

The factor is an inverse limit of ergodic nilsystems. Approximate g in L^1, to arbitrary δ, by continuous f ranging in $[0,1]$ on a finite nilfactor; the change in any K-correlation is at most $K\delta$. The preceding cyclic approximation gives functions f_i on one finite cyclic period

N divisible by p, of mean at least $a-2\delta$ and with each forbidden class K-average arbitrarily close to at most $K\delta$. Bernoulli-sample a mask with independent site probabilities $f_i(n)$. The expected normalized number of forbidden cyclic APs is bounded by $O_{K,p}(\delta)+o(1)$. Steps of order $< K$ can repeat sites, but at most $O_K(1)$ such steps exist in a cyclic group and their cost is $o(1)$ as N grows. Chebyshev for the density and Markov for the total violations give deterministic masks of density $\geq a-o(1)$ with $o(N^2)$ violations as $\delta \to 0$ and N increases.

Use the Král–Serra–Vena arithmetic removal lemma. For each pair $(a_0, v) \in Z/p \times S$ take X_j to be the mask points in class $a_0 + jv$. For $K \geq 3$ the AP system $x_j - 2x_{j+1} + x_{j+2} = 0$, $0 \leq j < K-2$, has determinantal divisor 1, from the triangular minor on the first $K-2$ columns. Its solutions with $x_j \in X_j$ are exactly the indicated labelled cyclic APs. There are $o(N^2)$, hence deleting $o(N)$ points from each of these finitely many sets destroys all the labelled solutions at once. For $K = 2$ use instead the trivial product bound $|X_0||X_1| = o(N^2)$, so the smaller set has size $o(N)$. This produces a truly K-avoiding mask at density $a-o(1)$.

Write $N = PQ$, where all prime factors of P divide p and Q is coprime to p. Since p|N, actually p|P. Restrict to the densest Q-coordinate slice under the CRT $Z/N \cong Z/P \times Z/Q$. Every step of Z/P has a lift with zero Q-coordinate, and its residue modulo p is the same, so the slice remains an exact avoider. P divides p^r for some r; pull back to that clock. This proves the upper bound.

Scope for Question 6.1 This criterion exactly identifies the torsion-free cap of a periodic class family with its finite prime-power lifts for each fixed K. An upper bound on just the base group Z/p is insufficient. For example, odd steps have a 3/4-density 4-avoider on $Z/4$ although the largest avoiding subset on $Z/2$ is a singleton. A potential counterexample using periods $p \to \infty$ needs uniform control through all lifts at the high test lengths. The theorem concerns periodic direction sets on a single integer axis. It does not equate arbitrary simultaneous H-actions with products of unrelated cyclic models. Separately, any profinitely open subset of Z meeting mZ for every m has $b = 0$: write the multicorrelation as a uniform limit of basic nilsequences plus an upper-Cesàro-null error. Its mean on mZ is bounded below uniformly in m by finitary Szemerédi. Approximate the structured part within one quarter of this bound by a basic nilsequence; let m be a multiple of all component periods of its orbit. Choose $d_0 \in D \cap mZ$ and q divisible by m with $d_0 + qN \subset D$. The basic nilsequence has the same mean on this class as on mZ by total equidistribution on the connected component. The error has mean zero on each fixed class. Hence D cannot have a positive avoiding witness at any K. The same proof works if D contains arbitrarily long segments of that one class, using uniform versions of the orbit and null-sequence averages.

There is nevertheless an exact existence reduction incorporating Cartesian incompatibility. An intermediate integer example with some finite-color monochromatic obstruction exists if and only if there are c,h_0,$\delta < 1$ such that for arbitrarily large K there is a finite cyclic group Z/p and a direction set S with these two properties: i the group has a c-coloring with no monochromatic h_0-run at steps S; (ii) every cyclic p^r-lift avoiding K-runs at all lifts of S has density $\leq \delta$ for every $r \geq 1$. The constants c,h_0 are not prescribed beforehand in this equivalence.

Given an integer example D, choose δ larger than $b(D)$ and use stationary compactness to find, for each K, a finite $F \subset D$ whose K-cap is $< \delta$. The allowed c-colorings on F form a nonempty one-dimensional finite-type shift, so one is periodic, of period p. The full periodic

lift of the residue images $S = F \bmod p$ is still c-colorable and enlarges F, so its K-cap is $< \delta$. The exact cyclic theorem supplies (ii), without changing c. Conversely for a pair of cyclic conditions, the corresponding periodic lift on Z has cap $\leq \delta$ and is c-colorable. By stationary compactness there are finite truncations with caps $< \delta'$ for any fixed $\delta < \delta' < 1$, retaining the low coloring. Put them on separate axes of $\oplus Z$: summing normalized coordinate colors modulo c gives the common h_0-obstruction with a cell of density at least $1/c$, and any K-avoider is bounded by the corresponding K-row. Integer transfer preserves b. The periods must grow: for any bounded p the periodic step family's b is 1 when $0 \notin S$, and only finitely many S occur. This equivalence is for finite-color obstructions; arbitrary single positive h-avoiders are not automatically of that form.

If the starting point is instead a joint low-cap obstruction in the universal free H family, for every K compactness first supplies finite rows I with a low joint K-cap. Encode them into $F = \bigcup_{i \in I} \lambda_i L F_i$ with the two-offset periodic c^2-coloring of universal-color-family.md [D21]. Any integer K-avoider for F pulls back to the joint model, so $F's$ cap is also low. The remainder of the cyclic reduction is then the same with c^2 colors. Thus the failure of distinct rows to have a common high-density realization would itself lead to a single finite cyclic template; no assumption that any of those original rows separately has low K-cap is needed.

D15. Exact periodic models via a connected root cover

Scope and status. Supporting proof and dependency material.

This is a shorter independent route to the periodic direction-set theorem in periodic-residue-lifts.md [D14]. It explains which periods are needed for an individual stationary system. Fix $h \geq 2$ and a periodic set D of positive differences, of period p.

Transfer to finite nilfactors without approximate pattern removal Take an ergodic h-avoider A for D. Use a fixed finite-order Host–Kra characteristic factor for the finitely many residue-class multiple averages (shift the slots and average under T^p; taking one additional seminorm order under T is sufficient). Its conditional function $g = E(1_A|\mathrm{Z})$ has zero class averages for D. For any $t > 0$, the set $\{g > t\}$ has zero class averages there too by positivity. This implies its correlations are zero individually on all of D. Indeed if B is any event and $B_0 = \bigcap_{j<h} T^{-jd}B$ has positive measure for $d \in D$, then the correlation of B at $d + pn$ dominates that of B_0 at time n for T^p. Ordinary multiple recurrence gives a positive lower Cesàro mean. Taking t↓0 shows $B = \{g > 0\}$ is an exact avoider and $\mu(B) \geq \mu(A)$.

Approximate this support on a finite ergodic nilfactor Y by the event E=$\{\mathrm{E}(1_B|\mathrm{Y})> 1 - 1/(2h)\}$. If an h-progression lies in E on a positive set of base starts, conditional fiber probabilities and a union bound show that all h translates of B have positive simultaneous intersection. Thus E exactly avoids D, and its mass can be made arbitrarily close to $\mu(B)$ by martingale convergence.

A finite cover of the return map has a root Suppose this nilsystem has P connected components, cyclically permuted by its translation. It is a height-P tower over the connected component return map b. A connected nilsystem can be written on a connected simply connected nilpotent Lie group quotient N/Γ with an affine map $b(y) = uA(y)$. Indeed the translating element is a product of an identity-component element and a lattice element, with A induced by conjugation by the latter. Lift to the universal cover of the identity component when necessary. The automorphism A is unipotent, preserves Γ, and is rational in Mal'cev coordinates. The real powers A^t act by automorphisms: the unipotent binomial formula is polynomial in t and the Lie-bracket identities hold at every integer. The resulting semidirect suspension $N \rtimes_{A^t} \mathbb{R}$ is simply connected nilpotent (use the lower central filtration of the original nilgroup, on which A acts trivially on successive quotients). Take the unique P-th root of $(u, 1)$ in this group. Its affine action s on N has linear part $A^{1/P}$. The finite-index sublattice

$$\Gamma' = \bigcap_{j=0}^{P-1} A^{j/P}\Gamma$$

is invariant under $A^{1/P}$, since A preserves Γ and its rational powers give commensurable lattices. Thus s acts on $Y' = N/\Gamma'$ and its P-th power lifts b. The action of s is totally ergodic. For each positive v, every orbit closure of $(s^P)^v$ projects onto N/Γ, since b^v is minimal on that connected ergodic nilsystem. The affine nilsystem orbit-closure theorem gives a finite union of closed subnilmanifolds. A finite cover projection forces some member to

have full dimension, and such a member equals Y' by connectedness. This proves minimality and unique ergodicity for these powers, and consequently for all powers of s.

The original system is a factor of $(Y' \times Z/P, (s, +1))$. With $\pi{:}Y' \to N/\Gamma$ the factor map, send (y, r) to level r of the original tower with return-coordinate $\pi(s^{-r}y)$, using $0 \le r < P$. This commutes with the transformations including at the wrap because $\pi s^P = b\pi$.

Read a finite residue mask Lift E to this product and denote its sections by E_r. Fix r and an allowed integer d. If $B' = \cap_{j<h} E_{r+jd \bmod P}$ has positive Haar measure, then multiple recurrence for the totally ergodic s along the residue d mod $\mathrm{lcm}(p, P)$ gives an allowed n in this class with $\cap_{j<h} s^{-jn} B'$ of positive measure. This yields a forbidden progression for E. Here the residue-class equality of connected totally ergodic averages follows, for example, from Moreira–Richter on arithmetic progressions (also author preprint); the separate erratum on another theorem of that paper is not needed.

Consequently for almost every $y \in Y'$ the cyclic subset $\{r : y \in E_r\}$ avoids every h-progression with step d mod P, $d \in D$. Its mean density is $\mu(E)$; some slice attains at least that value. All periods P so obtained occur in the rational spectrum of the original system since the component cycle is a factor. One can pass to $\mathrm{lcm}(P, p)$, take a densest slice on the factor coprime to p, and then pull back to Z/p^r as in the other proof. The reverse direction comes from periodic masks themselves.

In particular if the original system's rational eigenvalues have bounded exponent, exact avoidance at each fixed length admits finite masks of corresponding bounded periods and essentially the same density. This result is for a periodic direction set in Z. It gives no uniform bound on root depth across arbitrary witnessing systems or across the finite obstructions in Question 6.1.

D16. Polynomial translates of the weakly mixing reduction

Scope and status. Supporting proof and dependency material.

For an integer set R, write $abs*(R) = \{|r| : r \in R, r \neq 0\}$; common differences in this note are positive. The choice of cutting parameters in the weakly mixing reduction [D22] can ensure much more than the equality of critical avoiding densities. In particular every nonzero translate of the resulting single-stage family misses a one-dimensional Bohr neighborhood of zero after any zero difference is removed. This is compatible with any possible value of its original threshold.

Construction Fix nonempty $D \subset N$ and put $F_j = D \cap [1, j]$. Start with any positive integer H_1 and use the same heights as in the weakly mixing reduction:

$$H_{j+1} = 3B_jH_j + 1, D' = \bigcup_j H_jF_j.$$

Let S be the set of all finite signed sums using at most one block at each stage, including the empty $\sum 0$. Thus a summand at stage j has the form H_jf with $f \in F_j \cup (-F_j)$; a stage may be skipped. Write $S_{<}j$ for the finite set of prefixes before stage j and $L_j = 1 + \sum_{i<j} H_i \max(F_i \cup \{0\})$. All choices below depend only on D and previously chosen heights.

At stage j choose an integer $Q_j \geq j$, at least as large as every

$$|P(s + H_jf) - P(s)|/H_j$$

for $s \in S_{<}j$, $f \in \pm F_j$, and every integral polynomial P of degree and coefficient height at most j. This is a finite maximum of integers, since $P(s + H_jf) - P(s)$ is divisible by H_j. Choose the active cut B_j so large that, for example,

$$B_j \geq 2^{6j}(1 + L_{j+1})^{2j+4}(1 + Q_j)^2.$$

Here $L_{j+1} = L_j + H_j \max(F_j \cup \{0\})$ is known before B_j is chosen. This includes the summability condition of the weakly mixing reduction, so $b(D') = b(D)$, with the indicated zero-entropy weakly mixing witnesses. No delicate arithmetic properties of $3B_jH_j + 1$ are used below; only the growth of successive heights is required.

Separating a polynomial image of all block sums For every $P \in Z[x]$ and $r \in Z \backslash P(S)$ there is a single real θ and an $\eta > 0$ such that

$$||\theta(P(d) - r)|| \geq \eta \quad \text{for every} \quad d \in S.$$

In particular $P(S)$ is closed in the relative Bohr topology of Z, with rank one separation of each integer outside it. To prove the assertion, replace P by $P - r$, so that $P(s)$ is a nonzero integer on every finite prefix. Choose a sufficiently large N (in particular above the fixed degree and coefficient height). Put $K = \max(2, \max_{s \in S_{<N}} |P(s)|)$ and choose a prime $K < p \leq 2K$. For fixed P, K is at most a constant times $L_N^{\deg(P)}$. Our growth conditions, by taking N large, give

$$1/H_N < 1/(16pK),$$

$$H_{j+1}/H_j > 32pQ_j 2^j \quad \text{for all} \quad j \geq N.$$

Indeed the construction of H_N uses L_N^{2N+2}, and $L_j \geq L_N$ in every subsequent cut. The case of constant P causes no difficulty.

There is a real θ with

$$|\theta - 1/p| < 1/(8pK), ||H_j\theta|| \leq 2^{-j}/(8pQ_j)(j \geq N).$$

For clarity, put $\delta_j = 2^{-j}/(8pQ_j)$. At stage N take the closed interval of radius δ_N/H_N about a nearest multiple of $1/H_N$ to $1/p$. At stage $j+1$ take the interval of radius δ_{j+1}/H_{j+1} centered at a nearest multiple of $1/H_{j+1}$ to the preceding interval's center. The growth inequality makes every interval lie inside the previous one, and the initial interval has the required closeness to $1/p$. Choose θ in their intersection.

For a $\sum d \in S$ let s be its prefix before N. Telescope P on the growing prefixes to obtain

$$P(d) = P(s) + \sum_{j \geq N} H_j g_j, |g_j| \leq Q_j,$$

where only finitely many g_j are nonzero. Since $1 \leq |P(s)| < p$ we have $||P(s)/p|| \geq 1/p$. The two errors in replacing $\theta P(d)$ by $P(s)/p$ are bounded respectively by $1/(8p)$ and $\sum_{j \geq N} 2^{-j}/(8p)$. The total is less than $1/(2p)$. This proves the promised separation.

Single-stage differences For a polynomial P, if $r \neq P(0)$, then

$$b(abs * (P(D') - r)) = 1.$$

More precisely this difference set itself misses a rank-one Bohr neighborhood of zero, even if some of the finitely many earlier values equal r (those zero differences are deleted). For this assertion apply the same nested-interval construction to $P - r$, but take K larger than $|P(0) - r|$ and than the absolute values $|P(H_i f) - r|$ at $i < N$. This K still has at most fixed polynomial growth in L_N. At early nonzero values use the $1/p$ separation directly. At late one-block inputs the telescoping prefix is just 0, and $P(0) - r \neq 0$. Taking absolute values does not change circle norms. In particular choosing $P(x) = x$ gives $b(abs * (D' + r)) = 1$

for every nonzero r. If $r \notin P(D') \cup \{P(0)\}$, no zeros are removed, so the Bohr closure of $P(D')$ inside Z is contained in $P(D') \cup \{P(0)\}$. We have not used arbitrary interpolation of an h-avoiding D-event to prove anything about the threshold at $r = P(0)$.

Here and below rank-one Bohr separation gives the claim $b = 1$ by the elementary root-rotation sieve: if $||\theta u|| \geq \eta$ throughout an integer difference family U, divide θ by an integer multiple of all small denominators. For $\gamma = u\theta/M$ every rational orbit then has large enough denominator to meet any chosen short open interval, and irrational orbits are dense. Compactness in start point and allowed γ gives a common finite hitting time. This makes the complement of an arbitrarily short Haar interval a uniform avoider.

Exact classification for the enlarged finite-sums envelope For nonconstant P one has the full alternative, for every integer r,

$$b(abs * (P(S) - r)) = 0 \quad \text{if} \quad r \in P(S), \quad \text{and} \quad 1 \quad \text{if} \quad r \notin P(S).$$

The second assertion is the separation proved above. For the first choose a finite block $\sum s$ with $P(s) = r$. Take one positive nonzero step from each of infinitely many later nonempty F_j, sufficiently far that every nonempty $\sum t$ of them is beyond the real roots of the nonconstant polynomial $Q(t) = P(s + t) - P(s)$. All $s + t$ belong to S. The polynomial IP Szemerédi theorem of Bergelson–McCutcheon, Theorem 0.9 applied to the zero-constant polynomials $jQ(t)$, $0 \leq j < h$, gives simultaneous multiple recurrence along these finite sums for every positive event and each fixed h. None of the $Q(t)'s$ is zero and sign reversal only changes the initial point of a progression. For $P(x) = x$ the ordinary IP Szemerédi theorem suffices: a translated envelope is fully Szemerédi-large precisely when the translating offset is itself representable with opposite sign. The same reasoning works for the unsigned finite-sums envelope with digits 0 or F_j throughout.

Independent tower check and scope There is also a direct rigidity check on the weakly mixing base R of the companion construction. At stage j the complement of the pure supertower has summable measures in j: its measure is bounded by the sum of all future spacer measures, with one spacer at the succeeding three-copy cuts and heights growing at least geometrically. The excluded top and bottom Q_j active subcolumns have measure at most $2Q_j/B_j$. Therefore almost every initial point lies in safe subcolumns at every sufficiently late stage. Any shift $H_j g$, $|g| \leq Q_j$, at such a point moves to the parallel copy with exactly the same preceding tower level. For a sum of such shifts use descending stage order: each higher move preserves all lower-stage subcolumn labels, so the remaining lower moves stay safe. Thus on every finite tower cylinder the base returns pointwise to the same cylinder uniformly on all polynomial increments $P(d) - P(0)$ from sufficiently late finite sums. This is rigidity in this particular auxiliary system. The Bohr separation proof above additionally controls the early prefix shifts, which cannot be dropped when studying translates of the entire envelope.

The single-stage set retains exactly the unknown critical value of D; the all-sums envelope has the explicit alternative above. In particular the construction applies even when D is already fully Szemerédi-large (for example $D = N$), so rank-one Bohr separation of every nonzero translate, on its own, gives no upper bound for $b(D')$.

Optional nested heights give rational separation The same weakly mixing reduction admits a variant in which all the separations just obtained are **profinite**, in place of requiring the nested real-frequency construction. Keep the pure cut B_j as large as above. At the following spacer stage choose $2M_j$ parallel copies of the pure stack, with $M_j = \max(4, H_j \cdot j!)$ replaced by a multiple of $H_j \cdot j!$ at least 4. Add a single spacer after exactly M_j copies, for instance after the even-numbered copies. The next pure height is

$$H_{j+1} = 2M_j B_j H_j + M_j.$$

Thus H_j divides H_{j+1}, and each fixed positive integer divides all sufficiently late heights. The total spacer fraction at this step is at most $1/(2B_jH_j)$ up to an immaterial normalization, and tower complements are summable as before. In the eigenfunction check for the skew action, at least a fixed positive fraction of old levels are now paired with an identical copy at each of the two adjacent gaps B_jH_j and $B_jH_j + 1$: at most the last inter-copy comparison is unavailable. On these pairs the earlier labels and native fiber agree. Approximation of a unit-modulus eigenfunction by such labels forces its eigenvalue to be 1 as before. The pure active moves, the L^1 summability of the active cocycle, the entropy calculation, and the recovery of $b(D)$ are unchanged by this spacering. In particular divisibility of the *whole* heights does not impose a finite cyclic eigenfactor, because the spacers occur separately between individual copies.

For each fixed polynomial P and integer r, the growth condition gives a cutoff N with $H_N > \max_{s \in S_{<N}} |P(s) - r|$. Every finite $\sum d \in S$ reduces modulo H_N to its exact early prefix s, so $P(d) - r \equiv P(s) - r (\bmod H_N)$. When $r \notin P(S)$ the latter is nonzero throughout. Hence the single modulus H_N separates r from $P(S)$, and the polynomial images are profinitely closed inside Z. For the single-stage set when $r \neq P(0)$, all sufficiently late $P(d) - r$ reduce to $P(0) - r$ modulo H_N; choose N large enough that the nonzero early values have absolute value less than H_N as well. After deleting any actual zeros, one obtains uniform modular avoidance. For the finite-sums polynomial classification the positive case $r \in P(S)$ still uses the polynomial IP theorem as above. This variant improves the separation conclusions while the earlier argument applies to the original three-copy spacer choice without alteration.

More generally, for fixed integral polynomials in m variables, any simultaneous system $P_i(s) = r_i$ soluble on S^m modulo every integer is soluble there over Z. Choose N so large that every early-prefix value $P_i(s_1, \ldots, s_m) - r_i$ has absolute value $< H_N$ (each prefix coordinate is bounded by L_N, and the polynomials are fixed). Reduction modulo H_N therefore gives actual simultaneous equalities on the selected prefixes. This also shows that the same critical-density classification holds for any single **nonconstant multivariable** P. At an attained value r choose a tuple s with $P(s) = r$. Choose successive fresh vectors of positive single-block increments in Z^m whose supports are disjoint from the original prefixes and from every earlier choice. For a finite list of earlier IP sums u, the simultaneous conditions $P(s + u + b) \neq r$ exclude a finite union of proper polynomial zero sets for b. They can be met on a product of m infinite pools of as-yet unused positive single blocks (choose disjoint infinite stage reservoirs for the coordinates); a nonzero polynomial cannot vanish on a Cartesian product of infinite subsets of Z. Induction ensures the polynomial increment is nonzero on all nonempty subsequent vector-IP sums and every coordinate of s plus those sums belongs to S. Apply the multivariate polynomial IP Szemerédi theorem to these vector IP increments. The other case is the modular separation already proved.

What transferring a shifted avoider back to zero would require Put $E_J = \bigcup_{j \geq J} H_j F_j$. For a fixed $r \neq 0$ write $c_r(E_J + r)$ for the supremum, over all stationary systems U, positive integers K and events B K-avoiding $E_J + r$, of

$$\mu(B \cap U^{-r}B \cap \cdots \cap U^{-(K-1)r}B).$$

For negative r take J sufficiently large that the shifted steps are positive. Then

$$\lim_{J \to \infty} c_r(E_J + r) = b(D), although b(E_J + r) = 1 at \quad \text{every} \quad sufficiently large J.$$

For the upper bound the displayed r-block event C itself K-avoids E_J: a K-site d-line in C, $d \in E_J$, supplies a K-site $(d + r)$-line in B by choosing the jth r-offset at the jth point. Removing the finitely many early D' steps does not change b, so $\mu(C) \leq b(D') = b(D)$. The suprema are nondecreasing in J.

For the lower bound take an original h-avoiding A of mean a and use its skew system in weak-mixing-transfer.md [D22], reading the native fiber at stage J (subtract the contributions of all earlier active digits in the skew coordinate). Keep the ordinary height-H_J tower, drop a band of $2h|r|$ levels at both ends, and restrict to all late active digits at distance at least $h \max(F_j)$ from the ends of their subcolumns. The events not kept cost $o_J(1)$ for fixed h,r, by the tower growth and the summable active margins. Let B mark native-fiber membership in A on this kept region. At a marked start all h sampled pure moves with step $H_j f$, $j \geq J$, remain in parallel copies of that same stage-J relative level and advance the native fiber by the successive powers of T^f. Independently shifting the jth sampled point by any ordinary amount of absolute value $\leq h|r|$ keeps the same native fiber, since the point is interior to its stage-J tower copy. Consequently B h-avoids $E_J + r$ (indeed every such bounded per-site perturbation on the h samples is forbidden). To estimate its r-block event begin slightly farther from the stage-J endpoints, say by $4h|r|$. All the h ordinary shifts tr now remain in the same stage-J copy, so they retain the same native-fiber value and all higher-digit safety conditions. This start set has measure at least $a - o_J(1)$ and belongs to the r-block event. Let a approach the cap.

This reformulation pinpoints a limit of the direct translate idea. In any particular realization where the infinitely many later $d's$ tend strongly to the identity, all K-avoiding B for $E_J + r$ have **zero** r-block mass. Indeed for each fixed K, the intersection measures along $d + r$ tend to that along r by strong continuity and boundedness. The elementary near-full rotation sieves treating d as asymptotically zero therefore do not achieve the extra r-block property needed to raise the original cap. Even after the isolated translates have avoidance density one, obtaining high-density r-block events above remains equivalent, on tails, to the unknown value at shift zero.

D17. Every integer critical density occurs on one rational group with an integer period

Scope and status. Supporting proof and dependency material.

Here is an elementary reduction of the full question to prime-denominator rows. It uses the stationarity definition a_h of the fixed-length avoiding density and $b = \sup_h a_h$. No finite coloring is assumed. This is a reduction, not a solution of Question 6.1.

Fix once and for all distinct primes $p_j > 2j$ so large that $\sum_j j/p_j$ converges. For any $D \subset N$, let $F_j = D \cap [1, j]$. With $\pi{:}Q \to Q/Z$ the natural map, define

$$E_D = \pi^{-1}(\{f/p_j \bmod 1 : j \geq 1, f \in F_j\}).$$

Then $b_Q(E_D) = b_Z(D)$. More precisely $a_h{,}Q(E_D) \leq a_h{,}Z(D)$ at every h. The construction is the same for every D; its prime list prescribed in advance. The admissible cosets are supported on one prime component of Q/Z at a time. Since $p_j > 2j$, every listed quotient element is nonzero and has order p_j. One can symmetrize E_D by negation without changing any AP cap, by reversing the order of any progression.

For the upper inequality, restrict an E_D-avoiding Q-action to the single cyclic generator $1/p_j$. Its stationary binary sequence avoids every difference $f \in F_j$, preserving the same mean. Weak compactness of stationary binary laws as j grows gives a Z-process avoiding all of D with that mean. The upper bound for b follows.

For the reverse inequality hold any h, any A of mass a h-avoiding D under an invertible transformation T, and any integer J. Write $P = Q/Z = \oplus_p C_{p\infty}$, the direct sum of Prüfer groups. In each $C_{p_j\infty}$ fix a transversal for its order-p_j subgroup (take zero for the identity coset). Define $e_j(u) \in \{0, \ldots, p_j - 1\}$ by writing u as the chosen representative plus $e_j(u)/p_j$. Then $e_j(0) = 0$ and

$$e_j(u + f/p_j) \equiv e_j(u) + f (\mathrm{mod} p_j).$$

For $x \in P$ put $s(x) = \sum_j e_j(x_j)$, a finite sum. For a random original point z, mark a position x if $T^{s(x)}z \in A$ and, for every $j \geq J$, its digit satisfies $e_j(x_j) \leq p_j - 1 - (h-1)j$. Choose J sufficiently large that the upper bounds are nonnegative. For any retained start and any $f \in F_j$ with $j \geq J$, the first h digit evaluations then have no wrap, and the corresponding T-powers are $s(x){,}s(x)+f{,}\ldots{,}s(x)+(h-1)f$. Thus the mark has no h-run in the indicated tail directions, almost surely simultaneously over all (countably many) starts. Random z is used for a stationary density estimate, without any assumption on recurrence of the deterministic integer $s(x)$.

Average the translates of the resulting binary marking over increasing finite subgroups of P exhausting P and pass to a weak limit to get a stationary P-process. At a uniform point x in each subgroup, any p_j-component is either trivial (digit zero) or contains the order-p_j subgroup; in the latter case the digit is uniform modulo p_j. This is because in any finite subgroup of Q/Z the primes separate into their primary components, with uniform marginals. The probability of deleting a point at any tail digit is at most $(h-1)\sum_{j \geq J} j/p_j$.

Before the deletions the expected mark is a at each x by stationarity of T. The limiting process consequently has mean at least a minus this tail bound. Pulled back to Q via π, it still h-avoids all integer lifts of these tail classes.

There are finitely many earlier nonzero cosets mod$Z$. For clarity, these are generally infinite sets of actual rational directions, and a finite-difference patch alone would not suffice. Let their minimal nonzero circle distance from 0 be η. For an arbitrarily short open circle arc I choose M divisible by all denominators up to $2/|I|$. Let Q act on the Haar circle via the character $x \mapsto x/M$. Every early direction x has $||M(x/M)|| \geq \eta$. For any such velocity, every starting point hits I in a bounded number of iterates uniformly over the compact set of possible velocities: rational velocities have denominator $> 2/|I|$ and hence meet I, and irrational ones are dense. Covering compact starting-point/velocity space by the resulting finite-hit open sets gives uniformity. Thus the complement of I is a near-full event avoiding sufficiently long progressions at all early cosets. Take its independent product with the tail mark, allowing the common length to increase, and let J increase and $|I|$ decrease. The resulting avoiding densities approach a. Since the comparison holds for every input h and A, it proves the reverse b inequality.

Consequently Question 6.1 is equivalent to determining the endpoint dichotomy on this single class of Z-saturated subsets of Q. Every E_D is rank one as an ambient rational group, regardless of the complexity of D. Each individual finite subgroup trace (restriction to $(1/L)Z$) is an ordinary periodic set of integer steps; the exact periodic root formulas can be used at fixed h, but the lifting depths and L are not bounded uniformly by this result.

Exact tail limit in the unlifted torsion group Put $S_J = \{f/p_j \bmod 1 : j \geq J, f \in F_j\}$ in $P = Q/Z$ itself. Then

$$a_h, Z(D) - (h-1)\sum_{j \geq J} j/p_j \leq a_h, P(S_J) \leq a_h, Z(D)$$

whenever the tail buffers are nonnegative. The upper inequality again restricts to the generator $1/p_j$ for $j \to \infty$: the resulting stationary Z-law is p_j-periodic, and all F_j constraints still hold. For the lower inequality the digit proof above already gives a stationary P-law at the original length h; no early directions have to be patched. Thus a_h,$P(S_J)\uparrow a_h$,$Z(D)$ for every h. Since each fixed-h cap increases with J and with h, taking the two suprema gives $b_P(S_J)\uparrow b_Z(D)$. This differs from keeping a single initial bounded-order prime direction forever.

In this formula all permitted directions lie in the direct sum of the finite order-p_j subgroups. Restriction and coinduction to P identify its capacity with that on this direct sum. For each fixed h and J the latter is exactly the decreasing infimum of the finite extremal densities on the products $\prod_{J \leq j \leq n} Z/p_j$ with the single-axis steps $F_j e_j$. An invariant law restricts to a finite group action and yields a mask of at least its mean by averaging an orbit. Conversely a finite mask gives its Haar-translated law, to which the other coordinates can act trivially before compactness. No prime-power lifts enter this tail formula.

Let $p(J)$ be the least prime among active rows $F_j nonempty$ of the tail, and write $s = b_Z(D)$. For every such tail there are the all-length bounds

$$(1 - 1/p(J))s \leq b_P(S_J) \leq \min(s, 1 - 1/p(J)).$$

For the new upper bound a single allowed element of order $p(J)$ gives the orbit union bound. For the lower bound choose h and an original h-avoider of density arbitrarily close to s, and carry-safe tail rows $j \geq I$ for I large. Every step of an earlier active row $J \leq j < I$ moves the Haar circle under the natural character $x \in P \mapsto x \bmod 1$ with exact order p_j. A fixed open arc of length $1/p(J) + \eta$ therefore hits the whole orbit of every such step within their finite maximum order. Apply this sieve in an independent circle action; the common avoidance length can be increased to that maximum. Sending I to infinity and η to zero gives the result. In particular if $s = 1$ then $b_P(S_J) = 1 - 1/p(J)$ exactly. If $s = 0$ then $b_P(S_J) = 0$, whereas if $0 < s < 1$ then for all sufficiently late J the torsion cap lies strictly between 0 and its one-orbit ceiling. This is a reformulation of the question, not evidence deciding the endpoint.

Another quantitative choice of finite-row patch gives, for $s < 1$,

$$b_P(S_J) \geq s - \sum_{j \geq J, F_j \neq \varnothing} s^{p_j}/p_j.$$

Indeed let a be a stationary h-avoiding density on the rows from I onward. For each point of the finite earlier Cartesian product $G = \prod_{J \leq j < I} C_{p_j}$, use an independent copy of this entire stationary tail mask. For fixed tail position the marks at distinct points of G are independent with probability a. If an entire p_j-axis line is marked, delete one point on it. The expected normalized loss over all j-axis lines is at most a^{p_j}/p_j for that j. Use any deterministic choice of the deleted point and average over translations on G to enforce stationarity; the rule is translation covariant in the tail. The remaining event still h-avoids the tail and avoids every earlier full orbit at sufficiently large common length. The carry estimate supplies a approaching s as h is chosen and I then increased; one can thin to have $a \leq s$ when necessary. The displayed infinite sum converges by the fixed prime growth. This describes how little a large initial prime can cost even when the original threshold is intermediate.

Attaining the one-orbit ceiling is a polychromatic condition on the other rows Split off the active prime $p = p(J)$ from the direct sum, and write B for the group of the remaining rows and W for their steps. Then $b_P(S_J) = 1 - 1/p$ if and only if the following holds. For every $\epsilon > 0$ there is a stationary B-process of p colors together with a defect symbol of probability less than ϵ, and some finite K, such that every K-term W-line either visits a defect or visits all p colors.

For the forward implication take a stationary binary avoider of mean at least $1-1/p-\eta$ and view its configuration on $C_p \times B$. At every tail position x the p slots have at least one hole, since one selected nonzero p-step traverses the full p-orbit. By stationarity their expected number of holes is at most $1 + p\eta$, so the probability of two or more is at most $p\eta$. Mark the unique missing slot as the color when it exists, and use the defect symbol otherwise. Along a tail run with all sites good, any missing color would be a fixed p-slot which is retained throughout the run, contradicting avoidance. This new process is stationary under B-translations. Conversely, over a given colored B-process erase the slot matching the color on good fibers and all slots on a defect. Each early p-line then has a hole, and every K-run on a fixed p-slot in an allowed tail direction has one as well. An independent uniform cyclic offset of the p-slot names, or translation averaging over C_p, gives full stationarity. The

resulting mean is at least $(1-\epsilon)(1-1/p)$; its length is at most $\max(K,p)$. This proves the equivalence without requiring a deterministic periodic coloring of the infinite tail or a uniform K over ϵ.

D18. Uniform stability on escaping allowed step words implies $b(D) = 1$

Scope and status. Supporting proof and dependency material.

For a two-sided word $w \in \{0,1\}^Z$ call w k-stable when there are no integers $a_1,\ldots,a_k,b_1,\ldots,b_k$ such that $w(a_i + b_j) = 1$ exactly when $i \le j$. The property is closed under pointwise limits, and translations preserve it. A complement is $(k+1)$-stable whenever the word is k-stable: reverse the row/column indices of a complementary half graph and offset one list, dropping one index, to convert its strict triangular pattern to a weak one.

Sufficient criterion. Suppose there is a two-sided integer set A of positive upper Banach density, avoiding h-term D-progressions for a fixed h, such that for some fixed k, every pointwise limit of words $t \mapsto 1_A(n_i + td_i)$, $n_i \in Z$, $d_i \in D$ and $d_i \to \infty$, is k-stable. Then $b(D) = 1$. No restriction is placed on a particular word at any fixed step d. The same proof applies if limits are taken using arbitrary varying points from the support of a positive-mean stationary h-avoider. Thus any counterexample of intermediate $b(D)$ must realize half-graphs of arbitrarily large finite order in escaping D-word limits in each of its positive-density obstruction witnesses. There is no proof here that arbitrary witnesses possess stable replacements.

The stronger hypothesis that the actual step words at *every fixed* d be stable would already imply a periodic obstruction: fix one $d_0 \in D$; in each residue class modulo d_0 the word equals a periodic word up to zero upper Banach density by the theorem below. Thus A itself agrees with a periodic word modulo such a set. Any step in D divisible by the period would contradict avoidance on the positive-density approximating cycle.

Stable words admit uniformly bounded approximate periods For all k,$\delta > 0$ (k an integer) there is $P(k,\delta)$ such that every k-stable binary word x on Z differs from some P-periodic binary word z on a set of upper Banach density at most δ. This is a combination of Theorems 2.3 and 1.3 of Conant–Pillay–Terry.

In Theorem 2.3 take the invariant stable relation $x + y \in A$. Parts (iv),(vi) give a union Y of cosets of a finite-index subgroup of Z such that A$\triangle$Y is nongeneric and has zero measure under the unique left-invariant measure on the local Boolean algebra. Since every translation-invariant mean on all subsets restricts to such a measure, d*(A$\triangle$Y)=0: if the upper Banach density were positive, limits of interval sampling along a maximizing sequence would give a translation-invariant mean positive on the set. The index at this stage may depend on A. The periodic Y is k-stable, because any alleged bad finite pattern in its period group lifts to an integer pattern and all of the corresponding finitely many shifted null errors can be avoided by translating by a multiple of its period. Apply finite Theorem 1.3 to Y on its finite cyclic group. It is δ-close in normalized Hamming measure to a union of classes modulo an index $m \le n(k,\delta)$. Its lift is periodic with period dividing $P(k,\delta) = \mathrm{lcm}(1,\ldots,n(k,\delta))$; its discrepancy with A has upper Banach density at most δ.

Independent dilations and simultaneous activation Write a for the mean of a stationary two-sided h-avoiding measure satisfying the escaping hypothesis, and $p = 1-a$. Such a positive-mean measure follows from the stated integer A by empirical averaging on long intervals; every point in its support belongs to the ordinary shift-orbit closure of 1_A. Finite

tested configurations on varying support points can therefore be copied by integer translates of the original word, even when the tested coordinates are widely separated, and the escaping hypothesis passes to the stationary support. Fix $\epsilon > 0$; choose r with $p^r < \epsilon/2$. Choose $\delta > 0$ so that $rh^r\delta < 1$. Use $k' = k + 1$ for the complementary active symbols. Let $P = P(k', \delta)$, set $M_1 = 1$ and choose successive M_{j+1} divisible by $M_j P$ $onemaytake M_{j+1} = 2PM_j$.

Use r independent stationary samples ξ_j and r independent uniform phases s_j mod M_j. At each n let

$$g_j(n) = 1 - \xi_j(\lfloor (n + s_j)/M_j \rfloor).$$

This produces independent stationary row processes; adding one to n advances each phase and at wraparound translates its stationary ξ_j by one. First retain n when not every $g_j(n) = 1$, a process of mean $1 - p^r$. In addition, for each $d \in D$ take a fresh uniform phase modulo dL_d and delete d consecutive residue classes at that phase, with integers L_d so large that $\sum_{d\in D} 1/L_d < \epsilon/2$. This further stationary deletion has probability at most $\epsilon/2$, and every progression of step d and length L_d necessarily strikes its own deleted strip. It suffices to show a single uniform bound on lengths of surviving progressions, even when the chosen support words/phases vary.

Otherwise center a sequence of increasingly long surviving segments at n_N, with $d_N \in D$. The periodic strips force $d_N \to \infty$. Pass to a pointwise limit of all r row gate words in the progression index t. Write the rows as $x_j(t)$. No t has all $x_j(t) = 1$. For each residue u mod M_j, reading $t = u + M_j v$ turns the integer floor into a basepoint plus vd_N. Consequently $v \mapsto x_j(u + M_j v)$ is k'-stable by the escaping hypothesis, and every h consecutive v contain an active 1 because the original witnesses avoid h-term D-progressions. The stable regularity lemma above supplies a P-periodic approximant for each such residue subsequence. Interleaving gives a word z_j of period $M_j P$ with

$$d * \{t : x_j(t) \neq z_j(t)\} \leq \delta,$$

since each of the finitely many residue classes has upper Banach discrepancy at most δ in its own integer parameter.

Let U be the set of all sums $\sum_j m_j M_j$ with $0 \leq m_j < h$. Its size is at most h^r and it includes all partial sums using zero tails. The union of the $r|U|$ discrepancy sets shifted by the elements of U has upper Banach density < 1. Choose t off that union, so all tested row values coincide with the respective z_j at $t + u$, $u \in U$. At step j choose $m_j \in \{0, \ldots, h - 1\}$ making x_j active after updating the current time by $m_j M_j$; this is always possible by its h-option stencil. Earlier active z_i stay unchanged because M_j is divisible by $M_i P$ for $i < j$. At the final time all z_i and hence all x_i are active, a contradiction. The retained stationary process therefore avoids a fixed length. Taking a high-mean ergodic component and a generic forward sample supplies an ordinary upper-density witness, as needed for $b(D)$.

Fourier–Stieltjes gate criterion This argument also works for any family of independently realized diluted gate rows whose active probabilities are bounded by some $p < 1$ and whose limiting residue words have a bounded-size activation stencil of multiples of M_j. In place of k'-stability one can assume that such residue words are separated binary gates of

sequences $g(v) = \int e(v\theta) d\sigma(\theta)$, with $||\sigma|| \leq C$ uniformly for the row (independent of the limiting word) and a uniform positive gap between the two legal ranges in the compact disk $|g| \leq C$. This is a zeroth-order Fourier–Stieltjes condition. It does not follow merely from a bound on a higher difference of g.

Take a fixed polynomial in g and its conjugate that is within 1/4 of the binary gate on the separated compact ranges (Stone–Weierstrass); this polynomial has Fourier–Stieltjes norm bounded by a constant C' depending only on its coefficients and C. Its translates $g' a_i + b_j$ admit a Hilbert matrix factorization with products of the row and column norm bounds at most C', by the polar decomposition of the representing measure. In a half graph of order m, the paired upper/lower matrix entries would have real parts differing by at least 1/2. Test the matrix against $1/(j - i)$, zero on the diagonal. Its pairing is at least $(1/2) \sum_{i<j} 1/(j-i)$, of order $m \log m$. Its pairing against the Hilbert factorization is at most $\pi C' m$ by Cauchy–Schwarz and the discrete Hilbert transform bound (the two-sided Fourier multiplier $\sum_{n \neq 0} e(n\theta)/n$ is bounded by π in essential supremum). This bounds m in terms of C' and verifies uniform stability.

Positive density of a general h-avoider, zero entropy in ordinary integer time, or the absence of h consecutive ones in its step words does not furnish this stability bound by itself. For instance arbitrary data on the even positions, with all odd positions zero, meet the adjacent-pair restriction for step one and can exhibit arbitrary-order half graphs on even offsets. The difference set with just step one is elementary; this example concerns only the logical nonimplication for a given witness.

D19. Strict finite-length gaps and sparse samples at an intermediate threshold

Scope and status. Supporting proof and dependency material.

Write $a_h(D)$ for the maximum mean of a stationary h-avoider and $b(D) = \sup_h a_h(D)$. Equivalently $1 - a_h(D)$ is the minimum measure of an event meeting every h-slot D-progression almost surely. Call such an event a hitter of gap h. Conventions below assume $D \neq \emptyset$. The sole external input for this argument is the ordinary finite Szemerédi theorem.

Fix one hitter H_0 of gap k and measure $p_0 < 1$. For any hitter H with gap h and measure p, choose $W = W(h, k)$ so a subset of $[W]$ of density $1/(2h)$ contains a k-term AP. Increase W to be at least $2h$. On every W-slot d-line the H marks have density at least $\lfloor W/h \rfloor / W \geq 1/(2h)$, uniformly over all $d \in D$, modulo a common null set. Thus among them there is a k-term AP in the time index with some stride $1 \leq c \leq W$.

Attach independent stationary c-root systems of the system of H_0, one for each $1 \leq c \leq W$. The c-root is the suspension tower with uniform level modulo c, so one advances by exactly T^d on original states after cd time units. Denote the root mark by $H_{0,c}$; its measure is p_0. In the product retain as new hitting event

$$H' = H \cap \bigcup_{c=1}^{W} H_{0,c}.$$

On the selected k-term time-index AP the corresponding root necessarily contributes a mark at one of the sites, making H' a hitter of gap W. Its measure is $p[1 - (1 - p_0)^W]$.

Consequently, whenever $b(D) > 0$,

$$a_W(D) \;\geq\; a_h(D) + (1 - a_h(D))(1 - p_0)^W > a_h(D).$$

Indeed the single forbidden step bound gives $a_h(D) \leq 1 - 1/h$. The estimate applies even when there is no pair witness. In particular, if $0 < b(D) < 1$, the endpoint density $b(D)$ itself is Szemerédi-large for D: every finite-length avoidance cap is strictly below it. Neither compactness nor strictly increasing caps alone excludes an intermediate limiting value.

Necessary sampling defect Write $\rho = 1 - b(D) > 0$. Fix $W \geq k$, $L \geq W$, and let $G_{L,W}(H)$ be the starting points for which, for every $d \in D$, the first L sites of their d-line contain a k-term AP of H-marks in the time index with stride at most W. With the same product event above, $\bigcup_L G_{L,W}(H)$ has measure at most

$$\frac{p}{\rho}\,[1 - (1 - p_0)^W].$$

Indeed all points over $G_{L,W}(H)$ lie in $U_L(H')$. The hitting-operator bound $\rho\mu(U_L(H')) \leq \mu(H')$ from hitting-operator.md gives this for each L, and the sets increase.

If $\rho \in (0, 1)$, take p sufficiently close to ρ that this bound is strictly below 1. A positive-measure set of starting points then has, for every L, a d-line with no such time-index pattern. This refines a single exceptional-line obstruction to a positive-measure obstruction at common starts. The W and p quantifiers matter. One cannot pass to arbitrarily large W with a single fixed finite-gap H by this argument.

For a multiplicatively closed D, the corresponding obstruction is impossible. Every sufficiently long d-line of a fixed finite-gap H has relative hitting density at least $\rho/2$, uniformly in $d \in D$: otherwise empirical measures of increasingly long bad line segments give another stationary hitter of the same gap and measure at most $\rho/2$, since $de \in D$ for all $e \in D$. Choose W now using $\rho/4, k$, independently of H or its gap. Partitioning such long lines into W-blocks gives one block of density at least $\rho/4$. Thus $G_{L,W}(H)$ is conull for sufficiently large $L = L(H)$. Taking $p \downarrow \rho$ contradicts the displayed bound. This proves the zero–one dichotomy for multiplicative semigroups.

D20. Finite-order models, roots, and Hamming constructions

Scope and status. Supporting proof and dependency material.

Status: exploratory, no solution claimed.

Use $b(R)$ = sup over upper natural densities of sets avoiding all k-term R-APs for some fixed finite k. To disprove Q6.1 one needs $0 < b(R) < 1$.

Exact difference between groups In a group of exponent a prime p, even ONE nonzero direction v already forces arbitrarily long progressions in every set of density $> 1 - 1/p$, because every full coset of the cyclic subgroup has p points and an AP may make multiple passes around that coset. Taking all but one point per v-coset gives density exactly $1 - 1/p$ and no p-term progression. This is an intermediate threshold in the torsion group, not on Z. In contrast, for a fixed integer $v > 0$, deleting multiples of a sufficiently large prime not dividing v gives avoidance at densities arbitrarily close to one.

There are no nontrivial finite-order points in the universal Bohr compactification of Z: its Pontryagin dual is the discrete divisible group T. For a homomorphism x:$T_d$iscrete -> T, p $x = 0$ would give $x(\alpha) = px(\beta) = 0$ for every α by divisibility. Consequently even a sequence of step models approximating torsion in one fixed rotation has no universal torsion interpretation.

Elementary obstruction under finite-Bohr separation Suppose every d in R has some $i \le m$ with $||d\alpha_i||$>=eps>0. Given $\eta > 0$ take integer M divisible by every $1 \le q \le 2/\eta$. For a bad coordinate i the rotation increment $\gamma = d\alpha_i/M$ mod 1 lies in the compact set L=$\{\gamma: ||M\gamma||$>=eps$\}$. Every rotation in L hits any fixed open arc of length η from every starting point in a bounded number K of steps, uniformly over starts and γ. Indeed, an irrational increment has a dense orbit; a rational in L has denominator exceeding $2/\eta$. By compactness finitely many of the open sets in the hitting-time cover of T x L suffice. Translate the gap in each of the m coordinates and take A as the integers avoiding all m gaps. Averaging the shifts of the gaps and applying reverse Fatou gives some choice with upper density at least $1 - m\eta$. No K-term $R - AP$ can be in A. Thus $b(R) = 1$. This is the elementary Bohr case of Alweiss-Farhangi Theorem 1.8.

The calculation explains why strong continuity of an action of an ambient group with a finite-order displacement does not give a threshold over *all* Z-actions: the root rotations are legitimate Z-actions and need not extend continuously to the selected ambient group in the required way.

Potential Hamming model, not a resolution Griesmer, Australas. J. Combin. 91(2025), Lemma 4.1: https://ajc.maths.uq.edu.au/pdf/91/ajc_v91_p001.pdf. An antipodal Hamming ball of bounded radius k in high-dimensional $(Z/2)^m$ thickened slightly in T^m admits integer pullbacks with pair-avoiders of density near 1/2. Individual finite-dimensional models are finitely Bohr separated, so have $b = 1$, but the required avoiding length may grow with dimension. The source obtains finite models forcing multicolor *pairs*; this does not show any fixed high density forces arbitrarily long APs over Z. A possible finite-gadget goal is to prove that for each fixed length L, choosing the Hamming radius k then the dimension m sufficiently large forces an $L - AP$ in every set of density $> 1 - \epsilon$ $onefixed\epsilon > 0$. The pair

avoidance would then still need to be glued across stages. No such L-term Hamming-ball forcing theorem has been verified. The elementary bounded-order argument from the torsion group does not establish it after transfer to Z.

An apparent compactness shortcut fails: for every fixed m, the pullback based only on its m torus coordinates misses a Bohr neighborhood. In a sequence of finite stages, obtaining for each invariant measure a good d in some finite model gives no right to discard conditions on that d from all subsequent coordinates. Nor is 'eventually for every m there is some good d' an open condition in the invariant measure.

For the gluing at the level of the integers there is a relevant precise result: Griesmer, cited above, Lemma 3.6. A finite difference-nonrecurrence witness of density δ and one for a new finite F of density $\eta < 1/2$ can be joined after dilation with witness density $> 2\delta\eta$ (taking the hypotheses and boundary buffers in his lemma). If one had finite gadgets forcing k-APs at a *fixed* density but admitting pair-avoiders of density arbitrarily near 1/2 for each k, the inductive gluing in his Theorem 1.1 would furnish a counterexample to Question 6.1. His stated finite gadgets force pairs in many colors, not the needed k-APs. The construct track has a separate root-cosine argument that rules out straightforward near-antipodal Hamming gadgets when the normalized coordinate errors are small.

A two-copy improvement for pair obstructions If there is a pair-avoider of upper density α, then $b(R) \geq 1 - (1-\alpha)^2$. To see this rigorously, realize a stationary two-sided pair-avoiding binary process of mean at least α by the usual empirical-measure construction. Take two independent copies X,Y and an independent fair phase e in $\{0,1\}$. Set $Z(n) = \max(X(n), Y(\lfloor (n+e)/2 \rfloor))$. This law is stationary: a shift changes phase and applies an integer shift to Y at the carry, preserving its law. Its mean is at least $1-(1-\alpha)^2$. Every trajectory (outside a null set) avoids length-6 R-progressions. In fact, covering the indices 0,...,5 of such a progression by colors 0 (first-coordinate ones) and 1 (second-coordinate ones) would forbid adjacent $0's$ and distance-two $1's$. If an interior position 2 or 3 had color 1, its two positions 2 away would both be 0; their two inward neighbors must both be 1, violating distance two. Thus positions 2 and 3 would both be 0, also impossible. Ergodic decomposition/genericity yields a one-sided realization of upper density at least the mean. This gives an improved lower bound, not density one.

Merely taking an arbitrary union of many scaled copies and using only the separate constraints "color i is forbidden at distance t_i" cannot continue this proof. For any positive integers t_1,...,t_m with $m \geq 4$, the mean over a circle phase θ of $\sum_i ||t_i\theta||$ equals $m/4 \geq 1$. Choose a θ with sum ≥ 1 and partition the circle into consecutive half-open arcs of lengths at most the corresponding norms, permitting length 0. Color n by the arc of $n\theta$. This covers all integers, yet the i-th class has no pair at distance t_i, since translates of a half-open arc of length $\leq ||t_i\theta||$ are disjoint. Thus those constraints can admit infinite union-runs at any number of sampling strides ≥ 4.

In fact three strides suffice for colorability. Divide by gcd initially. If at least two t_i are odd, $\theta = 1/2$ works. Otherwise two are even, say 2b and 2c, and one odd. Some y has $||$by$||+||cy|| \geq 3/4$: remove the joint gcd using surjectivity of multiplication on the circle. If both are odd use $y = 1/2$, else let c be even and b odd; b is coprime to 2c. Choose $y = (2j+1)/(2c)$, making the odd residue $(2j+1)b$ equal to $c-1$ mod $2c$. Then $||cy|| = 1/2$ and $||$by$||=1/2 - 1/(2c) \geq 1/4$. Taking x with $2x = y$ and choosing x or $x + 1/2$ ensures the odd stride contributes at least 1/4 without changing the two even terms.

Kronecker density increment (positive necessary condition) Let (X, μ, T) be an ergodic stationary system and A an event with $\mu(\bigcap_{j=0}^{h-1} T^{-jd}A) = 0$ for every d in D. Let the compact rotation $\pi{:}X \to K$ be a factor, with K a closed subgroup of a finite torus T^m, stationary Haar law and rotation α. Put $g(y) = E(1_A|\mathrm{K})y$. Then

$$ess\sup g \le b(D).$$

Proof: translations of the positive-measure set $\{g > p\}$ cover Haar-almost all K whenever p<ess sup g, by ergodicity. Select finitely many translate times n_i to cover all but ϵ and a measurable selector of one available n_i on the covered part. Approximate the selector in measure by a function constant on cells of a fixed finite product grid on T^m; these partitions generate the Borel sets on K (choose continuity cuts if needed). The event C which uses $T^{-n_i}A$ on the π-preimage of each cell then has measure $\ge p - O(\epsilon)$. Choose $L = (m+1)h$, and let B_0 be a Bohr neighborhood where each coordinate of $d\alpha$ has norm less than the minimum grid spacing divided by 2L. On any L-progression with d in D intersect B_0, each coordinate of a lifted grid phase crosses at most one cell boundary during the samples; hence some h consecutive terms use a common selector value, impossible whenever the entire orbit segment lies in C, up to null sets. Thus b(D intersect B_0)$\ge$ $p - O(\epsilon)$ by the stationary correspondence. Because $b(B_0^c) = 1$ and b(R union S)$\ge$ $b(R)b(S)$ by taking independent stationary avoiders and intersecting their events, b(D intersect B_0)=$b(D)$. Let the error tend to zero and p tend to the essential supremum. Passing to the entire Kronecker factor uses an increasing sequence of finite-dimensional rotation factors and martingale convergence in L1.

In particular a witness that becomes arbitrarily dense conditioned on arithmetic residue classes or on finite-dimensional rotation factors forces $b(D) = 1$. A putative counterexample $b(D) = s < 1$ requires conditional densities $\le s$ uniformly across these factors. This argument has not been upgraded to arbitrary higher-order factors, where a d in a return neighborhood can still cause large fiber shear.

Approximate root/tower amplification Suppose X is a two-sided stationary binary process of mean $a > 0$ which has no h-term all-one $D - AP$ almost surely (simultaneously for all starts and D, by countability). Let $L \ge 2$ be an integer and put

$$e(d) = \Pr(X(0) \ne X(Ld)).$$

If $\sum_{d\in D} e(d)$<infinity, then $b(D) = 1$. More quantitatively, for a finite F subset of D, for any $r \ge 1$ let $K = 1 + (h-1)\sum_{s=0}^{r-1} L^s$ and $C = Kr^2h^2L^r$. Then $b_K(F) \ge 1 - (1-a)^r - C\sum_{d\in F} e(d)$, where b_K refers to avoidance at the stated length K.

Take mutually independent copies X_s for $0 \le s < r$ and independent uniform phases θ_s mod L^s. The processes $Z_s(n) = X_s(\lfloor (n+\theta_s)/L^s \rfloor)$ are stationary (translation permutes uniform phases and shifts a stationary copy at the appropriate carry), independent across s. Let $Z = \max_s Z_s$. Its mean is $1 - (1-a)^r$. Fix a step d and start n. From $t = 0$ successively choose j in $[0, h)$ so that $Z_s(n + (t + jL^s)d) = 0$, and update t:=$t + jL^s$. Such j exists by h-avoidance, since row s at these sampled times has quotient arguments advancing by d. To ensure previously found zeros persist it suffices that for all candidate times $0 \le t < K$,

indices 0<=ell<$s < r$ and $0 \le j < h$, the corresponding entries of Z_ℓ at t,$t + jL^s$ are equal. Failure of one equality has probability at most $jL^{s-\ell-1}e(d)$, by stationarity and telescoping. The union bound for these equalities is at most $Ce(d)$. If none fails, the construction reaches $0 \le t < K$ with $Z(n + td) = 0$, so the probability that n starts an all-one $K - AP$ of step d is at most $Ce(d)$.

Delete from Z every start of a bad F-progression. The resulting process has mean $\ge 1 - (1 - a)^r - C \sum_F e(d)$, and cannot contain a bad progression. A point of upper density at least this mean exists by ergodic decomposition and the individual ergodic theorem. This argument works equally for an infinite F whose error sum is finite. For D with summable e, remove a finite initial set of steps to make the tail error as small as desired. A finite set of step sizes does not change b: delete a single residue modulo a sufficiently large prime exceeding its largest step, then all sufficiently long APs at those steps hit it and the cost is arbitrarily small. Finally let r tend to infinity.

The same argument works verbatim for a sequence of finite gadgets when the sum of directional rigidity errors tends to zero with h,L and a fixed; just a bound tending to zero for the maximum individual error without controlling their sum does not justify the repair. Fixed-period abelian actions are included when all allowed translations act on the A-partition with common exponent L $thene(d) = 0$. This is a sufficient mechanism for amplification, not an automatic property of arbitrary obstructions.

Uniform amplification from a nearly extremal short witness **Proposition.** Suppose A is an h-term D-avoider in a stationary system, of mean a. Fix d_0 in D. Put $A_j = T^{-jd_0}A$ for $0 \le j < h$, and G={x: exactly $h - 1$ of the A_j contain x}. Write $g = \mu(G)$. For every positive r,

$$b_{h^r}(D) \ge g^r(1 - h^{-r}),$$

$$g \ge ha - (h - 2).$$

Here b_K is the supremum of upper densities of sets avoiding K-term D-APs. In particular the estimate is independent of the number, size or period of the allowed steps. For $h = 2$, one has exactly $g = 2a$, yielding $b_{2^r}D>=(2a)^r - a^r$.

Proof. Since d_0 lies in D, the sum of the indicators of the A_j is never h a.s. It is at most $h - 2$ off G, giving the estimate on g. On G label x by the unique index $m(x)$ of the missing set. On any h successive good points of a d-orbit for d in D all h labels occur: otherwise a translate A_j contains that h-progression, contrary to its avoidance. Thus along any run of good points the labels repeat with period h wherever that comparison is defined.

Take independent stationary copies at root scales 1,h,...,h^{r-1}, each sampled using a uniform independent phase modulo its scale as in the preceding section. Accept a position iff all r rows are good and their labels are not all a prescribed least frequent label. By independence the accepted process has mean $g^r - \mu(G \cap \{m = i\})^r \ge g^r(1 - h^{-r})$. If an h^r-term progression is all accepted, then all its rows are good throughout. In row s, on subsequences with index step h^s, all h successive label values occur and the label repeats after h such moves. Starting at time 0, choose a digit j in $[0, h)$ and add jh^s to attain label i in row s. All earlier chosen labels persist when a higher digit is added, because this amounts to a multiple

of h steps of the earlier row on its stride, inside the given all-good progression. Proceeding from $s = 0$ to $r - 1$ yields an index $\le h^r - 1$ with all labels i, a contradiction. Stationary correspondence yields the density conclusion.

Consequence. For every fixed h and target $\delta < 1$ there are r and $\eta > 0$ such that any finite or infinite F with an h-avoider of mean $> 1 - 1/h - \eta$ admits an h^r-avoider of mean $> \delta$. Choose r so $1 - h^{-r} > \delta$; then $g \ge 1 - h\eta$ may be chosen arbitrarily close to 1. In particular a counterexample to the all-density question cannot be constructed by the Griesmer near-half pair-gluing criterion above: the desired individual finite gadgets with pair-avoiders arbitrarily close to density 1/2 cannot force arbitrarily long APs at any fixed $\delta < 1$. This does not rule out other gluing schemes with low-density witnesses bounded away from the elementary cap $1 - 1/h$.

An intermediate example can always be made uniformly sparse If $b(D) > 0$, choose a positive-measure ergodic event A missing some h-term D-progression (for all d in D). Huang–Shao–Ye, *Nil Bohr-sets and almost automorphy of higher order*, §2.2, following their Theorem 2.2.3 [Bergelson–Host–Kra], https://arxiv.org/pdf/1407.1179#page=19, proves that the positive h-return set contains $B \setminus Z$ for a nil_{h-1}-Bohr_0 set B and an exceptional Z of zero upper Banach density. Thus $D \cap B \subset Z$. By Alweiss–Farhangi Theorem 1.8 the complement of B has $b = 1$. The independent product bound consequently gives $b(D \cap B) = b(D)$. In particular it would suffice to decide Question 6.1 for sets of zero upper Banach density. Conversely a set having positive upper Banach density in every finite-order nil-Bohr_0 set is fully Szemerédi-large.

Uniformly convergent powers Let T be a measure-preserving homeomorphism of a compact metric probability space. Suppose, as $d \to \infty$ in D, T^d converges uniformly to an invertible probability-preserving aperiodic map S (in particular to T^m with $m \neq 0$ and an aperiodic original action). Then $b(D) = 1$.

For any $\eta > 0$, take a Rohlin tower for S of height $h \gg 1/\eta$ covering all but $\eta/3$ of the space. Its levels other than the base have total measure exceeding $1-2\eta/3$, and h consecutive iterates cannot stay within the union of these levels: each non-top level advances to the next and no point can enter a non-base level from outside its unique preceding level. Take compact cores, of total measure exceeding $1 - \eta$, and write their union as K. The family of h points $x, Sx, \ldots, S^{h-1}x$ cannot lie in K for any x; the disjoint cores can be chosen within genuine tower levels after removing an invariant null set used in the measurable tower. By compactness this exclusion persists for any homeomorphism sufficiently uniformly close to S. Therefore the high-measure core K is an h-avoiding event for all sufficiently late d in D. Finite exceptional directions cost arbitrarily small density via the sparse periodic sieve.

The same reasoning permits any countable compact cluster family of aperiodic invertible transformations S_i, if the powers of T along D are relatively compact in the uniform topology with just these cluster maps. Allot $\eta 2^{-i}$ to S_i in the same tower construction. The neighborhoods where the resulting cores work cover the compact cluster family, hence finitely many suffice. All late T^d lie in their union. Intersect independent stationary witnesses for the finitely many corresponding step sets, with combined loss $< \eta$. This uses uniform convergence of transformations on the compact model, not merely convergence in measure or strong convergence of Koopman operators. Results producing uniformly rigid models on subsequences do not imply this hypothesis for the original D, see

https://www.impan.pl/shop/en/publication/transaction/download/product/91911.

Three-color pair obstructions If $Cay(Z, \pm D)$ is properly 3-colorable, there is a circle character mapping all $d \in D$ to $[1/3, 2/3]$; thus $b(D) = 1$ by Bohr separation. See Theorem W and its stated provenance in https://github.com/decalion89/chromatic-number-of-the-plane#three-colours-and-characters-3-october-2026 (an unrefereed public research note). The underlying reasoning can be checked directly. Regard the colors as $Z/3$; give the directed edge $(x, x + d)$ the unique sign $e_d(x) \in \{-1, 1\}$ representing the difference in colors. Each commuting square has oriented sign-sum zero (it is even, in $[-4, 4]$, and divisible by 3). Hence the sign-sum along any relation in the generators D, including inverses, is independent of the basepoint in the subgroup they generate: shift the loop by each generator, using squares and canceling the two comparison edges. It is a multiple of 3, and its parity equals the length of the loop. Average signs along this subgroup with a translation-invariant mean. If s_d is the resulting mean, for a relation $\Sigma k_d d = 0$ its winding sum $\Sigma k_d s_d$ is the constant just computed. Consequently $3\Sigma k_d + \Sigma k_d s_d$ is divisible by six. The assignment $d \mapsto (3 + s_d)/6$ thus gives a character modulo one on the generated subgroup, in $[1/3, 2/3]$ on all d. It extends to Z by divisibility of the circle. A positive-measure pair-avoider alone is only a fractional-color bound; it does not imply this proper 3-coloring hypothesis.

D21. The universal finite-color obstruction family

Scope and status. Supporting proof and dependency material.

Fix c,h_0. On $H = \oplus_F Ze_F$ take $V = \bigcup_F Fe_F$, over all nonempty finite $F \subset N$ admitting a c-coloring of Z with no monochromatic h_0-progression at the steps F. Its c-coloring is immediate: choose a coloring in each coordinate and sum the (basepoint-normalized) coordinate colors mod$c$. Thus $b_H(V) \geq 1/c$.

For any integer D with such a c-coloring the $F_n = D \cap [1, n]$ are among the allowed rows. Restrict a given K-avoiding H-system to the e_{F_n}-axis for each n and pass to a weak limit of the resulting stationary binary laws. One obtains a K-avoiding D-system of the same density. Consequently $b_H(V) \leq b(D)$ for every such D. The same result holds with b_K in place of b for each fixed K. A solution $b_H(V) = 1$ for this one family would settle the density-one conclusion for every c-colorable h_0-obstruction simultaneously.

The universal test also covers bounded-support multidimensional gadgets. Suppose $W \subset Z^m$ is finite, each $v \in W$ has at most s nonzero coordinates, and a coloring χ of Z^m prevents monochromatic h_0-runs at all v. Take distinct huge primes p_j $muchlargerthan(s+1)h_0 \max_v |v_j|$, and CRT integer weights L_j congruent to 1 modulo p_j and 0 modulo every other p-coordinate. Then $\phi(v) = \Sigma L_j v_j$ is a nonzero integer, since some nonzero coordinate v_j survives modulo p_j. Pick $s+1$ choices of interval representatives of Z/p_j (cyclic cuts), mutually separated by more than the h_0-run span in each coordinate. Color n by the tuple of χ-values of the $s+1$ representative vectors of n modulo the p_j. On a run of step $\phi(v)$, each moving coordinate spoils at most one cut by wrapping; hence with at most s changing coordinates at least one representative vector follows an actual h_0-term v-line with no wrap in any coordinate. This gives c^{s+1} colors preventing all $|\phi(W)|$ runs. Any K-avoider for the corresponding universal c^{s+1}, h_0 integer row pulls along ϕ to a K-avoider in Z^m for W. By invariant-law compactness this upper comparison for universal b extends to any countable family W in $\oplus Z$ satisfying the same support bound and admitting a single such multivariate coloring. No multidimensional periodicity of χ is required.

Integer realization preserving a finite coloring everywhere Enumerate F_i. Choose an avoiding periodic χ_i:$Z \to Z/c$ of period p_i. Such a coloring exists by the finite forbidden blocks and the periodic path argument in finite-color-gluing.md [D26]. Fix $L = 3$ and $a \in \{1, 2\}$. Choose arbitrarily large bases b_i divisible by Lp_i, with $b_i > (h_0 - 1)L \max F_i$ and $\sum_i L \max F_i / b_i$ finite. Define $\lambda_1 = 1$, $\lambda_{i+1} = b_i \lambda_i$, and $D = \bigcup \lambda_i L F_i$. Use ordinary digits $x_i(n) \in [0, b_i - 1]$ for $n \geq 0$. Put $g_{i,a}(x) = \chi_i(\lfloor (x + a)/L \rfloor)$, extended periodically in x mod b_i. Color n by the pair of sums mod$c$

$$(\sum_i (g_{i,a}(x_i(n)) - g_{i,a}(0)))_{a=1,2}.$$

The sums are finite. On a step λ_i Lf, $f \in F_i$, and h_0 consecutive sites, all lower digits stay fixed, and the ith digit contributes a whole $\chi_i f$-run in either offset, even if it wraps through b_i. Above i, the upper integer increases by one at most once through this run. Each full carry digit $b_j - 1 \to 0$ leaves BOTH upper offsets unchanged: each floor at $b_j - 1$ gives b_j/L while at zero it gives 0, equal modp_j. The terminal upper digit $x \to x + 1$ at most changes

one offset, because $L \geq 3$ and the offsets are distinct. Choose the other offset. All its higher colors have remained constant throughout the run, while its ith color varies. This proves a global c^2-coloring with no monochromatic h_0-run, with no boundary deletions. Translation compactness extends it from N to Z.

The balanced-digit transfer in free-abelian-transfer.md [D06] can use these radices even when they are even: use balanced digits $[-\lfloor b_i/2 \rfloor, \lceil b_i/2 \rceil - 1]$, symmetric up to endpoints. Alternatively choose the bases as above and use the ordinary complete digit intervals on $[0, \lambda_{j+1} - 1]$, imposing interior margins on all tail digits for signed additions as in that proof. Beyond the highest digit the ordinary expansion is zero, which is at the lower endpoint of the digit range; to avoid a loss on infinitely many inactive digits one translates the balanced allowed digits into ordinary coordinates or works directly in a full periodic digit box at each truncation. For this particular V (only pure-coordinate directions and positive f) ordinary digits suffice without lower margins: keep upper buffers $x_i \leq b_i - 1 - (h-1)L \max F_i$ for all $i \geq m$. On the nonnegative prefixes $[0, \lambda_{j+1} - 1]$, proportion outside these buffers $\leq \sum_{i \geq m} (h-1)L \max F_i / b_i$ after taking $b_i > (h-1)L \max F_i$ on that tail. If a kept n is a start with pure positive step at such a tail row, its h sites have no carry. Pull back a random H-witness there, then remove early rows with sparse periodic sieves as in the transfer proof. This gives $b(D) \geq b_H(LV)$; the homomorphism $H \to Z$ gives the reverse. Finally $b_H(LV) = b_H(V)$ since LH≅H and all LH-actions can be induced to H (or restrict the digit-root shift action). For an explicit extension, coinduce an LH-witness on probability copies indexed by H/LH with a cocycle from representatives, and evaluate the event at the identity coset; LH acts there by its original action.

In particular if the universal $b_H(V) < 1$ for $h_0 = 2$ we would obtain a single integer D with a finite proper coloring and with $b(D) < 1$, which forces Bohr recurrence of D. This would answer the classical Katznelson–Veech question negatively. If $b_H(V) = 1$ for all c,h_0, then any putative intermediate counterexample must be chromatically large.

Three pair colors are insufficient For every finite F admitting an edge-separating circle coloring ϕ of its integer distance graph with separation $a > 1/4$, there exists $\alpha \in T$ with $||d\alpha|| \geq a$ for every $d \in F$. Indeed put $b_d(x)$ equal to the unique lift in $[a, 1-a]$ of $\phi(x+d) - \phi(x)$. The difference of the two sums along the two sides of a commuting square is an integer of absolute value at most $2(1-2a) < 1$, hence zero. Restrict x to the single coset $\langle \mathrm{F} \rangle$, put $G = Z^F$, and $\pi(e_d) = d$. The lifted edge functions integrate to a real height on G, since squares commute and reversed steps backtrack. For each $k \in \ker \pi$ its height increment is an integer independent of the basepoint: the edge data are unchanged by k. Let v_d be the translation-invariant mean of b_d on that coset, taking common Cesàro subsequential limits for bounded translates. Averaging a path for k now gives $\Sigma k_d v_d \in Z$. Thus $d \mapsto v_d \bmod 1$ defines a homomorphism on $\langle \mathrm{F} \rangle$ and, since T is divisible, extends to Z. Every v_d lies in the required interval. The same conclusion for infinite sets follows by compactness of circle frequencies over finite subsystems. The square-winding ingredient is related to Krebs and Sankar, Section 3.3; no priority claim for the multiplier criterion.

A proper three-coloring gives $a = 1/3$ by using the three equally spaced phases. All these row frequencies define a single character ξ of the free H, uniformly separating every $v \in V$ from zero. To verify $b_H(V) = 1$, take one interval U of length ϵ. Choose Q divisible by all positive denominators $\leq 2/\epsilon$, and lift ξ to a character ζ with $Q\zeta = \xi$ on the free H. Each possible phase $z = \zeta(v)$ lies in the compact set with $||Qz|| \geq 1/3$, so any rational such z has denominator $> 2/\epsilon$. Every translate of every corresponding z-orbit hits the interior of

U: irrational orbits are dense, and rational orbits form a mesh of gap $< \epsilon/2$. Compactness jointly in starting point and z gives a common finite hit time. The complement of U under this Haar-circle action is thus a near-full avoider. More generally the same argument applies to all rows of circular chromatic number < 4 using common separation $\geq 1/4$, even if their individual best margins decrease to $1/4$. It does not treat arbitrary proper four-colorings, since a tight square can have winding ± 1 at equality.

Higher differences, a strong lift criterion and the binary triple boundary There is a precise higher-order version of the flat-lift argument. Let $s \geq 1$ and let a bounded circle function $\phi{:}Z \to T$ have all s-th differences $P_d\phi = (T^d - I)^s\phi$, $d \in F$, in a single closed arc of length $w < 2^{-s}$, represented by b_d in one common real interval of length w. Then for some $\alpha \in T$ all the phases $d^s\alpha$ lie in that same arc. Nontrivial separation means an arc disjoint from zero. It suffices to treat finite F and then use compactness for infinite F. First normalize $\gcd(F) = g$ by restricting to the coset gZ and using $u = T^g$, writing $h_d = d/g$. The integers $P_e b_d - P_d b_e$ vanish identically: their norms are at most $2^s w < 1$ after centering both $b's$ in the common interval, and the two circle differences commute. For $Q_d(u) = (1 + \cdots + u^{h_d - 1})^s$, there are integer polynomials A_d with $\Sigma A_d Q_d = 1$. For $s = 1$ this follows by Euclidean subtraction of the monic expressions $u^h - 1$, which gives $\gcd u - 1$ as an integer polynomial combination of them; divide by $u - 1$. For general s, expand a sufficiently large power of the $s = 1$ identity, so every summand contains an s-th power of one generator. Let $f = \Sigma A_d(u) b_d$. Then $f \equiv (u-1)^s\phi \bmod 1$. For every d the bounded integer function $e_d = b_d - Q_d(u)f$ is annihilated by every P_e (commute and use $\Sigma A_i P_i = (u-1)^s$). A bounded sequence annihilated by P_e is constant on each e-progression, since the restriction there is a bounded polynomial; hence e_d is constant on the coset. Taking an invariant mean yields $v_d = Eb_d \equiv h_d^s Ef \bmod 1$, proving the claim after dividing the phase by g^s.

For $s = 2$ a binary $c(n)$ with no monochromatic triple does give $\Delta_d^2(c/3) \in \{1/3, 2/3\}$, but the containing arc has length $1/3$ rather than $< 1/4$. The integer curvature can genuinely be nonzero: take a 3×3 array of bits with rows $110, 001, 011$. All rows and columns are nonconstant, but taking the lift in $[1/3, 2/3]$ separately for row and column second differences gives different mixed differences (difference 1). Thus the pair-circle theorem cannot be applied to the binary-triple family by this identification alone. There is nevertheless an exact necessary spectral restriction. For $f = (-1)^c$ and any stationary orbit law, with $\rho(n) = \int f \cdot T^n f$, the probability of monochromatic triples equals $(1 + 2\rho(d) + \rho(2d))/4$ by the elementary three-sign identity. It is therefore a positive-definite nonnegative function vanishing on F and its spectral measure has mass at least $1/4$ at zero. This gives a strict vector 4-coloring obstruction, not a finite character separating F.

Every binary 3-avoiding coloring is also nonconstant on each triangle of its difference Cayley graph. If 0,d,e have the same color and d,e,$e - d$ are allowed up to sign, then $-d$,$-e$,$2d - e$,$2e - d$ have the opposite color. Consequently the d-line $(-e, d - e, 2d - e)$ and e-line $(-d, e - d, 2e - d)$ force $d - e$ and $e - d$ back to the original color. The triple $(e - d, 0, d - e)$ is then forbidden. In particular five vertices with every pair difference allowed already force a binary monochromatic triple.

The sharp converse to the half-circle $1/4$ test is false. On $Z/53$ the explicit cyclic binary word

00100100110110110100100101101001100110100101101101001

has zero monochromatic triples at each of the residues 1,2,7,14, hence also at their negatives. Directly enumerating the 53 characters gives $\max_{\ell} \min_{d=1,2,7,14} |\ell \mathrm{d}|$*{53}=13, attained e.g. at ℓ=13, where* $|\cdot|\{53\}$ is least absolute residue. For the integer D comprising all lifts of these residue classes, an irrational circle frequency has zero gap by density. For a rational α with 53α of order q, the infimum over a residue class $d + 53Z$ equals $||q\alpha d||/q$, and $q\alpha{=}\ell/53$ modulo one; hence the overall best gap is exactly $13/53 < 1/4$. Distinguish the four bare integers, which do have a 1/3 frequency. This refutes a sharp uniform converse, not all positive rank-one gaps. The 53-digit word and character bound are independently verifiable by four cyclic scans.

D22. A reduction to weakly mixing zero-entropy systems

Scope and status. Supporting proof and dependency material.

For every $D \subset N$ there is a single $D' \subset N$ with $b(D') = b(D)$. Its supremum can be approached by avoiding events in invertible weakly mixing systems of zero entropy. The set D' depends only on D, not on the witnesses used to approach the supremum. If Question 6.1 fails, it therefore fails on a set whose positive avoiding density is witnessed within this much smaller dynamical class. This is not a weak-mixing modeling theorem for D itself.

Staircase heights and the upper bound Write $F_j = D \cap [1, j]$. Use a rank-one interval cutting-and-stacking transformation R. Alternate pure stacks and three-copy Chacon stacks. The height at the beginning of pure stage j is H_j. Cut into B_j equally wide copies and stack with no spacers, obtaining height $B_j H_j$. Choose $B_j \to \infty$ so fast that $\sum_j j/B_j < \infty$. The next stage stacks three copies of the pure tower, placing one spacer between the second and third; its height is $H_{j+1} = 3B_j H_j + 1$. The usual rank-one interval construction, normalized to total measure one, gives an invertible ergodic probability transformation whose towers increase to full measure and whose level partitions generate. The total spacer measure is finite by the height growth. For instance initial interval length one, with later levels having the successive cut widths, gives a finite total to normalize. Put

$$D' = \bigcup_j H_j F_j.$$

For any stationary K-avoider on D', its law under the H_j-th power is a K-avoider on F_j with the same mean. A weak limit of the resulting binary laws K-avoids D. Thus $a_K(D') \leq a_K(D)$ for every K, giving $b(D') \leq b(D)$.

An integrable cocycle Take any ergodic invertible (X, μ, T), with an h-avoiding event A for D. On the rank-one base let $e_j(b)$ be the index 0,...,$B_j - 1$ of the subcolumn within the pure stage-j stack, and set it to zero off that stack. Then

$$c(b) = \sum_j (e_j(Rb) - e_j(b))$$

converges absolutely in L^1 and almost everywhere. In fact the L^1 variation of the j-th term is at most $2/H_j$: there are $B_j - 1$ internal increases by one and the downward change at the summit is bounded by $B_j - 1$, each on at most one new-level width. Positions outside cannot enter except at the base, of digit zero. In particular c is integer-valued with integral zero. Define

$$S(b, z) = (Rb, T^{c(b)} z)$$

on the product probability space.

At a pure stack, moving H_j levels within the stack from any of its first $B_j - 1$ copies returns to the same lower-stage position, increases e_j by one and leaves all higher digits fixed. Telescoping the convergent sum gives $c_{H_j} = 1$ there. These sets have measure tending to one. Moreover R^{H_j} converges strongly to the identity, since membership in every fixed earlier stage level is preserved on these sets, and those levels approximate arbitrary base indicators. Thus the Koopman powers S^{H_j} converge strongly to the map $V(b, z) = (b, Tz)$. Every S-invariant function is V-invariant, hence base-measurable by ergodicity of T, and then constant by ergodicity of R.

For weak mixing, let f be an S-eigenfunction of eigenvalue λ, normalized to $|f| = 1$ by ergodicity. If $G = B_j H_j$ is the height before a Chacon stack, then R^G on the first copy and R^{G+1} on the second copy take every point to the identical level of the following copy. Every active digit stays fixed and so does the fiber. Each good domain has measure tending to $1/3$. Any earlier level function times a function on X is unchanged on both respective domains; approximation in L^2 gives $\lambda^G \to 1$ and $\lambda^{G+1} \to 1$. Hence $\lambda = 1$, proving weak mixing.

The same weakly mixing system is rigid: consecutive active powers S^{H_j} and $S^{H_{j+1}}$ converge strongly to the same unitary V, so $S^{H_{j+1}-H_j}$ converges strongly to the identity. These positive difference heights tend to infinity.

The extension has zero entropy, even if T has positive entropy. The base is rank-one of zero entropy. For a finite fiber partition Q, conditional on the full base, its first n observations are translates by T at the partial sums $c_t = \sum_{s<t} c(R^s b)$. With $M_n = \max_{t \le n} |c_t|$, the corresponding conditional word entropy is at most $2M_n + 1 \log|\mathrm{Q}|$ integrated over the base. We have $M_n/n \to 0$ almost surely by Birkhoff, since c has mean zero. The convergence is also in L^1: $M_n/n \le n^{-1} \sum_{s<n} |c| \circ R^s$, whose averages are uniformly integrable (truncate $|c|$ and bound the L^1 tail uniformly). Finite product partitions therefore have zero relative entropy. The entropy addition formula over the base gives $h(S) = 0$.

The avoiding event and the reverse bound Fix a starting stage J and retain only points whose base lies in its pure supertower. This costs a quantity tending to zero with J. For every $j \ge J$ delete the top $h \max(F_j)$ subcolumns of the pure stage-j stack $emptyF_jneedsnocut$. The total further mass loss is at most $h \sum_{j \ge J} j/B_j$. Mark a retained point if $z \in A$. A retained base start lies in each later pure tower because tower sets are nested. For $d = H_j f$, $f \in F_j$ and $j \ge J$, all of its first h sites stay inside that same pure stack. At each move all dependence on smaller and larger stage digits cancels, while e_j increases by f. Thus the successive fibers are z,$T^f z$,...,$T^{(h-1)f} z$. Almost surely these sites cannot all be marked.

The finitely many early steps can be forbidden inside this same weakly mixing system at arbitrarily small cost: for each such d take a Rokhlin tower for S^d of sufficiently large height and measure arbitrarily close to one. Add its base level and its complement as a hole. Within at most height+1 iterates every start hits that hole, since inside the tower the only entrance to a nonbase level is from the preceding level. Use the union of the finitely many holes and increase the common length. The remaining mean is at least $\mu(A)$ minus arbitrarily small error as J grows. Taking suprema proves the theorem, and weak mixing and zero entropy were properties of the ambient system, unaffected by the deletions.

One ambient system suffices for all the approximations. Select countably many ergodic input systems with avoiding events whose measures approach $b(D)$. On their countable prod-

uct with the diagonal transformation take an ergodic component of product measure. Almost every component projects to the prescribed measure on each coordinate: these marginals are barycenters of invariant measures and are ergodic, hence extreme. Intersect the countably many conull conditions. The resulting ergodic joining T has all the original events as factors. Using this T once in the cocycle construction gives one rigid weakly mixing zero-entropy S containing avoiding events on D' of densities arbitrarily close to $b(D)$.

The argument uses only ordinary rank-one cutting and stacking, Rokhlin towers, the ergodic theorem, and the entropy addition formula. The active return and spacer return relations give separate checks of ergodicity and the eigenvalue conclusion.

D23. The finite-color hierarchy

This supporting construction supplies the topological and finite-color inputs used in the density-one hierarchy. Its density conclusion is proved separately in R139. The proof below retains its stated polynomial return-time input.

Statement and the induction property

Write $\mathbb{N} = \{1, 2, \dots\}$. A set $D \subseteq \mathbb{N}$ is k-large if, for every coloring of $\mathbb{N}$ in at most k colors and every $h \geq 2$, there is a monochromatic progression $b, b + d, \dots, b + (h-1)d$ with $b \in \mathbb{N}$ and $d \in D$. The definition is due to Brown, Graham and Landman [16]. Farhangi explicitly asks about successive numbers of colors in [17, Section 2, after Definition 2.10].

Theorem D23.1. *For every $k \geq 1$ there is a k-large set D_k and a coloring of $\mathbb{N}$ in $k+1$ colors with no monochromatic D_k-progression of length $H_k = 400 \cdot 2^k$.*

For a finite-dimensional nilsystem (Y, T), a point y, and an open neighborhood V of y, put $N(y, V) = \{n \in \mathbb{N} : T^n y \in V\}$. Say that D has property P_s if for all these data, every coloring of $N(y, V)$ in at most s colors has arbitrarily long monochromatic D-progressions. The one-point system shows that P_s implies s-largeness. Nilrotations are invertible, and the orbit closure of each point is itself a minimal nilsystem [21]. We will use this fact also for products and powers of translations.

We shall construct D_s with P_s and an $(s+1)$-color obstruction. The construction can start with Alweiss's all-orders nil-recurrent example and its binary obstruction as in the preceding odd-step argument. To make the induction self-contained, we give the base case by the same construction as the induction step.

A finite-dimensional set of steps

Fix the size $q \geq 2$ of the desired obstructing palette. Put

$$Q = 100q, \qquad a = Q^{-1}, \qquad \eta = (10^6 Q^2)^{-1}.$$

In $\mathbb{T}^M$, let O_M consist of the vectors with centered coordinates θ_j for which some i satisfies

$$|\theta_i + a| < \eta, \qquad |\theta_j| < \eta \ (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2a.$$

This is an open set. Fix $1, \alpha_1, \dots, \alpha_M$ rationally independent, for example by taking α_j to be the square root of the jth prime. Write $G_M = \{d \in \mathbb{N} : d\alpha \in O_M\}$. The distinguished small velocity and the small total variation in the other coordinates are taken from Alweiss [18, Section 2].

Lemma D23.2. *Given $b \geq 0$, for all M sufficiently large in terms of b, q, every connected subtorus of $\mathbb{T}^M$ of codimension at most b meets O_M.*

Proof. Write its tangent space as the kernel of a real matrix whose columns are $v_1, \dots, v_M$ spanning a space of dimension at most b. If that space is zero the assertion is immediate. Give it the norm with unit ball $\operatorname{conv}\{\pm v_j : 1 \leq j \leq M\}$. For any $\epsilon > 0$ the ball is covered

by at most $(1 + 2/\epsilon)^b$ balls of radius ϵ, by the volume packing bound. For M sufficiently large, one ball contains v_i and L further indexed columns. Write $\bar{v}$ for their average. Then $v_i - \bar{v} = \sum c_j v_j$ with $\sum |c_j| \le 2\epsilon$. Moving $c_i v_i$ to the left expresses $v_i = \sum_{j \ne i} t_j v_j$ with

$$\sum_{j \ne i} |t_j| \le (1 + 2\epsilon)/(1 - 2\epsilon), \qquad \max_{j \ne i} |t_j| \le (L^{-1} + 2\epsilon)/(1 - 2\epsilon).$$

Choose ϵ small and L large enough that these bounds are strictly below 2 and η/a, respectively. The vector whose ith entry is $-a$ and whose other entries are at_j is in the tangent space. Its reduction modulo one gives the desired point. □

Lemma D23.3. *Fix $h \ge 2$ and a dimension bound C. Take M as in Lemma D23.2 with $b = hC$. For every nilmanifold of dimension at most C, translation T, point y and open $V \ni y$, there is a syndetic set of positive integers $d \in G_M$ with $T^{jd} y \in V$ for $1 \le j \le h$.*

Proof. Take the joint orbit of $(y, \dots, y, 0)$ under $(T, T^2, \dots, T^h, +\alpha)$. The joint transformation is minimal on this orbit closure, which is a finite union of connected subnilmanifolds [21]. The component through the starting point projects onto $\mathbb{T}^M$. Indeed the entire closure projects onto the torus, its components are cyclically permuted, and their projections are translates of a single connected subtorus. A finite union of cosets of a proper connected subtorus cannot cover $\mathbb{T}^M$.

After changing basepoint, this component is the orbit of a connected rational subgroup L of the product of the ambient nilpotent group with the torus (or its universal cover). Its Lie algebra projects onto the M torus directions and the first h coordinates together have dimension at most hC. Thus $\operatorname{Lie} L$ contains a pure torus subspace of dimension at least $M - hC$, the kernel of projection onto the first h coordinates. Exponentiating and taking closure gives a connected subtorus of at least this dimension in the fiber over $(y, \dots, y)$. Closure is permissible because the component is closed. By Lemma D23.2 this fiber meets the open target $V^h \times O_M$. Returns to a nonempty relatively open set in a minimal compact system are syndetic. □

The coloring at a single stage

Set $F(n) = \sum_{j=1}^{M} \operatorname{Re}(e(n\alpha_j) - 1)$, where $e(t) = \exp(2\pi i t)$, and $J(n) = \lfloor qF(n) \rfloor$. On every $2Q$-term G_M-progression, J visits all residues modulo q. We include the estimate because the length must be independent of M. A d-step changes qF by at most $2\pi q(3/Q + \eta) < 1$. For an initial position n, choose $0 \le t < Q$ so that $\|n\alpha_i - t/Q\| \le 1/(2Q)$. Then $\|n\alpha_i + t\theta_i\| \le 1/(2Q) + Q\eta$, so the real part of the corresponding phase exceeds 0.99. Set $m = Q/2$. The second difference of F at the indices $t, t + m, t + 2m$ of the progression has distinguished term

$$\operatorname{Re}\big(e(n\alpha_i + t\theta_i)(e(m\theta_i) - 1)^2\big).$$

It exceeds $3.96 - 4\pi Q\eta - \pi^2 Q^2 \eta^2$, since $|e(m\theta_i) + 1| \le \pi Q\eta$. The total error from the other coordinates is at most $\pi^2 Q^2 \sum_{j \ne i} \theta_j^2 < 2\pi^2 Q\eta$. The second difference exceeds 3.9, giving a range of width greater than 1.95 for F. The floors of qF move by at most one, and therefore visit at least q consecutive integers.

Let $u = q - 1$. A chart at an integer J injects the input labels $\{0, \dots, u - 1\}$ into $\mathbb{Z}/q\mathbb{Z}$. For input i, take the integer $t \in [J - u, J - 1]$ congruent to i modulo u and set $\iota_J(i) = t - 1 \bmod q$.

The missing color is $J-1 \bmod q$. On passing from J to $J+1$, exactly one input changes output, moving to the previously missing color. In particular

$$|J-J'|\le 1, \quad \iota_J(i)=\iota_{J'}(i') \quad \Longrightarrow \quad i=i'.$$

Suppose $B\subseteq\mathbb{N}$ has an obstructing u-coloring f with forbidden length $H\ge 2$, and let

$$B_M=\{d\in B:\textstyle\sum_{j=1}^{M}\|d\alpha_j\|<(4\pi qH)^{-1}\}, \qquad S_M=G_M\cup B_M.$$

Then $n\mapsto \iota_{J(n)}(f(n))$ avoids monochromatic S_M-progressions of length

$$L=\max(2Q,H).$$

Every G_M-progression of this length encounters a chart omitting each possible output. On the first H sites of a B_M-progression, the values of qF differ by less than $1/2$. Their floors lie in at most two consecutive charts, so constant output would force constant input. The bound L is independent of the torus dimension, the frequencies, and the number of colors in a lower-bound test.

Forcing one additional color on a prescribed window

Assume that B has P_s and a blocking coloring in $s+1$ colors. In the previous section take $u=s+1$, $q=s+2$. For the initial step $s=0$, take $B=\varnothing$, $H=2$ and constant f. Only the first case of the proof below can then occur.

Lemma D23.4. *Given h, C, choose M by Lemma D23.3. The set S_M forces a monochromatic h-term progression for every $(s+1)$-coloring of $N(y,V)$, for every window on a nilmanifold of dimension at most C.*

Proof. Extend the coloring arbitrarily to all of $\mathbb{Z}$. In the forward orbit limit set of the colored sequence joined to y, choose a nonempty minimal subsystem X. It projects onto the orbit closure Y_0 of y, since that closure is minimal. Write

$$X \xrightarrow{\ \pi\ } Z \xrightarrow{\ \rho\ } Y_0$$

for the factorization through the maximal infinite-step pronilfactor of X. Denote $V\cap Y_0$ by V_0. Let C_1 be the clopen cylinder of the first color in X.

Suppose first that $\pi(C_1)$ contains $W=\rho^{-1}V_0$. Then $U=C_1\cap\pi^{-1}W$ is open, and $\pi(U)=W$ is open. For an open set A write $\mathcal{R}_h(A)$ for the integers d with $\bigcap_{t=0}^{h-1}T^{-td}A\neq\varnothing$, using the transformation on the relevant system. On Z one has $\mathcal{R}_h(W)=\mathcal{R}_h(V_0)$ by surjectivity. Theorem A of Glasscock, Koutsogiannis, Le, Moreira, Richter and Robertson [20], for the distinct linear forms $n,2n,\dots,hn$, says that $\mathcal{R}_h(W)\setminus\mathcal{R}_h(U)$ is not piecewise syndetic in $\mathbb{Z}$. These forms give the same differences as the forms $0,n,\dots,(h-1)n$ by invertibility. Lemma D23.3 supplies a set of positive differences in $G_M\cap\mathcal{R}_h(V_0)$ syndetic on the positive half-line, hence piecewise syndetic in $\mathbb{Z}$. Some such difference belongs to $\mathcal{R}_h(U)$. We therefore have h visits of the first color inside the prescribed window.

Otherwise, $W\setminus\pi(C_1)$ is a nonempty open set because $\pi(C_1)$ is compact. It contains the full inverse image of a nonempty open set V' in a finite nilfactor Z' of Z, by the inverse limit description of the pronilfactor. Above V' only the remaining s colors occur and every point

projects into V_0. This case is impossible for $s = 0$. Pick a point of X over $z' \in V'$. Along its orbit use the s colors on visits to V'. Intersect these visits with the times n for which $n\alpha$ belongs to a neighborhood O of zero in $\mathbb{T}^M$. Choose O so small that $\sum_j \|v_j\| < (4\pi qH)^{-1}$ for every $v \in O - O$. The simultaneous visits form the return-time set of $(z', 0)$ to $V' \times O$ in a finite-dimensional product nilsystem. Apply P_s of B to obtain a monochromatic progression of length h. Its first two torus coordinates differ by $d\alpha$ with $d \in B$, so $d \in B_M$.

In either case the resulting configuration occurs in X, with all its Y_0-coordinates in the open set V_0. The defining finitely many color conditions and window conditions are open in the joined system. Since X lies in the forward orbit limit set, the original coloring has the same configuration at a positive starting position. □

Finite extraction and the integer construction

We record why one can exhaust the required nilsystem tests by countably many compact parameter families. A connected simply connected nilpotent group N with lattice Λ belongs to one of countably many isomorphism classes of such pairs. Indeed its rational Malcev algebra has rational structure constants and the lattice has a finite generating list of rational points in that structure [22]. The lattice-preserving automorphisms are also countable, since they are determined by the images of lattice generators. For each pair take all its lattice-preserving unipotent automorphisms A. On N/Λ the affine maps $R_b : z \mapsto bA(z)$, $b \in N$, are niltranslations in the presentation

$$(N \rtimes_A \mathbb{Z})/(\Lambda \rtimes_A \mathbb{Z}).$$

For clarity, this semidirect group is nilpotent. On the Lie algebra of N, the unipotent automorphism A embeds into the one-parameter group $A^t = \exp(t \log A)$ of automorphisms, as the bracket identity is polynomial in t and holds at all integers. The Lie algebra of $N \rtimes_{A^t} \mathbb{R}$ is solvable. In its adjoint representation the generator of $\mathbb{R}$ acts nilpotently, and each element of the nilpotent ideal $\operatorname{Lie} N$ acts nilpotently on the full algebra. Lie's triangularization theorem over $\mathbb{C}$ makes all these matrices simultaneously strictly upper triangular. Hence the connected simply connected semidirect group is nilpotent, as is its subgroup $N \rtimes_A \mathbb{Z}$.

These presentations suffice after restriction to a finite-index set of times. In G/Γ the component of any starting point has a finite period m under the translation. Conjugating the translation by a fixed left translation puts the initial point in the identity component. This component is $G^\circ/(G^\circ \cap \Gamma)$. If a is the translating element, write $a^m = b\gamma$ with $b \in G^\circ$, $\gamma \in \Gamma$. On the component its action is $b\,\mathrm{Ad}(\gamma)$. Pass to the simply connected cover N of G° and the inverse-image lattice. Conjugation lifts to a lattice-preserving automorphism A. It is unipotent because iterated commutators with γ vanish in G, and differentiating the iterated commutator map at the identity gives a power of $A - I$. Lifting b gives the claimed affine presentation for the return map.

List tests consisting of a presentation (N, Λ, A), a positive integer m, a compact set of parameters b from a countable exhaustion of N, and a window pair $K \subset V$ in N/Λ, with K closed and V open. Use closures of small basis balls inside larger basis balls. Such a list contains, for any prescribed open window and point in it, a pair with the point in K and V inside that window.

Set $\lambda_1 = 1$. At stage j take the initial j entries of the list and retain those with $m \mid \lambda_j$. Use target length $h_j = j + 2$ and dimension bound C_j the largest dimension tested (include

the one-point test at every stage). Choose M_j by Lemma D23.4. For each test the parameter space is

$$\{b \text{ in its compact set}\} \times K \times \{1, \dots, s+1\}^{\mathbb{Z}}.$$

Use the transformation $R_b^{\lambda_j/m}$. This exponent is already fixed. Lemma D23.4 applies at *every* parameter, including those for which the transformation is not minimal on the whole manifold. For fixed positive start a_0 and $d \in S_{M_j}$, monochromaticity and the conditions

$$(R_b^{\lambda_j/m})^{a_0+td} x \in V \quad (0 \le t < h_j)$$

define an open subset of parameter space. These sets cover it. Finite subcovers over the finitely many tests supply a finite nonempty packet $F_j \subset S_{M_j}$ that works throughout. No uniform return time in the arbitrary symbolic extension is assumed. Compactness is applied after the pointwise assertion of Lemma D23.4.

Choose an integer radix $b_j > 2L \max F_j$, large enough and divisible enough that $(j+1)! \mid \lambda_j b_j$, and put

$$\lambda_{j+1} = \lambda_j b_j, \qquad D_{s+1} = \bigcup_{j \ge 1} \lambda_j F_j.$$

This is a countable inductive construction by finite choices. One may take the least successful finite initial truncation of S_{M_j} and the least admissible radix after fixing the tests and rationally independent frequencies.

To see P_{s+1}, take any target window and coloring of its visiting times. Use the component return map with period m as above. Select a test containing its return translation parameter, with the initial point in K and the test window inside the original one on the component. Choose j so large that the test appears, $m \mid j! \mid \lambda_j$, and h_j is at least the desired length. Extend the coloring arbitrarily to all positive times and read it at the multiples of λ_j (extend to negative indices arbitrarily for the compact test). The finite packet gives a monochromatic progression at positive indices, all of whose visits under $R_b^{\lambda_j/m}$ lie in V. Multiplication of the indices by λ_j gives precisely the required visits in the original nilsystem with common difference in D_{s+1}.

For the upper bound take the blocking coloring c_j for S_{M_j} from Section 3, with its outputs in $\mathbb{Z}/q\mathbb{Z}$. Writing $z_j(n)$ for the mixed-radix digit of n at place λ_j, define the q-coloring

$$c(n) = \sum_{j \ge 1} \big(c_j(1 + z_j(n)) - c_j(1)\big) \pmod q.$$

The sum is finite at each n. On $2L$ sites of step $\lambda_j f$, $f \in F_j$, the lower digits stay fixed. The higher prefix $\lfloor n/\lambda_{j+1} \rfloor$ changes at most once over the sites, since $(2L-1)f < b_j$. On at least L consecutive sites both the higher and the lower digits are fixed and the jth digit increases by f each time. Monochromaticity would contradict the choice of c_j. Starting with $H_0 = 2$, the lengths obey $H_{s+1} = 2\max(200(s+2), H_s)$. Thus $H_k = 400 \cdot 2^k$ for $k \ge 1$. This proves the induction and Theorem D23.1.

The first strict separation in a simpler form

Let E be a set with P_1 and a binary obstruction f of length H, for instance D_1 above. One can also take the original fixed-parameter Alweiss example [18, 19]. Then $S = (\mathbb{N} \setminus 3\mathbb{N}) \cup 3E$ is 2-large. Indeed $3E$ has P_1, by applying P_1 to T^3. For a binary coloring take a minimal

subsystem and its maximal pronilfactor as in Lemma D23.4, now with the one-point target window. If the first color projects onto the whole factor, Theorem A in [20] supplies a monochromatic progression at some positive step outside $3\mathbb{N}$, since those steps are piecewise syndetic. Otherwise the second color contains the full inverse image of a nonempty open cylinder, and $3E$ supplies the progression there.

On the three residue classes use the palettes $\{1,2\},\{2,3\},\{3,1\}$, choosing from the appropriate pair at $3n+r$ by $f(n+1)$, $0 \le r < 3$. A difference prime to three visits all three palettes in its first three sites. A difference $3e$, $e \in E$, stays in one residue and gives quotient step e. Thus this three-coloring excludes length $\max(3, H)$. Simply iterating the same modular enlargement would collapse successive moduli into one. The finite-dimensional construction above allows one to force progressions inside prescribed nilsystem windows at the next induction step.

D24. Uniform binary avoidance for bounded signed power sums

Scope and status. Result with proof.

Result. For fixed integer base $q \geq 2$ and fixed coefficient/count bounds C,t, all positive steps representable as sums of at most t signed terms a q^e admit a binary coloring avoiding some common progression length. The same finite-trajectory argument gives, for each fixed r, a coloring in which every sufficiently long eligible progression meets every color; consequently this step family has unrestricted avoidance capacity one.

Scope. Bounds are fixed independently of the exponents and their gaps. Includes cancellations, repeated exponents, fixed dilations, and finite unions of bounded-rank q-arid languages after clearing denominators. The binary obstruction is not an assertion about arbitrary automatic sets.

Qualifications. The parameters q, t, C are positive integers, with $q \geq 2$; a real coefficient bound may be rounded upward. The density-one strengthening is proved in D25 from the same trajectory estimate.

Claim. Fix integers $q \geq 2$, $t \geq 1$, and $C \geq 1$. There is a binary coloring of the nonnegative integers and a length K such that no K-term monochromatic AP has positive step expressible as a sum of at most t numbers a q^e, where $e \geq 0$ and a is an integer of absolute value at most C. Cancellations, repeated exponents, and arbitrarily widely separated exponents are allowed.

Parameters and coloring Take a sufficiently large integer P, with q, t, C fixed. Set $m = 2t + 5$, and choose m distinct odd primes p_j between P and $2^m P$ by Bertrand's postulate. Let M be their product and set $K = (2t + 6) \max_j p_j$. Put

$$g = \lceil \log_q(4tCK) \rceil + 2, \qquad H = (t + 1)g, \qquad B = q^H.$$

Then B is polynomially bounded in P, $B > 4tCK$, and $(t+1)t^2C^2B^2 < M$ for all sufficiently large P, because $m > 2(t + 1)$.

For each $s = 0,\ldots,H-1$ partition the base-q digits of n into blocks with cuts at exponents s,$s + H$,$s + 2H$,..., and at 0. Ignore a repeated 0 cut. Thus there may be one initial short block and all later blocks have width H. Let $Q_s(n)$ be the sum of squares of the corresponding integer block digits reduced modulo M. Trailing zero digits contribute nothing. Our binary coloring will be $b((Q_s(n))_s)$, where b is one binary coloring of the finite space $(Z/M)^H$.

We show i each eligible K-progression visits at least $P/2$ distinct points of this finite space; (ii) there are only $\exp(O_{q,t,C}((\log P)^2))$ possible ordered K-tuples of visited points. A random b is constant on a given tuple with probability at most $2^{1-P/2}$, so the union bound then proves the claim.

Coordinate formulas Fix n, an eligible positive d, and a chosen signed representation of d. For any digit cut at place $Q = q^e$, write $d = d_{<Q} + QD_Q$ by splitting the represented summands at exponents $< e$ and $\geq e$. These are signed integers, not ordinary residues. Then

$$\lfloor (n+id)/Q \rfloor = \lfloor n/Q \rfloor + iD_Q + E_Q(i),$$

$$E_Q(i) = \lfloor (n \bmod Q)/Q + id_{<Q}/Q \rfloor.$$

Always |d_{$< Q$}|/$Q \le tC$. On any block with endpoints Q,$Q' = b_0 Q$, its digit trajectory is $u + iv + E_Q(i) - b_0 E_{Q'}(i)$, where u is its initial digit and v is the signed contribution of summands in this block after division by Q. Missing summands give $v = 0$. For a contiguous chunk of blocks containing no summand the whole packed integer changes from u to $u+E_Q(i)$ mod(Q'/Q). For the infinite tail above the last summand it changes to $u + E_Q(i)$ without a modulus.

Proof of i: a safe coordinate There is an s such that every represented exponent lies at least g below its next block cut in that coordinate. Each exponent excludes at most $g-1$ residue classes of s, and $H > t(g-1)$. At each actual cut Q the total contribution of lower summands satisfies |d_{$< Q$}|/$Q \le tCq^{-g}$. Hence $E_Q(i)$ is constant except for at most one jump over $0 \le i < K$.

Single out the $\le t$ blocks containing summands and also the initial short block, if any. Between them are runs of unsingled inert blocks, including the infinite tail. The boundary cuts of the singled blocks, together with the initial cut if needed, number at most $2(t+1)+1$. On a consecutive segment of at least $K/(2t+4)$ indices no indicated E_Q changes. On that segment each inert chunk is fixed, and each singled digit is affine in i with slope v. Hence $Q_s(n+id)$ is the reduction of an ordinary quadratic in i with leading coefficient $A = \sum v^2$. It is positive since $d > 0$, and $\le (t+1)t^2C^2B^2 < M$. Therefore some p_j does not divide A. The segment has at least p_j successive indices; modulo this odd prime the quadratic takes $p_j + 1/2$ distinct values. The full vectors therefore visit more than $P/2$ points.

Proof of (ii): the other coordinates cannot encode arbitrary tracks For one fixed s (safe or not), the words $E_Q(i) : 0 \le i < K$ over *all* possible cuts, d, and n number $O_{t,C}(K^4)$. Indeed the whole word is specified by (α, β) in $[-tC, tC]x[0,1)$, through $\lfloor \beta + i\alpha \rfloor$. There are $O_{t,C}(K^2)$ relevant lines $\beta + i\alpha = j$, so their planar arrangement, including lower-dimensional faces to retain endpoint conventions, has $O_{t,C}(K^4)$ faces.

Use the same singled-block and chunk decomposition. A singled digit's entire square trajectory mod M has only polynomially many in P possibilities: in the block formula above specify u,v,b_0 mod M and two E words. No absolute position data are needed. Each inert chunk contains only full width-H blocks (the short initial block was singled out). Write its initial integer as $u = u_0 + Bu_1$, $0 \le u_0 < B$. Its lowest digit becomes $u_0 + E_Q(i)$ mod$B$. Since $|E_Q(i)|$<=tCK+1 $< B$, the higher portion is one of $u_1 - 1$,u_1,$u_1 + 1$, reduced cyclically modulo its length in the case of a finite chunk. For the infinite tail only nonnegative actual values occur. The contribution of the higher digits to the sum of squares is just one of three constants mod$M$, indexed by $\lfloor (u_0 + E_Q(i))/B \rfloor$. Thus B choices of u_0, one E word, and three residues mod$M$ specify the chunk trajectory. An empty chunk contributes zero. In particular this count does not depend on the number of digits in a gap or the absolute location of that gap.

There are $\le t+1$ singled digits and $\le t+2$ inert chunks; ordering these constituents takes $O_t(1)$ possible templates. Since B,M,K are polynomially bounded in P, this gives at most

$P^{O_{q,t,C}(1)}$ possible coordinate trajectories. Multiply trivially over $H = O(\log P)$ offsets to obtain (ii).

Consequences and scope

- The difference set $\{q^a - q^b : a > b \geq 0\}$ is not 2-large. It is Bohr recurrent: for any finite torus translation α, two distinct points from the sequence $q^n\alpha$ are as close as desired by compactness.
- The same conclusion holds for D whenever hD is contained in such a bounded signed power-sum family for a fixed positive integer h: precompose the binary obstruction with $n \to hn$.
- Consequently any q-arid language of bounded rank gives a non-2-large step set. For each language of the form $w_0 u_1^{n_1} w_1 \ldots u_j^{n_j} w_j$, the base-q value, by a geometric series for each repeated word, is a sum of boundedly many rational multiples of powers of q, with coefficients from a fixed finite list. Clear denominators; a finite union only enlarges C,t and the denominator. This does not cover arbitrary automatic sets or arbitrary Morse atoms.

D25. The classical Chacon system and intermediate critical densities

Scope and status. Result with proof.

Result. If a positive-measure event in the classical weakly mixing three-cut Chacon system, or a measure-theoretic factor, avoids every h-term D-progression for one fixed $h \geq 2$, then $b(D) = 1$. The proof forces D into finitely many dilated bounded signed tower-height hosts and applies the signed-power construction.

Scope. Specific Chacon system and its factors. Pair avoidance uses the weak closure; higher-order avoidance additionally uses the pairwise-independent-joining property. The general sign-sensitive Markov-factor obstruction and singularity-to-product conclusion are separate lemmas.

Qualifications. The arithmetic subsequence lemma retains the decomposition in the proof of the cited Chacon weak-closure theorem; the operator-valued closure classification alone is insufficient for that deduction. The final Markov-factor bound assumes the stated factorization and supplies no universal factorization theorem.

Let T denote Chacon's classical three-cut weakly mixing transformation. If A has positive measure and, for a fixed $h \geq 2$,

$$\mu(A \cap T^{-d}A \cap \ldots \cap T^{-(h-1)d}A) = 0 \quad \text{for every} \quad d \in D,$$

then $b(D) = 1$. This also applies to any measure-theoretic factor of Chacon carrying such a witness. The proof uses the weak closure of the Koopman powers and, for $h > 2$, the pairwise-independent-joining property. The same conclusion does not follow below for arbitrary continuous-spectrum systems.

A subsequence form of the Chacon weak-closure theorem Let U be the Koopman operator and Θ the projection onto constants. The tower heights are $h_n = (3^{n+1} - 1)/2$, $n \geq 0$. Write S_b for all signed sums of at most b such heights with integer coefficients of absolute value at most b. The zero sum is allowed. These hosts are nested; every bounded integer set belongs to some S_b, since $h_0 = 1$.

Lemma. If $U^{-k_j} \rightharpoonup L \neq \Theta$, then some subsequence of the integers k_j belongs to one S_b.

This strengthens the operator-valued statement only by retaining the arithmetic decomposition already used in its proof. The primary source is Janvresse–Prikhod'ko–de la Rue–Ryzhikov, *Weak limits of powers of Chacon's automorphism*, §5, Proposition 5.6 and proof of Theorem 5.1. Here are the details of the deduction. A bounded subsequence needs no argument. For an unbounded positive sequence, pass to a subsequence with stabilized greedy leading digits in the tower heights. Proposition 5.6 gives $L = \Theta$ unless these limiting digits are eventually all 0 or eventually all 2. In either remaining case the proof of that proposition, specifically the two representations following its formula (10), writes

$$k_j = mh_{n_j} + k'_j, U^{-k'_j} \rightharpoonup L', L = P_m(U)L',$$

with fixed $m \geq 1$ (absorbing the fixed offset u_r from (10) into k'_j). The final all-2 carry can make k'_j negative, which is allowed. When peeling a negative sequence, apply the same argument to its absolute values and take adjoints. By symmetry of the probability law π_m (Proposition 4.2 of the same paper), $P_m(U)^*$ is an integer translate of $P_m(U)$. Absorb the translation into the residual integer sequence; this changes it by a fixed additive constant and preserves the signed-height-sum deduction.

As in the proof of Theorem 5.1 in the same source, let r_L be the supremum of the numbers of factors in decompositions $L = P_{m_1}(U) \dots P_{m_r}(U)L''$, L'' in the weak closure of all signed powers. Their Lemma 5.4 and the correspondence of weak limits with measures on Z show $r_L < \infty$ whenever $L \neq \Theta$. Finite integer translates of the residual weak limit remain in the weak closure. Each peeling reduces this rank by at least one, and the residual limit cannot be Θ since averaging-polynomial factors fix Θ. After finitely many subsequence extractions one reaches a bounded residual, by the same rank induction (an escaping sequence with rank zero would peel at least once). Consequently the original sequence on that subsequence is a finite sum of fixed integer multiples of heights with varying indices, and a bounded residual. This proves the lemma.

For an immediate consequence, any sequence with $k_j \notin S_j$ satisfies $U^{-k_j} \rightharpoonup \Theta$. Indeed on the bounded operators on separable L^2 the weak-operator topology on the unit ball is metrizable and compact. Any hypothetical other subsequential limit would contradict the lemma and nesting of the hosts.

Multiple correlations For pair avoidance, if D were not contained in any S_b, choose $d_j \in D \setminus S_j$; the last consequence would give $\mu(A \cap T^{-d_j}A) \to \mu(A)^2 > 0$.

For every fixed h, there is a b with

$$D \subset \bigcup_{i=1}^{h-1} d \in N : id \in S_b. \quad (1)$$

Otherwise choose $d_j \in D$ with $id_j \notin S_j$ for all $1 \leq i < h$. In the laws of $(x, T^{d_j}x, \dots, T^{(h-1)d_j}x)$, every two-coordinate marginal converges to product measure by the lemma applied to $(v-u)d_j$. Taking a subsequential limit as self-joinings gives an invariant pairwise-independent h-fold self-joining. The standard weak topology of joinings here is convergence on measurable rectangles, obtainable by a diagonal argument on a generating algebra and L^1 approximation (equivalently use a compact symbolic model with zero-boundary generating partitions).

Chacon has purely singular spectrum. For example the same source's weak limit $U^{-h_n} \rightharpoonup P_1(U) = (I+U)/2$ gives a direct argument: on an absolutely continuous spectral summand the Riemann–Lebesgue lemma would instead give zero weak limit, whereas $(I+U)/2$ cannot vanish on a nonzero absolutely continuous summand. Host's pairwise-independent-joinings theorem for weakly mixing singular-spectrum systems then ensures the limit is product measure Host, *Mixing of all orders and pairwise independent joinings of systems with singular spectrum*, Israel J. Math. 76(1991), 289–298. Its mass on A^h is $\mu(A)^h > 0$, contradicting our avoiding condition. This establishes (1).

From the arithmetic hosts to density one The double of S_b lies in a bounded signed sum of powers of 3 with bounded integer coefficients, since $2h_n = 3^{n+1} - 1$. The signed-power construction in bounded-power-steps.md [D24] gives the following density form as well. For any fixed number r of colors, the proof constructs a finite vector encoding of n for which each long eligible progression visits at least $P/2$ distinct vectors, and at most $\exp(O((\log P)^2))$ ordered encoded trajectories are possible. Independently color the vectors using r equiprobable colors. The probability of missing a specified color on a given trajectory is at most $(1 - 1/r)^{P/2}$. The union bound over trajectories and r colors tends to zero as P increases with r fixed. Therefore there is a coloring in which every eligible progression of the fixed constructed length meets every one of the r colors. Some color class has lower density at most $1/r$; its complement is an avoiding set of upper density at least $1 - 1/r$. This proves $b((2S_b) \cap N) = 1$, and scaling preserves b. More directly, pulling back a dense stationary avoiding event for $2iD_i$ to the power action gives a high-density event for $D_i = \{d \in D : id \in S_b\}$. The finite union of these $b = 1$ cells has $b = 1$, by intersections of independent stationary witnesses and the union bound for their density losses. This gives the claim.

Boundary of this mechanism The argument applies whenever escaping a nested sequence of $b = 1$ arithmetic hosts forces weak convergence of all relevant powers to Θ, with the PID conclusion or an explicit higher-order mixing input for $h > 2$. Weak mixing alone only gives escape from some zero-upper-Banach-density set. It does not by itself produce such arithmetic hosts. In fact arbitrarily strong *positive* Fourier coefficients of a continuous spectral measure can persist on an IP set: take a rigid weakly mixing unitary, choose n_j with $\Sigma||U^{n_j} f - f||$ small, and telescope on every finite sum. IP sets are fully multiply recurrent. Thus a proof treating the entire absolute nonmixing spectrum as a density-one-avoidance host cannot work in general; the sign/disjointness structure of the particular avoiding event matters.

A sign-sensitive bound valid for general weak Markov limits For any ergodic transformation T and any A of measure $a > 0$ choose R so that $\mu(\bigcup_{i=0}^{R} T^i A) > 1 - a$; ergodicity gives this for some finite R. Suppose a weak limit L of powers has $\langle L1_A, 1_A\rangle = 0$. It cannot factor as $L = P_1 \dots P_R L'$, with L' probability-preserving Markov and each P_j a nonnegative convex combination of integer powers of U whose support includes two consecutive integers. Indeed the product of these R polynomials has strictly positive weights on a consecutive block $[v, v + R]$. Positivity forces every $\langle U^{v+i} L' 1_A, 1_A\rangle = 0$ on that block. But $0 \le L'1_A \le 1$ and $\int L'1_A = a$, so its positive support cannot avoid a translated copy of $\bigcup_{i=0}^{R} T^i A$, whose complement has measure $< a$. The same argument works if the two powers differ by a fixed $q > 0$ and T^q is ergodic, and for finite palettes of gap sizes in a totally ergodic system by pigeonholing and commuting the factors. The bound is independent of the coefficients, provided the indicated weights are positive. It does not presume a factorization of an arbitrary weak operator limit; that existence is precisely the extra information supplied by rank-one peeling in the Chacon example.

For a weakly mixing T the product self-joining is ergodic. Hence any weak limit of the graph joinings along a pair-avoiding D is singular to product measure: a nonzero absolutely continuous component of an invariant probability relative to the invariant ergodic product is a positive multiple of that product, which would charge $A \times A$. Singular joinings need not be graphs or admit the peeling above.

D26. A conditional finite-color criterion for an intermediate critical density

Scope and status. Conditional result.

Result. A fixed finite palette and fixed forbidden length, together with finite difference menus forcing arbitrarily long progressions above one fixed density $\delta < 1$, imply the existence of one integer difference set D with $0 < b(D) \le \delta$. Mixed-radix color addition and stationary compactness glue the finite menus.

Scope. The implication is proved; existence of the finite menus satisfying both hypotheses is not proved. The number of colors, forbidden length, and density bound must remain uniform in k. Merely uniform positive independent-set densities do not give the required coloring gluing.

Qualifications. The parameters $c, h \ge 2$ and $0 < \delta < 1$ are fixed while $k \ge 2$ varies. The finite-menu hypothesis remains unproved.

Fix integers $c, h \ge 2$ and a real $0 < \delta < 1$. Suppose that for every integer $k \ge 2$ there is a nonempty finite $F_k \subset N$ such that every integer set of upper density $> \delta$ contains a k-term F_k-progression, and F_k admits a c-coloring of Z with no monochromatic h-term F_k-progression. Then Question 6.1 has a negative answer: one can build a single D retaining all the high-density conclusions and a positive-density h-term avoiding set.

Each finite F_k admits a periodic avoiding c-coloring χ_k. Indeed the colorings form a nonempty one-dimensional shift of finite type with forbidden patterns supported on intervals of fixed length $\le 1 + (h-1)\max F_k$. A bi-infinite path in the resulting finite block graph yields a directed cycle and hence a periodic coloring.

Let $\eta \in (0,1)$, and choose integer bases b_i, each a multiple of the period of χ_i, with $\sum_i h \max F_i / b_i < \eta$. Set $\lambda_1 = 1$ and $\lambda_{i+1} = b_i \lambda_i$, and define $D = \bigcup_i \lambda_i F_i$. Given a set B of upper density $> \delta$ and an index i, there is some residue s mod λ_i in which the sliced set $\{n : s + \lambda_i n \in B\}$ has upper density $> \delta$. Apply the forcing property for F_i in that slice and then undo the dilation.

For the lower-order obstruction consider a finite stage m and all nonnegative n in the full mixed-radix period λ_{m+1}. Color n by $\sum_{i \le m} \chi_i(x_i)$ modulo c, where x_i is its digit in place λ_i. Delete n if some digit x_i is within $h \max F_i$ of the top of its digit range. This removes measure $< \eta$ on the period, and some retained color cell has density at least $(1-\eta)/c$. On an h-term progression with step $\lambda_i f$, $f \in F_i$, all lower digits are fixed and the upper digit vector stays fixed whenever the starting digit survives the deletion: $x_i + (h-1)f < b_i$. The ith digits form an h-term f-progression, so their χ_i-colors cannot be equal. This works through the top of a period when $m = i$ as well, since the upper boundary excludes that wrap. The periodic cell excludes simultaneously all h-term progressions of steps in the first m stages.

Take the translation-invariant measures of these periodic indicators on $\{0,1\}^Z$. Any weak-star subsequential limit has cylinder density at least $(1-\eta)/c$ and gives measure zero to every fixed forbidden h-cylinder for the union D. Some ergodic component has coordinate-one density at least $(1-\eta)/c$ and is concentrated on configurations avoiding all countably many forbidden cylinders. A generic configuration gives the asserted integer set. Thus $0 < b(D) \le \delta$, with $b(D)$ the supremum over fixed-length avoiding densities.

This sufficient condition does not claim such finite F_k exist. A finite analog with uniformly positive independent-set densities alone lacks the common color addition used in the mixed-

radix argument. The general phenomenon is related to independence ratios of Cartesian graph products, where a direct intersection of independent cells can have vanishing density.

D27. An exact critical-density formula for the local hitting operator

Scope and status. Result with proof.

Result. For nonempty D, the optimal universal local-hitting ratio $\sup \mu(U_L(H))/\mu(H)$, over invertible stationary systems, positive-measure H and finite L, equals $1/(1 - b(D))$, with infinity allowed. The associated nonnegative-function operator has the same optimal strong L1 bound.

Scope. The infimum over d precedes the maximum along its progression. Both the length and system may vary in the supremum. The equality is a characterization of the unknown capacity, not a proof that its intermediate values exist or fail to exist.

Qualifications. The ratio uses $D \neq \varnothing$ and $\mu(H) > 0$. Since the time window includes zero, $H \subseteq U_L(H)$; this gives the measure identity used in the proof. The L^1 extension uses threshold margins for the countable infimum.

Assume D nonempty. In any invertible stationary probability system define

$$U_L(H) = \bigcap_{d \in D} \bigcup_{0 \leq t < L} T^{-td} H$$

for a measurable event H and a positive integer L. Uniformly over all systems, events of positive measure and L, the optimal bound is

$$\sup \mu(U_L(H))/\mu(H) = 1/(1 - b(D)),$$

with infinity allowed. Equivalently the same bound is the universal strong L^1 norm on nonnegative functions of $\sup_L \inf_{d \in D} \max_{0 \leq t < L} f \circ T^{td}$. The order of infimum and maximum matters.

For one direction, a k-avoider of density a has a complement H of mass $1 - a$ with $U_k(H)$ conull. For the reverse direction, fix H, put $\eta = \mu(H)$, $U = U_L(H)$ and $w = \mu(U \setminus H)$. Start with any stationary k-avoider B of mass b, independent of this system. In their product accept the points in $(B \times H^c) \cup (X_B \times (U \setminus H))$. The new mass is $b(1-\eta)+(1-b)w$. Within any $(k + L - 1)$-term all-accepted progression at step d, none of the points of the first k slots can be in $U \setminus H$ because all the subsequent L slots, including itself, avoid H. All first k slots must therefore be in B, which is impossible. Repeating with independent copies, even starting from the empty event, makes the mass converge to $w/(w+\eta)$. Since $w+\eta = \mu(U)$, this gives $b(D) \geq 1 - \eta/\mu(U)$. The L^1 version follows by integrating the level sets; for the countable infimum use threshold margins or the fact that exceptional threshold equalities are invisible under layer cake.

The operator is monotone, and $U_K(U_L(H)) \subset U_{K+L-1}(H)$. A finite universal norm strictly greater than one is equivalent to $0 < b(D) < 1$. To establish a positive answer by this route it would suffice to amplify any nontrivial local hitting ratio into unbounded ones. Taking products of the two events does not establish such amplification, since different coordinates can reach H at different times in the same allowed progression.

Chapter 14

Coverage and source concordance

Every mathematical entry in the dated investigation is represented below. Computational scripts and numerical output are identified separately; numerical searches are not proofs. Source digests and the complete editorial coverage data accompany the editable report source. This table supplies the source-to-report correspondence without requiring access to the original working directories.

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Computational supplements

The finite-prime searches (orders 17, 19, and 23) and the two-stream finite binary searches are retained in the source package. The statements in the technical entries distinguish explicit finite counterexamples from extrapolations. The experiments do not establish bounds uniform in the prime, rank, or progression length.

References appearing in the technical collection

The following index lists external references cited in the technical entries. Their original descriptions are retained at the point of use. Primary literature supplies the stated background inputs; the claims specific to this investigation are proved in the report or marked conditional.

1. `https://aif.centre-mersenne.org/item/10.5802/aif.1707.pdf`

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2. `https://annals.math.princeton.edu/2005/161-1/p08`

   Cited in R145, R237.

3. `https://annals.math.princeton.edu/2012/175-2/p02`

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